The Foreman and the serif family

Handbook of the general definitions, constructions and theorems

20 October 2026

How this handbook is written

This handbook states the general theory of the Foreman and the serif family: the members of the serif family and their relations; the tables and families they evaluate; the Foreman's classification of a family and his tree; the generalized integrals the Foreman sets; the closings, constructions and arithmetic built on them; and the family read as one transform and as a model of data. Proofs, worked applications and the catalogue of conventional names are in the reference, *The Foreman and the serif family* (16 October 2026); in the source file every statement is preceded by a comment naming its label and line there. A statement here that is more general than its counterpart in the reference, or that has no counterpart there, is proved in Part VIII, and the comment before it in the source file says so.

One governing definition per object. Each object of the theory is defined once, with all of its slots. Every later appearance of the object is that definition with particular data in its slots, written through the full signature of its governing symbol: a specialization displays the data it chooses, and it never receives a symbol of its own. Thus a Breta on one scale is a Breta with one bunch, a Trailmoment on a line is a Trailmoment whose cut region is a half-line, and a value of interest at particular exponents is the interest with those exponents in its slots.

Abbreviations are local. Inside one statement an abbreviation may be declared, once and in words, for data that the statement holds fixed; the declaration displays the full symbol it abbreviates, and it lapses at the end of the statement.

Relations are stated at their height. A relation holds on the values of the slots on which its proof establishes it (Definition 5.6). A hypothesis that restricts a slot is written into the statement that needs it, and nothing outside that statement restricts the object.

Final answers are written with exp, log, the stacked Gamma and the serif family (Definition 1.1(vi)).

Contents

How this handbook is written **1**

I The serif family **9**

1 Conventions, the stacked Gamma and the finite part **10**
1.1 The stacked Gamma . . . 11
1.2 The finite part . . . 11

2 The Gramma and the Zamma **13**
2.1 The Gramma . . . 13
2.2 The Zamma . . . 15

3 The Breta **18**
3.1 Definition and the tower . . . 18
3.2 Letters . . . 20
3.3 The shape integral . . . 21
3.4 Products of shape integrals . . . 22
3.5 The Beta port . . . 22

4 Fibre integrals and closing **25**

5 The Trailmoment **28**
5.1 Definition . . . 28
5.2 Slots, corners and heights . . . 29
5.3 Relations at free slots . . . 30
5.4 Lattices . . . 32
5.5 The Gamma read by Trailmoments . . . 32
5.6 Reflections of the index . . . 35
5.7 Joint sources . . . 36
5.8 Trails . . . 37
5.9 Products of Trailmoments . . . 38
5.10 Convolutions of inputs . . . 40

6 Rational words and live inputs **42**
6.1 Live inputs . . . 45

II Tables and families **47**

7 Tables **48**
7.1 Characters, tables, lists 48
7.2 Circuits and molecules 48
7.3 The Skeleton 49
7.4 Multivectors of any tensor rank 51

8 Families **53**
8.1 The interest and its walls 53
8.2 The native gauge 57

III Classification **60**

9 Realizations and the still **61**
9.1 Realizations 61
9.2 The potential and the still 62

10 The reading, classes, atoms **64**
10.1 Views and the reading 64
10.2 Identifying writings 65
10.3 What a class holds 66

11 Lean, gauge and label **67**
11.1 The lean 67
11.2 The gauge 67
11.3 Answers and the label 68
11.4 Maps as certificates 69

12 Restriction to a bunch **70**
12.1 Patterns and the Taylor cube 70
12.2 The Skeleton polynomials of a bunch 71
12.3 Levels 73
12.4 Shares and the native split 75

13 The order of classes and the Foreman bunch **77**
13.1 The rises of a class 77
13.2 The order 78
13.3 What the first stages read 79

14 Peel, children and the Foreman tree **81**
14.1 The peel and the tree 81
14.2 Patterns of one kind 82
14.3 What the room keeps 83

IV Evaluation **86**

15 The integral at a bunch **87**
15.1 The scale of a bunch . . . 87
15.2 Closing . . . 89
15.3 The port . . . 92
15.4 Signed letters . . . 93

16 Integrals, expectations and the Foreman procedure **94**
16.1 Forms, sections, completions . . . 94
16.2 Three integrals and their expectations . . . 95
16.3 Quasi-integrals as Laurent data . . . 100
16.4 The pseudointegral . . . 101

17 The Breta law and the port on the balance wall **107**
17.1 The laws . . . 107
17.2 The port on the balance wall . . . 110
17.3 Cycles . . . 112

18 Blocks: generation and cleaning **114**
18.1 Generated families . . . 114
18.2 Bunches of width one . . . 115
18.3 Pooling . . . 117
18.4 Walls glue . . . 118
18.5 Full leans . . . 118
18.6 Partial leans . . . 119
18.7 Closing at a wall and closing at the gauge . . . 122
18.8 The scale on the totals . . . 128
18.9 Width zero . . . 133
18.10 Pooling is a move . . . 133
18.11 Cleaning a molecule . . . 134

19 Words of a tree **135**
19.1 The two letters of a table at once . . . 135
19.2 The junction in pattern form . . . 136
19.3 The word of a tree . . . 137
19.4 Closing the tree by draws . . . 139
19.5 Devices at the centre . . . 142
19.6 The junction under level form . . . 144
19.7 The node in one scale . . . 146
19.8 The reach read on the law . . . 153
19.9 One tree . . . 158

V Construction and composition **162**

20 Gluing and what crosses it **163**
20.1 Gluing along a mark . . . 163
20.2 Splicing hinges in pairs . . . 166

20.3 What crosses a gluing 168

21 Periods of constructions **171**
21.1 The core reads its cells 171
21.2 Pinned cells 173
21.3 Blocks aggregate 174
21.4 Closures 175
21.5 Chains and transfer 176
21.6 A bunch of any width 178

22 The law of a cell **180**
22.1 Cells and their laws 180
22.2 Absorbing a cell 181
22.3 Two cells glued 182
22.4 What a cell retains 185
22.5 Cores and their markers 187
22.6 Series and parallel nodes 188
22.7 Hinge trees 189

23 Repetition and the change of query **193**
23.1 Repetition at a node 193
23.2 Chains 195
23.3 The weight of a hinge 196
23.4 Changing masses and weights 197

24 Holds and the lean convolution **199**
24.1 Holds and the periods at the full leans 199
24.2 The lean convolution 201
24.3 Holding some marks 202
24.4 Newton words 204

25 Periods at the full leans and networks along a hinge tree **207**
25.1 Molecules and marker loads 207
25.2 The periods of a substitution at its full leans 208
25.3 What a node costs 209
25.4 The network theorem 210
25.5 Nodes in hold form 211
25.6 The procedure and its cost 213
25.7 Changing the query in hold form 214

26 Finite interfaces and certified enclosures **216**
26.1 The joint law of the channels and its moments 216
26.2 Repetition is a finite matrix 218
26.3 A certified enclosure of the period 219
26.4 Changing the query with finite data 220
26.5 Certified parameter banks 220
26.6 Procedures 222

27 Bunches of any width **224**
27.1 The seam of a bunch . . . 224
27.2 Pieces, their laws, and the glue . . . 227
27.3 The Gramma of a set of pairs . . . 233
27.4 What a piece retains . . . 235
27.5 Storage, and what replaces it . . . 237

28 Composing bunches **239**
28.1 Cores, nested pieces and chains of bunches . . . 239
28.2 The procedure . . . 244
28.3 The correspondence . . . 245

29 Closing a glue along a direction **247**
29.1 Closing along a direction . . . 247
29.2 Closing along a path . . . 251
29.3 Retained crossings as joint words . . . 259
29.3.1 Completing the retained fibre before taking its size . . . 259
29.3.2 Coincident crossings and the boundary at root rate one . . . 265

30 Laws at their ends: regions of equality and reach **268**
30.1 Regions on which a law is represented . . . 268
30.2 The reach along a direction . . . 273
30.3 What the calculus of bunches settles . . . 282

31 Price and complexity **285**
31.1 Constructions and their price . . . 285

VI The arithmetic of the output **288**

32 Positive words **289**
32.1 Coefficient jets and basis contrasts . . . 289
32.2 Manufactured tables and their port words . . . 291
32.3 Mark relations with the room integrated . . . 293

33 The critical-line law **297**
33.1 Critical-line words . . . 298

34 The gap and the shape words **300**
34.1 Clearing and the gap . . . 300
34.2 Shape words with polynomial insertions . . . 301
34.3 Rows . . . 302
34.4 The Beta port ladder and the filters . . . 303
34.5 Growing Beta ports . . . 305

35 Rays, native gauges and tiers **307**
35.1 The relations along a ray . . . 307
35.2 Native gauges carry solutions . . . 308
35.3 The centre ray of a family . . . 309

35.4 What a wall carries . . . 311
35.5 What a value can contain . . . 313

36 Relations and separation **315**
36.1 Words and relations . . . 315
36.2 Relation blocks . . . 316
36.3 Separators . . . 317

37 Finite algebra for scale reads **319**
37.1 Complete relations for linear letters and unit columns . . . 319
37.2 Polynomial letters on one scale . . . 323

VII The family as one Laplace transform and as inference **327**

38 The family as one Laplace transform **328**
38.1 Questions and the law of the tilt . . . 328
38.2 The interaction of a bunch . . . 332

39 The family as inference **335**
39.1 Samples of bases . . . 335
39.2 Samples of tilts . . . 338
39.3 Draws, orders and races . . . 343

VIII Proofs of the general forms **347**

40 Proofs of the general forms **348**
40.1 Chapter 1: conventions, the stacked Gamma and the finite part . . . 348
40.2 Chapter 2: the Gramma and the Zamma . . . 351
40.3 Chapter 3: the Breta . . . 354
40.4 Chapter 4: fibre integrals and closing . . . 358
40.5 Chapter 5: the Trailmoment . . . 359
40.6 Chapter 6: rational words and live inputs . . . 362
40.7 Chapter 7: tables . . . 364
40.8 Chapter 8: families . . . 366
40.9 Chapter 9: realizations and the still . . . 371
40.10 Chapter 10: the reading, classes, atoms . . . 372
40.11 Chapter 11: lean, gauge and label . . . 373
40.12 Chapter 12: restriction to a bunch . . . 375
40.13 Chapter 13: the order of classes and the Foreman bunch . . . 377
40.14 Chapter 14: peel, children and the Foreman tree . . . 377
40.15 Chapter 15: the integral at a bunch . . . 380
40.16 Chapter 16: integrals, expectations and the Foreman procedure . . . 385
40.17 Chapter 17: the Breta law and the port on the balance wall . . . 389
40.18 Chapter 18: blocks: generation and cleaning . . . 396
40.19 Chapter 19: words of a tree . . . 410
40.20 Chapter 20: gluing and what crosses it . . . 423
40.21 Chapter 21: periods of constructions . . . 424

40.22Chapter 22: the law of a cell 432
40.23Chapter 23: repetition and the change of query 440
40.24Chapter 24: holds and the lean convolution 443
40.25Chapter 25: periods at the full leans and networks along a hinge tree 450
40.26Chapter 26: finite interfaces and certified enclosures 457
40.27Chapter 27: bunches of any width 458
40.28Chapter 28: composing bunches 467
40.29Chapter 29: closing a glue along a direction 468
40.30Chapter 30: laws at their ends: regions of equality and reach 479
40.31Chapter 32: positive words 482
40.32Chapter 34: the gap and the shape words 486
40.33Chapter 35: rays, native gauges and tiers 487
40.34Chapter 36: relations and separation 494
40.35Chapter 37: finite algebra for scale reads 495
40.36Chapter 38: the family as one Laplace transform 498
40.36.1 What the one-table proofs read, and what a family reads 498
40.36.2 The exponent of a question 499
40.36.3 The two ends of strength 500
40.36.4 How far a question can lean, and the tails 503
40.36.5 The interaction of a bunch 504
40.37Chapter 39: the family as inference 506
40.37.1 Samples of bases 506
40.37.2 Statistics from a centre 508
40.37.3 The Grey law, the dial and the sorted draw 509
40.38Receipts 511
40.38.1 Chapters 32 to 37 511
40.38.2 Chapters 38 and 39 511

Part I

The serif family

Chapter 1

Conventions, the stacked Gamma and the finite part

Definition 1.1 (Conventions)**.** (i) $\mathcal{E}$ and B are finite nonempty sets and $n = |\mathcal{E}|$. For $x \in \mathbb{R}^{\mathcal{E}}_{>0}$ and $v \in \mathbb{C}^{\mathcal{E}}$, $x^v = \prod_e x_e^{v_e}$ with $x_e^{v_e} = \exp(v_e \log x_e)$. For a vector v, $|v| = \sum_e v_e$. $\mathbf{1}$ is the all-ones vector, $\mathbf{1}_S$ the indicator of S, $x^S = x^{\mathbf{1}_S}$, $v_S = \sum_{e\in S} v_e$, $v|_S$ the restriction of v to S, and $x^{-1} = (x_e^{-1})_e$. For a set X and elements a, b, $X - a = X \setminus \{a\}$ and $X + b = X \cup \{b\}$. $\mathbb{N} = \{0, 1, 2, \dots\}$.

(ii) For $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$ the *logarithmic coordinates* are $y = \log \alpha$ and the *signed coordinates* are $z = \frac{1}{2}y$, coordinatewise, so $\alpha = e^y = e^{2z}$.

(iii) $\Delta_B = \{u \in \mathbb{R}^B_{>0} : |u| = 1\}$ is the *shape simplex* of B, and $d\omega_B(u)$ is Lebesgue measure in the coordinates $(u_e)_{e\neq e_0}$ for a fixed $e_0 \in B$; a change of e_0 is affine of determinant ± 1. For $|B| = 1$, ω_B is the Dirac measure at the point of Δ_B.

(iv) An integral *converges* when it converges absolutely.

(v) $(a)_j = \Gamma(a + j)/\Gamma(a)$ for $j \in \mathbb{N}$; $d_n = \operatorname{lcm}(1, \dots, n)$ with $d_0 = 1$; e_k is the elementary symmetric polynomial of degree k; for a coordinate x of the unit interval, $\bar{x} = 1 - x$; $[X]\, Y$ is the coefficient of X in Y.

(vi) *Final answers* are written with exp, log, the stacked Gamma and the serif family: the Gramma G and the Zamma Z (Chapter 2), the Breta B (Chapter 3) and the Trailmoment T (Chapter 5). A value of interest is left in another form only with a proof that it has no such expression. A *word* is an expression built from rationals, the data and the variables by sums, products, powers with rational exponents or exponents among the data, these functions, integrals over simplices, half-lines and fibres, and the iterate $\mathcal{Q}^M$ of an integral operator whose kernel is a word. A finite sequence $(w_1, \dots, w_k)$ is a *string* of length k. An *insertion* is a function multiplied into an integrand.

(vii) For masses $\mathbf{m}$, $d\alpha_{\mathbf{m}} := \alpha^{\mathbf{m}-\mathbf{1}} d\alpha$ is a measure on the orthant, and a form $f\, d\alpha_{\mathbf{m}}$ unchanged by $\alpha \mapsto \lambda\alpha$, f homogeneous of degree $-m_{\mathcal{E}}$, is integrated over the rays of the orthant, that is over any section (Lemma 16.2), as the form $W\, d\alpha_{\mathbf{m}}$ of a family word W (Definition 8.1) is on its balance wall; a mark or a set of marks as a subscript, as in $d\alpha_B$, denotes coordinates. The law of a random variable, or of a measure under a map, is its image measure; an identity between members of the serif family is a relation (Definition 5.6). An expectation names its random variable and that variable's law, $\mathbb{E}_{U\sim P}[g(U)] = \int g(u)\, P(du)$; inside a statement that has named them once, the subscript may be dropped for the rest of that statement. The

Gamma law of index $\nu > 0$ is $\mathrm{Gamma}_\nu(dx) = x^{\nu-1}e^{-x}dx/\Gamma(\nu)$ on $(0,\infty)$, and for $\mathbf{m} \in \mathbb{R}^B_{>0}$, $\mathrm{Gamma}_{\mathbf{m}} = \bigotimes_{e\in B} \mathrm{Gamma}_{m_e}$ is the law on $\mathbb{R}^B_{>0}$ of independent Gamma coordinates of indices m_e. The generalized integrals of Chapter 16 are unnormalized, and their expectations divide them by the period of the centre: on a connected family every expectation of a form is taken under the law of its centre, $\mathbb{P}_{\mathrm{c}}$, of the departure that its evaluation sets (Definition 16.8).

(viii) The *root rate* of terms a_η indexed by sizes $\eta \to \infty$, as the terms of a lattice by the real parts of their exponents or the terms of a closing by its steps, is $\limsup |a_\eta|^{1/\eta}$. When the number of terms of size at most N grows at most polynomially in N, terms of root rate below one are absolutely summable, and terms of root rate above one do not tend to zero.

1.1 The stacked Gamma

Definition 1.2 (The stacked Gamma)**.** For $\alpha \in \mathbb{C}^p$ and $\beta \in \mathbb{C}^q$,

$$\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \dots,\ \alpha_p\\ \beta_1,\ \dots,\ \beta_q\end{matrix}\right] := \frac{\Gamma(\alpha_1)\cdots\Gamma(\alpha_p)}{\Gamma(\beta_1)\cdots\Gamma(\beta_q)},$$

wherever no α_i is a pole of Γ; it is a meromorphic function of its entries. The stacked Gamma is *balanced* when $\sum_i \alpha_i = \sum_j \beta_j$.

The subscripts of the bracket record the numbers of its upper and lower entries, so a quotient of Gammas is read off the symbol with no translation. The stacked Gamma $\Gamma_{\substack{|B|\\1}}\!\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right] = \prod_{e\in B}\Gamma(m_e)/\Gamma(m_B)$ of masses $\mathbf{m} \in \mathbb{C}^B$ over their total m_B is, for $\mathrm{Re}\, m_e > 0$, the integral of $u^{\mathbf{m}-\mathbf{1}}$ over the shape simplex Δ_B (Theorem 3.9).

Proposition 1.3 (Relations of the stacked Gamma)**.** *(i)* Concatenation. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]\Gamma_{\substack{p'\\q'}}\!\left[\begin{matrix}\boldsymbol{\alpha}'\\ \boldsymbol{\beta}'\end{matrix}\right] = \Gamma_{\substack{p+p'\\q+q'}}\!\left[\begin{matrix}\boldsymbol{\alpha},\ \boldsymbol{\alpha}'\\ \boldsymbol{\beta},\ \boldsymbol{\beta}'\end{matrix}\right]$.

(ii) Cancellation. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha},\ c\\ \boldsymbol{\beta},\ c\end{matrix}\right] = \Gamma_{\substack{p-1\\q-1}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]$.

(iii) Shift. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1+1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right] = \alpha_1\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]$ *and* $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \beta_1+1,\ \boldsymbol{\beta}\end{matrix}\right] = \beta_1^{-1}\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \beta_1,\ \boldsymbol{\beta}\end{matrix}\right]$.

(iv) Residue. *For* $j \in \mathbb{N}$, $\operatorname*{Res}_{\alpha_1=-j}\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right] = \dfrac{(-1)^j}{j!}\Gamma_{\substack{p-1\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]\Big|_{\alpha_1=-j}$.

(v) Reflection. $\Gamma_{\substack{2\\0}}\big[a,\ 1-a\big] = \pi/\sin\pi a$.

(vi) Laurent data at a pole. *For* $j \in \mathbb{N}$,

$$\Gamma(-j+\varepsilon) = \frac{(-1)^j}{j!}\Big(\frac{1}{\varepsilon} + (\log\Gamma)'(j+1) + O(\varepsilon)\Big), \qquad (\log\Gamma)'(\nu+1) = (\log\Gamma)'(\nu) + \frac{1}{\nu}.$$

1.2 The finite part

Definition 1.4 (The finite part)**.** A function h on $(0,\infty)$ has a *log-power expansion* at 0 if for every N there are finitely many coefficients $c_{k,j}$, with exponents $k \in \mathbb{C}$ of real part below N and

$j \in \mathbb{N}$, such that $h(x) - \sum c_{k,j} x^k \log^j x = O(x^N)$ as $x \to 0$; at ∞ the same holds with x^{-k} and $O(x^{-N})$. The coefficients are determined by h. Then $\operatorname{Reg}_{x\to 0} h$ and $\operatorname{Reg}_{x\to\infty} h$ are the coefficients of $x^0 \log^0 x$ in these expansions, zero if absent. If f has a primitive h with log-power expansions at both ends,

$$\operatorname{Reg} \int_0^\infty f\,dx := \operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h,$$

which does not depend on the primitive; on $(0, b]$, $\operatorname{Reg} \int_0^b f := h(b) - \operatorname{Reg}_{x\to 0} h$. A sum is regularized the same way, $\operatorname{Reg}_{N\to\infty}$ of its partial sums. The coordinate is part of the data of a finite part, and for iterated finite parts so is the order.

Proposition 1.5 (A convergent integral is its own finite part)**.** *If $\int_0^\infty f\,dx$ converges and f has a primitive with log-power expansions at both ends, then $\operatorname{Reg} \int_0^\infty f\,dx = \int_0^\infty f\,dx$.*

Theorem 1.6 (A finite part is a constant term)**.** *If f has log-power expansions at 0 and at ∞, then*

$$\operatorname{Reg} \int_0^\infty f\,dx = \operatorname*{CT}_{\varepsilon=0} \int_0^\infty x^\varepsilon f(x)\,dx,$$

the right side continued meromorphically in ε from $\int_0^1$ and from $\int_1^\infty$ separately. Termwise, $x^b \log^j x$ in the expansion at 0 contributes $\int_0^1 x^{b+\varepsilon} \log^j x\,dx = (-1)^j j!/(b+1+\varepsilon)^{j+1}$, whose constant term is $(-1)^j j!/(b+1)^{j+1}$ for $b \neq -1$ and 0 for $b = -1$; the end at ∞ is the same after $x \mapsto 1/x$.

Corollary 1.7 (Finite parts at a pole of a Mellin integral)**.** *Let f have a log-power expansion at 0 and be integrable on $[\delta, 1]$ for every $\delta > 0$, and let $I(a) = \int_0^1 x^{a-1} f(x)\,dx$, continued meromorphically. Then $\operatorname{Reg} \int_0^1 x^{a-1} f(x)\,dx = \operatorname{CT}_{a'=a} I(a')$ for every $a \in \mathbb{C}$: it is $I(a)$ at every a that is not a pole, and the poles lie at the points $a = -k$ for the exponents k of the expansion, a term $x^k \log^j x$ giving a pole of order at most $j+1$.*

Lemma 1.8 (A finite part depends on the coordinate)**.** *Let h have a log-power expansion at ∞, and let a_j be its coefficient of $x^0 \log^j x$, $j \in \mathbb{N}$. For $c, p > 0$, $h(cx^p)$ has a log-power expansion at ∞, and*

$$\operatorname{Reg}_{x\to\infty} h(cx^p) = \sum_{j\geq 0} a_j\,(\log c)^j;$$

the same holds at 0. Consequently, if f has log-power expansions at 0 and at ∞, with coefficients f_j^0 and f_j^∞ of $x^{-1} \log^j x$, then for $c, p > 0$

$$\operatorname{Reg} \int_0^\infty f(cx^p)\,c\,p\,x^{p-1}\,dx = \operatorname{Reg} \int_0^\infty f(x)\,dx + \sum_{j\geq 0} \frac{(\log c)^{j+1}}{j+1} \big(f_j^\infty - f_j^0\big) :$$

a power of the coordinate leaves a finite part unchanged, and a dilation moves it by a polynomial in $\log c$ read from the terms $x^{-1} \log^j x$ of the integrand at its two ends.

A finite part is the difference of the regularized limits of one primitive at the two ends, not the integral of a measure. It is linear, it moves with the coordinate (Lemma 1.8), and iterated in several variables it moves with the order. The generalized integrals of Chapter 16 are built from it: a quasi-integral is an iterated finite part along a chart and an order, and the pseudointegral is the value of the native completion (Definition 16.26).

Chapter 2

The Gramma and the Zamma

2.1 The Gramma

Definition 2.1 (Gramma)**.** Let $\mathcal{E}$ be a finite set of *scales*. A *configuration* on $\mathcal{E}$ is a finite family $t = (t_1, \dots, t_N)$ of *exponents* $t_k \in \mathbb{R}^{\mathcal{E}}$; t also denotes the $\mathcal{E} \times N$ matrix with these columns. For *coefficients* $\boldsymbol{\xi} \in (0,\infty)^N$ and an *index* $\nu \in \mathbb{C}^{\mathcal{E}}$, the *Gramma* is

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) := \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}} \exp\Big(-\sum_{k=1}^{N} \xi_k\, x^{t_k} \Big)\, dx$$

wherever the integral converges. Put $\operatorname{cone}(t) = \{\sum_k c_k t_k :\ c \in [0,\infty)^N\}$ and $\mathcal{T}_t = \{\nu : \operatorname{Re}\nu \in \operatorname{int}\operatorname{cone}(t)\}$. In $y = \log x$,

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{\mathbb{R}^{\mathcal{E}}} e^{\langle \nu, y\rangle} \exp\Big(-\sum_{k} \xi_k e^{\langle t_k, y\rangle} \Big)\, dy,$$

the moment generating function, in the index, of the density $\exp\big(-\sum_k \xi_k e^{\langle t_k,y\rangle}\big)$. Exponents and coefficients combine by concatenation, $t \oplus t'$ and $\boldsymbol{\xi} \oplus \boldsymbol{\xi}'$ (configurations on two sets of scales extended by zero to their union), and a shape u dresses coefficients by $u^t\boldsymbol{\xi} := (u^{t_k}\xi_k)_k$. The *unit configuration* on $\mathcal{E}$ is $I_{\mathcal{E}} = (\mathbf{e}_e)_{e\in\mathcal{E}}$, one unit column per scale, and on one scale the configuration of the single exponent 1 is written (1).

The three slots of the Gramma (Definition 5.6) are its index, its configuration and its coefficients. Every structural relation of the Gramma is a statement about how these slots move: the index moves along the exponents, the coefficients are rescaled by the torus of the scales, and the configuration passes to its facets at its walls.

Theorem 2.2 (Domain)**.** $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ *converges if and only if* $\nu \in \mathcal{T}_t$*, and it is holomorphic on* $\mathcal{T}_t$*. If* $\operatorname{cone}(t) = \mathbb{R}^{\mathcal{E}}$ *it is entire; if* $\operatorname{cone}(t)$ *lies in a hyperplane it converges nowhere.*

Theorem 2.3 (Relations of the Gramma)**.** *Let* $\nu \in \mathcal{T}_t$*; then* $\nu + t_k \in \mathcal{T}_t$ *for every* k*.*

(i) Ladder. $\partial_{\xi_k}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = -\mathsf{G}_{\nu+t_k;t}(\boldsymbol{\xi})$*; for* $\mathbf{w} \in \mathbb{N}^N$*,* $\partial^{\mathbf{w}}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = (-1)^{|\mathbf{w}|}\mathsf{G}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi})$*.*

(ii) Exchange, one relation per scale. *For each* $e \in \mathcal{E}$*:* $\nu_e\, \mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_k t_{k,e}\, \xi_k\, \mathsf{G}_{\nu+t_k;t}(\boldsymbol{\xi})$*.*

(iii) Torus homogeneity. *For* $\lambda \in \mathbb{R}^{\mathcal{E}}_{>0}$*:* $\mathsf{G}_{\nu;t}\big((\lambda^{t_k}\xi_k)_k\big) = \lambda^{-\nu}\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ *(Proposition 2.4 at* $A = I_{\mathcal{E}}$*,* $\mathbf{h} = \log\lambda$*).*

(iv) Toric relation. *For every* $\mathbf{v} \in \mathbb{Z}^N$ *with* $t\mathbf{v} = 0$*, written* $\mathbf{v} = \mathbf{v}^+ - \mathbf{v}^-$ *with* $\mathbf{v}^\pm \in \mathbb{N}^N$ *of disjoint supports,*

$$\partial^{\mathbf{v}^+} \mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = (-1)^{|\mathbf{v}^+|-|\mathbf{v}^-|}\, \partial^{\mathbf{v}^-} \mathsf{G}_{\nu;t}(\boldsymbol{\xi}).$$

The ladder moves the index along one exponent; the exchange relation, one for each scale, is linear in the Grammas at the index and at its N neighbours $\nu + t_k$; the toric relation records the linear dependencies among the exponents themselves. On one scale with integer exponents the exchange relation is a recurrence on the indices $\nu + \mathbb{Z}$ linking the indices from $\nu + \min(0, \min_k t_k)$ to $\nu + \max(0, \max_k t_k)$.

Proposition 2.4 (Affine changes of the logarithmic coordinates)**.** *For* $A \in GL_{\mathcal{E}}(\mathbb{R})$ *and* $\mathbf{h} \in \mathbb{R}^{\mathcal{E}}$*:* $\mathcal{T}_{A^\mathsf{T} t} = A^\mathsf{T} \mathcal{T}_t$*, and for* $\nu \in \mathcal{T}_t$

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det A|\, e^{\langle \nu, \mathbf{h}\rangle}\, \mathsf{G}_{A^\mathsf{T}\nu;A^\mathsf{T}t}\big((e^{\langle t_k, \mathbf{h}\rangle}\xi_k)_k\big).$$

At $A = -I_{\mathcal{E}}$ *and* $\mathbf{h} = 0$ *this is the reflection,* $\mathcal{T}_{-t} = -\mathcal{T}_t$ *and* $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \mathsf{G}_{-\nu;-t}(\boldsymbol{\xi})$.

Theorem 2.5 (Product)**.** *If* $\mathcal{E} = \mathcal{E}_1 \sqcup \cdots \sqcup \mathcal{E}_q$ *and each* t_k *is supported in one* $\mathcal{E}_i$ *(zero vectors assigned to* $\mathcal{E}_1$*), then* $\mathcal{T}_t = \mathcal{T}_{t^{(1)}} \times \cdots \times \mathcal{T}_{t^{(q)}}$ *and*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \prod_{i=1}^{q} \mathsf{G}_{\nu|_{\mathcal{E}_i};t^{(i)}}(\boldsymbol{\xi}^{(i)}),$$

with $t^{(i)}$ *the exponents supported in* $\mathcal{E}_i$ *and* $\boldsymbol{\xi}^{(i)}$ *their coefficients.*

Corollary 2.6 (Square configurations)**.** *Let* $N = |\mathcal{E}|$ *and let the matrix* t *be invertible, and put* $\mathbf{b} = t^{-1}\nu$*. Then* $\mathcal{T}_t = \{\nu : \operatorname{Re}\mathbf{b} \in (0,\infty)^N\}$ *and*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1} \prod_{k=1}^{N} \Gamma(b_k)\, \xi_k^{-b_k}.$$

In particular $\mathsf{G}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \prod_e \Gamma(\nu_e)\, \xi_e^{-\nu_e}$*; on one scale* $\mathsf{G}_{\nu;(t_1)}(\xi) = t_1^{-1}\Gamma(\nu/t_1)\, \xi^{-\nu/t_1}$ *for* $t_1 > 0$*; and* $\Gamma(\nu) = \mathsf{G}_{\nu;(1)}(1)$*, whose exchange relation is* $\nu\Gamma(\nu) = \Gamma(\nu+1)$.

The Gamma function is thus the Gramma of one scale and one unit exponent, and at every invertible square configuration the Gramma is a stacked Gamma times powers, divided by $|\det t|$. A configuration with more exponents than scales is where the Gramma becomes more than a stacked Gamma times powers: the difference $N - |\mathcal{E}|$ is the dimension of the share side of Proposition 4.2.

Lemma 2.7 (Scale and shape)**.** *The map* $(0,\infty) \times \Delta_B \to \mathbb{R}^B_{>0}$*,* $(\rho, u) \mapsto \rho u$*, is a bijection with inverse* $x \mapsto (|x|, x/|x|)$*, and* $dx = \rho^{|B|-1}\, d\rho\, d\omega_B(u)$*. Consequently* $(\rho u)^\pi = \rho^{\langle \pi, \mathbf{1}_B\rangle} u^\pi$ *for* $\pi \in \mathbb{R}^B$*, and* $x^{\mathbf{m}-\mathbf{1}} dx = \rho^{m_B - 1} u^{\mathbf{m}-\mathbf{1}}\, d\rho\, d\omega_B(u)$ *for* $\mathbf{m} \in \mathbb{C}^B$.

Theorem 2.8 (Scale and shape along sets)**.** *Let* $B_1, \ldots, B_q$ *be pairwise disjoint nonempty subsets of* $\mathcal{E}$*,* $R = \mathcal{E} \setminus (B_1 \cup \cdots \cup B_q)$ *and* $\nu \in \mathcal{T}_t$*. With the* levels $\ell_{k,i} = \langle t_k, \mathbf{1}_{B_i}\rangle$*, the configuration* $\ell = (\ell_1, \ldots, \ell_N)$ *on* q *scales and the coefficients* $\mathbf{b}(u, x_R) = \big(\xi_k \prod_i u_i^{t_k|_{B_i}} x_R^{t_k|_R}\big)_k$ *for* $u = (u_i)_i \in \prod_i \Delta_{B_i}$*,*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{\mathbb{R}^R_{>0}} x_R^{\nu|_R - \mathbf{1}} \int_{\prod_i \Delta_{B_i}} \prod_i u_i^{\nu|_{B_i} - \mathbf{1}}\; \mathsf{G}_{(\nu_{B_i})_i;\ell}\big(\mathbf{b}(u, x_R)\big)\, d\omega(u)\, dx_R,$$

with $d\omega = \prod_i d\omega_{B_i}$*, the inner Gramma, on* q *scales at the index* $(\nu_{B_i})_i$*,* $\nu_{B_i} = \sum_{e \in B_i} \nu_e$*, converging for almost every* (u, x_R)*; when* R *is empty there is no outer integral.*

Passing to scale and shape on disjoint sets of scales replaces each set by one scale, its total, carrying the levels of the exponents on the set; what is left of the exponents dresses the coefficients by the shapes. The Breta of Chapter 3 is this passage made into a symbol.

Theorem 2.9 (The walls of a Gramma)**.** *Let $\mathcal{E} = \mathcal{E}' \sqcup \{e\}$, write $t_k = (t'_k, t_{k,e})$ and $\nu = (\nu', \nu_e)$, and suppose $t_{k,e} \geq 0$ for every k, with some $t_{k,e} > 0$. Let $H = \{k : t_{k,e} = 0\}$ be the* facet terms*, suppose* $\operatorname{cone}(t'_H)$ *has nonempty interior in $\mathbb{R}^{\mathcal{E}'}$ and contains t'_k for every $k \notin H$, and let $\operatorname{Re}\nu' \in \operatorname{int}\operatorname{cone}(t'_H)$. Then $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ converges for $0 < \operatorname{Re}\nu_e < c$, for some $c \in (0, \infty]$, and extends to a meromorphic function of $\nu_e \in \mathbb{C}$ with at most simple poles at $\nu_e = -p$, $p \in P = \{\sum_{k\notin H} n_k t_{k,e} : \mathbf{n} \in \mathbb{N}^{H^c}\}$, where*

$$\operatorname*{Res}_{\nu_e=-p} \mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_{\mathbf{n}\in\mathbb{N}^{H^c}:\ \sum_k n_k t_{k,e} = p} \prod_{k\notin H} \frac{(-\xi_k)^{n_k}}{n_k!}\ \mathsf{G}_{\nu'+\sum_{k\notin H} n_k t'_k;\ t'_H}(\boldsymbol{\xi}_H).$$

At $p = 0$ the residue is the Gramma of the facet terms: at its first wall a Gramma factors onto its facet. A linear change of the logarithmic coordinates, $y = Ay'$ with $A \in GL_{\mathcal{E}}(\mathbb{R})$, gives $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det A|\, \mathsf{G}_{A^\intercal\nu;A^\intercal t}(\boldsymbol{\xi})$ (Proposition 2.4 at $\mathbf{h} = 0$), and with the last column of A the inward normal of a facet of $\operatorname{cone}(t)$ *it puts that facet in the position of H.*

The poles of a Gramma sit on the walls of its cone, and each residue is a Gramma of the facet, of one scale fewer, at an index moved by the exponents that leave the facet. On one scale with positive exponents this is the continuation of the Gramma to the whole index line, with at most simple poles at the points $-p$, $p = \langle \mathbf{n}, t\rangle$ for $\mathbf{n} \in \mathbb{N}^N$, the residue at $-p$ being $\sum_{\langle \mathbf{n},t\rangle = p} \prod_k (-\xi_k)^{n_k}/n_k!$.

Proposition 2.10 (The expansion lattice)**.** *Let S be a set of terms such that the integral with the signs of those terms flipped converges,*

$$\int_{\mathbb{R}^{\mathcal{E}}} \exp\Big(\langle \operatorname{Re}\nu, y\rangle + \sum_{k\in S} \xi_k e^{\langle t_k, y\rangle} - \sum_{k\notin S} \xi_k e^{\langle t_k, y\rangle}\Big) dy < \infty.$$

Then, absolutely,

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_{\mathbf{n}\in\mathbb{N}^S} \prod_{k\in S} \frac{(-\xi_k)^{n_k}}{n_k!}\ \mathsf{G}_{\nu+\sum_{k\in S} n_k t_k;\ t_{S^c}}(\boldsymbol{\xi}_{S^c}).$$

Theorem 2.11 (The decay polytope)**.** *Let $\nu_0 \in \operatorname{int}\operatorname{cone}(t)$ be real, $\eta \in \mathbb{R}^{\mathcal{E}}$, and $D_t = \{y \in \mathbb{R}^{\mathcal{E}} : |\langle t_k, y\rangle| < \frac{\pi}{2}\ \forall k\}$. For every $y \in D_t$,*

$$\big|\mathsf{G}_{\nu_0+i\eta;t}(\boldsymbol{\xi})\big| \leq e^{-\langle \eta, y\rangle}\ \mathsf{G}_{\nu_0;t}\big((\xi_k \cos\langle t_k, y\rangle)_k\big).$$

Consequently, with $h_t(\eta) = \sup_{y\in D_t}\langle \eta, y\rangle$, for every $\varepsilon \in (0,1)$

$$\big|\mathsf{G}_{\nu_0+i\eta;t}(\boldsymbol{\xi})\big| \leq \mathsf{G}_{\nu_0;t}\big(\sin(\tfrac{\pi\varepsilon}{2})\boldsymbol{\xi}\big)\, e^{-(1-\varepsilon)h_t(\eta)},$$

and h_t is a norm when the t_k span $\mathbb{R}^{\mathcal{E}}$.

2.2 The Zamma

Definition 2.12 (Zamma)**.** Wherever $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ converges and no ν_e lies in $-\mathbb{N}$, the *Zamma* is the Gramma divided by the Gammas of its own index,

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) := \frac{\mathsf{G}_{\nu;t}(\boldsymbol{\xi})}{\prod_{e\in\mathcal{E}} \Gamma(\nu_e)}.$$

At the unit configuration it is $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) := \prod_e \xi_e^{-\nu_e}$ for every $\nu \in \mathbb{C}^{\mathcal{E}}$; where $\operatorname{Re}\nu_e > 0$ for every e this is the quotient $\mathsf{G}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi})/\prod_e \Gamma(\nu_e)$. The Zamma carries the continuations of the Gramma wherever the Gramma has them.

The Zamma divides out of the Gramma what is first order in its index, the Gammas that the unit configuration produces on its own; what remains is the content of the configuration. It has the three slots of the Gramma.

Proposition 2.13 (First facts about the Zamma)**.** *(i)* First order. *At the unit configuration the Zamma is the monomial of Definition 2.12; at an invertible square configuration t, $N = |\mathcal{E}|$, with $\mathbf{b} = t^{-1}\nu$,*

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\,\Gamma_{\!N}\!\begin{bmatrix}\mathbf{b}\\ \nu\end{bmatrix}\prod_k \xi_k^{-b_k}.$$

(ii) A power of a sum is a Zamma. *With N unit exponents on one scale, $\mathsf{Z}_{\nu;(1,\dots,1)}(\boldsymbol{\xi}) = \big(\sum_k \xi_k\big)^{-\nu}$.*

(iii) Relations. *Where the terms converge, with quotients of Gammas in place of the factors the index moves,*

$$\partial_{\xi_k}\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = -\prod_e \Gamma_1\!\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}),$$

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = \sum_k t_{k,e}\,\xi_k\,\Gamma_1\!\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e + 1\end{bmatrix}\prod_{f\neq e}\Gamma_1\!\begin{bmatrix}\nu_f + t_{k,f}\\ \nu_f\end{bmatrix}\mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}) \qquad (e \in \mathcal{E}).$$

The other relations of Theorem 2.3 are read with the same normalization, and identities persist on the meromorphic continuations.

(iv) As an expectation. *If a configuration on $\mathcal{E}$ contains the unit configuration $I_{\mathcal{E}}$ with unit coefficients, then for real $\nu \in \mathbb{R}^{\mathcal{E}}_{>0}$*

$$\mathsf{Z}_{\nu;I_{\mathcal{E}}\oplus t'}(\mathbf{1}\oplus\boldsymbol{\xi}') = \mathbb{E}_{U\sim\mathrm{Gamma}_\nu}\Big[\exp\Big(-\sum_k \xi'_k U^{t'_k}\Big)\Big];$$

it lies in $(0,1]$, decreases in every coefficient, and, where every $\nu + t'_k$ lies in $\mathbb{R}^{\mathcal{E}}_{>0}$,

$$\mathsf{Z}_{\nu;I_{\mathcal{E}}\oplus t'}(\mathbf{1}\oplus\boldsymbol{\xi}') \;\geq\; \exp\Big(-\sum_k \xi'_k \prod_{e\in\mathcal{E}}\Gamma_1\!\begin{bmatrix}\nu_e + t'_{k,e}\\ \nu_e\end{bmatrix}\Big).$$

Proposition 2.14 (Powers of a polynomial in the scales)**.** *Let t be a configuration and $\nu \in \mathcal{T}_t$ with no ν_e in $-\mathbb{N}$, or $t = I_{\mathcal{E}}$ and $\nu \in \mathbb{C}^{\mathcal{E}}$; let $\xi_k(\rho) = \sum_{l\in\mathbb{N}^d}\rho^l\xi_k^{(l)}$ be polynomials in $\rho \in \mathbb{R}^d$ with real coefficients and $\boldsymbol{\xi}^{(0)} \in (0,\infty)^N$, and let $c_q^\nu = [\rho^q]\,\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho))$ be the Taylor coefficients at $\rho = 0$. At the unit configuration, where $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \boldsymbol{\xi}^{-\nu}$ is read with the principal branch near $\boldsymbol{\xi}^{(0)}$, the coefficients $\xi_k^{(l)}$ with $l \neq 0$ may be complex and $\rho \in \mathbb{C}^d$.*

(i) $c_q^\nu = \sum_{\mathbf{n}}(-1)^{|\mathbf{n}|}\prod_{e\in\mathcal{E}}\Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})\prod_{l\neq0,\,k}\frac{(\xi_k^{(l)})^{n_{l,k}}}{n_{l,k}!}$, *the sum over the arrays $\mathbf{n} = (n_{l,k})_{l\neq0,\,k}$ in $\mathbb{N}$ with $\sum_{l,k} l\,n_{l,k} = q$, where $\mathbf{w} = (\sum_l n_{l,k})_k$; at the unit configuration the factor $\prod_e\Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}$ is the polynomial $\prod_e(\nu_e)_{w_e}$, and away from it each product $\prod_e\Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})$ is $\mathsf{G}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})/\prod_e\Gamma(\nu_e)$.*

(ii) *At one unit exponent, $t = (1)$, with $\boldsymbol{\xi}(\rho) = L(\rho)$ and the index λ: for every i, $\sum_l L^{(l)}\,(q_i - l_i + 1 + \lambda l_i)\,c^{\lambda}_{q-l+\mathbf{e}_i} = 0$, coefficients with a negative index being 0.*

(iii) $\partial_{\xi_k^{(l)}} c^{\nu}_{q} = -\prod_e \Gamma_1\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix} c^{\nu+t_k}_{q-l}$, *read in the same way.*

(iv) *For a configuration t, an index $\nu \in \mathbb{C}^{\mathcal{E}}$ and coefficients $\xi_k(\rho)$ depending smoothly on ρ (Proposition 2.13(iii) and the chain rule),*

$$\rho_i \partial_{\rho_i} \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho)) = -\sum_k \prod_e \Gamma_1\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix} \big(\rho_i \partial_{\rho_i} \xi_k\big) \mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}(\rho)).$$

Chapter 3

The Breta

3.1 Definition and the tower

Definition 3.1 (Breta)**.** The data of a Breta are:

- a finite set $\mathcal{E}$ of scales and a partition $\mathcal{P}$ of $\mathcal{E}$ into *bunches* Q; each bunch is read in scale and shape, $x_Q = \rho_Q u_Q$ with $\rho_Q > 0$ and $u_Q \in \Delta_Q$;
- a *shape index* $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ and a *scale index* $\boldsymbol{\nu} = (\nu_Q)_{Q\in\mathcal{P}}$;
- terms $k = 1, \dots, N$, each with a *shape exponent* $\pi_k \in \mathbb{R}^{\mathcal{E}}$, a *level* $\ell_k = (\ell_{k,Q})_Q \in \mathbb{R}^{\mathcal{P}}$ and a coefficient $\xi_k > 0$.

With $u^{\pi_k} = \prod_Q u_Q^{\pi_k|_Q}$ and $d\omega = \prod_Q d\omega_Q$, the *Breta* is

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) := \int_{\prod_Q \Delta_Q} u^{\mathbf{m}-\mathbf{1}}\ \mathsf{Z}_{\boldsymbol{\nu};\ell}\big((u^{\pi_k}\xi_k)_k\big)\, d\omega(u),$$

the Zamma on the bunch scales with configuration ℓ, wherever the integral converges. The data are *homogeneous* when every level is the degree of its shape exponent on each bunch, $\ell_{k,Q} = \langle \pi_k, \mathbf{1}_Q\rangle$; then the shape exponents form a configuration $t = (\pi_k)_k$ on $\mathcal{E}$ and the Breta is written $\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};t}(\boldsymbol{\xi})$. When $\mathcal{P} = \{B\}$ has one bunch, the scale index has one entry and is written as a number, $\mathsf{B}^{\{B\}}_{\mathbf{m},\nu;t}(\boldsymbol{\xi})$.

The Breta has five slots: the partition, the shape index, the scale index, the terms and the coefficients. It is the Gramma read in scale and shape on every bunch at once (Theorem 2.8), with the scale integrals closed into a Zamma and the shapes left as an integral; the scale index is free, and for homogeneous data fixing it at the bunch totals of the shape index, $\nu_Q = m_Q$, makes the Breta the Gramma itself up to Gammas.

Theorem 3.2 (The tower)**.** *Let the data be homogeneous and $\mathbf{m} \in \mathcal{T}_t$. For every partition $\mathcal{P}$ of the scales,*

$$\mathsf{G}_{\mathbf{m};t}(\boldsymbol{\xi}) = \prod_{Q\in\mathcal{P}} \Gamma(m_Q)\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m},(m_Q)_Q;t}(\boldsymbol{\xi}).$$

With the finest partition the Breta is $\mathsf{Z}_{\mathbf{m};t}(\boldsymbol{\xi})$; with the coarsest, $\Gamma(m_{\mathcal{E}})$ times the Breta is the integral of Theorem 2.8 with $R = \varnothing$. At the unit configuration I_B of a bunch B and the coarsest partition $\{B\}$, the Zamma inside is $\big(\sum_e \xi_e u_e\big)^{-m_B}$, and

$$\prod_{e\in B} \Gamma(m_e)\, \xi_e^{-m_e} = \Gamma(m_B)\, \mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\boldsymbol{\xi}) \qquad (\operatorname{Re} m_e > 0).$$

Corollary 3.3 (Coarsening)**.** *Let the data be homogeneous, and let $\mathcal{P}$ refine $\mathcal{Q}$. Then*

$$\mathsf{B}^{\mathcal{Q}}_{\mathbf{m},(m_Q)_Q;t}(\boldsymbol{\xi}) = \prod_{Q\in\mathcal{Q}} \Gamma_{1}^{k_Q}\begin{bmatrix}(m_P)_{P\subseteq Q}\\ m_Q\end{bmatrix}\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m},(m_P)_P;t}(\boldsymbol{\xi}),$$

with k_Q the number of bunches of $\mathcal{P}$ inside Q.

Proposition 3.4 (The Breta on the orthant)**.** *For any data,*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \frac{1}{\prod_Q \Gamma(\nu_Q)} \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{m}-\mathbf{1}} \prod_Q |x_Q|^{\nu_Q - m_Q} \exp\Big(-\sum_k \xi_k\, x^{\pi_k} \prod_Q |x_Q|^{\ell_{k,Q}-\langle \pi_k,\mathbf{1}_Q\rangle}\Big)\, dx,$$

either side converging when the other does: a free scale index is one more power of each bunch's total, a level that is not the degree of its shape exponent on a bunch is one more power of that bunch's total inside its term, and for homogeneous data at $\nu_Q = m_Q$ this is Theorem 3.2.

Theorem 3.5 (Relations of the Breta)**.** *Each identity holds wherever every Breta in it converges. The terms are (π,ℓ); homogeneous data have $t_k = \pi_k$ and $\ell_{k,Q} = \langle t_k, \mathbf{1}_Q\rangle$. In this statement write $g_k = \prod_Q \Gamma_1\begin{bmatrix}\nu_Q+\ell_{k,Q}\\ \nu_Q\end{bmatrix}$.*

(i) Shape split*, per bunch Q:*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_{e\in Q} \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}).$$

(ii) Ladder*, per term k:*

$$\partial_{\xi_k} \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = -g_k\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}).$$

(iii) Scale exchange*, per bunch Q:*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_k \ell_{k,Q}\,\xi_k\,\Gamma_1\begin{bmatrix}\nu_Q+\ell_{k,Q}\\ \nu_Q+1\end{bmatrix} \prod_{Q'\neq Q} \Gamma_1\begin{bmatrix}\nu_{Q'}+\ell_{k,Q'}\\ \nu_{Q'}\end{bmatrix}\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}).$$

(iv) Shape exchange*, for scales $e\neq f$ of one bunch, with $\operatorname{Re} m_e, \operatorname{Re} m_f > 0$:*

$$m_e\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_f,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) - m_f\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_k \xi_k\, g_k \Big[\pi_{k,e}\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_f,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}) - \pi_{k,f}\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\Big].$$

(v) Shape exchange on the orthant*, for scales $e\neq f$ of one bunch Q:*

$$\begin{aligned}(m_e - m_f)\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) + (\nu_Q - m_Q)\Big(\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) - \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_f,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})\Big)\\ = \sum_k \xi_k\, g_k \Big[(\pi_{k,e}-\pi_{k,f})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\\ + \big(\ell_{k,Q} - \langle \pi_k,\mathbf{1}_Q\rangle\big)\Big(\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}) - \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_f,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\Big)\Big];\end{aligned}$$

for homogeneous data the second line of the right side is absent.

For homogeneous data a term raises the shape total of each bunch and its scale index by the same level, so the ladder, the scale exchange and the shape exchange (v) keep the condition $\nu_Q = m_Q$; in (v) the shifts of the shape index alone carry the factor $\nu_Q - m_Q$, which vanishes there. The shape split and the shape exchange (iv) shift the shape index alone.

3.2 Letters

A *letter* is a function that enters an integrand through a power P^{-s}, s its *letter exponent*. On bunches read in scale and shape, a polynomial letter with positive coefficients is a sum of patterns, each a monomial in the scales and the shapes; the letter presentation writes the integral of a kernel against a product of letters as one Breta, by giving every letter a scale of its own.

Proposition 3.6 (Letter presentation)**.** *Let the scales B carry a partition $\mathcal{Q}$ into bunches, each read in scale and shape, $x_Q = \rho_Q u_Q$, with $\rho = (\rho_Q)_{Q\in\mathcal{Q}}$ and $u = (u_Q)_{Q\in\mathcal{Q}}$; let* kernel terms *$g_j \rho^{\lambda_j} u^{\pi_j}$ ($g_j > 0$, $\lambda_j \in \mathbb{R}^{\mathcal{Q}}$, $\pi_j \in \mathbb{R}^B$) and letters*

$$P_i(\rho, u) = \sum_p C_{i,p}\, \rho^{\ell_{i,p}} u^{\pi_{i,p}}, \qquad C_{i,p} > 0, \qquad i = 1, \dots, N,$$

be given, the sum over the patterns *p of the letter P_i, with $\ell_{i,p} \in \mathbb{R}^{\mathcal{Q}}$, $\pi_{i,p} \in \mathbb{R}^B$ and letter exponents s_i, $\operatorname{Re} s_i > 0$. The array $(C_{i,p})$ lists the coefficients of the patterns. Introduce one scale v_i per letter, put $V = \{v_1, \dots, v_N\}$, and take the partition of the scales $B \sqcup V$*

$$\mathcal{P} = \mathcal{Q} \cup \{\{v_1\}, \dots, \{v_N\}\}.$$

On $B \sqcup V$ the shape index is $(\mathbf{m}, \mathbf{s})$, and on the bunches the scale index is $(\boldsymbol{\nu}, \mathbf{s})$, $\boldsymbol{\nu} \in \mathbb{C}^{\mathcal{Q}}$. The term arrays $(\widetilde{\pi}, \widetilde{\ell})$ have one column per kernel term and one per pattern:

column	*shape exponent on $B \sqcup V$*	*level on $\mathcal{P}$*	*coefficient*
j	$(\pi_j, 0_V)$	$(\lambda_j, 0_V)$	g_j
(i,p)	$(\pi_{i,p}, \mathbf{e}_i)$	$(\ell_{i,p}, \mathbf{e}_i)$	$C_{i,p}$

with $\mathbf{e}_i$ the unit vector at v_i in the shape column and at the singleton bunch $\{v_i\}$ in the level column, and $\mathbf{g} \oplus \mathbf{C}$ lists the coefficients in this column order. Then

$$\int_{\prod_Q \Delta_Q} u^{\mathbf{m}-\mathbf{1}} \int_{\mathbb{R}^{\mathcal{Q}}_{>0}} \rho^{\boldsymbol{\nu}-\mathbf{1}} \exp\Big(-\sum_j g_j \rho^{\lambda_j} u^{\pi_j}\Big) \prod_i P_i(\rho, u)^{-s_i}\, d\rho\, d\omega(u) = \prod_{Q\in\mathcal{Q}} \Gamma(\nu_Q)\, \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

The shapes of the singleton bunches are points, so the entries of the shape index on V do not enter the integral; they move with the scale entries on V, which keeps the relations of Theorem 3.5 literal. In the rest of this statement the partition, the term arrays and the coefficients are fixed, and

$$b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) \quad \textit{stands for} \quad \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

The relations of Theorem 3.5 read: the scale exchange at $\{v_i\}$ is the letter split

$$b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) = \sum_p C_{i,p} \prod_{Q\in\mathcal{Q}} \Gamma_1\begin{bmatrix} \nu_Q + \ell_{i,p,Q} \\ \nu_Q \end{bmatrix} b(\mathbf{m} + \pi_{i,p}, \boldsymbol{\nu} + \ell_{i,p}; \mathbf{s} + \mathbf{e}_i);$$

the ladders are

$$\partial_{g_j} b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) = -\prod_{Q\in\mathcal{Q}} \Gamma_1\begin{bmatrix} \nu_Q + \lambda_{j,Q} \\ \nu_Q \end{bmatrix} b(\mathbf{m} + \pi_j, \boldsymbol{\nu} + \lambda_j; \mathbf{s}),$$

$$\partial_{C_{i,p}} b(\mathbf{m},\boldsymbol{\nu};\mathbf{s}) = -s_i \prod_{Q\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_Q+\ell_{i,p,Q}\\ \nu_Q\end{bmatrix}\, b(\mathbf{m}+\pi_{i,p},\boldsymbol{\nu}+\ell_{i,p};\mathbf{s}+\mathbf{e}_i);$$

and the scale exchange on each $Q\in\mathcal{Q}$ is

$$\nu_Q\, b(\mathbf{m},\boldsymbol{\nu};\mathbf{s}) = \sum_j \lambda_{j,Q}\, g_j \prod_{Q'\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_{Q'}+\lambda_{j,Q'}\\ \nu_{Q'}\end{bmatrix}\, b(\mathbf{m}+\pi_j,\boldsymbol{\nu}+\lambda_j;\mathbf{s}) + \sum_{i,p} s_i\, \ell_{i,p,Q}\, C_{i,p} \prod_{Q'\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_{Q'}+\ell_{i,p,Q'}\\ \nu_{Q'}\end{bmatrix}\, b(\mathbf{m}+\pi_{i,p},\boldsymbol{\nu}+\ell_{i,p};\mathbf{s}+\mathbf{e}_i).$$

Proposition 3.7 (Letter data on the orthant)**.** *Use the partition and the term arrays of Proposition 3.6, and for $\alpha\in\mathbb{R}^B_{>0}$ write $P_i(\alpha)$ for the letter at $\rho_Q=|\alpha_Q|$, $u_Q=\alpha_Q/|\alpha_Q|$:*

$$P_i(\alpha) = \sum_p C_{i,p}\,\alpha^{\pi_{i,p}} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\ell_{i,p,Q}-\langle\pi_{i,p},\mathbf{1}_Q\rangle}.$$

Then, either side converging when the other does,

$$\prod_{Q\in\mathcal{Q}} \Gamma(\nu_Q)\, \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g}\oplus\mathbf{C}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\nu_Q-m_Q} \exp\Big(-\sum_j g_j\,\alpha^{\pi_j} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\lambda_{j,Q}-\langle\pi_j,\mathbf{1}_Q\rangle}\Big) \prod_i P_i(\alpha)^{-s_i}\, d\alpha :$$

each bunch total $|\alpha_Q|$ is one more letter, at the exponent $m_Q-\nu_Q$, and it enters every kernel term and pattern whose level on Q is not its degree there. For homogeneous letter data, $\lambda_{j,Q}=\langle\pi_j,\mathbf{1}_Q\rangle$ and $\ell_{i,p,Q}=\langle\pi_{i,p},\mathbf{1}_Q\rangle$, the letters are $P_i(\alpha)=\sum_p C_{i,p}\alpha^{\pi_{i,p}}$, and at $\nu_Q=m_Q$ the totals are absent.

Proposition 3.8 (Equal levels collapse)**.** *In a Breta the dressed coefficients of all terms with one level add: the Zamma inside has one term per distinct level, whose coefficient is that level's polynomial in the shape. For homogeneous letters of a bunch the level of a pattern is its degree.*

3.3 The shape integral

Theorem 3.9 (The shape integral)**.** *For $\operatorname{Re} m_e>0$ ($e\in B$),*

$$\int_{\Delta_B} u^{\mathbf{m}-\mathbf{1}}\, d\omega_B(u) = \mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\mathbf{1}) = \Gamma_{|B|}\!\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix} :$$

the tower (Theorem 3.2) at the unit configuration I_B with unit coefficients, where the Zamma inside the Breta is $\big(\sum_e u_e\big)^{-m_B}=1$.

The stacked Gamma of the masses over their total is therefore the integral of the shape density $u^{\mathbf{m}-\mathbf{1}}$ on the simplex, and the Dirichlet law is that density normalised.

Definition 3.10 (Dirichlet law)**.** For $\mathbf{m}\in\mathbb{R}^B_{>0}$, $\operatorname{Dir}_{\mathbf{m}}(du) := u^{\mathbf{m}-\mathbf{1}}\, d\omega_B(u)/\Gamma_{|B|}\!\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}$, a probability measure on Δ_B.

Theorem 3.11 (Moments and steps)**.** *For* $\mathbf{m} \in \mathbb{R}^B_{>0}$ *and* $\mathbf{k} \in \mathbb{C}^B$ *with* $\mathrm{Re}(m_e + k_e) > 0$ *for every* $e \in B$*,*

$$\mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}\big[U^{\mathbf{k}}\big] = \frac{\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}+\mathbf{k}\\ m_B+|\mathbf{k}|\end{bmatrix}}{\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}},$$

which for $\mathbf{k} \in \mathbb{N}^B$ *is* $\prod_e (m_e)_{k_e}/(m_B)_{|\mathbf{k}|}$*. For every* $\mathbf{m} \in \mathbb{C}^B$ *off the poles the stacked Gamma* $\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}$ *satisfies the shape split and the shape exchange of its Breta* $\mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\mathbf{1})$*:*

$$\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix} = \sum_{e\in B}\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{bmatrix}, \qquad m_e\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}+\mathbf{e}_f\\ m_B+1\end{bmatrix} = m_f\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{bmatrix},$$

$$\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{bmatrix} = \frac{m_e}{m_B}\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}.$$

Theorem 3.12 (Aggregation)**.** *Let* $B = B_1 \sqcup \cdots \sqcup B_q$ *with nonempty blocks and* $J = \{1,\dots,q\}$*. The map* $(w, v^{(1)},\dots,v^{(q)}) \mapsto u$*,* $u|_{B_j} = w_j v^{(j)}$*, is a bijection* $\Delta_J \times \prod_j \Delta_{B_j} \to \Delta_B$*, and*

$$u^{\mathbf{m}-\mathbf{1}}d\omega_B = \prod_j w_j^{m_{B_j}-1}\, d\omega_J(w)\prod_j (v^{(j)})^{\mathbf{m}|_{B_j}-\mathbf{1}} d\omega_{B_j}(v^{(j)}).$$

Consequently $\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix} = \Gamma_{\substack{q\\1}}\begin{bmatrix}(m_{B_j})_j\\ m_B\end{bmatrix}\prod_j \Gamma_{\substack{|B_j|\\1}}\begin{bmatrix}\mathbf{m}|_{B_j}\\ m_{B_j}\end{bmatrix}$*, and under* $\mathrm{Dir}_{\mathbf{m}}$ *the block totals and the block shapes are independent, with laws* $\mathrm{Dir}_{(m_{B_j})_j}$ *and* $\mathrm{Dir}_{\mathbf{m}|_{B_j}}$*.*

3.4 Products of shape integrals

Theorem 3.13 (The Shruffle)**.** *For a bunch* Q *and a total order* ς *on* Q *let* $\Delta^{\varsigma}_Q = \{u \in \Delta_Q : u_{\varsigma(1)} > u_{\varsigma(2)} > \cdots\}$ *and* $I^{\varsigma}_Q(f;\mathbf{m}) = \int_{\Delta^{\varsigma}_Q} u^{\mathbf{m}-\mathbf{1}} f(u)\, d\omega_Q$*. For pairwise disjoint bunches* $A_1,\dots,A_q$ *with total orders* ς_i *on* A_i*, functions* f_i *on* Δ_{A_i}*,* $A = A_1 \sqcup \cdots \sqcup A_q$ *and* $\bigotimes_i f_i(w) = \prod_i f_i(w_{A_i}/|w_{A_i}|)$ *on* Δ_A*, with* $\mathrm{Re}\, m_{A_i} > 0$ *and the integrals absolutely convergent,*

$$\prod_{i=1}^{q} I^{\varsigma_i}_{A_i}(f_i;\mathbf{m}|_{A_i}) = \Gamma_{\substack{1\\q}}\begin{bmatrix} & m_A & \\ m_{A_1},& \dots,& m_{A_q}\end{bmatrix}\sum_{\varsigma} I^{\varsigma}_A\Big(\bigotimes_i f_i;\mathbf{m}\Big),$$

the sum over the orders ς *of* A *that restrict to every* ς_i*.*

The product of shape integrals on disjoint bunches $A_1,\dots,A_q$ is $\Gamma_{\substack{1\\q}}\begin{bmatrix} & m_A & \\ m_{A_1},& \dots,& m_{A_q}\end{bmatrix}$ times the sum of the shape integrals on the joined bunch over the interleavings of their orders: this is the Shruffle, the product rule of the shape side, whose counterpart for Trailmoments at a common cut is the Struffle (Theorem 5.25).

3.5 The Beta port

Definition 3.14 (Beta port)**.** For $q \in \mathbb{N}$ and $a, b \in \mathbb{C}$,

$$\Pi^{a,b}_q(x) = \frac{(-1)^q x^{1-a}(1-x)^{1-b}}{q!}\,\partial^q_x\big[x^{a+q-1}(1-x)^{b+q-1}\big]$$

$$= \sum_{k=0}^{q} (-1)^{q+k} \binom{a+q-1}{q-k} \binom{b+q-1}{k} x^k (1-x)^{q-k},$$

a polynomial of degree at most q. For positive integers a, b its coefficients are integers; for fixed q and a, b affine in an integer parameter, its coefficients are integers of polynomial growth in that parameter.

Theorem 3.15 (Relations of the Beta port)**.** *(i)* Action on a monomial. *For* $\operatorname{Re} c > 0$ *and* $\operatorname{Re} b > 0$*:* $\int_0^1 x^{c-1}(1-x)^{b-1}\,\Pi_q^{a,b}(x)\,dx = \binom{c-a}{q}\Gamma\,{}_2{}_1\!\left[\begin{matrix} c,\ b+q \\ c+b+q \end{matrix}\right]$*. With* $c = a + j$*,* $j \in \mathbb{N}$*, the binomial vanishes for* $j < q$*: the Beta port is orthogonal, for the density* $x^{a-1}(1-x)^{b-1}$*, to every polynomial of degree below* q*.*

(ii) Integration by parts. *For* $\operatorname{Re} a, \operatorname{Re} b > 0$ *and* g *with* q *continuous derivatives on* $[0,1]$*,*

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,\Pi_q^{a,b}(x)\,g(x)\,dx = \frac{1}{q!}\int_0^1 x^{a+q-1}(1-x)^{b+q-1}\,g^{(q)}(x)\,dx;$$

with one Beta port per scale, of orders $\mathbf{n} = (n_e)_{e\in\mathcal{E}}$*,*

$$\int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{a_e-1}(1-x_e)^{b_e-1}\Pi_{n_e}^{a_e,b_e}(x_e)\,g(\mathbf{x})\,d\mathbf{x} \\ = \frac{1}{\prod_e n_e!}\int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{a_e+n_e-1}(1-x_e)^{b_e+n_e-1}\,\partial^{\mathbf{n}} g(\mathbf{x})\,d\mathbf{x}.$$

(iii) Letter ladder. *For letters* $L_i = c_i + d_i x$ *positive on* $[0,1]$ *(*$c_i > 0$ *and* $c_i + d_i > 0$*) and exponents* $w_i \in \mathbb{C}$*,*

$$\frac{(-1)^q}{q!}\,\partial_x^q \prod_i L_i^{-w_i} = \prod_i L_i^{-w_i} \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!}\Big(\frac{d_i}{L_i}\Big)^{v_i},$$

so that, for $\operatorname{Re} a, \operatorname{Re} b > 0$*,*

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,(-1)^q \Pi_q^{a,b} \prod_i L_i^{-w_i}\,dx \\ = \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!}\int_0^1 x^{a-1}(1-x)^{b-1}\prod_i L_i^{-w_i}\prod_i \Big(\frac{x(1-x)d_i}{L_i}\Big)^{v_i}dx.$$

Let $a, b > 0$ *and every* $w_i \geq 0$*. If every* $d_i \geq 0$ *the value is nonnegative, and if every* $d_i \leq 0$ *it is* $(-1)^q$ *times a nonnegative number; whatever the signs of the* d_i*, if* $x(1-x)|d_i|/L_i \leq \varepsilon$ *on* $[0,1]$ *for every* i*, its modulus is at most* $\frac{(|\mathbf{w}|)_q}{q!}\varepsilon^q \int_0^1 x^{a-1}(1-x)^{b-1}\prod_i L_i^{-w_i}dx$*.*

(iv) Reflection. $\Pi_q^{a,b}(1-x) = (-1)^q \Pi_q^{b,a}(x)$*.*

(v) Three-term and derivative relations. *For* $q \geq 1$ *and* $a, b \in \mathbb{C}$ *with* $a + b + 2q - i \neq 0$ *for* $i = 0, 1, 2$*,*

$$x\,\Pi_q^{a,b} = \frac{(q+1)(a+b+q-1)}{(a+b+2q)(a+b+2q-1)}\,\Pi_{q+1}^{a,b} + \Big(\frac{(q+1)(a+q)}{a+b+2q} - \frac{q(a+q-1)}{a+b+2q-2}\Big)\Pi_q^{a,b}$$

$$+\frac{(a+q-1)(b+q-1)}{(a+b+2q-1)(a+b+2q-2)}\,\Pi_{q-1}^{a,b},$$

$$x(1-x)\,\partial_x\Pi_q^{a,b}=\Big(\frac{q(b+q-1)}{a+b+2q-2}-q\,x\Big)\Pi_q^{a,b}+\frac{(a+q-1)(b+q-1)}{a+b+2q-2}\,\Pi_{q-1}^{a,b};$$

for every $q\in\mathbb{N}$ the coefficient of x^j in $\Pi_q^{a,b}$ is $(-1)^{q+j}\binom{a+q-1}{q-j}\binom{a+b+q+j-2}{j}$, and that of x^q is $\binom{a+b+2q-2}{q}$.

The Beta port moves derivatives of an integrand onto its shape density: integrating against $\Pi_q^{a,b}$ is integrating the q-th derivative against the density with both exponents raised by q. Against a product of linear letters whose slopes have one sign it keeps the sign of the integrand, up to $(-1)^q$ when the slopes are negative, which is what the arithmetic of Part VI uses. The Beta port of order q is orthogonal, for the density $x^{a-1}(1-x)^{b-1}$, to every polynomial of lower degree (Theorem 3.15(i)), and at every pair of exponents multiplication by x and the operator $x(1-x)\partial_x$ move each Beta port by at most one step in its order (Theorem 3.15(v)): finite coefficient rules, which is what turns a mark relation (Theorem 32.10) into a recurrence of bounded width.

Chapter 4

Fibre integrals and closing

Definition 4.1 (Fibre integral)**.** Let $\mathbf{b} \in \mathbb{C}^N$ be *shares*, pinned by $P\mathbf{b} = \boldsymbol{\pi}$ with P a real $q \times N$ matrix with linearly independent rows and $\boldsymbol{\pi} \in \mathbb{C}^q$. Let f be holomorphic on a tube $\{\operatorname{Re}\mathbf{b} \in U\}$, and let $\mathbf{c} \in U$ be an *admissible real point*, $P\mathbf{c} = \operatorname{Re}\boldsymbol{\pi}$. The *fibre integral* is

$$\int_{P\mathbf{b}=\boldsymbol{\pi}} f := (2\pi)^{q-N} \int_{\mathbb{R}^N} f(\mathbf{c}+i\mathbf{y})\,\delta^q\big(P\mathbf{y} - \operatorname{Im}\boldsymbol{\pi}\big)\,d\mathbf{y} = \frac{(2\pi)^{q-N}}{\sqrt{\det PP^{\mathsf{T}}}} \int_{P\mathbf{y}=\operatorname{Im}\boldsymbol{\pi}} f(\mathbf{c}+i\mathbf{y})\,d\mathrm{vol}_{N-q}(\mathbf{y})$$

whenever it converges, vol_{N-q} being the $(N-q)$-dimensional volume on the affine subspace; by Cauchy's theorem it does not depend on $\mathbf{c}$ within one connected set of admissible real points. Its *dimension* is $N-q$. With one pin, $P = (1,\dots,1)$ and $\boldsymbol{\pi} = s$, it is the *share integral*

$$\int_{a_1+\cdots+a_k=s} f(\mathbf{a}) = \frac{1}{(2\pi i)^{k-1}} \int_{c_1+i\mathbb{R}} \cdots \int_{c_{k-1}+i\mathbb{R}} f\Big(a_1,\dots,a_{k-1},\, s - \sum_{i<k} a_i\Big)\,da_1\cdots da_{k-1}.$$

At dimension zero, $q = N$, the value is $|\det P|^{-1} f(P^{-1}\boldsymbol{\pi})$. When $\boldsymbol{\pi}$ is real, with $V = \{\mathbf{y} :\ P\mathbf{y} = 0\}$ and the measure $d\boldsymbol{\vartheta} = (\det PP^{\mathsf{T}})^{-1/2}\,d\mathrm{vol}_{N-q}$ on it, the fibre integral is written $\int_{\mathbf{c}+iV} f = (2\pi)^{q-N}\int_V f(\mathbf{c}+i\boldsymbol{\vartheta})\,d\boldsymbol{\vartheta}$, its admissible real point shown.

Proposition 4.2 (The share side of the Gramma)**.** *For $\nu \in \mathcal{T}_t$,*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{t\mathbf{b}=\nu} \prod_{k=1}^{N} \Gamma(b_k)\,\xi_k^{-b_k},$$

the fibre integral with one share per term, pinned by the configuration ($|\mathcal{E}|$ pins), of dimension $N - |\mathcal{E}|$; its admissible real points are the $\mathbf{c} \in (0,\infty)^N$ with $t\mathbf{c} = \operatorname{Re}\nu$, and they exist exactly on the domain. At dimension zero it is the value of Corollary 2.6.

On the share side the configuration is the matrix of the pins and the index their values, one pin per scale; each term carries one share, with its Gamma and its coefficient's power as factors.

Theorem 4.3 (The native gauge of a configuration)**.** *Let the t_k span $\mathbb{R}^{\mathcal{E}}$, $\nu \in \operatorname{int}\operatorname{cone}(t)$ real and $\boldsymbol{\xi} > 0$.*

(i) $\Phi_{\boldsymbol{\xi}}(y) = \sum_k \xi_k e^{\langle t_k, y\rangle} - \langle \nu, y\rangle$ is strictly convex and tends to $+\infty$ along every ray; its unique critical point $y^\star$ solves $\sum_k \xi_k e^{\langle t_k, y^\star\rangle} t_k = \nu$, and $e^{y^\star}$ is the native gauge *of the configuration.*

(ii) The shares $w_k^\star = \xi_k e^{\langle t_k, y^\star\rangle}$ are an admissible real point of the share side (Proposition 4.2). On the real fibre $\{\mathbf{w} > 0 :\ t\mathbf{w} = \nu\}$ the function $S(\mathbf{w}) = \sum_k \big(w_k - w_k \log(w_k/\xi_k)\big)$ is strictly concave with unique critical point $\mathbf{w}^\star$, and $S(\mathbf{w}^\star) = \Phi_{\boldsymbol{\xi}}(y^\star)$.

(iii) For $\nu' \in \mathbb{C}^{\mathcal{E}}$, as $\lambda \to \infty$,

$$\mathsf{G}_{\lambda\nu+\nu';t}(\lambda\boldsymbol{\xi}) = e^{\langle \nu', y^\star\rangle}\, e^{-\lambda\Phi_{\boldsymbol{\xi}}(y^\star)} \Big(\frac{2\pi}{\lambda}\Big)^{|\mathcal{E}|/2} \det\Big(\sum_k w_k^\star t_k t_k^{\mathsf{T}}\Big)^{-1/2} \big(1 + O(\lambda^{-1})\big).$$

The native gauge of a configuration is read on both of its sides at once: on the orthant side its logarithmic coordinate $y^\star$ minimizes the convex function $\Phi_{\boldsymbol{\xi}}$, on the share side the shares $\mathbf{w}^\star$ it determines maximize the entropy-like function S on the real fibre, and the minimum equals the maximum.

Corollary 4.4 (Splitting a coefficient)**.** *Let $k \geq 2$, $X \in (0,\infty)^k$, $\operatorname{Re} s > 0$ and $\mathbf{c} \in (0,\infty)^{k-1}$ with $\sum_i c_i < \operatorname{Re} s$. On this contour, every share having positive real part,*

$$\mathsf{Z}_{s;(1)}(X_1 + \cdots + X_k) = \Big(\sum_{i=1}^k X_i\Big)^{-s} = \int_{a_1+\cdots+a_k=s} \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix} \prod_{i=1}^k X_i^{-a_i},$$

equivalently $\Gamma(s)(X_1+\cdots+X_k)^{-s} = \int_{|\mathbf{a}|=s} \prod_i \Gamma(a_i) X_i^{-a_i}$: the share side of the one-scale Gamma $\mathsf{G}_{s;(1,\dots,1)}(X)$ (Proposition 4.2). The stacked Gamma $\Gamma_{\substack{k\\1}}\!\left[\begin{smallmatrix}\mathbf{a}\\ s\end{smallmatrix}\right]$ is the share kernel *of the letter $X_1 + \cdots + X_k$ at the exponent s.*

Theorem 4.5 (Closing)**.** *Let a fibre integral have the form $\int_{P\mathbf{b}=\boldsymbol{\pi}} \Gamma\big(\ell(\mathbf{b})\big) G(\mathbf{b})$ with ℓ affine and not constant on the fibre. Suppose that, moving the contour along a direction of the fibre in which ℓ decreases, $\Gamma(\ell)$ is the only factor with poles crossed, and the shifted contour's contribution tends to zero. Then*

$$\int_{P\mathbf{b}=\boldsymbol{\pi}} \Gamma\big(\ell(\mathbf{b})\big)\, G(\mathbf{b}) = \sum_{j\geq 0} \frac{(-1)^j}{j!} \int_{P\mathbf{b}=\boldsymbol{\pi},\ \ell(\mathbf{b})=-j} G(\mathbf{b}),$$

each term a fibre integral with one more pin. When several factors have poles crossed, each contributes its own lattice; where several share a pole hyperplane, the residue is a Laurent coefficient (Proposition 5.14).

Closing trades one dimension of the fibre for a sum over the poles crossed: the Gamma whose argument falls is replaced by its residues. Where closing can be repeated until the fibre has dimension zero, every term is the value $|\det P|^{-1} G(P^{-1}\boldsymbol{\pi})$ of the remaining factors at one point.

Theorem 4.6 (The share kernels form a semigroup)**.** *For a letter $\sum_{i=1}^k X_i$, shares $\mathbf{a}, \mathbf{b}, \mathbf{g} \in \mathbb{C}^k$ and $|\mathbf{g}| = s + s'$, on admissible contours,*

$$\int_{\mathbf{a}+\mathbf{b}=\mathbf{g},\ |\mathbf{a}|=s} \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix} \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{b}\\ s'\end{bmatrix} = \Gamma_{\substack{k\\1}}\!\begin{bmatrix}\mathbf{g}\\ s+s'\end{bmatrix}.$$

Splitting the letter at the exponent s and at the exponent s' and merging the shares is splitting it at $s + s'$: the share kernels of Corollary 4.4 compose like the powers they open.

Corollary 4.7 (Four Gammas on a line)**.** *On a contour separating the poles of $\Gamma(a+z)\Gamma(b+z)$ from those of $\Gamma(c-z)\Gamma(d-z)$,*

$$\frac{1}{2\pi i}\int \Gamma(a+z)\,\Gamma(b+z)\,\Gamma(c-z)\,\Gamma(d-z)\,dz = \Gamma_{\substack{4\\1}}\!\begin{bmatrix} a+c,\ b+d,\ a+d,\ b+c \\ a+b+c+d \end{bmatrix}:$$

Theorem 4.6 at $k=2$, multiplied by $\Gamma(a+c)\Gamma(b+d)$, with the shares $(a+z,\ c-z)$ at the exponent $a+c$ and $(d-z,\ b+z)$ at the exponent $b+d$, merged to $(a+d,\ b+c)$.

The four Gammas are the Gamma factors of the four shares of a Gramma on three scales, pinned by the three scales, and the one free share is the contour variable; integrating the orthant side scale by scale leaves $\Gamma(a+c)\Gamma(b+d)$ times the Beta corner ${}_{1}^{2}\Gamma\begin{bmatrix} a+d,\ b+c \\ a+b+c+d \end{bmatrix}$, the shape integral of two scales (Theorem 3.9).

Chapter 5

The Trailmoment

5.1 Definition

Definition 5.1 (Trailmoment)**.** The data of a Trailmoment are an index ν and a configuration t, those of a Zamma; a measure ϕ, the *input*, on a space Y, the *source*; a measurable *coefficient function* $\mathbf{c} : Y \to (0,\infty)^N$; and a measurable *cut region* $Y' \subseteq Y$. The *Trailmoment* is

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathsf{Z}_{\nu;t}\big(\mathbf{c}(y)\big)\,\phi(dy)$$

wherever it converges: a normalized Zamma kernel read against the input over the cut region.

The Trailmoment has five slots: its index and configuration, which are the kernel's, its coefficient function, its input and its cut region. On a line, $Y = \mathbb{R}$, the cut region (N,∞) is the *cut* at N, and the region $(N, N']$ between two cuts is a *head*; on $\mathbb{R}^d$ the region $X_N = \{N < y_1 < \cdots < y_d\}$ is a *trail region*, the nested tails that give the Trailmoment its name; the trails of Section 5.8 are read over it. The input is written as a density with its domain marked by an indicator, so that the measure is part of the data.

Definition 5.2 (Lattice inputs; the index-one value)**.** The *lattice input* on the coset $b+q\mathbb{Z}$, $b \in \mathbb{R}$, of step $q > 0$ and frequency $a \in \mathbb{C}$ is $e^{2\pi i a x}\,\mathbb{I}_{b+q\mathbb{Z}} = \sum_{m\in b+q\mathbb{Z}} e^{2\pi i a m}\,\delta_m$; at frequency 0 it is counting measure on $b+q\mathbb{Z}$, written $\mathbb{I}_{b+q\mathbb{Z}}$, and for $\operatorname{Im} a > 0$ the values $e^{2\pi i a m}$ decay geometrically toward $+\infty$. The input $f\,\mathbb{I}_{b+q\mathbb{Z}}$ has the density f on the lattice, and $\mathbb{I}_{\mathbb{Z}}$ is the *counting input*. At one unit exponent and the coefficient function x the kernel is $\mathsf{Z}_{\nu;(1)}(x) = x^{-\nu}$, so at a cut beyond which every lattice point is positive

$$(\mathsf{T}_{\nu;(1)}[x]\,f\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) = \sum_{m\in b+\mathbb{Z},\ m>N} f(m)\,m^{-\nu}.$$

At index one the lattice input is regularized at its upper end:

$$(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) := \lim_{M\to\infty}\Big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,N+M]) - \log M\Big),$$

which exists, the heads being $\log M + (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) + O(M^{-1})$: the regularized value is the coefficient of $M^0 \log^0 M$ in the heads.

Proposition 5.3 (The cut relation)**.** *The Trailmoment is additive in its cut region: for disjoint measurable $Y_1', Y_2' \subseteq Y$, $(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_1' \sqcup Y_2') = (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_1') + (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_2')$, either side converging when the other does; on a line, for $N < N'$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,\infty)) = (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,N']) + (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N',\infty)),$$

and for a lattice input at index one it holds with the regularized values.

Definition 5.4 (Inputs)**.** An input is a nonnegative σ-finite measure μ, or such a measure with a complex density, $\phi = f\mu$ with $|\phi| = |f|\mu$; a Trailmoment converges when $\int |\mathsf{Z}_{\nu;t}(\mathbf{c})|\, d|\phi|$ over the cut region is finite. The image $\tau_*\phi$ of an input with a complex density under a measurable map τ is used only in integrals $\int g\, d(\tau_*\phi) := \int g \circ \tau\, d\phi$ with $\int |g \circ \tau|\, d|\phi| < \infty$; $|\tau_*\phi|$ denotes $\tau_*|\phi|$, and tameness and convergence of $\tau_*\phi$ are meant with it. The *Lebesgue input* $\mathbb{I}_{\mathbb{R}}$ is Lebesgue measure on $\mathbb{R}$, $f\,\mathbb{I}_{\mathbb{R}}$ is the input with density f, $\mathbb{I}_D$ is Lebesgue measure on an interval D, and $\mathbb{I}_{\mathbb{R}^{\mathcal{E}}}$ is Lebesgue measure on $\mathbb{R}^{\mathcal{E}}$; the lattice inputs are those of Definition 5.2, and at integer frequency the lattice input on $\mathbb{Z}$ is the counting input.

An input is *tame beyond the cut N* when its part beyond N is carried by $[x_0,\infty)$ for some $x_0 > 0$ and $\int_{(N,\infty)} x^{-\beta}\, |\phi|(dx) < \infty$ for some real β. Its Laplace transform beyond the cut is

$$\mathcal{F}_N(u) = \int_{(N,\infty)} e^{-ux}\, \phi(dx),$$

which converges for $\operatorname{Re} u > 0$, with $|\mathcal{F}_N(u)| \le e^{-(u-1)x_0} \int_{(N,\infty)} e^{-x}\, |\phi|(dx)$ for $u \ge 1$. When $\mathcal{F}_N$ is moreover meromorphic at $u = 0$, $(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ denotes, for every ν off its poles, the continuation in which $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ is the meromorphic continuation of $\int_0^\infty u^{\nu-1}\mathcal{F}_N(u)\, du$, the Laplace side of Theorem 5.11 at one unit exponent. At index one on a lattice input it is the regularized value of Definition 5.2, which is the constant term of the continuation at $\nu = 1$.

Proposition 5.5 (The continued Trailmoment on a line)**.** *Let ϕ be tame beyond the cut N with $\mathcal{F}_N(u) = \sum_{k \ge -k_0} c_k u^k$ near $u = 0$. Then $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ is meromorphic in ν with at most simple poles, at $\nu = -k$, $k \ge -k_0$, with residue c_k, and at the nonpositive integers*

$$(\mathsf{T}_{-j;(1)}[x]\,\phi)((N,\infty)) = (-1)^j j!\, c_j \qquad (j \in \mathbb{N}).$$

At a nonpositive integer index the Trailmoment of a tame input is a Taylor coefficient of its Laplace transform, so a cut that releases a head changes these values by the head read at the same index.

5.2 Slots, corners and heights

Definition 5.6 (Slots, corners, height)**.** A *slot* is a place of a symbol of the serif family where one of its data stands. A Gramma or a Zamma has three slots: its index, its configuration and its coefficients. A Breta has five: its partition into bunches, its shape index, its scale index, its terms (shape exponents and levels) and its coefficients. A Trailmoment has five: its index ν, its configuration t, its coefficient function $\mathbf{c}$, its input ϕ and its cut region. A *corner* of a member of the serif family is that member with some of its slots given particular values. A *relation* is an identity between members of the serif family. Its *height* is the set of values of the slots on which one proof establishes it; a slot is *free* in a relation when the height does not restrict it and *pinned* otherwise. For a pinned slot, the *mechanism* is the step of the proof that uses the pinned value.

A relation recited at a corner is understood once its height is known: which slots the proof leaves free, which it uses, and at which step. A relation whose height is every value of every slot is a property of the operator. A pin records what a proof uses; it does not say that the relation fails at the other values of the slot, and where the converse is proved, so that the relation is a property of the pinned value, it is stated with the relation.

5.3 Relations at free slots

Theorem 5.7 (The gauge of the coefficient function)**.** *Let $(\nu, t, \mathbf{c}, \phi, Y')$ be the data of a Trailmoment on a source Y and let $\ell : Y \to (0,\infty)^{\mathcal{E}}$ be measurable. Then*

$$\big(\mathsf{T}_{\nu;t}[(\ell^{t_k} c_k)_k]\,\phi\big)(Y') = \big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\ell^{-\nu}\phi\big)(Y'),$$

either side converging when the other does. If the columns t_k, $k \in S$, form a basis of $\mathbb{R}^{\mathcal{E}}$, the choice $\log \ell = -(t_S^{\mathsf{T}})^{-1} \log \mathbf{c}_S$ makes $\ell^{t_k} c_k = 1$ for every $k \in S$: the gauge moves $|\mathcal{E}|$ of the N coefficient functions into the input, and the kernel then depends on the source point only through the $N - |\mathcal{E}|$ functions $\ell^{t_k} c_k$, $k \notin S$. At the unit configuration, $(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathbf{c}^{-\nu}\, d\phi$.

At the unit configuration the kernel is a density on the input: index, coefficient function and input enter only through the measure $\mathbf{c}^{-\nu}\phi$. With $N > |\mathcal{E}|$ terms the kernel depends on the source point through $N - |\mathcal{E}|$ gauge-invariant functions as well as through a density, and this count is the dimension of the share side (Theorem 5.10).

Theorem 5.8 (Exchange and ladder of the Trailmoment)**.** *Each identity holds wherever every Trailmoment in it converges.*

(i) Exchange, one relation per scale. *For each $e \in \mathcal{E}$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \sum_{k=1}^{N} t_{k,e}\ \Gamma_1^{\,1}\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e + 1\end{bmatrix} \prod_{f \neq e} \Gamma_1^{\,1}\begin{bmatrix}\nu_f + t_{k,f}\\ \nu_f\end{bmatrix}\ \big(\mathsf{T}_{\nu + t_k;t}[\mathbf{c}]\,c_k\phi\big)(Y').$$

(ii) Ladder. *If $\mathbf{c} = \mathbf{c}_\zeta$ depends differentiably on a real parameter ζ and the differentiated integrand is dominated by an integrable function locally uniformly in ζ,*

$$\partial_\zeta(\mathsf{T}_{\nu;t}[\mathbf{c}_\zeta]\,\phi)(Y') = -\sum_{k=1}^{N} \prod_{e} \Gamma_1^{\,1}\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix}\ \big(\mathsf{T}_{\nu + t_k;t}[\mathbf{c}_\zeta]\,(\partial_\zeta c_k)\,\phi\big)(Y').$$

(iii) The unit configuration. *At $t = I_{\mathcal{E}}$ the exchange relation at each scale has one term, and on the interior of the set of indices where the Trailmoment converges the index is differentiated through the input: for every $e \in \mathcal{E}$,*

$$(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = (\mathsf{T}_{\nu + \mathbf{e}_e;I_{\mathcal{E}}}[\mathbf{c}]\,c_e\phi)(Y'), \qquad \partial_{\nu_e}(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = -(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,(\log c_e)\,\phi)(Y').$$

At the unit configuration a step of the index at a scale is a factor of that coordinate of the coefficient function on the input, in both directions. With several terms on a scale the exchange relation is linear in $N + 1$ Trailmoments whose inputs differ by the factors c_k, one relation per scale, and a step of the index is no longer a density.

Theorem 5.9 (Transport)**.** *For a measurable map $\tau : Y \to Y_2$, an input ϕ on Y, a coefficient function $\mathbf{c}$ on Y_2 and a cut region $Y_2' \subseteq Y_2$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\tau_*\phi)(Y_2') = (\mathsf{T}_{\nu;t}[\mathbf{c}\circ\tau]\,\phi)(\tau^{-1}Y_2'),$$

whenever the right side converges; for a nonnegative input either side converges when the other does. In particular, transporting by the coefficient function itself, a convergent Trailmoment is

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \big(\mathsf{T}_{\nu;t}[\mathrm{id}]\,\mathbf{c}_*(\phi|_{Y'})\big)\big((0,\infty)^N\big)\,:$$

coefficient function, input and cut region enter only through the image $\mathbf{c}_*(\phi|_{Y'})$ of the input on the orthant of coefficients.

A Trailmoment of a nonnegative input is therefore a function of three things: an index, a configuration, and a measure on the orthant of coefficients. Two descriptions of that measure, by different coefficient functions, inputs and cut regions, give the same member of the serif family. An increasing bijection of the line carries half-lines to half-lines: with $\tau(x) = x + h$ it is the *shift* of the cut, and with $\tau(x) = qx$ at one unit exponent it gives $(\mathsf{T}_{\nu;(1)}[x]\,\tau_*\phi)((qN,\infty)) = q^{-\nu}(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$.

Theorem 5.10 (The share side of the Trailmoment)**.** *Let $\operatorname{Re}\nu \in \operatorname{int}\operatorname{cone}(t)$, and let $\mathbf{s} \in (0,\infty)^N$ be an admissible real point, $t\mathbf{s} = \operatorname{Re}\nu$, with $\int_{Y'}\prod_k c_k(y)^{-s_k}\,|\phi|(dy) < \infty$. On the contour through $\mathbf{s}$,*

$$\prod_{e\in\mathcal{E}}\Gamma(\nu_e)\ (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{t\mathbf{b}=\nu}\ \prod_{k=1}^{N}\Gamma(b_k)\ (\mathsf{T}_{\mathbf{b};I_N}[\mathbf{c}]\,\phi)(Y'),$$

$$(\mathsf{T}_{\mathbf{b};I_N}[\mathbf{c}]\,\phi)(Y') = \int_{Y'}\prod_{k=1}^{N} c_k(y)^{-b_k}\,\phi(dy),$$

a fibre integral of dimension $N - |\mathcal{E}|$ of Trailmoments at the unit configuration I_N, with one scale per term of t. Where some ν_e lies in $-\mathbb{N}$ the left side is read as $\int_{Y'}\mathsf{G}_{\nu;t}(\mathbf{c})\,d\phi$.

Every configuration is written through the unit configuration: a Trailmoment at any configuration is a superposition of Trailmoments at the unit configuration over the fibre of its configuration, each multiplied by the Gammas of its shares. Closing the fibre integral (Theorem 4.5) replaces the superposition by residues, and the residues are Trailmoments at the unit configuration at shifted indices; that is why their values appear in the closings of every configuration.

Theorem 5.11 (The Laplace side of the Trailmoment)**.** *The joint Laplace transform of the data $(\mathbf{c},\phi,Y')$ is $\mathcal{F}(v) = \int_{Y'} e^{-\langle v,\mathbf{c}(y)\rangle}\,\phi(dy)$ for $v \in (0,\infty)^N$. For $\nu \in \mathcal{T}_t$, wherever the double integral converges,*

$$\prod_{e\in\mathcal{E}}\Gamma(\nu_e)\ (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}}\,\mathcal{F}\big(x^{t_1},\dots,x^{t_N}\big)\,dx = \big(\mathsf{T}_{\mathbf{1}-\nu;I_{\mathcal{E}}}[x]\ \mathcal{F}(x^t)\,\mathbb{I}_{\mathbb{R}^{\mathcal{E}}}\big)\big(\mathbb{R}^{\mathcal{E}}_{>0}\big),$$

where $\mathcal{F}(x^t) = \mathcal{F}(x^{t_1},\dots,x^{t_N})$ and the kernel of the last Trailmoment is $\prod_e x_e^{\nu_e-1}$, the Zamma of the unit configuration at the index $\mathbf{1}-\nu$; where some ν_e lies in $-\mathbb{N}$ the left side is read as $\int_{Y'}\mathsf{G}_{\nu;t}(\mathbf{c})\,d\phi$. The exponential is the Zamma of a term at exponent zero, $\mathsf{Z}_{1;(1,0)}(1,\xi) = e^{-\xi}$, so the joint Laplace transform is a Trailmoment of the same input: $\mathcal{F}(v) = (\mathsf{T}_{1;(1,0)}[(1,\langle v,\mathbf{c}\rangle)]\,\phi)(Y')$.

Index ν on the input side is index $\mathbf{1}-\nu$ on the Laplace side. Besides the input itself, a Trailmoment has two sides: the share side, which resolves the configuration into unit configurations, and the Laplace side, which replaces the input by its Laplace transform; the reflections of the index (Theorems 5.18 and 5.19), the Shruffle (Theorem 5.26) and the Cremona (Theorem 5.28) are read on the Laplace side, and the lattice relations and the Struffle on the input itself.

5.4 Lattices

Theorem 5.12 (The lattice relations)**.** *Let $b\in\mathbb{R}$, let $q\geq 1$ be an integer, let $(\nu,t,\mathbf{c})$ be the data of a kernel on $\mathbb{R}$, f a density and N a cut.*

(i) Cosets. *Wherever the right side converges,*

$$\sum_{r=0}^{q-1}\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,f\,\mathbb{I}_{\frac{b+r}{q}+\mathbb{Z}}\big)((N,\infty))=\big(\mathsf{T}_{\nu;t}[\mathbf{c}(\,\cdot\,/q)]\,f(\,\cdot\,/q)\,\mathbb{I}_{b+\mathbb{Z}}\big)((qN,\infty)).$$

(ii) Frequencies. *For $a\in\mathbb{R}$, wherever the Trailmoments on the left converge,*

$$\sum_{r=0}^{q-1}\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,e^{2\pi i(a+\frac{r}{q})(x-b)}f\,\mathbb{I}_{b+\mathbb{Z}}\big)((N,\infty))=q\,\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,e^{2\pi ia(x-b)}f\,\mathbb{I}_{b+q\mathbb{Z}}\big)((N,\infty)),$$

the right side summing over the lattice of step q.

A lattice is assembled from a sublattice in two ways: by cosets, the q cosets of $\mathbb{Z}$ in $q^{-1}\mathbb{Z}$; by frequencies, the q exponentials $e^{2\pi irx/q}$, $0\leq r<q$. Both statements are free in the index, the configuration, the coefficient function, the density, the coset and the cut. At one unit exponent with the coefficient function x the rescaled coefficient function of (i) multiplies the kernel by the scalar q^{ν}, $\mathsf{Z}_{\nu;(1)}(x/q)=q^{\nu}x^{-\nu}$; at index one, with the regularized values, the same assembly carries the further term $q\log q$, the finite part read in two coordinates (Lemma 1.8).

5.5 The Gamma read by Trailmoments

Proposition 5.13 (Heads and higher orders at lattice data)**.** *For real $b>-1$ and $|z|<1+b$,*

$$\log\Gamma(1+b-z)=\log\Gamma(1+b)+\sum_{i\geq 1}\frac{z^i}{i}\,(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b,\infty)),$$

the index-one Trailmoment regularized; equivalently, for $j\in\mathbb{N}$,

$$(\log\Gamma)^{(j+1)}(1+b)=(-1)^{j+1}j!\,(\mathsf{T}_{j+1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b,\infty)).$$

Further $\Gamma(-k+z)=\dfrac{(-1)^k\,\Gamma(1+z)\,\Gamma(1-z)}{z\,\Gamma(k+1-z)}$ for $k\in\mathbb{N}$. In this statement write $h=\prod_a\Gamma(1+b_a-z_a)/\prod_c\Gamma(1+b'_c-z'_c)$, a stacked Gamma whose constants b_a,b'_c lie in one translate $b_0+\mathbb{N}$ with $b_0>-1$, with $\sum_a z_a=\sum_c z'_c$, $|z_a|<1+b_a$ and $|z'_c|<1+b'_c$. Then

$$\log\frac{h}{h|_{z=0}}=\sum_c z'_c\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b_0+\mathbb{Z}})((b_0,b'_c])$$

$$
\begin{aligned}
&- \sum_a z_a \, (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_0, b_a]) \\
&+ \sum_{i\geq 2} \frac{1}{i} \Big(\sum_a z_a^i \, (\mathsf{T}_{i;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_a, \infty)) \\
&\qquad\qquad - \sum_c z_c'^{\,i} \, (\mathsf{T}_{i;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b'_c, \infty)) \Big) :
\end{aligned}
$$

the regularized index-one values cancel, the first order is a finite sum over a head of the lattice, rational when b_0 is, and the orders $i \geq 2$ are Trailmoments of the lattice input cut at the data. At integer data, $b_0 = 0$, the first order is rational and the higher orders are Trailmoments of the counting input.

Proposition 5.14 (Laurent data of the Gamma at its poles)**.** *Let $j \in \mathbb{N}$. For $|\varepsilon| < 1$, with $(-1)^j \varepsilon\, \Gamma(-j+\varepsilon) = \Gamma(1+\varepsilon)\, \Gamma(1-\varepsilon)/\Gamma(1+j-\varepsilon)$,*

$$
\begin{aligned}
\log\big((-1)^j \varepsilon\, \Gamma(-j+\varepsilon)\big) = &- \log j! - (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))\, \varepsilon \\
&- \sum_{k\geq 2} \frac{(\mathsf{T}_{k;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))}{k}\, \varepsilon^k + \sum_{k\geq 1} \frac{(\mathsf{T}_{2k;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))}{k}\, \varepsilon^{2k},
\end{aligned}
$$

and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))$. Let h be holomorphic near $b = -j$, and let $k_1, \ldots, k_p$ be integers with $j_i = j - k_i \in \mathbb{N}$, so that every $\Gamma(b+k_i)$ has a pole at $-j$. Then

$$
\operatorname*{Res}_{b=-j} \Big(h(b) \prod_{i=1}^{p} \Gamma(b+k_i) \Big) = (-1)^{j_1+\cdots+j_p} \sum_{l=0}^{p-1} \frac{h^{(l)}(-j)}{l!}\, [\varepsilon^{p-1-l}] \prod_{i=1}^{p} \big((-1)^{j_i} \varepsilon\, \Gamma(-j_i+\varepsilon) \big),
$$

each factor of the product being the exponential of the display at j_i; at $p = 2$ and $k_1 = k_2 = 0$ it is $\frac{1}{(j!)^2} \big(h'(-j) + 2\, (\log\Gamma)'(j+1)\, h(-j) \big)$. Every Laurent coefficient of $\prod_i \Gamma(b+k_i)$ at $-j$ is $\prod_i (j_i!)^{-1}$ times a polynomial in Trailmoments of the counting input cut at the j_i and at 0.

Proposition 5.15 (Selecting an index inside a word)**.** *Let $w \geq 2$ be an integer and $h(z) = \prod_i \Gamma(1 + b_i - a_i z)^{n_i}$, a finite product with real $b_i > -1$, $a_i \in \mathbb{C}$ and $n_i \in \mathbb{Z}$. Its w-th multiplicative difference satisfies*

$$
\prod_{j=0}^{w} h(jz)^{(-1)^{w-j}\binom{w}{j}} = 1 + (w-1)! \Big[\sum_i n_i a_i^w \, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{b_i+\mathbb{Z}})((b_i,\infty)) \Big] z^w + O(z^{w+1}).
$$

At integer values $b_i = k_i \in \mathbb{N}$ with rational a_i, an integer $\ell \geq 1$ with every $\ell a_i \in \mathbb{Z}$ and $k_* = \max_i k_i$, the coefficient of z^w is $A\, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - B$ with $A = (w-1)! \sum_i n_i a_i^w$ and $B = (w-1)! \sum_i n_i a_i^w \, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,k_i])$, both cleared by $\ell^w d_{k_*}^w$. For the one factor $\Gamma(k+1-z)$, the coefficient of z^w in $\prod_{j=0}^{w} \Gamma(k+1-jz)^{(-1)^{w-j}\binom{w}{j}}$ is $(w-1)!\, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((k,\infty))$, and passing from k to $k+1$ multiplies this product by $\prod_{j=0}^{w} (k+1-jz)^{(-1)^{w-j}\binom{w}{j}}$. The difference acts multiplicatively on one word; it supplies no interchange with the sums of a closing.

The w-th difference annihilates every power of z below the w-th in the logarithm of a Gamma word, so it removes the regularized index-one values together with all Trailmoments of lower index and keeps exactly those of index w: from a word in the Gammas of the data it selects one Trailmoment index, with a rational head.

Lemma 5.16 (Two Gammas and their reflections). *The stacked Gamma* $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big] = \pi^2/\sin^2\pi a$ *has at every integer* m *the principal part* $(a-m)^{-2}$ *and no simple part, it equals* $\sum_{m\in\mathbb{Z}}(a+m)^{-2}$*, and on the line* $a=\frac12+iy$ *it equals* $4\pi^2/(e^{\pi y}+e^{-\pi y})^2$*.*

(i) Right closing. *For* $0<c<1$ *and* g *rational, holomorphic on* $\{\operatorname{Re} a\ge c\}$*, with* $g(a)=O(|a|^{-2})$*:* $\dfrac{1}{2\pi i}\displaystyle\int_{(c)}\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ g(a)\,da = -\sum_{m\ge1} g'(m)$.

(ii) Left closing. *For* $0<X<1$*:* $\dfrac{1}{2\pi i}\displaystyle\int_{(1/2)}\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ X^{-a}\,da = \frac{\log X}{X-1}$.

(iii) Fourier read. *For real* t*,*

$$\frac{1}{2\pi}\int_{\mathbb{R}}\Gamma_{\substack{4\\0}}\big[\tfrac12+iy,\ \tfrac12+iy,\ \tfrac12-iy,\ \tfrac12-iy\big]\ e^{ity}\,dy = \Gamma_{\substack{2\\0}}\big[1+\tfrac{it}{2\pi},\ 1-\tfrac{it}{2\pi}\big] = \frac{t}{e^{t/2}-e^{-t/2}}.$$

A closing whose poles coincide (Theorem 4.5) takes Laurent coefficients of its Gammas, and a balanced word near lattice data has heads and higher orders (Proposition 5.13): in both cases the Gammas contribute words in Trailmoments of lattice inputs, read at the cuts that the data set. The stacked Gamma of Lemma 5.16 is the simplest coincidence, two Gammas and their two reflections meeting at every integer: its double poles have no simple parts, so its right closing reads the derivative of the rest of the integrand at the positive integers, and on the line $\operatorname{Re} a=\frac12$ it is 2π times the density of the critical-line law (Chapter 33).

Theorem 5.17 (Reflected pairs and matched parity). *(i)* Cosets. *For* $0<b<1$ *and every integer* $n\ge1$*,*

$$(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b-1,\infty)) + (-1)^n\,(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{-b+\mathbb{Z}})((-b,\infty)) = \frac{(-1)^{n-1}}{(n-1)!}\,\partial_b^{n-1}\big(\pi\cot\pi b\big),$$

the index-one values regularized. The left side is $\sum_{m\in b+\mathbb{Z}} m^{-n}$ *over the whole lattice, in symmetric order at* $n=1$*, and the right side is* π^n *times a polynomial of degree* n *in* $\cot\pi b$ *with rational coefficients.*

(ii) Frequencies. *For* $0<b\le1$ *and every integer* $n\ge2$*, the Trailmoment* $(\mathsf{T}_{n;(1)}[x]\,\sin\big(2\pi bx-\frac{\pi(n-1)}{2}\big)\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *is* $\frac{(-1)^{n-1}(2\pi)^n}{2}$ *times the coefficient of* u^{n-1} *in* $e^{-bu}/(1-e^{-u})$*. The density is* $(-1)^{n/2}\cos(2\pi bx)$ *at even* n *and* $(-1)^{(n-1)/2}\sin(2\pi bx)$ *at odd* n*, and the value is* π^n *times a polynomial of degree* n *in* b *with rational coefficients.*

(iii) Periodic densities. *For integers* $0<r<q$ *and* $n\ge1$*,*

$$(\mathsf{T}_{n;(1)}[x]\,\big(\mathbb{I}_{r+q\mathbb{Z}}+(-1)^n\,\mathbb{I}_{-r+q\mathbb{Z}}\big))((0,\infty)) = \frac{(-1)^{n-1}}{q^n(n-1)!}\,\partial_b^{n-1}\big(\pi\cot\pi b\big)\Big|_{b=r/q},$$

at $n=1$ *with the points of the two lattices taken in increasing order.*

(iv) Folding. *At* $b=\frac12$ *the two lattices of (i) are one, and (i) reads*

$$\big(1+(-1)^n\big)(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty)) = \frac{(-1)^{n-1}}{(n-1)!}\,\partial_b^{n-1}(\pi\cot\pi b)\Big|_{b=1/2}:$$

at odd n *both sides vanish, and at even* $n=2j$*, with the lattice relation at the step* 2 *(Theorem 5.12(i)),*

$$(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \frac{-1}{2\,(2^{2j}-1)\,(2j-1)!}\ \partial_b^{2j-1}\big(\pi\cot\pi b\big)\Big|_{b=1/2}.$$

At $b = 1$, (ii) gives $(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (-1)^{j+1}\frac{(2\pi)^{2j}}{2}$ times the coefficient of u^{2j-1} in $1/(e^u-1)$, and the identity $0=0$ at odd n.

At an integer index n the kernel $\mathsf{Z}_{n;(1)}(x) = x^{-n}$ has the parity of n under $x \mapsto -x$. What closes, for every coset $b+\mathbb{Z}$ and every integer index, is the value of the whole coset, points of both signs: the reflection of the Gamma evaluates it, and at $n=2$ and $0<b<1$ it is $\Gamma_{\substack{4\\0}}\big[b,\ b,\ 1-b,\ 1-b\big]$ (Lemma 5.16). A Trailmoment reads the points beyond its cut only, so this relation gives it when the input, continued to the whole lattice, has the parity of the index: an even periodic density closes at even indices and an odd one at odd indices, by (iii), and on the coset $q\mathbb{Z}$ by (iv). On a lattice equal to its own reflection the pair of (i) folds onto one Trailmoment with the factor $1+(-1)^n$; at even n that Trailmoment closes, and at odd n the relation is the identity $0=0$ and carries no information. Statement (ii) is the same rule read on frequencies. The mismatched parities, the sine at an even index and the cosine at an odd index, are reached by neither statement on any coset or at any frequency; the counting input at an odd index is the cosine at frequency zero, and its Trailmoments are the ones that the positive words of Part VI carry.

5.6 Reflections of the index

Theorem 5.18 (Reflection through the poles of the Laplace transform)**.** *Let $\mathcal{F}$ be a function on $(0,\infty)$ that extends to a meromorphic function on $\mathbb{C}$ with no pole on $(0,\infty)$, such that*

(a) its poles other than $u=0$ are simple, with residue r_p at the pole p, and its pole at 0 has order $k_0 \geq 0$, with $k_0 = 0$ when there is none;

(b) there are radii $R_m \to \infty$ and a real A with $\sup_{|u|=R_m} |\mathcal{F}(u)| = O(R_m^A)$.

Write $\Sigma_m(\nu) = \sum_{0<|p|<R_m} r_p\,(-p)^{\nu-1}$, with $(-p)^{\nu-1} = \exp\big((\nu-1)\log(-p)\big)$ and $|\arg(-p)| < \pi$.

(i) On the strip. *If moreover $\mathcal{F}(u) = O(u^A)$ as $u \to +\infty$, then for $k_0 < \operatorname{Re}\nu < -A$ with $\nu \notin \mathbb{Z}$,*

$$\int_0^\infty u^{\nu-1}\mathcal{F}(u)\,du = \frac{\pi}{\sin\pi\nu}\lim_{m\to\infty}\Sigma_m(\nu).$$

(ii) Beyond the strip. *If $\mathcal{F} = \mathcal{F}_N$ is the Laplace transform of an input ϕ tame beyond the cut N, its poles lie in $\operatorname{Re} u \leq 0$, and for $\operatorname{Re}\nu < -A$ with $\nu \notin \{1,2,3,\dots\}$,*

$$(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty)) = \Gamma(1-\nu)\lim_{m\to\infty}\Sigma_m(\nu).$$

Reflection of the index replaces an input by the poles of its Laplace transform: each simple pole p becomes the point measure $r_p\,\delta_{-p}$, read by the kernel $\mathsf{Z}_{1-\nu;(1)}$. The relation is free in the input within (a) and (b), and pinned at one unit exponent with the coefficient function x; the mechanism is the Laplace side of the kernel $\mathsf{Z}_{\nu;(1)}$, where the index is the exponent of the scale and a full turn of the scale multiplies the integrand by $e^{2\pi i\nu}$. For the input $e^{-x}\,\mathbb{I}_{\mathbb{R}}$ cut at 0, whose Laplace transform $1/(1+u)$ has the one pole -1, statement (i) is the reflection of the Gamma (Proposition 1.3(v)); for a lattice input the strip of (i) is empty, and the reflection is a statement about the continued Trailmoment, which is (ii).

Theorem 5.19 (Reflection by inversion of the Laplace transform)**.** *Let ϕ and ψ be tame beyond the cut 0, with Laplace transforms $\mathcal{F}$ and $\mathcal{G}$, and suppose that for some $k \in \mathbb{R}$, $\lambda > 0$, $C \neq 0$ and $f_0, g_0 \in \mathbb{C}$,*

$$\mathcal{F}(u) + f_0 = C\,u^{-k}\Big(\mathcal{G}\big(\tfrac{\lambda}{u}\big) + g_0\Big) \qquad (u>0).$$

Then $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty))$*, defined for large* $\operatorname{Re}\nu$ *by the Laplace side, continues meromorphically to* $\mathbb{C}$*,*

$$\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty)) = \int_{\sqrt{\lambda}}^{\infty} u^{\nu-1}\mathcal{F}(u)\,du + C\lambda^{\nu-k}\int_{\sqrt{\lambda}}^{\infty} u^{k-\nu-1}\mathcal{G}(u)\,du + \frac{C\,g_0\,\lambda^{(\nu-k)/2}}{\nu-k} - \frac{f_0\,\lambda^{\nu/2}}{\nu},$$

whose only possible poles are simple, at $\nu = k$ *and* $\nu = 0$*. The same holds with* ψ *in the place of* ϕ*, and*

$$\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty)) = C\,\lambda^{\nu-k}\,\Gamma(k-\nu)\,(\mathsf{T}_{k-\nu;(1)}[x]\,\psi)((0,\infty)).$$

The inversion $u \mapsto \lambda/u$ of the scale is the reflection of the Gramma (Proposition 2.4) on the Laplace side. The relation is free in the two inputs, which need not be lattices and whose Laplace transforms need not be meromorphic; it uses one property of the pair, that the inversion of the scale carries the Laplace transform of one to that of the other. The constants f_0 and g_0 are the multiples of δ_0 that the two inputs would carry at the origin, where the kernel $\mathsf{Z}_{\nu;(1)}$ is not defined, and for $k \neq 0$ they give the residues of the continuation, $-f_0$ at $\nu = 0$ and Cg_0 at $\nu = k$.

Theorem 5.20 (Reflection at the coefficient function $x+\mu$)**.** *Let* ϕ*,* ψ*,* $\mathcal{F}$*,* $\mathcal{G}$*,* k*,* λ*,* C*,* f_0 *and* g_0 *be as in Theorem 5.19, and let* $\mu > 0$*. For* $\operatorname{Re} s > \max(k,0)$ *with* $\int_{(0,\infty)}(x+\mu)^{-\operatorname{Re} s}\,|\phi|(dx) < \infty$*,*

$$\Gamma(s)\Big[(\mathsf{T}_{s;(1)}[x+\mu]\,\phi)((0,\infty)) + f_0\,\mu^{-s}\Big] = C\,\lambda^{s-k}\int_{(0,\infty)} \mathsf{G}_{k-s;(1,-1)}(y,\lambda\mu)\,\psi(dy) + C\,g_0\,\Gamma(s-k)\,\mu^{k-s}.$$

The integral on the right converges for every $s \in \mathbb{C}$ *and is entire in* s*.*

5.7 Joint sources

Theorem 5.21 (Closure with the source kept joint)**.** *(i)* $\mathsf{Z}_{\nu;t}(\mathbf{c})\,\mathsf{Z}_{\eta;s}(\mathbf{d}) = \mathsf{Z}_{(\nu,\eta);\,t\oplus s}(\mathbf{c}\oplus\mathbf{d})$*, the configuration* $t \oplus s$ *acting on the disjoint union of the two sets of scales.*

(ii) A finite expression built from Zammas with positive indices by products and by nested reads against nonnegative measures is one Trailmoment on a source: its configuration is the concatenation $\oplus$ *of the factors' configurations, and its input is the joint measure* $\phi(dy)\,\psi_y(dw)$ *of the source variables, every shared variable, conditional dependence and cut retained. The same holds for complex data where every integral converges absolutely.*

Proposition 5.22 (The joint Laplace transform of Dirichlet inputs)**.** *Let* $U = (U_Q)_{Q\in\mathcal{P}}$ *have independent blocks* $U_Q \sim \operatorname{Dir}_{\mathbf{m}|_Q}$ *on a partition* $\mathcal{P}$ *of the scales, let* $P_1,\dots,P_q$ *be finite sums* $P_i(u) = \sum_{\mathbf{a}}[u^{\mathbf{a}}]P_i\,u^{\mathbf{a}}$ *over exponents* $\mathbf{a} \in [0,\infty)^{\mathcal{E}}$*, polynomials when the exponents are integers, and let* t *be a configuration of* q *columns on* d *scales and* $\nu \in \mathbb{C}^d$*. Then* $\mathcal{F}(v) = \mathbb{E}_{U\sim\bigotimes_Q \operatorname{Dir}_{\mathbf{m}|_Q}}\big[\exp\big(-\sum_i v_i P_i(U)\big)\big]$ *is entire and*

$$\mathcal{F}(v) = \sum_{\mathbf{k}\in\mathbb{N}^q} \frac{(-v)^{\mathbf{k}}}{\mathbf{k}!} M(\mathbf{k}),$$

$$M(\mathbf{k}) = \sum_{\mathbf{a}} \big[u^{\mathbf{a}}\big]\big(P_1(u)^{k_1}\cdots P_q(u)^{k_q}\big)\prod_e \Gamma_1\begin{bmatrix} m_e + a_e \\ m_e \end{bmatrix}\prod_Q \Gamma_1\begin{bmatrix} m_Q \\ m_Q + a_Q \end{bmatrix}, \qquad a_Q = \sum_{e\in Q} a_e;$$

for integer exponents the stacked Gammas are $\prod_e (m_e)_{a_e} / \prod_Q (m_Q)_{a_Q}$. *If every* P_i *is positive on the interior of the source and* $\operatorname{Re}\nu_j > 0$, *then, wherever the read converges,*

$$\mathbb{E}_{U\sim\bigotimes_Q \mathrm{Dir}_{\mathbf{m}|_Q}}\big[\mathsf{Z}_{\nu;t}\big(P_1(U),\dots,P_q(U)\big)\big] = \frac{1}{\prod_j \Gamma(\nu_j)} \int_{(0,\infty)^d} x^{\nu-\mathbf{1}}\, \mathcal{F}\big(x^{t_1},\dots,x^{t_q}\big)\, dx,$$

the Laplace side of the Trailmoment $(\mathsf{T}_{\nu;t}[(P_i)_i] \otimes_Q \mathrm{Dir}_{\mathbf{m}|_Q})\big(\prod_Q \Delta_Q\big)$. *For real coefficients, two choices of the data with equal joint moment arrays* M *give the vectors* $(P_i(U))_i$ *equal laws, hence equal reads of every kernel.*

5.8 Trails

Definition 5.23 (Trail data)**.** Take d scales, an index $\boldsymbol{\nu} \in \mathbb{C}^d$, the *trail columns* $I_d = (\mathbf{e}_1,\dots,\mathbf{e}_d)$, and *constant columns* $(t_k)_{k\in C}$ with constant coefficients $\lambda_k > 0$ (a constant column of coefficient zero is absent). Put

$$t = I_d \oplus (t_k)_{k\in C}, \qquad \mathbf{c}(y) = (y_1,\dots,y_d,(\lambda_k)_{k\in C}), \qquad X_N = \{N < y_1 < \cdots < y_d\}.$$

The *trail* at the cut N of an input ϕ on $\mathbb{R}^d$, carried where every trail coefficient is positive, is the Trailmoment of Definition 5.1 with this coefficient function and cut region,

$$(\mathsf{T}_{\boldsymbol{\nu};\, I_d\oplus(t_k)_{k\in C}}[(y_1,\dots,y_d,(\lambda_k)_{k\in C})]\,\phi)(X_N);$$

with no constant columns it is $(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\,\phi)(X_N)$. For lattices $\mathbb{L}_1,\dots,\mathbb{L}_d \subset \mathbb{R}$ the input $\mathbb{I}_{\mathbb{L}_1\times\cdots\times\mathbb{L}_d}$ is counting measure on their product, of which the cut region selects the points in X_N. The input is *given by increments* $\psi_1,\dots,\psi_d$, measures on $(0,\infty)$, when it is the image ϕ_N of $\bigotimes_j \psi_j$ under $s \mapsto y$, $y_i = N + s_1 + \cdots + s_i$; their Laplace transforms are $L_j(U) = \int e^{-sU}\psi_j(ds)$. By transport (Theorem 5.9) the trail of such an input is the Trailmoment on the increment source $(0,\infty)^d$ with the input $\bigotimes_j \psi_j$ and the coefficient function $s \mapsto (N+s_1,\dots,N+s_1+\cdots+s_d,(\lambda_k)_{k\in C})$, over the whole source: there the cut N is carried by the coefficient function.

Theorem 5.24 (The trail by its increments)**.** *Let the input be given by increments, put* $t = I_d \oplus (t_k)_{k\in C}$, *and let* $U_i = x_i + \cdots + x_d$ *be the sums of the scales from the* i*-th on,* $U_{d+1} = 0$. *Wherever the right side converges absolutely, with* $x_i = U_i - U_{i+1}$,

$$\begin{aligned}&(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N)\\ &\qquad = \int_{U_1>\cdots>U_d>0} \prod_{i=1}^{d} \frac{(U_i-U_{i+1})^{\nu_i-1}}{\Gamma(\nu_i)}\, \exp\Big(-\sum_{k\in C}\lambda_k x^{t_k}\Big) e^{-NU_1} \prod_{j=1}^{d} L_j(U_j)\, dU.\end{aligned}$$

Its relations hold wherever the reads in them converge, the derivatives under the integral being dominated:

(i) Moving the cut, *with the increments fixed:*

$$\partial_N(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N) = -\sum_i \nu_i\,(\mathsf{T}_{\boldsymbol{\nu}+\mathbf{e}_i;t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N).$$

(ii) Ladder.

$$\partial_{\lambda_k}(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N) = -\Gamma_d\begin{bmatrix}\boldsymbol{\nu}+t_k\\ \boldsymbol{\nu}\end{bmatrix}\,(\mathsf{T}_{\boldsymbol{\nu}+t_k;t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N).$$

(iii) Exchange, per scale.

$$(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N) = (\mathsf{T}_{\boldsymbol{\nu}+\mathbf{e}_i;t}[(y,(\lambda_k)_{k\in C})]\,y_i\phi_N)(X_N) + \frac{1}{\nu_i}\sum_{k\in C} t_{k,i}\lambda_k \Gamma_{\substack{d\\ d}}\begin{bmatrix}\boldsymbol{\nu}+t_k\\ \boldsymbol{\nu}\end{bmatrix}\,(\mathsf{T}_{\boldsymbol{\nu}+t_k;t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N).$$

(iv) Composition. *If every constant column is supported on one scale, partition C into sets C_i with $\mathrm{supp}(t_k)\subseteq\{i\}$ for $k\in C_i$, assigning every zero column to one scale, and put $t^{(i)} = (1)\oplus(t_{k,i})_{k\in C_i}$. Then*

$$\mathsf{Z}_{\boldsymbol{\nu};t}(y,(\lambda_k)_{k\in C}) = \prod_i \mathsf{Z}_{\nu_i;t^{(i)}}(y_i,(\lambda_k)_{k\in C_i}),$$

and the trail is the iterated read over $s_1,\dots,s_d>0$, in that order, of $\prod_i \mathsf{Z}_{\nu_i;t^{(i)}}(N+s_1+\cdots+s_i,(\lambda_k)_{k\in C_i})$ against $\bigotimes_i\psi_i$: each one-scale Trailmoment has the next read in its input. With counting increments, every ψ_j the counting input on the positive integers, no constant columns and $N\in b+\mathbb{Z}$, this is the nest of one-scale Trailmoments at successive running cuts, with indices $\nu_1,\dots,\nu_d$ and lattice input $\mathbb{I}_{b+\mathbb{Z}}$, and the trail is $(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\,\mathbb{I}_{(b+\mathbb{Z})^d})(\{N<y_1<\cdots<y_d\})$.

5.9 Products of Trailmoments

Theorem 5.25 (The Struffle)**.** *(i) Let ϕ,ψ be inputs on $\mathbb{R}$ with any indices, configurations and coefficient functions. If the product below converges with $|\phi|$, $|\psi|$ and the moduli of the kernels, then*

$$\begin{aligned}&(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,\infty))\,(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\psi)((N,\infty))\\ &\quad=\Big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\big[x\mapsto(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\psi)((x,\infty))\big]\phi\Big)((N,\infty))\\ &\qquad+\Big(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\big[x\mapsto(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((x,\infty))\big]\psi\Big)((N,\infty))\\ &\qquad+\big(\mathsf{T}_{(\nu,\nu');\,t\oplus t'}[\mathbf{c}\oplus\mathbf{c}']\,\phi\wedge\psi\big)((N,\infty)),\end{aligned}$$

where the diagonal input *is $\phi\wedge\psi=\sum_x\phi(\{x\})\psi(\{x\})\delta_x$, the sum over the points that both inputs charge, and the diagonal kernel is the Zamma of the concatenation $t\oplus t'$ (Theorem 5.21(i)).*

(ii) For inputs $\phi_1,\dots,\phi_r$ on $\mathbb{R}$ with kernel data $(\nu_i,t_i,\mathbf{c}_i)$, if $\prod_i(\mathsf{T}_{\nu_i;t_i}[\mathbf{c}_i]\,\phi_i)((N,\infty))$ converges with the total variations and the moduli of the kernels, it is the sum over the ordered partitions $(B_1,\dots,B_s)$ of $\{1,\dots,r\}$ into nonempty blocks of

$$\int_{N<x_1<\cdots<x_s}\prod_{j=1}^{s}\Big(\prod_{i\in B_j}\mathsf{Z}_{\nu_i;t_i}(\mathbf{c}_i(x_j))\Big)\Big(\bigwedge_{i\in B_j}\phi_i\Big)(dx_j),$$

with $\bigwedge_{i\in B}\phi_i=\sum_x\prod_{i\in B}\phi_i(\{x\})\delta_x$ for a block of at least two elements and ϕ_i for the block $\{i\}$.

The Struffle orders the points of the inputs along the line; the diagonal is there exactly when the inputs charge common points beyond the cut. It is free in the index, the configuration and

the coefficient function of both factors and in their inputs on the line; the cut region is pinned at one common cut. At one unit exponent with one coefficient function on both factors the diagonal kernel is $\mathsf{Z}_{\nu+\nu';(1)}$.

Theorem 5.26 (The Shruffle on a line)**.** *For a string $w = (w_1, \dots, w_n)$ of measurable functions on $(0,\infty)$ put* $\operatorname{chain}(w) = \int_{x_1>\dots>x_n>0} \prod_i w_i(x_i)\,dx$*, defined when* $\operatorname{chain}(|w|) < \infty$*.*

(i) If $\operatorname{chain}(|w^{(1)}|), \dots, \operatorname{chain}(|w^{(r)}|)$ *are finite,* $\prod_{i=1}^r \operatorname{chain}(w^{(i)}) = \sum_{\varsigma} \operatorname{chain}(\varsigma)$*, the sum over the $(n_1+\dots+n_r)!/(n_1!\cdots n_r!)$ interleavings of the r strings, n_i the length of $w^{(i)}$.*

(ii) Let a trail have its input given by nonnegative increments, with Laplace transforms $L_1, \dots, L_d$, and a cut $N \in \mathbb{R}$ such that the image of the first increment ψ_1 under $s \mapsto s+N$ is carried by $(0,\infty)$, as it is when $N \geq 0$; let it have no constant columns and positive integer indices $a_1, \dots, a_d$. Then, both sides finite or infinite together, with 1^a for a copies of the constant function 1, the trail is the chain integral of its string:

$$(\mathsf{T}_{\mathbf{a};I_d}[y]\,\phi_N)(X_N) = \operatorname{chain}\big(e^{-N\cdot}L_1,\ 1^{a_1-1},\ L_2,\ 1^{a_2-1},\ \dots,\ L_d,\ 1^{a_d-1}\big).$$

(iii) The product of finitely many such trails is the sum, over the interleavings of their strings, of trails of the same kind at the cut 0: every interleaving begins with a Laplace transform and is the string of the trail whose increments have the Laplace transforms of the interleaving in their order, each index being one more than the number of constant functions 1 that follow its Laplace transform, $e^{-N\cdot}L$ being the Laplace transform of the increment moved by N.

The Shruffle orders the scales on the Laplace side. It is free in the nonnegative increments, in the number of scales, and in the cut as long as the first point beyond it is positive; it is pinned at positive integer indices without constant columns, where $(U_i - U_{i+1})^{a-1}/\Gamma(a)$ is the volume of a chain of points. Where a nest of Trailmoments on the input side is a trail given by increments all equal to one measure at the cut 0, as for the counting input cut at 0, the Struffle and the Shruffle expand one product into the same members, and comparing the two expansions gives linear relations among the trails.

Theorem 5.27 (The sum over the splittings of an index)**.** *Let $b \in \mathbb{R}$, let N be a cut, and let $m_0 > 0$ be the first point of $b+\mathbb{Z}$ beyond N. For an integer $w \geq 3$,*

$$\sum_{a=1}^{w-2} (\mathsf{T}_{(a,w-a);I_2}[y]\,\mathbb{I}_{(b+\mathbb{Z})^2})(\{N < y_1 < \dots < y_2\}) = (\mathsf{T}_{w-1;(1)}[x]\,\eta\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)),$$

$$\eta(m) = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m-m_0,\infty)) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((m,\infty)) = \sum_{k\geq 1}\Big(\frac{1}{m-m_0+k} - \frac{1}{m+k}\Big),$$

the index-one values regularized. At every point m of the lattice beyond the cut,

$$\eta(m) - \frac{1}{m} = (\log\Gamma)'(m) - (\log\Gamma)'(m-m_0+1),$$

which is positive for $m_0 > 1$, zero for $m_0 = 1$ and negative for $m_0 < 1$. So the sum over the splittings is $(\mathsf{T}_{w;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty))$ if and only if $m_0 = 1$, that is, for the counting input at a cut in $[0,1)$; it is larger for $m_0 > 1$ and smaller for $m_0 < 1$. For the counting input cut at an integer $c \geq 0$, $m_0 = c+1$ and $\eta(m) = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m-c-1,m])$, a head of $c+1$ terms. For a lattice $b+q\mathbb{Z}$ of step $q > 0$ with a positive first point m_0 beyond the cut, the sum over $1 \leq a \leq w-2$ of the trails with the indices $(a, w-a)$ on its points is $(\mathsf{T}_{w;(1)}[x]\,\mathbb{I}_{b+q\mathbb{Z}})((N,\infty))$ if and only if $m_0 = q$.

Summing the trails on two scales over the splittings of an index telescopes in the inner variable, and what is left is one Trailmoment of index one less, with the density η. The index rises back by one exactly when $\eta(m) = \frac{1}{m}$ at every point, which is exactly when the first point beyond the cut is 1.

Theorem 5.28 (The Cremona of an increment)**.** *Let $L > 0$ be continuous on $(0,\infty)$ with $\int_x^\infty L < \infty$ for $x > 0$ and $\int_0^\infty L = \infty$, and let $\mathrm{tail}_L(x) = \int_x^\infty L(u)\,du$, the* tail function*, a decreasing bijection of $(0,\infty)$. For a string w whose entries are $L(x)\,dx$ or dx, let $\mathrm{chain}_L(w)$ be its chain integral over $x_1 > \cdots > x_n > 0$, largest variable first, and w^* the reversed string with $L(x)\,dx$ and dx swapped. Then $\mathrm{tail}_L \circ \mathrm{tail}_L = \mathrm{id}$ if and only if $L(\mathrm{tail}_L(x))\,L(x) = 1$ for every x, and in that case $\mathrm{chain}_L(w) = \mathrm{chain}_L(w^*)$ for every string, finite or infinite together.*

The Cremona uses one property of the increment: its tail function tail_L is an involution of the scale, so that the substitution $y = \mathrm{tail}_L(x)$ carries dx to $L(y)\,dy$ as well as $L(x)\,dx$ to dy. For the counting increments, $L(u) = 1/(e^u - 1)$ and the involution is the curve $e^{-x} + e^{-y} = 1$.

5.10 Convolutions of inputs

Definition 5.29 (The two convolutions)**.** Let ϕ and ψ be inputs carried by $[0,\infty)$, and $\phi \otimes \psi$ their product on $[0,\infty)^2$. The *additive convolution* $\phi \boxplus \psi$ is the image of $\phi \otimes \psi$ under $(x,y) \mapsto x + y$. When both inputs are carried by $(0,\infty)$, the *multiplicative convolution* $\phi \boxtimes \psi$ is the image of $\phi \otimes \psi$ under $(x,y) \mapsto xy$. Both are images in the sense of Definition 5.4: for inputs with complex densities they are used only in integrals that converge absolutely on the product $\phi \otimes \psi$. The multiplicative convolution of two σ-finite measures need not be σ-finite; it is used only where an integral of Theorem 5.30(ii) converges absolutely, and there it is σ-finite, its kernel being positive and finite on $(0,\infty)$. For lattice inputs with densities, $(f\,\mathbb{I}_\mathbb{Z}) \boxplus (g\,\mathbb{I}_\mathbb{Z})$ and $(f\,\mathbb{I}_\mathbb{Z}) \boxtimes (g\,\mathbb{I}_\mathbb{Z})$ denote the convolutions of the parts beyond the cut 0: for densities f and g on the positive integers, the first has the density $\sum_{j+k=n} f(j)g(k)$ at n and the second the density $\sum_{jk=n} f(j)g(k)$; likewise for lattice inputs on other cosets. For an input carried by $[0,\infty)$ write $\mathcal{F}_\phi(u) = \int_{[0,\infty)} e^{-ux}\,\phi(dx)$, the kernel being 1 at the point $x = 0$: it is the Laplace transform beyond every cut $N < 0$, and for an input carried by $(0,\infty)$ the Laplace transform beyond the cut 0.

Theorem 5.30 (Two convolutions)**.** *Let ϕ on Y_1 and ψ on Y_2 be inputs with coefficient functions c and c' and cut regions Y_1' and Y_2', and write $c + c'$ and $c\,c'$ for the functions $(y_1,y_2) \mapsto c(y_1) + c'(y_2)$ and $(y_1,y_2) \mapsto c(y_1)\,c'(y_2)$ on $Y_1 \times Y_2$.*

(i) The kernel $\mathsf{Z}_{1;(1,0)}(1,\cdot)$ on a sum. *If the two Trailmoments on the right converge,*

$$\big(\mathsf{T}_{1;(1,0)}[(1,c+c')]\ \phi \otimes \psi\big)(Y_1' \times Y_2') = (\mathsf{T}_{1;(1,0)}[(1,c)]\,\phi)(Y_1')\ (\mathsf{T}_{1;(1,0)}[(1,c')]\,\psi)(Y_2').$$

For inputs carried by $[0,\infty)$ and $u \in \mathbb{C}$ such that $e^{-x\,\mathrm{Re}\,u}$ is integrable for $|\phi|$ and for $|\psi|$, $\mathcal{F}_{\phi\boxplus\psi}(u) = \mathcal{F}_\phi(u)\,\mathcal{F}_\psi(u)$.

(ii) The kernel $\mathsf{Z}_{\nu;I_\mathcal{E}}$ on a product. *For coefficient functions $\mathbf{c}$ and $\mathbf{c}'$ with values in $(0,\infty)^\mathcal{E}$, $\mathbf{c}\,\mathbf{c}'$ their coordinatewise product and $\nu \in \mathbb{C}^\mathcal{E}$, if the two Trailmoments on the right converge,*

$$\big(\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}\,\mathbf{c}']\ \phi \otimes \psi\big)(Y_1' \times Y_2') = (\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}]\,\phi)(Y_1')\ (\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}']\,\psi)(Y_2').$$

For inputs carried by $(0,\infty)$,

$$(\mathsf{T}_{\nu;(1)}[x]\,\phi \boxtimes \psi)((0,\infty)) = (\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty))\ (\mathsf{T}_{\nu;(1)}[x]\,\psi)((0,\infty)),$$

$$\mathcal{F}_{\phi\boxtimes\psi}(u) = \int_{(0,\infty)} \mathcal{F}_{\psi}(ux)\,\phi(dx),$$

the first when its right side converges and the second when $\iint e^{-xy\,\mathrm{Re}\,u}\,|\phi|(dx)\,|\psi|(dy) < \infty$.

(iii) Converses. *Let g be a continuous complex-valued function on $(0,\infty)^{\mathcal{E}}$, not identically zero, with $g(s+s') = g(s)\,g(s')$ for all $s, s' \in (0,\infty)^{\mathcal{E}}$. Then $g(s) = e^{-\langle u,s\rangle}$ for one $u \in \mathbb{C}^{\mathcal{E}}$. Let h be a continuous complex-valued function on $(0,\infty)^{\mathcal{E}}$, not identically zero, with $h(x\,x') = h(x)\,h(x')$ for the coordinatewise product. Then $h = \mathsf{Z}_{\nu;I_{\mathcal{E}}}$ for one $\nu \in \mathbb{C}^{\mathcal{E}}$. The two hypotheses are the product relations of (i) and of (ii), for a kernel g or h in the place of the kernel there, on every pair of Dirac measures.*

The two convolutions add and multiply the points of two inputs, and each has one kernel that turns it into a product: on sums the kernel $\mathsf{Z}_{1;(1,0)}(1,\cdot)$ of a term at exponent zero beside a unit term, which is e^{-c}; on products the kernel $\mathsf{Z}_{\nu;I_{\mathcal{E}}}$ of the unit configuration, the power $\mathbf{c} \mapsto \mathbf{c}^{-\nu}$. Both statements are free in the two inputs, the two coefficient functions and the two cut regions, and each is pinned in its kernel; (iii) is the converse of that pin.

Chapter 6

Rational words and live inputs

A rational word acting on a Trailmoment through its input is moved, by partial fractions and the shift of the cut, onto Trailmoments of shifted inputs at the unit configuration; the pole columns of the word then separate what is a Trailmoment from what is a finite difference. An input is *live* when the values of its density are themselves read from the serif family, a Breta or a Gramma at a shifting slot, and the relations of that member give the input a recurrence of finite order. Every identity for rational words and live inputs is a finite identity first, at finite cuts; its infinite form uses the displayed convergence, or the common finite upper cut of the closing it comes from.

Definition 6.1 (Rational words and their transport)**.** A *rational word* is a function

$$R(x) = \sum_{k=0}^{k_{\max}} \sum_{j=1}^{J_k} c_{k,j}(x+k)^{-j},$$

kept with its partial-fraction presentation. For the data $(\nu, t, \mathbf{c}, \phi)$ of a Trailmoment on a line and the cut region (N, ∞), the rational word acts through the input,

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,(R\phi))((N,\infty)) = \int_{(N,\infty)} R(x)\,\mathsf{Z}_{\nu;t}(\mathbf{c}(x))\,\phi(dx).$$

Write $(\mathrm{Sh}^h f)(m) = f(m+h)$, $\Delta H(m) = H(m+1) - H(m)$, $\tau_k(x) = x + k$ and $\phi_k = (\tau_k)_*\phi$. On the counting input at index zero, where $\mathsf{Z}_{0;(1)}(x) = 1$, this is $(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} R(m)f(m)$ for an integer cut $N \geq 0$.

Theorem 6.2 (Shifted inputs)**.** *With* $g_k(y) = \mathsf{Z}_{\nu;t}(\mathbf{c}(y-k))f(y-k)$,

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,(Rf\phi))((N,\infty)) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(g_k\phi_k))((N+k,\infty)).$$

On the counting input at integer cuts $M > N \geq 0$*,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,M]) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(\mathrm{Sh}^{-k} f)\,\mathbb{I}_{\mathbb{Z}})((N+k,M+k]),$$

and when the Trailmoments converge individually

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(\mathrm{Sh}^{-k} f)\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)).$$

When only the combination converges, the common finite upper cut is kept and the limit is taken in the combination. For $f = \mathbf{1}$, $\sum_k c_{k,1} = 0$ and an integer $m_0 \geq 1$, with the index-one Trailmoments regularized,

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((m_0-1,\infty)) &= \sum_{k,s} c_{k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m_0-1+k,\infty)) \\ &= \sum_{s\geq 2}\sum_{k} c_{k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m_0-1+k,\infty)) - \sum_{k} c_{k,1}\, (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, m_0-1+k]) \\ &= \int_0^\infty \Big[\sum_{k,s} \frac{c_{k,s}}{\Gamma(s)} u^{s-1} e^{-ku}\Big] \frac{e^{-(m_0-1)u}}{e^u-1}\, du.\end{aligned}$$

Theorem 6.3 (Common cut and heads)**.** *For a common cut $N' \geq N + k_{\max}$,*

$$\begin{aligned}(\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) &= (\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + (\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N+k,N']),\end{aligned}$$

the head being $\sum_{N+k<\ell\leq N'} \ell^{-j} f(\ell-k)$ (Proposition 5.3). For $R = -g'$ with $g(a) = \sum_{h,s} c_{h,s}(a+h)^{-s} = O(a^{-2})$ and $C_s = \sum_h c_{h,s}$, under individual convergence,

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\, (-g')f\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) &= \sum_{s} s\, C_s\, (\mathsf{T}_{s+1;(1)}[x]\, f\, \mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + \sum_{h,s} s\, c_{h,s}\, (\mathsf{T}_{s+1;(1)}[x]\, (\mathrm{Sh}^{-h} f - f)\, \mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + \sum_{h,s} s\, c_{h,s}\, (\mathsf{T}_{s+1;(1)}[x]\, (\mathrm{Sh}^{-h} f)\, \mathbb{I}_{\mathbb{Z}})((N+h,N']).\end{aligned}$$

Theorem 6.4 (Pole columns are a rational difference quotient)**.** *Let $R(t) = \sum_{k=0}^{k_{\max}} \sum_{s=1}^{w} c_{k,s}(t+k)^{-s}$ with coefficients in a subfield $\mathbb{K}$ of $\mathbb{C}$, $C_s = \sum_k c_{k,s}$, and $H(t) = \sum_{k,s} c_{k,s} \sum_{j=0}^{k-1} (t+j)^{-s}$, a rational function with coefficients in $\mathbb{K}$. Then*

$$R(t) = \sum_{s=1}^{w} C_s t^{-s} + \Delta H(t),$$

and $R(t) = O(t^{-2})$ exactly when $C_1 = 0$. Conversely a difference ΔH of a proper rational function with poles on the nonpositive integer lattice has every column sum zero; modulo such differences, the rational words of maximum order w have the basis $t^{-1}, \ldots, t^{-w}$, and those that are $O(t^{-2})$ the basis $t^{-2}, \ldots, t^{-w}$. If the poles fall in several cosets of $\mathbb{Z}$, each coset has its own columns.

Theorem 6.5 (Summation by parts with a live input)**.** *Let H be a function on the integers above N and f a density on them.*

(i) $\Delta(Hf)(m) = \Delta H(m) f(m) + H(m+1)\Delta f(m)$, *so at finite cuts $M > N$*

$$\sum_{N<m\leq M} \Delta H(m) f(m) = H(M+1)f(M+1) - H(N+1)f(N+1) - \sum_{N<m\leq M} H(m+1)\Delta f(m).$$

(ii) If the upper boundary vanishes and the series converges, $\sum_{m>N} \Delta H(m) f(m) = -H(N+1)f(N+1) - \sum_{m>N} H(m+1)\Delta f(m)$.

(iii) For a Trailmoment read as a function of its cut, $\Delta\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((m,\infty)) = -(m+1)^{-p}g(m+1)$ *(Proposition 5.3), and hence*

$$\sum_{m>N}\Delta H(m)\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((m,\infty)) \\ = -H(N+1)\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)) + (\mathsf{T}_{p;(1)}[x]\,Hg\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)),$$

with one fewer Trailmoment in the input.

(iv) With R, C_s, H *as in Theorem 6.4 and the index-one Trailmoment regularized (Definition 5.2),*

$$\lim_{M\to\infty}\Big((\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N,N+M]) - C_1\log M\Big) = \sum_{s=1}^{w} C_s\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) - H(N+1);$$

when $C_1 = 0$*, that is when* $R = O(t^{-2})$*, the left side is* $(\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ *and the sum starts at* $s = 2$*. For a general density* f*, when the terms converge and the upper boundary vanishes,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{s=1}^{w} C_s\,(\mathsf{T}_{s;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) \\ - H(N+1)f(N+1) - \sum_{m>N} H(m+1)\Delta f(m).$$

Theorem 6.6 (Translation and the terminating descent)**.** *For the counting input, each inner Trailmoment below is read at its displayed running cut.*

(i) $(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y-k,\infty)) = (\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y,\infty)) + (\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y-k,y])$*, the head being* $\sum_{a=0}^{k-1}(y-a)^{-p}g(y-a)$ *(Proposition 5.3).*

(ii)

$$(\mathsf{T}_{j;(1)}[x]\,\big[x\mapsto(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((x-k,\infty))\big]\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) \\ = (\mathsf{T}_{j;(1)}[x]\,\big[x\mapsto(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((x,\infty))\big]\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) \\ + \sum_{a=0}^{k-1}(\mathsf{T}_{p;(1)}[x]\,(x+a)^{-j}g\,\mathbb{I}_{\mathbb{Z}})((N+k-a,\infty)).$$

(iii) For positive integers p, j*, each product* $z^{-p}(z+a)^{-j}$ *has finite rational partial fractions with poles at* 0 *and* $-a$ *only. The recursion therefore terminates on every finite nest of Trailmoments of the counting input, each taking the next as a function of its cut: it returns finitely many such nests, finite shifted copies of* g *and their heads, keeps the selected cuts, lowers at each correction the number of Trailmoments nested in the remaining input, and uses only rational operations on the pole data.*

(iv) If $R(-L-t) = -R(t)$*, then* $\sum_k c_{k,j} = 0$ *for every even* j*, and for a density* f *that is a nest of* d *Trailmoments of the counting input, at a common cut* C*,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{j}\Big(\sum_{k} c_{k,j}\Big)(\mathsf{T}_{j;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((C,\infty)) + E_d,$$

with E_d a sum of terms with at most d nested Trailmoments, so the terms with $d+1$ nested Trailmoments carry only odd outer indices, starting at three when $\sum_k c_{k,1} = 0$. For $f = \mathbf{1}$, an integer $m_0 \geq 1$ and $R(-k_{\max} - t) = -R(t)$, $c_{k_{\max}-k,s} = -(-1)^s c_{k,s}$, and at every even index s the Trailmoments combine into the finite sum

$$\sum_k c_{k,s}\Big[(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1 + k, \infty)) - (\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1, \infty))\Big] \\ = -\sum_k c_{k,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1, m_0 - 1 + k]).$$

Proposition 6.7 (Zamma kernels and general measures)**.** *For counting measure the product rule applies to the kernel times the density, $g(m) = \mathsf{Z}_{\nu;t}(\mathbf{c}(m))f(m)$, with $\Delta g(m) = \mathsf{Z}_{\nu;t}(\mathbf{c}(m+1))\Delta f(m) + f(m)\Delta[\mathsf{Z}_{\nu;t}(\mathbf{c}(m))]$. For a general measure μ restricted to $(N, M]$, $\int \Delta H\, g\, d\mu_{N,M} = \int H\, d\big((\tau_1)_*(g\mu_{N,M}) - g\mu_{N,M}\big)$.*

6.1 Live inputs

Theorem 6.8 (A Breta input with a recurrence of finite order)**.** *For $\operatorname{Re} a, \operatorname{Re} b > 0$, real $q_i < 1$ with $q_i \neq 0$, and $s_i \in \mathbb{C}$, let*

$$f(m) = \int_0^1 x^{m+a-1}(1-x)^{b-1} \prod_{i=1}^{r} (1 - q_i x)^{-s_i}\, dx \qquad (m \geq 0).$$

For $\operatorname{Re} s_i > 0$ this is a Breta in which m sits in the shape index: on the bunch $\{1, 2\}$ with the shape $u = (x, 1-x)$, the kernel terms ρu_1 and ρu_2 with unit coefficients and the letters $P_i(\rho, u) = \rho\big((1-q_i)u_1 + u_2\big)$, Proposition 3.6 at the scale index $1 + |\mathbf{s}|$ gives

$$f(m) = \Gamma(1 + |\mathbf{s}|)\, \mathsf{B}^{\mathcal{P}}_{((m+a,\,b),\,\mathbf{s}),\,(1+|\mathbf{s}|,\,\mathbf{s});\,(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

Then $f(m) = O(m^{-\operatorname{Re} b})$, and with $Q = \prod_i (1 - q_i x)$, $Q_i = Q/(1 - q_i x)$ and

$$B_m(x) = \big[(m+a)(1-x) - bx\big]Q(x) + x(1-x)\sum_i (s_i - 1) q_i Q_i(x) = \sum_{\ell=0}^{r+1} B_\ell(m) x^\ell,$$

the density obeys $\sum_{\ell=0}^{r+1} B_\ell(m) f(m+\ell) = 0$, with $B_0(m) = m + a$ and $B_{r+1}(m) = (-1)^{r+1}(\prod_i q_i)(m + a + b + r - \sum_i s_i)$. If $a + b + r - \sum_i s_i \notin -\mathbb{N}$, both extreme coefficients are nonzero for every $m \geq 0$, and all shifts reduce to $(f(m), \ldots, f(m+r))$ with coefficients rational in m.

Theorem 6.9 (A fixed set of input-dependent evaluations)**.** *With f, a, b, r, q_i and s_i as in Theorem 6.8, write in this statement $Q(x) = \prod_i (1 - q_i x)$, $C(x) = (1-x)Q(x)$, $\eta(x) = x^{a-1}(1-x)^{b-1}\prod_i (1 - q_i x)^{-s_i}$ and*

$$B(x) = \big[a(1-x) - bx\big]Q(x) + x(1-x)\sum_i (s_i - 1)\, q_i\, \frac{Q(x)}{1 - q_i x},$$

so that $(xC\eta)' = B\eta$; let B_ℓ, C_ℓ and Q_ℓ be the coefficients of x^ℓ.

(i) For each maximum index A, every shift reduces to $f(N+1),\dots,f(N+r+1)$ and the Trailmoments $(\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))=\sum_{m>N}f(m)(m+k)^{-p}$, $1\le p\le A$, $0\le k\le r$, together with $(\mathsf{T}_{p;(1)}[x+a]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ when a is an integer at least $r+1$.

(ii) For $k\ge r+1$ and $p\ge 2$,

$$\sum_{\ell=0}^{r+1}(B_\ell-kC_\ell)\,(\mathsf{T}_{p;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))+\sum_{\ell=0}^{r+1}C_\ell\,(\mathsf{T}_{p-1;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$$
$$=\sum_{\ell=0}^{r+1}\sum_{v=1}^{\ell}\Big[\frac{B_\ell-kC_\ell}{(N+v+k-\ell)^p}+\frac{C_\ell}{(N+v+k-\ell)^{p-1}}\Big]f(N+v),$$

and at $p=1$,

$$\sum_{\ell=0}^{r+1}(B_\ell-kC_\ell)\,(\mathsf{T}_{1;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$$
$$=\sum_{\ell=0}^{r+1}\sum_{v=1}^{\ell}\frac{B_\ell-kC_\ell}{N+v+k-\ell}f(N+v)-\sum_{\ell=0}^{r}Q_\ell f(N+1+\ell).$$

(iii) Cuts move by $(\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty))=(\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))-f(N+1)/(N+1+k)^p$ (Proposition 5.3).

(iv) For rational data, if c_* clears the coefficients of B, C and Q, then $d_{N+J}^{A}\prod_{r+1\le k\le J,\ k\ne a}|c_*(a-k)|^{A}$ clears all shifts through J and indices through A.

Theorem 6.10 (A Gramma input with a recurrence of finite order)**.** *Let a one-scale configuration t have positive integer exponents with greatest exponent L, the term at exponent L present. For $\lambda,c>0$ put $g(m)=\mathsf{G}_{\lambda+m;t}(\boldsymbol{\xi})/\Gamma(c+m)$. Then*

$$(m+\lambda)g(m)=\sum_{k}t_k\xi_k(m+c)_{t_k}\,g(m+t_k),$$

a shift relation of order L, the exchange relation of the Gramma (Theorem 2.3(ii)) divided by $\Gamma(c+m)$; for $L\ge 2$, $\log g(m)\le -(1-1/L)m\log m+O(m)$, so g decays faster than every inverse power.

Part II

Tables and families

Chapter 7

Tables

7.1 Characters, tables, lists

Definition 7.1 (Characters, tables, lists)**.** The *characters* are a finite set $\mathcal{E}$, whose elements are the *marks*; each mark e carries a positive variable α_e, and its name carries nothing. A *table* on $\mathcal{E}$ is a nonempty family $\mathcal{B}$ of r-element subsets of $\mathcal{E}$, the *bases*, with the *exchange property*: for bases T, T' and $a \in T \setminus T'$ there is $b \in T' \setminus T$ with $(T \setminus \{a\}) \cup \{b\}$ a basis. Each basis carries a weight $\theta_T > 0$; r is the *rank* and $r^* = n - r$ the *corank*. A *list* is a table whose weights are all 1; it is specified by its characters and its bases, $(\mathcal{E}, \mathcal{B})$, and by nothing else; for a set S and $0 \le k \le |S|$, $\binom{S}{k}$ is the list on S whose bases are the k-subsets of S. A mark in no basis is a *loop*, a mark in every basis a *coloop*. The *dual table* has the bases $\mathcal{E} \setminus T$ with the weights θ_T. The *Foreman polynomials* of a table are its basis and cobasis polynomials,

$$K(\alpha) = \sum_{T \in \mathcal{B}} \theta_T \, \alpha^T, \qquad \Psi(\alpha) = \sum_{T \in \mathcal{B}} \theta_T \, \alpha^{\mathcal{E} \setminus T} \qquad (\alpha \in \mathbb{R}^{\mathcal{E}}_{>0});$$

$K(\mathbf{1}) = \Psi(\mathbf{1}) = \sum_T \theta_T$ is the total weight. K and Ψ are the two letters of the table (Section 3.2): they enter a family word through powers (Definition 8.1). The *table's gauge* is $\alpha = \mathbf{1}$.

A table carries two descriptors of one object, what is there (the bases, read by K) and what is not (their complements, read by Ψ), and the marks on which both sit; any two of these determine the third, and the theory keeps both polynomials at every step.

Proposition 7.2 (One function read twice)**.** *For every table, $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$, and the Foreman polynomials of the dual table are $K^* = \Psi$, $\Psi^* = K$. Inverting the gauge swaps K and Ψ up to the factor $\alpha^{\mathcal{E}}$, and up to scale the gauges fixed by $\alpha \mapsto \alpha^{-1}$ are $\alpha \propto \mathbf{1}$.*

Definition 7.3 (Deletion and contraction)**.** For a mark e and $R = \mathcal{E} \setminus \{e\}$, the *deletion* has the bases $T \not\ni e$ and the *contraction* the bases $T \setminus \{e\}$, $T \ni e$, both on R with the inherited weights; the deletion is a table when e is not a coloop, and the contraction when e is not a loop. With $K_{\setminus e}, \Psi_{\setminus e}$ and $K_{/e}, \Psi_{/e}$ the sums over these bases, a sum without terms being 0,

$$K = K_{\setminus e} + \alpha_e K_{/e}, \qquad \Psi = \alpha_e \Psi_{\setminus e} + \Psi_{/e}.$$

7.2 Circuits and molecules

Definition 7.4 (Independence, circuits, linking, molecules)**.** A set $S \subseteq \mathcal{E}$ is *independent* in a table if it lies in a basis; its *rank* $r(S)$ is the largest size of an independent subset. A *circuit* is a

minimal set that is not independent, and a *cocircuit* is a minimal set that meets every basis. Two marks are *linked* if they lie in a common circuit of some table of a family on $\mathcal{E}$ (Definition 8.1). The *molecules* are the smallest nonempty sets of marks that contain, with each mark, every mark linked to it; a molecule with one mark is a *unit*, and the set of molecules is the *polycule*. A family whose marks form more than one molecule is a *compound*.

Definition 7.5 (Minors, restriction, contraction)**.** For disjoint $A, D \subseteq \mathcal{E}$ such that some basis contains A and avoids D, the *minor* contracting A and deleting D is the table on $\mathcal{E} \setminus (A \cup D)$ with the bases $T \setminus A$, $T \supseteq A$, $T \cap D = \varnothing$, and the weights θ_T; deletion and contraction of one mark are the minors with $D = \{e\}$ and with $A = \{e\}$. For $X \subseteq \mathcal{E}$ the *restriction* $\mathcal{B}|X$ is the table on X with the bases $T \cap X$, $|T \cap X| = r(X)$, and the *contraction* $\mathcal{B}/X$ is the table on $\mathcal{E} \setminus X$ with the bases $T \setminus X$ of the same T; when the weights of these T factor as $\theta_T = \theta'_{T\cap X}\theta''_{T\setminus X}$, $\mathcal{B}|X$ and $\mathcal{B}/X$ carry θ' and θ''. When e is not a loop, $\mathcal{B}/\{e\}$ is the contraction of e (Definition 7.3); at a loop $\mathcal{B}/\{e\}$ is the deletion of e, while $K_{/e} = \Psi_{/e} = 0$.

Lemma 7.6 (Linking read on the bases)**.** *Marks $e \neq f$ lie in a common circuit of a table exactly when some basis T has $e \in T$, $f \notin T$ and $(T \setminus \{e\}) \cup \{f\}$ a basis.*

Lemma 7.7 (The circuit lemma)**.** *If $w \in \mathbb{R}^{\mathcal{E}}$ has $\langle w, \mathbf{1}_T\rangle$ constant over the bases of a table, then $w_a = w_b$ for any two marks a, b linked by a circuit of that table. If $\langle w, \mathbf{1}_T\rangle$ is constant over the bases of every table of a family, w is constant on every molecule.*

Proposition 7.8 (Molecules split every table)**.** *Each molecule is a union of molecules of every table, those of a table being the molecules of that table alone, and every table is the* direct sum *of its restrictions to the molecules: its bases are exactly the unions of one basis of each restriction. When the weights factor across the molecules, $\theta_T = \prod_M \theta_M(T \cap \mathcal{E}_M)$, as they do for a list, the Foreman polynomials factor, $K(\alpha) = \prod_M K_M(\alpha_{\mathcal{E}_M})$ and $\Psi(\alpha) = \prod_M \Psi_M(\alpha_{\mathcal{E}_M})$.*

Theorem 7.9 (The Foreman polynomials of a linked table do not factor)**.** *If the circuits of a table link all of its $n \geq 2$ marks, then K and Ψ are irreducible polynomials.*

7.3 The Skeleton

Definition 7.10 (Skeleton counts)**.** The *cocircuit layer* of a table is its set of cocircuits and the *circuit layer* its set of circuits; the two layers together are the table's *Skeleton*. For $S \subseteq \mathcal{E}$, with $r^*(S) = |S| + r(\mathcal{E} \setminus S) - r$ the dual rank,

$$d(S) = |S| - r^*(S), \qquad d^*(S) = |S| - r(S), \qquad c(S) = |S| - d(S) - d^*(S), \qquad \varepsilon(S) = d(S) - d^*(S) :$$

$c(S)$ is the *width* and $\varepsilon(S)$ the *slant* of S.

Theorem 7.11 (The window)**.** *For every $S \subseteq \mathcal{E}$: $d(S) = \min_T |T \cap S|$, $d^*(S) = \min_T |S \setminus T|$, $r(S) = \max_T |T \cap S|$, and $\{|T \cap S| : T \in \mathcal{B}\} = \{d(S), d(S)+1, \ldots, r(S)\}$. Hence $c(S) = r(S) - d(S)$ is the width of this window. Moreover $c(\mathcal{E} \setminus S) = c(S)$ and $\varepsilon(S) + \varepsilon(\mathcal{E} \setminus S) = 2r - n$.*

The number of marks of S that a basis occupies ranges over a window of consecutive integers; its floor is $d(S)$, its width is the width $c(S)$, and the same window read on the complements has the floor $d^*(S)$. Every count that the integrals of a family need is read from these two layers.

Definition 7.12 (Signed indicators and the centred polynomial)**.** For $T \subseteq \mathcal{E}$ the *signed indicator* is $\chi_T = \mathbf{1}_T - \mathbf{1}_{\mathcal{E}\setminus T}$. For $\alpha = e^{2z}$ the *centred polynomial* is

$$\Lambda(z) = \sum_{T\in\mathcal{B}} \theta_T\, e^{\langle z,\chi_T\rangle}.$$

Theorem 7.13 (One function, two Foreman polynomials)**.** *For $\alpha = e^{2z}$:*

(i) $K(\alpha) = e^{\langle \mathbf{1},z\rangle}\Lambda(z)$ *and* $\Psi(\alpha) = e^{\langle \mathbf{1},z\rangle}\Lambda(-z)$*;*

(ii) the dual table has the centred polynomial $\Lambda^*(z) = \Lambda(-z)$*: duality and inversion of the gauge are* $z \mapsto -z$*;*

(iii) for every $v \in \mathbb{R}^{\mathcal{E}}$ *constant on each molecule of the table alone, with value* v_M *on the molecule* M*,* $\langle v, \chi_T\rangle = \sum_M v_M(2r(M) - |M|)$ *for every basis and*

$$\Lambda(z+v) = \exp\Big(\sum_M v_M\big(2r(M) - |M|\big)\Big)\Lambda(z);$$

these are the only v *for which* $\Lambda(z+\lambda v)/\Lambda(z)$ *does not depend on* z *for every* λ*, and at* $v = \lambda\mathbf{1}$ *the factor is* $e^{\lambda(2r-n)}$*.*

Theorem 7.14 (The inner moment generating function)**.** *For a gauge* $\alpha = e^{2z}$ *the* basis law *and the* cobasis law *at* α *are the laws on* $\mathcal{B}$

$$\mathbb{P}_\alpha(T) = \frac{\theta_T \alpha^T}{K(\alpha)} = \frac{\theta_T e^{\langle z,\chi_T\rangle}}{\Lambda(z)}, \qquad \mathbb{P}^*_\alpha(T) = \frac{\theta_T \alpha^{\mathcal{E}\setminus T}}{\Psi(\alpha)} = \mathbb{P}_{\alpha^{-1}}(T).$$

Then

$$\frac{\Lambda(z+w)}{\Lambda(z)} = \mathbb{E}_{T\sim\mathbb{P}_\alpha}\big[e^{\langle w,\chi_T\rangle}\big], \qquad \nabla \log\Lambda(z) = \mathbb{E}_{T\sim\mathbb{P}_\alpha}[\chi_T] = 2R(\alpha) - \mathbf{1},$$

where $R_e(\alpha) = \alpha_e\partial_e \log K(\alpha) = \mathbb{P}_\alpha(e \in T)$ *is the* occupation*, and the derivatives of* $\log\Lambda$ *of order* k *at* z *are the joint cumulants of order* k *of* χ_T *under* $\mathbb{P}_\alpha$*. The* even *and* odd statistics $E(z) = \log\Lambda(z) + \log\Lambda(-z)$ *and* $O(z) = \log\Lambda(z) - \log\Lambda(-z)$ *satisfy* $e^{E(z)} = K\Psi/\alpha^{\mathcal{E}}$*,* $e^{O(z)} = K/\Psi$*, and, for every* $w \in \mathbb{R}^{\mathcal{E}}$*,*

$$e^{E(z+w)} = e^{E(z)}\,\mathbb{E}_{(T,T')\sim\mathbb{P}_\alpha\otimes\mathbb{P}^*_\alpha}\big[e^{\langle 2w,\mathbf{1}_T - \mathbf{1}_{T'}\rangle}\big]:$$

$E(z+\cdot) - E(z)$ *is the cumulant generating function of* $\chi_T - \chi_{T'}$ *for independent* $T \sim \mathbb{P}_\alpha$ *and* $T' \sim \mathbb{P}^*_\alpha$*; at the table's gauge* $e^{E(0)} = K(\mathbf{1})^2$*. For table* j *of a family the two laws at* α *are* $\mathbb{P}_{j,\alpha}$ *and* $\mathbb{P}^*_{j,\alpha} = \mathbb{P}_{j,\alpha^{-1}}$*, the occupation is* $R_j(\alpha)$*, and the* dual occupation $R^*_j(\alpha)$*, with* $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T) = 1 - R_{j,e}(\alpha^{-1})$*, is the occupation of the dual table. At the table's gauge the two laws coincide,* $\mathbb{P}^*_{j,\mathbf{1}} = \mathbb{P}_{j,\mathbf{1}}$*.*

The centred polynomial is a moment generating function: its terms at z, divided by $\Lambda(z)$, are the law of a random basis, and its logarithm generates the cumulants of the signed indicator. The basis law and the cobasis law at a gauge are one law read at α and at α^{-1}, and at every gauge the product of the two evaluations generates the difference of a basis drawn from the basis law and an independent basis drawn from the cobasis law.

Theorem 7.15 (The Skeleton is the Newton data of Λ)**.** *For every* $S \subseteq \mathcal{E}$ *the values* $\langle \mathbf{1}_S, \chi_T\rangle$*,* $T \in \mathcal{B}$*, fill* $\{2j - |S| :\ d(S) \le j \le r(S)\}$*. The Newton polytope of* Λ *is* $2P - \mathbf{1}$ *with* $P = \mathrm{conv}\{\mathbf{1}_T\}$ *the base polytope; its width in the direction* $\mathbf{1}_S$ *is* $2c(S)$ *and the midpoint of its extent in that direction is* $\varepsilon(S)$*.*

Proposition 7.16 (Loops and coloops)**.** *If e is a loop, $\Lambda(z) = e^{-z_e}\Lambda'(z|_{\mathcal{E}\setminus e})$ with Λ' the centred polynomial of the same bases on $\mathcal{E}\setminus e$; if e is a coloop, $\Lambda(z) = e^{z_e}\Lambda'(z|_{\mathcal{E}\setminus e})$ with Λ' that of the bases $T\setminus e$.*

7.4 Multivectors of any tensor rank

Definition 7.17 (Multivectors and tensor rank)**.** Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. A *multivector* of a table of rank r is an $\boldsymbol{\omega} \in \bigwedge^r \mathbb{F}^{\mathcal{E}}$ with $|\boldsymbol{\omega}_T|^2 = \theta_T$ for $T \in \mathcal{B}$ and $\boldsymbol{\omega}_T = 0$ otherwise; $\boldsymbol{\omega}_T = \sqrt{\theta_T}$ is one. Its *tensor rank* $\mathrm{rk}(\boldsymbol{\omega})$ is the least number of decomposable elements of $\bigwedge^r \mathbb{F}^{\mathcal{E}}$ with sum $\boldsymbol{\omega}$, written $\boldsymbol{\omega} = \sum_{k=1}^{\mathrm{rk}(\boldsymbol{\omega})} \boldsymbol{\omega}^{(k)}$, $\boldsymbol{\omega}^{(k)}_T = \det A^{(k)}_T$ with *slices* $A^{(k)} \in \mathbb{F}^{r\times n}$. The table is *tensorminental of tensor rank q* over $\mathbb{F}$ when q is the least tensor rank of its multivectors, and *determinantal* when $q = 1$. The *dual multivector* is the Hodge complement $\boldsymbol{\omega}^* \in \bigwedge^{n-r} \mathbb{F}^{\mathcal{E}}$, $|\boldsymbol{\omega}^*_{\mathcal{E}\setminus T}| = |\boldsymbol{\omega}_T|$, and $\mathrm{rk}(\boldsymbol{\omega}^*) = \mathrm{rk}(\boldsymbol{\omega})$.

Every table has multivectors, and every multivector has a tensor rank; tensor rank one, the determinantal case, is where a single matrix carries the table, and it is one case of the statements on multivectors, which hold at every tensor rank.

Proposition 7.18 (The letters of a tensorminental table)**.** *For a decomposition $\boldsymbol{\omega} = \sum_k \boldsymbol{\omega}^{(k)}$ into decomposable elements with slices $A^{(k)}$ and $D = \mathrm{diag}(\alpha)$,*

$$K(\alpha) = \sum_{k,l} \det\big(A^{(k)} D A^{(l)\mathsf{H}}\big) = \mathbf{1}^{\mathsf{H}}\mathbf{G}(\alpha)\mathbf{1}, \qquad \mathbf{G}_{kl}(\alpha) = \det\big(A^{(k)} D A^{(l)\mathsf{H}}\big),$$

and Ψ is the same expression for the slices of the dual multivector. $\mathbf{G}(\alpha)$ is the Gram matrix of the $\boldsymbol{\omega}^{(k)}$ for $\langle \eta, \xi\rangle_\alpha = \sum_T \alpha^T \eta_T \overline{\xi_T}$, Hermitian and positive semidefinite.

Theorem 7.19 (Tensor rank is hereditary)**.** *For disjoint $A, D \subseteq \mathcal{E}$ with A extending to a basis avoiding D, $\boldsymbol{\omega}_{A|D} := (\mathbf{e}_A \lrcorner \boldsymbol{\omega})|_{\mathcal{E}\setminus(A\cup D)}$, the restriction of the interior product of $\boldsymbol{\omega}$ with the wedge $\mathbf{e}_A$ of the unit vectors of the marks of A, $(\mathbf{e}_A \lrcorner \boldsymbol{\omega})_X = \pm\boldsymbol{\omega}_{X\cup A}$, is a multivector of the minor that contracts A and deletes D, with the weight $\theta_{X\cup A}$ on each basis X of the minor, and $\mathrm{rk}(\boldsymbol{\omega}_{A|D}) \le \mathrm{rk}(\boldsymbol{\omega})$.*

Theorem 7.20 (The pattern law of a tensorminental table is a finite interference)**.** *Let the table have a multivector of tensor rank q with slices $A^{(1)}, \dots, A^{(q)}$, let $B \subseteq \mathcal{E}$ and $R = \mathcal{E}\setminus B$, and let α be a gauge at which every $M_{kl} = A^{(k)} D_\alpha A^{(l)\mathsf{H}}$, $D_\alpha = \mathrm{diag}(\alpha)$, with $\mathbf{G}_{kl} \not\equiv 0$ is invertible; pairs with $\mathbf{G}_{kl} \equiv 0$ as a polynomial in the gauge are omitted (Proposition 7.18). For $x \in \mathbb{C}^B$, with $x \circ \alpha_B = (x_e\alpha_e)_{e\in B}$,*

$$\mathbb{E}_{T\sim\mathbb{P}_\alpha}\Big[\prod_{e\in B} x_e^{\mathbf{1}[e\in T]}\Big] = \frac{K(x\circ\alpha_B, \alpha_R)}{K(\alpha)} = \sum_{k,l} w_{kl} \det\big(I + (D_x - I)\Pi^{(kl)}_{BB}\big),$$

where $\mathbb{P}_\alpha$ is the basis law at α, $D_x = \mathrm{diag}(x)$, $w_{kl} = \mathbf{G}_{kl}(\alpha)/K(\alpha)$ and $\Pi^{(kl)} = A^{(l)\mathsf{H}} M_{kl}^{-1} A^{(k)} D_\alpha$. Here $\sum_{k,l} w_{kl} = 1$, $w_{lk} = \overline{w_{kl}}$, every $\Pi^{(kl)}$ is idempotent, and $\Pi^{(kk)}$ is the D_α-orthogonal projection onto the row space of $A^{(k)}$. The hypothesis holds off a proper algebraic set of gauges.

Corollary 7.21 (Moments and cumulants)**.** *In the setting of Theorem 7.20:*

(i) $\mathbb{P}_\alpha(S \subseteq T) = \sum_{k,l} w_{kl} \det \Pi^{(kl)}_{SS}$;

(ii) the law of $T \cap B$ is the w-mixture of q^2 determinantal components, whose cumulants are $\kappa^{(kl)}(e) = \Pi_{ee}^{(kl)}$ and, for $p \geq 2$ distinct marks,

$$\kappa^{(kl)}(e_1, \dots, e_p) = (-1)^{p-1} \sum_{\varsigma} \prod_{i=1}^{p} \Pi^{(kl)}_{e_{\varsigma(i)} e_{\varsigma(i+1)}},$$

the sum over the cyclic orders ς of $\{1, \dots, p\}$, with $\varsigma(p+1) = \varsigma(1)$; the joint cumulants κ_α of the indicators $X_e = \mathbf{1}[e \in T]$ under $\mathbb{P}_\alpha$ (Definition 9.2) are their total cumulants over the mixing index: for distinct marks $e_1, \dots, e_p$ of B,

$$\kappa_\alpha(e_1, \dots, e_p) = \sum_{\mathfrak{p}} \kappa_w\Big(\big(\kappa^{(kl)}(e_C)\big)_{C \in \mathfrak{p}} \Big),$$

the sum over the partitions $\mathfrak{p}$ of $\{1, \dots, p\}$ into nonempty blocks, $e_C = (e_c)_{c \in C}$, where for finitely many functions $(Y_i)_{i \in I}$ of the index (k, l)

$$\kappa_w\big((Y_i)_{i \in I}\big) = \sum_{\mathfrak{q}} (-1)^{|\mathfrak{q}|-1} (|\mathfrak{q}| - 1)! \prod_{D \in \mathfrak{q}} \sum_{k,l} w_{kl} \prod_{i \in D} Y_i(k, l),$$

the sum over the partitions $\mathfrak{q}$ of I, is their joint cumulant over the index with the mixing coefficients w_{kl}; at one mark this is the occupation $R_e(\alpha) = \sum_{k,l} w_{kl} \Pi_{ee}^{(kl)}$. This is an identity of generating functions, valid for complex mixing coefficients summing to one.

A table with a multivector of tensor rank q reads the law of its patterns as an interference of q^2 determinantal components, mixed with complex coefficients that sum to one. At tensor rank one there is a single component and the mixture is a determinantal law; at every higher tensor rank the joint cumulants over the mixing index of the components' cumulants are the part that no single component carries.

Chapter 8

Families

8.1 The interest and its walls

Definition 8.1 (Family, interest, walls)**.** A *family* on $\mathcal{E}$ is a finite sequence $F = \big((\mathcal{B}_j, \theta_j);\, s_j, t_j\big)_{j=1}^{N}$ of tables on $\mathcal{E}$ with exponents $s_j, t_j \in \mathbb{C}$; $K_j, \Psi_j, \Lambda_j, E_j, O_j$, r_j and $d_j, d_j^*, c_j, \varepsilon_j$ belong to table j. A table is *active* when its exponents are not both zero, and a letter when its exponent is not zero. For *masses* $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ the *interest* is

$$J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j}\, d\alpha,$$

with its slots: the family's tables, the masses $\mathbf{m}$ and the exponents $\mathbf{s} = (s_j)_j$, $\mathbf{t} = (t_j)_j$. The *family word* $W = \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ is homogeneous of degree $-\sum_j (s_j r_j + t_j r_j^*)$, and with the measure $d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m}-\mathbf{1}} d\alpha$ (Definition 1.1(vii)), $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int W\, e^{-|\alpha|} d\alpha_{\mathbf{m}}$. The *pair coordinates* are the *strengths* $\sigma_j = \frac{s_j + t_j}{2}$ and the leans $\delta_j = \frac{s_j - t_j}{2}$ (Definition 9.1), with the total strength $\sigma = \sum_j \sigma_j$, so that $s_j = \sigma_j + \delta_j$ and $t_j = \sigma_j - \delta_j$; the *offsets* are $\boldsymbol{\mu} = \mathbf{m} - \sigma \mathbf{1}$. The *walls* and *top walls* are

$$\kappa_S = m_S - \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big),$$

$$\kappa_S^{\mathrm{top}} = \sum_j \big(s_j r_j(S) + t_j(|S| - d_j(S))\big) - m_S \qquad (\varnothing \neq S \subseteq \mathcal{E}).$$

The *imbalance* is $\kappa_{\mathcal{E}} = m_{\mathcal{E}} - \sum_j (s_j r_j + t_j r_j^*)$, and the *balance wall* is the hyperplane $\kappa_{\mathcal{E}} = 0$. On it the data are fixed by their pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$, with $\boldsymbol{\mu}_\perp = \boldsymbol{\mu} - \frac{\mu_{\mathcal{E}}}{n}\mathbf{1}$ the projection of the offsets to $\mathbf{1}^\perp$: the exponents are $(\boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta})$ and the masses are the *balanced masses*

$$\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}) = \boldsymbol{\mu}_\perp + \Big(\sigma + \frac{1}{n} \sum_j \delta_j (2r_j - n)\Big)\mathbf{1},$$

linear in $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$. F is *connected* if every nonempty proper S has $c_j(S) > 0$ for some active table j. A family of one table is named by its table: for a table $\mathcal{A}$ with its weights $\theta^{\mathcal{A}}$, every object of the family $((\mathcal{A}, \theta^{\mathcal{A}}); s, t)$ carries $\mathcal{A}$ in place of F, as $J_{\mathcal{A}}(\mathbf{m}; s, t)$. A table may carry two systems of positive weights, θ^K for its letter K and θ^Ψ for its letter Ψ; at (s, t) it names the family of the two tables $((\mathcal{B}, \theta^K); s, 0)$ and $((\mathcal{B}, \theta^\Psi); 0, t)$, whose word is $K_{\theta^K}^{-s} \Psi_{\theta^\Psi}^{-t}$.

The exponents of a table enter in two coordinates: σ_j, how much the family weighs the table, and δ_j, which side it leans to, toward the bases (K) or toward their complements (Ψ). The pair $(K_j \Psi_j, K_j / \Psi_j)$ carries σ_j and δ_j separately, and the walls are read in either coordinates.

Theorem 8.2 (The interest is a partition function in the pair coordinates)**.** *(i) In the offsets,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\boldsymbol{\mu}-\mathbf{1}} e^{-|\alpha|} \prod_j \Lambda_j(\tfrac{1}{2}\log\alpha)^{-s_j}\Lambda_j(-\tfrac{1}{2}\log\alpha)^{-t_j}\, d\alpha.$$

(ii) In signed coordinates,

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = 2^n \int_{\mathbb{R}^{\mathcal{E}}} \exp\Big(2\langle\boldsymbol{\mu},z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big) - \sum_e e^{2z_e}\Big)\, dz,$$

the partition function of the exponential family with natural parameters $(\boldsymbol{\mu};\sigma_j,\delta_j)$ and sufficient statistics $(2z;-E_j,-O_j)$.

(iii) $\kappa_S = \mu_S + \sum_j \big(\sigma_j\, c_j(S) - \delta_j\, \varepsilon_j(S)\big)$ *and* $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$.

Theorem 8.3 (The interest is a Gramma; its letter and mark relations)**.** *For* $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$*, on the scales* $\mathcal{E} \sqcup \{v_j\} \sqcup \{w_j\}$ *let* t_F *be the configuration of the columns* $(\mathbf{e}_e;0;0)$ *with coefficient* 1*, and* $(\chi_T/2;\mathbf{e}_j;0)$*,* $(-\chi_T/2;0;\mathbf{e}_j)$ *with coefficient* $\theta_{j,T}$ *for* $T \in \mathcal{B}_j$*. Then*

$$\prod_j \Gamma(s_j)\Gamma(t_j)\, J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \mathsf{G}_{(\boldsymbol{\mu};\mathbf{s};\mathbf{t});t_F}(\mathbf{1}\oplus\boldsymbol{\theta}\oplus\boldsymbol{\theta}),$$

either side converging with the other. As identities of meromorphic functions:

(i) (letter relations) for each j*,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_{T\in\mathcal{B}_j} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t}) = \sum_{T\in\mathcal{B}_j} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s},\mathbf{t}+\mathbf{e}_j);$$

(ii) (mark relations) for each e*,*

$$m_e J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = J_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \sum_j s_j \sum_{T\ni e} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t}) + \sum_j t_j \sum_{T\not\ni e} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s},\mathbf{t}+\mathbf{e}_j);$$

(iii) the same for the period $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *of Theorem 8.10, with* $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}})P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*: the letter relations hold verbatim, and in the mark relation the pure mass shift carries the factor* $\kappa_{\mathcal{E}}$*,* $m_e P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \kappa_{\mathcal{E}} P_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \cdots$.

The interest is not a new kind of object: it is one Gramma, on the marks and one scale for each letter, and its relations are that Gramma's exchange relations. The letter relation is the exchange relation on a letter's scale, and the mark relation the exchange relation on a mark's scale, rewritten with the letter relations.

Proposition 8.4 (The interest over latent vectors)**.** *Let every table* j *of a family be determinantal over* $\mathbb{F}_j \in \{\mathbb{R},\mathbb{C}\}$ *(Definition 7.17), put* $\beta_j = \dim_{\mathbb{R}} \mathbb{F}_j$*, and let* $2s_j/\beta_j$ *and* $2t_j/\beta_j$ *be nonnegative integers. Let* $V_j \subseteq \mathbb{F}_j^{\mathcal{E}}$ *be the row space of a slice of table* j *and* $V_j^{\perp}$ *its orthogonal complement, each with its Lebesgue measure, and for the* latent vectors $x_{j,i} \in V_j$*,* $1 \le i \le 2s_j/\beta_j$*, and* $x^*_{j,i} \in V_j^{\perp}$*,* $1 \le i \le 2t_j/\beta_j$*, put* $\mathfrak{q}_e = \sum_j \big(\sum_i |(x_{j,i})_e|^2 + \sum_i |(x^*_{j,i})_e|^2\big)$*. Then*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \frac{\prod_e \Gamma(m_e)}{\prod_j K_j(\mathbf{1})^{s_j+t_j}\pi^{r_j s_j + r_j^* t_j}} \int \prod_{e\in\mathcal{E}} \big(1+\mathfrak{q}_e\big)^{-m_e} \prod_{j,i} dx_{j,i}\, dx^*_{j,i},$$

both sides finite or infinite together. Where they are finite, the Breta law $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *(Definition 16.7) is the law of* α *under the joint law of* α *and the latent vectors with density proportional to* $\alpha^{\mathbf{m-1}}\exp\big(-\sum_e \alpha_e(1+\mathfrak{q}_e)\big)$*, under which, given the latent vectors, the scales* α_e *are independent and* $\alpha_e(1+\mathfrak{q}_e)$ *has the law* Gamma_{m_e}*. Over* $\mathbb{R}$ *with* $s_j = t_j$ *the sums* $u_{j,i} = x_{j,i} + x^*_{j,i}$ *run over* $\mathbb{R}^{\mathcal{E}}$*, with* $|(x_{j,i})_e|^2 + |(x^*_{j,i})_e|^2 = (\mathrm{pr}_j\, u_{j,i})_e^2 + \big((I-\mathrm{pr}_j)u_{j,i}\big)_e^2$*,* pr_j *the orthogonal projection onto* V_j*.*

The letters of a determinantal table open as Gaussian integrals: its basis polynomial is a Gram determinant, whose inverse powers are Gaussian integrals over the row space of its slice, real or complex, and its cobasis polynomial is the same over the orthogonal complement, each letter with as many latent vectors as its exponent asks; over $\mathbb{R}$ at equal exponents the two pair into Gaussian vectors on $\mathbb{R}^{\mathcal{E}}$ read through the two projections. The integrals over the scales are then Gamma integrals, and the interest becomes a rational integral in the latent vectors.

Lemma 8.5 (Greedy bases on a sector)**.** *Let* π *be a total order of* $\mathcal{E}$ *with tails* $U_k = \{\pi(k),\dots,\pi(n)\}$*. There are bases* $T^{\pm}_\pi$ *with* $|T^+_\pi \cap U_k| = d(U_k)$ *and* $|U_k \setminus T^-_\pi| = d^*(U_k)$ *for all* k*, and on the sector* $H_\pi = \{\alpha_{\pi(1)} \ge \cdots \ge \alpha_{\pi(n)}\}$ *every basis has* $\alpha^T \le \alpha^{T^+_\pi}$ *and* $\alpha^{\mathcal{E}\setminus T} \le \alpha^{\mathcal{E}\setminus T^-_\pi}$*. For a nonempty set of* r*-element subsets of* $\mathcal{E}$ *this conclusion holds exactly when the set has the exchange property.*

The sectors are where the exchange property is used: on each sector one basis dominates every monomial of K and one cobasis every monomial of Ψ, and that is what gives the walls their form. What does not use the exchange property is the opening of every monomial by its shares (Corollary 4.4).

Lemma 8.6 (Cube integrals)**.** *Let* $h(q,\lambda)$ *be smooth in* q *near* $[0,1]^m$*, with every* q*-derivative continuous in* (q,λ) *and holomorphic in* λ *on an open set. Then* $\mathrm{cube}(x,\lambda;h) = \int_{[0,1]^m}\prod_k q_k^{x_k-1}h(q,\lambda)\,dq$*, defined for* $\mathrm{Re}\,x_k > 0$*, extends meromorphically to* $\mathbb{C}^m$*, jointly in* λ*, with poles on the hyperplanes* $x_k = -j$*,* $j \in \mathbb{N}$*: with* $h = \sum_{j<M} h_j(q',\lambda)q_1^j + q_1^M r_M$*,*

$$\mathrm{cube}(x,\lambda;h) = \sum_{j<M}\frac{\mathrm{cube}(x',\lambda;h_j)}{x_1+j} + \mathrm{cube}\big((x_1+M,x'),\lambda;r_M\big),$$

repeated in each coordinate.

Theorem 8.7 (Existence and continuation)**.** *(i)* $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *converges if and only if* $\mathrm{Re}\,\kappa_S > 0$ *for every nonempty* $S \subseteq \mathcal{E}$*, whatever* $\mathbf{s},\mathbf{t} \in \mathbb{C}^N$*; it is holomorphic there.*

(ii) On that region

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}})\sum_\pi \int_{[0,1]^{n-1}}\prod_{k=2}^n q_k^{\kappa_{U_k}-1}h_\pi(q)\,dq, \qquad h_\pi = L_\pi^{-\kappa_{\mathcal{E}}}\prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j},$$

where in the sector coordinates $\alpha_{\pi(1)} = \rho$*,* $\alpha_{\pi(k)} = \rho q_2\cdots q_k$ *one has*

$$K_j = \rho^{r_j}\prod_k q_k^{d_j(U_k)}k_{j,\pi}(q), \qquad \Psi_j = \rho^{n-r_j}\prod_k q_k^{d^*_j(U_k)}\psi_{j,\pi}(q), \qquad |\alpha| = \rho L_\pi(q);$$

the sector letters $k_{j,\pi},\psi_{j,\pi},L_\pi$ *have nonnegative coefficients and positive constant terms* $\theta_{j,T^+_{j,\pi}}$*,* $\theta_{j,T^-_{j,\pi}}$*,* $T^{\pm}_{j,\pi}$ *the greedy bases of table* j *(Lemma 8.5), and* 1*, and* $L_\pi \in [1,n]$*.*

(iii) $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *extends to a meromorphic function of* $(\mathbf{m},\mathbf{s},\mathbf{t})$ *with poles on the hyperplanes* $\kappa_S = -j$, $S \neq \varnothing$, $j \in \mathbb{N}$*; identities valid on an open subset of the region persist.*

Corollary 8.8 (Past several walls at once)**.** *For an order* π *write* $\kappa_k = \kappa_{U_k}$ *and let*

$$I_\pi = \int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_k - 1} h_\pi(q)\, dq = \int_{[0,\infty)^{n-1}} e^{-\sum_k \kappa_k u_k} h_\pi(e^{-u})\, du$$

be the term of Theorem 8.7(ii), written also in the gaps $u_k = -\log q_k$*; in the log-ratios* $\eta_k = \log(\alpha_{\pi(1)}/\alpha_{\pi(k)}) = u_2 + \cdots + u_k$ *the sector is the trail region* $0 < \eta_2 < \cdots < \eta_n$*. Let no* κ_k *be a nonpositive integer and* $L_k \in \mathbb{N}$ *with* $\operatorname{Re}\kappa_k + L_k > 0$*. Then the continuation of* I_π *in the masses is*

$$\sum_{A \subseteq \{2,\dots,n\}} \sum_{\mathbf{l}:\ l_k < L_k\ (k \in A)} \prod_{k \in A} \frac{1}{\kappa_k + l_k} \int_{[0,1]^{A^c}} \prod_{k \notin A} q_k^{\kappa_k - 1} \Big[\prod_{k \notin A} (1 - \mathrm{Tay}_k)\, h_\pi^{(A,\mathbf{l})} \Big] dq_{A^c},$$

where $h_\pi^{(A,\mathbf{l})} = \partial_{q_A}^{\mathbf{l}} h_\pi / \mathbf{l}!$ *at* $q_A = 0$ *and* Tay_k *takes the Taylor polynomial in* q_k *of degree below* L_k*: the term* A *collapses the tails* U_k*,* $k \in A$*, together, at the levels* $\mathbf{l}$*.*

Definition 8.9 (The continued interest)**.** The continuation of Theorem 8.7(iii) keeps the symbol $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ away from its poles. Among them is the whole balance wall, where $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ has the residue $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.10). The factor $e^{-|\alpha|}$ damps the scaling mode, so no completion enters the continued interest; for data as in Definition 16.26 it is the pseudointegral of the family word against the damped measure (Definition 16.26(ii)).

Theorem 8.10 (Projective form; the period)**.** *Put*

$$\begin{aligned} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) &= \int_{\Delta_\mathcal{E}} u^{\boldsymbol{\mu}-\mathbf{1}} \prod_j \Lambda_j(\tfrac{1}{2}\log u)^{-s_j} \Lambda_j(-\tfrac{1}{2}\log u)^{-t_j}\, d\omega_\mathcal{E}(u) \\ &= \int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j}\, d\omega_\mathcal{E}(u). \end{aligned}$$

It converges if and only if $\operatorname{Re}\kappa_S > 0$ *for every nonempty proper* S*, and* $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_\mathcal{E}) P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *as meromorphic functions;* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *is the* period*. On the balance wall the period is the residue of* $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *there; for every positive* h *homogeneous of degree one, the integral of* $\alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\alpha$ *over* $\{h = 1\}$ *(Definition 1.1(vii)) is the period.*

Proposition 8.11 (The interest under Gamma draws)**.** *Let the data be real with* $\mathbf{m} > 0$*.*

(i) Where every proper wall is positive, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_\mathcal{E} \end{bmatrix} \mathbb{E}_{U \sim \mathrm{Dir}_\mathbf{m}}[W(U)]$*. With* $\phi_\mathbf{m}$ *the joint law of* $\big(K_1(U),\dots,K_N(U),\Psi_1(U),\dots,\Psi_N(U)\big)$ *for* $U \sim \mathrm{Dir}_\mathbf{m}$*, an input on the source* $(0,\infty)^{2N}$*,* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *is a Trailmoment at the unit configuration* I_{2N}*:*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_\mathcal{E} \end{bmatrix} \big(\mathsf{T}_{(\mathbf{s},\mathbf{t});I_{2N}}[\mathrm{id}]\, \phi_\mathbf{m}\big)\big((0,\infty)^{2N}\big).$$

(ii) Where moreover $\kappa_\mathcal{E} > 0$*, so that every wall is positive, let* $\alpha \sim \mathrm{Gamma}_\mathbf{m}$*, with total* $|\alpha|$ *and shape* $U = \alpha/|\alpha|$*. The scale and the shape separate:* $|\alpha| \sim \mathrm{Gamma}_{m_\mathcal{E}}$ *and* $U \sim \mathrm{Dir}_\mathbf{m}$ *are independent, and*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_e \Gamma(m_e)\, \mathbb{E}_{\alpha \sim \mathrm{Gamma}_\mathbf{m}}[W(\alpha)],$$

$$\mathbb{E}_{\alpha \sim \mathrm{Gamma}_\mathbf{m}}[W(\alpha)] = \frac{\Gamma(\kappa_\mathcal{E})}{\Gamma(m_\mathcal{E})}\, \mathbb{E}_{U \sim \mathrm{Dir}_\mathbf{m}}[W(U)].$$

(iii) In the exponents, $J_F(\mathbf{m};\mathbf{s},\mathbf{t})/\prod_e \Gamma(m_e)$ is the moment generating function of the letters' logarithms $-\log K_j(\alpha)$, $-\log \Psi_j(\alpha)$ for $\alpha \sim \mathrm{Gamma}_{\mathbf{m}}$, and $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ that of $-\log K_j(U)$, $-\log \Psi_j(U)$ for $U \sim \mathrm{Dir}_{\mathbf{m}}$. Their ratio $\Gamma(\kappa_{\mathcal{E}})/\Gamma(m_{\mathcal{E}})$ has its pole on the balance wall, where the moment generating function under $\mathrm{Gamma}_{\mathbf{m}}$ diverges and that under $\mathrm{Dir}_{\mathbf{m}}$ is the period over $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$.

Theorem 8.12 (Walls under Cremona)**.** *As functions of $(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\delta})$: $\kappa_S^{\mathrm{top}}(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\delta}) = \kappa_S(-\boldsymbol{\mu};\boldsymbol{\sigma},-\boldsymbol{\delta})$, $\kappa_S+\kappa_S^{\mathrm{top}} = 2\sum_j \sigma_j c_j(S)$, and $\kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}}+\kappa_S^{\mathrm{top}}$. In pair coordinates $\kappa_S^{\mathrm{top}} = \sum_j \big(\sigma_j c_j(S)+\delta_j \varepsilon_j(S)\big)-\mu_S$.*

Theorem 8.13 (Cremona on the balance wall)**.** *For a connected family, on the hyperplane $\kappa_{\mathcal{E}} = 0$,*

$$P_F(\boldsymbol{\mu}+\sigma\mathbf{1};\boldsymbol{\sigma}+\boldsymbol{\delta},\boldsymbol{\sigma}-\boldsymbol{\delta}) = P_F(-\boldsymbol{\mu}+\sigma\mathbf{1};\boldsymbol{\sigma}-\boldsymbol{\delta},\boldsymbol{\sigma}+\boldsymbol{\delta}),$$

the walls of the right side at S being those of the left at $\mathcal{E}\setminus S$; the identity continues meromorphically along the hyperplane.

On the balance wall the period is unchanged when the offsets and the leans change sign together: Cremona (Definition 9.1) swaps the two exponents of each table, $\delta_j \mapsto -\delta_j$, negates the offsets, and swaps every wall with the wall of the complement.

Proposition 8.14 (Dual tables)**.** *Replacing every table by its dual, with the same exponents, gives the interest at the exponents (t_j, s_j): in pair coordinates $\delta_j \mapsto -\delta_j$.*

Proposition 8.15 (The wall of a compound)**.** *For every nonempty proper S, $\kappa_S + \kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}} + 2\sum_j \sigma_j c_j(S)$. So if $c_j(S) = 0$ in every table with $\sigma_j \neq 0$, then $\kappa_S + \kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}}$, and on the balance wall a compound split along S has a wall with real part at most zero, and its period diverges.*

8.2 The native gauge

Definition 8.16 (Target; native gauge)**.** For data with $m_{\mathcal{E}} \neq 0$ the *target* is $\boldsymbol{\tau} = \frac{\sum_j (s_j r_j + t_j r_j^*)}{m_{\mathcal{E}}}\mathbf{m}$; on the balance wall it is $\mathbf{m}$. A point $\alpha^\star \in (\mathbb{C}^\times)^{\mathcal{E}}$ with $K_j(\alpha^\star)\Psi_j(\alpha^\star) \neq 0$ for every j is *native* when its *usage* equals the target,

$$\sum_j \big(s_j\, R_j(\alpha^\star) + t_j\, R_j^*(\alpha^\star)\big) = \boldsymbol{\tau},$$

where $R_{j,e} = \alpha_e \partial_e K_j / K_j$ is the occupation of table j and $R^*_{j,e} = \alpha_e \partial_e \Psi_j/\Psi_j$ that of its dual (Theorem 7.14), rational functions homogeneous of degree zero; so nativity is a property of the point up to scale. A native point in $\mathbb{R}^{\mathcal{E}}_{>0}$, written $\alpha^\star = e^{2z^\star}$, is a *native gauge*; the other native points are the *native gauges at complex points.*

Definition 8.17 (Potential; its even and odd parts)**.** In logarithmic coordinates the *potential* of a family with its exponent pairs is

$$\Phi(y) = \sum_j \Big(s_j \log K_j(e^y) + t_j \log \Psi_j(e^y)\Big) = \Phi_{\mathbf{E}}(y) + \Phi_{\mathbf{O}}(y),$$

$$\Phi_{\mathbf{E}}(y) = \sum_j \sigma_j \log \mathbf{E}_j(e^y), \qquad \Phi_{\mathbf{O}}(y) = \sum_j \delta_j \log \mathbf{O}_j(e^y), \qquad \mathbf{E}_j = K_j\Psi_j, \quad \mathbf{O}_j = K_j/\Psi_j,$$

the *even* and *odd letters* of table j, so that $\mathbf{E}_j(e^{2z}) = e^{2\langle \mathbf{1}, z\rangle} e^{E_j(z)}$ and $\mathbf{O}_j(e^{2z}) = e^{O_j(z)}$. The pair $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$, either part written $\Phi_{\mathbf{X}}$ with $\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}$, is read whole: neither part is switched off at any lean, and the two are never added before they are read.

Theorem 8.18 (The native gauge is the saddle)**.** *The potential satisfies* $\partial_{y_e}\Phi = \sum_j \big(s_j R_{j,e} + t_j R^*_{j,e}\big)$*, the usage at* $\alpha = e^y$*, and the integrand of the interest is* $e^{\langle \mathbf{m}, y\rangle - \Phi(y) - |\alpha|}$ *in* y*. In the signed coordinates* $z = \frac{1}{2}y$*,* $\Phi(y) = \sigma\langle \mathbf{1}, y\rangle + \Phi^\circ(z)$ *with the* centred potential $\Phi^\circ(z) = \sum_j \big(s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z)\big) = \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)$*.*

(i) A gauge $\alpha^\star$ *is native iff* $\nabla\Phi(\log \alpha^\star) = \boldsymbol{\tau}$*; on the balance wall, iff* $\log \alpha^\star$ *is a critical point of* $\langle \mathbf{m}, y\rangle - \Phi(y)$*.*

(ii) For real data with $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$*,* Φ *is convex, and a native gauge exists iff* $\boldsymbol{\tau}$ *lies in the relative interior of the Newton sum* $\mathcal{N}_F = \sum_j \big(s_j P_j + t_j P^*_j\big)$*, with* P_j *the base polytope of table* j *(Theorem 7.15) and* P^*_j *that of its dual; it is then unique modulo the space of directions along which* Φ *is affine.*

(iii) For such data $\mathcal{N}_F = \{x : \ x_{\mathcal{E}} = g(\mathcal{E}),\ x_S \geq g(S)\ \forall S\}$ *with the set function* $g = \sum_j (s_j d_j + t_j d^*_j)$*. For a connected family that space is* $\mathbb{R}\mathbf{1}$*, and the condition is* $\tau_S - g(S) > 0$ *for every nonempty proper* S*: the walls at the target are positive. The native gauge is then the unique point, up to scale, at which* $\alpha^{\boldsymbol{\tau}} W(\alpha)$*, constant along each ray, is largest; on the balance wall* $\boldsymbol{\tau} = \mathbf{m}$*.*

A native gauge is a gauge at which the family uses each mark exactly as much as the target asks, and the same equation at a complex point gives the native gauges at complex points that the rays of Chapter 35 use. With real data, nonnegative exponents and positive masses, for a connected family it is unique up to scale and exists exactly when every proper wall at the target is positive (Theorem 8.18(iii)); on a compound it exists exactly when every molecule of the active family is balanced at the target with its proper walls positive there, with one free scale for each such molecule (Proposition 8.19). It is a point at which $\alpha^{\boldsymbol{\tau}} W(\alpha)$ is largest, the saddle of the potential, and it is the measuring stick of every computation on the family.

Proposition 8.19 (The native gauges of a compound)**.** *Let* F *have real data,* $s_j, t_j \geq 0$ *not all zero,* $\mathbf{m} > 0$*, and target* $\boldsymbol{\tau}$*, and let* $\mathcal{E}_1, \ldots, \mathcal{E}_q$ *be the molecules of its active family. A native gauge exists exactly when every* $\mathcal{E}_i$ *is balanced at the target,* $\tau_{\mathcal{E}_i} = \sum_j \big(s_j r_j(\mathcal{E}_i) + t_j r^*_j(\mathcal{E}_i)\big)$*, and every proper wall of every* $\mathcal{E}_i$ *is positive there,* $\tau_S > \sum_j \big(s_j d_j(S) + t_j d^*_j(S)\big)$ *for* $\varnothing \neq S \subsetneq \mathcal{E}_i$*. The native gauges are then the gauges* β *with* $\beta/\alpha^\star$ *constant on every* $\mathcal{E}_i$*,* $\alpha^\star$ *any one of them: one free scale for each molecule. When the weights factor across the* $\mathcal{E}_i$*, the native gauges are the products of native gauges of the molecules. A mark that is a loop or a coloop in every table is a unit, whose letters are monomials: its still is zero, and its rung one is its mass (Definition 9.4).*

Proposition 8.20 (A realization sits on a wall only along its molecules)**.** *Let a family carry real exponents* $s_j, t_j \geq 0$*,* $s_j + t_j > 0$*, and a gauge* α*, with masses its usage* $\mathbf{m} = \sum_j \big(s_j R_j(\alpha) + t_j R^*_j(\alpha)\big)$*, that is, a realization (Definition 9.1). With* $N_S = |T \cap S|$*,*

$$\kappa_S = \sum_j \Big(s_j \big(\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[N_S] - d_j(S)\big) + t_j \big(r_j(S) - \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}[N_S]\big)\Big) \ \geq\ 0 \qquad (\varnothing \neq S \subseteq \mathcal{E}),$$

with equality exactly when S *is a union of molecules (Definition 7.4). In particular* $\kappa_{\mathcal{E}} = 0$*, and every proper wall of a connected family is positive.*

Proposition 8.21 (The centre of an arbitrary family)**.** *Take tables on $\mathcal{E}$, $s_j = t_j = \sigma_j \geq 0$ with $\sigma = \sum_j \sigma_j > 0$, and $\mathbf{m} = \sigma\mathbf{1}$. Then $\delta_j = 0$, $\boldsymbol{\mu} = 0$, the centred potential $\Phi^\circ(z) = \sum_j \sigma_j E_j(z)$ is even, and:*

(i) the gauge $\mathbf{1}$ is native, unique modulo one scale for each molecule of the active family (Proposition 8.19);

(ii) the walls are $\kappa_S = \sum_j \sigma_j c_j(S)$, positive on every proper nonempty S exactly when the active family is connected;

(iii) every odd derivative of Φ° vanishes at $z = 0$, so every odd derivative of order at least three of the potential Φ vanishes at $y = 0$.

The family at the centre, every $\delta_j = 0$ (Definition 9.1), with the masses $\sigma\mathbf{1}$ has its offsets zero. It exists for every family, its native gauge is the table's gauge, and its walls are the combinations $\kappa_S = \sum_j \sigma_j c_j(S)$ of the widths; it is where the family is read before any lean is asked of it.

Proposition 8.22 (Prescribing a gauge)**.** *For any positive gauge a and positive exponent pairs, the usage $\tau_e = \sum_j \big(s_j R_{j,e}(a) + t_j R^*_{j,e}(a)\big)$ has total $\sum_j (s_j r_j + t_j r^*_j)$, and every mass vector on the ray $c\boldsymbol{\tau}$, $c > 0$, makes a native. Fixing values $K^0_j > 0$ and $\Psi^0_j > 0$ of the letters at a, the constraints*

$$\sum_T \theta_{j,T} a^T = K^0_j, \qquad \sum_T \theta_{j,T} a^{\mathcal{E}\setminus T} = \Psi^0_j, \qquad \sum_{j,T} \theta_{j,T} \Big[\frac{s_j a^T}{K^0_j} \mathbf{1}_T + \frac{t_j a^{\mathcal{E}\setminus T}}{\Psi^0_j} \mathbf{1}_{\mathcal{E}\setminus T} \Big] = \boldsymbol{\tau}$$

are linear in the weights, and every positive deformation in their homogeneous kernel preserves them exactly.

Part III

Classification

Chapter 9

Realizations and the still

9.1 Realizations

Definition 9.1 (Realization, lean, carrier)**.** Let a family carry tables j on its marks $\mathcal{E}$, $n = |\mathcal{E}|$, its potential read as the pair $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of Definition 8.17. A *realization* of the family is $\Upsilon = \big((s_j, t_j)_j;\ \alpha\big)$: exponents $s_j, t_j \geq 0$ with $s_j + t_j > 0$, and a gauge $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$. Its pair coordinates are $\sigma_j = \frac{1}{2}(s_j + t_j) > 0$ and $\delta_j = \frac{1}{2}(s_j - t_j)$, $|\delta_j| \leq \sigma_j$, and in them the realization is written

$$\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha} = \big((\sigma_j + \delta_j,\ \sigma_j - \delta_j)_j;\ \alpha\big).$$

$\boldsymbol{\delta} = (\delta_j)_j$ is the *lean*, and the *centre* is $\boldsymbol{\delta} = 0$; at the table's gauge the realization at the centre has the masses $\sigma\mathbf{1}$ (Proposition 8.21), and a family at the centre is taken with these masses. Table j has a *basis lean* when $\delta_j > 0$, a *cobasis lean* when $\delta_j < 0$, and *rides the line* when $\delta_j = 0$; its lean is *full* when $|\delta_j| = \sigma_j$ (then the factor of table j in the family word is $K_j^{-s_j}$ alone, or $\Psi_j^{-t_j}$ alone) and *partial* when $0 < |\delta_j| < \sigma_j$. The family has a basis lean when every $\delta_j \geq 0$ and some $\delta_j > 0$, a cobasis lean when every $\delta_j \leq 0$ and some $\delta_j < 0$, and otherwise, off the centre, a mixed lean. The realization is *at one pair* (σ, δ), $\sigma = \sum_j \sigma_j$ and $\delta = \sum_j \delta_j$, when its pairs lie on one ray, $(\sigma_j, \delta_j) = \frac{\sigma_j}{\sigma}(\sigma, \delta)$ for every j; its *lean sign* is the sign of δ. A family with one table is at one pair. The *masses* of Υ are its usage,

$$\tau_e(\Upsilon) = \sum_j \Big(s_j\, \frac{\alpha_e \partial_e K_j}{K_j}(\alpha) + t_j\, \frac{\alpha_e \partial_e \Psi_j}{\Psi_j}(\alpha) \Big),$$

for which α is the native gauge (Proposition 8.22). The *carrier* of Υ is $(\alpha;\ K_j, \Psi_j;\ \boldsymbol{\tau};\ s_j, t_j)$. A gauge is written as a deviation from the table's gauge, $\alpha = e^{y_0}$. Inverting the gauge and swapping the two exponents of every table, $\big((s_j, t_j)_j; \alpha\big) \mapsto \big((t_j, s_j)_j; \alpha^{-1}\big)$, sends the offsets and the lean to $-\boldsymbol{\mu}$ and $-\boldsymbol{\delta}$, the change of data of Theorem 8.13; so *Cremona* swaps basis and cobasis leans, and the table's gauge is its fixed point up to scale (Proposition 7.2).

A realization is read off its exponents: the masses are the usage at its gauge, so the gauge is native by definition, and a realization sits on a wall only along a union of molecules, so that a realization of a molecule never does (Proposition 8.20). A molecule is read at all of its realizations, and a particular integral is one of them, fixed by its exponents and its native gauge.

Definition 9.2 (Weight law, indicators, levels, cumulants)**.** The *weight law* of table j is its basis law at the table's gauge, $\mathbb{P}_{j,\mathbf{1}}(T) = \theta_{j,T}/K_j(\mathbf{1})$ on $\mathcal{B}_j$; for a list it is uniform. For a random basis T the *indicators* are $X_e = \mathbf{1}[e \in T]$, and for a set C of marks the *level* is $N_C = |T \cap C| = \sum_{e \in C} X_e$.

For functions $Y_1, \dots, Y_k$ of T, $\kappa_{j,\alpha}(Y_1, \dots, Y_k)$ is their joint cumulant under $\mathbb{P}_{j,\alpha}$, and $Y^{\times a}$ in its argument lists Y a times; along a string $w = (e_1, \dots, e_k)$ of marks, $\kappa_{j,\alpha}(w) = \kappa_{j,\alpha}(X_{e_1}, \dots, X_{e_k}) = \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \log K_j(e^y)$ at $y = \log \alpha$. With one table the index j is dropped: $\mathbb{P}_\alpha$, $R(\alpha)$, κ_α.

Proposition 9.3 (Masses at a gauge)**.** *At $\alpha = e^{y_0}$, in this statement write $R^{\pm}_{j,e} = \frac{1}{2}\big(R_{j,e}(e^{y_0}) \pm R_{j,e}(e^{-y_0})\big)$; then*

$$\tau_e = \sum_j \big(s_j R_{j,e}(e^{y_0}) + t_j(1 - R_{j,e}(e^{-y_0}))\big) = \sum_j \big(\sigma_j(1 + 2R^-_{j,e}) + \delta_j(2R^+_{j,e} - 1)\big),$$

and at $\alpha = \mathbf{1}$, where $R^-_{j,e} = 0$ and $R^+_{j,e} = R_{j,e}(\mathbf{1})$, $\tau_e = \sum_j \big(\sigma_j + \delta_j(2R_{j,e}(\mathbf{1}) - 1)\big)$.

9.2 The potential and the still

Definition 9.4 (Still, rungs, tallies, line functions)**.** The *still* of Υ is the sequence of symmetric tensors

$$\kappa_\Upsilon(e_1, \dots, e_k) = 2^k\, \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \Phi(\log \alpha), \qquad k \geq 2,$$

the Taylor tensor of Φ in the signed coordinates $z = \frac{1}{2}y$ from order two on; its entries of order k form *rung k*. *Rung one* is the gradient $\partial_{y_e}\Phi(\log\alpha) = \tau_e(\Upsilon)$, the masses (Theorem 8.18), taken without the factor 2. Each part has its own still, $\kappa^{\mathbf{E}}_\Upsilon$ and $\kappa^{\mathbf{O}}_\Upsilon$, with $\kappa_\Upsilon = \kappa^{\mathbf{E}}_\Upsilon + \kappa^{\mathbf{O}}_\Upsilon$. The *tally* of a mark e at rung k is the multiset $\mathrm{tally}^{\mathbf{X}}_k(e) = \{\{\kappa^{\mathbf{X}}_\Upsilon(e, w) : \ w \in \mathcal{E}^{k-1}\}\}$. The *self-line* and *partner-line functions* of e are

$$\varphi^{\mathbf{X}}_e(v) = \partial_{y_e}\Phi_{\mathbf{X}}\big(\log\alpha + v\mathbf{1}_e\big), \qquad \varphi^{\mathbf{X}}_{e,g}(v) = \partial_{y_e}\Phi_{\mathbf{X}}\big(\log\alpha + v\mathbf{1}_g\big) \quad (g \neq e).$$

The still is the Taylor tensor of the potential at the realization from rung two on, one object, recording how the marks covary under the two laws; beside it, rung one is the masses. The letters are affine in every variable, so they have finitely many nontrivial derivatives, and every rung is read from those through the logarithm; the readings of Chapter 10 take the potential as one function, of which the still is the expansion.

Proposition 9.5 (The still at the table's gauge)**.** *At every realization $\Upsilon = \big((s_j, t_j)_j; \mathbf{1}\big)$, for $k \geq 2$ and every string w of length k,*

$$\kappa_\Upsilon(w) = 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w) = \begin{cases} 2^{k+1} \sum_j \sigma_j\, \kappa_{j,\mathbf{1}}(w), & k \text{ even}, \\ 2^{k+1} \sum_j \delta_j\, \kappa_{j,\mathbf{1}}(w), & k \text{ odd}. \end{cases}$$

This is Proposition 9.7 at $y_0 = 0$. At the table's gauge $\Phi_{\mathbf{E}}$ is even in y and $\Phi_{\mathbf{O}}$ is odd, each up to a linear term; so the even rungs are those of $\kappa^{\mathbf{E}}_\Upsilon$, independent of the lean, and the odd rungs are those of $\kappa^{\mathbf{O}}_\Upsilon$, linear in the lean and zero at the centre.

Proposition 9.6 (Balanced tallies)**.** *At every realization Υ, for each part, every rung $k \geq 2$ and every mark e, the tally of e sums to 0: $\sum_{w \in \mathcal{E}^{k-1}} \kappa^{\mathbf{X}}_\Upsilon(e, w) = 0$.*

Proposition 9.7 (A deviation reweights the bases)**.** *At the realization $\Upsilon = \big((s_j, t_j)_j; e^{y_0}\big)$, for $k \geq 2$ and every string w of length k,*

$$\kappa_\Upsilon(w) = 2^k \sum_j \Big(s_j\, \kappa_{j,e^{y_0}}(w) + t_j(-1)^k \kappa_{j,e^{-y_0}}(w)\Big).$$

In this statement write $\kappa_j^{\pm}(w) = \frac{1}{2}\big(\kappa_{j,e^{y_0}}(w) \pm \kappa_{j,e^{-y_0}}(w)\big)$*; then*

$$\kappa_{\Upsilon}(w) = \begin{cases} 2^{k+1} \sum_j \big(\sigma_j \kappa_j^+(w) + \delta_j \kappa_j^-(w)\big), & k \text{ even}, \\ 2^{k+1} \sum_j \big(\sigma_j \kappa_j^-(w) + \delta_j \kappa_j^+(w)\big), & k \text{ odd}, \end{cases}$$

and the cumulants at the deviation are read from those at the table's gauge, $\kappa_{j,e^{y_0}}(w) = \sum_{i \geq 0} \frac{1}{i!} \sum_{u \in \mathcal{E}^i} \kappa_{j,\mathbf{1}}(w,u) \prod_{l=1}^{i} y_{0,u_l}$*, for* y_0 *in the domain of convergence of the Taylor series of* $y \mapsto \log K_j(e^y)$ *at* 0*.*

Proposition 9.8 (The two occupations of a mark)**.** *For a mark* e *of table* j*, neither a loop nor a coloop, with room* $R = \mathcal{E} - e$*, let* $K_{j,/e}(\alpha_R)$ *and* $K_{j,\setminus e}(\alpha_R)$ *be the basis polynomials of its contraction and deletion at* e *(Definition 7.3), and*

$$\mathrm{D}_{j,e}(y_R) = \log K_{j,/e}(e^{y_R}) - \log K_{j,\setminus e}(e^{y_R}), \qquad \mathrm{D}_{j,e}^{\pm}(y_R) = \tfrac{1}{2}\big(\mathrm{D}_{j,e}(y_R) \pm \mathrm{D}_{j,e}(-y_R)\big).$$

At the gauge e^y *the occupation and the dual occupation of* e *(Theorem 7.14) are*

$$\frac{\alpha_e \partial_e K_j}{K_j} = \frac{1}{1 + e^{-\lambda_{j,e} - \eta_{j,e}}}, \qquad \frac{\alpha_e \partial_e \Psi_j}{\Psi_j} = \frac{1}{1 + e^{-\lambda_{j,e} + \eta_{j,e}}},$$
$$\lambda_{j,e} = y_e + \mathrm{D}_{j,e}^-(y_R), \qquad \eta_{j,e} = \mathrm{D}_{j,e}^+(y_R),$$

and at the table's gauge $\lambda_{j,e} = 0$*: the two occupations are* $(1 + e^{-\eta_{j,e}})^{-1}$ *and* $(1 + e^{\eta_{j,e}})^{-1}$*.*

Proposition 9.9 (Occupations at a gauge)**.** *At* $\alpha = e^{y_0}$*, with* $R_{j,e}^{\pm}$ *as in Proposition 9.3,* $\varphi_e^{\mathbf{E}}(0) = \sum_j \sigma_j (1 + 2R_{j,e}^-)$ *and* $\varphi_e^{\mathbf{O}}(0) = \sum_j \delta_j (2R_{j,e}^+ - 1)$*. At* $\alpha = \mathbf{1}$*,* $\varphi_e^{\mathbf{E}}(0) = \sum_j \sigma_j$ *at every mark and* $\varphi_e^{\mathbf{O}}(0) = \sum_j \delta_j (2R_{j,e}(\mathbf{1}) - 1)$*.*

Chapter 10

The reading, classes, atoms

10.1 Views and the reading

Definition 10.1 (Views; the reading; classes; atoms)**.** Let M be a molecule with n marks carrying tables j, and Υ a realization of M (Definition 9.1). The room of a mark e is $\mathcal{E} \setminus \{e\}$ (the room of the bunch $\{e\}$, Definition 12.1), and an *order* of a set is a listing of its elements. For a mark e and distinct marks $\mathbf{g} = (g_1, \dots, g_k)$ of its room, $0 \le k \le n-1$, the *view* of e with the partners $\mathbf{g}$ is the pair $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}})$,

$$\Phi^{\mathbf{X}}_{e,\mathbf{g}}(v_0, v_1, \dots, v_k) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_e + \sum_{i=1}^{k} v_i \mathbf{1}_{g_i}\Big),$$

the potential on the scales of e and its partners with every other mark at the gauge; the variable v_i sits at *position* i. The *reading of e with k partners* is the multiset of its views over all k-tuples of distinct partners,

$$\mathrm{read}^{(k)}_{\Upsilon}(e) = \{\!\{(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}}) :\ \mathbf{g} \in (\mathcal{E}\setminus e)^{(k)}\}\!\} :$$

the functions are kept and the names of the partners are not. With $k = n-1$, $\mathbf{g}$ is an order of the room and the pair $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}})$ is the potential *written from* e in that order; $\mathrm{read}^{(n-1)}_{\Upsilon}(e)$ is the *reading* of e. The *classes* of Υ are the level sets of $e \mapsto \mathrm{read}^{(n-1)}_{\Upsilon}(e)$; they form the partition $\mathrm{Cl}(\Upsilon)$. The realization is *atomic*, an *atom*, when $\mathrm{Cl}(\Upsilon) = \{\mathcal{E}\}$. For a family whose weights factor across its molecules, the classes of a realization are, by definition, the classes of the realizations of its molecules, each read on its own marks. With the gauge α and $\boldsymbol{\sigma}$ fixed, the partition at the lean $\boldsymbol{\delta}$ is $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$.

The reading of a mark is the whole potential seen from that mark, its partners unnamed, and the reading is a function on the marks whose values are functions. Classes are its level sets, and an atom is a realization at which it is constant, so that the integral cannot tell its marks apart at any order.

Proposition 10.2 (A view through the patterns)**.** *For table j and positions $0, \dots, k$ held by $e, g_1, \dots, g_k$, let the pattern of a basis be the set $p \subseteq \{0, \dots, k\}$ of positions it holds (its trace on the bunch $\{e, g_1, \dots, g_k\}$, Definition 12.1, read in positions), $\bar{p}$ its complement, and*

$$K_{j;e,\mathbf{g}}(x) = \sum_{p} \vartheta^{K}_{j,p}\, x^{p}, \qquad \Psi_{j;e,\mathbf{g}}(x) = \sum_{p} \vartheta^{\Psi}_{j,p}\, x^{\bar{p}}, \qquad \vartheta^{K}_{j,p} = \sum_{T:\, p} \theta_{j,T}\alpha^{T}, \quad \vartheta^{\Psi}_{j,p} = \sum_{T:\, p} \theta_{j,T}\alpha^{\mathcal{E}\setminus T},$$

sums over the bases of pattern p. With $x_i = e^{v_i}$,

$$\Phi^{\mathbf{E}}_{e,\mathbf{g}} = \sum_j \sigma_j \log\big(K_{j;e,\mathbf{g}}\Psi_{j;e,\mathbf{g}}\big), \qquad \Phi^{\mathbf{O}}_{e,\mathbf{g}} = \sum_j \delta_j \log\big(K_{j;e,\mathbf{g}}/\Psi_{j;e,\mathbf{g}}\big).$$

At every gauge

$$K_{j;e,\mathbf{g}}(\mathbf{1}+x) = K_j(\alpha) \sum_{A\subseteq\{0,\dots,k\}} \mathbb{P}_{j,\alpha}\big(p(T)\supseteq A\big)\, x^A,$$
$$\Psi_{j;e,\mathbf{g}}(\mathbf{1}+x) = \Psi_j(\alpha) \sum_{A\subseteq\{0,\dots,k\}} \mathbb{P}^*_{j,\alpha}\big(p(T)\cap A=\varnothing\big)\, x^A,$$

with $p(T)$ the pattern of T, so the view with k partners is read from the joint inclusion laws of $e, g_1, \dots, g_k$ under the basis and cobasis laws. At the table's gauge $\vartheta^K_{j,p} = \vartheta^\Psi_{j,p} = \vartheta_{j,p}$, the weight of the bases of pattern p, both laws are the weight law $\mathbb{P}_{j,\mathbf{1}}$, and $\Psi_{j;e,\mathbf{g}}(x) = x^{\{0,\dots,k\}}K_{j;e,\mathbf{g}}(x^{-1})$. For one table, two views of one realization agree at the centre exactly when the polynomials $K_{e,\mathbf{g}}\Psi_{e,\mathbf{g}}$ agree, and at a lean exactly when $K_{e,\mathbf{g}}$ and $\Psi_{e,\mathbf{g}}$ both agree; at the table's gauge $K_{e,\mathbf{g}}$ alone decides at a lean.

Proposition 10.3 (More partners refine; the tallies are inside)**.** *(i) $\mathrm{read}^{(k)}_\Upsilon(e)$ determines $\mathrm{read}^{(k')}_\Upsilon(e)$ for every $k' \le k$; so the level sets of the reading with k partners refine those with k'.*

(ii) For $k \ge 1$, $\mathrm{read}^{(k)}_\Upsilon(e)$ determines the self-line functions $\varphi^{\mathbf{X}}_e$, the multisets $\{\{\varphi^{\mathbf{X}}_{e,g}\}\}_{g\ne e}$ of partner-line functions, and for every rung the multiset of the entries $\kappa^{\mathbf{X}}_\Upsilon(e,w)$ over the strings w of marks with at most k distinct marks other than e; in particular the tallies of e at the rungs $2,\dots,k+1$.

(iii) The reading determines the tallies of e at every rung $k \ge 2$, its self-line and partner-line functions for both parts, and its mass $\tau_e(\Upsilon)$.

Proposition 10.4 (One writing determines the reading)**.** *The reading of e is determined by any one writing of the potential from e. So the reading is constant on a set C of marks exactly when one pair of functions is the potential written from every mark of C: there are orders $\mathbf{g}_e$ of the rooms, $e \in C$, with $\Phi^{\mathbf{X}}_{e,\mathbf{g}_e} = \Phi^{\mathbf{X}}_{f,\mathbf{g}_f}$ for all $e, f \in C$ and both parts. In particular Υ is an atom exactly when one pair of functions is the potential written from every mark.*

10.2 Identifying writings

Theorem 10.5 (One table)**.** *Let M carry one table, $n \ge 2$, at the table's gauge, and let C be a set of marks. The reading is constant on C*

(i) at a lean, exactly when one polynomial is K written from every mark of C;

(ii) at the centre, exactly when one polynomial is, from every mark of C, either K or Ψ written from it.

Here K written from e in the order $\mathbf{g}$ is $K_{e,\mathbf{g}}$ with $k = n-1$, a polynomial in the position variables.

Theorem 10.6 (Several tables: the potential is read factor by factor)**.** *At a realization Υ write every $K_j(\alpha\circ x)$ and $\Psi_j(\alpha\circ x)$ as a positive constant times a product of irreducible polynomials in*

x, each positive on the orthant and equal to 1 at $x = \mathbf{1}$. Give each distinct such polynomial L the exponents

$$w_{\mathbf{E}}(L) = \sum \sigma_j > 0, \qquad w_{\mathbf{O}}(L) = \sum \pm\delta_j,$$

summed over its occurrences with multiplicity, with $+$ in a K_j and $-$ in a Ψ_j. Then $\Phi_{\mathbf{X}}(\log\alpha + \log x) = \Phi_{\mathbf{X}}(\log\alpha) + \sum_L w_{\mathbf{X}}(L)\log L(x)$, and two pairs $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of this form, in the same variables, are equal exactly when they have the same polynomials with the same pairs of exponents $(w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$. So the reading is constant on C exactly when one set of polynomials with exponents $\{(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))\}$, with the variables placed in positions, is the family written from every mark of C. For tables whose weights factor across their molecules the polynomials L are the normalized Foreman polynomials of their restrictions to those molecules.

Corollary 10.7 (Moving blocks of partners reads alike)**.** *Let $\mathbf{g}, \mathbf{g}'$ be orders of the rooms of e and f with $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}}) = (\Phi^{\mathbf{E}}_{f,\mathbf{g}'}, \Phi^{\mathbf{O}}_{f,\mathbf{g}'})$, and let β be the bijection of $\mathcal{E}$ with $\beta(e) = f$ and $\beta(g_i) = g'_i$. Then for all disjoint sets $S_1, \dots, S_m$ of the room of e,*

$$\Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_e + \sum_{i=1}^{m} v_i\mathbf{1}_{S_i}\Big) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_f + \sum_{i=1}^{m} v_i\mathbf{1}_{\beta(S_i)}\Big).$$

So on a class every move of the gauge that shifts a mark and several sets of partners, each by one amount, reads alike: for every partition $S_1, \dots, S_m$ of the room, the potential moved by v_0 on the mark and by v_i on S_i reads alike from every mark of the class.

Proposition 10.8 (Computing the classes)**.** *Read every mark with $k = 0, 1, 2, \dots$ partners and group the marks by value. The level sets only refine as k grows (Proposition 10.3). A level set is a class as soon as one pair of functions is written from each of its marks (Proposition 10.4); the orders are built one partner at a time, keeping every view equal to the restriction of that one function, which loses nothing since a view with fewer partners is a restriction of a writing. Where no one function serves a whole level set, the marks from which a given function can be written form one class and the rest are read again. The computation ends by $k = n - 1$, and at a gauge α every step is an identity of polynomials with coefficients in the field generated by the weights and the coordinates of α; at the table's gauge that is the field generated by the weights, rational for rational weights and integers for lists (Proposition 10.2, Theorem 10.6).*

10.3 What a class holds

Proposition 10.9 (Masses and occupations on a class)**.** *At every realization Υ the masses $\tau_e(\Upsilon)$ and each of their two parts $\varphi^{\mathbf{E}}_e(0)$ and $\varphi^{\mathbf{O}}_e(0)$ (Proposition 9.9) are constant on every class. At the table's gauge:*

(i) $\sum_j \delta_j(2R_{j,e}(\mathbf{1}) - 1)$ is constant on every class, and $\kappa^{\mathbf{E}}_{\Upsilon}(e, e) = 8\sum_j \sigma_j R_{j,e}(\mathbf{1})(1 - R_{j,e}(\mathbf{1}))$;

(ii) with one table, the occupation $R_e(\mathbf{1})$ is constant on every class at a lean, and at the centre two marks e, f of one class have $R_f(\mathbf{1}) = R_e(\mathbf{1})$ or $R_f(\mathbf{1}) = 1 - R_e(\mathbf{1})$.

Chapter 11

Lean, gauge and label

11.1 The lean

Theorem 11.1 (The lean only splits)**.** *At every gauge and every $\boldsymbol{\sigma}$, the partition at a lean $\boldsymbol{\delta}$ refines the partition at the centre. An atom at a lean is an atom at the centre, and a realization that is not an atom at the centre is an atom at no lean.*

Proposition 11.2 (What of the lean is read)**.** *With the gauge α fixed, $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$ depends on $(\boldsymbol{\sigma},\boldsymbol{\delta})$ only through the rays $\{c\boldsymbol{\sigma}\}$ and $\{c'\boldsymbol{\delta}\}$, $c,c'>0$, and $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\alpha})=\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$. So at one pair, $\boldsymbol{\delta}=\frac{\delta}{\sigma}\boldsymbol{\sigma}$, there are two partitions: $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},0;\alpha})$ at the centre, and $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$, the same for every $\delta\neq 0$ and every $\boldsymbol{\sigma}$ on its ray, which refines it.*

Proposition 11.3 (A lean reads one side)**.** *Let Υ be at one pair (σ,δ), at the table's gauge, and put*

$$\Phi_K(y)=\frac{1}{\sigma}\sum_j \sigma_j \log K_j(e^y), \qquad \Phi_\Psi(y)=\frac{1}{\sigma}\sum_j \sigma_j \log \Psi_j(e^y)=\Phi_K(-y)+\langle \mathbf{1},y\rangle.$$

Then $\Phi_{\mathbf{E}}=\sigma(\Phi_K+\Phi_\Psi)$ and $\Phi_{\mathbf{O}}=\delta(\Phi_K-\Phi_\Psi)$, and:

(i) at every lean the reading of a mark determines, and is determined by, the multiset of writings of Φ_K from it; so for $\delta\neq 0$, $\mathrm{Cl}(\Upsilon)$ is the partition of the marks into the level sets of the multiset of writings of Φ_K, and equally of Φ_Ψ;

(ii) with the tensors $\mathcal{K}_k(w)=\frac{1}{\sigma}\sum_j \sigma_j \kappa_{j,\mathbf{1}}(w)$ on the strings of length k, rung k is $2^{k+1}\sigma\mathcal{K}_k$ for even k and $2^{k+1}\delta\mathcal{K}_k$ for odd k: at a basis lean every rung is a positive multiple of the cumulant sequence $(\mathcal{K}_k)_k$ of the bases, at a cobasis lean of the cumulant sequence $((-1)^k\mathcal{K}_k)_k$ of the cobases, and at the centre only the even rungs, common to the two sequences, are read.

A lean reads the molecule with a preference: at a basis lean it reads the cumulant sequence of the bases, at a cobasis lean that of the cobases, and at the centre only what the two sequences share. The lean can only split the classes of the centre, never join them.

11.2 The gauge

Theorem 11.4 (The gauge step)**.** *Let $\Upsilon=((s_j,t_j)_j;\alpha)$ be a realization of a molecule. If the self-line functions $\varphi_e^{\mathbf{E}}$ of all marks agree up to additive constants, in particular if Υ is an atom, then $\alpha\propto\mathbf{1}$.*

Corollary 11.5 (Scale changes nothing)**.** *At $\alpha = c\mathbf{1}$ every writing is the writing at $\mathbf{1}$ plus a constant that depends on neither the mark nor the order. So $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};c\mathbf{1}}) = \mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}})$, and every atom is an atom at the table's gauge.*

11.3 Answers and the label

Proposition 11.6 (A compound is read molecule by molecule)**.** *Let the weights factor across the molecules. Then $\Phi_{\mathbf{X}}(y) = \sum_M \Phi_{\mathbf{X},M}(y_{\mathcal{E}_M})$ with $\Phi_{\mathbf{X},M}$ the potential of the restriction to M, and every entry of the still on a string that meets two molecules is 0.*

Proposition 11.7 (Assembly from arbitrary seed tables)**.** *Let H be a set of bijections of $\mathcal{E}$ closed under composition and inverses that takes every mark to every other. Start from seed tables on $\mathcal{E}$, and for each seed include its image under every $h \in H$, weights transported with the bases and repetitions kept. Give every image of seed a the exponents (s_a, t_a); let q_a be the number of its images with multiplicity, r_a its rank, $D = \sum_a q_a(s_a r_a + t_a(n - r_a))$, $\mathbf{m} = \frac{D}{n}\mathbf{1}$, and suppose the active assembled family is connected. Then for every positive exponent pairs the gauge $\mathbf{1}$ is native, every proper wall is positive, and the realization $((s_a, t_a)_a; \mathbf{1})$ of the assembled family is an atom.*

(i) At $\mathbf{1}$ the occupations summed over the images of seed a are the same at every mark, $q_a r_a/n$, and the dual occupations summed likewise are $q_a(n - r_a)/n$; their sum with the coefficients s_a and t_a is D/n.

(ii) The images under all permutations of a seed with at least two bases form a connected family.

(iii) The seed weights may vary over any positive values before assembly, giving atoms at the one gauge $\mathbf{1}$.

Definition 11.8 (Answers and the label)**.** A molecule *answers yes* when some realization of it is atomic, and *no* otherwise. Let k be the number of molecules of a family with at least two marks and i the number of them that answer yes; units are atomic at every realization and are not counted. The family is *Foreman k-symmetric* when $i = k$ and *Foreman asymmetric* when $i < k$; an asymmetric family with $i > 0$ is *sub-symmetric (i of k)*. The *label* of the family is the pair (i, k).

Theorem 11.9 (The answer is read at the centre)**.** *A molecule answers yes if and only if some centre $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ is atomic. With one table this holds if and only if one polynomial is, from every mark, either K or Ψ written from it; the condition does not involve σ.*

Corollary 11.10 (The label is read at the centres)**.** *By Theorem 11.9 the label of a family is read at the centres $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ of the table's gauge of its molecules, and a molecule carrying one table is decided by reading it once, at any one such centre.*

Corollary 11.11 (The leaning realizations at one pair)**.** *Let a molecule be read at one pair at the table's gauge, with the strengths $\boldsymbol{\sigma}$. Its leaning realizations $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, $\boldsymbol{\delta} = \frac{\delta}{\sigma}\boldsymbol{\sigma}$, $\delta \neq 0$, are atomic all together or not at all; if they are, so is the centre, and an atomic centre need not make them atomic.*

Whether a molecule can be atomic and whether a particular realization of it is atomic are read differently. The first is read at the centres of the table's gauge, at any one centre when the molecule carries one table; the second at the realization itself, and a lean may split what the centre held together.

Lemma 11.12 (No degenerate pair)**.** *At every realization Υ of a family and for marks $e \neq f$, $\kappa^{\mathbf{E}}_{\Upsilon}(e,f)^2 \leq \kappa^{\mathbf{E}}_{\Upsilon}(e,e)\,\kappa^{\mathbf{E}}_{\Upsilon}(f,f)$, with equality exactly when e or f is a unit or $\{e,f\}$ is a molecule. In particular the inequality is strict for any two marks of a molecule with at least three marks.*

Lemma 11.13 (Equal diagonals on a class)**.** *At every realization Υ, for each part and every $k \geq 2$, the diagonal entry $\kappa^{\mathbf{X}}_{\Upsilon}(e,\dots,e) = 2^k(\varphi^{\mathbf{X}}_e)^{(k-1)}(0)$ of rung k is the same at every mark of a class, the reading determining the self-line functions (Proposition 10.3(iii)); in particular $\kappa^{\mathbf{E}}_{\Upsilon}(e,e) = 4\,(\varphi^{\mathbf{E}}_e)'(0)$ is constant on every class, and at an atom it is the same at every mark.*

11.4 Maps as certificates

Definition 11.14 (Carrying and exchanging)**.** Let F and F' be families on $\mathcal{E}$ and $\mathcal{E}'$ with tables indexed by $\{1,\dots,N\}$ and $\{1,\dots,N'\}$. A bijection $g : \mathcal{E} \to \mathcal{E}'$ *carries* F to F' when some bijection $h : \{1,\dots,N\} \to \{1,\dots,N'\}$ has $g(\mathcal{B}_j) = \mathcal{B}'_{h(j)}$, $\theta'_{h(j),g(T)} = \theta_{j,T}$ and $(\sigma'_{h(j)},\delta'_{h(j)}) = (\sigma_j,\delta_j)$; it *exchanges* F with F' when instead $T \mapsto \mathcal{E}' \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}'_{h(j)}$ with $\theta'_{h(j),\mathcal{E}'\setminus g(T)} = \theta_{j,T}$ and the same condition on the pairs. With $F' = F$, g carries or exchanges the family. Write $(g^*y)_e = y_{g(e)}$ for $y \in \mathbb{R}^{\mathcal{E}'}$.

Proposition 11.15 (What a map certifies)**.** *At gauges α of F and α' of F' with $g^*\log\alpha' = \log\alpha$, the tables' gauges in particular, and in this statement writing $\Phi^{\boldsymbol{\delta}}_{\mathbf{O}}$ for the odd potential at the lean $\boldsymbol{\delta}$: if g carries F to F' then $\Phi_{\mathbf{X}} \circ g^* = \Phi'_{\mathbf{X}}$, and the reading of $g(e)$ in F' equals that of e in F; if g exchanges F with F' then $\Phi_{\mathbf{E}} \circ g^* = \Phi'_{\mathbf{E}}$ and $\Phi^{-\boldsymbol{\delta}}_{\mathbf{O}} \circ g^* = \Phi'^{\boldsymbol{\delta}}_{\mathbf{O}}$, and the reading of $g(e)$ in F' at $-\boldsymbol{\delta}$ equals that of e in F at $\boldsymbol{\delta}$. So a map carrying the family takes classes onto classes, and an exchange of the family takes the classes at $\boldsymbol{\delta}$ onto those at $-\boldsymbol{\delta}$ and the classes of the centre onto the classes of the centre.*

A map is a certificate, not a definition: classes are the level sets of the reading, and a map carrying or exchanging the family is one way to exhibit that two marks read alike. With one table the writings and the maps say the same thing (Corollary 11.17); with several tables the reading is the definition.

Corollary 11.16 (An exchange negates odd cumulants)**.** *If g exchanges a family, with h the bijection of its tables (Definition 11.14), then under the weight laws $\kappa_{h(j),\mathbf{1}}(g(w)) = (-1)^k\kappa_{j,\mathbf{1}}(w)$ for every table j and every string w of length $k \geq 2$; so the stills at the table's gauge satisfy $\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(g(w)) = \kappa_{\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}}(w)$, and g maps every class of the centre onto itself.*

Corollary 11.17 (With one table the writings are certificates)**.** *In a one-table molecule at the table's gauge, e and f lie in one class at a lean exactly when some bijection taking e to f carries the table, and in one class at the centre exactly when some bijection taking e to f carries or exchanges it. At every realization of every molecule, e and f lie in one class exactly when some bijection keeping the potential takes e to f (Definition 14.17, Lemma 14.18).*

Theorem 11.18 (A class across the centre)**.** *In a one-table molecule at the table's gauge, a class C of the centre is, at every lean, one class or two classes with equally many marks. In the second case the two are C_K and C_Ψ, the marks of C from which K, respectively Ψ, is written as one and the same polynomial, and an exchange β built from two such writings has $\beta(C) = C$ and $\beta(C_K) = C_\Psi$.*

Leaning away from the centre, a class of a one-table molecule either stays whole or splits into the half from which the basis polynomial is written and the half from which the cobasis polynomial is: the lean sign decides which half the Foreman meets first (Corollary 13.11).

Chapter 12

Restriction to a bunch

12.1 Patterns and the Taylor cube

Definition 12.1 (Patterns and children)**.** Let $(\mathcal{B}, \theta)$ be a table of rank r on $\mathcal{E}$. A bunch (Definition 3.1) is here any nonempty set $B \subseteq \mathcal{E}$, and $R = \mathcal{E} \setminus B$ is its *room*; for a family the objects of a bunch in table j carry the index j. The *patterns* of B are the traces of the bases on it, $\mathrm{pat}(B) = \{T \cap B : \ T \in \mathcal{B}\}$, and $\mathrm{pat}_j(B)$ for table j of a family. For $p \in \mathrm{pat}(B)$ the *signed pattern* is $\chi_p = \mathbf{1}_p - \mathbf{1}_{B \setminus p} \in \mathbb{R}^B$ and the *signed level* is $\ell_p = |p| - \frac{|B|}{2}$. The *child* at p is the table on R with the bases $\mathcal{B}_{(p)} = \{T \setminus p : \ T \cap B = p\}$ and the weights $\theta_{(p),X} = \theta_{X \cup p}$; its Foreman polynomials and its centred polynomial are $K_{(p)}$, $\Psi_{(p)}$ and $\Lambda_{(p)}$.

Lemma 12.2 (Children are tables)**.** *$\mathcal{B}_{(p)}$ is the set of bases of the minor contracting p and deleting $B \setminus p$ (Definition 7.5); it has the exchange property and rank $r - |p|$.*

Definition 12.3 (Communicatroid; interchange)**.** The *communicatroid* of B is the indexed family of its children $(\mathcal{B}_{(p)}, \theta_{(p)})_{p \in \mathrm{pat}(B)}$, tables on the room indexed by their patterns; its *gluing* is $\{U \sqcup p : p \in \mathrm{pat}(B),\ U \in \mathcal{B}_{(p)}\}$ with the weights $\theta_{(p),U}$. An indexed family $(\mathcal{A}_p)_p$ of tables on R, indexed by sets $p \subseteq B$ with $\mathcal{A}_p$ of rank $r - |p|$, has *interchange* when, for $U \sqcup p$ and $U' \sqcup p'$ in its gluing and $x \in (U \sqcup p) \setminus (U' \sqcup p')$, there is $y \in (U' \sqcup p') \setminus (U \sqcup p)$ with

$$U - x + y \in \mathcal{A}_p \ \ (x, y \in R), \qquad U + y \in \mathcal{A}_{p-x} \ \ (x \in B,\ y \in R),$$
$$U - x \in \mathcal{A}_{p+y} \ \ (x \in R,\ y \in B), \qquad U \in \mathcal{A}_{p-x+y} \ \ (x, y \in B) :$$

a step of the exchange property inside a child, one level down, one level up, or along a level.

Proposition 12.4 (Interchange is the parent's exchange property)**.** *The gluing of the communicatroid of B is the table, weights included, and the communicatroid has interchange. Conversely, an indexed family of tables on R with interchange is the communicatroid of B in its gluing, which is a table: the children remember the parent because the parent is the glued children.*

Proposition 12.5 (Adjacent siblings)**.** *If p and $p + b$ are patterns of B, the children at p and at $p + b$ are the deletion and the contraction of b in one table on $R + b$, the minor contracting p and deleting $B \setminus (p + b)$. So the child at $p + b$ is a quotient of the child at p of rank one less: every set of room marks that is maximal for its rank in the child at $p + b$ is maximal for its rank in the child at p, every basis of the child at $p + b$ extends by one room mark to a basis of the child at p, and every basis of the child at p less one suitable mark is a basis of the child at $p + b$.*

Theorem 12.6 (The Taylor cube)**.** *For $\alpha = (\alpha_B, \alpha_R)$ and $z = (z_B, z_R)$,*

$$K(\alpha) = \sum_{p \in \mathrm{pat}(B)} \alpha_B^p K_{(p)}(\alpha_R), \qquad \Psi(\alpha) = \sum_{p \in \mathrm{pat}(B)} \alpha_B^{B \setminus p} \Psi_{(p)}(\alpha_R),$$
$$\Lambda(z) = \sum_{p \in \mathrm{pat}(B)} e^{\langle z_B, \chi_p \rangle} \Lambda_{(p)}(z_R),$$

with $K_{(p)} = (\partial^p K)|_{\alpha_B = 0}$. Along the scale of B, $\alpha_B = e^{2z_B} = \rho u$ with $u \in \Delta_B$, the factor of p in Λ is $e^{\langle z_B, \chi_p \rangle} = \rho^{\ell_p} u^{\chi_p/2}$; the values $|p|$ fill $\{d(B), \dots, r(B)\}$, and the signed levels fill a window of width $c(B)$ centred at $\varepsilon(B)/2$. For $B = \{g\}$ this is the expansion $K = K_{\setminus g} + \alpha_g K_{/g}$, $\Psi = \alpha_g \Psi_{\setminus g} + \Psi_{/g}$ of Definition 7.3.

The Taylor cube is the bunch read on its own coordinates: each pattern contributes a monomial on the bunch, at the level $|p|$, times the polynomial of its child on the room. The levels that occur fill the window of the bunch (Theorem 7.11), and nothing about the room is lost, since the children glued back are the table.

Theorem 12.7 (The pattern tensor)**.** *Under the basis law at α, $\mathbb{P}_\alpha(T \cap B = p) = \alpha_B^p K_{(p)}(\alpha_R)/K(\alpha)$, and under the cobasis law $\mathbb{P}^*_\alpha(T \cap B = p) = \alpha_B^{B \setminus p} \Psi_{(p)}(\alpha_R)/\Psi(\alpha)$. The moments of the bunch are the upper sums $\mathbb{P}_\alpha(T \cap B \supseteq S) = \sum_{p \supseteq S} \mathbb{P}_\alpha(T \cap B = p)$, and the joint cumulants $\kappa_\alpha(w)$ on the strings w of marks of B are their image under the moment–cumulant relation. Scaling α_B multiplies the entry of p by its own monomial, and inverting the gauge swaps the two laws pattern by pattern (Proposition 12.25(ii)).*

Proposition 12.8 (Tinkers along a bunch; selecting patterns)**.** *For a* tinker *$v = (v_p)_{p \in \mathrm{pat}(B)}$ of positive numbers, the weights $v_{T \cap B} \theta_T$ define the* tinkered table*, with the same bases and the Foreman polynomials*

$$K_v(\alpha) = \sum_p v_p \, \alpha_B^p K_{(p)}(\alpha_R), \qquad \Psi_v(\alpha) = \sum_p v_p \, \alpha_B^{B \setminus p} \Psi_{(p)}(\alpha_R).$$

Selecting the pattern p on B, that is keeping the bases with $T \cap B = p$, and then p' on B' selects $p \cup p'$ on $B \cup B'$ when $p \cap B' = p' \cap B$, and nothing otherwise; so the selections on a table with n marks are at most the 3^n conditions that require each mark present, absent, or nothing. At the centre every tinkered table keeps **1** *native (Proposition 8.21).*

12.2 The Skeleton polynomials of a bunch

Definition 12.9 (Skeleton polynomials, values and coins)**.** At a gauge α, with the level $N_B = |T \cap B|$ and the *colevel* $N_B^* = |B \setminus T|$, the *Skeleton polynomials* of B are

$$\mathsf{Q}_B^K(x; \alpha) = \mathbb{E}_{T \sim \mathbb{P}_\alpha}\big[x^{N_B - d(B)}\big] = x^{-d(B)} \frac{K(x\alpha_B, \alpha_R)}{K(\alpha)},$$
$$\mathsf{Q}_B^\Psi(x; \alpha) = \mathbb{E}_{T \sim \mathbb{P}^*_\alpha}\big[x^{N_B^* - d^*(B)}\big] = x^{-d^*(B)} \frac{\Psi(x\alpha_B, \alpha_R)}{\Psi(\alpha)},$$

and $\mathsf{Q}^K_{j,B}$, $\mathsf{Q}^\Psi_{j,B}$ for table j of a family. Each has degree $c(B)$, positive coefficients and value 1 at $x = 1$, so its roots avoid $[0, \infty)$; writing them $-\xi_1, \dots, -\xi_{c(B)}$,

$$\mathsf{Q}_B^K(x; \alpha) = \prod_{i=1}^{c(B)} \frac{x + \xi_i}{1 + \xi_i} = \prod_{i=1}^{c(B)} (1 - p_i + p_i x), \qquad p_i = \frac{1}{1 + \xi_i}.$$

The ξ_i are the *Skeleton values*, their multiset $\operatorname{spec}_B^K(\alpha) \subset \mathbb{C} \setminus (-\infty, 0]$ is the *Skeleton spectrum*, and the p_i are the *coins*; likewise $\operatorname{spec}_B^\Psi(\alpha)$ and the coins of Q_B^Ψ. The *cumulants of a coin* are $\varkappa_1(p) = p$ and $\varkappa_{k+1}(p) = p(1-p)\,\varkappa_k'(p)$.

The Skeleton polynomial is the law of the level of the bunch, shifted to start at its floor: its degree is the width of the bunch, and its roots are the negatives of the Skeleton values. When the spectrum is positive the level less its floor $d(B)$ is a sum of $c(B)$ independent $\{0,1\}$-valued variables whose means are the coins.

Proposition 12.10 (Relations of the Skeleton polynomials)**.** *(i)* Scaling. *Scaling α_B by $\lambda > 0$ divides the spectrum by λ:*

$$\mathsf{Q}_B^K\big(x; (\lambda\alpha_B, \alpha_R)\big) = \mathsf{Q}_B^K(\lambda x; \alpha)/\mathsf{Q}_B^K(\lambda; \alpha);$$

likewise for Ψ; scaling all of α changes nothing.

(ii) Complement. $\mathsf{Q}_R^K(x;\alpha) = x^{c(B)}\mathsf{Q}_B^K(x^{-1};\alpha)$*: the spectrum of the room is the reciprocal of that of the bunch, and each coin p_i becomes $1-p_i$; likewise for Ψ.*

(iii) Cremona. $\mathsf{Q}_B^\Psi(x;\alpha) = x^{c(B)}\mathsf{Q}_B^K(x^{-1};\alpha^{-1})$*; at the table's gauge,* $\mathsf{Q}_B^\Psi(x;\mathbf{1}) = x^{c(B)}\mathsf{Q}_B^K(x^{-1};\mathbf{1})$.

(iv) Coins. $\log \mathsf{Q}_B^K(e^v;\alpha) = \sum_i \log(1-p_i+p_ie^v)$*; under $\mathbb{P}_\alpha$ the level N_B has mean $d(B)+\sum_i p_i = \sum_{e\in B} R_e(\alpha)$ and k-th cumulant $\sum_i \varkappa_k(p_i)$ for $k \geq 2$. When the spectrum is positive the coins lie in $(0,1)$ and $N_B - d(B)$ is distributed as a sum of independent $\{0,1\}$-valued variables whose means are the coins.*

Proposition 12.11 (The still read along a bunch)**.** *Let a family be at a gauge α, let $p_{j,i}$ be the coins of $\mathsf{Q}_{j,B}^K(\cdot;\alpha)$ and $p_{j,i}^*$ those of $\mathsf{Q}_{j,B}^\Psi(\cdot;\alpha)$, and put $\partial_B = \sum_{e\in B}\partial_{y_e}$. Then*

$$\sum_j \Big(s_j \sum_i p_{j,i} + t_j \sum_i p_{j,i}^*\Big) = \partial_B\Phi(\log\alpha) - \sum_j \big(s_j d_j(B) + t_j d_j^*(B)\big),$$
$$\partial_B^k\Phi(\log\alpha) = \sum_j \Big(s_j \sum_i \varkappa_k(p_{j,i}) + t_j \sum_i \varkappa_k(p_{j,i}^*)\Big) \qquad (k \geq 2).$$

At a native gauge the first total is the wall κ_B at the masses $\boldsymbol{\tau}$, and on the balance wall it is κ_B; the entries of rung k of the still of $\big((s_j,t_j)_j;\alpha\big)$, summed over B^k, are 2^k times the second total.

Proposition 12.12 (Coins of a tensorminental table)**.** *In the setting of Theorem 7.20, let $p_i^{(kl)}$, $1 \leq i \leq |B|$, be the eigenvalues of $\Pi_{BB}^{(kl)}$. Then*

$$x^{d(B)}\mathsf{Q}_B^K(x;\alpha) = \sum_{k,l} w_{kl} \prod_{i=1}^{|B|} \big(1 - p_i^{(kl)} + p_i^{(kl)}x\big) :$$

the Skeleton polynomial is the w-mixture of the coin products of the components. At tensor rank one over $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$ there is one component, and $\Pi_{BB}^{(11)}$ is similar to the compression to $\mathbb{F}^B$ of the orthogonal projection onto the row space of $AD_\alpha^{1/2}$, A the slice; its eigenvalues are $\cos^2\vartheta_i$ for the principal angles ϑ_i between $\mathbb{F}^B$ and that row space. Of these, $d(B)$ are 1 and $d^*(B)$ are 0, and the $c(B)$ coins are the values $\cos^2\vartheta_i \in (0,1)$, with the positive Skeleton values $\tan^2\vartheta_i$.

Definition 12.13 (The pair of Skeleton functions)**.** The *even* and *odd Skeleton functions* of B are $\mathsf{Q}^{\mathbf{E}}_B(x;\alpha) = \mathsf{Q}^K_B(x;\alpha)\,\mathsf{Q}^\Psi_B(x;\alpha)$ and $\mathsf{Q}^{\mathbf{O}}_B(x;\alpha) = \mathsf{Q}^K_B(x;\alpha)/\mathsf{Q}^\Psi_B(x;\alpha)$. So

$$\mathsf{Q}^{\mathbf{E}}_B(x;\alpha) = \mathbb{E}_{(T,T')\sim\mathbb{P}_\alpha\otimes\mathbb{P}^*_\alpha}\Big[x^{|T\cap B|-d(B)+|B\setminus T'|-d^*(B)}\Big],$$

and its roots are the negatives of the *joint spectrum* $\operatorname{spec}^K_B(\alpha)\cup\operatorname{spec}^\Psi_B(\alpha)$.

Proposition 12.14 (A set on its own scale is read by its Skeleton functions)**.** *For a family at a gauge α, a set C of marks and $x = e^v$,*

$$\Phi_{\mathbf{E}}(\log\alpha + v\mathbf{1}_C) - \Phi_{\mathbf{E}}(\log\alpha) = \sum_j \sigma_j\Big(\log \mathsf{Q}^{\mathbf{E}}_{j,C}(x;\alpha) + \big(d_j(C) + d^*_j(C)\big)v\Big),$$
$$\Phi_{\mathbf{O}}(\log\alpha + v\mathbf{1}_C) - \Phi_{\mathbf{O}}(\log\alpha) = \sum_j \delta_j\Big(\log \mathsf{Q}^{\mathbf{O}}_{j,C}(x;\alpha) + \big(d_j(C) - d^*_j(C)\big)v\Big).$$

Along the scale of a set the K-half of each table's joint spectrum carries the exponent $s_j = \sigma_j + \delta_j$ and the Ψ-half carries $t_j = \sigma_j - \delta_j$.

12.3 Levels

Definition 12.15 (Levels)**.** For $d(B) \le k \le r(B)$ the *level-k patterns* are $\operatorname{pat}^{(k)}(B) = \{p \in \operatorname{pat}(B) :\ |p| = k\}$, the *level-$k$ bases* are $\mathcal{B}^{(k)}_B = \{T \in \mathcal{B} :\ |T\cap B| = k\}$, and the *level letter* is

$$K^{(k)}_B(\alpha) = \sum_{T\in\mathcal{B}^{(k)}_B} \theta_T\alpha^T = \sum_{p\in\operatorname{pat}^{(k)}(B)} \alpha^p_B K_{(p)}(\alpha_R).$$

For $d^*(B) \le k \le |B| - d(B)$ the *colevel letter* is $\Psi^{(k)}_B(\alpha) = \sum_{T:\,|B\setminus T|=k}\theta_T\alpha^{\mathcal{E}\setminus T}$. The *excess* of the level k, of its patterns and of its bases is $k - d(B)$, and that of the colevel k is $k - d^*(B)$. Then $K = \sum_k K^{(k)}_B$, $\Psi = \sum_k \Psi^{(k)}_B$,

$$x^{d(B)}\mathsf{Q}^K_B(x;\alpha) = \sum_k x^k\,\frac{K^{(k)}_B(\alpha)}{K(\alpha)}, \qquad x^{d^*(B)}\mathsf{Q}^\Psi_B(x;\alpha) = \sum_k x^k\,\frac{\Psi^{(k)}_B(\alpha)}{\Psi(\alpha)}.$$

Proposition 12.16 (Pattern levels are lifts)**.** *Let $\mathcal{B}|B$ be the restriction to B and $\mathcal{B}/R$ the contraction of the room, a table on B with rank function $r_{\mathcal{B}/R}(X) = r(X\cup R) - r(R)$ and rank $d(B)$.*

(i) $p \subseteq B$ is a pattern if and only if p is independent and $r_{\mathcal{B}/R}(p) = d(B)$.

(ii) For $i = k - d(B)$, the independent sets $X \subseteq B$ with $|X| - r_{\mathcal{B}/R}(X) \le i$ are the independent sets of a table on B of rank k, the i-th lift of $\mathcal{B}/R$ toward $\mathcal{B}|B$, whose bases are the level-k patterns. So every pattern level $\operatorname{pat}^{(k)}(B)$ is a table on B.

(iii) The room sides $T\cap R$ of the level-k bases are the bases of the $(r(B)-k)$-th lift of $\mathcal{B}/B$ toward $\mathcal{B}|R$.

Theorem 12.17 (Levels are sums of children)**.** $\mathcal{B}^{(k)}_B = \bigsqcup_{p\in\operatorname{pat}^{(k)}(B)}\mathcal{B}_{[p]}$, *where $\mathcal{B}_{[p]} = \{T \in \mathcal{B} :\ T\cap B = p\}$ is the set of bases of the table on $\mathcal{E}$ that has p as coloops, $B\setminus p$ as loops and the child*

at p on R, with the parent's weights; its basis polynomial is $\alpha_B^p K_{(p)}(\alpha_R)$. A bijection g of $\mathcal{E}$ that carries the table and keeps B takes $\mathcal{B}_{[p]}$ to $\mathcal{B}_{[g(p)]}$. The extreme levels have constant children:

$$\mathcal{B}_B^{(d(B))} = \{p \sqcup X :\ p \in \mathcal{B}/R,\ X \in \mathcal{B}|R\}, \qquad \mathcal{B}_B^{(r(B))} = \{p \sqcup X :\ p \in \mathcal{B}|B,\ X \in \mathcal{B}/B\};$$

for a table determinantal over $\mathbb{R}$ or $\mathbb{C}$ the weights of the extreme levels factor along these products, and with them the extreme letters, $K_B^{(d(B))}(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}|R}(\alpha_R)$ and $K_B^{(r(B))}(\alpha) = K_{\mathcal{B}|B}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)$.

The bases of one level need not form a table: a step of the exchange property may leave the level, and it then moves to an adjacent one (Proposition 12.4). What a level keeps is its patterns, which form a table (Proposition 12.16); its bases are the glued children of those patterns, and children on adjacent levels are linked by Proposition 12.5. The level is the coordinate the bunch keeps for itself, and the children hand the rest down to the room.

Theorem 12.18 (Shared walls)**.** *Let $L_i = \sum_h K_{i,h}$, $i = 1, \dots, q$, each a sum of the basis polynomials $K_{i,h}$ of tables $\mathcal{B}_{i,h}$ on $\mathcal{E}$, with exponents $w_i > 0$, and put $d_{i,h}(S) = \min_{T \in \mathcal{B}_{i,h}} |T \cap S|$. Then*

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m-1}} e^{-|\alpha|} \prod_i L_i(\alpha)^{-w_i}\, d\alpha$$

converges if and only if there are $w_{i,h} > 0$ with $\sum_h w_{i,h} = w_i$ and $m_S > \sum_{i,h} w_{i,h}\, d_{i,h}(S)$ for every nonempty $S \subseteq \mathcal{E}$; equivalently $\mathbf{m} \in \operatorname{int}\big(\sum_i w_i \operatorname{conv} \bigcup_h P_{i,h} + \mathbb{R}^{\mathcal{E}}_{\geq 0}\big)$, with $P_{i,h}$ the base polytope of $\mathcal{B}_{i,h}$.

Proposition 12.19 (The walls of one level are exact)**.** *With P the base polytope, $P_B^{(k)} = \operatorname{conv}\{\mathbf{1}_T : T \in \mathcal{B}_B^{(k)}\}$ and $d_B^{(k)}(S) = \min_{T \in \mathcal{B}_B^{(k)}} |T \cap S|$:*

$$P_B^{(k)} = P \cap \{x_B = k\}, \qquad P_B^{(k)} + \mathbb{R}^{\mathcal{E}}_{\geq 0} = \{x :\ x_S \geq d_B^{(k)}(S) \text{ for all } S\}.$$

So for $w > 0$ the integral $\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m-1}} e^{-|\alpha|} K_B^{(k)}(\alpha)^{-w} d\alpha$ converges exactly when $m_S > w\, d_B^{(k)}(S)$ for every nonempty S. The one-dimensional faces of $P_B^{(k)}$ that do not join two bases differing by a single step of the exchange property are diagonals of square two-dimensional faces of P, one step into B and one out of it.

For several levels together the 0/1 walls need not be exact: the facets of $\sum_k w_k P_B^{(k)} + \mathbb{R}^{\mathcal{E}}_{\geq 0}$ are normal to rays of the common refinement of the normal fans of the $P_B^{(k)}$, which can lie outside $\{0,1\}^{\mathcal{E}}$, and the exact domain of several levels together is then the one of Theorem 12.18.

Corollary 12.20 (The native level split is interior)**.** *Let a family with real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, have a native gauge $\alpha^\star$, and give the level letters $K_{j,B}^{(k)}$ of table j the exponents $s_j\, \mathbb{P}_{j,\alpha^\star}(N_B = k)$ and its colevel letters $\Psi_{j,B}^{(k)}$ the exponents $t_j\, \mathbb{P}^*_{j,\alpha^\star}(N^*_B = k)$. Then $\boldsymbol{\tau}$ lies in the relative interior of the Minkowski sum of the level and colevel polytopes, each multiplied by the exponent of its letter, and every wall of these letters at the masses $\boldsymbol{\tau}$,*

$$\tau_S - \sum_{j,k} \Big(s_j\, \mathbb{P}_{j,\alpha^\star}(N_B = k)\, d_{j,B}^{(k)}(S) + t_j\, \mathbb{P}^*_{j,\alpha^\star}(N^*_B = k)\, d_{j,B}^{*(k)}(S) \Big),$$

$d^{(k)}_{j,B}$ the level counts of Proposition 12.19 and $d^{*(k)}_{j,B}(S)$ the least of $|S \setminus T|$ over the colevel-k bases, is positive except on the sets S on which x_S is constant over every level and colevel polytope, among them B and R. If the marks of B are joined to one another, and the marks of R to one another, by chains in which consecutive marks lie in a common circuit of a table with a positive exponent, as they are when such a table is connected, then the directions v with $\langle v, x\rangle$ constant on that Minkowski sum are spanned by $\mathbf{1}_B$ and $\mathbf{1}_R$, there are no exceptions besides these and $\varnothing, \mathcal{E}$, and

$$\int_{\Delta_B\times\Delta_R} u^{\boldsymbol{\tau}|_B-\mathbf{1}} v^{\boldsymbol{\tau}|_R-\mathbf{1}} \prod_{j,k} K^{(k)}_{j,B}(u,v)^{-s_j \mathbb{P}_{j,\alpha^\star}(N_B=k)} \Psi^{(k)}_{j,B}(u,v)^{-t_j \mathbb{P}^*_{j,\alpha^\star}(N^*_B=k)}\, d\omega_B(u)\, d\omega_R(v)$$

converges.

12.4 Shares and the native split

Definition 12.21 (Shares, loads, children of a family)**.** For a family and a bunch B, *shares* are $\mathbf{a}_j = (a_{j,p})_{p\in \mathrm{pat}_j(B)}$ with $|\mathbf{a}_j| = s_j$ and $\mathbf{b}_j = (b_{j,p})_{p\in \mathrm{pat}_j(B)}$ with $|\mathbf{b}_j| = t_j$; the *pair shares* are $\sigma_{j,p} = \frac{a_{j,p}+b_{j,p}}{2}$ and $\delta_{j,p} = \frac{a_{j,p}-b_{j,p}}{2}$. A K-share of pattern p loads the marks of p, and a Ψ-share loads those of $B \setminus p$: the *load* and the *odd load* on $e \in B$ are

$$\mathrm{load}_e = \sum_j \Big(\sum_{p\ni e} a_{j,p} + \sum_{p\not\ni e} b_{j,p} \Big) = \sigma + \langle \boldsymbol{\delta}\rangle_e, \qquad \langle \boldsymbol{\delta}\rangle_e = \sum_{j,p} \delta_{j,p}\chi_p(e).$$

The *children* are the family

$$F_B(\mathbf{a},\mathbf{b}) = \big((\mathcal{B}_{j,(p)}, \theta_{j,(p)});\, a_{j,p}, b_{j,p} \big)_{j,p} \quad \text{on } R.$$

Definition 12.22 (Pattern probabilities; the native split)**.** At a gauge β the *pattern probabilities* of table j are $\pi_{j,p}(\beta) = \mathbb{P}_{j,\beta}(T\cap B = p)$ and $\pi^*_{j,p}(\beta) = \mathbb{P}^*_{j,\beta}(T\cap B = p)$. At a native gauge $\alpha^\star$ the *native split* is the choice of shares $\mathbf{a}^\star$, $\mathbf{b}^\star$ with $a^\star_{j,p} = s_j \pi_{j,p}(\alpha^\star)$ and $b^\star_{j,p} = t_j \pi^*_{j,p}(\alpha^\star)$, and $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$ are the children at the native split.

Theorem 12.23 (Nativity is hereditary)**.** *$\alpha^\star|_R$ is a native gauge for the children $F_B(\mathbf{a}^\star,\mathbf{b}^\star)$ with target $\boldsymbol{\tau}|_R$, and they lie on their own balance wall at the masses $\boldsymbol{\tau}|_R$; every set $S \subseteq R$ with $c_{j,(p)}(S) = 0$ for every child table is balanced, $\tau_S = \sum_{j,p} \big(a^\star_{j,p} r_{j,(p)}(S) + b^\star_{j,p} r^*_{j,(p)}(S) \big)$.*

Proposition 12.24 (The centre is hereditary)**.** *Let $\Upsilon = ((s_j,t_j)_j; \mathbf{1})$ be at the table's gauge, with its masses. The native split gives the child (j,p) the pair*

$$\sigma_{j,p} = \pi_{j,p}(\mathbf{1})\,\sigma_j, \qquad \delta_{j,p} = \pi_{j,p}(\mathbf{1})\,\delta_j, \qquad \pi_{j,p}(\mathbf{1}) = \mathbb{P}_{j,\mathbf{1}}(T\cap B = p) > 0,$$

and the gauge $\mathbf{1}$ on R, the children's own table's gauge, native for them. At a centre every child sits at a centre; at one pair the children are at the same pair; a full lean is handed down full.

Proposition 12.25 (The split is forced)**.** *(i) For every gauge β and every probability vector w on $\mathrm{pat}_j(B)$ with every $w_p > 0$,*

$$K_j(\beta) \;\ge\; \prod_p \Big(\frac{\beta_B^p K_{j,(p)}(\beta_R)}{w_p} \Big)^{w_p},$$

with equality if and only if $\pi_{j,p}(\beta) = w_p$ for every p; likewise for Ψ_j with the pattern terms $\beta_B^{B\setminus p}\Psi_{j,(p)}(\beta_R)$ and $\pi^*_{j,p}(\beta)$. So the shares $a_{j,p} = s_j w_p$ replace $K_j^{-s_j}$ by the product of its pattern terms exactly at the gauges where the basis law puts the proportions w on the patterns; at the gauge of a realization the only such shares are its native split.

(ii) $\beta_B^{-p}K_{j,(p)}(\beta_R^{-1}) = \beta^{-\mathcal{E}}\beta_B^{B\setminus p}\Psi_{j,(p)}(\beta_R)$ *and* $\pi_{j,p}(\beta^{-1}) = \pi^*_{j,p}(\beta)$*: Cremona turns the K-term of each pattern into its Ψ-term, acting on each child as the child's own Cremona. At the table's gauge* $\pi^*_{j,p}(\mathbf{1}) = \pi_{j,p}(\mathbf{1})$*: the forced shares split s_j and t_j in the same proportions.*

A bunch hands its room a family of children, one for each pattern of each table, and the native split is the one choice of shares under which the room inherits everything: the gauge stays native, the masses restrict, and a centre hands down centres. The inequality of (i) is why: at a gauge, the pattern terms of a letter can stand in for the letter with no loss only in the proportions in which the law at that gauge already uses them.

Chapter 13

The order of classes and the Foreman bunch

13.1 The rises of a class

Definition 13.1 (The rises of a class)**.** Let a molecule M with n marks be read at realizations $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ at the table's gauge with $\boldsymbol{\sigma}$ fixed, and let C be a class of the realization at hand, with m marks, and R its room. For a function f of $v=(v_0,\dots,v_k)$ analytic at 0, $\mathring{f}(v)=f(v)-f(0)-\sum_i v_i\,\partial_{v_i}f(0)$ is f less its value and its tangent.

(i) For $0\le k\le n-m$ and distinct partners $\mathbf{g}=(g_1,\dots,g_k)$ in R, the *rise of C with the partners* $\mathbf{g}$ is the increase of the potential along the class, each partner on its own scale:

$$V^{\mathbf{X}}_{C,\mathbf{g}}(v_0,\dots,v_k)=\Phi_{\mathbf{X}}\Big(v_0\mathbf{1}_C+\sum_{i=1}^{k}v_i\mathbf{1}_{g_i}\Big)-\Phi_{\mathbf{X}}\Big(\sum_{i=1}^{k}v_i\mathbf{1}_{g_i}\Big).$$

Stage k is the multiset $\mathcal{V}_k(C)$ of the pairs $(\mathring{V}^{\mathbf{E}}_{C,\mathbf{g}},\mathring{V}^{\mathbf{O}}_{C,\mathbf{g}})$ over the k-tuples of distinct marks of R; stage 0 is the class on its own scale.

(ii) For an order $w=(c_1,\dots,c_m;r_1,\dots,r_{n-m})$ of the marks that lists those of C first and then those of R, the *rise from the class* is

$$\overline{V}^{\mathbf{X}}_{C,w}(u_1,\dots,u_m,v_1,\dots,v_{n-m})=\Phi_{\mathbf{X}}\Big(\sum_i u_i\mathbf{1}_{c_i}+\sum_j v_j\mathbf{1}_{r_j}\Big)-\Phi_{\mathbf{X}}(0),$$

with positions $0,\dots,n-1$ in the order of w, and the *last stage* is the multiset $\mathcal{V}_*(C)$ of the pairs $(\overline{V}^{\mathbf{E}}_{C,w},\overline{V}^{\mathbf{O}}_{C,w})$ over the $m!\,(n-m)!$ such orders.

$\mathcal{V}_k(C)$ has $(n-m)!/(n-m-k)!$ elements; $v_k=0$ turns the rise with $(g_1,\dots,g_k)$ into the rise with $(g_1,\dots,g_{k-1})$; for a single mark an element of $\mathcal{V}_*(\{e\})$ is a writing of the potential from e, less $\Phi_{\mathbf{X}}(0)$.

A class is first read on its own scale, then with one partner of the room at a time, then with more, and last from the class with every mark on its own scale. Every stage before the last is taken less its value and tangent; the last stage holds the whole potential.

Proposition 13.2 (Every rise is read by the Skeleton functions of the class)**.** *With $x=e^{v_0}$ and $\alpha=e^{\sum_i v_i\mathbf{1}_{g_i}}$, the gauge the partners move to,*

$$V^{\mathbf{E}}_{C,\mathbf{g}}(v_0,\dots,v_k)=\sum_j\sigma_j\Big(\log\mathtt{Q}^{\mathbf{E}}_{j,C}(x;\alpha)+\big(d_j(C)+d_j^*(C)\big)v_0\Big),$$

$$V^{\mathbf{O}}_{C,\mathbf{g}}(v_0,\dots,v_k)=\sum_j \delta_j\Big(\log \mathsf{Q}^{\mathbf{O}}_{j,C}(x;\alpha)+\big(d_j(C)-d^*_j(C)\big)v_0\Big).$$

Stage 0 reads the Skeleton functions of C at the table's gauge, and stage k reads them at every gauge that k partners move to.

Proposition 13.3 (The rises hold the cumulants of the class with its partners)**.** *For $\mathbf{a}=(a_0,\dots,a_k)\in\mathbb{N}^{k+1}$ write, in this statement, $\kappa_j(\mathbf{a})=\kappa_{j,\mathbf{1}}\big(N_C^{\times a_0},X_{g_1}^{\times a_1},\dots,X_{g_k}^{\times a_k}\big)$, the joint cumulant, under the weight law $\mathbb{P}_{j,\mathbf{1}}$, of the sequence of variables in which the level N_C appears a_0 times and the indicator X_{g_i} appears a_i times. For $a_0\geq 1$ and $|\mathbf{a}|\geq 2$,*

$$[v^{\mathbf{a}}]\,V^{\mathbf{E}}_{C,\mathbf{g}}=\frac{1+(-1)^{|\mathbf{a}|}}{\mathbf{a}!}\sum_j\sigma_j\kappa_j(\mathbf{a}),\qquad [v^{\mathbf{a}}]\,V^{\mathbf{O}}_{C,\mathbf{g}}=\frac{1-(-1)^{|\mathbf{a}|}}{\mathbf{a}!}\sum_j\delta_j\kappa_j(\mathbf{a}),$$

and the coefficients with $a_0=0$ vanish. By multilinearity $\kappa_j(\mathbf{a})$ is the sum, over the strings $u\in C^{a_0}$, of the cumulants $\kappa_{j,\mathbf{1}}$ along the strings that continue u with each g_i taken a_i times: the entries of the still on those strings, summed over the class and read table by table.

Corollary 13.4 (The first two stages in coins)**.** *(i) For $a\geq 2$, $[v_0^a]\,\mathring{V}^{\mathbf{E}}_{C,()}=\frac{2}{a!}\sum_j\sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ for even a and $[v_0^a]\,\mathring{V}^{\mathbf{O}}_{C,()}=\frac{2}{a!}\sum_j\delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ for odd a, the other coefficients being zero; and $\kappa_{j,\mathbf{1}}(N_C^{\times a})=\sum_i\varkappa_a(p_{j,i})$ over the coins $p_{j,i}$ of $\mathsf{Q}^K_{j,C}(\cdot;\mathbf{1})$.*

(ii) For a partner g and $q\geq 1$, with $R_{j,g}(e^{\pm v_0\mathbf{1}_C})$ the occupation of g in table j at the gauge $e^{\pm v_0\mathbf{1}_C}$,

$$[v_1^q]\,V^{\mathbf{E}}_{C,(g)}=\frac{1}{q!}\sum_j\sigma_j\Big(\varkappa_q\big(R_{j,g}(e^{v_0\mathbf{1}_C})\big)+(-1)^q\varkappa_q\big(R_{j,g}(e^{-v_0\mathbf{1}_C})\big)-\big(1+(-1)^q\big)\varkappa_q\big(R_{j,g}(\mathbf{1})\big)\Big),$$

$$[v_1^q]\,V^{\mathbf{O}}_{C,(g)}=\frac{1}{q!}\sum_j\delta_j\Big(\varkappa_q\big(R_{j,g}(e^{v_0\mathbf{1}_C})\big)-(-1)^q\varkappa_q\big(R_{j,g}(e^{-v_0\mathbf{1}_C})\big)-\big(1-(-1)^q\big)\varkappa_q\big(R_{j,g}(\mathbf{1})\big)\Big).$$

13.2 The order

Definition 13.5 (Comparing)**.** (i) *Exponents.* For $\mathbf{a},\mathbf{a}'\in\mathbb{N}^{k+1}$, $\mathbf{a}\sqsubset\mathbf{a}'$ when $a_i<a'_i$ at the largest position i with $a_i\neq a'_i$.

(ii) *Functions.* For functions $f\neq f'$ of the positions $0,\dots,k$, analytic at 0: $f\lhd f'$ when $[v^{\mathbf{a}}]f<[v^{\mathbf{a}}]f'$ at the $\sqsubset$-least exponent at which their coefficients differ.

(iii) *Pairs.* $(f^{\mathbf{E}},f^{\mathbf{O}})\lhd(f'^{\mathbf{E}},f'^{\mathbf{O}})$ when $f^{\mathbf{E}}\lhd f'^{\mathbf{E}}$, or $f^{\mathbf{E}}=f'^{\mathbf{E}}$ and $f^{\mathbf{O}}\lhd f'^{\mathbf{O}}$.

(iv) *Multisets.* Two different multisets of pairs with equally many elements are read from the least element upward in $\lhd$, and the one that first holds the smaller element comes first.

(v) *Classes.* For classes $C\neq D$ of one realization with m marks each, $C\prec D$ when $\mathcal{V}_s(C)\lhd\mathcal{V}_s(D)$ at the first s in $0,1,\dots,n-m,*$ with $\mathcal{V}_s(C)\neq\mathcal{V}_s(D)$.

Definition 13.6 (The Foreman bunch)**.** At the centre, the *Foreman bunch* $B(M;\boldsymbol{\sigma},0)$ is the class of $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ with the fewest marks and, among those, the $\prec$-least. At a lean $\boldsymbol{\delta}$ each class C of $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ lies in one class C° of the centre (Theorem 11.1), and $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ is the class with the least

$$\mathrm{key}_{\boldsymbol{\sigma},\boldsymbol{\delta}}(C)=\big(|C|,\ |C^\circ|,\ C^\circ,\ C\big),$$

compared entry by entry: sizes as integers, C° by $\prec$ at the centre, C by $\prec$ at the lean. At every realization the Foreman bunch is a class with the fewest marks among the classes of that realization; the classes of the centre and the order only choose among those.

Theorem 13.7 (What the order is)**.** *(i)* Total. *For every m, $\prec$ is a strict total order on the classes with m marks; $B(M;\boldsymbol{\sigma},0)$ and $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ are each one class.*

(ii) Directions. *With the classes given, $\prec$ depends on the realization only through the rays of $\boldsymbol{\sigma}$ and $\boldsymbol{\delta}$; at one pair only through the lean sign.*

(iii) Mirror. *If g exchanges the family (Definition 11.14), every stage of $g(C)$ at $-\boldsymbol{\delta}$ equals that stage of C at $\boldsymbol{\delta}$; so $C \prec D$ at $\boldsymbol{\delta}$ exactly when $g(C) \prec g(D)$ at $-\boldsymbol{\delta}$, and $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta}) = g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big)$. If g carries the family, the same holds at one lean.*

(iv) Nesting. *When some class at $\boldsymbol{\delta}$ inside $B(M;\boldsymbol{\sigma},0)$ has the fewest marks among the classes at $\boldsymbol{\delta}$, $B(M;\boldsymbol{\sigma},\boldsymbol{\delta}) \subseteq B(M;\boldsymbol{\sigma},0)$, and it is the class at $\boldsymbol{\delta}$ inside $B(M;\boldsymbol{\sigma},0)$ with the fewest marks and, among those, the $\prec$-least. In a one-table molecule this is so whenever $B(M;\boldsymbol{\sigma},0)$ splits at $\boldsymbol{\delta}$. Otherwise $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ need not lie in $B(M;\boldsymbol{\sigma},0)$.*

Every rise is a linear combination of logarithms of polynomials in $e^{v_0},\dots,e^{v_k}$, less the same combination at $v_0 = 0$ (Proposition 13.2), so equality of two rises is an identity of polynomials with exponents (Theorem 10.6). When two rises differ, the first differing exponent is found column by column: $v_k = 0$ is the rise with one partner fewer, the coefficient of v_k^b is a rational function of the scales at the other positions (Corollary 13.4(ii) for $k = 1$), each column is certified equal by a polynomial identity, and the deciding coefficient is a joint cumulant of Proposition 13.3, multiplied by the σ_j or the δ_j, and rational when the weights of the tables are.

Proposition 13.8 (Rung one enters only at the last stage)**.** *Every stage before the last reads the still from rung two on; the masses, rung one, enter only at the last stage, and at a centre they are the same at every mark.*

13.3 What the first stages read

Proposition 13.9 (At the centre the first number is how far the class is from a scale of its own)**.** $\operatorname{Var}_{T\sim\mathbb{P}_{j,\mathbf{1}}} |T\cap U| = 0$ *in every table exactly when U is a union of molecules. So for a class C of a molecule other than the whole,*

$$[v_0^2]\,\mathring{V}^{\mathbf{E}}_{C,()} = \sum_j \sigma_j \operatorname{Var}_{T\sim\mathbb{P}_{j,\mathbf{1}}} |T\cap C| > 0.$$

Proposition 13.10 (Mirror classes)**.** *Let C, D be classes with equally many marks whose levels are mirrored across the tables: there are a bijection h of the tables with $(\sigma_{h(j)},\delta_{h(j)}) = (\sigma_j,\delta_j)$ and integers k_j with $\mathbb{P}_{h(j),\mathbf{1}}(|T\cap D| = a) = \mathbb{P}_{j,\mathbf{1}}(|T\cap C| = k_j - a)$ for every j and every a. This holds with h the identity for two classes that together make up the molecule ($k_j = r_j$), and for C and $g(C)$ under an exchange g of the family, h the bijection of its tables ($k_j = |C|$). Then:*

(i) $\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a \kappa_{j,\mathbf{1}}(N_C^{\times a})$ *for $a \geq 2$, so $\sum_j \sigma_j \kappa_{j,\mathbf{1}}(N_D^{\times a}) = (-1)^a \sum_j \sigma_j \kappa_{j,\mathbf{1}}(N_C^{\times a})$, and likewise with the δ_j: the $\mathbf{E}$ parts of $\mathcal{V}_0(C)$ and $\mathcal{V}_0(D)$ are equal and their $\mathbf{O}$ parts are negatives;*

(ii) at a lean, if some odd $a \geq 3$ has $\sum_j \delta_j \kappa_{j,\mathbf{1}}(N_C^{\times a}) \neq 0$ and a is the least such, then $C \prec D$ exactly when $\sum_j \delta_j \kappa_{j,\mathbf{1}}(N_C^{\times a}) < 0$, and at the opposite lean the order is the other one;

(iii) at the centre stage 0 *does not separate* C *and* D*, and a later stage does.*

Corollary 13.11 (A bunch across the centre)**.** *In a one-table molecule, if* $B(M;\sigma,0)$ *splits at a lean* δ *into* C_K *and* C_Ψ *(Theorem 11.18), then* $B(M;\sigma,\delta) \in \{C_K, C_\Psi\}$ *and* $B(M;\sigma,-\delta) = \beta\big(B(M;\sigma,\delta)\big)$ *is the other one,* β *the exchange of Theorem 11.18; the two are mirror classes, ordered at a lean by Proposition 13.10(ii).*

The Foreman bunch is the least of the smallest classes, and the order that chooses it reads the class first on its own scale. At the centre the first number read is the spread of the level of the class under the weight laws, which vanishes only for a union of molecules; a lean adds the odd cumulants of that level, and two mirror classes are told apart by the first of them that does not vanish, the opposite lean reversing the choice.

Chapter 14

Peel, children and the Foreman tree

14.1 The peel and the tree

Definition 14.1 (Peel; childcules)**.** Families are read at realizations at the table's gauge, and for a molecule M of a family F, $F|M$ is the family of the restrictions of its tables to M. The *peel* of a family F at a bunch B is the family of its children at the native split of the table's gauge (Proposition 12.24),

$$F_B(\mathbf{a}^\star, \mathbf{b}^\star), \qquad a^\star_{j,p} = \pi_{j,p}(\mathbf{1})\, s_j, \quad b^\star_{j,p} = \pi_{j,p}(\mathbf{1})\, t_j,$$

on the room R; children with the same bases and weights are one table, with their exponents added. A *childcule* is a molecule of the peel.

Definition 14.2 (The record of a peel)**.** The *record* of the peel of F at B is the bunch B with, for every table j, the communicatroid of B (Definition 12.3) and the pattern probabilities $\pi_{j,p}(\mathbf{1})$: the indexed children $\big(j,p;\ \mathcal{B}_{j,(p)}, \theta_{j,(p)};\ \pi_{j,p}(\mathbf{1})s_j,\ \pi_{j,p}(\mathbf{1})t_j\big)$. The peel is the record with the indices (j,p) forgotten and equal children merged.

Proposition 14.3 (The record keeps the parent)**.** *The record determines F: table j is the gluing of its communicatroid (Proposition 12.4), and its exponents are the sums of its children's,* $\sum_p \pi_{j,p}(\mathbf{1})(s_j, t_j) = (s_j, t_j)$.

Merging equal children forgets which pattern each share came from, and the Gammas $\Gamma\big(\mu_e - \langle\boldsymbol{\delta}\rangle_e\big)$ of the port (Theorem 15.15) depend on it. The record keeps it. A bunch of width one whose communicatroid is that of one mark, two children on two levels, may instead be pooled into one mark of the room, its shape integrated and its scale kept (Theorem 18.8).

Definition 14.4 (The Foreman tree)**.** For a family F read at $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, the *Foreman tree* $\mathrm{Tree}(F; \boldsymbol{\sigma}, \boldsymbol{\delta})$ splits F into molecules. A molecule with one mark is a unit (Definition 7.4), and a molecule M whose realization is atomic is an atom (Definition 10.1); otherwise the tree records the Foreman bunch $B(M; \boldsymbol{\sigma}_M, \boldsymbol{\delta}_M)$, $(\boldsymbol{\sigma}_M, \boldsymbol{\delta}_M)$ the pair coordinates of the tables of M, and continues with the peel of $F|M$ at that bunch, on its room.

The tree asks each molecule whether it is an atom; where it is not, the tree takes the least of its smallest classes as the bunch, hands the room the children of the bunch at the native split, and asks again. Nothing is lost on the way down: the record keeps the parent, and the native split keeps the gauge native and the centre a centre.

Proposition 14.5 (Peeling in two steps is peeling at once)**.** *For $B = B_1 \sqcup B_2$, the peel at B_2 of the peel of F at B_1 is the peel of F at B: the child of F at (j,p) and the child at $\big((j, p\cap B_1), p\cap B_2\big)$ of the peel at B_1 are one table on $\mathcal{E}\setminus B$ with the same exponents. So if a molecule is peeled at B_1, its room is one childcule and the bunch there is B_2, the family reached is the peel at $B_1 \sqcup B_2$.*

Theorem 14.6 (Three trees)**.** $\mathrm{Tree}(F;\boldsymbol{\sigma},\boldsymbol{\delta})$ *depends on $(\boldsymbol{\sigma},\boldsymbol{\delta})$ only through their rays; at one pair only through the lean sign. So a molecule at one pair has three Foreman trees,* $\mathrm{Tree}(F;\boldsymbol{\sigma},0)$ *and* $\mathrm{Tree}(F;\boldsymbol{\sigma},\pm\boldsymbol{\delta})$ *for a basis lean $\boldsymbol{\delta}$, fixed by its tables.*

Lemma 14.7 (Maps descend)**.** *Let $g:\mathcal{E}\to\mathcal{E}'$ be a bijection, $B\subseteq\mathcal{E}$ with room R, and $B'=g(B)$.*

(i) If g carries F to F', then $g|_R$ carries the peel of F at B to the peel of F' at B', the child (j,p) going to $(h(j),g(p))$, h the bijection of the tables of Definition 11.14.

(ii) If g exchanges F with F', then $g|_R$ exchanges the peel of F at B with the peel of F' at B', the child (j,p) going to $(h(j),B'\setminus g(p))$.

(iii) In both cases g maps the molecules of F onto those of F', and on a molecule M its restriction carries, respectively exchanges, $F|M$ with $F'|g(M)$.

Theorem 14.8 (The mirror)**.** *If g exchanges F with F', then $g\big(\mathrm{Tree}(F;\boldsymbol{\sigma},\boldsymbol{\delta})\big)=\mathrm{Tree}(F';\boldsymbol{\sigma},-\boldsymbol{\delta})$: molecules go onto molecules, units onto units, atoms onto atoms, each recorded bunch onto the bunch recorded at the same place, rooms onto rooms. If g carries F to F' the same holds at the same lean.*

Corollary 14.9 (A molecule with an exchange)**.** *If g exchanges a molecule M (Definition 11.14), then g maps* $\mathrm{Tree}(M;\boldsymbol{\sigma},0)$ *onto itself, every class of the centre onto itself (Corollary 11.16), and* $\mathrm{Tree}(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ *onto* $\mathrm{Tree}(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$*: $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})=g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big)$ and $g\big(B(M;\boldsymbol{\sigma},0)\big)=B(M;\boldsymbol{\sigma},0)$. If g carries M, it maps every class of $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ onto itself and each of the trees at 0, $\boldsymbol{\delta}$ and $-\boldsymbol{\delta}$ onto itself.*

Lemma 14.10 (An exchange turns a window over)**.** *If g exchanges a family F with F', with the bijection h of the tables (Definition 11.14), and $S'=g(S)$, then for every table j the window of S' in table $h(j)$ of F' is $\{|S|-r_j(S),\dots,|S|-d_j(S)\}$: S in table j and S' in table $h(j)$ have the same width, and the bases of table j at level k on S correspond to those of table $h(j)$ at level $|S|-k$ on S'.*

Corollary 14.11 (Below a no)**.** *A molecule that answers no is peeled at its Foreman bunch read at a centre of the table's gauge; each childcule is a molecule at a centre of the table's gauge, and whether it answers yes is read there in turn.*

14.2 Patterns of one kind

Proposition 14.12 (Level form)**.** *A function of the patterns of a bunch B has* level form *when it depends on $|p|$ alone, and B is in level form in a table when its children are, and in a family when it is in level form in every table. B is in level form in table j, $K_{j,(p)}=K_{j,(p')}$ whenever $|p|=|p'|$, if and only if every transposition of two marks of B, fixing every other mark, maps the bases of table j onto themselves with their weights; so B is in level form in a family exactly when every such transposition carries the family with the identity on its tables.*

Definition 14.13 (Kinds of pattern)**.** Let a one-table molecule be read at the table's gauge and B a set of marks. A *writing from a pattern p* is K written in an order whose first $|p|$ positions hold

p, the next $|B \setminus p|$ hold $B \setminus p$, and the rest the room; a writing of Ψ from $B \setminus p$ likewise. Patterns p, p' are of one *kind at a lean* when one polynomial is written from p and from p', and of one *kind at the centre* when moreover a writing from p may equal a writing of Ψ from $B \setminus p'$.

Theorem 14.14 (One child per kind)**.** *(i) p, p' are of one kind at a lean exactly when a bijection keeping B and carrying the table maps p to p', and of one kind at the centre exactly when, in addition, an exchanging bijection keeping B may map p to $B \setminus p'$. Being of one kind is reflexive, symmetric and transitive.*

(ii) Patterns of one kind have one child between them: the child at p' is the image of the child at p under the bijection restricted to the room, or the dual of that image when the bijection exchanges; their pattern probabilities are equal.

(iii) So the level letters $K_B^{(k)}(\alpha) = \sum_{p \in \mathrm{pat}^{(k)}(B)} \alpha_B^p K_{(p)}(\alpha_R)$ are determined by one child per kind, the patterns of each kind, and the bijections.

Corollary 14.15 (A family with maps in hand)**.** *Let $\mathcal{G}$ be any set of bijections keeping B that carry or exchange a family. Patterns reached from one another by the moves $p \mapsto g(p)$, $g \in \mathcal{G}$ carrying, and $p \mapsto B \setminus g(p)$, $g \in \mathcal{G}$ exchanging, have children that are one table up to the bijections on the room and duality, with equal pattern probabilities.*

Kinds are read on the writings of the two polynomials of one table; with several tables there is no single pair of polynomials to write, and the corollary takes the bijections as given.

Theorem 14.16 (The pattern tensor of a class)**.** *Let B be a class of a one-table molecule at a realization at the table's gauge, $e, f \in B$, and β the bijection taking an order in which the class's one function is written from e to one in which it is written from f (Theorem 10.5). Then:*

(i) β carries the table, or at the centre possibly exchanges it, and maps every class onto itself; in particular $\beta(B) = B$ and $\beta(R) = R$;

(ii) the child at $\beta(p)$ is the child at p with the room mapped by β, or, when β exchanges, the child at $B \setminus \beta(p)$ is the dual of that image; so p and $\beta(p)$, respectively p and $B \setminus \beta(p)$, are of one kind;

(iii) when β carries, the patterns through e correspond to those through f; when it exchanges, the patterns through e correspond to those avoiding f; in both cases with equal pattern probabilities and with children equal up to the bijection on the room and duality.

So the pattern tensor of a class is the same from each of its marks, at the centre up to complements.

14.3 What the room keeps

Definition 14.17 (Keeping the potential; twins; descents; copies)**.** Let a family be read at a realization $\Upsilon = ((s_j, t_j)_j; \alpha)$, with the potential $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of Definition 8.17 and $(g^* y)_e = y_{g(e)}$ as in Definition 11.14. A bijection g of the marks of a molecule M *keeps the potential* when $\Phi_{\mathbf{X}}(\log \alpha + g^* y) = \Phi_{\mathbf{X}}(\log \alpha + y)$ for every y and both parts; at the table's gauge this is $\Phi_{\mathbf{X}} \circ g^* = \Phi_{\mathbf{X}}$. A bijection $g : M \to M'$ onto the marks of a molecule M', read at a realization with the gauge α', *takes the potential* of M to that of M' when $\Phi_{\mathbf{X}}(\log \alpha + g^* y) = \Phi'_{\mathbf{X}}(\log \alpha' + y) + c_{\mathbf{X}}$ for every y and both parts, with constants $c_{\mathbf{X}}$. Twins, descents and copies are read at the table's gauge, $\alpha = \mathbf{1}$: two childcules of the peel of M are *twins* when a bijection keeping the potential of M maps one onto the other. A *descent* of a Foreman tree is a sequence $M_0 \supset M_1 \supset \cdots \supset M_d$ of its nodes from the molecule M_0 to a leaf M_d, each M_{k+1} a childcule of the peel of M_k at its bunch.

On a descent, with B_i the bunch of M_i, the *copies inside* M_k of $Y = M_i$ or $Y = B_i$, $k \le i$, are the sets $g_k(g_{k+1}(\cdots g_{i-1}(Y)\cdots))$, each g_l a bijection keeping the potential of M_l; for $k = i$ they are Y alone.

Lemma 14.18 (The writings certify the potential)**.** *Let a molecule be read at a realization* $\Upsilon = ((s_j, t_j)_j; \alpha)$.

(i) If e and f read alike, the bijection β that takes, position by position, an order with e first to an order with f first in which one pair of functions is written from both (Proposition 10.4) keeps the potential, and $\beta(e) = f$.

(ii) A bijection keeping the potential maps every class onto itself, and a bijection taking the potential of M to that of M' maps the classes of M onto those of M', at the centre and at the realization's lean.

So two marks read alike exactly when a bijection keeping the potential takes one to the other.

Lemma 14.19 (The peel reads the factors of the letters)**.** *Write the potential of a family by its irreducible factors with their exponents $(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$, each normalized by $L(\mathbf{1}) = 1$ (Theorem 10.6), and for a bunch B let $L_{[q]}$ be the coefficient of α_B^q in L, a polynomial on the room R. In this statement write $\Phi_{\mathbf{X}}^{\mathrm{peel}}$ for the potential of the peel $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$. Up to a constant,*

$$\Phi_{\mathbf{X}}^{\mathrm{peel}}(y_R) = \sum_L w_{\mathbf{X}}(L) \sum_q L_{[q]}(\mathbf{1}) \log L_{[q]}\big(e^{y_R}\big),$$

the inner sum over the q with $L_{[q]} \neq 0$, whose values $L_{[q]}(\mathbf{1})$ add to 1.

Theorem 14.20 (The room keeps its classes)**.** *Let a molecule M be read at a realization at the table's gauge, let B be a union of its classes, and let the children's weights factor across the childcules of the peel. Every bijection g keeping the potential of M keeps B; its restriction to the room keeps the potential of the peel, maps each childcule Y onto a childcule $g(Y)$, and takes the potential of Y to that of $g(Y)$, the constant being 0 when $g(Y) = Y$. Consequently, for a class C of M inside the room:*

(i) the childcules that C meets are twins of one another, and C meets each of them in equally many marks;

(ii) inside each childcule, C lies in one class.

Proposition 14.21 (The tree is read off the potential)**.** *Let a molecule M of F and a molecule M' of F' be read at realizations at the table's gauge, with pair coordinates $(\boldsymbol{\sigma}, \boldsymbol{\delta})$ and $(\boldsymbol{\sigma}', \boldsymbol{\delta}')$, the children's weights factoring across the childcules at every node of their trees, and let $g : M \to M'$ take the potential of M to that of M'. Then g maps* $\mathrm{Tree}(F|M; \boldsymbol{\sigma}, \boldsymbol{\delta})$ *onto* $\mathrm{Tree}(F'|M'; \boldsymbol{\sigma}', \boldsymbol{\delta}')$*: classes onto classes, units onto units, atoms onto atoms, each bunch onto the bunch recorded at the same place, and each childcule onto a childcule, its potential to that of its image.*

Theorem 14.22 (The leaf bounds the bunches)**.** *Let a molecule F be read at a realization at the table's gauge, the centre or a lean, the children's weights factoring across the childcules at every node, and let $M_0 \supset M_1 \supset \cdots \supset M_d$ be a descent of its Foreman tree at that realization, B_k the bunch of M_k and $R_k = M_k \setminus B_k$. Let h_k be the number of twins of M_{k+1} among the childcules of M_k, M_{k+1} itself included, and $h_{<k} = h_0 h_1 \cdots h_{k-1}$.*

(i) A class of M_k inside R_k that meets M_{k+1} meets the twins of M_{k+1} and no other childcule, each in equally many marks, and lies in one class of each. The copies inside M_k of M_i,

$k \le i \le d$, respectively of B_i, $k \le i < d$, are $h_{<i}/h_{<k}$ disjoint nodes, respectively bunches, of the subtree at M_k, leaves when $i = d$; and M_d together with its copies inside M_k is a union of whole classes of M_k.

(ii) A class C of M_k that meets M_{k+1} has at least $|B_k|$ marks and meets M_{k+1} in $|C|/h_k \ge |B_k|/h_k$ of them, all in one class of M_{k+1}; the bunch B_{k+1}, and the leaf when $k+1 = d$, contains such a part. So

$$|B_0| \le h_{<1}|B_1| \le h_{<2}|B_2| \le \cdots \le h_{<d-1}|B_{d-1}| \le h_{<d}|M_d|,$$

the k-th term being the number of marks of the copies of B_k inside M_0, and the last that of the copies of M_d: down the descent the bunches with their copies never shrink, and the leaf with its copies has at least as many marks as every bunch with its copies.

Corollary 14.23 (A linked molecule)**.** *Let some active table of F be connected, as for a molecule of one table. Every peel then leaves one childcule, so the tree is a chain (Corollary 19.8), there are no twins, and the room of each node is the next node. For every chain that peels a class at each node, every class of a node other than its bunch lies inside one class of the next node, and the leaf is a union of whole classes of every node above it. For the Foreman tree, which is then a chain, at every realization, the centre and every lean,*

$$|B_0| \le |B_1| \le \cdots \le |B_{d-1}| \le |M_d| :$$

the leaf alone has at least as many marks as every bunch.

Corollary 14.24 (What each node of a chain integrates)**.** *Let the tree of a molecule on n marks be a chain $M_0 \supset M_1 \supset \cdots \supset M_d$ with bunches B_k and rooms $R_k = M_{k+1}$. The simplex of the molecule is the product*

$$\Delta_{M_0} \cong \prod_{k=0}^{d-1} \left(\Delta_{B_k} \times \Delta_{\{B_k, R_k\}} \right) \times \Delta_{M_d}$$

of the shape of each bunch, the totals of the bunch and its room at each node, and the shape of the leaf, and under $\mathrm{Dir}_{\mathbf{m}}$ these coordinates are independent: they are the coordinates of the tree of Theorem 19.14, which use no weights. The node of B_k has $|B_k|$ coordinates, $|B_k| - 1$ of shape and one total, the leaf has $|M_d| - 1$, and they add up to $n - 1$. When some active table is connected, the Foreman tree, which is then a chain, has, at the centre and at every lean, no node with more coordinates than the leaf has marks.

Theorem 14.22 rests on two things: the peel splits a class only across twin childcules, each part inside one class (Theorem 14.20), and the bunch is a class with the fewest marks of the realization at which its node is read. Peeled at another bunch, a molecule can end at a leaf smaller than the bunch above it; at a lean the count is read on the classes of the lean, which can split a class of the centre into parts smaller than the least class of the centre; and the copies are needed when a room falls into twins.

Part IV

Evaluation

Chapter 15

The integral at a bunch

15.1 The scale of a bunch

Proposition 15.1 (A scale integral is a Breta of one bunch)**.** *Let $J = \{0, \dots, L\}$ carry coefficients $\boldsymbol{\xi} = (\xi_\ell)_{\ell \in J}$ and letter exponents $\mathbf{w} = (w_\ell)_{\ell \in J}$ with total $|\mathbf{w}| = \sum_\ell w_\ell$, and let I_J be the unit configuration on J. For $\boldsymbol{\xi} > 0$, $\operatorname{Re} w_\ell > 0$ and $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$,*

$$\int_0^\infty x^{\kappa-1} \prod_{\ell \in J} (x + \xi_\ell)^{-w_\ell}\, dx = \Gamma_{|J|}^{\,2} \begin{bmatrix} \kappa, & |\mathbf{w}| - \kappa \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w},\, |\mathbf{w}|-\kappa;\, I_J}(\boldsymbol{\xi}) :$$

the Breta of Definition 3.1 with the one bunch J, the shape index $\mathbf{w}$, the scale index $|\mathbf{w}| - \kappa$, the unit configuration as its terms and $\boldsymbol{\xi}$ as its coefficients. For $\xi_\ell \in \mathbb{C} \setminus (-\infty, 0]$ the left side, with principal branches, converges exactly when $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$ and is holomorphic in its data there, and it continues the right side from positive coefficients and from letter exponents with positive real part; wherever the left side converges, the product on the right denotes its value. Coincident coefficients merge into one with the summed letter exponent.

The scale of a bunch is integrated against a product of linear letters, one for each point of the joint spectrum, and that integral is a Breta: the points of the spectrum are the coefficients, the letter exponents the shape index, and the index of the scale integral sits in the scale index as $|\mathbf{w}| - \kappa$. Nothing beyond the general Breta is needed to write it. With one coefficient the Breta is the power $\xi_0^{\kappa - w_0}$, and the identity is the integral of a linear letter: for $c_0, c_1 > 0$ and $0 < \operatorname{Re}\nu < \operatorname{Re} s$, $\int_0^\infty \rho^{\nu-1}(c_0 + c_1\rho)^{-s}\, d\rho = \Gamma_1^{\,2} \begin{bmatrix} \nu, & s - \nu \\ & s \end{bmatrix} c_0^{\nu - s} c_1^{-\nu}$. A stacked Gamma $\Gamma_1^{\,2} \begin{bmatrix} a, & b \\ & a+b \end{bmatrix}$ is a *Beta corner*.

Theorem 15.2 (The scale of a bunch is one Breta)**.** *Let a family F on $\mathcal{E}$ be at a gauge α with masses $\mathbf{m}$, let B be a proper bunch with room R, its Skeleton polynomials, values and coins those of Definition 12.9, and let $\xi_0, \dots, \xi_L$ be the distinct points of the joint spectrum of the letters with nonzero exponent, the union of the spectra $\operatorname{spec}^K_{j,B}(\alpha)$ with $s_j \neq 0$ and $\operatorname{spec}^\Psi_{j,B}(\alpha)$ with $t_j \neq 0$, with the letter exponents*

$$w_\ell = \sum_j \big(s_j \operatorname{mult}^K_{j,\ell} + t_j \operatorname{mult}^\Psi_{j,\ell}\big) = \sum_j \big(\sigma_j(\operatorname{mult}^K_{j,\ell} + \operatorname{mult}^\Psi_{j,\ell}) + \delta_j(\operatorname{mult}^K_{j,\ell} - \operatorname{mult}^\Psi_{j,\ell})\big),$$

$\operatorname{mult}^K_{j,\ell}$ *and* $\operatorname{mult}^\Psi_{j,\ell}$ *the multiplicities of ξ_ℓ in $\operatorname{spec}^K_{j,B}(\alpha)$ and $\operatorname{spec}^\Psi_{j,B}(\alpha)$, and $J = \{0, \dots, L\}$. The*

scale integral

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = \int_0^\infty x^{m_B-1}\prod_j K_j(x\alpha_B,\alpha_R)^{-s_j}\Psi_j(x\alpha_B,\alpha_R)^{-t_j}\,dx$$

converges exactly when $\operatorname{Re}\kappa_B > 0$ *and* $\operatorname{Re}\kappa_B^{\mathrm{top}} > 0$*, and then*

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\prod_\ell(1+\xi_\ell)^{w_\ell}\ \Gamma_{\substack{2\\|J|}}\!\left[\begin{matrix}\kappa_B,\ \kappa_B^{\mathrm{top}}\\ \mathbf{w}\end{matrix}\right]\ \mathsf{B}^{\{J\}}_{\mathbf{w},\,\kappa_B^{\mathrm{top}};\,I_J}(\boldsymbol{\xi}),$$

with $|\mathbf{w}| = \sum_j (s_j+t_j)c_j(B) = \kappa_B + \kappa_B^{\mathrm{top}}$*, so that the scale index* $|\mathbf{w}| - \kappa_B$ *of Proposition 15.1 is the top wall; the Breta is read by that proposition at complex spectra.*

Corollary 15.3 (The scale of a bunch of width one)**.** *Let* B *be a bunch with room* R*. Along* B *each letter of a table* j *with* $c_j(B) = 1$ *has one Skeleton value, the level letter at the floor over the level letter at the largest level, colevel letters for* Ψ_j *(Definition 12.15), and its coin is the probability of the largest level, respectively colevel; the letters of a table with* $c_j(B) = 0$ *have none:*

$$\mathrm{spec}^K_{j,B}(\alpha) = \Big\{\frac{K^{(d_j(B))}_{j,B}(\alpha)}{K^{(d_j(B)+1)}_{j,B}(\alpha)}\Big\},\ \textit{coin}\ \mathbb{P}_{j,\alpha}\big(N_B = d_j(B)+1\big);$$

$$\mathrm{spec}^\Psi_{j,B}(\alpha) = \Big\{\frac{\Psi^{(d^*_j(B))}_{j,B}(\alpha)}{\Psi^{(d^*_j(B)+1)}_{j,B}(\alpha)}\Big\},\ \textit{coin}\ \mathbb{P}^*_{j,\alpha}\big(N^*_B = d^*_j(B)+1\big).$$

At $B = \{e\}$*,* e *neither a loop nor a coloop of table* j*, the two levels are* $K_{j,\setminus e}(\alpha_R)$ *and* $\alpha_e K_{j,/e}(\alpha_R)$ *and the two colevels* $\Psi_{j,/e}(\alpha_R)$ *and* $\alpha_e\Psi_{j,\setminus e}(\alpha_R)$*: each Skeleton value is a deletion–contraction ratio, and the coins are the occupation* $R_{j,e}(\alpha)$ *and the dual occupation* $R^*_{j,e}(\alpha)$*.*

Corollary 15.4 (One share per coefficient beyond the first)**.** *Let* $\operatorname{Re} w_\ell > 0$ *for all* ℓ*,* $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$*, and choose* $c_\ell \in (0, \operatorname{Re} w_\ell)$*,* $\ell = 1,\dots,L$*, with* $\operatorname{Re}\kappa - \operatorname{Re} w_0 < \sum_\ell c_\ell < \operatorname{Re}\kappa$*, possible exactly under these conditions. With the share integrals on* $\operatorname{Re} a_\ell = c_\ell$ *and* $|\mathbf{a}| = a_1 + \cdots + a_L$*,*

$$\Gamma_{\substack{2\\|J|}}\!\left[\begin{matrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{matrix}\right]\ \mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi})$$
$$= \int_{a_1+b_1=w_1}\cdots\int_{a_L+b_L=w_L}\ \prod_{\ell=1}^{L}\Gamma_{\substack{2\\1}}\!\left[\begin{matrix}a_\ell,\ b_\ell\\ w_\ell\end{matrix}\right]\xi_\ell^{-b_\ell}\cdot\xi_0^{\kappa-|\mathbf{a}|-w_0}\,\Gamma_{\substack{2\\1}}\!\left[\begin{matrix}\kappa-|\mathbf{a}|,\ w_0-\kappa+|\mathbf{a}|\\ w_0\end{matrix}\right].$$

Any coefficient may serve as ξ_0*;* $L+1$ *distinct values need* L *shares.*

Theorem 15.5 (The scale in its spectral gaps)**.** *Let* $\xi_\ell \in \mathbb{C}\setminus(-\infty,0]$ *and* $w_\ell \in \mathbb{C}$*,* $\ell \in J$*, with* $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$*, and let* $r > 0$ *have* $|\xi_\ell - r| < r$ *for every* ℓ*; such an* r *exists exactly when every* ξ_ℓ *has positive real part (Proposition 15.10). Put* $g_\ell = 1 - \xi_\ell/r$*, the* spectral gaps *at* r*, so that* $|g_\ell| < 1$*. Then*

$$\Gamma_{\substack{2\\|J|}}\!\left[\begin{matrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{matrix}\right]\ \mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}) = r^{\kappa-|\mathbf{w}|}\sum_{\mathbf{n}\in\mathbb{N}^J}\prod_{\ell\in J}\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{n_\ell}\ \Gamma_{\substack{2\\1}}\!\left[\begin{matrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{matrix}\right],$$

a convergent series whose terms, grouped by the total order $|\mathbf{n}|$*, have root rate at most* $\max_\ell|g_\ell|$*. A coefficient equal to* r *has spectral gap zero and drops out. For* $0 < \xi_\ell \le r$*,* $w_\ell > 0$ *and* $0 < \kappa < |\mathbf{w}|$ *the gaps lie in* $[0,1)$ *and every term is positive. For positive coefficients the bound* $\max_\ell|g_\ell|$ *is least at* $r = \frac{1}{2}(\max_\ell\xi_\ell + \min_\ell\xi_\ell)$*, where it is* $\frac{\max_\ell\xi_\ell - \min_\ell\xi_\ell}{\max_\ell\xi_\ell + \min_\ell\xi_\ell}$*, against* $1 - \frac{\min_\ell\xi_\ell}{\max_\ell\xi_\ell}$ *at* $r = \max_\ell\xi_\ell$*.*

Corollary 15.6 (Inversion is Cremona on the scale)**.** *In Theorem 15.2:*

$$\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}) = \prod_{\ell} \xi_\ell^{-w_\ell}\, \mathsf{B}^{\{J\}}_{\mathbf{w},\,\kappa;\,I_J}(\boldsymbol{\xi}^{-1}),$$

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = (\alpha^{\mathcal{E}})^{-2\sigma}\ \mathrm{Sc}_B(\alpha^{-1};2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})\,:$$

the joint spectrum at $(\alpha^{-1};\mathbf{t},\mathbf{s})$ is the reciprocal of that at $(\alpha;\mathbf{s},\mathbf{t})$ with the same letter exponents, and the index of the scale integral moves from κ_B to $\kappa_B^{\mathrm{top}} = |\mathbf{w}| - \kappa_B$.

Corollary 15.7 (The period fibres over the scale of a bunch)**.** *Let B be a proper bunch of a family F on $\mathcal{E}$, with room R, and let $h > 0$ be a function of α_R homogeneous of degree one. Wherever the integral of $W\, d\alpha_{\mathbf{m}}$ over the section $\{h(\alpha_R) = 1\}$ of the orthant converges, both sections taken with the measures of Lemma 16.2,*

$$\int_{h(\alpha_R)=1} W\, d\alpha_{\mathbf{m}} = \int_{\{\alpha_R\in\mathbb{R}^R_{>0}:\ h(\alpha_R)=1\}} \int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}}\alpha_R^{\mathbf{m}|_R-\mathbf{1}}\ \mathrm{Sc}_B(u,\alpha_R;\mathbf{m};\mathbf{s},\mathbf{t})\, d\omega_B(u),$$

and at every point Sc_B is the product of Theorem 15.2, a stacked Gamma times a Breta of one bunch. On the balance wall the left side is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. At the chart $h = \alpha_g$, $g \in R$, the left side is $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ (Proposition 16.4(i)) and the outer integral is over α_{R-g} at $\alpha_g = 1$; at $h = |\alpha_R|$ it is over Δ_R against $d\omega_R$.

15.2 Closing

Definition 15.8 (Exceedance polynomials)**.** For a bunch B of a table the *exceedance polynomials* are $\mathsf{Q}^K_{B,>}(x;\alpha) = \sum_{k\geq 0}\mathbb{P}_\alpha(N_B - d(B) > k)\, x^k$ and $\mathsf{Q}^\Psi_{B,>}(x;\alpha) = \sum_{k\geq 0}\mathbb{P}^*_\alpha(N^*_B - d^*(B) > k)\, x^k$. They have nonnegative coefficients and degree $c(B)-1$, and $1-\mathsf{Q}^K_B = (1-x)\mathsf{Q}^K_{B,>}$, $1-\mathsf{Q}^\Psi_B = (1-x)\mathsf{Q}^\Psi_{B,>}$.

Theorem 15.9 (The scale closes on its two sectors)**.** *Let B be a proper bunch of a family with room R, and let the data be real with $s_j, t_j \geq 0$, $\kappa_B > 0$ and $\kappa_B^{\mathrm{top}} > 0$. For $S \in \{B, R\}$ write the polynomials $\mathsf{Q}^K_{j,S}(\cdot;\alpha)$, with the exponents s_j, and $\mathsf{Q}^\Psi_{j,S}(\cdot;\alpha)$, with the exponents t_j, as $\mathsf{Q}_{S,i}(\cdot;\alpha)$ with exponents w_i, $1 \leq i \leq 2N$ for a family of N tables, $\mathbf{w} = (w_i)_i$, and their exceedance polynomials as $\mathsf{Q}_{S,i,>}$.*

(i) The scale integral is the sum of the Trailmoments of its two sectors, with the coefficient functions $x \mapsto (\mathsf{Q}_{S,i}(x;\alpha))_i$ and the inputs $x^{\kappa-1}\mathbb{I}_{(0,1)}$:

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\Big(\big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{B,i}(x;\alpha))_i]\, x^{\kappa_B-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big) + \big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{R,i}(x;\alpha))_i]\, x^{\kappa_B^{\mathrm{top}}-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big)\Big).$$

(ii) With $\lambda_S = \max_i\big(1-\mathsf{Q}_{S,i}(0;\alpha)\big) < 1$ and $\kappa > 0$,

$$\big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{S,i}(x;\alpha))_i]\, x^{\kappa-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big) = \int_0^1 x^{\kappa-1}\prod_i \mathsf{Q}_{S,i}(x;\alpha)^{-w_i}\, dx$$
$$= \sum_{\mathbf{n}\in\mathbb{N}^{2N}}\prod_i \frac{(w_i)_{n_i}}{n_i!}\sum_{k\geq 0}\Big([x^k]\prod_i \mathsf{Q}_{S,i,>}(x;\alpha)^{n_i}\Big)\Gamma^{\,2}_{\,1}\left[\begin{matrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{matrix}\right],$$

a series of nonnegative terms, those with $|\mathbf{n}| = M$ summing to at most $\frac{(|\mathbf{w}|)_M}{M!}\lambda_S^M\kappa^{-1}$.

For complex exponents with $\operatorname{Re}\kappa > 0$ *the series converges absolutely to the same value. No Skeleton value is used, and a complex spectrum closes as a real one does.*

Proposition 15.10 (Closing around a point of the axis)**.** *The expansion* $x+\xi = (x+r)\big(1-\frac{r-\xi}{x+r}\big)$, $r>0$, *converges uniformly on* $x \ge 0$ *exactly when* $|\xi - r| < r$*; one* r *serves every value of a finite spectrum exactly when every value has positive real part, that is, when every irreducible real factor of every Skeleton polynomial has positive coefficients.*

Corollary 15.11 (The period closes on the ordering sectors)**.** *Let the data be complex with* $\kappa_{\mathcal{E}} = 0$ *and* $\operatorname{Re}\kappa_S > 0$ *for every nonempty proper* S*. For an order* π *with tails* U_k *and sector letters* $k_{j,\pi}, \psi_{j,\pi}$ *(Theorem 8.7), with* $N_S = |T \cap S|$*, on* $[0,1]^{n-1}$

$$1 - \frac{k_{j,\pi}(q)}{K_j(\mathbf{1})} = \sum_{k=2}^{n} (1-q_k)\, \mathbb{E}_{T\sim\mathbb{P}_{j,\mathbf{1}}}\Big[\prod_{k'<k} q_{k'}^{\,N_{U_{k'}} - d_j(U_{k'})} \sum_{0 \le i < N_{U_k} - d_j(U_k)} q_k^{\,i} \Big],$$

the same for $\psi_{j,\pi}$ *with the colevels and* d_j^**; both sides lie in* $[0,\lambda]$*,* $\lambda < 1$ *the largest of the numbers* $1 - \theta_{j,T^{\pm}_{j,\pi}}/K_j(\mathbf{1})$*. Consequently*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(\mathbf{1})^{-s_j-t_j} \sum_{\pi} \sum_{\mathbf{n},\mathbf{n}'} \prod_j \frac{(s_j)_{n_j}(t_j)_{n_j'}}{n_j!\, n_j'!} \times \int_{[0,1]^{n-1}} \prod_k q_k^{\,\kappa_{U_k}-1} \prod_j \Big(1 - \frac{k_{j,\pi}(q)}{K_j(\mathbf{1})}\Big)^{n_j} \Big(1 - \frac{\psi_{j,\pi}(q)}{K_j(\mathbf{1})}\Big)^{n_j'} dq,$$

a convergent series, each integral a finite positive combination of products

$$\prod_k \Gamma \begin{bmatrix} \kappa_{U_k} + \beta_k,\ \gamma_k + 1 \\ \kappa_{U_k} + \beta_k + \gamma_k + 1 \end{bmatrix}, \qquad \beta_k, \gamma_k \in \mathbb{N};$$

for real data with $s_j, t_j \ge 0$ *every term is nonnegative. With* $w = \sum_j (|s_j| + |t_j|)$*, the sum of the moduli of the exponents, the terms with* $|\mathbf{n}| + |\mathbf{n}'| = M$ *have moduli summing to at most* $\frac{(w)_M}{M!}\lambda^M \prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)} \sum_\pi \prod_k (\operatorname{Re}\kappa_{U_k})^{-1}$*.*

Proposition 15.12 (An enclosure of the period in the field of its data)**.** *In Corollary 15.11 let the data be real, and let the weights, masses and exponents lie in a subfield* k *of* $\mathbb{R}$*. For an order* π *put* $a_{j,\pi} = 1 - k_{j,\pi}/K_j(\mathbf{1})$ *and* $b_{j,\pi} = 1 - \psi_{j,\pi}/K_j(\mathbf{1})$*, let* $\lambda_\pi < 1$ *be the largest of the numbers* $1 - \theta_{j,T^{\pm}_{j,\pi}}/K_j(\mathbf{1})$*, and let* $f_{\pi,M}(q)$ *be the coefficient of* ζ^M *in the power series* $\prod_j (1-\zeta a_{j,\pi})^{-s_j}(1-\zeta b_{j,\pi})^{-t_j}$*.*

(i) $\prod_j K_j(\mathbf{1})^{s_j+t_j}\, P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi \sum_{M\ge 0} I_{\pi,M}$ *with* $I_{\pi,M} = \int_{[0,1]^{n-1}} \prod_k q_k^{\,\kappa_{U_k}-1} f_{\pi,M}(q)\, dq$*, an element of* k*, nonnegative when* $s_j, t_j \ge 0$*, by* $\int \prod_k q_k^{\,\kappa_{U_k}-1+\beta_k} dq = \prod_k (\kappa_{U_k}+\beta_k)^{-1}$*.*

(ii) With $w = \sum_j (|s_j| + |t_j|)$*, if* $r_{\pi,Q} = \lambda_\pi \max\big(1, \frac{w+Q+1}{Q+2}\big) < 1$*, then* $\big|\sum_{M>Q} I_{\pi,M}\big| \le \operatorname{err}_{\pi,Q} = \prod_k \kappa_{U_k}^{-1} \frac{(w)_{Q+1}}{(Q+1)!} \frac{\lambda_\pi^{Q+1}}{1-r_{\pi,Q}}$*, the sum being nonnegative when* $s_j, t_j \ge 0$*, and* $\operatorname{err}_{\pi,Q} \to 0$ *as* $Q \to \infty$*: the partial sums minus and plus* $\sum_\pi \operatorname{err}_{\pi,Q}$*, and for* $s_j, t_j \ge 0$ *the partial sums and the partial sums plus* $\sum_\pi \operatorname{err}_{\pi,Q}$*, are elements of* k *enclosing the left side of (i) within any prescribed width.*

(iii) The coefficients $f_{\pi,M}$ satisfy a linear recurrence in M with coefficients polynomial in M and in the $a_{j,\pi}, b_{j,\pi}$: the power series $f(\zeta)=\prod_i(1-\zeta A_i)^{-w_i}$, with (A_i,w_i) running over the pairs $(a_{j,\pi},s_j)$ and $(b_{j,\pi},t_j)$, satisfies $\prod_i(1-\zeta A_i)\,f'(\zeta)=\big(\sum_i w_iA_i\prod_{i'\neq i}(1-\zeta A_{i'})\big)f(\zeta)$.

Theorem 15.13 (The peak closing of a period)**.** *Let the data be real with $s_j,t_j\geq 0$, $\kappa_{\mathcal{E}}=0$ and every proper wall positive. Write the letters K_j (exponent s_j, degree r_j, counts d_j) and Ψ_j (exponent t_j, degree $n-r_j$, counts d_j^*) as L_i with exponents w_i, degrees g_i and counts $d_i(S)$, choose references $M_i\geq\max_{\Delta_{\mathcal{E}}}L_i$, and put $b_{\mathbf{n}}=\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}\prod_i(M_i-L_i(u))^{n_i}d\omega_{\mathcal{E}}(u)\geq 0$.*

(i) $P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}M_i^{-w_i-n_i}b_{\mathbf{n}}$, a series of nonnegative terms. With the references $M_i=\max_T\theta_{j,T}/g_i!$, j the table of the letter L_i, every $b_{\mathbf{n}}$ is a finite positive combination of the stacked Gammas $\Gamma_{\,n}^{\,1}\!\left[\begin{matrix}\mathbf{m}+\boldsymbol{\gamma}\\ m_{\mathcal{E}}+|\boldsymbol{\gamma}|\end{matrix}\right]$, $\boldsymbol{\gamma}\in\mathbb{N}^{\mathcal{E}}$; the same series, with every $b_{\mathbf{n}}$ such a finite positive combination, holds for $\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}\prod_i L_i^{-w_i}\,d\omega_{\mathcal{E}}$ with any letters L_i homogeneous of degree g_i with nonnegative coefficients and squarefree monomials, M_i being the largest coefficient of L_i over $g_i!$.

(ii) For each letter let ϖ_i be the least of $\kappa_S/d_i(S)$ over nonempty proper S with $d_i(S)>0$, and k_i the largest length of a chain $S_1\subsetneq\cdots\subsetneq S_{k_i}$ with $\kappa_{S_l}=\varpi_i d_i(S_l)$; put $\varpi=\min_i\varpi_i$ and $k=\max\{k_i:\varpi_i=\varpi\}$. The partial sums $P^{\leq Q}$ of (i) over $\max_i n_i\leq Q$ satisfy $P_F(\mathbf{m};\mathbf{s},\mathbf{t})-P^{\leq Q}=\Theta\big(Q^{-\varpi}(\log Q)^{k-1}\big)$, and if a single letter attains ϖ,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})-P^{\leq Q}\sim\frac{c_i\,\Gamma(w_i+\varpi)}{\varpi\,\Gamma(k)\,\Gamma(w_i)}\,Q^{-\varpi}(\log Q)^{k-1},$$

with $c_i>0$ the coefficient of $(z+\varpi_i)^{-k_i}$ in $M_i^{-z}\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}L_i^z\prod_{i'}L_{i'}^{-w_{i'}}d\omega_{\mathcal{E}}$, whose pole at $-\varpi_i$ has order exactly k_i. A letter that vanishes on no proper face closes geometrically.

(iii) The fastest start is the sharp reference $M_i=\max_{\Delta_{\mathcal{E}}}L_i$; where it is attained inside the simplex, $u_e^\star\partial_e\log L_i(u^\star)=g_iu_e^\star$ for every mark. It can be attained on a proper face of the simplex.

(iv) At the target of a molecule with a native gauge every proper wall is positive (Theorem 8.18(iii)), so the series applies there and its rate ϖ is positive.

Theorem 15.14 (Continued Beta corners)**.** *In Theorem 15.9 and Corollary 15.11 let the data be complex and let no index κ, respectively no κ_{U_k}, be a nonpositive integer; for an index with $\operatorname{Re}\kappa\leq 0$ let $h\in\mathbb{N}$ have $-h-1<\operatorname{Re}\kappa\leq -h$. Then the series converge, with each Beta corner $\Gamma_{\,2}^{\,1}\!\left[\begin{matrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{matrix}\right]$ in Theorem 15.9 and $\Gamma_{\,2}^{\,1}\!\left[\begin{matrix}\kappa_{U_k}+\beta_k,\ \gamma_k+1\\ \kappa_{U_k}+\beta_k+\gamma_k+1\end{matrix}\right]$ in Corollary 15.11 read as the meromorphic stacked Gamma, to the* Reg *of the integral, its continuation in the indices; the terms of total order M are at most $\frac{(w)_M}{M!}\lambda^M$ times a polynomial in M whose degree is the sum of the integers $h+1$ over the indices with real part at most 0 (over the proper walls with $\operatorname{Re}\kappa_S\leq 0$ in Corollary 15.11), with $w=\sum_j(|s_j|+|t_j|)$ the sum of the moduli of the exponents. The sector series needs no positivity of the walls and computes the continued value at every point off the pole hyperplanes.*

The closings expand each letter around a reference and integrate term by term: on the two sectors of a bunch's scale the reference is the letter's value at the end of the sector, on the ordering sectors of the simplex it is the total weight, and at the peak it is a bound on the simplex. For real data with nonnegative exponents every term is a positive combination of stacked Gammas, nonnegative where the walls are positive, and the rate at which they fall is read from the walls.

15.3 The port

Theorem 15.15 (The port)**.** *Let B be a bunch of a family F on $\mathcal{E}$, with room R, and suppose there are reals $c_{j,p}, c'_{j,p} > 0$, one for each pattern p of B in table j, with $\sum_p c_{j,p} = \operatorname{Re} s_j$ and $\sum_p c'_{j,p} = \operatorname{Re} t_j$, such that (a) $\operatorname{Re}\mu_e - \sum_{j,p} \frac{c_{j,p}-c'_{j,p}}{2}\chi_p(e) > 0$ for $e \in B$, and (b) the children $F_B(\mathbf{c},\mathbf{c}')$ converge at the masses $\operatorname{Re}\mathbf{m}|_R$. Then $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converges and, with the share integrals on $\operatorname{Re} a_{j,p} = c_{j,p}$, $\operatorname{Re} b_{j,p} = c'_{j,p}$,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j} \int_{|\mathbf{b}_j|=t_j} \prod_j \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} \times \prod_{e\in B} \Gamma\big(\mu_e - \langle\boldsymbol{\delta}\rangle_e\big)\, J_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

with the odd loads $\langle\boldsymbol{\delta}\rangle_e$ of the shares (Definition 12.21). For $B = \mathcal{E}$ this is the atom formula*: every child is $\{\varnothing\}$ with the weight $\theta_{j,T}$, and $J_{F_{\mathcal{E}}(\mathbf{a},\mathbf{b})}(\varnothing;\mathbf{a},\mathbf{b}) = \prod_{j,T} \theta_{j,T}^{-a_{j,T}-b_{j,T}}$.*

The port opens every letter over the patterns of the bunch, one share for each pattern and letter. On the bunch the shares leave a product of Gammas, one per mark, whose arguments are the offsets less the odd loads; on the room they leave the interest of the children at the shares as their exponents.

Corollary 15.16 (A loop or coloop in every table)**.** *Let every mark of a set A be in every table a loop or a coloop, with $\eta_{j,e} = -1$ for a loop and $\eta_{j,e} = +1$ for a coloop. Then $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_{e\in A}\Gamma\big(\mu_e - \sum_j \eta_{j,e}\delta_j\big)\, J_{F'}(\mathbf{m}|_{\mathcal{E}\setminus A};\mathbf{s},\mathbf{t})$, with F' the family on $\mathcal{E}\setminus A$ of the deletions of its tables at the loops and the contractions at the coloops (Definition 7.3).*

Corollary 15.17 (The damped bunch integral is a Breta)**.** *In Theorem 15.15 fix α_R, let $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$ (a letter with exponent zero is 1 and gets no scale), put $z = \frac{1}{2}\log\alpha$, introduce one singleton scale per letter, $V = \{v_{j,+}, v_{j,-}\}_j$, put $\mathbf{q} = (\mathbf{s},\mathbf{t})$ on V, and take the partition $\mathcal{P}_{\text{port}} = \{B\} \cup \{\{v\} : v \in V\}$. The term arrays $(\pi^{\text{port}}, \ell^{\text{port}})$ of the letter presentation (Proposition 3.6) have the columns:*

- *for each $e \in B$, the kernel column with shape exponent $(\mathbf{e}_e, 0_V)$, level $(1, 0_V)$ and coefficient 1;*
- *for every (j,p), a column with shape exponent $(\chi_p/2, \mathbf{e}_{j,+})$, level $(\ell_p, \mathbf{e}_{j,+})$ and coefficient $\Lambda_{j,(p)}(z_R)$;*
- *for every (j,p), a column with shape exponent $(-\chi_p/2, \mathbf{e}_{j,-})$, level $(-\ell_p, \mathbf{e}_{j,-})$ and coefficient $\Lambda_{j,(p)}(-z_R)$;*

and $\boldsymbol{\xi}_{\text{port}}(\alpha_R)$ lists the coefficients in this order. Then, either side converging with the other,

$$\int_{\mathbb{R}^B_{>0}} \alpha_B^{\boldsymbol{\mu}|_B - \mathbf{1}} e^{-|\alpha_B|} \prod_j \Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}\, d\alpha_B = \Gamma(\mu_B)\, \mathsf{B}^{\mathcal{P}_{\text{port}}}_{(\boldsymbol{\mu}|_B,\mathbf{q}),\,(\mu_B,\mathbf{q});\,(\pi^{\text{port}},\ell^{\text{port}})}\big(\boldsymbol{\xi}_{\text{port}}(\alpha_R)\big).$$

The scale of Theorem 15.2 keeps the letters closed and integrates the scale alone, one share per distinct Skeleton value beyond the first (Corollary 15.4).

Theorem 15.18 (The native split is an admissible contour)**.** *For real data with $s_j, t_j \geq 0$, not all zero, $\mathbf{m} > 0$ and $\kappa_{\mathcal{E}} > 0$, a native gauge $\alpha^\star$ and a bunch B, the real parts $c_{j,p} = s_j\pi_{j,p}(\alpha^\star)$, $c'_{j,p} = t_j\pi^*_{j,p}(\alpha^\star)$ of the native split satisfy the hypotheses of Theorem 15.15; a letter with exponent zero is 1 and is not opened.*

Corollary 15.19 (The port decides convergence)**.** *For real data with $s_j, t_j \geq 0$ and any bunch B, $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ converges if and only if some real parts satisfy the hypotheses of Theorem 15.15 at B.*

15.4 Signed letters

Definition 15.20 (Signed letters)**.** A *signed letter* is $L = L_+ - L_-$ with $L_\pm$ positive combinations of monomials of one degree g; adjoining $\zeta|u|^g$ gives $L+\zeta$ on the simplex, positive for $\zeta > -\min_{\Delta_\mathcal{E}} L$.

Proposition 15.21 (A signed letter is a boundary value)**.** *Let a signed letter L enter a form with exponent s beside positive letters L_i with exponents w_i, and for $\zeta > -\min_{\Delta_\mathcal{E}} L$ put $G(\zeta) = \int_{\Delta_\mathcal{E}} u^{\mathbf{m-1}} (L+\zeta)^{-s} \prod_i L_i^{-w_i}\, d\omega_\mathcal{E}$.*

(i) With ν the image of $u^{\mathbf{m-1}} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$ under $u \mapsto L(u)$, $G(\zeta) = \int (y+\zeta)^{-s}\, \nu(dy)$ is holomorphic off $(-\infty, -\min_{\Delta_\mathcal{E}} L]$, continues across that half-line except at the negatives of the critical values of L on the simplex and its faces, and satisfies a linear differential equation in ζ with polynomial coefficients.

(ii) $G(0 \pm i0) = \int_{L>0} u^{\mathbf{m-1}} L^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E} + e^{\mp i\pi s} \int_{L<0} u^{\mathbf{m-1}} |L|^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$, the region integrals convergent for $\operatorname{Re} s < 1$ near regular zeros and taken by Lemma 8.6 in the coordinate $y = L(u)$ otherwise.

(iii) $G(0+i0) - G(0-i0) = -2i \sin(\pi s) \int_{L<0} u^{\mathbf{m-1}} |L|^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$.

On the scale of a bunch a zero of a signed letter is a split point: on each of the intervals it bounds, rescaled to $[0,1]$, the zero is at an endpoint, where it is a wall of the integral over that interval and Theorem 15.14 applies, and the phase $e^{\mp i\pi s}$ attaches to the interval on which the letter is negative.

Chapter 16

Integrals, expectations and the Foreman procedure

16.1 Forms, sections, completions

Definition 16.1 (Differential forms, charts, schedules)**.** The *differential form* of a family at masses $\mathbf{m}$ is

$$\omega = W\, d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m-1}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} d\alpha,$$

the family word against the measure $d\alpha_{\mathbf{m}}$ (Definition 8.1). It is *projective* when $\kappa_{\mathcal{E}} = 0$, and then it is integrated over the rays of the orthant, over any section (Lemma 16.2). The *chart* at a mark g is the section $\alpha_g = 1$ with the coordinates α_e, $e \neq g$, and the chart integral is $\int_{\alpha_g=1} \omega = \int_{\mathbb{R}_{>0}^{\mathcal{E}-g}} (\cdots)|_{\alpha_g=1} \prod_{e\neq g} d\alpha_e$. Passing to the chart at g is the *snip* at g. A *schedule* $\mathfrak{s} = (g; \varsigma)$ is the chart at a mark g and an order $\varsigma(1), \dots, \varsigma(n-1)$ of the other marks.

Lemma 16.2 (Every section gives the period)**.** *If ω is projective with convergent period, f its integrand, $h > 0$ homogeneous of degree one and φ a function on $(0,\infty)$ with $\int_0^\infty |\varphi(\rho)|\, d\rho/\rho < \infty$, then $\int_{\mathbb{R}_{>0}^{\mathcal{E}}} f\, \varphi(h)\, d\alpha = \big(\int_0^\infty \varphi(\rho) \frac{d\rho}{\rho}\big) P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$. In particular every chart integral converges and equals $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$.*

Definition 16.3 (Completions)**.** For $w \in \mathbb{R}^{\mathcal{E}}$ with $w_{\mathcal{E}} = 1$ the *completion* of $\omega = W\, d\alpha_{\mathbf{m}}$ along w is $\omega_{\mathbf{m}'} = \alpha^{-\kappa_{\mathcal{E}} w} \omega = W\, d\alpha_{\mathbf{m}'}$, $\mathbf{m}' = \mathbf{m} - \kappa_{\mathcal{E}} w$, which is projective: a completion acts on the measure and leaves the word. The *chart completion* at g has $w = \mathbf{e}_g$ and the masses $\mathbf{m}^{(g)} = \mathbf{m} - \kappa_{\mathcal{E}} \mathbf{e}_g$. For real data the *native completion* has $w = \mathbf{m}/m_{\mathcal{E}}$; its measure is the differential

$$d\alpha_{\boldsymbol{\tau}} = \alpha^{-(\kappa_{\mathcal{E}}/m_{\mathcal{E}})\mathbf{m}}\, d\alpha_{\mathbf{m}}, \qquad \boldsymbol{\tau} = \frac{n\sigma + \sum_j \delta_j (2r_j - n)}{n\sigma + \mu_{\mathcal{E}}}\, \mathbf{m},$$

at the target, with $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j (2r_j - n)$ and $m_{\mathcal{E}} = n\sigma + \mu_{\mathcal{E}}$ in pair coordinates (Theorem 8.2(iii)), and the completed form is $\omega_{\boldsymbol{\tau}} = W\, d\alpha_{\boldsymbol{\tau}}$.

Proposition 16.4 (Snips are completions)**.** *(i) $\int_{\alpha_g=1} \omega = \int_{\alpha_g=1} \omega_{\mathbf{m}^{(g)}}$, the period $P_F(\mathbf{m}^{(g)}; \mathbf{s}, \mathbf{t})$ wherever it converges.*

(ii) $K_j|_{\alpha_g=1} = K_{j,\backslash g} + K_{j,/g}$ and $\Psi_j|_{\alpha_g=1} = \Psi_{j,\backslash g} + \Psi_{j,/g}$, and for $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$, on the contours of Corollary 4.4,

$$\prod_j K_j^{-s_j}\Psi_j^{-t_j}\Big|_{\alpha_g=1} = \int\cdots\int \prod_j \Gamma_{\,1}^{\,2}\begin{bmatrix} a_j,\ a_j' \\ s_j \end{bmatrix} \Gamma_{\,1}^{\,2}\begin{bmatrix} b_j,\ b_j' \\ t_j \end{bmatrix} K_{j,\setminus g}^{-a_j} K_{j,/g}^{-a_j'} \Psi_{j,\setminus g}^{-b_j} \Psi_{j,/g}^{-b_j'}$$

over $a_j + a_j' = s_j$, $b_j + b_j' = t_j$: a snip opens every letter into its deletion and its contraction, with stacked-Gamma shares.

(iii) For a bunch B, the integral of ω over the bunch section $|\alpha_B| = 1$ is the integral over the rays of the projective form $|\alpha_B|^{-\kappa_\mathcal{E}}\omega$. In this statement write F' for F with one table appended at the exponents $(\kappa_\mathcal{E}, 0)$: the list $\binom{B}{1}$ read on $\mathcal{E}$, whose bases are the singletons of B, with the marks outside B as loops, and whose letter is $|\alpha_B|$. Then $|\alpha_B|^{-\kappa_\mathcal{E}}\omega$ is the form of F' at the masses $\mathbf{m}$, F' is balanced there, its proper walls are $\kappa_S - \kappa_\mathcal{E}\,\mathbf{1}[B \subseteq S]$, and

$$\int_{|\alpha_B|=1} \omega = P_{F'}\big(\mathbf{m}; (\mathbf{s}, \kappa_\mathcal{E}), (\mathbf{t}, 0)\big)$$

wherever $\operatorname{Re}\kappa_S - \operatorname{Re}\kappa_\mathcal{E}\,\mathbf{1}[B \subseteq S] > 0$ for every nonempty proper S. At $B = \{g\}$ the appended letter is the monomial α_g, and this is (i); at $B = \mathcal{E}$ the appended table is the list $\binom{\mathcal{E}}{1}$ (Proposition 38.8). For $k = -\kappa_\mathcal{E} \in \mathbb{N}$, $k \geq 1$, $|\alpha_B|^k = \sum_{|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\alpha_B^{\mathbf{n}}$, so $\int_{|\alpha_B|=1}\omega = \sum_{\mathbf{n}\in\mathbb{N}^B,\,|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\int_{|\alpha_B|=1}\omega_{\mathbf{m}+\mathbf{n}}$, each term the period $P_F(\mathbf{m}+\mathbf{n};\mathbf{s},\mathbf{t})$ wherever it converges.

Proposition 16.5 (The scaling mode)**.** *Along $z \mapsto z + c\mathbf{1}$, $E_j(z + c\mathbf{1}) = E_j(z)$, $O_j(z + c\mathbf{1}) = O_j(z) + 2c(2r_j - n)$, and the exponent $2\langle\boldsymbol{\mu}, z\rangle - \sum_j(\sigma_j E_j + \delta_j O_j)$ of the pair form moves by $2c\kappa_\mathcal{E}$. Off the balance wall the form is not integrable along this direction; it is* damped *by $e^{-|\alpha|}$ (the interest, $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_\mathcal{E})P_F(\mathbf{m};\mathbf{s},\mathbf{t})$),* conditioned *by a section (at a chart a completion, Proposition 16.4; in general another projective form, Proposition 16.29), or* slid *by moving the masses along their ray to the balance wall (the native completion).*

Theorem 16.6 (The slide)**.** *Let ω have real data on a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, native gauge $\alpha^\star$ and target $\boldsymbol{\tau}$.*

(i) If $\kappa_\mathcal{E} \neq 0$, exactly one completion of ω has $\alpha^\star$ as a native gauge: $\omega_{\boldsymbol{\tau}}$. If $\kappa_\mathcal{E} = 0$, every completion is ω and $\boldsymbol{\tau} = \mathbf{m}$.

(ii) As the masses slide along their ray, $\mathbf{m} \mapsto \lambda\mathbf{m}$, the data $(\lambda\mathbf{m};\mathbf{s},\mathbf{t})$ lie on the balance wall exactly when $\lambda\mathbf{m} = \boldsymbol{\tau}$.

(iii) Every proper wall of $\omega_{\boldsymbol{\tau}}$ is positive, so its period converges.

(iv) $\boldsymbol{\tau}$, $\omega_{\boldsymbol{\tau}}$ and $\alpha^\star$ are the same at every point of the ray of $\mathbf{m}$.

Each of the three treatments of the free scaling mode (Proposition 16.5) is an evaluation, and sliding keeps the native gauge.

16.2 Three integrals and their expectations

Definition 16.7 (Breta shape law; Breta law)**.** For real data with every proper wall positive, the *Breta shape law* $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ is the probability measure

$$u^{\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-s_j}\Psi_j(u)^{-t_j}\,d\omega_\mathcal{E}(u)\big/P_F(\mathbf{m};\mathbf{s},\mathbf{t}) \quad \text{on } \Delta_\mathcal{E}.$$

With every wall positive, the *Breta law* $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ is the law on $\mathbb{R}_{>0}^{\mathcal{E}}$ with density $\alpha^{\mathbf{m}-\mathbf{1}}e^{-|\alpha|}\prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

Definition 16.8 (Three integrals and their expectations)**.** Let $\omega = W\,d\alpha_{\mathbf{m}}$ have real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, so that W is homogeneous of degree $-\tau_{\mathcal{E}}$, $\tau_{\mathcal{E}} = n\sigma + \sum_j \delta_j(2r_j - n)$ with $\sigma = \sum_j \sigma_j$. On a connected family the centre of ω is the form of the family at the centre (Definition 9.1), at the exponents $(\boldsymbol{\sigma},\boldsymbol{\sigma})$ and the masses $\sigma\mathbf{1}$,

$$\omega_{\mathrm{c}} = \prod_j \left(K_j\Psi_j\right)^{-\sigma_j} d\alpha_{\sigma\mathbf{1}}.$$

It is projective, its proper walls $\sum_j \sigma_j c_j(S)$ are positive and the table's gauge is native for it (Proposition 8.21), so its period $P_{\mathrm{c}} = P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$ is a positive convergent integral, and $\mathbb{P}_{\mathrm{c}} = \mathrm{Breta}^{\circ}(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, the measure $\omega_{\mathrm{c}}/P_{\mathrm{c}}$ read on the rays, is the *law of the centre*. For masses $\mathbf{m}'$ of total $\tau_{\mathcal{E}}$ the *departure* of $W\,d\alpha_{\mathbf{m}'}$ from the centre is the ratio of the two forms,

$$\frac{W\,d\alpha_{\mathbf{m}'}}{\omega_{\mathrm{c}}}(\alpha) = \alpha^{\mathbf{m}'-\sigma\mathbf{1}}\prod_j\Big(\frac{K_j(\alpha)}{\Psi_j(\alpha)}\Big)^{-\delta_j},$$

homogeneous of degree 0: an *offset factor* $\alpha^{\mathbf{m}'-\sigma\mathbf{1}}$ and a *lean factor* $\prod_j(K_j/\Psi_j)^{-\delta_j}$. Each integral below is unnormalized. On a connected family its expectation divides it by P_{c}: it is the expectation, under the law of the centre, of the departure that its evaluation sets.

(i) The *integral* $\int W\,d\alpha_{\mathbf{m}}$ is defined when ω is projective with convergent period: it is the integral of ω over any section, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, and $\int W\,d\alpha_{\mathbf{m}} = P_{\mathrm{c}}\,\mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\big[(\omega/\omega_{\mathrm{c}})(U)\big]$.

(ii) For a schedule $\mathfrak{s} = (g;\varsigma)$ the *quasi-integral* is the iterated finite part

$$\mathrm{Reg}^{\mathfrak{s}}\int W\,d\alpha_{\mathbf{m}} = \mathrm{Reg}\int_0^{\infty} dx_{n-1}\cdots\mathrm{Reg}\int_0^{\infty} dx_1\Big(\alpha^{\mathbf{m}-\mathbf{1}}\prod_j K_j^{-s_j}\Psi_j^{-t_j}\Big)\Big|_{\alpha_g=1}, \qquad x_k = \alpha_{\varsigma(k)},$$

defined whenever the integrand of every one of these iterated integrals has log-power expansions at 0 and at ∞. It sets the departure of the chart completion, and the *quasi-expectation* is $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}\big[(\omega_{\mathbf{m}^{(g)}}/\omega_{\mathrm{c}})(U)\big] = \mathrm{Reg}^{\mathfrak{s}}\!\int W\,d\alpha_{\mathbf{m}}\big/P_{\mathrm{c}}$.

(iii) For a connected family of at least two marks, the *pseudointegral* is

$$\Big\backslash W\,d\alpha_{\mathbf{m}} = \operatorname*{CT}_{\eta=0} P_F(\boldsymbol{\tau}+\eta\mathbf{m};\mathbf{s},\mathbf{t}),$$

the constant term at the target along the ray of the masses, P_F being meromorphic in the masses with poles only where a proper wall is a nonpositive integer, on the *pole hyperplanes* $\kappa_S = -q$, S proper, $q\in\mathbb{N}$ (Theorem 16.15). Off those hyperplanes it is $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, the value at the target of the meromorphic function P_F on the balance wall. With a native gauge every proper wall is positive at $\boldsymbol{\tau}$ and $\Big\backslash W\,d\alpha_{\mathbf{m}} = \int W\,d\alpha_{\boldsymbol{\tau}}$: the Foreman sets the measure to the native completion. It sets the departure of the native completion, and the *pseudoexpectation* is

$$\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \Big\backslash W\,d\alpha_{\mathbf{m}}\Big/P_{\mathrm{c}};$$

with a native gauge it is the expectation $\mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$: the Foreman sets the departure and the centre sets the law. On a family whose weights factor across its molecules, the pseudointegral and the pseudoexpectation are products over the molecules, each at its own

target and with its own centre; a one-mark factor contributes the value of its integrand at $\alpha = \mathbf{1}$ to the pseudointegral and 1 to the pseudoexpectation, its integrand and its centre's agreeing there.

A point $(\boldsymbol{\mu}; \boldsymbol{\sigma}, \boldsymbol{\delta})$ splits into an even part, the strengths σ_j, and an odd part, the offsets and the leans; the fold (Theorem 16.16) negates the odd part and keeps the even part. The centre is the point with the same even part and no odd part. It lies on the balance wall, it is inside its walls at every strength on a connected family, and its native gauge is the table's gauge, so its period is a convergent integral of the family itself, the same for every evaluation. Dividing by it prices the odd part of a point against its even part: with a native gauge the pseudoexpectation is the exponential of a divergence from the law of the centre, at least 1, equal to 1 exactly when the native completion is the centre's form, and unchanged by the fold (Proposition 16.31).

Theorem 16.9 (Quasi-integrals extend the integral)**.** *Each* $\mathrm{Reg}^{\mathfrak{s}}\int$ *is linear in the integrand, and* $\mathrm{Reg}^{\mathfrak{s}}\int W\, d\alpha_{\mathbf{m}} = \int W\, d\alpha_{\mathbf{m}}$ *for every schedule wherever* $\int W\, d\alpha_{\mathbf{m}}$ *is defined; there every chart completion is* ω, $\mathbf{m}^{(g)} = \mathbf{m}$, *and on a connected family* $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\mathbf{m}^{(g)}}/\omega_{\mathrm{c}})(U)] = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega/\omega_{\mathrm{c}})(U)]$. *Elsewhere the value can depend on the chart, on the order and on the coordinates (Lemma 1.8).*

Proposition 16.10 (The form of a compound at one point)**.** *Let* F *be the direct sum of families* $F_1, \dots, F_q$ *on* $\mathcal{E}_1, \dots, \mathcal{E}_q$, $q \geq 2$, $\mathcal{E} = \mathcal{E}_1 \sqcup \dots \sqcup \mathcal{E}_q$, *every table a direct sum across the split with factoring weights, so that* $K_j = \prod_i K_{j,i}$ *and* $\Psi_j = \prod_i \Psi_{j,i}$*; let the data be real, the masses* $\mathbf{m} = (\mathbf{m}_1, \dots, \mathbf{m}_q)$ *on the balance wall of* F*, and* κ_i *the imbalance of* F_i *at* $\mathbf{m}_i$*, so* $\sum_i \kappa_i = 0$. *On the section* $|\alpha_q| = 1$ *with* $\alpha_i = \rho_i u_i$, $u_i \in \Delta_{\mathcal{E}_i}$, *for* $i < q$ *and* $u_q = \alpha_q$, *the form of* F *is*

$$\prod_{i<q} \rho_i^{\kappa_i - 1} d\rho_i \;\cdot\; \prod_{i=1}^{q} u_i^{\mathbf{m}_i - \mathbf{1}} \prod_j K_{j,i}^{-s_j} \Psi_{j,i}^{-t_j}\, d\omega_{\mathcal{E}_i}.$$

(i) Its integral diverges for all $\kappa_1, \dots, \kappa_q$.

(ii) The finite part of each ρ_i*-integral is* 0 *for every* κ_i.

(iii) On the masses $\mathbf{m}(\boldsymbol{\eta}) = (\mathbf{m}_i^0 + i\eta_i w_i)_{i=1}^q$, $\boldsymbol{\eta} \in \mathbb{R}^q$ *with* $\sum_i \eta_i = 0$, *with* $\mathbf{m}_i^0$ *balanced for* F_i *inside its walls and* $w_i \in \mathbb{R}^{\mathcal{E}_i}$ *of total* 1, *so that* $\kappa_i = i\eta_i$, *the integral of the form is, as a distribution in* $(\eta_1, \dots, \eta_{q-1})$, $(2\pi)^{q-1} \prod_{i=1}^q P_{F_i}(\mathbf{m}_i^0; \mathbf{s}, \mathbf{t})$ *times the Dirac measure at* 0.

So the compound carries nothing beyond the points of its parts, and only where each molecule is balanced on its own.

Theorem 16.11 (Presentations)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$ *have real data on a connected family,* $s_j, t_j \geq 0$ *not all zero and* $\mathbf{m} > 0$. *A* presentation *of* ω *is a bijection* φ *of the marks together with coordinates* $x = \lambda^{-1}\alpha$, $\lambda \in \mathbb{R}^{\mathcal{E}}_{>0}$, *in which the letters are* $K_j(\lambda x)$, $\Psi_j(\lambda x)$ *and the form carries the constant* $\lambda^{\mathbf{m}}$. *The split, the targets, the native gauges as points of the torus, the stills, the classes, the native splits and the children computed from a presentation are the images under* φ *of those computed from* ω, *and, when the target lies on no pole hyperplane, the pseudointegral and the pseudoexpectation of the presentation, its centre presented with it, are* $(\lambda^{\mathbf{m}})^{\kappa\varepsilon/m_{\mathcal{E}}}$ *times those of* ω, *the factor being* 1 *on the balance wall; on a pole hyperplane the constant term moves with the coordinates as well (Proposition 16.28). In the coordinates* x *the table's gauge is* $x = \lambda^{-1}$, *and the Foreman tree is the same. On a family whose weights factor across its molecules the factor is the product of the molecules' factors.*

Definition 16.12 (The Foreman procedure)**.** *Classification*, from the tables alone:

(1) *Split* the marks into molecules and units (Proposition 11.6).

(2) *Read* each molecule M at a centre $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ of the table's gauge (Chapter 10).

(3) *Answer*: a constant reading is an atom; otherwise take the Foreman bunch $B(M;\boldsymbol{\sigma},0)$ (Definition 13.6).

(4) *Peel* at $B(M;\boldsymbol{\sigma},0)$; the childcules sit at the centres they inherit (Proposition 12.24).

(5) *Recurse* to atoms and units: the output is the Foreman tree and, at the root, the label.

Evaluation at a real point $(\mathbf{m};\mathbf{s},\mathbf{t})$ with $s_j,t_j\geq 0$, not all zero, and $\mathbf{m}>0$:

(6) *Native gauge* of each molecule, and its target $\boldsymbol{\tau}$ (Theorem 8.18).

(7) *Slide* the masses to the target (Theorem 16.6).

(8) *Ports* along the tree, every share contour through the native split (Theorems 15.15, 15.18 and 12.23); on the balance wall the port of Theorem 17.12.

(9) *Atoms*: the atom formula (Theorem 15.15 with $B=\mathcal{E}$; on the balance wall Corollary 17.15).

On the orthant side the ports integrate the Foreman bunch of each node inside the integral over its room, all its marks at once in scale and shape (Corollary 15.17): the root's bunch first and the atoms last. This is the *Foreman's order*.

Theorem 16.13 (The Foreman computes the pseudointegral)**.** *Let $\omega = W\,d\alpha_{\mathbf{m}}$ have real data on a connected family, with $s_j,t_j\geq 0$, not all zero, and $\mathbf{m}>0$, and a native gauge. For every $\eta>0$ the procedure at the masses $\boldsymbol{\tau}+\eta\mathbf{m}$ runs on admissible contours and computes $J_F(\boldsymbol{\tau}+\eta\mathbf{m};\mathbf{s},\mathbf{t})$, and*

$$\backslash\!\!\!\int W\,d\alpha_{\mathbf{m}} = \lim_{\eta\to 0^+}\frac{J_F(\boldsymbol{\tau}+\eta\mathbf{m};\mathbf{s},\mathbf{t})}{\Gamma(\eta m_{\mathcal{E}})}.$$

No finite part, chart, order or closing rule enters: the regulator η leaves by a limit.

Corollary 16.14 (One member among the generalized integrals)**.** *Wherever $\int W\,d\alpha_{\mathbf{m}}$ is defined for real data on a connected family, $\backslash\!\!\!\int W\,d\alpha_{\mathbf{m}} = \int W\,d\alpha_{\mathbf{m}}$, and with Theorem 16.9 every evaluation sets the one departure $\omega/\omega_{\mathrm{c}}$ on the balance wall. Off the balance wall, with $s_j,t_j\geq 0$, not all zero, and $\mathbf{m}>0$, the period at the target converges exactly when the family has a native gauge (Theorem 8.18(iii)), and otherwise $\backslash\!\!\!\int W\,d\alpha_{\mathbf{m}}$ is its continuation in the masses, on a pole hyperplane the constant term of the continuation along the ray.*

Theorem 16.15 (The pseudointegral off the domain)**.** *For every family, P_F is meromorphic in the masses and the exponents, with poles only where a proper wall is a nonpositive integer, and so is its restriction to the balance wall. If some nonempty proper S has $c_j(S)=0$ in every active table, then, first where every proper wall has positive real part and then as meromorphic functions,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma{}^{\,2}_{\,1}\!\begin{bmatrix}\kappa_S,\ \kappa_{\mathcal{E}\setminus S}\\ \kappa_{\mathcal{E}}\end{bmatrix}\int_{\Delta_S\times\Delta_{\mathcal{E}\setminus S}}\alpha^{\mathbf{m}-\mathbf{1}}\prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\,d\omega_S\,d\omega_{\mathcal{E}\setminus S},$$

and the restriction of P_F to the balance wall is 0. Let now F be connected with real data, $s_j,t_j\geq 0$ not all zero; on the balance wall the value of P_F at a target depends on no path.

(i) At a wall $\kappa_S=-q$, $q\in\mathbb{N}$, where every other proper wall is positive, the residue in κ_S is that of Theorem 16.23.

(ii) The continuation in the masses of $f \mapsto \int_{\Delta_{\mathcal{E}}} f\, u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\omega_{\mathcal{E}}$*, taken by Lemma 8.6 on each ordering sector, is a distribution on* $\Delta_{\mathcal{E}}$ *with normaliser* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and monomial moments* $P_F(\mathbf{m}+\boldsymbol{\gamma};\mathbf{s},\mathbf{t})/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*; where every proper wall is positive it is* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *times the Breta shape law. Just past the first wall crossed it is not a positive measure:* P_F *passes from* $+\infty$ *to* $-\infty$ *there, while* $P_F(\mathbf{m}+\boldsymbol{\gamma};\mathbf{s},\mathbf{t})$ *is a positive convergent integral for* γ_S *large.*

(iii) Theorem 17.6, Theorem 8.13 and the recurrences of Theorem 8.3 hold for these values as identities of meromorphic functions.

Theorem 16.16 (The fold: Cremona off the balance wall)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$*, with the exponents* $(\mathbf{s},\mathbf{t})$*, have real data on a connected family, with* $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$*, native gauge* $\alpha^\star$ *and target* $\boldsymbol{\tau}$*. Then* $2\sigma\mathbf{1} - \boldsymbol{\tau}$ *is positive and balanced for the exponents* $(\mathbf{t},\mathbf{s})$*, with native gauge* $(\alpha^\star)^{-1}$*, and for every* $\lambda > 0$

$$\int \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\alpha_{\mathbf{m}} = \int \prod_j K_j^{-t_j} \Psi_j^{-s_j}\, d\alpha_{\lambda(2\sigma\mathbf{1}-\boldsymbol{\tau})}.$$

The map from the ray of $\mathbf{m}$ *to the ray of* $2\sigma\mathbf{1} - \boldsymbol{\tau}$ *is an involution on rays.*

The fold swaps the two letters of every table and reflects the target through $\sigma\mathbf{1}$, negating the offsets and the leans; it keeps the strengths, so the two forms have one centre, and it carries uniform masses to uniform masses.

Proposition 16.17 (Points and realizations)**.** *A real point* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of a connected family, with* $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$ *and native gauge* $\alpha^\star$*, determines the realization* $((s_j,t_j)_j;\alpha^\star)$*, whose masses are the target; every realization arises so (Proposition 8.22). The reading of a point is the reading of its realization and may differ from point to point; the label is read at centres of the table's gauge, the Foreman tree at the realizations of the table's gauge at the centre and at the leans (Theorem 14.6), and both are functions of the tables. An atomic point is at the table's gauge up to scale (Theorem 11.4), and its centre is atomic (Theorem 11.1).*

Theorem 16.18 (Parity)**.** *Let* g *be an exchange of a connected family, with the bijection* h *of its tables (Definition 11.14); then* $r_{h(j)} = n - r_j$ *for every* j*. With* $(g^*z)_e = z_{g(e)}$ *(Definition 11.14), let* Φ° *be the centred potential (Theorem 8.18).*

(i) $\Lambda_j(g^*z) = \Lambda_{h(j)}(-z)$*; so* $E_j(g^*z) = E_{h(j)}(z)$*,* $O_j(g^*z) = -O_{h(j)}(z)$ *and* $\Phi^\circ(-g^*z) = \Phi^\circ(z)$*.*

(ii) If the data are real with $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$*, have a native gauge, and* $\tau_{g(e)} = 2\sigma - \tau_e$ *for every* e*, the native gauge* $e^{2z^\star}$ *with* $\langle \mathbf{1}, z^\star \rangle = 0$ *has* $-g^*z^\star = z^\star$*, and the still of the realization* $\Upsilon = ((s_j,t_j)_j; e^{2z^\star})$ *has* $\kappa_\Upsilon(g(e_1),\dots,g(e_k)) = (-1)^k \kappa_\Upsilon(e_1,\dots,e_k)$ *for every* k*.*

(iii) Under (ii) even rungs never separate e *from* $g(e)$*, and at odd rungs their tallies (Definition 9.4) are negatives of each other.*

(iv) If moreover $\boldsymbol{\delta} = 0$ *and* $\boldsymbol{\tau} = \sigma\mathbf{1}$*, then* $z^\star = 0$*, every odd rung vanishes, and no rung separates* e *from* $g(e)$*.*

(v) At $(\mathbf{t},\mathbf{s})$ *and the target* $2\sigma\mathbf{1} - \boldsymbol{\tau}$ *the native gauge is* $e^{-2z^\star}$ *and rung* k *is* $(-1)^k$ *times rung* k *at* $(\mathbf{s},\mathbf{t})$*; under (ii) this is the still at* $(\mathbf{s},\mathbf{t})$ *carried by* g*, and the Foreman bunch at* $(\mathbf{t},\mathbf{s})$ *is the image under* g^{-1} *of the Foreman bunch at* $(\mathbf{s},\mathbf{t})$*.*

16.3 Quasi-integrals as Laurent data

Definition 16.19 (Regular schedules)**.** For a schedule $\mathfrak{s} = (g;\varsigma)$ put $x_k = \alpha_{\varsigma(k)}$ and $v_k = \mathbf{e}_{\varsigma(k)} - \mathbf{e}_g$. Let $\mathcal{J}_0$ be the chart integrand and $\mathcal{J}_k(\varepsilon_1,\dots,\varepsilon_k;x_{>k}) = \int_0^\infty x_k^{\varepsilon_k}\,\mathcal{J}_{k-1}\,dx_k$, with $\int_0^1$ and $\int_1^\infty$ each continued meromorphically in the *regulator* ε_k. The schedule is *regular* when for every k, as $x_k \to 0$ and as $x_k \to \infty$, $\mathcal{J}_{k-1}$ is a finite sum of terms $c(\varepsilon,x_{>k})\,x_k^{a(\varepsilon)}$ plus $O(x_k^{\pm M})$ for every M, the exponents affine in $\varepsilon_{<k}$ and independent of $x_{>k}$, uniformly for the regulators in a small polydisc away from their poles.

Theorem 16.20 (Bridge)**.** *For real data on a connected family and a regular schedule,* $\operatorname{Reg}^{\mathfrak{s}}\!\int W\,d\alpha_{\mathbf{m}}$ *is defined and*

$$\operatorname{Reg}^{\mathfrak{s}}\!\int W\,d\alpha_{\mathbf{m}} = \operatorname*{CT}_{\varepsilon_{n-1}}\cdots\operatorname*{CT}_{\varepsilon_1} P_F\Big(\mathbf{m}^{(g)} + \sum_{k=1}^{n-1}\varepsilon_k v_k;\mathbf{s},\mathbf{t}\Big),$$

the constant terms taken innermost first, $|\varepsilon_1| \ll \cdots \ll |\varepsilon_{n-1}|$.

Theorem 16.21 (A wall is closed by its last variable)**.** *For a schedule* $\mathfrak{s} = (g;\varsigma)$ *put* $\mathbf{p} = \mathbf{m}^{(g)}$ *and* $I(\varepsilon) = P_F\big(\mathbf{p} + \sum_k \varepsilon_k v_k;\mathbf{s},\mathbf{t}\big)$*. A proper* U *is* polar *when* $\kappa_U(\mathbf{p}) \in -\mathbb{N}$*; its* members *are the* k *with* $\varsigma(k) \in U$ *if* $g \notin U$*, and with* $\varsigma(k) \notin U$ *if* $g \in U$*;* $\operatorname{last}(U)$ *is its last member.*

(i) Near 0, $I(\varepsilon) = \sum_C f_C(\varepsilon)\big/\prod_{U\in C}\ell_U(\varepsilon) + H(\varepsilon)$, $\ell_U = \pm\sum_{k\ member}\varepsilon_k$*, over the nonempty sets* C *of polar walls that are tails of one order, with* f_C, H *holomorphic; the member sets in one* C *are nested or disjoint.*

(ii) Under $\operatorname{CT}_{\varepsilon_k}$, $1/\ell_U$ *is holomorphic at* $\varepsilon_k = 0$ *unless* $k = \operatorname{last}(U)$*; for one polar wall with numerator* f_C *the iterated constant term of* f_C/ℓ_U *is* $\pm\partial_{\varepsilon_a} f_C(0)$, $a = \operatorname{last}(U)$.

(iii) For a fixed chart, the quasi-integral of a regular schedule depends on the order only through $U \mapsto \operatorname{last}(U)$ *on the polar walls.*

With one regulator ε_D for each class D of a realization, taken in some order of the classes, $\ell_U = \pm\sum_D |U\cap D|\,\varepsilon_D$; a wall is closed by the last class it meets, which contributes $|U\cap D|^{-1}\partial_{\varepsilon_D}$, and the value depends only on the order of the classes, which can be taken from the order of classes (Theorem 13.7).

Proposition 16.22 (The coordinates set the log-scale)**.** *In the coordinates* $u = \alpha/\beta$, $\beta_g = 1$*, with the regulators* ε_k *entering as* $u_{\varsigma(k)}^{\varepsilon_k}$*, the regulated function is* $e^{-\sum_k \varepsilon_k\lambda_k} I(\varepsilon)$, $\lambda_k = \log\beta_{\varsigma(k)}$*, and for one polar wall with numerator* f_C *and last member* a *the iterated constant term becomes* $\pm\big(\partial_{\varepsilon_a} f_C(0) - \lambda_a f_C(0)\big)$*: changing coordinates shifts the finite part by* $-\log\beta_{\varsigma(a)}$ *times the signed residue of each wall closed by a mark whose gauge differs from the chart's.*

Theorem 16.23 (Residue at a wall)**.** *Let* ω *be projective,* S *a proper bunch with room* R, $g \in R$ *and* $\alpha_S = \rho u$*. In this statement write* $K_j^{[l]} = K_{j,S}^{(d_j(S)+l)}(u,\alpha_R)$ *and* $\Psi_j^{[l]} = \Psi_{j,S}^{(d_j^*(S)+l)}(u,\alpha_R)$ *for the level and colevel letters of Definition 12.15 counted from their floors, so that* $K_j(\rho u,\alpha_R) = \rho^{d_j(S)}\sum_{l\ge0}\rho^l K_j^{[l]}$ *and* $\Psi_j(\rho u,\alpha_R) = \rho^{d_j^*(S)}\sum_{l\ge0}\rho^l\Psi_j^{[l]}$*. At points of* $\kappa_S = -q$, $q \in \mathbb{N}$*, where every other proper wall is positive,*

$$\operatorname*{Res}_{\kappa_S=-q} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\alpha_g=1}\int_{\Delta_S} u^{\mathbf{m}|_S-\mathbf{1}}\alpha_R^{\mathbf{m}|_R-\mathbf{1}}\prod_j\big(K_j^{[0]}\big)^{-s_j}\big(\Psi_j^{[0]}\big)^{-t_j}\,[\rho^q]h\;d\omega_S\,d\alpha_{R-g},$$

where

$$h = \prod_j \Big(\sum_l \rho^l K_j^{[l]}/K_j^{[0]} \Big)^{-s_j} \Big(\sum_l \rho^l \Psi_j^{[l]}/\Psi_j^{[0]} \Big)^{-t_j},$$

$$[\rho^q] \Big(1 + \sum_{l \ge 1} \rho^l X_l\Big)^{-s} = \sum_{\sum_l l n_l = q} (-1)^{|\mathbf{n}|} \frac{\Gamma(s+|\mathbf{n}|)}{\Gamma(s)} \prod_l \frac{X_l^{n_l}}{n_l!}.$$

If the weights factor on the bottom levels, then with the faces *of the wall, the families*

$$F^S = \big((\mathcal{B}_j/R;\, s_j, 0),\ (\mathcal{B}_j|S;\, 0, t_j)\big)_j \text{ on } S, \qquad F^R = \big((\mathcal{B}_j|R;\, s_j, 0),\ (\mathcal{B}_j/S;\, 0, t_j)\big)_j \text{ on } R$$

(Definition 7.5), the residue at $\kappa_S = 0$ is the product of their periods at the masses $\mathbf{m}|_S$ and $\mathbf{m}|_R$; both faces are balanced, the walls of F^S are the κ_U, $U \subseteq S$, and those of F^R are the $\kappa_{S \cup U}$, $U \subseteq R$, on the wall.

Proposition 16.24 (A molecule at a nonpositive wall)**.** *Let a molecule carry real data with $s_j, t_j \ge 0$ and target $\boldsymbol{\tau}$, and let S be a proper bunch with room R, $\alpha_S = \rho u$ on a chart $\alpha_g = 1$, $g \in R$, with $H(\rho) = H(\rho; u, \alpha_R)$ the native completion $W\, d\alpha_{\boldsymbol{\tau}}$ divided by $\rho^{\kappa_S(\boldsymbol{\tau})-1} d\rho\, d\omega_S\, d\alpha_{R-g}$ and $H_l = [\rho^l]H$ as in Theorem 16.23.*

(i) Along the collapse of S, $\Phi(y_0 - \lambda \mathbf{1}_S) - \langle \boldsymbol{\tau}, y_0 - \lambda \mathbf{1}_S \rangle = \lambda\, \kappa_S(\boldsymbol{\tau}) + O(1)$ as $\lambda \to \infty$, the left side being the negative logarithm of the density in $y = \log \alpha$ (Theorem 8.18); so the density does not decay along the collapse exactly when $\kappa_S(\boldsymbol{\tau}) \le 0$, and then the form is not integrable at $\rho = 0$.

(ii) With $\langle \cdot \rangle$ the integral over u and α_R, if every other proper wall at $\boldsymbol{\tau}$ is positive, $\kappa_S(\boldsymbol{\tau})$ is not a nonpositive integer, and $L \in \mathbb{N}$ has $\kappa_S(\boldsymbol{\tau}) + L > 0$,

$$P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t}) = \sum_{l<L} \frac{\langle H_l \rangle}{\kappa_S(\boldsymbol{\tau}) + l} + \int_0^1 \rho^{\kappa_S(\boldsymbol{\tau})-1} \Big(\langle H \rangle - \sum_{l<L} \rho^l \langle H_l \rangle \Big) d\rho + \int_1^\infty \rho^{\kappa_S(\boldsymbol{\tau})-1} \langle H \rangle\, d\rho,$$

the continuation of the period past the wall of S; its poles at $\kappa_S = -q$ have the residues $\langle H_q \rangle$ of Theorem 16.23. When the weights factor on the bottom levels, $\langle H_0 \rangle$ is the product of the integrals of the faces F^S at $\boldsymbol{\tau}|_S$ and F^R at $\boldsymbol{\tau}|_R$, whose imbalances are $\kappa_S(\boldsymbol{\tau})$ and $-\kappa_S(\boldsymbol{\tau})$: the leading term is the compound of Proposition 16.10.

Corollary 16.25 (The vertex)**.** *At $S = \mathcal{E} - e$, with e neither a loop nor a coloop in any table, the face F^R is the single mark e and contributes* 1*, and*

$$\langle H_0 \rangle = \int_{\Delta_{\mathcal{E}-e}} u^{\boldsymbol{\tau}|_{\mathcal{E}-e} - \mathbf{1}} \prod_j K_{j,/e}(u)^{-s_j} \Psi_{j,\backslash e}(u)^{-t_j}\, d\omega_{\mathcal{E}-e}(u),$$

the integral of the vertex child*: the contractions at e on the basis side and the deletions on the cobasis side. Proposition 16.24(ii) at $S = \mathcal{E} - e$ removes the leading terms of the vertex from the period exactly.*

16.4 The pseudointegral

Definition 16.26 (The pseudointegral over a region)**.** Let F be a family with real data, $s_j, t_j \ge 0$ not all zero, masses $\mathbf{m} > 0$ and weights factoring across its molecules, with form $\omega = W\, d\alpha_{\mathbf{m}}$.

(i) A *region* of ω is a domain $\mathcal{R}$ in coordinates x with a change of variables mapping it bijectively onto a section $\{h = 1\}$ of the orthant, $h > 0$ homogeneous of degree one: a chart $\alpha_g = 1$, the simplex, a bunch section $|\alpha_B| = 1$. The *density* of ω on $\mathcal{R}$ is the function f for which $f(x)\,dx$ is the form that ω induces on the section (Lemma 16.2). A region of the damped form $e^{-|\alpha|}\omega$ is the same with the orthant in place of the section. For a family of tables with positive rational weights and rational data on the balance wall the regions reached from these by moves (Definition 18.34) are domains cut out by polynomial inequalities over $\mathbb{Q}$.

(ii) The *pseudointegral* of ω over a region is

$$\int\!\!\cdots\!\!\int_{\mathcal{R}} f(x)\,dx := \int W\,d\alpha_{\mathbf{m}}$$

(Definition 16.8(iii)), and over a region of the damped form, away from the poles of J_F, it is the continued interest, $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int W\,e^{-|\alpha|}d\alpha_{\mathbf{m}}$ (Definition 8.9), written $\int \alpha^{\mathbf{m-1}}e^{-|\alpha|}\prod_j K_j^{-s_j}\Psi_j^{-t_j}d\alpha$ when the integrand is displayed. The value belongs to ω: the native completion and the ray of Definition 16.8(iii) are taken in the table's coordinates α, and a presentation (Theorem 16.11) declares them anew and is another form; the region and its coordinates say how the integration is presented, and a density determines ω only together with the family, a chart $\alpha_g = 1$ not seeing the mass m_g and the simplex not seeing a power of $|\alpha|$, the letter of the list $\binom{\mathcal{E}}{1}$ (Proposition 38.8). Divided by the period of the centre, P_{c} on a molecule, it is the pseudoexpectation $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.

(iii) The pseudointegral of ω is read in the Foreman's order (Definition 16.12). The tree is drawn from the tables at the centre of the table's gauge; on each molecule the measure $d\alpha_{\mathbf{m}}$ is set to $d\alpha_{\boldsymbol{\tau}}$; the bunches are integrated in the Foreman's order, each at once in scale and shape; where the target lies outside a wall the value is the continuation in the masses, and on a pole hyperplane its constant term along the ray of the masses. The damped form takes no completion, and the Foreman computes it on the ray beyond the target (Theorem 16.13).

Theorem 16.27 (The pseudointegral and the native completion)**.** *Let $\omega = W\,d\alpha_{\mathbf{m}}$ have real data on a connected family, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, with target $\boldsymbol{\tau}$.*

(i) With a native gauge $\alpha^\star$ the pseudointegral is read at the realization $\Upsilon = ((s_j,t_j)_j;\alpha^\star)$, whose masses are $\boldsymbol{\tau}$ and whose proper walls are positive. The form $W\,d\alpha_{\boldsymbol{\tau}}$ is positive with convergent period, and every iterated integral of it over every section, in any coordinates, in the Foreman's order and in every other, converges absolutely to

$$\int W\,d\alpha_{\mathbf{m}} = \int W\,d\alpha_{\boldsymbol{\tau}} = P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})\,:$$

no finite part is taken, and the pseudoexpectation is the expectation of a positive function, $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.

(ii) If $\boldsymbol{\tau}$ lies on no pole hyperplane, every regular schedule of the native completion returns the pseudointegral,

$$\mathrm{Reg}^{\mathfrak{s}}\int W\,d\alpha_{\boldsymbol{\tau}} = \int W\,d\alpha_{\mathbf{m}} = P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}),$$

each iterated integral a finite part (Definition 1.4), and the value does not depend on the order; its quasi-expectation is the pseudoexpectation, $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.

(iii) On a pole hyperplane $\int\!\!\!\!\!\!\diagup W\, d\alpha_{\mathbf{m}} = \mathrm{CT}_{\eta=0}\, P_F(\boldsymbol{\tau}+\eta\mathbf{m};\mathbf{s},\mathbf{t})$ *exists and depends on* $\mathbf{m}$ *only through its ray. It takes no chart and no order: its regulator is* η*, along the ray of the masses, and its section is the simplex that the damping singles out,* $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = J_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma(\kappa_{\mathcal{E}})$ *(Theorem 8.10).*

Proposition 16.28 (A pole target in other coordinates)**.** *In Theorem 16.27(iii) let* $x = \lambda^{-1}\alpha$ *be the coordinates of a presentation (Theorem 16.11),* $W^{\lambda}(x) = W(\lambda x)$*, and*

$$G_k(\eta) = \int_{\Delta_{\mathcal{E}}} \big(\log|\lambda u|\big)^k\, u^{\boldsymbol{\tau}+\eta\mathbf{m}-\mathbf{1}} W^{\lambda}(u)\, d\omega_{\mathcal{E}}(u) \qquad (k\in\mathbb{N}),$$

continued in η*, so that* $G_0(\eta)$ *is the integral over the simplex of the presented word* W^{λ} *at the masses* $\boldsymbol{\tau}+\eta\mathbf{m}$*. Each* G_k *has at* $\eta = 0$ *a pole of order at most* $n-1$*, and the presentation has pseudointegral*

$$(\lambda^{\mathbf{m}})^{\kappa_{\mathcal{E}}/m_{\mathcal{E}}}\Big(\int\!\!\!\!\!\!\diagup W\, d\alpha_{\mathbf{m}} - \lambda^{\boldsymbol{\tau}} \sum_{1\le i+k\le n-1} \frac{\langle\mathbf{m},\log\lambda\rangle^i}{i!}\,\frac{(-m_{\mathcal{E}})^k}{k!}\,[\eta^{-i-k}]\,G_k\Big),$$

$[\eta^{-i-k}]$ *the Laurent coefficient at* $\eta = 0$*. Off the pole hyperplanes every* G_k *is holomorphic at* 0 *and this is Theorem 16.11; on them the finite part is taken in the coordinates of the table, as every finite part is taken in its own coordinate (Lemma 1.8).*

Proposition 16.29 (Sections off the balance wall)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$ *have real data,* $s_j, t_j \ge 0$ *not all zero,* $\mathbf{m} > 0$ *and imbalance* $\kappa_{\mathcal{E}} \neq 0$*.*

(i) The integral of ω *over a section* $\{h = 1\}$ *is the integral over every section of the projective form* $h^{-\kappa_{\mathcal{E}}}\omega$*, wherever it converges. Over a chart it is the period at the chart completion (Proposition 16.4(i)), over a bunch section* $|\alpha_B| = 1$ *the period of the family with the list* $\binom{B}{1}$*, the marks outside* B *as loops, appended at the exponents* $(\kappa_{\mathcal{E}}, 0)$*, a multinomial sum of completions when* $-\kappa_{\mathcal{E}} \in \mathbb{N}$ *(Proposition 16.4(iii)), and over the simplex, where the proper walls at* $\mathbf{m}$ *are positive,* $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)]$ *(Proposition 8.11(i)). Different sections integrate different projective forms; the pseudointegral integrates the native completion over all of them.*

(ii) The walls of the chart completion at g *and of the native completion are*

$$\kappa_S(\mathbf{m}^{(g)}) = \kappa_S(\mathbf{m}) - \kappa_{\mathcal{E}}\,\mathbf{1}_S(g), \qquad \kappa_S(\boldsymbol{\tau}) = \kappa_S(\mathbf{m}) - \frac{\kappa_{\mathcal{E}}}{m_{\mathcal{E}}}\,m_S :$$

the snip moves the walls through g *by the whole imbalance and leaves the others, and the native completion moves each wall by its share. For* $\kappa_{\mathcal{E}} > 0$ *a proper wall through* g *is lower at the chart completion than at the target by* $\kappa_{\mathcal{E}}(1 - m_S/m_{\mathcal{E}})$*, and one avoiding* g *is higher by* $\kappa_{\mathcal{E}} m_S/m_{\mathcal{E}}$*. So a snip can put the walls through* g *on pole hyperplanes, or past them, while the target is inside every wall.*

Theorem 16.30 (Integrals over sections)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$ *have real data,* $s_j, t_j \ge 0$ *not all zero and* $\mathbf{m} > 0$*, and for* $h > 0$ *homogeneous of degree one let* $\int_{h=1}\omega$ *denote the integral over the section* $\{h = 1\}$ *with the measure of Lemma 16.2.*

(i) By Proposition 16.29(i) the integral of ω *over a section* $\{h = 1\}$ *is the integral over the rays of* $h^{-\kappa_{\mathcal{E}}}\omega$*, and for* $\kappa_{\mathcal{E}} \neq 0$*,* $h \mapsto h^{-\kappa_{\mathcal{E}}}$ *carries the sections bijectively onto the positive multipliers of degree* $-\kappa_{\mathcal{E}}$*. The monomial sections* $\{\alpha^w = 1\}$*,* $w_{\mathcal{E}} = 1$*, are the completions,* $(\alpha^w)^{-\kappa_{\mathcal{E}}}\omega = W\, d\alpha_{\mathbf{m}-\kappa_{\mathcal{E}}w}$*: they act on the measure and keep the word.*

(ii) The native completion is the section $\{\alpha^{\mathbf{m}} = 1\}$, the hyperplane $\langle \mathbf{m}, y\rangle = 0$ in $y = \log\alpha$, where the density $e^{\langle \mathbf{m},y\rangle}$ of $d\alpha_{\mathbf{m}}$ against dy is 1; it is the one monomial section with this property. With a native gauge on a connected family of at least two marks, for every mark g,

$$\Big\backslash W\, d\alpha_{\mathbf{m}} = \frac{m_{\mathcal{E}}}{m_g}\int_{\mathbb{R}^{\mathcal{E}-g}} W(e^y)\, dy_{\mathcal{E}-g}, \qquad y_g = -\frac{1}{m_g}\sum_{e\neq g} m_e y_e,$$

and $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ is this integral over the integral of ω_{c} over the same section, which is P_{c}.

(iii) The simplex $\{|\alpha| = 1\}$ is not a monomial section for $n \geq 2$: its multiplier is $|\alpha|^{-\kappa_{\mathcal{E}}}$, which keeps the measure and multiplies the word by a power of the letter $K_0 = |\alpha|$ of the list $\binom{\mathcal{E}}{1}$. The integral over it is the integral of the projective form $K_0^{-\kappa_{\mathcal{E}}} W\, d\alpha_{\mathbf{m}}$, the family with that table added at the exponents $(\kappa_{\mathcal{E}}, 0)$ (Proposition 38.8, Proposition 16.4(iii) at $B = \mathcal{E}$), and wherever every proper wall at $\mathbf{m}$ is positive it is the period $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ of Proposition 8.11(i), the law $\mathrm{Dir}_{\mathbf{m}}$ there being the law of $\binom{\mathcal{E}}{1}$ at a full basis lean.

(iv) On the balance wall every section reads $\int W\, d\alpha_{\mathbf{m}}$ (Theorem 8.10).

Proposition 16.31 (The centre sets the law)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data on a connected family of $n \geq 2$ marks, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, with target $\boldsymbol{\tau}$, centre ω_{c} and law of the centre $\mathbb{P}_{\mathrm{c}}$ (Definition 16.8). For laws $\mathbb{P}, \mathbb{P}'$ with $\mathbb{P}$ absolutely continuous with respect to $\mathbb{P}'$ write $\mathrm{KL}(\mathbb{P} \,\|\, \mathbb{P}') = \mathbb{E}_{U\sim\mathbb{P}}\big[\log \frac{d\mathbb{P}}{d\mathbb{P}'}(U)\big]$, the Kullback–Leibler divergence.*

(i) The native completion is the centre times the departure, $\omega_{\boldsymbol{\tau}} = \frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}\,\omega_{\mathrm{c}}$, and with a native gauge the Foreman's law, the shape law $\mathrm{Breta}^{\circ}(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ of the native completion, is the law of the centre reweighted by the departure:

$$\mathrm{Breta}^{\circ}(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})(du) = \frac{(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(u)}{\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]}\,\mathbb{P}_{\mathrm{c}}(du).$$

(ii) With a native gauge,

$$\log \mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \mathrm{KL}\big(\mathbb{P}_{\mathrm{c}} \,\big\|\, \mathrm{Breta}^{\circ}(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})\big) \;\geq\; 0,$$

with equality exactly when $\mathrm{Breta}^{\circ}(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t}) = \mathbb{P}_{\mathrm{c}}$. So every pseudointegral with a native gauge is at least the period of its centre, $\big\backslash W\, d\alpha_{\mathbf{m}} \geq P_{\mathrm{c}}$, and among the forms of given strengths σ_j with a native gauge the centre has the least value.

(iii) The pseudoexpectation is kept by the fold: the two forms of Theorem 16.16, at the masses $\mathbf{m}$ with the exponents $(\mathbf{s}, \mathbf{t})$ and at the masses $\lambda(2\sigma\mathbf{1} - \boldsymbol{\tau})$ with $(\mathbf{t}, \mathbf{s})$, have one centre and one pseudoexpectation, and the same holds for continued values off the pole hyperplanes where $2\sigma\mathbf{1} - \boldsymbol{\tau} > 0$.

(iv) Let ω be the form of a realization at the table's gauge, $\alpha^{\star} = \mathbf{1}$ (Definition 9.1). In signed coordinates on $\mathbf{1}^{\perp}$ the departure is

$$\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(e^{2z}) = \exp\Big(-\sum_j \delta_j \tilde{O}_j(z)\Big), \qquad \tilde{O}_j(z) = O_j(z) - \langle \nabla O_j(0), z\rangle,$$

each $\tilde{O}_j$ odd with value, gradient and Hessian 0 at 0. The Foreman's law and the law of the centre have their mode at the table's gauge and the same Hessian of the logarithm of the

density there, and the logarithm of their ratio is, up to a constant, $-\sum_j \delta_j \tilde{O}_j$: minus the odd part of the centred potential less its linear term, whose Taylor coefficients at the table's gauge are minus the odd rungs of order at least three of the potential, the rungs that vanish at the centre (Proposition 8.21(iii)).

Proposition 16.32 (The list $\binom{\mathcal{E}}{1}$ as a normaliser)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data on a connected family of $n \geq 2$ marks, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, so that W is homogeneous of degree $-\tau_{\mathcal{E}}$; let $K_0 = |\alpha|$ be the letter of the list $\binom{\mathcal{E}}{1}$, whose bases are the singletons.*

(i) Every completion $W\, d\alpha_{\mathbf{m}'}$, $\mathbf{m}' = \mathbf{m} - \kappa_{\mathcal{E}} w$, has total mass $m'_{\mathcal{E}} = \tau_{\mathcal{E}}$. The native completion is the one whose masses keep the proportions of $\mathbf{m}$, $\boldsymbol{\tau}/\tau_{\mathcal{E}} = \mathbf{m}/m_{\mathcal{E}}$, and they are positive, while the chart completion at g has the mass $m_g - \kappa_{\mathcal{E}}$ at g.

(ii) For $\mathbf{m}' > 0$ with $m'_{\mathcal{E}} = \tau_{\mathcal{E}}$ the form $K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}'}$ is projective, with integral $\Gamma_1^n \begin{bmatrix} \mathbf{m}' \\ \tau_{\mathcal{E}} \end{bmatrix}$ over every section: the period of $\binom{\mathcal{E}}{1}$ at the full basis lean $(\tau_{\mathcal{E}}, 0)$, whose law is $\mathrm{Dir}_{\mathbf{m}'}$ (Proposition 38.8). So $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} = \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}}$; for a schedule with $\mathbf{m}^{(g)} > 0$, $\Gamma_1^n \begin{bmatrix} \mathbf{m}^{(g)} \\ \tau_{\mathcal{E}} \end{bmatrix} = \mathrm{Reg}^{\mathfrak{s}} \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}}$, a convergent integral; and with a native gauge

$$\int W\, d\alpha_{\mathbf{m}} \Big/ \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}} = \mathbb{E}_{U \sim \mathrm{Dir}_{\boldsymbol{\tau}}}[W(U)].$$

(iii) With a native gauge, and for continued values off the pole hyperplanes where $2\sigma\mathbf{1} - \boldsymbol{\tau} > 0$, the fold keeps the pseudointegral and carries the target $\boldsymbol{\tau}$ to $2\sigma\mathbf{1} - \boldsymbol{\tau}$ (Theorem 16.16), so the quotients of (ii) for ω and for its image are $P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ divided by $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix}$ and by $\Gamma_1^n \begin{bmatrix} 2\sigma\mathbf{1} - \boldsymbol{\tau} \\ \tau^*_{\mathcal{E}} \end{bmatrix}$, $\tau^*_{\mathcal{E}} = 2n\sigma - \tau_{\mathcal{E}}$. The second quotient over the first is $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} / \Gamma_1^n \begin{bmatrix} 2\sigma\mathbf{1} - \boldsymbol{\tau} \\ \tau^*_{\mathcal{E}} \end{bmatrix}$, which is not 1 in general. The period of $\binom{\mathcal{E}}{1}$ is that of a full basis lean of a table outside the family, and it prices a lean and its mirror differently; the centre, which the fold fixes, prices them alike (Proposition 16.31(iii)).

(iv) Along the ray of the masses, wherever the proper walls at $\lambda\mathbf{m}$ are positive, the integral of the projective form $K_0^{-(\lambda m_{\mathcal{E}} - \tau_{\mathcal{E}})} W\, d\alpha_{\lambda\mathbf{m}}$ is $\Gamma_1^n \begin{bmatrix} \lambda\mathbf{m} \\ \lambda m_{\mathcal{E}} \end{bmatrix} \mathbb{E}_{U \sim \mathrm{Dir}_{\lambda\mathbf{m}}}[W(U)]$ (Theorem 16.30(iii), Proposition 8.11(i)). The power of K_0 vanishes exactly at $\lambda = \tau_{\mathcal{E}}/m_{\mathcal{E}}$, where the form is the native completion $W\, d\alpha_{\boldsymbol{\tau}}$.

Corollary 16.33 (The pseudointegral is the quasi-integral that always exists)**.** *For real data with $s_j, t_j \geq 0$ not all zero, $\mathbf{m} > 0$ and weights factoring across the molecules, the pseudointegral $\int W\, d\alpha_{\mathbf{m}}$ and the pseudoexpectation $\mathsf{E}_{U \sim \mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ are defined. On a molecule whose target lies on no pole hyperplane the pseudointegral is the quasi-integral of the native completion and the pseudoexpectation its quasi-expectation, the same along every regular schedule; on a pole hyperplane, where the regular schedules of the native completion can depend on the order (Theorem 16.21), it is the one constant term along the ray of the masses. On each molecule every generalized expectation divides by the one period of its centre, a positive convergent integral (Proposition 8.21); when the molecule has a native gauge, the native completion is the one completion that keeps it (Theorem 16.6(i)). The quasi-integrals of ω itself are Laurent data at its chart completions (Theorem 16.20), which off the balance wall can depend on the chart and on the order.*

The pseudointegral names a procedure as well as a value. Written for a family whose realization is specified later, $\wr\cdots\wr_{\mathcal{R}}$ commits to the Foreman's order at every realization: the tree is drawn once from the tables, and the realization $((s_j,t_j)_j;\alpha^\star)$ of the point fixes the measure $d\alpha_{\boldsymbol{\tau}}$ and where the contours of the ports run. An iterated integral in another order, or after a snip, computes a quasi-integral and not the pseudointegral, even where the numbers agree; when they agree, that is Theorem 16.27(i)–(ii), not the order that was followed.

On a pole hyperplane the regulator η of the pseudointegral enters as $u^{\eta\mathbf{m}}$, as in Theorem 1.6. Past a wall the pseudointegral normalizes the distribution of Theorem 16.15(ii).

Chapter 17

The Breta law and the port on the balance wall

17.1 The laws

Theorem 17.1 (The pseudointegral is a normalizer)**.** *Let F be connected on $n \geq 2$ marks with real data, $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, a native gauge $\alpha^\star$, target $\boldsymbol{\tau}$ and form $\omega = W\, d\alpha_{\mathbf{m}}$.*

(i) $\int W\, d\alpha_{\mathbf{m}} = P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ is the normalizer of $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$, and the pseudoexpectation $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \int W\, d\alpha_{\mathbf{m}}/P_{\mathrm{c}}$ is the normalizer of its density against the law of the centre (Proposition 16.31(i)). With $\phi_{\mathbf{m}'}$ the joint law of $\big(K_1(U), \dots, K_N(U), \Psi_1(U), \dots, \Psi_N(U)\big)$ for $U \sim \mathrm{Dir}_{\mathbf{m}'}$, both periods are stacked Gammas of masses over their total times Trailmoments (Proposition 8.11(i)):

$$\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\big[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)\big] = \frac{\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} \big(\mathsf{T}_{(\mathbf{s},\mathbf{t});I_{2N}}[\mathrm{id}]\, \phi_{\boldsymbol{\tau}}\big)\big((0,\infty)^{2N}\big)}{\Gamma_1^n \begin{bmatrix} \sigma\mathbf{1} \\ n\sigma \end{bmatrix} \big(\mathsf{T}_{(\boldsymbol{\sigma},\boldsymbol{\sigma});I_{2N}}[\mathrm{id}]\, \phi_{\sigma\mathbf{1}}\big)\big((0,\infty)^{2N}\big)}.$$

(ii) In the coordinates $y = \log u$ modulo $\mathbb{R}\mathbf{1}$ the law $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ has density proportional to $e^{\langle \boldsymbol{\tau}, y\rangle - \Phi(y)}$, whose unique mode is $\log \alpha^\star$.

(iii) (Cremona) If $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ on the balance wall, then $U^{-1}/|U^{-1}| \sim \mathrm{Breta}^\circ(F; 2\sigma\mathbf{1} - \mathbf{m}; \mathbf{t}, \mathbf{s})$.

(iv) (The even–odd pair) On the balance wall the inversion of (iii) carries $\mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ at the data $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ to the law at $(-\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, -\boldsymbol{\delta})$; at $(0; \boldsymbol{\sigma}, \mathbf{0})$ it fixes the law, and with Z the signed coordinate of U the odd statistics $(O_j(Z))_j$ have a law symmetric under $O \mapsto -O$. The joint Laplace transform of the even and odd statistics is Corollary 38.2.

Proposition 17.2 (The mark relation of the Breta shape law)**.** *Let F have real data, masses $\mathbf{m}$ on the balance wall and every proper wall positive, and in this statement write $\mathbf{u}(\alpha) = \sum_j \big(s_j R_j(\alpha) + t_j R_j^*(\alpha)\big)$ for the usage at α and $\mathbb{E}$ for the expectation of $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$. For $Y = \log U$ modulo $\mathbb{R}\mathbf{1}$, every mark e and every observable g, a smooth function of degree 0 in α such that g and $\partial_{y_e} g$ grow at most exponentially at a rate below the walls,*

$$\mathbb{E}\big[(m_e - \mathbf{u}_e(U))\, g(U) + \partial_{y_e} g(U)\big] = 0.$$

At $g=1$ the masses are the mean usage, $\mathbb{E}[\mathbf{u}(U)]=\mathbf{m}$; with a native gauge $\alpha^\star$ the usage equals the target $\boldsymbol{\tau}=\mathbf{m}$ at the mode $\log\alpha^\star$ (Theorem 17.1(ii)).

Proposition 17.3 (Scale and shapes along a bunch)**.** *For a proper bunch B with room R, masses $\mathbf{m}>0$ and every proper wall of positive real part,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\Gamma_{1}^{|B|}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}\Gamma_{1}^{|R|}\begin{bmatrix}\mathbf{m}|_R\\ m_R\end{bmatrix}$$
$$\times\,\mathbb{E}_{(U'_B,U'_R)\sim \mathrm{Dir}_{\mathbf{m}|_B}\otimes \mathrm{Dir}_{\mathbf{m}|_R}}\Big[\int_0^\infty x^{m_B-1}(1+x)^{-\kappa_\mathcal{E}}\prod_j K_j(xU'_B,U'_R)^{-s_j}\Psi_j(xU'_B,U'_R)^{-t_j}\,dx\Big],$$

the inner integral being the scale integral of the family with the list $\binom{\mathcal{E}}{1}$ appended at the exponents $(\kappa_\mathcal{E},0)$, whose letter at (xU'_B,U'_R) is $1+x$ (Proposition 38.8); on the balance wall it is $\mathrm{Sc}_B(U'_B,U'_R;\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 15.2), and the display is Corollary 15.7 at $h=|\alpha_R|$. For real data with every proper wall positive and $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$, with the internal shapes $V_B=U_B/|U_B|$ and $V_R=U_R/|U_R|$ and the totals $|U_B|+|U_R|=1$ (Theorem 3.12), the scale $\rho=|U_B|/|U_R|$ of B given the internal shapes has density proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}}\prod_j K_j(\rho V_B,V_R)^{-s_j}\Psi_j(\rho V_B,V_R)^{-t_j}$.

Theorem 17.4 (Structure of the Breta law)**.** *Let $X\sim\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$; in this statement $\mathbb{E}$ is the expectation of X.*

(i) $|X|\sim\mathrm{Gamma}_{\kappa_\mathcal{E}}$ *and* $X/|X|\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, independently.*

(ii) The laws form an exponential family with respect to $e^{-|\alpha|}d\alpha/\alpha^{\mathbf{1}}$, with statistics $(\log X_e)_e$ and $(-\log K_j(X),-\log\Psi_j(X))_j$, natural parameters $(\mathbf{m},\mathbf{s},\mathbf{t})$ and log-partition $\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$: $\mathbb{E}\log X_e=\partial_{m_e}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, $\mathbb{E}\log(K_j\Psi_j)(X)=-D_{(\mathbf{0};\mathbf{e}_j,\mathbf{e}_j)}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, $\mathbb{E}\log(K_j/\Psi_j)(X)=-D_{(\mathbf{0};\mathbf{e}_j,-\mathbf{e}_j)}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, D_v the derivative along the direction v of the data, the masses fixed, and the covariances are second derivatives.

(iii) $\mathbb{E}[X^{\mathbf{k}}]=J_F(\mathbf{m}+\mathbf{k};\mathbf{s},\mathbf{t})/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{E}\big[\prod_j K_j(X)^{-a_j}\Psi_j(X)^{-b_j}\big]=J_F(\mathbf{m};\mathbf{s}+\mathbf{a},\mathbf{t}+\mathbf{b})/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

Theorem 17.5 (The race representation)**.** *Let $X\sim\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$. Given $X=\alpha$, run for each table j a* basis race*, the basis T ringing with intensity $\theta_{j,T}\alpha^T$, and a* cobasis race*, with intensity $\theta_{j,T}\alpha^{\mathcal{E}\setminus T}$, all clocks independent; let Y_j and Y_j^* be the first ringing times and T_j, T'_j the winners. In this statement $\mathbb{E}$ and $\mathbb{P}$ are taken over the clocks of the races at α.*

(i) In each race the time and the winner are independent: $K_j(\alpha)Y_j\sim\mathrm{Gamma}_1$ and $T_j\sim\mathbb{P}_{j,\alpha}$; $\Psi_j(\alpha)Y_j^*\sim\mathrm{Gamma}_1$ and $T'_j\sim\mathbb{P}^*_{j,\alpha}$.

(ii) $K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}=\mathbb{E}\big[Y_j^{s_j}(Y_j^*)^{t_j}\big]/\big(\Gamma(1+s_j)\Gamma(1+t_j)\big)$.

(iii) With $\mathrm{g}_j=-\log(K_jY_j)$ and $\mathrm{g}'_j=-\log(\Psi_jY_j^*)$, independent, $-\log(Y_jY_j^*)=\log(K_j\Psi_j)+\mathrm{g}_j+\mathrm{g}'_j$ and $-\log(Y_j/Y_j^*)=\log(K_j/\Psi_j)+\mathrm{g}_j-\mathrm{g}'_j$, and, for real a,b with $1+a\pm b>0$, $\mathbb{E}\big[e^{-a(\mathrm{g}_j+\mathrm{g}'_j)-b(\mathrm{g}_j-\mathrm{g}'_j)}\big]=\Gamma(1+a+b)\,\Gamma(1+a-b)$.

(iv) $\mathbb{P}(Y_j<Y_j^*)=K_j/(K_j+\Psi_j)=1/(1+e^{-O_j(z)})$, $\alpha=e^{2z}$.

(v) $D=\chi_{T_j}-\chi_{T'_j}$ has $\mathbb{E}[e^{\langle w,D\rangle}]=e^{E_j(z+w)-E_j(z)}$: the even statistic E_j is the cumulant function of the difference of the two winners (by (i), Theorem 7.14 for table j).

The two exponents of a table are the two clocks of its races: its even letter reads the sum $\mathrm{g}_j+\mathrm{g}'_j$ of the two logarithmic clocks of (iii) and its odd letter their difference, so that the pair coordinates (σ_j,δ_j) are the exponents of $\mathrm{g}_j+\mathrm{g}'_j$ and $\mathrm{g}_j-\mathrm{g}'_j$.

Theorem 17.6 (Conjugacy)**.** *Let* $X \sim \mathrm{Breta}(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$.

(i) Given the winner T *of the* j*-th basis race, its time unobserved,* $X \sim \mathrm{Breta}(F; \mathbf{m}+\mathbf{1}_T; \mathbf{s}+\mathbf{e}_j, \mathbf{t})$*; given the winner* T' *of the* j*-th cobasis race,* $X \sim \mathrm{Breta}(F; \mathbf{m} + \mathbf{1}_{\mathcal{E}\setminus T'}; \mathbf{s}, \mathbf{t} + \mathbf{e}_j)$*. Both keep* $\kappa_{\mathcal{E}}$*.*

(ii) Under the joint law of X *and the races of Theorem 17.5,* $\mathbb{P}(T_j = T) = \theta_{j,T} P_F(\mathbf{m} + \mathbf{1}_T; \mathbf{s} + \mathbf{e}_j, \mathbf{t})/P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *and* $\mathbb{P}(T'_j = T') = \theta_{j,T'} P_F(\mathbf{m} + \mathbf{1}_{\mathcal{E}\setminus T'}; \mathbf{s}, \mathbf{t} + \mathbf{e}_j)/P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$*, each summing to one.*

(iii) Given the time $Y_j = \lambda$ *as well,* X *has the density of* $\mathrm{Breta}(F; \mathbf{m} + \mathbf{1}_T; \mathbf{s}, \mathbf{t})$ *multiplied by* $e^{-\lambda K_j(\alpha)}$ *and renormalized: the kernel terms* $\lambda\theta_{j,T}\alpha^T$ *join those of* $e^{-|\alpha|}$*, and the law exists only on the orthant.*

Theorem 17.7 (The winners of a tensorminental table)**.** *If table* j *is tensorminental of tensor rank* q *with slices* $A^{(k)}$*, then for a bunch* B *with room* R *and at a gauge as in Theorem 7.20, the pattern* $T_j \cap B$ *of the basis winner has the generating function* $\sum_{k,l} w_{kl} \det\big(I + (D_x - I)\Pi^{(kl)}_{BB}\big)$ *of that theorem, and the cobasis winner's is the same with the slices of the dual multivector. Conditioning on the winner's pattern leaves tables with multivectors of tensor rank at most* q *on the room.*

Proposition 17.8 (The period along a ray of data)**.** *Let* F *be connected, let* $(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ *be real data on its balance wall with* $\boldsymbol{\tau} > 0$*,* $s_j, t_j \geq 0$ *not all zero, and a native gauge* $\alpha^\star$*, let* $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$ *be complex data on the balance wall, and in this statement write* $W_0 = \prod_j K_j^{-s_{0,j}} \Psi_j^{-t_{0,j}}$*. As* $\lambda \to +\infty$ *the data* $(\mathbf{m}_0 + \lambda\boldsymbol{\tau}; \mathbf{s}_0 + \lambda\mathbf{s}, \mathbf{t}_0 + \lambda\mathbf{t})$ *have every proper wall of positive real part, and, with* Φ *the potential at* $(\mathbf{s}, \mathbf{t})$ *and a chart* g*,*

$$P_F(\mathbf{m}_0 + \lambda\boldsymbol{\tau}; \mathbf{s}_0 + \lambda\mathbf{s}, \mathbf{t}_0 + \lambda\mathbf{t}) \sim (\alpha^\star)^{\mathbf{m}_0} W_0(\alpha^\star) \Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}} \det\big(\nabla^2\Phi(\log\alpha^\star)\big)^{-1/2}_{(g)} e^{-\lambda(\Phi(\log\alpha^\star) - \langle\boldsymbol{\tau}, \log\alpha^\star\rangle)},$$

(g*) deleting the row and the column of* g*, where* $\nabla^2\Phi$ *is* $\frac{1}{4}$ *times rung two of the still and* $(\alpha^\star)^{\mathbf{m}_0} W_0(\alpha^\star)$ *does not depend on the scale of* $\alpha^\star$*. At* $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0) = 0$ *the display is the leading term of Theorem 38.5(ii) at real* λ *for the question of the data* $(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$*. For real* $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$*, under* $\mathrm{Breta}^\circ(F; \mathbf{m}_0 + \lambda\boldsymbol{\tau}; \mathbf{s}_0 + \lambda\mathbf{s}, \mathbf{t}_0 + \lambda\mathbf{t})$*,* $\sqrt{\lambda}(Y - \log\alpha^\star)$ *tends to the centred normal law on* $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ *with covariance* $(\nabla^2\Phi)^{-1}$*,* $Y = \log U$*.*

Definition 17.9 (Skeleton law)**.** For coefficients $\xi_\ell \in \mathbb{C}\setminus(-\infty, 0]$, $\ell \in J = \{0, \ldots, L\}$, closed under conjugation, real nonzero letter exponents w_ℓ, equal on conjugate coefficients, with total $|\mathbf{w}|$, and $0 < \kappa < |\mathbf{w}|$, the *Skeleton law* $\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})$ is the law on $(0, \infty)$ with density

$$y^{\kappa-1} \prod_\ell (y + \xi_\ell)^{-w_\ell} \Big/ \Gamma_{|J|}\begin{bmatrix} \kappa, & |\mathbf{w}| - \kappa \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w}, |\mathbf{w}|-\kappa;\, I_J}(\boldsymbol{\xi}),$$

its normalizer being Proposition 15.1; the density is positive, and integrable exactly when $0 < \kappa < |\mathbf{w}|$.

Theorem 17.10 (The Skeleton law)**.** *(i) Under* $\mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$*, the scale of a proper bunch* B *given the internal shapes of* B *and of its room has the law* $\mathrm{Skel}_{\mathbf{w},\kappa_B}(\boldsymbol{\xi})$*:* $\boldsymbol{\xi}$ *the joint spectrum of* B *at that point and* $\mathbf{w}$ *its letter exponents of Theorem 15.2, and off the balance wall the coefficient* 1 *besides, with the letter exponent* $\kappa_{\mathcal{E}}$*, added to the letter exponent of* 1 *when* 1 *is a point of the joint spectrum; coefficients of letter exponent zero are discarded.*

(ii) For $Y \sim \mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})$ and $-\kappa < \operatorname{Re} k < |\mathbf{w}| - \kappa$,

$$\mathbb{E}_{Y\sim\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})}[Y^k] = \Gamma_{\,2}^{\,2}\!\begin{bmatrix}\kappa+k,\ |\mathbf{w}|-\kappa-k\\ \kappa,\ |\mathbf{w}|-\kappa\end{bmatrix}\frac{\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa-k;\,I_J}(\boldsymbol{\xi})}{\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi})},$$

the quotient of the values of Proposition 15.1 at the indices $\kappa+k$ and κ, read as that quotient when a letter exponent is not positive; and $1/Y \sim \mathrm{Skel}_{\mathbf{w},|\mathbf{w}|-\kappa}(\boldsymbol{\xi}^{-1})$.

(iii) With real coefficients, positive letter exponents and a point $r \geq \max_\ell \xi_\ell$ of the axis, Y is the mixture over $\mathbf{n} \in \mathbb{N}^J$ of $rY_{\mathbf{n}}$, $Y_{\mathbf{n}} \sim \mathrm{Skel}_{|\mathbf{w}|+|\mathbf{n}|,\kappa}(1)$ the law with the one coefficient 1, mixed in proportion to $\prod_{\ell\in J}\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{n_\ell}\Gamma_{\,2}^{\,2}\!\begin{bmatrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{bmatrix}$, $g_\ell = 1-\xi_\ell/r$ the spectral gaps at r of Theorem 15.5; at $r = \max_\ell \xi_\ell$ the largest coefficient drops out.

(iv) In the situation of (i) on the balance wall, with $s_j, t_j \geq 0$: given $Y \leq 1$, Y is the mixture of the first coordinates of the laws $\mathrm{Dir}_{(\kappa_B+k,|\mathbf{n}|+1)}$, and given $Y \geq 1$, $1/Y$ is the mixture of the first coordinates of $\mathrm{Dir}_{(\kappa_B^{\mathrm{top}}+k,|\mathbf{n}|+1)}$, mixed in proportion to the terms of Theorem 15.9(ii); the two sectors carry probabilities proportional to the sector Trailmoments of that theorem at the indices κ_B and κ_B^{top}.

17.2 The port on the balance wall

Definition 17.11 (Wall contour; joint period)**.** On the balance wall, with $\operatorname{Re} s_j > 0$ or $s_j = 0$, and $\operatorname{Re} t_j > 0$ or $t_j = 0$, for every j, a *wall contour* at a bunch B puts the shares of Theorem 15.15 of the letters with nonzero exponent on real parts $c_{j,p}, c'_{j,p} > 0$ with $\sum_p c_{j,p} = \operatorname{Re} s_j$, $\sum_p c'_{j,p} = \operatorname{Re} t_j$ and real odd load equal to the real part of the offset on the bunch, $\sum_{j,p}\frac{c_{j,p}-c'_{j,p}}{2}\chi_p(e) = \operatorname{Re}\mu_e$; equivalently the real part of every mass of the bunch is carried exactly, $\operatorname{Re}\mathrm{load}_e = \operatorname{Re} m_e$. The *fibre integral at B* is the fibre integral (Definition 4.1) with the pins $|\mathbf{a}_j| = s_j$, $|\mathbf{b}_j| = t_j$ and $\langle\boldsymbol{\delta}\rangle_e = \mu_e$, $e \in B$. For a family $G = ((\mathcal{B}_i,\theta_i);a_i,b_i)_i$ whose molecules are $R_1,\dots,R_q$, the *joint period* $P_G^{\otimes}(\mathbf{m};\mathbf{a},\mathbf{b})$ is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}\prod_i K_i(\alpha)^{-a_i}\Psi_i(\alpha)^{-b_i}$ over $\alpha \in \prod_{k=1}^q \Delta_{R_k}$ against $\prod_k d\omega_{R_k}$, wherever it converges; when the weights factor across the molecules it is the product of the molecules' periods.

Theorem 17.12 (The port on the balance wall)**.** *Let F have data on the balance wall with $\operatorname{Re} s_j > 0$ or $s_j = 0$, and $\operatorname{Re} t_j > 0$ or $t_j = 0$, for every j, let $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converge, and let B be a proper bunch with room R; put $z = \frac{1}{2}\log\alpha$.*

(i) The bunch factor *$\Xi_B(\alpha_R;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t}) = \int_{\mathbb{R}^B_{>0}}\alpha_B^{\boldsymbol{\mu}|_B-\mathbf{1}}\prod_j\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}\,d\alpha_B$ converges for every α_R, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R} v^{\boldsymbol{\mu}|_R-\mathbf{1}}\,\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})\,d\omega_R(v)$, and on every wall contour*

$$\Xi_B(\alpha_R;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ \langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_B}\ \prod_j \Gamma_{|\operatorname{pat}_j(B)|}\!\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_j(B)|}\!\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}$$
$$\times\prod_{j,p}\Lambda_{j,(p)}(z_R)^{-a_{j,p}}\Lambda_{j,(p)}(-z_R)^{-b_{j,p}}.$$

(ii) Let $R_1,\dots,R_q$ be the molecules of the children and $\kappa_{R_i}(\mathbf{a},\mathbf{b})$ the wall at R_i of the children $F_B(\mathbf{a},\mathbf{b})$ with the masses $\mathbf{m}|_R$ (Definition 8.1); as R_i is a molecule, $\kappa_{R_i}(\mathbf{a},\mathbf{b}) = m_{R_i} - \sum_{j,p}\big(a_{j,p}r_{j,(p)}(R_i) + b_{j,p}(|R_i| - r_{j,(p)}(R_i))\big)$, with $r_{j,(p)}$ the rank function of the child (j,p).

Let $\mathcal{X}$ be the fibre of the shares cut by the pins $|\mathbf{a}_j| = s_j$, $|\mathbf{b}_j| = t_j$, $\langle\boldsymbol{\delta}\rangle = \boldsymbol{\mu}|_B$ and $\kappa_{R_1} = \cdots = \kappa_{R_{q-1}} = 0$, with $|B|+q-1$ pins beyond the share sums; $\kappa_{R_q} = 0$ holds on it. On a wall contour with $\operatorname{Re}\kappa_{R_i} = 0$ for $i < q$ at which $P^{\otimes}_{F_B(\mathbf{c},\mathbf{c}')}(\operatorname{Re}\mathbf{m}|_R; \mathbf{c}, \mathbf{c}')$ converges,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathcal{X}} \prod_j \Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} P^{\otimes}_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}).$$

(iii) The native split of the real parts $(\operatorname{Re}\mathbf{m}; \operatorname{Re}\mathbf{s}, \operatorname{Re}\mathbf{t})$, which have a native gauge, is a wall contour whose real parts are those of the points of $\mathcal{X}$, and $P^{\otimes}$ converges there; for real data it is the native split, and it lies on the fibre.

The bunch is integrated on its orthant, scale and shape at once: writing $\alpha_B = \rho u$ before opening the letters would split the $|B|$ pins into a pin on the scale and a stacked Gamma over the shape whose indices sum to zero, carrying $1/\Gamma(0) = 0$, and neither has a value alone. For the same reason $B = \mathcal{E}$ is excluded: the imaginary part of the odd load then lies in the hyperplane $\sum_{e\in\mathcal{E}} \operatorname{Im}\langle\boldsymbol{\delta}\rangle_e = \operatorname{Im}\mu_{\mathcal{E}}$, the scaling mode, and one mark stays in the room.

Lemma 17.13 (Connected tables have connected children)**.** *If some table with $s_j + t_j > 0$ is connected, then at every bunch the children at a split with positive shares form a connected family.*

At the masses $\boldsymbol{\tau}+\eta\mathbf{m}$ the port on the orthant carries the stacked Gamma with the upper entries $\nu_e = \mu_e - \langle\boldsymbol{\delta}\rangle_e$, $e \in B$, and κ_{R_i}, and the lower entry $\eta m_{\mathcal{E}}$; its $|B|+q$ upper entries sum to $\eta m_{\mathcal{E}}$ and have real parts positive multiples of η on the wall contour of the native split (Theorem 17.12(iii)); as $\eta \to 0$ it tends to $(2\pi)^{|B|+q-1}$ times the delta of the pins. So the port on the balance wall is the limit of the Foreman procedure on the ray (Theorem 16.13): the Gammas are the pins.

Proposition 17.14 (Peeling a molecule)**.** *In Theorem 17.12 with a native gauge, let R_1 be a molecule of the children and $g \in R_1$. Each child letter is $\sum_X \alpha^X_{R_1} K_{[X]}(\alpha_{R\setminus R_1})$ over the bases X of its restriction to R_1, with $K_{[X]}$ the letter on $R \setminus R_1$ with the bases Y and the weights $\theta_{X\cup Y}$ (for Ψ, $\alpha^{R_1\setminus X}_{R_1}\Psi_{[X]}$). Opening these sums with one share per part, $P^{\otimes}$ is the fibre integral with the $|R_1|-1$ pins $\operatorname{load}_e = m_e$, $e \in R_1 - g$, of the stacked Gammas of these shares against the joint period of the new children on the remaining molecules; the pin at g holds by the balance of R_1, the native split lies on the fibre, and the joint period converges there. With weights multiplicative across R_1 every $K_{[X]}$ is a multiple of one letter and this is the factorization of Theorem 17.12(ii).*

Corollary 17.15 (Atoms on the balance wall)**.** *In Theorem 17.12, fix g. On a wall contour for $\mathcal{E} - g$, with one share per basis and letter,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ \langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_{\mathcal{E}-g}} \prod_j \Gamma_{|\mathcal{B}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \Gamma_{|\mathcal{B}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} \prod_{j,T} \theta_{j,T}^{-a_{j,T}-b_{j,T}},$$

independent of g; the pin at g holds by itself.

Running the ports on the balance wall down to rooms of one mark writes $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ as an iterated fibre integral with $n-1$ pins in all; with one share per pattern and letter its dimension is $\sum(|\mathcal{B}_j|-1) - (n-1)$, one term for each active letter, j its table, independent of the tree when no letter is split across molecules.

Corollary 17.16 (Closure)**.** *If the weights of a form $W\,d\alpha_{\mathbf{m}}$ with real data, $s_j, t_j \geq 0$ not all zero, factor across its molecules and every molecule has a native gauge, $\int W\,d\alpha_{\mathbf{m}}$ is a product over the molecules of iterated fibre integrals over affine fibres of share space, with integrands products of stacked Gammas of the shares and monomials in the weights, $n-1$ pins on a molecule of n marks. The period of each molecule at its target is also the nonnegative series of Corollary 15.11 and of Theorem 15.13. Every quasi-integral is an iterated constant term of the continuation of the share integral of Theorem 15.15 at its chart completion.*

Corollary 17.17 (The marginal law of the shape)**.** *In Theorem 17.12 let $U \sim \mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$, with the internal shapes $V_B = U_B/|U_B|$ and $V_R = U_R/|U_R|$ of B and R (Proposition 17.3).*

(i) $\mathbb{E}_{U' \sim \mathrm{Dir}_{\mathbf{m}|_B}}\big[\,\mathrm{Sc}_B(U', v; \mathbf{m}; \mathbf{s}, \mathbf{t})\big] = v^{-\sigma\mathbf{1}}\,\Xi_B(v; \boldsymbol{\mu}|_B; \mathbf{s}, \mathbf{t}) / \Gamma_{|B|}\begin{bmatrix}\mathbf{m}|_B \\ m_B\end{bmatrix}$.

(ii) Given $V_R = v$, for $\mathbf{k} \in \mathbb{C}^B$ with $|\mathbf{k}| = 0$, $\mathbb{E}_{U \sim \mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})}[V_B^{\mathbf{k}} \mid V_R = v] = \Xi_B(v; \boldsymbol{\mu}|_B + \mathbf{k}; \mathbf{s}, \mathbf{t}) / \Xi_B(v; \boldsymbol{\mu}|_B; \mathbf{s}, \mathbf{t})$, the bunch factor at shifted offsets, whose fibre is $\langle \boldsymbol{\delta} \rangle = \boldsymbol{\mu}|_B + \mathbf{k}$.

(iii) V_R has density proportional to $v^{\boldsymbol{\mu}|_R - \mathbf{1}}\,\Xi_B(v; \boldsymbol{\mu}|_B; \mathbf{s}, \mathbf{t})$, the fibre mixture of the children's shape laws in proportion to the stacked Gammas times $P_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b})$ when the children are connected (Lemma 17.13).

(iv) Given both shapes, the scale of B has the Skeleton law of Theorem 17.10(i).

17.3 Cycles

Definition 17.18 (Cycle of a family)**.** A *cycle* of a family is a bunch C that is a circuit of one of its tables. That table's cobasis polynomial has floor one on C, and on its bottom colevel the letter is linear on C; a letter linear on a bunch gives the bunch the *kernel* e^{-h}, h the letter divided by the sum of its coefficients (Theorem 17.19(ii)).

Theorem 17.19 (A cycle carries its own kernel)**.** *Let F have real data with $s_j, t_j \geq 0$, $\kappa_{\mathcal{E}} = 0$, $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ convergent, and $C \neq \mathcal{E}$ a circuit of table j_0 with $t_{j_0} > 0$ and room R.*

(i) The patterns of table j_0 on C of colevel one are exactly $p_e = C - e$, $e \in C$, and the child at p_e is the contraction $\mathcal{B}_{j_0}/C$ on R with the weights $\theta_{j_0, p_e \cup X}$. So $\Psi_{j_0} = \Psi^{\flat} + \Psi^{\sharp}$ with $\Psi^{\flat} = \sum_{e \in C} \alpha_e \Psi_{j_0,(p_e)}(\alpha_R)$ linear in α_C. The circuits are the only bunches whose colevel-one patterns are one per mark.

(ii) With $G(\alpha_R) = \sum_{e \in C} \Psi_{j_0,(p_e)}(\alpha_R)$ and $h = \Psi^{\flat}/G$, if the family with $\Psi_{j_0}^{-t_{j_0}}$ replaced by $(\Psi^{\flat})^{-c}(\Psi^{\sharp})^{-(t_{j_0}-c)}$ converges, then on $\mathrm{Re}\,\beta = c$

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{1}{\Gamma(t_{j_0})} \int_{\beta+\beta'=t_{j_0}} \Gamma(\beta') \times \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-h(\alpha)} G(\alpha_R)^{-\beta} \Psi^{\sharp}(\alpha)^{-\beta'} K_{j_0}^{-s_{j_0}} \prod_{j \neq j_0} K_j^{-s_j} \Psi_j^{-t_j}\, d\alpha.$$

When the colevel-one children are multiples ϑ_e of one letter, as for unit weights and for determinantal tables, the kernel is $e^{-\langle \vartheta, \alpha_C \rangle / |\vartheta|}$.

(iii) On a wall contour at C the pins at the marks of C hold exactly when $b_{j_0,p_e} = m_e - \lambda_e$, λ_e the load on e of all the other shares, and $\sum_{j,p} \big(a_{j,p}(|p| - d_j(C)) + b_{j,p}(|C \setminus p| - d_j^*(C))\big) = \kappa_C$, the coefficients being the excesses of the level and the colevel of p in table j (Definition 12.15);

the fibre integral at C is over the other shares with the single pin that their total excess is the wall of the cycle, the Gammas $\Gamma(m_e-\lambda_e)$ of the cycle standing in the places of the $\Gamma(b_{j_0,p_e})$.

Dually, a cocircuit has floor one in the basis polynomial, and its bottom level gives its kernel.

Corollary 17.20 (A table that is one circuit)**.** *Let $F=((\mathcal{B},\theta);s,t)$ with $s,t>0$, where the bases of $\mathcal{B}$ are the sets $\mathcal{E}-e$, $e\in\mathcal{E}$, with the weights θ_e; let the masses lie on the balance wall, $m_{\mathcal{E}}=(n-1)s+t$, with $m_S>s(|S|-1)$ for every nonempty proper S, and put $\nu_e=m_e-s$. Then*

$$\begin{aligned}P_F(\mathbf{m};s,t)&=\frac{\prod_e\theta_e^{-\nu_e}}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{s-1}\prod_{e\in\mathcal{E}}\mathsf{G}_{\nu_e;(1,-1)}\big(1,\theta_e^2y\big)\,dy\\&=\frac{\prod_e\theta_e^{-\nu_e}}{\Gamma(s)\Gamma(t)}\int_{\sum_e a_e=s}\prod_{e\in\mathcal{E}}\Gamma(a_e)\,\Gamma(a_e+\nu_e)\,\theta_e^{-2a_e}:\end{aligned}$$

the cobasis polynomial $\Psi(\alpha)=\langle\theta,\alpha\rangle$ is linear on the whole set, which so carries its own kernel (Definition 17.18), and the fibre of the atom formula (Corollary 17.15), of dimension $n-1$, closes to one scale.

Chapter 18

Blocks: generation and cleaning

A bunch whose bases meet it on two levels only is seen by its room as one mark: the room reads the bunch through two ratios of polynomials on its marks, one for each Foreman polynomial, and the bunch is a table on its own marks and one more. Blocks generate families by replacing marks; a bunch of width one is read as a mark of its room, its shape integrated while its scale is kept, its walls glued, and its scale closed at a wall, at the gauge and on the totals of the bunch and the room. Cleaning, the repeated pooling of such bunches, is a move between period data and has one result.

18.1 Generated families

A family can be generated rather than found: seed tables assembled under a set of bijections give atoms (Proposition 11.7), and a mark that is neither a loop nor a coloop, replaced by a block of new marks, gives a bunch of width one, which the room reads as the mark it replaced.

Theorem 18.1 (Replacement of marks by blocks)**.** *Let $(\mathcal{B},\theta)$ be a table of rank r on $\mathcal{E}=\mathcal{E}_1\sqcup\mathcal{E}_2$, and replace every mark h by a* block *X_h of $q_h\geq 1$ new marks, by the* first rule *at the marks of $\mathcal{E}_1$ and by the* complementary rule*, the first rule read on the dual table, at those of $\mathcal{E}_2$: a basis T gives the sets that contain one mark of X_h for each $h\in T\cap\mathcal{E}_1$, none for $h\in\mathcal{E}_1\setminus T$, all of X_h for $h\in T\cap\mathcal{E}_2$ and all of X_h but one for $h\in\mathcal{E}_2\setminus T$, each with the weight of T. They are the bases of a table on $\bigsqcup_h X_h$ of rank $r+\sum_{h\in\mathcal{E}_2}(q_h-1)$; a block of one mark leaves its mark in place, and when no mark is a loop or a coloop the table is the substitution into $(\mathcal{B},\theta)$ of the parallel blocks $\binom{X_h\sqcup\{g_h\}}{1}$ at $\mathcal{E}_1$ and the series blocks $\binom{X_h\sqcup\{g_h\}}{q_h}$ at $\mathcal{E}_2$ (Definition 20.1, Corollary 20.5(i)). Its Foreman polynomials are, each e_k read on the marks of its block,*

$$K'=\prod_{h\in\mathcal{E}_2} e_{q_h-1}\cdot K\Big(\big(e_1\big)_{h\in\mathcal{E}_1},\big(\tfrac{e_{q_h}}{e_{q_h-1}}\big)_{h\in\mathcal{E}_2}\Big),\qquad \Psi'=\prod_{h\in\mathcal{E}_1} e_{q_h-1}\cdot\Psi\Big(\big(\tfrac{e_{q_h}}{e_{q_h-1}}\big)_{h\in\mathcal{E}_1},\big(e_1\big)_{h\in\mathcal{E}_2}\Big);$$

for one block $X=\{x_1,\dots,x_q\}$ at the mark h, the other marks left in place, the first rule gives $K'=K_{\setminus h}+e_1(\alpha_X)\,K_{/h}$ and $\Psi'=e_q(\alpha_X)\,\Psi_{\setminus h}+e_{q-1}(\alpha_X)\,\Psi_{/h}$, and the complementary rule gives $K'=e_q(\alpha_X)\,K_{/h}+e_{q-1}(\alpha_X)\,K_{\setminus h}$ and $\Psi'=\Psi_{/h}+e_1(\alpha_X)\,\Psi_{\setminus h}$ (Definition 7.3). In scale and shape, $\alpha_X=\rho u$ with $u\in\Delta_X$: $e_1(\alpha_X)=\rho$, $e_q(\alpha_X)=\rho^q e_q(u)$, $e_{q-1}(\alpha_X)=\rho^{q-1}e_{q-1}(u)$, and $\alpha_X^{\mathbf{m}|_X-\mathbf{1}}d\alpha_X=\rho^{m_X-1}d\rho\;u^{\mathbf{m}|_X-\mathbf{1}}d\omega_X(u)$ (Lemma 2.7).

Theorem 18.2 (Counts, walls and atoms under replacement)**.** *Let $(\mathcal{B},\theta)$ be a table of rank r on $\mathcal{E}$, $n=|\mathcal{E}|$, and replace every mark h by a block X_h of $q_h\geq 1$ new marks as in Theorem 18.1, by the first rule at the marks of $\mathcal{E}_1$ and by the complementary rule at those of $\mathcal{E}_2$, $\mathcal{E}=\mathcal{E}_1\sqcup\mathcal{E}_2$.*

(i) The marks become $\sum_h q_h$; the rank becomes $r + \sum_{h\in\mathcal{E}_2}(q_h - 1)$; the total weight becomes

$$\sum_{T\in\mathcal{B}} \theta_T \prod_{h\in T\cap\mathcal{E}_1} q_h \prod_{h\in\mathcal{E}_2\setminus T} q_h = \prod_{h\in\mathcal{E}_2} q_h \cdot K\Big((q_h)_{h\in\mathcal{E}_1}, (q_h^{-1})_{h\in\mathcal{E}_2}\Big),$$

which is $K((q_h)_h)$ under the first rule alone and $\Psi((q_h)_h)$ under the complementary rule alone.

(ii) For a set Y of new marks let $\lceil Y\rceil = \{h:\ X_h\cap Y\neq\varnothing\}$ and $\lfloor Y\rfloor = \{h:\ X_h\subseteq Y\}$, and let r', d', $d^{*\prime}$, c' be the counts of the new table (Definition 7.10). Then

$$d'(Y) = \sum_{h\in\mathcal{E}_2} \max\big(|X_h\cap Y| - 1, 0\big) + d\big((\lfloor Y\rfloor\cap\mathcal{E}_1)\sqcup(\lceil Y\rceil\cap\mathcal{E}_2)\big),$$
$$d^{*\prime}(Y) = \sum_{h\in\mathcal{E}_1} \max\big(|X_h\cap Y| - 1, 0\big) + d^*\big((\lceil Y\rceil\cap\mathcal{E}_1)\sqcup(\lfloor Y\rfloor\cap\mathcal{E}_2)\big),$$

and $c'(Y) = |Y| - d'(Y) - d^{*\prime}(Y)$. Under the first rule alone $r'(Y) = r(\lceil Y\rceil)$, $d'(Y) = r - r(\mathcal{E}\setminus\lfloor Y\rfloor)$ and $c'(Y) = r(\lceil Y\rceil) + r(\mathcal{E}\setminus\lfloor Y\rfloor) - r$; under the complementary rule alone $c'(Y) = |\lceil Y\rceil| - |\lfloor Y\rfloor| + r(\mathcal{E}\setminus\lceil Y\rceil) + r(\lfloor Y\rfloor) - r$. If $Y = \bigcup_{h\in S} X_h$, then $c'(Y) = c(S)$; when the seed table is connected on at least two marks, a proper nonempty part of one block has width one.

(iii) Let a family of seed tables with positive exponent pairs be connected on at least two marks and carried to itself by a set H of bijections as in Proposition 11.7, and replace every mark by a block of the same q marks, all by the first rule or all by the complementary rule. Then the bijections of H lift to the blocks, the permutations inside a block carry every replaced table to itself, the replaced family is connected, and for every q its realization at the table's gauge is an atom.

The shape of a block is integrated by Theorem 18.8, and its scale is the mark of the seed.

18.2 Bunches of width one

Definition 18.3 (Parallel and series bunches)**.** A *parallel bunch* of a table (a parallel class) is a set of at least two marks, none a loop, no two of which lie in a common basis; a *series bunch* (a series class) is a set of at least two marks, none a coloop, every basis containing all but at most one of them. A bunch B of width $c(B) = r(B) - d(B)$ (Definition 7.10, Theorem 7.11) has its patterns on the $c(B)+1$ levels $d(B),\dots,r(B)$ (Theorem 12.6), and its room has the same width.

Definition 18.4 (Hinge values; their ratio)**.** Let a bunch B of a table $(\mathcal{B},\theta)$ have width one, the weights factoring across B and its room R, $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$, so that $\mathcal{B}|B$ and $\mathcal{B}/R$ carry θ' and $\mathcal{B}|R$ and $\mathcal{B}/B$ carry θ'' (Definition 7.5). The *hinge value of B by K* and its *hinge value by Ψ* are

$$\gamma_B^K = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}}, \qquad \gamma_B^\Psi = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}},$$

functions of α_B homogeneous of degree one, the values at which K and Ψ see the bunch replaced by one mark, its hinge (Theorem 18.6), and their *ratio*

$$\varrho_B = \frac{\gamma_B^\Psi}{\gamma_B^K} = \frac{\mathbf{E}_{\mathcal{B}/R}}{\mathbf{E}_{\mathcal{B}|B}},$$

the quotient of the even letters $\mathbf{E} = K\Psi$ (Definition 8.17) of the two tables on B, is homogeneous of degree zero, a function of the shape $u = \alpha_B/|\alpha_B|$ alone. The room has width one as well, and its hinge values are those of R as a bunch: $\gamma_R^K = K_{\mathcal{B}|R}/K_{\mathcal{B}/B}$, $\gamma_R^\Psi = \Psi_{\mathcal{B}/B}/\Psi_{\mathcal{B}|R}$ and $\varrho_R = \mathbf{E}_{\mathcal{B}/B}/\mathbf{E}_{\mathcal{B}|R}$. The factorization is fixed up to one positive constant on each of the two levels: multiplying θ' by λ_d on the patterns of the level $d(B)$ and by λ_r on those of the level $r(B)$, and dividing θ'' by λ_d and by λ_r on the room sides of the bases of these levels, multiplies γ_B^K and γ_R^K by $c = \lambda_r/\lambda_d$, γ_B^Ψ and γ_R^Ψ by c^{-1}, and ϱ_B and ϱ_R by c^{-2}, and leaves γ_R^K/γ_B^K, $\gamma_R^\Psi/\gamma_B^\Psi$ and ϱ_B/ϱ_R unchanged; statements about hinge values and their ratios are read at one fixed factorization. For table j of a family they are $\gamma_{j,B}^K$, $\gamma_{j,B}^\Psi$ and $\varrho_{j,B}$, each table with its own two constants.

Proposition 18.5 (Hinge values of parallel and series bunches)**.** *A parallel or series bunch has width at most one, and width zero exactly when it is a union of molecules. Let a parallel bunch B of width one have its weights factoring across B and R (Definition 18.4); then $\mathcal{B}/R$ has the single basis $\varnothing$, with the weight $\theta'_\varnothing$, and $\mathcal{B}|B$ the singletons, with the weights $\theta'_{\{e\}}$, and in this statement write $w_e = \theta'_{\{e\}}/\theta'_\varnothing$. So $K_{\mathcal{B}/R} = \theta'_\varnothing$, $K_{\mathcal{B}|B} = \theta'_\varnothing \sum_{e\in B} w_e\alpha_e$, $\Psi_{\mathcal{B}|B} = \theta'_\varnothing\, e_{|B|}(\alpha_B) \sum_{e\in B} w_e\alpha_e^{-1}$ and $\Psi_{\mathcal{B}/R} = \theta'_\varnothing\, e_{|B|}(\alpha_B)$, and*

$$\gamma_B^K = \sum_{e\in B} w_e\alpha_e, \qquad \gamma_B^\Psi = \Big(\sum_{e\in B} w_e\alpha_e^{-1}\Big)^{-1}:$$

K adds the marks of the class with the factors w_e and Ψ combines them as reciprocals with the same factors. A series bunch B of width one, its weights factoring, swaps the two roles: $\mathcal{B}|B$ has the single basis B, with the weight θ'_B, and $\mathcal{B}/R$ the sets $B-e$, with the weights θ'_{B-e}; with $w_e = \theta'_{B-e}/\theta'_B$, $K_{\mathcal{B}|B} = \theta'_B\, e_{|B|}(\alpha_B)$, $K_{\mathcal{B}/R} = \theta'_B\, e_{|B|}(\alpha_B) \sum_e w_e\alpha_e^{-1}$, $\Psi_{\mathcal{B}|B} = \theta'_B$ and $\Psi_{\mathcal{B}/R} = \theta'_B \sum_e w_e\alpha_e$, so $\gamma_B^\Psi = \sum_e w_e\alpha_e$ and $\gamma_B^K = \big(\sum_e w_e\alpha_e^{-1}\big)^{-1}$. In a list, the case of cleaning (Definition 18.36), with the factorization $\theta' = \theta'' = 1$, every w_e is 1, and a pair with the shape $(x, 1-x)$ has $\varrho_B = x(1-x)$ when it is parallel and $\varrho_B = 1/(x(1-x))$ when it is in series.

A block X_h of Theorem 18.1 with $q_h \geq 2$ whose mark is neither a loop nor a coloop of the seed is a parallel bunch of width one under the first rule and a series bunch of width one under the complementary rule.

Theorem 18.6 (A bunch of width one is one mark to its room)**.** *Let a bunch B of a table $(\mathcal{B}, \theta)$ have width one, the weights factoring across B and R, $\theta_T = \theta'_{T\cap B}\, \theta''_{T\cap R}$, and let g be a new mark. The* cell of the bunch $\mathcal{B}_B^+$ *is the table on $B+g$ whose bases avoiding g are those of $\mathcal{B}|B$ and whose bases through g are the $X+g$, $X \in \mathcal{B}/R$, with the weights θ'; the* cell of the room $\mathcal{B}_R^+$ *is the table on $R+g$ whose bases avoiding g are those of $\mathcal{B}|R$ and whose bases through g are the $Y+g$, $Y \in \mathcal{B}/B$, with the weights θ''. Both are tables, lists when $(\mathcal{B}, \theta)$ is one, in which g, the* hinge*, is neither a loop nor a coloop, and the bases of $\mathcal{B}$ are theirs glued along g with g on exactly one side:*

$$\mathcal{B} = \{X \sqcup Y:\ X \in \mathcal{B}|B,\ Y \in \mathcal{B}/B\} \ \sqcup\ \{X \sqcup Y:\ X \in \mathcal{B}/R,\ Y \in \mathcal{B}|R\}.$$

With the first argument of the polynomials of a cell the variable of its hinge and the hinge values of Definition 18.4,

$$K(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}_R^+}\big(\gamma_B^K(\alpha_B), \alpha_R\big), \qquad \Psi(\alpha) = \Psi_{\mathcal{B}|B}(\alpha_B)\, \Psi_{\mathcal{B}_R^+}\big(\gamma_B^\Psi(\alpha_B), \alpha_R\big).$$

Conversely, if $K(\alpha) = \sum_{l=0}^k A_l(\alpha_B)\, C_l(\alpha_R)$ with every C_l homogeneous, then $c(B) \leq k$; in particular $K(\alpha) = A(\alpha_B)\big(C_0(\alpha_R) + G(\alpha_B)\, C_1(\alpha_R)\big)$ with C_0 and C_1 homogeneous gives $c(B) \leq 1$.

The deletions and the contractions of the hinge are the four minors $\mathcal{B}_B^+ \setminus g = \mathcal{B}|B$ and $\mathcal{B}_B^+/g = \mathcal{B}/R$, $\mathcal{B}_R^+ \setminus g = \mathcal{B}|R$ and $\mathcal{B}_R^+/g = \mathcal{B}/B$ of the table. So the hinge values of B are those of B in its own cell, and across a bunch of width one the bunch and the room see each other as one mark: the room reads the hinge at $\alpha_g = \gamma_B^K(\alpha_B)$ in K and at $\alpha_g = \gamma_B^\Psi(\alpha_B)$ in Ψ.

Corollary 18.7 (The scale of a bunch of width one is that of one mark)**.** *Along the scale of a bunch B of width one of a table whose weights factor across B and R, each Foreman polynomial has one Skeleton value, the room's hinge value over the bunch's:*

$$\operatorname{spec}_B^K(\alpha) = \Big\{ \frac{\gamma_R^K(\alpha_R)}{\gamma_B^K(\alpha_B)} \Big\} = \Big\{ \frac{K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}|R}(\alpha_R)}{K_{\mathcal{B}|B}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)} \Big\},$$
$$\operatorname{spec}_B^\Psi(\alpha) = \Big\{ \frac{\gamma_R^\Psi(\alpha_R)}{\gamma_B^\Psi(\alpha_B)} \Big\} = \Big\{ \frac{\Psi_{\mathcal{B}|B}(\alpha_B)\, \Psi_{\mathcal{B}/B}(\alpha_R)}{\Psi_{\mathcal{B}/R}(\alpha_B)\, \Psi_{\mathcal{B}|R}(\alpha_R)} \Big\},$$

the Skeleton value of the hinge as a one-mark bunch of the cell of the room (Corollary 15.3), read at $\alpha_g = \gamma_B^K(\alpha_B)$, respectively at $\alpha_g = \gamma_B^\Psi(\alpha_B)$. The value of K over that of Ψ is $\varrho_B(\alpha_B)/\varrho_R(\alpha_R)$, the even letters of the minors of the level $d(B)$ over those of the level $r(B)$, and their product is the same quotient of the odd letters.

18.3 Pooling

Theorem 18.8 (Pooling integrates the shape and keeps the scale)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family and B a bunch of width one in every table, the weights of every table factoring across B and R, and read $\alpha_B = \rho u$, $\rho > 0$, $u \in \Delta_B$ (Lemma 2.7). For $u \in \Delta_B$ the* pooled family *$F_{B\to g}(u)$ is the family on $R + g$ with, for every j, two tables with the bases of the cell of the room $\mathcal{B}_{j,R}^+$ (Theorem 18.6): the first with its weights multiplied by $\gamma_{j,B}^K(u)$ on its bases through g and the exponents $(s_j, 0)$, the second with its weights multiplied by $\gamma_{j,B}^\Psi(u)$ on its bases avoiding g and the exponents $(0, t_j)$. Its letters are $K_{\mathcal{B}_{j,R}^+}\big(\gamma_{j,B}^K(u)\alpha_g, \alpha_R\big)^{-s_j}$ and $\Psi_{\mathcal{B}_{j,R}^+}\big(\gamma_{j,B}^\Psi(u)\alpha_g, \alpha_R\big)^{-t_j}$; listing the N first tables and then the N second ones, its exponents are $(\mathbf{s}, \mathbf{0})$ and $(\mathbf{0}, \mathbf{t})$. To* pool *B is to pass from F to its pooled families $F_{B\to g}(u)$, keeping the cell kernel of B (Definition 21.2) and the hinge values of every table. Give the hinge the mass κ_B, the wall of B. Then $F_{B\to g}(u)$ has the imbalance of F, and*

$$J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j} \Psi_{\mathcal{B}_j|B}(u)^{-t_j} \times J_{F_{B\to g}(u)}\big((\kappa_B, \mathbf{m}|_R); (\mathbf{s}, \mathbf{0}), (\mathbf{0}, \mathbf{t})\big)\, d\omega_B(u),$$

both sides converging together; the same holds with P for J, on the balance wall as well.

The port (Theorem 15.15) integrates the scale and the shape of a bunch together; pooling integrates the shape of a bunch of width one and keeps its scale, which becomes the hinge of the room. For a list the pooled families again have weights that factor across every bunch of R and across every bunch containing g, so poolings can be repeated. Proposition 17.3 is the same split with the scale integral left whole. Read backwards, pooling is gluing: the table is its cell of the bunch substituted into its cell of the room at the hinge (Proposition 20.3), and every cell of a substitution is pooled at once by Theorem 21.3.

18.4 Walls glue

Theorem 18.9 (Walls glue)**.** *Let a bunch B of a table $(\mathcal{B},\theta)$ have width one, the weights factoring across B and R, with the cells $\mathcal{B}_B^+$ and $\mathcal{B}_R^+$ of Theorem 18.6. For $S_B\subseteq B$ and $S_R\subseteq R$ the counts of the table glue from those of the cells, the hinge counted on exactly one side, as in the gluing of the bases:*

$$d(S_B\sqcup S_R)=\min\big\{d_{\mathcal{B}_B^+}(S_B+g)+d_{\mathcal{B}_R^+}(S_R),\ d_{\mathcal{B}_B^+}(S_B)+d_{\mathcal{B}_R^+}(S_R+g)\big\},$$
$$d^*(S_B\sqcup S_R)=\min\big\{d^*_{\mathcal{B}_B^+}(S_B+g)+d^*_{\mathcal{B}_R^+}(S_R),\ d^*_{\mathcal{B}_B^+}(S_B)+d^*_{\mathcal{B}_R^+}(S_R+g)\big\}.$$

For a family $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ with masses $\mathbf{m}$, B having width one in every table and the weights of every table factoring across B and R:

(i) the walls of the pooled family $F_{B\to g}(u)$ at the masses $(\kappa_B,\mathbf{m}|_R)$ are walls of F: its wall at S_R is κ_{S_R} and its wall at S_R+g is $\kappa_{B\sqcup S_R}$;

(ii) the density $u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\Psi_{\mathcal{B}_j|B}(u)^{-t_j}$ of the cell kernel of B has at the face $u_{S_B}=0$ of Δ_B the wall of F, $m_{S_B}-\sum_j\big(s_j\,d_{\mathcal{B}_j/R}(S_B)+t_j\,d^*_{\mathcal{B}_j|B}(S_B)\big)=\kappa_{S_B}$;

(iii) on the balance wall, with the cell families $G_B=((\mathcal{B}^+_{j,B},\theta'_j);s_j,t_j)_j$ at the masses $(\mathbf{m}|_B,2\sigma-\kappa_B)$ and $G_R=((\mathcal{B}^+_{j,R},\theta''_j);s_j,t_j)_j$ at the masses $(\kappa_B,\mathbf{m}|_R)$, σ the total strength, whose walls are written $\kappa_S^{G_B}$ and $\kappa_S^{G_R}$, both cell families lie on their balance walls and every wall of a cell family is a wall of F,

$$\kappa^{G_B}_{S_B}=\kappa_{S_B},\qquad \kappa^{G_B}_{S_B+g}=\kappa_{S_B\sqcup R},\qquad \kappa^{G_R}_{S_R}=\kappa_{S_R},\qquad \kappa^{G_R}_{S_R+g}=\kappa_{B\sqcup S_R};$$

if moreover $s_j,t_j\geq 0$ and every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B, as when one letter is active, every wall of F is glued from walls of the cell families,

$$\kappa_{S_B\sqcup S_R}=\min\big\{\kappa^{G_B}_{S_B}+\kappa^{G_R}_{S_R},\ \kappa^{G_B}_{S_B+g}+\kappa^{G_R}_{S_R+g}\big\},$$

and F has every proper wall positive exactly when both cell families do.

18.5 Full leans

Theorem 18.10 (Pooling factors at one hinge value)**.** *Let $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights of every table factoring across B and R. Suppose that every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B, the factorizations being chosen so that every constant is 1: every active letter has the one hinge value γ. (A change of the factorization of table j multiplies $\gamma^K_{j,B}$ by a positive constant and $\gamma^\Psi_{j,B}$ by its inverse (Definition 18.4). So a table with one active letter can always be so normalized, and when every table is at a full lean the hypothesis is a condition across the tables alone; one table with both letters active can be so normalized exactly when ϱ_B is constant.) Then $0<\kappa_B<2\sigma$, σ the total strength (Definition 8.1), and, with the cell families $G_B=((\mathcal{B}^+_{j,B},\theta'_j);s_j,t_j)_j$ and $G_R=((\mathcal{B}^+_{j,R},\theta''_j);s_j,t_j)_j$,*

$$\Gamma^{\,2}_{\,1}\begin{bmatrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{bmatrix}P_F(\mathbf{m};\mathbf{s},\mathbf{t})=P_{G_B}\big((\mathbf{m}|_B,2\sigma-\kappa_B);\mathbf{s},\mathbf{t}\big)\,P_{G_R}\big((\kappa_B,\mathbf{m}|_R);\mathbf{s},\mathbf{t}\big),$$

both cell families on their balance walls with every proper wall positive. One table at the exponents $(0,t)$ or $(s,0)$ has one active letter, and then 2σ is t, respectively s. Corollary 21.6 is the same at every cell of a core.

When every active letter has one hinge value, as for one table at a full lean, where one letter is active, the hinge carries the mass κ_B in the room and $2\sigma - \kappa_B$ in the bunch, and the period of the family is the product of the periods of its two cell families divided by the Beta corner of the hinge.

Proposition 18.11 (Pooling a realization)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family, B a bunch of width one in every table, the weights of every table factoring across B and R, and let $\Upsilon = ((s_j, t_j)_j; \alpha)$ be a realization of F with masses $\boldsymbol{\tau}$ (Definition 9.1); write $\alpha_B = \rho u$, $\rho > 0$, $u \in \Delta_B$ (Lemma 2.7), and read the two tables of index j of the pooled family $F_{B\to g}(u)$ of Theorem 18.8 as one table with two systems of weights at the exponents (s_j, t_j) (Definition 8.1). Then $((s_j, t_j)_j; (\rho, \alpha_R))$ is a realization of $F_{B\to g}(u)$, with the masses $\big(\tau_B - \sum_j (s_j d_j(B) + t_j d_j^*(B)),\ \boldsymbol{\tau}|_R\big)$, the first at the hinge: pooling sends a realization to a realization, and the mass of the hinge is the wall κ_B at the masses $\boldsymbol{\tau}$. With one table at the exponents $(0, t)$ the letter Ψ of $F_{B\to g}(u)$ at (ρ, α_R) is that of the cell of the room $(\mathcal{B}_R^+, \theta'')$ at $(\gamma_B^\Psi(\alpha_B), \alpha_R)$, and $((0, t); (\gamma_B^\Psi(\alpha_B), \alpha_R))$ is a realization of $(\mathcal{B}_R^+, \theta'')$ with the masses $(\tau_B - t\, d^*(B),\ \boldsymbol{\tau}|_R)$.*

Proposition 18.12 (Hinge values that agree)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights of every table factoring across B and R. Suppose that every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B: $\gamma_{j,B}^K = c_j \gamma$ whenever $s_j \neq 0$ and $\gamma_{j,B}^\Psi = c_j^* \gamma$ whenever $t_j \neq 0$, with constants $c_j, c_j^* > 0$; put $c_j = 1$ when $s_j = 0$ and $c_j^* = 1$ when $t_j = 0$. Let $G_R^{\mathbf{c}}$ be the pooled family of Theorem 18.8 with these constants in place of the hinge values: for every j the table $\mathcal{B}_{j,R}^+$ with its weights multiplied by c_j on its bases through g, at the exponents $(s_j, 0)$, and with its weights multiplied by c_j^* on its bases avoiding g, at the exponents $(0, t_j)$. Then*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = P_{G_R^{\mathbf{c}}}\big((\kappa_B, \mathbf{m}|_R); (\mathbf{s}, \mathbf{0}), (\mathbf{0}, \mathbf{t})\big) \times \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j} \Psi_{\mathcal{B}_j|B}(u)^{-t_j}\, \gamma(u)^{-\kappa_B}\, d\omega_B(u) :$$

the shape and the room separate at every pair of exponents. It is Theorem 21.5 at the one cell $\mathcal{B}_B^+$ of the gluing of Proposition 20.3(iii). For one table with ϱ_B constant, a factorization with $\varrho_B = 1$ (Definition 18.4) gives $c = c^* = 1$ and $\gamma = \gamma_B^K$, and $G_R^{\mathbf{c}}$ is then $((\mathcal{B}_R^+, \theta''); s, t)$.

18.6 Partial leans

Theorem 18.13 (Gluing across a bunch of width one)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real data with $s_j, t_j \geq 0$, $s_j + t_j > 0$, and masses $\mathbf{m}$ on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights factoring across B and R. In table j call $d_j(B)$ the* level d *and $r_j(B) = d_j(B) + 1$ the* level r*.* Level shares $\mathbf{a} = (a_j^d, a_j^r)_j$ *and* $\mathbf{b} = (b_j^d, b_j^r)_j$ *split the exponents over the two levels, $a_j^d + a_j^r = s_j$ and $b_j^d + b_j^r = t_j$, a letter of exponent zero being opened by nothing, its two level shares 0; they are the shares of Definition 12.21 summed over the patterns of each level, and their pair shares (Definition 12.21) are the* even share $\sigma_j^\ell = \frac{1}{2}(a_j^\ell + b_j^\ell)$ *and the* odd share $\delta_j^\ell = \frac{1}{2}(a_j^\ell - b_j^\ell)$*, $\ell \in \{d, r\}$. The* bunch family *and the* room family *are*

$$F^{\text{bunch}}(\mathbf{a}, \mathbf{b}) = \big((\mathcal{B}_j/R;\, a_j^d, b_j^d),\ (\mathcal{B}_j|B;\, a_j^r, b_j^r)\big)_j \quad \text{on } B,$$
$$F^{\text{room}}(\mathbf{a}, \mathbf{b}) = \big((\mathcal{B}_j|R;\, a_j^d, b_j^d),\ (\mathcal{B}_j/B;\, a_j^r, b_j^r)\big)_j \quad \text{on } R :$$

each level carries one table of the bunch and one of the room, at the same shares. The scale of B pins the shares of the minors it multiplies,

$$\sum_j \big(a_j^r + b_j^d\big) = \kappa_B, \qquad \textit{equivalently} \qquad \sum_j \delta_j^d = \tfrac{1}{2}\Big(\sum_j s_j - \kappa_B\Big),$$

and, for pattern shares with these level sums, equivalently $\sum_{e\in B}\langle\boldsymbol{\delta}\rangle_e = \mu_B$, the sum over the bunch of the pins $\langle\boldsymbol{\delta}\rangle_e = \mu_e$ of a wall contour (Definition 17.11). On this pin both families lie on their balance walls at $\mathbf{m}|_B$ and $\mathbf{m}|_R$, and on every contour on which the stacked Gammas below have arguments of positive real part and both periods converge absolutely,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = 2^{N-1}\int_{\sum_j \delta_j^d = \frac{1}{2}(\sum_j s_j - \kappa_B)} \prod_j \Gamma_{\substack{4\\2}}\!\begin{bmatrix} a_j^d,\ b_j^d,\ a_j^r,\ b_j^r \\ s_j,\ t_j \end{bmatrix} \times P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

the fibre integral (Definition 4.1) in the pair coordinates $(\sigma_j^d,\delta_j^d)_j$, with $\sigma_j^r = \sigma_j - \sigma_j^d$ and $\delta_j^r = \delta_j - \delta_j^d$: its dimension is $2N-1$, and its N even shares σ_j^d are free. This display needs every s_j and t_j positive; in general, with p the number of active letters,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\sum_j (a_j^r + b_j^d) = \kappa_B} \prod_{s_j\neq 0} \Gamma_{\substack{2\\1}}\!\begin{bmatrix} a_j^d,\ a_j^r \\ s_j \end{bmatrix} \prod_{t_j\neq 0} \Gamma_{\substack{2\\1}}\!\begin{bmatrix} b_j^d,\ b_j^r \\ t_j \end{bmatrix} \times P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

the share integral (Definition 4.1) over the shares a_j^r, $s_j \neq 0$, and b_j^d, $t_j \neq 0$, of the minors that the scale of B multiplies, pinned by their sum κ_B, of dimension $p-1$; with every exponent positive it is the display above, and at $p=1$ it is the value at the one admissible point. For one table with $s,t>0$ the odd share is fixed, $2\delta^d = s - \kappa_B = \delta\,(1+\varepsilon(B)) - \mu_B$, and the even share alone is free: on a line $\operatorname{Re}\sigma^d = \sigma_*^d$ of such a contour,

$$P_F(\mathbf{m};s,t) = \frac{1}{2\pi i}\int_{\operatorname{Re}\sigma^d = \sigma_*^d} \Gamma_{\substack{4\\2}}\!\begin{bmatrix} a^d,\ b^d,\ a^r,\ b^r \\ s,\ t \end{bmatrix} P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})\, d\sigma^d.$$

Swapping B and R swaps the two levels and leaves the formula unchanged.

The scale of the bunch is opened letter by letter, one share on each level of each letter; the scale integrates to the pin, and what is left is the form of the bunch family on its simplex times the form of the room family on its own. The level shares are the pattern shares of the port (Theorem 15.15) summed over each level: at width one the children of the level d are all $\mathcal{B}_j|R$ and those of the level r all $\mathcal{B}_j/B$, up to constant factors of their weights (Theorem 12.17), so the room family is the children of the port with equal children merged. At the level shares $(a_j^d, b_j^d, a_j^r, b_j^r) = (s_j, 0, 0, t_j)$, on the wall $\kappa_B = 0$, the two families are the faces of Theorem 16.23. Left unopened on the room, as in Theorem 18.8, the shape enters the pooled family only through the hinge values of B by its letters, and a rescaling of the hinge variable absorbs one of them: the shares that separate the shape from the room are one fewer than the hinge values of B by the active letters counted up to constant factors, none when every active letter reads B alike (Proposition 18.12), as for one table at a full lean (Theorem 18.10).

Corollary 18.14 (One Skeleton value pins the share)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch B have width one in every table, the weights factoring across B and R. Suppose that every active letter reads B alike, its hinge value a constant multiple of one function γ on Δ_B, and reads R alike with the same constant, its hinge value of R that constant times one function γ_R on Δ_R: the active letters have the one Skeleton value $\gamma_R(v)/\gamma(u)$ on the scale of B at every gauge $(u, v) \in \Delta_B \times \Delta_R$ (Corollary 18.7). Then $0 < \kappa_B < 2\sigma$, σ the total strength, and for all level shares $(\mathbf{a}, \mathbf{b})$ of Theorem 18.13, those of the inactive letters 0, with $a_j^d + a_j^r = s_j$, $b_j^d + b_j^r = t_j$ and $\sum_j (a_j^r + b_j^d) = \kappa_B$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa_B, & 2\sigma - \kappa_B \\ & 2\sigma\end{bmatrix} P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

the product of the two periods not depending on the level shares: the Beta corner of the hinge times the periods of the bunch family and the room family. One active letter has one Skeleton value: for one table at $(0, t)$, at the level shares $(a^d, b^d, a^r, b^r) = (0, \kappa_B, 0, t - \kappa_B)$, the two periods are those of the forms $u^{\mathbf{m}|_B - \mathbf{1}} \Psi_{\mathcal{B}/R}^{-\kappa_B} \Psi_{\mathcal{B}|B}^{-(t-\kappa_B)}$ on Δ_B and $v^{\mathbf{m}|_R - \mathbf{1}} \Psi_{\mathcal{B}|R}^{-\kappa_B} \Psi_{\mathcal{B}/B}^{-(t-\kappa_B)}$ on Δ_R, and at $(s, 0)$, at the level shares $(s - \kappa_B, 0, \kappa_B, 0)$, the same holds with the letters of K.

Lemma 18.15 (Slopes and the range of the ratios)**.** *In Theorem 18.13 with every s_j and t_j positive, the even share σ_j^d of table j enters the two families only through $\big(\varrho_{j,B}(u)/\varrho_{j,R}(v)\big)^{-\sigma_j^d}$, $(u, v) \in \Delta_B \times \Delta_R$, the ratio of the hinge values of the bunch over that of the room in table j (Definition 18.4). The wall at S of either family is affine in the shares, with the* slope *in σ_j^d the width of S in the table of the level d of table j less its width in the table of the level r of table j: with r_j^+ the rank function of the cell of table j on the side of S,*

$$\begin{aligned} &\big[r_j^+(S + g) - r_j^+(S)\big] + \big[r_j^+(B \setminus S + g) - r_j^+(B \setminus S)\big] - 1 \qquad (S \subseteq B), \\ &1 - \big[r_j^+(S + g) - r_j^+(S)\big] - \big[r_j^+(R \setminus S + g) - r_j^+(R \setminus S)\big] \qquad (S \subseteq R). \end{aligned}$$

$\sup_{\Delta_B} \varrho_{j,B} < \infty$ exactly when no slope of table j on B is positive and $\inf_{\Delta_B} \varrho_{j,B} > 0$ exactly when none on B is negative; $\inf_{\Delta_R} \varrho_{j,R} > 0$ exactly when no slope of table j on R is positive and $\sup_{\Delta_R} \varrho_{j,R} < \infty$ exactly when none on R is negative. So a wall of either family falls as σ_j^d moves toward the level d only if $\sup \varrho_{j,B} / \inf \varrho_{j,R} = \infty$, and as it moves toward the level r only if $\inf \varrho_{j,B} / \sup \varrho_{j,R} = 0$. These conditions, like the quotient $\varrho_{j,B}/\varrho_{j,R}$, do not depend on the factorization. For one table the index j is omitted.

A wall that falls as the even share moves toward the level being closed on is therefore the same thing as the two levels coming to overlap at that wall (Proposition 18.25): the quotient ϱ_B/ϱ_R of the even letters of the two levels, below one where the closing starts, is unbounded near the face where the wall sits.

Corollary 18.16 (Closing the even share)**.** *For one table in Theorem 18.13 let $f(\sigma^d)$ be the integrand of its one-table display.*

(i) If $\sup_{\Delta_B} \varrho_B < \inf_{\Delta_R} \varrho_R$, then $P_F(\mathbf{m}; s, t) = \sum \mathrm{Res}\, f$ over the poles of the Gamma pair $\Gamma(a^d)\Gamma(b^d)$ of the level d, at $\sigma^d = \delta^d - n$ and $\sigma^d = -\delta^d - n$, $n \in \mathbb{N}$, the n-th terms of the size $n^{s+t-2}\big(\sup \varrho_B / \inf \varrho_R\big)^n$; every period at these points converges, and when $2\delta^d \notin \mathbb{Z}$ the n-th terms are at most $C(1 + n)^D \big(\sup \varrho_B / \inf \varrho_R\big)^n P_F(\mathbf{m}; s, t)$, with C and D depending only on that ratio, the exponents and κ_B (Theorem 18.21(i),(iii) with one table, its letters listed as K, then Ψ). If $\inf_{\Delta_B} \varrho_B > \sup_{\Delta_R} \varrho_R$, then $P_F(\mathbf{m}; s, t) = -\sum \mathrm{Res}\, f$ over the poles of the Gamma pair of the level r, at $\sigma^r = \pm\delta^r - n$. The share closes toward the level whose even letter is the smaller.

(ii) At $\sigma^d = \delta^d - n$ the level shares are $(a^d, b^d, a^r, b^r) = (2\delta^d - n,\ -n,\ s - 2\delta^d + n,\ t + n)$, and the residue is

$$\frac{(-1)^n}{n!}\Gamma^{\,3}_{\,2}\begin{bmatrix} 2\delta^d - n, & s - 2\delta^d + n, & t + n \\ & s, & t \end{bmatrix} P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

where $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})} = \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}/R}^{-2\delta^d} K_{\mathcal{B}|B}^{-(s-2\delta^d)} \Psi_{\mathcal{B}|B}^{-t}\, \varrho_B^n\, d\omega_B$ is the n-th moment of ϱ_B against a form that does not depend on n, and $P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}$ likewise the n-th moment of ϱ_R^{-1}. At $\sigma^d = -\delta^d - n$ the level shares are $(-n,\ -2\delta^d - n,\ s + n,\ t + 2\delta^d + n)$, and the same holds with the form $\Psi_{\mathcal{B}/R}^{2\delta^d} K_{\mathcal{B}|B}^{-s} \Psi_{\mathcal{B}|B}^{-t-2\delta^d}$ on B and its counterpart on R. Closing pairs the moments of the ratios ϱ_B and ϱ_R of the bunch and of the room. The two lattices of a level meet exactly when $2\delta^\ell \in \mathbb{Z}$, at the centre exactly when $\mu_B \in \mathbb{Z}$. At a common point, where $\Gamma(a^d)$ sits at $-k$ and $\Gamma(b^d)$ at $-n$, the pole is double: with $h = \Gamma^{\,2}_{\,2}\begin{bmatrix} s + k, & t + n \\ s, & t \end{bmatrix} P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})} P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}$,

$$\operatorname{Res} f = \frac{(-1)^{n+k}}{n!\,k!}\, h \Big((\log\Gamma)'(n+1) - (\log\Gamma)'(t+n) + (\log\Gamma)'(k+1) - (\log\Gamma)'(s+k) + \partial_{\sigma^d} \log\left(P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})} P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})} \right) \Big).$$

At integer exponents the two differences of $(\log\Gamma)'$ are heads, finite sums of reciprocals (Proposition 5.13), and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ for $j \in \mathbb{N}$ (Proposition 5.14).

Off resonance every pole is simple and the closed value is a sum over two lattices of products of moments of the two ratios ϱ_B and ϱ_R, with stacked Gamma coefficients: no heads. Each lattice's series has a pole where the two lattices of the level meet, from $\Gamma(2\delta^d - n)$ and $\Gamma(-2\delta^d - n)$, and their sum tends to the double-pole residues of (ii); the heads at the centre are this meeting. When the ranges of the even letters of the two levels overlap, the quotient $\varrho_B(u)/\varrho_R(v)$ taking values on both sides of 1, neither closing of (i) applies; the residues of a family at one of its walls are those of Theorem 16.23, periods of the family's faces.

18.7 Closing at a wall and closing at the gauge

A share between the levels of a letter can be closed in two places. *At the gauge* every letter is expanded about its value at a point of the scale; the terms are nonnegative and nothing is asked of the Skeleton values. That closing is Theorem 15.9, and it closes every letter on the scale at once. *At a wall* a letter is expanded about one of its two extreme coefficients, the other coefficients entering as integer powers; the terms then separate over the bunch and the room, and the expansion holds as far as the letters are separated along the scale. The closing at a wall holds for any number of letters and any width; across a bunch of width one it is a lattice of products of periods of the bunch family and the room family, and it stops where the values no longer stand in one order, which for one table is where the two levels overlap (Proposition 18.25).

Definition 18.17 (Letters on the scale)**.** Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real masses and real exponents of either sign, let B be a proper bunch with room R, and read $\alpha_B = \rho u$ with $u \in \Delta_B$. At the gauge (u, α_R) the *letters on the scale* of B are the $2N$ polynomials in ρ

$$\rho^{-d_j(B)} K_j(\rho u, \alpha_R) = \sum_{l=0}^{c_j(B)} \rho^l\, K_{j,B}^{(d_j(B)+l)}(u, \alpha_R), \qquad \rho^{-d_j^*(B)} \Psi_j(\rho u, \alpha_R) = \sum_{l=0}^{c_j(B)} \rho^l\, \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R),$$

with the letter exponents s_j and t_j: their coefficients are the level and colevel letters of Definition 12.15 counted from their floors, and divided by their values at $\rho = 1$ they are the Skeleton polynomials $\mathsf{Q}^K_{j,B}(\rho;(u,\alpha_R))$ and $\mathsf{Q}^\Psi_{j,B}(\rho;(u,\alpha_R))$ (Definition 12.9). Listed as $L_i(\rho) = \sum_{l=0}^{c_i} \rho^l L_i^{(l)}$, $i = 1,\dots,2N$, with the letter exponent w_i and the width $c_i = c_j(B)$ of their table, every coefficient is positive (Theorem 12.6), a letter of width zero is constant, and the scale integral of Theorem 15.2 at the gauge (u,α_R) is

$$\mathrm{Sc}_B(u,\alpha_R;\mathbf{m};\mathbf{s},\mathbf{t}) = \int_0^\infty \rho^{\kappa_B-1} \prod_i L_i(\rho)^{-w_i}\, d\rho,$$

convergent exactly when $0 < \kappa_B < \sum_i w_i c_i$, where $\sum_i w_i c_i = \kappa_B + \kappa_B^{\mathrm{top}}$.

Definition 18.18 (Letters read from an end of the scale; separated groups)**.** Fix $\lambda \in (0,1)$. A letter $L(\rho) = \sum_{l=0}^{c} \rho^l L^{(l)}$ with positive coefficients is *read from the bottom up to* a when $\sum_{l\ge 1} a^l L^{(l)} \le \lambda\, L^{(0)}$, and *read from the top down to* a when $\sum_{l<c} a^l L^{(l)} \le \lambda\, a^c L^{(c)}$. A partition of the letters on the scale of B into *groups* $\mathcal{G}_1,\dots,\mathcal{G}_q$ is *separated* by the scales $0 < a_1 < \cdots < a_{q-1}$ when every letter of $\mathcal{G}_k$ is read from the bottom up to a_{k-1} (for $k \ge 2$) and from the top down to a_k (for $k \le q-1$); a letter whose exponent is a nonpositive integer is a polynomial factor of the integrand, may be put in any group, and nothing is asked of it. A group $\mathcal{G}$ has the product $f_{\mathcal{G}} = \prod_{i\in\mathcal{G}} L_i^{-w_i}$, the *scale degree* $w_{\mathcal{G}} = \sum_{i\in\mathcal{G}} w_i c_i$, and the *bottom* and *top coefficients* $\beta^{\downarrow}_{\mathcal{G},n}$ and $\beta^{\uparrow}_{\mathcal{G},n}$,

$$f_{\mathcal{G}}(\rho) = \sum_{n\ge 0} \beta^{\downarrow}_{\mathcal{G},n}\, \rho^n \quad (\rho \text{ small}), \qquad f_{\mathcal{G}}(\rho) = \rho^{-w_{\mathcal{G}}} \sum_{n\ge 0} \beta^{\uparrow}_{\mathcal{G},n}\, \rho^{-n} \quad (\rho \text{ large}).$$

Below the scale a a letter read from the bottom is its lowest coefficient up to the factor $1+\lambda$, and above a a letter read from the top is its top term $\rho^c L^{(c)}$ up to the same factor. At width one, $L(\rho) = L^{(0)} + \rho L^{(1)}$ has the single Skeleton value $\xi = L^{(0)}/L^{(1)}$ (Corollary 18.7); it is read from the bottom up to a exactly when $a \le \lambda\xi$ and from the top down to a exactly when $a \ge \xi/\lambda$. So single letters of width one with the values $\xi_1,\dots,\xi_q$ are separated, by $a_k = \sqrt{\xi_k \xi_{k+1}}$, exactly when $\xi_k \le \lambda^2 \xi_{k+1}$ for every k: their values stand in one order with room between them.

Theorem 18.19 (The scale closes between separated groups)**.** *At the gauge (u,α_R) let the letters on the scale of B be separated into groups $\mathcal{G}_1,\dots,\mathcal{G}_q$, let $0 < \kappa < \sum_i w_i c_i$, and suppose $\kappa - \sum_{k'\le k} w_{\mathcal{G}_{k'}} \notin \mathbb{Z}$ for $1 \le k < q$. For $1 \le k \le q$ and $\mathbf{n} = (n_{k'})_{k'\ne k} \in \mathbb{N}^{q-1}$ put*

$$\nu_k(\mathbf{n}) = \kappa - \sum_{k'<k} \big(w_{\mathcal{G}_{k'}} + n_{k'}\big) + \sum_{k'>k} n_{k'}.$$

(i)

$$\int_0^\infty \rho^{\kappa-1} \prod_i L_i(\rho)^{-w_i}\, d\rho = \sum_{k=1}^{q} \sum_{\mathbf{n}} \prod_{k'<k} \beta^{\uparrow}_{\mathcal{G}_{k'},n_{k'}} \prod_{k'>k} \beta^{\downarrow}_{\mathcal{G}_{k'},n_{k'}}\ \mathrm{Reg}\int_0^\infty \rho^{\nu_k(\mathbf{n})-1} f_{\mathcal{G}_k}(\rho)\, d\rho;$$

at $\kappa = \kappa_B$ the left side is $\mathrm{Sc}_B(u,\alpha_R;\mathbf{m};\mathbf{s},\mathbf{t})$.

(ii) The series converges absolutely, and the sum of the absolute values of its terms is at most C times the left side, where C depends only on λ, the letter exponents, the widths, κ and the distances of the numbers $\kappa - \sum_{k'\le k} w_{\mathcal{G}_{k'}}$ from $\mathbb{Z}$.

(iii) $\beta^{\downarrow}_{\mathcal{G},n} = \prod_{i\in\mathcal{G}} \left(L_i^{(0)}\right)^{-w_i} [\rho^n] \prod_{i\in\mathcal{G}} \left(1+\sum_{l\geq 1} \rho^l L_i^{(l)}/L_i^{(0)}\right)^{-w_i}$*: the bottom coefficients of a group are the integrand of Theorem 16.23 for the residue at $\kappa_B = -n$, taken for the letters of the group alone; the top coefficients are the same with the coefficients $L_i^{(c_i)}$ in place of $L_i^{(0)}$, for the residue at $\kappa_B^{\mathrm{top}} = -n$.*

Each group in turn keeps the scale. The groups that lie above it on the scale are read from the bottom, the groups below it from the top, and what is left to integrate is the keeping group alone, at an index shifted by the integers of the others, and as a finite part when that index has left the interval $(0, w_{\mathcal{G}})$ in which the group's own integral converges. The coefficients are the wall residues the family already has: for each letter read, a power of its extreme coefficient times a polynomial in its other coefficients. Every coefficient of K_j is a sum of the pattern terms $u^p K_{j,(p)}(\alpha_R)$ and every coefficient of Ψ_j of the terms $u^{B\setminus p}\Psi_{j,(p)}(\alpha_R)$, so the children enter with their pattern index, as in the record of a peel (Definition 14.2).

Corollary 18.20 (Two groups: the walls of one against the continuation of the other)**.** *Let the letters be separated into a lower group $\mathcal{L}$ and an upper group $\mathcal{U}$, with $0 < \kappa < \sum_i w_i c_i$ and $\kappa - w_{\mathcal{L}} \notin \mathbb{Z}$. Then*

$$\int_0^\infty \rho^{\kappa-1} f_{\mathcal{L}} f_{\mathcal{U}}\, d\rho = \sum_{n\geq 0} \beta^{\downarrow}_{\mathcal{U},n}\, \mathrm{Reg}\int_0^\infty \rho^{\kappa+n-1} f_{\mathcal{L}}\, d\rho + \sum_{n\geq 0} \beta^{\uparrow}_{\mathcal{L},n}\, \mathrm{Reg}\int_0^\infty \rho^{\kappa-w_{\mathcal{L}}-n-1} f_{\mathcal{U}}\, d\rho.$$

For any group, $\mathrm{Reg}\int_0^\infty \rho^{\nu-1} f_{\mathcal{G}}\, d\rho$ is the value at ν of a meromorphic function, and at a pole of that function its constant term there (Theorem 1.6); the poles are at most simple and lie among the points $\nu = -n$, of residue $\beta^{\downarrow}_{\mathcal{G},n}$, and $\nu = w_{\mathcal{G}} + n$, of residue $-\beta^{\uparrow}_{\mathcal{G},n}$, $n \in \mathbb{N}$; the two families of points meet only when $w_{\mathcal{G}} \in -\mathbb{N}$, and at a common point the two residues add.

The scale integral of a product closes on the walls of each factor. The residues of the upper group at the wall of B are paired with the scale integral of the lower group continued past its top, and the residues of the lower group at the top wall with the scale integral of the upper group continued past its bottom. No bound on a moved contour is used: both expansions hold on the stretch of the scale between the groups, and the scale is divided there.

Theorem 18.21 (Joint closing across a bunch of width one)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real data, its masses on the balance wall with $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ convergent, and let a bunch B have width at most one in every table, the weights factoring across B and R. A table in which B has width zero is the direct sum of its restrictions to B and to R (Proposition 7.8); its two letters on the scale are constants, and it enters the bunch family and the room family of Theorem 18.13 as those restrictions with its own exponents. For the N' tables in which B has width one the letters on the scale (Definition 18.17) are, on $\Delta_B \times \Delta_R$,*

$$K_{\mathcal{B}_j/R}(u) K_{\mathcal{B}_j|R}(v) + \rho\, K_{\mathcal{B}_j|B}(u) K_{\mathcal{B}_j/B}(v), \qquad \Psi_{\mathcal{B}_j|B}(u)\Psi_{\mathcal{B}_j/B}(v) + \rho\, \Psi_{\mathcal{B}_j/R}(u)\Psi_{\mathcal{B}_j|R}(v),$$

of the letter exponents s_j and t_j, with the Skeleton values of Corollary 18.7; in each the coefficient multiplying ρ is its scale coefficient*, the level letter of the level r for K_j and the colevel letter of the level d for Ψ_j, and the other its* constant coefficient*. Suppose the $2N'$ letters can be listed as $\ell = 1, \dots, 2N'$, with letter exponents w_ℓ and Skeleton values ξ_ℓ, so that for some $\vartheta < 1$*

$$\xi_\ell \leq \vartheta\, \xi_{\ell+1} \quad \text{on } \Delta_B \times \Delta_R \qquad (1 \leq \ell < 2N'),$$

and that $\kappa_B - \sum_{\ell'\leq \ell} w_{\ell'} \notin \mathbb{Z}$ for $1 \leq \ell < 2N'$. For a letter i and $\mathbf{n} = (n_\ell)_{\ell\neq i} \in \mathbb{N}^{2N'-1}$ put $\nu_i(\mathbf{n}) = \kappa_B - \sum_{\ell<i}(w_\ell + n_\ell) + \sum_{\ell>i} n_\ell$, and let $(\mathbf{a}, \mathbf{b})_{i,\mathbf{n}}$ be the level shares in which

- *a letter $\ell < i$ has its constant coefficient at $-n_\ell$ and its scale coefficient at $w_\ell + n_\ell$,*
- *a letter $\ell > i$ has its scale coefficient at $-n_\ell$ and its constant coefficient at $w_\ell + n_\ell$,*
- *the letter i has its scale coefficient at $\nu_i(\mathbf{n})$ and its constant coefficient at $w_i - \nu_i(\mathbf{n})$.*

Then:

(i) at every $(\mathbf{a},\mathbf{b})_{i,\mathbf{n}}$ the bunch family and the room family lie on their balance walls and their periods converge;

(ii) with both periods at $(\mathbf{a},\mathbf{b}) = (\mathbf{a},\mathbf{b})_{i,\mathbf{n}}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_{i=1}^{2N'} \sum_{\mathbf{n}\in\mathbb{N}^{2N'-1}} \prod_{\ell\neq i} \frac{(-1)^{n_\ell}(w_\ell)_{n_\ell}}{n_\ell!}\, \Gamma_{\substack{2\\1}}\!\begin{bmatrix}\nu_i(\mathbf{n}),\ w_i-\nu_i(\mathbf{n})\\ w_i\end{bmatrix} \times P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

where the stacked Gamma is 0 when $w_i \in -\mathbb{N}$; when no letter exponent lies in $-\mathbb{N}$ the coefficient is $\prod_{\ell\neq i}\frac{(-1)^{n_\ell}}{n_\ell!}\,\Gamma_{\substack{2N'+1\\2N'}}\!\begin{bmatrix}(w_\ell+n_\ell)_{\ell\neq i},\ \nu_i(\mathbf{n}),\ w_i-\nu_i(\mathbf{n})\\ (w_\ell)_\ell\end{bmatrix}$;

(iii) the series converges absolutely, the absolute values of all its terms add up to at most $C\,P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, and those with $|\mathbf{n}| = M$ to at most $C'(1+M)^D\vartheta^M\,P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, where C, C' and D depend only on ϑ, the exponents and κ_B.

The $2N'$ Skeleton values stand in one order on the whole shape domain, and each letter in turn keeps the scale. A letter whose value lies below the keeper's is read from the top of the scale, so its constant coefficient is opened; a letter whose value lies above is read from the bottom, so its scale coefficient is opened: each letter opens the one of its two terms that is the smaller at the keeper's place on the scale. An opened coefficient sits at the share $-n$, which puts the n-th power of that coefficient into the forms of the two families, a moment; the keeper carries the Beta corner $\Gamma_{\substack{2\\1}}\!\begin{bmatrix}\nu,\ w-\nu\\ w\end{bmatrix}$ of a linear letter (Proposition 15.1 with one coefficient), continued when ν has left $(0,w)$. So closing across the bunch pairs moments of the bunch family with moments of the room family on $2N'$ lattices of dimension $2N'-1$, the free shares of Theorem 18.13 closed at once. Nothing is asked of the walls of the two families, (i) being a conclusion, and nothing uniform in $\mathbf{n}$ is needed beyond (iii), which bounds the whole lattice by the period itself.

The values are quotients of hinge values, the room's over the bunch's by each letter (Corollary 18.7). The value of K_j over that of Ψ_j is $\varrho_{j,B}(u)/\varrho_{j,R}(v)$, the even letters of its two levels, and their product is the corresponding quotient of the odd letters: the two values of a table sit on either side of the geometric mean set by its odd letters, at the distance set by its even letters. The N' quotients of even letters and the $N'-1$ quotients of the products of one table by those of another are what the N' even shares and the $N'-1$ free odd shares of Theorem 18.13 carry; for one table only ϱ_B/ϱ_R is left.

Corollary 18.22 (Joint closing, nested)**.** *In Theorem 18.21 let B' be a bunch of the bunch family, of width at most one in each of its tables, with factoring weights, and let the letters of the bunch family on the scale of B' have their Skeleton values in one order with a ratio $\vartheta' < 1$. Then at every $(\mathbf{a},\mathbf{b})_{i,\mathbf{n}}$ at which the non-integrality condition of Theorem 18.21 holds for the bunch family at B', $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ is given by Theorem 18.21 for the family $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ at B'; likewise for the room family at a bunch of R. Continuing down a tree of splits of width one, each node splitting its family at a bunch of width one into the bunch family and the room family, with the*

values ordered at every node, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is a nested lattice sum of stacked Gammas and of the periods of the families at the leaves, each leaf at exponents shifted by the integers of the lattices above it. The sum is taken from the leaves up, and at each node it converges absolutely at that node's own root rate.

The tree of the nesting is a tree of splits of width one and not the Foreman tree of Definition 14.4: a node may be any bunch of width one at which the values are ordered. At a lattice point of the node above, a letter of the family below sits at an exponent $-n$ or $w+n$; the letters below are the same letters at shifted powers, and the next split applies to them because whether the values are in one order is read off the tables. Nothing uniform in the outer integers is used: at each lattice point the inner sum is an identity. The constant C of Theorem 18.21(iii) at an inner node depends on the exponents there and grows at most geometrically in the outer integers; that matters only for summing in another order. The multiple series converges absolutely when the outer ratio ϑ beats that growth, and otherwise the sum is the iterated one.

Proposition 18.23 (Values in one order without room)**.** *In Theorem 18.21 let $\xi_\ell < \xi_{\ell+1}$ on $\Delta_B \times \Delta_R$ for every ℓ, with no $\vartheta < 1$. For $0 < \zeta < 1$ let $P^{(\zeta)}$ be the integral obtained from $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ by multiplying the constant coefficient of the ℓ-th letter by $\zeta^{-\ell}$; it is the period of the family in which K_j and Ψ_j are read from two copies of table j whose level weights are scaled accordingly. Then every period of Theorem 18.21(i) converges, and with $A_{i,\mathbf{n}}$ the term of index $(i,\mathbf{n})$ of the series of Theorem 18.21(ii) and $x_i = \sum_{\ell>i} \ell w_\ell + i\sum_{\ell\le i} w_\ell - i\kappa_B$,*

$$P^{(\zeta)} = \sum_{i,\mathbf{n}} \zeta^{\,x_i + \sum_\ell |\ell - i|\, n_\ell} A_{i,\mathbf{n}} \quad \textit{absolutely,} \qquad P^{(\zeta)} \to P_F(\mathbf{m};\mathbf{s},\mathbf{t}) \ \ (\zeta \uparrow 1).$$

So the lattice sum of Theorem 18.21(ii) is summable to the period by scaling the level weights, and equals the period whenever it converges absolutely.

Proposition 18.24 (The same shares closed at the gauge)**.** *Let $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ have complex exponents and masses $\mathbf{m}$ and let a bunch B have width one in every table, the weights factoring across B and R, with $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa_B^{\mathrm{top}} > 0$. Write the Skeleton polynomials at the gauge $(u,v) \in \Delta_B \times \Delta_R$ as*

$$\mathsf{Q}^K_{j,B}\big(x;(u,v)\big) = \pi_{j,d} + \pi_{j,r}\,x, \qquad \mathsf{Q}^\Psi_{j,B}\big(x;(u,v)\big) = \pi^*_{j,r} + \pi^*_{j,d}\,x,$$

the chances of the two levels under the basis law and the cobasis law at (u,v), each the sum over its level of the pattern probabilities of Definition 12.22. Then, with $M = |\mathbf{n}| + |\mathbf{n}'|$,

$$\operatorname{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(u,v)^{-s_j}\Psi_j(u,v)^{-t_j} \sum_{\mathbf{n},\mathbf{n}'\in\mathbb{N}^N} \prod_j \frac{(s_j)_{n_j}(t_j)_{n'_j}}{n_j!\,n'_j!}$$
$$\times \Big(\prod_j \pi_{j,r}^{n_j}\pi_{j,d}^{*\,n'_j}\ {}_2\Gamma_1\!\left[\begin{matrix}\kappa_B,\ M+1\\ \kappa_B+M+1\end{matrix}\right] + \prod_j \pi_{j,d}^{n_j}\pi_{j,r}^{*\,n'_j}\ {}_2\Gamma_1\!\left[\begin{matrix}\kappa_B^{\mathrm{top}},\ M+1\\ \kappa_B^{\mathrm{top}}+M+1\end{matrix}\right] \Big),$$

a series converging absolutely, of nonnegative terms when the data are real with $s_j, t_j \ge 0$, whatever the order of the Skeleton values: it is Theorem 15.9 at width one, whose exceedance polynomials are the constants $\pi_{j,r}$, $\pi^*_{j,d}$ on the bunch and $\pi_{j,d}$, $\pi^*_{j,r}$ on the room. Without the prefactor the terms with M fixed add up in absolute value to at most $\frac{(\sum_j(|s_j|+|t_j|))_M}{M!}\big(\lambda_B^M/\operatorname{Re}\kappa_B + \lambda_R^M/\operatorname{Re}\kappa_B^{\mathrm{top}}\big)$, with $\lambda_B = \max_j(\pi_{j,r},\pi^*_{j,d}) < 1$ and $\lambda_R = \max_j(\pi_{j,d},\pi^*_{j,r}) < 1$.

At the gauge every letter is closed about its value at one point of the scale, both levels together. The lattice has one index for each letter, and what is raised to the powers are the chances of the levels at that point. Nothing is asked of the order of the values, and the root rate is the largest chance of a level. The price is that the closed letters $K_j(u,v)$ and $\Psi_j(u,v)$ stay in front, so the terms do not separate over the bunch and the room; integrated over the shapes on their ordering sectors the same closing is Corollary 15.11, with Theorem 15.14 past the walls.

Proposition 18.25 (Overlapping levels)**.** *In Theorem 18.13 with one table let $s,t>0$ and $2\delta^d = s-\kappa_B \notin \mathbb{Z}$, and let $\mathcal{M}(x)$ be the integral over $\Delta_B\times\Delta_R$ of*

$$u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}\big(K_{\mathcal{B}/R}(u)K_{\mathcal{B}|R}(v)\big)^{\kappa_B-s}\big(K_{\mathcal{B}|B}(u)K_{\mathcal{B}/B}(v)\big)^{-\kappa_B}\big(\Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(v)\big)^{-t}\Big(\frac{\varrho_B(u)}{\varrho_R(v)}\Big)^{x}.$$

The terms of the two lattices of the level d in Corollary 18.16(ii) are $A_n = a_n\mathcal{M}(n)$ and $A'_n = a'_n\mathcal{M}(n+2\delta^d)$, with

$$a_n = \Gamma_{\!2}\begin{bmatrix}2\delta^d,\ 1-2\delta^d\\ s,\ t\end{bmatrix}\Gamma_{\!2}\begin{bmatrix}t+n,\ \kappa_B+n\\ 1+n,\ 1-2\delta^d+n\end{bmatrix},$$
$$a'_n = -\Gamma_{\!2}\begin{bmatrix}2\delta^d,\ 1-2\delta^d\\ s,\ t\end{bmatrix}\Gamma_{\!2}\begin{bmatrix}s+n,\ t+2\delta^d+n\\ 1+n,\ 1+2\delta^d+n\end{bmatrix}.$$

Suppose these integrals converge for every n. If $\varrho_B(u)>\varrho_R(v)$ on a set of positive measure, then $|A_n+A'_n|\to\infty$; likewise for the level r if $\varrho_B(u)<\varrho_R(v)$ on a set of positive measure. So when the two levels overlap*, both sets having positive measure, the lattices of neither level sum to the period by convergent periods.*

The two lattices of a level have opposite signs and coefficients whose ratio tends to -1. When $\sup\varrho_B<\inf\varrho_R$ each of them converges by itself. When $\varrho_B(u)<\varrho_R(v)$ throughout with $\sup\varrho_B=\inf\varrho_R$ neither need converge; the moments at n and at $n+2\delta^d$ then agree to leading order, so the two lattices cancel to that order, and Proposition 18.23 sums them to the period by scaling the level weights. Past that, the two lattices part and the series diverges. By Lemma 18.15 the overlap reaches a wall when ϱ_B/ϱ_R is unbounded, and then the periods themselves stop converging along the lattice. An overlap is closed on the totals of the bunch and the room (Section 18.8), and, read in one scale, on the lattice of continued periods of a level whenever the reach of the bunch on its side lies below the decay of the other (Theorem 19.38).

Proposition 18.26 (Between groups at the wall, inside a group at the gauge)**.** *In Theorem 18.19 fix a group $\mathcal{G}$, an index ν with $\nu\notin-\mathbb{N}$ and $\nu-w_{\mathcal{G}}\notin\mathbb{N}$, and a point ρ_0 of the scale. For the letter i of K_j in $\mathcal{G}$ put $\mathsf{Q}_i=\mathsf{Q}^K_{j,B}(\cdot;(\rho_0u,\alpha_R))$ and $\mathsf{Q}^\top_i=\mathsf{Q}^K_{j,R}(\cdot;(\rho_0u,\alpha_R))$, and for that of Ψ_j the same with Ψ; then $\mathsf{Q}_i(x)=L_i(x\rho_0)/L_i(\rho_0)$ and $\mathsf{Q}^\top_i(x)=x^{c_i}\mathsf{Q}_i(1/x)$ (Proposition 12.10(i),(ii)), and*

$$\operatorname{Reg}\int_0^\infty\rho^{\nu-1}f_{\mathcal{G}}(\rho)\,d\rho=\rho_0^\nu f_{\mathcal{G}}(\rho_0)\Big(\operatorname{Reg}\int_0^1 x^{\nu-1}\prod_{i\in\mathcal{G}}\mathsf{Q}_i(x)^{-w_i}dx$$
$$+\operatorname{Reg}\int_0^1 x^{w_{\mathcal{G}}-\nu-1}\prod_{i\in\mathcal{G}}\mathsf{Q}^\top_i(x)^{-w_i}dx\Big).$$

Each of the two finite parts is the series of Theorem 15.9(ii) for these polynomials, absolutely convergent, with its Beta corners continued as in Theorem 15.14. So for every separated partition every share of the scale is closed: between the groups on the lattices of Theorem 18.19, inside a group at the gauge. With one group this is Theorem 15.9; with single letters of width one in one order it is Theorem 18.21.

A group that cannot be split, because the values of its letters do not stand in one order, is kept as one Breta of one bunch (Proposition 15.1); it is closed about a point of the scale, where no order is needed, and the letters outside it are closed about its ends. The two closings do not compete: the wall closing does as much as the scale separates, and the gauge closing does the rest. When its index lies in $(0, w_{\mathcal{G}})$ and its letter exponents are nonnegative, a group read on the totals of the bunch and the room closes about constant references, its terms pairing the bunch with the room at width one (Section 18.8).

18.8 The scale on the totals

The scale of a bunch is the quotient of two totals, that of the bunch and that of the room, and the two totals lie on the simplex $\Delta_{\{B,R\}}$ of the two blocks (Theorem 3.12). Read there, a letter on the scale is homogeneous in the totals, with its level or colevel letters as coefficients. At width one it is linear, with its constant coefficient at one vertex of the simplex and its scale coefficient at the other, and its Skeleton value is the point of the scale where its two terms are equal. The free shares of Theorem 18.13 are the share side of this integral over the totals, the pin being the shape integral over the totals (Theorem 3.9), and the lattices of Section 18.7 close them. The integral over the totals can also be closed as it stands. On the simplex a linear letter is at most the larger of its two coefficients, whatever their ratio, so it closes about a constant reference with nonnegative terms, which pair moments of the bunch with moments of the room. No order of the values is asked, and the overlap that stops the lattices of a level (Proposition 18.25) is not seen.

Definition 18.27 (The totals and their letters)**.** The *totals* of a bunch B and its room R are the point $x = (x_B, x_R)$ of the shape simplex $\Delta_{\{B,R\}}$ of the two blocks. Let $\mathbf{m}^{\mathrm{tot}} = (m_B^{\mathrm{tot}}, m_R^{\mathrm{tot}})$ have positive entries, let ϕ be a nonnegative measure on $\Delta_B \times \Delta_R$, and let finitely many *letters on the totals*

$$\widehat{L}_i(x) = \sum_{l=0}^{c_i} x_B^{\,l}\, x_R^{\,c_i - l} L_i^{(l)}, \qquad i = 1, \dots, q,$$

with letter exponents $w_i \in \mathbb{R}$, have coefficients $L_i^{(l)} \geq 0$ that are measurable functions of the shapes $(u, v) \in \Delta_B \times \Delta_R$, each letter positive on the interior of $\Delta_{\{B,R\}}$ at ϕ-almost every shape. The *integral over the totals* at the shapes (u, v) is the inner integral below, over $\Delta_{\{B,R\}}$, and its integral against ϕ is the Trailmoment on the source $\Delta_{\{B,R\}} \times \Delta_B \times \Delta_R$ (Definition 5.1) with the coefficient function $(x, u, v) \mapsto (\widehat{L}_i(x))_i$ and the input $x^{\mathbf{m}^{\mathrm{tot}} - \mathbf{1}} d\omega(x)\, \phi(du, dv)$, read over the whole source,

$$\begin{aligned}\big(\mathsf{T}_{\mathbf{w};I_q}[(\widehat{L}_i)_i]\; x^{\mathbf{m}^{\mathrm{tot}} - \mathbf{1}} d\omega(x)\, \phi(du, dv)\big)\big(\Delta_{\{B,R\}} \times \Delta_B \times \Delta_R\big)& \\ = \int_{\Delta_B \times \Delta_R} \int_{\Delta_{\{B,R\}}} x^{\mathbf{m}^{\mathrm{tot}} - \mathbf{1}} \prod_i \widehat{L}_i(x)^{-w_i}\, d\omega(x)\, \phi(du, dv),&\end{aligned}$$

wherever it converges; its kernel $\mathsf{Z}_{\mathbf{w};I_q}((\widehat{L}_i)_i) = \prod_i \widehat{L}_i^{-w_i}$ is the product of the one-scale kernels $\mathsf{Z}_{w_i;(1)}(\widehat{L}_i)$ (Definition 2.12, Theorem 5.21(i)). For one linear letter $x_R L^{(0)} + x_B L^{(1)}$ the integral over the totals is the Breta with the one bunch $\{B, R\}$, the shape index $\mathbf{m}^{\mathrm{tot}}$, the scale index w and the unit configuration (Definition 3.1):

$$\int_{\Delta_{\{B,R\}}} x^{\mathbf{m}^{\mathrm{tot}} - \mathbf{1}} \big(x_B L^{(1)} + x_R L^{(0)}\big)^{-w} d\omega(x) = \mathsf{B}^{\{\{B,R\}\}}_{\mathbf{m}^{\mathrm{tot}},\, w;\, I_{\{B,R\}}}\big(L^{(1)}, L^{(0)}\big).$$

Theorem 18.28 (The scale on the totals)**.** *At the gauge (u,α_R) let $\kappa_B>0$ and $\kappa_B^{\mathrm{top}}>0$, and read every letter on the scale of B (Definition 18.17) on the totals,*

$$\widehat{L}_i(x)=x_R^{c_i}L_i(x_B/x_R)=\sum_{l=0}^{c_i}x_B^{\,l}\,x_R^{\,c_i-l}L_i^{(l)},$$

a letter of width zero being a constant on the totals.

(i) At every point $\rho_0>0$ of the scale, which may depend on the shapes,

$$\mathrm{Sc}_B(u,\alpha_R;\mathbf{m};\mathbf{s},\mathbf{t})=\rho_0^{\kappa_B}\int_{\Delta_{\{B,R\}}}x_B^{\kappa_B-1}x_R^{\kappa_B^{\mathrm{top}}-1}\prod_i\widehat{L}_i(\rho_0x_B,\ x_R)^{-w_i}\,d\omega(x),$$

both sides converging together. At $\rho_0=1$ and $\alpha_R=v\in\Delta_R$ the point $(\rho u,v)$ is projectively $(x_Bu,x_Rv)\in\Delta_{\mathcal{E}}$, whose totals on the blocks B and R are x_B and x_R (Theorem 3.12), with $\rho=x_B/x_R$. On the balance wall $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the Trailmoment of Definition 18.27 for the letters $\widehat{L}_i$ at $\mathbf{m}^{\mathrm{tot}}=(\kappa_B,\kappa_B^{\mathrm{top}})$ with $\phi=u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B(u)\,d\omega_R(v)$; both sides converge together.

(ii) Let B have width at most one in every table, the weights factoring across B and R; write the letters of width one $L_i(\rho)=L_i^{(0)}+\rho L_i^{(1)}$, with their constant and scale coefficients and their Skeleton values $\xi_i=L_i^{(0)}/L_i^{(1)}$ (Theorem 18.21), and fix one of them, i_0. At the point $\rho_0=\xi_{i_0}$ the two terms of i_0 are equal and its letter on the totals is the constant $L_{i_0}^{(0)}$. The products in the displays run over the letters of width one, and both right sides carry the further factor $\prod L_i^{-w_i}$ over the constant letters of the tables of width zero:

$$\begin{aligned}\mathrm{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t})&=\big(L_{i_0}^{(0)}\big)^{\kappa_B-w_{i_0}}\big(L_{i_0}^{(1)}\big)^{\kappa_B^{\mathrm{top}}-w_{i_0}}\\&\qquad\times\int_{\Delta_{\{B,R\}}}x_B^{\kappa_B-1}x_R^{\kappa_B^{\mathrm{top}}-1}\prod_{i\neq i_0}\big(x_RL_i^{(0)}L_{i_0}^{(1)}+x_BL_{i_0}^{(0)}L_i^{(1)}\big)^{-w_i}\,d\omega(x)\\&=\xi_{i_0}^{\kappa_B}\prod_i\big(L_i^{(0)}\big)^{-w_i}\int_{\Delta_{\{B,R\}}}x_B^{\kappa_B-1}x_R^{\kappa_B^{\mathrm{top}}-1}\prod_{i\neq i_0}\Big(x_R+x_B\,\frac{\xi_{i_0}}{\xi_i}\Big)^{-w_i}\,d\omega(x).\end{aligned}$$

On the balance wall $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the Trailmoment of Definition 18.27 for these linear letters at $\mathbf{m}^{\mathrm{tot}}=(\kappa_B,\kappa_B^{\mathrm{top}})$, with ϕ of (i) multiplied by $(L_{i_0}^{(0)})^{\kappa_B-w_{i_0}}(L_{i_0}^{(1)})^{\kappa_B^{\mathrm{top}}-w_{i_0}}$ and by the further factor of the constant letters: the coefficients of each linear letter are the constant coefficient of one letter times the scale coefficient of the other, and their ratio is the ratio of the two Skeleton values.

(iii) In (ii) let every $w_i>0$, and open each linear letter by Corollary 4.4, with the share ν_i on $x_BL_{i_0}^{(0)}L_i^{(1)}$ and $w_i-\nu_i$ on $x_RL_i^{(0)}L_{i_0}^{(1)}$, on a contour with $0<\mathrm{Re}\,\nu_i<w_i$ and $\kappa_B-w_{i_0}<\sum_{i\neq i_0}\mathrm{Re}\,\nu_i<\kappa_B$. The integral over the totals is then the Beta corner $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\nu_{i_0},\ w_{i_0}-\nu_{i_0}\\ w_{i_0}\end{smallmatrix}\right]$ of the letter i_0 at $\nu_{i_0}=\kappa_B-\sum_{i\neq i_0}\nu_i$, so that the shares on the scale coefficients add to κ_B identically, and, with the further factor of (ii) on the right,

$$\mathrm{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\sum_i\nu_i=\kappa_B}\ \prod_i\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\nu_i,\ w_i-\nu_i\\ w_i\end{smallmatrix}\right]\big(L_i^{(0)}\big)^{-(w_i-\nu_i)}\big(L_i^{(1)}\big)^{-\nu_i},$$

the share side of the scale integral with one share per letter (Corollary 15.4). Integrated over the shapes it is Theorem 18.13: the pin of the shares is the shape integral over the totals.

The scale integral is the integral over the totals, and the shares of Theorem 18.13 are its share side: opening every letter but one leaves over the totals the Beta corner of the last, and the pin is what that integral asks of the shares. The point ρ_0 is free. At $\rho_0 = 1$ the totals are those of Theorem 3.12, the coordinates of a node in Theorem 19.14; at the value of a letter that letter leaves the integral, and for one table a single linear letter is left.

Corollary 18.29 (One table: the even share on the totals)**.** *For one table in Theorem 18.13 write, on* $\Delta_B \times \Delta_R$*,*

$$\mathbf{E}^d = \mathbf{E}_{\mathcal{B}/R}(u)\,\mathbf{E}_{\mathcal{B}|R}(v), \qquad \mathbf{E}^r = \mathbf{E}_{\mathcal{B}|B}(u)\,\mathbf{E}_{\mathcal{B}/B}(v)$$

for the even letters of the two levels, so that $\mathbf{E}^d/\mathbf{E}^r = \varrho_B(u)/\varrho_R(v)$*, and let* $2\delta^d = s - \kappa_B$ *and* $2\delta^r = \kappa_B - t$*; here* $\kappa_B^{\mathrm{top}} = s + t - \kappa_B$*. Then*

$$\begin{aligned}\mathrm{Sc}_B(u,v;\mathbf{m};s,t) &= \big(K_{\mathcal{B}/R}(u)K_{\mathcal{B}|R}(v)\big)^{-2\delta^d}\big(K_{\mathcal{B}|B}(u)K_{\mathcal{B}/B}(v)\big)^{-2\delta^r}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),\,t;\,I_{\{B,R\}}}\big(\mathbf{E}^d,\mathbf{E}^r\big)\\ &= \big(\Psi_{\mathcal{B}/R}(u)\Psi_{\mathcal{B}|R}(v)\big)^{2\delta^d}\big(\Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(v)\big)^{2\delta^r}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),\,s;\,I_{\{B,R\}}}\big(\mathbf{E}^r,\mathbf{E}^d\big),\end{aligned}$$

and $\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),t;I_{\{B,R\}}}(\mathbf{E}^d,\mathbf{E}^r) = (\mathbf{E}^r)^{-t}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),t;I_{\{B,R\}}}\big(\varrho_B(u)/\varrho_R(v),1\big)$*. So* $P_F(\mathbf{m};s,t)$ *is the Trailmoment of Definition 18.27 for the single linear letter* $x_R\mathbf{E}^r + x_B\mathbf{E}^d$ *at the exponent* t *and at* $\mathbf{m}^{\mathrm{tot}} = (\kappa_B, \kappa_B^{\mathrm{top}})$*, with* $\phi = \big(K_{\mathcal{B}/R}K_{\mathcal{B}|R}\big)^{-2\delta^d}\big(K_{\mathcal{B}|B}K_{\mathcal{B}/B}\big)^{-2\delta^r}u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B\,d\omega_R$*, and the shapes enter the Breta only through the ratio* $\varrho_B(u)/\varrho_R(v)$ *of its coefficients. Opened by Corollary 4.4, the linear letter has the share* $b^d = \sigma^d - \delta^d$ *on* $x_B\mathbf{E}^d$*, and the integrand is that of Theorem 18.13: the free even share is the share of this letter, and the integral over the totals is the even share integrated.*

Theorem 18.30 (Closing at the peaks, one reference per coefficient)**.** *In the Trailmoment of Definition 18.27 let every* $w_i > 0$*, and choose constants* $\bar L_{i,l} \ge L_i^{(l)}$*, one for each coefficient, at* ϕ*-almost every shape; the* reference letters *are* $\bar L_i(x) = \sum_{l=0}^{c_i} x_B^l x_R^{c_i-l}\bar L_{i,l}$*. For an array* $\mathbf{k} = (k_{i,l})$ *with the rows* $\mathbf{k}_i = (k_{i,l})_{0\le l\le c_i}$ *put*

$$\boldsymbol{\beta}(\mathbf{k}) = \big(\beta_B(\mathbf{k}),\beta_R(\mathbf{k})\big) = \Big(m_B^{\mathrm{tot}} + \sum_{i,l} l\,k_{i,l},\ m_R^{\mathrm{tot}} + \sum_{i,l}(c_i-l)\,k_{i,l}\Big), \qquad \mathbf{w}(\mathbf{k}) = \big(w_i + |\mathbf{k}_i|\big)_i.$$

Introduce one singleton scale $\mathfrak{v}_i$ *per letter,* $\mathfrak{V} = \{\mathfrak{v}_1,\dots,\mathfrak{v}_q\}$*, take the partition* $\mathcal{P}_{\mathrm{tot}} = \{\{B,R\}\}\cup\{\{\mathfrak{v}_i\} : 1\le i\le q\}$*, and let* $(\pi^{\mathrm{tot}},\ell^{\mathrm{tot}})$ *be the term arrays of the letter presentation (Proposition 3.6) with the kernel column of shape exponent* 0*, level* $(1,0_{\mathfrak{V}})$ *and coefficient* 1*, and for every* (i,l) *with* $\bar L_{i,l} > 0$ *a column of shape exponent* $\big((l,c_i-l),\mathbf{e}_i\big)$*, level* $(0,\mathbf{e}_i)$ *and coefficient* $\bar L_{i,l}$*;* $1\oplus\bar{\mathbf{L}}$ *lists the coefficients in this order. Then, with the arrays* $\mathbf{k}$ *supported on these columns,*

$$\begin{aligned}&\int_{\Delta_B\times\Delta_R}\int_{\Delta_{\{B,R\}}} x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}}\prod_i \widehat{L}_i(x)^{-w_i}\,d\omega(x)\,\phi(du,dv)\\ &\qquad = \sum_{\mathbf{k}}\prod_i\frac{(w_i)_{|\mathbf{k}_i|}}{\mathbf{k}_i!}\;\mathsf{B}^{\mathcal{P}_{\mathrm{tot}}}_{(\boldsymbol{\beta}(\mathbf{k}),\mathbf{w}(\mathbf{k})),\,(1,\mathbf{w}(\mathbf{k}));\,(\pi^{\mathrm{tot}},\ell^{\mathrm{tot}})}\big(1\oplus\bar{\mathbf{L}}\big)\int\prod_{i,l}\big(\bar L_{i,l}-L_i^{(l)}\big)^{k_{i,l}}\,d\phi,\end{aligned}$$

a series of nonnegative terms, both sides finite or infinite together. The Breta coefficient is a number: the kernel integrates its scale to 1*, and*

$$\mathsf{B}^{\mathcal{P}_{\mathrm{tot}}}_{(\boldsymbol{\beta}(\mathbf{k}),\mathbf{w}(\mathbf{k})),\,(1,\mathbf{w}(\mathbf{k}));\,(\pi^{\mathrm{tot}},\ell^{\mathrm{tot}})}\big(1\oplus\bar{\mathbf{L}}\big) = \int_{\Delta_{\{B,R\}}} x^{\boldsymbol{\beta}(\mathbf{k})-\mathbf{1}}\prod_i \bar L_i(x)^{-w_i-|\mathbf{k}_i|}\,d\omega(x).$$

Each moment is a read of ϕ *against the one-scale kernels* $\mathsf{Z}_{-k_{i,l};(1)}\big(\bar L_{i,l}-L_i^{(l)}\big) = \big(\bar L_{i,l}-L_i^{(l)}\big)^{k_{i,l}}$*.*

Corollary 18.31 (Closing at the peak on the totals)**.** *In Theorem 18.30 take the references* $\bar L_{i,l} = \binom{c_i}{l}\bar L_i$ *with constants* $\bar L_i > 0$ *such that the deviations* $h_{i,l} = \binom{c_i}{l}\bar L_i - L_i^{(l)}$ *are nonnegative. The reference letter is then the constant* $\bar L_i(x_B + x_R)^{c_i} = \bar L_i$ *on the totals, the Breta coefficient is* $\prod_i \bar L_i^{-w_i-|\mathbf{k}_i|}$ *times the Beta corner* $\Gamma_{\substack{2\\1}}\!\left[\begin{matrix}\beta_B(\mathbf{k}),\ \beta_R(\mathbf{k})\\ \beta_B(\mathbf{k})+\beta_R(\mathbf{k})\end{matrix}\right]$*, and that Trailmoment is*

$$\sum_{\mathbf{k}} \prod_i \frac{(w_i)_{|\mathbf{k}_i|}}{\mathbf{k}_i!}\, \bar L_i^{-w_i-|\mathbf{k}_i|}\ \Gamma_{\substack{2\\1}}\!\left[\begin{matrix}\beta_B(\mathbf{k}),\ \beta_R(\mathbf{k})\\ \beta_B(\mathbf{k})+\beta_R(\mathbf{k})\end{matrix}\right] \int \prod_{i,l} h_{i,l}^{k_{i,l}}\, d\phi.$$

For linear letters $\widehat L_i(x) = x_R L_i^{(0)} + x_B L_i^{(1)}$ *with* $\bar L_i \ge L_i^{(0)}, L_i^{(1)}$*, with* n_i *counting the constant coefficient and* n_i' *the scale coefficient of the letter* i*, it is*

$$\sum_{\mathbf{n},\mathbf{n}'} \prod_i \frac{(w_i)_{n_i+n_i'}}{n_i!\, n_i'!}\, \bar L_i^{-w_i-n_i-n_i'}\ \Gamma_{\substack{2\\1}}\!\left[\begin{matrix}m_B^{\mathrm{tot}}+|\mathbf{n}'|,\ m_R^{\mathrm{tot}}+|\mathbf{n}|\\ |\mathbf{m}^{\mathrm{tot}}|+|\mathbf{n}|+|\mathbf{n}'|\end{matrix}\right] \int \prod_i \big(\bar L_i - L_i^{(0)}\big)^{n_i}\big(\bar L_i - L_i^{(1)}\big)^{n_i'}\, d\phi.$$

If a coefficient is a product $f'(u)f''(v)$ *and its reference a product* $\bar f'\bar f''$ *of constants* $\bar f' \ge f'$ *and* $\bar f'' \ge f''$*, then* $\bar f'\bar f'' - f'f'' = \bar f'(\bar f'' - f'') + f''(\bar f' - f')$*, both terms nonnegative and each a function on the bunch times a function on the room; so when every coefficient, every reference and* ϕ *are products, every moment is a finite nonnegative combination of products of moments over* Δ_B *and over* Δ_R*.*

No order of the values is used, only bounds on the coefficients, and the terms are nonnegative. Applied to every letter on the whole simplex with the references $M_i = \max_T \theta_{j,T}/g_i!$, Theorem 15.13 is the draw series (Proposition 19.11); applied on the totals it closes the scale of the bunch alone, and the shapes of the bunch and the room enter only through the moments of the deviations of the coefficients from their references. Across a bunch of width one every coefficient is a letter of a minor on the bunch times a letter of a minor on the room (Theorem 18.21), so with references that are products the moments pair the bunch with the room, as the lattices of Theorem 18.21 do, with nonnegative terms. If every $\widehat L_i$ is at least $(1-\lambda)\bar L_i$ on the totals at every shape, with $\lambda < 1$, the terms of total order $\sum_i |\mathbf{k}_i| = k$ add up to at most $\frac{(|\mathbf{w}|)_k}{k!}\lambda^k$ times the term of order 0; when a coefficient vanishes on a face of a shape simplex the terms need not decay geometrically.

For one table read at $\rho_0 = 1$ the two letters on the totals are linear, and so are their reference letters, one constant for each coefficient. Rescaling the scale by the ratio of the references of the letter K carries its reference letter to a multiple of $x_R + x_B$, and that of Ψ to a multiple of the linear letter whose scale coefficient over its constant coefficient is the reference of the even letter of the level d over that of the level r; the rescaling multiplies the Breta coefficient by a power of the ratio of the references of K. Two letters on the line of the totals have two zeros besides its two endpoints, and four points of a line have one modulus, the ratio of the references of the even letters of the two levels: the Breta coefficients depend on the references only through their scales and this modulus, defined up to its inverse by the naming of the two levels. What one reference for each coefficient adds is that every deviation can vanish at its own peak, at the price of the one modulus in the coefficients. When the modulus is 1 both reference letters are multiples of $x_R + x_B$, which is 1 on the simplex, and Corollary 18.31 applies. With N tables there are $2N$ letters on the totals and $2N-1$ such moduli.

Corollary 18.32 (The ladder on the totals)**.** *In Theorem 18.28(ii) let every* $w_i > 0$*, put* $w' = \sum_{i\ne i_0} w_i$*, and let* $\zeta > 0$*.*

(i) At shapes where every ratio ξ_{i_0}/ξ_i lies in $(0,2\zeta)$,

$$\int_{\Delta_{\{B,R\}}} x_B^{\kappa_B-1}x_R^{\kappa_B^{\mathrm{top}}-1}\prod_{i\neq i_0}\Big(x_R+x_B\,\frac{\xi_{i_0}}{\xi_i}\Big)^{-w_i}d\omega(x)$$
$$=\sum_{\mathbf{n}}\prod_{i\neq i_0}\frac{(w_i)_{n_i}}{n_i!}\Big(\zeta-\frac{\xi_{i_0}}{\xi_i}\Big)^{n_i}\ \mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+|\mathbf{n}|,\kappa_B^{\mathrm{top}}),\,w'+|\mathbf{n}|;\,I_{\{B,R\}}}(\zeta,1),$$

absolutely: the linear letters are moved to the common ratio ζ, where they merge into one, and the coefficients do not depend on the shapes. At $\zeta=1$ they are the Beta corners $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\kappa_B+|\mathbf{n}|,\ \kappa_B^{\mathrm{top}}\\ \kappa_B+\kappa_B^{\mathrm{top}}+|\mathbf{n}|\end{smallmatrix}\right]$. For one linear letter this is the ladder of the Breta in its first coefficient (Theorem 3.5(ii)), $\partial_\zeta\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w;I_{\{B,R\}}}(\zeta,1)=-w\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+1,\kappa_B^{\mathrm{top}}),w+1;I_{\{B,R\}}}(\zeta,1)$, summed as a Taylor series.

(ii) Let $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converge and let every ratio ξ_{i_0}/ξ_i lie in $[\zeta_-,\zeta_+]\subset(0,2\zeta)$ on $\Delta_B\times\Delta_R$. Then (i) integrates term by term against ϕ of Theorem 18.28(i) multiplied by $\xi_{i_0}^{\kappa_B}\prod_i(L_i^{(0)})^{-w_i}$ and by the further factor of Theorem 18.28(ii), and the terms of $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ with $|\mathbf{n}|=n$ add up in absolute value to at most $C\frac{(w')_n}{n!}\vartheta^nP_F(\mathbf{m};\mathbf{s},\mathbf{t})$, where $\vartheta=\max(1-\zeta_-/\zeta,\ \zeta_+/\zeta-1)<1$ and $C=\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w';I_{\{B,R\}}}(\zeta,1)\big/\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w';I_{\{B,R\}}}(\zeta_+,1)$. If $\zeta\geq\sup\xi_{i_0}/\xi_i$ for every i the terms are nonnegative and no lower bound on the ratios is needed; both sides are then finite or infinite together.

(iii) For one table with i_0 the letter of K, under the hypotheses of (ii) the only other ratio is $\varrho_B(u)/\varrho_R(v)$, and

$$P_F(\mathbf{m};s,t)=\sum_{n\geq0}\frac{(t)_n}{n!}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+n,\kappa_B^{\mathrm{top}}),\,t+n;\,I_{\{B,R\}}}(\zeta,1)\int_{\Delta_B\times\Delta_R}\big(\zeta\mathbf{E}^r-\mathbf{E}^d\big)^n\,(\mathbf{E}^r)^{-t-n}\,\phi(du,dv),$$

with $\mathbf{E}^d$, $\mathbf{E}^r$ and ϕ of Corollary 18.29. With i_0 the letter of Ψ the ratio is $\varrho_R(v)/\varrho_B(u)$, and the same holds with s for t, the moments $\int(\zeta\mathbf{E}^d-\mathbf{E}^r)^n(\mathbf{E}^d)^{-s-n}$ and the measure $\phi=\big(\Psi_{\mathcal{B}/R}\Psi_{\mathcal{B}|R}\big)^{2\delta^d}\big(\Psi_{\mathcal{B}|B}\Psi_{\mathcal{B}/B}\big)^{2\delta^r}u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B\,d\omega_R$. So one table closes on the totals with nonnegative terms about any $\zeta\geq\sup\varrho_B/\inf\varrho_R$ when no slope is positive, and with i_0 the letter of Ψ about any point at least $\sup\varrho_R/\inf\varrho_B$ when no slope is negative (Lemma 18.15). With $\zeta=\zeta'\zeta''$, $\zeta'\geq\sup_{\Delta_B}\varrho_B$ and $\zeta''\geq1/\inf_{\Delta_R}\varrho_R$,

$$\zeta\mathbf{E}^r-\mathbf{E}^d=\zeta'\mathbf{E}_{\mathcal{B}|B}(u)\big(\zeta''\mathbf{E}_{\mathcal{B}/B}(v)-\mathbf{E}_{\mathcal{B}|R}(v)\big)+\mathbf{E}_{\mathcal{B}|R}(v)\big(\zeta'\mathbf{E}_{\mathcal{B}|B}(u)-\mathbf{E}_{\mathcal{B}/R}(u)\big),$$

both terms nonnegative and each a function on the bunch times a function on the room.

(iv) Since $x_R+x_B\xi_{i_0}/\xi_i=(\xi_{i_0}/\xi_i)\big(x_B+x_R\xi_i/\xi_{i_0}\big)$, the integral of (i) is $\prod_{i\neq i_0}(\xi_i/\xi_{i_0})^{w_i}$ times the same integral with κ_B and κ_B^{top} swapped and every ratio replaced by its inverse. So (i) and (ii) hold in the inverse ratios, the factor $\prod_{i\neq i_0}(\xi_i/\xi_{i_0})^{w_i}$ joining ϕ, with nonnegative terms about any common point at least $\sup\xi_i/\xi_{i_0}$: the ladder closes whenever the ratios are bounded on one side.

The three closings of the even share read the same linear letters. At a wall the even share closes toward the level whose even letter is the smaller, which needs ϱ_B/ϱ_R on one side of 1 with room (Corollary 18.16(i)); the ladder moves the linear letters to a common ratio, which needs the

quotient bounded on one side, and its coefficients are constants of the scale; the peak closing moves each linear letter to a constant reference, which needs the coefficients bounded and asks nothing of their ratio. In all three the terms pair moments of the bunch with moments of the room. With $\varrho_B(u) < \varrho_R(v)$ throughout and $\sup \varrho_B = \inf \varrho_R$, the ladder about $\zeta = 1$ is a series of nonnegative terms whose coefficients are Beta corners, where Proposition 18.23 sums the lattices by scaling the level weights. Where the quotient takes every positive value, of the three only the peak closing applies; the gauge closing applies as well, its terms not separating over the bunch and the room (Proposition 18.24).

18.9 Width zero

Proposition 18.33 (A loop or a coloop on the simplex)**.** *Let a mark e be in every table a loop ($\eta_j = -1$) or a coloop ($\eta_j = +1$), let F' be the family of Corollary 15.16 on $\mathcal{E} - e$, and put $\nu_e = \mu_e - \sum_j \eta_j \delta_j$, so that $\kappa' = \kappa_{\mathcal{E}} - \nu_e$ is the imbalance of F' at the masses $\mathbf{m}|_{\mathcal{E}-e}$. Where $\nu_e > 0$ and $\kappa' > 0$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \Gamma_{\,1}^{\,2}\begin{bmatrix} \nu_e, & \kappa' \\ & \kappa_{\mathcal{E}} \end{bmatrix}\, P_{F'}(\mathbf{m}|_{\mathcal{E}-e}; \mathbf{s}, \mathbf{t}),$$

the mark rescaling every other mark by the common factor $1 - u_e$; it is the compound formula of Theorem 16.15 at $S = \{e\}$, with the walls $\kappa_{\{e\}} = \nu_e$ and $\kappa_{\mathcal{E}\setminus e} = \kappa'$. Marks $e_1, \dots, e_k$ of this kind combine first among themselves and then with the rest, $\Gamma_{\,1}^{\,k+1}\begin{bmatrix} (\nu_{e_i})_i, & \kappa' \\ & \kappa_{\mathcal{E}} \end{bmatrix} = \Gamma_{\,1}^{\,k}\begin{bmatrix} (\nu_{e_i})_i \\ a \end{bmatrix}\, \Gamma_{\,1}^{\,2}\begin{bmatrix} a, & \kappa' \\ & \kappa_{\mathcal{E}} \end{bmatrix}$ *with $a = \sum_i \nu_{e_i}$ and κ' the imbalance once all of them are removed (Theorem 3.12). On the balance wall $\kappa' = -\nu_e$ and the period of the compound diverges (Proposition 8.15): the Foreman tree splits molecules first (Definition 14.4).*

The bunch $\{e\}$ has width zero: the room does not see it at all, and the period integrates it out as the Beta corner $\Gamma_{\,1}^{\,2}\begin{bmatrix} \nu_e, & \kappa' \\ & \kappa_{\mathcal{E}} \end{bmatrix}$, whose Gamma $\Gamma(\nu_e)$ is the Gamma $\Gamma(\mu_e - \langle \boldsymbol{\delta} \rangle_e)$ of the port at e (Theorem 15.15).

18.10 Pooling is a move

Definition 18.34 (Period datum; moves)**.** Let a family of tables with positive rational weights have rational exponents and masses and lie on the balance wall. Its *period datum* is the positive orthant of $\mathbb{R}^{\mathcal{E}}$, read projectively, with the form $W\, d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\alpha$, algebraic over $\mathbb{Q}$ and homogeneous of degree zero, the hypersurface $\{\prod_j K_j \Psi_j = 0\}$ and the coordinate hyperplanes; the face $\{\alpha_S = 0\}$ carries the wall κ_S (Theorem 8.7). A *move* between period data is a change of variables by a birational map defined over $\mathbb{Q}$ carrying the open domain bijectively onto the open domain, faces to faces and form to form; a product of data; a linear combination; or the boundary term of an exact form.

Theorem 18.35 (Pooling is a move)**.** *Let a bunch B have width one in every table of a family of tables with positive rational weights, the weights of every table factoring across B and R with rational factors, at rational exponents and masses on the balance wall.*

(i) Theorem 18.8 is one change of variables, $(\rho, u, \alpha_R) \mapsto (\rho u, \alpha_R)$, the scale and the shape of B: it is birational over $\mathbb{Q}$ with the inverse $\alpha \mapsto (|\alpha_B|, \alpha_B/|\alpha_B|, \alpha_R)$, bijective on the open domains, and it carries the face of $S \subseteq \mathcal{E}$ to $\{\rho = 0\} \times \{\alpha_{S\cap R} = 0\}$ if $S \supseteq B$ and to

$\{u_{S\cap B}=0\}\times\{\alpha_{S\cap R}=0\}$ *otherwise, with the walls matched by Theorem 18.9: by (i) there when* $S\supseteq B$ *or* $S\subseteq R$*, by (ii) there when* $S\subseteq B$*, and by the gluing of the counts otherwise.*

(ii) When the factorizations, with rational factors, give every active letter the one hinge value γ *of Theorem 18.10, that theorem is a composite of moves: the change of variables (i); the change* $\rho\mapsto\rho\,\gamma(u)$*, birational over* $\mathbb{Q}$ *and positive on the open domain; the product of the datum of the shape with that of the cell family of the room; and the integral of the hinge, the Beta corner of a linear letter (Proposition 15.1 with one coefficient), itself a change of variables and a product with the datum of the list on two marks whose bases are the two singletons, at the exponents* $(0,2\sigma)$*. Every integral in the chain converges exactly when* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *does (Theorem 18.10).*

So the identities of Theorems 18.8 and 18.10 hold between period data, not only between their values: when every active letter has one hinge value the period datum of the family times that of the list on two marks is the product of the data of its two cell families.

In the interior Theorem 18.8 writes the period as an integral over the shape simplex, whose faces carry the walls of Theorem 18.9(ii), of the periods of the pooled families $F_{B\to g}(u)$, whose walls are those of Theorem 18.9(i).

18.11 Cleaning a molecule

Definition 18.36 (Cleaning)**.** A *cleaning* of a list is a sequence of poolings of parallel and series bunches of width one, each in the list left by the previous ones, until at most two marks or no such class remain.

Proposition 18.37 (Poolings of disjoint classes commute)**.** *Let* X *and* Y *be disjoint parallel or series bunches of width one of a list. Pooling* X *keeps* Y *a class of the same kind and width, with the same restriction to* Y *and the same contraction of its room, and the two orders of pooling give the same list on* $(\mathcal{E}\setminus(X\cup Y))\sqcup\{g_X,g_Y\}$*, the same cell kernels and the same hinge values. A cleaning ends after at most* $|\mathcal{E}|-2$ *poolings.*

Proposition 18.38 (Cleaning commutes with maps)**.** *A bijection that carries one table to another (Definition 11.14) maps the parallel and series bunches of the first onto those of the second, with the same widths and, the weights factoring across a bunch and its room and the factorization carried along, the same hinge values. An exchange maps parallel bunches onto series bunches and swaps the hinge values by* K *and by* Ψ*, the weight* θ'_p *of a pattern* p *of a bunch* B *becoming that of the pattern* $g(B\setminus p)$ *of its image. Between lists it maps the cleaning of the first list at the exponents* (s,t) *onto the cleaning of the second at* (t,s)*.*

Theorem 18.39 (Cleaning has one result)**.** *Let a list be one molecule. Every cleaning of it ends at the same list up to the names of its marks. If that list has more than two marks, every cleaning ends with the same marks, each the same set of marks of the original list; a list that ends with two marks may divide its marks between them in more than one way. Pooling the parallel bunches first, then the series bunches, and repeating, is one cleaning, and every other order ends at the same place.*

Pooling keeps a molecule a molecule, and two poolings at one list can be continued to one list; so the end of a cleaning does not depend on the order of the poolings. The cleaned list has no parallel or series bunch of width one left, each of its marks gathers a set of marks of the molecule, and the period of the molecule is the integral, over the shape simplices of the pooled bunches against their cell kernels, of the periods of the pooled families of the cleaned list.

Chapter 19

Words of a tree

A period is read back from what the Foreman procedure leaves: the bunches peeled at the nodes of a tree of bunches and the leaves at its bottom. The period of a family is the word of such a tree, an iterated fibre integral of stacked Gammas and of the periods of the leaves; opened at the atoms along a chain, or wherever no table is opened on two childcules, the word is one Gramma on the marks and the letters. The word closes in three ways: by draws, with nonnegative terms at nonnegative exponents; at the centre, through the half-sum and the difference of the two letters of a table; and node by node in one scale, where each side of a node of width one is an interval from the decay of its transform to its reach. Read on one tree, the period shows what each node adds, what the leaves contain, and how the integrations of the simplex are shared out.

19.1 The two letters of a table at once

Proposition 19.1 (The pair ladder is the expansion about the midpoint)**.** *For $a, b > 0$, $s, t > 0$ and $x = \frac{a-b}{a+b} \in (-1, 1)$,*

$$a^{-s} b^{-t} = \Big(\frac{a+b}{2}\Big)^{-s-t} (1+x)^{-s} (1-x)^{-t},$$

and the coefficients of the power series $(1+x)^{-s}(1-x)^{-t}$ are

$$[x^k](1+x)^{-s}(1-x)^{-t} = \sum_{i+i'=k} (-1)^i \frac{(s)_i\,(t)_{i'}}{i!\,i'!} = \frac{(s+t)_k}{k!} \Gamma_{\begin{smallmatrix}1\\2\end{smallmatrix}}\begin{bmatrix} s+t \\ s,\ t \end{bmatrix} \int_0^1 w^{s-1}(1-w)^{t-1}(1-2w)^k\, dw.$$

So $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} s,\ t \\ s+t+k \end{bmatrix} [x^k](1+x)^{-s}(1-x)^{-t} = \frac{1}{k!}\int_0^1 w^{s-1}(1-w)^{t-1}(1-2w)^k\, dw$; swapping s and t multiplies the k-th coefficient by $(-1)^k$; and at $s = t$ only even k occur, $[x^{2l}](1-x^2)^{-s} = (s)_l/l!$.

The two letters of a table enter a period as $\Lambda(z)^{-s}\Lambda(-z)^{-t}$. The pair ladder expands such a pair about its midpoint; under one power, the shape of a two-element bunch in the tower decides how much of each letter is taken (Proposition 19.3).

Definition 19.2 (Side shape; half-sum and difference; pattern coefficients)**.** For table j of a family and $w \in [0,1]$ the letter of the *side shape* w is $L_{j,w}(z) = w\Lambda_j(z) + (1-w)\Lambda_j(-z)$; $L_{j,1/2}(z) = \frac{1}{2}\big(\Lambda_j(z) + \Lambda_j(-z)\big)$ is the *half-sum* and $\partial_w L_{j,w}(z) = \Lambda_j(z) - \Lambda_j(-z)$ the *difference* of the two letters of table j. For a bunch B with room R and the Taylor cube $\Lambda_j(z) = \sum_{p \subseteq B} e^{\langle z_B, \chi_p\rangle} \Lambda_{j,(p)}(z_R)$

(Theorem 12.6), $\Lambda_{j,(p)} = 0$ when p is not a pattern of table j, the *pattern coefficients* of the side shape are

$$c_{j,p}(w; z_R) = w\,\Lambda_{j,(p)}(z_R) + (1-w)\,\Lambda^*_{j,(p)}(z_R), \qquad \Lambda^*_{j,(p)}(z_R) = \Lambda_{j,(B\setminus p)}(-z_R),$$

so that $L_{j,w}(z) = \sum_p e^{\langle z_B, \chi_p\rangle} c_{j,p}(w; z_R)$. For one table the index j is omitted.

Proposition 19.3 (The side shape, and what an exchange does to it)**.** *Let a bijection g carry or exchange a family F, with the bijection h of its tables (Definition 11.14). If g carries F, then $L_{h(j),w}(z) = L_{j,w}(g^*z)$; if it exchanges F, then $L_{h(j),w}(z) = L_{j,1-w}(g^*z)$ (Definition 19.2). For $s_j, t_j > 0$, by the tower at the unit configuration on a two-element set J (Theorem 3.2),*

$$\begin{aligned}\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j} &= \Gamma_{\frac12}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j\end{bmatrix}\, \mathsf{B}^{\{J\}}_{(s_j,t_j),\, s_j+t_j;\, I_J}\big(\Lambda_j(z), \Lambda_j(-z)\big)\\ &= \Gamma_{\frac12}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j\end{bmatrix} \int_0^1 w^{s_j-1}(1-w)^{t_j-1}\, \mathsf{Z}_{s_j+t_j;(1)}\big(L_{j,w}(z)\big)\, dw,\end{aligned}$$

and for one table an exchange together with $w \mapsto 1-w$ keeps $\mathsf{Z}_{s+t;(1)}(L_w(z))$, keeping the factor $w^{s-1}(1-w)^{t-1}$ exactly when $s = t$.

The side shape weighs the basis letter against the cobasis letter, and an exchange of the table turns it over. The half-sum is even in z and the difference odd: $\Lambda(z)\Lambda(-z) = L_{1/2}(z)^2 - \frac14\big(\partial_w L_w(z)\big)^2$.

19.2 The junction in pattern form

Theorem 19.4 (The junction in pattern form)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a molecule on n marks with real exponents $s_j, t_j \geq 0$, its masses on the balance wall with every proper wall positive, and let B be a proper bunch with room R and $g \in R$. For a table at a full lean put $w_j = 1$ when $t_j = 0$ and $w_j = 0$ when $s_j = 0$. With the pattern coefficients $c_{j,p}(w; z_R)$ of the side shapes (Definition 19.2),*

$$\begin{aligned}P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{2^{n-1}}{\prod_{j:\, s_j>0}\Gamma(s_j)\prod_{j:\, t_j>0}\Gamma(t_j)} \int \prod_{j:\, s_j, t_j>0} w_j^{s_j-1}(1-w_j)^{t_j-1} \int_{\mathbb{R}^{R-g}} e^{2\langle \boldsymbol{\mu}|_R, z_R\rangle}\\ \times \int \prod_{j\ \mathit{active}} y_j^{s_j+t_j-1}\, \mathsf{G}_{2\boldsymbol{\mu}|_B;(\chi_p)_p}\Big(\Big(\sum_{j\ \mathit{active}} y_j\, c_{j,p}(w_j; z_R)\Big)_p\Big)\, d\mathbf{y}\, dz_{R-g}\Big|_{z_g=0}\, d\mathbf{w},\end{aligned}$$

with $d\mathbf{y} = \prod_j dy_j$ over the active tables, $y_j \in (0,\infty)$, and $d\mathbf{w} = \prod_j dw_j$ over the tables with $s_j, t_j > 0$, $w_j \in [0,1]$; the Gramma on the scales e^{z_B} with one term for each $p \subseteq B$ whose coefficient $\sum_j y_j c_{j,p}(w_j; z_R)$ is not 0. A table at a full lean has no integral over its side shape: at $(s_j, 0)$ its coefficients are $y_j\Lambda_{j,(p)}(z_R)$, at $(0, t_j)$ they are $y_j\Lambda^*_{j,(p)}(z_R)$.

The two letters of every table become one letter, the side shapes are integrated last, and the bunch is integrated on its own orthant against the patterns of all the tables at once: each pattern is one term of one Gramma on the bunch, whose coefficient adds the pattern coefficients of the tables, each at its own scale y_j, and the room sits in the coefficients. A map that keeps the bunch moves the coefficients with the side shapes and the tables.

Proposition 19.5 (What a map keeping the bunch gives the coefficients)**.** *In Theorem 19.4 let a bijection $\tilde{g}$ of $\mathcal{E}$ keep B, with $\tilde{g}_R$ its restriction to the room, and carry or exchange F with the bijection h of its tables. If $\tilde{g}$ carries F, then $c_{h(j),\tilde{g}(p)}(w;z_R) = c_{j,p}(w;\tilde{g}_R^* z_R)$ for every j, p and w; if it exchanges F, then $c_{h(j),\tilde{g}(p)}(1-w;z_R) = c_{j,p}(w;\tilde{g}_R^* z_R)$.*

Proposition 19.6 (Level form of the half-sum)**.** *In Theorem 19.4 the half-sum and the difference of table j have the pattern coefficients $c_{j,p}(\frac{1}{2};z_R) = \frac{1}{2}\big(\Lambda_{j,(p)} + \Lambda^*_{j,(p)}\big)(z_R)$ and $\partial_w c_{j,p}(w;z_R) = \big(\Lambda_{j,(p)} - \Lambda^*_{j,(p)}\big)(z_R)$.*

(i) For every table, $c_{j,B\setminus p}(\frac{1}{2};-z_R) = c_{j,p}(\frac{1}{2};z_R)$ and $\partial_w c_{j,B\setminus p}(w;-z_R) = -\partial_w c_{j,p}(w;z_R)$.

(ii) The half-sum of table j has level form on B (Proposition 14.12), $c_{j,p}(\frac{1}{2};\cdot)$ depending on $|p|$ alone, exactly when every transposition of two marks of B fixing the rest carries the multiset $\mathcal{B}_j \uplus \{\mathcal{E}\setminus T :\ T\in\mathcal{B}_j\}$, with its weights, onto itself; if $2r_j \neq n$ this is the condition that every such transposition carries table j (Proposition 14.12).

(iii) If every such transposition carries or exchanges table j, its half-sum has level form and the square of its difference is unchanged by every permutation of B.

19.3 The word of a tree

Definition 19.7 (Tree of bunches)**.** A *tree of bunches* of a molecule F has nodes v with marks M_v, the root having $M_v = \mathcal{E}$. A node v that is not a leaf carries a proper bunch $B_v \subsetneq M_v$; the tables at v have their children at B_v, the *tables made at* v, and the molecules $R_{v,1},\dots,R_{v,q_v}$ of these children on $M_v \setminus B_v$, in a fixed order, are the childcules of v (Definition 14.1), each a node whose tables are the restrictions to it of the tables made at v. A leaf v has $B_v = M_v$. The tables at a node are thus the children of the tables of F at the patterns shown on the bunches above it, restricted to its marks (Proposition 14.5); with a native gauge they carry as exponents the shares of the native splits at the nodes above (Theorem 12.23). Its descents are the sequences of nodes from the root to a leaf, each a childcule of the one before, as on the Foreman tree (Definition 14.17), which is a tree of bunches (Definition 14.4).

Corollary 19.8 (The tree of a linked molecule is a chain)**.** *Let some active table of a molecule F be connected, as for a molecule of one table. Every tree of bunches of F is then a chain $M_0 \supset M_1 \supset \cdots \supset M_d$ with $M_{k+1} = M_k \setminus B_k$, above one leaf: the tables at M_k are the children of the tables of F at $S_k = B_0 \sqcup \cdots \sqcup B_{k-1}$ (Proposition 14.5), which form one molecule (Lemma 17.13). With a native gauge their exponents are the shares of the native split, and on the Foreman tree the family at M_k is the peel $F_{S_k}(\mathbf{a}^\star,\mathbf{b}^\star)$ (Definition 14.1).*

Corollary 19.9 (The word of a tree)**.** *Let F have real data with $s_j, t_j \geq 0$, a native gauge and its masses on the balance wall, and let the children's weights factor across the childcules at every node of a tree of bunches. Run the port of Theorem 17.12(ii) at every node, on the native splits. A table $\mathcal{B}'$ made at a node, with the exponents $(a_{\mathcal{B}'}, b_{\mathcal{B}'})$, is the direct sum of its restrictions to the childcules there (Proposition 7.8); let $q_{\mathcal{B}'}$ be the number of those childcules, other than leaves, on which its letter is not a monomial. Then*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int_{\mathcal{X}} \prod_{\mathcal{B}'} \big(\Gamma(a_{\mathcal{B}'})\Gamma(b_{\mathcal{B}'})\big)^{1-q_{\mathcal{B}'}} \prod_{\text{leaves } v} P_{F_v}(\mathbf{m}|_{M_v};\mathbf{a}_v,\mathbf{b}_v),$$

the word of the tree*: an iterated fibre integral through the native splits, over the fibre $\mathcal{X}$ cut out by the share sums and by the $|B_v|+q_v-1$ pins of every node v that is not a leaf (Theorem 17.12(ii)),*

a monomial restriction entering the pins of its node as a load; F_v is the family of the tables at the leaf v with their exponents $(\mathbf{a}_v, \mathbf{b}_v)$, and a letter with exponent zero carries no shares and no Gamma. A table opened on one childcule carries no factor, and when an active table is connected every $q_{\mathcal{B}'}$ is 0 *or* 1 *(Corollary 19.8).*

Theorem 19.10 (Building back up)**.** *In Corollary 19.9 open every leaf by the atom formula on the balance wall (Corollary 17.15), a unit being a monomial, and write the shares (a, b) of one pattern of one table in pair coordinates, $\sigma = \frac{a+b}{2}$ and $\delta = \frac{a-b}{2}$.*

(i) Polycules multiply. *At every node and at every point of its fibre on the wall contour of the native split (Theorem 17.12(iii)), the integral over the shares below the node is the product over its childcules of their words: the polycule of the childcules is evaluated as the product of its molecules, a unit contributing a monomial in the weights.*

(ii) The pins are odd. *If every table has $s_j, t_j > 0$, every pin other than the share sums involves the odd shares alone: the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$ of the bunch, the pins of the childcules,*

$$\kappa_{R_i} = \mu_{R_i} - \sum_{j,p} \delta_{j,p} \big(2r_{j,(p)}(R_i) - |R_i| \big) = 0,$$

the sum over the tables made at the node, indexed as in Theorem 17.12 by the table j at the node and the pattern p, and the pins of the atoms. The share sums are $\sum_p \sigma_{j,p} = \sigma_j$ and $\sum_p \delta_{j,p} = \delta_j$, so the fibre is the product of an even fibre, cut by the even sums alone, and an odd fibre.

(iii) One Gramma. *If at every node every table made there is a monomial on all its childcules but at most one, in particular along a chain (Corollary 19.8), the leaf shares are one for each basis $T \in \mathcal{B}_j$ and letter, the pins of the tree are, for a fixed mark g, the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$, $e \in \mathcal{E} - g$, of Corollary 17.15 after a change of determinant ± 1, and the word is the atom formula, one Gramma on the marks and the letters:*

$$\prod_{j:\, s_j > 0} \Gamma(s_j) \prod_{j:\, t_j > 0} \Gamma(t_j)\, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \mathsf{G}_{(\boldsymbol{\mu}|_{\mathcal{E}-g}; \mathbf{s}; \mathbf{t});\, t_F^{(g)}}(\boldsymbol{\theta} \oplus \boldsymbol{\theta}),$$

with $t_F^{(g)}$ the configuration t_F of Theorem 8.3 without its unit columns and without the row of g: on the scales $(\mathcal{E} - g) \sqcup \{v_j\} \sqcup \{w_j\}$, the columns $(\frac{1}{2}\chi_T|_{\mathcal{E}-g}; \mathbf{e}_j; 0)$ and $(-\frac{1}{2}\chi_T|_{\mathcal{E}-g}; 0; \mathbf{e}_j)$, $T \in \mathcal{B}_j$, each with the coefficient $\theta_{j,T}$, a letter with exponent zero omitted: one scale for each mark but g, which carries an odd pin, and one for each letter.

(iv) The even shares close jointly. *In (iii) let every table have $s_j, t_j > 0$, and write $c_{j,T} = b_{j,T} - a_{j,T} = -2\delta_{j,T}$. At fixed differences the K-shares of table j close into one scale,*

$$\int_{\sum_T a_{j,T} = s_j} \prod_{T \in \mathcal{B}_j} \Gamma(a_{j,T})\, \Gamma(a_{j,T} + c_{j,T})\, \theta_{j,T}^{-2a_{j,T}} = \int_0^\infty y^{s_j - 1} \prod_{T \in \mathcal{B}_j} \mathsf{G}_{c_{j,T};(1,-1)}\big(1, \theta_{j,T}^2 y\big)\, dy,$$

and $\prod_j \Gamma(s_j)\Gamma(t_j) P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is the fibre integral over the differences, with the pins $\sum_T c_{j,T} = t_j - s_j$ and $-\frac{1}{2}\sum_{j,T} c_{j,T}\chi_T(e) = \mu_e$, $e \in \mathcal{E} - g$, of $\prod_j \prod_T \theta_{j,T}^{-c_{j,T}}$ times these scale integrals; on the wall contour of the native split (Theorem 17.12(iii)) every integral converges absolutely. The fibre of differences has dimension $\sum_j (|\mathcal{B}_j| - 1) - (n - 1)$. At the centre, every table riding the line, $s_j = t_j$, with the masses $\sigma\mathbf{1}$, the pins are $\sum_{T \in \mathcal{B}_j} c_{j,T} = 0$ for every j and $\sum_j M_j^{\mathsf{T}} \mathbf{c}_j = 0$, with $\mathbf{c}_j = (c_{j,T})_{T \in \mathcal{B}_j}$ and the incidence matrices $(M_j)_{Te} = \mathbf{1}_T(e)$, and the contour is $\operatorname{Re} \mathbf{c} = 0$.

Built back up, the word is the period read from its atoms. At every node the childcules multiply and the pins see only the odd shares; along a chain nothing is opened twice, and the word is the atom formula itself, one Gramma on the marks and the letters whose configuration is that of the interest with the masses' own columns removed. The even shares are free but for their sums, and they close table by table into one scale of Grammas $\mathsf{G}_{\nu;(1,-1)}$, as at the centre of a molecule (Theorem 19.10(iv)) and for the table that is one circuit (Corollary 17.20).

19.4 Closing the tree by draws

Proposition 19.11 (The draw series)**.** *Let $U=(U_Q)_{Q\in\mathcal{P}}$ have independent blocks $U_Q\sim\mathrm{Dir}_{\mathbf{m}|_Q}$ on a partition $\mathcal{P}$ of the marks, and let $P_1,\dots,P_q$ be nonzero polynomials with nonnegative coefficients, P_i homogeneous of degree $g_{i,Q}$ in the variables of each block Q. For an exponent vector $\mathbf{a}$ with block sums $g_{i,Q}$ put $b_{\mathbf{a}}=\prod_Q g_{i,Q}!/\prod_e a_e!$ and $C_i=\max_{\mathbf{a}}\big([u^{\mathbf{a}}]P_i\big)/b_{\mathbf{a}}$. Then $\prod_Q\big(\sum_{e\in Q}u_e\big)^{g_{i,Q}}-P_i/C_i$ has nonnegative coefficients and equals $1-P_i/C_i\in[0,1)$ on the interior of the source, and for $\nu_i\ge 0$*

$$\mathbb{E}_{U\sim\bigotimes_Q\mathrm{Dir}_{\mathbf{m}|_Q}}\Big[\prod_i P_i(U)^{-\nu_i}\Big]=\prod_i C_i^{-\nu_i}\sum_{\mathbf{k}\in\mathbb{N}^q}\prod_i\frac{(\nu_i)_{k_i}}{k_i!}\times\mathbb{E}_{U\sim\bigotimes_Q\mathrm{Dir}_{\mathbf{m}|_Q}}\Big[\prod_i\Big(1-\frac{P_i(U)}{C_i}\Big)^{k_i}\Big],$$

every term nonnegative, both sides finite or infinite together. The left side is the Trailmoment of Proposition 5.22,

$$\Big(\mathsf{T}_{\boldsymbol{\nu};I_q}[(P_i)_i]\bigotimes_Q\mathrm{Dir}_{\mathbf{m}|_Q}\Big)\Big(\prod_Q\Delta_Q\Big),$$

and each expectation on the right is a finite nonnegative combination of the moments of Theorem 3.11. For a family $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ with $s_j,t_j\ge 0$, one block, $P_{2j-1}=K_j$ and $P_{2j}=\Psi_j$ with the indices (s_j,t_j), the constants are $C_{2j-1}=\theta_{j,\max}/r_j!$ and $C_{2j}=\theta_{j,\max}/(n-r_j)!$, $\theta_{j,\max}=\max_T\theta_{j,T}$; the left side times $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ is $\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-s_j}\Psi_j(u)^{-t_j}\,d\omega_{\mathcal{E}}(u)$, which on the balance wall is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$; and the expectation of the term $\mathbf{k}=(k_j,l_j)_j$ is the chance that, in the urn that draws a mark with probability proportional to its current mass and then adds one to that mass, for $j=1,\dots,N$ in turn k_j rounds of r_j draws and then l_j rounds of $n-r_j$ draws all fail for table j, a round failing when its marks are not a basis, respectively not a cobasis, of table j, and, when they are the basis T, respectively the cobasis $\mathcal{E}\setminus T$, with probability $1-\theta_{j,T}/\theta_{j,\max}$: a finite nonnegative combination of Dirichlet moments, a finite sum of them for lists. This series is Theorem 15.13(i) with the references $M_{2j-1}=C_{2j-1}$ and $M_{2j}=C_{2j}$.

Corollary 19.12 (At $\delta=0$ and at a full lean the period reads the law of one letter for each table)**.** *Let a family $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$, every table active, carry its masses $\mathbf{m}$ on the balance wall with every proper wall positive, every table riding the line, $(s_j,t_j)=(\sigma_j,\sigma_j)$, or at a full lean, $(s_j,0)$ or $(0,t_j)$. Put $L_j=\mathbf{E}_j=K_j\Psi_j$ and $\nu_j=\sigma_j$ when table j rides the line, $L_j=K_j$ and $\nu_j=s_j$ at $(s_j,0)$, and $L_j=\Psi_j$ and $\nu_j=t_j$ at $(0,t_j)$, and let $\phi_{\mathbf{m}}$ be the joint law of $(L_1(U),\dots,L_N(U))$ for $U\sim\mathrm{Dir}_{\mathbf{m}}$ (Proposition 8.11). The kernel $\prod_j a_j^{-s_j}b_j^{-t_j}$ of the letters $a_j=K_j$ and $b_j=\Psi_j$ is the kernel $\mathsf{Z}_{\boldsymbol{\nu};I_N}\big((L_j)_j\big)$, one scale for each table, so*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\,\big(\mathsf{T}_{\boldsymbol{\nu};I_N}[\mathrm{id}]\,\phi_{\mathbf{m}}\big)\big((0,\infty)^N\big);$$

with one table $\phi_{\mathbf{m}}$ is a law on the half-line and the right side is $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} (\mathsf{T}_{\nu;(1)}[x]\,\phi_{\mathbf{m}})((0,\infty))$, at $\delta = 0$ the law $\mathbf{E}_* \operatorname{Dir}_{\mathbf{m}}$ of the even letter $\mathbf{E}(U) = K(U)\Psi(U)$ (Theorem 5.9). The law $\phi_{\mathbf{m}}$ is carried by $\prod_j (0, \max_{\Delta_{\mathcal{E}}} L_j]$, it is determined by its moments, and its moments are finite sums of Dirichlet moments (Theorem 3.11).

Theorem 19.13 (The draw series along a chain)**.** *Let a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have the exponents $s_j, t_j \geq 0$ and masses $\mathbf{m} \in \mathbb{R}^{\mathcal{E}}_{>0}$, and let $M_0 = \mathcal{E} \supset M_1 \supset \cdots \supset M_d$ be a chain of sets with $B_i = M_i \setminus M_{i+1}$, $i < d$, and $S_i = B_0 \sqcup \cdots \sqcup B_{i-1}$, $S_0 = \varnothing$, $S_{d+1} = \mathcal{E}$. A round of draws* extends on *S for table j when its draws inside S are distinct and are $T \cap S$ for some basis T of table j. In this statement write $\theta_j(S; P)$ for the largest weight $\theta_{j,T}$ over the bases T of table j with $T \cap S = P$, and say that a round that extends on S_i and on S_{i+1}, with the traces P_i and P_{i+1}, is* accepted at *S_{i+1} with probability $\theta_j(S_{i+1}; P_{i+1})/\theta_j(S_i; P_i)$, independently for each i. Let $\mathfrak{f}_{j,i}(u)$, $0 \leq i \leq d$, be the chance that a round of r_j draws from u extends on S_i and is accepted at $S_1, \ldots, S_i$ but does not both extend on S_{i+1} and get accepted there, and $\mathfrak{f}^*_{j,i}$ the same for rounds of $n - r_j$ draws and the cobases $\mathcal{E} \setminus T$ with the weights $\theta_{j,T}$. Then, with $\theta_{j,\max}$ the largest weight of table j, $1 - r_j!\, K_j/\theta_{j,\max} = \sum_i \mathfrak{f}_{j,i}$, $1 - (n - r_j)!\, \Psi_j/\theta_{j,\max} = \sum_i \mathfrak{f}^*_{j,i}$, and*

$$\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\omega_{\mathcal{E}} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_j \Big(\frac{r_j!}{\theta_{j,\max}}\Big)^{s_j} \Big(\frac{(n-r_j)!}{\theta_{j,\max}}\Big)^{t_j} \times \sum_{\mathbf{k},\mathbf{l}} \prod_j \frac{(s_j)_{|\mathbf{k}_j|}\,(t_j)_{|\mathbf{l}_j|}}{\mathbf{k}_j!\,\mathbf{l}_j!}\; \mathbb{E}_{U \sim \operatorname{Dir}_{\mathbf{m}}} \Big[\prod_{j,i} \mathfrak{f}_{j,i}(U)^{k_{j,i}}\, \mathfrak{f}^*_{j,i}(U)^{l_{j,i}} \Big],$$

over $\mathbf{k}_j, \mathbf{l}_j \in \mathbb{N}^{d+1}$, every term a finite nonnegative combination of Dirichlet moments, a finite sum of them for lists, whose rounds are always accepted. The numbers $\sum_{i' \geq i} k_{j,i'}$ of failing basis rounds of table j that pass $B_0, \ldots, B_{i-1}$ decrease with i: they are nested, as the coordinates of the region X_N of a trail are (Definition 5.23). Let a bijection g keep each B_i and the masses, let h be a bijection of the tables, and put $(g_* u)_{g(e)} = u_e$. If g maps the bases of every table j onto those of table $h(j)$ with their weights, then $\mathfrak{f}_{h(j),i}(g_* u) = \mathfrak{f}_{j,i}(u)$ and $\mathfrak{f}^*_{h(j),i}(g_* u) = \mathfrak{f}^*_{j,i}(u)$; if $T \mapsto \mathcal{E} \setminus g(T)$ does, then $\mathfrak{f}^*_{h(j),i}(g_* u) = \mathfrak{f}_{j,i}(u)$ and $\mathfrak{f}_{h(j),i}(g_* u) = \mathfrak{f}^*_{j,i}(u)$, and the array of expectations is unchanged when every $(\mathbf{k}_j, \mathbf{l}_j)$ is replaced by $(\mathbf{l}_{h^{-1}(j)}, \mathbf{k}_{h^{-1}(j)})$; for one table, where an exchange forces $2r = n$, this is the symmetry under $\mathbf{k} \leftrightarrow \mathbf{l}$.

Theorem 19.14 (The draw series along a tree)**.** *Let a molecule $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ on $\mathcal{E}$ carry $s_j, t_j \geq 0$ and masses $\mathbf{m} \in \mathbb{R}^{\mathcal{E}}_{>0}$, and take a tree of bunches (Definition 19.7) at every node of which the children's weights factor across the childcules, as they do for lists: for a table $\mathcal{B}'$ at a node v and a pattern p of $\mathcal{B}'$ at B_v, the restrictions $\mathcal{B}'_{(p)}|R_{v,i}$ of the child to the childcules (Proposition 7.8) carry weights whose products are the weights of the child, $\theta_{(p),X} = \prod_i \theta_{(p),i}(X \cap R_{v,i})$. Write $\theta_{\max}(\mathcal{B}')$ for the largest weight of a table $\mathcal{B}'$, and say that a round of draws that is a basis T of $\mathcal{B}'$ is* accepted *with probability $\theta_T/\theta_{\max}(\mathcal{B}')$. A cobasis round of table j is a basis round of its dual table, whose children are the duals of the children, with the same weights (Lemma 14.7(ii) with the identity). The* coordinates of the tree *are, at each node v, the totals $x^{(v)} \in \Delta_{\{B_v, R_{v,1}, \ldots, R_{v,q_v}\}}$ of the shape of M_v on its blocks and the shape $u^{(v)} \in \Delta_{B_v}$ of the bunch; under $\operatorname{Dir}_{\mathbf{m}}$ they are independent, with the laws $\operatorname{Dir}_{(m_{B_v}, m_{R_{v,1}}, \ldots, m_{R_{v,q_v}})}$ and $\operatorname{Dir}_{\mathbf{m}|_{B_v}}$ (Theorem 3.12). For a table $\mathcal{B}'$ at a node write its chances as functions of the coordinates of the subtree of that node.*

(i) A round passes a node, then each childcule. *For a table $\mathcal{B}'$ of rank r' at a node v, the chance $\mathfrak{s}_{\mathcal{B}'} = r'!\, K_{\mathcal{B}'}/\theta_{\max}(\mathcal{B}')$ that r' draws from the shape of M_v are a basis of $\mathcal{B}'$ and are accepted*

is

$$\mathfrak{s}_{\mathcal{B}'} = \sum_{p} \mathfrak{p}_{\mathcal{B}',p} \prod_{i} \mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i}},$$

$$\mathfrak{p}_{\mathcal{B}',p} = \frac{\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_{v,i}\big)}{\theta_{\max}(\mathcal{B}')} \, \frac{r'!}{\prod_i r'_{p,i}!} \, \big(x^{(v)}_{B_v}\big)^{|p|} \big(u^{(v)}\big)^{p} \prod_i \big(x^{(v)}_{R_{v,i}}\big)^{r'_{p,i}},$$

over the patterns p of $\mathcal{B}'$ at B_v, with $r'_{p,i}$ the rank on $R_{v,i}$ of the child $\mathcal{B}'_{(p)}$: $\mathfrak{p}_{\mathcal{B}',p}$ is the chance that the round passes *v showing p, its draws in B_v being the marks of p and its numbers of draws in the childcules the $r'_{p,i}$, and is accepted there, with the probability given by its first factor, which is at most 1; and $\mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i}}$ is the chance that its draws in $R_{v,i}$ are a basis of the restriction of the child there and are accepted. At a leaf $\mathfrak{s}_{\mathcal{B}'} = r'!\, K_{\mathcal{B}'}/\theta_{\max}(\mathcal{B}')$.*

(ii) A failure is met at one node. *The chance $1 - \mathfrak{s}_{\mathcal{B}'}$ that the round fails, by not being a basis or by not being accepted, is $\sum_w \mathfrak{f}_{\mathcal{B}'\to w}$ over the nodes and leaves w of the subtree of v, where $\mathfrak{f}_{\mathcal{B}'\to v} = 1 - \sum_p \mathfrak{p}_{\mathcal{B}',p}$ and, for w in the subtree of $R_{v,i}$,*

$$\mathfrak{f}_{\mathcal{B}'\to w} = \sum_{p} \mathfrak{p}_{\mathcal{B}',p} \prod_{i'<i} \mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i'}} \; \mathfrak{f}_{\mathcal{B}'_{(p)}|R_{v,i}\to w},$$

the chance that the round first fails at w. Each $\mathfrak{f}_{\mathcal{B}'\to w}$ is a finite sum of products, over the blocks of the tree, of polynomials with nonnegative coefficients in one block each.

(iii) The series. *With $\mathfrak{f}_{j,w}$ and $\mathfrak{f}^*_{j,w}$ these chances for the basis and the cobasis rounds of table j of F from the root, and $\theta_{j,\max}$ the largest weight of table j,*

$$\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} \, d\omega_{\mathcal{E}} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_j \Big(\frac{r_j!}{\theta_{j,\max}}\Big)^{s_j} \Big(\frac{(n-r_j)!}{\theta_{j,\max}}\Big)^{t_j}$$
$$\times \sum_{\mathbf{k},\mathbf{l}} \prod_j \frac{(s_j)_{|\mathbf{k}_j|}\,(t_j)_{|\mathbf{l}_j|}}{\mathbf{k}_j!\,\mathbf{l}_j!} \; \mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}\Big[\prod_{j,w} \mathfrak{f}_{j,w}(U)^{k_{j,w}} \, \mathfrak{f}^*_{j,w}(U)^{l_{j,w}}\Big],$$

every term nonnegative, both sides finite or infinite together; on the balance wall the left side is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. Each expectation is a finite nonnegative combination of products over the blocks of Dirichlet moments, a finite sum of them for lists: at every node, the node's own moments times the product over its childcules of theirs.

(iv) Nesting along the tree. *The numbers $k_{j,\ge w} = \sum_{w'} k_{j,w'}$, over the w' in the subtree of w, of the basis rounds of table j that first fail in the subtree of w, satisfy $k_{j,\ge w} = k_{j,w} + \sum_{w'} k_{j,\ge w'}$ over the childcules w' of w, and likewise for the cobasis rounds: along every descent they are nested, as the coordinates of the region X_N of a trail are (Definition 5.23). On a chain this is Theorem 19.13.*

At nonnegative exponents every share of the word closes at once. The draw series closes each letter about its largest coefficient on the simplex, and the tree only sorts its failures: a round that fails is charged to the first node at which it stops passing, the chances of passing are monomials in the coordinates of that node, and the coordinates of different nodes are independent. So every term is a nonnegative combination of products, over the blocks of the tree, of Dirichlet moments, and the counts of failures nest along the descents as the coordinates of the region X_N of a trail do.

Proposition 19.15 (Rates at the centre)**.** *Let a family on $n \geq 3$ marks be at the centre, every table with the exponents (σ_j, σ_j), with the masses $\sigma\mathbf{1}$, $\sigma = \sum_j \sigma_j$, and let every active table be connected.*

(i) The walls are $\kappa_S = \sum_j \sigma_j c_j(S)$ (Proposition 8.21(ii)), so $\kappa_S = \kappa_{\mathcal{E}\setminus S}$.

(ii) The rate of the peak closing (Theorem 15.13(ii)) is $\varpi = \sigma/\max_j \max(r_j - 1, n - r_j - 1)$, the maximum over the active tables, attained at the faces $S = \mathcal{E} - e$, where every active table has $c_j = 1$, $d_j = r_j - 1$ and $d_j^* = n - r_j - 1$.

(iii) Near the vertex of e, with $u_e = 1 - x$ and $u_{\mathcal{E}-e} = xv$, $v \in \Delta_{\mathcal{E}-e}$, every active table has

$$K_j(u) = (1-x)\, x^{r_j - 1} K_{j,/e}(v) + x^{r_j} K_{j,\setminus e}(v),$$
$$\Psi_j(u) = (1-x)\, x^{n - r_j - 1} \Psi_{j,\setminus e}(v) + x^{n - r_j} \Psi_{j,/e}(v);$$

the vertex product *of e in table j is $K_{j,/e}\Psi_{j,\setminus e}$, the product of the letters of the two minors that dominate as $x \to 0$.*

(iv) For two marks e, f and a bijection β with $\beta(e) = f$ that carries or exchanges the family, with the bijection h of its tables, the restriction $\beta : \mathcal{E} - e \to \mathcal{E} - f$ takes the vertex product of e in table j to that of f in table $h(j)$. For one table such a β exists exactly when e and f lie in one class at the centre (Corollary 11.17).

19.5 Devices at the centre

Proposition 19.16 (The half-sum and the difference of a table with $2r = n$)**.** *Let $F = ((\mathcal{B}, \theta); \sigma, \sigma)$, $\sigma > 0$, have $2r = n$ and masses $\mathbf{m}$ on its balance wall. The half-sum $\frac{1}{2}(K + \Psi) = \frac{1}{2}\sum_T \theta_T\big(\alpha^T + \alpha^{\mathcal{E}\setminus T}\big)$ is homogeneous of degree r and unchanged by $T \mapsto \mathcal{E} \setminus T$; the difference $K - \Psi$ vanishes at the table's gauge; in signed coordinates they are $e^{\langle \mathbf{1}, z\rangle} L_{1/2}(z)$ and $e^{\langle \mathbf{1}, z\rangle}\partial_w L_w(z)$ (Definition 19.2). Then $K\Psi = \big(\frac{1}{2}(K+\Psi)\big)^2 - \frac{1}{4}(K-\Psi)^2$ with $|K - \Psi| < K + \Psi$ on the orthant, and by Proposition 19.1*

$$(K\Psi)^{-\sigma} = \sum_{l \geq 0} \frac{(\sigma)_l}{l!\, 4^l} (K - \Psi)^{2l} \Big(\frac{K + \Psi}{2}\Big)^{-2\sigma - 2l},$$

a series of nonnegative terms. For a mark e with room $R = \mathcal{E} - e$ write $\frac{1}{2}(K + \Psi) = h_0(\alpha_R) + \alpha_e h_1(\alpha_R)$, so that, by Definition 7.3, $h_0 = \frac{1}{2}(K_{\setminus e} + \Psi_{/e})$, $h_1 = \frac{1}{2}(K_{/e} + \Psi_{\setminus e})$ and $h_1(u) = u^R h_0(u^{-1})$. Expanding $(K - \Psi)^{2l} = \sum_{\mathbf{a}} \big([\alpha^{\mathbf{a}}](K - \Psi)^{2l}\big)\, \alpha^{\mathbf{a}}$ and integrating the mark e of each term as the Beta corner of a linear letter (Proposition 15.1 with one coefficient), with $w_l = 2\sigma + 2l$,

$$P_F(\mathbf{m}; \sigma, \sigma) = \sum_{l \geq 0} \frac{(\sigma)_l}{l!\, 4^l} \sum_{\mathbf{a}} \big([\alpha^{\mathbf{a}}](K - \Psi)^{2l}\big)\, \Gamma_{\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}} \begin{bmatrix} m_e + a_e, & w_l - m_e - a_e \\ & w_l \end{bmatrix}$$
$$\times \int_{\Delta_R} u^{(\mathbf{m}+\mathbf{a})|_R - \mathbf{1}} h_0(u)^{-(w_l - m_e - a_e)} h_1(u)^{-(m_e + a_e)}\, d\omega_R(u),$$

whenever the terms converge absolutely.

The half-sum is a positive polynomial whose support is closed under complements, but it is not in general the basis polynomial of a table: the bases together with the cobases need not have the exchange property, even for connected tables, so the integrals left over the room are integrals of a positive polynomial against its complement and not periods of a family. When complementation maps the bases onto themselves the difference vanishes and the room is a period.

Corollary 19.17 (Merged Skeleton values)**.** *If $T \mapsto \mathcal{E} \setminus T$ maps the bases of a table $(\mathcal{B}, \theta)$ onto themselves with $\theta_{\mathcal{E}\setminus T} = \theta_T$, then $2r = n$, no mark is a loop or a coloop, $\Psi = K$, $\Psi_{\setminus e} = K_{/e}$ and $\Psi_{/e} = K_{\setminus e}$ for every mark e, and the two Skeleton values of the scale of e coincide at every gauge (Corollary 15.3). For $F = ((\mathcal{B}, \theta); s, t)$ with $s, t \geq 0$, $s + t > 0$, its masses on the balance wall with every proper wall positive, and a mark e, one has $0 < m_e < s + t$ and*

$$P_F(\mathbf{m}; s, t) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} m_e, & s+t-m_e \\ & s+t \end{bmatrix} P_{\mathcal{B}\setminus e}\big(\mathbf{m}|_{\mathcal{E}-e};\ s+t-m_e,\ m_e\big),$$

the period of the deletion $(\mathcal{B}\setminus e, \theta)$ on $\mathcal{E}-e$ at the exponents $(s+t-m_e, m_e)$, on its balance wall; so $P_F(\mathbf{m}; s, t)$ depends on the exponents only through $s+t$, and at the centre, $s = t = \sigma$ and $\mathbf{m} = \sigma\mathbf{1}$, the identity reads $P_F(\sigma\mathbf{1}; \sigma, \sigma) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} \sigma, & \sigma \\ & 2\sigma \end{bmatrix} P_{\mathcal{B}\setminus e}(\sigma\mathbf{1}|_{\mathcal{E}-e}; \sigma, \sigma)$.

Theorem 19.18 (The scale integral of one mark)**.** *Let a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ carry real exponents $s_j, t_j \geq 0$, its masses on the balance wall with every proper wall positive, and let e be a mark that is neither a loop nor a coloop in any active table, with room $R = \mathcal{E} - e$; write $\Lambda_{j,\setminus e}$ and $\Lambda_{j,/e}$ for the centred polynomials of the deletion and the contraction of table j, so that $\Lambda_j(z) = e^{z_e}\Lambda_{j,/e}(z_R) + e^{-z_e}\Lambda_{j,\setminus e}(z_R)$ (Theorem 12.6).*

(i) At the gauge $(1, e^{2z_R})$ the scale integral of the bunch $\{e\}$ (Theorem 15.2) is

$$\begin{aligned}&\mathrm{Sc}_{\{e\}}\big((1, e^{2z_R}); \mathbf{m}; \mathbf{s}, \mathbf{t}\big)\\ &= e^{-2\sigma\langle\mathbf{1}, z_R\rangle}\int_0^\infty x^{m_e-1}\prod_j \big(\Lambda_{j,\setminus e}(z_R) + x\Lambda_{j,/e}(z_R)\big)^{-s_j}\big(\Lambda_{j,/e}(-z_R) + x\Lambda_{j,\setminus e}(-z_R)\big)^{-t_j}dx,\end{aligned}$$

$\sigma = \sum_j \sigma_j$ the total strength: up to the factor $e^{-2\sigma\langle\mathbf{1}, z_R\rangle}$ it depends only on m_e, the exponents and the functions $\Lambda_{j,\setminus e}$ and $\Lambda_{j,/e}$ on the room, and Corollary 15.7 at the bunch $\{e\}$ reads, for every $g \in R$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-2}\int_{\mathbb{R}^{R-g}} e^{2\langle\mathbf{m}|_R, z_R\rangle}\ \mathrm{Sc}_{\{e\}}\big((1, e^{2z_R}); \mathbf{m}; \mathbf{s}, \mathbf{t}\big)\, dz_{R-g}\Big|_{z_g=0}.$$

(ii) At an atom $\Upsilon = ((s_j, t_j)_j; \alpha^\star)$ with its target $\boldsymbol{\tau}$, under $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ the law of the log-ratios $(y_{g_k} - y_e)_{k=1}^{n-1}$, read in an order $\mathbf{g}_e$ of the room that writes the potential from e as the one pair of functions of Proposition 10.4, does not depend on e.

(iii) For one table at the centre, $s = t = \sigma$ and $\boldsymbol{\mu} = 0$, with the even statistics $E_{\setminus e}$ and $E_{/e}$ of the deletion and the contraction (Theorem 7.14) and a two-element set J, the functions of z_R understood,

$$\begin{aligned}&\int_0^\infty x^{\sigma-1}\big(\Lambda_{\setminus e} + x\Lambda_{/e}\big)^{-\sigma}\big(\Lambda_{/e}(-\cdot) + x\Lambda_{\setminus e}(-\cdot)\big)^{-\sigma}dx\\ &\qquad = e^{-\frac{\sigma}{2}(E_{\setminus e}+E_{/e})}\,\mathsf{B}^{\{J\}}_{(\sigma,\sigma),\sigma;\,I_J}\big(e^{\frac{1}{2}(E_{\setminus e}-E_{/e})},\ e^{\frac{1}{2}(E_{/e}-E_{\setminus e})}\big),\end{aligned}$$

and the Breta is $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} \sigma, & \sigma \\ & 2\sigma \end{bmatrix}$ where the two even statistics agree.

For one table at the centre the scale integral of a mark is the factor $e^{-\frac{\sigma}{2}(E_{\setminus e}+E_{/e})}$ times a Breta that sees the room through one number, the difference of the even statistics of the deletion and the contraction; at an atom the law of the log-ratios read from a mark does not depend on the mark. Proposition 19.19 reads the scale integral of the mark with the exponents in the shape index.

Proposition 19.19 (A mark is the Breta of its children's even letters)**.** *For one table, a mark e that is neither a loop nor a coloop, exponents $s,t>0$, $0<m_e<s+t$ and the top wall $\kappa^{\mathrm{top}}_{\{e\}}=s+t-m_e$, integrating $\alpha_e=x$ with the room fixed,*

$$\int_0^\infty x^{m_e-1}K^{-s}\Psi^{-t}\,dx = K_{/e}^{\,t-m_e}\Psi_{\backslash e}^{\,s-m_e}\,\Gamma_2^2\begin{bmatrix} m_e,\ \kappa^{\mathrm{top}}_{\{e\}} \\ s,\ t \end{bmatrix}\ \mathsf{B}^{\{J\}}_{(s,t),\,\kappa^{\mathrm{top}}_{\{e\}};\,I_J}\big(\mathbf{E}_{\backslash e},\mathbf{E}_{/e}\big),$$

with $\mathbf{E}_{\backslash e}=K_{\backslash e}\Psi_{\backslash e}$ and $\mathbf{E}_{/e}=K_{/e}\Psi_{/e}$ the even letters of the deletion and the contraction, J a two-element set, and

$$\mathsf{B}^{\{J\}}_{(s,t),\,\kappa;\,I_J}(a,b)=\int_0^1 w^{s-1}(1-w)^{t-1}\,\mathsf{Z}_{\kappa;(1)}\big(wa+(1-w)b\big)\,dw,$$

the shape w of J weighing the deletion against the contraction. At the centre, $s=t=m_e=\sigma$, the prefactor and the stacked Gamma are 1.

The same scale integral is the Breta of Corollary 18.29 for the bunch $\{e\}$, whose levels d and r have the children $\mathcal{B}\setminus e$ and $\mathcal{B}/e$: there the walls $(m_e,\kappa^{\mathrm{top}}_{\{e\}})$ fill the shape index and the exponent t the scale index, here the exponents (s,t) fill the shape index and the top wall the scale index. Both read the scale integral of the mark through the even letters of the two children, the deletion and the contraction.

19.6 The junction under level form

Lemma 19.20 (An even power of e^v-e^{-v} on a symmetric two-sided scale)**.** *For $\xi>0$ and $l\in\mathbb{N}$,*

$$\int_{\mathbb{R}}\big(e^v-e^{-v}\big)^{2l}e^{-\xi(e^v+e^{-v})}\,dv = 4^l\big(\tfrac12\big)_l\,\xi^{-l}\,\mathsf{G}_{l;(1,-1)}(\xi,\xi).$$

Theorem 19.21 (The junction under level form)**.** *Let $F=((\mathcal{B}_j,\theta_j);\sigma_j,\sigma_j)_{j=1}^N$, every $\sigma_j>0$, be a molecule on n marks with the masses $\sigma\mathbf{1}$, $\sigma=\sum_j\sigma_j$, let B, $b=|B|$, be a proper bunch with room R such that every transposition of two marks of B fixing the rest carries or exchanges every table, and let $g\in R$. By Proposition 19.6 every half-sum has level form; in this statement write $q_{j,k}(z_R)$ for the common value of $c_{j,p}(\frac12;z_R)$ over $|p|=k$ and $\mathbf{q}_j=(q_{j,0},\dots,q_{j,b})$, so that $q_{j,b-k}(-z_R)=q_{j,k}(z_R)$ and*

$$L_{j,1/2}(z)=\sum_{k=0}^{b}e^{-\langle\mathbf{1},z_B\rangle}e_k\big(e^{2z_B}\big)\,q_{j,k}(z_R),\qquad e^{-\langle\mathbf{1},z_B\rangle}e_k\big(e^{2z_B}\big)=\sum_{|p|=k}e^{\langle z_B,\chi_p\rangle},$$

the centred polynomial of the list on B whose bases are the k-subsets of B; and for each $\mathbf{l}=(l_j)_j\in\mathbb{N}^N$

$$\prod_j\big(\partial_w L_{j,w}(z)\big)^{2l_j}=e^{-2|\mathbf{l}|\langle\mathbf{1},z_B\rangle}\sum_{|\mathbf{h}|=2|\mathbf{l}|}\prod_{k=0}^{b}e_k\big(e^{2z_B}\big)^{h_k}\,c^{(\mathbf{l})}_{\mathbf{h}}(z_R),$$

with $c^{(\mathbf{l})}_{\mathbf{h}}$ Laurent polynomials in e^{z_R}.

(i) With the Gramma $\mathsf{G}_{0;(\chi_p)_{p\subseteq B}}\big((\upsilon_{|p|})_{p\subseteq B}\big)$, $\boldsymbol{\upsilon}=(\upsilon_0,\dots,\upsilon_b)$, whose coefficients depend on $|p|$ alone and which depends on B through b alone,

$$P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma}) = 2^{n-1}\sum_{\mathbf{l}\in\mathbb{N}^N}\prod_j \frac{(\sigma_j)_{l_j}}{l_j!\,4^{l_j}\,\Gamma(2\sigma_j+2l_j)}\int_{\mathbb{R}^{R-g}}\int_{(0,\infty)^N}\prod_j y_j^{2\sigma_j+2l_j-1}$$
$$\times\sum_{|\mathbf{h}|=2|\mathbf{l}|} c_{\mathbf{h}}^{(\mathbf{l})}(z_R)\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\,\mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)\Big|_{\boldsymbol{\upsilon}=\sum_j y_j\mathbf{q}_j(z_R)}\,d\mathbf{y}\,dz_{R-g}\Big|_{z_g=0},$$

the inner sum being $\int_{\mathbb{R}^B}\prod_j\big(\partial_w L_{j,w}(z)\big)^{2l_j}e^{-\sum_j y_j L_{j,1/2}(z)}\,dz_B\ge 0$*, a level with* $\sum_j y_j q_{j,k}=0$ *absent.*

(ii) One mark at a time. *For* $e\in B$*,* $B'=B-e$ *and* $\boldsymbol{\upsilon}>0$*,*

$$\mathsf{G}_{0;(\chi_p)_{p\subseteq B}}\big((\upsilon_{|p|})_p\big)=\int_0^\infty \mathsf{G}_{0;(\chi_p)_{p\subseteq B'}}\big((x\,\upsilon_{|p|+1}+x^{-1}\upsilon_{|p|})_p\big)\,\frac{dx}{x},$$

and for $B=\{e\}$ *this Gramma is* $\mathsf{G}_{0;(1,-1)}(\upsilon_1,\upsilon_0)$*.*

(iii) The share side. *Let* G *be the family on* B *of the lists whose bases are the* k*-subsets of* B*,* $1\le k\le b-1$*, with the exponents* $(a'_k,0)$*, at the masses* $\frac{1}{b}\sum_{k=1}^{b-1}ka'_k\,\mathbf{1}_B$ *on its balance wall. Where every* $\operatorname{Re}a'_k>0$ *its period converges, is unchanged when the exponents of the lists of ranks* k *and* $b-k$ *are swapped, and is read on the hyperplane* $\langle\mathbf{1},w\rangle=0$*:*

$$P_G\Big(\tfrac{1}{b}\sum_{k=1}^{b-1}ka'_k\,\mathbf{1}_B;(a'_k)_{k=1}^{b-1},\mathbf{0}\Big)=2^{b-1}\sqrt{b}\int_{\langle\mathbf{1},w\rangle=0}\prod_{k=1}^{b-1}e_k\big(e^{2w}\big)^{-a'_k}\,\mathrm{dvol}_{b-1}(w).$$

For $\boldsymbol{\upsilon}>0$ *and* $\mathbf{h}\in\mathbb{N}^{b+1}$*, on a contour with* $\operatorname{Re}a_k>h_k$ *for every* k *and with* $a'_k=a_k-h_k$*,*

$$\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\,\mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)(\boldsymbol{\upsilon})$$
$$=2^{1-b}\int_{\sum_{k=0}^b(2k-b)(a_k-h_k)=0}\prod_{k=0}^{b}\Gamma(a_k)\,\upsilon_k^{-a_k}\;P_G\Big(\tfrac{1}{b}\sum_{k=1}^{b-1}ka'_k\,\mathbf{1}_B;(a'_k)_{k=1}^{b-1},\mathbf{0}\Big).$$

(iv) Two marks. *If* $B=\{e,f\}$ *and the swap of* e *and* f *exchanges every table, then with* $z_\pm=z_e\pm z_f$*, for every* j*,*

$$L_{j,1/2}(z)=e^{-z_+}c_{j,\varnothing}(\tfrac12;z_R)+e^{z_+}c_{j,\varnothing}(\tfrac12;-z_R)+\big(e^{z_-}+e^{-z_-}\big)c_{j,\{e\}}(\tfrac12;z_R),$$
$$\partial_w L_{j,w}(z)=\big(e^{z_-}-e^{-z_-}\big)\,\partial_w c_{j,\{e\}}(w;z_R),$$

the functions $c_{j,\{e\}}(\frac12;\cdot)$ *and* $\partial_w c_{j,\{e\}}$ *are even, and, writing in this statement* $C_\varnothing(z_R)=\sum_j y_j c_{j,\varnothing}(\frac12;z_R)$ *and* $C_e(z_R)=\sum_j y_j c_{j,\{e\}}(\frac12;z_R)$*,*

$$P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})=2^{n-2}\sum_{\mathbf{l}\in\mathbb{N}^N}\big(\tfrac12\big)_{|\mathbf{l}|}\prod_j\frac{(\sigma_j)_{l_j}}{l_j!\,\Gamma(2\sigma_j+2l_j)}\int_{\mathbb{R}^{R-g}}\prod_j\big(\partial_w c_{j,\{e\}}\big)^{2l_j}$$
$$\times\int_{(0,\infty)^N}\prod_j y_j^{2\sigma_j+2l_j-1}\,C_e^{-|\mathbf{l}|}\,\mathsf{G}_{0;(1,-1)}\big(C_\varnothing(-z_R),\,C_\varnothing(z_R)\big)$$
$$\times\,\mathsf{G}_{|\mathbf{l}|;(1,-1)}\big(C_e(z_R),\,C_e(z_R)\big)\,d\mathbf{y}\,dz_{R-g}\Big|_{z_g=0},$$

a series of nonnegative terms.

Under level form the bunch enters only through its levels: one Gramma with one coefficient for each level, which depends on the number of its marks alone and is computed one mark at a time, is differentiated at the level coefficients of the half-sums, added over the tables with their scales, and its share side is a period of the lists of k-subsets of the bunch's own marks. For a pair whose swap exchanges every table the bunch splits along $z_e + z_f$ and $z_e - z_f$ into two Grammas $\mathsf{G}_{\nu;(1,-1)}$.

19.7 The node in one scale

At a bunch of width one the integrand of Theorem 18.13 is, in the even share, the product of two Mellin transforms, one carrying the bunch and one the room. Read back on the half-line, the period is one integral over one scale of two functions, the bunch transform and the room transform, each an integral of a Gramma $\mathsf{G}_{\nu;(1,-1)}$ over its own shape. A closing toward a level is the expansion of one of them in powers of the scale, integrated term by term against the other. The expansion of a bunch is measured by one number, its reach: every bunch has a reach, no reach lies below the least value of the bunch's ratio, and a node closes on its lattice toward every level at which the growths of its two families multiply to less than one. Each side of a node is an interval, from the decay of its transform to its reach, and, against a room and a bunch bounded below, a node closes on a geometric lattice exactly when the two intervals are disjoint.

Definition 19.22 (Node of width one; its two transforms)**.** A *node of width one*, a node of a tree of bunches (Definition 19.7) whose bunch has width one, is a family $F = ((\mathcal{B}, \theta); s, t)$ with real data, $s, t > 0$, its masses on the balance wall with every proper wall positive, and a bunch B of width one with room R, the weights factoring across them: the setting of Theorem 18.13 for one table. Along the level d its level shares are $(a^d, b^d, a^r, b^r) = (s - \kappa_B + b^d,\ b^d,\ \kappa_B - b^d,\ t - b^d)$, the odd share being fixed at $2\delta^d = s - \kappa_B$; its ratios ϱ_B on Δ_B and ϱ_R on Δ_R are those of Definition 18.4; and σ_*^d is a real value of the even share at which both families converge and the four arguments of its stacked Gamma are positive, so that the line $\operatorname{Re}\sigma^d = \sigma_*^d$ is a contour of Theorem 18.13, with $b_* = \sigma_*^d - \delta^d > 0$ the real part of b^d there. The node carries the nonnegative measures

$$\phi_B(du) = u^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}/R}^{-(s-\kappa_B)} K_{\mathcal{B}|B}^{-\kappa_B} \Psi_{\mathcal{B}|B}^{-t}\, d\omega_B(u) \quad \text{on } \Delta_B,$$
$$\phi_R(dv) = v^{\mathbf{m}|_R - \mathbf{1}} K_{\mathcal{B}|R}^{-(s-\kappa_B)} K_{\mathcal{B}/B}^{-\kappa_B} \Psi_{\mathcal{B}/B}^{-t}\, d\omega_R(v) \quad \text{on } \Delta_R,$$

the forms of the bunch family and of the room family at the level share $b^d = 0$; ϕ_B is the base form of the cell of the bunch $\mathcal{B}_B^+$ at the exponents (s, t) (Definition 22.1). At the share b^d the periods of the two families are

$$P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b}) = \int_{\Delta_B} \varrho_B^{-b^d}\, d\phi_B, \qquad P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}) = \int_{\Delta_R} \varrho_R^{b^d}\, d\phi_R,$$

continued meromorphically in b^d by their sector forms (Theorem 8.7, Lemma 8.6): the *continued periods along the level d*. For $y > 0$ the *bunch transform* and the *room transform* are

$$f_B(y) = \int_{\Delta_B} \mathsf{G}_{s-\kappa_B;(1,-1)}\big(1, \varrho_B(u)\, y\big)\, \phi_B(du),$$
$$f_R(y) = \int_{\Delta_R} \big(\varrho_R(v)\, y\big)^{\kappa_B}\, \mathsf{G}_{t-\kappa_B;(1,-1)}\big(1, \varrho_R(v)\, y\big)\, \phi_R(dv),$$

with values in $[0, \infty]$. Swapping B and R gives the *swapped node*, with the bunch R, the room B and the two levels swapped (Theorem 18.13).

The bunch family at the share b^d has the form ϕ_B times $(\mathbf{E}_{\mathcal{B}/R}/\mathbf{E}_{\mathcal{B}|B})^{-b^d} = \varrho_B^{-b^d}$, and the room family the form ϕ_R times $(\mathbf{E}_{\mathcal{B}|R}/\mathbf{E}_{\mathcal{B}/B})^{-b^d} = \varrho_R^{b^d}$: the even share enters through the ratio of the hinge values of each side alone (Lemma 18.15).

Lemma 19.23 (Two letters in one scale)**.** *For linear letters $L_i(\rho) = L_i^{(0)} + \rho L_i^{(1)}$ with positive coefficients and the Skeleton values $\xi_i = L_i^{(0)}/L_i^{(1)}$, $i = 1, 2$, exponents $s, t > 0$ and $0 < \kappa < s + t$,*

$$\int_0^\infty \rho^{\kappa-1} L_1(\rho)^{-s} L_2(\rho)^{-t}\, d\rho \\ = \frac{(L_1^{(0)})^{\kappa-s}(L_1^{(1)})^{-\kappa}(L_2^{(0)})^{-t}}{\Gamma(s)\Gamma(t)} \int_0^\infty x^{\kappa-1}\, \mathsf{G}_{s-\kappa;(1,-1)}\Big(1, \frac{\xi_1}{\xi_2}\, x\Big)\, \mathsf{G}_{t-\kappa;(1,-1)}(1, x)\, dx.$$

Theorem 19.24 (The node in one scale)**.** *Let F and B be a node of width one (Definition 19.22).*

(i) The period is one integral over one scale of the two transforms,

$$P_F(\mathbf{m}; s, t) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty f_B(y)\, f_R(y)\, \frac{dy}{y},$$

the integrand nonnegative.

(ii) With the hinge values of Definition 18.4,

$$f_B(y) = \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}|B}^{-s} \Psi_{\mathcal{B}|B}^{-t}\, \mathsf{G}_{s-\kappa_B;(1,-1)}\big(\gamma_B^K(u)^{-1},\, y\, \gamma_B^\Psi(u)\big)\, d\omega_B(u) :$$

the bunch meets the room through its two hinge values, as in pooling (Theorem 18.8).

(iii) For real $b^d > 0$ with $a^d > 0$, and for real b^d with $a^r > 0$ and $b^r > 0$, respectively, in $[0, \infty]$,

$$\int_0^\infty y^{b^d - 1} f_B(y)\, dy = \Gamma(b^d)\, \Gamma(a^d) \int_{\Delta_B} \varrho_B^{-b^d}\, d\phi_B,$$
$$\int_0^\infty y^{-b^d - 1} f_R(y)\, dy = \Gamma(a^r)\, \Gamma(b^r) \int_{\Delta_R} \varrho_R^{b^d}\, d\phi_R.$$

Where the integrals over the shapes are finite they are the continued periods of the bunch family and of the room family at the share b^d, with their Gamma pairs, and (i), written through these two integrals in the share b^d, is Theorem 18.13 for one table.

(iv) If no slope on the room is positive, so that $\inf \varrho_R > 0$ (Lemma 18.15), then $\phi_R(\Delta_R) < \infty$, and with $a = \frac{t+\kappa_B}{2} - \frac{1}{4}$

$$f_R(y) \le \sqrt{\pi}\, e^{(t-\kappa_B)^2/4}\, \phi_R(\Delta_R)\, \big(y \inf \varrho_R\big)^a\, e^{-2\sqrt{y \inf \varrho_R}} \qquad \Big(y \ge \frac{\max(1, a^2)}{\inf \varrho_R}\Big) :$$

the room transform decays at the least ratio of the room.

Lemma 19.25 (The bunch transform on the orthant)**.** *For a node of width one and $y > 0$,*

$$f_B(y) = \int_{\mathbb{R}^B_{>0}} x^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}|B}(x)^{-s} \Psi_{\mathcal{B}|B}(x)^{-t} \exp\Big(- \frac{1}{\gamma_B^K(x)} - y\, \gamma_B^\Psi(x)\Big)\, dx.$$

The bunch transform is a Laplace transform in the hinge value γ_B^Ψ, damped by the hinge value γ_B^K: the room meets the bunch through its two hinge values and nothing else. Its expansion in powers of y is measured by its reach.

Definition 19.26 (Lattice expansion; reach)**.** A function f on $(0,\infty)$ has a *lattice expansion of reach* $x \ge 0$ when there are finitely many reals $\eta_1,\dots,\eta_q$, an integer $J \ge 0$ and coefficients $\beta_{\eta,j}$, for η in $A = \bigcup_i (\eta_i + \mathbb{N})$ and $0 \le j \le J$, such that

$$f(y) = \sum_{\eta\in A}\sum_{j=0}^{J} \beta_{\eta,j}\, y^{\eta}(-\log y)^j \quad (y>0), \qquad |\beta_{\eta,j}| \le C(1+\eta)^D \,\frac{x^\eta}{\Gamma(1+\eta)^2} \quad (\eta\in A,\ \eta\ge 0),$$

with constants C and D; the series then converges absolutely at every $y>0$, and $\sum_{\eta,j}|\beta_{\eta,j}|<\infty$. Its *least exponent* is the least η with some $\beta_{\eta,j}\neq 0$. A function has at most one lattice expansion. The *reach of the bunch* of a node of width one, $\mathrm{reach}(B)$, is the infimum of the x for which the bunch transform has a lattice expansion of reach x, and the *reach of the room*, $\mathrm{reach}(R)$, is the reach of R as the bunch of the swapped node.

Lemma 19.27 (The lattice of one mark)**.** *Let ν be real.*

(i) For $x>0$, $\mathsf{G}_{\nu;(1,-1)}(1,x)$ is the sum of the residues of $\Gamma(z)\Gamma(z+\nu)x^{-z}$ at the poles of $\Gamma(z)\Gamma(z+\nu)$. Off the integers

$$\mathsf{G}_{\nu;(1,-1)}(1,x) = \sum_{k\ge 0}\frac{(-1)^k}{k!}\Big(\Gamma(\nu-k)\,x^k + \Gamma(-\nu-k)\,x^{\nu+k}\Big);$$

at $\nu = l\in\mathbb{N}$

$$\mathsf{G}_{l;(1,-1)}(1,x) = \sum_{k<l}\frac{(-1)^k}{k!}\Gamma(l-k)\,x^k + (-1)^l\sum_{k\ge l}\frac{x^k}{k!\,(k-l)!}\Big((\log\Gamma)'(1+k)+(\log\Gamma)'(1+k-l)-\log x\Big),$$

and $\mathsf{G}_{-l;(1,-1)}(1,x) = x^{-l}\mathsf{G}_{l;(1,-1)}(1,x)$.

(ii) For $x\ge 1$, $0<\mathsf{G}_{\nu;(1,-1)}(1,x) \le \sqrt{\pi}\,e^{\nu^2/4}\,x^{\nu/2-1/4}e^{-2\sqrt{x}}$.

(iii) For $w>0$, $y\mapsto \mathsf{G}_{\nu;(1,-1)}(1,w^2y)$ has a lattice expansion of reach w^2, with exponents in $\mathbb{N}\cup(\nu+\mathbb{N})$, in $-l+\mathbb{N}$ at $\nu=-l$, and at most one power of $-\log y$.

Lemma 19.28 (Products of lattice expansions)**.** *If f_1 and f_2 have lattice expansions of reaches x_1 and x_2, then f_1f_2 has one of reach $(\sqrt{x_1}+\sqrt{x_2})^2$, whose terms are the products of theirs collected by exponent, and for real a, $y^a f_1$ has one of reach x_1.*

Lemma 19.29 (Closing a lattice expansion against a decaying function)**.** *Let f have a lattice expansion $\sum_{\eta,j}\beta_{\eta,j}y^\eta(-\log y)^j$ of reach x with least exponent η_0, and let g be measurable on $(0,\infty)$ and integrable on its compact subsets, with $\int_0^1 y^{\eta_0-1-\epsilon}|g(y)|\,dy<\infty$ for some $\epsilon>0$ and $|g(y)|\le C'y^a e^{-2\sqrt{x'y}}$ for $y\ge y_0$, where $x<x'$. Then*

$$\int_0^\infty f(y)\,g(y)\,\frac{dy}{y} = \sum_{\eta,j}\beta_{\eta,j}\int_0^\infty y^{\eta-1}(-\log y)^j\,g(y)\,dy,$$

every integral on the right and the series converging absolutely, and the terms with $\eta\ge N$ add up in absolute value to at most $C''(1+N)^{D''}(x/x')^N$, with C'' and D'' independent of N.

Lemma 19.30 (Continued cube integrals grow at most exponentially)**.** *Let G and h be analytic on a neighbourhood of $[0,1]^m$ in $\mathbb{C}^m$, with $h > 0$ on $[0,1]^m$, and let a_k, b_k be real with $a_k > 0$ wherever $b_k = 0$. The integral*

$$I(z) = \int_{[0,1]^m} \prod_k q_k^{a_k + b_k z - 1}\, G(q)\, h(q)^{-z}\, dq,$$

convergent where every $a_k + b_k \operatorname{Re} z > 0$, continues meromorphically to $z \in \mathbb{C}$, with poles only in the set $\mathcal{W}$ of the real points at which some $a_k + b_k z$ lies in $-\mathbb{N}$ (Lemma 8.6). For every $\vartheta > 0$ there is M, and then for every $\epsilon > 0$ there are C and D, with

$$|I(z)| \le C(1 + |z|)^D\, M^{|\operatorname{Re} z|}\, e^{\vartheta |\operatorname{Im} z|}$$

at every z at distance at least ϵ from $\mathcal{W}$.

Definition 19.31 (Growth along a level; growth at the integers)**.** For a node of width one, the *growth* $\operatorname{growth}(B)$ of the bunch family along the level d is the infimum of the $M \ge 0$ for which, for every $\epsilon > 0$, there are C, D and $\vartheta < \pi$ with

$$\left| P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b}) \right| \le C(1 + |b^d|)^D\, M^{-\operatorname{Re} b^d}\, e^{\vartheta |\operatorname{Im} b^d|}$$

for the continued period at the share b^d, at every b^d with $\operatorname{Re} b^d \le 0$ at distance at least ϵ from the set $\mathcal{W}_B$ of the real points at which a proper wall of the bunch family lies in $-\mathbb{N}$, which holds its poles; the growth $\operatorname{growth}(R)$ of the room family is the same with $P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b})$ and the set $\mathcal{W}_R$ of the room family. Along the level r the growths are those of the swapped node. The *growth at the integers* of the bunch is

$$\operatorname{growth}_{\mathbb{Z}}(B) = \limsup_{k \to \infty} \left| P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b}) \right|^{1/k},$$

the continued period of the bunch family taken at $b^d = -k$, $k \in \mathbb{N}$.

Theorem 19.32 (A node closes at the growth of its two families)**.** *Let F and B be a node of width one.*

(i) Every bunch has a reach. *The growths are finite;* $\operatorname{growth}(B) \le \sup \varrho_B$ *and* $\operatorname{growth}(R) \le 1/\inf \varrho_R$ *when these are finite; and* $\operatorname{reach}(B) \le \operatorname{growth}(B)$. *The exponents of the lattice expansion of f_B are the η for which $b^d = -\eta$ is a pole of $\Gamma(b^d)\Gamma(a^d) P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})$ to the left of the line $\operatorname{Re} b^d = b_*$, and $\sum_j j!\, \beta_{\eta,j} (b^d + \eta)^{-j-1}$ is its principal part there: the lattice of the bunch is the lattice of its continued period with its Gamma pair, the wall residues joining.*

(ii) Against a room bounded below. *Let no slope on the room be positive, and let* $\operatorname{reach}(B) < \inf \varrho_R$. *Then*

$$P_F(\mathbf{m}; s, t) = \frac{1}{\Gamma(s)\Gamma(t)} \sum_{\eta, j} \beta_{\eta,j}\, \partial_{b^d}^j \Big[\Gamma(a^r)\Gamma(b^r)\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}) \Big]_{b^d = -\eta},$$

absolutely, every room period in it convergent, and for every $x > \operatorname{reach}(B)$ the terms with $\eta \ge N$ add up in absolute value to at most $C(1+N)^D (x/\inf \varrho_R)^N$. The terms are the residues of the integrand of Theorem 18.13 at its poles toward the level d: the even share closes on the lattice of continued periods of the bunch family at the root rate $\operatorname{reach}(B)/\inf \varrho_R$.

(iii) Both families continued. *Whatever the slopes, if* $\operatorname{growth}(B)\operatorname{growth}(R)<1$*, then* $P_F(\mathbf{m};s,t)$ *is the sum of the residues of the integrand*

$$\Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\begin{bmatrix}a^d,\ b^d,\ a^r,\ b^r\\ s,\ t\end{bmatrix}\,P_{F^{\mathrm{bunch}(\mathbf{a},\mathbf{b})}}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\,P_{F^{\mathrm{room}(\mathbf{a},\mathbf{b})}}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$$

of Theorem 18.13, continued in b^d*, at its poles with* $\operatorname{Re} b^d<b_*$*; for every* $M_B>\operatorname{growth}(B)$ *and* $M_R>\operatorname{growth}(R)$ *the residues at the poles with real part in* $(-N-1,-N]$ *add up in absolute value to at most* $C(1+N)^D(M_BM_R)^N$*. The even share closes on the lattices of continued periods of both families, the wall residues of both joining.*

(iv) Toward the level r*. Swapping* B *and* R *leaves the period and Theorem 18.13 unchanged; the swapped node has the bunch* R *with the ratio* ϱ_R *and the room* B *with the ratio* ϱ_B*. Applied to it, (i) to (iii) close the even share toward the level* r*: against a bunch with* $\inf\varrho_B>0$ *at the root rate* $\operatorname{reach}(R)/\inf\varrho_B$*, and in general at a root rate at most the product of the growths of the two families along the level* r*.*

A node is therefore closed on its lattice toward the level at which the growths of its two families multiply to less than one, each growth read on its family alone; where neither level has a product below one, the peak closing (Theorem 18.30) closes the node with nonnegative terms and needs no root rate. Against a room bounded below the bunch enters through its reach alone, and the domination is a reach against a decay.

Proposition 19.33 (No reach is below the least ratio)**.** *Let* F *and* B *be a node of width one.*

(i) If $\inf\varrho_B>0$*, there are* C *and* a *with* $f_B(y)\le Cy^a\,e^{-2\sqrt{y\inf\varrho_B}}$ *for* $y\ge1$*, and for every* $x'>\inf\varrho_B$ *there is* $C_->0$ *with* $f_B(y)\ge C_-e^{-2\sqrt{x'y}}$ *for* $y\ge1$*: the bunch transform decays at the least ratio of the bunch.*

(ii) $\operatorname{reach}(B)\ge\inf\varrho_B$*: no lattice expansion of the bunch transform has a reach below the least ratio.*

(iii) When ϱ_B *is bounded,* $\operatorname{reach}(B)=\operatorname{growth}(B)=\sup\varrho_B$*: the reach of a bounded bunch is its largest ratio.*

Against a room bounded below, the root rate $\operatorname{reach}(B)/\inf\varrho_R$ of Theorem 19.32(ii) therefore lies between $\inf\varrho_B/\inf\varrho_R$ and $\sup\varrho_B/\inf\varrho_R$. The series bunches take the lower end, and the bounded bunches the upper, which is $\sup(\varrho_B/\varrho_R)$, the root rate of Corollary 18.16(i).

Proposition 19.34 (A series bunch is its marks)**.** *Let the bunch* B *of a node of width one be a series bunch,* θ'_B *the weight of its pattern* B *and* θ'_{B-e} *that of its pattern* $B-e$ *(Definition 18.4, Proposition 18.5), and in this statement write* $w_e=\theta'_{B-e}/\theta'_B$*.*

(i) The bunch transform is the product over the marks of B*:*

$$\begin{aligned}f_B(y)&=(\theta'_B)^{-s-t}\prod_{e\in B}\mathsf{G}_{m_e-s;(1,-1)}(w_ey,\,w_e)\\&=(\theta'_B)^{-s-t}\,y^{s-\kappa_B}\prod_{e\in B}w_e^{s-m_e}\,\mathsf{G}_{m_e-s;(1,-1)}\big(1,\,w_e^2y\big).\end{aligned}$$

(ii) $\varrho_B(u)=\big(\sum_e w_e/u_e\big)\big(\sum_e w_eu_e\big)\ge\big(\sum_e w_e\big)^2$*, with equality exactly at the point of equal coordinates of* Δ_B*, and* ϱ_B *is unbounded: whatever the weights, the least ratio of a series bunch is taken at the point of equal coordinates of its shape.*

(iii) f_B *has the lattice expansion of reach* $\big(\sum_e w_e\big)^2 = \inf \varrho_B$ *that is the product of the lattices of its marks, so* $\operatorname{reach}(B) = \inf \varrho_B$. *With* $\boldsymbol{\nu} = \mathbf{m}|_B - s\mathbf{1}_B$ *and no* ν_e *an integer, its exponents lie in the lattices* $-\nu_S + \mathbb{N}$, $S \subseteq B$: $S = \varnothing$ *and* $S = B$ *are the lattices of the Gamma pair, and for every other* S *the lattice is where the wall of the bunch family at* $B \setminus S$, *continued, meets* $-\mathbb{N}$. *Finally* $f_B(y) \le C\, y^a\, e^{-2(\sum_e w_e)\sqrt{y}}$ *for* $y \ge \max_e w_e^{-2}$, *with constants* C *and* a.

Proposition 19.35 (A bunch in series is the product of its parts)**.** *Let the bunch of a node of width one be* $B = B_1 \sqcup \cdots \sqcup B_q$, *and on each* B_i *let* $\mathcal{X}_i$ *and* $\mathcal{Y}_i$ *be tables,* $\mathcal{Y}_i$ *of rank one less than* $\mathcal{X}_i$, *with the hinge values* $\gamma_i^K = K_{\mathcal{X}_i}/K_{\mathcal{Y}_i}$ *and* $\gamma_i^\Psi = \Psi_{\mathcal{Y}_i}/\Psi_{\mathcal{X}_i}$ *and the ratio* $\varrho_i = \mathbf{E}_{\mathcal{Y}_i}/\mathbf{E}_{\mathcal{X}_i}$, *such that for constants* $C, C' > 0$

$$K_{\mathcal{B}|B} = C \prod_i K_{\mathcal{X}_i}, \qquad \Psi_{\mathcal{B}|B} = C' \prod_i \Psi_{\mathcal{X}_i}, \qquad \frac{1}{\gamma_B^K} = \sum_i \frac{1}{\gamma_i^K}, \qquad \gamma_B^\Psi = \sum_i \gamma_i^\Psi :$$

B *is the* series connection *of the parts* B_i. *A series bunch is the series connection of its marks, with* $K_{\mathcal{X}_e} = \alpha_e$, $\Psi_{\mathcal{X}_e} = 1$, $K_{\mathcal{Y}_e} = w_e$, $\Psi_{\mathcal{Y}_e} = w_e\alpha_e$ *and* $\varrho_e = w_e^2$ *(Proposition 19.34).*

(i) $f_B = C^{-s}C'^{-t} \prod_i f_i^\circ$, *with the* orthant transforms *of the parts*

$$f_i^\circ(y) = \int_{\mathbb{R}_{>0}^{B_i}} x^{\mathbf{m}|_{B_i} - \mathbf{1}} K_{\mathcal{X}_i}(x)^{-s} \Psi_{\mathcal{X}_i}(x)^{-t} \exp\Big(- \frac{1}{\gamma_i^K(x)} - y\, \gamma_i^\Psi(x) \Big)\, dx;$$

and $\sqrt{\inf \varrho_B} = \sum_i \sqrt{\inf \varrho_i}$.

(ii) If every f_i° *has a lattice expansion of reach* x_i, *then* f_B *has one of reach* $\big(\sum_i \sqrt{x_i}\big)^2$.

(iii) Let every part be a single mark, or have a bounded ratio and a finite measure $u^{\mathbf{m}|_{B_i} - \mathbf{1}} K_{\mathcal{X}_i}^{-s} \Psi_{\mathcal{X}_i}^{-t} (\gamma_i^\Psi)^{-a_i}\, d\omega_{B_i}$ *on* Δ_{B_i}, *where* $a_i = m_{B_i} - s\, r_i - t\,(|B_i| - r_i)$ *with* r_i *the rank of* $\mathcal{X}_i$; *and let no sum of the* a_i *over a nonempty set of parts be an integer. Then* f_i° *has the reach* $\sup \varrho_i$, *and* $\operatorname{reach}(B) = \big(\sum_i \sqrt{\sup \varrho_i}\big)^2$: *in a series connection the square roots of the reaches add.*

Theorem 19.36 (A series connection closes at its reach)**.** *(i) Let the bunch* B *of a node of width one be the series connection of parts* $B_1, \dots, B_q$ *(Proposition 19.35), every orthant transform* f_i° *having a lattice expansion of reach* x_i, *and let no slope on the room be positive. If* $\big(\sum_i \sqrt{x_i}\big)^2 < \inf \varrho_R$, *the even share closes toward the level* d *on the lattice of continued periods of the bunch family (Theorem 19.32(ii)): for every* $x > \big(\sum_i \sqrt{x_i}\big)^2$ *its terms with* $\eta \ge N$ *add up in absolute value to at most a constant times* $(1+N)^D (x/\inf \varrho_R)^N$, *and its coefficients* $\beta_{\eta,j}$ *are those of* $C^{-s}C'^{-t} \prod_i f_i^\circ$, *finite sums of products, over the parts, of the coefficients of their lattice expansions. Under the hypotheses of Proposition 19.35(iii) the root rate of the terms is* $\big(\sum_i \sqrt{\sup \varrho_i}\big)^2 / \inf \varrho_R$ *(Theorem 19.38(ii)). For a series bunch, the series connection of its marks with* w_e *as in Proposition 19.34 and* $x_e = w_e^2$, *the root rate is* $\inf \varrho_B / \inf \varrho_R = \big(\sum_e w_e\big)^2 / \inf \varrho_R$ *and the coefficients are those of* $(\theta'_B)^{-s-t} y^{s-\kappa_B} \prod_e w_e^{s-m_e} \mathsf{G}_{m_e - s;(1,-1)}(1, w_e^2 y)$: *finite sums of products, over the marks of* B, *of the coefficients of Lemma 19.27, Laurent coefficients of the Gamma pairs* $\Gamma(z)\Gamma(z + m_e - s)$.

(ii) Let B *be the series connection of parts* $B_1, \dots, B_q$ *in the dual table at the exponents* (t, s) *(Proposition 8.14), its orthant transforms there having lattice expansions of reaches* x_i^*, *and let no slope on the room be negative. If* $\big(\sum_i \sqrt{x_i^*}\big)^2 \sup \varrho_R < 1$, *the even share closes toward the level* r *on the lattice of continued periods of the bunch family, at a root rate at most*

$\big(\sum_i \sqrt{x_i^*}\big)^2 \sup \varrho_R$. *A parallel bunch B is a series bunch of the dual table (Proposition 18.38), with $w_e = \theta'_{\{e\}}/\theta'_{\varnothing}$, $\theta'_{\varnothing}$ and $\theta'_{\{e\}}$ the weights of its patterns $\varnothing$ and $\{e\}$; then $x_e^* = w_e^2$ and $\sup \varrho_B = \big(\sum_e w_e\big)^{-2}$, taken at the point of equal coordinates of Δ_B, so the share closes when $\sup \varrho_R < \sup \varrho_B$, at the root rate $\sup \varrho_R / \sup \varrho_B$.*

No order of the values is asked: the room's ratio need only be bounded on the side away from the closing, and the bunch's ratio may be unbounded, its walls falling toward the level closed on. When the bunch and the room are both series bunches, the root rate $\inf \varrho_B / \inf \varrho_R$ is read at the points of equal coordinates of both shapes. When the extreme ratio of the room sits at a vertex of its shape simplex the root rate is not read at the points of equal coordinates, which are only stationary.

Lemma 19.37 (The point of equal coordinates is stationary)**.** *Let $\mathcal{X}$ be a table on a set S, with any weights. At every point u of $(0,\infty)^S$ with equal coordinates, $u_e\, \partial_{u_e} \log \mathbf{E}_{\mathcal{X}}(u) = 1$ for every $e \in S$. So a ratio of even letters of tables on S is stationary at the point of equal coordinates of Δ_S: ϱ_B on Δ_B, ϱ_R on Δ_R, and ϱ_B/ϱ_R at the pair of these points.*

Theorem 19.38 (The two intervals of a node)**.** *For a node of width one, the bunch transform of the swapped node is $y^{-t} f_R(y)$, so that $\inf \varrho_B$ and $\inf \varrho_R$ are the decays of the two transforms, and $\operatorname{reach}(B)$ and $\operatorname{reach}(R)$ their reaches (Proposition 19.33(i), Lemma 19.28).*

(i) No transform reaches less far than it decays*: $\inf \varrho_B \le \operatorname{reach}(B)$ and $\inf \varrho_R \le \operatorname{reach}(R)$.*

(ii) Toward the level d. *Let no slope on the room be positive, so that $\inf \varrho_R > 0$, and index the terms of the lattice of Theorem 19.32(ii) by (η, j). Their root rate is $\operatorname{reach}(B)/\inf \varrho_R$: for every $x > \operatorname{reach}(B)$ they are at most $C(1+\eta)^D (x/\inf \varrho_R)^\eta$ in modulus, and for every $\epsilon > 0$ infinitely many of them exceed $\big(\operatorname{reach}(B)/\inf \varrho_R - \epsilon\big)^\eta$. So the lattice converges absolutely when $\operatorname{reach}(B) < \inf \varrho_R$, and its terms do not tend to zero when $\operatorname{reach}(B) > \inf \varrho_R$.*

(iii) Toward the level r. *Let no slope on the bunch be negative, so that $\inf \varrho_B > 0$. The terms of the lattice of continued periods toward the level r have the root rate $\operatorname{reach}(R)/\inf \varrho_B$ in the same sense: the lattice converges absolutely when $\operatorname{reach}(R) < \inf \varrho_B$, and its terms do not tend to zero when $\operatorname{reach}(R) > \inf \varrho_B$.*

(iv) One level at most. *$\operatorname{reach}(B) < \inf \varrho_R$ and $\operatorname{reach}(R) < \inf \varrho_B$ never hold together. Under both hypotheses of (ii) and (iii) the node therefore closes on a geometrically convergent lattice exactly when the intervals $[\inf \varrho_B, \operatorname{reach}(B)]$ of the bunch and $[\inf \varrho_R, \operatorname{reach}(R)]$ of the room are disjoint, toward the level of the lower one, at the ratio of the upper end of the lower interval to the lower end of the upper. When each interval extends beyond the lower end of the other, neither lattice converges, and the share closes at its peaks (Theorem 18.30); when they touch, the root rate is one and domination decides nothing.*

Each side of a node is read by two numbers of its own transform, its decay and the reach of its expansion, and no expansion reaches less far than its transform decays. A series bunch has both at its least ratio, and its interval is the point $(\sum_e w_e)^2$ (Proposition 19.34); a bounded bunch spreads over its whole range $[\inf \varrho_B, \sup \varrho_B]$ (Proposition 19.33(iii)); in a series connection of marks and bounded parts the square roots of the reaches add (Proposition 19.35(iii)). When the bunch and the room are both series bunches the two intervals are points: the node closes toward the level of the smaller point at the ratio of the two points, its terms toward the other level grow at the inverse ratio, and where the points meet domination decides nothing. Where a far side is unbounded, the level toward it can still close with both families continued, at a root rate at most the product of

their growths (Theorem 19.32(iii)). By Lemma 19.37 the value of ϱ_B/ϱ_R at the points of equal coordinates of Δ_B and Δ_R is a stationary value of the node's ratio, but the root rate is read at the extremes of the two transforms. The reach of a bunch that is neither bounded nor built from parallel and series bunches is read on the law of its ratio (Section 19.8).

Theorem 19.39 (Products of Grammas $\mathsf{G}_{\nu;(1,-1)}$ close on nested lattices)**.** *For a finite set S, numbers $w_e > 0$ and real indices ν_e, put $w_S = \sum_{e\in S} w_e$ and $\nu_S^- = \sum_{e\in S} \min(0,\nu_e)$, and in this statement write, for $\operatorname{Re} z > -\nu_S^-$,*

$$\mathcal{M}_S(z) = \int_0^\infty y^{z-1} \prod_{e\in S} \mathsf{G}_{\nu_e;(1,-1)}\big(1, w_e^2 y\big)\, dy.$$

(i) $\mathcal{M}_{\{e\}}(z) = w_e^{-2z}\,\Gamma(z)\Gamma(z+\nu_e)$, and for two marks with $w_e = w_f = w$

$$\mathcal{M}_{\{e,f\}}(z) = w^{-2z}\,{}_4\Gamma_1\!\left[\begin{matrix} z,\ z+\nu_e,\ z+\nu_f,\ z+\nu_e+\nu_f \\ 2z+\nu_e+\nu_f \end{matrix}\right].$$

(ii) For $S = S_1 \sqcup S_2$ with $w_{S_1} < w_{S_2}$, let $\prod_{e\in S_1} \mathsf{G}_{\nu_e;(1,-1)}(1, w_e^2 y) = \sum \beta_{\eta,j} y^\eta (-\log y)^j$ be the product of the lattices of the marks of S_1 (Lemmas 19.27 and 19.28). For $\operatorname{Re} z > -\nu_S^-$ and $i \in \mathbb{N}$,

$$(-\partial_z)^i \mathcal{M}_S(z) = \sum_{\eta,j} \beta_{\eta,j}\,(-\partial_z)^{i+j} \mathcal{M}_{S_2}(z+\eta),$$

every $z+\eta$ in the domain of $\mathcal{M}_{S_2}$, absolutely, the terms with $\eta \geq N$ adding up in absolute value to at most $C(1+N)^D (w_{S_1}/w_{S_2})^{2N}$.

(iii) Removing at each step a mark e of least w_e, until one mark is left or two with equal w_e, writes $\mathcal{M}_S(z)$ and its derivatives as nested lattice sums of products of Laurent coefficients of Gamma pairs at the points of the lattices, ending at (i): at most $|S|-1$ nested lattices, summed from the inside out, the one that removes e from S' converging absolutely at every point of the lattices above it, at the root rate $\big(w_e/(w_{S'} - w_e)\big)^2$, its constant depending on that point.

Corollary 19.40 (A table that is one circuit is a nested lattice of Gamma data)**.** *In Corollary 17.20 the integral over y is the integral of Theorem 19.39 with $S = \mathcal{E}$, $w_e = \theta_e$ and the indices $\nu_e = m_e - s$, taken at $z = s$, and $s > -\sum_e \min(0,\nu_e)$. So the period of every table that is one circuit, at all exponents $s,t > 0$, all masses of Corollary 17.20 and all weights, is a nested lattice sum of products of Laurent coefficients of Gamma pairs, with at most $n-1$ nested lattices, each converging absolutely at every point of the lattices above it and summed from the inside out. With unit weights, removing one mark at a time, the root rates are $(n-1)^{-2}, (n-2)^{-2}, \dots, \frac{1}{4}$, and the last two marks give four Gammas over one; splitting the marks into a lighter part S_1 and a heavier part S_2 converges at the root rate $(\theta_{S_1}/\theta_{S_2})^2$. At integer data with unit weights every coefficient is a rational combination of products of values $(\log\Gamma)^{(i)}(k)$, $i,k \geq 1$, which are Trailmoments of the counting input (Propositions 5.13 and 5.14).*

19.8 The reach read on the law

The reach of a bunch is defined by the expansion of its transform (Definition 19.26) and is bounded below by the least ratio and above by the growth of the continued family (Proposition 19.33,

Theorem 19.32). Read on one measure on the half-line, the law of the ratio under the base form, the reach is the point beyond which the law has a density equal to its own wall series (Definition 19.41), the series whose terms are the poles of the continued family; every point at which the law fails to be analytic therefore bounds the reach from below, and the least ratio and the largest ratio of a bounded bunch are such points. From above the reach is bounded by the ceiling of a continuation (Definition 19.45): in each sector the falling variables are carried around zero on circles, as far as the sector letters allow, and the largest modulus of the ratio met there bounds the reach. From below it is bounded by the growth of the continued family at the integers; a continuation whose largest values sit at simple tops that do not cancel has that growth as its ceiling, so that the reach is at least the modulus of a stationary value of the ratio on a flat face, and equal to it when the ceilings come down to it.

Definition 19.41 (The law of a bunch; wall series)**.** The *law* of the bunch of a node of width one is the image $(\varrho_B)_*\phi_B$ of its base form under its ratio, a measure on $(0,\infty)$ (Definition 5.4). The continued period of the bunch family at the share b^d is $\int_0^\infty x^{-b^d}\,d\big((\varrho_B)_*\phi_B\big)(x)$ wherever the integral converges, which is for $\operatorname{Re} b^d$ in an open interval containing b_*, the *domain interval* of the law, the walls being affine in b^d and the conditions of Theorem 8.10 strict. A function g on (x_0,∞), $x_0 \geq 0$, is a *wall series of radius* x_0 when there are finitely many reals η_i, an integer $J \geq 0$ and coefficients $a_{\eta,j}$, for $\eta \in A = \bigcup_i (\eta_i + \mathbb{N})$ and $0 \leq j \leq J$, with $\sum_{\eta,j} |a_{\eta,j}| x'^{-\eta} < \infty$ for every $x' > x_0$, such that

$$g(x) = \sum_{\eta,j} a_{\eta,j}\, x^{-\eta-1} (\log x)^j \qquad (x > x_0).$$

The series then converges absolutely and locally uniformly on $\{x \in \mathbb{C} :\ |x| > x_0,\ |\arg x| < \pi\}$, so a wall series is real-analytic.

Theorem 19.42 (The reach is read on the law)**.** *Let F and B be a node of width one.*

(i) On $(\operatorname{reach}(B), \infty)$ *the law $(\varrho_B)_*\phi_B$ has a density, and the density is a wall series of radius* $\operatorname{reach}(B)$*. Its terms are the poles of the continued period of the bunch family to the left of the domain interval: a pole $b^d = -\eta$ with the principal part $\sum_j j!\, a_{\eta,j}(b^d+\eta)^{-j-1}$ gives the terms $a_{\eta,j}x^{-\eta-1}(\log x)^j$, and there are no others.*

(ii) If on some (x_0,∞) the law has a density that is a wall series of radius x_0, then $\operatorname{reach}(B) \leq x_0$*.*

(iii) So the reach is the least $x_0 \geq 0$ for which the law has on (x_0,∞) a density that is a wall series of radius x_0. In particular $\operatorname{reach}(B) \geq x_1$ *at every $x_1 > 0$ near which $(\varrho_B)_*\phi_B$ has no real-analytic density: at the least ratio when it is positive, and at the largest ratio of a bounded bunch.*

So no reach lies below a point where the law of its ratio is singular, and Proposition 19.33(ii), with the lower half of (iii), is the case of the two ends of the law. The law of a family at a node is likewise its own wall series beyond its growth, so that $\operatorname{reach}(B) \leq \operatorname{growth}(B)$ of Theorem 19.32(i) is an equality (Corollary 30.11); along a direction of the shares the law reads the reach of a piece of any width (Theorem 30.10). From above the reach is bounded by the ceilings of continuations.

Lemma 19.43 (Finite parts on a keyhole)**.** *Let $0 < r \leq 1$, $r' > r$, a real, and let G be holomorphic on $U \times \mathbb{C}$ for an open U containing $\{|q| \leq r'\} \cup [r,1]$. Then $I(z) = \int_0^1 q^{a+z-1}G(q,z)\,dq$, defined for $\operatorname{Re} z > -a$, continues meromorphically to $\mathbb{C}$ with poles only where $a+z \in -\mathbb{N}$, and*

$$I(z) = \sum_{l\geq 0} g_l(z)\,\frac{r^{a+z+l}}{a+z+l} + \int_r^1 q^{a+z-1}G(q,z)\,dq, \qquad g_l(z) = \frac{1}{2\pi i}\oint_{|\zeta|=r'} \zeta^{-l-1}G(\zeta,z)\,d\zeta.$$

Where $|a+z+l| \ge \epsilon$ *for every* $l \in \mathbb{N}$,

$$|I(z)| \le \frac{r'}{\epsilon(r'-r)}\, r^{a+\operatorname{Re} z} \max_{|\zeta|=r'} |G(\zeta,z)| + \int_r^1 q^{a+\operatorname{Re} z-1} |G(q,z)|\, dq.$$

If $0 < q_0 \le 1$ *and* U *contains the closed disc* D *with diameter* $[-q_0, 1]$*, then off the points with* $a+z \in \mathbb{Z}$

$$I(z) = \frac{1}{e^{2\pi i(a+z)}-1} \oint_{\partial D} q^{a+z-1} G(q,z)\, dq,$$

the circle run once counterclockwise from 1*, with* q^{a+z-1} *continued along it from its positive value at* 1.

Definition 19.44 (Falling and base variables; keyholes)**.** For a node of width one and an order π of its bunch B let $q = (q_k)_k$ be its sector coordinates and U_k its tails, and let $k_{\mathcal{B}|B,\pi}$, $\psi_{\mathcal{B}|B,\pi}$, $k_{\mathcal{B}/R,\pi}$, $\psi_{\mathcal{B}/R,\pi}$ be the sector letters of the two tables of the bunch family (Theorem 8.7(ii)). Along the level d the wall of the bunch family at U_k is $\kappa_k + \operatorname{sl}_k b^d$ at the share b^d, with its *slope* $\operatorname{sl}_k \in \{-1, 0, 1\}$ (Lemma 18.15). On the sector of π the base form is

$$\phi_\pi = \prod_k q_k^{\kappa_k - 1}\, k_{\mathcal{B}/R,\pi}^{-(s-\kappa_B)}\, k_{\mathcal{B}|B,\pi}^{-\kappa_B}\, \psi_{\mathcal{B}|B,\pi}^{-t}\, dq,$$

and the ratio is

$$\varrho_\pi = \frac{k_{\mathcal{B}/R,\pi}\, \psi_{\mathcal{B}/R,\pi}}{k_{\mathcal{B}|B,\pi}\, \psi_{\mathcal{B}|B,\pi}} \prod_k q_k^{-\operatorname{sl}_k},$$

continued to the complex points of the coordinates at which the sector letters do not vanish; so the continued period of the bunch family at the share b^d is $\sum_\pi \int_{[0,1]^{|B|-1}} \varrho_\pi^{-b^d}\, \phi_\pi$. The q_k with $\operatorname{sl}_k = 1$ are the *falling variables* of π, whose walls fall as $\operatorname{Re} b^d$ decreases, and the others its *base variables*, *flat* when $\operatorname{sl}_k = 0$ and *rising* when $\operatorname{sl}_k = -1$. Where some flat variables vanish the sector reaches a *flat face* of the shape, on which every wall crossed has slope 0 and ϱ_π has the finite limit given by the leading parts of the letters; where some falling variable vanishes it reaches a *falling face*. For $0 < r \le b \le 1$ the *keyhole* and its *outline* are $Y_r[0,b] = \{q \in \mathbb{C} : |q| \le r\} \cup [r,b]$ and $\mathcal{O}_r[0,b] = \{|q| = r\} \cup [r,b]$, with $Y_r = Y_r[0,1]$ and $\mathcal{O}_r = \mathcal{O}_r[0,1]$.

Definition 19.45 (Keyhole continuations; ceiling)**.** A *keyhole continuation* of the bunch family of a node of width one assigns to every order π of B finitely many closed boxes Ω with disjoint interiors covering the cube of its base variables, and to every such box and falling variable q_k a radius $r_{\pi,\Omega,k} \in (0,1]$, such that none of the four sector letters vanishes on the set $Y = \Omega \times \prod_k Y_{r_{\pi,\Omega,k}}$, the product over the falling variables. The boxes may also divide the sides of the falling variables: they then cover the whole cube of the order, a box being the product of its part in the base variables and of its sides in the falling variables, with radii of its own; a falling variable with the side $[0, b_k]$ carries a radius $r \le b_k$ and the sets $Y_r[0,b_k]$ and $\mathcal{O}_r[0,b_k]$, and one with the side $[a_k, b_k]$, $a_k > 0$, is kept real, its sets Y and $\mathcal{O}$ being $[a_k, b_k]$. A box, divided or not, is written Ω with its radii and its sets Y and $\mathcal{O} = \Omega \times \prod_k \mathcal{O}_{r_{\pi,\Omega,k}}$, where Ω stands for its part in the variables without keyholes and the products run over the falling variables with keyholes. The *ceiling* $H_\mathcal{O}$ of the continuation is the largest value of $|\varrho_\pi|$ on the sets $\mathcal{O}$ of its boxes, an order without falling variables contributing the largest value of ϱ_π on its cube.

The sector letters are positive on the closed cube, with positive constant terms (Theorem 8.7(ii)), so small radii give keyhole continuations, and every bunch has one of finite ceiling. Whether a letter vanishes on a keyhole is decided on its outline and at one point.

Lemma 19.46 (Zeros on a keyhole are decided on its outline and at one point)**.** *Let Ω be a connected set, J a finite set, $0 < r_j \le b_j \le 1$ for $j \in J$, and L a continuous function on $\Omega \times \mathbb{C}^J$, a polynomial in $q = (q_j)_{j\in J}$. If L has no zero on $\Omega \times \prod_j \mathcal{O}_{r_j}[0, b_j]$, and for some $v_0 \in \Omega$ and each j the polynomial in q_j obtained from $L(v_0, \cdot)$ by putting $q_k = r_k$ for $k \neq j$ has no zero in the disc $|q_j| < r_j$, then L has no zero on $\Omega \times \prod_j Y_{r_j}[0, b_j]$.*

So a box and its radii keep a letter from vanishing on the set Y of the box when the letter has no zero on its set $\mathcal{O}$, where the ceiling is read, and, at one point of Ω with the other falling coordinates at their radii, none in the disc of each keyhole; the falling variables kept real count among the coordinates of Ω.

Theorem 19.47 (No reach exceeds the ceiling)**.** *For a node of width one, the reach of its bunch is at most the ceiling of every keyhole continuation:* $\inf \varrho_B \le \operatorname{reach}(B) \le H_{\mathcal{O}}$ *for the ceiling $H_{\mathcal{O}}$ of each. When ϱ_B is bounded, every keyhole continuation has the ceiling* $\sup \varrho_B$*, and* $\operatorname{reach}(B) = \sup \varrho_B$*.*

The letters of the contraction enter the keyhole continuations because the family is continued at points off the integers, where $\varrho_\pi^{-b^d}$ branches at their zeros; at the integers $-k$, $k \in \mathbb{N}$, when $s - \kappa_B$ is zero or a negative integer, only the letters of the restriction $\mathcal{B}|B$ matter. When a zero of a letter of the contraction is met by the keyholes that would bring $|\varrho_\pi|$ down, the ceiling can exceed the reach. From below, the reach is read at the integers, where $\psi_{\mathcal{B}/R,\pi}$ does not enter, nor $k_{\mathcal{B}/R,\pi}$ when $s - \kappa_B$ is an integer, and a continuation whose largest values sit at simple tops that do not cancel cannot be lowered.

Definition 19.48 (Integer keyhole continuations; simple tops)**.** For a node of width one let $\nu = s - \kappa_B$. An *integer keyhole continuation* is defined as in Definition 19.45 with only $k_{\mathcal{B}|B,\pi}$ and $\psi_{\mathcal{B}|B,\pi}$, and $k_{\mathcal{B}/R,\pi}$ when ν is not an integer, required not to vanish on the sets Y of its boxes; its ceiling is defined in the same way. For an order π and a box Ω write J for its falling variables with keyholes and r_j for their radii, and let C_j be the circle $|q_j| = r_j$ run once counterclockwise from r_j. The *keyhole contour* of the continuation is the formal sum over the orders and their boxes of

$$\Omega \times \prod_{j\in J} \Big(\frac{C_j}{e^{2\pi i \kappa_j} - 1} + [r_j, 1] \Big),$$

the segment being $[r_j, b_j]$ on a divided side $[0, b_j]$, a falling variable kept real on $[a_j, b_j]$ entering with that segment, and an order without falling variables contributing its cube; on the keyhole contour of π it carries the form ϕ_π, the powers continued along the circles from their positive values at r_j. A *top* is a point of the keyhole contour at which $|\varrho_\pi|$ equals the ceiling. A top p is *simple* when it lies on no face of its box other than faces $q_j = 0$ of flat variables, its falling coordinates lie on the circles away from r_j or inside the segments, and, with $\mathbf{x}$ the coordinates that vanish at p and $\mathbf{y}$ real coordinates of the keyhole contour for the others, vanishing at p (the angle on a circle),

$$\log \frac{\varrho_\pi}{\varrho_\pi(p)} = -\langle \boldsymbol{\lambda}, \mathbf{x} \rangle - \tfrac{1}{2}\, \mathbf{y}^{\mathsf{T}} \mathbf{Q}\, \mathbf{y} + O\big(|\mathbf{x}|^2 + |\mathbf{x}|\,|\mathbf{y}| + |\mathbf{y}|^3\big)$$

on the keyhole contour near p, with $\operatorname{Re} \lambda_j > 0$ for every j and $\operatorname{Re} \mathbf{Q}$ positive definite. The *exponent* of p is $\operatorname{ex}(p) = \sum_j \kappa_j + \frac{1}{2} \dim \mathbf{y}$, the sum over the coordinates $\mathbf{x}$, and its *coefficient* is

$$A(p) = \phi_p \prod_j \Gamma(\kappa_j)\, \lambda_j^{-\kappa_j}\, (2\pi)^{\dim \mathbf{y}/2} (\det \mathbf{Q})^{-1/2},$$

ϕ_p the value at p of the density of ϕ_π with the coefficient $1/(e^{2\pi i\kappa} - 1)$ of each circle of the keyhole contour through p, κ the wall of its falling variable, against $\prod_j x_j^{\kappa_j - 1}\, d\mathbf{x}\, d\mathbf{y}$, the powers and the root continued from real $\boldsymbol{\lambda}$ and real $\mathbf{Q}$ along segments.

The coordinates $\mathbf{x}$ belong to flat variables, whose walls κ_j are positive, so the Gammas are finite.

Theorem 19.49 (A simple top is the growth at the integers)**.** *For a node of width one let the walls κ_j of the falling variables of every order be off the integers, and let* $\mathrm{growth}_{\mathbb{Z}}(B)$ *be the growth at the integers (Definition 19.31).*

(i) For every integer keyhole continuation and every integer $k \geq \max(0, s - \kappa_B)$, the continued period at $b^d = -k$ is the integral of $\varrho_\pi^k \phi_\pi$ over its keyhole contour. So its modulus is at most $CH_{\mathcal{O}}^k$, $H_{\mathcal{O}}$ the ceiling, and $\mathrm{growth}_{\mathbb{Z}}(B) \leq H_{\mathcal{O}}$.

(ii) $\mathrm{reach}(B) \geq \mathrm{growth}_{\mathbb{Z}}(B)$.

(iii) Let an integer keyhole continuation of ceiling $H_{\mathcal{O}}$ have finitely many tops, all simple, let ex_0 *be the least of their exponents, and for $|\zeta| = 1$ let A_ζ be the sum of the coefficients $A(p)$ of the tops with* $\mathrm{ex}(p) = \mathrm{ex}_0$ *and $\varrho_\pi(p) = H_{\mathcal{O}}\zeta$. Each $A(p)$ is nonzero, and each top is a stationary point of the restriction of ϱ_π to the flat face $\mathbf{x} = 0$, the sector itself when there are no coordinates $\mathbf{x}$. If some $A_\zeta \neq 0$, then* $\mathrm{growth}_{\mathbb{Z}}(B) = H_{\mathcal{O}} \leq \mathrm{reach}(B)$*: the reach is at least the modulus of a stationary value of the ratio on a flat face.*

(iv) If moreover there are keyhole continuations with ceilings arbitrarily close to $H_{\mathcal{O}}$, then $\mathrm{reach}(B) = H_{\mathcal{O}}$*. In particular, if a keyhole continuation, with all four letters kept from vanishing, has finitely many tops, all simple, and some $A_\zeta \neq 0$, the reach is its ceiling.*

So $\mathrm{growth}_{\mathbb{Z}}(B) \leq \mathrm{reach}(B)$ always, with equality to the ceiling whenever the tops of an integer keyhole continuation are finitely many, all simple, and do not cancel, and the keyhole continuations come down to it: the reach is then the modulus of a stationary value of the ratio on a flat face, the one at which the best continuation is pinned. The stationary point need not be a positive point of the face.

The upper end needs the letters of the contraction only where they meet the falling faces. Elsewhere a zero of $k_{\mathcal{B}/R,\pi}$ or $\psi_{\mathcal{B}/R,\pi}$ is a zero of ϱ_π, and the law is read near infinity on the fibres $\varrho_\pi = x$ with $|x|$ large, which never pass through it.

Lemma 19.50 (A falling variable takes each large value once)**.** *Let g be meromorphic on a neighbourhood of the closed disc $|q| \leq r$, with a simple pole at 0 as its only pole there, and $|g| \leq M$ on the circle $|q| = r$. Then g takes every value x with $|x| > M$ exactly once in the disc, with $g' \neq 0$ there, and maps the open set $\{|q| < r : \ |g(q)| > M\} \cup \{0\}$ biholomorphically onto $\{|x| > M\} \cup \{\infty\}$.*

Lemma 19.51 (A wall series continued is a wall series)**.** *If the sum of a wall series of radius x_0 continues analytically along every path in $\{|x| > x_1\}$, with $x_1 < x_0$, then the series is a wall series of radius x_1.*

Definition 19.52 (Fibre continuations; stationary bound)**.** A *fibre continuation* is defined as in Definition 19.45, divided sides allowed, with only $k_{\mathcal{B}|B,\pi}$ and $\psi_{\mathcal{B}|B,\pi}$ required not to vanish on the sets Y of its boxes, and $k_{\mathcal{B}/R,\pi}$ and $\psi_{\mathcal{B}/R,\pi}$ only at their points with a falling coordinate 0; its ceiling $H_{\mathcal{O}}$ is defined in the same way. Its *stationary bound* S is the supremum of $|\varrho_\pi|$ over the points of these sets with no falling coordinate 0 at which at least two falling coordinates lie inside their discs, $|q_j| < r_j$, and the derivatives of ϱ_π in all the falling coordinates inside their discs vanish; $S = 0$ when there is no such point.

With one falling coordinate inside its disc and the others on their outlines, ϱ_π is in that coordinate as in Lemma 19.50, with the bound $H_{\mathcal{O}}$ on the circle, or identically 0 when a rising base coordinate is 0; so it has no stationary point there with $|\varrho_\pi| > H_{\mathcal{O}}$, and the stationary bound concerns two or more falling coordinates at once.

Theorem 19.53 (The contraction enters only on the falling faces)**.** *For a node of width one and every fibre continuation, of ceiling $H_{\mathcal{O}}$ and stationary bound S,* $\operatorname{reach}(B) \le \max(H_{\mathcal{O}}, S)$*. So, under the hypotheses of Theorem 19.49, if a fibre continuation that is also an integer keyhole continuation has $S \le H_{\mathcal{O}}$ and finitely many tops, all simple, with some $A_\zeta \neq 0$, then* $\operatorname{reach}(B) = \operatorname{growth}_{\mathbb{Z}}(B) = H_{\mathcal{O}}$*: the reach is read at the integers.*

The letters of the contraction, which Theorem 19.47 keeps from vanishing on whole keyholes, matter for the upper bound only on the falling faces, where the coefficient of the pole of the ratio vanishes, and through the stationary points of the ratio in two or more falling variables. The two bounds from above work box by box, so they can share one continuation.

Theorem 19.54 (Keyhole boxes and fibre boxes mix)**.** *Let a continuation assign to every order π of the bunch B of a node of width one boxes and radii as in Definition 19.45, the sides of the falling variables divided or not, each box either a* keyhole box*, on whose sets Y none of the four sector letters vanishes, or a* fibre box*, on whose sets Y the conditions of Definition 19.52 hold. Let $H_{\mathcal{O}}$ be its ceiling and S the stationary bound of Definition 19.52 taken over its fibre boxes alone. Then* $\operatorname{reach}(B) \le \max(H_{\mathcal{O}}, S)$*.*

Corollary 19.55 (One falling variable needs only the restriction)**.** *In an order with at most one falling variable, a box on whose sets Y the letters of the restriction $\mathcal{B}|B$ do not vanish is a fibre box, and it adds nothing to the stationary bound. So in a continuation of Theorem 19.54 such boxes need only the letters of the restriction, and the reach is at most* $\max(H_{\mathcal{O}}, S)$ *with S taken over the fibre boxes of the orders with two or more falling variables.*

So the stationary bound enters only where a letter of the contraction vanishes inside the keyholes of a sector with two or more falling variables. The zeros of the contraction on the falling faces can still keep every such bound above the reach.

These theorems bound the right end of each interval $[\inf \varrho_B, \operatorname{reach}(B)]$ of Theorem 19.38 from both sides. The reach need not be read on the positive shape: the ratio is real at the real points of a flat face whatever their signs, and a keyhole of a falling variable crosses the negative axis nearer to 0 than every pole of the ratio on it, so a simple top can sit at a point with a negative coordinate.

19.9 One tree

Read on one tree, the period of a point, built back up, is the periods of the leaves multiplied by the share kernels of the nodes and integrated over the free shares; its $n-1$ integrations are shared out among the bunches and the leaves; at integer data the share kernels bring the Trailmoments of the counting input order by order; and where the free shares close on integer points the period is the sum of the residues along the lattice of the tree.

Definition 19.56 (Share kernels, contents, integration count)**.** Let a molecule of lists with a native gauge be read on a tree of bunches, its Foreman tree or another, with the port of the balance wall run at every node on the native splits (Corollary 19.9). The share kernels of the tree (Corollary 4.4) are what the ports place on the shares: one stacked Gamma $\Gamma_{\substack{k\\1}}\begin{bmatrix}\mathbf{a}\\ a\end{bmatrix}$ for each letter opened at a node, k the number of its patterns, $\mathbf{a}$ its shares and a its exponent, which in Corollary 19.9 leave the factors $\big(\Gamma(a_{\mathcal{B}'})\Gamma(b_{\mathcal{B}'})\big)^{1-q_{\mathcal{B}'}}$ and $\prod_j \Gamma(s_j)^{-1}\Gamma(t_j)^{-1}$; each is balanced (Definition 1.2), its shares adding to its exponent on the fibre. The *free shares* are coordinates of the fibre of the word. Let $\mathcal{Z}$ be the $\mathbb{Q}$-algebra generated by the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $k \ge 2$, of the counting

input. The *content* $\mathcal{C}_M$ of a childcule M of a node, a leaf in particular, is the $\mathcal{Z}$-module spanned by the values, at the integer exponents and masses at which it converges, of the period of a family on M whose tables are those of M, and of its derivatives in the exponents and masses along the balance wall; the content of the molecule is the same with its own tables. The content $\mathcal{C}_B$ of the bunch B of a node with room R is the same for the families on B whose tables, each with both letters, are the minors $\mathcal{B}'|B$ and $\mathcal{B}'/R$ of the tables $\mathcal{B}'$ at the node: those of the face of B (Theorem 16.23), at width one those of the bunch family (Theorem 18.13). The content $\mathcal{C}_R$ of the room is the same for the families on R whose tables are the $\mathcal{B}'|R$ and $\mathcal{B}'/B$.

The *integration count* of an integral as written is the number of its integrations, assigned to the written expression by these rules. The period of a family on k marks has the integration count $k-1$, its simplex. A trail $(\mathsf{T}_{\mathbf{a};I_d}[y]\,\mathbb{I}_{(b+\mathbb{Z})^d})(\{b<y_1<\cdots<y_d\})$ of a lattice input with integer indices, $b>-1$ (Definition 5.23), has the integration count $a_1+\cdots+a_d$, its total index; the counting trails are the case $b=0$. A Gamma has the integration count one, $\Gamma(a)=\int_0^\infty x^{a-1}e^{-x}\,dx$ being one integration, so a stacked Gamma with k Gammas above and l below has the integration count $k-l$. A derivative in an exponent or a mass adds one, the logarithm of a letter or of a mark. A product has the sum of the integration counts of its factors, and a fibre integral of dimension D (Definition 4.1) the integration count of its integrand less D, each share integrated inverting one Mellin transform. A real number has *integration count at most* w when it is a rational combination of values, at rational data with unit weights, of integrals of integration count at most w.

The openings keep the integration count. The power $(X_1+\cdots+X_k)^{-s}$ is the share integral of $\Gamma_{\substack{k\\1}}\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix}\prod_i X_i^{-a_i}$, of integration count $(k-1)-(k-1)=0$ (Corollary 4.4), and e^{-y} is the integral of $\Gamma(b)y^{-b}$ over one share, of integration count $1-1=0$. A Beta corner $\Gamma_{\substack{2\\1}}\begin{bmatrix}a,\ b\\ a+b\end{bmatrix}$ has the integration count one, as the period of the list on two marks whose bases are the two singletons; the stacked Gamma $\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}$ of the shape integral has the integration count $|B|-1$, its simplex (Theorem 3.9); the interest $J_F=\Gamma(\kappa_{\mathcal{E}})P_F$ has the integration count n, its orthant (Theorem 8.10). The atom formula of a family on n marks (Corollary 17.15) has the integration count $\sum(|\mathcal{B}_j|-1)$ of its stacked Gammas, over the active letters, less its dimension $\sum(|\mathcal{B}_j|-1)-(n-1)$: $n-1$, the integration count of the period. Taking the residue at a pole of one Gamma removes that Gamma and adds one pin (Theorem 4.5), so every term of a closing has the integration count of the integral closed.

Theorem 19.57 (One tree suffices)**.** *Let F be a molecule of lists on n marks with real exponents $s_j,t_j\ge 0$ and masses $\mathbf{m}$ on the balance wall, with $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ convergent, and let its Foreman tree be read at its exponent pairs, at the centre or at a lean (Proposition 16.17, Theorem 14.6). Run the port of the balance wall at every node on the native splits of the point, as in Corollary 19.9. Parts (i), (ii), (iv) and (v) hold as well for every tree of bunches in place of the Foreman tree.*

(i) The word starts at the leaves. *$P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the word of the tree (Corollary 19.9): the periods of the leaves, each read at its shares and each entering once on its descent, multiplied by the share kernels and integrated over the free shares. At a node the words of the childcules multiply, and along a chain, its leaf opened by the atom formula, the word is one Gramma on the marks and the letters (Theorem 19.10).*

(ii) The integrations are counted. *The word has the integration count $n-1$, that of the period, and at rational data $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ has integration count at most $n-1$. At a node with bunch B and q childcules the integrations of the simplex are the $|B|+q-1$ pins of the node, $|B|-1$ for*

the shape of the bunch and q for the totals of the bunch and of the childcules, $q+1$ numbers that add to one; in a leaf L they are the $|L|-1$ integrations of its period.

(iii) The leaf bounds the bunches. *Along every descent of the Foreman tree, at the centre and at every lean, the copies of every bunch have at most as many marks as the copies of the leaf (Theorem 14.22). When some active list is connected, as for a molecule of one list, the tree is a chain and $|B_0| \le \cdots \le |B_{d-1}| \le |L|$ (Corollary 14.23): the shape of every bunch has at most as many integrations as the leaf, and its node adds one, the total of the bunch against its room.*

(iv) At integer data the share kernels bring $\mathcal{Z}$. *Let the exponents and masses be integers and let x be a point of the fibre at which every share is an integer. The pins and the share sums have integer coefficients, so the directions of the fibre have a rational basis; write the fibre near x as $x + V\mathbf{v}$ with V rational. Near x the share kernels are*

$$c \prod_i \ell_i(\mathbf{v})^{-\epsilon_i} \exp h(\mathbf{v}),$$

with c rational, $\ell_i(\mathbf{v})$ the deviation from its value at x of each argument of the share kernels that is a nonpositive integer at x, ϵ_i equal to 1 for a Gamma above and to -1 for one below, and h a power series without constant term whose part of degree k has its coefficients in $\mathbb{Q} + \mathbb{Q}\,(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ for $k \ge 2$ and in $\mathbb{Q}$ for $k = 1$: the terms of order k of the share kernels carry total index at most k. Where the periods of the leaves converge at x, every residue of the word at x, taken in turn along rational directions spanning the fibre, is therefore a finite $\mathcal{Z}$-combination of products, over the leaves, of elements of their contents.

(v) Summed along the tree. *A free share closed on the poles of the share kernels, where they alone have poles on the way and the closing converges, writes the word as a sum over a lattice of terms of its own integration count, each a fibre integral with one more pin (Theorem 4.5); where every free share closes so, on points at which every share is an integer and the periods of the leaves converge, the period at integer data is the sum along that lattice of the residues of (iv), finite $\mathcal{Z}$-combinations of products of the contents of the leaves. Across a bunch of width one, grouping its patterns by level puts the period of the bunch family in the word in place of the bunch (Theorem 18.13); for one table the even share then closes on its lattice whenever $\sup \varrho_B < \inf \varrho_R$ or $\inf \varrho_B > \sup \varrho_R$, and at integer data its residues are finite $\mathcal{Z}$-combinations of products of an element of $\mathcal{C}_B$ and one of $\mathcal{C}_R$ (Corollary 18.16).*

Whatever the order of the even letters, with $s, t > 0$, the even share of a node of width one closes on the lattices of continued periods of the bunch family and the room family toward a level at which their growths multiply to less than one (Theorem 19.32); a series bunch closes toward the level d against every room with $\inf \varrho_R > \inf \varrho_B$, and a series connection against every room with $\sqrt{\inf \varrho_R}$ above the sum of the square roots of the reaches of its parts, the room's ratio bounded only on the side away from the closing (Theorem 19.36); and against a room bounded below and a bunch bounded below the node closes on a geometric lattice exactly when the intervals of the bunch and of the room are disjoint, at its peaks otherwise (Theorem 19.38). Off the integers the level shares of several tables close in the same way across bunches of width one whose values stand in one order, node after node down a tree of splits of width one (Theorem 18.21, Corollary 18.22). At nonnegative exponents the period is in every case the sum, along the draws nested on the descents of the tree, of products over its blocks of Dirichlet moments (Theorem 19.14).

The pins need not be unimodular, so a closing can reach points at which some share is not an integer, even at integer data. There the share kernels have rational arguments, and their expansion

runs in heads and Trailmoments of the lattice inputs of those arguments (Proposition 5.13); that is why (iv) and (v) speak of integer points. Where a node has no free share its sum has one term. For a list whose bunch has width one, at a full lean, $(s, 0)$ or $(0, t)$, the level shares are pinned and the period is a Beta corner times the periods of the bunch family and the room family (Corollary 18.14); at rational data this is an identity of period data (Theorem 18.35), at integer data the Beta corner is rational, and with the room a leaf the period is a rational multiple of a product of an element of $\mathcal{C}_B$ and one of the content of the leaf. When the two hinge values of a bunch of width one agree the shape and the room separate at every pair of exponents (Proposition 18.12), and at the wall of a set, with the weights factoring on its bottom levels, the residue of the period is the product of the periods of its two faces (Theorem 16.23).

The integration count of a period bounds that of its value, and the theorem is stated for lists because the weights of a table can carry integrations of their own, which the integration count of a period leaves out. With unit weights and rational data nothing is carried in. On an atom the tree is the atom and the theorem says no more than its own content; its atom formula is a word of the same kind whose leaves are single marks, and the argument of (iv) puts each of its residues at integer points in $\mathcal{Z}$. The lattice can make what no bunch or leaf shows: a tree whose bunches and leaves have their contents in $\mathcal{Z}$ can sum, along the lattice of one free share, to a Trailmoment of a lattice input with a frequency (Definition 5.4) that none of them carries. The finite form of the theorem, the period in the $\mathcal{Z}$-algebra generated by the contents of the bunches and the leaves, holds where the shares are pinned at an integer point; where they are free the tree gives the sum along its lattice, and whether such a sum also lies in that algebra is a problem about relations between values, which the tree leaves open.

Theorem 19.57 is the Foreman procedure read, step by step, as a statement about numbers. *Split first*: a compound is split into its polycule before any molecule is asked whether it answers yes, and at every node the polycule of the childcules is evaluated as the product of its molecules (Theorem 19.10(i)). *Read at the table's gauge, then ask*: each molecule is read at its table's gauge, native for the masses it uses and inherited by its children (Proposition 12.24), whether it answers yes is read at the centre (Theorem 11.9), and a realization that is not atomic is peeled. *Peel small*: the peel is taken at a class with the fewest marks, it splits a class only across twin childcules, each part inside one class (Theorem 14.20), so the leaf with its copies holds whole classes of every node above it and is at least as large as every bunch with its copies (Theorem 14.22). *The share kernels reapply*: what a node adds is its port, stacked Gammas of the shares of its bunch and the pins of its shape and its totals; near integer data a balanced stacked Gamma brings $\mathcal{Z}$ order by order in total index, and in the draw series each node contributes its own Dirichlet moments to every product over the blocks. *One tree*: every tree of bunches computes the same period and shares out the same $n - 1$ integrations, and the Foreman tree is one in which the copies of no bunch have more marks than the copies of the leaf, and when some active list is connected no bunch has more marks than the leaf. *Pinned shares*: where the shares of a node are pinned, as for one list at a full lean across a bunch of width one, the node contributes an identity of period data (Theorem 18.35).

Part V

Construction and composition

Chapter 20

Gluing and what crosses it

Every table read at a bunch is the gluing of what its bunch and its room carry (Proposition 12.4), and at a bunch of width one the gluing is along one mark, the hinge (Theorem 18.6). Gluing is an operation with its own calculus. A table is built from cells, tables with one marked hinge, substituted into the marks of a core; the bases, the weights, the multivectors, the letters and the walls of the result are read off the cells and the core; the blocks of Chapter 18 and the series connections of Section 19.7 are the cells and cores of the simplest kinds; two cell families glued along their hinges have hinge walls that add to 2σ on the balance wall; and every connected table whose weights factor at its bunches of width one, every list and every determinantal table among them, is a tree of such substitutions whose nodes are prime tables and series and parallel nodes. Cells with several hinges splice in pairs along any graph of pairs, closed paths included, and stay tables, their channels composing multilinearly. What a room receives from its bunch, at any width, for any family and any multivector, is the joint law of finitely many bunch letters: as many as the ranks of the level weights, which for a multivector are bounded by the squares of its interface ranks and for a determinantal table by $\binom{c(B)}{i}^2$ at the level $d(B)+i$, whatever the numbers of marks; for a determinantal table the bunch crosses its gluing as one positive definite matrix of size $c(B)$, whose determinant at width one is the hinge value γ_B^K, and one scalar.

20.1 Gluing along a mark

Definition 20.1 (Cells, cores, substitution)**.** A *cell* is a table $\mathcal{A}$ on $B \sqcup \{g\}$ in which the mark g, its *hinge*, is neither a loop nor a coloop; it may carry two systems of positive weights on its bases, θ^K for its letter $K_{\mathcal{A}}$ and θ^Ψ for its letter $\Psi_{\mathcal{A}}$ (Definition 8.1), and it is *plain* when they agree, as for one table. Its *hinge minors* are the deletion $\mathcal{A} \setminus g$ and the contraction $\mathcal{A}/g$, tables on B of ranks $r_{\mathcal{A}}$ and $r_{\mathcal{A}} - 1$ with the inherited weights (Definition 7.3), so that

$$K_{\mathcal{A}} = K_{\mathcal{A}\setminus g} + \alpha_g K_{\mathcal{A}/g}, \qquad \Psi_{\mathcal{A}} = \alpha_g \Psi_{\mathcal{A}\setminus g} + \Psi_{\mathcal{A}/g};$$

its hinge values and its ratio, as for a bunch of width one (Definition 18.4), are

$$\gamma_{\mathcal{A}}^K = \frac{K_{\mathcal{A}\setminus g}}{K_{\mathcal{A}/g}}, \qquad \gamma_{\mathcal{A}}^\Psi = \frac{\Psi_{\mathcal{A}/g}}{\Psi_{\mathcal{A}\setminus g}}, \qquad \varrho_{\mathcal{A}} = \frac{\gamma_{\mathcal{A}}^\Psi}{\gamma_{\mathcal{A}}^K} = \frac{\mathbf{E}_{\mathcal{A}/g}}{\mathbf{E}_{\mathcal{A}\setminus g}},$$

functions of α_B homogeneous of the degrees 1, 1 and 0. The cell of a bunch $\mathcal{B}_B^+$ of Theorem 18.6 is a cell, with the hinge minors $\mathcal{B}|B$ and $\mathcal{B}/R$, the hinge values γ_B^K, γ_B^Ψ and the ratio ϱ_B of Definition 18.4. A *core* is a table $\mathcal{H}$ on marks $h_1, \dots, h_q$, none of them a loop or a coloop, plain or

with two systems of weights as a cell. For cells $\mathcal{A}_i$ on $B_i \sqcup \{g_i\}$, the B_i disjoint, the *substitution* $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is the family of sets $\bigsqcup_i X_i \subseteq \bigsqcup_i B_i$ for which some basis T of $\mathcal{H}$ has $X_i \in \mathcal{A}_i \setminus g_i$ when $h_i \in T$ and $X_i \in \mathcal{A}_i/g_i$ when $h_i \notin T$, with the weight, in each letter,

$$\theta_{\sqcup_i X_i} = \theta^{\mathcal{H}}_T \prod_{h_i \in T} \theta^{\mathcal{A}_i}_{X_i} \prod_{h_i \notin T} \theta^{\mathcal{A}_i}_{X_i + g_i},$$

every θ read as the weight θ^K of the letter K throughout, or as θ^Ψ throughout; T is determined by the sizes, $h_i \in T$ exactly when $|X_i| = r_{\mathcal{A}_i}$. A mark h is *left in place* by the cell $\binom{\{h,g\}}{1}$, the list on $\{h, g\}$ with the bases $\{h\}$ and $\{g\}$, whose hinge minors have the single bases $\{h\}$ and $\varnothing$ and whose hinge values are both α_h. One cell $\mathcal{A}$ at one mark h of $\mathcal{H}$, the other marks left in place, is the *gluing of $\mathcal{A}$ and $\mathcal{H}$ along h*: the gluing (Definition 12.3) of the communicatroid of B whose children are $\theta^{\mathcal{A}}_X$ times $\mathcal{H}/h$ at the bases X of $\mathcal{A} \setminus g$ and $\theta^{\mathcal{A}}_{X+g}$ times $\mathcal{H} \setminus h$ at the bases X of $\mathcal{A}/g$. A family is *substituted* table by table, $(\mathcal{B}_j, \theta_j) = \mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$, the cells of all tables at the mark h_i sharing the marks $B_i \sqcup \{g_i\}$; its masses are on $\mathcal{E} = \bigsqcup_i B_i$.

Lemma 20.2 (Gluing along a mark gives a table)**.** *Let $\mathcal{A}$ be a cell on $B \sqcup \{g\}$ and $\mathcal{A}'$ a cell on $R \sqcup \{g'\}$. The sets $X \sqcup Y$ with $X \in \mathcal{A} \setminus g$ and $Y \in \mathcal{A}'/g'$, or $X \in \mathcal{A}/g$ and $Y \in \mathcal{A}' \setminus g'$, are the bases of a table on $B \sqcup R$ of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, in which B is a bunch of width one whose cell of the bunch and cell of the room (Theorem 18.6) are $\mathcal{A}$ and $\mathcal{A}'$, the hinges read as the one mark g.*

Proposition 20.3 (Substitution)**.** *(i) $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is a table of rank $r_{\mathcal{H}} + \sum_i (r_{\mathcal{A}_i} - 1)$; it is the gluing of the cells to $\mathcal{H}$ along its marks one at a time, in any order; each B_i is a bunch of it of width one whose cell of the bunch is $\mathcal{A}_i$, its weights factoring across B_i and its room.*

(ii) Substituting into a mark of a cell is substituting the substituted cell: for a mark x of B_1 and a cell $\mathcal{A}'$ at x,

$$\big(\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle\big)\langle \mathcal{A}' \text{ at } x\rangle = \mathcal{H}\big\langle \mathcal{A}_1\langle \mathcal{A}' \text{ at } x\rangle, \mathcal{A}_2, \dots, \mathcal{A}_q\big\rangle.$$

(iii) Conversely, a table with a bunch B of width one, the weights factoring across B and R, is the gluing of its cell of the bunch $\mathcal{B}^+_B$ and its cell of the room $\mathcal{B}^+_R$ along g (Theorem 18.6): every gluing along a mark is that decomposition read backwards, and every such decomposition is a gluing.

(iv) If $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is connected, so are $\mathcal{H}$ and every cell. A connected table with a bunch B of width one, $|B|, |R| \geq 2$, the weights factoring across B and R, is the gluing along a mark of two connected tables on fewer marks.

Theorem 20.4 (The letters of a substitution)**.** *Let K and Ψ be the letters of $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$, each with its own weights, and read the hinge minors and the hinge values of each cell on its own marks. Then*

$$K(\alpha) = \prod_i K_{\mathcal{A}_i/g_i}(\alpha_{B_i})\ K_{\mathcal{H}}\big(\gamma^K_{\mathcal{A}_1}(\alpha_{B_1}), \dots, \gamma^K_{\mathcal{A}_q}(\alpha_{B_q})\big),$$

$$\Psi(\alpha) = \prod_i \Psi_{\mathcal{A}_i \setminus g_i}(\alpha_{B_i})\ \Psi_{\mathcal{H}}\big(\gamma^\Psi_{\mathcal{A}_1}(\alpha_{B_1}), \dots, \gamma^\Psi_{\mathcal{A}_q}(\alpha_{B_q})\big).$$

For one cell, the other marks left in place, this is Theorem 18.6.

The core reads each of its cells as one mark, at the hinge value of the cell by the letter: in K the mark h_i stands at $\gamma^K_{\mathcal{A}_i}$ and in Ψ at $\gamma^\Psi_{\mathcal{A}_i}$, and what is left over is a product of one letter of a hinge minor of every cell. Nothing in the core sees more of a cell than its two hinge values.

Corollary 20.5 (Blocks and closures). *(i) The* series block *on a set B of ℓ marks is the cell $\binom{B\sqcup\{g\}}{\ell}$, whose hinge minors have $K_{\backslash g} = e_\ell(\alpha_B)$, $K_{/g} = e_{\ell-1}(\alpha_B)$, $\Psi_{\backslash g} = 1$ and $\Psi_{/g} = e_1(\alpha_B)$; the* parallel block *is the cell $\binom{B\sqcup\{g\}}{1}$, with $K_{\backslash g} = e_1(\alpha_B)$, $K_{/g} = 1$, $\Psi_{\backslash g} = e_{\ell-1}(\alpha_B)$ and $\Psi_{/g} = e_\ell(\alpha_B)$. Series blocks at the marks h_i, $i \in S$, of a core $\mathcal{H}$ and parallel blocks at the others replace the marks of $\mathcal{H}$ by blocks as in Theorem 18.1, the series blocks by its complementary rule and the parallel blocks by its first rule, and give the letters of that theorem, each e_k read on the marks of its block: a series block is seen by Ψ through its total and by K through the hinge value $e_\ell/e_{\ell-1} = \big(\sum_x \alpha_x^{-1}\big)^{-1}$, a parallel block the other way round.*

(ii) The series closure *of cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ is the substitution $\binom{\{h_1,\dots,h_q\}}{q-1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ into the core of unit weights; its letters are*

$$K = \prod_i K_{\mathcal{A}_i\backslash g_i} \cdot \sum_i \frac{K_{\mathcal{A}_i/g_i}}{K_{\mathcal{A}_i\backslash g_i}}, \qquad \Psi = \prod_i \Psi_{\mathcal{A}_i\backslash g_i} \cdot \sum_i \frac{\Psi_{\mathcal{A}_i/g_i}}{\Psi_{\mathcal{A}_i\backslash g_i}},$$

so that the bunch $B_1 \sqcup \dots \sqcup B_{q-1}$ is the series connection of the parts B_i with the tables $\mathcal{A}_i \setminus g_i$ and $\mathcal{A}_i/g_i$ (Proposition 19.35, with $C = C' = 1$). The parallel closure $\binom{\{h_1,\dots,h_q\}}{1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ *has the letters $K = \prod_i K_{\mathcal{A}_i/g_i} \cdot \sum_i K_{\mathcal{A}_i\backslash g_i}/K_{\mathcal{A}_i/g_i}$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i/g_i} \cdot \sum_i \Psi_{\mathcal{A}_i\backslash g_i}/\Psi_{\mathcal{A}_i/g_i}$; it is the dual of the series closure of the dual cells.*

(iii) The gluing of two cells along their hinges is $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}_1, \mathcal{A}_2\rangle$, with the letters $K = K_{\mathcal{A}_1\backslash g_1} K_{\mathcal{A}_2/g_2} + K_{\mathcal{A}_1/g_1} K_{\mathcal{A}_2\backslash g_2}$ and $\Psi = \Psi_{\mathcal{A}_1/g_1}\Psi_{\mathcal{A}_2\backslash g_2} + \Psi_{\mathcal{A}_1\backslash g_1}\Psi_{\mathcal{A}_2/g_2}$.

Proposition 20.6 (The multivector of a substitution). *Let $\mathcal{H}$ and the cells carry multivectors $\boldsymbol{\omega}_\mathcal{H}$ and $\boldsymbol{\omega}_i$ (Definition 7.17), the hinge first in the order of the marks of each cell, and order the marks of the substitution by putting B_i, in its order in $\mathcal{A}_i$, in the place of h_i in the order of $\mathcal{H}$. Then*

$$\boldsymbol{\omega}\Big(\bigsqcup_i X_i\Big) = \boldsymbol{\omega}_\mathcal{H}(T) \prod_{h_i\in T} \boldsymbol{\omega}_i(X_i) \prod_{h_i\notin T} \boldsymbol{\omega}_i(g_i + X_i)$$

is a multivector of $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$; it is multilinear in $(\boldsymbol{\omega}_\mathcal{H}, \boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_q)$, a substitution of decomposable multivectors is decomposable, and

$$\operatorname{rk}(\boldsymbol{\omega}) \le \operatorname{rk}(\boldsymbol{\omega}_\mathcal{H}) \prod_i \operatorname{rk}(\boldsymbol{\omega}_i).$$

A substitution of determinantal tables is determinantal.

Theorem 20.7 (The walls of a substitution). *For $S = \bigsqcup_i S_i$, $S_i \subseteq B_i$, let $D(S)$ be the set of the marks h_i of the core with $d_{\mathcal{A}_i}(S_i+g_i) > d_{\mathcal{A}_i}(S_i)$ and $D^*(S)$ that of the h_i with $d^*_{\mathcal{A}_i}(S_i+g_i) > d^*_{\mathcal{A}_i}(S_i)$, the two differences being 0 or 1. The counts of the substitution are*

$$d(S) = \sum_i d_{\mathcal{A}_i}(S_i) + d_\mathcal{H}\big(D(S)\big), \qquad d^*(S) = \sum_i d^*_{\mathcal{A}_i}(S_i) + d^*_\mathcal{H}\big(D^*(S)\big),$$

and for a substituted family, with $D_j(S)$ and $D^*_j(S)$ read in table j, its walls are

$$\kappa_S = \sum_i \Big(m_{S_i} - \sum_j \big(s_j d_{\mathcal{A}_{ij}}(S_i) + t_j d^*_{\mathcal{A}_{ij}}(S_i)\big)\Big) - \sum_j \Big(s_j d_{\mathcal{H}_j}\big(D_j(S)\big) + t_j d^*_{\mathcal{H}_j}\big(D^*_j(S)\big)\Big).$$

At $S = B_i$, $D(S) = D^*(S) = \{h_i\}$ and the wall is $\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big)$, the mass that pooling gives the hinge of B_i (Theorem 18.8).

Lemma 20.8 (Two-sums of cells; hinge walls)**.** *Let $\mathcal{A}$ be a cell on $B \sqcup \{g\}$ and $\mathcal{A}'$ a cell on $R \sqcup \{g\}$ with the one hinge g, $B \cap R = \varnothing$, and let $\mathcal{A} \oplus_g \mathcal{A}'$ be their gluing along the hinge, $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}, \mathcal{A}' \rangle$ (Corollary 20.5(iii)), each basis carrying in each letter the product of the weights of the two bases of $\mathcal{A}$ and $\mathcal{A}'$ it is made of: a table on $B \sqcup R$ of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, plain when $\mathcal{A}$ and $\mathcal{A}'$ are, in which B is a bunch of width one with the cell of the bunch $\mathcal{A}$ and the cell of the room $\mathcal{A}'$ (Lemma 20.2); every table with a bunch of width one, the weights factoring across it, is the two-sum of its two cells (Proposition 20.3(iii)). Let the cell families $F_B = ((\mathcal{A}_j, \theta^{\mathcal{A}_j}); s_j, t_j)_{j=1}^N$ on $B \sqcup \{g\}$ and $F_R = ((\mathcal{A}'_j, \theta^{\mathcal{A}'_j}); s_j, t_j)_{j=1}^N$ on $R \sqcup \{g\}$ be glued table by table into the family $F = ((\mathcal{A}_j \oplus_g \mathcal{A}'_j); s_j, t_j)_j$ on $B \sqcup R$, with masses $\mathbf{m}$. The walls of F at B and at R are the hinge walls of the two cell families (Definition 22.1), for one table those of the two cells, $\kappa_{\mathcal{A}}$ and $\kappa_{\mathcal{A}'}$,*

$$\kappa_{F_B} = m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big), \qquad \kappa_{F_R} = m_R - \sum_j \big(s_j(r_{\mathcal{A}'_j} - 1) + t_j(|R| - r_{\mathcal{A}'_j})\big),$$

and $\kappa_{F_B} + \kappa_{F_R} = 2\sigma + \kappa_{\mathcal{E}}$, σ the total strength and $\kappa_{\mathcal{E}}$ the imbalance of F (Definition 8.1): on the balance wall the two hinge walls add to $2\sigma = \sum_j (s_j + t_j)$, and the mass $2\sigma - \kappa_{F_B}$ on g puts the cell family F_B itself on its balance wall.

Each letter of a two-sum is a letter of a hinge minor of one cell times the letter of the other cell read at one hinge value of the first (Theorem 20.4): K sees $\mathcal{A}$ through $\gamma^K_{\mathcal{A}}$ and Ψ through $\gamma^\Psi_{\mathcal{A}}$. The hinge wall of a cell is a number of the cell alone, its masses and its exponents, and balance makes the two hinge walls of a two-sum complementary; the period of a two-sum is read through them in Theorem 22.4.

Theorem 20.9 (A connected table is a tree of prime tables)**.** *Call a connected table on at least four marks* prime *when it has no bunch of width one with at least two marks on each side. Every connected table on at least three marks whose weights factor across each bunch of width one at which it is decomposed, a list or a determinantal table for instance (Theorem 12.17), is obtained by substitutions along a tree from prime tables and the series and parallel nodes $\binom{S}{|S|-1}$ and $\binom{S}{1}$, $|S| \geq 3$ (Definition 22.12), the weights carried along. With no two series nodes and no two parallel nodes adjacent in the tree, the tree, the* canonical hinge tree *(Definition 22.14), is unique up to the names of the markers. The prime nodes of a list are therefore invariants of the list.*

20.2 Splicing hinges in pairs

A cell has one hinge. Gluing along several marks at once, or around a closed path of cells, needs cells with several hinges and a splicing of hinges in pairs. The splicing stays inside tables because the levels of a bunch are tables (Proposition 12.16); one pair joining two cells is the gluing of Lemma 20.2.

Definition 20.10 (Cells with several hinges; channels)**.** A *cell with the hinges G* is a table $\mathcal{A}$ on $B \sqcup G$ with two systems of positive weights θ^K, θ^Ψ on its bases; $n_{\mathcal{A}} = |B|$ is its number of marks other than its hinges. Its *channels* are the letters of the children of the bunch G (Definition 12.1): for a pattern $p \in \mathrm{pat}(G)$, the basis polynomial $K_{\mathcal{A},(p)}$ of the child at p with the weights θ^K and its cobasis polynomial $\Psi_{\mathcal{A},(p)}$ with the weights θ^Ψ, polynomials on B, so that by the Taylor cube (Theorem 12.6)

$$K_{\mathcal{A}} = \sum_{p \in \mathrm{pat}(G)} \alpha_G^p \, K_{\mathcal{A},(p)}(\alpha_B), \qquad \Psi_{\mathcal{A}} = \sum_{p \in \mathrm{pat}(G)} \alpha_G^{G \setminus p} \, \Psi_{\mathcal{A},(p)}(\alpha_B) :$$

p is the set of hinges a basis occupies, in both letters. A channel has positive coefficients on its support and the degree

$$\deg K_{\mathcal{A},(p)} = r_{\mathcal{A}} - |p|, \qquad \deg \Psi_{\mathcal{A},(p)} = n_{\mathcal{A}} - r_{\mathcal{A}} + |p|.$$

The *channel vector* $\mathrm{ch}(\mathcal{A})$ lists the channels by their *tags*, the pairs of a letter and a set $p \subseteq G$, $2^{|G|+1}$ of them; a tag whose set is not a pattern is an absent channel, counted as zero. A cell of Definition 20.1 is the case of one hinge g that is neither a loop nor a coloop, with $K_{\mathcal{A},(\varnothing)} = K_{\mathcal{A}\setminus g}$, $K_{\mathcal{A},(\{g\})} = K_{\mathcal{A}/g}$, $\Psi_{\mathcal{A},(\varnothing)} = \Psi_{\mathcal{A}\setminus g}$ and $\Psi_{\mathcal{A},(\{g\})} = \Psi_{\mathcal{A}/g}$. *Splicing* a pair $\{a, b\}$ of hinges keeps the bases that occupy exactly one of them and forgets the pair; on either letter it is

$$L \longmapsto (\partial_{\alpha_a} + \partial_{\alpha_b}) L\big|_{\alpha_a = \alpha_b = 0}.$$

Cells are spliced along pairs of their hinges by splicing the pairs in their direct sum (Proposition 7.8).

Theorem 20.11 (Splicing hinges in pairs)**.** *(i) Splicing a pair $\{a, b\}$ of hinges of a cell $\mathcal{A}$ at which some basis occupies exactly one of a, b gives a cell on $B \sqcup (G \setminus \{a, b\})$ of rank $r_{\mathcal{A}} - 1$: its bases are the sets $T \setminus \{a, b\}$ with $|T \cap \{a, b\}| = 1$, each carrying in each letter the sum of the weights of the bases T that give it.*

(ii) Let a direct sum of cells carry a set of disjoint pairs of hinges, forming any graph on the cells: closed paths, a pair inside one cell and several pairs between two cells are allowed. If some choice of one basis in each cell occupies exactly one hinge of every pair, splicing the pairs one at a time, in any order, gives one cell; its rank is the sum of the ranks less the number of pairs, its bases are the unions of the unpaired parts of such choices, and its weight in each letter at a basis is the sum, over the choices that give it, of the products of the weights chosen. Plain cells give a plain cell.

(iii) Under the hypothesis of (ii), let the cells be $\mathcal{A}_1, \dots, \mathcal{A}_m$ with the hinges $G_1, \dots, G_m$, and in this statement let G' be the set of the hinges that lie in no pair, k the number of pairs and, for $p \subseteq G'$, let J run over the sets of paired hinges that contain exactly one hinge of every pair. The spliced cell $\mathcal{A}$ has the channels

$$K_{\mathcal{A},(p)} = \sum_J \prod_{i=1}^{m} K_{\mathcal{A}_i,((p \sqcup J) \cap G_i)}, \qquad \Psi_{\mathcal{A},(p)} = \sum_J \prod_{i=1}^{m} \Psi_{\mathcal{A}_i,((p \sqcup J) \cap G_i)},$$

absent channels counting as zero; it has $\sum_i n_{\mathcal{A}_i}$ marks besides its hinges G' and the rank $\sum_i r_{\mathcal{A}_i} - k$, and every product on the right has the degree of the channel on the left. The map $\big(\mathrm{ch}(\mathcal{A}_1), \dots, \mathrm{ch}(\mathcal{A}_m)\big) \mapsto \mathrm{ch}(\mathcal{A})$ is multilinear. For two cells $\mathcal{A}_1$ and $\mathcal{A}_2$ spliced along k pairs, each pair named once, so that the paired hinges of either cell form one set I of k names and the other hinges form G'_1 and G'_2, it is the bilinear map

$$K_{\mathcal{A}_{12},(p_1 \sqcup p_2)} = \sum_{I' \subseteq I} K_{\mathcal{A}_1,(p_1 \sqcup I')}\, K_{\mathcal{A}_2,(p_2 \sqcup (I \setminus I'))},$$

$$\Psi_{\mathcal{A}_{12},(p_1 \sqcup p_2)} = \sum_{I' \subseteq I} \Psi_{\mathcal{A}_1,(p_1 \sqcup I')}\, \Psi_{\mathcal{A}_2,(p_2 \sqcup (I \setminus I'))},$$

$p_1 \subseteq G'_1$, $p_2 \subseteq G'_2$; with $k = 1$ and no other hinges it is Corollary 20.5(iii).

Chapter 26 turns the bilinearity of Theorem 20.11(iii) into finite moment vectors and transfer matrices.

20.3 What crosses a gluing

Definition 20.12 (Level weights, level factorizations, interface)**.** Let $(\mathcal{B},\theta)$ be a table of rank r with a bunch B, of any width, and room R. For a level k, $d(B)\le k\le r(B)$, the *level weights* are the matrix $\Theta_B^{(k)}=(\theta_{p\sqcup X})_{p,X}$ indexed by the k-sets $p\subseteq B$ and the $(r-k)$-sets $X\subseteq R$, with $\theta_{p\sqcup X}=0$ when $p\sqcup X$ is not a basis. A *level factorization* writes every level letter (Definition 12.15) as a sum of products over a finite set C_k,

$$K_B^{(k)}(\alpha)=\sum_{c\in C_k} V_{k,c}(\alpha_B)\,L_{k,c}(\alpha_R),$$

each $V_{k,c}$ a real linear combination of the monomials α_B^p, $|p|=k$, and each $L_{k,c}$ a real linear combination of the monomials α_R^X, $|X|=r-k$: equivalently, a writing of $\Theta_B^{(k)}$ as a sum of $|C_k|$ products of a column and a row, so that $|C_k|\ge\operatorname{rank}\Theta_B^{(k)}$, with equality for some factorization. The $V_{k,c}$ are the *bunch letters* and the $L_{k,c}$ the *room letters*. With $V_{k,c}^*(\alpha_B)=\alpha_B^B V_{k,c}(\alpha_B^{-1})$ and $L_{k,c}^*(\alpha_R)=\alpha_R^R L_{k,c}(\alpha_R^{-1})$ the same factorization writes the colevel letter of the colevel $|B|-k$,

$$\Psi_B^{(|B|-k)}(\alpha)=\sum_{c\in C_k} V_{k,c}^*(\alpha_B)\,L_{k,c}^*(\alpha_R)\,:$$

each level is one function read twice (Proposition 7.2). The *stations* of B are the maximal sets of patterns whose children are positive multiples of one another, each with a fixed representative, the *child of the station*; every station lies in one level. The *children factorization* lets c run over the stations of level k, with $V_{k,c}=\sum_{p\in c} w_p\,\alpha_B^p$ for the child at p equal to w_p times the child of the station, and $L_{k,c}$ the basis polynomial of the child of the station: all its terms are nonnegative. The *interface dimension* of B is $\sum_k\operatorname{rank}\Theta_B^{(k)}$; for a family it is summed over the tables.

Along the scale of B, $\alpha_B=\rho u$ with $u\in\Delta_B$ (Lemma 2.7), $V_{k,c}(\rho u)=\rho^k V_{k,c}(u)$ and $V_{k,c}^*(\rho u)=\rho^{|B|-k}V_{k,c}^*(u)$; since $|B|-r(B)=d^*(B)$,

$$K(\rho u,\alpha_R)=\rho^{d(B)}\sum_k\rho^{k-d(B)}\sum_c V_{k,c}(u)\,L_{k,c}(\alpha_R),$$
$$\Psi(\rho u,\alpha_R)=\rho^{d^*(B)}\sum_k\rho^{r(B)-k}\sum_c V_{k,c}^*(u)\,L_{k,c}^*(\alpha_R),$$

each letter counted from its own bottom level, both powers running over $\rho^0,\dots,\rho^{c(B)}$.

Theorem 20.13 (What crosses a gluing)**.** *Let $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a family on $\mathcal{E}$ with real data and masses $\mathbf{m}$ on its balance wall, B a bunch with room R, and fix a level factorization of every table, with the bunch letters $V_{j,k,c}$, the room letters $L_{j,k,c}$ and the starred letters $V_{j,k,c}^*$, $L_{j,k,c}^*$. Let $\mathbf{V}(u)=\big(V_{j,k,c}(u),V_{j,k,c}^*(u)\big)_{j,k,c}$ be the* letter vector *of $u\in\Delta_B$. Then, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_R} v^{\mathbf{m}|_R-\mathbf{1}}\int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}}\int_0^\infty \rho^{\kappa_B-1}\prod_j\Big(\sum_{k,c}\rho^{k-d_j(B)}V_{j,k,c}(u)\,L_{j,k,c}(v)\Big)^{-s_j}$$
$$\times\Big(\sum_{k,c}\rho^{r_j(B)-k}V_{j,k,c}^*(u)\,L_{j,k,c}^*(v)\Big)^{-t_j}d\rho\,d\omega_B(u)\,d\omega_R(v),$$

κ_B the wall of B, and the integrand, apart from the measure $u^{\mathbf{m}|_B-\mathbf{1}}d\omega_B(u)$, depends on u only through $\mathbf{V}(u)$. So the period sees the bunch only through its wall κ_B, its widths $c_j(B)$ and the image $\mathbf{V}_*\big(u^{\mathbf{m}|_B-\mathbf{1}}d\omega_B(u)\big)$, the joint law of its bunch letters, unnormalized: two families that agree on the room, with the same room letters, and whose bunches have equal walls, widths and laws of their letter vectors, have equal periods, whatever the numbers of marks of the bunches; the bunches are interchangeable in every room. The same holds for the interest $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ at any real masses, with $\alpha_R \in \mathbb{R}^R_{>0}$ in place of $v \in \Delta_R$ and the factor $e^{-\rho-|\alpha_R|}$ in the integrand.

At width one with factoring weights the children factorization has one set at each level, and the letter vector is $(K_{\mathcal{B}/R}, K_{\mathcal{B}|B}; \Psi_{\mathcal{B}|B}, \Psi_{\mathcal{B}/R})$, each letter from its bottom level: Theorem 18.8, which keeps the scale as the hinge g, keeps of it the cell kernel (Definition 21.2) and the two hinge values γ^K_B, γ^Ψ_B. The theorem holds whatever the weights; what the weights decide is how many bunch letters there are.

Theorem 20.14 (Multivectors bound what crosses)**.** *Let the table carry a multivector $\boldsymbol{\omega}$ of any tensor rank (Definition 7.17), the marks of B ordered before those of R, so that $\boldsymbol{\omega}_{p\sqcup X}$ is the coefficient of $\mathbf{e}_p \wedge \mathbf{e}_X$, $\mathbf{e}_p$ the wedge of the unit vectors of the marks of p in their order. At the level k let the* interface rank $\mathrm{ir}_k(\boldsymbol{\omega})$ *of $\boldsymbol{\omega}$ at B be the rank of the matrix $(\boldsymbol{\omega}_{p\sqcup X})_{p,X}$, and write $\boldsymbol{\omega}_{p\sqcup X} = \sum_{i=1}^{\mathrm{ir}_k(\boldsymbol{\omega})} f_i(p)\, f'_i(X)$.*

(i) $\operatorname{rank}\Theta^{(k)}_B \le \mathrm{ir}_k(\boldsymbol{\omega})^2$*, and* $\operatorname{rank}\Theta^{(k)}_B \le \frac{1}{2}\mathrm{ir}_k(\boldsymbol{\omega})\big(\mathrm{ir}_k(\boldsymbol{\omega})+1\big)$ *when $\boldsymbol{\omega}$ is real. With the positive semidefinite Hermitian matrices*

$$\mathbf{C}_k(\alpha_B) = \sum_p \alpha_B^p\, f(p)f(p)^*, \qquad \mathbf{C}'_k(\alpha_R) = \sum_X \alpha_R^X\, f'(X)f'(X)^*,$$

with $f(p) = \big(f_i(p)\big)_i$ and $f'(X) = \big(f'_i(X)\big)_i$, the level letter is $K^{(k)}_B(\alpha) = \operatorname{tr}\big(\mathbf{C}_k(\alpha_B)\,\overline{\mathbf{C}'_k(\alpha_R)}\big)$, and the real and imaginary parts of the entries of $\mathbf{C}_k(\alpha_B)$ are the bunch letters of a real level factorization with at most $\mathrm{ir}_k(\boldsymbol{\omega})^2$ terms.

(ii) If $\boldsymbol{\omega} = \sum_{l=1}^{\mathrm{rk}(\boldsymbol{\omega})} \boldsymbol{\omega}^{(l)}$ with slices $A^{(l)}$, let the width of the slice *c_l be the dimension of the intersection of the column spaces of the columns $A^{(l)}_B$ of B and $A^{(l)}_R$ of R, and its* floor *$d_l = r - \dim\big($column space of $A^{(l)}_R\big)$, which at tensor rank one are $c(B)$ and $d(B)$. Then $\mathrm{ir}_k(\boldsymbol{\omega}) \le \sum_l \binom{c_l}{k-d_l}$.*

(iii) So $\operatorname{rank}\Theta^{(k)}_B \le \big(\sum_l \binom{c_l}{k-d_l}\big)^2$, and the interface dimension of B is at most $\sum_k \big(\sum_l \binom{c_l}{k-d_l}\big)^2$, a bound in the widths and floors of the slices alone, whatever the numbers of marks of B and R; for a real multivector each square may be replaced by $\frac{1}{2}i_k(i_k+1)$, $i_k = \sum_l \binom{c_l}{k-d_l}$. For a determinantal table, of tensor rank one with $c_1 = c(B)$ and $d_1 = d(B)$, $\mathrm{ir}_k(\boldsymbol{\omega}) \le \binom{c(B)}{k-d(B)}$, $\operatorname{rank}\Theta^{(k)}_B \le \binom{c(B)}{k-d(B)}^2$ and the interface dimension is at most $\binom{2c(B)}{c(B)}$; at width one each level has one bunch letter, as Theorem 12.17 says.

Theorem 20.15 (The interface of a determinantal table)**.** *Let a table be determinantal over $\mathbb{F}$ with slice A, $\theta_T = |\det A_T|^2$, and let B have width $c(B) = w$, the minors $\mathcal{B}|B$ and $\mathcal{B}/R$ on B and $\mathcal{B}|R$ and $\mathcal{B}/B$ on R carrying the factoring weights of Theorem 12.17. There are Hermitian $w \times w$ matrices $\mathbf{S}_B(\alpha_B)$ and $\mathbf{S}_R(\alpha_R)$, positive definite on the open orthant, each entry a rational function homogeneous of degree one, with*

$$K(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)\, \det\big(\mathbf{S}_B(\alpha_B) + \mathbf{S}_R(\alpha_R)\big), \qquad \det\mathbf{S}_B = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}}, \qquad \det\mathbf{S}_R = \frac{K_{\mathcal{B}|R}}{K_{\mathcal{B}/B}},$$

and $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$. *Along the scale of* B, $K(\rho\alpha_B, \alpha_R) = \rho^{d(B)} K_{\mathcal{B}/R}(\alpha_B) K_{\mathcal{B}/B}(\alpha_R) \det\big(\rho\,\mathbf{S}_B + \mathbf{S}_R\big)$, *so the Skeleton values of* B *(Definition 12.9) are the eigenvalues of* $\mathbf{S}_B^{-1/2}\mathbf{S}_R\mathbf{S}_B^{-1/2}$, *real and positive. At width one* $\mathbf{S}_B = \gamma_B^K$, *the hinge value by* K *(Definition 18.4). So a bunch of width* w *of a determinantal table crosses its gluing as one positive definite* $w \times w$ *matrix and one scalar, and the same read at* α^{-1} *for* Ψ.

For a determinantal table a bunch B is thus read by its room through one positive definite matrix, $\binom{c(B)+1}{2}$ real entries when it is symmetric and $c(B)^2$ real parameters when it is Hermitian, and one scalar for each letter, and the bunch letters of Theorem 20.14 are the compound determinants of that matrix. For a multivector of tensor rank $\mathrm{rk}(\boldsymbol{\omega})$ the level letters carry the entries of the matrices $\mathbf{C}_k$ of Theorem 20.14(i), whose size is fixed by the interface ranks. In every case Theorem 20.13 says what crosses, and Theorems 20.14 and 20.15 say how much.

Chapter 21

Periods of constructions

A construction (Definition 31.1) hands its period a structure: the period of a substitution is read from the core and the cells, for any family. The core reads each cell through the joint law of its hinge values, which is pooling done at every cell at once; opened on the core's own shares, the cells ride on the core's fibre, each entering through the period of its own family at the exponents the shares assign it; and where every active letter reads a cell alike, the cell's share is pinned; where the active letters read every cell by one hinge value, as at a full lean of one table, the period factors through the core, and along a hinge tree it is the product of the periods of the nodes over Beta corners at the edges: at a full lean of a table whose weights factor at its bunches of width one, a product of stacked Gammas, monomials in the weights and the periods of the prime nodes of its canonical hinge tree, and k copies glued along any tree cost one period and a power. Blocks aggregate exactly where every active letter reads them by one hinge value, their total or the inverse of the sum of the reciprocals of their marks. The series closure of any cells, at any positive exponents, is one integral over the shared scales of the product of the transforms of the cells, so identical cells enter as one transform to a power; chains of cells are the powers of one operator; and the letters of any chain of bunches, and their integer joint moments, are products of finite matrices. A bunch of any width opens on one share per term of its level factorization; in level form, one term per level, one table at a full lean keeps $c(B)-1$ free shares. The same periods are read through the laws of the cell families, one measure on $(0,\infty)^{2N-1}$ for each (Chapter 22), and through the periods at the full leans of the nodes (Chapters 24 and 25).

21.1 The core reads its cells

Definition 21.1 (Twists)**.** A *twist* of a family F on $\mathcal{E}$ assigns to every table j two vectors $\mathbf{x}_j^K, \mathbf{x}_j^\Psi \in (0,\infty)^{\mathcal{E}}$. The *twisted family* $F[\mathbf{x}]$, $\mathbf{x} = (\mathbf{x}_j^K, \mathbf{x}_j^\Psi)_j$, has the bases and the exponents of F, the basis weights $\theta_{j,T}^K(\mathbf{x}_j^K)^T$ and the cobasis weights $\theta_{j,T}^\Psi(\mathbf{x}_j^\Psi)^{\mathcal{E}\setminus T}$: its letters are $K_j(\mathbf{x}_j^K \circ \alpha)$ and $\Psi_j(\mathbf{x}_j^\Psi \circ \alpha)$, with $\mathbf{x}\circ\alpha = (x_e\alpha_e)_e$. A twist *at a mark* g has all its coordinates 1 off g and is written by its values $\mathbf{x} = (x_j^K, x_j^\Psi)_j$ at g; then $F[\mathbf{x}@g]$ multiplies, in table j, the weights in K_j of the bases through g by x_j^K and the weights in Ψ_j of the bases avoiding g by x_j^Ψ. A twist of one table $\mathcal{H}$ in its cobasis letter alone, $\mathbf{x}^K = \mathbf{1}$, is written $\mathcal{H}[\mathbf{x}]$ with $\mathbf{x} = \mathbf{x}^\Psi$: $K_{\mathcal{H}[\mathbf{x}]} = K_{\mathcal{H}}$ and $\Psi_{\mathcal{H}[\mathbf{x}]}(\alpha) = \Psi_{\mathcal{H}}(\mathbf{x}\circ\alpha)$. On the balance wall the period obeys the *torus relation*: for every twist $\mathbf{x}$, every $\boldsymbol{\lambda} \in (0,\infty)^{\mathcal{E}}$ and all constants $c_j, c_j^* > 0$,

$$P_{F[(c_j\boldsymbol{\lambda}\circ\mathbf{x}_j^K,\ c_j^*\boldsymbol{\lambda}\circ\mathbf{x}_j^\Psi)_j]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \boldsymbol{\lambda}^{-\mathbf{m}} \prod_j c_j^{-r_j s_j}\, c_j^{*\,-r_j^* t_j}\, P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}),$$

both sides converging together, in $[0,\infty]$ for real data; at the scalar $\boldsymbol{\lambda}=\mu\mathbf{1}$ with $c_j=c_j^*=\mu^{-1}$ the twist is unchanged and the factor is 1, by balance. So a twist at a mark g, $\boldsymbol{\lambda}$ equal to 1 off g, has $P_{F[\lambda\mathbf{x}@g]}(\mathbf{m};\mathbf{s},\mathbf{t})=\lambda^{-m_g}P_{F[\mathbf{x}@g]}(\mathbf{m};\mathbf{s},\mathbf{t})$ and is read with $x_1^K=1$, a point of $(0,\infty)^{2N-1}$; for one table it is one number, $x=x_1^\Psi$, and only the cobasis letter moves. A twist of every mark of one letter by one factor multiplies the period by that factor to the power $-r_js_j$ or $-r_j^*t_j$; for one table in its cobasis letter, $P_{\mathcal{H}[\mu\mathbf{x}]}(\mathbf{m};s,t)=\mu^{-r^*t}P_{\mathcal{H}[\mathbf{x}]}(\mathbf{m};s,t)$.

Definition 21.2 (Cell kernels; the core family read at a shape)**.** Let $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a substituted family (Definition 20.1), table j being $\mathcal{H}_j\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$ with the cells $\mathcal{A}_{ij}$ on $B_i\sqcup\{g_i\}$. The *cell kernel* of B_i is the measure

$$u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K_{\mathcal{A}_{ij}/g_i}(u)^{-s_j}\Psi_{\mathcal{A}_{ij}\setminus g_i}(u)^{-t_j}\,d\omega_{B_i}(u)\qquad\text{on }\Delta_{B_i}.$$

The *core family* is $F_{\mathcal{H}}=((\mathcal{H}_j,\theta^{\mathcal{H}_j});s_j,t_j)_j$ on the marks $h_1,\dots,h_q$. For $u=(u_i)_i\in\prod_i\Delta_{B_i}$ the *core family read at* u is its twist $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ (Definition 21.1) by the hinge values, $\boldsymbol{\gamma}(u)$ putting $\gamma^K_{\mathcal{A}_{ij}}(u_i)$ in K_j and $\gamma^\Psi_{\mathcal{A}_{ij}}(u_i)$ in Ψ_j at the mark h_i: table j has the basis weights $\theta_T^{\mathcal{H}_j}\prod_{h_i\in T}\gamma^K_{\mathcal{A}_{ij}}(u_i)$ and the cobasis weights $\theta_T^{\mathcal{H}_j}\prod_{h_i\notin T}\gamma^\Psi_{\mathcal{A}_{ij}}(u_i)$, and the letters $K_{\mathcal{H}_j}\big((\gamma^K_{\mathcal{A}_{ij}}(u_i)\,\alpha_{h_i})_i\big)$ and $\Psi_{\mathcal{H}_j}\big((\gamma^\Psi_{\mathcal{A}_{ij}}(u_i)\,\alpha_{h_i})_i\big)$.

Theorem 21.3 (The core reads its cells)**.** *For a substituted family F the core family read at u has the imbalance of F, and*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\prod_i\Delta_{B_i}}\prod_i\Big(u_i^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K_{\mathcal{A}_{ij}/g_i}(u_i)^{-s_j}\Psi_{\mathcal{A}_{ij}\setminus g_i}(u_i)^{-t_j}\,d\omega_{B_i}(u_i)\Big)\\ \times P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_1},\dots,\kappa_{B_q});\mathbf{s},\mathbf{t}\big),$$

both sides converging together; the same holds with J for P, and on the balance wall. So the period of a substitution depends on each cell only through its wall κ_{B_i} and the joint law, under its cell kernel, of its hinge values $\big(\gamma^K_{\mathcal{A}_{ij}},\gamma^\Psi_{\mathcal{A}_{ij}}\big)_j$: two cells with equal walls and equal laws are interchangeable in every core. For one cell this is Theorem 18.8.

Pooling, done at every cell at once, leaves the core: each cell is integrated over its shape with its kernel and collapses to one mark of the core, which keeps the mass κ_{B_i} and is read through the cell's hinge values as weights. The cells meet only in the core.

Theorem 21.4 (The cells ride on the core's fibre)**.** *Let F be a substituted family with complex data, every exponent* 0 *or of positive real part. Open $K_{\mathcal{H}_j}$ with one share $a_{j,T}$ and $\Psi_{\mathcal{H}_j}$ with one share $b_{j,T}$ per basis T of $\mathcal{H}_j$ (in this statement $|\mathcal{H}_j|$ is the number of bases of $\mathcal{H}_j$), $\sum_T a_{j,T}=s_j$ and $\sum_T b_{j,T}=t_j$, a letter with exponent zero opened by nothing, and let the* loads *of the mark h_i be $a_j(h_i)=\sum_{T\ni h_i}a_{j,T}$ and $b_j(h_i)=\sum_{T\not\ni h_i}b_{j,T}$. The* load family *$F_i(\mathbf{a},\mathbf{b})$ is the family on B_i with, for every j, the table $\mathcal{A}_{ij}\setminus g_i$ at the exponents $\big(a_j(h_i),\,t_j-b_j(h_i)\big)$ and the table $\mathcal{A}_{ij}/g_i$ at $\big(s_j-a_j(h_i),\,b_j(h_i)\big)$; its imbalance is $\kappa_{B_i}-\sum_j\big(a_j(h_i)+b_j(h_i)\big)$. In this statement write $\mathbf{s}^{(i)}=\big((a_j(h_i))_j,(s_j-a_j(h_i))_j\big)$ and $\mathbf{t}^{(i)}=\big((t_j-b_j(h_i))_j,(b_j(h_i))_j\big)$ for its exponents, the tables $\mathcal{A}_{ij}\setminus g_i$ listed first. On every contour of positive real parts of the shares at which all load families converge absolutely,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j}\prod_j\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\prod_{j,T}\big(\theta_T^{\mathcal{H}_j}\big)^{-a_{j,T}-b_{j,T}}$$

$$\times \prod_i J_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big),$$

with $J_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big)=\Gamma\big(\kappa_{B_i}-\sum_j(a_j(h_i)+b_j(h_i))\big)P_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big)$. *The integrand without the periods of the load families is that of the interest* $J_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)$ *of the core family (Definition 21.2) opened with one share per basis (Corollary 4.4).*

The cells ride on the core's share space: the core's letters are opened over its bases, every K-share of a basis loads the marks it occupies and every Ψ-share the marks it leaves out, and each cell enters through the period of its own two hinge minors at the exponents that the loads of its mark assign them.

21.2 Pinned cells

Theorem 21.5 (A cell read alike is pinned)**.** *Let F be a substituted family at every cell of which every active letter reads alike: for every i there are a function γ_i on Δ_{B_i} and constants $c_{ij},c^*_{ij}>0$ with $\gamma^K_{\mathcal{A}_{ij}}=c_{ij}\gamma_i$ whenever $s_j\neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}}=c^*_{ij}\gamma_i$ whenever $t_j\neq 0$, and $c_{ij}=1$ when $s_j=0$, $c^*_{ij}=1$ when $t_j=0$. Let $F_{\mathcal{H}}[\mathbf{c}]$ be the core family twisted by these constants at the marks h_i, c_{ij} in K_j and c^*_{ij} in Ψ_j (Definitions 21.1 and 21.2). Then on the balance wall*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=P_{F_{\mathcal{H}}[\mathbf{c}]}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i\int_{\Delta_{B_i}}\gamma_i^{-\kappa_{B_i}}\,u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\,d\omega_{B_i}(u),$$

the core at the walls of the cells times the pinned shape integrals *of the cells. In general the free shares at the cell i are one fewer than the distinct hinge values of B_i by the active letters, as at a bunch of width one (Theorem 18.13).*

Corollary 21.6 (Read by one hinge value, the period factors through the core)**.** *Let F be a substituted family with real data on its balance wall (Definition 20.1), table j being $\mathcal{H}_j\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$, at every cell of which the active letters read the cell by one hinge value: for every i there is a function γ_i on Δ_{B_i} with $\gamma^K_{\mathcal{A}_{ij}}=\gamma_i$ whenever $s_j\neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}}=\gamma_i$ whenever $t_j\neq 0$, as at a mark left in place, whose two hinge values are both α_h. Let σ be the total strength (Definition 8.1) and $F_i=((\mathcal{A}_{ij},\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ the cell family of the cells at B_i, on $B_i\sqcup\{g_i\}$ (Definition 22.1). Then, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i\int_{\Delta_{B_i}}\gamma_i^{-\kappa_{B_i}}\,u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\,d\omega_{B_i}(u),$$

and at a cell with $0<\kappa_{B_i}<2\sigma$ the family F_i at the masses $(\mathbf{m}|_{B_i},2\sigma-\kappa_{B_i})$ is on its balance wall and its pinned shape integral is $P_{F_i}\big((\mathbf{m}|_{B_i},2\sigma-\kappa_{B_i});\mathbf{s},\mathbf{t}\big)/\Gamma^{\,2}_{1}\!\left[\begin{matrix}\kappa_{B_i},\ 2\sigma-\kappa_{B_i}\\ 2\sigma\end{matrix}\right]$. If F has every proper wall positive, then $0<\kappa_{B_i}<2\sigma$ for every cell, and

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})\prod_i\Gamma^{\,2}_{1}\!\left[\begin{matrix}\kappa_{B_i},\ 2\sigma-\kappa_{B_i}\\ 2\sigma\end{matrix}\right]=P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i P_{F_i}\big((\mathbf{m}|_{B_i},2\sigma-\kappa_{B_i});\mathbf{s},\mathbf{t}\big),$$

every factor on its balance wall with every proper wall positive, the hinge g_i at the mass $2\sigma-\kappa_{B_i}$ and the mark h_i of the core family at κ_{B_i}. For one cell this is Theorem 18.10, and the first display is the shape-integral form of Proposition 18.12.

Corollary 21.7 (Periods along a hinge tree read by one hinge value)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real data on its balance wall with every proper wall positive, let its tables have a common hinge tree $\mathfrak{T}$ rooted at o, with the node families $F_{\mathcal{H}_v}$ (Definition 22.14), and let the active letters read every subtree cell by one hinge value: for every $v \neq o$ there is a function γ_v with $\gamma^K_{\mathcal{A}_{v,j}} = \gamma_v$ whenever $s_j \neq 0$ and $\gamma^\Psi_{\mathcal{A}_{v,j}} = \gamma_v$ whenever $t_j \neq 0$. With σ the total strength (Definition 8.1),*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{\prod_v P_{F_{\mathcal{H}_v}}(\mathbf{m}_v; \mathbf{s}, \mathbf{t})}{\prod_{edges\ \mathfrak{e}} \Gamma_1^2\begin{bmatrix} \kappa_\mathfrak{e},\ 2\sigma - \kappa_\mathfrak{e} \\ 2\sigma \end{bmatrix}},$$

each node family on its balance wall with every proper wall positive, its masses $\mathbf{m}_v$ those of its marks and, on the marker at an end of the edge $\mathfrak{e}$, the wall of the marks across $\mathfrak{e}$: $\kappa_\mathfrak{e}$ on one side and $2\sigma - \kappa_\mathfrak{e}$ on the other. For one table at a full lean, $(0, t)$ or $(s, 0)$, one letter is active, the hypothesis holds on every hinge tree, and this is Theorem 25.6 at the loads $\mathbf{m}$. With the canonical hinge tree of a connected table whose weights factor across each bunch of width one at which it is decomposed, a list or a determinantal table for instance (Theorem 20.9), a series or a parallel node, with any weights on its bases, contributes at a full lean a stacked Gamma times a monomial in its weights (Proposition 25.10), so the period at a full lean of such a table is a product of stacked Gammas, monomials in the weights and the periods of its prime nodes. In particular connected tables $\mathcal{A}_1, \dots, \mathcal{A}_k$ at $(0, t)$, each on its balance wall with every proper wall positive, glued along any tree, every gluing joining a mark of mass $\kappa_\mathfrak{e}$ of one to a mark of mass $t - \kappa_\mathfrak{e}$ of another, have the period

$$\frac{\prod_v P_{\mathcal{A}_v}(\mathbf{m}_v; 0, t)}{\prod_\mathfrak{e} \Gamma_1^2\begin{bmatrix} \kappa_\mathfrak{e},\ t - \kappa_\mathfrak{e} \\ t \end{bmatrix}},$$

whatever the shape of the tree; for k copies of one table glued at the masses $(\kappa, t - \kappa)$ this is $\Gamma_1^2\begin{bmatrix} \kappa,\ t - \kappa \\ t \end{bmatrix} \big(P_\mathcal{A}(\mathbf{m}; 0, t)/\Gamma_1^2\begin{bmatrix} \kappa,\ t - \kappa \\ t \end{bmatrix}\big)^k$.

Where the active letters read every cell through one hinge value, as at a full lean of one table, where only one letter sees a hinge, Ψ at $(0, t)$ and K at $(s, 0)$, every cell is pinned, every hinge integral is the Beta corner of a linear letter (Proposition 15.1 with one coefficient), and the period of the whole construction is the product of the periods of its nodes over the Beta corners of its hinges. Nothing is integrated against anything.

21.3 Blocks aggregate

Theorem 21.8 (Blocks read by one hinge value aggregate exactly)**.** *Let F be a family substituted with blocks (Corollary 20.5(i)) into the tables $\mathcal{H}_j$ of a core family $F_\mathcal{H} = ((\mathcal{H}_j, \theta^{\mathcal{H}_j}); s_j, t_j)_j$, with unit weights in the blocks, the block B_i of ℓ_i marks. The active letters read B_i* through its total *when every table with $s_j \neq 0$ has a parallel block and every table with $t_j \neq 0$ a series block at B_i, and through $e_{\ell_i}/e_{\ell_i - 1} = \big(\sum_{x \in B_i} \alpha_x^{-1}\big)^{-1}$ when every table with $s_j \neq 0$ has a series block and every table with $t_j \neq 0$ a parallel block (Corollary 20.5(i)). Let $U \sim \mathrm{Dir}_\mathbf{m}$ (Definition 3.10), with the block totals $|U_{B_i}|$.*

(i) If the active letters read every block through its total, the joint law of all the active letters, $\big(K_j(U)\big)_{s_j \neq 0}$ and $\big(\Psi_j(U)\big)_{t_j \neq 0}$, with the totals $(|U_{B_i}|)_i$ themselves, is that of $\big(K_{\mathcal{H}_j}(U')\big)_{s_j \neq 0}$,

$\big(\Psi_{\mathcal{H}_j}(U')\big)_{t_j\neq 0}$ *and* U' *for* $U' \sim \mathrm{Dir}_{(m_{B_i})_i}$*: every nonnegative function of these variables, and every absolutely integrable one, has the same expectation on both sides.*

(ii) Let the active letters read B_i *through its total for* $i \in I$ *and through* e_{ℓ_i}/e_{ℓ_i-1} *for* $i \notin I$*, and let* σ *be the total strength (Definition 8.1). The wall of a block is* $\kappa_{B_i} = m_{B_i}$ *for* $i \in I$ *and* $\kappa_{B_i} = m_{B_i} - (\ell_i - 1)\,2\sigma$ *for* $i \notin I$*, and on the balance wall, in* $[0,\infty]$*,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_{i\in I} \Gamma_1^{\ell_i}\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix} \prod_{i\notin I} \Gamma_1^{\ell_i}\begin{bmatrix}(2\sigma - m_x)_{x\in B_i}\\ 2\sigma - \kappa_{B_i}\end{bmatrix}\; P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big),$$

a stacked Gamma read as $+\infty$ *when one of its upper arguments is not positive; both sides are finite together when these are positive.*

(iii) If the active letters read every block through its total, then at fixed pooled masses m_{B_i}*, dividing each among any number of marks leaves the expectation* $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)]$ *of the family word unchanged and multiplies the period by the stacked Gammas* $\Gamma_1^{|B_i|}\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix}$ *alone: lengthening the blocks creates no new integral.*

A block read through its total by some active letters and through $e_\ell/e_{\ell-1}$ *by others keeps the block as one mark of Theorem 21.3, with its kernel and its hinge values.*

21.4 Closures

Theorem 21.9 (The series closure at positive exponents)**.** *Let every table of* F *be a series closure,* $(\mathcal{B}_j,\theta_j) = \binom{\{h_1,\dots,h_q\}}{q-1}\langle \mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$ *(Corollary 20.5(ii)), with exponents* $s_j, t_j > 0$*, and let* f_{F_i} *be the transform of the cell family* $F_i = ((\mathcal{A}_{ij},\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ *of the cells at* B_i*, on* $B_i \sqcup \{g_i\}$ *(Definition 22.1). On the balance wall, in* $[0,\infty]$*,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_j x_j^{s_j-1} \prod_{j\geq 2} y_j^{t_j-1} \prod_{i=1}^{q} f_{F_i}\big(\mathbf{x},(1,y_2,\dots,y_N)\big)\; d\mathbf{x}\,dy_2\cdots dy_N,$$

the integral over $(0,\infty)^N \times (0,\infty)^{N-1}$*: the cells communicate only through* $2N-1$ *shared scales. A cell entering* k *times enters as its transform to the power* k*: one transform per kind of cell, its multiplicity a power, the order of the cells immaterial; for one table the closure is* $\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}\prod_i f_{\mathcal{A}_i}(y)\,dy$*. With* $|\mathbf{s}| = \sum_j s_j$ *and the wall* κ_{B_i} *of the cell* B_i*, its hinge wall* κ_{F_i} *(Theorem 20.7), each transform is one Gramma integrated over the shape of the cell,*

$$f_{F_i}(\mathbf{x},\mathbf{y}) = \int_{\Delta_{B_i}} \mathsf{G}_{|\mathbf{s}|-\kappa_{B_i};(1,-1)}\Big(\sum_j \frac{x_j}{\gamma^K_{\mathcal{A}_{ij}}(u)},\; \sum_j y_j\,\gamma^{\Psi}_{\mathcal{A}_{ij}}(u)\Big) \times u^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K_{\mathcal{A}_{ij}\setminus g_i}(u)^{-s_j}\,\Psi_{\mathcal{A}_{ij}\setminus g_i}(u)^{-t_j}\,d\omega_{B_i}(u),$$

against the form of the family of the hinge minors $F_i \setminus g_i = ((\mathcal{A}_{ij}\setminus g_i,\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ *at the masses* $\mathbf{m}|_{B_i}$*. When that family is on its balance wall,* $\kappa_{B_i} = |\mathbf{s}|$*, as at every cell of one table at the centre,* $s = t = \sigma$ *with every mass* σ*, the index is* 0*, the Gramma depends on its two coefficients through their product,* $\mathsf{G}_{0;(1,-1)}(a,b) = \mathsf{G}_{0;(1,-1)}(1,ab)$ *(Theorem 2.3(iii)), and, when the family of the hinge minors has every proper wall positive,*

$$f_{F_i}(\mathbf{x},\mathbf{y}) = P_{F_i\setminus g_i}(\mathbf{m}|_{B_i};\mathbf{s},\mathbf{t})\;\mathbb{E}_{U\sim\mathrm{Breta}^\circ(F_i\setminus g_i;\mathbf{m}|_{B_i};\mathbf{s},\mathbf{t})}\Big[\mathsf{G}_{0;(1,-1)}\Big(1,\;\sum_j \frac{x_j}{\gamma^K_{\mathcal{A}_{ij}}(U)}\sum_j y_j\,\gamma^{\Psi}_{\mathcal{A}_{ij}}(U)\Big)\Big],$$

for one table $P_{\mathcal{A}_i\setminus g_i}(\mathbf{m}|_{B_i};s,t)\,\mathbb{E}\big[\mathsf{G}_{0;(1,-1)}\big(1,xy\,\varrho_{\mathcal{A}_i}(U)\big)\big]$*: the image under the ratio of the form of the hinge minor, read through* $\mathsf{G}_{0;(1,-1)}$*. So* $k\ge 2$ *copies of one cell family closed in series have the period* $\prod_j(\Gamma(s_j)\Gamma(t_j))^{-1}\int\prod_j x_j^{s_j-1}\prod_{j\ge2}y_j^{t_j-1}f_{F_1}\big(\mathbf{x},(1,y_2,\dots,y_N)\big)^k\,d\mathbf{x}\,dy_2\cdots dy_N$*.*

For one table the closure of q cells is a node of width one (Definition 19.22) with the bunch $B_1\sqcup\cdots\sqcup B_{q-1}$, a series connection (Proposition 19.35), against the room B_q. Theorem 19.24 reads it as $\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty f_B(y)f_R(y)\frac{dy}{y}$, and Theorem 21.9 is the same integral written symmetrically in all the cells, $\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}\prod_i f_{\mathcal{A}_i}(y)\,dy$ with the transforms of the cells (Definition 22.1), which are the orthant transforms f_i° of Proposition 19.35; for k equal cells at the centre the formula for k copies is Theorem 23.2(i), with $k-1$ copies at a series node and one copy as the room. The table that is one circuit, $\binom{\mathcal{E}}{n-1}$, is the series closure of n single marks, each with the transform $\mathsf{G}_{s-m_e;(1,-1)}(1,y)$ up to its weight (Corollary 17.20). The parallel closure is the dual statement (Proposition 8.14).

21.5 Chains and transfer

Theorem 21.10 (A chain is the power of one operator)**.** *Let* $F=((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ *be a family on* $\mathcal{E}$ *with two marks* $g_{\mathrm{in}}\ne g_{\mathrm{out}}$*, each neither a loop nor a coloop in every table, real exponents, masses on the balance wall with* $m_{g_{\mathrm{in}}}+m_{g_{\mathrm{out}}}=\sum_j(s_j+t_j)$*, and let* $F^{\triangleright k}$ *be* k *copies of* F *glued table by table along marks,* $g_{\mathrm{out}}^{(i)}$ *with* $g_{\mathrm{in}}^{(i+1)}$ *(Lemma 20.8). For* $\mathbf{x}=(x_j^K,x_j^\Psi)_j\in(0,\infty)^{2N}$ *let* $F^{\triangleright k}[\mathbf{x}@g_{\mathrm{in}}^{(1)}]$ *be* $F^{\triangleright k}$ *twisted at* $g_{\mathrm{in}}^{(1)}$ *(Definition 21.1), and in this statement write* $P_k(\mathbf{x})$ *for its period at the masses of the copies and* $\mathcal{A}_j^{\mathbf{x}}$ *for table* j *of* $F[\mathbf{x}@g_{\mathrm{in}}]$*, read as a cell with the hinge* g_{out}*. With* $B=\mathcal{E}\setminus\{g_{\mathrm{out}}\}$ *define on nonnegative functions* H *of the twist*

$$(\mathcal{Q}H)(\mathbf{x})=\int_{\Delta_B}H\Big(\big(\gamma^K_{\mathcal{A}_j^{\mathbf{x}}}(u),\gamma^\Psi_{\mathcal{A}_j^{\mathbf{x}}}(u)\big)_j\Big)\,u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{A}_j^{\mathbf{x}}/g_{\mathrm{out}}}(u)^{-s_j}\Psi_{\mathcal{A}_j^{\mathbf{x}}\setminus g_{\mathrm{out}}}(u)^{-t_j}\,d\omega_B(u),$$

$$P_0(\mathbf{x})=\int_0^1 v^{m_{g_{\mathrm{in}}}-1}(1-v)^{m_{g_{\mathrm{out}}}-1}\prod_j\big(1-v+x_j^Kv\big)^{-s_j}\big(1-v+x_j^\Psi v\big)^{-t_j}\,dv.$$

Then $P_k=\mathcal{Q}P_{k-1}$*, and in* $[0,\infty]$

$$P_{F^{\triangleright k}}(\mathbf{m};\mathbf{s},\mathbf{t})=(\mathcal{Q}^kP_0)(\mathbf{1}).$$

The operator acts on functions of the $2N$ *twists, of which one is removed by the scale of the hinge. Where the twist is pinned, the active letters reading the cell by one hinge value,* $\gamma^K_{\mathcal{A}_j}$ *for* $s_j\ne0$ *and* $\gamma^\Psi_{\mathcal{A}_j}$ *for* $t_j\ne0$ *all equal, as at a full lean of one table, every* P_k *is a multiple of one power of the twist along the diagonal* $\mathbf{x}=\lambda\mathbf{1}$*,* $\mathcal{Q}$ *multiplies it by* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}m_{g_{\mathrm{in}}},\ m_{g_{\mathrm{out}}}\\ m_{g_{\mathrm{in}}}+m_{g_{\mathrm{out}}}\end{matrix}\right]$*, and*

$$P_{F^{\triangleright k}}(\mathbf{m};\mathbf{s},\mathbf{t})=\frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})^k}{\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}m_{g_{\mathrm{in}}},\ m_{g_{\mathrm{out}}}\\ m_{g_{\mathrm{in}}}+m_{g_{\mathrm{out}}}\end{matrix}\right]^{k-1}};$$

the product of Corollary 21.7 on a path; hinge values that are only proportional, with constants other than one, still pin each cell (Theorem 21.5) but carry the twist off the diagonal.

A chain is pooled one copy at a time: the first copy, read as the cell of its bunch, collapses to its hinge and twists the next copy by its hinge values, so the period of the chain is one operator on functions of the twist applied once per copy, and the two-mark table left at the end supplies the starting function. Where the hinge values pin the twist the operator is a multiplication, and the chain costs a power.

Theorem 21.11 (Letters by transfer)**.** *Let $(\mathcal{B},\theta)$ be a table with any weights and $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$ a chain of bunches, with the steps $B_{l-1} = S_l \setminus S_{l-1}$. Group the patterns of each S_l into its stations (Definition 20.12), the children being tables on $\mathcal{E}\setminus S_l$ and the child at p being w_p times the child of its station c. For a station c of S_{l-1} and a station c' of S_l let*

$$\mathbf{M}_l^K(c,c') = \sum_{p'} w(c,p')\,\alpha_{B_{l-1}}^{p'}, \qquad \mathbf{M}_l^\Psi(c,c') = \sum_{p'} w(c,p')\,\alpha_{B_{l-1}}^{B_{l-1}\setminus p'},$$

the sums over the patterns p' of B_{l-1} in the child of the station c whose child is $w(c,p')$ times the child of the station c'. Then, with $\mathbf{e}$ the unit vector of the station of the empty pattern of S_0 and $\boldsymbol{\theta}_\varnothing$ the vector of the weights of the one basis $\varnothing$ of the children of the stations of S_L,

$$K = \mathbf{e}^{\mathsf{T}}\mathbf{M}_1^K\mathbf{M}_2^K\cdots\mathbf{M}_L^K\,\boldsymbol{\theta}_\varnothing, \qquad \Psi = \mathbf{e}^{\mathsf{T}}\mathbf{M}_1^\Psi\cdots\mathbf{M}_L^\Psi\,\boldsymbol{\theta}_\varnothing.$$

The number of states at S_l is the number of its stations, at least the interface dimension of S_l (Definition 20.12), and the states can be reduced to that many linearly independent ones; at a bunch S_l of width one with factoring weights it is two. For a table carrying a multivector $\boldsymbol{\omega}$ of any tensor rank (Definition 7.17), with the widths c_i and the floors d_i of its slices at S_l (Theorem 20.14(ii)), the interface dimension of S_l is at most $\sum_k \big(\sum_i \binom{c_i}{k-d_i}\big)^2$, each square q^2 replaced by $\frac{1}{2}q(q+1)$ when $\boldsymbol{\omega}$ is real; for a determinantal table it is at most $\binom{2c(S_l)}{c(S_l)}$ (Theorem 20.14).

Theorem 21.12 (Integer joint moments by fixed matrices)**.** *Let the letters of a family $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be products of matrix functions of the variables of the steps $B_0,\dots,B_{L-1}$ of one chain of bunches, $K_j = \mathbf{e}_j^{\mathsf{T}}\mathbf{M}_{j,0}^K(\alpha_{B_0})\cdots\mathbf{M}_{j,L-1}^K(\alpha_{B_{L-1}})\boldsymbol{\theta}_{j,\varnothing}$ and $\Psi_j = \mathbf{e}_j^{\mathsf{T}}\mathbf{M}_{j,0}^\Psi(\alpha_{B_0})\cdots\mathbf{M}_{j,L-1}^\Psi(\alpha_{B_{L-1}})\boldsymbol{\theta}_{j,\varnothing}$ with constant vectors $\mathbf{e}_j$ and $\boldsymbol{\theta}_{j,\varnothing}$, as Theorem 21.11 gives for every table along any chain of bunches; and let the steps carry independent positive measures $\phi_0,\dots,\phi_{L-1}$ with the required moments finite. For integers $a_j,b_j\ge 0$ put*

$$\mathbf{A}_l = \int \bigotimes_j \Big(\mathbf{M}_{j,l}^K(\alpha)^{\otimes a_j}\otimes \mathbf{M}_{j,l}^\Psi(\alpha)^{\otimes b_j}\Big)\,\phi_l(d\alpha),$$

one α in all the factors. Then

$$\int \prod_j K_j^{a_j}\Psi_j^{b_j}\prod_{l<L}\phi_l(d\alpha_{B_l}) = \Big(\bigotimes_j \mathbf{e}_j^{\otimes(a_j+b_j)}\Big)^{\mathsf{T}}\mathbf{A}_0\mathbf{A}_1\cdots\mathbf{A}_{L-1}\Big(\bigotimes_j \boldsymbol{\theta}_{j,\varnothing}^{\otimes(a_j+b_j)}\Big).$$

For a chain of identical steps, with one pair of matrix functions $\mathbf{M}_j^K,\mathbf{M}_j^\Psi$ for each table, the states of all the S_l indexed by one set, $\boldsymbol{\theta}_{j,\varnothing}$ vanishing at the states that do not occur at S_L, and one measure ϕ, every $\mathbf{A}_l$ is one matrix $\mathbf{A}$: the moment sequence along L is the power of one matrix, of size $\prod_j(\text{number of states of table } j)^{a_j+b_j}$, and satisfies a linear recurrence with constant coefficients of that order. With independent laws Gamma_{m_x} on the marks (Definition 1.1(vii)) the moment $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\big[\prod_j K_j(U)^{a_j}\Psi_j(U)^{b_j}\big]$ is that integral divided by the rising factorial $(m_{\mathcal{E}})_{\sum_j(a_jr_j+b_jr_j^*)}$; with the law $\mathrm{Dir}_{\mathbf{m}|_{B_l}}$ on each step the entries are the Dirichlet moments $\prod_{x\in B_l}(m_x)_{k_x}/(m_{B_l})_{|\mathbf{k}|}$ of the monomials $\alpha_{B_l}^{\mathbf{k}}$ (Theorem 3.11).

21.6 A bunch of any width

Theorem 21.13 (Gluing across a bunch of any width)**.** *Let F be a family with real data on its balance wall with every proper wall positive, B a bunch of any width with room R, and fix level factorizations with nonnegative terms (Definition 20.12), the children factorization for instance, with the bunch letters $V_{j,k,c}$ and the room letters $L_{j,k,c}$; drop the terms whose bunch letter or room letter vanishes identically, and let C_j be the set of the terms (k,c) left for table j. Open every active letter with one share per term, $a_{j,k,c}$ for the term $V_{j,k,c}L_{j,k,c}$ of K_j with $\sum_{k,c} a_{j,k,c} = s_j$ and $b_{j,k,c}$ for the term $V^*_{j,k,c}L^*_{j,k,c}$ of Ψ_j with $\sum_{k,c} b_{j,k,c} = t_j$, pinned by the scale of B,*

$$\sum_{j,k,c} \Big(\big(k - d_j(B)\big) a_{j,k,c} + \big(r_j(B) - k\big) b_{j,k,c} \Big) = \kappa_B .$$

On every contour of positive real parts at which the two integrals below converge,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ pin} \prod_j \Gamma_{|C_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix} \Gamma_{|C_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix}$$
$$\times \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_{j,k,c} V_{j,k,c}^{-a_{j,k,c}} {V^*_{j,k,c}}^{-b_{j,k,c}}\, d\omega_B(u) \int_{\Delta_R} v^{\mathbf{m}|_R - \mathbf{1}} \prod_{j,k,c} L_{j,k,c}^{-a_{j,k,c}} {L^*_{j,k,c}}^{-b_{j,k,c}}\, d\omega_R(v),$$

the two inner integrals projective integrals of degree zero, the periods of the bunch word *and the* room word *at the shares. Their letters are the bunch and room letters of the level factorization, which need not be the letters of tables, so their convergence is that of the integrals themselves. The free shares are the terms less the active letters less one. For one table at a full lean $(0,t)$ in level form, one station per level, with the share b_k and the bunch and room letters V^*_k, L^*_k of the station of the level k, $d(B) \le k \le r(B)$, the pins are $\sum_k b_k = t$ and $\sum_k (r(B)-k)\, b_k = \kappa_B$, the free shares are the $c(B)-1$ shares of the levels strictly between $d(B)$ and $r(B)$, and on a contour of such shares*

$$P_F(\mathbf{m};0,t) = \frac{1}{(2\pi i)^{c(B)-1}} \int \frac{\prod_{k=d(B)}^{r(B)} \Gamma(b_k)}{c(B)\,\Gamma(t)} \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_k V_k^{*-b_k}\, d\omega_B(u)$$
$$\times \int_{\Delta_R} v^{\mathbf{m}|_R - \mathbf{1}} \prod_k L_k^{*-b_k}\, d\omega_R(v) \prod_{d(B)<k<r(B)} db_k,$$

with $b_{d(B)} = \frac{1}{c(B)}\big(\kappa_B - \sum_{d(B)<k<r(B)} (r(B)-k)\, b_k\big)$ and $b_{r(B)} = t - b_{d(B)} - \sum_{d(B)<k<r(B)} b_k$. At $c(B) = 2$, with $b_{d(B)+1} = z$, the shares are $\frac{\kappa_B - z}{2}$, z and $\frac{2t - \kappa_B - z}{2}$; at $c(B) = 1$ no share is free, and the period is $\Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa_B,\ t-\kappa_B \\ t\end{bmatrix}$ times the two inner periods (Corollary 18.14). At width one with factoring weights, every letter active, the free shares are $2N-1$, and this is Theorem 18.13.

For one table in level form at a full lean $(0,t)$ the scale can also be integrated before the shapes. Writing $\Psi(\rho u, v) = \rho^{d^*(B)} \sum_{i=0}^{c(B)} \rho^i V^*_{r(B)-i}(u) L^*_{r(B)-i}(v)$ and rescaling ρ so that the two extreme terms become 1 and $x^{c(B)}$, the period is the integral over $\Delta_B \times \Delta_R$ of a form that factors into a bunch part and a room part, times the scale integral $\int_0^\infty x^{\kappa_B - 1}\big(1 + \sum_{i=1}^{c(B)-1} \zeta_i x^i + x^{c(B)}\big)^{-t} dx$, whose coefficients

$$\zeta_i = \frac{V^*_{r(B)-i}(u)}{V^*_{r(B)}(u)^{1-i/c(B)}\, V^*_{d(B)}(u)^{i/c(B)}} \cdot \frac{L^*_{r(B)-i}(v)}{L^*_{r(B)}(v)^{1-i/c(B)}\, L^*_{d(B)}(v)^{i/c(B)}}$$

are each a bunch part times a room part: the room receives $c(B)-1$ ratios. At $c(B)=1$ the scale integral is ${}_{1}^{2}\Gamma\left[\begin{matrix}\kappa_B,\ t-\kappa_B\\ t\end{matrix}\right]$, which is Theorem 18.10; its Mellin transform in $(\zeta_1,\dots,\zeta_{c(B)-1})$ at $(z_1,\dots,z_{c(B)-1})$ is $\prod_{k=d(B)}^{r(B)}\Gamma(b_k)/\big(c(B)\Gamma(t)\big)$ at the shares $b_{r(B)-i}=z_i$ of Theorem 21.13, its share kernel with the factor $1/c(B)$ of the pin, at $c(B)=2$ the kernel $\Gamma(z)\Gamma\big(\frac{\kappa_B-z}{2}\big)\Gamma\big(\frac{2t-\kappa_B-z}{2}\big)/\big(2\Gamma(t)\big)$. With the factorization taken to be the canonical one of the seam of the bunch, the bunch word is written through one law, which with the wall is exactly what the bunch contributes (Chapter 27).

Chapter 22

The law of a cell

A table with a bunch of width one across which its weights factor is two cells glued along a hinge that neither of them shows to the other (Lemma 20.8). What a cell family must carry so that its contribution to every family built on it is fixed is its hinge wall and one measure on $(0,\infty)^{2N-1}$, the law of its ratio vector under its base form, for one cell a measure on the half-line, and nothing less (Theorem 22.7). A cell family enters a period only through that law (Theorem 22.4); the law is read in two ways, by the transform of the cell family and by its dual, and the two transforms expand one function on the two sides of one strip (Proposition 22.5). The step that makes all of this compose is Lemma 22.2: a cell sitting at a mark of any family can be integrated away, leaving the family with that mark twisted by the ratio vector of the cell family; pooling (Theorem 18.8) and Theorem 21.3, in which the core sees each cell as one mark, are its cases. A core with cells at its markers is read the same way: the law of a substitution is the mixture, over the laws of its cell families, of the laws of the twisted core family (Theorem 22.11); at a series node the transforms multiply and at a parallel node the dual transforms multiply (Theorem 22.13); and the period of a family built along a common hinge tree is read from the leaves up, with one law per edge, computed once for each distinct subtree (Theorem 22.15, Proposition 22.16). The calculus extends to bunches of every width (Chapter 27).

22.1 Cells and their laws

Definition 22.1 (The law of a cell family)**.** A *cell family* is a family $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ on $B \sqcup \{g\}$ whose tables are cells with the one hinge g (Definition 20.1), with real exponents, carrying masses $\mathbf{m}$ on B; one cell $\mathcal{A}$ at the exponents (s,t) is the cell family $((\mathcal{A},\theta); s, t)$, and its objects carry the subscript $\mathcal{A}$. Its *hinge wall* is

$$\kappa_F = m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big),$$

for one cell the hinge wall $\kappa_{\mathcal{A}}$ of Lemma 20.8. Its *base form* and its *ratio vector* are

$$\phi_F(du) = u^{\mathbf{m-1}} \prod_j K_{\mathcal{A}_j/g}(u)^{-s_j}\, \Psi_{\mathcal{A}_j\setminus g}(u)^{-t_j}\, \gamma^K_{\mathcal{A}_1}(u)^{-\kappa_F}\, d\omega_B(u) \quad \text{on } \Delta_B,$$

$$\boldsymbol{\varrho}_F = \Big(\frac{\gamma^K_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}},\ \frac{\gamma^\Psi_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)_j :\ \Delta_B \to (0,\infty)^{2N-1},$$

the coordinate $\gamma^K_{\mathcal{A}_1}/\gamma^K_{\mathcal{A}_1} = 1$ omitted, and its *law* is the image $(\boldsymbol{\varrho}_F)_*\phi_F$, a positive measure on $(0,\infty)^{2N-1}$. For $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$, also written as the pair $(\mathbf{z}_K, \mathbf{z}_\Psi)$ of $\mathbf{z}_K = (z_{j,K})_j$ and $\mathbf{z}_\Psi = (z_{j,\Psi})_j$, write

$$\boldsymbol{\varrho}_F^{-\mathbf{z}} = \prod_{j\geq 2} \Big(\frac{\gamma^K_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)^{-z_{j,K}} \prod_j \Big(\frac{\gamma^\Psi_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)^{-z_{j,\Psi}},$$

the coordinates of the ratio vector to the powers $-z_{j,K}$ $(j \geq 2)$ and $-z_{j,\Psi}$; the *Mellin transform* of the law, $\int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F$, lies in $[0,\infty]$ for real $\mathbf{z}$ and is defined for complex $\mathbf{z}$ at whose real part it is finite, with modulus at most that value. For one cell the ratio vector is the ratio $\varrho_{\mathcal{A}}$ (Definition 20.1), the base form is

$$\phi_{\mathcal{A}}(du) = u^{\mathbf{m-1}} K_{\mathcal{A}/g}(u)^{\kappa_{\mathcal{A}}-s} K_{\mathcal{A}\setminus g}(u)^{-\kappa_{\mathcal{A}}} \Psi_{\mathcal{A}\setminus g}(u)^{-t}\, d\omega_B(u),$$

the law $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$ is a measure on $(0,\infty)$, and, with $z = z_{1,\Psi}$, the Mellin transform is $\int_{\Delta_B} \varrho_{\mathcal{A}}^{-z}\, d\phi_{\mathcal{A}} = \int_0^\infty x^{-z}\, d\big((\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}\big)(x)$. For a cell family with $s_j, t_j > 0$ the *transform*, the *dual transform* and the two *hinge measures* are, for $\mathbf{x}, \mathbf{y} \in (0,\infty)^N$,

$$f_F(\mathbf{x},\mathbf{y}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j\setminus g} \Psi^{-t_j}_{\mathcal{A}_j\setminus g} \exp\Big(-\sum_j \frac{x_j}{\gamma^K_{\mathcal{A}_j}} - \sum_j y_j\, \gamma^\Psi_{\mathcal{A}_j}\Big)\, d\alpha,$$

$$\check{f}_F(\mathbf{x},\mathbf{y}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j/g} \Psi^{-t_j}_{\mathcal{A}_j/g} \exp\Big(-\sum_j \frac{x_j}{\gamma^\Psi_{\mathcal{A}_j}} - \sum_j y_j\, \gamma^K_{\mathcal{A}_j}\Big)\, d\alpha,$$

with values in $[0,\infty]$, the measure $\mathfrak{r}_F$, the image of $\alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j\setminus g} \Psi^{-t_j}_{\mathcal{A}_j\setminus g}\, d\alpha$ under $\alpha \mapsto \big((1/\gamma^K_{\mathcal{A}_j}(\alpha))_j, (\gamma^\Psi_{\mathcal{A}_j}(\alpha))_j\big)$, and the measure $\check{\mathfrak{r}}_F$, the image of $\alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j/g} \Psi^{-t_j}_{\mathcal{A}_j/g}\, d\alpha$ under $\alpha \mapsto \big((1/\gamma^\Psi_{\mathcal{A}_j}(\alpha))_j, (\gamma^K_{\mathcal{A}_j}(\alpha))_j\big)$, positive measures on $(0,\infty)^{2N}$. For $c > 0$, $f_F(\mathbf{x},\mathbf{y}) = c^{\kappa_F - |\mathbf{s}|} f_F(\mathbf{x}/c, c\mathbf{y})$ and $\check{f}_F(\mathbf{x},\mathbf{y}) = c^{\kappa_F - |\mathbf{t}|} \check{f}_F(\mathbf{x}/c, c\mathbf{y})$, so each is fixed by its values at $x_1 = 1$. For one cell they are read there: the transform and the dual transform are $f_{\mathcal{A}}(y) = f_F(1,y)$ and $\check{f}_{\mathcal{A}}(y) = \check{f}_F(1,y)$, $y > 0$, and the hinge measures $\mathfrak{r}^\Psi_{\mathcal{A}}$ and $\mathfrak{r}^K_{\mathcal{A}}$ are the images of $e^{-\lambda}\mathfrak{r}_F(d\lambda, d\mu)$ and of $e^{-\lambda}\check{\mathfrak{r}}_F(d\lambda, d\mu)$ under $(\lambda,\mu) \mapsto \mu$, positive measures on $(0,\infty)$.

The base form of a cell is the restriction to Δ_B of the orthant form $x^{\mathbf{m-1}} K^{\kappa_{\mathcal{A}}-s}_{\mathcal{A}/g} K^{-\kappa_{\mathcal{A}}}_{\mathcal{A}\setminus g} \Psi^{-t}_{\mathcal{A}\setminus g} dx$, homogeneous of degree $m_B + (\kappa_{\mathcal{A}} - s)(r_{\mathcal{A}} - 1) - \kappa_{\mathcal{A}} r_{\mathcal{A}} - t(|B| - r_{\mathcal{A}}) = 0$, so it is read on any section of a positive function of degree one. When $\mathcal{A}$ is the cell of a bunch B of width one in a table, its hinge minors are the restriction $\mathcal{B}|B$ and the contraction $\mathcal{B}/R$, $\kappa_{\mathcal{A}}$ is the wall κ_B, $\phi_{\mathcal{A}}$ is the base form ϕ_B, $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$ is the law of the bunch, $\int \varrho_{\mathcal{A}}^{-b^d} d\phi_{\mathcal{A}}$ is the continued period of the bunch family at the share b^d, and $f_{\mathcal{A}}$ is the bunch transform f_B (Definitions 19.22 and 19.41, Lemma 19.25); the cell kernel of Definition 21.2 is $(\gamma^K_{\mathcal{A}})^{\kappa_{\mathcal{A}}}\phi_{\mathcal{A}}$, and for a cell family $(\gamma^K_{\mathcal{A}_1})^{\kappa_F}\phi_F$. None of these objects mentions a room: a cell carries them by itself, and they are all that a room ever sees of it (Theorem 22.7). The hinge wall is read from the other side of the hinge: with the mass $s + t - \kappa_{\mathcal{A}}$ on g the table $\mathcal{A}$ itself is on its balance wall, and in a two-sum on its balance wall that mass is the hinge wall of the other cell (Lemma 20.8).

22.2 Absorbing a cell

Lemma 22.2 (Absorbing a cell)**.** *Let $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a cell family on $B \sqcup \{g\}$ with masses $\mathbf{m}$ on B, its hinge wall κ_F, base form ϕ_F and ratio vector $\boldsymbol{\varrho}_F$, whose coordinates are*

written $x_{j,K}$ and $x_{j,\Psi}$, $x_{1,K}=1$. Let Z be a finite set disjoint from $B\sqcup\{g\}$, $\mathbf{m}_Z$ masses on Z, and $f_1,\dots,f_p$ positive continuous functions on $(0,\infty)^Z\times(0,\infty)$, each assigned to one letter of F, with real exponents a_i such that the exponents assigned to each letter add to its exponent, s_j for $K_{\mathcal{A}_j}$ and t_j for $\Psi_{\mathcal{A}_j}$. In this statement write L_i, γ_i and x_i for the letter of the hinge minor, the hinge value and the coordinate of the letter of f_i: $L_i=K_{\mathcal{A}_j/g}$, $\gamma_i=\gamma^K_{\mathcal{A}_j}$ and $x_i=x_{j,K}$ when it is $K_{\mathcal{A}_j}=K_{\mathcal{A}_j/g}\,(\gamma^K_{\mathcal{A}_j}+\alpha_g)$, and $L_i=\Psi_{\mathcal{A}_j\setminus g}$, $\gamma_i=\gamma^\Psi_{\mathcal{A}_j}$ and $x_i=x_{j,\Psi}$ when it is $\Psi_{\mathcal{A}_j}=\Psi_{\mathcal{A}_j\setminus g}\,(\gamma^\Psi_{\mathcal{A}_j}+\alpha_g)$. Let $H\ge 0$ be a Borel function on $(0,\infty)^p$. If the form

$$\omega=\alpha_B^{\mathbf{m}-\mathbf{1}}\alpha_Z^{\mathbf{m}_Z-\mathbf{1}}\,H\Big(\big(f_i(\alpha_Z,\gamma_i(\alpha_B))\big)_i\Big)\prod_{i=1}^{p}\Big[L_i(\alpha_B)\,f_i\big(\alpha_Z,\gamma_i(\alpha_B)\big)\Big]^{-a_i}d\alpha_B\,d\alpha_Z$$

is homogeneous of degree zero on $(0,\infty)^{B\sqcup Z}$, then, in $[0,\infty]$,

$$\int_{\{h=1\}}\omega=\int_{(0,\infty)^{2N-1}}\Big(\int_{\{h'=1\}}\omega_{\mathbf{x}}\Big)\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{x}),$$
$$\omega_{\mathbf{x}}=\alpha_Z^{\mathbf{m}_Z-\mathbf{1}}\alpha_g^{\kappa_F-1}\,H\Big(\big(f_i(\alpha_Z,x_i\alpha_g)\big)_i\Big)\prod_i f_i(\alpha_Z,x_i\alpha_g)^{-a_i}\,d\alpha_Z\,d\alpha_g,$$

the integrals taken over any sections of positive functions h, h' of degree one on the two orthants.

For one table the ratio vector is the ratio of the cell. The lemma says that whatever the rest of a family does with a cell, provided it sees the cell through the cell's letters at the hinge, it sees it only through the law: the cell collapses to its hinge, which keeps the mass κ_F, and the letters that see the hinge see it stretched by the coordinates of the ratio vector. Pooling (Theorem 18.8) is the case of one cell family read on the shape before the law is formed; Theorem 21.3, in which the core sees each cell as one mark, is the lemma at every cell of a core, with the hinge values of the cells kept as weights of the core instead of their ratios.

22.3 Two cells glued

Lemma 22.3 (Two marks)**.** *For $s,t>0$, $\kappa,\kappa'>0$ with $\kappa+\kappa'=s+t$, and $x,x'>0$, let $\binom{\{g,g'\}}{1}[(x,x')]$ be the list $\binom{\{g,g'\}}{1}$ twisted in its cobasis letter by x at g and by x' at g' (Definition 21.1). With the masses κ on g and κ' on g' its period is*

$$\begin{aligned}P_{\binom{\{g,g'\}}{1}[(x,x')]}\big((\kappa,\kappa');s,t\big)&=\int_0^\infty v^{\kappa-1}(1+v)^{-s}(xv+x')^{-t}\,dv\\&=x^{-t}\,\Gamma_2\begin{bmatrix}\kappa,&\kappa'\\ s,&t\end{bmatrix}\,\mathsf{B}^{\{\{0,1\}\}}_{(s,t),\,\kappa';\,I_{\{0,1\}}}\big((1,x'/x)\big)\\&=\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}\,\mathsf{G}_{s-\kappa;(1,-1)}(1,xy)\,\mathsf{G}_{s-\kappa';(1,-1)}(1,x'y)\,dy\end{aligned}$$

(Proposition 15.1, the two coefficients merging into one of letter exponent $s+t$ when $x=x'$), unchanged when (κ,x) and (κ',x') are swapped. For a family of tables the kernel is the period of the twisted core family of Theorem 22.11(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table.

Theorem 22.4 (Two cell families glued)**.** *Let F_1 and F_2 be cell families with plain tables on $B\sqcup\{g\}$ and $R\sqcup\{g\}$, at the exponents $(s_j,t_j)_j$ with $s_j,t_j>0$, and let $F_1\oplus_g F_2$, whose table j is the two-sum of the tables j of F_1 and F_2 (Lemma 20.8), carry masses $\mathbf{m}$ on its balance wall, so that $\kappa_{F_1}+\kappa_{F_2}=$*

2σ, σ *the total strength (Definition 8.1); let* $0<\kappa_{F_1}<2\sigma$. *In this statement the points of the laws of* F_1 *and* F_2 *are written* $\mathbf{z}$ *and* $\mathbf{z}'$, $\mathcal{G}(\mathbf{z},\mathbf{z}')=\int_0^\infty v^{\kappa_{F_1}-1}\prod_j(z_{j,K}v+z'_{j,K})^{-s_j}(z_{j,\Psi}v+z'_{j,\Psi})^{-t_j}\,dv$ *is the period of the twisted core family of Theorem 22.11(i) with the core* $\binom{\{g_1,g_2\}}{1}$ *in every table, the transforms are read at* $x_1=1$, *with* $\mathbf{x}'=(x_2,\dots,x_N)$, *and the hinge measures of* F_1 *act in the variables* $(\boldsymbol{\lambda},\boldsymbol{\mu})$, *those of* F_2 *in* $(\boldsymbol{\lambda}',\boldsymbol{\mu}')$. *Then, in* $[0,\infty]$,

$$
\begin{aligned}
P_{F_1\oplus_g F_2}(\mathbf{m};\mathbf{s},\mathbf{t}) &= \iint \mathcal{G}(\mathbf{z},\mathbf{z}')\,d\big((\boldsymbol{\varrho}_{F_1})_*\phi_{F_1}\big)(\mathbf{z})\,d\big((\boldsymbol{\varrho}_{F_2})_*\phi_{F_2}\big)(\mathbf{z}') && \text{(i)}\\
&= \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)}\int\prod_{j\ge2}x_j^{s_j-1}\prod_j y_j^{t_j-1}\,f_{F_1}f_{F_2}\,d\mathbf{x}'\,d\mathbf{y}\\
&= \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)}\int\prod_{j\ge2}x_j^{t_j-1}\prod_j y_j^{s_j-1}\,\check f_{F_1}\check f_{F_2}\,d\mathbf{x}'\,d\mathbf{y} && \text{(ii)}\\
&= \frac{1}{\Gamma(s_1)}\iint e^{-\lambda_1-\lambda'_1}\prod_{j\ge2}(\lambda_j+\lambda'_j)^{-s_j}\prod_j(\mu_j+\mu'_j)^{-t_j}\,d\mathfrak{r}_{F_1}\,d\mathfrak{r}_{F_2}\\
&= \frac{1}{\Gamma(t_1)}\iint e^{-\lambda_1-\lambda'_1}\prod_{j\ge2}(\lambda_j+\lambda'_j)^{-t_j}\prod_j(\mu_j+\mu'_j)^{-s_j}\,d\check{\mathfrak{r}}_{F_1}\,d\check{\mathfrak{r}}_{F_2}, && \text{(iii)}
\end{aligned}
$$

and these are finite exactly when every proper wall of $F_1\oplus_g F_2$ *is positive. On every contour of the level shares of Theorem 18.13 on which its stacked Gammas have arguments of positive real part and the two Mellin transforms below converge absolutely (Definition 22.1),*

$$
\begin{aligned}
P_{F_1\oplus_g F_2}(\mathbf{m};\mathbf{s},\mathbf{t}) = 2^{N-1}\int_{\sum_j\delta_j^d=\frac12(|\mathbf{s}|-\kappa_{F_1})}\ \prod_j\Gamma_{\frac42}\begin{bmatrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{bmatrix}\\
\times\int_{\Delta_B}\boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r,\mathbf{b}^d)}\,d\phi_{F_1}\int_{\Delta_R}\boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d,\mathbf{b}^r)}\,d\phi_{F_2}, \qquad \text{(iv)}
\end{aligned}
$$

the formula of Theorem 18.13, the periods of its bunch family and room family at the level shares being these two Mellin transforms. For one table the laws are $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$ *and* $(\varrho_{\mathcal{A}'})_*\phi_{\mathcal{A}'}$ *and the kernel of (i) is the period of Lemma 22.3; (ii) reads* $\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}f_{\mathcal{A}}f_{\mathcal{A}'}\,dy=\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{s-1}\check f_{\mathcal{A}}\check f_{\mathcal{A}'}\,dy$, *its first form Theorem 19.24(i) with the room transform* $f_R(y)=y^tf_{\mathcal{A}'}(y)$; *(iii) reads* $\frac{1}{\Gamma(s)}\iint(\lambda+\lambda')^{-t}\,d\mathfrak{r}^\Psi_{\mathcal{A}}\,d\mathfrak{r}^\Psi_{\mathcal{A}'}=\frac{1}{\Gamma(t)}\iint(\lambda+\lambda')^{-s}\,d\mathfrak{r}^K_{\mathcal{A}}\,d\mathfrak{r}^K_{\mathcal{A}'}$; *and (iv) holds on a line* $\operatorname{Re}z=z_0$ *with* $\max(0,\kappa_{\mathcal{A}}-s)<z_0<\min(t,\kappa_{\mathcal{A}})$ *on which* $\int\varrho_{\mathcal{A}}^{-z_0}d\phi_{\mathcal{A}}$ *and* $\int\varrho_{\mathcal{A}'}^{z_0-t}d\phi_{\mathcal{A}'}$ *are finite, as*

$$
\begin{aligned}
P_{\mathcal{A}\oplus_g\mathcal{A}'}(\mathbf{m};s,t) = \frac{1}{\Gamma(s)\Gamma(t)}\,\frac{1}{2\pi i}\int_{\operatorname{Re}z=z_0}\Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}})\Gamma(t-z)\Gamma(\kappa_{\mathcal{A}}-z)\\
\times\int_{\Delta_B}\varrho_{\mathcal{A}}^{-z}\,d\phi_{\mathcal{A}}\int_{\Delta_R}\varrho_{\mathcal{A}'}^{z-t}\,d\phi_{\mathcal{A}'}\,dz.
\end{aligned}
$$

The period of a two-sum is a pairing of two laws by one kernel, the period of the twisted core family on two marks, for one table the period of two marks. Read through the Grammas the pairing is one integral over the scales of the two transforms, symmetric in the two sides; for one table, read through the hinge measures, it is the moment of order $-t$ of a sum: under $\mathfrak{r}^\Psi_{\mathcal{A}}$ the variable is the hinge value $\gamma^\Psi_{\mathcal{A}}$ of the cell (Proposition 22.5(ii)), and gluing the two cells into one table adds their hinge values by Ψ, as $\Psi=\Psi_{\mathcal{A}\setminus g}\Psi_{\mathcal{A}'\setminus g}(\gamma^\Psi_{\mathcal{A}}+\gamma^\Psi_{\mathcal{A}'})$ says.

Proposition 22.5 (Faces of one law)**.** *Let $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a cell family on $B \sqcup \{g\}$ with plain tables, $s_j, t_j > 0$, masses $\mathbf{m}$ on B and the hinge wall $\kappa = \kappa_F$; in this statement the points of $(0,\infty)^{2N-1}$ are written $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$, $z_{1,K} = 1$, and $\mathbf{x}' = (x_2, \dots, x_N)$.*

(i) Law forms. *For $\mathbf{x}, \mathbf{y} \in (0,\infty)^N$,*

$$f_F(\mathbf{x},\mathbf{y}) = \int \mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j \frac{x_j}{z_{j,K}},\ \sum_j y_j z_{j,\Psi}\Big) \prod_j z_{j,K}^{-s_j}\, d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{z}),$$

$$\check{f}_F(\mathbf{x},\mathbf{y}) = \int \mathsf{G}_{|\mathbf{t}|-\kappa;(1,-1)}\Big(\sum_j x_j \frac{z_{1,\Psi}}{z_{j,\Psi}},\ \sum_j y_j \frac{z_{j,K}}{z_{1,\Psi}}\Big) \prod_j \Big(\frac{z_{j,\Psi}}{z_{1,\Psi}}\Big)^{-t_j} z_{1,\Psi}^{-\kappa}\, d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{z}) :$$

the two transforms depend on F only through its hinge wall and its law. For one cell, for $y > 0$,

$$f_{\mathcal{A}}(y) = \int_{\Delta_B} \mathsf{G}_{s-\kappa;(1,-1)}\big(1, \varrho_{\mathcal{A}} y\big)\, d\phi_{\mathcal{A}}, \qquad \check{f}_{\mathcal{A}}(y) = \int_{\Delta_B} \mathsf{G}_{t-\kappa;(1,-1)}\big(1, y/\varrho_{\mathcal{A}}\big)\, \varrho_{\mathcal{A}}^{-\kappa}\, d\phi_{\mathcal{A}}.$$

(ii) Hinge measures.

$$f_F(\mathbf{x},\mathbf{y}) = \int e^{-\langle \mathbf{x},\boldsymbol{\lambda}\rangle - \langle \mathbf{y},\boldsymbol{\mu}\rangle}\, d\mathfrak{r}_F(\boldsymbol{\lambda},\boldsymbol{\mu}), \qquad \check{f}_F(\mathbf{x},\mathbf{y}) = \int e^{-\langle \mathbf{x},\boldsymbol{\lambda}\rangle - \langle \mathbf{y},\boldsymbol{\mu}\rangle}\, d\check{\mathfrak{r}}_F(\boldsymbol{\lambda},\boldsymbol{\mu}),$$

and $\mathfrak{r}_F$ is the image of the product of $\prod_j z_{j,K}^{-s_j}\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})$ and $p^{\kappa-|\mathbf{s}|-1}dp$ under $(\mathbf{z},p) \mapsto \big((1/(pz_{j,K}))_j, (pz_{j,\Psi})_j\big)$. So f_F and $\check{f}_F$ are completely monotone in their $2N$ variables, $(-1)^{|\mathbf{n}|}\partial^{\mathbf{n}} f_F \geq 0$ for every multi-index $\mathbf{n}$ on the interior of the set where f_F is finite and the same for $\check{f}_F$, and log-convex along every segment on which they are finite. For one cell, $f_{\mathcal{A}}(y) = \int e^{-y\lambda} d\mathfrak{r}_{\mathcal{A}}^{\Psi}(\lambda)$ and $\check{f}_{\mathcal{A}}(y) = \int e^{-y\lambda} d\mathfrak{r}_{\mathcal{A}}^{K}(\lambda)$; $\mathfrak{r}_{\mathcal{A}}^{\Psi}$ is the image of $\alpha^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}\setminus g}^{-s} \Psi_{\mathcal{A}\setminus g}^{-t} e^{-1/\gamma_{\mathcal{A}}^{K}} d\alpha$ under $\gamma_{\mathcal{A}}^{\Psi}$ and of $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx) \otimes p^{s-\kappa-1}e^{-p}dp$ under $(x,p) \mapsto x/p$, and $\mathfrak{r}_{\mathcal{A}}^{K}$ the image of $\alpha^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}/g}^{-s} \Psi_{\mathcal{A}/g}^{-t} e^{-1/\gamma_{\mathcal{A}}^{\Psi}} d\alpha$ under $\gamma_{\mathcal{A}}^{K}$ and of $x^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx) \otimes p^{t-\kappa-1}e^{-p}dp$ under $(x,p) \mapsto 1/(xp)$; so $f_{\mathcal{A}}$ and $\check{f}_{\mathcal{A}}$ are decreasing and log-convex wherever finite.

(iii) Mellin transforms. *For real $\mathbf{a}, \mathbf{b} \in (0,\infty)^N$ with $\sum_j b_j - \sum_j a_j = \kappa - |\mathbf{s}|$, in $[0,\infty]$,*

$$\int_{(0,\infty)^{2N-1}} \prod_{j\geq 2} x_j^{a_j-1} \prod_j y_j^{b_j-1}\, f_F\big((1,\mathbf{x}'),\mathbf{y}\big)\, d\mathbf{x}'\, d\mathbf{y} = \prod_j \Gamma(a_j)\Gamma(b_j) \int_{\Delta_B} \boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})}\, d\phi_F,$$

and for real $\mathbf{a}, \mathbf{b} \in (0,\infty)^N$ with $\sum_j b_j - \sum_j a_j = \kappa - |\mathbf{t}|$ the same holds for $\check{f}_F$ with $\int_{\Delta_B} \boldsymbol{\varrho}_F^{-(\mathbf{b},\mathbf{t}-\mathbf{a})}\, d\phi_F$ on the right; both Mellin transforms of the law are taken on the hyperplane $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa$ of their arguments. For one cell, with $z = b$, these are

$$\int_0^\infty y^{z-1} f_{\mathcal{A}}(y)\, dy = \Gamma(z)\Gamma(z+s-\kappa) \int_{\Delta_B} \varrho_{\mathcal{A}}^{-z}\, d\phi_{\mathcal{A}} \qquad \big(z > \max(0, \kappa - s)\big),$$

$$\int_0^\infty y^{z-1} \check{f}_{\mathcal{A}}(y)\, dy = \Gamma(z)\Gamma(z+t-\kappa) \int_{\Delta_B} \varrho_{\mathcal{A}}^{z-\kappa}\, d\phi_{\mathcal{A}} \qquad \big(z > \max(0, \kappa - t)\big).$$

(iv) The dual cell family. *The dual tables $\mathcal{A}_j^*$, with the hinge g, at the exponents (t_j, s_j) form a cell family F^* with the hinge wall κ, the hinge values $\gamma^K_{\mathcal{A}_j^*} = \gamma^\Psi_{\mathcal{A}_j}$ and $\gamma^\Psi_{\mathcal{A}_j^*} = \gamma^K_{\mathcal{A}_j}$, the base form $(\gamma^\Psi_{\mathcal{A}_1}/\gamma^K_{\mathcal{A}_1})^{-\kappa}\phi_F$, the law the image of $z_{1,\Psi}^{-\kappa}\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})$ under $\mathbf{z} \mapsto (z_{j,\Psi}/z_{1,\Psi},\, z_{j,K}/z_{1,\Psi})_j$, the transform $\check{f}_F$ and the dual transform f_F. For one cell the ratio is $1/\varrho_{\mathcal{A}}$, the base form $\varrho_{\mathcal{A}}^{-\kappa}\phi_{\mathcal{A}}$, the law the image of $x^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx)$ under $x \mapsto 1/x$, and $\int \varrho_{\mathcal{A}^*}^{-z} d\phi_{\mathcal{A}^*} = \int \varrho_{\mathcal{A}}^{z-\kappa} d\phi_{\mathcal{A}}$.*

(v) Change of transform. *For real $\mathbf{a}, \mathbf{b}$ with $0 < a_j < s_j$, $0 < b_j < t_j$ and $\sum_j b_j - \sum_j a_j = \kappa - |\mathbf{s}|$, the left side of (iii) for $\check{f}_F$ at $(\mathbf{t} - \mathbf{b}, \mathbf{s} - \mathbf{a})$ times $\prod_j \Gamma(a_j)\Gamma(b_j)$ equals that for f_F at $(\mathbf{a}, \mathbf{b})$ times $\prod_j \Gamma(t_j - b_j)\Gamma(s_j - a_j)$, in $(0, \infty]$. For one cell, with $z = s - a$: for real z with $\max(0, \kappa - t) < z < \min(s, \kappa)$, in $(0, \infty]$,*

$$\int_0^\infty y^{z-1} \check{f}_{\mathcal{A}}(y)\, dy = \frac{\Gamma(z)\Gamma(z + t - \kappa)}{\Gamma(\kappa - z)\Gamma(s - z)} \int_0^\infty y^{\kappa - z - 1} f_{\mathcal{A}}(y)\, dy.$$

When $\mathcal{A}$ is the cell of the bunch of a node of width one (Definition 19.22) and $\int \varrho_{\mathcal{A}}^{-b} d\phi_{\mathcal{A}} < \infty$ at some b with $\max(0, \kappa_{\mathcal{A}} - s) < b < \min(t, \kappa_{\mathcal{A}})$, the expansion of $f_{\mathcal{A}}$ at $y \to 0$ is the sum of the residues of $y^{-z}\Gamma(z)\Gamma(z + s - \kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}}$ to the left of the line $\operatorname{Re} z = b$ (Theorem 19.32(i)), and that of $\check{f}_{\mathcal{A}}$, the transform of $\mathcal{A}^*$ read as the bunch of the dual node of width one at the exponents (t, s) (Proposition 8.14), is the sum of the residues of $y^{-z}\Gamma(z)\Gamma(z + t - \kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{z - \kappa_{\mathcal{A}}} d\phi_{\mathcal{A}}$ to the left of the line $\operatorname{Re} z = \kappa_{\mathcal{A}} - b$, at the poles of the Gamma pair and at the points $\kappa_{\mathcal{A}} - w$, w a pole of the continued $\int \varrho_{\mathcal{A}}^{-w} d\phi_{\mathcal{A}}$ to the right of the line $\operatorname{Re} w = b$: the two transforms expand one function on the two sides of its strip.

So one law has four faces. The base form and its image, the law, are where the cell family lives; the transform reads the law through Grammas of index $|\mathbf{s}| - \kappa_F$ and is the Laplace transform of the hinge measure $\mathfrak{r}_F$, carried by the reciprocals of the hinge values by K and the hinge values by Ψ; the dual transform reads the law turned over through Grammas of index $|\mathbf{t}| - \kappa_F$ and is the Laplace transform of $\check{\mathfrak{r}}_F$, carried by the reciprocals of the hinge values by Ψ and the hinge values by K; and on the Mellin side both are the one function $\int \boldsymbol{\varrho}_F^{-\mathbf{z}} d\phi_F$ with two Gamma products. Passing from one transform to the other costs nothing on the Mellin side, a reflection and a ratio of Gammas, and costs integral transforms on the side of the transforms; for one cell, on the half-line, the two expansions at $y \to 0$ are read on opposite sides of the strip: the transform expands where the law is large, the dual transform where it is small.

22.4 What a cell retains

Definition 22.6 (One weighted mark)**.** For $\mathbf{x}_0 = (x_{0,j,K}, x_{0,j,\Psi})_{j=1}^N$ with $x_{0,1,K} = 1$, a point of $(0, \infty)^{2N-1}$, and a mark e with the mass m_e, the *one-mark cell family of ratio vector* $\mathbf{x}_0$ has in table j the bases $\{e\}$ and $\{g\}$, with the weights $x_{0,j,K}$ and 1 in its letter K_j and the weights 1 and $x_{0,j,\Psi}$ in its letter Ψ_j (Definition 20.1): $K_j = x_{0,j,K}\alpha_e + \alpha_g$ and $\Psi_j = \alpha_g + x_{0,j,\Psi}\alpha_e$. Its hinge values are $x_{0,j,K}\alpha_e$ and $x_{0,j,\Psi}\alpha_e$, its ratio vector is the constant $\mathbf{x}_0$, its hinge wall is m_e, and its law is $\delta_{\mathbf{x}_0}$. For one table and $x_0 > 0$, the *one-mark cell of ratio* x_0 is the plain table $\left(\binom{\{e,g\}}{1}, (1, x_0^{1/2})\right)$ on $\{e, g\}$, the bases $\{e\}$ and $\{g\}$ with the weights 1 and $x_0^{1/2}$. Its hinge minors have $K_{\backslash g} = \alpha_e$, $K_{/g} = x_0^{1/2}$, $\Psi_{\backslash g} = 1$ and $\Psi_{/g} = x_0^{1/2}\alpha_e$, its ratio is the constant x_0, its hinge wall is m_e, and its law is $x_0^{(m_e - s)/2}\,\delta_{x_0}$.

Theorem 22.7 (The law is what a cell family retains)**.** *Let F and F' be cell families with plain tables, at the exponents $(s_j, t_j)_j$ with $s_j, t_j > 0$, with masses on their marks other than the hinge*

and one hinge wall $\kappa \in (0, 2\sigma)$, σ the total strength (Definition 8.1), and let there be a real $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ with $0 < z_{j,K} < s_j$, $0 < z_{j,\Psi} < t_j$ and $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa$ at which $\int \boldsymbol{\varrho}_F^{-\mathbf{z}} d\phi_F$ and $\int \boldsymbol{\varrho}_{F'}^{-\mathbf{z}} d\phi_{F'}$ are finite (Definition 22.1). The following are equivalent.

(a) $(\boldsymbol{\varrho}_F)_* \phi_F = (\boldsymbol{\varrho}_{F'})_* \phi_{F'}$.

(b) $P_{F \oplus_g F''} = P_{F' \oplus_g F''}$, *in* $[0, \infty]$, *for every cell family F'' with the hinge wall $2\sigma - \kappa$, the masses of F, F' and F'' kept.*

(c) (b) holds for the one-mark cell families F'' of ratio vector $\mathbf{x}_0$ and mass $2\sigma - \kappa$ (Definition 22.6), for every $\mathbf{x}_0$ in a nonempty open subset of $(0, \infty)^{2N-1}$.

(d) $\int \boldsymbol{\varrho}_F^{-\mathbf{w}} d\phi_F = \int \boldsymbol{\varrho}_{F'}^{-\mathbf{w}} d\phi_{F'}$ *at every complex $\mathbf{w}$ with real part $\mathbf{z}$.*

So the hinge wall and the law are the whole of what a cell family contributes to the families built on it, and the one-mark cell families, whose laws are single points, already separate any two laws. For one table the plain one-mark cells of ratio x_0 suffice in (c), and (d) is the agreement of $\int \varrho_{\mathcal{A}}^{-w} d\phi_{\mathcal{A}}$ and $\int \varrho_{\mathcal{A}'}^{-w} d\phi_{\mathcal{A}'}$ on the line $\operatorname{Re} w = z_{1,\Psi}$, $\max(0, \kappa - s) < z_{1,\Psi} < \min(t, \kappa)$.

What determines the contribution of a cell family to a larger construction is its hinge wall, which balance hands to the other side of the hinge, and its law, one positive measure on $(0, \infty)^{2N-1}$, on the half-line for one cell. Anything finer about the cells, their marks, their masses one by one, their letters, is invisible from outside, and anything coarser fails already against one weighted mark. A cell family is therefore replaceable, in every family it can sit in, by any other cell family of the same wall and the same law, and a cell family whose law is one point is a mark. Theorem 20.13 says the same at any width for the joint law of the bunch letters; at width one the joint law collapses to the law of the ratio vector.

Corollary 22.8 (Cells that are marks)**.** *Let F be a cell family with plain tables and $s_j, t_j > 0$, and $\mathcal{A}$ a plain cell with exponents $s, t > 0$.*

(i) The law of F is carried by a single point, $(\boldsymbol{\varrho}_F)_* \phi_F = \phi_F(\Delta_B)\, \delta_{\mathbf{x}_0}$, exactly when the ratio vector is constant, $\boldsymbol{\varrho}_F \equiv \mathbf{x}_0$; then $\phi_F(\Delta_B) \in (0, \infty]$, finite whenever F sits in a gluing of finite period, and in every gluing F acts as $\phi_F(\Delta_B)$ times the one-mark cell family of ratio vector $\mathbf{x}_0$ and mass κ_F. For one cell with $\varrho_{\mathcal{A}} \equiv x_0$ this is $\phi_{\mathcal{A}}(\Delta_B)\, x_0^{(s - \kappa_{\mathcal{A}})/2}$ times the one-mark cell of ratio x_0 and mass $\kappa_{\mathcal{A}}$.

(ii) If $\mathcal{A}/g$ is the dual table of $\mathcal{A} \setminus g$ with the identity on B and the same weights, then $\varrho_{\mathcal{A}} \equiv 1$ and $\phi_{\mathcal{A}}(\Delta_B) = P_{\mathcal{A} \setminus g}(\mathbf{m}; \kappa_{\mathcal{A}}, s + t - \kappa_{\mathcal{A}})$.

(iii) If a table $(\mathcal{B}, \theta)$ is identically self-dual*, its bases closed under complement with $\theta_{\mathcal{E} \setminus T} = \theta_T$, then for every mark g and masses on the balance wall with every proper wall positive,*

$$P_F(\mathbf{m}; s, t) = \Gamma_{\!\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}} \begin{bmatrix} m_g, & s + t - m_g \\ & s + t \end{bmatrix} P_{F \setminus g}\big(\mathbf{m}|_{\mathcal{E} \setminus g}; s + t - m_g, m_g\big),$$

$F = ((\mathcal{B}, \theta); s, t)$ and $F \setminus g = ((\mathcal{B} \setminus g, \theta); s + t - m_g, m_g)$ the family of the deletion.

So (iii) splits off any mark of an identically self-dual table, by one Beta corner, at every pair of exponents, and Proposition 18.12, every letter active, is (i) read across a bunch of width one, the ratio vector being constant. The relation is a consequence of structure alone: the mark g and the rest of the table see each other through a ratio that is identically one, so the hinge integral separates. For a table that is its own dual only up to a relabelling of its marks the hypothesis of (iii) fails, and (iii) does not apply.

22.5 Cores and their markers

A table is built from cells by two-sums at the markers of cores.

Definition 22.9 (Markers; the twisted core)**.** Let the exponents be $s, t > 0$. A *core with markers* is a plain table $\mathcal{H}$ on $\mathcal{E}_\mathcal{H} \sqcup G_\mathcal{H}$ none of whose marks is a loop or a coloop, a core of Definition 20.1, whose *markers* are the marks of $G_\mathcal{H}$, the marks at which cells may be substituted (the marks h_i of Definition 20.1). *Cells at the markers* are plain cells $\mathcal{A}_1, \dots, \mathcal{A}_k$ on $B_i \sqcup \{g_i\}$ whose hinges are identified with distinct markers $g_1, \dots, g_k$, the sets $\mathcal{E}_\mathcal{H}$ and B_i disjoint, and $\mathcal{H}\langle\mathcal{A}_1, \dots, \mathcal{A}_k\rangle$ is their substitution (Definition 20.1), the marks of $\mathcal{E}_\mathcal{H}$ and the markers without a cell left in place. For $\mathbf{x} \in (0,\infty)^k$ the *twisted core* $\mathcal{H}[\mathbf{x}]$ is $\mathcal{H}$ twisted in its cobasis letter by x_i at every g_i (Definition 21.1), carrying the masses $\kappa_{\mathcal{A}_i}$ on the markers g_i and the given masses $\mathbf{m}|_{\mathcal{E}_\mathcal{H}}$ on $\mathcal{E}_\mathcal{H}$; it has the basis weights θ_T and the cobasis weights $\theta_T \prod_{g_i \notin T} x_i$, and its period is that of a table with two systems of weights (Definition 8.1). With the cells at the shapes $u = (u_i)_i$, the twisted core $\mathcal{H}[(\varrho_{\mathcal{A}_i}(u_i))_i]$ is the core read at u of Definition 21.2 with the scale of every marker g_i divided by $\gamma^K_{\mathcal{A}_i}(u_i)$. A core with one marker g_0 left free is a *hinged core*; then $\mathcal{H}[\mathbf{x}]$ is a cell with the hinge g_0, not plain, and its hinge wall, base form, ratio and law are those of Definition 22.1. A core placed at a vertex of a hinge tree (Definition 22.14) is a node of the tree.

Lemma 22.10 (Substitution at the markers)**.** *The substitution $\mathcal{H}\langle\mathcal{A}_1, \dots, \mathcal{A}_k\rangle$ does not depend on the order of the two-sums (Proposition 20.3(i)). Its bases are the sets $(T \cap \mathcal{E}_\mathcal{H}) \sqcup \bigsqcup_i C_i$ with T a basis of $\mathcal{H}$ and, for each i, C_i a basis of $\mathcal{A}_i$ avoiding g_i when $g_i \in T$ and $C_i + g_i$ a basis of $\mathcal{A}_i$ when $g_i \notin T$, with the product weights. Its rank is $r_\mathcal{H} + \sum_i (r_{\mathcal{A}_i} - 1)$, and with G' the markers left free (Theorem 20.4)*

$$K_{\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle} = \prod_i K_{\mathcal{A}_i/g_i} \cdot K_\mathcal{H}\big(\alpha_{\mathcal{E}_\mathcal{H}}, (\gamma^K_{\mathcal{A}_i})_i, \alpha_{G'}\big),$$
$$\Psi_{\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle} = \prod_i \Psi_{\mathcal{A}_i\setminus g_i} \cdot \Psi_\mathcal{H}\big(\alpha_{\mathcal{E}_\mathcal{H}}, (\gamma^\Psi_{\mathcal{A}_i})_i, \alpha_{G'}\big).$$

For a hinged core the letters of the two hinge minors $\mathcal{H}\langle\mathcal{A}_1, \dots, \mathcal{A}_k\rangle \setminus g_0$ and $\mathcal{H}\langle\mathcal{A}_1, \dots, \mathcal{A}_k\rangle/g_0$ are the same products with the letters of $\mathcal{H} \setminus g_0$ and $\mathcal{H}/g_0$ in place of $K_\mathcal{H}$ and $\Psi_\mathcal{H}$.

Theorem 22.11 (The law of a substitution)**.** *Let F be a family with real data whose table j is $\mathcal{H}_j\langle\mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$, the cores $\mathcal{H}_j$ plain on $\mathcal{E}_\mathcal{H} \sqcup G_\mathcal{H}$ with $\{g_1, \dots, g_k\} \subseteq G_\mathcal{H}$, the cells of all tables at g_i forming the cell family $F_i = ((\mathcal{A}_{ij}, \theta^{\mathcal{A}_{ij}}); s_j, t_j)_j$ on $B_i \sqcup \{g_i\}$ (Definition 22.1), with masses on $\mathcal{E}_\mathcal{H}$ and on the marks of the cells. For $\mathbf{x} = (\mathbf{x}_i)_i$, $\mathbf{x}_i = (x_{i,j,K}, x_{i,j,\Psi})_j \in (0,\infty)^{2N-1}$ with $x_{i,1,K} = 1$, let $F_\mathcal{H}[\mathbf{x}]$ be the family $((\mathcal{H}_j, \theta^{\mathcal{H}_j}); s_j, t_j)_j$ twisted at every g_i by $\mathbf{x}_i$, $x_{i,j,K}$ in K_j and $x_{i,j,\Psi}$ in Ψ_j (Definition 21.1), with the masses κ_{F_i} on the markers g_i and the given masses on $\mathcal{E}_\mathcal{H}$.*

(i) If $G_\mathcal{H} = \{g_1, \dots, g_k\}$ and F is on its balance wall, then, in $[0,\infty]$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int P_{F_\mathcal{H}[\mathbf{x}]}\big((\mathbf{m}|_{\mathcal{E}_\mathcal{H}}, \kappa_{F_1}, \dots, \kappa_{F_k}); \mathbf{s}, \mathbf{t}\big) \prod_i d\big((\boldsymbol{\varrho}_{F_i})_* \phi_{F_i}\big)(\mathbf{x}_i).$$

With the core $\binom{\{g_1,g_2\}}{1}$ in every table this is the gluing of two cell families F_1 and F_2, $\kappa_{F_1} + \kappa_{F_2} = 2\sigma$, σ the total strength (Definition 8.1): the period of the twisted core family is $\int_0^\infty v^{\kappa_{F_1}-1} \prod_j (x_{1,j,K} v + x_{2,j,K})^{-s_j} (x_{1,j,\Psi} v + x_{2,j,\Psi})^{-t_j}\, dv$, a Breta of one bunch with at most $2N$ coefficients (Proposition 15.1), unchanged when $(\kappa_{F_1}, \mathbf{x}_1)$ and $(\kappa_{F_2}, \mathbf{x}_2)$ are swapped, and for one table it is the period of Lemma 22.3.

(ii) If $G_{\mathcal{H}} = \{g_0, g_1, \dots, g_k\}$, F is a cell family with the hinge g_0; its hinge wall is that of $F_{\mathcal{H}}[\mathbf{x}]$, which does not depend on $\mathbf{x}$,

$$\kappa_F = m_{\mathcal{E}_{\mathcal{H}}} + \sum_i \kappa_{F_i} - \sum_j \big(s_j(r_{\mathcal{H}_j} - 1) + t_j(|\mathcal{E}_{\mathcal{H}}| + k - r_{\mathcal{H}_j})\big),$$

and its law is the mixture of the twisted laws: for every Borel $H \geq 0$ on $(0,\infty)^{2N-1}$,

$$\int H\, d\big((\boldsymbol{\varrho}_F)_*\phi_F\big) = \int \Big(\int H\, d\big((\boldsymbol{\varrho}_{F_{\mathcal{H}}[\mathbf{x}]})_*\phi_{F_{\mathcal{H}}[\mathbf{x}]}\big)\Big) \prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i).$$

In particular $\int \boldsymbol{\varrho}_F^{-\mathbf{z}} d\phi_F = \int \big(\int \boldsymbol{\varrho}_{F_{\mathcal{H}}[\mathbf{x}]}^{-\mathbf{z}} d\phi_{F_{\mathcal{H}}[\mathbf{x}]}\big) \prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i)$ for every real $\mathbf{z}$.

For one table with $s, t > 0$ the laws are measures on $(0,\infty)$ and the twisted core family is the twisted core $\mathcal{H}[\mathbf{x}]$ of Definition 22.9.

The core is the only place where cells interact. Below a core each cell family is one measure on $(0,\infty)^{2N-1}$, for one table on a half-line; at the core the cell families are fed into the core family as twists of both letters of every table at the markers, of the cobasis letter alone for one table (Lemma 22.2), and the core family, with the masses κ_{F_i} on its markers, is integrated once for each value of the twists. So the work at a core is governed by the core itself and by the number of its markers, not by the sizes of the cells: a cell with a thousand marks and a cell with one mark cost the same once their laws are known. Theorem 18.13 is the gluing of two cell families read through Mellin transforms (Theorem 22.4(iv)).

22.6 Series and parallel nodes

Definition 22.12 (Series and parallel cores)**.** For $q \geq 2$ and the markers $G = \{g_0, g_1, \dots, g_q\}$ a *series core* is a table with the bases of the list $\binom{G}{q}$ and a *parallel core* a table with the bases of the list $\binom{G}{1}$, with any positive weights. With cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at $g_1, \dots, g_q$ and the hinge g_0, $\binom{G}{q}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ and $\binom{G}{1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$, the cores with unit weights, are the *series connection* of the cells, whose bunch $\bigsqcup_i B_i$ is the series connection of the parts B_i of Proposition 19.35, and their *parallel connection*; at the vertices of a hinge tree series and parallel cores are *series nodes* and *parallel nodes*.

A series bunch of marks is the series connection of its marks, each a one-mark cell, and a parallel bunch the parallel connection (Proposition 18.5); and the series closure $\binom{\{h_1,\dots,h_q\}}{q-1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ of Corollary 20.5(ii) is the series connection of $\mathcal{A}_1, \dots, \mathcal{A}_{q-1}$ glued to $\mathcal{A}_q$ along g_0.

Theorem 22.13 (Series and parallel nodes)**.** *Let $F_1, \dots, F_q$, $q \geq 2$, be cell families with plain tables at the exponents $(s_j, t_j)_j$, $s_j, t_j > 0$, the table j of F_i being $\mathcal{A}_{ij}$, and $G = \{g_0, \dots, g_q\}$.*

(i) Series. *The cell family F whose table j is $\mathcal{A}_j = \binom{G}{q}\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$ has $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{s}|$, its hinge values combine in every table as $1/\gamma^K_{\mathcal{A}_j} = \sum_i 1/\gamma^K_{\mathcal{A}_{ij}}$ and $\gamma^\Psi_{\mathcal{A}_j} = \sum_i \gamma^\Psi_{\mathcal{A}_{ij}}$, and*

$$f_F = \prod_{i=1}^q f_{F_i}, \qquad \mathfrak{r}_F = \mathfrak{r}_{F_1} * \cdots * \mathfrak{r}_{F_q},$$

the convolution on $(0,\infty)^{2N}$. For one table, $\mathcal{A} = \binom{G}{q}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ has $f_{\mathcal{A}} = \prod_i f_{\mathcal{A}_i}$ and $\mathfrak{r}^\Psi_{\mathcal{A}} = \mathfrak{r}^\Psi_{\mathcal{A}_1} * \cdots * \mathfrak{r}^\Psi_{\mathcal{A}_q}$, the convolution on $(0,\infty)$, and the series core twisted by $\mathbf{x}$ at $g_1, \dots, g_q$

has, on the simplex of its markers $g_1,\dots,g_q$ with the coordinates v, the ratio and the base form

$$\Big(\sum_i \frac{1}{v_i}\Big)\Big(\sum_i x_i v_i\Big), \qquad \prod_i v_i^{\kappa_{\mathcal{A}_i}-1}\, e_{q-1}(v)^{\kappa_{\mathcal{A}}-s}\Big(\prod_i v_i\Big)^{-\kappa_{\mathcal{A}}} d\omega_{\{g_1,\dots,g_q\}}(v).$$

(ii) Parallel. *The cell family F whose table j is $\mathcal{A}_j = \binom{G}{1}\langle \mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$ has $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{t}|$, its hinge values combine as $\gamma^K_{\mathcal{A}_j} = \sum_i \gamma^K_{\mathcal{A}_{ij}}$ and $1/\gamma^\Psi_{\mathcal{A}_j} = \sum_i 1/\gamma^\Psi_{\mathcal{A}_{ij}}$, and*

$$\check f_F = \prod_{i=1}^q \check f_{F_i}, \qquad \check{\mathfrak{r}}_F = \check{\mathfrak{r}}_{F_1} * \cdots * \check{\mathfrak{r}}_{F_q};$$

for one table $\check f_{\mathcal{A}} = \prod_i \check f_{\mathcal{A}_i}$ and $\mathfrak{r}^K_{\mathcal{A}} = \mathfrak{r}^K_{\mathcal{A}_1} * \cdots * \mathfrak{r}^K_{\mathcal{A}_q}$.

(iii) Extremes of the ratio. *For one table, with plain cells $\mathcal{A}_1,\dots,\mathcal{A}_q$ at the exponents (s,t): in a series connection $\sqrt{\inf \varrho_{\mathcal{A}}} = \sum_i \sqrt{\inf \varrho_{\mathcal{A}_i}}$ (Proposition 19.35(i)) and $\sup \varrho_{\mathcal{A}} = \infty$; in a parallel connection $1/\sqrt{\sup \varrho_{\mathcal{A}}} = \sum_i 1/\sqrt{\sup \varrho_{\mathcal{A}_i}}$ and $\inf \varrho_{\mathcal{A}} = 0$, with $1/\sqrt{\infty} = 0$.*

(iv) Reaches. *For one table: if every $f_{\mathcal{A}_i}$ has a lattice expansion of reach ζ_i (Definition 19.26), the series connection has $f_{\mathcal{A}}$ with a lattice expansion of reach $\big(\sum_i \sqrt{\zeta_i}\big)^2$ (Proposition 19.35(ii)); if every $\check f_{\mathcal{A}_i}$ has one of reach ζ_i, the parallel connection has $\check f_{\mathcal{A}}$ with one of reach $\big(\sum_i \sqrt{\zeta_i}\big)^2$.*

A series node adds the hinge values by Ψ of its cells and a parallel node their hinge values by K; in the matching transform each is a product, and in the matching hinge measure a convolution. In the other transform neither node is a product: the dual orthant form of a series connection does not factor, since $K_{\mathcal{A}/g_0} = \prod_i K_{\mathcal{A}_i/g_i}\, e_{q-1}\big((\gamma^K_{\mathcal{A}_i})_i\big)$ and $1/\gamma^K_{\mathcal{A}} = \sum_i 1/\gamma^K_{\mathcal{A}_i}$ mix the cells. What changes at the edges where the kind of node changes is the transform, not the cells (Proposition 22.5(v)). The ceilings of Section 19.8 are read on a bunch as a whole, where the letters $K_{\mathcal{A}/g_0}$ and $\Psi_{\mathcal{A}/g_0}$ of the hinge minors of a series connection mix the cells and their zeros can raise a ceiling above the reach; at a node the cells are separate factors and the reach is read cell by cell (Lemma 19.28). So ceilings are to be applied to the prime nodes of a hinge tree, and reaches combined across series and parallel nodes by (iv).

22.7 Hinge trees

Definition 22.14 (Hinge tree)**.** A *hinge tree* of a table on $\mathcal{E}$ is a finite tree $\mathfrak{T}$ with a core $\mathcal{H}_v$ on $\mathcal{E}_v \sqcup G_v$ at each vertex v, its *node*, where the $\mathcal{E}_v$ partition $\mathcal{E}$ and $G_v = \{g_{v,\mathfrak{e}}\}$ has one marker for each edge $\mathfrak{e}$ at v, such that the table is obtained from the $\mathcal{H}_v$ by the two-sums identifying $g_{v,\mathfrak{e}}$ with $g_{w,\mathfrak{e}}$ for every edge $\mathfrak{e} = \{v,w\}$. Rooted at a vertex o, every vertex $v \neq o$ has a parent edge $\mathfrak{e}_v$, and the *subtree cell* $\mathcal{A}_v$ is the substitution of the $\mathcal{H}_w$ over the vertices w below v, v included, with the hinge $g_{v,\mathfrak{e}_v}$; its cells are the subtree cells of the vertices directly below v. The tables of a family $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_j$ have a *common hinge tree* when one tree $\mathfrak{T}$, with the same sets $\mathcal{E}_v$ and markers G_v, is a hinge tree of every table, table j with the nodes $\mathcal{H}_{v,j}$; the nodes then form the *node families* $F_{\mathcal{H}_v} = ((\mathcal{H}_{v,j},\theta^{\mathcal{H}_{v,j}});s_j,t_j)_j$ and, rooted at o, the subtree cells $\mathcal{A}_{v,j}$ the cell families $F_v = ((\mathcal{A}_{v,j},\theta^{\mathcal{A}_{v,j}});s_j,t_j)_j$ of the vertices $v \neq o$ (Definition 22.1).

By Lemma 22.10, applied leaf by leaf, the substitution along a tree does not depend on the order of the two-sums: its bases are the sets $\bigsqcup_v (T_v \cap \mathcal{E}_v)$ with T_v a basis of $\mathcal{H}_v$ and every edge's marker in exactly one of the two bases at its ends. Every connected table has a hinge tree with one

vertex, and decomposing at bunches of width one with at least two marks on each side, the weights factoring across each such bunch, markers counted as marks, refines it until no node has such a bunch. A node on at least four marks without one is prime (Theorem 20.9); the nodes on three marks, with the bases of $\binom{G}{2}$ or of $\binom{G}{1}$, $|G| = 3$, are series and parallel nodes, and merging adjacent series nodes, and adjacent parallel nodes, gives the series and parallel nodes of any size; with no two series and no two parallel nodes adjacent this is the canonical hinge tree of Theorem 20.9.

Theorem 22.15 (The period along a hinge tree)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real data on its balance wall and its tables a common hinge tree rooted at o, with the node families $F_{\mathcal{H}_v}$ and the cell families F_v of the vertices $v \neq o$ (Definition 22.14), and let $F_{\mathcal{H}_v}[\mathbf{x}]$ be $F_{\mathcal{H}_v}$ twisted at the markers of the edges to the vertices w directly below v by $\mathbf{x}_w \in (0,\infty)^{2N-1}$, with the masses κ_{F_w} there and $\mathbf{m}|_{\mathcal{E}_v}$ on $\mathcal{E}_v$ (Theorem 22.11).*

(i) For $v \neq o$, κ_{F_v} is the hinge wall of $F_{\mathcal{H}_v}[\mathbf{x}]$, and

$$\int H\, d\big((\boldsymbol{\varrho}_{F_v})_*\phi_{F_v}\big) = \int \Big(\int H\, d\big((\boldsymbol{\varrho}_{F_{\mathcal{H}_v}[\mathbf{x}]})_*\phi_{F_{\mathcal{H}_v}[\mathbf{x}]}\big)\Big) \prod_w d\big((\boldsymbol{\varrho}_{F_w})_*\phi_{F_w}\big)(\mathbf{x}_w)$$

for every Borel $H \geq 0$ on $(0,\infty)^{2N-1}$.

(ii) The period is

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int P_{F_{\mathcal{H}_o[\mathbf{x}]}}\big((\mathbf{m}|_{\mathcal{E}_o}, (\kappa_{F_w})_w);\mathbf{s},\mathbf{t}\big) \prod_w d\big((\boldsymbol{\varrho}_{F_w})_*\phi_{F_w}\big)(\mathbf{x}_w),$$

the product over the vertices w directly below the root.

(iii) The law of F_v is determined by the tables $\mathcal{H}_{w,j}$ and the masses on $\mathcal{E}_w$ for the vertices w below v, and by the exponents; nothing above v enters it.

For one table the cell families are the subtree cells $\mathcal{A}_v$, with laws on $(0,\infty)$.

Where the active letters read every subtree cell by one hinge value, as at a full lean of one table, every cell of the tree is pinned (Theorem 21.5), and the period along the tree is the product of Corollary 21.7; Chapter 25 carries the product to every load.

Proposition 22.16 (What the work depends on)**.** *In Theorem 22.15:*

(i) Two vertices whose subtree cells are carried onto each other, in every table, by one bijection keeping the tables, the weights, the masses and the hinge have equal hinge walls and equal laws. So the laws to be computed are one for each orbit of subtree cells under such bijections.

(ii) Let $s_j, t_j > 0$. At a series node v whose cell families directly below fall into orbits $c_1, \dots, c_p$ with multiplicities $k_1, \dots, k_p$, $f_{F_v} = \prod_l f_{c_l}^{k_l}$ and $\kappa_{F_v} = \sum_l k_l \kappa_{c_l} - \big(\sum_l k_l - 1\big)|\mathbf{s}|$, f_{c_l} and κ_{c_l} the transform and the hinge wall of a member of c_l; at a parallel node $\check{f}_{F_v} = \prod_l \check{f}_{c_l}^{k_l}$ and $\kappa_{F_v} = \sum_l k_l \kappa_{c_l} - \big(\sum_l k_l - 1\big)|\mathbf{t}|$. The multiplicities enter as exponents.

(iii) A change at a vertex v_0 of the weights, of the masses keeping their total, or of the tables $\mathcal{H}_{v_0,j}$ keeping their ranks and their numbers of marks, changes no hinge wall and changes the laws of the cell families F_v only for the vertices v on the path from v_0 to the root. A change of masses at two vertices v_0, v_1 that keeps the total mass changes the hinge wall κ_{F_v} exactly for the vertices v whose subtree contains one of v_0, v_1 and not the other, the edges on the path between v_0 and v_1, by the change of the total mass below $\mathfrak{e}_v$.

A hinge tree is read from the leaves up by a computation whose reusable data are the laws on the edges. Its cost is governed by the prime nodes, where cells interact through the node's own integral, and by the number of distinct subtrees, since equal subtrees have equal laws; series and parallel nodes cost no integration in the matching transform, and repeated subtrees enter as powers, as the hinge tree procedure makes explicit.

Definition 22.17 (The hinge tree procedure)**.** *Given*: a common hinge tree of the tables of a family (Definition 22.14), the masses on the marks of its nodes, exponents $s_j, t_j > 0$, and a root. *Reusable data*: for every orbit of subtree cells (Proposition 22.16(i)) its hinge wall κ and its law, kept in one of three representations: (a) the transform f and the dual transform $\check f$ on a grid of $(\mathbf{x}, \mathbf{y})$ with $x_1 = 1$, of y for one table; (b) the Mellin transform $\int \boldsymbol{\varrho}^{-\mathbf{z}} d\phi$ of the law on a vertical subspace of its domain of convergence, for one table $\int \varrho^{-z} d\phi$ on a vertical line of its strip; (c) for one table, the lattice expansion of f at $y \to 0$, the residues of $y^{-z}\Gamma(z)\Gamma(z+s-\kappa)\int \varrho^{-z} d\phi$ to the left of the line (Theorem 19.32(i)). *Operations*, from the leaves up:

(1) a one-mark leaf, the one-mark cell family of ratio vector $\mathbf{x}_0$ (Definition 22.6), has $\kappa = m_e$, the law $\delta_{\mathbf{x}_0}$ and the transforms

$$f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,K}^{-s_j}\, \mathsf{G}_{|\mathbf{s}|-m_e;(1,-1)}\Big(\sum_j \frac{x_j}{x_{0,j,K}},\ \sum_j y_j x_{0,j,\Psi}\Big),$$
$$\check f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,\Psi}^{-t_j}\, \mathsf{G}_{|\mathbf{t}|-m_e;(1,-1)}\Big(\sum_j \frac{x_j}{x_{0,j,\Psi}},\ \sum_j y_j x_{0,j,K}\Big);$$

for one table the one-mark cell of ratio x_0 has the law $x_0^{(m_e-s)/2}\,\delta_{x_0}$ and

$$f(y) = x_0^{(m_e-s)/2}\mathsf{G}_{s-m_e;(1,-1)}(1, x_0 y), \qquad \check f(y) = x_0^{(m_e-s)/2-m_e}\mathsf{G}_{t-m_e;(1,-1)}(1, y/x_0);$$

(2) at a series node multiply the transforms of the vertices directly below, in (a) pointwise, in (b) by Mellin convolution of the Mellin transforms of Proposition 22.5(iii), for one table $\Gamma(z)\Gamma(z+s-\kappa)\int \varrho^{-z} d\phi$, in (c) by the product of lattice expansions (Lemma 19.28); at a parallel node the same with the dual transforms;

(3) along an edge where the kind of node changes, change the transform by Proposition 22.5(v): in (b) a reflection and a ratio of Gammas, in (a) a Mellin transform followed by an inverse Mellin transform, each one integral on $(0,\infty)^{2N-1}$, on the half-line for one table;

(4) at a prime node evaluate the Mellin transform of the law of the twisted node family, or its transform, as a period of the twisted node family, and integrate it against the laws of the vertices directly below (Theorem 22.11(ii));

(5) at the root evaluate Theorem 22.11(i); for a series root, with the cell families F_i directly below and their transforms read at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$, the period is

$$\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j\ge 2} x_j^{s_j-1} \prod_j y_j^{t_j-1} \prod_i f_{F_i}\, d\mathbf{x}'\, d\mathbf{y},$$
$$\text{for one table} \quad \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} \prod_i f_{\mathcal{A}_i}(y)\, dy,$$

and for a parallel root the same with s_j and t_j swapped and the dual transforms (Theorem 22.4(ii)).

Output: the period, and with it every law on every edge, ready to be reused when the tree is changed (Proposition 23.6).

Corollary 22.18 (Series-parallel families)**.** *Let the tables of a family with exponents $s_j, t_j > 0$ have a common hinge tree whose nodes are series and parallel nodes and whose leaves are one-mark cell families, and at every edge joining a series node to a parallel node let the law of the subtree below have a Mellin transform finite at a real $\mathbf{z}$ with $0 < z_{j,K} < s_j$ and $0 < z_{j,\Psi} < t_j$ for every j (Proposition 22.5(v)). Then its period is obtained from the transforms of the leaves, Grammas of the marks (Definition 22.17(1)), by pointwise products, by two integral transforms on $(0,\infty)^{2N-1}$ for each edge joining a series node to a parallel node, and by one final integral on $(0,\infty)^{2N-1}$; the number of integrations depends on the alternations of the tree and not on the number of marks, and subtrees repeated at a node enter as powers. For one table the integral transforms are on the half-line.*

Chapter 23

Repetition and the change of query

A construction that repeats a cell should not repeat the work, and a query that changes one weight or one mass should not restart the computation. A cell family repeated k times at a series node enters its period as the k-th power of its transform, and repetition without bound forces its hinge wall to be $\sum_j s_j$; for one cell the growth in k is then factorial, with the base $A_1/\min(s,t)$, A_1 the period of the cell with its hinge deleted, and a prefactor read from one further constant of the cell and one leading coefficient of the room (Theorems 23.1 and 23.2); a chain of repeated cell families is a power of one positive operator (Proposition 23.4); the weight of a hinge enters through a Mellin convolution whose two expansions are the two closings of the node of Section 19.7 (Theorem 23.5); and a change of data at one node moves only the laws on one path of the tree (Proposition 23.6).

23.1 Repetition at a node

For a cell family F with plain tables $\mathcal{A}_j$ and $k \geq 1$, F_k^{s} and F_k^{p} are the cell families whose table j is the series and the parallel connection of k copies of $\mathcal{A}_j$ (Definition 22.12), with $F_1^{\mathrm{s}} = F_1^{\mathrm{p}} = F$; for one plain cell $\mathcal{A}$ they are written $\mathcal{A}_k^{\mathrm{s}}$ and $\mathcal{A}_k^{\mathrm{p}}$. The cell family is *series-balanced* when $\kappa_F = |\mathbf{s}|$ and *parallel-balanced* when $\kappa_F = |\mathbf{t}|$; by Definition 22.1 the first says that the deletions $\mathcal{A}_j \setminus g$, at the exponents (s_j, t_j), form a family on its balance wall at the masses $\mathbf{m}$, $m_B = \sum_j \big(s_j r_{\mathcal{A}_j} + t_j(|B| - r_{\mathcal{A}_j})\big)$, and the second that the contractions $\mathcal{A}_j/g$ do; for one cell they say $\kappa_{\mathcal{A}} = s$ and $\kappa_{\mathcal{A}} = t$.

Theorem 23.1 (How often a cell family repeats)**.** *Let F be a cell family with plain tables and real exponents.*

(i) $\kappa_{F_k^{\mathrm{s}}} = |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|)$ *and* $\kappa_{F_k^{\mathrm{p}}} = |\mathbf{t}| + k(\kappa_F - |\mathbf{t}|)$.

(ii) If F' is a cell family with $F_k^{\mathrm{s}} \oplus_g F'$ on its balance wall and of convergent period, then $0 < |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|) < 2\sigma$, σ the total strength (Definition 8.1). So series connections of any number of copies of F occur in convergent periods only if F is series-balanced; otherwise $k < |\mathbf{s}|/(|\mathbf{s}| - \kappa_F)$ when $\kappa_F < |\mathbf{s}|$ and $k < |\mathbf{t}|/(\kappa_F - |\mathbf{s}|)$ when $\kappa_F > |\mathbf{s}|$. For parallel connections the same holds with $|\mathbf{t}|$ in place of $|\mathbf{s}|$ and $|\mathbf{s}|$ in place of $|\mathbf{t}|$.

The hinge wall of a series connection drifts by $\kappa_F - |\mathbf{s}|$ for every copy, and the room must absorb the drift; only a series-balanced cell family repeats without limit. It is the cell family whose deletions at the hinge form a balanced family, and for one cell the period of that deletion is the number that governs the repetition.

Theorem 23.2 (Series repetition)**.** *Let F be a series-balanced cell family on $B \sqcup \{g\}$ and $F' = ((\mathcal{A}'_j, \theta'_j); s_j, t_j)_j$ a cell family on $R \sqcup \{g\}$ with $\kappa_{F'} = |\mathbf{t}|$, plain tables, exponents $s_j, t_j > 0$, so that $F \overset{\mathrm{s}}{_k}\!\oplus_g F'$ is on its balance wall for every k; in this statement write $P(k)$ for its period, at the masses of the copies and of F', and read the transforms at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$.*

(i) In $[0, \infty]$,

$$P(k) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, f_F^k f_{F'} \, d\mathbf{x}' \, d\mathbf{y},$$

the k-th moment of the image under f_F of the measure

$$\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, f_{F'}(\mathbf{x}, \mathbf{y}) \, d\mathbf{x}' \, d\mathbf{y}.$$

Where finite, $P(k)^2 \leq P(k-1)P(k+1)$, and for $z \geq 0$, with $P(0)$ the period at $\mathbf{m}|_R$ of the family of the contractions $((\mathcal{A}'_j/g, \theta'_j); s_j, t_j)_j$, the hinge of F' contracted,

$$\sum_{k \geq 0} P(k) \frac{z^k}{k!} = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, e^{z f_F} f_{F'} \, d\mathbf{x}' \, d\mathbf{y}.$$

For one cell, $P(k) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}}(y)^k f_{\mathcal{A}'}(y) \, dy$ and $P(0) = \int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'} = P_{\mathcal{A}'/g}(\mathbf{m}|_R; s, t)$.

(ii) For one cell $\mathcal{A}$ and one cell $\mathcal{A}'$, at the exponents (s, t), let $\mathcal{A} \backslash g$ have every proper wall positive at the masses of $\mathcal{A}$ and the exponents (s, t), and put

$$A_1 = \int_{\Delta_B} d\phi_{\mathcal{A}} = P_{\mathcal{A}\backslash g}(\mathbf{m}|_B; s, t), \qquad A_0 = 2A_1(\log\Gamma)'(1) - \int_{\Delta_B} \log \varrho_{\mathcal{A}} \, d\phi_{\mathcal{A}}.$$

Then $\int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}}$ is finite on a neighbourhood of $z = 0$, and for every $\varepsilon \in (0, 1)$ at which $\int \varrho_{\mathcal{A}}^{\varepsilon} d\phi_{\mathcal{A}} < \infty$ there is C with $\left| f_{\mathcal{A}}(y) - A_1(-\log y) - A_0 \right| \leq C y^{\varepsilon}$ for all $y > 0$.

(iii) Let moreover $\int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'}$ be finite on a neighbourhood of $z = \max(0, t - s)$, and let the polynomial of the room *be*

$$\mathcal{R}(v) = \begin{cases} \Gamma(s-t) \int d\phi_{\mathcal{A}'} & (s > t), \\ \Gamma(t-s) \int \varrho_{\mathcal{A}'}^{s-t} d\phi_{\mathcal{A}'} & (s < t), \\ v \int d\phi_{\mathcal{A}'} + 2(\log\Gamma)'(1) \int d\phi_{\mathcal{A}'} - \int \log \varrho_{\mathcal{A}'} \, d\phi_{\mathcal{A}'} & (s = t), \end{cases}$$

with the coefficients $\mathcal{R}(v - A_0/A_1) = \sum_i \mathcal{R}_i v^i$; let $P(1) < \infty$. Then there is $\varepsilon > 0$ such that for every $\vartheta \in \big(\min(s,t)/(\min(s,t) + \varepsilon), 1\big)$

$$P(k) = \frac{A_1^k \, e^{\min(s,t) A_0/A_1}}{\Gamma(s)\Gamma(t)} \sum_i \mathcal{R}_i \, \frac{\Gamma(k + i + 1)}{\min(s,t)^{k+i+1}} \big(1 + O(\vartheta^k)\big).$$

In particular $P(k) \sim \frac{c \, e^{\min(s,t) A_0/A_1}}{\Gamma(s)\Gamma(t)} \Gamma(k + 1 + j) A_1^k \min(s,t)^{-k-1-j}$, with $j = 1$ and $c = \int d\phi_{\mathcal{A}'}$ when $s = t$ and $j = 0$ and $c = \mathcal{R}$ otherwise: the growth is factorial, with the base $A_1/\min(s,t)$.

(iv) Under the hypotheses of (iii) the series of (i) converges for $0 \leq z < \min(s,t)/A_1$ and diverges at $z = \min(s,t)/A_1$.

So a repeated cell family enters its construction as a power, and the whole sequence $P(k)$ is the sequence of moments of one positive measure: log-convex, with an exponential generating function that is one integral. For one cell the factorial growth is not a property of the room: the transform of a series-balanced cell grows like $A_1 \log(1/y)$ at small scales, its k-th power concentrates where $\log(1/y)$ is of the order $k/\min(s,t)$, and there only two numbers of the cell survive, the period A_1 of the cell with its hinge deleted and the constant A_0, and only the first lattice term of the room, through $\min(s,t)$ and $\mathcal{R}$. The error is exponentially small, not a power of $1/k$: the next terms of both lattice expansions sit at distance ε to the right and contribute in the ratio $\big(\min(s,t)/(\min(s,t)+\varepsilon)\big)^k$. Theorem 21.9 reads the series closure of any cells, for any family, through their transforms; for k equal cells at the centre it is (i) with $k-1$ copies at the series node and one copy as the room.

Corollary 23.3 (Parallel repetition)**.** *Let F be a parallel-balanced cell family on $B \sqcup \{g\}$ and $F' = ((\mathcal{A}'_j, \theta'_j); s_j, t_j)_j$ a cell family on $R \sqcup \{g\}$ with $\kappa_{F'} = |\mathbf{s}|$, plain tables, exponents $s_j, t_j > 0$, and in this statement write $P(k)$ for the period of $F_k^{\mathrm{p}} \oplus_g F'$, the transforms read at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$. Then*

$$P(k) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \ge 2} x_j^{t_j - 1} \prod_j y_j^{s_j - 1}\, \check{f}_F^{\,k} \check{f}_{F'}\, d\mathbf{x}'\, d\mathbf{y},$$

with $P(0)$ the period at $\mathbf{m}|_R$ of the family of the deletions $((\mathcal{A}'_j \setminus g, \theta'_j); s_j, t_j)_j$; for one cell, $P(k) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{s-1} \check{f}_{\mathcal{A}}(y)^k \check{f}_{\mathcal{A}'}(y)\, dy$. For one cell $\mathcal{A}$ and one cell $\mathcal{A}'$ let $\mathcal{A}/g$ have every proper wall positive at the masses of $\mathcal{A}$ and the exponents (s,t), and put

$$A_1 = \int \varrho_{\mathcal{A}}^{-t} d\phi_{\mathcal{A}} = P_{\mathcal{A}/g}(\mathbf{m}|_B; s, t), \qquad A_0 = 2A_1 (\log\Gamma)'(1) + \int \varrho_{\mathcal{A}}^{-t} \log \varrho_{\mathcal{A}}\, d\phi_{\mathcal{A}}.$$

Then Theorem 23.2(ii) to (iv) hold with s and t swapped and with these A_1 and A_0, the dual transform $\check{f}_{\mathcal{A}}$ in place of the transform $f_{\mathcal{A}}$, the finiteness of $\int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}}$ near $z = t$ in place of near $z = 0$, the condition $\int \varrho_{\mathcal{A}}^{-t-\varepsilon} d\phi_{\mathcal{A}} < \infty$ in place of $\int \varrho_{\mathcal{A}}^{\varepsilon} d\phi_{\mathcal{A}} < \infty$, the finiteness of $\int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'}$ near $z = \min(s,t)$ as the hypothesis on $\mathcal{A}'$, and the polynomial of the room

$$\mathcal{R}(v) = \begin{cases} \Gamma(t-s) \int \varrho_{\mathcal{A}'}^{-s} d\phi_{\mathcal{A}'} & (s < t), \\ \Gamma(s-t) \int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'} & (s > t), \\ v \int \varrho_{\mathcal{A}'}^{-s} d\phi_{\mathcal{A}'} + 2(\log\Gamma)'(1) \int \varrho_{\mathcal{A}'}^{-s} d\phi_{\mathcal{A}'} + \int \varrho_{\mathcal{A}'}^{-s} \log \varrho_{\mathcal{A}'}\, d\phi_{\mathcal{A}'} & (s = t), \end{cases}$$

whose leading coefficient is the constant c of the equivalent in (iii).

23.2 Chains

Proposition 23.4 (A chain is a power of one operator)**.** *Let a* chain cell family *U have in every table a hinged core $\mathcal{U}_j$ (Definition 22.9) with one further marker g_{in}, its hinge g_{out}, own marks $\mathcal{E}_{\mathcal{U}}$ with masses, and real exponents (s_j, t_j); further fixed cell families may be substituted into U beforehand. For a cell family F_0 let $F_{l+1} = U\langle F_l \rangle$, table by table, the cell at g_{in}.*

(i) $\kappa_{F_{l+1}} = \kappa_{F_l} + \kappa_U$, with $\kappa_U = m_{\mathcal{E}_{\mathcal{U}}} - \sum_j \big(s_j(r_{\mathcal{U}_j} - 1) + t_j(|\mathcal{E}_{\mathcal{U}}| + 1 - r_{\mathcal{U}_j})\big)$ the hinge wall of U with the mass 0 on g_{in}. For a fixed cell family F', the families $F_l \oplus_g F'$ are all on their balance walls exactly when $F_0 \oplus_g F'$ is and $\kappa_U = 0$; then $\kappa_{F_l} = \kappa_{F_0}$.

(ii) Let $\kappa_U = 0$, and in this statement write λ_l for the law of F_l and $\lambda^U_{\mathbf{x}}$ for the law of U twisted at g_{in} by $\mathbf{x} \in (0,\infty)^{2N-1}$ (Definition 21.1) with the mass κ_{F_0} there. On positive measures μ and nonnegative functions H on $(0,\infty)^{2N-1}$ put

$$\mathcal{Q}\mu = \int \lambda^U_{\mathbf{x}}\, d\mu(\mathbf{x}), \qquad (\mathcal{Q}^* H)(\mathbf{x}) = \int H\, d\lambda^U_{\mathbf{x}},$$

$$h_{F'}(\mathbf{x}) = \int \Big(\int_0^\infty v^{\kappa_{F_0}-1} \prod_j (x_{j,K} v + x'_{j,K})^{-s_j} (x_{j,\Psi} v + x'_{j,\Psi})^{-t_j}\, dv \Big)\, d\big((\boldsymbol{\varrho}_{F'})_* \phi_{F'}\big)(\mathbf{x}'),$$

the inner integral the period of the twisted core family of Theorem 22.11(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table. Then $\lambda_l = \mathcal{Q}^l \lambda_0$ and $P_{F_l \oplus_g F'} = \int (\mathcal{Q}^{*l} h_{F'})\, d\lambda_0$; for $z \ge 0$, $\sum_l z^l P_{F_l \oplus_g F'} = \int \sum_l z^l (\mathcal{Q}^{*l} h_{F'})\, d\lambda_0$, in $[0,\infty]$.

(iii) If $H > 0$ and $M > 0$ have $\mathcal{Q}^* H \le MH$ and $h_{F'} \le CH$, then $P_{F_l \oplus_g F'} \le C M^l \int H\, d\lambda_0$; if $H' \ge 0$ and $M' > 0$ have $\mathcal{Q}^* H' \ge M' H'$ and $h_{F'} \ge c\, H'$, then $P_{F_l \oplus_g F'} \ge c\, M'^l \int H'\, d\lambda_0$.

(iv) Let the tables be plain and $s_j, t_j > 0$. If U is the series core on $\{g_{\text{out}}, g_{\text{in}}, g'\}$ with a fixed cell family F'' at g', then $f_{F_{l+1}} = f_{F''} f_{F_l}$ and $\kappa_U = \kappa_{F''} - |\mathbf{s}|$; if it is the parallel core, then $\check{f}_{F_{l+1}} = \check{f}_{F''} \check{f}_{F_l}$ and $\kappa_U = \kappa_{F''} - |\mathbf{t}|$.

For one table the laws are measures on $(0,\infty)$, the twist is one number, and the inner integral of $h_{F'}$ is the period of Lemma 22.3.

A chain is one kernel applied l times: the law of the chain cell family twisted at g_{in}, computed once, carries the law of the chain from one length to the next, and the period at length l pairs the l-th iterate with the room. The work for all lengths up to l is l applications of one operator, and for one length $\lceil \log_2 l \rceil$ compositions of kernels. A series chain cell family is diagonal in the transform and a parallel one in the dual transform, so for them the operator is a multiplication and the length is an exponent; a chain cell family that alternates the two kinds mixes the two hinge values, and its operator is not diagonal in either. Theorem 21.10 is the same chain read on functions of the twists, a family glued to copies of itself at two marks, with the pinned case $P_F(\mathbf{m};\mathbf{s},\mathbf{t})^k / \Gamma_2 \binom{m_{g_{\text{in}}},\ m_{g_{\text{out}}}}{m_{g_{\text{in}}} + m_{g_{\text{out}}}}^{k-1}$; Proposition 23.4 reads it on laws, and positive test functions bound it from both sides. Every polynomial Gamma moment of a chain is the power of one finite matrix (Theorem 26.5).

23.3 The weight of a hinge

Theorem 23.5 (The weight of a hinge)**.** *For plain cells $\mathcal{A}$ on $B \sqcup \{g\}$ and $\mathcal{A}'$ on $R \sqcup \{g\}$ with exponents $s, t > 0$, and $w > 0$, let $w\mathcal{A}$ be $\mathcal{A}$ with the weights of its bases through g multiplied by w, so that in $(w\mathcal{A}) \oplus_g \mathcal{A}'$ the bases made of a basis of $\mathcal{A}/g$ and a basis of $\mathcal{A}' \setminus g$, the second kind of Lemma 20.2, carry the factor w; let it be on its balance wall, put $q = w^2$, and in this statement write $\Pi(q) = q^{(s-\kappa_{\mathcal{A}})/2} P_{(w\mathcal{A}) \oplus_g \mathcal{A}'}(\mathbf{m}; s, t)$.*

(i) For a cell family F with plain tables $\mathcal{A}_j$ and $\mathbf{w} \in (0,\infty)^N$, let $\mathbf{w}F$ be F with the weights of the bases through g multiplied by w_j in table j. Its hinge wall is κ_F, its ratio vector has the coordinates $(w_1/w_j)\, \gamma^K_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}$ and $w_1 w_j\, \gamma^\Psi_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}$, the hinge values of F being read, $\phi_{\mathbf{w}F} = w_1^{\kappa_F} \prod_j w_j^{-s_j}\, \phi_F$, and, for $s_j, t_j > 0$, $f_{\mathbf{w}F}(\mathbf{x},\mathbf{y}) = f_F(\mathbf{w} \circ \mathbf{x}, \mathbf{w} \circ \mathbf{y})$ and $\check{f}_{\mathbf{w}F}(\mathbf{x},\mathbf{y}) = \prod_j w_j^{-s_j - t_j}\, \check{f}_F(\mathbf{x}/\mathbf{w}, \mathbf{y}/\mathbf{w})$, the products and quotients taken coordinatewise. For one cell, $\varrho_{w\mathcal{A}} = w^2 \varrho_{\mathcal{A}}$, $\phi_{w\mathcal{A}} = w^{\kappa_{\mathcal{A}} - s} \phi_{\mathcal{A}}$, $f_{w\mathcal{A}}(y) = w^{\kappa_{\mathcal{A}} - s} f_{\mathcal{A}}(w^2 y)$ and $\check{f}_{w\mathcal{A}}(y) = w^{-\kappa_{\mathcal{A}} - s} \check{f}_{\mathcal{A}}(y/w^2)$.

(ii) $\Pi(q) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}}(qy) f_{\mathcal{A}'}(y)\, dy$, *a decreasing function of* q, *and for real* z *with* $\max(0, \kappa_{\mathcal{A}} - s) < z < \min(t, \kappa_{\mathcal{A}})$, *in* $(0, \infty]$,

$$\int_0^\infty q^{z-1} \Pi(q)\, dq = \frac{\Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}} \cdot \Gamma(t-z)\Gamma(\kappa_{\mathcal{A}} - z) \int \varrho_{\mathcal{A}'}^{z-t} d\phi_{\mathcal{A}'}}{\Gamma(s)\Gamma(t)}.$$

(iii) If this is finite at a real c *in that interval, then* Π *is finite and continuous on* $(0, \infty)$ *and is the value there of the function* $\frac{1}{2\pi i} \int_{(c)} q^{-z} (\cdots)\, dz$, *with* $(\cdots)$ *the right side of the display in (ii), holomorphic on the sector* $|\arg q| < 2\pi$ *of the logarithmic cover of* $\mathbb{C} \setminus \{0\}$.

(iv) In the setting of Section 19.7, with $\mathcal{A}$ *the cell of the bunch and* $\mathcal{A}'$ *that of the room, let* $f_{\mathcal{A}}$ *have a lattice expansion* $\sum_{\eta,j} \beta_{\eta,j} y^\eta (-\log y)^j$ *of every reach above* $\operatorname{reach}(B)$ *(Definition 19.26), and let* $\inf \varrho_{\mathcal{A}'} > 0$. *Then for* $0 < q < \inf \varrho_{\mathcal{A}'} / \operatorname{reach}(B)$, *absolutely,*

$$\Pi(q) = \frac{1}{\Gamma(s)\Gamma(t)} \sum_{\eta,j} \beta_{\eta,j}\, q^\eta \sum_{i=0}^{j} \binom{j}{i} (-\log q)^{j-i} \times (-\partial_z)^i \Big[\Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}'}) \int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'} \Big]_{z=t+\eta};$$

at $q = 1$ *this is the closing of Theorem 19.32(ii) toward the level* d. *Swapping* $\mathcal{A}$ *and* $\mathcal{A}'$, q^{-t} *times the same expansion in* $1/q$ *holds for* $q > \operatorname{reach}(R)/\inf \varrho_{\mathcal{A}}$, *and is the closing toward the level* r. *The two ranges never overlap.*

The closings of a node of Section 19.7 are therefore not two separate devices but the two expansions, at $q \to 0$ and at $q \to \infty$, of one function of the weight of its hinge, holomorphic on two sheets around the origin and given everywhere by the contour integral of (iii), whose integrand is the product of the Mellin transforms of the two sides over $\Gamma(s)\Gamma(t)$ (Proposition 22.5(iii)). The unweighted period is the value at $q = 1$; a closing reaches it when 1 lies inside its range, its terms do not tend to zero when 1 lies beyond it, and at the common end of the two ranges the root rate is one and domination decides nothing (Theorem 19.38(ii) to (iv), the ranges being read at the reaches of the two sides); and the coefficients of each expansion are the reusable lattice data of one side paired with the Mellin data of the other, the same for every weight.

23.4 Changing masses and weights

Proposition 23.6 (Carrying a calculation through a change)**.** *Let the tables of a family have a common hinge tree rooted at* o *(Definition 22.14), with every law computed by Definition 22.17.*

(i) A change at a vertex v' *of the weights, of the masses keeping their total, or of the tables* $\mathcal{H}_{v',j}$ *keeping their ranks and their numbers of marks, is carried through by recomputing the laws at the vertices on the path from* v' *to* o, *each from the new law of the vertex directly below it on the path and the old laws of the other vertices directly below it, and then the root.*

(ii) A change of the weight of the hinge of the edge $\mathfrak{e}_v$ *by the factor* w_j *in table* j *replaces the law of* F_v *by* $w_1^{\kappa_{F_v}} \prod_j w_j^{-s_j}$ *times its image under* $(x_{j,K}, x_{j,\Psi})_j \mapsto \big((w_1/w_j)\, x_{j,K},\, w_1 w_j\, x_{j,\Psi}\big)_j$ *(Theorem 23.5(i)), for one table by* $w^{\kappa_{\mathcal{A}_v} - s}$ *times its image under* $x \mapsto w^2 x$; *nothing below* v *changes, and above* v *(i) applies along the path from* v *to* o.

(iii) A change of masses at two vertices v', v'' *keeping the total mass changes the hinge walls named in Proposition 22.16(iii), those on the edges of the path between* v' *and* v'', *and is carried through by recomputing the laws on the paths from* v' *and from* v'' *to* o.

A query about one part of a construction is answered by the part of the computation that sees it. The laws on the edges off the path are kept, and with them the work that made them; a construction with many equal subtrees, or one asked many queries about one of its nodes, pays for each subtree once and for each query along one path.

Chapter 24

Holds and the lean convolution

A table built along a hinge tree is read from the laws of its subtrees, and at a prime node the twisted core is integrated over its own simplex once for every value of the twists (Definition 22.17(4)). Holding the marks removes that integral, and with it the dependence of the reusable data on the query. Every mark of every letter of a family can be twisted, and a twist can be *held* at a load, by a Mellin transform. Holding every mark separates the letters of the family completely: the hold is the product of the periods of its letters alone, the periods of its tables at their full leans, each read at its own load (Lemma 24.4). Undoing the hold writes the period at all exponents, every mass and every twist as one fibre integral of dimension $(2N-1)(n-1)$ over the splits of the masses among the letters (Theorem 24.5). The periods at the full leans of a table built along a hinge tree are the products of those of its nodes over Beta corners at its edges, with no integral at all (Theorem 25.6), so the period of a family of such tables is one fibre integral of node data (Theorem 25.12).

24.1 Holds and the periods at the full leans

Definition 24.1 (Holding)**.** Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family on $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$, with exponents $s_j, t_j > 0$, whose tables are each one molecule, table j of rank r_j and corank $r_j^* = n - r_j$, so that $0 < r_j < n$ and the bases of every table span $\mathbb{R}^{\mathcal{E}}$: a vector orthogonal to every $\mathbf{1}_T$, $T \in \mathcal{B}_j$, is constant by Lemma 7.7, and then zero since $r_j > 0$. $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is the period (Theorem 8.10) on the balance wall $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$, and $F[\mathbf{x}]$ is F twisted by $\mathbf{x} = (\mathbf{x}_j^K, \mathbf{x}_j^\Psi)_j$ (Definition 21.1). For $G \subseteq \mathcal{E}$, masses $\mathbf{m}$ and real loads $\boldsymbol{\beta} = (\boldsymbol{\beta}_j^K, \boldsymbol{\beta}_j^\Psi)_j$, $\boldsymbol{\beta}_j^K, \boldsymbol{\beta}_j^\Psi \in \mathbb{R}^G$, with $\boldsymbol{\beta}_1^K$ omitted, *holding the marks of G at $\boldsymbol{\beta}$* is the Mellin transform in the twists of every letter but K_1 at the marks of G, the twist of K_1 and the twists at the other marks kept at $\mathbf{1}$,

$$\mathrm{Hold}_G(\boldsymbol{\beta}) = \int_{((0,\infty)^G)^{2N-1}} \prod_{j \geq 2} (\mathbf{x}_j^K)^{\boldsymbol{\beta}_j^K - \mathbf{1}} \prod_{j=1}^N (\mathbf{x}_j^\Psi)^{\boldsymbol{\beta}_j^\Psi - \mathbf{1}} \, P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) \, d\mathbf{x},$$

in $[0, \infty]$ for real masses and loads; for complex masses or loads it is taken where it converges absolutely, that is where it is finite at the real parts.

The twists of every letter but K_1 are held. By the torus relation of Definition 21.1, a twist of every mark of one letter by one factor multiplies the period by that factor to the power $-r_j s_j$, or $-r_j^* t_j$ for a cobasis letter, so that a hold of every mark is taken with one twist of each held letter fixed; and a twist of every letter by one vector $\boldsymbol{\lambda}$ multiplies the period by $\boldsymbol{\lambda}^{-\mathbf{m}}$, so that the twist

of K_1 is kept at $\mathbf{1}$. A twist is a change of weights: Proposition 25.19 carries product weights on the bases into twists.

Definition 24.2 (Loads; admissible points)**.** For a split of the masses among the letters of F, $\mathbf{m} = \sum_j (\mathbf{a}_j + \mathbf{b}_j)$, $\mathbf{a}_j$ is the *K_j-load* and $\mathbf{b}_j$ the *Ψ_j-load*, as a K-share of a pattern loads its marks and a Ψ-share the marks it leaves out (Definition 12.21); for one table, $\mathbf{m} = \mathbf{a} + \mathbf{b}$ with the K-load $\mathbf{a}$ and the Ψ-load $\mathbf{b}$. The periods of a table $\mathcal{H}$ of F, with the weights $\boldsymbol{\theta}$, of rank r and corank r^*, at its two full leans are read as functions of their loads, $P_{\mathcal{H}}(\mathbf{a}; |\mathbf{a}|/r, 0)$ and $P_{\mathcal{H}}(\mathbf{b}; 0, |\mathbf{b}|/r^*)$, the exponent of the one active letter being the total load divided by its degree. Its letters give the Grammas (Definition 2.1)

$$\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_{T\in\mathcal{H}}}(\boldsymbol{\theta}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{a}-\mathbf{1}} e^{-K_{\mathcal{H}}(x)} dx, \qquad \mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_{T\in\mathcal{H}}}(\boldsymbol{\theta}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{b}-\mathbf{1}} e^{-\Psi_{\mathcal{H}}(x)} dx.$$

A real load $\mathbf{a}$ with $|\mathbf{a}| = rs$, $s > 0$, is *admissible for the full basis lean* when $\mathbf{a} \in s \cdot \operatorname{ri} P(\mathcal{H})$, $P(\mathcal{H})$ the base polytope of $\mathcal{H}$ (Theorem 7.15) and ri the relative interior, and a real $\mathbf{b}$ with $|\mathbf{b}| = r^* t$, $t > 0$, is *admissible for the full cobasis lean* when $\mathbf{b} \in t \cdot \operatorname{ri} P(\mathcal{H}^*)$, $P(\mathcal{H}^*)$ the base polytope of the dual table. A real point $\mathbf{c} = (\mathbf{c}_j^K, \mathbf{c}_j^\Psi)_j$ is *admissible* for $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ when $\sum_j (\mathbf{c}_j^K + \mathbf{c}_j^\Psi) = \operatorname{Re} \mathbf{m}$, every $\mathbf{c}_j^K$ is admissible for the full basis lean of table j at s_j and every $\mathbf{c}_j^\Psi$ for its full cobasis lean at t_j; for one table it is named by its Ψ-load $\mathbf{c} = \mathbf{c}_1^\Psi$, with $\operatorname{Re} \mathbf{m} - \mathbf{c}$ admissible for the full basis lean and $\mathbf{c}$ for the full cobasis lean.

Proposition 24.3 (The periods at the two full leans)**.** *In (i), (ii) and (iv) $\mathcal{H}$ is a table of F, with the weights $\boldsymbol{\theta}$, of rank r and corank r^*.*

(i) For $\mathbf{a} \in \mathbb{C}^{\mathcal{E}}$ with $\operatorname{Re}|\mathbf{a}| > 0$ and $s = |\mathbf{a}|/r$, $P_{\mathcal{H}}(\mathbf{a}; s, 0) = r\, \mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta})/\Gamma(s)$, both sides converging together; likewise $P_{\mathcal{H}}(\mathbf{b}; 0, t) = r^* \mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta})/\Gamma(t)$ with $t = |\mathbf{b}|/r^*$. By Cremona, $P_{\mathcal{H}}(\mathbf{b}; 0, t) = P_{\mathcal{H}}(t\mathbf{1} - \mathbf{b}; t, 0)$.

(ii) $P_{\mathcal{H}}(\mathbf{a}; s, 0)$ converges exactly when $\operatorname{Re} a_S > (\operatorname{Re} s)\, d(S)$ for every nonempty proper S, which for real $s > 0$ is $\operatorname{Re} \mathbf{a} \in s \cdot \operatorname{ri} P(\mathcal{H})$; $P_{\mathcal{H}}(\mathbf{b}; 0, t)$ converges exactly when $\operatorname{Re} b_S > (\operatorname{Re} t)\, d^*(S)$ for every nonempty proper S, which for real $t > 0$ is $\operatorname{Re} \mathbf{b} \in t \cdot \operatorname{ri} P(\mathcal{H}^*)$.

(iii) At every gauge α the occupation $R_j(\alpha)$ of table j lies in $\operatorname{ri} P(\mathcal{B}_j)$ and its dual occupation $R_j^*(\alpha)$ in $\operatorname{ri} P(\mathcal{B}_j^*)$ (Theorem 7.14). At a native gauge $\alpha^\star$ of real masses $\mathbf{m}$ on the balance wall (Definition 8.16), which exists when every proper wall is positive, the native loads *$\mathbf{a}_j^\star = s_j R_j(\alpha^\star)$ and $\mathbf{b}_j^\star = t_j R_j^*(\alpha^\star)$ are a split of $\mathbf{m}$ among the letters of F, and $\mathbf{c} = (\mathbf{a}_j^\star, \mathbf{b}_j^\star)_j$ is admissible.*

(iv) On a vertical line the periods at the full leans decay exponentially: for admissible real $\mathbf{a}$, every $\mathbf{y} \in \mathbb{R}^{\mathcal{E}}$ with $|\mathbf{y}| = 0$ and every $\varepsilon \in (0, 1)$,

$$\left|P_{\mathcal{H}}(\mathbf{a} + i\mathbf{y}; s, 0)\right| \le \min\left(1,\ \left(\sin \tfrac{\pi\varepsilon}{2}\right)^{-s} e^{-(1-\varepsilon)h(\mathbf{y})}\right) P_{\mathcal{H}}(\mathbf{a}; s, 0),$$

with h the support function of the set $\{y :\ |\langle \mathbf{1}_T, y\rangle| < \frac{\pi}{2}$ for every basis $T\}$, a norm on $\mathbb{R}^{\mathcal{E}}$; likewise at the full cobasis lean with the vectors $\mathbf{1}_{\mathcal{E}\setminus T}$.

Items (i), (ii) and (iv) hold for every positive letter with a Newton polytope of dimension $n-1$, in the form of Proposition 24.13.

Lemma 24.4 (Holding every mark separates the letters)**.** *Fix $e_0 \in \mathcal{E}$ and let $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ have $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$. For real loads $\boldsymbol{\beta} = (\boldsymbol{\beta}_j^K, \boldsymbol{\beta}_j^\Psi)_j$ on $\mathcal{E} \setminus e_0$, $\boldsymbol{\beta}_1^K$ omitted, let $\mathbf{a}_j$ ($j \ge 2$) extend*

$\boldsymbol{\beta}_j^K$ *by* $a_{j,e_0} = r_j s_j - |\boldsymbol{\beta}_j^K|$, *let* $\mathbf{b}_j$ *extend* $\boldsymbol{\beta}_j^\Psi$ *by* $b_{j,e_0} = r_j^* t_j - |\boldsymbol{\beta}_j^\Psi|$, *and put* $\mathbf{a}_1 = \mathbf{m} - \sum_{j\geq 2} \mathbf{a}_j - \sum_j \mathbf{b}_j$. *Then, in* $[0,\infty]$ *for real* $\mathbf{m}$,

$$\mathrm{Hold}_{\mathcal{E}\setminus e_0}(\boldsymbol{\beta}) = \prod_{j=1}^{N} P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j),$$

the twists at e_0 *being* 1*; for complex* $\mathbf{m}$ *and complex* $\boldsymbol{\beta}$ *the same holds whenever the integral converges at the real parts.*

The lemma is the whole mechanism. Every letter but K_1 sees its own twists and K_1 sees none; a Mellin transform in all of them at once absorbs every mark of each twisted letter into its own load, and what is left of K_1 is its own period at the remaining load. Nothing is integrated against anything: the hold is a product.

24.2 The lean convolution

Theorem 24.5 (The lean convolution)**.** *Let* $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ *have* $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$.

(i) Admissible points for $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *exist exactly when every proper wall of* $\mathrm{Re}\,\mathbf{m}$ *is positive; they form a convex set, and for real* $\mathbf{m}$ *the native loads (Proposition 24.3(iii)) are one of them.*

(ii) On every admissible point and for every twist $\mathbf{x} = (\mathbf{x}_j^K, \mathbf{x}_j^\Psi)_j$ *(Definition 21.1),*

$$P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\substack{\sum_j (\mathbf{a}_j + \mathbf{b}_j) = \mathbf{m} \\ |\mathbf{a}_j| = r_j s_j \ (j\geq 2),\ |\mathbf{b}_j| = r_j^* t_j}} \prod_{j=1}^{N} P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j},$$

the fibre integral of Definition 4.1, of dimension $(2N-1)(n-1)$*, absolutely convergent; in the coordinates of the loads of every letter but* K_1 *at the marks other than a fixed mark* e_0 *it is* $(2\pi i)^{-(2N-1)(n-1)} \int \cdots$ *over the imaginary translate through the admissible point, the loads at* e_0 *and* $\mathbf{a}_1$ *being fixed by the pins. At* $\mathbf{x} = \mathbf{1}$ *it is the period.*

(iii) The modulus of the integrand is at most its value at the real point,

$$\prod_j P_{\mathcal{B}_j}(\mathbf{c}_j^K; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{c}_j^\Psi; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{c}_j^K} (\mathbf{x}_j^\Psi)^{-\mathbf{c}_j^\Psi},$$

and it decays exponentially in the imaginary part of the load of every letter but K_1*, at the rate of Proposition 24.3(iv) for its table.*

With exponents $s_j, t_j \geq 0$*, not all zero, the same holds over the active letters,* p *of them, any one of them in the place of* K_1*, in dimension* $(p-1)(n-1)$.

For one table, at $\mathcal{H} = \binom{\{1,2\}}{1}$ the lean convolution is Corollary 4.7, and at the circuit $\binom{\mathcal{E}}{n-1}$ it is Corollary 17.20, the periods at the full leans being those of Proposition 25.10. Opening each of them on its share side (Proposition 4.2) gives the atom formula of Corollary 17.15, one share per basis and letter: the atom formula opens every letter over its bases and every mass over the letters at once, while the lean convolution keeps each letter closed, read at its full lean, and opens only the split of each mass among the letters.

Corollary 24.6 (What a family retains)**.** *The Grammas* $\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_{T\in\mathcal{B}_j}}(\boldsymbol{\theta}_j)$*,* $j = 1, \ldots, N$*, each one function of the* n *variables* $\mathbf{a}$*, determine every period of* F *and of all its twists: at all exponents*

$s_j, t_j \geq 0$, not all zero, at all masses on the balance wall with every proper wall positive, and at every twist. At the exponents $(\mathbf{s}, \mathbf{t})$ only the values of the Gramma of table j on the hyperplanes $|\mathbf{a}| = r_j s_j$ and $|t_j \mathbf{1} - \mathbf{b}| = r_j t_j$ enter, through $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0) = r_j \, \mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(s_j)$ and $P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j) = r_j \, \mathsf{G}_{t_j\mathbf{1}-\mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(t_j)$. Conversely the periods of table j at its full basis leans determine its Gramma, through Proposition 24.3(i).

A cell of a hinge tree retains, for one pair of exponents, one measure on the half-line (Theorem 22.7); a family retains, for all exponents and all masses at once, one function of the loads for each of its tables. The two retentions serve different queries; Lemma 24.9 relates them, and Section 25.5 compares them.

24.3 Holding some marks

Theorem 24.7 (Pinned holds)**.** *Let every proper wall of $\operatorname{Re}\mathbf{m}$ be positive, e_0 a fixed mark, $\varnothing \neq G \subseteq \mathcal{E} \setminus e_0$, and $\boldsymbol{\beta} = (\boldsymbol{\beta}_j^K, \boldsymbol{\beta}_j^\Psi)_j$, $\boldsymbol{\beta}_1^K$ omitted, the restriction to G of the loads of an admissible point. Then for every real $\boldsymbol{\eta} = (\boldsymbol{\eta}_j^K, \boldsymbol{\eta}_j^\Psi)_j$ of the same shape the hold $\operatorname{Hold}_G(\boldsymbol{\beta} + i\boldsymbol{\eta})$ converges absolutely and equals the fibre integral of Theorem 24.5(ii) at $\mathbf{x} = \mathbf{1}$ with the $(2N-1)|G|$ further pins $\mathbf{a}_j|_G = \boldsymbol{\beta}_j^K + i\boldsymbol{\eta}_j^K$ $(j \geq 2)$ and $\mathbf{b}_j|_G = \boldsymbol{\beta}_j^\Psi + i\boldsymbol{\eta}_j^\Psi$,*

$$\operatorname{Hold}_G(\boldsymbol{\beta} + i\boldsymbol{\eta}) = \int_{\sum_j (\mathbf{a}_j + \mathbf{b}_j) = \mathbf{m},\ |\mathbf{a}_j| = r_j s_j \ (j \geq 2),\ |\mathbf{b}_j| = r_j^* t_j,\ \text{the pins on } G} \ \prod_{j=1}^{N} P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j),$$

of dimension $(2N-1)(n-1-|G|)$, in the coordinates of the loads of every letter but K_1 at the marks off $G \cup e_0$ the integral $(2\pi i)^{-(2N-1)(n-1-|G|)} \int \cdots$. At $G = \mathcal{E} \setminus e_0$ it is Lemma 24.4.

For one mark the hold has a closed form in the minors at that mark. For a mark g that is neither a loop nor a coloop of any table, the patterns of the bunch $\{g\}$ in table j are $\varnothing$ and $\{g\}$, with the children the deletion $\mathcal{B}_j \setminus g$, of rank r_j and corank $r_j^* - 1$, and the contraction $\mathcal{B}_j/g$, of rank $r_j - 1$ and corank r_j^* (Definition 12.1). For K-loads a_j and Ψ-loads b_j at g the children family of $\{g\}$ (Definition 12.21) at the K-shares $(s_j - a_j, a_j)$ and the Ψ-shares $(b_j, t_j - b_j)$ of the patterns $\varnothing$ and $\{g\}$ is

$$F_{\{g\}}\big((s_j - a_j, a_j), (b_j, t_j - b_j)\big)_j = \Big(\big((\mathcal{B}_j \setminus g, \theta_j);\ s_j - a_j,\ b_j\big),\ \big((\mathcal{B}_j/g, \theta_j);\ a_j,\ t_j - b_j\big)\Big)_j,$$

a family of $2N$ tables on $\mathcal{E} \setminus g$ whose word is $\prod_j K_{j\setminus g}^{a_j - s_j} \Psi_{j\setminus g}^{-b_j} K_{j/g}^{-a_j} \Psi_{j/g}^{b_j - t_j}$. At masses $\mathbf{m}|_{\mathcal{E}\setminus g}$ with $\sum_j (a_j + b_j) = m_g$ it is on its balance wall: its degree is $\sum_j \big((s_j - a_j) r_j + b_j(r_j^* - 1) + a_j(r_j - 1) + (t_j - b_j) r_j^*\big) = \sum_j (r_j s_j + r_j^* t_j) - m_g$.

Proposition 24.8 (Holding one mark)**.** *Let g be a mark that is neither a loop nor a coloop of any table, $\mathbf{m}$ real masses on the balance wall, a_j $(j \geq 2)$ and b_j $(j \leq N)$ real, and $a_1 = m_g - \sum_{j \geq 2} a_j - \sum_j b_j$. The hold*

$$\operatorname{Hold}_{\{g\}}\big((a_j)_{j\geq 2}, (b_j)_j\big) = \int_{(0,\infty)^{2N-1}} \prod_{j \geq 2} (x_j^K)^{a_j - 1} \prod_{j=1}^{N} (x_j^\Psi)^{b_j - 1}\, P_{F[\mathbf{x}@g]}(\mathbf{m}; \mathbf{s}, \mathbf{t})\, d\mathbf{x}, \qquad x_1^K = 1,$$

is finite exactly when $0 < a_j < s_j$ and $0 < b_j < t_j$ for every j and every proper wall of $F_{\{g\}}\big((s_j - a_j, a_j), (b_j, t_j - b_j)\big)_j$ at $\mathbf{m}|_{\mathcal{E}\setminus g}$ is positive, and then

$$\operatorname{Hold}_{\{g\}}\big((a_j)_{j\geq 2}, (b_j)_j\big) = \prod_{j=1}^{N} \Gamma\!\begin{bmatrix} a_j,\ s_j - a_j \\ s_j \end{bmatrix} \Gamma\!\begin{bmatrix} b_j,\ t_j - b_j \\ t_j \end{bmatrix} P_{F_{\{g\}}((s_j - a_j, a_j), (b_j, t_j - b_j))_j}\big(\mathbf{m}|_{\mathcal{E}\setminus g}\big);$$

the identity holds as well for complex loads and complex masses whose real parts are such data. Where the loads are those at g of an admissible point, Theorem 24.7 writes the same value as the lean convolution with the pins at g. If $m_g > 0$, then for every letter of F whose exponent u exceeds m_g, as the twists at g of all the other letters decrease to 0, the twist of that letter staying 1, $P_{F[\mathbf{x}@g]}(\mathbf{m};\mathbf{s},\mathbf{t})$ increases, in $(0,\infty]$, to ${}_1^{}\Gamma_2\begin{bmatrix} m_g, & u-m_g \\ & u \end{bmatrix}$ times the period at $\mathbf{m}|_{\mathcal{E}\setminus g}$ of the children family of $\{g\}$ at the shares that put the whole mass m_g on that letter: $(s_j - m_g, m_g)$ if it is K_j, $(m_g, t_j - m_g)$ if it is Ψ_j, $(s_{j'}, 0)$ for every other letter $K_{j'}$ and $(0, t_{j'})$ for every other letter $\Psi_{j'}$.

The ends of the twist are the faces of the children family at which every letter but one has the mark g switched off: the chosen letter keeps both its minors and every other letter keeps one. For one table the two limits are those of $P_{\mathcal{H}[x@g]}$ as $x \downarrow 0$ and of $x^{m_g} P_{\mathcal{H}[x@g]}$ as $x \uparrow \infty$ (Definition 21.1), and a weight on a mark driven to one end of its range leaves the deletion and the contraction of the mark, two tables carrying three active letters between them. With Proposition 25.19 and Definition 21.1, a weight w on the bases through g is the twist w^{-2} up to the factor w^{-m_g-t}, so the period behaves like w^{-m_g-t} times the first limit as $w \to \infty$ and like w^{m_g-t} times the second as $w \to 0$, where these are finite.

Lemma 24.9 (The Mellin transform of a law is the period of a children family)**.** *Let $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a cell family on $B \sqcup \{g\}$ with masses $\mathbf{m}$ on B, hinge wall κ_F, base form ϕ_F and ratio vector $\boldsymbol{\varrho}_F$ (Definition 22.1). In this statement let $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ have $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa_F$, and write $\boldsymbol{\varrho}_F^{-\mathbf{z}} = \prod_{j\geq 2} \left(\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}\right)^{-z_{j,K}} \prod_j \left(\gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}\right)^{-z_{j,\Psi}}$, the coordinates of the ratio vector to the powers $-z_{j,K}$ $(j \geq 2)$ and $-z_{j,\Psi}$. For real $\mathbf{z}$, in $[0,\infty]$,*

$$\int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F = P_{F_{\{g\}}((z_{j,K},\, s_j - z_{j,K}),(t_j - z_{j,\Psi},\, z_{j,\Psi}))_j}(\mathbf{m}),$$

the period of the children family of the bunch $\{g\}$ of F (Definition 12.21), its tables $\mathcal{A}_j \setminus g$ at the exponents $(z_{j,K}, t_j - z_{j,\Psi})$ and $\mathcal{A}_j/g$ at $(s_j - z_{j,K}, z_{j,\Psi})$, on its balance wall at $\mathbf{m}$. Let moreover $s_j, t_j > 0$, and put the mass $\sum_j (s_j + t_j) - \kappa_F$ on g, which puts the family of the tables $\mathcal{A}_j$ on its balance wall. For complex $\mathbf{z}$ whose real part has $0 < \operatorname{Re} z_{j,K} < s_j$ and $0 < \operatorname{Re} z_{j,\Psi} < t_j$ for every j and is a point at which $\int \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}} d\phi_F$ is finite,

$$\int_{(0,\infty)^{2N-1}} \prod_{j\geq 2} (x_j^K)^{s_j - z_{j,K} - 1} \prod_{j=1}^N (x_j^\Psi)^{t_j - z_{j,\Psi} - 1}\, P_{F[\mathbf{x}@g]}\Big(\big(\mathbf{m}, \textstyle\sum_j (s_j + t_j) - \kappa_F\big); \mathbf{s}, \mathbf{t}\Big)\, d\mathbf{x}$$
$$= \prod_{j=1}^N {}_1^{}\Gamma_2\begin{bmatrix} s_j - z_{j,K}, & z_{j,K} \\ & s_j \end{bmatrix} {}_1^{}\Gamma_2\begin{bmatrix} t_j - z_{j,\Psi}, & z_{j,\Psi} \\ & t_j \end{bmatrix} \int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F,$$

$F[\mathbf{x}@g]$ the family of the tables $\mathcal{A}_j$ twisted at g, $x_1^K = 1$ (Definition 21.1). So the Mellin transform of the law of a cell family is the Mellin transform of its own period in the twists of its hinge, divided by $2N$ Beta corners.

Corollary 24.10 (Identically self-dual tables twisted at one mark)**.** *Let the family be one table $\mathcal{H}$ at the exponents (s,t), identically self-dual (Corollary 22.8(iii)), g a mark, and $\mathbf{m}$ on the balance wall with every proper wall positive. For every $x > 0$,*

$$P_{\mathcal{H}[x@g]}(\mathbf{m}; s, t) = P_{\mathcal{H}\setminus g}\big(\mathbf{m}|_{\mathcal{E}\setminus g};\ s + t - m_g,\ m_g\big)\, P_{{}_1^{}\{g,g'\}_{[(x,1)]}}\big((m_g, s + t - m_g); s, t\big),$$

the second factor the period $\int_0^\infty v^{m_g-1}(1+v)^{-s}(1+xv)^{-t}dv$ *of the twisted list on two marks (Lemma 22.3); and the hold at* g *is* $\Gamma_{\,1}^{\,2}\!\left[\begin{matrix} m_g-b,\ s-m_g+b \\ s \end{matrix}\right]\Gamma_{\,1}^{\,2}\!\left[\begin{matrix} b,\ t-b \\ t \end{matrix}\right]$ *times the same period of* $\mathcal{H}\setminus g$ *for every* b *in its strip: the children family does not depend on the split. At* $x=1$ *this is Corollary 22.8(iii), and Corollary 4.7 recovers it from the hold,*

$$\frac{1}{2\pi i}\int \Gamma_{\,1}^{\,2}\!\left[\begin{matrix} m_g-b,\ s-m_g+b \\ s \end{matrix}\right]\Gamma_{\,1}^{\,2}\!\left[\begin{matrix} b,\ t-b \\ t \end{matrix}\right]\,db = \Gamma_{\,1}^{\,2}\!\left[\begin{matrix} m_g,\ s+t-m_g \\ s+t \end{matrix}\right].$$

24.4 Newton words

Nothing in Lemma 24.4 uses the exchange property: twists separate any product of powers of positive letters into its factors. The convergence criterion of Theorem 8.7, whose sectors rest on the greedy bases that need the exchange property (Lemma 8.5), extends to such products through the Gramma and its domain (Theorem 24.12), and on its domain the separation holds (Theorem 24.14).

Definition 24.11 (Positive letters; Newton words and sums)**.** A *positive letter* on $\mathcal{E}$ is a nonzero homogeneous polynomial $L=\sum_k c_k x^{t_k}$ of degree $\deg L\ge 1$ with positive coefficients c_k on distinct exponents $t_k\in\mathbb{N}^{\mathcal{E}}$, its *support*; its Newton polytope is $\mathcal{N}_L=\operatorname{conv}\{t_k\}$, in the hyperplane $|x|=\deg L$. A *product of powers* of positive letters $W=\prod_i L_i^{-w_i}$, with exponents $w_i\in\mathbb{C}$, has the degree $D=\sum_i w_i\deg L_i$, and for masses $\mathbf{m}\in\mathbb{C}^{\mathcal{E}}$ with $m_{\mathcal{E}}=D$ its period $P_W(\mathbf{m})$ is the integral of the form $\alpha^{\mathbf{m}-\mathbf{1}}W\,d\alpha$ over a section (Lemma 16.2), wherever it converges absolutely; it does not depend on the section. It is a *Newton word* when every $w_i>0$, with the *Newton sum* $\mathcal{N}(W)=\sum_i w_i\mathcal{N}_{L_i}$, in the hyperplane $|x|=D$. A Newton word is *split into factors* when it is written $W=W_1\cdots W_p$, each factor a product of some of its letters with their exponents. The family word of a family is a product of powers of its letters, with $P_W(\mathbf{m})=P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, and a Newton word in its active letters when its exponents are nonnegative; for a table $\mathcal{N}_K=P(\mathcal{H})$ and $\mathcal{N}_\Psi=P(\mathcal{H}^*)$, and for a family $\mathcal{N}(W)$ is the Newton sum $\mathcal{N}_F$ of Theorem 8.18.

Theorem 24.12 (Convergence of a Newton word)**.** $P_W(\mathbf{m})$ *converges exactly when* $\mathcal{N}(W)$ *has dimension* $n-1$ *and* $\operatorname{Re}\mathbf{m}\in\operatorname{ri}\mathcal{N}(W)$*; it is then holomorphic in* $\mathbf{m}$ *on the balance hyperplane. For a family of tables this is the condition that every proper wall be positive (Theorems 8.10 and 8.18(iii)).*

Proposition 24.13 (One positive letter)**.** *Let* $L=\sum_k c_k x^{t_k}$ *be a positive letter with* $\dim\mathcal{N}_L=n-1$, $\mathbf{a}\in\mathbb{C}^{\mathcal{E}}$ *with* $\operatorname{Re}|\mathbf{a}|>0$, *and* $w=|\mathbf{a}|/\deg L$.

(i) $P_{L^{-w}}(\mathbf{a})=\deg L\,\mathsf{G}_{\mathbf{a};(t_k)_k}\big((c_k)_k\big)/\Gamma(w)$, *both sides converging together.*

(ii) $P_{L^{-w}}(\mathbf{a})$ *converges exactly when* $\operatorname{Re}\mathbf{a}\in(\operatorname{Re}w)\operatorname{ri}\mathcal{N}_L$.

(iii) For real $\mathbf{a}$ *at which it converges, every* $\boldsymbol{\eta}\in\mathbb{R}^{\mathcal{E}}$ *with* $|\boldsymbol{\eta}|=0$ *and every* $\varepsilon\in(0,1)$,

$$\big|P_{L^{-w}}(\mathbf{a}+i\boldsymbol{\eta})\big|\le\min\Big(1,\ \big(\sin\tfrac{\pi\varepsilon}{2}\big)^{-w}e^{-(1-\varepsilon)h_L(\boldsymbol{\eta})}\Big)\,P_{L^{-w}}(\mathbf{a}),$$

with h_L *the support function of* $\{y:\ |\langle t_k,y\rangle|<\frac{\pi}{2}$ *for every* $k\}$, *a norm on* $\mathbb{R}^{\mathcal{E}}$.

Theorem 24.14 (Separation into factors)**.** *Let* $W=W_1\cdots W_p$ *be split into factors with* $\dim\mathcal{N}(W_i)=n-1$ *for every* i, *of degrees* D_i, *and let* $\mathbf{m}\in\mathbb{C}^{\mathcal{E}}$ *have* $m_{\mathcal{E}}=D$. *Points* $(\mathbf{c}_i)_i$ *with*

$\mathbf{c}_i \in \operatorname{ri}\mathcal{N}(W_i)$ *and* $\sum_i \mathbf{c}_i = \operatorname{Re}\mathbf{m}$ *exist exactly when* $\operatorname{Re}\mathbf{m} \in \operatorname{ri}\mathcal{N}(W)$*, and on each of them, for a fixed mark* e_0 *and all twists* $\mathbf{x}_i \in (0,\infty)^{\mathcal{E}}$ *of the factors,*

$$\int_{\alpha_{e_0}=1} \alpha^{\mathbf{m}-\mathbf{1}} \prod_i W_i(\mathbf{x}_i \circ \alpha)\, d\alpha' = \int_{\mathbf{a}_1+\cdots+\mathbf{a}_p=\mathbf{m},\ |\mathbf{a}_i|=D_i\ (i<p)} \prod_i P_{W_i}(\mathbf{a}_i)\, \mathbf{x}_i^{-\mathbf{a}_i},$$

a fibre integral of dimension $(p-1)(n-1)$*, absolutely convergent,* P_{W_i} *the period of the factor* W_i *at its own masses. For a family of tables that are each one molecule, one factor per active letter gives Theorem 24.5, of dimension* $(p-1)(n-1)$*,* p *the number of active letters,* $2N$ *when every exponent is positive.*

Corollary 24.15 (Exponents add on one letter)**.** *For a positive letter* L *with* $\dim\mathcal{N}_L = n-1$*, exponents* $w, w' > 0$ *and* $\operatorname{Re}\mathbf{m} \in (w+w')\operatorname{ri}\mathcal{N}_L$*,*

$$P_{L^{-w-w'}}(\mathbf{m}) = \int_{\mathbf{a}+\mathbf{a}'=\mathbf{m},\ |\mathbf{a}|=w\deg L} P_{L^{-w}}(\mathbf{a})\, P_{L^{-w'}}(\mathbf{a}').$$

At $L = e_1(x) = \sum_{e\in\mathcal{E}} x_e$*,* $P_{L^{-w}}(\mathbf{a}) = \Gamma_1^n \begin{bmatrix}\mathbf{a}\\ w\end{bmatrix}$ *and this is Theorem 4.6; for an identically self-dual table* $\Psi = K$*, and the lean convolution is this identity at* $L = K$*,* $w = s$*,* $w' = t$*,* $P_{\mathcal{H}}(\mathbf{m}; s, t) = P_{\mathcal{H}}(\mathbf{m}; s+t, 0)$*.*

Proposition 24.16 (Simplicial letters)**.** *If the support of a positive letter* L *consists of* n *linearly independent exponents* $t_1, \dots, t_n$*, so that* L *is* simplicial*, with the configuration* $t = (t_1, \dots, t_n)$*, then for* $\mathbf{a}$ *with* $|\mathbf{a}| = w\deg L$ *and the shares* $\mathbf{b} = t^{-1}\mathbf{a}$*,*

$$P_{L^{-w}}(\mathbf{a}) = \frac{\deg L}{|\det t|}\, \Gamma_1^n \begin{bmatrix}\mathbf{b}\\ w\end{bmatrix} \prod_k c_k^{-b_k}, \qquad |\mathbf{b}| = w,$$

converging exactly when $\operatorname{Re}\mathbf{b} > 0$*: the Gramma of Proposition 24.13(i) at a square configuration (Corollary 2.6). A table that is one molecule has a simplicial basis letter exactly when it has* n *bases.*

Lemma 24.17 (Concatenation of Grammas)**.** *Let* $t^{(1)}, \dots, t^{(p)}$*,* $p \ge 2$*, be configurations on* $\mathcal{E}$ *with* $t^{(2)}, \dots, t^{(p)}$ *spanning* $\mathbb{R}^{\mathcal{E}}$*,* $\boldsymbol{\xi}^{(1)}, \dots, \boldsymbol{\xi}^{(p)}$ *coefficients, and* $\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(i)})$ *for* $i \ge 2$ *with* $\operatorname{Re}\nu - \sum_{i\ge2}\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(1)})$*. Then*

$$\begin{aligned}&\mathsf{G}_{\nu;t^{(1)}\oplus\cdots\oplus t^{(p)}}\big(\boldsymbol{\xi}^{(1)}\oplus\cdots\oplus\boldsymbol{\xi}^{(p)}\big)\\ &\qquad = \frac{1}{(2\pi i)^{(p-1)n}} \int_{\mathbf{c}_2+i\mathbb{R}^{\mathcal{E}}} \cdots \int_{\mathbf{c}_p+i\mathbb{R}^{\mathcal{E}}} \mathsf{G}_{\nu-\sum_{i\ge2}\mathbf{w}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)}) \prod_{i\ge2} \mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})\, d\mathbf{w}_2 \cdots d\mathbf{w}_p,\end{aligned}$$

absolutely convergent: a Gramma of a concatenated configuration is the Mellin convolution of the Grammas of its parts.

For one table the concatenation $(\mathbf{1}_T)_T \oplus (\mathbf{1}_{\mathcal{E}\setminus T})_T$ with the coefficients $\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta}$ is the Gramma $\mathsf{G}_{\mathbf{m};(\mathbf{1}_T)_T\oplus(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta}) = \int x^{\mathbf{m}-\mathbf{1}} e^{-K(x)-\lambda\Psi(x)}dx$, and Lemma 24.17 writes it as $(2\pi i)^{-n}\int \mathsf{G}_{\mathbf{m}-\mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta})\, \mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta})\, \lambda^{-|\mathbf{b}|/r^*} d\mathbf{b}$, by the torus homogeneity of $\mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}$. The Mellin transform in λ at t pins $|\mathbf{b}| = r^*t$, the pin contributing the factor r^*, and through Proposition 24.3(i) returns $\Gamma(s)\Gamma(t)/r$ times the lean convolution of Theorem 24.5 at

$\mathbf{x} = \mathbf{1}$; Proposition 25.21 states the resulting identity $\int_0^\infty \lambda^{t-1} \mathsf{G}_{\mathbf{m};(\mathbf{1}_T)_T \oplus (\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta} \oplus \lambda\boldsymbol{\theta})\, d\lambda = \Gamma(s)\Gamma(t) P_{\mathcal{H}}(\mathbf{m}; s, t)/r$. So at fixed masses the lean is the Mellin variable of the relative scale λ of the two letters, and the two-step integration separates what the masses fix, the Gramma of the concatenated configuration, from what the exponents choose, one Mellin transform in one variable.

Chapter 25

Periods at the full leans and networks along a hinge tree

At a full lean the period of a two-sum is the product of the periods of its two cells over one Beta corner (Theorem 18.10), and along any hinge tree this is Corollary 21.7, which for one table at a full lean is Theorem 25.6 at the loads $\mathbf{m}$. Along a hinge tree this needs only bookkeeping, and the bookkeeping is affine: every marker receives a load that is a fixed affine function of the loads of the marks, the same function at every pair of exponents. So the periods of a substitution at its full leans are products of those of its nodes, read at affine loads, over Beta corners at its edges, with no integral. The nodes whose periods at the full leans cost nothing are characterized by counting bases. The period of a family of such tables, at all exponents, every mass and every twist, is then one fibre integral of node data, of dimension $(2N-1)(n-1)$: the lean convolution of Chapter 24 with the periods of every letter at its full lean read along the hinge tree of its table (Theorem 25.12).

25.1 Molecules and marker loads

Lemma 25.1 (Unions of molecules)**.** *For a table on $\mathcal{E}$ of rank r and $S \subseteq \mathcal{E}$ the following are equivalent: (a) $r(S) + r(\mathcal{E} \setminus S) = r$; (b) every basis meets S in the same number of marks; (c) S is a union of molecules of the table (Definition 7.4).*

Lemma 25.2 (Substitutions of molecules)**.** *For a core $\mathcal{H}$ and cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at its marks (Definition 20.1), the substitution $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q \rangle$ is one molecule exactly when $\mathcal{H}$ and the $\mathcal{A}_i$, as tables on their marks and hinges, are each one molecule. In particular the two-sum $\mathcal{A} \oplus_g \mathcal{A}'$ of two cells is one molecule exactly when $\mathcal{A}$ and $\mathcal{A}'$ are.*

So a substitution of nodes that are molecules is one molecule, and a table with every proper wall positive, which is one molecule by Proposition 8.15, has every cell of its hinge trees one molecule.

Definition 25.3 (Branches of a hinge tree)**.** Let $\mathcal{B}$ be a table on $\mathcal{E}$, $|\mathcal{E}| = n$, of rank r and corank r^*, with a hinge tree $\mathfrak{T}$ (Definition 22.14) whose nodes $\mathcal{H}_v$ on $\mathcal{E}_v \sqcup G_v$ are each one molecule, of rank r_v and corank $r_v^* = |\mathcal{E}_v| + |G_v| - r_v$; the weights of $\mathcal{B}$ are the products of the weights of the nodes (Lemma 20.8). For an edge $\mathfrak{e} = \{v, w\}$ of $\mathfrak{T}$ and its end v, the *branch* $\mathfrak{T}_v^{\mathfrak{e}}$ is the component of $\mathfrak{T} - \mathfrak{e}$ containing v, $\mathcal{E}_v^{\mathfrak{e}}$ the union of the $\mathcal{E}_u$ over its vertices, and $\mathcal{A}_v^{\mathfrak{e}}$ the substitution of its nodes, a cell on $\mathcal{E}_v^{\mathfrak{e}} \sqcup \{g_{v,\mathfrak{e}}\}$ of rank $r_v^{\mathfrak{e}}$ and corank $r_v^{*\mathfrak{e}} = |\mathcal{E}_v^{\mathfrak{e}}| + 1 - r_v^{\mathfrak{e}}$; $\mathcal{B}$ is the two-sum of $\mathcal{A}_v^{\mathfrak{e}}$ and $\mathcal{A}_w^{\mathfrak{e}}$.

Definition 25.4 (Marker loads; node loads)**.** For a table with a hinge tree as in Definition 25.3 and loads $\mathbf{a}, \mathbf{b} \in \mathbb{C}^{\mathcal{E}}$ with $|\mathbf{a}| = rs$ and $|\mathbf{b}| = r^*t$, the *marker loads* at the end v of the edge $\mathfrak{e}$ are

$$k_{v,\mathfrak{e}} = s\, r_v^{\mathfrak{e}} - a_{\mathcal{E}_v^{\mathfrak{e}}}, \qquad l_{v,\mathfrak{e}} = t\, r_v^{*\mathfrak{e}} - b_{\mathcal{E}_v^{\mathfrak{e}}},$$

and the *node loads* of v are $\mathbf{a}_v = \big(\mathbf{a}|_{\mathcal{E}_v}, (k_{v,\mathfrak{e}})_{\mathfrak{e}\ni v}\big)$ and $\mathbf{b}_v = \big(\mathbf{b}|_{\mathcal{E}_v}, (l_{v,\mathfrak{e}})_{\mathfrak{e}\ni v}\big)$, the load of the marker $g_{v,\mathfrak{e}}$ being $k_{v,\mathfrak{e}}$, respectively $l_{v,\mathfrak{e}}$.

Lemma 25.5 (Node loads)**.** *In Definition 25.4:*

(i) $k_{v,\mathfrak{e}} + k_{w,\mathfrak{e}} = s$ *and* $l_{v,\mathfrak{e}} + l_{w,\mathfrak{e}} = t$ *on every edge.*

(ii) $|\mathbf{a}_v| = r_v s$ *and* $|\mathbf{b}_v| = r_v^* t$ *at every node.*

(iii) If $\mathbf{a} + \mathbf{b} = \mathbf{m}$ *with* $\mathbf{m}$ *on the balance wall of* $\mathcal{B}$*, then* $k_{v,\mathfrak{e}} + l_{v,\mathfrak{e}} = s + t - \kappa_{\mathcal{A}_v^{\mathfrak{e}}} = \kappa_{\mathcal{A}_w^{\mathfrak{e}}}$*: the node masses* $\mathbf{a}_v + \mathbf{b}_v$ *are* $\mathbf{m}|_{\mathcal{E}_v}$ *on the marks and the hinge walls of the branches on the markers, the masses of the twisted core in Theorem 22.11.*

25.2 The periods of a substitution at its full leans

Theorem 25.6 (Periods at the full leans along a hinge tree)**.** *In Definition 25.4, for real loads* $\mathbf{a}$ *with* $|\mathbf{a}| = rs$*, if every marker load* $k_{v,\mathfrak{e}}$ *lies in* $(0, s)$*, then, in* $(0, \infty]$*,*

$$P_{\mathcal{B}}(\mathbf{a}; s, 0) = \frac{\prod_v P_{\mathcal{H}_v}(\mathbf{a}_v; s, 0)}{\prod_{\mathfrak{e}=\{v,w\}} \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix} k_{v,\mathfrak{e}},\ k_{w,\mathfrak{e}} \\ s \end{smallmatrix}\right]},$$

and if some marker load lies outside $(0, s)$*, then* $P_{\mathcal{B}}(\mathbf{a}; s, 0) = +\infty$*. So* $P_{\mathcal{B}}(\mathbf{a}; s, 0)$ *is finite exactly when every marker load lies in* $(0, s)$ *and the period of every node at its full basis lean is finite at its node loads. The same holds at the full cobasis leans, with* $\mathbf{b}$*, the loads* $l_{v,\mathfrak{e}}$ *and* t*. For complex loads both identities hold on the tube over the real loads at which they are finite, both sides being holomorphic there.*

In the language of bunches this is Theorem 18.10 at every bunch of width one of the tree at once: the bunch $\mathcal{E}_v^{\mathfrak{e}}$ has, at $(s, 0)$, the wall $k_{w,\mathfrak{e}}$, and its two cells are $\mathcal{A}_v^{\mathfrak{e}}$ and $\mathcal{A}_w^{\mathfrak{e}}$ (Lemma 20.8). What the tree adds is that the loads of all markers are affine in the loads of the marks, fixed once for every pair of exponents, so that the periods of the substitution at its full leans are evaluated at any loads by evaluating each node once.

Corollary 25.7 (Convergence is decided at the nodes)**.** *Let* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ *have exponents* $s_j, t_j > 0$*, and let table* j *carry a hinge tree* $\mathfrak{T}_j$ *as in Definition 25.3, its nodes* $\mathcal{H}_{j,v}$ *on* $\mathcal{E}_{j,v} \sqcup G_{j,v}$ *each one molecule, with the node loads of Definition 25.4 read in table* j*. For masses* $\mathbf{m}$ *on the balance wall of* F *every proper wall of* F *is positive exactly when there is a real split* $\mathbf{m} = \sum_j (\mathbf{a}_j + \mathbf{b}_j)$ *with* $|\mathbf{a}_j| = r_j s_j$ *and* $|\mathbf{b}_j| = r_j^* t_j$ *at which every node of every table has*

$$(\mathbf{a}_{j,v})_S > s_j\, d_{j,v}(S) \quad \text{and} \quad (\mathbf{b}_{j,v})_S > t_j\, d_{j,v}^*(S) \qquad (\varnothing \neq S \subsetneq \mathcal{E}_{j,v} \sqcup G_{j,v}),$$

$d_{j,v}$ and $d_{j,v}^*$ the counts of $\mathcal{H}_{j,v}$. This is a system of $\sum_j \sum_v 2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ strict linear inequalities in $(2N-1)(n-1)$ unknowns, a number linear in the number of nodes when the nodes have bounded size, against the $2^n - 2$ walls of F; the native loads of F solve it.

25.3 What a node costs

Lemma 25.8 (Bases outnumber marks)**.** *A table on $n \geq 2$ marks that is one molecule has at least n bases, and exactly n only when its rank is 1 or $n-1$, that is, as a list, when it is $\binom{\mathcal{E}}{1}$ or $\binom{\mathcal{E}}{n-1}$.*

Corollary 25.9 (The cost of a node)**.** *For a node $\mathcal{H}_v$ that is one molecule on $n_v = |\mathcal{E}_v| + |G_v|$ marks with $|\mathcal{H}_v|$ bases, the share side of its period at each of its two full leans (Proposition 4.2) is a fibre integral of dimension $|\mathcal{H}_v| - n_v \geq 0$. The dimension is zero exactly when $\mathcal{H}_v$ has the bases of $\binom{\mathcal{E}_v \sqcup G_v}{n_v - 1}$ or of $\binom{\mathcal{E}_v \sqcup G_v}{1}$, with any weights: the series and the parallel nodes up to their weights, whose periods at the full leans are stacked Gammas times monomials in the weights (Propositions 24.16 and 25.10); a prime node on at least four marks has $|\mathcal{H}_v| > n_v$.*

Proposition 25.10 (Series and parallel nodes at their full leans)**.** *On a set $\mathcal{X}$ of $q+1$ marks let $\binom{\mathcal{X}}{q}$ carry the weight θ_e on its basis $\mathcal{X} \setminus e$ and $\binom{\mathcal{X}}{1}$ the weight θ_e on its basis $\{e\}$, the series and the parallel node with weights (Definition 22.12). Then*

$$P_{\binom{\mathcal{X}}{q}}(\mathbf{a}; s, 0) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(s-a_e)_e\\ s\end{bmatrix}\prod_e \theta_e^{a_e - s}, \qquad P_{\binom{\mathcal{X}}{q}}(\mathbf{b}; 0, t) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(b_e)_e\\ t\end{bmatrix}\prod_e \theta_e^{-b_e},$$

$$P_{\binom{\mathcal{X}}{1}}(\mathbf{a}; s, 0) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(a_e)_e\\ s\end{bmatrix}\prod_e \theta_e^{-a_e}, \qquad P_{\binom{\mathcal{X}}{1}}(\mathbf{b}; 0, t) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(t-b_e)_e\\ t\end{bmatrix}\prod_e \theta_e^{b_e - t},$$

with $|\mathbf{a}| = qs$, $|\mathbf{b}| = t$ at the series node and $|\mathbf{a}| = s$, $|\mathbf{b}| = qt$ at the parallel node; each converges exactly when every Gamma above has an argument of positive real part. These are Proposition 24.16 at the configurations $(\mathbf{e}_x)_{x\in\mathcal{X}}$, of the letters K of $\binom{\mathcal{X}}{1}$ and Ψ of $\binom{\mathcal{X}}{q}$, and $(\mathbf{1} - \mathbf{e}_x)_{x\in\mathcal{X}}$, of the letters K of $\binom{\mathcal{X}}{q}$ and Ψ of $\binom{\mathcal{X}}{1}$.

Corollary 25.11 (Series-parallel tables at their full leans)**.** *In Definition 25.3 let every node of $\mathfrak{T}$ be a series or a parallel node (Definition 22.12), the node v on $\mathcal{E}_v \sqcup G_v$ carrying the weight $\theta_{v,x}$ on its basis that omits x if it is a series node and on its basis $\{x\}$ if it is a parallel node, and write $(\mathbf{a}_v)_x$ and $(\mathbf{b}_v)_x$ for the node loads at $x \in \mathcal{E}_v \sqcup G_v$ (Definition 25.4). Then*

$$P_{\mathcal{B}}(\mathbf{a}; s, 0) = \frac{1}{\Gamma(s)} \prod_{e \text{ at series}} \Gamma(s - a_e) \prod_{e \text{ at parallel}} \Gamma(a_e) \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(s - k_{\mathfrak{e}})}{\Gamma(k_{\mathfrak{e}})}$$
$$\times \prod_{v \text{ series}} \prod_x \theta_{v,x}^{(\mathbf{a}_v)_x - s} \prod_{v \text{ parallel}} \prod_x \theta_{v,x}^{-(\mathbf{a}_v)_x},$$

$$P_{\mathcal{B}}(\mathbf{b}; 0, t) = \frac{1}{\Gamma(t)} \prod_{e \text{ at series}} \Gamma(b_e) \prod_{e \text{ at parallel}} \Gamma(t - b_e) \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(l_{\mathfrak{e}})}{\Gamma(t - l_{\mathfrak{e}})}$$
$$\times \prod_{v \text{ series}} \prod_x \theta_{v,x}^{-(\mathbf{b}_v)_x} \prod_{v \text{ parallel}} \prod_x \theta_{v,x}^{(\mathbf{b}_v)_x - t},$$

the products over the marks by the kind of their node, over the edges joining a series node to a parallel node, with $k_{\mathfrak{e}}$, $l_{\mathfrak{e}}$ the marker loads at the series end, and over the nodes and their marks and markers x; an edge joining two nodes of one kind contributes no Gamma. Each of these periods is finite exactly when every Gamma above its line has an argument with positive real part.

The period of a series-parallel table at a full lean is written down from its tree in time linear in the size of the tree, one Gamma for each mark, two for each change of kind along an edge and a monomial in the weights of each node, while the table may have a number of bases exponential in n. The atom formula of the same period (Corollary 17.15) has one share for each basis.

25.4 The network theorem

The lean convolution of a family of substitutions, the periods of its letters at their full leans written along the hinge trees of its tables, is one fibre integral whose integrand is a product of node data at affine loads. Read as one integral over all the loads, it has dimension $(2N-1)(n-1)$ for every family whose tables carry hinge trees, and for series-parallel tables its integrand is a written-down stacked Gamma times an exponential of the loads. Integrated from the leaves up, when the tables share one hinge tree, the same integral is computed node by node with one rule at every node, prime or not (Theorem 25.15): a node enters through the periods of its tables at their full leans and its cell families through the Mellin transforms of their laws; for one table this is the hinge tree procedure of Chapter 22.

Theorem 25.12 (The network)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ be a family on $\mathcal{E}$ with exponents $s_j, t_j > 0$ whose table j carries a hinge tree $\mathfrak{T}_j$ as in Definition 25.3, its nodes $\mathcal{H}_{j,v}$ each one molecule, and let $\mathbf{m}$ be on the balance wall of F with every proper wall positive. Then every table is one molecule, and on every admissible point (Definition 24.2) and for every twist $\mathbf{x}$ (Definition 21.1),*

$$P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \prod_{j=1}^{N} \frac{\prod_{v\in\mathfrak{T}_j} P_{\mathcal{H}_{j,v}}(\mathbf{a}_{j,v}; s_j, 0)\, P_{\mathcal{H}_{j,v}}(\mathbf{b}_{j,v}; 0, t_j)}{\prod_{\mathfrak{e}=\{v,w\}\in\mathfrak{T}_j} {}_2\Gamma_1\!\left[\begin{matrix} k_{j;v,\mathfrak{e}},\ k_{j;w,\mathfrak{e}} \\ s_j \end{matrix}\right] {}_2\Gamma_1\!\left[\begin{matrix} l_{j;v,\mathfrak{e}},\ l_{j;w,\mathfrak{e}} \\ t_j \end{matrix}\right]} \ (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j},$$

the fibre integral of Theorem 24.5 over the loads $(\mathbf{a}_j, \mathbf{b}_j)_j$ of the marks, of dimension $(2N-1)(n-1)$, with the node loads $\mathbf{a}_{j,v}, \mathbf{b}_{j,v}$ and the marker loads $k_{j;v,\mathfrak{e}}, l_{j;v,\mathfrak{e}}$ of Definition 25.4 read in table j at its loads $(\mathbf{a}_j, \mathbf{b}_j)$ and its exponents (s_j, t_j). At every admissible point the periods of every node at both full leans are finite and every marker load of table j lies in $(0, s_j)$, respectively $(0, t_j)$; the native loads of F are an admissible point. The nodes enter only through their periods at the two full leans, as functions of their loads, the same functions for all exponents, all masses and all twists.

Corollary 25.13 (Series-parallel periods)**.** *In Theorem 25.12 let every node of every table be a series or a parallel node with the weights of Corollary 25.11, $\theta_{j,v,x}$ at the node v of $\mathfrak{T}_j$. Then*

$$\begin{aligned}\prod_j \Gamma(s_j)\Gamma(t_j)\, P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \prod_j \Bigg[&\prod_{e \text{ at series}} \Gamma(s_j - a_{j,e})\Gamma(b_{j,e}) \prod_{e \text{ at parallel}} \Gamma(a_{j,e})\Gamma(t_j - b_{j,e}) \\ &\times \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(s_j - k_{j,\mathfrak{e}})\,\Gamma(l_{j,\mathfrak{e}})}{\Gamma(k_{j,\mathfrak{e}})\,\Gamma(t_j - l_{j,\mathfrak{e}})} \prod_{v \text{ series}} \prod_x \theta_{j,v,x}^{(\mathbf{a}_{j,v})_x - (\mathbf{b}_{j,v})_x - s_j} \\ &\times \prod_{v \text{ parallel}} \prod_x \theta_{j,v,x}^{(\mathbf{b}_{j,v})_x - (\mathbf{a}_{j,v})_x - t_j} \, (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j} \Bigg],\end{aligned}$$

the kinds of the nodes and the marker loads $k_{j,\mathfrak{e}}$, $l_{j,\mathfrak{e}}$ at the series end read in $\mathfrak{T}_j$, the fibre integral of Theorem 25.12, of dimension $(2N-1)(n-1)$, on any point at which every Gamma above the line has an argument of positive real part. Every period of a family of series-parallel tables with any weights on their nodes, at all exponents, at all masses with positive walls and at every twist, is a fibre integral of dimension $(2N-1)(n-1)$ of an explicit stacked Gamma times an exponential of the loads, whatever the numbers of bases, and at a full lean of one table no integral is left (Corollary 25.11).

Proposition 25.14 (Pooling a pair)**.** *In Corollary 25.13, for one table twisted in its cobasis letter by* $\mathbf{x}$ *(Definition 21.1), let* e, f *be two marks of one series node* v *whose twists times the squares of their weights agree,* $x_e\theta_{v,e}^2 = x_f\theta_{v,f}^2$*, and* $c_e = s - m_e$*,* $c_f = s - m_f$*. Their loads enter the integrand through* $\Gamma(c_e + b_e)\Gamma(b_e)\Gamma(c_f + b_f)\Gamma(b_f)$ *and otherwise through* $b_{\{e,f\}} = b_e + b_f$ *alone, and at fixed* $b_{\{e,f\}}$

$$\frac{1}{2\pi i}\int \Gamma(c_e + b_e)\Gamma(b_e)\Gamma\big(c_f + b_{\{e,f\}} - b_e\big)\Gamma\big(b_{\{e,f\}} - b_e\big)\, db_e \\ = \Gamma_{\,1}^{\,4}\begin{bmatrix} c_e + c_f + b_{\{e,f\}},\ c_e + b_{\{e,f\}},\ c_f + b_{\{e,f\}},\ b_{\{e,f\}} \\ c_e + c_f + 2b_{\{e,f\}} \end{bmatrix}.$$

For two marks of one parallel node v *with* $x_e\theta_{v,e}^{-2} = x_f\theta_{v,f}^{-2}$*, at fixed* $b_{\{e,f\}}$*, the pair gives*

$$\Gamma_{\,1}^{\,4}\begin{bmatrix} m_e + m_f - b_{\{e,f\}},\ m_e + t - b_{\{e,f\}},\ m_f + t - b_{\{e,f\}},\ 2t - b_{\{e,f\}} \\ m_e + m_f + 2t - 2b_{\{e,f\}} \end{bmatrix}.$$

Each pooled pair lowers the dimension of the fibre by one, and disjoint pairs can be pooled at once.

The pooled pair is a series pair read as one cell: its factor is $\Gamma_{\,1}^{\,2}\begin{bmatrix} c_e + b_{\{e,f\}},\ c_f + b_{\{e,f\}} \\ c_e + c_f + 2b_{\{e,f\}} \end{bmatrix}\Gamma(c_e + c_f + b_{\{e,f\}})\Gamma(b_{\{e,f\}})$, the Beta corner being the Mellin transform of the law of the pair at $b_{\{e,f\}}$ and the two Gammas those of its hinge in the periods of the node at its full leans (Theorem 25.15). A series bunch of three or more marks at one node can be pooled pair by pair only once: the pooled factor of a pair is no longer of the form $\Gamma(c+b)\Gamma(b)$, and the class is a circuit of its marks (Corollary 17.20).

25.5 Nodes in hold form

Theorem 25.15 (Nodes in hold form)**.** *Let* F *be a substituted family (Definition 20.1) whose tables are each one molecule, with exponents* $s_j, t_j > 0$*, table* j *being* $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$ *with the cores* $\mathcal{H}_j$ *on* $\mathcal{E}_{\mathcal{H}} \sqcup G_{\mathcal{H}}$ *and the cells* $\mathcal{A}_{ij}$ *on* $B_i \sqcup \{g_i\}$ *at the markers* $g_1, \dots, g_k$ *(Definition 22.9), and let* $F_i = ((\mathcal{A}_{ij}, \theta^{\mathcal{A}_{ij}}); s_j, t_j)_j$ *be the cell family at* g_i *(Definition 22.1). For loads* $(\mathbf{a}_j, \mathbf{b}_j)_j$ *on the marks and markers of the cores write* $\mathbf{z}_i = (a_{j,g_i}, b_{j,g_i})_j$ *for their values at* g_i*, and in this statement write* $\int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i}$ *for the Mellin transform of the law of* F_i *of Lemma 24.9 at* $\mathbf{z}_i$*.*

(i) Root. *If* $G_{\mathcal{H}} = \{g_1, \dots, g_k\}$ *and* F *is on its balance wall with every proper wall positive, let* $\mathcal{F}$ *be the fibre of the loads of the cores with the pins* $\sum_j(\mathbf{a}_j + \mathbf{b}_j) = (\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}, \kappa_{F_1}, \dots, \kappa_{F_k})$*,* $|\mathbf{a}_j| = r_{\mathcal{H}_j} s_j$ *(*$j \geq 2$*) and* $|\mathbf{b}_j| = r_{\mathcal{H}_j}^* t_j$*. On the node loads at the cores (Definition 25.4, in every table) of any admissible point of* F *and for twists* $\mathbf{x}$ *of* F *with every coordinate* 1 *off* $\mathcal{E}_{\mathcal{H}}$*,*

$$P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\mathcal{F}} \prod_{j=1}^{N} P_{\mathcal{H}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_j; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{a}_j|_{\mathcal{E}_{\mathcal{H}}}} (\mathbf{x}_j^{\Psi})^{-\mathbf{b}_j|_{\mathcal{E}_{\mathcal{H}}}} \prod_{i=1}^{k} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i}\, d\phi_{F_i},$$

of dimension $(2N-1)(|\mathcal{E}_{\mathcal{H}}| + k - 1)$*, every* $\int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i}$ *finite at the real point.*

(ii) Hinged core. *If* $G_{\mathcal{H}} = \{g_0, g_1, \dots, g_k\}$*, let* F' *be the cell family with the hinge* g_0 *whose table* j *is* $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$*, with the hinge wall* $\kappa_{F'}$*, and for* $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ *with* $\sum_j(z_{j,K} + z_{j,\Psi}) = \kappa_{F'}$ *let* $\mathcal{F}_{\mathbf{z}}$ *be the fibre of the loads of the cores with the pins* $\sum_j(\mathbf{a}_j + \mathbf{b}_j) =$

$\big(\mathbf{m}|\mathcal{E}_{\mathcal{H}},\kappa_{F_1},\dots,\kappa_{F_k},\sum_j(s_j+t_j)-\kappa_{F'}\big)$, $|\mathbf{a}_j|=r_{\mathcal{H}_j}s_j$ $(j\ge 2)$, $|\mathbf{b}_j|=r^*_{\mathcal{H}_j}t_j$, $a_{j,g_0}=s_j-z_{j,K}$ $(j\ge 2)$ *and* $b_{j,g_0}=t_j-z_{j,\Psi}$. *For* $\mathbf{z}$ *whose real part is* $(s_j-c^K_{j,g_0},t_j-c^\Psi_{j,g_0})_j$, $\mathbf{c}$ *the node loads at the cores of an admissible point of the family of the tables of* F' *with the mass* $\sum_j(s_j+t_j)-\kappa_{F'}$ *on* g_0,

$$\prod_{j=1}^N \Gamma_{\binom{2}{1}}\begin{bmatrix} s_j-z_{j,K},\ z_{j,K}\\ s_j\end{bmatrix}\Gamma_{\binom{2}{1}}\begin{bmatrix} t_j-z_{j,\Psi},\ z_{j,\Psi}\\ t_j\end{bmatrix}\int \boldsymbol{\varrho}_{F'}^{-\mathbf{z}}\,d\phi_{F'}$$
$$=\int_{\mathcal{F}_{\mathbf{z}}}\prod_{j=1}^N P_{\mathcal{H}_j}(\mathbf{a}_j;s_j,0)\,P_{\mathcal{H}_j}(\mathbf{b}_j;0,t_j)\prod_{i=1}^k\int\boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i}\,d\phi_{F_i},$$

of dimension $(2N-1)(|\mathcal{E}_{\mathcal{H}}|+k-1)$.

The cores and the cells are molecules by Lemma 25.2.

Corollary 25.16 (The glue and the series and parallel rules in hold form)**.** *On the lines of Theorem 25.15:*

(i) For one table with $\mathcal{H}=\binom{\{g_1,g_2\}}{1}$ *and the cells* $\mathcal{A}$ *and* $\mathcal{A}'$, *Theorem 25.15(i) is the integral of Theorem 22.4(iv), with the same integrand; for a family with the core* $\binom{\{g_1,g_2\}}{1}$ *in every table it is the integral of Theorem 18.13 over the scale shares.*

(ii) For one table and a series node $\binom{G}{q}$, *Theorem 25.15(ii) reads*

$$\Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}})\int\varrho_{\mathcal{A}}^{-z}d\phi_{\mathcal{A}}=\int_{z_1+\cdots+z_q=z}\prod_{i=1}^q\Gamma(z_i)\Gamma(z_i+s-\kappa_{\mathcal{A}_i})\int\varrho_{\mathcal{A}_i}^{-z_i}d\phi_{\mathcal{A}_i},$$

the Mellin transform of $f_{\mathcal{A}}$ *as the Mellin convolution of those of the* $f_{\mathcal{A}_i}$ *(Proposition 22.5(iii)): the Mellin transform, on these lines, of the product rule* $f_{\mathcal{A}}=\prod_i f_{\mathcal{A}_i}$ *of Theorem 22.13(i).*

(iii) For one table and a parallel node $\binom{G}{1}$ *it reads, with* $z'=\kappa_{\mathcal{A}}-z$ *and* $z'_i=\kappa_{\mathcal{A}_i}-z_i$,

$$\Gamma(z')\Gamma(z'+t-\kappa_{\mathcal{A}})\int\varrho_{\mathcal{A}}^{z'-\kappa_{\mathcal{A}}}d\phi_{\mathcal{A}}=\int_{z'_1+\cdots+z'_q=z'}\prod_{i=1}^q\Gamma(z'_i)\Gamma(z'_i+t-\kappa_{\mathcal{A}_i})\int\varrho_{\mathcal{A}_i}^{z'_i-\kappa_{\mathcal{A}_i}}d\phi_{\mathcal{A}_i},$$

the Mellin convolution of the transforms of the $\check f_{\mathcal{A}_i}$: *the Mellin transform, on these lines, of the product rule* $\check f_{\mathcal{A}}=\prod_i\check f_{\mathcal{A}_i}$ *of Theorem 22.13(ii).*

So the three rules of the hinge tree procedure, the glue and the series and parallel products (Definition 22.17), are one rule read at three nodes, on the lines that the network certifies, the rules themselves holding on their own wider ranges; and the same rule applies at a prime node, where the procedure integrates the twisted core over its simplex for every value of the twists (Definition 22.17(4)). In Theorem 25.15 the prime node enters through the periods of its tables at their full leans, functions of its loads that do not depend on the twists, on the exponents or on the masses, and the cell families enter through the Mellin transforms of their laws on imaginary translates, for one table the representation (b) of Definition 22.17; Theorem 25.15(i) is Theorem 22.11(i) with the twisted core replaced by its lean convolution and the integration against the laws carried out through Mellin transforms.

25.6 The procedure and its cost

Definition 25.17 (The network procedure)**.** *Given*: a hinge tree of every table of a family (Definition 22.14) with the weights and the marks of its nodes; exponents $s_j, t_j > 0$ and masses $\mathbf{m}$ on the balance wall, or several of each; optionally twists $\mathbf{x}$, or product weights (Proposition 25.19). *Reusable data*: for every orbit of nodes, over all the tables, under bijections keeping the table and its weights, the Gramma $\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta})$ of its table, a function of the loads alone (Corollary 24.6), kept as (a) the closed form at series and parallel nodes (Proposition 25.10); (b) the share side, a fibre integral of dimension $|\mathcal{H}_v| - n_v$ (Corollary 25.9); (c) for an identically self-dual node, the reduction to the deletion of one mark (Corollaries 22.8(iii) and 24.10); (d) values on the imaginary translates in use. *Operations*:

(1) *Contour.* Find a split satisfying the node inequalities of Corollary 25.7, a linear feasibility problem with $\sum_j \sum_v 2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ strict inequalities in $(2N-1)(n-1)$ unknowns; the native loads solve it, and infeasibility proves divergence.

(2) *Assemble.* The node loads of every table by Definition 25.4, the integrand of Theorem 25.12, at series-parallel nodes that of Corollary 25.13, with $\prod_j (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j}$ for twists.

(3) *Pool*, for one table, disjoint pairs of marks of one series or parallel node whose twists and weights satisfy the condition of Proposition 25.14, by that proposition.

(4) *Integrate* the remaining loads on the contour; in the imaginary parts the integrand is analytic on a strip about the contour, the admissible points forming an open set, and decays exponentially there (Theorem 24.5(iii)), so the trapezoidal rule with step ς has an error of the order $e^{-2\pi d/\varsigma}$, d the half-width of the strip; or close contours on lattices (Theorem 4.5).

(5) *Recursive Mellin evaluation*, an alternative to (2) to (4) when the hinge trees of the tables are one rooted tree with the same marks $\mathcal{E}_v$ at every vertex, so that at every vertex v other than the root the subtree cells of the tables (Definition 22.14) form a cell family, the *subtree cell family* of v (Definition 22.1): from the leaves up, compute the Mellin transform of the law of each orbit of subtree cell families on an imaginary translate by Theorem 25.15(ii), one fibre integral of dimension $(2N-1)(|\mathcal{E}_v| + |G_v| - 2)$ per point, and finish at the root by Theorem 25.15(i); for one table, at series and parallel nodes with repeated subtrees, equal and with equal masses, pass to the transforms, where repetition is a power (Proposition 22.16(ii)).

Output: the period, and with the same reusable data the period at other exponents, masses and twists (Proposition 25.20).

Proposition 25.18 (What the network costs)**.** *In Definition 25.17:*

(i) The fibre of the loads in (1)–(4) has dimension $(2N-1)(n-1)$, for one table less one for each pooled pair; opening every prime node on the share sides of the periods of its table at the two full leans adds $2(|\mathcal{H}_{j,v}| - n_{j,v})$ for each, and series and parallel nodes add nothing. The dimension does not depend on the numbers $|\mathcal{B}_j|$ of bases of the tables, against $\sum_j 2|\mathcal{B}_j| - 2N - n + 1$ for the atom formula (Corollary 17.15) and $\sum_j |\mathcal{B}_j| - n + 1$ after its even shares close (Theorem 19.10(iv)).

(ii) The reusable data are one function of n_v variables per orbit of nodes, the same for every node of the orbit, in every table, and for every query; a value of the period of a node at a full lean at given loads, once computed, serves every node of the orbit and every query at which those loads occur.

(iii) In the recursive Mellin evaluation the work is one Mellin transform on an imaginary translate for each orbit of subtree cell families (as in Proposition 22.16(i)), each point costing one

fibre integral of dimension $(2N-1)(|\mathcal{E}_v|+|G_v|-2)$ *at its top node, and at the root one of dimension* $(2N-1)(|\mathcal{E}_o|+|G_o|-1)$*, whose integrand needs the periods of the tables of the node at their full leans, at the loads of the fibre: closed forms at series and parallel nodes, and at a prime node an integral of its own, of dimension* $|\mathcal{H}_{j,v}|-n_v$ *on its share side, unless another representation of Definition 25.17 is cheaper.*

In the recursive Mellin evaluation equal subtrees at a series node enter as a power of one transform (Theorems 22.13(i) and 22.4(ii)), whose growth in the number of copies Theorem 23.2 reads when the subtree is series-balanced.

25.7 Changing the query in hold form

A query about a period is a choice of exponents, of masses, of weights and of the kind of answer asked for. In the hold form each of them enters at one place: exponents and masses move the pins and the affine marker loads, twists and product weights multiply the integrand by an exponential of the loads, and a replaced node, or a general change of the weights of one node, replaces that node's two functions and nothing else. Product weights are read as twists, and the exponents at fixed masses as the Mellin variables of the relative scales of the letters.

Proposition 25.19 (Product weights are twists)**.** *For a family* F *on* $\mathcal{E}$ *and* $\boldsymbol{\zeta}=(\boldsymbol{\zeta}_j)_j$*,* $\boldsymbol{\zeta}_j\in(0,\infty)^{\mathcal{E}}$*, let* $F^{\boldsymbol{\zeta}}$ *be* F *with the weights* $\theta_{j,T}\boldsymbol{\zeta}_j^T$ *in table* j*. Then* $K_{F^{\boldsymbol{\zeta}},j}(\alpha)=K_j(\boldsymbol{\zeta}_j\circ\alpha)$ *and* $\Psi_{F^{\boldsymbol{\zeta}},j}(\alpha)=\boldsymbol{\zeta}_j^{\mathcal{E}}\Psi_j(\alpha/\boldsymbol{\zeta}_j)$*:* $F^{\boldsymbol{\zeta}}$ *has the letters of the twist* $F[\mathbf{x}]$ *with* $\mathbf{x}_j^K=\boldsymbol{\zeta}_j$ *and* $\mathbf{x}_j^\Psi=\boldsymbol{\zeta}_j^{-1}$ *(Definition 21.1), its cobasis letters multiplied by the constants* $\boldsymbol{\zeta}_j^{\mathcal{E}}$*, and*

$$P_{F^{\boldsymbol{\zeta}}}(\mathbf{m};\mathbf{s},\mathbf{t})=\prod_j\boldsymbol{\zeta}_j^{-t_j\mathbf{1}}\,P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}),$$

$$P_{\mathcal{B}_j^{\boldsymbol{\zeta}_j}}(\mathbf{a};s_j,0)=\boldsymbol{\zeta}_j^{-\mathbf{a}}P_{\mathcal{B}_j}(\mathbf{a};s_j,0),\qquad P_{\mathcal{B}_j^{\boldsymbol{\zeta}_j}}(\mathbf{b};0,t_j)=\boldsymbol{\zeta}_j^{\mathbf{b}-t_j\mathbf{1}}P_{\mathcal{B}_j}(\mathbf{b};0,t_j),$$

so that the lean convolution of $F^{\boldsymbol{\zeta}}$ *is that of* F *with the exponential* $\prod_j\boldsymbol{\zeta}_j^{-t_j\mathbf{1}}\boldsymbol{\zeta}_j^{\mathbf{b}_j-\mathbf{a}_j}$ *of the loads. With one* $\boldsymbol{\zeta}$ *for every table,* $P_{F^{\boldsymbol{\zeta}}}(\mathbf{m};\mathbf{s},\mathbf{t})=\boldsymbol{\zeta}^{-\mathbf{m}-|\mathbf{t}|\mathbf{1}}P_{F[\mathbf{x}']}(\mathbf{m};\mathbf{s},\mathbf{t})$*,* $\mathbf{x}'$ *the twist of every cobasis letter by* $\boldsymbol{\zeta}^{-2}$ *and* $|\mathbf{t}|=\sum_j t_j$*. Weights* $\zeta_{j,g}$ *on the bases through one mark* g*, in each table* j*, are the twist at* g *with* $x_j^K=\zeta_{j,g}$ *and* $x_j^\Psi=\zeta_{j,g}^{-1}$*, up to the factor* $\prod_j\zeta_{j,g}^{-t_j}$*: the weight of a hinge of Theorem 23.5, at any mark.*

Proposition 25.20 (Carrying a calculation through a change, in hold form)**.** *In Theorem 25.12, with its node data computed:*

(i) a change of the exponents (s_j,t_j) *changes the pins* $|\mathbf{a}_j|=r_js_j$*,* $|\mathbf{b}_j|=r_j^*t_j$ *and the marker loads of table* j*, which are affine in* (s_j,t_j)*, and nothing else; the native loads of the new data are a new contour;*

(ii) a change of the masses changes the load $\mathbf{a}_1=\mathbf{m}-\sum_{j\geq2}\mathbf{a}_j-\sum_j\mathbf{b}_j$ *in the coordinates of Theorem 24.5(ii), and through it the marker loads* $k_{1;v,\mathfrak{e}}$ *of table* 1*, and in the integrand nothing else; in (i) and (ii) the contour must move when the old one is no longer admissible;*

(iii) a change of twists, or of product weights on the bases of the tables, multiplies the integrand by an exponential of the loads (Proposition 25.19); product weights on the bases of one node multiply its periods at the two full leans by exponentials of its node loads;

(iv) replacing a node of a table by another table on the same marks and markers, of the same rank, one molecule with its markers neither loops nor coloops, or changing the weights of a node in any way, replaces the periods of that node at its two full leans and nothing else;

(v) in the recursive Mellin evaluation (Definition 25.17(5)), a change at one node changes the Mellin transforms of the laws of the subtree cell families on the path from it to the root only, each recomputed by Theorem 25.15(ii) from the new one of the vertex directly below it on the path and the kept ones of the other vertices directly below it, as Proposition 23.6 says of laws.

The reusable data of Chapter 22, the laws of the subtrees, depend on the exponents and on the masses below each edge (Theorem 22.15(iii)); the reusable data of the network do not, and a sweep over exponents, over masses, over twists or over product weights recomputes no node. What does change is the contour, and Corollary 25.7 decides at the nodes whether the new query has an answer.

Proposition 25.21 (The exponents are moment indices)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ be a family on $\mathcal{E}$, fix real masses $\mathbf{m}$, and let I be the set of exponents $(\mathbf{s}, \mathbf{t}) \in (0,\infty)^{2N}$ with $\sum_j (r_j s_j + r_j^* t_j) = m_{\mathcal{E}}$ at which every proper wall of $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is positive: a relatively open convex subset of that hyperplane, possibly empty. In this statement write $L_1, \dots, L_{2N}$ for the letters K_j and Ψ_j of F, w_i for their exponents and g_i for their degrees, and for each i and $\boldsymbol{\lambda} \in (0,\infty)^{2N-1}$ put*

$$\mathsf{G}_i(\mathbf{m}; \boldsymbol{\lambda}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{m}-\mathbf{1}} \exp\Big(- L_i(x) - \sum_{i' \neq i} \lambda_{i'} L_{i'}(x) \Big)\, dx,$$

the Gramma (Definition 2.1) of the concatenated configuration of all the letters, with the coefficients of L_i and $\lambda_{i'}$ times those of $L_{i'}$. For $(\mathbf{s}, \mathbf{t}) \in I$ and every i,

$$\begin{aligned} \Gamma(w_i)\, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) &= g_i \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{m}-\mathbf{1}} e^{-L_i(x)} \prod_{i' \neq i} L_{i'}(x)^{-w_{i'}}\, dx \\ &= \frac{g_i}{\prod_{i' \neq i} \Gamma(w_{i'})} \int_{(0,\infty)^{2N-1}} \prod_{i' \neq i} \lambda_{i'}^{w_{i'}-1}\, \mathsf{G}_i(\mathbf{m}; \boldsymbol{\lambda})\, d\boldsymbol{\lambda}. \end{aligned}$$

At fixed masses the numbers $\Gamma(w_i) P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})/g_i$ on I, functions of the exponents of the other letters, are the integrals of $\prod_{i' \neq i} L_{i'}^{-w_{i'}}$ against the one positive measure $x^{\mathbf{m}-\mathbf{1}} e^{-L_i(x)} dx$, and $\prod_{i'} \Gamma(w_{i'})\, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})/g_i$ is the Mellin transform at $(w_{i'})_{i' \neq i}$ of the one function $\mathsf{G}_i(\mathbf{m}; \cdot)$ of the relative scales of the other letters to L_i, completely monotone in each of them. On I the period $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is log-convex, by Proposition 8.11(iii) and Hölder's inequality.

So the period of a family at fixed masses is one integral over the orthant with any one of its letters closed: the closed letter leaves one positive measure on the orthant, the integrand of its Gramma at the index $\mathbf{m}$, and the period is a joint moment of the other letters under it, their exponents with the sign changed being its orders.

Chapter 26

Finite interfaces and certified enclosures

A cell enters every construction through its hinge wall and its law (Theorem 22.7), and a cell with several hinges through the joint law of its channels (Theorem 20.13, Definition 20.10). A law is not a finite list of numbers. Finitely many numbers are retained for every prescribed polynomial moment, the joint moments of the channels under independent Gamma draws on the marks, up to a degree D: they compose by the bilinear map of Theorem 20.11, a repeated chain cell acts on them by one finite matrix, so that every polynomial Gamma moment of a repetition satisfies a linear recurrence with constant coefficients, and the walls turn a moment of finite degree into a certified enclosure of the period. The period is retained as well over a neighbourhood of the query: the walls give the neighbourhood explicitly, the Taylor coefficients in the weights are periods with walls at least as high, and a finite parameter bank with two explicit errors answers every query inside it. Exact representation size, arithmetic work and accuracy are separate quantities.

26.1 The joint law of the channels and its moments

Let $\mathcal{A}$ be a cell with hinges G on its marks B, or cells with the hinges G on the same marks B kept together (Definition 26.1), with channel vector $\mathrm{ch}(\mathcal{A})$ (Definition 20.10), with positive masses $\mathbf{m}$ on B; the hinges carry no mass. Its total mass m_B and the product $\prod_{e\in B}\Gamma(m_e)$ are kept as data: at a splicing the total masses add and these products multiply. For a multi-index $\mathbf{k}$ on the channels, $\mathrm{ch}(\mathcal{A})^{\mathbf{k}}$ is the product of the channels with these multiplicities, a polynomial of the degree $\deg\mathbf{k}=\sum_c k_c\deg c$, the sum over the channels c.

Definition 26.1 (The channel law; moment vectors)**.** Cells $\mathcal{A}_1,\dots,\mathcal{A}_N$ with the hinges G on the same marks B are *kept together* when their channels are read jointly: $\mathcal{A}=(\mathcal{A}_1,\dots,\mathcal{A}_N)$ has the channel vector $\mathrm{ch}(\mathcal{A})=(\mathrm{ch}(\mathcal{A}_j))_j$, listing the channels of all of them by the tags (j,letter,p), $N2^{|G|+1}$ of them, absent channels counting as zero; one cell is the case $N=1$, and the tables of a family, with $G=\varnothing$, are kept together with their letters as channels. The *channel law* of $\mathcal{A}$ is the image $\mathrm{ch}(\mathcal{A})_*\,\mathrm{Dir}_{\mathbf{m}}$ of the Dirichlet law of the shape (Definition 3.10) under the channel vector. The *Gamma draws* on the marks are $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$, independent Gamma_{m_e} variables (Proposition 8.11), and the *moments* and the *moment vector of degree* D of $\mathcal{A}$ are

$$\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})=\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}\big[\,\mathrm{ch}(\mathcal{A})(\alpha)^{\mathbf{k}}\big],\qquad \mathrm{Mom}_{\mathcal{A}}^{\le D}=\big(\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})\big)_{|\mathbf{k}|\le D},$$

at most $\binom{q+D}{D}$ numbers, q the number of channels. A *polynomial Gamma moment* of degree at

most D of $\mathcal{A}$ is $\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}\big[p(\mathrm{ch}(\mathcal{A})(\alpha))\big]$ for a polynomial p of degree at most D in the channels, and a *shape moment* is the same with $U\sim\mathrm{Dir}_{\mathbf{m}}$ in place of α. A cell without marks has constant channels, a point as its law and the products of its channels as its moments, with total mass 0 and the empty product of Gammas.

The channels must be kept jointly: the moments of the channels one at a time do not determine a product of different channels exposed to the same room.

Proposition 26.2 (The channel law composes)**.** *Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be cells, or cells kept together, with disjoint marks B_1, B_2 and masses $\mathbf{m}_1$, $\mathbf{m}_2$, spliced table by table along pairs of their hinges to $\mathcal{A}_{12}$ (Theorem 20.11), both with marks. Let $U_1\sim\mathrm{Dir}_{\mathbf{m}_1}$, $U_2\sim\mathrm{Dir}_{\mathbf{m}_2}$ and $(V_1,V_2)\sim\mathrm{Dir}_{(m_{B_1},m_{B_2})}$ be independent; replace every channel c of $\mathcal{A}_1$ by $V_1^{\deg c}c(U_1)$ and every channel c of $\mathcal{A}_2$ by $V_2^{\deg c}c(U_2)$. The law of the image of the two channel vectors so replaced under the bilinear map of Theorem 20.11(iii), applied table by table, is the channel law of $\mathcal{A}_{12}$. For the tables of a family, kept together without hinges, the channels are their letters, and the period of the family is Proposition 8.11(i) read on the channel law: one Trailmoment of the channel law on its source space (Definition 5.1, Theorem 5.21). For a cell family $F=((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ with the one hinge g (Definition 22.1), its cells kept together, with the coordinates $(x_{j,0},x_{j,1},x^*_{j,0},x^*_{j,1})$ of the channels $K_{\mathcal{A}_j\setminus g}$, $K_{\mathcal{A}_j/g}$, $\Psi_{\mathcal{A}_j\setminus g}$, $\Psi_{\mathcal{A}_j/g}$, the law of F is the weighted image*

$$(\boldsymbol{\varrho}_F)_*\phi_F=\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\ (R_F)_*\Big[\prod_j x_{j,1}^{-s_j}x_{j,0}^{*\,-t_j}\Big(\frac{x_{1,0}}{x_{1,1}}\Big)^{-\kappa_F}\ \mathrm{ch}(\mathcal{A})_*\,\mathrm{Dir}_{\mathbf{m}}\Big],$$

with $R_F=\big((x_{j,0}x_{1,1}/(x_{j,1}x_{1,0}))_{j\ge2},(x^*_{j,1}x_{1,1}/(x^*_{j,0}x_{1,0}))_j\big)$, an equality of positive measures, possibly infinite: the channel law, which knows nothing of the exponents, gives the law of the cell family at all exponents by a weight and an image.

Theorem 26.3 (Gamma draws close the moments)**.** *(i)* Leaves. *If $\mathrm{ch}(\mathcal{A})(x)^{\mathbf{k}}=\sum_{\mathbf{h}}c_{\mathbf{h}}x^{\mathbf{h}}$, then $\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})=\sum_{\mathbf{h}}c_{\mathbf{h}}\prod_e(m_e)_{h_e}$, rational at rational masses and weights.*

(ii) Assembly. *Let $\mathrm{ch}(\mathcal{A}_{12})=\mathrm{sp}\big(\mathrm{ch}(\mathcal{A}_1),\mathrm{ch}(\mathcal{A}_2)\big)$ with the bilinear map* sp *of Theorem 20.11(iii), applied table by table to cells kept together, and write $\mathrm{sp}(c,c')^{\mathbf{k}}=\sum_{\mathbf{k}_1,\mathbf{k}_2}b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}c^{\mathbf{k}_1}c'^{\mathbf{k}_2}$, a sum over $|\mathbf{k}_1|=|\mathbf{k}_2|=|\mathbf{k}|$. Then*

$$\mathrm{Mom}_{\mathcal{A}_{12}}(\mathbf{k})=\sum_{\mathbf{k}_1,\mathbf{k}_2}b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}\,\mathrm{Mom}_{\mathcal{A}_1}(\mathbf{k}_1)\,\mathrm{Mom}_{\mathcal{A}_2}(\mathbf{k}_2),$$

and no moment of degree above $|\mathbf{k}|$ of either cell is needed; a splicing inside one cell is linear and keeps the degree.

(iii) Return to the shape. *For a cell with marks, $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\big[\mathrm{ch}(\mathcal{A})(U)^{\mathbf{k}}\big]=\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})/(m_B)_{\deg\mathbf{k}}$: the joint moments of the channel law through degree D are read off $\mathrm{Mom}_{\mathcal{A}}^{\le D}$, the rank, the sizes and the tags fixing the denominators.*

(iv) Sufficiency. *Two cells with the same hinges, channel tags and degrees and the same $\mathrm{Mom}^{\le D}$ give every final polynomial Gamma moment of degree at most D in the channels, in any assembly with cells on disjoint marks, the same value, even where an intermediate cell exposes more hinges than the replaced one; with the same total mass as well, they give every final shape moment of degree at most D of the channels of the root, under its Dirichlet law, the same value.*

(v) The whole law. *For a cell with marks, the* $\mathrm{Mom}_{\mathcal{A}}^{\le D}$ *for all* D*, with* m_B *and the degrees, determine its channel law, and with it the expectation of every integrable function of the channels (Proposition 26.2).*

The draws are used for nonnegative moments of the channels. A negative moment of a period integrand under the draws diverges on the balance wall: the scale contributes $\int_0^\infty S^{-1}e^{-S}dS$, the pole of $\Gamma(\kappa_{\mathcal{E}})$ in Proposition 8.11. The return to the shape, followed by Proposition 26.2 or by the peak closing (Theorem 15.13(i)) with its stopping rule (Theorem 26.7), is the change of representation that the period needs.

Corollary 26.4 (The work follows the distinct interfaces)**.** *Fix a degree* D*, an assembly that exposes at most* h *hinges at every cell it keeps, and identifications of its repeated subassemblies keeping their data and hinges. Every requested moment of the root is computed exactly by keeping at most* $\binom{2^{h+1}+D}{D}$ *numbers for each distinct cell,* $\binom{N2^{h+1}+D}{D}$ *for* N *tables kept together. The arithmetic is* $O\big(\sum_v E_v(D)\big)$ *operations once the local expansions and the leaf moments are built,* $E_v(D)$ *the number of nonzero coefficients* $b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}$ *at a distinct splicing* v *of two cells, at most the product of the sizes of its three moment vectors, or of two for a splicing inside one cell. Equal copies share one moment vector and remain independent draws: their moments are multiplied as in Theorem 26.3(ii), never identified.*

This counts operations. Building the local expansions, finding an assembly of small width, the bit lengths of exact coefficients and the degree a moment needs are separate parts of the cost; finding the cheapest assembly is a search, as for constructions (Section 31.1).

26.2 Repetition is a finite matrix

Theorem 26.5 (A finite matrix for every polynomial Gamma moment)**.** *Let a chain cell* $\mathcal{U}$ *(Proposition 23.4, here with several hinges) be a cell with two disjoint sets of hinges* G_{in} *and* G_{out}*, each in a fixed bijection with the hinges of a cell* $\mathcal{A}_0$*, and let* $\mathcal{A}_{l+1}$ *be* $\mathcal{U}$*, with fresh marks and the same masses and weights at every use, spliced onto* $\mathcal{A}_l$ *by pairing* G_{in} *with the hinges of* $\mathcal{A}_l$*; so* $\mathcal{A}_{l+1}$ *exposes* G_{out}*, named through the bijection as the hinges of* $\mathcal{A}_l$*, with the same channel tags, and its channel vector is the image of* $\mathrm{ch}(\mathcal{A}_l)$ *and that of the copy of* $\mathcal{U}$ *under the map* sp*. Keep one list of channel tags for every* l*, absent channels being zero. For every* D *there is a matrix* $\mathbf{T}_D$*, independent of* l *and computed from* $\mathcal{U}$ *by finite expansion and Theorem 26.3(i), with*

$$\mathrm{Mom}_{\mathcal{A}_l}^{\le D} = \mathbf{T}_D^{\,l}\ \mathrm{Mom}_{\mathcal{A}_0}^{\le D}\,.$$

Every fixed polynomial Gamma moment of degree at most D *of* $\mathcal{A}_l$*, a fixed terminal cell spliced on it included, is* $\mathbf{e}^{\mathsf{T}}\mathbf{T}_D^{\,l}\,\mathrm{Mom}_{\mathcal{A}_0}^{\le D}$ *for a fixed row* $\mathbf{e}^{\mathsf{T}}$*; so its generating series in* l *is* $\mathbf{e}^{\mathsf{T}}(I-\zeta\mathbf{T}_D)^{-1}\,\mathrm{Mom}_{\mathcal{A}_0}^{\le D}$*, rational in* ζ*, the sequence satisfies the linear recurrence with constant coefficients given by the characteristic polynomial of* $\mathbf{T}_D$*, of order at most* $\binom{q+D}{D}$*, and one length* l *costs* $O\big(\binom{q+D}{D}^3\log(l{+}1)\big)$ *operations by repeated squaring.* $\mathbf{T}_D$ *is block diagonal by the degree, and a moment of one degree uses its block alone; a repeated sequence of different chain cells is the power of the product of their matrices.*

On a cell with one hinge g, a chain cell $\mathcal{U}$ with the hinges i and o, i spliced to g, and the channels $K_{\mathcal{U},(P)}$, $\Psi_{\mathcal{U},(P)}$, $P\subseteq\{i,o\}$, acts by

$$\begin{pmatrix} K_{\mathcal{A}_{l+1},(\varnothing)} \\ K_{\mathcal{A}_{l+1},(\{o\})} \end{pmatrix} = \begin{pmatrix} K_{\mathcal{U},(\{i\})} & K_{\mathcal{U},(\varnothing)} \\ K_{\mathcal{U},(\{i,o\})} & K_{\mathcal{U},(\{o\})} \end{pmatrix} \begin{pmatrix} K_{\mathcal{A}_l,(\varnothing)} \\ K_{\mathcal{A}_l,(\{g\})} \end{pmatrix},$$

and by the same matrix of the $\Psi_{\mathcal{U},(P)}$ on the Ψ channels, which is Theorem 20.11(iii) with one pair. A moment of degree a in the K channels and b in the Ψ channels lives on $(a+1)(b+1)$ monomials, and the block of $\mathbf{T}_D$ on them is the joint expectation over the draws of $\mathcal{U}$ of the expanded actions: the two matrices share the marks of $\mathcal{U}$ and are not independent. For a family at integer exponents, the joint moments of its letters along a chain of bunches with matrix letters are the case of Theorem 21.12. An iterate of an integral operator gives no finite closing relation by itself (Theorem 21.10, Proposition 23.4); every polynomial Gamma moment of a repetition does. The period is recovered from a sequence of such moments with certified tails, in Section 26.3.

Corollary 26.6 (Shape moments have recurrences)**.** *For the repetition of Theorem 26.5 let the total masses be $m_{\mathcal{A}_l} = m_{\mathcal{A}_0} + l\, m_{\mathcal{U}}$, $m_{\mathcal{U}}$ that of $\mathcal{U}$, let p be a polynomial in the channels, the same for every l, homogeneous of degree $d_l = d_0 + lv$ in the marks, v an integer, and let M_l be the polynomial Gamma moment of $\mathcal{A}_l$ for p. The shape moment for p is $M_l/(m_{\mathcal{A}_l})_{d_l}$, and every recurrence $\sum_i c_i M_{l+i} = 0$ gives $\sum_i c_i \frac{(m_{\mathcal{A}_{l+i}})_{d_{l+i}}}{(m_{\mathcal{A}_l})_{d_l}} \frac{M_{l+i}}{(m_{\mathcal{A}_{l+i}})_{d_{l+i}}} = 0$. If the masses are rational and $h \geq 1$ is an integer with $h m_{\mathcal{U}} \in \mathbb{Z}$, then on each residue of l modulo h, with $\mathbf{T}_D^h$ for $\mathbf{T}_D$, the coefficients are rational functions of l over the field of the weights, polynomials after clearing denominators, in $\mathbb{Q}[l]$ at rational weights.*

26.3 A certified enclosure of the period

Theorem 26.7 (Explicit stopping from the walls)**.** *Let a family F on $n \geq 2$ marks have real masses on its balance wall and exponents $s_j, t_j \geq 0$, with every proper wall positive. Write its active letters, the K_j with $s_j > 0$ and the Ψ_j with $t_j > 0$, as L_i with exponents $w_i > 0$, degrees g_i and counts $d_i(S)$, as in Theorem 15.13, so that $m_{\mathcal{E}} = \sum_i g_i w_i$ and $\kappa_S = m_S - \sum_i w_i d_i(S)$. Choose references $M_i \geq \max_{\Delta_{\mathcal{E}}} L_i$, and in this statement put $Y_i = L_i/M_i \in (0,1]$ on the open simplex. For an order π of the marks let $U_{\pi,k}$ be its tails, $L_{i,\pi}$ the sector letter of L_i, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.7(ii), and $\underline{L}_{i,\pi} > 0$ its constant term. For integers $Q_i \geq 0$ put*

$$P^{\leq \mathbf{Q}} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_i M_i^{-w_i} \sum_{0 \leq k_i \leq Q_i} \prod_i \frac{(w_i)_{k_i}}{k_i!}\ \mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}} \Big[\prod_i \big(1 - Y_i(U)\big)^{k_i} \Big],$$

the partial sums of the peak closing (Theorem 15.13(i)), computed exactly from the moments of the letters through the degree $\sum_i Q_i$, each monomial with its own rising factorial (Theorem 26.3(iii)). For every choice of $\epsilon_i > 0$ with

$$\kappa_S - \epsilon_i d_i(S) > 0 \qquad (\varnothing \neq S \subsetneq \mathcal{E}),$$

which is exactly $\epsilon_i < \varpi_i$, ϖ_i the least of the ratios $\kappa_S/d_i(S)$ in Theorem 15.13(ii) (any $\epsilon_i > 0$ when no proper S has $d_i(S) > 0$), put

$$C_i = M_i^{\epsilon_i}\, n^{g_i \epsilon_i} \sum_{\pi} \underline{L}_{i,\pi}^{-\epsilon_i} \prod_{i'} \underline{L}_{i',\pi}^{-w_{i'}} \prod_{k=2}^{n} \frac{1}{\kappa_{U_{\pi,k}} - \epsilon_i d_i(U_{\pi,k})}.$$

Then

$$0 \leq P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) - P^{\leq \mathbf{Q}} \leq \sum_i C_i \frac{\Gamma(w_i + \epsilon_i)\, \Gamma(w_i + Q_i + 1)}{\Gamma(w_i)\, \Gamma(w_i + Q_i + 1 + \epsilon_i)},$$

and for $\epsilon_i \leq 1$ the last factor is at most $(w_i/(w_i + Q_i + 1))^{\epsilon_i}$. So finitely many moments and finitely many walls enclose the period to any prescribed accuracy; the same moments answer every other

choice of exponents w_i with $\sum_i g_i w_i = m_{\mathcal{E}}$ at the same masses, only the coefficients of $P^{\leq \mathbf{Q}}$, the walls and the certificate changing.

At rational data the constants of the certificate are positive rationals and their rational powers, and no period other than the one sought enters it; the stacked Gamma $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ and the Gamma ratios stay exact products and quotients of values of the Gamma function, so the enclosure is exact without being rational. The certificate can itself be expensive, C_i being a sum over all $n!$ orders, and it is crude: it replaces every sector letter by its least value. The true tail is of the order $Q^{-\varpi}(\log Q)^{k-1}$ (Theorem 15.13(ii)); the certified decay $Q_i^{-\epsilon_i}$ tends to it as $\epsilon_i \to \varpi_i$, at the cost of the factor $\prod_k (\kappa_{U_{\pi,k}} - \epsilon_i d_i(U_{\pi,k}))^{-1}$, which grows without bound there. Sharper references and sharper sector bounds lower the degree needed without changing the moments. The degree needed can grow with the size of the assembly, so this is a terminating certified procedure, not a uniform fast approximation.

26.4 Changing the query with finite data

Proposition 26.8 (Parameters and insertions)**.** *Fix the local supports and the assembly.*

(i) For each D the leaf moments and the splicing and transfer coefficients are polynomials in the local weights and in the masses; they can be kept symbolic and specialized at any positive values.

(ii) Carrying the leaf moments, Theorem 26.3(ii) and Theorem 26.5 in the truncated ring $\mathbb{Q}(\boldsymbol{\theta}, \mathbf{m})[\boldsymbol{\zeta}]/(\boldsymbol{\zeta})^{h+1}$ of p local parameters $\boldsymbol{\zeta}$, over the field generated by the weights $\boldsymbol{\theta}$ and the masses, gives every derivative of the moments through the order h, at most $\binom{p+h}{h}$ coefficients per entry.

(iii) A further polynomial observable, a mark total for instance, can be made a channel when every local update is separately affine in each independent cell vector, a constant channel adjoined.

(iv) Raising the masses by nonnegative integers $\boldsymbol{\gamma}$ is a polynomial insertion: for every observable g,

$$\mathbb{E}_{\alpha \sim \mathrm{Gamma}_{\mathbf{m}+\boldsymbol{\gamma}}}[g(\alpha)] = \mathbb{E}_{\alpha \sim \mathrm{Gamma}_{\mathbf{m}}}\left[\alpha^{\boldsymbol{\gamma}} g(\alpha)\right] \Big/ \prod_e (m_e)_{\gamma_e}.$$

The moments do not depend on the exponents asked for: changing the exponents reuses every moment, and changing a local weight or mass changes the local data and their ancestors in the assembly only, as Proposition 23.6 says of laws. Shared parameters may appear in many copies, whose draws stay independent.

26.5 Certified parameter banks

A family F on its balance wall, its letters K_j and Ψ_j written L_i, $i = 1, \ldots, \ell$, with the exponents w_i (the s_j and t_j), the degrees g_i and the counts $d_i(S) = \min\{\gamma_S : [u^\gamma] L_i > 0\}$, so that $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i L_i(u)^{-w_i} d\omega_{\mathcal{E}}(u)$, with $m_{\mathcal{E}} = \sum_i g_i w_i$ and the walls $\kappa_S = m_S - \sum_i w_i d_i(S) > 0$ at the starting data, is asked a neighbourhood of queries: p complex parameters $\boldsymbol{\zeta}$, and a bank of Taylor coefficients through the total order h, at most $\binom{h+p}{p}$ numbers. The order of a bank is the number of differentiations, not the degree of a polynomial moment of Section 26.1.

Theorem 26.9 (Taylor coefficients in the weights are periods with walls at least as high)**.** *Let the exponents w_i be real and change the coefficients of the letters, $L_i(\boldsymbol{\zeta}) = L_i + \sum_{j=1}^{p} \zeta_j D_{ij}$, each D_{ij} homogeneous of degree g_i with support in that of L_i and coefficients of either sign, and in this statement write $P(\boldsymbol{\zeta}) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i L_i(\boldsymbol{\zeta})(u)^{-w_i}\, d\omega_{\mathcal{E}}(u)$, so that $P(\mathbf{0}) = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. Then, with $l_i = \sum_j l_{ij}$,*

$$[\boldsymbol{\zeta}^{\mathbf{k}}]P = \sum_{\substack{l_{ij}\geq 0 \\ \sum_i l_{ij} = k_j}} \prod_i \frac{(-1)^{l_i}(w_i)_{l_i}}{\prod_j l_{ij}!} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i \Big(L_i^{-w_i - l_i} \prod_j D_{ij}^{l_{ij}} \Big)\, d\omega_{\mathcal{E}},$$

and after the numerators are expanded every term is a convergent period of the family at masses $\mathbf{m}+\boldsymbol{\gamma}$ and exponents $w_i + l_i$, balanced, with every proper wall $\kappa'_S = \kappa_S + \gamma_S - \sum_i l_i d_i(S) \geq \kappa_S$. Theorem 32.1(iii) is the case of order two, once every table j' is doubled into the two copies with the weights $\theta_{j',T} \pm \varepsilon v^{\mathsf{T}} \mathbf{b}_{j',T}$ of Theorem 32.1(i) and the equal letters are merged.

Theorem 26.10 (A uniform certificate for weights)**.** *In the setting of Theorem 26.9, with $P(\boldsymbol{\zeta})$ as there, write $L_i = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} u^{\boldsymbol{\gamma}}$ with $c_{i,\boldsymbol{\gamma}} > 0$ and $D_{ij} = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} v_{i,\boldsymbol{\gamma},j} u^{\boldsymbol{\gamma}}$, $V_{ij} = \max_{\boldsymbol{\gamma}} |v_{i,\boldsymbol{\gamma},j}|$, and choose radii $R_j > 0$ with $r_i = \sum_j R_j V_{ij} < 1$. On $|\zeta_j| \leq R_j$, $|P(\boldsymbol{\zeta})| \leq P(\mathbf{0}) \prod_i \max\big((1-r_i)^{-w_i}, (1+r_i)^{-w_i}\big) =: M$, made explicit by $P(\mathbf{0}) \leq \sum_\pi \prod_i \max\big(\underline{L}_{i,\pi}^{-w_i}, \overline{L}_{i,\pi}^{-w_i}\big) \prod_{k=2}^{n} \kappa_{U_{\pi,k}}^{-1}$, with the sector constants $\underline{L}_{i,\pi}$ and $\overline{L}_{i,\pi}$ of Theorem 26.11, or by the upper end of the enclosure of Proposition 15.12; P is holomorphic near the closed polydisc, $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \leq M\mathbf{R}^{-\mathbf{k}}$, and its Taylor polynomial $P^{[h]}$ through the order h has, on $|\zeta_j| \leq \vartheta R_j$ with $0 \leq \vartheta < 1$,*

$$|P(\boldsymbol{\zeta}) - P^{[h]}(\boldsymbol{\zeta})| \leq M \sum_{k>h} \binom{k+p-1}{p-1} \vartheta^k \leq M\, \frac{\binom{h+p}{p-1}\vartheta^{h+1}}{1 - \vartheta(h+p+1)/(h+2)},$$

the second bound when $\vartheta(h+p+1) < h+2$.

Theorem 26.11 (Wall-certified parameter banks)**.** *Keep the letters and change the masses and exponents affinely, $m_e(\boldsymbol{\zeta}) = m_e^0 + \sum_j a_{ej}\zeta_j$ and $w_i(\boldsymbol{\zeta}) = w_i^0 + \sum_j b_{ij}\zeta_j$ with real slopes and balance in every direction, so that $\kappa_S(\boldsymbol{\zeta}) = \kappa_S^0 + \sum_j c_{Sj}\zeta_j$ with $c_{Sj} = \sum_{e\in S} a_{ej} - \sum_i b_{ij} d_i(S)$, and in this statement write $P(\boldsymbol{\zeta}) = P_F(\mathbf{m}(\boldsymbol{\zeta});\mathbf{s}(\boldsymbol{\zeta}),\mathbf{t}(\boldsymbol{\zeta}))$, the exponents $w_i(\boldsymbol{\zeta})$ on the letters. Choose $R_j > 0$ with*

$$\underline{\kappa}_S := \kappa_S^0 - \sum_j |c_{Sj}| R_j > 0 \qquad (\varnothing \neq S \subsetneq \mathcal{E}).$$

On the sector of an order π every letter is its monomial times its sector letter $L_{i,\pi}$, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.7(ii), with nonnegative coefficients in the sector coordinates; put $\underline{L}_{i,\pi} = L_{i,\pi}(\mathbf{0})$, $\overline{L}_{i,\pi} = L_{i,\pi}(\mathbf{1})$, $\eta_{i,\pi} = \max(|\log \underline{L}_{i,\pi}|, |\log \overline{L}_{i,\pi}|)$, $\nu_i = \sum_j |b_{ij}| R_j$,

$$B_\pi = \prod_i \max\big(\underline{L}_{i,\pi}^{-w_i^0}, \overline{L}_{i,\pi}^{-w_i^0}\big) e^{\eta_{i,\pi}\nu_i}, \qquad M_{\mathbf{R}} = \sum_\pi B_\pi \prod_{k=2}^{n} \underline{\kappa}_{U_{\pi,k}}^{-1}.$$

Then P is holomorphic near $|\zeta_j| \leq R_j$ with $|P| \leq M_{\mathbf{R}}$, every $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \leq M_{\mathbf{R}}\mathbf{R}^{-\mathbf{k}}$, the bound of Theorem 26.10 holds with $M_{\mathbf{R}}$ for M, and

$$[\boldsymbol{\zeta}^{\mathbf{k}}]P = \frac{1}{\mathbf{k}!} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}^0-\mathbf{1}} \prod_i L_i^{-w_i^0} \prod_j \Big(\sum_e a_{ej} \log u_e - \sum_i b_{ij} \log L_i \Big)^{k_j} d\omega_{\mathcal{E}},$$

absolutely integrable and of degree zero. The starting exponents may be any reals at which the proper walls are positive.

Theorem 26.12 (A finite compiler with two errors)**.** *In the setting of Theorem 26.11, with $P(\boldsymbol{\zeta})$ as there, put $y_{i,\pi} = 1 - L_{i,\pi}/\overline{L}_{i,\pi} \in [0, \lambda_\pi]$, $\lambda_\pi = \max_i(1 - \underline{L}_{i,\pi}/\overline{L}_{i,\pi}) < 1$, $\beta_i = |w_i^0| + \nu_i$ and $A_\pi = \prod_i \overline{L}_{i,\pi}^{-w_i^0} e^{|\log \overline{L}_{i,\pi}|\nu_i} \prod_{k=2}^n \underline{\kappa}_{U_{\pi,k}}^{-1}$. Expand $\prod_i L_{i,\pi}^{-w_i(\boldsymbol{\zeta})} = \prod_i \overline{L}_{i,\pi}^{-w_i(\boldsymbol{\zeta})} \sum_{\mathbf{l}} \prod_i \frac{(w_i(\boldsymbol{\zeta}))_{l_i}}{l_i!} y_{i,\pi}^{l_i}$, truncate at $|\mathbf{l}| \le Q$, and integrate the finitely many terms exactly, each $\prod_i y_{i,\pi}^{l_i} = \sum_{\boldsymbol{\gamma}} c_{\boldsymbol{\gamma}} q^{\boldsymbol{\gamma}}$ in the sector coordinates q contributing*

$$\prod_i \overline{L}_{i,\pi}^{-w_i(\boldsymbol{\zeta})} \prod_i \frac{(w_i(\boldsymbol{\zeta}))_{l_i}}{l_i!} \sum_{\boldsymbol{\gamma}} c_{\boldsymbol{\gamma}} \prod_{k=2}^n \frac{1}{\kappa_{U_{\pi,k}}(\boldsymbol{\zeta}) + \gamma_k},$$

to obtain $\widetilde{P}_Q(\boldsymbol{\zeta})$. Then on the whole polydisc

$$|P(\boldsymbol{\zeta}) - \widetilde{P}_Q(\boldsymbol{\zeta})| \le \mathrm{err}_Q := \sum_\pi A_\pi \sum_{k>Q} \frac{(|\boldsymbol{\beta}|)_k}{k!} \lambda_\pi^k \le \sum_\pi A_\pi \frac{(|\boldsymbol{\beta}|)_{Q+1} \lambda_\pi^{Q+1}}{(Q+1)!\,(1 - r_{\pi,Q})},$$

the second bound when $r_{\pi,Q} = \lambda_\pi \max\left(1, \frac{|\boldsymbol{\beta}|+Q+1}{Q+2}\right) < 1$ for every π; and the Taylor polynomial $\widetilde{P}_Q^{[h]}$ of $\widetilde{P}_Q$ through the order h, computed by finite arithmetic in a truncated ring, has on $|\zeta_j| \le \vartheta R_j$

$$|P(\boldsymbol{\zeta}) - \widetilde{P}_Q^{[h]}(\boldsymbol{\zeta})| \le \frac{\mathrm{err}_Q}{(1-\vartheta)^p} + M_{\mathbf{R}} \sum_{k>h} \binom{k+p-1}{p-1} \vartheta^k.$$

With no parameters, $p = 0$, so that $\nu_i = 0$ and $\underline{\kappa}_S = \kappa_S$, and with every $w_i^0 \ge 0$, the first bound is that of Proposition 15.12(ii): there $\overline{L}_{i,\pi} = L_i(\mathbf{1})$ for every π, $\widetilde{P}_Q = \prod_i L_i(\mathbf{1})^{-w_i^0} \sum_\pi \sum_{M \le Q} I_{\pi,M}$, and the explicit tail is $\prod_i L_i(\mathbf{1})^{-w_i^0} \sum_\pi \mathrm{err}_{\pi,Q}$, the letters being the K_j and Ψ_j with the exponents s_j and t_j.

For fixed data and $\vartheta < 1$ the orders Q and h grow like the logarithm of the inverse error, up to the polynomial factors of the scalar tails; expanding the sector polynomials still costs work that depends on the dimension and the supports. The walls also cover paths: along a segment on which every proper wall is positive, with κ_- the least proper wall at the two ends and a the largest change of a proper wall along it, every point of the segment, parametrized by $\lambda \in [0,1]$, has the disc of radius $\kappa_-/(2a)$ in the complex parameter on which every proper wall keeps a real part at least $\kappa_-/2$, the walls being affine, and a compact path has a finite cover by such discs. A bank at one point does not give the coefficients at another; what is reused everywhere is the compiled sector data, the coefficient procedure and the wall test.

26.6 Procedures

Definition 26.13 (The interface procedure)**.** *Given*: cells with their weights and masses; the pairs of hinges to splice and an assembly order with its identifications of repeated parts; a polynomial moment, a peak truncation or an accuracy, with, for an accuracy, references and a wall certificate. *Steps*: (1) read the channels of the leaves by coefficient extraction, with their degrees, the sizes, the total masses and the products $\prod_e \Gamma(m_e)$; (2) choose D, the degree of the moment or $\sum_i Q_i$ from Theorem 26.7; (3) compute the leaf moments by Theorem 26.3(i), only at the multi-indices the moment pulls back to; (4) compile every distinct splicing by Theorem 26.3(ii), and every repeated chain cell once into $\mathbf{T}_D$ and its power (Theorem 26.5), reusing a vector only when its tags and data match; (5) splice the pairs at the root, return to the shape by Theorem 26.3(iii),

and evaluate $P^{\leq \mathbf{Q}}$ with its certificate. *Output*: an exact polynomial moment, a recurrence or rational generating series for a repetition, or a certified enclosure of the period, with the vectors, the splicing coefficients, the parameter expressions and the certificate kept for the next query.

Definition 26.14 (The bank procedure)**.** *Given*: a family on its balance wall with positive walls, and affine directions of change of its masses and exponents, or of the weights of its letters. *Steps* along masses and exponents: compute the walls and a box of radii satisfying the strict inequalities of Theorem 26.11; form the sector constants and $M_{\mathbf{R}}$; choose Q and h until the bound of Theorem 26.12 is below the tolerance; compile the retained sector terms in the truncated ring. Along weights, where the sector letters move with $\boldsymbol{\zeta}$ and Theorem 26.12 does not apply: choose radii by Theorem 26.10, with its bound made explicit there; each coefficient through the order h is the finite signed sum of periods with walls at least as high of Theorem 26.9, each enclosed by Proposition 15.12, and the tail beyond h is bounded by Theorem 26.10. *Output*: the parameter bank, its domain and its certified error; every query inside the smaller box is a polynomial evaluation, and the exact coefficients remain available as the periods of Theorem 26.9 or the integrals of Theorem 26.11.

Chapter 27

Bunches of any width

A bunch of width one is one mark to its room (Theorem 18.6): its patterns fill two levels, each level is one child of the bunch against one table of the room, and what the room sees of the bunch is its wall and the law of its ratio vector, one ratio for one table (Lemma 22.2, Theorem 22.7). At every width a room sees its bunch through the wall, the widths and the joint law of the bunch letters of a level factorization (Theorem 20.13); that statement has a canonical form, and the calculus of width one carries to every width. At the width c the patterns fill $c + 1$ levels, and a level between the extremes pairs several stations of patterns with several stations of room sides: which pairs meet, and with which constants, is the seam of the bunch (Definition 27.1), graded by the excess, which no description by marks or by level totals contains. The room sees the bunch through the letters of the tables it presents to it, one for each station of room sides, and the law of these letters is what the bunch retains: it suffices against every partner, and the tinkers of one partner already separate any two laws (Theorem 27.18). A bunch of width c is absorbed into its partner as one stand-in mark of degree c (Lemma 27.8), a mark of a table exactly at width one (Theorem 27.22), and a nonempty family of sets of one size is the family of bases of a table exactly when, at one bunch, its children and room children are tables, adjacent ones are quotients, the extreme ones are one, and its levels are lifts (Theorem 27.26). The circuits of the seam are the directions of the shares of the compatible pairs that neither side sees, and the period integrates over them through the Gramma of the compatible pairs (Proposition 27.13). Width one is the case of the hinge throughout: the seam of a bunch of width one is its hinge (Proposition 27.3(ii)), and its law is the law of its ratio vector read through one fixed change of scale (Proposition 27.10). Bunches compose along chains and trees (Chapter 28), the glue closes along the directions of its shares (Chapter 29), and a law has regions of equality and a reach at its ends (Chapter 30).

27.1 The seam of a bunch

Definition 27.1 (Seam)**.** Let a table $(\mathcal{B}, \theta)$ have the bunch B and the room R. A level k of B is named by its excess $i = k - d(B)$ (Definition 12.15), $0 \le i \le c(B)$: its patterns have $d(B) + i$ marks and its room sides $d(R) + c(B) - i$, the room having the width of the bunch (Definition 18.3). Fix an excess i.

(i) Two patterns of excess i lie in one station when their children (Definition 12.1) are positive multiples of one another, weights included, as at the bunches S_l of Theorem 21.11. The *room child* at a room side X is the child of the bunch R at its pattern X: the table on B with the bases $\mathcal{B}_{(X)} = \{p : \ p \sqcup X \in \mathcal{B}\}$ and the weights $\theta_{(X),p} = \theta_{p \sqcup X}$ (Lemma 12.2). Two room sides of excess i lie in one station when their room children are positive multiples of one another.

The stations of patterns of excess i form the set $\mathcal{K}_i$ and those of room sides the set $\mathcal{D}_i$: the stations of the bunch and of the room are $\mathcal{K} = \bigsqcup_i \mathcal{K}_i$ and $\mathcal{D} = \bigsqcup_i \mathcal{D}_i$, each station carrying its excess $i(C)$ or $i(D)$.

(ii) Within a station C the child at $p \in C$ is w_p times one weighted table on R, the *child of the station* C, with the bases $\mathcal{B}_{(C)}$ and the weights $\theta_{(C)}$, the *multipliers* $w_p > 0$ being normalized by $\max_{p\in C} w_p = 1$; likewise the room child at $X \in D$ is w'_X times the *room child of the station* D, a table on B with the bases $\mathcal{B}_{(D)}$ and the weights $\theta_{(D)}$, with $\max_{X\in D} w'_X = 1$. The *pair constant* of $C \in \mathcal{K}_i$ and $D \in \mathcal{D}_i$ is $h_{CD} = \theta_{p\sqcup X}/(w_p w'_X)$ for any $p \in C$ and $X \in D$ with $p \sqcup X \in \mathcal{B}$, which does not depend on the choice (Proposition 27.2(ii)), and $h_{CD} = 0$ when there is none; (C, D) is *compatible* when $h_{CD} > 0$. The *seam matrix* of the excess i is $H_i = (h_{CD})_{C\in\mathcal{K}_i,\, D\in\mathcal{D}_i}$, and the *seam* of B is $(H_0, \dots, H_{c(B)})$ with its stations; a family has one seam in each table.

(iii) The *station letters* are $\mathsf{x}_C = \sum_{p\in C} w_p\, \alpha_B^p$ and $\mathsf{x}^*_C = \sum_{p\in C} w_p\, \alpha_B^{B\setminus p}$ on B, and $\mathsf{y}_D = \sum_{X\in D} w'_X\, \alpha_R^X$ and $\mathsf{y}^*_D = \sum_{X\in D} w'_X\, \alpha_R^{R\setminus X}$ on R. The *seam letters* of B are the letters $K_{(D)}$ and $\Psi_{(D)}$ of the room children of the stations $D \in \mathcal{D}$, polynomials on B; those of R are the letters $K_{(C)}$ and $\Psi_{(C)}$ of the children of the stations $C \in \mathcal{K}$, polynomials on R.

For a list every multiplier and every pair constant is 1. The normalization by the largest multiplier chooses nothing and commutes with every bijection of the marks; any other normalization multiplies x_C and y_D by positive constants and h_{CD} by their inverses, and leaves every product $h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ unchanged. In table j of a family the stations, the letters and the constants carry the index j: $\mathcal{K}_j$, $\mathcal{D}_j = \bigsqcup_i \mathcal{D}_{j,i}$, x_{jC}, y_{jD}, $h_{j,CD}$, $K_{j,(D)}$ and so on.

Proposition 27.2 (The seam identity)**.** *In Definition 27.1:*

(i) The seam letters are the station letters through the seam:

$$K_{(D)} = \sum_C h_{CD}\,\mathsf{x}_C, \qquad \Psi_{(D)} = \sum_C h_{CD}\,\mathsf{x}^*_C,$$
$$K_{(C)} = \sum_D h_{CD}\,\mathsf{y}_D, \qquad \Psi_{(C)} = \sum_D h_{CD}\,\mathsf{y}^*_D :$$

the seam letters of a side are the letters of the tables that the other side presents to it.

(ii) If (C, D) is compatible, every $p \in C$ and $X \in D$ make a basis, with $\theta_{p\sqcup X} = h_{CD}\, w_p\, w'_X$, and the bases of excess i are the disjoint union of these rectangles *$C \times D$ over the compatible pairs.*

(iii) With x_i, x^*_i, y_i, y^*_i the vectors of the station letters of excess i,

$$K = \sum_{i=0}^{c(B)} \mathsf{x}_i^{\mathsf{T}} H_i\, \mathsf{y}_i = \sum_{D\in\mathcal{D}} K_{(D)}\, \mathsf{y}_D = \sum_{C\in\mathcal{K}} \mathsf{x}_C\, K_{(C)},$$
$$\Psi = \sum_{i=0}^{c(B)} \mathsf{x}_i^{*\mathsf{T}} H_i\, \mathsf{y}^*_i = \sum_{D\in\mathcal{D}} \Psi_{(D)}\, \mathsf{y}^*_D = \sum_{C\in\mathcal{K}} \mathsf{x}^*_C\, \Psi_{(C)},$$

and along the scale of B, $\alpha_B = \rho u$ with $u \in \Delta_B$ (Lemma 2.7),

$$K(\rho u, \alpha_R) = \rho^{d(B)} \sum_{i=0}^{c(B)} \rho^i \sum_{D\in\mathcal{D}_i} K_{(D)}(u)\, \mathsf{y}_D(\alpha_R),$$

$$\Psi(\rho u, \alpha_R) = \rho^{d^*(B)} \sum_{i=0}^{c(B)} \rho^{c(B)-i} \sum_{D \in \mathcal{D}_i} \Psi_{(D)}(u)\, \mathsf{y}^*_D(\alpha_R).$$

The writing by the stations of the room, with the bunch letters $K_{(D)}$ and the room letters y_D, is a level factorization with nonnegative terms (Definition 20.12), the room-station factorization*; the writing by the stations of the bunch is the children factorization.*

(iv) Within one excess no two stations of $\mathcal{K}_i$ have proportional rows of H_i, and no two stations of $\mathcal{D}_i$ proportional columns.

(v) At the excesses 0 and $c(B)$ every pattern meets every room side in a basis (Theorem 12.17). H_0, respectively $H_{c(B)}$, is a single positive entry exactly when the weights of that level factor along these products, as for unit weights, for weights factoring across B and R and for tables determinantal over $\mathbb{R}$ or $\mathbb{C}$; when both do, the room child of the one station of excess 0 is $\mathcal{B}/R$ and that of excess $c(B)$ is $\mathcal{B}|B$ (Proposition 12.16), these tables carrying the weights of the room children of the two stations, and the seam letters at the two extremes are $K_{\mathcal{B}/R}$, $\Psi_{\mathcal{B}/R}$ and $K_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}|B}$.

(vi) Swapping B and R transposes every H_i and reverses the excesses, $i \mapsto c(B) - i$; the seam letters of R are then those of the bunch R.

A compatible pair fills its rectangle: inside it every tuple of the bunch meets every tuple of the room, with weights of rank one. What the marks do not show is the pattern of the rectangles, which stations of the bunch meet which stations of the room. At an extreme excess every station of the bunch meets every station of the room, in one rectangle when the weights of that level factor; between the extremes a station need not meet every station of the other side, and the seam matrix H_i, not the marks, says how they meet.

Proposition 27.3 (The width ladder)**.** *In Definition 27.1:*

(i) $c(B) = 0$ exactly when B is a union of molecules of the table; there is one excess, every pattern meets every room side, and $K = \mathsf{x}_0^{\mathsf{T}} H_0\, \mathsf{y}_0$, $\Psi = \mathsf{x}_0^{*\mathsf{T}} H_0\, \mathsf{y}_0^*$. The seam is one rectangle $C \times D$ exactly when the weights factor across B and R, as for lists; then $K = h_{CD}\, \mathsf{x}_C\, \mathsf{y}_D$ and $\Psi = h_{CD}\, \mathsf{x}_C^*\, \mathsf{y}_D^*$: the letters factor over the bunch and the room.

(ii) $c(B) = 1$ exactly when there are two excesses, both extreme. Each is one rectangle exactly when the weights factor across B and R; then for the stations D_0 and D_1 of the excesses 0 and 1 the seam letters are $K_{(D_0)} = K_{\mathcal{B}/R}$, $K_{(D_1)} = K_{\mathcal{B}|B}$, $\Psi_{(D_0)} = \Psi_{\mathcal{B}/R}$ and $\Psi_{(D_1)} = \Psi_{\mathcal{B}|B}$, the room station letters are $\mathsf{y}_{D_0} = K_{\mathcal{B}|R}$ and $\mathsf{y}_{D_1} = K_{\mathcal{B}/B}$, these two tables carrying the multipliers, the largest 1, and the room-station factorization is the writing

$$K = K_{\mathcal{B}/R}\, K_{\mathcal{B}|R} + K_{\mathcal{B}|B}\, K_{\mathcal{B}/B}$$

of Theorem 18.6: the letter adds over two levels, and the seam is the hinge.

(iii) For $c(B) \geq 2$ the excesses $0 < i < c(B)$ can carry several rectangles.

At every width an excess with one station on one side has one station on the other, and the children at the patterns of each excess are equal, B being in level form, exactly when every transposition of two marks of B, fixing every other mark, carries the table (Proposition 14.12); then every excess is one rectangle.

27.2 Pieces, their laws, and the glue

The seam identity writes a family as a pairing of what the bunch presents to the room and what the room presents to the bunch, station by station. The pairing uses nothing of the exchange property and holds for pieces of any kind.

Definition 27.4 (Pieces with stations; their glue)**.** Let $N \geq 1$ and $c_1, \dots, c_N \geq 0$ be integers. A set of *stations* for N tables of the widths c_j is a family $\mathcal{D} = (\mathcal{D}_j)_{j=1}^{N}$ of finite sets graded by excesses, $\mathcal{D}_j = \bigsqcup_{i=0}^{c_j} \mathcal{D}_{j,i}$, with $\mathcal{D}_{j,0}$ and $\mathcal{D}_{j,c_j}$ nonempty; a station $D \in \mathcal{D}_{j,i}$ has the excess $i(D) = i$. A *piece* with the stations $\mathcal{D}$ on a finite set B, at exponents $(s_j, t_j)_j$ with $s_j, t_j \geq 0$, carries for every j and $D \in \mathcal{D}_j$ two nonzero polynomials V_{jD} and V^*_{jD} on B with nonnegative coefficients, its *seam letters* (positive letters, Definition 24.11, where their degree is positive), homogeneous of the degrees $d_j + i(D)$ and $d^*_j + c_j - i(D)$ for integers $d_j, d^*_j \geq 0$, and masses $\mathbf{m}_B$; its *wall* and its *top wall* are

$$\kappa_B = m_B - \sum_j \big(s_j d_j + t_j d^*_j\big), \qquad \kappa_B^{\mathrm{top}} = \sum_j (s_j + t_j) c_j - \kappa_B.$$

A *partner* is a piece with the same stations on a set R disjoint from B, read with the reversed excesses $c_j - i(D)$: its letters, the *room letters* L_{jD} and L^*_{jD} (as in Definition 20.12), have the degrees $e_j + c_j - i(D)$ and $e^*_j + i(D)$, and its wall is $\kappa_R = m_R - \sum_j (s_j e_j + t_j e^*_j)$. A *family of Newton words* is a sequence $(K_j, \Psi_j; s_j, t_j)_{j=1}^{N}$ of pairs of positive letters on one set with exponents $s_j, t_j \geq 0$, carrying masses; its word is $\prod_j K_j^{-s_j} \Psi_j^{-t_j}$ and its period is the period of that word (Definition 24.11). The *glue* of the piece and the partner is the family of Newton words on $B \sqcup R$ with the letters

$$K_j = \sum_{D \in \mathcal{D}_j} V_{jD}\, L_{jD}, \qquad \Psi_j = \sum_{D \in \mathcal{D}_j} V^*_{jD}\, L^*_{jD},$$

the exponents (s_j, t_j) and the masses $(\mathbf{m}_B, \mathbf{m}_R)$; its period is written $P_{B \sqcup R}(\mathbf{m})$, and its balance wall is $\kappa_R = \kappa_B^{\mathrm{top}}$. A piece is named by its set, or by a calligraphic letter $\mathcal{X}$ when several pieces are compared, and its objects carry that name.

By Proposition 27.2(iii) a bunch of a family of tables, with its seam letters $V_{jD} = K_{j,(D)}$ and $V^*_{jD} = \Psi_{j,(D)}$, is a piece with the stations of its room, $d_j = d_j(B)$ and $d^*_j = d^*_j(B)$; its room, with the room station letters $L_{jD} = \mathsf{y}_{jD}$ and $L^*_{jD} = \mathsf{y}^*_{jD}$, is a partner, $e_j = d_j(R)$ and $e^*_j = d^*_j(R)$; and their glue is the family. By (vi) the same holds with B and R swapped, on the stations $\mathcal{K}$. The walls are the walls κ_B and κ_R of the bunch and of the room, and the top wall is the top wall κ_B^{top} of the bunch (Definition 8.1). A glue need not be a family of tables; its period converges exactly when its Newton sum has the dimension $|B \sqcup R| - 1$ and the masses lie in its relative interior (Theorem 24.12), and every statement about pieces is a statement about the period.

Definition 27.5 (The law of a piece)**.** Let a piece on B with the stations $\mathcal{D}$ have exponents with $\sum_j (s_j + t_j) c_j > 0$. Its active letters (Definition 8.1) are K_j when $s_j > 0$ and Ψ_j when $t_j > 0$; every coordinate, share, pin, product and count indexed by a letter runs over the active letters only, an inactive letter entering no period. The *excess sums* are $V_j^{(i)} = \sum_{D \in \mathcal{D}_{j,i}} V_{jD}$ and $V_j^{*(i)} = \sum_{D \in \mathcal{D}_{j,i}} V^*_{jD}$, the superscript counting excesses. On Δ_B put

$$\gamma_B^{\rho} = \prod_j \Big(\frac{V_j^{(c_j)}}{V_j^{(0)}}\Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big(\frac{V_j^{*(0)}}{V_j^{*(c_j)}}\Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

$$\phi_B^\rho(du) = u^{\mathbf{m}_B-\mathbf{1}} \prod_j \big(V_j^{(0)}\big)^{-s_j} \big(V_j^{*(c_j)}\big)^{-t_j} \, (\gamma_B^\rho)^{-\kappa_B} \, d\omega_B(u),$$

$$\varrho_{jD}^\rho = \frac{V_{jD}}{V_j^{(0)} \, (\gamma_B^\rho)^{i(D)}}, \qquad \varrho_{jD}^{\rho *} = \frac{V_{jD}^*}{V_j^{*(c_j)} \, (\gamma_B^\rho)^{c_j - i(D)}},$$

the *scale of the piece*, homogeneous of degree one, the *scale form*, and the coordinates of the *seam vector* $\boldsymbol{\varrho}_B^\rho = (\varrho_{jD}^\rho, \varrho_{jD}^{\rho *})_{j,D}$, homogeneous of degree zero, with a coordinate ϱ_{jD}^ρ for each active K_j and $D \in \mathcal{D}_j$ and a coordinate $\varrho_{jD}^{\rho *}$ for each active Ψ_j and $D \in \mathcal{D}_j$. Its values, the *seam points*, lie on the *seam torus*: the set of the points $\mathbf{x} = (x_{jD}, x_{jD}^*)_{j,D}$ with positive coordinates on which $\sum_{D \in \mathcal{D}_{j,0}} x_{jD} = 1$ for every active K_j, $\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD}^* = 1$ for every active Ψ_j, and

$$\prod_j \Big(\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} \Big)^{s_j} \Big(\sum_{D \in \mathcal{D}_{j,0}} x_{jD}^* \Big)^{t_j} = 1,$$

of dimension $n_{\mathcal{D}}$, the number of the coordinates less the number of the active letters less one, which is $2|\mathcal{D}| - 2N - 1$, $|\mathcal{D}| = \sum_j |\mathcal{D}_j|$, when every letter is active. The *law* of the piece is the image $(\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$, a positive measure carried by the seam torus. Its Mellin transform, at shares $(\mathbf{a}, \mathbf{b}) = (a_{jD}, b_{jD})_{j,D}$ of the active letters,

$$\int_{\Delta_B} (\boldsymbol{\varrho}_B^\rho)^{-(\mathbf{a},\mathbf{b})} \, d\phi_B^\rho = \int \mathbf{x}^{-(\mathbf{a},\mathbf{b})} \, d\big((\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho\big)(\mathbf{x}), \qquad \mathbf{x}^{-(\mathbf{a},\mathbf{b})} = \prod_{j,D} x_{jD}^{-a_{jD}} \, {x_{jD}^*}^{-b_{jD}},$$

is the *seam function*, in $[0, \infty]$ at real shares and defined at complex shares wherever it converges absolutely. A partner has its scale, scale form, seam vector, law and seam function by the same definition, with its room letters and the reversed excesses.

The scale form is projective: its orthant form has the degree $m_B - \sum_j (s_j d_j + t_j d_j^*) - \kappa_B = 0$, so it is read on any section (Lemma 16.2). The seam points lie on the seam torus: the sums at the excesses 0 and c_j are 1 by definition, and the product is $(\gamma_B^\rho)^{\kappa_B + \kappa_B^{\mathrm{top}}} (\gamma_B^\rho)^{-\sum_j (s_j + t_j) c_j} = 1$. The scale of the piece is the geometric mean of the oriented extreme ratios of the active letters, each taken to the root of its width: K_j measures the scale by $(V_j^{(c_j)} / V_j^{(0)})^{1/c_j}$ and Ψ_j by $(V_j^{*(0)} / V_j^{*(c_j)})^{1/c_j}$, in the proportions $s_j c_j$ and $t_j c_j$; a letter of width zero does not see the scale. Any other positive scale $q \, \gamma_B^\rho$ of degree one, q a positive function of the seam vector, multiplies ϕ_B^ρ by $q^{-\kappa_B}$ and the coordinates of the seam vector by powers of q: a fixed bijection of the tori with a density, so the law is determined up to it. The scale named is canonical: it uses the seam letters of the active letters alone, treats the two letters alike, and is kept at the centre by the bijections exchanging the family (Proposition 27.6(iii)). With weights factoring at the extreme levels, $V_j^{(0)} = K_{\mathcal{B}_j/R}$, $V_j^{(c_j)} = K_{\mathcal{B}_j|B}$, $V_j^{*(0)} = \Psi_{\mathcal{B}_j/R}$ and $V_j^{*(c_j)} = \Psi_{\mathcal{B}_j|B}$ (Proposition 27.2(v)); so

$$\gamma_B^\rho = \prod_j \Big(\frac{K_{\mathcal{B}_j|B}}{K_{\mathcal{B}_j/R}} \Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big(\frac{\Psi_{\mathcal{B}_j/R}}{\Psi_{\mathcal{B}_j|B}} \Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

$$\phi_B^\rho(du) = u^{\mathbf{m}_B-\mathbf{1}} \prod_j K_{\mathcal{B}_j/R}^{-s_j} \Psi_{\mathcal{B}_j|B}^{-t_j} \, (\gamma_B^\rho)^{-\kappa_B} \, d\omega_B(u).$$

The law is read off the measure of Theorem 20.13. For the room-station factorization the letter vector is $\mathbf{V}(u) = \big(K_{j,(D)}(u), \Psi_{j,(D)}(u)\big)_{j,D}$; the density of ϕ_B^ρ against $u^{\mathbf{m}_B-\mathbf{1}} d\omega_B$ and the seam vector are functions of $\mathbf{V}(u)$, so the law is an image of the measure $\mathbf{V}_*\big(u^{\mathbf{m}_B-\mathbf{1}} d\omega_B(u)\big)$ with

a density. The seam vector forgets the seam letters of the inactive letters and, of the others, one constant for each active letter and the scale, $V_{jD} \mapsto \lambda^{i(D)} V_{jD}$ and $V_{jD}^* \mapsto \lambda^{c_j - i(D)} V_{jD}^*$: the directions that the pins of Lemma 27.7 fix, on which the scale form changes by constant factors.

Proposition 27.6 (Symmetries of a seam)**.** *Let a bijection g of the marks $\mathcal{E}$ of a family map B onto B' and R onto R'.*

(i) If g carries table j to table j' (Definition 11.14), it maps the patterns and the room sides of B of each excess in table j onto those of B' of the same excess in table j', stations onto stations and compatible pairs onto compatible pairs, and the station letters and seam letters of the image stations are those of the stations read at $g^*\alpha$, the normalization of the multipliers commuting with g.

(ii) If g exchanges table j with table j', the windows of B and B' correspond as in Lemma 14.10, and $p \mapsto B' \setminus g(p)$, $X \mapsto R' \setminus g(X)$ map the stations of B of excess i in table j onto those of B' of excess $c(B) - i$ in table j' and compatible pairs onto compatible pairs, the station letter x_C and the seam letter $K_{(D)}$ going to the station letter $\mathsf{x}^*_{C'}$ and the seam letter $\Psi_{(D')}$ of the image stations C', D', and back.

(iii) So the bijections that keep B and carry or exchange the family permute the stations of its seams, with their station letters and seam letters, up to the bijection. Let such a bijection also keep the masses. A carrying one maps the seam torus of B onto itself, permuting the coordinates, and keeps the law $(\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$. An exchanging one, sending table j to table j', maps the seam torus of B at the lean $\boldsymbol{\delta}$ onto the seam torus of B at the lean $-\boldsymbol{\delta}$, the coordinates ϱ_{jD}^ρ of table j going to the coordinates $\varrho_{j'D'}^{\rho *}$ of table j' and back, and maps the law of B at $\boldsymbol{\delta}$ onto the law of B at $-\boldsymbol{\delta}$; at the centre it keeps the law. At the centre the potential is kept by the carrying and the exchanging maps alike, at a lean by the carrying maps (Proposition 11.15): the centre does not tell apart stations joined by a bijection carrying or exchanging the family, nor a lean stations joined by a carrying one.

At the centre two stations can lie in one orbit through an exchange and in different orbits at every lean: the centre sees them as one, a lean separates them. These are the confusable stations of the seam, the analogue for stations of the halves C_K, C_Ψ of a class of marks (Theorem 11.18); the seam carries them down the tree with the classes of marks (Theorem 14.20).

Lemma 27.7 (The seam function is the period of the seam word)**.** *On the* pins of the seam*, the letter totals $\sum_D a_{jD} = s_j$ for every active K_j and $\sum_D b_{jD} = t_j$ for every active Ψ_j and the* scale pin

$$\sum_{j,D} \Big(i(D)\, a_{jD} + \big(c_j - i(D)\big)\, b_{jD} \Big) = \kappa_B,$$

the seam function is the period at the masses $\mathbf{m}_B$ of the seam word $W_B(\mathbf{a}, \mathbf{b}) = \prod_{j,D} V_{jD}^{-a_{jD}} {V_{jD}^*}^{-b_{jD}}$ *(Definition 24.11):*

$$\int_{\Delta_B} (\boldsymbol{\varrho}_B^\rho)^{-(\mathbf{a},\mathbf{b})}\, d\phi_B^\rho = P_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B) = \int_{\Delta_B} u^{\mathbf{m}_B - \mathbf{1}} \prod_{j,D} V_{jD}(u)^{-a_{jD}} V_{jD}^*(u)^{-b_{jD}}\, d\omega_B(u).$$

For the piece of a bunch of a family the seam word is the bunch word of the room-station factorization (Theorem 21.13), and its period on the pins is the Mellin transform of one law.

Lemma 27.8 (Absorbing a piece of any width)**.** *Let a piece on B with the stations $\mathcal{D}$ carry masses $\mathbf{m}_B$ and exponents $(s_j, t_j)_j$ with $\sum_j (s_j + t_j) c_j > 0$, its wall κ_B and its seam vector $\boldsymbol{\varrho}_B^\rho$, whose*

values are written $\mathbf{x} = (x_{jD}, x^*_{jD})_{j,D}$. *Let* g *be a mark outside* B, *the* stand-in *of the piece,* Z *a finite set disjoint from* $B \sqcup \{g\}$, $\mathbf{m}_Z$ *masses on* Z, *and* $f_1, \dots, f_p$ *positive continuous functions, each assigned to one active letter of the piece,* f_l *defined on* $(0,\infty)^Z \times (0,\infty)^{\mathcal{D}_j}$ *when its letter is* K_j *or* Ψ_j *and homogeneous of degree one in its second argument,* $f_l(\alpha_Z, \lambda\mathbf{y}) = \lambda\, f_l(\alpha_Z, \mathbf{y})$ *for* $\lambda > 0$, *with real exponents* a_l *such that the exponents assigned to each letter add to its exponent. In this statement write* $\mathbf{V}_l$ *for the vector of the seam letters of the letter of* f_l, $(V_{jD})_{D\in\mathcal{D}_j}$ *for* K_j *and* $(V^*_{jD})_{D\in\mathcal{D}_j}$ *for* Ψ_j; V_l *for its extreme excess sum,* $V^{(0)}_j$ *for* K_j *and* $V^{*(c_j)}_j$ *for* Ψ_j; *and* $\mathbf{x}_l(\alpha_g)$ *for the vector* $(x_{jD}\,\alpha_g^{i(D)})_D$ *for* K_j *and* $(x^*_{jD}\,\alpha_g^{c_j - i(D)})_D$ *for* Ψ_j. *Let* $H \geq 0$ *be a Borel function on* $(0,\infty)^p$. *If the form*

$$\omega = \alpha_B^{\mathbf{m}_B - \mathbf{1}} \alpha_Z^{\mathbf{m}_Z - \mathbf{1}}\, H\Big(\Big(\frac{f_l\big(\alpha_Z, \mathbf{V}_l(\alpha_B)\big)}{V_l(\alpha_B)}\Big)_l\Big) \prod_{l=1}^{p} f_l\big(\alpha_Z, \mathbf{V}_l(\alpha_B)\big)^{-a_l}\, d\alpha_B\, d\alpha_Z$$

is homogeneous of degree zero on $(0,\infty)^{B\sqcup Z}$, *then, in* $[0,\infty]$,

$$\int_{\{h=1\}} \omega = \int \Big(\int_{\{h'=1\}} \omega_{\mathbf{x}} \Big)\, d\big((\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B\big)(\mathbf{x}),$$
$$\omega_{\mathbf{x}} = \alpha_Z^{\mathbf{m}_Z - \mathbf{1}} \alpha_g^{\kappa_B - 1}\, H\Big(\big(f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))\big)_l\Big) \prod_l f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big)^{-a_l}\, d\alpha_Z\, d\alpha_g,$$

the integrals taken over any sections of positive functions h, h' *of degree one on the two orthants.*

The piece collapses to its stand-in g, which carries the mass κ_B; a letter of table j that sees the piece sees the stand-in with the degree c_j, at its stations twisted by the seam point. At width one this is Lemma 22.2 read through Proposition 27.10.

Theorem 27.9 (The glue across a seam)**.** *Let a piece on* B *and a partner on* R *with the stations* $\mathcal{D}$ *be glued, with masses on the balance wall of the glue and* $\sum_j (s_j + t_j) c_j > 0$.

(i) The piece absorbed. *In* $[0,\infty]$,

$$P_{B\sqcup R}(\mathbf{m}) = \int P_{R^{(\mathbf{x})}}(\mathbf{m}_R, \kappa_B)\, d\big((\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B\big)(\mathbf{x}),$$

with $R^{(\mathbf{x})}$ *the* partner with its stand-in at $\mathbf{x}$*: the family of Newton words on* $R \sqcup \{g\}$ *with the letters* $\sum_D x_{jD}\, \alpha_g^{i(D)} L_{jD}$ *and* $\sum_D x^*_{jD}\, \alpha_g^{c_j - i(D)} L^*_{jD}$, *the exponents* (s_j, t_j) *and the mass* κ_B *on* g.

(ii) Both absorbed. *On the scales* $\{g\}$ *and one scale for each active letter let* $t_{\mathcal{D}}$ *be the configuration whose column at the coordinate* (j, D) *of* K_j *is* $i(D)\,\mathbf{e}_g + \mathbf{e}_{K_j}$ *and at that of* Ψ_j *is* $(c_j - i(D))\,\mathbf{e}_g + \mathbf{e}_{\Psi_j}$. *In* $[0,\infty]$,

$$P_{B\sqcup R}(\mathbf{m}) = \iint \Gamma(\kappa_B)\, \mathsf{Z}_{(\kappa_B, \mathbf{s}, \mathbf{t}); t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{x}')\, d\big((\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B\big)(\mathbf{x})\, d\big((\boldsymbol{\varrho}^\rho_R)_* \phi^\rho_R\big)(\mathbf{x}'),$$

$\mathbf{x} \circ \mathbf{x}'$ *the coordinatewise product and the index* $(\kappa_B, \mathbf{s}, \mathbf{t})$ *read on the active letters (Definition 2.12). The kernel is the period of the* stand-in pair*, the marks* g *and* g' *with the masses* κ_B *and* κ_R *and the letters* $\sum_D x_{jD} x'_{jD}\, \alpha_g^{i(D)} \alpha_{g'}^{c_j - i(D)}$ *and* $\sum_D x^*_{jD} x'^*_{jD}\, \alpha_g^{c_j - i(D)} \alpha_{g'}^{i(D)}$, *read on one scale:*

$$\Gamma(\kappa)\, \mathsf{Z}_{(\kappa, \mathbf{s}, \mathbf{t}); t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{x}') = \int_0^\infty v^{\kappa - 1} \prod_j \Big(\sum_{D\in\mathcal{D}_j} x_{jD} x'_{jD}\, v^{i(D)} \Big)^{-s_j} \Big(\sum_{D\in\mathcal{D}_j} x^*_{jD} x'^*_{jD}\, v^{c_j - i(D)} \Big)^{-t_j} dv.$$

(iii) One share per station. *On every contour of shares a_{jD}, b_{jD} of the active letters with positive real parts, on the pins of the seam (Lemma 27.7), which on the balance wall are the pins of the partner as well, at which the seam functions of the piece and of the partner converge,*

$$P_{B\sqcup R}(\mathbf{m}) = \int_{pins} \prod_{j:\, s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} \int_{\Delta_B} (\boldsymbol{\varrho}_B^{\rho})^{-(\mathbf{a},\mathbf{b})}\, d\phi_B^{\rho} \int_{\Delta_R} (\boldsymbol{\varrho}_R^{\rho})^{-(\mathbf{a},\mathbf{b})}\, d\phi_R^{\rho},$$

a fibre integral of the dimension $n_{\mathcal{D}}$ of the seam torus: the two seam functions are evaluated at the same shares, one per station and letter.

Parts (i) and (ii) hold in every case; (iii) needs a point of the pins at which both seam functions are finite. Full dimension of the Newton sums of the seam word and of the room word is necessary for it (Theorem 24.12), and when the glue is a family of tables with a native gauge, every proper wall positive, it is sufficient, the native split at the stations being such a point (Lemma 28.8); without it the share side carries one pin more for each direction that a side does not see (Theorem 27.11).

The glue is one pairing at every width: the bunch is a law on its seam torus, the room another, and the kernel is the period of two stand-in marks whose letters pair the stations. At width one the seam torus has one dimension less than the number of active letters, a half-line for one table with $s, t > 0$ and $(0,\infty)^{2N-1}$ when every letter is active (Proposition 27.10), and the kernel is the period of two marks in every table (Lemma 22.3, Theorem 22.11(i)); at the width c the seam torus has the dimension $n_{\mathcal{D}}$ and the kernel is one scale with at most c_j+1 levels in the letters of table j, exactly c_j+1 for a table. Opened on its shares, the glue is the product of the two seam functions against the share kernels of Corollary 4.4, one for each active letter.

Proposition 27.10 (Width one is the hinge)**.** *Let B have width one in every table of a family with exponents $s_j, t_j > 0$, the weights factoring across B and R, $\theta_{j,T} = \theta'_{j,T\cap B}\,\theta''_{j,T\cap R}$, with θ''_j of largest value* 1 *on the room sides of each excess, so that θ''_j gives the multipliers of the room sides (Definition 27.1), with the stations D_0 and D_1 of the excesses* 0 *and* 1 *in every table (Proposition 27.3(ii)), the hinge values $\gamma^K_{j,B}$, $\gamma^\Psi_{j,B}$ of Definition 18.4, and the cell family $F_B^+ = ((\mathcal{B}^+_{j,B}, \theta'_j); s_j, t_j)_j$ of the bunch (Definition 22.1), with its hinge wall κ_B, base form $\phi_{F_B^+}$ and ratio vector $\boldsymbol{\varrho}_{F_B^+}$, whose values are written $\mathbf{x} = (x_{j,K}, x_{j,\Psi})_j$, $x_{1,K} = 1$. In this statement write $\psi(\mathbf{x}) = \prod_j x_{j,K}^{s_j/(2\sigma)} x_{j,\Psi}^{t_j/(2\sigma)}$, $\sigma = \sum_j \sigma_j$ the total strength (Definition 8.1).*

(i) $\gamma_B^{\rho} = \prod_j (\gamma^K_{j,B})^{s_j/(2\sigma)} (\gamma^\Psi_{j,B})^{t_j/(2\sigma)} = \gamma^K_{1,B}\, \psi(\boldsymbol{\varrho}_{F_B^+})$ *and* $\phi_B^{\rho} = \psi(\boldsymbol{\varrho}_{F_B^+})^{-\kappa_B}\, \phi_{F_B^+}$.

(ii) The seam vector of the piece has $\varrho^{\rho}_{jD_0} = \varrho^{\rho*}_{jD_1} = 1$, $\varrho^{\rho}_{jD_1} = x_{j,K}/\psi(\mathbf{x})$ and $\varrho^{\rho*}_{jD_0} = x_{j,\Psi}/\psi(\mathbf{x})$ at $\mathbf{x} = \boldsymbol{\varrho}_{F_B^+}$; the map $\mathbf{x} \mapsto \big(x_{j,K}/\psi(\mathbf{x}),\, x_{j,\Psi}/\psi(\mathbf{x})\big)_j$ is a bijection of $(0,\infty)^{2N-1}$ onto the seam torus, and under it the law $(\boldsymbol{\varrho}_B^{\rho})_* \phi_B^{\rho}$ is the image of $\psi^{-\kappa_B}$ times the law $(\boldsymbol{\varrho}_{F_B^+})_* \phi_{F_B^+}$ of the cell family.

(iii) The partner with its stand-in at the seam point of $\mathbf{x}$ is, after $\alpha_g \mapsto \psi(\mathbf{x})\,\alpha_g$, the family $((\mathcal{B}^+_{j,R}, \theta''_j); s_j, t_j)_j$ of the cells of the room twisted at g by $\mathbf{x}$, $x_{j,K}$ in K_j and $x_{j,\Psi}$ in Ψ_j (Definition 21.1); the change of variable multiplies $\alpha_g^{\kappa_B - 1} d\alpha_g$ by $\psi(\mathbf{x})^{\kappa_B}$, which cancels the density of (ii), and Lemma 27.8 is Lemma 22.2 for the cell family F_B^+.

(iv) The room, read as the partner, has the same description through the cell family $F_R^+ = ((\mathcal{B}^+_{j,R}, \theta''_j); s_j, t_j)_j$ of the room, with its hinge wall κ_R and ratio vector $\boldsymbol{\varrho}_{F_R^+}$, whose values are written $\mathbf{x}'$, $x'_{1,K} = 1$: its scale is $\gamma^K_{1,R}\, \psi(\boldsymbol{\varrho}_{F_R^+})$, its scale form $\psi(\boldsymbol{\varrho}_{F_R^+})^{-\kappa_R} \phi_{F_R^+}$, its seam vector has $\varrho^{\rho}_{jD_1} = \varrho^{\rho*}_{jD_0} = 1$, $\varrho^{\rho}_{jD_0} = x'_{j,K}/\psi(\mathbf{x}')$ and $\varrho^{\rho*}_{jD_1} = x'_{j,\Psi}/\psi(\mathbf{x}')$ at $\mathbf{x}' = \boldsymbol{\varrho}_{F_R^+}$, and

its law is the image of $\psi^{-\kappa_R}$ *times the law* $(\boldsymbol{\varrho}_{F_R^+})_*\phi_{F_R^+}$ *under the bijection of (ii). At the seam points the coordinate ratios* $\varrho^{\rho*}_{jD_0}/\varrho^{\rho}_{jD_1}$ *of the piece and* $\varrho^{\rho*}_{jD_1}/\varrho^{\rho}_{jD_0}$ *of the partner are the ratios* $\varrho_{j,B}$ *and* $\varrho_{j,R}$ *of Definition 18.4. At the seam points* $\mathbf{p}$ *of the piece and* $\mathbf{p}'$ *of the partner that are the images of* $\mathbf{x}$ *and* $\mathbf{x}'$*,*

$$\begin{aligned}\Gamma(\kappa_B)\,\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{p}\circ\mathbf{p}') &= \int_0^\infty v^{\kappa_B-1}\prod_j\Big(\frac{x_{j,K}\,v}{\psi(\mathbf{x})}+\frac{x'_{j,K}}{\psi(\mathbf{x}')}\Big)^{-s_j}\Big(\frac{x_{j,\Psi}\,v}{\psi(\mathbf{x})}+\frac{x'_{j,\Psi}}{\psi(\mathbf{x}')}\Big)^{-t_j}dv\\ &= \psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}\,P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big),\end{aligned}$$

$F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]$ the core $\binom{\{g,g'\}}{1}$ *in every table twisted at g by $\mathbf{x}$ and at g' by $\mathbf{x}'$, with the masses κ_B and κ_R (Theorem 22.11(i)). So Theorem 27.9(i) is Lemma 22.2 at the hinge, by part (iii); Theorem 27.9(ii) is Theorem 22.11(i) with that core and the cell families F_B^+ at g and F_R^+ at g', the factor $\psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}$ cancelling the densities of the two laws; and Theorem 27.9(iii) is Theorem 18.13.*

At width one the scale of the piece is the geometric mean of the hinge values in the proportions of the exponents, and the general law is the law of the cell family of the bunch with a fixed power of its ratio vector as density. Every statement about pieces therefore specializes to its statement about cells through this one fixed change, and the calculus of cells (Chapter 22) is the case of width one of the calculus of pieces.

Theorem 27.11 (The share side with transverse pins)**.** *In Theorem 27.9(iii), the period of the glue converging, so that its Newton sum has full dimension (Theorem 24.12), let* $\vec{\mathcal{N}}_B\subseteq\mathbb{R}^B$ *and* $\vec{\mathcal{N}}_R\subseteq\mathbb{R}^R$ *be the spaces of the directions of the Newton sums of the seam word of the piece and of the room word of the partner at shares with positive real parts, which do not depend on the shares, and* $M=\vec{\mathcal{N}}_B^{\perp}\oplus\vec{\mathcal{N}}_R^{\perp}\subseteq\mathbb{R}^{B\sqcup R}$*, which contains* $\mathbf{1}$*; let* $\mathbf{v}_1,\dots,\mathbf{v}_q$ *complete* $\mathbf{1}$ *to a basis of the lattice* $M\cap\mathbb{Z}^{B\sqcup R}$*. For every active letter and station fix the exponents* $\mathbf{n}_{jD}$*,* $\mathbf{n}^*_{jD}$*,* $\mathbf{n}'_{jD}$ *and* $\mathbf{n}'^*_{jD}$ *of one monomial of each of* V_{jD}*,* V^*_{jD}*,* L_{jD} *and* L^*_{jD}*, and in this statement write*

$$\mathbf{n}(\mathbf{a},\mathbf{b})=\sum_{j,D}\Big(a_{jD}\big(\mathbf{n}_{jD}+\mathbf{n}'_{jD}\big)+b_{jD}\big(\mathbf{n}^*_{jD}+\mathbf{n}'^*_{jD}\big)\Big),$$

which modulo $\vec{\mathcal{N}}_B\oplus\vec{\mathcal{N}}_R$ *does not depend on the monomials fixed. For a word W on B whose Newton sum has the space of directions* $\vec{\mathcal{N}}_B$*, its* reduced period *is*

$$P^\flat_W(\mathbf{m}_B)=\int_{\mathbb{R}^B/\vec{\mathcal{N}}_B^{\perp}}e^{\langle\mathbf{m}_B,y\rangle}\,W(e^y)\,dy,$$

dy the quotient of Lebesgue measure by the measure on $\vec{\mathcal{N}}_B^{\perp}$ *that gives its integer lattice the covolume one; likewise on R. On every contour of shares of the active letters with positive real parts, pinned by the letter totals and by* $\langle\mathbf{n}(\mathbf{a},\mathbf{b}),\mathbf{v}_l\rangle=\langle\mathbf{m},\mathbf{v}_l\rangle$*, $l\le q$, pins that are independent by the full dimension, at which the reduced periods of the two words converge,*

$$P_{B\sqcup R}(\mathbf{m})=\int_{pins}\prod_{j:\,s_j>0}\Gamma_{\frac{|\mathcal{D}_j|}{1}}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\prod_{j:\,t_j>0}\Gamma_{\frac{|\mathcal{D}_j|}{1}}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\;P^\flat_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B)\,P^\flat_{W_R(\mathbf{a},\mathbf{b})}(\mathbf{m}_R),$$

$W_R(\mathbf{a},\mathbf{b})=\prod_{j,D}L_{jD}^{-a_{jD}}{L^*_{jD}}^{-b_{jD}}$ the room word of the partner; on the pins the integrands of the reduced periods are invariant under $\vec{\mathcal{N}}_B^{\perp}$ *and* $\vec{\mathcal{N}}_R^{\perp}$*. When* $\vec{\mathcal{N}}_B$ *and* $\vec{\mathcal{N}}_R$ *are the hyperplanes*

of sum zero, $q = 1$, the pin is the scale pin, the reduced periods are the periods, and this is Theorem 27.9(iii). When every room letter of the partner is a monomial, as for the room of a table whose stations of room sides are single room sides, with the coefficients 1, $\vec{\mathcal{N}}_R = 0$, the pins on R say that the exponent of the partner's monomials is $\mathbf{m}_R$, and the reduced period of the partner is the product of the coefficients of these monomials raised to minus the shares.

So a degenerate seam word or room word costs pins, not convergence: the directions that a side does not see are fixed by the masses, one pin each, and the integral over them is gone.

27.3 The Gramma of a set of pairs

The station letters are what the bunch is; the seam letters are what the room sees. The seam joins the two, and its compatible pairs enter the period through one Gramma.

Definition 27.12 (The Gramma of a set of pairs)**.** Let $\mathfrak{R} \subseteq \mathcal{K} \times \mathcal{D}$ be a set of pairs of stations of two finite sets, every station in some pair, with coefficients $h_{CD} > 0$ on its pairs. Its *station map* sends $\mathbf{e} = (e_{CD}) \in \mathbb{C}^{\mathfrak{R}}$ to $\partial\mathbf{e}$, with $(\partial\mathbf{e})_C = \sum_D e_{CD}$ and $(\partial\mathbf{e})_D = \sum_C e_{CD}$. Two pairs lie in one *part* when a chain of pairs, each sharing a station with the next, joins them; a *circuit* is a minimal set of pairs whose columns $\mathbf{e}_C + \mathbf{e}_D$ are linearly dependent, which is the set of pairs of a closed chain $C_1D_1C_2D_2\cdots C_mD_mC_1$, $m \geq 2$, of distinct stations with (C_l, D_l) and (C_{l+1}, D_l) in $\mathfrak{R}$. Choose one station of $\mathcal{D}$ in each part, and call the other stations *kept*. For *loads* $(\mathbf{a}, \mathbf{a}') \in \mathbb{C}^{\mathcal{K}} \oplus \mathbb{C}^{\mathcal{D}}$ whose totals on the two sides of every part agree, the *Gramma of* $\mathfrak{R}$ is the Gramma $\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})$ (Definition 2.1) on the scales of the kept stations whose configuration $\partial\mathfrak{R}$ has at the pair (C, D) the column $\mathbf{e}_C + \mathbf{e}_D$, a station that is not kept read as 0, whose index is the loads of the kept stations, and whose coefficients are the h_{CD}. By Proposition 4.2 it is the fibre integral

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h}) = \int_{\partial\mathbf{e}=(\mathbf{a},\mathbf{a}')} \prod_{(C,D)\in\mathfrak{R}} \Gamma(e_{CD})\, h_{CD}^{-e_{CD}},$$

pinned by the sums at the kept stations, which imply the others. Another choice of the stations not kept changes the pins by an integer matrix of determinant ± 1, which leaves the fibre integral unchanged; its dimension is $|\mathfrak{R}| - |\mathcal{K}| - |\mathcal{D}|$ plus the number of parts. The *seam Gramma* of the excess i is the Gramma of the set $\mathfrak{R}_i$ of the compatible pairs of H_i, with the pair constants as its coefficients.

Proposition 27.13 (The seam transfers the station loads of the bunch to those of the room)**.** *For a bunch of a family let the* station word *of the bunch be $\prod_{j,C} \mathsf{x}_{jC}^{-a_{jC}} {\mathsf{x}^*_{jC}}^{-b_{jC}}$, with the station loads a_{jC} of the active K_j and b_{jC} of the active Ψ_j on the pins $\sum_C a_{jC} = s_j$, $\sum_C b_{jC} = t_j$ and $\sum_{j,C} \big(i(C)a_{jC} + (c_j - i(C))b_{jC}\big) = \kappa_B$, and let the station loads a_{jD} and b_{jD} of the room be shares on the pins of the seam (Lemma 27.7). On contours of positive real parts at which the station word of the bunch converges,*

$$\prod_{j:\, s_j>0} \prod_{D\in\mathcal{D}_j} \Gamma(a_{jD}) \prod_{j:\, t_j>0} \prod_{D\in\mathcal{D}_j} \Gamma(b_{jD})\; P_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B)$$
$$= \int \prod_{j:\, s_j>0} \prod_{i=0}^{c_j} \mathsf{G}_{(\mathbf{a}_{j,\mathcal{K}_i},\mathbf{a}_{j,\mathcal{D}_i});\partial\mathfrak{R}_{j,i}}(\mathbf{h}_j) \prod_{j:\, t_j>0} \prod_{i=0}^{c_j} \mathsf{G}_{(\mathbf{b}_{j,\mathcal{K}_i},\mathbf{b}_{j,\mathcal{D}_i});\partial\mathfrak{R}_{j,i}}(\mathbf{h}_j)\; P_{\prod_{j,C} \mathsf{x}_{jC}^{-a_{jC}} {\mathsf{x}^*_{jC}}^{-b_{jC}}}(\mathbf{m}_B),$$

the integral over the station loads of the bunch with equal totals on the two sides of every part of every set $\mathfrak{R}_{j,i}$ of the compatible pairs of table j at the excess i, $\mathbf{h}_j$ its pair constants.

Along a circuit the pair terms $h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ of a level have the alternating product $\prod_l h_{C_lD_l}/h_{C_{l+1}D_l}$, a constant: a direction $\mathbf{e}$ of the pair shares with $\partial\mathbf{e}=0$ moves neither the station loads of the bunch, which the bunch sees, nor those of the room, which the room sees, and the period integrates over it through the seam Gramma alone. A part without circuits has a fibre of dimension zero, and its Gramma is the product $\prod\Gamma(e_{CD})h_{CD}^{-e_{CD}}$ at the one solution of $\partial\mathbf{e}=(\mathbf{a},\mathbf{a}')$ (Corollary 2.6).

Proposition 27.14 (The Gramma of a set of pairs in the Foreman's order)**.** *Let $\mathfrak{R}$ have the parts $\mathfrak{R}_1,\dots,\mathfrak{R}_\nu$, the part $\mathfrak{R}_l$ on the stations $\mathcal{K}_l\sqcup\mathcal{D}_l$, and loads with $|\mathbf{a}_{\mathcal{K}_l}|=|\mathbf{a}'_{\mathcal{D}_l}|$ for every l; in this statement write $L_C(v)=\sum_{D\colon(C,D)\in\mathfrak{R}}h_{CD}\,v_D$ for $C\in\mathcal{K}_l$ and $v\in\Delta_{\mathcal{D}_l}$. Then*

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})=\prod_{C\in\mathcal{K}}\Gamma(a_C)\prod_{l=1}^{\nu}\int_{\Delta_{\mathcal{D}_l}}v^{\mathbf{a}'_{\mathcal{D}_l}-\mathbf{1}}\prod_{C\in\mathcal{K}_l}L_C(v)^{-a_C}\,d\omega_{\mathcal{D}_l}(v)$$
$$=\prod_{C\in\mathcal{K}}\Gamma(a_C)\prod_{l=1}^{\nu}P_{\prod_{C\in\mathcal{K}_l}L_C^{-a_C}}(\mathbf{a}'_{\mathcal{D}_l}),$$

both sides converging together, the period of the part l being that of the word $\prod_{C\in\mathcal{K}_l}L_C^{-a_C}$ on $\mathcal{D}_l$ at the masses $\mathbf{a}'_{\mathcal{D}_l}$; with $\operatorname{Re}a_C>0$ it converges exactly when, for every l, $\operatorname{Re}\mathbf{a}'_{\mathcal{D}_l}$ lies in the relative interior of $\sum_{C\in\mathcal{K}_l}\operatorname{Re}a_C\,\operatorname{conv}\{\mathbf{e}_D:\ (C,D)\in\mathfrak{R}\}$ (Theorem 24.12).

The stations of the bunch are integrated first, each on its half-line, and those of the room last, on the simplex of each part: the Foreman's order of Definition 16.12, the bunch inside the integral over its room. For a seam the inner step returns, at each station C of the bunch, its child letter $K_{(C)}=\sum_D h_{CD}\,\mathsf{y}_D$ with every room station letter y_D replaced by a scale v_D. No station is singled out and no order is chosen.

Proposition 27.15 (Closing the Gramma of a set of pairs)**.** *In Proposition 27.14 let the loads be real and positive in the domain for (i) and (iii); in this statement write, for a part $\mathfrak{R}_l$, $L_C(v)=\sum_{D\colon(C,D)\in\mathfrak{R}}h_{CD}\,v_D$ for $C\in\mathcal{K}_l$ and $v\in\Delta_{\mathcal{D}_l}$, $M_C=\max_D h_{CD}$ for its largest coefficient, its largest value on the simplex, and $\mathsf{G}_{(\mathbf{a},\mathbf{a}')}$ for $\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})$.*

(i) Peak closing. *For every part $\mathfrak{R}_l$, with $\bar L_C=1-L_C/M_C\ge0$ on $\Delta_{\mathcal{D}_l}$,*

$$\int_{\Delta_{\mathcal{D}_l}}v^{\mathbf{a}'_{\mathcal{D}_l}-\mathbf{1}}\prod_{C\in\mathcal{K}_l}L_C(v)^{-a_C}\,d\omega_{\mathcal{D}_l}(v)$$
$$=\prod_{C\in\mathcal{K}_l}M_C^{-a_C}\,\Gamma_{|\mathcal{D}_l|}\begin{bmatrix}\mathbf{a}'_{\mathcal{D}_l}\\ |\mathbf{a}'_{\mathcal{D}_l}|\end{bmatrix}\sum_{\mathbf{n}\in\mathbb{N}^{\mathcal{K}_l}}\prod_{C\in\mathcal{K}_l}\frac{(a_C)_{n_C}}{n_C!}\ \mathbb{E}_{V\sim\mathrm{Dir}_{\mathbf{a}'_{\mathcal{D}_l}}}\Big[\prod_{C\in\mathcal{K}_l}\bar L_C(V)^{n_C}\Big],$$

a series of nonnegative terms, each a finite nonnegative combination of Dirichlet moments (Theorem 3.11), rational at rational loads and coefficients; it is Theorem 15.13(i) with the sharp references M_C, and $\mathsf{G}_{(\mathbf{a},\mathbf{a}')}$ is $\prod_C\Gamma(a_C)$ times the product of these right sides over the parts.

(ii) Relations at shifted loads. *At all loads in the domain, real or complex, for every C and D,*

$$a_C\,\mathsf{G}_{(\mathbf{a},\mathbf{a}')}=\sum_{D'\colon(C,D')\in\mathfrak{R}}h_{CD'}\,\mathsf{G}_{(\mathbf{a}+\mathbf{e}_C,\,\mathbf{a}'+\mathbf{e}_{D'})},$$
$$a'_D\,\mathsf{G}_{(\mathbf{a},\mathbf{a}')}=\sum_{C'\colon(C',D)\in\mathfrak{R}}h_{C'D}\,\mathsf{G}_{(\mathbf{a}+\mathbf{e}_{C'},\,\mathbf{a}'+\mathbf{e}_D)}.$$

(iii) Residues. *A part that is one circuit $C_1D_1\cdots C_mD_m$ has one share z moving round the circuit with alternating signs: for a real point $\mathbf{q}$ of the fibre the pairs (C_l,D_l) carry q_l+z and the pairs (C_{l+1},D_l) carry q'_l-z. With $x=\prod_l h_{C_lD_l}/h_{C_{l+1}D_l}$ and $\Sigma=\sum_l(q_l+q'_l)$, the total load of the part, its Gramma is a constant times*

$$\frac{1}{2\pi i}\int_{(0)} x^{-z}\prod_{l=1}^{m}\Gamma(q_l+z)\,\Gamma(q'_l-z)\,dz.$$

If $x<1$, or $x=1$ and $\Sigma<m$, the contour moves to the left across the poles of the $\Gamma(q_l+z)$, and if $x>1$, or $x=1$ and $\Sigma<m$, to the right across those of the $\Gamma(q'_l-z)$ (Theorem 4.5): the Gramma is the limit of the sums of the residues at the poles crossed, a series over m lattices, absolutely convergent when $x\neq1$ or $\Sigma<m-1$. The residue at a simple pole is a product of the other Gammas at the shifted point, and where poles of several factors meet, the q_l being congruent modulo $\mathbb{Z}$, it is a Laurent coefficient. At $x=1$ with $\Sigma\geq m$ the shifted contours need not tend to zero and the residues need not sum.

Theorem 27.16 (Shares inside the rectangles close)**.** *Fix a table and an excess i, and give every basis T of excess i a share a_T. For pattern loads $\mathbf{a}^B=(a^B_p)_p$ and room-side loads $\mathbf{a}^R=(a^R_X)_X$ whose station loads $a^B_C=\sum_{p\in C}a^B_p$ and $a^R_D=\sum_{X\in D}a^R_X$ have equal totals on every part of $\mathfrak{R}_i$, on contours of positive real parts,*

$$\int_{\sum_X a_{p\sqcup X}=a^B_p,\ \sum_p a_{p\sqcup X}=a^R_X}\ \prod_T\Gamma(a_T)\,\theta_T^{-a_T}$$
$$=\prod_{C\in\mathcal{K}_i}\Gamma_{|C|}\begin{bmatrix}\mathbf{a}^B|_C\\ a^B_C\end{bmatrix}\prod_p w_p^{-a^B_p}\prod_{D\in\mathcal{D}_i}\Gamma_{|D|}\begin{bmatrix}\mathbf{a}^R|_D\\ a^R_D\end{bmatrix}\prod_X {w'_X}^{-a^R_X}\ \mathsf{G}_{(\mathbf{a}^B_{\mathcal{K}_i},\mathbf{a}^R_{\mathcal{D}_i});\partial\mathfrak{R}_i}(\mathbf{h}),$$

$\mathbf{a}^B_{\mathcal{K}_i}$ and $\mathbf{a}^R_{\mathcal{D}_i}$ the vectors of the station loads, the fibre on the left pinned by these sums at all patterns and at all room sides but one in each part of $\mathfrak{R}_i$, which imply the others.

Inside the rectangle of a compatible pair the shares spread freely and close to stacked Gammas, one for each station; what does not close is how the stations share one another, the seam Gramma.

27.4 What a piece retains

Definition 27.17 (Tinkers)**.** A *tinker* of a partner $\mathcal{Y}$ with the stations $\mathcal{D}$ is a vector $\mathbf{v}=(v_{jD},v^*_{jD})_{j,D}$ of positive numbers, one for each coordinate of the seam vector; the *tinkered partner* $\mathcal{Y}_\mathbf{v}$ has the room letters $v_{jD}L_{jD}$ and $v^*_{jD}L^*_{jD}$. A tinker is *plain* when $v_{jD}=v^*_{jD}$ for every coordinate. On the room of a table a plain tinker multiplies the weights of the room sides of each station by one constant, which keeps the bases and gives a table (Proposition 12.8, read for the bunch R); a tinker that is not plain gives the room two systems of weights, one for each letter, as a cell may carry (Definition 20.1).

Theorem 27.18 (The law is what a piece retains, at every width)**.** *Let $\mathcal{X}$ and $\mathcal{X}'$ be pieces with the stations $\mathcal{D}$, on sets B and B' of any sizes, with the exponents $(s_j,t_j)_j$, $\sum_j(s_j+t_j)c_j>0$, and one wall κ. Let $\mathcal{Y}_0$ be a partner of both with the wall $\sum_j(s_j+t_j)c_j-\kappa$, and let some real point $\mathbf{c}$ of the pins of the seam, with positive coordinates, lie in the interior, relative to the pins, of the domains of the seam functions of $\mathcal{X}$, $\mathcal{X}'$ and $\mathcal{Y}_0$. The following are equivalent.*

(a) $(\boldsymbol{\varrho}^\rho_\mathcal{X})_*\phi^\rho_\mathcal{X}=(\boldsymbol{\varrho}^\rho_{\mathcal{X}'})_*\phi^\rho_{\mathcal{X}'}$.

(b) $P_{\mathcal{X}\sqcup\mathcal{Y}} = P_{\mathcal{X}'\sqcup\mathcal{Y}}$*, in* $[0,\infty]$*, for every partner* $\mathcal{Y}$ *with the wall* $\sum_j (s_j+t_j)c_j - \kappa$*.*

(c) $P_{\mathcal{X}\sqcup\mathcal{Y}_{0,\mathbf{v}}} = P_{\mathcal{X}'\sqcup\mathcal{Y}_{0,\mathbf{v}}}$ *for every tinker* $\mathbf{v}$ *of* $\mathcal{Y}_0$*.*

(d) The seam functions of $\mathcal{X}$ *and* $\mathcal{X}'$ *agree on the contour of the pins through* $\mathbf{c}$*.*

When every table j *with both letters active has* $|\mathcal{D}_j| = 1$*, except at most one, which has* $c_j \geq 1$ *and* $|\mathcal{D}_j| = 2$*, the plain tinkers suffice in (c). For one table at width one with factoring weights,* $|\mathcal{D}| = 2$*, and the theorem is Theorem 22.7 read through Proposition 27.10: the one-mark cell of ratio* 1 *(Definition 22.6), glued at the hinge, is a partner whose plain tinkers* $(1, q^{1/2})$ *are the one-mark cells of ratio* q*.*

So the wall and the law are the whole of what a piece contributes to the families glued on it, and the tinkers of one partner already separate any two laws. A plain tinker multiplies the two room letters of a station by one constant, so with every letter active, and some table of positive width, the plain tinkers of one partner move only $|\mathcal{D}| - N$ of the $2|\mathcal{D}| - 2N - 1$ directions of the seam torus. The other directions are reached by partners with one system of weights whose weights and masses vary.

Definition 27.19 (Plain partners)**.** A partner on R with the stations $\mathcal{D}$ is *plain* when for every table j with an active letter and every $D \in \mathcal{D}_j$ there are a nonempty family $\mathcal{S}_{jD}$ of subsets of R of $e_j + c_j - i(D)$ marks, its *room sides*, and positive weights $w'_{jD,X}$, $X \in \mathcal{S}_{jD}$, with

$$L_{jD} = \sum_{X\in\mathcal{S}_{jD}} w'_{jD,X}\, \alpha_R^X, \qquad L^*_{jD} = \sum_{X\in\mathcal{S}_{jD}} w'_{jD,X}\, \alpha_R^{R\setminus X},$$

one system of weights for the two letters of a station; then $e_j^* = |R| - e_j - c_j$. The room of a family of tables is plain (Proposition 27.2(iii)), its stations of room sides being disjoint, with $w'_{jD,X}$ the multiplier of X in table j (Definition 27.1).

Definition 27.20 (The station transform of a piece)**.** For a piece $\mathcal{X}$ on B with the stations $\mathcal{D}$, $\sum_j (s_j+t_j)c_j > 0$, the wall $\kappa_{\mathcal{X}}$ and the masses $\mathbf{m}_B$, and for $\mathbf{p} = (p_{jD}, p^*_{jD})_{j,D}$ with positive coordinates, one for each coordinate of the seam vector, the *station transform* is

$$\begin{aligned}\mathcal{I}_{\mathcal{X}}(\mathbf{p}) &= \int \Gamma(\kappa_{\mathcal{X}})\, \mathsf{Z}_{(\kappa_{\mathcal{X}},\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}\circ\mathbf{p})\, d\big((\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})_*\phi^{\rho}_{\mathcal{X}}\big)(\mathbf{x}) \\ &= \int_{(0,\infty)^B} \alpha^{\mathbf{m}_B - \mathbf{1}} \prod_j \Big(\sum_D p_{jD}V_{jD}\Big)^{-s_j} \Big(\sum_D p^*_{jD}V^*_{jD}\Big)^{-t_j} d\alpha,\end{aligned}$$

in $[0,\infty]$: the kernel of Theorem 27.9(ii) read against the law, the period of the piece against the constant room letters $\mathbf{p}$, the second expression by the computation of Lemma 27.8 with $\alpha = \rho u$ and $v = \rho\,\gamma^{\rho}_{\mathcal{X}}(u)$.

Theorem 27.21 (Plain partners separate laws)**.** *Let* $\mathcal{X}$ *and* $\mathcal{X}'$ *be pieces as in Theorem 27.18, with a real point* $\mathbf{c}$ *of the pins, with positive coordinates, in the interior of the domains of their seam functions on the pins. Fix room sides* $\mathcal{S}_{jD}$ *on a set* R*, every family with at least two members, such that the room word of the plain partners with these room sides has at* $\mathbf{c}$ *a Newton sum of the dimension* $|R|-1$*, as when one family of an active letter has a Newton polytope of that dimension, and masses* $\mathbf{m}^0_R$ *in its relative interior, with* $\kappa_R = \sum_j (s_j+t_j)c_j - \kappa$*. Then* $(\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})_*\phi^{\rho}_{\mathcal{X}} = (\boldsymbol{\varrho}^{\rho}_{\mathcal{X}'})_*\phi^{\rho}_{\mathcal{X}'}$ *exactly when* $P_{\mathcal{X}\sqcup\mathcal{Y}} = P_{\mathcal{X}'\sqcup\mathcal{Y}}$ *for every plain partner* $\mathcal{Y}$ *with these room sides, at every positive weights and every masses in a neighbourhood of* $\mathbf{m}^0_R$ *on the hyperplane* $\kappa_R = \sum_j (s_j+t_j)c_j - \kappa$*. Such room sides exist for all stations and degrees:* $e_j = 1$*,* $|R| \geq \max_j c_j + 2$*, and every* $\mathcal{S}_{jD}$ *all the subsets of* $1 + c_j - i(D)$ *marks.*

The room of a family of tables is a plain partner, and multiplying the weights of the bases of each table by positive numbers that depend on the room side alone keeps the bases and the stations, and multiplies the members of each station of room sides by these numbers. So when a bunch of a family has at least two room sides in every station of every table with an active letter and the room word of its room has a Newton sum of full dimension, these families, with the masses of the room near the family's and those of the bunch kept, separate its law from every other law whose seam function converges, as its own does, at the native split (Lemma 28.8(i), which puts the masses of the room in the relative interior there).

27.5 Storage, and what replaces it

Theorem 27.22 (A bunch is stored on one mark exactly at width one)**.** *In Theorem 27.9(i) let the partner be the room of a family of tables. The partner with its stand-in at* $\mathbf{x}$*,* $R^{(\mathbf{x})}$*, is a family of tables on* $R \sqcup \{g\}$*, with a system of weights for each letter as a cell may carry (Definition 20.1), at every point of the seam torus exactly when every table with an active letter has* $c_j \le 1$*, and* $c_j = 1$ *when both of its letters are active. At width one with factoring weights, normalized as in Proposition 27.10, it is the family of the cells* $\mathcal{B}^{+}_{j,R}$ *of the room with* g *twisted in both letters, which the change of variable of Proposition 27.10(iii) turns into the twist of Definition 21.1, and Lemma 27.8 is Lemma 22.2 with the scale changed from* $\gamma^{K}_{1,B}$ *to* γ^{ρ}_{B}*, a fixed bijection of the tori with a density (Proposition 27.10). A table with* $c_j = 0$ *does not see the stand-in, in either letter, and at the width* $c_j \ge 2$ *the stand-in enters the letters of table* j *with the degree* c_j*, the coefficient of* $\alpha_g^{c_j}$ *in* K_j *being* $\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} L_{jD} \neq 0$*: the partner with its stand-in is a family of Newton words, whose period converges exactly when its Newton sum has the dimension* $|R|$ *and the masses lie in its relative interior (Theorem 24.12).*

So the single mark of width one is the case $c = 1$ of a stand-in of degree c. What replaces storage at every width is the stand-in itself, the station-wise pairing of Theorem 27.9: every statement about pieces is made on pieces with stations, whose glue is a family of Newton words, and table structure is never used to compose.

Lemma 27.23 (The level cell of a bunch)**.** *Let* B *have the width* $c = c(B) \ge 1$ *in a table, and let* G *be a set of* c *new marks. The sets* $p \sqcup F$*,* $F \subseteq G$ *and* p *a pattern of excess* $c - |F|$*, are the bases of a table* $\mathcal{B}^{\natural}_{B}$ *on* $B \sqcup G$ *of rank* $r(B)$*, the* level cell *of* B*, with any positive weights;* $\mathcal{B}^{\natural}_{B} \setminus G = \mathcal{B}|B$ *and* $\mathcal{B}^{\natural}_{B}/G = \mathcal{B}/R$*. At width one* $\mathcal{B}^{\natural}_{B}$ *is the cell of the bunch* $\mathcal{B}^{+}_{B}$ *of Theorem 18.6.*

Theorem 27.24 (The level envelope)**.** *Let the weights of a table factor across* B *and* R*,* $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$*, with* $c = c(B) \ge 1$*, and let* $\mathcal{B}^{\natural}_{B}$ *on* $B \sqcup G$ *and* $\mathcal{B}^{\natural}_{R}$ *on* $R \sqcup G'$ *be the level cells of the bunch and of the room,* G' *a copy of* G*, with the weights* θ'_p *on* $p \sqcup F$ *and* $\theta''_X \binom{c}{|F'|}^{-1}$ *on* $X \sqcup F'$*. Splicing the* c *pairs of a mark of* G *and its copy (Theorem 20.11) gives the table* $\mathcal{B}^{\mathrm{env}}$ *on* $\mathcal{E}$*, the* level envelope *of the bunch, whose bases are the sets* $p \sqcup X$ *of a pattern* p *and a room side* X *of one excess, with the weights* $\theta'_p\,\theta''_X$*. It has the patterns and the room sides of the table at every excess, its seam has one rectangle at every excess, and* $\mathcal{B} \subseteq \mathcal{B}^{\mathrm{env}}$*, with equality exactly when every excess of the seam of* B *is one rectangle.*

A bunch in level form is therefore stored on $c(B)$ pairs of hinges, the table being the splicing of its two level cells, and every bunch is read off its envelope by its seam: the table keeps, at each excess, the rectangles of the compatible pairs.

Proposition 27.25 (When a glue of seamed pieces is a table: what is necessary)**.** *Let $\mathcal{B}$ be a nonempty family of r-sets of $B \sqcup R$, with children and room children as in Definition 27.1. If $\mathcal{B}$ is the family of bases of a table, then:*

(a) every child and every room child is a table;

(b) the children at adjacent patterns p and $p+b$ are quotients, the child at $p+b$ of the child at p, and likewise the room children at adjacent room sides;

(c) at the least and at the largest number of marks of a pattern all the children are one table, and likewise all the room children;

(d) with d and $d+c$ the least and the largest numbers of marks of a pattern, for every k with $d \le k \le d+c$ the patterns of k marks are the k-sets independent in the table of the patterns of $d+c$ marks that span the table of the patterns of d marks, the bases of the $(k-d)$-th lift of the latter toward the former (Proposition 12.16); and likewise for the room sides.

Theorem 27.26 (The gluing criterion)**.** *Conditions (a)–(d) of Proposition 27.25 suffice: a nonempty family of r-sets of $B \sqcup R$ satisfying them is the family of bases of a table. So a nonempty family of sets of one size is the family of bases of a table exactly when, at one bunch, its children and room children are tables, adjacent ones are quotients, the extreme ones are one, and its levels are lifts (Proposition 12.16).*

Each of (c) and (d) is needed, and (d) on each side. The criterion is what storage of a bunch on new marks of a table asks, and nothing in the periods depends on it: the glue of Theorem 27.9 composes pieces whether or not their glue is a table.

Chapter 28

Composing bunches

The pieces of Chapter 27 compose. A family with several bunches is read through a core in which every bunch is replaced by its stand-in at its seam point, and its period is the mixture of the periods of these cores over the laws of the bunches (Theorem 28.1); a bunch inside a bunch makes the law of the larger one a mixture of the laws of its nodes (Theorem 28.2); and along a chain of bunches the letters are products of transfer matrices indexed by the stations of room sides, the laws are carried from bunch to bunch by one transfer each, and the period opens as a flow of shares through the stations (Theorem 28.3, Corollary 28.4). Every operation is a rule on the seam functions on contours or on the station transforms (Theorem 28.5), a closed chain of cells is read the same way (Theorem 28.7), and the Foreman tree with nodes of any width is evaluated by one procedure whose reusable data are the laws on its bunches (Definition 28.9). Read backward, the period as a function of the masses determines the potential, so the classes of marks and the tree are read off the periods, and with retention at every node this is the correspondence between a realization and its word (Theorems 28.11 and 28.12).

28.1 Cores, nested pieces and chains of bunches

Theorem 28.1 (The core reads its pieces, at any width)**.** *Let $B_1, \dots, B_q$ be disjoint bunches of a family F on its balance wall, each with $\sum_j (s_j + t_j) c_j(B_l) > 0$, with their scales $\gamma^\rho_{B_l}$, scale forms $\phi^\rho_{B_l}$ and seam vectors $\boldsymbol{\varrho}^\rho_{B_l}$ (Definition 27.5), and let Z be the marks outside them. For $u = (u_l)_l \in \prod_l \Delta_{B_l}$ read the letters of F at $\alpha_{B_l} = \alpha_{g_l} u_l / \gamma^\rho_{B_l}(u_l)$, for new marks g_l, and divide each letter of table j by the product over l of the extreme excess sums of the bunches B_l in that letter (Definition 27.5), $V_j^{(0)}$ for K_j and $V_j^{*(c_j(B_l))}$ for Ψ_j, read at the same point. The letters obtained depend on u only through the seam points $\mathbf{x}_l = \boldsymbol{\varrho}^\rho_{B_l}(u_l)$; with them, the exponents of F and the masses κ_{B_l} on g_l and $\mathbf{m}_Z$ on Z they form the* stand-in core $F^{(\mathbf{x}_1, \dots, \mathbf{x}_q)}$*, a family of Newton words on $Z \sqcup \{g_1, \dots, g_q\}$ (Definition 27.4) in which the stand-in g_l enters the letters K_j and Ψ_j with the degree $c_j(B_l)$. In $[0, \infty]$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\prod_l \Delta_{B_l}} \prod_l \phi^\rho_{B_l}(du_l)\; P_{F^{(\boldsymbol{\varrho}^\rho_{B_1}(u_1), \dots, \boldsymbol{\varrho}^\rho_{B_q}(u_q))}}\big(\mathbf{m}_Z, \kappa_{B_1}, \dots, \kappa_{B_q}\big)$$
$$= \int P_{F^{(\mathbf{x}_1, \dots, \mathbf{x}_q)}}\big(\mathbf{m}_Z, \kappa_{B_1}, \dots, \kappa_{B_q}\big) \prod_l d\big((\boldsymbol{\varrho}^\rho_{B_l})_* \phi^\rho_{B_l}\big)(\mathbf{x}_l).$$

At width one with factoring weights the stand-in core is the twisted family $F_{\mathcal{H}}[\mathbf{x}]$ of Theorem 22.11(i), up to the change of scale of Proposition 27.10, and this is Theorem 22.11(i); before

the laws are formed it is Theorem 21.3 with the scale of every marker divided by the scale $\gamma^{\rho}_{B_l}$ of its piece.

The core meets its pieces only through their seam points: every bunch collapses to one stand-in mark carrying the mass of its wall, and enters each letter with the degree of its width, at its stations twisted by its seam point. The cores of Chapter 21 are the case in which every piece is a cell.

Theorem 28.2 (The law of a piece from its sub-pieces)**.** *Let S be a bunch of a family with $\sum_j (s_j + t_j) c_j(S) > 0$, and $S_1, \dots, S_k \subseteq S$ disjoint bunches with $\sum_j (s_j + t_j) c_j(S_l) > 0$.*

(i) The seam letters of S are linear in those of each S_l, with nonnegative coefficients: in each table, for S_1, every station D of room sides of S and any member $X \in D$,

$$K_{(D)} = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{\,q}\, K_{(D_1(q))}, \qquad \Psi_{(D)} = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{(S \setminus S_1) \setminus q}\, \Psi_{(D_1(q))},$$

the sums over the sets $q \subseteq S \setminus S_1$ for which $q \sqcup X$ is a room side of S_1, $D_1(q)$ its station, $w'_{q \sqcup X}$ its multiplier there and w'_X that of X at S; the coefficients and the stations $D_1(q)$ do not depend on the member X.

(ii) Let the node $S^{(\mathbf{x}_1, \dots, \mathbf{x}_k)}$*, the node at S of the tree of the bunches $S \supseteq S_1, \dots, S_k$, be the piece on $(S \setminus \bigcup_l S_l) \sqcup \{g_1, \dots, g_k\}$ whose seam letters are those of S with each S_l replaced by its stand-in at $\mathbf{x}_l$, as in Theorem 28.1, with the masses κ_{S_l} on g_l. The node has the wall of S, and, in this statement writing λ_S, λ_{S_l} and $\lambda_{\mathbf{x}}$ for the laws of S, of S_l and of the node at $\mathbf{x} = (\mathbf{x}_l)_l$, the law of S is the mixture of the laws of its nodes:*

$$\int H \, d\lambda_S = \int \Big(\int H \, d\lambda_{\mathbf{x}} \Big) \prod_l d\lambda_{S_l}(\mathbf{x}_l) \qquad \text{for every Borel } H \geq 0,$$

and so are the seam functions, $\int \mathbf{z}^{-(\mathbf{a},\mathbf{b})}\, d\lambda_S(\mathbf{z}) = \int \big(\int \mathbf{z}^{-(\mathbf{a},\mathbf{b})}\, d\lambda_{\mathbf{x}}(\mathbf{z}) \big) \prod_l d\lambda_{S_l}(\mathbf{x}_l)$ at all shares $(\mathbf{a}, \mathbf{b})$.

At width one, with factoring weights, this is Theorem 22.11(ii), up to the change of scale of Proposition 27.10.

Theorem 28.3 (A chain of bunches)**.** *Let $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$, $L \geq 2$, be a chain of bunches of a family with the steps $B_{l-1} = S_l \setminus S_{l-1}$, and for $0 < l < L$ let $\mathcal{D}(S_l)$ be the stations of room sides of the bunch S_l, in each table, the index j dropped.*

(i) Letters. *For $2 \leq l \leq L-1$, $D' \in \mathcal{D}(S_{l-1})$ and $D \in \mathcal{D}(S_l)$ put, for any $X \in D$,*

$$\mathbf{N}^K_l(D', D) = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{\,q}_{B_{l-1}}, \qquad \mathbf{N}^{\Psi}_l(D', D) = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{B_{l-1} \setminus q}_{B_{l-1}},$$

the sums over the $q \subseteq B_{l-1}$ for which $q \sqcup X$ is a room side of S_{l-1} in the station D'. With $\mathbf{k} = (K_{(D)})_{D \in \mathcal{D}(S_1)}$ and $\mathbf{k}^* = (\Psi_{(D)})_{D \in \mathcal{D}(S_1)}$ the seam letters of S_1 and $\mathsf{y} = (\mathsf{y}_D)_{D \in \mathcal{D}(S_{L-1})}$ and $\mathsf{y}^* = (\mathsf{y}^*_D)_{D \in \mathcal{D}(S_{L-1})}$ the room station letters of S_{L-1},

$$K = \mathbf{k}^{\mathsf{T}} \mathbf{N}^K_2 \cdots \mathbf{N}^K_{L-1}\, \mathsf{y}, \qquad \Psi = \mathbf{k}^{*\mathsf{T}} \mathbf{N}^{\Psi}_2 \cdots \mathbf{N}^{\Psi}_{L-1}\, \mathsf{y}^*,$$

products of matrices of polynomials with nonnegative coefficients, indexed by the stations of room sides of the bunches S_l: Theorem 21.11 with these stations as the states.

(ii) Laws. *Let $\sum_j (s_j + t_j) c_j(S_l) > 0$ for every $0 < l < L$, and in this statement write λ_l for the law of S_l. For $2 \le l \le L-1$ let $S_l^{(\mathbf{x})}$ be the node of S_l with its sub-bunch S_{l-1} at $\mathbf{x}$ (Theorem 28.2(ii)), the piece on $B_{l-1} \sqcup \{g\}$ with the seam letters $\sum_{D'} x_{D'}\, \alpha_g^{i(D')} \mathbf{N}_l^K(D', D)$ and $\sum_{D'} x^*_{D'}\, \alpha_g^{c(S_{l-1}) - i(D')} \mathbf{N}_l^\Psi(D', D)$, $D \in \mathcal{D}(S_l)$, and the mass $\kappa_{S_{l-1}}$ on g, and let $\lambda_{\mathbf{x}}^{(l)}$ be its law. Then λ_l is the image of λ_{l-1} under the* transfer $\mathcal{Q}_l$,

$$\lambda_l = \mathcal{Q}_l \lambda_{l-1} = \int \lambda_{\mathbf{x}}^{(l)}\, d\lambda_{l-1}(\mathbf{x}),$$

and $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int P_{B_{L-1}^{(\mathbf{x})}}(\mathbf{m}_{B_{L-1}}, \kappa_{S_{L-1}})\, d\lambda_{L-1}(\mathbf{x})$, the room B_{L-1} of the last bunch S_{L-1} with its stand-in at $\mathbf{x}$ (Theorem 27.9(i)).

(iii) Repetition. *If the nodes $S_l^{(\cdot)}$, $2 \le l \le L-1$, are one piece up to bijections of marks keeping seam letters and masses, the transfers are one operator $\mathcal{Q}$ and $\lambda_l = \mathcal{Q}^{l-1} \lambda_1$: the chain is the power of one operator, whatever the widths of its bunches. At width one with factoring weights this is Theorem 21.10 and Proposition 23.4, the seam point being the twist up to the change of scale of Proposition 27.10.*

Corollary 28.4 (The word of a chain of bunches)**.** *In Theorem 28.3 read the room B_{L-1} of the last bunch S_{L-1} as a last node whose transfer matrix from $\mathcal{D}(S_{L-1})$ to one terminal station is the column* y*, and* y^* *for Ψ, and open every active letter with one share per nonzero entry of every transfer matrix: $e_{jD'D}^{(l)}$ for the entry (D', D) of $\mathbf{N}_l^K$ and $f_{jD'D}^{(l)}$ for that of $\mathbf{N}_l^\Psi$, $2 \le l \le L$. A station D of a bunch S_l, $2 \le l \le L-1$, carries the load $a_{jD}^{(l)} = \sum_{D'} e_{jD'D}^{(l)}$ in and $\sum_{D''} e_{jDD''}^{(l+1)}$ out, and a station D of the first bunch S_1 the load $a_{jD}^{(1)} = \sum_{D''} e_{jDD''}^{(2)}$; likewise b with f. In this statement write $W_l(\mathbf{e}, \mathbf{f}) = \prod_{D', D} \mathbf{N}_l^K(D', D)^{-e_{D'D}}\, \mathbf{N}_l^\Psi(D', D)^{-f_{D'D}}$ for the word of the node l on B_{l-1}, the products over the active letters of every table understood. On contours of positive real parts at which the seam function of S_1 and the words of the nodes converge, which needs their Newton sums to have full dimension and, for a family with a native gauge, holds as soon as they do (Lemma 28.8),*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_{j:\, s_j > 0} \frac{\prod_{l=2}^{L} \prod_{D', D} \Gamma(e_{jD'D}^{(l)})}{\Gamma(s_j) \prod_{l=2}^{L-1} \prod_D \Gamma(a_{jD}^{(l)})} \prod_{j:\, t_j > 0} \frac{\prod_{l=2}^{L} \prod_{D', D} \Gamma(f_{jD'D}^{(l)})}{\Gamma(t_j) \prod_{l=2}^{L-1} \prod_D \Gamma(b_{jD}^{(l)})} \times P_{W_{S_1}(\mathbf{a}^{(1)}, \mathbf{b}^{(1)})}(\mathbf{m}_{B_0}) \prod_{l=2}^{L} P_{W_l(\mathbf{e}^{(l)}, \mathbf{f}^{(l)})}(\mathbf{m}_{B_{l-1}}),$$

W_{S_1} the seam word of the first bunch S_1 (Lemma 27.7), the fibre pinned by the letter totals, the equality of the loads in and out at every station of the bunches $S_2, \dots, S_{L-1}$, and the homogeneity of the first bunch and of every node but one. At a bunch or a node whose Newton sum is degenerate the pins and the reduced period of Theorem 27.11 take the place of the period. At $L = 2$ this is Theorem 27.9(iii).

The word of a chain of bunches is a flow: one share per transfer entry, conserved at every station, each node contributing the period of its own word at the loads of its entries. Every station through which the flow passes is paid once by a Gamma in the denominator, except at the first bunch, where the seam function of the piece takes the loads. At width one with factoring weights every bunch S_l has two stations, and this is the word of the tree (Corollary 19.9) grouped by the two stations of each bunch.

Theorem 28.5 (Two transforms of a law, and their rules)**.** *Let a piece $\mathcal{X}$ on B with the stations $\mathcal{D}$ have $\sum_j (s_j+t_j)c_j > 0$, and a real point $\mathbf{c}$ of the pins of the seam, with positive coordinates, in the interior of the domain of its seam function on the pins.*

(i) Two transforms. *The law $(\boldsymbol{\varrho}^\rho_{\mathcal{X}})_*\phi^\rho_{\mathcal{X}}$ is determined by its seam function on the contour through $\mathbf{c}$, and by its station transform $\mathcal{I}_{\mathcal{X}}$ (Definition 27.20) on any nonempty open set of points $\mathbf{p}$; the two are related by*

$$\mathcal{I}_{\mathcal{X}}(\mathbf{p}) = \int_{pins} \prod_{j:\, s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\ \mathbf{p}^{-(\mathbf{a},\mathbf{b})} \int_{\Delta_B} (\boldsymbol{\varrho}^\rho_{\mathcal{X}})^{-(\mathbf{a},\mathbf{b})}\, d\phi^\rho_{\mathcal{X}},$$

$\mathbf{p}^{-(\mathbf{a},\mathbf{b})} = \prod_{j,D} p_{jD}^{-a_{jD}} {p^*_{jD}}^{-b_{jD}}$.

(ii) The glue. *For a partner $\mathcal{Y}$ with the wall $\sum_j (s_j+t_j)c_j - \kappa_{\mathcal{X}}$, $P_{\mathcal{X}\sqcup\mathcal{Y}} = \int \mathcal{I}_{\mathcal{X}}(\mathbf{x}')\, d\big((\boldsymbol{\varrho}^\rho_{\mathcal{Y}})_*\phi^\rho_{\mathcal{Y}}\big)(\mathbf{x}')$, the station transform read against the law of the partner; on the pins, at a point where the seam function of the partner converges too, it is the share side of Theorem 27.9(iii). It extends the node of width one in one scale (Theorem 19.24) to any width.*

(iii) Nesting. *In Theorem 28.2, $\mathcal{I}_S(\mathbf{p}) = \int \mathcal{I}_{S^{(\mathbf{x}_1,\dots,\mathbf{x}_k)}}(\mathbf{p}) \prod_l d\big((\boldsymbol{\varrho}^\rho_{S_l})_*\phi^\rho_{S_l}\big)(\mathbf{x}_l)$. With one sub-piece S_1, in this statement write $\mathbf{N}^K_{2,j}(\alpha)$ and $\mathbf{N}^\Psi_{2,j}(\alpha)$, $\alpha = \alpha_{S\setminus S_1}$, for the matrices $\mathbf{N}^K_2$ and $\mathbf{N}^\Psi_2$ in table j of the chain $\varnothing \subset S_1 \subset S \subset \mathcal{E}$ (Theorem 28.3(i)), the transfer matrices from the bunch S_1 to the bunch S. Then*

$$\mathcal{I}_S(\mathbf{p}) = \int_{(0,\infty)^{S\setminus S_1}} \alpha^{\mathbf{m}_{S\setminus S_1}-\mathbf{1}}\, \mathcal{I}_{S_1}\big(\big(\mathbf{N}^K_{2,j}(\alpha)\,\mathbf{p}_j\big)_j, \big(\mathbf{N}^\Psi_{2,j}(\alpha)\,\mathbf{p}^*_j\big)_j\big)\, d\alpha,$$

$\mathbf{p}_j = (p_{jD})_D$ *and* $\mathbf{p}^*_j = (p^*_{jD})_D$*; and at shares $(\mathbf{a},\mathbf{b})$ on the pins of the seam of S, on contours of positive real parts at which the seam function of S_1 and the word of the node converge,*

$$\int_{\Delta_S} (\boldsymbol{\varrho}^\rho_S)^{-(\mathbf{a},\mathbf{b})}\, d\phi^\rho_S = \int \prod_{j:\, s_j>0} \frac{\prod_{D_1,D}\Gamma(e_{jD_1D})}{\prod_D \Gamma(a_{jD})} \prod_{j:\, t_j>0} \frac{\prod_{D_1,D}\Gamma(f_{jD_1D})}{\prod_D \Gamma(b_{jD})} \times \int_{\Delta_{S_1}} (\boldsymbol{\varrho}^\rho_{S_1})^{-(\mathbf{a}^{(1)},\mathbf{b}^{(1)})}\, d\phi^\rho_{S_1}\ P_{W(\mathbf{e},\mathbf{f})}(\mathbf{m}_{S\setminus S_1}),$$

the shares e_{jD_1D} and f_{jD_1D} one per nonzero entry of the transfer matrices, pinned by $\sum_{D_1} e_{jD_1D} = a_{jD}$, $\sum_{D_1} f_{jD_1D} = b_{jD}$ and the homogeneity of the node, with the loads $a^{(1)}_{jD_1} = \sum_D e_{jD_1D}$ and $b^{(1)}_{jD_1} = \sum_D f_{jD_1D}$, and $W(\mathbf{e},\mathbf{f})$ the word of the entries on $S\setminus S_1$ as in Corollary 28.4.

(iv) Repetition. *The rule of (iii) for a node repeated along a chain is one operator on seam functions, and on station transforms one operator on functions of $\mathbf{p}$; a chain of repeated nodes is its power applied to the first bunch.*

So every operation along a chain is a rule on the seam functions on contours, or on the station transforms: the representations (b) and (a) of Definition 22.17, at every width.

Definition 28.6 (Cyclic chains)**.** A family on $\mathcal{E} = B_1 \sqcup \cdots \sqcup B_k$, $k \geq 2$, the *segments* B_i indexed by $\mathbb{Z}_k$, is a *cyclic chain* with the finite sets of *states* $\mathcal{J}_0, \dots, \mathcal{J}_{k-1}$, $\mathcal{J}_k = \mathcal{J}_0$, and the *top layer* $h \in \mathbb{N}$, when every active letter L is

$$L = [\varepsilon^h] \operatorname{tr} \prod_{i=1}^{k} \mathbf{M}_{L,i}(\alpha_{B_i};\varepsilon),$$

the coefficient of ε^h in the trace of a product of matrices $\mathbf{M}_{L,i}$ indexed by $\mathcal{J}_{i-1} \times \mathcal{J}_i$, the *segment matrices*, whose entries are polynomials in α_{B_i} and ε with nonnegative coefficients.

Theorem 28.7 (Cyclic chains)**.** *(i) Let every table j of a family be a closed chain of cells $\mathcal{A}_{j,1}, \dots, \mathcal{A}_{j,k}$ with several hinges spliced along hinge pairs (Theorem 20.11(ii)), the cells $\mathcal{A}_{j,i}$ of all tables having the marks B_i besides their hinges, and the pairs I_i joining the cell i to the cell $i+1$ named alike in every table, each pair named once. The family is a cyclic chain with the top layer 0: its states at the boundary i are the sets $J \subseteq I_i$ of the pairs whose hinge on the side of the cell i is occupied, and the segment matrices of the letters of table j are the channels of its cells (Definition 20.10),*

$$\mathbf{M}_{K_j,i}[J, J'] = K_{\mathcal{A}_{j,i},((I_{i-1} \setminus J) \sqcup J')}, \qquad \mathbf{M}_{\Psi_j,i}[J, J'] = \Psi_{\mathcal{A}_{j,i},((I_{i-1} \setminus J) \sqcup J')}.$$

(ii) Widen the states by layers, $\mathcal{J}_i \times \{0, \dots, h\}$, a term in ε^μ of an entry raising the layer by μ, and write $\widetilde{\mathbf{M}}_{L,i}$ for the widened segment matrices. Then $L = \sum_{J \in \mathcal{J}_0} \mathbf{e}_{(J,0)}^{\mathsf{T}} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$: for every starting state a chain in the sense of Theorem 28.3(i), the segment matrices as transfer matrices. In this statement write w_L for the exponent of the active letter L. Open every active letter first over the starting states J at which its term $\mathbf{e}_{(J,0)}^{\mathsf{T}} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$ is not zero, with shares $c_{L,J}$, $\sum_J c_{L,J} = w_L$, and then every chain as in Corollary 28.4: one share $e_{L,i}^{(J)}$ per starting state J and per nonzero entry of a widened segment matrix that lies on a path from $(J,0)$ to (J,h), the shares of each starting state a flow from $(J,0)$, with the total $c_{L,J}$, to (J,h), conserved at every boundary, $a_{L,i}^{(J)}$ the loads it leaves at the boundary after the segment i, and $e_{L,1}^{(J)} = a_{L,1}^{(J)}$. On contours of positive real parts at which the words of the segments converge,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_L \frac{1}{\Gamma(w_L)} \prod_{L,J} \frac{\prod_{i=2}^{k} \prod \Gamma(e_{L,i}^{(J)})}{\prod_{i=2}^{k-1} \prod \Gamma(a_{L,i}^{(J)})} \prod_{i=1}^{k} P_{W_i}(\mathbf{m}_{B_i}),$$

the inner products over the entries and over the states, W_i the word on B_i of the entries of the segment i at their total shares $\sum_J e_{L,i}^{(J)}$, and the fibre pinned by the conservation and by the homogeneity of every segment but one.

(iii) For k copies of one segment the integrand of (ii), read at each boundary as a function of the loads of all the starting states, is the kernel of one operator $\mathcal{C}$, restricted at each boundary to the loads of the states that lie on a path there, which integrates the loads out over the fibre pinned by the homogeneity of the segment. With $\mathcal{C}_\circ$ the same kernel without that pin, which for one segment is implied by those of the others, the letter totals and the balance, the period is

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_L \frac{\prod_J \Gamma(c_{L,J})}{\Gamma(w_L)} \left(\mathcal{C}^{k-1} \mathcal{C}_\circ\right)(\mathbf{c}, \mathbf{c})\, d\mathbf{c},$$

a power of one operator taken from the loads $\mathbf{c}$ at the starting states back to them. Its price is that of one segment and the binary digits of k (Chapter 31); evaluated, the power takes at most $\lfloor \log_2 k \rfloor$ squarings of $\mathcal{C}$ and as many products.

A closed chain is a chain whose last boundary is glued back to its first: the trace sums over the states at one boundary, and each starting state carries its own flow around the closed chain. Cells spliced along any graph of hinge pairs stay tables (Theorem 20.11); around a closed chain their letters are traces of products of channel matrices, and repeated cells give the power of one operator.

28.2 The procedure

Lemma 28.8 (The native split at the stations)**.** *Let a family of tables have real data on its balance wall, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, and a native gauge $\alpha^\star$ (Definition 8.16), B a bunch with $\sum_j (s_j + t_j) c_j(B) > 0$, and put, for the active letters,*

$$a_{jD} = s_j \, \mathbb{P}_{j,\alpha^\star}(T \cap R \in D), \qquad b_{jD} = t_j \, \mathbb{P}^*_{j,\alpha^\star}(T \cap R \in D), \qquad D \in \mathcal{D}_j,$$

the native split (Definition 12.22) aggregated to the stations of room sides; along a chain of bunches (Theorem 28.3) put likewise

$$e^{(l)}_{jD'D} = s_j \, \mathbb{P}_{j,\alpha^\star}\big(T \setminus S_{l-1} \in D',\ T \setminus S_l \in D\big), \qquad f^{(l)}_{jD'D} = t_j \, \mathbb{P}^*_{j,\alpha^\star}\big(T \setminus S_{l-1} \in D',\ T \setminus S_l \in D\big).$$

(i) $(\mathbf{a}, \mathbf{b})$ *is a point of the pins of the seam with positive coordinates, at which $\mathbf{m}_B$ and $\mathbf{m}_R$ lie in the relative interiors of the Newton sums of the bunch word and of the room word (Theorem 21.13); the two words converge there exactly when these Newton sums have the full dimensions $|B| - 1$ and $|R| - 1$ (Theorem 24.12).*

(ii) With $a^{(l)}_{jD} = s_j \, \mathbb{P}_{j,\alpha^\star}(T \setminus S_l \in D)$ and $b^{(l)}_{jD} = t_j \, \mathbb{P}^*_{j,\alpha^\star}(T \setminus S_l \in D)$, the native split at the stations of the bunch S_l, the entry shares conserve the loads, $\sum_{D'} e^{(l)}_{jD'D} = a^{(l)}_{jD} = \sum_{D''} e^{(l+1)}_{jDD''}$, and likewise with f and b; and $\mathbf{m}_{B_{l-1}}$ lies in the relative interior of the Newton sum of the word of the node l at its shares, which converges there exactly when that Newton sum has the dimension $|B_{l-1}| - 1$.

The Newton sums can be degenerate. When every station of room sides is a single room side, every room station letter is a monomial, the room word is a monomial, and its period diverges at every share; the period is then read by Theorem 27.9(i) or (ii), or on the share side of Theorem 27.11 with $\vec{\mathcal{N}}_R = 0$.

Definition 28.9 (The seam procedure)**.** *Given*: a family F with real data on its balance wall, every proper wall positive. *Classification*, from the tables alone: the molecules, the readings at the centre and the Foreman tree (Definition 16.12, steps (1)–(5)); for a molecule with a connected active table the tree is a chain whose bunches $B_0, \ldots, B_{d-1}$ (Corollary 19.8) are the steps of the chain of bunches $S_k = B_0 \sqcup \cdots \sqcup B_{k-1}$, the leaf being the last step. *Seams*, from the tables alone, at every bunch S_k:

(1) group the patterns of S_k of each excess by their weighted children up to positive multiples, and the room sides by their weighted room children, with the multipliers normalized by their largest value (Definition 27.1); the stations, the station letters and the pair constants are determined;

(2) record the seam matrices H_i and the transfer matrices $\mathbf{N}^K_k$, $\mathbf{N}^\Psi_k$ of Theorem 28.3(i);

(3) record the orbits of the stations under the bijections that keep S_k and keep the potential (Proposition 27.6, Lemma 14.18).

Evaluation at a point $(\mathbf{m}; \mathbf{s}, \mathbf{t})$:

(4) the law of $S_1 = B_0$ (Definition 27.5), with its canonical scale;

(5) the transfers of the laws from bunch to bunch (Theorem 28.3(ii)), one operator for repeated nodes (Theorem 28.3(iii)), the transfer entries of pairs of stations in one orbit being images of one another (Proposition 27.6(i));

(6) the period at the leaf with its stand-in at the seam point of the last bunch S_d, integrated against the law of that bunch (Theorem 27.9(i)); equivalently the word of Corollary 28.4 on the contour through the native shares of Lemma 28.8, with the transverse pins of Theorem 27.11 at every bunch or node whose Newton sum is degenerate, every node a Newton word with multilinear letters closed by its peak series with the references of Theorem 15.13(i) and every seam Gramma by its peak closing with the sharp references (Proposition 27.15(i)).

Output: the period, with the seams, the transfer matrices and the law on every bunch, ready to be reused when a node is changed. At a general Foreman tree the bunches of a node with several childcules are composed by Theorem 28.1, the childcules of a peel being the pieces of the core.

Nothing in the procedure is chosen. The tree is the Foreman tree (Definition 14.4); the stations are fixed by the children, and their multipliers by the normalization of the largest one; the scale of a piece is the geometric mean of the oriented extreme ratios of the active letters, each taken to the root of its width, kept by the symmetries of the family (Proposition 27.6(iii)); inside a seam Gramma the stations of the bunch are integrated before those of the room, the Foreman's order (Proposition 27.14), and another choice of the stations that are not kept changes the pins by a unimodular matrix; the real point of the share contour, where the word is used, is the native split at the stations and the entries (Lemma 28.8); the references of the closings are those of Theorem 15.13(i) and the sharp ones, read off the coefficients. Repetition is paid once: stations in one orbit carry data that are images of one another, and repeated nodes one operator.

Proposition 28.10 (What the work depends on)**.** *(i) The law of a piece depends on the piece only through the joint law of the seam letters of its active letters under the measure $u^{\mathbf{m}_B-\mathbf{1}}d\omega_B$, with its exponents and wall: two pieces, on sets of any sizes, whose seam letters have the same joint law have equal laws and are interchangeable in every glue (Theorem 27.18); in particular so are two pieces whose seam letters and masses correspond under a bijection of their marks.*

(ii) Along a chain the letters cost one product of matrices of the sizes $|\mathcal{D}(S_l)|$, and a seam Gramma of a part without circuits costs no integration (Definition 27.12, Corollary 2.6).

(iii) The law of a bunch depends on the room only through the room children of the bunch, the tables on the bunch that the room presents to it: changing the room while keeping its room children and their stations changes no law below the bunch.

28.3 The correspondence

The procedure reads a realization forward, from its tables to its word. The other direction asks what the periods distinguish, and it has two answers: as a function of the masses the period determines the potential, which carries the classes of marks and the tree; at one realization each bunch is seen by its room through its wall and its law, and only through them.

Theorem 28.11 (The period function determines the potential)**.** *Let a family have real exponents $s_j, t_j \geq 0$ with $s_j + t_j > 0$ for every j and a period that converges at some masses on its balance wall, and let U be the set of the complex masses on the balance wall at which it converges.*

(i) With the measure dy on the hyperplane $\mathbf{1}^{\perp} = \{y \in \mathbb{R}^{\mathcal{E}} : \langle \mathbf{1}, y\rangle = 0\}$ of the section $\prod_e \alpha_e = 1$ (Lemma 16.2),

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathbf{1}^{\perp}} e^{\langle \mathbf{m},y\rangle - \Phi(y)}\, dy, \qquad \mathbf{m} \in U,$$

Φ *the potential at the table's gauge (Definition 8.17). The function* $P_F(\cdot;\mathbf{s},\mathbf{t})$ *on* U *determines* Φ*, and is determined by it.*

(ii) Two families on the same marks with the same exponents have the same period function exactly when they have the same irreducible factors with their exponents $(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$ *(Theorem 10.6) and the same constant* $\Phi(0) = \sum_j (s_j + t_j) \log K_j(\mathbf{1})$*; at the centre, every* $\delta_j = 0$*, exactly when they have the same* $(L, w_{\mathbf{E}}(L))$ *and the same constant. In particular a family and the family of its dual tables (Proposition 8.14), the bases replaced by their complements, have the same period function at the centre.*

(iii) A bijection g *of the marks keeps the period function,* $P_F(g^*\mathbf{m};\mathbf{s},\mathbf{t}) = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *on* U*, exactly when it keeps the potential, both parts. For a molecule, two marks lie in one class exactly when a bijection keeping the period function takes one to the other (Lemma 14.18), and when its children's weights factor across the childcules at every node, every such bijection carries the Foreman tree onto itself, with its classes, units, atoms, bunches and childcules (Proposition 14.21).*

(iv) At the centre the bijections that carry the family and those that exchange it keep the period function; at a lean the carrying ones keep it, and an exchanging one sends it to the period function of the opposite lean (Proposition 11.15). For a molecule of one table two marks lie in one class at the centre exactly when a bijection carrying or exchanging the table takes one to the other, and at a lean exactly when a carrying one does (Corollary 11.17), and a class of the centre that is not a class at a lean is there two classes of equal size, which an exchange swaps (Theorem 11.18).

So two marks of a molecule are told apart by the period function of the centre, or of a lean, exactly when no bijection keeping that potential joins them; for one table these are, at the centre, the bijections carrying or exchanging the table, and at a lean the carrying ones. The confusable classes, the classes of a lean that share a class of the centre, are exactly what the period function of the centre does not separate and that of a lean does. The tree is read off the period function but does not determine it; the irreducible factors with their exponents, and the constant $\Phi(0)$, do.

Theorem 28.12 (The correspondence)**.** *Let a molecule be read at a realization with masses on its balance wall and every proper wall positive, its children's weights factoring across the childcules at every node.*

(i) Forward. *Its period is the reassembly of the laws of the bunches of its Foreman tree with the leaves, by the seam procedure: the law of the first bunch on its seam torus, transferred along the chain by the later bunches as nodes, the childcules of a node composed at its core, and glued to the leaves by the stand-ins; nothing in it is chosen (Definition 28.9, Theorems 28.1 and 28.3).*

(ii) Backward at the realization. *At the node reached after* k *peels the room sees the bunch* S_k*, the union of the bunches peeled down to it, through its wall and its law, and only through them: bunches with the same stations, exponents, walls and laws are interchangeable in every family glued on that room, and under the hypotheses of Theorem 27.18 two different laws are told apart by the tinkers of one room and by plain partners (Theorems 27.18 and 27.21); the laws of nested bunches compose by Theorems 28.2 and 28.3.*

Whether the molecule has an atomic realization is read at a centre: it has one exactly when some centre is atomic, and an atom at a lean is an atom at the centre (Theorems 11.9 and 11.1). The word of the tree is the forward procedure written out, and retention is the statement that it loses nothing that a room could detect.

Chapter 29

Closing a glue along a direction

A glue opened on its shares is a fibre integral of Gammas and two seam functions (Theorem 27.9(iii)). It is closed by moving the contour along a direction of the shares, collecting a fibre integral of residues at every polar hyperplane crossed and showing that what is left tends to zero. The glue closes on the lattices of its Gammas along every direction whose rate, read off the two laws through the largest values of the monomials of their letter vectors, is negative (Theorem 29.2), and past the walls of the piece, or of the piece and the partner both, at the growths of the continued seam functions (Theorem 29.3). Moving the contour is exact at every step, and the limit needs only a bound that travels with it (Lemma 29.4): the series depends on the direction only through its chamber (Theorem 29.6), the laws can be closed one source point at a time and averaged, at the rate read on the support of the source measure rather than on its convex hull (Theorem 29.7), and past the walls of a piece the bound is read on the outline of a keyhole continuation, the modulus and the phase of the sector letters taken at one point (Theorem 29.11). At simple nonresonant crossings of a rational ray, a crossing term kept with its falling shares integrated on their strips is a finite sum of joint Trailmoments of fixed functions, a falling Gamma becoming a positive share and one Beta variable and a continued source factor a segment and circle integrals (Theorem 29.18); for a positive kernel its root rate is the exponential of the least Gamma rate of its chamber in every fibre dimension and, averaged, of the essential supremum over the source of that least rate, whose sign, when it is not zero, decides whether the terms tend to zero (Theorem 29.26, Corollary 29.27); and an averaged closing can converge where no direction of its chamber certifies it, at a family of tables whose seam has single stations (Theorem 29.8). Coincident crossings are packets of factors polar on one crossed hyperplane, assembled before a sign or a top is read (Theorem 29.29), and at the root rate one a positive moment sequence is decided by its law near its top (Theorem 29.31).

29.1 Closing along a direction

Definition 29.1 (Letter vectors)**.** Let a piece on B and a partner on R with the stations $\mathcal{D}$ be glued with masses on the balance wall of the glue (Definition 27.4), the glue read on its shares as in Theorem 27.9(iii) or Theorem 27.11. The *letter vectors* of the piece and of the partner are $\mathbf{V}_B = (V_{jD}, V_{jD}^*)_{j,D}$ and $\mathbf{V}_R = (L_{jD}, L_{jD}^*)_{j,D}$, and for a vector $\mathbf{v}$ of the coordinates of the shares $\mathbf{V}_B^{\mathbf{v}} = \prod_{j,D} V_{jD}^{v_{jD}} V_{jD}^{*\,v_{jD}^*}$, and likewise $\mathbf{V}_R^{\mathbf{v}}$.

Theorem 29.2 (Closing the glue along a direction)**.** *In Theorem 27.11 let V be the space of the free directions of the shares, so that for $\mathbf{v} \in V$ the monomials $\mathbf{V}_B^{\mathbf{v}}$ and $\mathbf{V}_R^{\mathbf{v}}$ (Definition 29.1) are*

homogeneous of degree zero, and their suprema over the orthants do not change under $\vec{\mathcal{N}}_B^{\perp}$ *and* $\vec{\mathcal{N}}_R^{\perp}$. *When the seam word of the piece has a Newton sum of full dimension,* $\sup \mathbf{V}_B^{\mathbf{v}} = \sup_{u\in\Delta_B} \boldsymbol{\varrho}_B^{\rho}(u)^{\mathbf{v}}$, *the largest value of the monomial* $\mathbf{x}^{\mathbf{v}}$ *on the support of the law of the piece, and when the room word of the partner has a Newton sum of full dimension,* $\sup \mathbf{V}_R^{\mathbf{v}} = \sup_{v\in\Delta_R} \boldsymbol{\varrho}_R^{\rho}(v)^{\mathbf{v}}$, *the largest value on the support of the law of the partner; where the room letters of the partner are monomials,* $\sup \mathbf{V}_R^{\mathbf{v}} = \boldsymbol{\xi}^{\mathbf{v}}$, $\boldsymbol{\xi}$ *their coefficients. For* $\boldsymbol{\sigma} \in V$ *put*

$$h(\boldsymbol{\sigma}) = \sum_{j,D} \big(\sigma_{jD} \log|\sigma_{jD}| + \sigma^*_{jD} \log|\sigma^*_{jD}|\big) + \log\sup \mathbf{V}_B^{-\boldsymbol{\sigma}} + \log\sup \mathbf{V}_R^{-\boldsymbol{\sigma}},$$

with $0\log 0 = 0$, *the first sum over the shares of the active letters, the* rate *of the direction; it is homogeneous of degree one. Let* $h(\boldsymbol{\sigma}) < 0$, *and let the real point* $\mathbf{c}$ *of the pins lie off the null set of the points whose ray* $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \geq 0$, *meets two polar hyperplanes of the Gammas that are not parallel on the fibre at one point, or, for arbitrarily large* τ, *at points less than* τ^{-3} *apart. Then the ray meets only the hyperplanes* $a_{jD} = -n$ *with* $\sigma_{jD} < 0$ *and* $b_{jD} = -n$ *with* $\sigma^*_{jD} < 0$, $n \in \mathbb{N}$, *and*

$$P_{B\sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R},$$

the sum, in the order of crossing, of the fibre integrals $\mathcal{R}$ *of the residues of the integrand at the hyperplanes crossed, each pinned also by its hyperplane, of the dimension* $\dim V - 1$ *and of the same kind, at the point where the ray meets it; a hyperplane of the fibre on which several Gammas have poles is crossed once, for every* $\mathbf{c}$, *and its residue is a Laurent coefficient (Theorem 4.5). The sum converges absolutely: the residue at a crossing at* τ, *and the integral left after it, are* $O\big(e^{\tau(h(\boldsymbol{\sigma})+\epsilon)}\big)$ *for every* $\epsilon > 0$.

So the glue closes on the lattices of its Gammas along every direction of the cone $\{h < 0\}$ of V, the terms falling at least by the factor $e^{h(\boldsymbol{\sigma})}$ per unit of $\boldsymbol{\sigma}$. The cone and the rate are read off the laws through the largest values of the monomials of their letter vectors, and the term $\sum \sigma \log|\sigma|$ of the rate is the cost of moving shares between stations: it vanishes at width one with the stations of one excess merged, every letter then moving its two shares oppositely, and along the one direction of a node of width one the condition is that the ranges of the two ratios be separated, as in Section 19.7. A direction along which a law is unbounded, its supremum infinite, crosses the walls of its seam function as well. The seam function of a piece continues past its walls with a growth that its sector integrals bound, and the partner can be read in its monomials, which have no walls, or continued past its own walls in the same way.

Theorem 29.3 (Closing past the walls)**.** *Let the seam function of the piece converge at a real point of the pins of the seam with positive shares, so that its seam word has a Newton sum of full dimension (Theorem 24.12). Read the partner in its monomials,* $L_{jD} = \sum_k \xi_{jDk}\, \alpha_R^{\mathbf{n}_{jDk}}$ *and* $L^*_{jD} = \sum_k \xi^*_{jDk}\, \alpha_R^{\mathbf{n}^*_{jDk}}$, *with one share* a_{jDk} *or* b_{jDk} *for each monomial, and write* $\bar{a}_{jD} = \sum_k a_{jDk}$ *and* $\bar{b}_{jD} = \sum_k b_{jDk}$; *bars mark sums over the monomials.*

(i) On every contour of shares with positive real parts, pinned by the letter totals and by $\sum_{j,D,k}(a_{jDk}\mathbf{n}_{jDk} + b_{jDk}\mathbf{n}^*_{jDk}) = \mathbf{m}_R$, *at which the seam function of the piece converges, the period of the glue converging,*

$$P_{B\sqcup R}(\mathbf{m}) = \int_{pins} \prod_{j:\, s_j>0} \frac{\prod_{D,k} \Gamma(a_{jDk})\, \xi_{jDk}^{-a_{jDk}}}{\Gamma(s_j)} \prod_{j:\, t_j>0} \frac{\prod_{D,k} \Gamma(b_{jDk})\, {\xi^*_{jDk}}^{-b_{jDk}}}{\Gamma(t_j)}\; P_{W_B(\bar{\mathbf{a}},\bar{\mathbf{b}})}(\mathbf{m}_B).$$

(ii) Fix $e_0 \in B$ and a fan of the seam word*: finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each contained, for every seam letter V of an active letter, in a set $\{\mathbf{v} :\ \langle \mathbf{t},\mathbf{v}\rangle \le \langle \mathbf{t}',\mathbf{v}\rangle$ for every exponent $\mathbf{t}'$ of the support of $V\}$, $\mathbf{t}$ one exponent of that support, the pairings taken on $B\setminus e_0$. On a cone π of it, a* sector*, with the primitive integral generators $\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1}$, write the logarithmic coordinates $x = \log \alpha_{B\setminus e_0}$ of the section $\alpha_{e_0}=1$ as $x = -\sum_k u_k \mathbf{v}_{\pi,k}$, $u_k \ge 0$; the $q_k = e^{-u_k} \in (0,1]$ are the* sector coordinates*. On the section a seam letter V of an active letter is $V = \prod_k q_k^{\langle \mathbf{t},\mathbf{v}_{\pi,k}\rangle}\, k_{V,\pi}(q)$, $\mathbf{t}$ the exponent of its support at which every generator of π pairs least, and its* sector letter *$k_{V,\pi}$ is a polynomial in q with nonnegative coefficients and a positive constant term. Write $k_{jD,\pi}$ and $\psi_{jD,\pi}$ for the sector letters of V_{jD} and V^*_{jD}, $\mathbf{t}_{jD,\pi}$ and $\mathbf{t}^*_{jD,\pi}$ for their exponents, and for a generator $\mathbf{v}$ of π call*

$$\kappa_{\mathbf{v}}(\mathbf{z}) = \Big\langle \mathbf{m}_B - \sum_{j,D} \big(\bar a_{jD}\,\mathbf{t}_{jD,\pi} + \bar b_{jD}\,\mathbf{t}^*_{jD,\pi}\big),\ \mathbf{v}\Big\rangle, \qquad \mathbf{z} = (\bar{\mathbf{a}},\bar{\mathbf{b}}),$$

the pairing on $B\setminus e_0$, the wall *of $\mathbf{v}$: affine in the shares with a rational linear part, and the same for every sector of which $\mathbf{v}$ is a generator. On the pins of the seam, at every real point at which every wall is positive,*

$$P_{W_B(\mathbf{z})}(\mathbf{m}_B) = \sum_\pi |\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})| \int_{[0,1]^{|B|-1}} \prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})-1} \prod_{j,D} k_{jD,\pi}(q)^{-\bar a_{jD}}\,\psi_{jD,\pi}(q)^{-\bar b_{jD}}\,dq,$$

and the seam function converges at a real point exactly when every wall is positive there. It continues meromorphically to all complex shares, with poles, of order at most $|B|-1$, only on the hyperplanes on which a wall lies in $-\mathbb{N}$, and for every $\theta>0$ there are M, C_0 and n_1 with

$$\big|P_{W_B(\mathbf{z})}(\mathbf{m}_B)\big| \le C_0\,\delta^{-(|B|-1)}(1+|\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta|\operatorname{Im}\mathbf{z}|_1},$$

δ the smaller of 1 and the distance of $\mathbf{z}$ from these hyperplanes. The piece is of tables *when at every station the seam letters of the active letters are the basis and the cobasis polynomials of one table on B with the bases $\mathcal{B}_{jD}$ and positive weights, as for a bunch of a family of tables (Proposition 27.2(i)). Its seam word then has a Newton sum of full dimension exactly when the family of these tables is connected; the sectors of the orders π of B (Theorem 8.7(ii)) form a fan of it, every determinant being 1, the sector letters being those of the tables $\mathcal{B}_{jD}$ and the generators $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$ on $B\setminus e_0$, U_k the tails; the wall of the generator of U is the proper wall $\kappa_U = m_U - \sum_{j,D}\big(\bar a_{jD}\,d_{jD}(U) + \bar b_{jD}\,d^*_{jD}(U)\big)$, d_{jD} and d^*_{jD} the counts of $\mathcal{B}_{jD}$.*

(iii) For a direction $\boldsymbol{\sigma}$ of the shares of (i) and $\theta \ge 0$ let $g(\boldsymbol{\sigma};\theta)$, at the real point $\bar{\mathbf{c}}$ of the pins of the seam, be the infimum of the M for which there are C, n_1 and n_2 with

$$\big|P_{W_B(\bar{\mathbf{c}}+\tau\bar{\boldsymbol{\sigma}}+i\bar{\boldsymbol{\vartheta}})}(\mathbf{m}_B)\big| \le C\,\delta^{-n_2}(1+\tau+|\boldsymbol{\vartheta}|)^{n_1} M^{\tau} e^{\theta|\boldsymbol{\vartheta}|_1}$$

for $\tau \ge 0$ and $\boldsymbol{\vartheta}$ in the directions of the shares, δ the smaller of 1 and the distance from the polar hyperplanes: the growth *of the continued seam function along $\boldsymbol{\sigma}$ at the angle θ, the growth of Definition 19.31 read along any direction with the angle fixed. It is finite for $\theta>0$, and $g(\boldsymbol{\sigma};0) = \sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$ when no wall of the seam word falls along $\bar{\boldsymbol{\sigma}}$, which is exactly when that supremum is finite. Put*

$$h_\Gamma(\boldsymbol{\sigma}) = \sum_l \sigma_l \log|\sigma_l| - \sum_l \sigma_l \log \xi_l, \qquad h_\theta(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log g(\boldsymbol{\sigma};\theta) + \frac{2\theta^2}{\pi}\sum_{l:\,\sigma_l>0}\sigma_l,$$

l running over the shares and ξ_l being the coefficient of the monomial of the share l; h_Γ is the Gamma rate *of the direction. Let the real point $\mathbf{c}$ of the pins lie off the null set of Theorem 29.2, taken with the polar hyperplanes of the seam function as well, and let g and h_θ be taken at its sums $\bar{\mathbf{c}}$. If $h_\theta(\boldsymbol{\sigma}) < 0$ for some $\theta \in [0, \pi/8)$, then*

$$P_{B \sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R},$$

the sum, in the order of crossing, of the fibre integrals of the residues of the integrand at the polar hyperplanes met by the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$: the hyperplanes $a_l = -n$ of the shares with $\sigma_l < 0$, $n \in \mathbb{N}$, and the hyperplanes on which a wall of (ii) that falls along $\bar{\boldsymbol{\sigma}}$ lies in $-\mathbb{N}$, a hyperplane polar for several factors being crossed once, with its full residue. The sum converges absolutely, the terms at τ and the integrals left after them being $O\big(e^{\tau(h_\theta(\boldsymbol{\sigma})+\epsilon)}\big)$ for every $\epsilon > 0$.

(iv) Both sides continued. *Let the glue be read on the share side of Theorem 27.9(iii), one share a_{jD} or b_{jD} per station, in place of the reading of (i), and let the seam functions of the piece and of the partner both converge at a real point $\mathbf{c}$ of the pins with positive shares. By (ii), applied to the piece and to the partner, a piece on R read with the reversed excesses (Definition 27.4), both continue meromorphically to all complex shares of the pins, with their poles on the hyperplanes on which a wall of the seam word of the piece or of the room word of the partner lies in $-\mathbb{N}$, each with the bound of (ii). For a direction $\boldsymbol{\sigma}$ of the shares and $\theta \geq 0$ let $g_B(\boldsymbol{\sigma}; \theta)$ and $g_R(\boldsymbol{\sigma}; \theta)$ be the growths of the two continued seam functions along $\boldsymbol{\sigma}$ at $\mathbf{c}$, defined as in (iii) with the shares in place of their sums, and put*

$$h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) = \sum_l \sigma_l \log|\sigma_l| + \log g_B(\boldsymbol{\sigma}; \theta_B) + \log g_R(\boldsymbol{\sigma}; \theta_R) + \frac{2(\theta_B + \theta_R)^2}{\pi} \sum_{l:\, \sigma_l > 0} \sigma_l.$$

Let $\mathbf{c}$ lie off the null set of Theorem 29.2, taken with the polar hyperplanes of both seam functions as well. If $h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) < 0$ for some $\theta_B, \theta_R \geq 0$ with $\theta_B + \theta_R < \pi/8$, then $P_{B \sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R}$, the sum, in the order of crossing, of the fibre integrals of the residues of the integrand at the polar hyperplanes met by the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$: the hyperplanes $a_l = -n$ of the shares with $\sigma_l < 0$, $n \in \mathbb{N}$, and those on which a wall of either word that falls along $\boldsymbol{\sigma}$ lies in $-\mathbb{N}$, a hyperplane polar for several factors being crossed once, with its full residue. The sum converges absolutely, the terms at τ and the integrals left after them being $O\big(e^{\tau(h_{\theta_B, \theta_R}(\boldsymbol{\sigma})+\epsilon)}\big)$ for every $\epsilon > 0$. A side along whose direction no wall of its word falls has the growth $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$ of its letter vector at every angle, the largest value of the monomial on the support of its law (Theorem 29.2).

Along a direction in which the seam function converges, (iii) at $\theta = 0$ is the bound of Theorem 29.2 for the glue read in the monomials of the partner, and past a wall the residues of the seam function join those of the Gammas at the cost $\log g$ of its continuation, as the walls of the two families join the lattice at width one (Theorem 19.32(iii)). Read in its monomials as well, any piece contributes only its coefficients: the integrand is then Gammas and coefficients, every supremum of a monomial of the letter vectors is a monomial in the coefficients, and every direction of the shares with $h_\Gamma(\boldsymbol{\sigma}) < 0$ closes, by Theorem 29.2. In which representation to close is a choice of rate: a law bounded along the direction closes at the largest value of its monomial, a side past its walls at the growth of its continuation, a side read in monomials at its coefficients, at the price of one pin for each of its marks. So (iii) and (iv) read each side at its own rate, the coefficients

$-\sum_l \sigma_l \log \xi_l$ for a side read in its monomials and the logarithm of the growth of its continued seam function for a side read in its seam letters; at width one, where the share side has one free share, (iv) continues both families of the node, as Theorem 19.32(iii) does.

Expanding each active letter L, with its exponent e_L, in some of its terms, $L = Y_L + X_L$ with $X_L = \sum_k X_{L,k}$ the chosen terms, gives the multiple lattice whose terms are $(-1)^{|\mathbf{n}_L|}(e_L)_{|\mathbf{n}_L|}/\mathbf{n}_L!$ times the periods of $\alpha^{\mathbf{m-1}} \prod_L Y_L^{-e_L-|\mathbf{n}_L|} \prod_k X_{L,k}^{n_{L,k}}$, positive words up to sign. Summed term by term in absolute value, it converges absolutely exactly when $X_L \le Y_L$ on the orthant for every active letter and the integral of $\alpha^{\mathbf{m-1}} \prod_L (Y_L - X_L)^{-e_L}$ over a section is finite (Proposition 2.10). This needs the Newton polytope of X_L inside that of Y_L, so it never holds for a letter of a table, all of whose monomials are vertices of its Newton polytope: for families of tables the lattices close only in the order of a direction.

29.2 Closing along a path

The closings of Theorems 29.2 and 29.3 move the contour of the share fibre along a ray, collect a fibre integral of residues at every polar hyperplane crossed, and show that what is left tends to zero. The two steps come apart. Moving the contour is exact at every step: the period is the sum of the crossing terms met so far and of the fibre integral at the current point, by Cauchy's theorem on bounded contours. Only the last step is a limit, and it asks for a bound on the remaining fibre integral at the points that the path passes through, not for a domination of the double family of crossings and fibres, which is what interchanging the sum over the crossings with the fibre integrals would ask, and what fails past the walls. The step is Lemma 29.4, with three consequences: a closing depends on its direction only through its chamber (Theorem 29.6); the laws can be closed one source point at a time and averaged (Theorem 29.7); and past the walls of a piece the bound that travels is read on the outline of a keyhole continuation (Theorem 29.11). The bounds multiply over products, survive averaging and pass to the crossing terms (Proposition 29.13).

Lemma 29.4 (Closing along a path)**.** *Let A be the affine space of the real points of the real pins of a fibre integral and V its space of directions, of dimension $n \ge 1$, with the measure $d\boldsymbol{\vartheta}$, the fibre integral through a real point p being $\int_{p+iV} I$ (Definition 4.1) wherever it converges absolutely; and let $\Omega \subseteq A$ be open. Let I be meromorphic on the tube $\Omega + iV$, with its poles on the complex hyperplanes $\{\ell_H = 0\}$ of a family $\mathfrak{H}$ of real hyperplanes $H = \{\ell_H = 0\}$ of A, locally finite in Ω, each ℓ_H real, affine and not constant. Let $\gamma : [0,\infty) \to \Omega$ be continuous and piecewise linear, with finitely many breaks on every bounded interval and $\gamma(0) = \mathbf{c}$ off $\bigcup \mathfrak{H}$, and assume:*

(H0) Clean crossings. *γ meets $\bigcup \mathfrak{H}$ at a discrete set of times, its* crossings*, none of them a break of γ; at a crossing τ^* the hyperplanes of $\mathfrak{H}$ through $\gamma(\tau^*)$ are one hyperplane H, and $\ell_H \circ \gamma$ changes sign at τ^* with a nonzero slope.*

(H1) Local decay. *For every compact $Q \subset \Omega$ and $\delta > 0$ there are $C, c > 0$ with $|I(x+i\boldsymbol{\vartheta})| \le Ce^{-c|\boldsymbol{\vartheta}|}$ whenever $x \in Q$ and $x + i\boldsymbol{\vartheta}$ lies at distance at least δ from every hyperplane $\{\ell_H = 0\}$ of the tube.*

(H2) A travelling bound. *There are times $\tau_1 < \tau_2 < \cdots$ tending to infinity, none a crossing, at which*

$$\beta(\tau_k) = \int_V \big|I(\gamma(\tau_k) + i\boldsymbol{\vartheta})\big|\, d\boldsymbol{\vartheta}$$

is finite and tends to zero.

Then for every k

$$\int_{\mathbf{c}+iV} I = \sum_{\tau^*<\tau_k} \mathcal{R}_{\tau^*} + \int_{\gamma(\tau_k)+iV} I, \qquad \Big|\int_{\gamma(\tau_k)+iV} I\Big| \le (2\pi)^{-n}\beta(\tau_k),$$

so that $\int_{\mathbf{c}+iV} I = \lim_k \sum_{\tau^*<\tau_k} \mathcal{R}_{\tau^*}$. *The* crossing term *at a crossing τ^* of the hyperplane H is*

$$\mathcal{R}_{\tau^*} = \pm(2\pi)^{1-n} \int_{V_H} \mathrm{Res}_{w=0}\, I\big(\gamma(\tau^*) + w\,\mathbf{e} + i\boldsymbol{\vartheta}'\big)\, d\mu_H(\boldsymbol{\vartheta}'),$$

with $V_H \subset V$ the kernel of the linear part ℓ_H^{lin} of ℓ_H, μ_H the measure on V_H with $d\boldsymbol{\vartheta} = d\mu_H\, d\ell_H^{\mathrm{lin}}(\boldsymbol{\vartheta})$, $\mathbf{e} \in V$ any vector with $\ell_H^{\mathrm{lin}}(\mathbf{e}) = 1$, on which the term does not depend, and the sign $+$ when ℓ_H decreases along γ and $-$ when it increases. The residue is that of the whole integrand, a Laurent coefficient where several of its factors are polar on H, and the crossing term is its fibre integral, pinned also by $\ell_H = 0$. If some τ_k lies between any two consecutive crossings and $\sum_k \beta(\tau_k) < \infty$, the crossing terms are absolutely summable, a crossing between τ_k and τ_{k+1}, the only one there, having $|\mathcal{R}_{\tau^*}| \le (2\pi)^{-n}\big(\beta(\tau_k) + \beta(\tau_{k+1})\big)$.

Theorems 29.2 and 29.3 are the lemma along a ray: (H0) holds off a null set of starting points, (H1) from the exponential decay of the Gamma factors in the imaginary directions (Theorem 2.11 at one scale, carried to negative real parts by the shift of Proposition 1.3(iii)), and (H2) from the largest values of the monomials on the supports of the laws or from the growth of the continued seam function. Two closings of one fibre integral along two paths agree at all times up to their remainders,

$$\sum_{\tau^*<\tau_k} \mathcal{R}_{\tau^*} - \sum_{\tau'^*<\tau'_k} \mathcal{R}'_{\tau'^*} = \int_{\gamma'(\tau'_k)+iV} I - \int_{\gamma(\tau_k)+iV} I,$$

and one of them may converge while the other does not. Each crossing term is a fibre integral of one dimension less, of an integrand of the same kind, which the lemma can close again along a path of its own when a bound travels with it there; this is how multiple lattices arise.

Definition 29.5 (Chambers of a closing)**.** Let I be meromorphic on the tube $\Omega + iV$ of Lemma 29.4 over an open convex $\Omega \subseteq A$, with its poles on the complex hyperplanes on which one of finitely many real affine functions ℓ on A, not constant, lies in $-\mathbb{N}$: its *polar functions*. A direction $\boldsymbol{\sigma} \in V$ *falls* a polar function ℓ when $\partial_{\boldsymbol{\sigma}}\ell < 0$, at the *falling rate* $-\partial_{\boldsymbol{\sigma}}\ell$. For a real point $\mathbf{c} \in \Omega$ at which every polar function is positive, the *chamber* $C(\boldsymbol{\sigma})$ of a direction $\boldsymbol{\sigma}$ whose ray $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \ge 0$, lies in Ω is the set of the $\boldsymbol{\sigma}' \in V$ whose ray lies in Ω and which fall exactly the polar functions that $\boldsymbol{\sigma}$ falls, each at the same falling rate; it is convex. In Theorem 29.2 the polar functions are the shares of the active letters, and Ω is the interior of the set of the real points of the pins at which the seam functions of the piece and of the partner converge, their reduced periods in Theorem 27.11, which is convex by Hölder's inequality and contains the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \ge 0$, from a point $\mathbf{c} \in \Omega$ whenever $\sup \mathbf{V}_B^{-\boldsymbol{\sigma}}$ and $\sup \mathbf{V}_R^{-\boldsymbol{\sigma}}$ are finite. In Theorem 29.3(iii) they are the shares and the walls of the seam word, read as affine functions of the shares through their sums (Theorem 29.3(ii)), and $\Omega = A$. In a fibre integral of Gammas and coefficients, as the share side of a Gramma (Proposition 4.2) and the kernel of Theorem 29.7, they are the shares and $\Omega = A$, so that

$$C(\boldsymbol{\sigma}) = \{\boldsymbol{\sigma}' \in V :\ \sigma'_l = \sigma_l \text{ where } \sigma_l < 0,\ \ \sigma'_l \ge 0 \text{ elsewhere}\},$$

a compact convex polytope when the pins fix the total of the shares of every letter; the walls of source factors that are cube integrals (Definition 29.16) are polar functions as well.

Theorem 29.6 (A closing depends only on its chamber)**.** *Let I, $\mathbf{c}$ and $\boldsymbol{\sigma}$ be as in Definition 29.5, let I have the local decay (H1) of Lemma 29.4 on Ω, and let $\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})$. The rays $\mathbf{c}+\tau\boldsymbol{\sigma}$ and $\mathbf{c}+\tau\boldsymbol{\sigma}'$ meet the same polar hyperplanes at the same times, and (H0) holds along one of them exactly when it holds along the other; at every other time the fibre integrals through their two points are equal, and at a crossing their crossing terms are equal. So the closings along the two directions are one series, with the same remainders: the closing along $\boldsymbol{\sigma}$ converges, with the bounds of its terms and of its remainders, as soon as the closing along some direction of its chamber does. In Theorem 29.2, or in Theorem 29.3(iii), if $\mathbf{c}$ lies off the null set of Theorem 29.2, taken for Theorem 29.3(iii) with the polar hyperplanes of the seam function as well, the closing along $\boldsymbol{\sigma}$ converges absolutely as soon as*

$$\inf_{\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})} h(\boldsymbol{\sigma}') < 0,$$

h the rate of Theorem 29.2, the infimum being taken for Theorem 29.3(iii) over the rates h_θ, $\theta \in [0, \pi/8)$, as well; its terms and the integrals left after them are then $O(e^{\tau(h'+\epsilon)})$ for every $\epsilon > 0$, h' that infimum. On the chamber the rate h of Theorem 29.2 is convex.

So the closing along a direction converges as soon as some direction of its chamber is in the cone of Theorem 29.2, and its best certified rate is a convex problem in the free coordinates of the chamber. A chamber has positive dimension when the shares that do not fall have free directions on the pins, which stations of one excess supply. Theorem 29.2 reads both laws along one direction, through the largest values of the monomials of their letter vectors. The glue is also the average over the laws of the stand-in kernel (Theorem 27.9(ii)), whose integrand at one source pair has no walls, and, with the partner read in its monomials, the average over the law of the piece alone of an integrand without walls; each source point can therefore close along its own best direction of the chamber, and the averages of the crossing terms close the glue.

Theorem 29.7 (Averaging the laws)**.** *Take a share side, on pins that fix the total of the shares of every active letter, with V the space of the directions of the shares on the pins, whose integrand is*

$$\prod_{j\colon s_j>0} \frac{\prod \Gamma(a_l)}{\Gamma(s_j)} \prod_{j\colon t_j>0} \frac{\prod \Gamma(a_l)}{\Gamma(t_j)} \int e^{-\langle \mathbf{a}, y(\zeta)\rangle}\, d\lambda(\zeta),$$

the products over the shares l of K_j and of Ψ_j, $\mathbf{a}$ the vector of all the shares, λ a positive measure on a source, the source measure*, whose points are the* source points *ζ, and $y = (y_l)_l$ a continuous map of the source to the space of the shares. In Theorem 27.9 two share sides are of this kind, their shares l those of the active letters:*

- *(A) that of Theorem 27.9(iii), with one share per station; $\lambda = (\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho \otimes (\boldsymbol{\varrho}_R^\rho)_* \phi_R^\rho$, a source point $\zeta = (\mathbf{x}, \mathbf{x}')$ is a seam point of the piece and one of the partner, and $y_l(\zeta) = \log(x_{jD} x'_{jD})$ or $\log(x^*_{jD} x'^*_{jD})$ for the share of the station D of the letter;*
- *(B) that of Theorem 27.11 with the partner read in its monomials, one share per monomial, written as in Theorem 29.3(i), the seam word of the piece having a Newton sum of full dimension; $\lambda = (\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$ is the law of the piece, a source point is a seam point $\zeta = \mathbf{x}$ of the piece, and $y_l(\zeta) = \log x_{jD} + \log \xi_l$ or $\log x^*_{jD} + \log \xi_l$ for the share of the monomial of the coefficient ξ_l at the station D.*

Let S_λ be the support of the image of λ under y, the log-support *of the source measure, $\mathbf{c}$ a real point of the pins with positive coordinates at which $\int e^{-\langle \mathbf{c}, y\rangle}\, d\lambda$ converges (in (A) the seam functions of the piece and of the partner converge there, in (B) that of the piece), $\boldsymbol{\sigma} \in V$, and $C = C(\boldsymbol{\sigma})$ the*

chamber of Definition 29.5 of the kernel below, whose polar functions are the shares:

$$C = \{\boldsymbol{\sigma}' \in V :\ \sigma'_l = \sigma_l \text{ where } \sigma_l < 0,\quad \sigma'_l \geq 0 \text{ elsewhere}\},$$

a compact convex polytope. Two shares are in one bundle *when they belong to one letter and have one column in the pins: in (A) when their stations have one excess, in (B) when moreover they carry one monomial of the partner. For y in the space of the shares put*

$$h(\boldsymbol{\sigma}'; y) = \sum_l \sigma'_l \log|\sigma'_l| - \langle \boldsymbol{\sigma}', y\rangle, \qquad h_C(y) = \min_{\boldsymbol{\sigma}' \in C} h(\boldsymbol{\sigma}'; y), \qquad \hat{h}_C = \sup_{y \in S_\lambda} h_C(y) \in (-\infty, \infty],$$

the rate *of $\boldsymbol{\sigma}'$ at y, the* chamber rate *of the kernel at y, concave in y, and the* averaged rate.

Let $\mathbf{c}$ be interior, relative to the pins, to the set of the real points at which the Laplace transform $\int e^{-\langle \mathbf{a}, y\rangle}\, d(y_*\lambda)(y)$ of $y_*\lambda$ converges; equivalently, for a basis $\mathbf{v}_1, \dots, \mathbf{v}_d$ of V there is $\eta > 0$ with

$$\sum_{\boldsymbol{\epsilon} \in \{-1,1\}^d} \int \exp\Big(- \Big\langle \mathbf{c} - \eta \sum_i \epsilon_i \mathbf{v}_i,\, y \Big\rangle \Big)\, d(y_*\lambda)(y) < \infty, \tag{29.1}$$

a finite certificate that $\mathbf{c}$ is interior to the set where that Laplace transform converges, that set being convex. Put $|y|_V = \sum_i |\langle \mathbf{v}_i, y\rangle|$, let p be the largest total order of the Gamma factors polar on one hyperplane that the ray of $\boldsymbol{\sigma}$ crosses, at most their number, and in this statement write $w(y) = e^{-\langle \mathbf{c}, y\rangle}(1 + |y|_V)^{p-1}$. Every power of $1 + |y|_V$ is at most a constant times $e^{\eta |y|_V} \leq \sum_{\boldsymbol{\epsilon}} \exp(\eta \sum_i \epsilon_i \langle \mathbf{v}_i, y\rangle)$, so (29.1) gives $\int w\, d(y_*\lambda) < \infty$.

(i) The kernel. *The* kernel *at a source point ζ is the fibre integral on the pins*

$$\int_{pins} \prod_{j:\, s_j > 0} \frac{\prod \Gamma(a_l)}{\Gamma(s_j)} \prod_{j:\, t_j > 0} \frac{\prod \Gamma(a_l)}{\Gamma(t_j)}\ e^{-\langle \mathbf{a}, y(\zeta)\rangle},$$

of an integrand without walls, and the fibre integral of the share side is its integral against λ, which in (A) and (B) is $P_{B \sqcup R}(\mathbf{m})$; in (A) the kernel is $\Gamma(\kappa_B)\, \mathsf{Z}_{(\kappa_B, \mathbf{s}, \mathbf{t}); t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{x}')$ of Theorem 27.9(ii). Let $\mathbf{c}$ lie off the null set of Theorem 29.2 for the direction $\boldsymbol{\sigma}$. For every $\epsilon > 0$ there is C_ϵ, not depending on the source point, such that the crossing terms $\mathcal{R}_k(\zeta)$ of the closing of the kernel along $\boldsymbol{\sigma}$, at their crossing times τ_k, satisfy

$$|\mathcal{R}_k(\zeta)| \leq C_\epsilon\, w(y)\, e^{\tau_k (h_C(y) + \epsilon)}, \qquad y = y(\zeta),$$

and the integrals left at times separating the crossings satisfy the same bound at the time τ with $e^{-\langle \mathbf{c}, y\rangle}$ in place of $w(y)$. When every crossed hyperplane carries a simple pole, $p = 1$ and $w(y) = e^{-\langle \mathbf{c}, y\rangle}$.

(ii) The averaged closing. *If $\hat{h}_C < 0$, the fibre integral of the share side is*

$$\sum_k \int \mathcal{R}_k\, d\lambda,$$

the sum in the order of crossing, absolutely convergent, its terms $O(e^{\tau_k(\hat{h}_C + \epsilon)})$ for every $\epsilon > 0$. At a crossing whose point along some $\boldsymbol{\sigma}' \in C$ lies in the interior of the set where $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ converges, the averaged term is the crossing term of the share side there; so when it converges along the ray of $\boldsymbol{\sigma}$, the averaged closing is the closing of the share side along $\boldsymbol{\sigma}$, in (A) and (B) that of Theorem 29.2.

(iii) One direction reads the convex hull. $\min_{\boldsymbol{\sigma}'\in C}\sup_{y\in S_\lambda} h(\boldsymbol{\sigma}';y)=\sup_{y\in\operatorname{conv} S_\lambda} h_C(y)\geq\hat h_C$, *and* $\sup_{y\in S_\lambda} h(\boldsymbol{\sigma}';y)$ *is the rate of the closing of the share side along* $\boldsymbol{\sigma}'$ *read through the largest value of* $e^{-\langle\boldsymbol{\sigma}',y\rangle}$ *on the log-support, in (A) and (B) the rate* $h(\boldsymbol{\sigma}')$ *of Theorem 29.2: the best single direction of the chamber reads the convex hull of the log-support, the averaged closing the log-support itself. In a bundle* G *of shares that* $\boldsymbol{\sigma}$ *does not fall, the least value of* $\sum_{l\in G}(\sigma'_l\log\sigma'_l-\sigma'_l y_l)$ *over the* $\sigma'_l\geq 0$ *with* $\sum_G\sigma'_l=g$ *is* $g\log g-g\log\sum_{l\in G}e^{y_l}$*, attained at* $\sigma'_l=g\,e^{y_l}/\sum_{l'\in G}e^{y_{l'}}$*: at each source point the best direction merges the bundle into the sum* $e^{y_G}=\sum_{l\in G}e^{y_l}$*, which in (A) is the sum* $\sum_D x_{jD}x'_{jD}$ *that the kernel reads at the excess, splitting* g *in the proportions of its terms.*

(iv) Sharpness. *Let the pins be rational, as in (A) and (B), let* $\boldsymbol{\sigma}$ *have rational coordinates, let the crossings of its ray be simple and nonresonant (Definition 29.16), and let* $\hat h_C<\infty$*. Along each cohort of crossings, with its integer step of length* Δ *in* τ*, the averaged terms have one sign at each index and* $\lim_{n\to\infty}\left|\int\mathcal{R}_n\,d\lambda\right|^{1/n}=e^{\Delta\hat h_C}$ *(Theorem 29.26, Corollary 29.27); so for every* $\epsilon>0$*,* $\left|\int\mathcal{R}_k\,d\lambda\right|\geq e^{\tau_k(\hat h_C-\epsilon)}$ *for all large* k*, and the terms of the averaged closing, which are those of the closing of the share side along* $\boldsymbol{\sigma}$ *when* $\int e^{-\langle\mathbf{a},y\rangle}\,d\lambda$ *converges along it, tend to zero when* $\hat h_C<0$ *and do not when* $\hat h_C>0$*. When every share that* $\boldsymbol{\sigma}$ *falls is alone in its bundle and the share side read with one share per bundle (Corollary 4.4 merging each bundle) has a fibre of dimension one, a positive multiple of* $\boldsymbol{\sigma}$ *has a direction with rational coordinates in its chamber, along which the closing is the same series after a change of scale of* τ *(Theorem 29.6); the crossings are simple and nonresonant as soon as no two lattices of the falling shares meet; and every crossing term is*

$$\mathcal{R}_k(\zeta)=r_k\prod_G e^{-\bar a_{k,G}\,y_G(\zeta)},\qquad y_G=\log\sum_{l\in G}e^{y_l},$$

with r_k *real and not depending on the source point and* $\bar{\mathbf{a}}_k$ *the crossing point of the merged shares.*

At width one the share side of a node merged by its bundles has at most one free share and the seam torus is a half-line, so the log-support is an interval, its own hull, and one direction loses nothing. A gap between the averaged rate and the best single direction needs a chamber of positive dimension and a log-support that is not convex in its directions: the best split at each source point moves with it, and one direction must take one split for all of them. Averaging can reach a rate that no single direction certifies, and an averaged closing is in general no closing of one share side along one ray: its terms are averages of the crossing terms of the kernel, and it can converge past walls that every ray of the chamber meets. Where the averaged rate is infinite, a closing past the walls can still converge, at the rate of the best source point: past its walls the law is read through its expansion at its falling end, not through its moments, as at width one (Theorem 19.42). Past the walls of a piece the bound that travels is read on the sector letters (Theorem 29.11).

Theorem 29.8 (An averaging gap at a family of tables)**.** *There are three tables with the bases* $\binom{\{1,2,3,4\}}{2}$*, at the exponents* $s_j=1$*,* $t_j=0$ *and the masses* $\frac{3}{2}$ *on every mark, whose bunch* $\{1,2\}$*, with the room* $\{3,4\}$*, read through its seam with the product of the two laws as the source measure, the kind (A) of Theorem 29.7, has one station at each excess of each letter, so that every bundle of Theorem 29.7 is a single share, and a chamber* C *with*

$$\min_{\boldsymbol{\sigma}\in C}\sup_{y\in S_\lambda} h(\boldsymbol{\sigma};y)=0,\qquad \hat h_C=\sup_{y\in S_\lambda}\min_{\boldsymbol{\sigma}\in C} h(\boldsymbol{\sigma};y)<0:$$

no direction of the chamber certifies the closing (Theorem 29.2), and the averaged closing converges geometrically (Theorem 29.7(ii)). Two directions of the chamber, the better one taken at each source point, already give the bound.

Remark 29.9 (What the gap separates). The gap lies between the averaged closing and the directions of one share side, the seam of the bunch with its bundles merged, and no merging of stations removes it, every bundle being a single share. It is not a gap against every closing of the period: closing each source point along its own direction, as the bound does with two directions, is itself a closing of the period, and so is completing another inner read before closing. A gap in every closing is a statement about a specified set of closings; for the closings of one share side by the directions of its chamber, the family of Theorem 29.8 has one, while at a seam whose two stations of one excess multiply the same monomial of the partner, merging the two stations can remove such a gap.

Definition 29.10 (Keyhole continuations along a direction; ceiling and phase)**.** In Theorem 29.3 let $\boldsymbol{\sigma}$ be a direction of the shares and $\bar{\boldsymbol{\sigma}}$ its sums over the monomials of the partner, and fix a fan of the seam word (Theorem 29.3(ii)). For a sector π of it let $q = (q_k)_k$ be its sector coordinates, $\kappa_k = \kappa_{\mathbf{v}_{\pi,k}}$ the walls of its generators and $\mathbf{k}_\pi = (k_{jD,\pi}, \psi_{jD,\pi})_{j,D}$ the vector of its sector letters, so that the seam function on the pins is the sum over the sectors of

$$|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \int_{[0,1]^{|B|-1}} \prod_k q_k^{\kappa_k(\bar{\mathbf{a}},\bar{\mathbf{b}})-1}\, \mathbf{k}_\pi(q)^{-(\bar{\mathbf{a}},\bar{\mathbf{b}})}\, dq,$$

the walls affine in the shares; for a piece of tables and the sectors of the orders π of B these are the walls κ_{U_k} of the tails and the sector letters of the tables $\mathcal{B}_{jD}$ (Theorem 8.7(ii)). Put $\dot\kappa_k = \partial_{\bar{\boldsymbol{\sigma}}}\kappa_k$, the derivative of the wall along $\bar{\boldsymbol{\sigma}}$, and

$$\varrho_{\pi,\bar{\boldsymbol{\sigma}}}(q) = \prod_k q_k^{\dot\kappa_k}\ \mathbf{k}_\pi(q)^{-\bar{\boldsymbol{\sigma}}},$$

the *directional ratio in the sector* of π; its *falling variables* along $\bar{\boldsymbol{\sigma}}$ are the q_k with $\dot\kappa_k < 0$. A *keyhole continuation along* $\bar{\boldsymbol{\sigma}}$ is defined as in Definition 19.45, with the sectors in place of the orders, with these falling variables, and with the sector letters of $\mathbf{k}_\pi$ required not to vanish on the sets Y of its boxes; its *outline* $\mathcal{O}$ is the union of the sets $\mathcal{O}$ of its boxes, and its *ceiling along* $\bar{\boldsymbol{\sigma}}$ is the largest value of $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}|$ on the outline, π the sector of the box. Each set Y is a product of a box and of keyholes, simply connected, and on a neighbourhood of it the sector letters have logarithms real at the real points; the *phase* of the continuation at a point q of the outline of a box of π is the linear function $\boldsymbol{\vartheta} \mapsto \langle \bar{\boldsymbol{\vartheta}}, \arg \mathbf{k}_\pi(q)\rangle$ on the directions of the shares, arg the imaginary part of the logarithm and the bars marking sums over the monomials of the partner.

The modulus $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}(q)|$ is the factor by which the part of the integrand that grows with τ along $\bar{\boldsymbol{\sigma}}$ is multiplied per unit of τ at the point; the phase is the linear function by which the imaginary parts of the shares enter it. With no falling variables the outline is the real cube, the letters are positive there, the phase vanishes, and the ceiling is $\sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$, the largest value of the monomial of Theorem 29.2; at width one, along the fall of the share b^d of Section 19.7, the directional ratio in the sector is the ratio ϱ_π of Definition 19.44.

Theorem 29.11 (The travelling bound past the walls)**.** *In Theorem 29.3 let $\boldsymbol{\sigma}$ be a direction of the shares and take a keyhole continuation along $\bar{\boldsymbol{\sigma}}$ (Definition 29.10) with the outline $\mathcal{O}$.*

(i) For every $\epsilon > 0$ there is C such that for $\tau \geq 0$ and $\boldsymbol{\vartheta}$ in the directions of the shares

$$\big|P_{W_B(\bar{\mathbf{c}}+\tau\bar{\boldsymbol{\sigma}}+i\bar{\boldsymbol{\vartheta}})}(\mathbf{m}_B)\big| \leq C\,\delta^{-(|B|-1)} \sup_{q\in\mathcal{O}} \Big(|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}(q)|^{\tau}\, e^{\tau\epsilon+\langle\bar{\boldsymbol{\vartheta}},\arg \mathbf{k}_\pi(q)\rangle}\Big),$$

δ the smaller of 1 and the distance of $\bar{\mathbf{c}}+\tau\bar{\boldsymbol{\sigma}}$ from the hyperplanes on which a wall of the fan of Definition 29.10 lies in $-\mathbb{N}$, which hold the poles of the seam function.

(ii) Let $\boldsymbol{\upsilon}$ be a bounded function on $\mathcal{O}$ with values in the vectors indexed by the shares, with $\langle\bar{\boldsymbol{\vartheta}},\arg \mathbf{k}_\pi(q)\rangle = \sum_l \upsilon_l(q)\vartheta_l$ for $\boldsymbol{\vartheta}$ in the directions of the shares and $|\upsilon_l(q)| \leq \frac{\pi}{2} - \epsilon_0$ for some $\epsilon_0 > 0$; representatives differing by a vector orthogonal to the directions give the same phase on the fibre. Put

$$h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \sup_{q\in\mathcal{O}} \Big(\log|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}(q)| - \sum_{l:\,\sigma_l>0} \sigma_l \log\cos\upsilon_l(q)\Big),$$

h_Γ the Gamma rate of Theorem 29.3(iii). If $h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma}) < 0$ and the real point $\mathbf{c}$ of the pins lies off the null set of Theorem 29.2, taken with these hyperplanes as well, the conclusion of Theorem 29.3(iii) holds along $\boldsymbol{\sigma}$, its terms and the integrals left after them being $O(e^{\tau(h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma})+\epsilon)})$ for every $\epsilon > 0$.

Corollary 29.12 (When the phase costs nothing)**.** *In Theorem 29.11(ii) let $H_\mathcal{O}$ be the ceiling of the continuation along $\bar{\boldsymbol{\sigma}}$. Then $h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log H_\mathcal{O}$ in each of two cases:*

(i) $\upsilon_l(q) = 0$ for every $q \in \mathcal{O}$ and every share with $\sigma_l > 0$: the phase sits on shares that fall or stay, whose Gammas absorb it;

(ii) $-\sum_{l:\,\sigma_l>0} \sigma_l \log\cos\upsilon_l(q) \leq \log H_\mathcal{O} - \log|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}(q)|$ for every $q \in \mathcal{O}$: the modulus lost away from the highest points pays for the phase.

With no falling variables (i) holds with $\boldsymbol{\upsilon} = 0$, and the rate is that of Theorem 29.2 for the glue read in the monomials of the partner.

Theorem 29.3(iii) pays for an allowed growth $e^{\theta|\operatorname{Im}\mathbf{z}|_1}$ of the seam function with the term $\frac{2\theta^2}{\pi}\sum_{\sigma_l>0}\sigma_l$; Theorem 29.11 pays only for the phase that the letters have at the points where the ceiling is read, and at the exact price $-\log\cos\upsilon_l$ of the rotation. Enlarging the radii of the keyholes, as far as the sector letters do not vanish, can lower the ceiling; it comes down to the reach at simple tops that do not cancel (Theorem 30.15(iii)). The joint words of Section 29.3 complete the retained fibre on its strips before a modulus is taken, and their bound $r_G c_{\max}$ has no phase term; the ceiling $c_{\max}$ of the joint word need not be the ceiling of this outline.

Proposition 29.13 (How travelling bounds compose)**.** *(i)* Products. *When the share integrand carries several continued factors, each bounded as in Theorem 29.11(i) on its own outline, their product is bounded on the product of the outlines by the products of their moduli and of the exponentials of their phases, and Theorem 29.11(ii) holds with these on the product outline: the phases of different factors at one point of it may cancel.*

(ii) Averaging. *Let integrands I_ζ, for ζ in a source with a measure λ, have fibre integrals that close along one path with their crossings at the same times, with the crossing terms $\mathcal{R}_k(\zeta)$, and with crossing terms and remainders, at times τ_k separating the crossings, at most $M(\zeta)\beta(\tau_k)$, where M is integrable against λ and $\sum_k \beta(\tau_k) < \infty$. Then $\int \big(\int_{\mathbf{c}+iV} I_\zeta\big)\, d\lambda(\zeta) = \sum_k \int \mathcal{R}_k\, d\lambda$, absolutely convergent, the k-th term at most $\|M\|_1\beta(\tau_k)$. When for some $\hat{h} < 0$ the bound holds with $\beta(\tau) = e^{\tau(\hat{h}+\epsilon)}$ for every small $\epsilon > 0$, M depending on ϵ, the terms are $O(e^{\tau(\hat{h}+\epsilon)})$:*

the rate is the bound $\hat{h}$ shared by the source points, read on the support of λ, the uniformity over the source being the hypothesis on M. Where the fibre integral and the source integral of a crossing term converge absolutely together, $\int \mathcal{R}_k \, d\lambda$ is the crossing term of the averaged integrand $\int I_\zeta \, d\lambda$.

(iii) Crossing. *The crossing terms of the closings of Theorems 29.3(iii) and 29.11 are fibre integrals of one dimension less of integrands of the same kind. At a share crossing $a_l = -n$ the Gamma becomes $(-1)^n/n!$ and the other factors are read on the hyperplane. At a wall crossing where the pole of the seam function is simple, its residue is the* face coefficient *of the walls through the hyperplane: the sum, over the generators $\mathbf{v}$ of a fan of the seam word (Theorem 29.3(ii)) whose wall $\kappa_{\mathbf{v}}$ takes a value $-n_{\mathbf{v}} \in -\mathbb{N}$ on the hyperplane and over the sectors π of which they are generators, of $|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})|$ times the cube integral over the other coordinates of the sector of the $n_{\mathbf{v}}$-th Taylor coefficient at $q_{\mathbf{v}} = 0$ of the rest of the integrand (Lemma 8.6), $q_{\mathbf{v}}$ the sector coordinate of $\mathbf{v}$, each divided by the derivative of $\kappa_{\mathbf{v}}$ along the vector $\mathbf{e}$ of Lemma 29.4; its letters are the sector letters at $q_{\mathbf{v}} = 0$, positive on the face, and its numerators are polynomials in the shares and in the derivatives of the letters. For a piece of tables and the sectors of the orders the generators are those of the tails U, and the sectors the orders of which they are tails. Where the pole is multiple, because several factors, or two walls of one sector, are polar on the hyperplane, the residue is a Laurent coefficient, a finite sum of derivatives in the shares of products of such terms, which brings powers of the logarithms of the letters and of the q_k. A keyhole continuation of the seam function restricts to one of each face coefficient, $q_{\mathbf{v}} = 0$ lying in the disc of the keyhole of $q_{\mathbf{v}}$, and Cauchy's estimate on its circle bounds the Taylor coefficients by the values on the outline; so each crossing term can be closed again by Lemma 29.4 whenever a travelling bound of its own tends to zero.*

Completing an inner read is not a crossing. Merging the stations of one excess by Corollary 4.4, integrating a source before closing, or finishing a transverse fibre replaces the integrand by a completed one, with a smaller fibre; the identity between the two integrands holds where the completed read converges and persists under continuation, but the travelling bound does not pass from one to the other, and the completed object can grow along the outer path where the uncompleted integrand does not. Completing an inner read is free when the completed object obeys, along the outer path, a bound already in hand. Merging the stations of one excess reads the law through the sum of their seam letters, whose expansion at the falling end of the law can stop at an edge of the support nearer than the far edge that the stations read, and the merged closing can then diverge where the station closing converges. The two closings agree at every step up to their remainders (Lemma 29.4), and only the remainder of the station closing need tend to zero. On the shares, merging the stations is conditioning on their merged seam letter: integration commutes with completing the fibre, and continuation commutes with it only while the level sets of the merged seam letter stay inside the region where the expansion of the law at its falling end represents it. Completing the inner fibre before closing and keeping its inherited cut are two representations, whose finite heads agree together with their boundary words; the regions on which a law equals its expansion, with their cuts and boundary words, are built in Section 30.1, and a convergent germ alone does not equal the law.

29.3 Retained crossings as joint words

29.3.1 Completing the retained fibre before taking its size

Theorem 29.11 bounds the share integrand before its fibre is completed, and pays for the phase of the letters where it reads the ceiling. The procedure for joint words completes the retained shares first. It keeps the sectors and coordinate cuts of every continued source and its finite heads, and its auxiliary variables are part of one joint input of a Trailmoment (Definition 5.1), their number not growing along a retained cohort (Definition 29.16): a falling Gamma kept inside a crossing term becomes a positive share and one Beta variable, and a continued source strip becomes a segment and a few circle integrals whose powers on the circles have integral exponents.

Lemma 29.14 (A Gamma strip as a Beta input)**.** *Let $\delta \geq 1$ be an integer, $b = a + \delta$, and $0 < \operatorname{Re} b < 1$. For a nonzero coefficient ξ, with one fixed logarithm,*

$$\Gamma(a)\,\xi^{-a} = \frac{(-1)^{\delta}\xi^{\delta}}{\Gamma(\delta)} \int_0^1 (1-u)^{\delta-1}\,\Gamma(b)\,(\xi u)^{-b}\,du.$$

The identity is pointwise in every remaining share. So on a fibre $P\mathbf{a} = \boldsymbol{\pi}$ replacing a_l by $b_l = a_l + \delta_l$ changes the pins to $P\mathbf{b} = \boldsymbol{\pi} + P(\delta_l)_l$ and does not move the contour. A share with positive real part is left unchanged, with $\delta_l = 0$ and no auxiliary variable. In a family $\operatorname{Re} a_l(n) = a_l^0 - d_l n$ with $d_l \in \mathbb{N}$ and $a_l^0 \notin \mathbb{Z}$ one may take $\delta_l(n) = d_l n + \delta_l(0)$ with $0 < a_l^0 + \delta_l(0) < 1$, the lift being used at the n with $\delta_l(n) \geq 1$, all but finitely many when $d_l \geq 1$: one Beta input serves at every n.

Lemma 29.15 (A continued source strip in six integrals)**.** *Let g be holomorphic near $\{|q| \leq r_2\} \cup [r, 1]$, and choose $0 < r < r_1 < rT < r_2$ and $T > 1$. If $\ell \geq 1$ is an integer and $0 < \operatorname{Re}(\kappa + \ell) < 1$, the continued integral $I_\kappa = \operatorname{Reg} \int_0^1 q^{\kappa-1} g(q)\,dq$ (Definition 1.4) is*

$$I_\kappa = \int_r^1 q^{\kappa-1} g(q)\,dq + R_\ell + R'_\ell,$$

all circles run counterclockwise, with

$$R_\ell = \frac{1}{2\pi i} \int_0^r q^{\kappa+\ell-1} \oint_{|\zeta|=r_1} \frac{g(\zeta)\,\zeta^{-\ell}}{\zeta - q}\,d\zeta\,dq,$$

$$R'_\ell = -\frac{r^\kappa}{2\pi i}\Big(\int_1^T u^{\kappa-1} \oint_{|\zeta|=r_2} + \int_T^\infty u^{\kappa-1} \oint_{|\zeta|=r_1} \Big)\, g(\zeta)\,\frac{1 - (ru/\zeta)^\ell}{\zeta - ru}\,d\zeta\,du.$$

Splitting the numerator of R'_ℓ gives four absolutely convergent integrals, six in all. This remains true with holomorphic parameters and other source variables on compact sets, uniformly when the strip and the radius inequalities have positive margins. For $\kappa(\mathbf{a}, n) = \kappa_0(\mathbf{a}) - dn$ and $\ell(n) = \ell_0 + dn$, $d \in \mathbb{N}$, each of the six has a fixed domain and is a fixed function times an n-th power; the exponent of ζ that contains $\ell(n)$ is integral and independent of the shares that move. At a simple wall $\kappa = -N$, $N \in \mathbb{N}$, the residue is the single coefficient input

$$[q^N]g(q) = \frac{1}{2\pi i} \oint_{|\zeta|=r_1} g(\zeta)\,\zeta^{-N-1}\,d\zeta.$$

A cut upper end $q_0 > 0$ is kept by applying the same formula after $q = q_0 v$, the factor q_0^κ included; a positive lower cut is an ordinary segment throughout.

Two of the six integrals simplify by Cauchy's formula: the integral of R'_ℓ over $[1,T]$ with the numerator 1 is $-\int_r^{rT} q^{\kappa-1}g(q)\,dq$, and the integral over (T,∞) with the numerator 1 vanishes, its pole ru lying outside the smaller circle. So I_κ is the segment $\int_{rT}^1 q^{\kappa-1}g$, the remainder R_ℓ on the circle of radius r_1, and the two integrals of R'_ℓ with the numerator $(ru/\zeta)^\ell$; along a family $\kappa_0 - dn$ the segment carries the n-th power of q^{-d}, at most $(rT)^{-d}$, and the circle integrals that of ζ^{-d}, of modulus r_1^{-d} or r_2^{-d}. The radius r splits the contour, not the source: the sum does not depend on it.

Definition 29.16 (A rational retained cohort)**.** A *retained cohort* is one sequence of simple crossing terms, or one sequence of travelling fibre integrals, of a fibre integral of Gammas, coefficients and source factors whose pins carry a *grading*, a rational vector $\mathbf{v}$ with $\langle t_l, \mathbf{v}\rangle = 1$ for the column t_l of every share in the pins, such as the share side of Theorem 29.3 and the kernel of Theorem 29.7, graded by their letter totals, and the share side of a Gramma whose configuration has a grading (Proposition 4.2), with the following data retained: the pins; the inherited contour; the crossed hyperplane and its orientation, if any; the source sectors and their coordinate cuts. The contours have the real points $\mathbf{a}^0 + n\mathbf{d}$ with $\mathbf{d}$ rational. Nonresonance is tested against the full displayed arrangement of the polar hyperplanes of the Gammas and of the source walls, including candidate walls which might cancel after the sectors are added: the real points lie on none of these hyperplanes other than the one being crossed, and at a source crossing each contributing sector has just one wall on that hyperplane. The constant terms of the shares and walls and the indices are real. Finitely many initial terms may be removed and kept as a finite head. Passing to finitely many arithmetic subsequences makes the changes of all falling shares and walls integral; this is the *integer step* of the cohort, and its index n always refers to that step. The same definition permits a finite product of source factors that are cube integrals in their charts (Lemma 8.6)

$$\operatorname{Reg}\int_{(0,1)^k} q^{\boldsymbol{\kappa}(\mathbf{a})-\mathbf{1}}\prod_i P_i(q)^{-w_i(\mathbf{a})}\,dq,$$

the P_i with nonnegative coefficients and positive constant terms and the affine functions κ_i, w_i with rational linear parts. A growing wall is eventually positive; a stationary wall may be negative and off the integers, and is then continued with a constant integer ℓ in Lemma 29.15. Finite coordinate box cuts are part of the retained data. The sector integrals of a fan of a seam word (Theorem 29.3(ii)), those of tables among them (Corollary 8.8), have these properties.

Proposition 29.17 (Computing the cohorts of a rational ray)**.** *Let the pins and the finitely many candidate polar functions ℓ_i be rational, and let a rational open box in the affine space of the pins be given on which the required initial walls and shares are positive. For a prescribed rational direction a rational starting point in that box can be chosen such that distinct polar hyperplanes never meet the ray at the same time; factors on the same hyperplane are kept together. After a finite head, the crossing order and the separating gaps are a finite repeated pattern, and its crossing cohorts, endpoint cohorts, the integers ℓ of Lemma 29.15 and a uniform positive separation are computed by finite rational arithmetic. More precisely, with the direction multiplied by a common denominator so that every falling polar function decreases at an integer falling rate $d_i > 0$, there are at most $\sum_i d_i$ crossings in each interval of length one of this parameter; neither that number nor the stored source sectors depend on how many such intervals are read.*

Theorem 29.18 (A finite procedure for a retained cohort)**.** *For a rational retained cohort there is a finite procedure, using the pins and the monomials of the source factors, with the following*

output, the joint word *of the cohort, after a finite head and a possible change to the integer step:*

$$\mathcal{R}_n = G_n \sum_i \int_{Y_i} c_i(x)^n \, \phi_i(dx) = G_n \sum_i \big(\mathsf{T}_{-n;(1)}[c_i]\, \phi_i\big)(Y_i), \qquad n \in \mathbb{N}, \tag{29.2}$$

a finite sum over i. Here Y_i is a compact cube times finitely many circles, the coefficient function c_i is continuous on it, and the input ϕ_i is a complex measure of finite total variation; the coefficients, their branches, the orientations and every cut are retained. On every Y_i each coefficient of a letter is a real Laurent monomial times a coefficient factor*, holomorphic there, with its real part bounded below by a positive constant and its modulus bounded above, uniformly on the compact variables; these two bounds are the* margins *of the coefficient factors, and the radii $r < r_1 < rT < r_2$ of Lemma 29.15 are admissible while the margins hold. At the integer index $-n$ the Zamma of the one-term configuration is the polynomial $\mathsf{Z}_{-n;(1)}(c) = c^n$, at $c = 0$ as well, so the second expression is the Trailmoment of Definition 5.1 extended to a continuous complex coefficient function and a complex input of finite variation, as Theorem 5.21(ii) extends its reads to complex data where every integral converges absolutely.*

The scalar is explicit. Let $\mathbf{v}$ be the grading of the pins (Definition 29.16) and $w = \langle \mathbf{v}, \boldsymbol{\nu} \rangle$, $\boldsymbol{\nu}$ the values of the pins, the total of the shares on the pins, which on the share sides of Theorems 29.3 and 29.7 is $|\mathbf{s}| + |\mathbf{t}|$, the sum of the exponents of the active letters. Let $\mathcal{S}_-$ be the shares treated by Lemma 29.14, with $\delta_l(n) = \delta_l(0) + d_l n$. At a Gamma crossing let $-N(n)$, with $N(n) = N_0 + d_0 n$, be the pole of its Gamma that is crossed; at a source crossing or for a travelling fibre omit N and the factor $\Gamma(N+1)$ of G_n, and put $d_0 = 0$. Up to fixed nonzero factors of the share side, as $1/\Gamma(s_j)$ and $1/\Gamma(t_j)$, and the inherited crossing sign,

$$G_n = \frac{(-1)^{N(n)+\sum_{l\in\mathcal{S}_-}\delta_l(n)}\,\Gamma\big(w + N(n) + \sum_{l\in\mathcal{S}_-}\delta_l(n)\big)}{\Gamma(N(n)+1)\prod_{l\in\mathcal{S}_-}\Gamma(\delta_l(n))},$$

fixed nonzero constants may equally be put in the inputs. With $\bar{d} = d_0 + \sum_{l\in\mathcal{S}_-} d_l$,

$$\lim_{n\to\infty} |G_n|^{1/n} = r_G = \frac{\bar{d}^{\,\bar{d}}}{d_0^{d_0}\prod_{l\in\mathcal{S}_-} d_l^{d_l}}, \qquad 0^0 = 1;$$

when $\bar{d} = 0$ the scalar is constant and $r_G = 1$, and in all cases $|G_n| \le C(n+1)^p r_G^n$ for computable C and p. The procedure introduces at most six integrals per continued source wall and one Beta variable per negative retained Gamma, whatever n. Its remaining work is a finite common Newton fan and simplicial subdivision; this fan, the inputs and the coefficient functions c_i serve for every n, and the procedure does not expand the Taylor head of degree $\ell(n) - 1$ of Lemma 29.15 or a residue coefficient of degree $N(n)$ into its monomials.

Corollary 29.19 (Finite bounds and the exact positive root rate)**.** *For the joint word of Theorem 29.18 put $c_{\max} = \max_i \max_{Y_i} |c_i|$ and $M = \sum_i \|\phi_i\|$. Then*

$$|\mathcal{R}_n| \le M\,|G_n|\,c_{\max}^n \le CM(n+1)^p (r_G c_{\max})^n,$$

and $c_{\max}$ is the ceiling *of the joint word. Every quantity in this bound has a finite certificate: the margins of the coefficient factors and the sector walls bound M, and compact subdivision bounds $c_{\max}$; with computable input coefficients a strict inequality $r_G c_{\max} < r < 1$ is certified by finite subdivision and gives a computable geometric tail. If the original source has a convergent positive representation on every contour of the cohort, no source continuation or coefficient input is needed,*

and the remaining constant terms are real, then the output of the procedure has, after its common scalar sign is removed, one positive input ϕ and a coefficient function $c \geq 0$, and

$$\lim_{n\to\infty} |\mathcal{R}_n|^{1/n} = r_G \operatorname*{ess\,sup}_{\phi} c.$$

For a positive read on an unbounded domain, before it is compactified, the same holds with an infinite essential supremum whenever the moments exist. So averaging over a positive source measure takes the essential supremum on the joint input of the source, Beta and shape variables; it does not reverse the order of an infimum over directions and a supremum over source points.

Definition 29.20 (Simple tops of the joint word)**.** For (29.2) with $c_{\max} > 0$, tops are read as in Definition 19.48, with the coefficient function c_i in place of ϱ_π, the oriented Y_i in place of the keyhole contour and the input ϕ_i in place of ϕ_π: a simple top is a point p with $|c_i(p)| = c_{\max}$ for which local real coordinates on the oriented Y_i are $(\mathbf{x}, \mathbf{y}) \in [0,\epsilon)^{b_1} \times (-\epsilon,\epsilon)^{b_2}$, and

$$\log \frac{c_i(\mathbf{x},\mathbf{y})}{c_i(p)} = -\langle \boldsymbol{\lambda}, \mathbf{x}\rangle - \tfrac{1}{2}\,\mathbf{y}^{\mathsf{T}}\mathbf{Q}\,\mathbf{y} + O\big(|\mathbf{x}|^2 + |\mathbf{x}|\,|\mathbf{y}| + |\mathbf{y}|^3\big), \qquad \phi_i = \prod_{j=1}^{b_1} x_j^{\kappa_j - 1}\, g(\mathbf{x},\mathbf{y})\, d\mathbf{x}\, d\mathbf{y},$$

with $\kappa_j > 0$, $g(0,0) \neq 0$, $\operatorname{Re}\lambda_j > 0$ and $\operatorname{Re}\mathbf{Q}$ positive definite, $\mathbf{Q}$ complex symmetric, and g, continuous and bounded near $(0,0)$, including the orientation and every factor of the circle integrals. Its exponent and its coefficient are

$$\operatorname{ex}(p) = \sum_j \kappa_j + \tfrac{b_2}{2}, \qquad A(p) = g(0,0)\,(2\pi)^{b_2/2} (\det \mathbf{Q})^{-1/2} \prod_j \Gamma(\kappa_j)\, \lambda_j^{-\kappa_j},$$

the square root continued from positive definite real matrices. The subdivisions of the procedure, the splits of the Beta intervals and of the scales and the cones of the fan, divide one domain into parts on which the integrand is the restriction of one function; tops are read with these parts reassembled, a point on a common boundary of the parts being a simple top when the expansion holds on its reassembled neighbourhood, in the coordinates before the subdivision. Parts that are different integrals, as the segment and the circle integrals of Lemma 29.15, keep their own coefficients, and the contributions at one point are added before noncancellation is tested.

Theorem 29.21 (The joint tops give the root rate of a complex word)**.** *Let the largest values of $|c_i|$ on the joint word of Theorem 29.18 sit at finitely many simple tops. Let ex_0 be the least of their exponents, and for $|\zeta| = 1$ let A_ζ be the sum of the coefficients $A(p)$ of the tops with $\operatorname{ex}(p) = \operatorname{ex}_0$ and $c_i(p) = c_{\max}\,\zeta$. If some A_ζ is not zero, then*

$$\limsup_{n\to\infty} |\mathcal{R}_n|^{1/n} = r_G\, c_{\max}.$$

No separate price of the phase is added to $c_{\max}$: the completed fibre and its source phases belong to the one coefficient function and the oriented input. A largest value of $|c_i|$ alone is not the hypothesis; the expansion includes the stationarity of the complex phase in every tangential variable.

This is Theorem 19.49 for joint words: the coordinates $\mathbf{x}$ that vanish at a top carry the Gammas of the boundary, the coordinates $\mathbf{y}$ the factor $(2\pi)^{b_2/2}(\det\mathbf{Q})^{-1/2}$, and the coefficients of the tops of least exponent and one phase add before the root rate is read.

Remark 29.22 (Where the tops of a continued source factor sit). In the integrals of Lemma 29.15 a falling wall carries the n-th power of q^{-d} on the segment and of ζ^{-d} on the circles. On a circle the modulus of ζ^{-d} is constant and its phase turns, so a largest value of $|c_i|$ there is a simple top only where the other factors make the phase stationary, a saddle in the radius. The radii r, r_1, r_2 and T are free within the margins of the coefficient factors, and the sum does not depend on them; Theorem 29.21 is applied after the radii are chosen. When the largest values of $|c_i|$ sit at simple tops of the segment, of the shape or of the Beta inputs, or at saddles of the circles, and do not cancel, the root rate is $r_G c_{\max}$; a largest value spread over a circle, or one at the end $q = rT$ of the segment, which moves with the radii, certifies the upper bound only, and the radii may be moved, within the margins, to lower it. Where the law has log expansions the continuation is made on the law, and its bound has no phase and no margin (Proposition 30.14).

Corollary 29.23 (Closing along a path under an endpoint condition). *In Lemma 29.4 keep (H0) and (H1) and replace (H2) by the weaker condition $\int_{\gamma(\tau_k)+iV} I \to 0$ along times $\tau_k \to \infty$ off the poles. Then the sums of the signed crossing terms before τ_k converge to $\int_{\mathbf{c}+iV} I$. If the crossing terms are absolutely summable, this is their absolutely convergent ordered sum; (H2) remains a sufficient condition for the endpoint condition.*

Corollary 29.24 (Closing with the joint words). *For a rational ray with simple nonresonant crossings, compute the finitely many arithmetic cohorts of crossings and one cohort of endpoints off the poles in each repeating gap. If each crossing cohort and each required endpoint cohort has a certified upper root rate $r_G c_{\max} < 1$, the crossing terms are absolutely summable and their ordered sum is the initial fibre integral. The finite head is retained, and the bounds of Corollary 29.19 give an effective stopping rule. A cohort with the positive hypothesis of Corollary 29.19, or the noncancelling simple tops of Theorem 29.21, has the stated exact root rate and cannot tend to zero when it exceeds one.*

The procedure keeps the falling rates, which fix the crossed heads, and the cut source; within the chamber the growing shares can still be changed by Theorem 29.6, the completed word being the same. The bound $r_G c_{\max}$ has no phase term: $c_{\max}$ is the ceiling of this joint word, which need not equal the ceiling of a keyhole continuation read before the fibre is completed, nor the least ceiling over all joint words.

Lemma 29.25 (Positive capacity and the growing shares). *Let $t_g \in \mathbb{R}^r$, $g \in G$, be a finite nonempty configuration with a grading $\langle t_g, \mathbf{v}\rangle = 1$, let $\xi_g > 0$, $w > 0$ and $\langle \mathbf{v}, \nu\rangle = w$, and in this statement write $L(X) = \sum_{g\in G} \xi_g X^{t_g}$ and $\Pi(\nu, w) = \{\mathbf{b} \in [0,\infty)^G : \ \sum_g b_g = w, \ \sum_g b_g t_g = \nu\}$. With $0^0 = 1$ and the infimum of the empty set equal to $+\infty$,*

$$\sup_{X\in(0,\infty)^r} \frac{X^\nu}{L(X)^w} = \inf_{\mathbf{b}\in\Pi(\nu,w)} \prod_{g\in G} \Big(\frac{b_g}{w\,\xi_g}\Big)^{b_g},$$

the left side being the capacity *of L at (ν, w). The supremum may equally be taken on any logarithmic section transverse to $\mathbf{v}$. When ν/w lies in the relative interior of the convex hull of the t_g, the infimum has a unique positive minimizer and is attained by a gauge X, unique modulo the directions constant on all the t_g; on a proper face the same computation on that face gives the value, approached by suppressing the other monomials.*

The supremum on the left is the largest value of a monomial of the letter vector, and the logarithm of the infimum on the right is the least Gamma rate over the polytope $\Pi(\nu, w)$ of shares, less $w \log w$: the lemma is the duality that turns the chamber rate of a positive kernel into a capacity of its letter.

Theorem 29.26 (The exact positive rate of a retained chamber)**.** *Take a Gramma* $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ *with rational configuration and real index read on its share side (Proposition 4.2), the pins* $t\mathbf{a} = \nu$ *with the grading of the letter totals and positive coefficients* $\boldsymbol{\xi}$*. Let a nonzero rational ray start at a positive real point of the pins, and consider any simple nonresonant Gamma crossing cohort of that ray, or any cohort of travelling fibres off the poles, with its integer step as in Definition 29.16. Write its direction per step as* $\boldsymbol{\sigma}$*, and put*

$$\mathcal{S}_- = \{l :\ \sigma_l < 0\}, \qquad d_l = -\sigma_l \ \ (l \in \mathcal{S}_-), \qquad \mathcal{S}_+ = \mathcal{S}_-^c,$$

t_l *the columns of* t*, with the chamber of* $\boldsymbol{\sigma}$ *(Definition 29.5), the polar functions of the share side being the shares,*

$$C = \{\boldsymbol{\sigma}' :\ t\boldsymbol{\sigma}' = 0,\ \sigma'_l = -d_l\ (l \in \mathcal{S}_-),\ \sigma'_l \geq 0\ (l \in \mathcal{S}_+)\}.$$

Then the retained terms have a common sign at each index, independent of $\boldsymbol{\xi}$*, after their crossing orientation is fixed, and*

$$\lim_{n\to\infty} |\mathcal{R}_n(\boldsymbol{\xi})|^{1/n} = \exp \min_{\boldsymbol{\sigma}' \in C} h_\Gamma(\boldsymbol{\sigma}'),$$

h_Γ *the Gamma rate of Theorem 29.3(iii), the coefficients being those of the Gramma: the exponential of the least Gamma rate of the chamber. This minimizes over the growing shares, the falling rates* d_l *being fixed; if the integer step has the length* Δ *in the parameter of the ray, the logarithm is* Δ *times the rate per unit of that parameter.*

Corollary 29.27 (Exact positive averaging at every retained dimension)**.** *In Theorem 29.26 let the coefficients be measurable positive functions* $\xi_l(\zeta)$ *of a source with a nonzero positive measure* λ*, the configuration, the index, the chamber and the integer step fixed across the source, and put*

$$h_C(\zeta) = \min_{\boldsymbol{\sigma}' \in C} \Big(\sum_l \sigma'_l \log|\sigma'_l| - \sum_l \sigma'_l \log \xi_l(\zeta) \Big), \qquad \hat{h}_C = \operatorname*{ess\,sup}_\lambda h_C.$$

Let one averaged retained term be absolutely integrable, so that after multiplication by $\prod_l \xi_l(\zeta)^{-a_l^0}$*,* $\mathbf{a}^0$ *the real point at which the cohort starts (Definition 29.16), the joint positive input is finite. If* $\hat{h}_C < \infty$*, every later averaged term is integrable and*

$$\lim_{n\to\infty} \Big| \int \mathcal{R}_n(\zeta)\, d\lambda(\zeta) \Big|^{1/n} = e^{\hat{h}_C};$$

if all these terms are finite and $\hat{h}_C = +\infty$*, their root rate is* $+\infty$*. No estimate uniform over the source and no retained fibre of dimension one is needed. This applies to the kernel of Theorem 29.7 on rational pins along a rational ray whose crossings are simple and nonresonant, under a finite averaged rate and the certificate* (29.1)*, whose bound supplies the finite first term. For a continuous coefficient map the essential supremum is the supremum on the support of the image of* λ*; it is* $\operatorname{ess\,sup}_\zeta \min_C$*, with no interchange to* $\min_C \operatorname{ess\,sup}_\zeta$*.*

So the sharp part of Theorem 29.7, its part (iv), holds in every retained dimension, with no condition on the bundles: the falling shares need not be alone in them, and the share side merged by its bundles may have any number of free shares. The chamber minimum is a finite convex problem, shared by all the cohorts of the ray once their common integer step is chosen, and the completed source average is one joint Trailmoment. The positivity belongs to the completed source average; a continued source factor, with its oriented circle integrals, is read by Theorem 29.21, its phases not being replaced by a supremum over a positive support. A completed crossing and the tail it inherits are different series: closing a retained share only through finitely many of its poles leaves an exact Taylor remainder whose rate depends on how many poles are taken with each crossing, and it can grow while the completed crossings fall.

29.3.2 Coincident crossings and the boundary at root rate one

A residue is taken after the factors on its hyperplane have been assembled. A deformation may separate those factors for computation, but its separate residues are then one packet. The finite operation of Theorem 29.29 retains that packet, including its vanishing factors, and extends Theorem 29.18 to coincident crossings.

Lemma 29.28 (A source pole with a uniform parameter margin)**.** *Use the radii of Lemma 29.15. Near $\kappa = -N$, $N \in \mathbb{N}$, put $z = \kappa + N$ and fix $0 < \eta < 1$. On $|\operatorname{Re} z| \le \eta$ the same continued source factor is exactly*

$$I_\kappa = \int_r^1 q^{\kappa-1} g(q)\, dq + R_{N+1} + R'_N + g_N \frac{r^z}{z}, \qquad g_N = \frac{1}{2\pi i} \oint g(\zeta)\, \zeta^{-N-1}\, d\zeta,$$

R_{N+1} and R'_N given by the formulas of that lemma with these indices, $R'_0 = 0$. The first three terms are uniformly absolutely convergent on the whole strip; splitting R'_N gives at most seven integrals in all, the circle integral g_N carrying the explicit pole r^z/z. If $N = N_0 + dn$, the domains stay fixed and their dependence on n is again in fixed powers; in particular a small complex circle of the parameter about $z = 0$ has a common integrable source representation after its displayed pole is removed.

Theorem 29.29 (A finite jet for a coincident crossing)**.** *Consider a rational affine family of crossing fibres, indexed by n, with n in one fixed arithmetic progression. Fix the inherited contour and the whole packet of factors polar on its crossed hyperplane, let the other polar hyperplanes be separated from this packet on the real contour by a positive constant, and let the total order of the packet be at most p, independent of n. Perturb the constant terms of all shares and walls by one parameter ε, with a fixed rational vector of perturbations, keeping the affine pins and their prescribed continuation; the vector can be chosen outside finitely many rational hyperplanes so that distinct members of the packet separate, factors forced to remain coincident being kept as one member. Let $\mathcal{R}_{n,j}(\varepsilon)$ be the inherited crossing terms of the separated members. Then*

$$\mathcal{R}_n = [\varepsilon^0] \sum_j \mathcal{R}_{n,j}(\varepsilon).$$

Each member has a pole of order at most $p - 1$ in ε if its separated pole is simple, and the same bound suffices after keeping forced multiple members intact; so the constant coefficient uses only the Taylor coefficients through the degree $p - 1$ of the holomorphic parts after their poles are removed. Further meromorphic normalizing factors raise this order by their known pole orders. Every coefficient is a jet of the Gammas, of the source or of the parameter; no limit in n is part of the computation. More generally, whenever a finite exact procedure returns terms $\mathcal{R}_{n,j} = \varepsilon^{-o_j} \widetilde{\mathcal{R}}_{n,j}$ with $\widetilde{\mathcal{R}}_{n,j}$ holomorphic, the required coefficients are exactly $[\varepsilon^{o_j}]\widetilde{\mathcal{R}}_{n,j}$; positive orders of a factor that vanishes at $\varepsilon = 0$ are retained just as negative orders of a polar factor are. The local decay hypothesis of Lemma 29.4 is used on the common enclosing contour, uniformly on a small disc of the parameter. For the Gamma words and the continued sector integrals a uniform procedure is available: factors whose Gamma poles lie on the crossed hyperplane are constant on its residual fibre and are put in the scalar Gamma word, each source wall coincident on that hyperplane is treated by Lemma 29.28, and the other walls and Gamma strips have strict margins.

The auxiliary parameter separates the poles of one assembled crossing; it does not identify the independent regulators of a product of words, which keep their separate constant terms (Proposition 36.2). If the value asked for is a finite part, its prescribed Mellin regulator is retained and its

constant coefficient taken as in Theorem 1.6 and Corollary 1.7; finite parts in different coordinates are not interchangeable (Lemma 1.8: a dilation of the coordinate moves a finite part by a polynomial in the logarithm of the dilation). Several such regulators give finitely many multivariate Laurent coefficients, with the coefficient order fixed by the value asked for, the retention of finitely many numbers at every corner (Theorem 37.10) applied to a whole crossing fibre. The separation hypothesis concerns the actual residual contour: a transverse meeting of different hyperplanes needs the finite ordered packet of the chosen crossing flag, with its iterated transverse contours, and does not allow an arbitrary order of point residues. Nor does a Laurent coefficient keep a sharp root rate by itself: the coefficient can vanish.

Proposition 29.30 (Bounds and top coefficients pass to jets)**.** *In Theorem 29.29 let the procedure of Theorem 29.18 for the separated terms give a uniform bound on a circle of the parameter,* $|\varepsilon^{o_j}\mathcal{R}_{n,j}(\varepsilon)| \le C_\delta(1+n)^{p'}(q+\delta)^n$ *for every* $\delta > 0$*, the circle allowed to shrink with* δ*. Then* $\limsup_n |\mathcal{R}_n|^{1/n} \le q$*. For perturbations of the constant terms only, the slopes in* n*, the linear parts of the pins, the letters and the coefficients fixed, Lemma 29.28 and the extraction of the scalar give more: one common fan serves on a small disc of the parameter, each coefficient function* c_i *is independent of the parameter, only the fixed amplitudes, the explicit polar factors and the constant terms of the Gammas varying, and their endpoint powers have positive margins once the explicit poles of Lemma 29.28 are removed; so the finitely many Laurent coefficients of each order have the same upper root rate as their common collection of coefficient functions and Gamma slopes, with only polynomial and logarithmic losses in* n*. If the joint word has finitely many simple tops (Definition 29.20), uniformly separated from the rest of its domain, its finitely many Laurent coefficients supply every prescribed finite order of the expansion of* $\mathcal{R}_n$ *at the tops: differentiate the exact integrand to the required order in the parameter, expand its phase and amplitude to the required Taylor order at each top, use* $\mathbf{x} = \mathbf{X}/n$ *in its decaying boundary coordinates and* $\mathbf{y} = \mathbf{Y}/\sqrt{n}$ *in its coordinates* $\mathbf{y}$*, integrate the resulting monomials against the Gamma densities in* $\mathbf{X}$ *and the density* $e^{-\frac{1}{2}\mathbf{Y}^{\mathsf{T}}\mathbf{Q}\mathbf{Y}}$ *in* $\mathbf{Y}$*, and expand the Gamma prefactors to the same order. Equal top phases are added before any test of nonvanishing. A surviving coefficient at the root rate* q *proves* $\limsup_n |\mathcal{R}_n|^{1/n} = q$*; if the tested coefficients cancel, the upper bound remains and no sharp lower bound follows.*

At the root rate one the boundary is decided exactly, without simple or isolated tops, by the law of the coefficient function that the procedure retains in the positive case; its geometry is computed once and answers several queries at the boundary.

Theorem 29.31 (The test at root rate one for a positive moment sequence)**.** *Let* ϕ *be a finite positive measure,* $0 \le c \le 1$ *on its support,* $m_n = \int c^n \, d\phi$*, and*

$$b_n = G_n \, m_n, \qquad G_n > 0, \qquad G_n \sim C \, n^\beta, \quad C > 0, \ \beta \in \mathbb{R} :$$

the form of Corollary 29.19 at the root rate one, the coefficient function divided by its positive largest value $c_{\max}$ *and the scalar multiplied by* $c_{\max}^n$*, the exponent* β *read from the explicit Gamma quotient and the scalar sign a fixed number of modulus one to the power* n*. Finitely many initial terms have no effect.*

(i) The series $\sum_n b_n$ *is finite exactly when:* $\beta < -1$*, always;* $\beta = -1$ *and* $\int \log \frac{1}{1-c} \, d\phi < \infty$*;* $\beta > -1$ *and* $\int (1-c)^{-\beta-1} \, d\phi < \infty$*; the integrands of the last two conditions having the value* $+\infty$ *at* $c = 1$*. These are also the exact conditions for absolute convergence after multiplication by any* ζ^n*,* $|\zeta| = 1$*.*

(ii) Let moreover G_n be a balanced Gamma quotient with its exponential factor removed, or more generally let its forward differences satisfy $\Delta^k G_n = O(n^{\beta-k})$ for every fixed k. For $|\zeta| = 1$, $\zeta \neq 1$, the ordered series $\sum_n \zeta^n b_n$ converges exactly when $b_n \to 0$, and with $M(t) = \phi\{c > 1-t\}$ this is: $\beta < 0$, always; $\beta = 0$ and $\phi\{c = 1\} = 0$; $\beta > 0$ and $M(t) = o(t^\beta)$ as $t \downarrow 0$.

So (i) and (ii) tell absolute convergence, conditional convergence and the failure of the term test apart with one retained law.

A certified law near its largest value of the form $M(t) \asymp t^a(\log(1/t))^j$ turns these criteria into comparisons of powers and logarithms; the criteria stay exact for nonisolated or degenerate maxima, Dirac measures and mixtures. They do not change the crossing order: a periodic family must be reassembled in its original order before a conditional verdict is used, since splitting an alternating series into its even and odd subseries loses exactly that information.

Proposition 29.32 (Finite certificates at the boundary from the jets of the tops)**.** *Let the assembled terms at the root rate one have a certified finite expansion*

$$\mathcal{R}_n = \sum_{\zeta \in Z} \zeta^n \sum_{\beta \geq -1} \sum_{j=0}^{J_\beta} c_{\zeta,\beta,j}\, n^\beta (\log n)^j + O(n^{-1-\delta}), \qquad \delta > 0,$$

Z a finite subset of the unit circle, the real exponents β a finite set and equal triples combined; the jets of the tops of Proposition 29.30 produce such a certificate to every order needed when its hypotheses on simple tops and uniformity hold. Then:

(i) the series is absolutely convergent if and only if every displayed coefficient vanishes;

(ii) it converges in its given order if and only if every coefficient with $\zeta = 1$ vanishes and every coefficient with $\beta \geq 0$ vanishes;

(iii) when (ii) holds and (i) does not, the convergence is conditional.

The verdict uses exact certificates of zero or nonzero for the finitely many assembled coefficients; where those are unavailable, the finitely many coefficients are the output, a numerical near cancellation not being a certificate of zero.

Chapter 30

Laws at their ends: regions of equality and reach

A closing past the walls reads a law where its moments diverge, at its falling ends, through its expansion there. The regions on which a law equals its expansion at an end are built from the sector letters, with the map from the motion of the shares to the exponents, the coefficients, the truncation bounds and exact boundary words (Theorems 30.2 and 30.4); the domain where a series converges and the region where it equals the law are kept apart (Proposition 30.6). The reach is read on the law along every rational direction of the shares, as at width one: along a rational direction the seam function is the Mellin transform of one measure on the half-line, the image of the law, multiplied by $\mathbf{x}^{-\mathbf{w}_0}$ for the starting point $\mathbf{w}_0$, under the directional ratio, and the reach is the far boundary of the region on which that measure is its own wall series (Theorem 30.10). Run on the laws of the two families at a node of width one, the same argument reads the growth of each family on its law and makes that of the bunch its reach (Corollary 30.11). For a piece the reach is bounded by the ceilings of the keyhole continuations along the direction, with no phase, and attained at their simple tops that do not cancel (Theorem 30.15). A closing past the walls converges when the exponential of the Gamma rate times the reach of its tube, the imaginary translates of the line through its starting point, is less than one, its terms then having at most that root rate, and a law with log expansions bounds that reach in every direction with nothing paid for the phase; with one free share the tube is the line, and the terms reach that root rate under conditions that keep the poles apart (Theorems 30.16 and 30.17, Corollary 30.18).

30.1 Regions on which a law is represented

A region of equality belongs to an expansion of a law: to the coordinates advanced, the coordinates retained, and the grouping of its terms. A disc in a sector variable is not by itself a region of equality of the law: the map to the seam torus can identify points, have critical values, or have an image of smaller dimension than its torus. The procedure of Theorem 30.4 keeps that map, and gives a law, with its tangential source retained, before any assertion about a density on the whole torus.

Definition 30.1 (A chart at an end of a law)**.** Let a part of the source of a law have the coordinates $(q, z) \in (0, r_1) \times \cdots \times (0, r_k) \times Z$ and the form

$$q^{\boldsymbol{\kappa}-\mathbf{1}} A(q, z)\, dq\, \phi(dz), \qquad \kappa_i > 0,$$

ϕ a finite positive measure retained as an input and A holomorphic in q on a neighbourhood of a closed polydisc, uniformly bounded there for $z \in Z$; complex densities are allowed, and a fixed

monomial in the seam coordinates can be included in the integrand and removed after integration. The selected seam coordinates of the law at the end are $\check{x}_i^{-1}$, with

$$\check{x}_i = \prod_{j=1}^{k} q_j^{N_{ij}}\, \mathfrak{u}_i(q,z), \qquad 1 \le i \le k,$$

$\mathbf{N} = (N_{ij})$ an invertible integer matrix and the *chart factors* $\mathfrak{u}_i$ positive on the real source and holomorphic and nonzero on the complex neighbourhood; the other coordinates of the seam point are a retained map $\mathbf{C}(q,z)$, and branches are continued from the positive source. The *chart coordinates* are

$$\mathfrak{t}_j = \prod_i \check{x}_i^{(\mathbf{N}^{-1})_{ji}} = q_j \prod_i \mathfrak{u}_i(q,z)^{(\mathbf{N}^{-1})_{ji}}.$$

A certificate of the chart supplies positive radii R_i and a uniform analytic inverse $q_j = \mathfrak{t}_j\, g_j(\mathfrak{t},z)$ for $|\mathfrak{t}_i| < R_i$, with bounds for the inverse, its Jacobian and the holomorphic functions used in Theorem 30.2 and Corollary 30.3; the cuts on Z are retained. Write $\mathbf{1}_{\mathrm{cut}}(\mathfrak{t},z)$ for the indicator of any further retained cut of the source, $\mathbf{1}_{\mathrm{cut}} = 1$ when there is none; if a cut of the source moves in these coordinates, its indicator is retained in the integrand and is neither differentiated nor included in a holomorphic coefficient.

Theorem 30.2 (Computing and using an end chart)**.** *For a chart of Definition 30.1 the following computations are effective.*

(i) A smaller certified polydisc exists whenever the holomorphy and nonvanishing conditions hold uniformly on a compact tangential source. In its coordinates the source form is

$$\mathfrak{t}^{\boldsymbol{\kappa}-\mathbf{1}}\, \widetilde{A}(\mathfrak{t},z)\, d\mathfrak{t}\, \phi(dz), \qquad \widetilde{A} = A(\mathfrak{t}g,z) \prod_j g_j^{\kappa_j - 1} \det \frac{\partial(\mathfrak{t}g)}{\partial \mathfrak{t}},$$

the determinant positive on a small enough positive chart. The seam map is $(\mathfrak{t},z) \mapsto \big(\mathfrak{t}^{\mathbf{N}}, \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)$; with $\mathbf{C}_{\mathfrak{t}}(z) = \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)$, its density in the normal coordinates $\check{x}$, a measure in the tangential ones, is

$$\frac{\check{x}^{-\mathbf{1}}\mathfrak{t}^{\boldsymbol{\kappa}}}{|\det \mathbf{N}|}\, (\mathbf{C}_{\mathfrak{t}})_* \big(\mathbf{1}_{\mathrm{cut}}(\mathfrak{t},\cdot)\widetilde{A}(\mathfrak{t},\cdot)\,\phi\big)\, d\check{x},$$

a density only in the selected normal coordinates, whose tangential value may be a singular measure.

(ii) Write $\widetilde{A} = \sum_{\mathbf{n}\ge 0} \widetilde{A}_{\mathbf{n}}(z)\, \mathfrak{t}^{\mathbf{n}}$. The coefficients are tangential words computed by finite jet arithmetic on the original letters; the inverse certifies that they represent the source and is not evaluated. If $|\widetilde{A}| \le M$ on $|\mathfrak{t}_i| = R_i$, then $|\widetilde{A}_{\mathbf{n}}(z)| \le M \prod_i R_i^{-n_i}$. For every bounded measurable test function H, the law on the chart is, for real $0 < \mathfrak{t}_i < R_i$,

$$\mathfrak{t}^{\boldsymbol{\kappa}-\mathbf{1}} \sum_{\mathbf{n}\ge 0} \mathfrak{t}^{\mathbf{n}} \int_Z \widetilde{A}_{\mathbf{n}}(z)\, \mathbf{1}_{\mathrm{cut}}(\mathfrak{t},z)\, H\big(\mathfrak{t}^{\mathbf{N}}, \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)\, \phi(dz)\, d\mathfrak{t}. \tag{30.1}$$

The same seam map and the same retained cut occur in every term. With rational data the inverse branch is algebraic, specified by the positive branch and the certificate; alternatively every exact pairing with a test function is kept on the original source (q,z) with that cut pulled back, and an inverse function need not be added to the operations. The series converges in

total variation on compact subcharts, the normal power included. If the tangential map and the cut do not depend on $\mathfrak{t}$*, its coefficients are the fixed face laws* $\mathbf{C}_*(\widetilde{A}_{\mathbf{n}}\phi)$*; otherwise* (30.1) *is a series of pairings on the one tangential source, not an assertion that those images are fixed.*

(iii) With $\vartheta_i = |\mathfrak{t}_i|/R_i < 1$*, truncating to* $0 \le n_i < k_i$ *has, before the normal power and for* $\|H\|_\infty \le 1$*, the error at most*

$$M\,\phi(Z)\prod_i (1-\vartheta_i)^{-1}\sum_i \vartheta_i^{k_i}. \tag{30.2}$$

It also has an exact boundary word: with T_i *the Taylor operator through the degree* $k_i - 1$ *in* $\mathfrak{t}_i$*,*

$$\widetilde{A} = \prod_i T_i\widetilde{A} + \sum_{\varnothing\neq I\subseteq\{1,\dots,k\}}\prod_{i\in I}(1-T_i)\prod_{j\notin I}T_j\widetilde{A}, \tag{30.3}$$

$$(1-T_i)\widetilde{A} = \frac{\mathfrak{t}_i^{k_i}}{2\pi i}\oint_{|\zeta_i|=R_i}\frac{\widetilde{A}(\mathfrak{t}_1,\dots,\zeta_i,\dots,\mathfrak{t}_k,z)}{\zeta_i^{k_i}(\zeta_i-\mathfrak{t}_i)}\,d\zeta_i :$$

there are $2^k - 1$ *boundary words, whatever the truncation orders* k_i*.*

(iv) The chart records the map from the motion of the shares to the exponents. The following statements about walls and continuation concern normal Mellin powers with an analytic tangential composite, or a further word with its own sector certificate; the bounded measurable test functions of (ii) have only the convergent identity and the error bound, and no continuation is asserted for them. For the Mellin power in the seam coordinates $\breve{x}_i^{-1}$ *at the shares* $\mathbf{z}$*, the normal exponent of a term is* $\boldsymbol{\kappa} + \mathbf{N}^{\mathsf{T}}\mathbf{z} + \mathbf{n} - \mathbf{1}$*. For a fixed tangential cut and a normal Mellin power, integrating these powers gives the walls* $\kappa_j + (\mathbf{N}^{\mathsf{T}}\mathbf{z})_j + n_j = 0$*; a moving cut keeps its indicator in the integrand, with no such set of poles. A motion* $\boldsymbol{\sigma}$ *of the shares in the coordinates of these Mellin powers has the wall slopes* $\mathbf{N}^{\mathsf{T}}\boldsymbol{\sigma}$*; for the original coordinates of the shares one first applies to* $\boldsymbol{\sigma}$ *the recorded linear change of the exponents of the Mellin powers. Other requested coordinates remain in the tangential word, and are not replaced by their values on the face unless that replacement is part of the stated test function.*

Corollary 30.3 (Face jets for an analytic test function)**.** *Let the tangential cut be fixed, the map* $\mathbf{C}(\mathrm{tg}(\mathfrak{t},z),z)$ *holomorphic on a specified polydisc about* $\mathfrak{t} = 0$*, that point included, and* H *a bounded holomorphic tangential test function on a neighbourhood of its image. Expand* $\widetilde{A}(\mathfrak{t},z)\,H\big(\mathbf{C}(\mathrm{tg}(\mathfrak{t},z),z)\big)$ *instead of* $\widetilde{A}$ *alone: its coefficient of degree* $\mathbf{n}$ *is a finite sum of tangential source integrals of derivatives of* H *at* $\mathbf{C}(0,z)$*, multiplied by jets of* $\widetilde{A}$*,* $\mathbf{C}$ *and* g *of degrees at most* $\mathbf{n}$*. These are fixed face functionals, and their series and boundary words obey the same coefficient bounds as in Theorem 30.2(ii), with the bound of this composite. A change of analytic tangential test function needs its jets, not a new continuation of the source. The normal monomial* $\mathfrak{t}^{\mathbf{N}}$ *remains unexpanded in the integrand; a normal test function may be expanded as well when its composite is holomorphic and bounded on the polydisc and its jets are supplied by finite operations, a further condition, since negative entries of* $\mathbf{N}$ *can give poles at* $\mathfrak{t} = 0$*.*

Theorem 30.4 (Certified regions of equality and composition)**.** *For rational positive polynomial letters and rational monomial seam coordinates the following is a procedure.*

(i) The finite sector data determine the exponent matrices and the chart factors of Definition 30.1. Every invertible normal minor gives a proposed end chart, with the other variables

and test functions retained; enumerating rational tangential boxes and radii and verifying the inequalities of the certificate and of nonvanishing produces certified charts. Every compact tangential set on the end face $q = 0$ on which the proposed chart has nonvanishing chart factors and an invertible normal matrix has a small enough normal neighbourhood covered by finitely many charts produced in this way. Extending a chart away from that face needs its certificate, or a new certificate of inverse functions for the full Jacobian; invertibility of $\mathbf{N}$ alone does not supply it.

(ii) The region of equality supplied by one chart is the image of its certified polydisc in $\mathfrak{t}$ under $\check{x} = \mathfrak{t}^{\mathbf{N}}$, with the specified branches and cuts; its positive part satisfies $\sum_i (\mathbf{N}^{-1})_{ji} \log \check{x}_i < \log R_j$, $1 \le j \le k$, and need not be a rectangle in the original seam coordinates. A finite atlas is certified for a requested region of equality only after its source patches cover the inverse image of that region; a patch not yet covered is retained as its original source word. So at every step the output is an exact finite head together with the boundary words (30.3) *and the uncovered source word, with bounds for the parts certified small; an uncovered word is never omitted.*

(iii) Atlases may be restricted to common source patches, added, transported by their seam maps, and composed with a further test function. On compact certified subregions every order of the absolutely convergent reads has the same value. At finite truncations the equality uses the boundary words (30.3)*, and moving a cut retains the source between the old and the new cut. For words whose sector certificate supplies a meromorphic continuation, the same finite identities hold there, the full Laurent coefficient taken before pairing with a test function at coincident poles; an arbitrary bounded test function is not thereby continued. An infinite read in another grouping is asserted only with its own bound of convergence or as the continuation of the retained word.*

The retained data are the sector syntax, the matrix $\mathbf{N}$, the cuts, the recursion for the inverse jets and its certificate, and the tangential coefficient words, computed once per distinct source chart. For a prescribed error on $|\mathfrak{t}_i| \le \vartheta_i R_i$ the orders k_i are read off (30.2); the finite head has $\prod_i k_i$ coefficient words and its exact remainder $2^k - 1$ boundary words. A repeated chart reuses its coefficient words, a new analytic test function supplies its finite face jets, and the certificate and the retained boundary remain valid when only the truncation changes.

Proposition 30.5 (When a retained chart carries a density)**.** *A retained chart is in general not a chart of an ambient density. It becomes one in the selected normal coordinates and further tangential coordinates $\mathbf{y}$ when their image has a certified local product parametrization over $(\mathfrak{t}, \mathbf{y})$, with compact fibres, fixed cuts, a holomorphic form and an integrable uniform bound; integration over those fibres then gives a holomorphic coefficient function, and the series of Theorem 30.2, or of Corollary 30.3, represents that density with the same bounds. For a source form made of integrable powers of the boundary coordinates of a cube times a holomorphic density, such parametrizations are built near every compact regular fibre of a rational map $\mathbf{C}$ on the cube, when the map is defined on a neighbourhood of the closed cube, its restriction to every face meeting the fibre has a surjective differential, and its letters are nonzero. To assert a density for the whole law, the target patch must have no other or uncovered preimages, and in particular no omitted pole face may supply further source patches; otherwise the procedure describes the covered part of the law only, and the further preimages remain uncovered source words under Theorem 30.4. With the cube coordinates v_i put*

$$\mathbf{J}_{ji} = v_i(1 - v_i)\, \partial_{v_i} C_j, \qquad X_j = \sum_i v_i(1 - v_i) \big[\mathbf{J}^{\mathsf{T}} (\mathbf{J}\mathbf{J}^{\mathsf{T}})^{-1}\big]_{ij} \partial_{v_i};$$

then $D\mathbf{C}(X_j) = \mathbf{e}_j$. On a small enough neighbourhood the flows of these vector fields carry the reference fibre to the neighbouring fibres, keep all faces and give the product parametrization. Uniform nonvanishing of $\det(\mathbf{J}\mathbf{J}^{\mathsf{T}})$*, of the letters and of the relevant Jacobians, together with a bound for the vector fields on a larger neighbourhood, is a finite certificate; the iteration of the integral equations of the flows, certified to converge geometrically, computes the local flows and their holomorphic extensions.*

The possible failures of this procedure are explicit: zeros of letters, loss of properness at an omitted face of the source, or loss of surjectivity of the differential of the seam map on a face. Their images are candidate boundaries, not automatically singularities of the law, since different contributions can cancel. If the source has smaller dimension than the number of requested seam coordinates, the retained law and its atlas still exist but no ambient density is asserted. A genuine edge of the support is an obstruction: if the law vanishes on an open set on one side of an edge and has positive measure arbitrarily close on the other, it has no analytic density across that edge, by the identity theorem, as in Theorem 19.42(iii).

Proposition 30.6 (Convergence domains and equality regions)**.** *Let the fixed face laws of a certified chart, or the face functionals for a fixed bounded set of analytic test functions, be $\widetilde{A}_{\mathbf{n}}$ in a Banach space. The interior Ω_{abs} of*

$$\Big\{\mathfrak{t} : \sum_{\mathbf{n}\geq 0} \|\widetilde{A}_{\mathbf{n}}\|\,|\mathfrak{t}^{\mathbf{n}}| < \infty\Big\}$$

is the logarithmically convex domain of absolute convergence *of that series, which contains with each of its points $\mathfrak{t}$ every $\mathfrak{t}'$ with $|\mathfrak{t}'_i| \leq |\mathfrak{t}_i|$ for all i, and every certified polydisc lies in it. Its* region of equality *is retained separately: the chart patches covered by the source on which the series is proved to read the law, together with the patches reached by a chain of transports of charts that keep the equality. The series represents the law only on the intersection of the domain Ω_{abs} and the region of equality; analytic continuation of the sum alone is not a transport of the equality across an edge of the support or a singular value of the law. For a positive integer vector $\mathbf{d}$ and a fixed point $\mathfrak{t}$ of the chart retain the grouping*

$$\widetilde{A}_{(N)}(\mathfrak{t}) = \sum_{\langle \mathbf{d},\mathbf{n}\rangle = N} \widetilde{A}_{\mathbf{n}}\,\mathfrak{t}^{\mathbf{n}};$$

the grouped analytic germ in one variable v on the curve $(\mathfrak{t}_i v^{d_i})_i$ has the exact radius R with $R^{-1} = \limsup_N \|\widetilde{A}_{(N)}(\mathfrak{t})\|^{1/N}$: inside it the grouped series converges, and outside it its grouped terms do not tend to zero. These statements concern the germ, and they become statements about the law only where its region of equality is certified as well; on the boundary of convergence the coefficients decide. Completing another read replaces these grouped terms by their images, whose radius can be larger, the equality with the new expansion of the law still using the transported source and cuts. Neither radius is a reach of the law by definition, nor is it replaced by the supremum of an unintegrated modulus of the source.

The comparison of radii for a bounded read coefficient by coefficient concerns the same normal coordinates and the same bank of coefficients. Merging a law and then choosing a new coordinate is another operation: the source is carried by that map and its expansion is regraded in the new coordinate, whose coefficients and grouped terms need not be bounded images of the old ones, and whose boundary of equality can come earlier. The old and the new expansion agree through their common source, with the cuts, the finite heads and the boundary words transported; their

domains of convergence and atlases of equality are recorded in their own coordinates. Maximal analytic continuation, and deciding every cancellation at a singular boundary, are neither used by the procedure nor concluded from a ceiling. In one selected coordinate with a fixed tangential test function the procedure certifies local patches of equality with the wall series, and the global inversion of Theorem 30.10 (Theorem 19.42 at width one) identifies the reach of the law along that coordinate; in several coordinates the certified atlas of equality, the domain of convergence of the series and its retained grouping are stored separately.

30.2 The reach along a direction

At width one the reach of a bunch is read on the law of its ratio, as the point beyond which the law is its own wall series (Theorem 19.42); it is bounded above by the ceilings of the keyhole continuations and attained at their simple tops that do not cancel (Theorems 19.47 and 19.49), and against a room bounded below the terms of the node have the root rate of the reach over the least ratio of the room, which infinitely many of them reach (Theorem 19.38). A closing past the walls at any width advances along one direction of the shares, and along a rational direction the seam function is the Mellin transform of one measure on the half-line, the directional law (Definition 30.9). The theory of width one applies to that measure line by line, and its counterparts are stated along a direction.

Definition 30.7 (Walls and growth bounds)**.** Let a piece have its seam function $P_{W_B(\mathbf{w})}(\mathbf{m}_B)$ on the pins of the seam, an affine space A_B of real shares $\mathbf{w} = (\mathbf{a}, \mathbf{b})$ of the active letters with the space of directions $\vec{A}_B$, of dimension $n_{\mathcal{D}}$; for $\mathbf{w} \in A_B + i\vec{A}_B$ it is $\int \mathbf{x}^{-\mathbf{w}}\, d\big((\boldsymbol{\varrho}_B^{\rho})_*\phi_B^{\rho}\big)(\mathbf{x})$ where this converges, the law being carried by the closure of the image of the open simplex under the seam vector. The seam function has *walls* when

(W1) there are finitely many affine functions κ_U on A_B with rational linear parts, its *walls*, such that it converges at a real point exactly when every wall is positive there, which happens on an open convex set $\Omega_B \subseteq A_B$, and it continues meromorphically to $A_B + i\vec{A}_B$ with its poles on the hyperplanes on which a wall lies in $-\mathbb{N}$;

and it has *growth bounds* when

(W2) for every $\theta > 0$ there are M, C_0, n_1 and n_2 with $|P_{W_B(\mathbf{z})}(\mathbf{m}_B)| \le C_0\, \delta^{-n_2}(1 + |\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta|\operatorname{Im}\mathbf{z}|_1}$, δ the smaller of 1 and the distance of $\mathbf{z}$ from these hyperplanes.

The same definitions apply to the period of any word whose exponents are affine functions of the shares.

Every piece whose seam function converges at a real point has walls and growth bounds, and so does the seam function of its seam letters merged by bundles: the seam word of a piece is a word in positive letters, and so is the word of the merged seam letters of Theorem 29.7, sums of seam letters of several stations of one excess, which are no longer letters of tables; such a word continues past its walls as the seam word of every piece does (Theorem 29.3(ii)).

Lemma 30.8 (A word in positive letters continues past its walls)**.** *(i) Let $W = \prod_l L_l^{-w_l}$ be a product of powers of positive letters on a set B (Definition 24.11), whose exponents w_l and masses $\mathbf{m}_B$ are affine functions of $\mathbf{z} \in \mathbb{C}^p$ with rational linear parts, the masses on the balance hyperplane of W for every $\mathbf{z}$, and let its period converge at a real point $\mathbf{z}_0$, the exponents there being of any sign. Then there are finitely many affine functions κ_U on $\mathbb{C}^p$ with rational linear parts, its walls, such that the period converges at a real point exactly when every κ_U is*

positive there, and $P_W(\mathbf{z})$ continues meromorphically to $\mathbb{C}^p$, with its poles, of order at most $|B|-1$, on the hyperplanes $\kappa_U(\mathbf{z}) \in -\mathbb{N}$; and for every $\theta > 0$ there are M, C_0 and n_1 with

$$|P_W(\mathbf{z})| \le C_0\, \delta^{-(|B|-1)} (1+|\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta |\operatorname{Im}\mathbf{z}|_1},$$

δ the smaller of 1 and the distance of $\mathbf{z}$ from these hyperplanes.

(ii) For a piece and a partner read in its monomials, as in Theorem 29.7(B), merge the shares of each bundle G of Theorem 29.7 into $\bar{a}_G$, and put $L_G = \sum_{l\in G} \xi_l\, V_{jD(l)}$ for a bundle of shares of K_j, and L_G alike with the seam letters V^*_{jD} for a bundle of shares of Ψ_j. On the pins of the merged shares the merged seam word $W^{\mathrm{m}}_B(\bar{\mathbf{a}}) = \prod_G L_G^{-\bar{a}_G}$ *is such a product at the masses $\mathbf{m}_B$, and its period, the* merged seam function*,*

$$P_{W^{\mathrm{m}}_B(\bar{\mathbf{a}})}(\mathbf{m}_B) = \int \prod_G e^{-\bar{a}_G\, y_G}\, d\lambda, \qquad y_G = \log \sum_{l\in G} e^{y_l},$$

reads the merged values y_G at the source points, against the source measure λ (Theorem 29.7(iv)); it converges at the sums over the bundles of every real point with positive coordinates at whose sums over the monomials the seam function of the piece converges.

For the seam word of a piece, at the masses $\mathbf{m}_B$ and with the shares as its exponents, Lemma 30.8(i) gives walls and growth bounds whenever the seam function converges at a real point, the functions κ_U of the lemma serving as the walls, with $n_2 = |B|-1$; when the shares are positive at that point the seam word has a Newton sum of full dimension (Theorem 24.12), and the walls of the generators of a fan of the seam word serve, with the sector integrals of Theorem 29.3(ii), the proper walls for a piece of tables. A piece whose seam function on the pins is a product of Beta corners $\Gamma^{\,2}_{\,1}\begin{bmatrix} x,\ y \\ x+y \end{bmatrix}$ of affine functions of the shares with rational linear parts and of real exponentials of the shares, converging exactly where those affine functions are positive, has walls, those functions, and growth bounds, by the reflection formula (Proposition 1.3(v)) and the asymptotic expansion of $\log\Gamma$ away from the negative real axis, and growth bounds even at $\theta = 0$ when for each Beta corner the linear parts of its two arguments on $\vec{A}_B$ are proportional, as when one argument does not depend on the shares.

Definition 30.9 (Directional ratios and laws; the reach along a direction)**.** Let a piece have walls and growth bounds (Definition 30.7), and let $\bar{\boldsymbol{\sigma}} \in \vec{A}_B \setminus \{0\}$ have rational coordinates. Its *directional ratio* is the positive function $\mathbf{x}^{-\bar{\boldsymbol{\sigma}}}$ on the seam torus. For $\mathbf{w}_0 \in \Omega_B$ and $\bar{\boldsymbol{\vartheta}} \in \vec{A}_B$ the *directional law* at $\mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}}$ is the image

$$\lambda_{\bar{\boldsymbol{\sigma}},\mathbf{w}_0+i\bar{\boldsymbol{\vartheta}}} = \big(\mathbf{x}^{-\bar{\boldsymbol{\sigma}}}\big)_*\Big(\mathbf{x}^{-\mathbf{w}_0 - i\bar{\boldsymbol{\vartheta}}}\, (\boldsymbol{\varrho}^{\rho}_B)_* \phi^{\rho}_B\Big),$$

a complex measure on $(0,\infty)$ whose variation is at most the positive directional law $\lambda_{\bar{\boldsymbol{\sigma}},\mathbf{w}_0}$, which integrates to the seam function at $\mathbf{w}_0$; it is *oscillating* when $\bar{\boldsymbol{\vartheta}} \ne 0$, and the line $\{\mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}} + z\bar{\boldsymbol{\sigma}} : z \in \mathbb{R}\}$ is the *imaginary translate* by $\bar{\boldsymbol{\vartheta}}$ of the line through $\mathbf{w}_0$. Its *line function* is

$$\ell(z) = P_{W_B(\mathbf{w}_0+i\bar{\boldsymbol{\vartheta}}+z\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B) = \int_0^\infty X^z\, d\lambda_{\bar{\boldsymbol{\sigma}},\mathbf{w}_0+i\bar{\boldsymbol{\vartheta}}}(X),$$

absolutely convergent for $\operatorname{Re} z$ in the open interval of the x with $\mathbf{w}_0 + x\bar{\boldsymbol{\sigma}} \in \Omega_B$, which contains 0, and continued meromorphically by (W1). A wall falls along $\bar{\boldsymbol{\sigma}}$ when its falling rate $-\dot{\kappa}_U$,

$\dot\kappa_U = \partial_{\bar{\boldsymbol\sigma}}\kappa_U$, is positive (Definition 29.5). The poles of ℓ with $\operatorname{Re} z \ge 0$ lie among the *candidate poles*

$$z = \big(\kappa_U(\mathbf{w}_0) + n + i\,\partial_{\bar{\boldsymbol\vartheta}}\kappa_U\big)/(-\dot\kappa_U), \qquad n \in \mathbb{N},$$

of the falling walls: on finitely many horizontal lines, with real parts in finitely many cosets of one lattice $N_0^{-1}\mathbb{Z}$, the falling rates being rational. Let Σ be the half-strip of the z with $\operatorname{Re} z \ge 0$ and $\operatorname{Im} z$ within 1 of the interval spanned by 0 and these lines. The *reach along* $\bar{\boldsymbol\sigma}$ at $\mathbf{w}_0 + i\bar{\boldsymbol\vartheta}$ is

$$r(\bar{\boldsymbol\sigma}; \mathbf{w}_0 + i\bar{\boldsymbol\vartheta}) = \inf\big\{M \ge 0 :\ \text{for every } \delta > 0 \text{ there are } C, n' \text{ with } |\ell(z)| \le C(1+|z|)^{n'} M^{\operatorname{Re} z}\big\},$$

the bound asked at every $z \in \Sigma$ at distance at least δ from the candidate poles; it is finite by (W2), and by the maximum modulus principle on circles about the candidates it does not change when the actual poles are used. A *wall series* is as in Definition 19.41, the η_i allowed to be complex and its condition read as $\sum_{\eta,j} |a_{\eta,j}| x'^{-\operatorname{Re}\eta} < \infty$ for every $x' > x_0$, x_0 its radius. Asked on the whole half-plane $\operatorname{Re} z \ge 0$, with an allowance $e^{\theta|\operatorname{Im} z|}$ for any $\theta \ge 0$, the bound defines the same number (Theorem 30.10(v)). At $\bar{\boldsymbol\vartheta} = 0$ the directional law is the *law along* $\bar{\boldsymbol\sigma}$. The same definitions apply to the merged seam function of Lemma 30.8(ii) on the pins of the merged shares, the merged values e^{y_G} in place of the seam coordinates and the source measure in place of the law.

The falling end of the directional law is $X \to \infty$, where the walls that fall along $\bar{\boldsymbol\sigma}$ make its moments diverge. In an end chart of Theorem 30.2 the Mellin power at the shares $\mathbf{z}$ is $\mathfrak{t}^{\mathbf{N}^{\mathsf{T}}\mathbf{z}}$, so the directional ratio is $\mathfrak{t}^{-\mathbf{d}}$, $\mathbf{d} = -\mathbf{N}^{\mathsf{T}}\bar{\boldsymbol\sigma}$ the falling rates of the walls of the chart, times a power of the tangential coordinates: the wall series of the directional law groups the terms of the chart by their exponents $(\kappa_j + n_j)/d_j$ along the direction, and the reach is the far boundary, in X, of the region on which the law equals that expansion. At width one with one table and factoring weights, normalized as in Proposition 27.10, the seam torus is a curve, $\vec{A}_B$ is the line of $\bar{\boldsymbol\sigma} = (-1, 1, -1, 1)$ in the shares $(a_{D_0}, a_{D_1}, b_{D_0}, b_{D_1})$, and the directional ratio is $x^*_{D_0}/x_{D_1}$, which is ϱ_B at the seam points (Proposition 27.10); every coordinate of the torus is a power of ϱ_B, so the directional law is a power of ϱ_B times the law of Definition 19.41, and by Theorems 30.10 and 19.42 the reach along $\bar{\boldsymbol\sigma}$ is the reach $\operatorname{reach}(B)$ of the bunch.

Theorem 30.10 (The reach along a direction is read on the law)**.** *Let $\bar{\boldsymbol\sigma}$, $\mathbf{w}_0$, $\bar{\boldsymbol\vartheta}$, the directional law $\lambda = \lambda_{\bar{\boldsymbol\sigma}, \mathbf{w}_0 + i\bar{\boldsymbol\vartheta}}$, its line function ℓ and its reach r be as in Definition 30.9.*

(i) On (r, ∞) the measure λ has a density which is a wall series of radius r. Its terms are the poles of ℓ with $\operatorname{Re} z > 0$: a pole η with the principal part $\sum_j (-1)^{j+1} j!\, a_{\eta,j} (z-\eta)^{-j-1}$ gives the terms $a_{\eta,j} X^{-\eta-1} (\log X)^j$, and there are no others.

(ii) If on some (x_0, ∞) the measure λ has a density which is a wall series of radius x_0, then $r \le x_0$; and for every $x' > x_0$ there is C with $|\ell(z)| \le C\delta^{-J-1} x'^{\operatorname{Re} z}$ at every z with $\operatorname{Re} z \ge 0$ at distance at least $\delta \le 1$ from the poles, whatever $\operatorname{Im} z$, J the largest power of the logarithm in the series.

(iii) So r is the least $x_0 \ge 0$ for which λ has on (x_0, ∞) a density which is a wall series of radius x_0. It does not change when $\mathbf{w}_0$ moves along $\bar{\boldsymbol\sigma}$, $r(c\bar{\boldsymbol\sigma}; \cdot) = r(\bar{\boldsymbol\sigma}; \cdot)^c$ for rational $c > 0$, and r is at least every point near which λ has no real-analytic density. At $\bar{\boldsymbol\vartheta} = 0$ the law λ is positive and carried by the closure of the values of $\mathbf{x}^{-\bar{\boldsymbol\sigma}}$ on the support of the law of the piece: $r \ge X_{\min}$, the least of these values, when it is positive, the least value of the directional ratio taking the place of the least ratio (Proposition 19.33); and no wall falls along $\bar{\boldsymbol\sigma}$ exactly when $\mathbf{x}^{-\bar{\boldsymbol\sigma}}$ is bounded on that support, and then $r = X_{\max}$, the largest. When the seam word has a Newton sum of full dimension, as for a piece of tables whose tables form a connected

family (Theorem 29.3(ii)), $X_{\min} = (\sup \mathbf{V}_B^{\bar\sigma})^{-1}$ and $X_{\max} = \sup \mathbf{V}_B^{-\bar\sigma}$, the suprema of the monomials of the letter vector of Theorem 29.2 (Definition 29.1).

(iv) At $\bar\vartheta = 0$ let $\mathbb{L} = \{z_0 + kz_1 : \ k \in \mathbb{N}\}$, $z_1 > 0$ rational, be a progression at positive distance from the poles of ℓ, and put $r_{\rm res} = \limsup_\eta \max_j |a_{\eta,j}|^{1/\eta}$ over the poles and $r_{\mathbb{L}} = \limsup_k |\ell(z_0 + kz_1)|^{1/(z_0+kz_1)}$. Then $r = \max(r_{\rm res}, r_{\mathbb{L}})$: the reach is carried by the residues of the line or by its values on every progression.

(v) For every $\theta \geq 0$, r is the infimum of the $M \geq 0$ for which, for every $\delta > 0$, there are C and n' with $|\ell(z)| \leq C(1 + |z|)^{n'} M^{\operatorname{Re} z} e^{\theta |\operatorname{Im} z|}$ at every z with $\operatorname{Re} z \geq 0$ at distance at least δ from the candidate poles: the growth of the line function on its whole half-plane, at any allowance in the imaginary direction, is the reach.

At width one the same argument, run on the law of a family at a node, reads the growth of the family on its law, and that of the bunch is its reach.

Corollary 30.11 (At width one the growth of a family is read on its law)**.** *In Section 19.7 let λ_B be the image of the base form ϕ_B under ϱ_B and λ_R the image of ϕ_R under $1/\varrho_R$. For $X \in \{B, R\}$ the growth* $\operatorname{growth}(X)$ *along the level d (Definition 19.31) is the least $x_0 \geq 0$ for which λ_X has on (x_0, ∞) a density that is a wall series of radius x_0. In particular* $\operatorname{growth}(B) = \operatorname{reach}(B)$*: the inequality* $\operatorname{reach}(B) \leq \operatorname{growth}(B)$ *of Theorem 19.32(i) is an equality, and Theorem 19.32(iii) closes a node on the lattices of both families when the two points beyond which their laws are their own wall series multiply to less than one.*

A closing of the share side moves the imaginary parts of the shares over a whole fibre, so its bounds read the seam function on every imaginary translate of a line as well as on the line itself.

Corollary 30.12 (The reach of a tube)**.** *Let $\bar{\boldsymbol\sigma}$ and $\mathbf{w}_0$ be as in Definition 30.9 and $V' \subseteq \vec{A}_B$ a subspace containing $\bar{\boldsymbol\sigma}$. For $\theta \geq 0$ let $g_{V'}(\bar{\boldsymbol\sigma}; \theta)$ be the infimum of the M for which there are C, n_1 and n_2 with*

$$\left|P_{W_B(\mathbf{w}_0 + \tau\bar{\boldsymbol\sigma} + i\bar{\boldsymbol\vartheta})}(\mathbf{m}_B)\right| \leq C\,\delta^{-n_2}(1 + \tau + |\bar{\boldsymbol\vartheta}|)^{n_1} M^\tau e^{\theta|\bar{\boldsymbol\vartheta}|_1} \qquad (\tau \geq 0,\ \bar{\boldsymbol\vartheta} \in V'),$$

δ the smaller of 1 and the distance from the hyperplanes of (W1), and let $r^\sharp_{V'}(\bar{\boldsymbol\sigma}; \mathbf{w}_0) = \lim_{\theta \downarrow 0} g_{V'}(\bar{\boldsymbol\sigma}; \theta)$ be the reach of the tube *over V'.*

(i) $r(\bar{\boldsymbol\sigma}; \mathbf{w}_0 + i\bar{\boldsymbol\vartheta}) \leq r^\sharp_{V'}$ for every $\bar{\boldsymbol\vartheta} \in V'$; in particular $r(\bar{\boldsymbol\sigma}; \mathbf{w}_0) \leq r^\sharp_{V'}$, with equality when $V' = \mathbb{R}\bar{\boldsymbol\sigma}$.

(ii) A certificate on the real fibres. *Let $\mathbf{v}_1, \dots, \mathbf{v}_{n_{\mathcal{D}}-1}$ complete $\bar{\boldsymbol\sigma}$ to a basis of $\vec{A}_B$, and $y_i = \langle \mathbf{v}_i, \log \mathbf{x} \rangle$ on the torus. If the image of the measure $\mathbf{x}^{-\mathbf{w}_0}(\boldsymbol\varrho_B^\rho)_* \phi_B^\rho$ under $(\mathbf{x}^{-\bar{\boldsymbol\sigma}}, \mathbf{y})$, on the part where $\mathbf{x}^{-\bar{\boldsymbol\sigma}} > x_0$, is $\sum_{\eta,j} X^{-\eta-1}(\log X)^j dX \otimes \phi_{\eta,j}(d\mathbf{y})$, $j \leq J$, with finite signed measures $\phi_{\eta,j}$, the η in finitely many sets $\eta_i + \mathbb{N}$, and $\sum_{\eta,j} \|\phi_{\eta,j}\| x'^{-\eta} < \infty$ for every $x' > x_0$, then $g_{\vec{A}_B}(\bar{\boldsymbol\sigma}; 0) \leq x_0$, so that $r^\sharp_{\vec{A}_B} \leq x_0$: an expansion at the face over the real fibres certifies the tube, with no allowance for the phase.*

The certificate asks the expansion to hold over whole fibres of the directional ratio, with fixed measures on them, along one direction. Read in the coordinates in which the shares pair with the law linearly, expansions with fixed tangential measures at the falling ends certify the tube in every direction at once: in those coordinates a complex share multiplies the law by a factor of modulus one, and the continuation is made on the law itself.

Definition 30.13 (Log expansions of a law)**.** For a piece, with the space $\vec{A}_B$ of the directions of its shares of Definition 30.7 and its dual $\vec{A}_B^*$, the *log-ratio map* sends a point $\mathbf{x}$ of the seam torus to the linear function $\mathbf{z} \mapsto \langle \mathbf{z}, \log \mathbf{x} \rangle$ on $\vec{A}_B$, written $\log \mathbf{x}|_{\vec{A}_B}$; for a real point $\bar{\mathbf{c}}$ of the pins at which the seam function converges, $\lambda_{\bar{\mathbf{c}}} = (\log \mathbf{x}|_{\vec{A}_B})_* \big(\mathbf{x}^{-\bar{\mathbf{c}}} (\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho\big)$ is a finite positive measure on $\vec{A}_B^*$, and the seam function at $\bar{\mathbf{c}} + \mathbf{z}$ is $\int e^{-\langle \mathbf{z}, f \rangle}\, d\lambda_{\bar{\mathbf{c}}}(f)$ for $\mathbf{z} \in \vec{A}_B + i\vec{A}_B$ wherever the integral converges absolutely. A *log expansion* is the image of the complex measure

$$\sum_{\boldsymbol{\eta}, \mathbf{j}} \mathbf{p}^{\mathbf{j}} e^{-\langle \boldsymbol{\eta}, \mathbf{p} \rangle}\, d\mathbf{p} \otimes \phi_{\boldsymbol{\eta}, \mathbf{j}}(df) \qquad \text{on } [0, \infty)^k \times Y_Q$$

under $(\mathbf{p}, f) \mapsto f - \sum_{h \le k} p_h u_h$, where $u_1, \dots, u_k \in \vec{A}_B^*$ are linearly independent, $\boldsymbol{\eta}$ runs over the union of finitely many translates of $\mathbb{N}^k$ by positive vectors, $\mathbf{j} \le \mathbf{J}$, and the $\phi_{\boldsymbol{\eta}, \mathbf{j}}$ are complex measures on a compact set $Y_Q \subset \vec{A}_B^*$, its *inner face*, with $\sum \|\phi_{\boldsymbol{\eta}, \mathbf{j}}\| < \infty$. The law *has log expansions at* $\bar{\mathbf{c}}$ when $\lambda_{\bar{\mathbf{c}}}$ is the sum of finitely many log expansions Q and of a complex measure ϕ_0 carried by a compact set Y_0. Its *inner bound* along a direction $\bar{\boldsymbol{\sigma}} \in \vec{A}_B$ is the largest value of the directional ratio $e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle}$ on Y_0 and on the inner faces,

$$X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}) = \max \Big(\max_{f \in Y_0} e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle},\ \max_{Q} \max_{f \in Y_Q} e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle} \Big),$$

whose logarithm is convex in $\bar{\boldsymbol{\sigma}}$.

An end chart of Theorem 30.2 on a compact tangential source, whose tangential map and cut do not depend on $\mathfrak{t}$ and whose normal coordinates enter the log-ratio map through k linearly independent elements of $\vec{A}_B^*$, gives finitely many log expansions and a compact part: the log-ratio map is linear in $\log \mathfrak{t}$ and in the logarithms of the tangential coordinates; with $p_h = \log(R_h'/\mathfrak{t}_h)$ on the polydisc of the radii $R_h' < R_h$ the exponents are $\boldsymbol{\kappa} + \mathbf{n}$ and the coefficients $\widetilde{A}_{\mathbf{n}} R'^{\boldsymbol{\kappa} + \mathbf{n}}$, summable by Theorem 30.2(ii); where some $\mathfrak{t}_h$ lie in $[R_h', R_h)$ these join the tangential coordinates, the others keeping their expansions, summable uniformly by Cauchy's estimate on the closed polydisc; and where all of them do, the part is compact. A law with independent coordinates has such log expansions from the expansions of its factors at their ends.

Proposition 30.14 (A law with log expansions continues without phase)**.** *Let the law of a piece have log expansions at* $\bar{\mathbf{c}}$ *(Definition 30.13). Then for* $\mathbf{z} \in \vec{A}_B + i\vec{A}_B$ *off the hyperplanes* $\langle \mathbf{z}, u_h \rangle = \eta_h$ *of the log expansions,*

$$P_{W_B(\bar{\mathbf{c}} + \mathbf{z})}(\mathbf{m}_B) = \int e^{-\langle \mathbf{z}, f \rangle}\, d\phi_0(f) + \sum_{Q} \sum_{\boldsymbol{\eta}, \mathbf{j}} \prod_{h \le k} \frac{j_h!}{(\eta_h - \langle \mathbf{z}, u_h \rangle)^{j_h + 1}} \int_{Y_Q} e^{-\langle \mathbf{z}, f \rangle}\, d\phi_{\boldsymbol{\eta}, \mathbf{j}}(f),$$

continued: the seam function at $\bar{\mathbf{c}} + \cdot$ *is meromorphic on* $\vec{A}_B + i\vec{A}_B$ *with its poles on these hyperplanes, and there are* C *and* n *with*

$$\big| P_{W_B(\bar{\mathbf{c}} + \mathbf{z})}(\mathbf{m}_B) \big| \le C\, \delta^{-n} \max \Big(\max_{f \in Y_0} e^{-\langle \mathrm{Re}\, \mathbf{z}, f \rangle},\ \max_{Q} \max_{f \in Y_Q} e^{-\langle \mathrm{Re}\, \mathbf{z}, f \rangle} \Big),$$

δ *the smaller of* 1 *and the distance of* $\mathbf{z}$ *from these hyperplanes; the same holds, with another constant, for* δ *the smaller of* 1 *and the distance from the polar hyperplanes of the seam function alone. On the contour through a point* $\bar{\mathbf{c}} + \tau\bar{\boldsymbol{\sigma}}$, $\tau \ge 0$, *the bound is* $C\delta^{-n} X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})^\tau$, *with no factor in the imaginary parts and no power of* τ. *So, for a piece whose seam word has a Newton sum of full dimension and a rational* $\bar{\boldsymbol{\sigma}}$, $r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) \le r^\sharp_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) \le g_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; 0) \le X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})$, *and the growth of Theorem 29.3(iii) is* $g(\boldsymbol{\sigma}; 0) \le X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})$ *for every direction* $\boldsymbol{\sigma}$ *of the shares of Theorem 29.3(i) whose sums over the monomials are* $\bar{\boldsymbol{\sigma}}$.

The certificate of Corollary 30.12(ii) is one expansion along the direction itself, with fibre measures of any extent and no condition on the rest of the law: on the contours through the points of a ray of $\bar{\boldsymbol{\sigma}}$ the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ has modulus one and only the values of the directional ratio enter, not the compactness of the faces. Neither certificate applies as stated where the tangential coordinates of a face move with the normal ones, as when the support of the law curves in its log-ratio coordinates, or where a cut of the fibres moves, as in Theorem 30.2: the measures on the fibres are then not fixed, and the imaginary translates are read one at a time by Theorem 30.10, or bounded by the ceilings of Theorem 30.15, which carry the phase of the letters on these lines through their constants.

Without such a certificate a tube can reach farther than every one of its lines. There are sector integrals in two shares, with a single wall and growth bounds, for which every line of the tube along the fall of the wall has a finite reach while the reach of the tube over all the directions of the shares is infinite: the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ enters the Taylor coefficients of a letter at the falling face through the powers of $\bar{\boldsymbol{\vartheta}}$. The seam function of a piece can do the same: when its seam function on the pins is a Beta corner whose two arguments move independently on the pins, it has growth bounds only at positive θ, and on an imaginary translate along which one argument alone moves in the imaginary direction the Beta corner grows like a power of the imaginary part whose degree grows with the real parts, so that its tube reaches infinitely far while each of its lines has a finite reach. The sector letters of a piece of tables are never monomials in the falling variables, through which alone the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ would enter without reaching the Taylor coefficients at the falling face. Theorem 30.16(i) is the limit $\theta \to 0$ of Theorem 29.3(iii); when the reach of the tube is infinite, the bound of that theorem at a positive θ, which pays for the phase, is the one that travels.

Theorem 30.15 (Ceilings, simple tops and fibres along a direction)**.** *Let the seam function of the piece converge at a real point of the pins with positive shares, let $\bar{\boldsymbol{\sigma}}$ and $\mathbf{w}_0$ be as in Definition 30.9, the walls being those of a fan of the seam word (Theorem 29.3(ii)), and for a sector π of it let $\varrho_{\pi,\bar{\boldsymbol{\sigma}}} = \prod_k q_k^{\dot{\kappa}_k}\, \mathbf{k}_\pi^{-\bar{\boldsymbol{\sigma}}}$ be the directional ratio in the sector (Definition 29.10), the sector letters L of $\mathbf{k}_\pi$ entering with the coordinates $\bar{\sigma}_L$ of $\bar{\boldsymbol{\sigma}}$ at their stations and $w_{0,L}$ of $\mathbf{w}_0$. The keyhole continuations along a direction $\boldsymbol{\sigma}$ depend on $\boldsymbol{\sigma}$ only through $\bar{\boldsymbol{\sigma}}$, and the falling variables of π are those with $\dot{\kappa}_k < 0$; for a piece of tables the sectors of the orders π of B may be taken.*

(i) No reach exceeds the ceiling. *For every keyhole continuation along $\bar{\boldsymbol{\sigma}}$, of ceiling $H_{\mathcal{O}}$, and every $\bar{\boldsymbol{\vartheta}} \in \vec{A}_B$, $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}}) \le H_{\mathcal{O}}$: the phase of the letters does not enter the reach.*

(ii) The values at the integers. *Let $\bar{\boldsymbol{\sigma}}$ be scaled so that every $\dot{\kappa}_k$ is an integer, and let the walls of the falling variables of every sector be off the integers at $\mathbf{w}_0$. An* integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$ *is defined as in Definition 29.10 with every sector letter kept from vanishing except those L whose exponent $-w_{0,L} - m\bar{\sigma}_L$ at $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ is a nonnegative integer for every large integer m, its ceiling defined in the same way; its keyhole contour, its tops and its simple tops, with their exponents* $\exp(p)$ *and coefficients $A(p)$, are those of Definition 19.48 with $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of ϱ_π, the sector form $\phi_{\pi,\mathbf{w}_0} = |\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\kappa_k(\mathbf{w}_0)-1} \mathbf{k}_\pi^{-\mathbf{w}_0} dq$ in place of ϕ_π, the walls $\kappa_j(\mathbf{w}_0)$ of the sector in place of κ_j, and the faces $q_j = 0$ with $\dot{\kappa}_j = 0$ in place of the faces of the level variables. Then for every large integer m the seam function at $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ is the integral of $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}^m\, \phi_{\pi,\mathbf{w}_0}$ over the keyhole contour, and the growth at the integers* $\mathrm{growth}_{\mathbb{Z}}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = \limsup_m |P_{W_B(\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)|^{1/m}$ *is at most the ceiling of every integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$, and at most $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0)$.*

(iii) Simple tops. *If an integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$ of ceiling $H_{\mathcal{O}}$ has finitely many tops, all simple, and some $A_\zeta \neq 0$ (Theorem 19.49(iii)), then* $\mathrm{growth}_{\mathbb{Z}}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = H_{\mathcal{O}} \le r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0)$*; if moreover some keyhole continuation along $\bar{\boldsymbol{\sigma}}$ has the ceiling $H_{\mathcal{O}}$, then $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = H_{\mathcal{O}}$.*

(iv) Fibres. *Let $\bar{\boldsymbol{\sigma}}$ be scaled so that its coordinates and every $\dot{\kappa}_k$ are integers. Call a box of a continuation along $\bar{\boldsymbol{\sigma}}$ a* fibre box *when on its sets Y the letters with $\bar{\sigma}_L \geq 0$ do not vanish, and those with $\bar{\sigma}_L < 0$ do not vanish at the points with a falling coordinate 0; its stationary bound, as in Definition 19.52 with $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of ϱ_π, is the supremum of $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}|$ over the points of these sets with no falling coordinate 0 at which some falling coordinates lie inside their discs and the derivatives of $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in them vanish, the points with only one such coordinate, of falling rate one, left out. For every continuation along $\bar{\boldsymbol{\sigma}}$ whose boxes are keyhole boxes or fibre boxes, of ceiling $H_{\mathcal{O}}$, with the stationary bound S of its fibre boxes, $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) \leq \max(H_{\mathcal{O}}, S)$.*

So the reach along a direction sits between the growth at the integers and the ceiling of every keyhole continuation along it, and the two meet at simple tops that do not cancel when a keyhole continuation along the direction has the same ceiling, as at width one; and, by Theorem 30.10, without any top it is the far boundary of the region on which the directional law is its own wall series. The phase of the letters, which Theorem 29.11 pays for at the points where the ceiling is read, does not enter the reach: on a line the imaginary parts stay bounded. It enters only through the transverse fibre of a closing, which Theorem 30.16 reads.

The integer scaling in (iv) is needed: a sector with one falling variable of falling rate one and a letter entering with a coordinate that is not an integer can have a fibre box of stationary bound 0 and a reach above its ceiling, the directional ratio not being single valued in the disc, so that Lemma 19.50 does not apply; scaled to integers, the stationary point that the bound left out is counted.

Theorem 30.16 (The closing past the walls is read along its direction)**.** *Read the glue as in Theorem 29.3(i), the partner in its monomials, for a piece with walls and growth bounds whose seam word has a Newton sum of full dimension. Let V be the space of the directions of the shares, of dimension n, the number of free shares, $\boldsymbol{\sigma} \in V$ a direction with rational coordinates, $\bar{\boldsymbol{\sigma}}$ its sums, $V' \subseteq \vec{A}_B$ the space of the sums of the directions in V, and $\mathbf{c}$ the real starting point, with $\bar{\mathbf{c}} \in \Omega_B$, off the null set of Theorem 29.2 taken with the polar hyperplanes of the seam function. Let h_Γ be the Gamma rate of Theorem 29.3(iii), so that the rate of that theorem, which holds for such a piece (Theorem 29.3, its seam function converging at the sums $\bar{\mathbf{c}} \in \Omega_B$ of the positive shares of the starting point), is $h_\theta(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log g(\boldsymbol{\sigma}; \theta) + \frac{2\theta^2}{\pi} \sum_{\sigma_l > 0} \sigma_l$, and let I be the integrand of the share side.*

(i) The tube bounds the closing. *If $h_\Gamma(\boldsymbol{\sigma}) + \log r^\sharp_{V'}(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) < 0$, the closing along $\boldsymbol{\sigma}$ converges absolutely, its terms and the integrals left after them being $O\big(e^{\tau(h_\Gamma(\boldsymbol{\sigma}) + \log r^\sharp_{V'} + \epsilon)}\big)$ for every $\epsilon > 0$; the closing along $\boldsymbol{\sigma}$ is the closing along every direction of its chamber (Theorem 29.6), whose rates bound it as well. For the share side of Theorem 29.3(iv), with $\boldsymbol{\sigma}$ a direction of its shares with rational coordinates and $\mathbf{c}$ off the null set of that part, the closing along $\boldsymbol{\sigma}$ converges absolutely when $\sum_l \sigma_l \log|\sigma_l| + \log r^\sharp_B + \log r^\sharp_R < 0$, in this sentence $r^\sharp_B$ and $r^\sharp_R$ being the reaches of the tubes over the space of the directions of its shares (Corollary 30.12) of the seam functions of the piece and of the partner along $\boldsymbol{\sigma}$ at $\mathbf{c}$; its terms and the integrals left after them are then $O\big(e^{\tau(\sum_l \sigma_l \log|\sigma_l| + \log r^\sharp_B + \log r^\sharp_R + \epsilon)}\big)$ for every $\epsilon > 0$.*

(ii) One free share. *Let $n = 1$, and put $r = r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}})$ and $\ell(z) = P_{W_B(\bar{\mathbf{c}} + z\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)$. Then $r^\sharp_{V'} = r$, and for every $r' > r$ the terms are $O\big((1+\tau)^{n'} e^{\tau(h_\Gamma(\boldsymbol{\sigma}) + \log r')}\big)$; so the closing converges absolutely when $h_\Gamma(\boldsymbol{\sigma}) + \log r < 0$. Let $r > 0$, let $\mathbb{L}$ be the progression of the crossings of a falling share l_0, and let the data keep the poles apart: from some point on, $\mathbb{L}$ meets no pole of ℓ and no pole of another Gamma, and no pole of ℓ is a pole of a Gamma. If the poles*

of ℓ crossed by the ray are simple from some point on, or the values of ℓ on $\mathbb{L}$ carry the reach (Theorem 30.10(iv)), then for every $\epsilon > 0$ infinitely many terms have modulus at least $e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r-\epsilon)}$: the rate $h_\Gamma(\boldsymbol{\sigma}) + \log r$ is exact, and the terms do not tend to zero when it is positive. Without either proviso, if those poles have order at most p, the same holds for all but at most $p-1$ of the partners obtained by multiplying the coefficients by $e^{\tau_0 v_l}$, $\tau_0 \in \mathbb{R}$, for a fixed $\mathbf{v}$ with $\langle \mathbf{v}, \boldsymbol{\sigma}\rangle = 1$, the rate of the partner of τ_0 being $h_\Gamma(\boldsymbol{\sigma}) - \tau_0 + \log r$. The same holds for the share side of Theorem 27.9(iii) with one free share and a partner whose seam function converges along the ray, with $\sum_l \sigma_l \log|\sigma_l| + \log\sup \mathbf{V}_R^{-\bar{\boldsymbol{\sigma}}}$ in place of $h_\Gamma(\boldsymbol{\sigma})$, the partners being the tinkered partners that multiply the room letters of each station D by $e^{\tau_0 v_D}$ (Definition 27.17); at a node of width one, where the share side has one free share, this rate is $\log\big(\operatorname{reach}(B)/\inf \varrho_R\big)$, the logarithm of the root rate of Theorem 19.38(ii).

(iii) Several free shares. *Let $n \geq 2$. Fix a complement $V_\perp$ of $\mathbb{R}\boldsymbol{\sigma}$ in V, a Lebesgue measure $d\boldsymbol{\vartheta}_\perp$ on it and $k_\sigma > 0$ with $d\boldsymbol{\vartheta} = k_\sigma\, dy\, d\boldsymbol{\vartheta}_\perp$ for $\boldsymbol{\vartheta} = y\boldsymbol{\sigma} + \boldsymbol{\vartheta}_\perp$, $d\boldsymbol{\vartheta}$ the measure of Definition 4.1. Let T_σ be the space of the* transverse tinkers, *the tinkers of the partner read in its monomials (Definition 27.17, with one multiplier e^{v_l} for each monomial in place of each coordinate of the seam vector), multiplying the coefficients read on V as $\xi_l \mapsto \xi_l e^{v_l}$, with $\langle \mathbf{v}, \boldsymbol{\sigma}\rangle = 0$, identified with the dual of $V_\perp$ and carrying the dual Lebesgue measure $d\mathbf{v}$. The crossings of the closing along $\boldsymbol{\sigma}$ are the same for every tinker, and at a crossing at the time τ^**

$$\int_{T_\sigma} e^{\langle \mathbf{v}, \mathbf{c}\rangle}\, \mathcal{R}_{\tau^*}\big(\boldsymbol{\xi} e^{\mathbf{v}}\big)\, d\mathbf{v} = -k_\sigma \operatorname*{Res}_{z=\tau^*} I(\mathbf{c} + z\boldsymbol{\sigma}),$$

$\mathcal{R}_{\tau^*}(\boldsymbol{\xi} e^{\mathbf{v}})$ the crossing term of the partner with the coefficients $\boldsymbol{\xi} e^{\mathbf{v}}$. Integrated over the transverse tinkers against the factor $e^{\langle \mathbf{v}, \mathbf{c}\rangle}$, the terms of the closing are $-k_\sigma$ times the residues of the integrand on the real line through $\mathbf{c}$ in the direction $\boldsymbol{\sigma}$; these are $O\big((1+\tau)^{n'} e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r')}\big)$ for every $r' > r = r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}})$, and under the conditions of (ii) on that line infinitely many of them have modulus at least $e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r-\epsilon)}$.

Theorem 30.17 (A closing with one free share is read on the law along it)**.** *Take, in Theorem 29.7, the share side of the kind (B), its bundles merged by Corollary 4.4, the bars marking sums over the bundles, and in this statement write*

$$I(\bar{\mathbf{a}}) = \frac{\prod_G \Gamma(\bar{a}_G)}{\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)}\; P_{W_B^{\mathrm{m}}(\bar{\mathbf{a}})}(\mathbf{m}_B),$$

the products over the bundles and over the active letters, with the merged seam function of Lemma 30.8. Let the pins of the merged shares leave a fibre of dimension one, spanned by a rational direction $\bar{\boldsymbol{\sigma}}$, and let $\bar{\mathbf{c}}$ be a real point of the merged pins with positive coordinates at which the merged seam function converges. Let the merged law along $\bar{\boldsymbol{\sigma}}$, the law along $\bar{\boldsymbol{\sigma}}$ of the merged seam function (Definition 30.9), be the image of $\prod_G e^{-\bar{c}_G y_G}\lambda$ under $X = \prod_G e^{-\bar{\sigma}_G y_G}$, λ the source measure, its line function $\ell(\zeta) = P_{W_B^{\mathrm{m}}(\bar{\mathbf{c}}+\zeta\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)$ and its reach r, the least $x_0 \geq 0$ for which it has on (x_0, ∞) a density that is a wall series of radius x_0 (Definition 30.9, Theorem 30.10(iii)); and in this statement write $h_\Gamma^{\mathrm{m}}(\bar{\boldsymbol{\sigma}}) = \sum_G \bar{\sigma}_G \log|\bar{\sigma}_G|$, the Gamma rate of the merged share side, whose coefficients are inside the merged values.

(i) There is $k_P > 0$, depending only on the pins, with $P_{B\sqcup R}(\mathbf{m}) = \frac{k_P}{2\pi i}\int_{\operatorname{Re}\zeta=0} I(\bar{\mathbf{c}} + \zeta\bar{\boldsymbol{\sigma}})\, d\zeta$, and the crossing term (Lemma 29.4) at a crossing ζ^* of the ray is $-k_P \operatorname{Res}_{\zeta=\zeta^*} I(\bar{\mathbf{c}} + \zeta\bar{\boldsymbol{\sigma}})$: at the crossing $\bar{a}_f = -n$ of the falling share f, at which no other factor of I is polar, it is

$$k_P\, \frac{(-1)^n}{n!\,|\bar{\sigma}_f|}\, \frac{\prod_{G\neq f} \Gamma(\bar{a}_G)}{\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)}\; P_{W_B^{\mathrm{m}}(\bar{\mathbf{a}})}(\mathbf{m}_B)$$

at $\bar{\mathbf{a}} = \bar{\mathbf{c}} + \zeta^*\bar{\boldsymbol{\sigma}}$*, and at a simple pole of the merged seam function at which no Gamma is polar it is* $-k_P \prod_G \Gamma(\bar{a}_G)/\big(\prod_j \Gamma(s_j)\prod_j \Gamma(t_j)\big)$ *there times* $\operatorname{Res}_{\zeta^*} \ell$*.*

(ii) For every $r' > r$ *there are* C *and* n' *such that the crossing terms at their times* τ*, and the integrals left at the times at a fixed distance from the crossings, are at most* $C(1+\tau)^{n'} e^{\tau h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r'^{\tau}$*.*

(iii) Let the ray $\bar{\mathbf{c}} + \tau\bar{\boldsymbol{\sigma}}$*,* $\tau \geq 0$*, cross its polar points simply: no crossing point of a falling merged share, one with* $\bar{\sigma}_G < 0$*, is a pole of the merged seam function or a crossing point of another falling share, and the poles of the merged seam function on the ray are simple. Then for each falling share* f *the crossing terms of* f *and those at the poles of the merged seam function have together the root rate* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r$*: the upper limit of* $|\mathcal{R}|^{1/\tau}$ *over them is* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r$*. The same holds when the data keep the poles apart only from some point on, as in Theorem 30.16(ii), read on* ℓ *and on the progression of the crossings of* f*, and the poles of* ℓ *crossed by the ray are simple from some point on or the values of* ℓ *on that progression carry the reach; without either proviso, if those poles have order at most* p*, it holds for all but at most* $p-1$ *of the partners obtained by multiplying the coefficients of the shares of each bundle* G *by* $e^{\tau_0 v_G}$*,* $\tau_0 \in \mathbb{R}$*, for a fixed* $\mathbf{v}$ *with* $\sum_G v_G \bar{\sigma}_G = 1$*, whose merged laws are constant multiples of the images of the merged law under* $X \mapsto e^{-\tau_0} X$*, with the root rates* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}}) - \tau_0} r$*.*

(iv) So the closing of the merged share side along $\bar{\boldsymbol{\sigma}}$*,* $P_{B \sqcup R}(\mathbf{m}) = \sum \mathcal{R}$ *over the crossings of the ray in their order, converges absolutely when* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r < 1$*, and under the conditions of (iii) its terms do not tend to zero when* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r > 1$*. No hypothesis on tops or on the phase of the letters enters, only the separations of the poles in (iii): the rate is read on the merged law along the direction. Where* X *is bounded on the support of the law of the piece,* r *is its largest value there, and when moreover the falling shares are alone in their bundles,* $h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}}) + \log r$ *is the averaged rate* $\hat{h}_C$ *of Theorem 29.7(iv); where* X *is unbounded,* $\hat{h}_C$ *is infinite and the averaged terms diverge past the walls, and (iii) carries the sharpness of that theorem past them, with the reach of the merged law in place of the largest value of* X*. When every bundle is a single share,* $\boldsymbol{\sigma} = \bar{\boldsymbol{\sigma}}$*, the merged seam function is the product of* $\boldsymbol{\xi}^{-\mathbf{a}}$ *and of the seam function at the sums over the monomials (Theorem 29.3(i)),* r *is* $e^{-\sum_l \sigma_l \log \xi_l}$ *times the reach of the piece along the sums of* $\boldsymbol{\sigma}$ *(Definition 30.9), and* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r = e^{h_0(\boldsymbol{\sigma})}$*,* h_0 *the rate of Theorem 29.3(iii) at* $\theta = 0$*, whose growth* $g(\boldsymbol{\sigma}; 0)$ *is on a fibre of dimension one that reach: on such a fibre the rate of Theorem 29.3(iii) at* $\theta = 0$ *is exact under the conditions of (iii), and (iii) is Theorem 30.16(ii).*

The merged share side and the share side at the stations, before the stations of one excess are merged (Corollary 4.4), are two representations of one period, with two laws along their directions; completing the splits of a bundle can move the edge of the support at which the law stops being its own wall series. The merged law can stop being its own wall series at the image of a nearer edge of the support, where a level set of the merged seam letter ends the expansion of the law at its falling end, while the station closings, with more free shares, can read a farther edge: the merged closing can diverge where the station closing converges.

Corollary 30.18 (A law with log expansions bounds a closing of any fibre dimension)**.** *In Theorem 29.3(iii), with the starting point* $\mathbf{c}$ *off its null set, let the law of the piece have log expansions at the sums* $\bar{\mathbf{c}}$ *of* $\mathbf{c}$ *(Definition 30.13), with the inner bound* X_{in}*. The function* $h_\Gamma(\boldsymbol{\sigma}') + \log X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}')$ *is convex on the chamber* $C(\boldsymbol{\sigma})$ *of Definition 29.5. If its infimum* h' *on the chamber is negative, the closing along* $\boldsymbol{\sigma}$ *converges absolutely to* $P_{B \sqcup R}(\mathbf{m})$*, its terms and the integrals left after them being* $O(e^{\tau(h'+\epsilon)})$ *for every* $\epsilon > 0$*. The fibre may have any dimension, and no margin of coefficient*

factors and no phase enter: the rate is read on the compact part of the law and on the inner faces of its log expansions. In Theorem 29.3(iv), with $\mathbf{c}$ *off the null set of that part, if the laws of the piece and of the partner have log expansions at* $\mathbf{c}$*, with the inner bounds written* X^B_{in} *and* X^R_{in} *in this sentence, the same holds with* $\sum_l \sigma'_l \log|\sigma'_l| + \log X^B_{\mathrm{in}}(\boldsymbol{\sigma'}) + \log X^R_{\mathrm{in}}(\boldsymbol{\sigma'})$*, convex on the chamber* $C(\boldsymbol{\sigma})$*, in place of* $h_\Gamma(\boldsymbol{\sigma'}) + \log X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}')$*.*

So the counterparts of the reach theorems of width one hold along every rational direction. The tinkers of Definition 27.17 play two parts: all the tinkers of one partner separate the laws (Theorem 27.18), and the integral over the transverse tinkers of a closing, the partner read in its monomials, reads the reach of the real line (Theorem 30.16(iii)).

For a positive kernel the root rate of a crossing term is the exponential of the least Gamma rate of its chamber in every fibre dimension (Theorem 29.26), and averaging reads the log-support of the source measure (Theorem 29.7); Corollary 30.18 reads, past the walls, the compact part and the inner faces of the log expansions. It is an upper bound: with two or more free shares a crossing term is a fibre integral of positive dimension, whose size the law along one direction does not fix, and the best direction of a chamber can lose against the averaged closing of the same source (Theorem 29.8); the exact rate is then read on a joint word at a noncancelling simple top (Theorem 29.21). One partner does not read the real line: it averages over its transverse fibre, and its terms can fall below the rate of every real line of its chamber, the integral of the terms over the transverse tinkers then being carried by the other tinkered partners. Whatever the closing, the reach along the direction of the closing is at least the least value of the directional ratio on the support of the law, and at least every point near which the directional law, or after merging the merged law, has no real-analytic density (Theorem 30.10(iii)).

30.3 What the calculus of bunches settles

Remark 30.19 (What the calculus of bunches settles, and what it leaves open)*.* A closing of the glue along a direction of its share fibre is exact at every step and depends only on its chamber (Lemma 29.4, Theorem 29.6); Theorems 29.2, 29.3 and 29.11 give sufficient conditions for it, and the closings can be averaged over the source (Theorem 29.7). The cases that these conditions leave are settled as follows.

(i) *Crossing terms of positive dimension.* Along a rational ray, at simple nonresonant crossings and after a finite head, a crossing term kept with its falling shares integrated on their strips is, along its cohort, a scalar G_n times a finite sum $\sum_i \big(\mathsf{T}_{-n;(1)}[c_i]\,\phi_i\big)(Y_i)$ of joint Trailmoments of fixed functions on fixed compact domains, the joint word (Theorem 29.18): a falling Gamma becomes a positive share and one Beta variable (Lemma 29.14), a continued source strip a segment and circle integrals (Lemma 29.15), and nothing is extracted from the term. For a positive kernel its rate is the least Gamma rate of the chamber, in every fibre dimension and with no hypothesis on the tops (Theorem 29.26); averaged over a positive source it is the averaged rate $\hat{h}_C$ (Corollary 29.27), so a nonzero averaged rate decides by its sign whether the averaged terms tend to zero, with no condition on the bundles or the dimension (Theorem 29.7(iv)), and at $\hat{h}_C = 0$ Theorem 29.31 decides. The joint word has the root rate $r_G c_{\max}$ at noncancelling simple tops, r_G the root rate of the scalar and $c_{\max}$ the ceiling of the word (Theorem 29.21). Coincident crossings are packets of factors polar on one crossed hyperplane, assembled before a sign or a top is read (Theorem 29.29, Proposition 29.30); at the root rate one a positive moment sequence is decided by its law near the top (Theorem 29.31), and an expansion by its coefficients (Proposition 29.32).

(ii) *The phase.* A joint word is bounded by $|G_n| \sum_i \|\phi_i\| \, c_{\max}^n$ (Corollary 29.19): the fibre is completed before a modulus is taken, and the phase of the letters enters the coefficient functions, where it can cancel, and costs no term of the bound. It confines the radii of the procedure to the margins of the coefficient factors (Theorem 29.18, Remark 29.22). Read on the law, the phase costs nothing where the law has log expansions: in its log-ratio coordinates the shares pair with the law linearly, and the compact part and the inner faces of its log expansions bound the continued seam function on whole contours (Proposition 30.14). Where the tangential coordinates move with the normal coordinates at a falling end, the phase enters the coefficients of the expansion there, and a tube can reach infinitely far while each of its lines has a finite reach.

(iii) *Regions of equality and bounds at any width.* The regions on which a law equals its expansion at a falling end are certified from the sector letters, with the map from the motion of the shares to the exponents, the coefficients, the truncation bounds and exact boundary words (Theorems 30.2 and 30.4); the domain where a series converges and the region where it equals the law are kept apart (Proposition 30.6), and density charts are built at regular fibres (Proposition 30.5). The terms of a closing have at most the root rate $r_G c_{\max}$ of every joint word whose radii keep within the holomorphy of its source and the margins of its coefficient factors, $c_{\max}$ the ceiling of the word, and have that root rate when the word has a noncancelling simple top.

(iv) *The reach at any width.* Along every rational direction of the shares the reach is read on the law: it is the far boundary of the region on which the directional law, the law multiplied by $\mathbf{x}^{-\mathbf{w}_0}$ for the starting point $\mathbf{w}_0$ and read through the directional ratio, is its own wall series, whose exponents lie among the values $(\kappa_U(\mathbf{w}_0) + n)/(-\dot{\kappa}_U)$ of the walls divided by their falling rates, and it is at least every point at which that law is not analytic, the least value of the directional ratio among them (Theorem 30.10). The walls and the growth bounds that this uses hold for every seam function of a piece that converges at a real point and for its seam letters merged by bundles (Lemma 30.8), and at width one the same argument reads the growth of a family on its law, that of the bunch being its reach (Corollary 30.11). For a piece the reach is at most the ceiling of every keyhole continuation along the direction, with no phase, equal to it at simple tops that do not cancel when a keyhole continuation has the same ceiling, and bounded by fibres where letters vanish inside the keyholes, the direction scaled to integers (Theorem 30.15). A closing past the walls converges when the exponential of the Gamma rate times the reach of its tube is less than one, its terms then having at most that root rate (Theorem 30.16(i)); an expansion at the face over the real fibres bounds the reach of the tube along one direction and log expansions in every direction, with no allowance for the phase (Corollary 30.12, Proposition 30.14), and otherwise the tube can reach farther than every one of its lines; a law with log expansions bounds a closing of any fibre dimension by the rate read on the compact part and the inner faces of its log expansions (Corollary 30.18). With one free share, as at a node of width one, the tube is the line, and the terms attain the rate that the reach along the direction gives when the poles of the line and of the Gammas keep apart and the poles of the line are eventually simple or its values on a progression of crossings carry the reach (Theorem 30.16); the closing of a share side merged by its bundles to one free share, a series other than the station closings, has under the same conditions the root rate of Theorem 30.17(iii), read on the merged law along its direction, its terms not tending to zero beyond it, which carries the sharpness of Theorem 29.7(iv) past the walls (Theorem 30.17). In every number of free shares the terms integrated over the tinkers of the partner transverse to the direction are the residues of the

real line, which attain its rate under the same conditions (Theorem 30.16), while the terms of one partner average over its transverse fibre and can fall below every real line of its chamber.

(v) *A gap at a family of tables.* A family of three tables whose seam has single stations has a chamber in which no direction certifies the closing while the averaged closing converges geometrically (Theorem 29.8); the gap is relative to the closings of one share side by the directions of its chamber (Remark 29.9).

At width one the reach is read on the law, as the point beyond which the law is its own wall series, and it is the ceiling of a keyhole continuation whose tops are simple and do not cancel (Theorems 19.42 and 19.49). At any width the first holds along every rational direction (Theorem 30.10), and the second along every rational direction for every piece, under the conditions of Theorem 30.15(ii) and (iii), and for joint words (Theorem 29.21). What a node of width one does not have is a transverse fibre. With two or more free shares the crossing terms of one partner integrate over the transverse fibre the imaginary translates of the line through its starting point. Theorem 30.16(i) bounds them by the reach of the tube when the exponential of the Gamma rate times that reach is less than one; a law with log expansions bounds the tube by the directional ratio on the compact part and the inner faces of its log expansions, and a closing of any fibre dimension with nothing paid for the phase (Corollary 30.18), while without log expansions the reach of the tube can be infinite though every line of the tube has a finite reach, and the bound that travels pays for the phase at a positive θ (Theorem 29.3(iii)). Theorem 30.16(iii) gives the integral of the terms over the transverse tinkers the rate of the real line, under the conditions of its part (ii) on that line; Theorem 29.21 gives the rate of one partner at the noncancelling simple tops of its joint words. The real line does not give the rate of one partner, whose terms can fall below every real line of its chamber, and the best direction of a chamber can lose against averaging (Theorem 29.8). Before the walls, under the hypotheses of Theorem 29.7(iv), the rate of one partner is averaged over the source, and past them the closing of a share side merged by its bundles to one free share has, under the conditions of Theorem 30.17(iii), the rate read on the merged law along its direction. Two problems are open. The first is the rate past the walls of one partner whose joint words have no noncancelling simple top: at two or more free shares, where it asks for an averaged closing of the continued law, and at one free share when the data do not keep the poles apart or no proviso of Theorems 30.16(ii) and 30.17(iii) holds for that partner. The second is to read the growth $g(\boldsymbol{\sigma};\theta)$ on the law, as Theorem 30.10 reads the reach of a line. At $\theta = 0$ the reach of the line bounds it below and log expansions bound it above by their inner bound (Corollary 30.12, Proposition 30.14), and the two meet when, for every $\epsilon > 0$, the law has log expansions whose compact part and inner faces lie where the directional ratio is at most the reach plus ϵ; in general, and without log expansions, where it can be infinite at $\theta = 0$, it is known only through these bounds and those of Theorems 29.3(iii) and 29.11 and of Corollary 29.19, within the margins of the coefficient factors (Theorem 29.18). Degenerate and cancelling tops are open at width one as well.

Chapter 31

Price and complexity

A construction is a description of a family, and the exact theorems of Chapter 21 turn it into a description of the period; both have a price. The price of a construction is the length of its description, a repeated subtree described once and its multiplicity written in binary; the work of a word is the dimension of the integrals it asks for, an integral that recurs counted once. Repetition then costs what it should: M copies of a cell glued along a tree at a full lean, or closed in series at any exponents, or chained, have periods whose words cost the word of the cell and the binary length of M, and a departure from the repetition costs the cells that depart. The laws of cells (Chapter 22), the networks of holds (Chapters 24 and 25) and the finite interfaces (Chapter 26) are priced the same way: a law is computed once for each distinct subtree, a node of a network costs its periods at the two full leans whatever the number of bases of the whole table, and a polynomial Gamma moment of a repetition is the power of one finite matrix. What a value can contain is bounded the same way along a ray (Theorem 35.19).

31.1 Constructions and their price

Definition 31.1 (Construction)**.** A *construction* $\mathfrak{C}$ is a finite rooted tree with a family at each leaf, its marks, bases, weights, masses and exponents given, and at each inner node one of the operations: the direct sum of its subtrees; the substitution of its subtrees, each with a named hinge, into the marks of a core family named at the node, a mark left without a subtree being left in place (Definition 20.1); and the *powers* of a subtree $\mathfrak{C}_1$ with an integer M: $\mathfrak{C}_1^{\oplus M}$, M copies in direct sum; $\circ(\mathfrak{C}_1^{M})$, the series closure of M copies of the cell $\mathfrak{C}_1$ (Corollary 20.5(ii)); and $\mathfrak{C}_1^{\triangleright M}$, the chain of M copies glued along two named marks (Theorem 21.10). A subtree may be given by a reference to one already described. A construction builds one family, by Proposition 20.3 at each substitution.

Definition 31.2 (Price; work)**.** The *price* price($\mathfrak{C}$) of a construction is the number of bits of its description as written: for every leaf its marks, bases, weights, masses and exponents, the numbers written as rationals in binary; for every node its operation, its core and its attachments; for every power the $\lceil \log_2(M+1) \rceil$ bits of M; a subtree may be written once and referred to by an index wherever it recurs. The price of a word (Definition 1.1(vi)) is defined in the same way, and its *work* work(w) is the sum of the dimensions of its distinct integrals, an integral that recurs, under a power or an iterate in particular, counted once. The integration count of Definition 19.56 counts every occurrence: the work of w^M is that of w, and its integration count is M times that of w.

The work counts the distinct integrals a word is written with, and the price the length of what

has to be written down; neither counts the digits of an evaluated value, nor the cost of evaluating: a scalar power costs one exponentiation, a power of a $k \times k$ matrix costs $O(k^3 \log M)$ operations by repeated squaring, and an iterate $\mathcal{Q}^M$ of an operator costs M applications of $\mathcal{Q}$, nested integrals of M times its dimension, or $O(\log M)$ matrix products when $\mathcal{Q}$ leaves a finite-dimensional space of functions invariant. Every polynomial Gamma moment of degree D of a repetition (Chapter 26) is such a matrix power, a fixed row times a power of the transfer matrix $\mathbf{T}_D$ applied to the moment vector of the start (Theorem 26.5), and the period itself is reached through such moments with the certified stopping rule of Theorem 26.7, at a degree that depends on the accuracy asked.

Theorem 31.3 (Pricing by repetition)**.** *Let $\mathcal{A}$ be a family, a cell where a hinge is needed, described by a construction of price* $\mathrm{price}(\mathcal{A})$*, with the period $P_{\mathcal{A}}$ at its data, and let $M \geq 1$.*

(i) Direct sum. *The molecules of $\mathcal{A}^{\oplus M}$, $\mathcal{A}$ connected, are its M copies, and the product of their periods is $P_{\mathcal{A}}^{M}$ (Proposition 7.8; the period of the direct sum itself diverges on the balance wall, Proposition 8.15): price at most* $\mathrm{price}(P_{\mathcal{A}}) + \lceil \log_2(M+1) \rceil + O(1)$*, work that of $P_{\mathcal{A}}$.*

(ii) A tree of copies at a full lean. *For a table $\mathcal{A}$ at the exponents $(0,t)$, M copies glued along any tree at the masses $(\kappa, t-\kappa)$ on the two marks of every gluing have the word* $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa,\ t-\kappa\\ t\end{bmatrix} \big(P_{\mathcal{A}}(\mathbf{m};0,t)/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa,\ t-\kappa\\ t\end{bmatrix}\big)^{M}$ *(Corollary 21.7): price at most* $\mathrm{price}(P_{\mathcal{A}}) + \lceil \log_2(M+1) \rceil + O(1)$*, work that of $P_{\mathcal{A}}$, whatever the shape of the tree.*

(iii) A series closure. *The series closure of n_k copies of cells $\mathcal{A}_k$, $k \leq d$, at positive exponents of N tables on the balance wall has the word of Theorem 21.9,* $\prod_j (\Gamma(s_j)\Gamma(t_j))^{-1} \int \prod_j x_j^{s_j-1} \prod_{j\geq 2} y_j^{t_j-1} \prod_k \mathfrak{L}_k(x,y)^{n_k}\, dx\, dy$ *with the responses $\mathfrak{L}_k$ of the cells: price $O\big(\sum_k \mathrm{price}(\mathcal{A}_k)\big) + \sum_k \lceil \log_2(n_k+1) \rceil + O(N)$, the letters of a constructed cell being written through Theorem 20.4, and work at most $\sum_k |B_k| + 2N - 1$, B_k the marks of $\mathcal{A}_k$ besides its hinge, independent of the multiplicities and of the order of the cells.*

(iv) A chain. *$\mathcal{A}^{\triangleright M}$ has the word $(\mathcal{Q}^M P_0)(\mathbf{1})$ of Theorem 21.10, with the operator $\mathcal{Q}$ and the function P_0 of that theorem: price at most* $\mathrm{price}(\mathcal{A}) + \lceil \log_2(M+1) \rceil + O(1)$*, work at most $|\mathcal{E}_{\mathcal{A}}| - 1$; evaluated, the iterate unrolls to nested integrals of total dimension $M(|\mathcal{E}_{\mathcal{A}}| - 2) + 1$, that of the expanded period, unless $\mathcal{Q}$ leaves a finite-dimensional space of functions invariant; where the twist is pinned the word is* $P_{\mathcal{A}}^{M}/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} m_{g_{\mathrm{in}}},\ m_{g_{\mathrm{out}}} \\ m_{g_{\mathrm{in}}} + m_{g_{\mathrm{out}}}\end{bmatrix}^{M-1}$*, $m_{g_{\mathrm{in}}}, m_{g_{\mathrm{out}}}$ the masses of the two named marks $g_{\mathrm{in}}, g_{\mathrm{out}}$, of work $|\mathcal{E}_{\mathcal{A}}| - 1$ evaluated as well.*

(v) Integer joint moments *of a repeated transfer matrix, with Gamma laws on the marks or Dirichlet laws on the steps, have as word a fixed row times the power $\mathbf{A}^M$ of the one matrix of Theorem 21.12 times a fixed column, with rational entries at rational masses (Theorem 21.12): price at most that of the matrices of the cell plus $\lceil \log_2(M+1) \rceil + O(1)$, work zero.*

(vi) Blocks *read through their totals cost one stacked Gamma each,* $\Gamma_{\begin{smallmatrix}|B_i|\\1\end{smallmatrix}}\!\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix}$ *at the block B_i (Theorem 21.8): the work of the period of the family with blocks is that of the period of its core family, whatever the lengths of the blocks.*

(vii) A series repetition. *M copies of a series-balanced cell $\mathcal{A}$ at a series node against a cell $\mathcal{A}'$, at the exponents (s,t), have the word* $(\Gamma(s)\Gamma(t))^{-1} \int_0^\infty y^{t-1} f_{\mathcal{A}}(y)^M f_{\mathcal{A}'}(y)\, dy$*, with the transforms $f_{\mathcal{A}}, f_{\mathcal{A}'}$ of Theorem 23.2: price at most* $\mathrm{price}(\mathcal{A}) + \mathrm{price}(\mathcal{A}') + \lceil \log_2(M+1) \rceil + O(1)$*, work $|B| + |R| - 1$ for the marks B of $\mathcal{A}$ and R of $\mathcal{A}'$ besides the hinge, whatever M; at a*

series or parallel node of a hinge tree the multiplicities of equal subtrees enter as exponents (Proposition 22.16).

(viii) Series-parallel tables. *A table with a hinge tree of series and parallel nodes over marks has periods at the full leans that are stacked Gammas times monomials in the weights of its nodes, with one Gamma for each mark and two for each change of kind along an edge, and every period, at every lean and every twist, a fibre integral of dimension $n-1$ of an explicit stacked Gamma times an exponential of the loads, and a family of N such tables one of dimension $(2N-1)(n-1)$ (Corollaries 25.11 and 25.13): price linear in the size of the tree, work at most $n-1$, less one for each pooled pair (Proposition 25.14), and at most $(2N-1)(n-1)$ for the family, whatever the numbers of bases, which can be exponential in n.*

(ix) Polynomial Gamma moments of a repetition *have as word a fixed row times a power of the transfer matrix $\mathbf{T}_D$ applied to a fixed vector, with rational entries at rational data (Theorem 26.5): price that of the matrix of the chain cell plus $\lceil \log_2(M+1) \rceil + O(1)$, work zero.*

In the cases with a multiplicity M the expanded family has of the order of M times as many marks, and its period written as one integral over its simplex has the work of one less than its number of marks.

Theorem 31.4 (Repeated subtrees and departures)**.** *(i) In a construction, replacing identical subtrees, with matching hinges and data, by one subtree and references leaves the family built, the period and every exact word of Chapter 21 unchanged, and lowers the price by the price of the copies removed, less that of the references.*

(ii) In (ii), (iii) and (iv) of Theorem 31.3, replacing q of the M copies by cells $\mathcal{A}'_1, \dots, \mathcal{A}'_q$ changes the word to $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa,\ t-\kappa\\ t\end{bmatrix} \left(P_{\mathcal{A}}/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa,\ t-\kappa\\ t\end{bmatrix}\right)^{M-q} \prod_i \left(P_{\mathcal{A}'_i}/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa_i,\ t-\kappa_i\\ t\end{bmatrix}\right)$ at a full lean, to the product of $\mathfrak{L}_{\mathcal{A}}^{M-q}$ and of the responses of the new cells in a series closure, and to the composite of $\mathcal{Q}^{M-q}$ with the q operators of the new cells in a chain: the price grows by $\sum_i \operatorname{price}(\mathcal{A}'_i) + O(q \log M)$ and the work by at most the dimensions of the integrals of the new cells. A change of q leaves in a tree with at most h nodes on every path from its root to a leaf touches at most qh of its nodes.

Repetition is thus priced at the binary length of its count, as 2^M is written with M rather than with its digits, and the work follows the distinct subtrees. The construction is the certificate of the compression: a family given expanded, without its construction, can still be read for its bunches of width one, its prime nodes (Theorem 20.9) and its twins (Theorem 14.22), but finding its cheapest construction is a search. What is not settled is the other side of the comparison with the integers: a lower bound on the price or the work of the period of a family that is not a repetition. The prime nodes of its canonical hinge tree are invariants of a list (Theorem 20.9), at a full lean its period is a product of stacked Gammas and the periods at the full leans of those nodes (Corollary 21.7, Theorem 25.6), and at every lean a fibre integral of their periods at the full leans, of dimension $n-1$ (Theorem 25.12); a prime node $\mathcal{H}_v$ on n_v marks with $|\mathcal{H}_v|$ bases has periods at the full leans of dimension $|\mathcal{H}_v| - n_v > 0$ on their share sides (Corollary 25.9), and whether they cannot be written with less work is open. Symmetry is a second source of cheapness, distinct from repetition: a table assembled from a seed by a set of bijections (Proposition 11.7) is described by the seed and the bijections, and the table that is one circuit on n marks, prime for no n, is a series closure of single marks whose period is one integral over one scale of a product of n Grammas (Corollary 17.20), of work one whatever n; the list $\binom{\mathcal{E}}{r}$ with $2 \le r \le |\mathcal{E}| - 2$ is prime and cheap to describe, and how much work its period needs is the first case of that open problem.

Part VI

The arithmetic of the output

Chapter 32

Positive words

The output of the theory is a value: a period, a Breta, a closing of fibre integrals, and such values are read arithmetically. A closing at rational data writes a value as a rational combination of Trailmoments of lattice inputs and rational heads (Proposition 5.13, Lemma 35.20); a sequence of such values that do not vanish, cleared by integers that grow more slowly than the values decay, shows that some Trailmoment it carries is irrational (Theorem 34.3, Chapter 36). What keeps a value from vanishing is positivity, and positive words are made in four ways: second jets of a period in its weights, even powers of the contrast of two bases, Beta ports on the pairs of a manufactured family, and Beta ports read against one mark with the room integrated. The jets and the contrasts are combinations of periods of the same family at shifted data on its balance wall, bounded by the period times a factor that their statements control; the port words of a manufactured family are combinations of its interests at shifted masses over one normalization, and the reads of the Beta ports are bounded by the read of the constant insertion.

32.1 Coefficient jets and basis contrasts

Theorem 32.1 (Positive coefficient jets)**.** *Let F have real exponents $s_j, t_j \geq 0$ and real masses on its balance wall, every proper wall positive. For an integer $n \geq 1$, the* depth*, the data $(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ are again on the balance wall, with walls $n\kappa_S$; in this statement $\mathbb{E}$ is the expectation of $U \sim \mathrm{Breta}^{\circ}(F; n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$. Choose* weight directions $\mathbf{b}_{j,T} \in \mathbb{Q}^k$*, one for each table j and basis $T \in \mathcal{B}_j$, and put*

$$\partial_{\mathbf{b}} K_j = \sum_{T \in \mathcal{B}_j} \mathbf{b}_{j,T}\, \alpha^{T}, \qquad \partial_{\mathbf{b}} \Psi_j = \sum_{T \in \mathcal{B}_j} \mathbf{b}_{j,T}\, \alpha^{\mathcal{E}\setminus T},$$

the derivatives of the letters of table j along the weights in these directions.

(i) For $v \in \mathbb{R}^k$ and small real ε, replace every table j by the two tables $(\mathcal{B}_j, \theta_j \pm \varepsilon\, v^{\mathsf{T}}\mathbf{b}_j)$, each with the exponents $(ns_j/2, nt_j/2)$. The period of this doubled family *at the masses $n\mathbf{m}$ is*

$$P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})\ \mathbb{E}\Big[\Big(\prod_j \Big(1 - \varepsilon^2\Big(\frac{v^{\mathsf{T}}\partial_{\mathbf{b}} K_j}{K_j}\Big)^2\Big)^{-\frac{ns_j}{2}} \Big(1 - \varepsilon^2\Big(\frac{v^{\mathsf{T}}\partial_{\mathbf{b}} \Psi_j}{\Psi_j}\Big)^2\Big)^{-\frac{nt_j}{2}}\Big)(U)\Big],$$

whose coefficient of ε^2 is $v^{\mathsf{T}}\mathbf{H}_n v$, with the jet matrix

$$\mathbf{H}_n = \frac{n}{2}\, P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})\ \mathbb{E}\Big[\sum_j \Big(s_j\, \frac{\partial_{\mathbf{b}} K_j\, \partial_{\mathbf{b}} K_j^{\mathsf{T}}}{K_j^2} + t_j\, \frac{\partial_{\mathbf{b}} \Psi_j\, \partial_{\mathbf{b}} \Psi_j^{\mathsf{T}}}{\Psi_j^2}\Big)(U)\Big].$$

$\mathbf{H}_n$ *is positive semidefinite, and positive definite when the directions* $\mathbf{b}_{j,T}$ *of the active tables span* $\mathbb{R}^k$.

(ii) For a rational positive semidefinite $k\times k$ *matrix* Q, $\operatorname{tr}(Q\mathbf{H}_n) > 0$ *exactly when* $\mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T} > 0$ *for some active table* j *and basis* T. *Completing squares writes* $Q = \sum_l c_l v_l v_l^{\mathsf{T}}$ *with rational* $c_l > 0$ *and rational* v_l, *and so writes* $\operatorname{tr}(Q\mathbf{H}_n) = \sum_l c_l\, v_l^{\mathsf{T}} \mathbf{H}_n v_l$ *as a sum of directional second jets with positive rational coefficients.*

(iii) In periods of F *(Theorem 26.9 at order two, the two copies of each table merged),*

$$\operatorname{tr}(Q\mathbf{H}_n) = \frac{n}{2} \sum_j \sum_{T,T'\in\mathcal{B}_j} \mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T'} \Big[s_j\, P_F\big(n\mathbf{m} + \mathbf{1}_T + \mathbf{1}_{T'};\, n\mathbf{s} + 2\mathbf{e}_j,\, n\mathbf{t}\big) + t_j\, P_F\big(n\mathbf{m} + \mathbf{1}_{\mathcal{E}\setminus T} + \mathbf{1}_{\mathcal{E}\setminus T'};\, n\mathbf{s},\, n\mathbf{t} + 2\mathbf{e}_j\big) \Big] :$$

every term stays on its balance wall, a basis insertion changes the wall of S *by* $|T\cap S| + |T'\cap S| - 2d_j(S) \geq 0$, *and a cobasis insertion by* $|S\setminus T| + |S\setminus T'| - 2d_j^*(S) \geq 0$.

(iv) Whenever (ii) gives positivity, $0 < \operatorname{tr}(Q\mathbf{H}_n) \leq n\, C(Q)\, P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, *with* $C(Q) = \frac{1}{2}\sum_j (s_j + t_j) \max_T \mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T}/\theta_{j,T}^2$.

(v) With the basis weights and the cobasis weights of every table taken as independent variables, the part of $\mathbf{H}_n$ *carried by* K_j *is the Hessian of* $P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ *in the basis weights of table* j, *paired with the directions* $\mathbf{b}_{j,T}$, *divided by* $2(ns_j + 1)$, *and the part carried by* Ψ_j *is the same in the cobasis weights divided by* $2(nt_j + 1)$. *On the share side of the period (Corollary 17.15) a weight derivative multiplies the integrand by minus the share divided by its weight, so every pin and every candidate pole hyperplane stays fixed.*

(vi) At the centre, $s_j = t_j$ *for every* j *and* $\mathbf{m} = \sigma\mathbf{1}$, *the doubled family is again at its centre, and the table's gauge* $\mathbf{1}$ *stays native (Proposition 8.21).*

A weight direction moves the two copies of a table apart symmetrically, so the period moves at second order only, by the expectation of a square under the Breta shape law: a jet of the period in its weights is a positive word. Under the basis law $\partial_{\mathbf{b}} K_j/K_j$ is the mean of the directions $\mathbf{b}_{j,T}/\theta_{j,T}$, $\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[\mathbf{b}_{j,T}/\theta_{j,T}]$, so the jet is the second moment, under the shape law, of the mean weight direction that the bases drawn at the shape carry. Table by table it is the period's own Hessian in the weights divided by $2(ns_j + 1)$ or $2(nt_j + 1)$, it is a finite combination of periods of the same family on the balance wall at data whose walls are no smaller, and it costs at most a factor linear in the depth against the period.

Lemma 32.2 (Rounding inside a rational subspace)**.** *Let* V *be the subspace of the real symmetric* $k\times k$ *matrices on which finitely many linear functionals* $Q \mapsto \operatorname{tr}(MQ)$, M *rational, vanish, spanned over* $\mathbb{R}$ *by integer symmetric matrices* $G_1, \dots, G_r$, *and let* $Q_* = \sum_j u_j G_j \in V$ *with* $Q_* \succeq \varepsilon I$, $\varepsilon > 0$. *For an integer* $\ell \geq 1$ *put* $Q = \sum_j \operatorname{round}(\ell u_j)\, G_j$ *and* $\beta = \frac{1}{2}\sum_j \big(\operatorname{tr} G_j^2\big)^{1/2}$. *Then* Q *is an integer matrix in* V *with* $Q \succeq (\ell\varepsilon - \beta)I$ *and* $\operatorname{tr} Q \leq \ell \operatorname{tr} Q_* + k\beta$; *for* $\ell > \beta/\varepsilon$ *it is positive definite.*

The jet word $\operatorname{tr}(Q\mathbf{H}_n)$ is linear in Q, so conditions on its coefficients of Trailmoments, such as the vanishing of chosen ones, define a rational subspace of matrices Q. A positive definite matrix in that subspace rounds to an integer one in the same subspace, and the jet word keeps its positivity and the vanishing of the chosen coefficients together at integer data.

Theorem 32.3 (Two bases generate positive contrast words of any depth)**.** *Let F have real data $(\mathbf{m};\mathbf{s},\mathbf{t})$ on its balance wall with every proper wall positive, with the pair coordinates $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ (Definition 8.1) and the centred potential Φ° (Theorem 8.18). Let $T \neq T'$ be two bases of a table j, and for integers $n,h \geq 1$ let the* contrast word *be*

$$\frac{1}{2}\,P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})\;\mathbb{E}_{U\sim\mathrm{Breta}^\circ(F;n\mathbf{m};n\mathbf{s},n\mathbf{t})}\Big[\big(\mathbb{P}_{j,U}(T)-\mathbb{P}_{j,U}(T')\big)^{2h}+\big(\mathbb{P}^*_{j,U}(T)-\mathbb{P}^*_{j,U}(T')\big)^{2h}\Big]>0;$$

the two contrasts have modulus below one on the open orthant.

(i) The first half of the contrast word is

$$\frac{1}{2}\sum_{l=0}^{2h}(-1)^l\binom{2h}{l}\theta_{j,T}^{2h-l}\theta_{j,T'}^{l}\,P_F\big(n\mathbf{m}+(2h-l)\mathbf{1}_T+l\mathbf{1}_{T'};\,n\mathbf{s}+2h\mathbf{e}_j,\,n\mathbf{t}\big),$$

and the second is the same with $\mathbf{1}_{\mathcal{E}\setminus T}$, $\mathbf{1}_{\mathcal{E}\setminus T'}$ in the masses and $2h\mathbf{e}_j$ added to the cobasis exponents; every term is balanced with its walls weakly raised. The inversion $\alpha\mapsto\alpha^{-1}$ carries the second half at $(\mathbf{m};\mathbf{s},\mathbf{t})$ to the first half at $(2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})$, the data of Theorem 8.13; at the centre the two halves are equal.

(ii) For rational $\eta>0$ and $h=\eta n$, along the integers n with $\eta n\in\mathbb{N}$, $\frac{1}{n}\log$ of the contrast word tends to

$$-\inf_{z\in\mathbf{1}^\perp}\Big(\Phi^\circ(z)-2\langle\boldsymbol{\mu}_\perp,z\rangle-2\eta\log\max\Big(\big|\mathbb{P}_{j,e^{2z}}(T)-\mathbb{P}_{j,e^{2z}}(T')\big|,\;\big|\mathbb{P}^*_{j,e^{2z}}(T)-\mathbb{P}^*_{j,e^{2z}}(T')\big|\Big)\Big);$$

the infimum exceeds the minimum of $\Phi^\circ-2\langle\boldsymbol{\mu}_\perp,\cdot\rangle$ on $\mathbf{1}^\perp$ and tends to infinity with η; that minimum is $-\lim_n\frac{1}{n}\log P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$.

(iii) Let the exponents be nonnegative, and write $q=2h$ and $H_q(x,y)=\sum_{l=0}^{q}(-1)^l\binom{q}{l}(x)_{q-l}(y)_l$. When $s_j>0$, the first half is the share side of $P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ (Corollary 17.15) with its integrand multiplied by $H_q(a_{j,T},a_{j,T'})/\big(2(ns_j)_q\big)$, a the basis shares; when $t_j>0$, the second half is the same with $H_q(b_{j,T},b_{j,T'})/\big(2(nt_j)_q\big)$, b the cobasis shares: polynomial insertions that keep every pin and every pole hyperplane.

Two bases of one table give a contrast of modulus below one, which stays away from one on the compact sets where the Breta shape law concentrates; so its even powers are positive words that become exponentially small against the period, at a rate that grows without bound with the ratio of the power to the depth. At the centre with unit weights the basis law gives every basis the same chance at the table's gauge, where the law concentrates, and the contrast vanishes there. On the share side the contrast is a polynomial in the shares, so the contrast words of every depth keep the fibre, the pins and the poles of the period.

32.2 Manufactured tables and their port words

Definition 32.4 (Manufactured tables)**.** Let *seed letters* on $\mathcal{E}$ be polynomials $L_i(x)=\sum_{P\in\operatorname{supp}L_i}\theta_{i,P}\,x^P$, $i\in I$, with supports $\operatorname{supp}L_i\subseteq 2^{\mathcal{E}}$, weights $\theta_{i,P}>0$ and exponents $w_i\geq 0$; a table $(\mathcal{B}_j,\theta_j)$ of a family with exponents (s_j,t_j) gives the two seed letters K_j, with exponent s_j, and Ψ_j, with exponent t_j. Choose $\beta_i\in(0,1)$ and replace every mark e by a pair $\{e^+,e^-\}$ of new marks, on $\mathcal{E}^\pm=\{e^+,e^-:e\in\mathcal{E}\}$. The polynomials

$$\widetilde{K}_i(\alpha)=L_i(\mathbf{1})\prod_{e\in\mathcal{E}}(\alpha_{e^+}+\alpha_{e^-})-\beta_i\sum_P\theta_{i,P}\prod_{e\in P}\alpha_{e^+}\prod_{e\notin P}(\alpha_{e^+}+\alpha_{e^-}),$$

$$\widetilde{\Psi}_i(\alpha) = L_i(\mathbf{1}) \prod_{e\in\mathcal{E}} (\alpha_{e^+} + \alpha_{e^-}) - \beta_i \sum_P \theta_{i,P} \prod_{e\in P} \alpha_{e^-} \prod_{e\notin P} (\alpha_{e^+} + \alpha_{e^-})$$

are the basis and cobasis polynomials of a table on $\mathcal{E}^{\pm}$ of rank $|\mathcal{E}|$, whose bases $U^+ \cup (\mathcal{E}\backslash U)^-$, $U \subseteq \mathcal{E}$, take one mark of each pair, with the weights $\widetilde{\theta}_{i,U} = L_i(\mathbf{1}) - \beta_i \sum_{P\subseteq U} \theta_{i,P} \geq L_i(\mathbf{1})(1-\beta_i) > 0$; the seed weights are recovered from $\big(L_i(\mathbf{1}) - \widetilde{\theta}_{i,U}\big)/\beta_i = \sum_{P\subseteq U} \theta_{i,P}$ by inversion over the subsets. The *manufactured family* $\widetilde{F}$ has these tables with the exponents $(w_i, 0)$; for masses $\mathbf{m}$ on $\mathcal{E}^{\pm}$ its *pair walls* are $\kappa_{\{e^+,e^-\}} = m_{e^+} + m_{e^-} - |\mathbf{w}|$, $|\mathbf{w}| = \sum_i w_i$, and its *pair shapes* are $x_e = \alpha_{e^+}/(\alpha_{e^+} + \alpha_{e^-})$. For orders $\mathbf{n} \in \mathbb{N}^{\mathcal{E}}$, positive masses and positive pair walls, its *port word* is

$$M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = \frac{J_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0})}{\prod_e \Gamma\big(\kappa_{\{e^+,e^-\}}\big)}\ \mathbb{E}_{\alpha\sim \mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{w},\mathbf{0})}\Big[\prod_{e\in\mathcal{E}} \Pi_{n_e}^{m_{e^+},m_{e^-}}(x_e)\Big],$$

and $M^*_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ is the same at the exponents $(0, w_i)$, with $J_{\widetilde{F}}(\mathbf{m};\mathbf{0},\mathbf{w})$ and $\mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{0},\mathbf{w})$. An *occupancy* is a family $\mathbf{o} = (o_{i,P})$ of integers $o_{i,P} \geq 0$, zero when $w_i = 0$, with the totals $|o_i| = \sum_P o_{i,P}$ and $|\mathbf{o}| = \sum_i |o_i|$, the *loads* $\mathrm{load}_e(\mathbf{o}) = \sum_i \sum_{P\ni e} o_{i,P}$, and the *coefficient*

$$\mathrm{cf}(\mathbf{o}) = \prod_i \Big[\frac{(w_i)_{|o_i|}}{L_i(\mathbf{1})^{w_i} \prod_P o_{i,P}!} \prod_P \Big(\frac{\beta_i\, \theta_{i,P}}{L_i(\mathbf{1})}\Big)^{o_{i,P}}\Big].$$

Theorem 32.5 (Interest of a manufactured family; the occupancy sum)**.** *In the pair scales and shapes,* $(\alpha_{e^+}, \alpha_{e^-}) = \rho_e(x_e, 1-x_e)$,

$$\widetilde{K}_i = \prod_e \rho_e\big(L_i(\mathbf{1}) - \beta_i L_i(x)\big), \qquad \widetilde{\Psi}_i = \prod_e \rho_e\big(L_i(\mathbf{1}) - \beta_i L_i(\mathbf{1}-x)\big).$$

Let the masses be positive and the pair walls positive.

(i) Every wall of $\widetilde{F}$ *at the exponents* $(w_i, 0)$ *is positive, and*

$$J_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0}) = \prod_e \Gamma\big(\kappa_{\{e^+,e^-\}}\big) \int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{m_{e^+}-1}(1-x_e)^{m_{e^-}-1} \prod_i \big(L_i(\mathbf{1}) - \beta_i L_i(x)\big)^{-w_i}\, dx,$$

the integral convergent at all positive masses; the simplex integral $P_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0})$ *is* $\Gamma_{1\,|\mathcal{E}|}\begin{bmatrix}(\kappa_{\{e^+,e^-\}})_e \\ \sum_e \kappa_{\{e^+,e^-\}}\end{bmatrix}$ *times the same integral. Under* $\mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{w},\mathbf{0})$ *the pair scales are independent,* $\rho_e \sim \mathrm{Gamma}_{\kappa_{\{e^+,e^-\}}}$*, and independent of the pair shapes, whose density on the cube is proportional to the integrand. So* $M_{\widetilde{F},\mathbf{0}}(\mathbf{m})$ *is the integral displayed, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ *is that integral with* $\prod_e \Pi_{n_e}^{m_{e^+},m_{e^-}}(x_e)$ *inserted; with* $L_i(\mathbf{1}-x)$ *in place of* $L_i(x)$ *the same holds for* $M^*_{\widetilde{F},\mathbf{n}}(\mathbf{m})$.

(ii) The integral is the series of nonnegative terms

$$M_{\widetilde{F},\mathbf{0}}(\mathbf{m}) = \sum_{\mathbf{o}} \mathrm{cf}(\mathbf{o}) \prod_e \Gamma_{2\,1}\begin{bmatrix} m_{e^+} + \mathrm{load}_e(\mathbf{o}),\ m_{e^-} \\ m_{e^+} + m_{e^-} + \mathrm{load}_e(\mathbf{o})\end{bmatrix}$$

over the occupancies.

Theorem 32.6 (The port word of a manufactured family)**.** *Let the masses be positive, the pair walls positive and* $\mathbf{n} \in \mathbb{N}^{\mathcal{E}}$*. Then*

$$M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = \sum_{\mathbf{o}} \mathrm{cf}(\mathbf{o}) \prod_{e} \binom{\mathrm{load}_e(\mathbf{o})}{n_e} \Gamma_{1}^{2}\begin{bmatrix} m_{e^+} + \mathrm{load}_e(\mathbf{o}), & m_{e^-} + n_e \\ m_{e^+} + m_{e^-} + \mathrm{load}_e(\mathbf{o}) + n_e \end{bmatrix}$$
$$= \frac{1}{\prod_e n_e!} \int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{m_{e^+}+n_e-1}(1-x_e)^{m_{e^-}+n_e-1}\, \partial_x^{\mathbf{n}} \prod_i \big(L_i(\mathbf{1}) - \beta_i L_i(x)\big)^{-w_i}\, dx.$$

Every occupancy with $\mathrm{load}_e(\mathbf{o}) < n_e$ *for some* e *is annihilated, every surviving term is positive, every mixed derivative of* $\prod_i (L_i(\mathbf{1}) - \beta_i L_i(x))^{-w_i}$ *is nonnegative on the cube, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) > 0$ *exactly when every mark* e *with* $n_e > 0$ *lies in a member of the support of a seed letter with* $w_i > 0$*. At integer masses the Beta ports have integer coefficients, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ *is an integer combination of the* $M_{\widetilde{F},\mathbf{0}}$ *at the masses shifted by* $(k_e, n_e - k_e)$ *on the pair of* e*,* $0 \le k_e \le n_e$*: of the interests of* $\widetilde{F}$ *at these masses, all of pair walls* $\kappa_{\{e^+,e^-\}} + n_e$*, over the one normalization* $\prod_e \Gamma(\kappa_{\{e^+,e^-\}} + n_e)$*.*

Theorem 32.7 (Suppression selected by incidence)**.** *In Theorem 32.6 write* $\mathrm{occ}(\mathbf{n}) = \min\{|\mathbf{o}| : \mathrm{load}_e(\mathbf{o}) \ge n_e$ *for every* $e\}$ *over the occupancies, and* $\beta = \max_i \beta_i < 1$ *over the seed letters with* $w_i > 0$*. Then*

$$0 \le M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) \le \prod_e \Big[\Gamma_{1}^{2}\begin{bmatrix} m_{e^+}, & m_{e^-} \\ m_{e^+} + m_{e^-} \end{bmatrix} \frac{(m_{e^-})_{n_e}}{n_e!}\Big] \prod_i L_i(\mathbf{1})^{-w_i} \sum_{l \ge \mathrm{occ}(\mathbf{n})} \frac{(|\mathbf{w}|)_l}{l!}\, \beta^l.$$

For an integer demand $\mathbf{d} \in \mathbb{N}^{\mathcal{E}}$ *let* $\mathrm{fc}(\mathbf{d}) = \min\big\{\sum_{i,P} \lambda_{i,P} : \ \lambda \ge 0,\ \sum_{i,P} \lambda_{i,P} \mathbf{1}_P \ge \mathbf{d}\big\}$*, the* fractional cover *of the demand by the members of the supports of the seed letters with* $w_i > 0$*. Then* $\mathrm{occ}(q\mathbf{d}) = q\,\mathrm{fc}(\mathbf{d}) + O(1)$ *as* $q \to \infty$*, and* $\limsup_q \frac{1}{q} \log M_{\widetilde{F},q\mathbf{d}}(\mathbf{m}) \le \mathrm{fc}(\mathbf{d}) \log \beta < 0$ *for a nonzero demand that the supports cover. When the seed letters are the letters* K_j, Ψ_j *of the tables of a family, every surviving occupancy satisfies* $\sum_j \big(r_j(S)\,|o_{K_j}| + (|S| - d_j(S))\,|o_{\Psi_j}|\big) \ge \sum_{e \in S} n_e$ *for every* $S \subseteq \mathcal{E}$*,* $|o_{K_j}|$ *and* $|o_{\Psi_j}|$ *the total occupancies of the two letters of table* j*.*

Theorem 32.8 (Reflection of the port words of a manufactured family)**.** *The swap* $\alpha_{e^+} \leftrightarrow \alpha_{e^-}$ *on every pair,* $x_e \leftrightarrow 1 - x_e$*, swaps* $\widetilde{K}_i$ *with* $\widetilde{\Psi}_i$ *and* m_{e^+} *with* m_{e^-}*, and keeps every seed weight and every* β_i*: with* $\mathbf{m}'$ *the masses swapped on every pair,* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = (-1)^{|\mathbf{n}|} M^*_{\widetilde{F},\mathbf{n}}(\mathbf{m}')$*.*

A manufactured table turns any positive polynomials, the two letters of every table of a family among them, into a table whose bases are the transversals of the pairs. On the pair shapes its letter is the constant $L_i(\mathbf{1})$ less a copy of the seed scaled by β_i, so the port word is a series of nonnegative terms indexed by occupancies of the members of the seed, and a Beta port of order n_e on a pair keeps only the occupancies that load the mark at least n_e times. The port words are therefore positive exactly when the incidence of the supports can meet the demand, and they are suppressed at the exponential rate of the cheapest fractional cover of the demand: the incidence structure of the seed decides which port words survive and how small they are.

32.3 Mark relations with the room integrated

Definition 32.9 (Mark reads)**.** Select a mark with coordinate x, and let the coordinates α_R of its room range over a measure space with a density $\Omega(\alpha_R) \ge 0$. The letters (Section 3.2) in the selected mark are polynomials $L_i(x, \alpha_R)$ in x, $i = 1, \dots, h$, positive where x ranges, with exponents w_i; they are *linear* when $L_i = c_i(\alpha_R) + d_i(\alpha_R)\, x$ with $c_i > 0$ and $d_i \ge 0$, and a linear letter *sees*

the mark when $w_i d_i > 0$ on a set of positive measure. For $a, b > 0$ the *mark read* of an insertion $g(x, \alpha_R)$ on the box is

$$\langle g \rangle_{a,b;\mathbf{w}} = \int\!\!\int_0^1 g\, x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i}\, dx\; \Omega(\alpha_R)\, d\alpha_R,$$

and for $a, \lambda > 0$ its read on the half-line is

$$\langle g \rangle^{\lambda}_{a;\mathbf{w}} = \int\!\!\int_0^\infty g\, x^{a-1} e^{-\lambda x} \prod_i L_i^{-w_i}\, dx\; \Omega(\alpha_R)\, d\alpha_R,$$

wherever these converge; at a fixed point of the room, $\Omega(\alpha_R)\, d\alpha_R$ is the Dirac measure there. A sequence of reads at the data $a_n + j$, b_n, $\mathbf{w}(n)$ and Ω_n keeps the depth n (Theorem 32.1) apart from the *moment index* j of the selected mark.

Theorem 32.10 (The mark relation)**.** *Let the letters be polynomial in the selected mark, with exponents w_i, and let $A(x, \alpha_R)$ be a polynomial in x with coefficients depending on the room.*

(a) On the box, for $a, b > 0$:

$$\Big\langle x(1-x)\,\partial_x A + \big(a - (a+b)x\big) A - x(1-x)\, A \sum_i w_i\, \partial_x \log L_i \Big\rangle_{a,b;\mathbf{w}} = 0.$$

(b) On the half-line, for $a, \lambda > 0$, whenever the reads converge:

$$\Big\langle x\,\partial_x A + (a - \lambda x) A - x\, A \sum_i w_i\, \partial_x \log L_i \Big\rangle^{\lambda}_{a;\mathbf{w}} = 0.$$

(c) On a restricted interval the difference of the flux at its two ends is retained, and a kernel coupled to x contributes its product-rule term.

Multiplying by $\prod_i L_i$ gives a polynomial numerator; on a box of several marks the relation holds in each coordinate with the others as room.

Theorem 32.11 (The moments of a selected mark close in a finite span)**.** *On a family select a mark e, $x = \alpha_e$, with room $R = \mathcal{E} - e$; splitting the bases and the cobases by e gives the two coefficients of every letter in x. Pull the letters free of x into the room, absorb pure powers of x into its mass, and write the remaining letters $L_i = c_i(\alpha_R) + d_i(\alpha_R)\, x$, $c_i, d_i > 0$, $i = 1, \dots, h$, with exponents w_i; read them on the half-line at a fixed point of the room (Definition 32.9), with $m, \lambda > 0$. With $\prod_i L_i = \sum_{k=0}^{h} p_k x^k$, $p_{-1} = 0$, $q_0 = 0$ and $q_k = \sum_i w_i d_i\, [x^{k-1}] \prod_{i' \neq i} L_{i'}$,*

$$\lambda p_h\, \langle x^{n+h+1} \rangle^{\lambda}_{m;\mathbf{w}} = \sum_{k=0}^{h} \big[(m+n+k)\, p_k - q_k - \lambda p_{k-1}\big]\, \langle x^{n+k} \rangle^{\lambda}_{m;\mathbf{w}},$$

a recurrence whose coefficients are polynomials in the room coordinates, with leading coefficient λp_h, $p_h = \prod_i d_i > 0$. For a mark f of the room,

$$\partial_{\alpha_f} \langle x^n \rangle^{\lambda}_{m;\mathbf{w}} = -\sum_i w_i \Big[(\partial_{\alpha_f} d_i)\, \langle x^{n+1} \rangle^{\lambda}_{m;\mathbf{w}+\mathbf{e}_i} + (\partial_{\alpha_f} c_i)\, \langle x^n \rangle^{\lambda}_{m;\mathbf{w}+\mathbf{e}_i} \Big],$$

and the moments with at most ℓ such derivatives lie in the span, over rational functions of the room, of $\{\langle x^k \rangle^{\lambda}_{m;\mathbf{w}+v} : 0 \le k \le h,\ v \in \mathbb{N}^h,\ |v| \le \ell\}$: at most $(h+1)\binom{h+\ell}{\ell}$ states.

Theorem 32.12 (Positive reads of the Beta ports)**.** *Let the letters be linear, $L_i = c_i + d_i x$, with exponents $w_i \geq 0$, let $a, b > 0$ and $q \in \mathbb{N}$, and let J be a set of letters, with $w_J = \sum_{i\in J} w_i$.*

(i) The Beta port of order q reads (Theorem 3.15(iii), integrated over the room)

$$\big\langle (-1)^q \Pi_q^{a,b} \big\rangle_{a,b;\mathbf{w}} = \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!} \Big\langle \big(x(1-x)\big)^q \prod_i \Big(\frac{d_i}{L_i}\Big)^{v_i} \Big\rangle_{a,b;\mathbf{w}} \in \Big[0, \frac{(|\mathbf{w}|)_q}{q!} \langle 1\rangle_{a,b;\mathbf{w}}\Big].$$

(ii) The part of this sum over the v supported in J lies in $\big[0, \frac{(w_J)_q}{q!}\langle 1\rangle_{a,b;\mathbf{w}}\big]$, and for $q > 0$ it is positive when some letter of J sees the mark.

(iii) At $q = 1$,

$$a\,\langle 1\rangle_{a,b;\mathbf{w}} - (a+b)\,\langle x\rangle_{a,b;\mathbf{w}} - \sum_{i\notin J} w_i \Big\langle \frac{x(1-x)\,d_i}{L_i} \Big\rangle_{a,b;\mathbf{w}} = \sum_{i\in J} w_i \Big\langle \frac{x(1-x)\,d_i}{L_i} \Big\rangle_{a,b;\mathbf{w}} \in \big[0,\, w_J \langle 1\rangle_{a,b;\mathbf{w}}\big].$$

Along a sequence at the data $a_n + j$, b_n, $\mathbf{w}(n)$ the bounds hold with $\langle 1\rangle_{a_n,b_n;\mathbf{w}(n)} \geq \langle 1\rangle_{a_n+j,b_n;\mathbf{w}(n)}$ in place of the read at the moment index j, uniformly in j.

Proposition 32.13 (The finite-band polynomial)**.** *In Theorem 32.12 write $L_{J^c} = \prod_{i\notin J} L_i$. The polynomial*

$$A = \big[a - (a+b)x\big] L_{J^c} + x(1-x)\Big(\partial_x L_{J^c} - L_{J^c} \sum_{i\notin J} \frac{w_i d_i}{L_i}\Big),$$

of degree at most $h - |J| + 1$ in x, has the read

$$\langle A\rangle_{a,b;\mathbf{w}} = \sum_{i\in J} w_i \Big\langle \frac{x(1-x)\,d_i\,L_{J^c}}{L_i} \Big\rangle_{a,b;\mathbf{w}},$$

positive when a letter of J sees the mark and the reads are finite. On the half-line, $A = a\,L_{J^c} + x\,\partial_x L_{J^c} - \lambda x L_{J^c} - x L_{J^c} \sum_{i\notin J} w_i d_i/L_i$ has $\langle A\rangle^{\lambda}_{a;\mathbf{w}} = \sum_{i\in J} w_i \big\langle x\,d_i\,L_{J^c}/L_i\big\rangle^{\lambda}_{a;\mathbf{w}}$.

Theorem 32.14 (A positive sequence of Beta ports on an interval of a selected mark)**.** *In a projective chart take a selected mark at a mass $m_e \in (0,1]$, restricted to an interval $0 < \lambda_0 < \alpha_e < \lambda_1$, and write $\alpha_e = \lambda_0 + (\lambda_1 - \lambda_0)x$, $x \in [0,1]$; its letters are linear in α_e with nonnegative coefficients, $L_i = c_i + d_i x$ with $d_i \geq 0$ and $c_i \geq \lambda_0 d_i/(\lambda_1 - \lambda_0)$, and exponents $w_i \geq 0$, the letters constant in x going into the room density, and the factor $\alpha_e^{m_e - 1}$ of the mass is counted among them as the letter $\lambda_0 + (\lambda_1 - \lambda_0)x$ at the exponent $1 - m_e$. So $x(1-x)\,d_i/L_i \leq (\sqrt{\lambda_1} - \sqrt{\lambda_0})/(\sqrt{\lambda_1} + \sqrt{\lambda_0})$ on $[0,1]$ for every letter. Let $\langle 1\rangle_{1,1;\mathbf{w}}$ be finite. By Theorem 32.12(i) at $a = b = 1$ and the bound of Theorem 3.15(iii), integrated over the room, for $n \in \mathbb{N}$,*

$$0 < \big\langle (-1)^n \Pi_n^{1,1} \big\rangle_{1,1;\mathbf{w}} \leq \frac{(|\mathbf{w}|)_n}{n!} \Big(\frac{\sqrt{\lambda_1} - \sqrt{\lambda_0}}{\sqrt{\lambda_1} + \sqrt{\lambda_0}}\Big)^n \langle 1\rangle_{1,1;\mathbf{w}},$$

the first inequality when a letter sees the mark, as the letter of the mass does when $m_e < 1$.

Theorem 32.15 (The finite-band recurrence in the order of the Beta ports)**.** *At a fixed point of the room let the letters be linear, $L_i = c_i + d_i x$ with $c_i > 0$ and $d_i \geq 0$, $i = 1, \dots, h$, with exponents $w_i(n)$ depending on $n \in \mathbb{N}$, let $a_n, b_n > 0$, and in this statement write $L = \prod_i L_i = \sum_k p_k x^k$ and*

$$A_n = \big(a_n - (a_n + b_n)x\big)\, L + x(1-x)\, \partial_x L - x(1-x)\, L \sum_i \frac{w_i(n)\, d_i}{L_i}.$$

For every polynomial g, $\big\langle L\, x(1-x)\, \partial_x g + A_n g \big\rangle_{a_n, b_n; \mathbf{w}(n)} = 0$ (Theorem 32.10(a) with the polynomial Lg). Expanding $L\, x(1-x)\, \partial_x g + A_n g$ at $g = (-1)^r \Pi_r^{a_n, b_n}$ in the Beta ports $(-1)^{r+s} \Pi_{r+s}^{a_n, b_n}$ by the finite rules of Theorem 3.15(v) gives the relation

$$\sum_{s=-h-1}^{h+1} \mathfrak{b}_{n,r,s} \left\langle (-1)^{r+s} \Pi_{r+s}^{a_n, b_n} \right\rangle_{a_n, b_n; \mathbf{w}(n)} = 0$$

of width at most $2h + 3$, whose coefficients, written $\mathfrak{b}_{n,r,s}$ in this statement, are rational in n and r when a_n, b_n and the $w_i(n)$ are, and polynomial in the room data. With $|\mathbf{w}(n)| = \sum_i w_i(n)$ and $p_h = \prod_i d_i$, the furthest forward coefficient is

$$\mathfrak{b}_{n,r,h+1} = \big[|\mathbf{w}(n)| - a_n - b_n - h - r\big]\, p_h\, (-1)^{h+1} \binom{a_n + b_n + 2r - 2}{r} \Big/ \binom{a_n + b_n + 2r + 2h}{r + h + 1}.$$

Lemma 32.16 (A rational eventual sign certificate)**.** *Let $Q_n(X) = n^d q_d(X) + \sum_{k<d} n^k q_k(X)$ with polynomials q_k, and let rationals $X_1 \geq X_{\max} > 0$ and $\delta > 0$ satisfy $q_d \geq \delta$ on $[0, X_1]$. With $B_k = \sum_i \big|[X^i] q_k\big|\, X_1^i$, $Q_n \geq \delta n^d / 2$ on $[0, X_{\max}]$ for every $n \geq \max\big(1, 2 \sum_{k<d} B_k / \delta\big)$. Composition of recurrences with variable coefficients keeps its shifts: with* Sh *the shift $n \mapsto n + 1$, $\big(\sum_j b_j(n)\, \mathrm{Sh}^j\big)\big(\sum_i a_i(n)\, \mathrm{Sh}^i\big) = \sum_{i,j} b_j(n)\, a_i(n + j)\, \mathrm{Sh}^{i+j}$.*

With the room held fixed, a mark is integrated alone against its letters, and the mark relation is the integration by parts that this allows: the flux of the density $x^{a-1}(1-x)^{b-1}$ times a polynomial vanishes at both ends of the box, and that of the Gamma density at both ends of the half-line. Two uses follow. Against the Gamma density a mark of a family closes its moments into a recurrence of bounded order whose coefficients are polynomials in the room, and the derivatives in the room stay in a finite span: one mark closes all its moments. Against a Beta port the same relation gives positive reads, sums of the ratios $x(1-x)d_i/L_i$ that a letter seeing the mark keeps positive and the total exponent bounds, and at every pair $a_n, b_n > 0$ the Beta ports $\Pi_r^{a_n, b_n}$ obey a recurrence of bounded width in their order r whose furthest coefficient is explicit: when every d_i is positive it vanishes at the one order $r = |\mathbf{w}(n)| - a_n - b_n - h$ at most.

Chapter 33

The critical-line law

Let $\mathcal{E}$ have two marks, and let F be the family of the one list $\binom{\mathcal{E}}{1}$, whose bases are its two singletons, at the exponents $(1,1)$ and the masses $\mathbf{1}$ (Theorem 33.2). Its period is one, $P_F(\mathbf{1};1,1)=1$. On the fibre of the atom formula on the balance wall (Corollary 17.15), one share per basis and letter, the pins leave one free share: the basis share and the cobasis share of one basis are equal, $a=b=\frac{1}{2}+iy$, and the fibre is the line $\operatorname{Re} a=\frac{1}{2}$, the critical line. The law of y, the imaginary part of that share, is the critical-line law $\mathbb{P}_{\mathrm{crit}}$ (Definition 33.1). Its characteristic function is a stacked Gamma, its moments are rational, and its negative moments, read as finite parts, are rational multiples of the odd Trailmoments of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$ (Theorem 33.2). A function R of the share, read against an odd derivative of $\mathbb{P}_{\mathrm{crit}}$ along the critical line, gives the critical-line word $\mathrm{crit}_k[R]$ (Definition 33.3): it carries one Trailmoment of odd index with a rational part, and it is positive whenever R is the Fourier transform of a positive measure (Theorem 33.6).

Definition 33.1 (The critical-line law)**.** The *critical-line law* is the probability law $\mathbb{P}_{\mathrm{crit}}(dy)=p_{\mathrm{crit}}(y)\,dy$ on $\mathbb{R}$ with

$$p_{\mathrm{crit}}(y)=\frac{1}{2\pi}\,\Gamma_{\,0}^{\,4}\big[\tfrac{1}{2}+iy,\ \tfrac{1}{2}+iy,\ \tfrac{1}{2}-iy,\ \tfrac{1}{2}-iy\big]=\frac{2\pi}{(e^{\pi y}+e^{-\pi y})^2},$$

the stacked Gamma of Lemma 5.16 on the line $\operatorname{Re} a=\frac{1}{2}$ divided by 2π.

Theorem 33.2 (The critical-line law and its moments)**.** *Let $\mathcal{E}$ have two marks and let F be the family of the one list $\binom{\mathcal{E}}{1}$, whose bases are its two singletons, at the exponents $(1,1)$ and the masses $\mathbf{1}$. On the fibre of Corollary 17.15 the basis and cobasis shares of one basis are equal, $a=b=\frac{1}{2}+iy$, the fibre integral is $P_F(\mathbf{1};1,1)=1$, and the imaginary part Y of the share has the law $\mathbb{P}_{\mathrm{crit}}$. In this statement $\mathbb{E}$ is the expectation of $Y\sim\mathbb{P}_{\mathrm{crit}}$.*

(i) $\mathbb{E}[e^{i\xi Y}]$ *is the Fourier read of Lemma 5.16(iii) at $t=\xi$; every odd moment of Y vanishes and every even moment is rational.*

(ii) For $j\geq 1$ the finite part at $y=0$ (Definition 1.4) of $\mathbb{E}[Y^{-2j}]$ is

$$\begin{aligned}\operatorname{Reg}\mathbb{E}\big[Y^{-2j}\big]&=(-1)^j\,2j\,(\mathsf{T}_{2j+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty))\\&=(-1)^j\,2j\,(2^{2j+1}-1)\,(\mathsf{T}_{2j+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),\end{aligned}$$

the second line by the lattice relation (Theorem 5.12(i)) at the step $q=2$, the lattice $\frac{1}{2}\mathbb{Z}$ splitting into $\mathbb{Z}$ and $\frac{1}{2}+\mathbb{Z}$.

(iii) With $X=\frac{1}{2}+iY$, $\mathbb{E}[e^{tX}]=\dfrac{te^t}{e^t-1}=t+\dfrac{t}{e^t-1}$, and $d_{m+1}\,\mathbb{E}[X^m]\in\mathbb{Z}$ for every $m\in\mathbb{N}$.

The list $\binom{\mathcal{E}}{1}$ on two marks is the smallest table with a share that moves: at the centre the pins hold its basis and cobasis shares equal, on the critical line, and the period, one, is the integral of the law $\mathbb{P}_{\mathrm{crit}}$ of their imaginary part. Everything about that law is read off the stacked Gamma $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16. Its characteristic function is the stacked Gamma $\Gamma_{\substack{2\\0}}\big[1+\frac{i\xi}{2\pi},\ 1-\frac{i\xi}{2\pi}\big]$, its moments are rational with denominators bounded by the clearing integers, and its negative moments, which diverge at the origin, have finite parts that are rational multiples of the odd Trailmoments of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$, and so of the counting input.

33.1 Critical-line words

Definition 33.3 (Critical-line words)**.** For an integer $k \geq 1$ and a function R on the line $\operatorname{Re} u = \frac{1}{2}$, the *critical-line word* of order k is

$$\mathrm{crit}_k[R] = -\frac{2}{(2k)!}\,\frac{1}{2\pi i}\int_{(1/2)} \frac{R(u)}{1-2u}\,\partial_u^{2k-1}\Gamma_{\substack{4\\0}}\big[u,\ u,\ 1-u,\ 1-u\big]\,du,$$

wherever it converges; the integrand is regular at $u=\frac{1}{2}$, where the odd derivative of the stacked Gamma, which is even about $\frac{1}{2}$, vanishes. On the line, $u=\frac{1}{2}+iy$,

$$\mathrm{crit}_k[R] = \frac{(-1)^k}{(2k)!}\int_{\mathbb{R}} R(\tfrac{1}{2}+iy)\,\frac{p_{\mathrm{crit}}^{(2k-1)}(y)}{y}\,dy.$$

Proposition 33.4 (Critical-line words of polynomials)**.** *For an integer $k \geq 1$ and a polynomial $R(u) = (1-2u)\,Q(u) + R(\frac{1}{2})$, with $\mathbb{E}$ the expectation of $Y \sim \mathbb{P}_{\mathrm{crit}}$,*

$$\begin{aligned}\mathrm{crit}_k[R] &= R(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) + \frac{2}{(2k)!}\,\mathbb{E}\big[Q^{(2k-1)}(\tfrac{1}{2}+iY)\big]\\ &= (2^{2k+1}-1)\,R(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \frac{2}{(2k)!}\,\mathbb{E}\big[Q^{(2k-1)}(\tfrac{1}{2}+iY)\big],\end{aligned}$$

the second line by Theorem 5.12(i), as in Theorem 33.2(ii).

Corollary 33.5 (The rational part of an integer polynomial argument)**.** *For $P \in \mathbb{Z}[x]$ of degree D and $R(u) = P(2u)$, the rational part of $\mathrm{crit}_k[R]$ (Definition 34.1) is cleared by $\frac{(2k)!}{2}\,d_{\max(1,D-2k+1)}$.*

Theorem 33.6 (Positive arguments for the critical-line word)**.** *For an integer $k \geq 1$ and real t, with $\mathbb{E}$ the expectation of $Y \sim \mathbb{P}_{\mathrm{crit}}$,*

$$\mathrm{crit}_k\left[e^{t(u-\frac{1}{2})}\right] = \frac{1}{(2k)!}\int_{|t|}^{\infty} x^{2k-1}\,\mathbb{E}\big[e^{ixY}\big]\,dx > 0,$$

$$\mathrm{crit}_k[1] = (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = (2^{2k+1}-1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \quad \textit{(Theorem 5.12(i)).}$$

If $R(\frac{1}{2}+iy) = \int e^{ity}\,\phi(dt)$ for a finite nonzero positive measure ϕ on $\mathbb{R}$, then $\mathrm{crit}_k[R] = \int \mathrm{crit}_k\left[e^{t(u-\frac{1}{2})}\right]\phi(dt) > 0$.

On the critical line the word reads its argument against an odd derivative of the density of the critical-line law divided by the imaginary part of the share, a kernel without a fixed sign. Its Fourier transform, the word of an exponential argument, is the integral beyond $|t|$ of the characteristic function times a power, a positive function, so an argument that is the Fourier transform of a positive measure has a positive word. This is how positive words carrying one odd Trailmoment are made.

Theorem 33.7 (A positive rational argument with one odd index)**.** *Let $M \in \mathbb{N}$ and $p(v) = \sum_{j=0}^{D} b_j v^j \in \mathbb{Q}[v]$, nonnegative on $[0,1]$ and not identically zero, and let*

$$R_{M,p}(u) = \sum_{j=0}^{D} b_j \Big(\frac{1}{u+M+j} + \frac{1}{1-u+M+j} \Big), \qquad R_{M,p}(\tfrac{1}{2}) = 4 \sum_j \frac{b_j}{2M+2j+1} > 0.$$

Then for every $k \geq 1$

$$\begin{aligned} \mathrm{crit}_k[R_{M,p}] &= \frac{2}{(2k)!} \int_0^1 \int_0^1 \frac{v^M x^{-1/2}\, p(v)\, [-\log(vx)]^{2k}}{1-vx}\, dx\, dv \\ &= R_{M,p}(\tfrac{1}{2})\, (\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) \\ &\qquad - 4 \sum_j \frac{b_j}{2M+2j+1}\, (\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((M+j,\infty)) \\ &= (2^{2k+1}-2)\, R_{M,p}(\tfrac{1}{2})\, (\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) \\ &\qquad + 4 \sum_j \frac{b_j}{2M+2j+1}\, (\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,M+j]) > 0. \end{aligned}$$

If $C b_j \in \mathbb{Z}$ for every j, then $C\, d_{2M+2D+1}\, d_{M+D}^{2k+1}$ clears both rational coefficients; the Trailmoments of $\mathbb{I}_{\frac{1}{2}+\mathbb{Z}}$ are read in those of $\mathbb{I}_{\mathbb{Z}}$ by Theorem 5.12(i).

Corollary 33.8 (Arguments of two Beta corners)**.** *For $p(v) = (1-v)^q$, $q \in \mathbb{N}$, the argument of Theorem 33.7 is*

$$R_{M,(1-v)^q}(u) = \Gamma \begin{matrix} 2 \\ 1 \end{matrix} \begin{bmatrix} u+M,\ q+1 \\ u+M+q+1 \end{bmatrix} + \Gamma \begin{matrix} 2 \\ 1 \end{matrix} \begin{bmatrix} 1-u+M,\ q+1 \\ 2-u+M+q \end{bmatrix}.$$

Its words are positive and carry one Trailmoment of the counting input, of the odd index $2k+1$, at every k. For fixed M, $\mathrm{crit}_k[R_{M,(1-v)^q}] \sim \frac{4\,\Gamma(M+1)}{(2k)!}\, q^{-M-1} (\log q)^{2k}$ as $q \to \infty$.

Proposition 33.9 (The decay of the word of an exponential argument)**.** *For an integer $k \geq 1$ and $t > 0$, with the polynomial $c_k(t) = \sum_{l=0}^{2k} \frac{(2k)!}{l!}\, 2^{2k-l+1}\, t^l$,*

$$\frac{e^{-t/2}}{(2k)!}\, c_k(t) \leq \mathrm{crit}_k \left[e^{t(u-\frac{1}{2})} \right] \leq \frac{e^{-t/2}}{(2k)!\,(1-e^{-t})}\, c_k(t).$$

A rational argument whose positive measure is the exponential profile of a polynomial nonnegative on $[0,1]$ gives a positive word with one odd Trailmoment of the counting input and a rational part cleared by $d_{2M+2D+1} d_{M+D}^{2k+1}$. The words of exponential arguments decay like $e^{-t/2}$ times a polynomial of degree $2k$, so the words of the arguments of two Beta corners decay like a power of their exponent q: they are positive, carry one Trailmoment and are controlled, the raw material of the relation blocks of Chapter 36.

Chapter 34

The gap and the shape words

A value read at rational data has a row, its coefficients on Trailmoments of the counting input and a rational part (Definition 34.1). The gap theorem turns a sequence of nonzero values whose rows are cleared by integers that grow more slowly than the values decay into irrationality, and the shape words supply such sequences. They are integrals over a cube against one letter; opened by one share they close on the stacked Gamma $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16, so their rows are explicit, cleared by powers of the clearing integers d_n and carried by the Trailmoments $(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of index at most one more than the number of shapes. Beta ports inserted on one shape keep the words positive while they move the row, filters built from recurrences remove unwanted coordinates, and Beta ports whose order grows with the depth trade a slower decay for a selected coordinate.

34.1 Clearing and the gap

Definition 34.1 (Rows; clearing)**.** A value written as $x = \sum_{i=1}^{m} A_i v_i - B$, a rational combination of fixed real numbers v_i, its *coordinates*, such as Trailmoments of the counting input, less a rational number B, its *rational part*, has the *row* $(A_1, \dots, A_m; B)$; a positive integer *clears* the row when it makes all these coefficients integers.

Lemma 34.2 (The growth of the clearing integers)**.** *For integers $n \geq 0$ and $s \geq 1$, d_n^s is the least positive integer D with $D\,(\mathsf{T}_{s;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,n]) \in \mathbb{Z}$ for every integer density f, and*

$$\limsup_{n\to\infty} \frac{\log d_n}{n} \leq c_{\mathrm{lcm}} = \frac{14}{25}\log 2 + \frac{9}{25}\log 3 + \frac{1}{5}\log 5 = 1.10555\ldots < 1.106.$$

Theorem 34.3 (The gap)**.** *Let $v_1, \dots, v_m$ be real, and let real numbers $x_n = \sum_i A_{n,i} v_i - B_n$, with $A_{n,i}, B_n \in \mathbb{Q}$, have positive integers D_n clearing all these coefficients, with $0 < |D_n x_n| \to 0$.*

(i) Some v_i is irrational.

(ii) For $m = 1$, $A_{n,1} \neq 0$ for all large n.

The hypothesis $x_n \neq 0$ holds in particular when x_n is the value of a word with a strictly positive read.

An integer of modulus below one is zero; nothing else is used. What the theory contributes is the two other hypotheses: positivity, which keeps the cleared values from vanishing, and rows whose clearing integers grow more slowly than the values decay.

34.2 Shape words with polynomial insertions

Definition 34.4 (Shape words)**.** For integers $n \ge 0$ and $k \ge 2$ let $L = \bar{x}_0 + x_0 x_1 \cdots x_k$ on the cube $[0,1]^{k+1}$ with coordinates $x_0, \dots, x_k$. The *shape word* of depth n with k shapes of a polynomial insertion $P \in \mathbb{Q}[x_0, \dots, x_k]$ is

$$I_n(k;P) = \int_{[0,1]^{k+1}} P(x)\, \frac{\prod_{i=0}^{k} (x_i \bar{x}_i)^n}{L^{n+1}}\, dx = \int_{[0,1]^{k+1}} P\, f_k^n\, \frac{dx}{L}, \qquad f_k = \frac{\prod_{i=0}^{k} x_i \bar{x}_i}{L},$$

with the *base* f_k. For a monomial x^p, $p = (p_0, \dots, p_k)$, the *share function* is

$$g_{n,p}(a) = \frac{(n!)^{k-2}\, (1-a)_n^2\, (a)_{p_0}}{(n+1)_{p_0} \prod_{i=1}^{k} (a+p_i)_{n+1}},$$

and for $P = \sum_p \big([x^p]P\big) x^p$ it is $g_{n,P} = \sum_p \big([x^p]P\big)\, g_{n,p}$, with the coefficients $c_{n,h,s}(P)$ of its principal parts $c_{n,h,s}(P)(a+h)^{-s}$ at the poles $a = -h$ and the *column sums* $C_{n,s}(P) = \sum_h c_{n,h,s}(P)$. The *port factor* is $\mathfrak{a}_k = x_0\, x_1 \bar{x}_1\, x_2 \cdots x_k / L$, and $M_k(\eta) = \sup_{(0,1)^{k+1}} f_k \mathfrak{a}_k^{\eta}$ for $\eta \ge 0$.

Theorem 34.5 (Shape words: the one-share closing)**.** *Let P have degree at most D, with $p_0 \le (k-2)(n+1)$ on every monomial x^p of P.*

(i) Share side. *For a monomial, on $0 < \operatorname{Re} a = c < n+1$,*

$$\begin{aligned} I_n(k;x^p) &= \frac{1}{2\pi i} \int_{(c)} \Gamma^{\,2}_{\,1}\!\begin{bmatrix} a,\ n+1-a \\ n+1 \end{bmatrix} \Gamma^{\,2}_{\,1}\!\begin{bmatrix} a+p_0,\ n+1-a \\ n+1+p_0 \end{bmatrix} \prod_{i=1}^{k} \Gamma^{\,2}_{\,1}\!\begin{bmatrix} a+p_i,\ n+1 \\ a+p_i+n+1 \end{bmatrix} da \\ &= \frac{1}{2\pi i} \int_{(c)} \Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big]\ g_{n,p}(a)\, da. \end{aligned}$$

(ii) Closing. $I_n(k;P) = -\sum_{m \ge 1} g'_{n,P}(m)$ *(Lemma 5.16(i)). Every pole of $g_{n,P}$ lies among $0, -1, \dots, -n-D$ with order at most k; the integers $1, \dots, n$ are zeros of order at least two of each $g_{n,p}$; $g_{n,p}(a) = O(|a|^{-2})$.*

(iii) The row. $C_{n,1}(P) = 0$ *and*

$$\begin{aligned} I_n(k;P) &= \sum_{h=0}^{n+D} \sum_{s=1}^{k} s\, c_{n,h,s}(P)\, (\mathsf{T}_{s+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((h,\infty)) \\ &= \sum_{s=2}^{k} s\, C_{n,s}(P)\, (\mathsf{T}_{s+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - \sum_{h,s} s\, c_{n,h,s}(P)\, (\mathsf{T}_{s+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,h]). \end{aligned}$$

Proposition 34.6 (An integer certificate for inserted words)**.** *In Theorem 34.5 let P have integer coefficients. Then $(n+1)_D\, d_{n+D}^{k-s}\, c_{n,h,s}(P) \in \mathbb{Z}$, and $(n+1)_D\, d_{n+D}^{k+1}$ clears every coefficient of the row, its rational part included.*

Theorem 34.7 (Integer-valued insertion clearing)**.** *In Theorem 34.5 let P have integer coefficients, let $p_{\max}$ be its largest exponent of a shape x_i, $i \ge 1$, and $p_{0,\max}$ its largest exponent of x_0, and let $b_P = \operatorname{lcm}\big\{\binom{n+j}{j} :\ j \text{ an exponent of } x_0 \text{ in } P\big\}$. Then $b_P\, d^{k-s}_{\max(n+p_{\max},\, p_{0,\max})}\, c_{n,h,s}(P) \in \mathbb{Z}$ and $b_P\, d^{k+1}_{\max(n+p_{\max},\, p_{0,\max})}$ clears the whole row, and b_P divides $d_{n+p_{0,\max}}$; insertions free of x_0 are cleared by $d^{k+1}_{n+p_{\max}}$ whatever their degrees. For $k \ge 3$ the insertion x_0^n retains the top coordinate:*

$$C_{n,k}(x_0^n) = (-1)^{n(k+1)} \binom{2n}{n}, \qquad \big[(\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))\big]\, I_n(k;x_0^n) = k\, (-1)^{n(k+1)} \binom{2n}{n}.$$

Proposition 34.8 (The root rate of the shape words)**.** *For $k \geq 2$:*

(i) $0 < I_n(k;1) \leq M_k(0)^n\, I_0(k;1)$ *(Theorem 34.19 at $q = 0$);*

(ii) $\lim_n I_n(k;1)^{1/n} = M_k(0)$, *where*

$$M_k(0) = \max_{(0,1)^{k+1}} f_k = \max_{0<a<1} \frac{a^k(1-a)^k}{(1+a^{k/2})^2} \in \Big[\frac{4^{-k}}{(1+2^{-k/2})^2},\ 4^{-k}\Big].$$

The maximum over a is at the root $a \in (0, \frac{1}{2})$ of $a^{k/2+1} + 2a = 1$, the root of Theorem 34.19 at $\eta = 0$. For every $k \geq 2$, $-\log M_k(0) > (k+1)\, c_{\mathrm{lcm}}$, so that $d_n^{k+1} I_n(k;1) \to 0$.

Theorem 34.9 (An irrational Trailmoment of the counting input)**.** *The shape word $I_n(2;1)$ is positive, and with $a_l = (-1)^l \binom{n}{l}\binom{n+l}{l}$, the coefficient of x^l in $(-1)^n \Pi_n^{1,1}(x)$ (Theorem 3.15(v)), and $b_l = 2a_l \sum_{j\neq l} \frac{a_j}{j-l}$,*

$$\begin{aligned} I_n(2;1) &= \sum_{l=0}^{n} \Big(2a_l^2\, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((l,\infty)) + b_l\, (\mathsf{T}_{2;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((l,\infty))\Big) \\ &= 2\big(A_n\, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - B_n\big), \\ A_n &= \sum_l a_l^2 \in \mathbb{Z}, \qquad d_n^3 B_n \in \mathbb{Z}, \end{aligned}$$

$$B_n = \sum_l a_l^2\, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,l]) + \sum_l a_l\, (\mathsf{T}_{2;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,l]) \sum_{j\neq l} \frac{a_j}{j-l},$$

$I_n(2;1) \leq 2\, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))\, (\sqrt{2}-1)^{4n}$, *and* $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) \notin \mathbb{Q}$ *for every integer cut $N \geq 0$.*

A shape word is the cube against one letter, and one share opens the letter: the Beta corners of the shapes and of x_0 leave a single line integral of stacked Gammas, the stacked Gamma $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]$ times a rational share function, which closes on the positive integers. The share function has poles of order at most k at the nonpositive integers, so the row carries the Trailmoments of the counting input of index at most $k+1$ with rational parts cleared by powers of d_{n+D}, and the coefficients of the simple poles sum to zero because the share function decays quadratically. At two shapes the row has one coordinate besides the rationals, the decay $(\sqrt{2}-1)^{4n}$ beats the clearing d_n^3, and the gap gives the irrationality of $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N,\infty))$ at every integer cut N. With more shapes the decay still beats the clearing, but the row carries several coordinates, and the filters and the growing Beta ports select one (Proposition 34.18, Theorem 34.20).

34.3 Rows

Proposition 34.10 (The row of a closing in rational terms)**.** *Let a closing give $x_n = \sum_{t\geq t_0} r_n(t)$, the sum over the integers $t \geq t_0$, with $r_n(t) = \sum_{k,s} \tilde{c}_{n,k,s}(t+k)^{-s} = O(t^{-2})$, $\tilde{c}_{n,k,s} \in \mathbb{Q}$, integer pole positions and $h_k = t_0 + k - 1 \geq 0$. With $\tilde{C}_{n,s} = \sum_k \tilde{c}_{n,k,s}$,*

$$\begin{aligned} x_n &= \sum_{k,s} \tilde{c}_{n,k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((h_k,\infty)) = \sum_{s\geq 2} \tilde{C}_{n,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - B_n, \\ B_n &= \sum_{k,s} \tilde{c}_{n,k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,h_k]), \end{aligned}$$

the index-one Trailmoments regularized, and $D_n = \mathrm{lcm}_{k,s}\, \big(\mathrm{den}(\tilde{c}_{n,k,s})\, d_{h_k}^s\big)$ clears the row.

Proposition 34.11 (The primitive row)**.** *For a nonzero rational pair (A, B), a common clearing integer D and $g = \gcd(DA, DB)$, the pair $g^{-1}(DA, DB) \in \mathbb{Z}^2$ is, up to sign, independent of D and of every nonzero rational rescaling of (A, B), and with $(a, b) = g^{-1}(DA, DB)$, $av - b = (D/g)(Av - B)$ for every real v.*

Corollary 34.12 (The top-index criterion)**.** *Let $k \geq 3$, and suppose there are integer polynomials P_n of degree at most a fixed D with $C_{n,s}(P_n) = 0$ for $2 \leq s < k$ and $I_n(k; P_n) > 0$ for all large n, P_n possibly changing sign on the cube. If the sums $\|P_n\|_1$ of the absolute values of the coefficients of P_n satisfy*

$$\limsup_n \frac{1}{n} \log \|P_n\|_1 < -\log M_k(0) - (k+1) \limsup_n \frac{1}{n} \log d_n,$$

then $(\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, \infty)) \notin \mathbb{Q}$ for every integer cut $N \geq 0$.

A closing in rational terms has a row whose clearing is read off its pole positions and orders, and a row is a primitive integer pair up to the clearing. With k shapes the row of $I_n(k; P)$ carries $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, \infty)), \ldots, (\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, \infty))$; an insertion that kills the lower column sums and keeps the word positive at a cost below the margin between the decay and the clearing separates $(\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, \infty))$. The insertion may change sign: positivity is asked of the word, not of the insertion.

34.4 The Beta port ladder and the filters

Theorem 34.13 (The Beta port ladder of the shape words)**.** *For the shape word with $k \geq 2$ shapes at depth $n \geq 1$ write its letter as $L = \bar{x}_0 + x_0 x_2 \cdots x_k\, x_1$, and in this statement write $P_r = (-1)^r \Pi_r^{n+1,n+1}(x_1)$, $g_{n,r} = g_{n,P_r}$ and $g_n = g_{n,0}$, the share function of the insertion 1 (Definition 34.4).*

(i) $0 < I_n(k; P_r) \leq \binom{n+r}{r} I_n(k; 1)$ *(Theorem 34.19, the port factor being at most one).*

(ii) $g_{n,r}(a) = \binom{n+r}{r} \dfrac{(n+1-a)_r}{(n+1+a)_r}\, g_n(a)$ *with* $g_n(a) = \dfrac{(n!)^{k-2}(1-a)_n^2}{(a)_{n+1}^k}$, *and with* $b = n + r + 1$, $(r+1)(a+b)\, g_{n,r+1}(a) = b\,(b-a)\, g_{n,r}(a)$.

(iii) For a function φ on the integers and a finite set $\mathcal{M}$ of positive integers write $S_{n,r}[\varphi] = \sum_{m \in \mathcal{M}} g_{n,r}(m)\varphi(m)$ and $S'_{n,r}[\varphi] = -\sum_{m \in \mathcal{M}} g'_{n,r}(m)\varphi(m)$, the sums over the common finite set $\mathcal{M}$, with functions of the summation variable m inside the brackets. Then, with $b = n + r + 1$,

$$(r+1)\, S'_{n,r+1}\big[(m+b)\varphi\big] = b\, S'_{n,r}\big[(b-m)\varphi\big] + (r+1)\, S_{n,r+1}[\varphi] + b\, S_{n,r}[\varphi].$$

(iv) $\dfrac{g_{n+1,r}(a)}{g_{n,r}(a)} = \dfrac{(n+1)^{k-3}(n+r+1)(n+1-a)(n+r+1-a)}{(a+n+1)^{k-1}(a+n+r+1)}$, *and with* $U(m) = (m+n+1)^{k-1}(m+n+r+1)$ *and* $V(m) = (n+1)^{k-3}(n+r+1)(n+1-m)(n+r+1-m)$,

$$S_{n+1,r}[U\varphi] = S_{n,r}[V\varphi], \qquad S'_{n+1,r}[U\varphi] - S_{n+1,r}[U'\varphi] = S'_{n,r}[V\varphi] - S_{n,r}[V'\varphi].$$

(v) For $0 \leq r \leq h$ the insertions P_r have degree at most h, their rows are cleared by d_{n+h}^{k+1} (Theorem 34.7), and their positive reads cost at most $\binom{n+h}{h}$ times the base word $I_n(k; 1)$.

(vi) In the selected mark x_1 the furthest forward coefficient of Theorem 32.15, for the one letter L of exponent $n+1$, is nonzero on the open cube.

Proposition 34.14 (The insertion index supplies auxiliary states)**.** *Fix $k \ge 3$ and $n \ge 1$ and, in this statement, write $P_r = (-1)^r \Pi_r^{n+1,n+1}(x_1)$, an insertion of the shape word with k shapes. As functions of r, $C_{n,k}(P_r) \in \mathbb{Q}[r]$ and $C_{n,k-1}(P_r) = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r) + q_n(r)$ with $q_n \in \mathbb{Q}(r)$. Consequently a scalar finite shift recurrence over $\mathbb{Q}(r)$ annihilating $C_{n,k-1}(P_r)$ for all large r also annihilates $C_{n,k}(P_r)$.*

Proposition 34.15 (The shape step is a two-state operator relation)**.** *For $k \ge 2$ and $n \ge 0$ let g_n be the share function of the insertion 1, $g_n(a) = (n!)^{k-2}(1-a)_n^2/(a)_{n+1}^k$, and for integer cuts $0 \le N_0 < N_1$ write $S_n[\varphi] = \sum_{N_0<m\le N_1} g_n(m)\varphi(m)$ and $S_n'[\varphi] = -\sum_{N_0<m\le N_1} g_n'(m)\varphi(m)$. Then*

$$S_{n+1}\big[(m+n+1)^k\varphi\big] = (n+1)^{k-2}\,S_n\big[(m-n-1)^2\varphi\big],$$
$$S_{n+1}'\big[(m+n+1)^k\varphi\big] = k\,S_{n+1}\big[(m+n+1)^{k-1}\varphi\big] + (n+1)^{k-2}\,S_n'\big[(m-n-1)^2\varphi\big] - 2(n+1)^{k-2}\,S_n\big[(m-n-1)\varphi\big],$$

and the same holds over all $m > N_0$ whenever the sums converge one by one: for bounded φ, when $(k-2)(n+1) > 1$ in the first identity and $(k-2)(n+1) > 0$ in the second.

Proposition 34.16 (A polynomial filter from a bounded local recurrence)**.** *Let unwanted closing coefficients satisfy $c_{n,j} = \ell(n)\,y_{n,j}$ and $y_{n+1,j} = \mathbf{A}(n)\,y_{n,j}$, $1 \le j \le r$, with state vectors $y_{n,j}$ of a fixed length h, a $1\times h$ matrix $\ell(n)$ and an $h\times h$ matrix $\mathbf{A}(n)$ over $\mathbb{Q}(n)$. Write $\mathbf{A} = \widehat{\mathbf{A}}/q$ and $\ell = \hat{\ell}/s$ with integer polynomial matrices $\widehat{\mathbf{A}}$, $\hat{\ell}$ and integer polynomials q, s, put $\mathbf{A}^{[0]} = I$, $\mathbf{A}^{[i]}(n) = \mathbf{A}(n+i-1)\cdots\mathbf{A}(n)$, $Q(n) = \prod_{t=0}^{h-1} q(n+t)\prod_{t=0}^{h} s(n+t)$ and the $1\times h$ matrices $\mathbf{r}_i(n) = Q(n)\,\ell(n+i)\,\mathbf{A}^{[i]}(n)$, $0 \le i \le h$. If the $\mathbf{r}_i$ span $\mathbb{Q}(n)^{1\times h}$ and $a_i(n)$ is $(-1)^i$ times the determinant of the $h\times h$ matrix stacking the $\mathbf{r}_{i'}$, $i' \ne i$, in order, then the a_i are integer polynomials, not all zero, and*

$$\sum_{i=0}^{h} a_i(n)\,c_{n+i,j} = 0 \qquad (1 \le j \le r)$$

away from the integer zeros of the denominators; a target sequence x_n is retained at n exactly when $\sum_i a_i(n)\,x_{n+i} \ne 0$. With $D_{\mathbf{A}} = \max(\deg\widehat{\mathbf{A}}, \deg q)$, $D_s = \deg s$ and $D_\ell = \deg\hat{\ell}$: $\deg a_i \le h(hD_{\mathbf{A}} + hD_s + D_\ell)$. For two states, with $\ell = (1,0)$ and $\mathbf{A} = (\beta_{il})$, the filter is $\beta_{12}c_{n+2} - (\beta_{11}^+\beta_{12} + \beta_{12}^+\beta_{22})c_{n+1} + \beta_{12}^+ \det\mathbf{A}(n)\,c_n = 0$, $\beta_{il}^+ = \beta_{il}(n+1)$; for one state, $q(n)c_{n+1} = p(n)c_n$ gives the filter $q(n)\,\mathrm{Sh} - p(n)$, Sh the shift $n \mapsto n+1$.

Proposition 34.17 (The finite jet recurrence)**.** *If a state has the recurrence $U_{n+1}(\varepsilon) = \mathbf{A}(n,\varepsilon)\,U_n(\varepsilon)$, regular at $\varepsilon = 0$, then with $U_n(\varepsilon) = \sum_j U_{n,j}\varepsilon^j$ and $\mathbf{A}(n,\varepsilon) = \sum_j \mathbf{A}_j(n)\varepsilon^j$, $U_{n+1,j} = \sum_{i=0}^{j} \mathbf{A}_i(n)\,U_{n,j-i}$ for $0 \le j \le J$: a lower block-triangular recurrence for the jets of order at most J, with diagonal blocks $\mathbf{A}_0(n)$.*

Proposition 34.18 (A recurrence filters the positive word)**.** *Let $x_n = \int X^n\,d\mu$ with $0 < X \le X_{\max} < 1$ on the support of a finite positive measure μ, and $x_n = \sum_{j=1}^{r} \tilde{C}_{n,j}v_j - B_n$ with $\tilde{C}_{n,j}, B_n \in \mathbb{Q}$. Let $a_0(n),\dots,a_h(n) \in \mathbb{Z}[n]$, $Q_n(X) = \sum_i a_i(n)X^i$ and $\tilde{x}_n = \sum_i a_i(n)\,x_{n+i}$, and suppose that for all large n, $\sum_i a_i(n)\tilde{C}_{n+i,j} = 0$ for $j \ne j_*$ and Q_n has one sign on the support of μ, strictly on a set of positive measure. Then $\tilde{x}_n = \int X^n Q_n\,d\mu = A_n v_{j_*} - \sum_i a_i(n)B_{n+i}$ with $A_n = \sum_i a_i(n)\tilde{C}_{n+i,j_*}$, and $0 < |\tilde{x}_n| \le O(n^{\max_i \deg a_i})\,X_{\max}^n\mu(\mathbb{R})$. If positive integers $D_{n,h}$ clear the rows of $x_n,\dots,x_{n+h}$, and $\limsup_n \frac{1}{n}\log D_{n,h} < -\log X_{\max}$, then $v_{j_*} \notin \mathbb{Q}$ and $A_n \ne 0$ for all large n. The sign of Q_n on $(0, X_{\max}]$ is certified when $Q_n(X) = \sum_{l=0}^{h} \beta_l(n)X^l(X_1 - X)^{h-l}$ for a rational $X_1 > X_{\max}$ with all $\beta_l(n) \ge 0$ and one of them positive.*

The Beta ports of one shape move the row of a shape word along a ladder: the share function of the Beta port of order $r+1$ is that of order r times a rational factor, so the sums over a common finite set of integers satisfy finite operator relations in the order of the Beta ports and in the depth. Unwanted coordinates whose coefficients follow a bounded local recurrence in the depth are annihilated by a filter with integer polynomial coefficients, read off the determinants of that recurrence, and jets in a regulator follow a block-triangular recurrence with the same diagonal. A filter applied to a positive word stays positive as long as its polynomial keeps one sign on the support of the word's measure, which a nonnegative expansion in the products $X^l(X_1-X)^{h-l}$ certifies, and then the filtered word separates the coordinate it keeps, at the cost of clearing the rows of $h+1$ consecutive values.

34.5 Growing Beta ports

Theorem 34.19 (The exact maximum under a growing Beta port)**.** *Fix $k \ge 2$, with the base f_k and the port factor $\mathfrak{a}_k$ of Definition 34.4, so that $I_n(k;P) = \int P f_k^n \, dx/L$. For $q \in \mathbb{N}$,*

$$I_n\big(k;(-1)^q \Pi_q^{n+1,n+1}(x_1)\big) = \binom{n+q}{q} \int_{[0,1]^{k+1}} f_k^n \, \mathfrak{a}_k^q \, \frac{dx}{L} > 0.$$

For every real $\eta \ge 0$, $f_k \mathfrak{a}_k^\eta$ has a unique maximum on the open cube: with a the unique root in $0 < a < 1/(2+\eta)$ of

$$(1-2a)\big(1-(2+\eta)a\big)(1+\eta a)^{k-1} = (1+\eta)^k a^{k+2},$$

it is at $x_1 = a$, $x_2 = \cdots = x_k = \frac{(1+\eta)a}{1+\eta a}$, $x_0 = \frac{(1+\eta)a}{1-a}$, with value

$$M_k(\eta) = \frac{(1-a)^{k-2}}{(1+\eta a)^{k-1}} \, (1-2a)^{1+\eta} \, \big(1-(2+\eta)a\big),$$

and $0 < I_n\big(k;(-1)^q \Pi_q^{n+1,n+1}(x_1)\big) \le \binom{n+q}{q} M_k(q/n)^n \, I_0(k;1)$. At $\eta = 0$ this is the maximum of Proposition 34.8.

Theorem 34.20 (An arithmetic budget for growing Beta ports)**.** *For $k \ge 2$, $\eta > 0$ and $q_n = \lfloor \eta n \rfloor$, in this statement write $\ell_k(\eta) = \log M_k(\eta) + (1+\eta)\log(1+\eta) - \eta \log \eta$. Then*

$$\lim_n \frac{1}{n} \log I_n\big(k;(-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big) = \ell_k(\eta),$$

$$\limsup_n \frac{1}{n} \log \Big(d_{n+q_n}^{k+1} \, I_n\big(k;(-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big)\Big) \le \ell_k(\eta) + (k+1)(1+\eta)\, c_{\mathrm{lcm}}.$$

The integer $d_{n+q_n}^{k+1}$ clears the whole row, which stays on the coordinates $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, …, $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and a rational part, and the coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is

$$k \binom{n+q_n}{q_n} \sum_{j=0}^{n} (-1)^{jk} \binom{n}{j}^k \binom{n+j}{n}^2 \frac{(n+1+j)_{q_n}}{(n+1-j)_{q_n}},$$

positive for even k.

Corollary 34.21 (A growing Beta port with a certified gap)**.** *For every $k \geq 2$ there is a rational $\eta_0 > 0$ such that $\ell_k(\eta) + (k+1)(1+\eta)\, c_{\mathrm{lcm}} < 0$ for every rational $\eta \in (0, \eta_0]$ (Theorem 34.20), the root of Theorem 34.19 at each such η being certified as a root of an integer polynomial in a rational interval. So with $q_n = \lfloor \eta n \rfloor$ the positive words $I_n\big(k; (-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big)$ satisfy*

$$\limsup_n \frac{1}{n} \log \Big(d_{n+q_n}^{k+1}\, I_n\big(k; (-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big)\Big) < 0,$$

on the fixed coordinates $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, …, $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and a rational part, with a positive coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ when k is even.

A Beta port whose order grows in proportion to the depth pulls the word toward the port factor: the word decays at the maximum value $M_k(\eta)$ of $f_k \mathfrak{a}_k^{\eta}$, the binomial of the Beta port adds its entropy $(1+\eta)\log(1+\eta) - \eta \log \eta$, and the clearing grows to $d_{n+q_n}^{k+1}$. Where the sum is negative the cleared words tend to zero on a fixed set of coordinates, and for even k the top coordinate keeps a positive coefficient. For every number of shapes the margin is positive at small rational ratios, by continuity from the base word (Proposition 34.8), each certified by an explicit algebraic root, and the cleared positive words decay exponentially on the coordinates $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \ldots, (\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ with a rational part.

Chapter 35

Rays, native gauges and tiers

A ray moves the data of a family along an integral direction. Along it the values of every insertion obey two relations, integration by parts in every mark and the shift by the stride, and these relations make the values a holonomic sequence: they satisfy a recurrence of order at most the dimension of the ray. The complex points at which the direction is native, the native gauges of the ray, carry formal solutions of the relations and bound the dimension from below, and the frame of any recurrence is a power times a stacked Gamma. Every ray with real origin is one function, carried by a pencil, with a branch at the native gauge of its direction, and on the centre ray of a family every sign is a candidate native gauge; a ray along a wall carries the product of the rays of its two faces. What a value can contain is bounded: finitely many values span every value along a ray, and a closing in rational terms carries Trailmoments of index at most the largest order of its poles (Theorem 35.19, Lemma 35.20).

35.1 The relations along a ray

Definition 35.1 (Ray, stride, insertions, values)**.** Let F be a family on $\mathcal{E}$. A *ray* of F is the sequence of data $(\mathbf{m}_0 + k\mathbf{m}_1;\ \mathbf{s}_0 + k\mathbf{s}_1;\ \mathbf{t}_0 + k\mathbf{t}_1)$, $k \in \mathbb{N}$, the points at the integer times of the half-line from its *origin* $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$, which is complex, along its *direction* $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$, which is integral, $\mathbf{m}_1 \in \mathbb{Z}^{\mathcal{E}}$ and $s_{1j}, t_{1j} \in \mathbb{N}$, both on the balance wall; the walls along it are $\kappa_{0,S} + k\kappa_{1,S}$, with $\kappa_{0,S}$ and $\kappa_{1,S}$ the walls of the origin and of the direction. On the chart $y_{e_0} = 0$ of a fixed mark e_0, with $dy = \prod_{e \neq e_0} d\alpha_e/\alpha_e$, the *stride* and the *origin form* are

$$f = \alpha^{\mathbf{m}_1} \prod_j K_j^{-s_{1j}} \Psi_j^{-t_{1j}}, \qquad \omega_0 = \alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}\, dy,$$

f a rational function of degree 0. The *letter complement* of F is $X_F = \{\alpha \in (\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times : K_j(\alpha)\Psi_j(\alpha) \neq 0 \text{ for every } j\}$, and the insertions along the ray are its regular functions, the algebra $\mathbb{C}[X_F]$, which is generated by the monomials α^v, $v \in \mathbb{Z}^{\mathcal{E}}$ with $v_{\mathcal{E}} = 0$, and the chances of the bases under the basis and cobasis laws, $\alpha \mapsto \mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ and $\alpha \mapsto \mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$, $T \in \mathcal{B}_j$. The *values* along the ray are $\langle g\rangle_k = \int_{y_{e_0}=0} g\, f^k \omega_0$ for $g \in \mathbb{C}[X_F]$, and over a compact real $(n-1)$-dimensional chain D in X_F, $\langle g\rangle_{k;D} = \int_D g\, f^k \omega_0$. The *defects* of the origin and of the direction are the masses less the usages,

$$c_e^{(i)} = m_{i,e} - \sum_j \big(s_{ij} R_{j,e} + t_{ij} R^*_{j,e}\big), \qquad i = 0, 1,$$

with the occupation $R_{j,e}$ and the dual occupation $R^*_{j,e}$ of table j (Theorem 7.14).

Lemma 35.2 (Insertions). *(i) $\mathbb{C}[X_F]$ is closed under every ∂_{y_e}: $\partial_{y_e}\alpha^v = v_e\alpha^v$, $\partial_{y_e}\mathbb{P}_{j,\alpha}(T) = (\mathbf{1}_T(e) - R_{j,e})\,\mathbb{P}_{j,\alpha}(T)$ and $\partial_{y_e}\mathbb{P}^*_{j,\alpha}(T) = (\mathbf{1}_{\mathcal{E}\setminus T}(e) - R^*_{j,e})\,\mathbb{P}^*_{j,\alpha}(T)$; the occupations and the defects lie in $\mathbb{C}[X_F]$.*

(ii) The stride lies in $\mathbb{C}[X_F]$, and $c_e^{(1)} = \partial_{y_e}\log f$, $c_e^{(0)} + k\,c_e^{(1)} = \partial_{y_e}\log(f^k\omega_0/dy)$.

(iii) Every value $\langle g\rangle_k$ is a finite combination, with products of weights as coefficients, of periods P_F at data $(\mathbf{m}_0 + k\mathbf{m}_1 + v;\ \mathbf{s}_0 + k\mathbf{s}_1 + \mathbf{a};\ \mathbf{t}_0 + k\mathbf{t}_1 + \mathbf{b})$ on the balance wall, $v \in \mathbb{Z}^{\mathcal{E}}$ and $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$.

Theorem 35.3 (The relations along a ray). *If $\kappa_{1,S} > 0$ for every nonempty proper S, then for every insertion g the value $\langle g\rangle_k$ converges for all large k, and for every mark e*

$$\big\langle \big(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e}\big)g\big\rangle_k = 0, \qquad \langle fg\rangle_k = \langle g\rangle_{k+1}.$$

The first relation is integration by parts in the mark e against the form $f^k\omega_0$, whose logarithmic derivative is the defect; the second is the definition of the stride. Both act on the insertions, which are closed under the derivatives, so the values of all insertions along a ray form one module over the operators in k, and every relation among insertions modulo these two is a recurrence among values.

35.2 Native gauges carry solutions

Definition 35.4 (Native gauges of a ray; the dimension). A *native gauge* of a ray is a point α_c of X_F at which the direction is native, $c^{(1)}(\alpha_c) = 0$: a native gauge of the direction at a complex point (Definition 8.16); its *native value* is $f(\alpha_c)$, and it is *nondegenerate* when the Hessian of $\log f$ in y modulo $\mathbb{C}\mathbf{1}$ is nonsingular there. Near a nondegenerate native gauge there are holomorphic coordinates $v \in \mathbb{C}^{n-1}$ with $\log f = \log f(\alpha_c) - \frac{1}{2}\sum_i v_i^2$; the image D_c of a real ball $\{v \in \mathbb{R}^{n-1} : |v| \le \varepsilon\}$ is a *steepest-descent disc*. The *relation space* $\mathcal{L} \subseteq \mathbb{C}[X_F] \otimes \mathbb{C}(k)$ is the $\mathbb{C}(k)$-span of the elements $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g$, and the *dimension* of the ray is $\dim_{\mathbb{C}(k)}\big(\mathbb{C}[X_F] \otimes \mathbb{C}(k)\big)/\mathcal{L}$.

Theorem 35.5 (A native gauge carries a solution). *Let α_c be a nondegenerate native gauge of a ray and D_c a steepest-descent disc through it.*

(i) As $k \to \infty$,

$$\langle g\rangle_{k;D_c} \sim f(\alpha_c)^k\,k^{-(n-1)/2}\sum_{i\ge 0} a_i(g)\,k^{-i},$$

where $a_0(g) = (2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c)\,g(\alpha_c)$ with $(\omega_0/dv)(\alpha_c) \neq 0$, and each $a_i(g)$ is a finite combination of Taylor coefficients of $g\,\omega_0/dv$ and of $\log f$ at α_c. Further $\langle fg\rangle_{k;D_c} = \langle g\rangle_{k+1;D_c}$, and $\big\langle (c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g\big\rangle_{k;D_c}$ is at most a constant times $|f(\alpha_c)|^k e^{-k\varepsilon^2/2}$; so the expansions satisfy both relations to all orders.

(ii) For a finite set C of distinct nondegenerate native gauges α_c, the maps $A^{(c)} : \mathbb{C}[X_F] \otimes \mathbb{C}(k) \to \mathbb{C}((1/k))$, $g \mapsto \sum_l a_l^{(c)}(g)\,k^{-l}$, extended $\mathbb{C}(k)$-linearly, vanish on $\mathcal{L}$ and are linearly independent over $\mathbb{C}((1/k))$.

(iii) The dimension of the ray is at least the number of its nondegenerate native gauges.

Corollary 35.6 (The symbol). *Let g be an insertion and $p_0, \dots, p_h \in \mathbb{C}[k]$, not all zero, with $\sum_j p_j(k)\,f^j g \in \mathcal{L}$. Under the hypothesis of Theorem 35.3, $\sum_j p_j(k)\,\langle g\rangle_{k+j} = 0$ for all large k. With d the largest degree of the p_j and c_j the coefficient of k^d in p_j, the* symbol $\sum_j c_j\lambda^j$ *vanishes*

at the native value of every nondegenerate native gauge α_c with $g(\alpha_c) \neq 0$. So h is at least the number of distinct native values of such native gauges.

Proposition 35.7 (The native value bounds the order)**.** *Let a ray of a family have $\kappa_{1,S} > 0$ for every nonempty proper S. Then $\langle 1\rangle_k \neq 0$ for all large k, positive when the origin is real, and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$, the native value of the native gauge $\alpha^\star$ of the direction on $\mathbb{R}^{\mathcal{E}}_{>0}$, which is the largest value of f there. If $\sum_{j=0}^{h} p_j(k)\,\langle 1\rangle_{k+j} = 0$ for all large k, with $p_j \in \mathbb{Q}[k]$ not all zero, then the symbol vanishes at $f(\alpha^\star)$: $f(\alpha^\star)$ is algebraic, $h \geq [\mathbb{Q}(f(\alpha^\star)) : \mathbb{Q}]$, and every conjugate of $f(\alpha^\star)$ is a root of the symbol. In particular, when $f(\alpha^\star)$ is irrational the values $\langle 1\rangle_k$ satisfy no recurrence of order one over $\mathbb{Q}(k)$, and when it is transcendental none at all.*

Proposition 35.8 (The frame of a recurrence is a power times a stacked Gamma)**.** *Let $\sum_{j=0}^{h} p_j(k)\,x_{k+j} = 0$ for $k \geq k_0$, with $p_j \in \mathbb{C}[k]$ and p_0, p_h without zeros at the integers $k \geq k_0$, and let $x^{(1)}, \dots, x^{(h)}$ be solutions. Their* frame *$C_k = \det\left[x^{(i)}_{k+l}\right]_{1\leq i\leq h,\ 0\leq l<h}$ satisfies $C_{k+1} = (-1)^h \frac{p_0(k)}{p_h(k)}\,C_k$, and writing $\frac{p_0(k)}{p_h(k)} = c\,\frac{\prod_{i=1}^{\deg p_0}(k-a_i)}{\prod_{i=1}^{\deg p_h}(k-b_i)}$,*

$$C_k = C_{k_0}\big((-1)^h c\big)^{k-k_0}\,\Gamma_{\substack{\deg p_0\\ \deg p_h}}\begin{bmatrix} k-a_1,\ \dots \\ k-b_1,\ \dots \end{bmatrix} \Big/ \Gamma_{\substack{\deg p_0\\ \deg p_h}}\begin{bmatrix} k_0-a_1,\ \dots \\ k_0-b_1,\ \dots \end{bmatrix}.$$

The tier *of a sequence over a field $\Bbbk \subseteq \mathbb{C}$ is the least order of a recurrence with coefficients in $\Bbbk[k]$ and $p_0 p_h \neq 0$ that it satisfies for large k; a sequence of tier one is a power times a stacked Gamma, and so is every frame.*

A native gauge of a ray is a complex point at which the direction uses every mark exactly as its masses ask, the complex counterpart of the native gauge of a family. Through it a steepest-descent disc carries a value that satisfies both relations up to exponentially small terms, and its asymptotic expansion is a formal solution; distinct native gauges give independent solutions, so their number bounds the dimension, and their native values are roots of the symbol of every recurrence that follows from the relations. On the real orthant the native gauge of the direction governs the growth of the values, and an irrational native value forbids a recurrence of order one. Whatever the order, the frame of a recurrence is a power times a stacked Gamma: the sequences of tier one are the bottom of the hierarchy of sequences that the relations produce.

35.3 The centre ray of a family

Definition 35.9 (The centre ray)**.** Let F be a family of N tables on $n \geq 2$ marks, and let strengths $\sigma_j \in \mathbb{N}$, not all zero, make it connected: every nonempty proper S has $c_j(S) > 0$ for some j with $\sigma_j > 0$ (Definition 8.1); put $\boldsymbol{\sigma} = (\sigma_j)_j$ and $\sigma = \sum_j \sigma_j$. The *centre ray* of F at the strengths $\boldsymbol{\sigma}$ is its ray with origin 0 and direction $(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$. Its stride is

$$f = \alpha^{\sigma\mathbf{1}} \prod_j \big(K_j \Psi_j\big)^{-\sigma_j} = \prod_j \big(K_j(e^{y}) K_j(e^{-y})\big)^{-\sigma_j},$$

its values are $\langle 1\rangle_k = \int_{y_{e_0}=0} f^k\,dy = P_F(k\sigma\mathbf{1}; k\boldsymbol{\sigma}, k\boldsymbol{\sigma})$, $k \geq 1$, and its direction walls are $\kappa_{1,S} = \sum_j \sigma_j c_j(S) > 0$ (Proposition 8.21).

Theorem 35.10 (A ray is one function)**.** *Let a ray of a family on $n \geq 2$ marks have real origin and $\kappa_{1,S} > 0$ for every nonempty proper S, let $\alpha^\star$ be the native gauge of its direction on $\mathbb{R}^{\mathcal{E}}_{>0}$, with $\alpha^\star_{e_0} = 1$ (Theorem 8.18), and let $k_0 \geq 0$ be an integer with $\kappa_{0,S} + k_0\kappa_{1,S} > 0$ for every nonempty proper S.*

(i) $0 < f \le f(\alpha^\star)$ *on* $\mathbb{R}^{\mathcal{E}}_{>0}$*, with equality only at* $\alpha \propto \alpha^\star$*, and* $f \le C' e^{-c'\|y\|}$ *on the chart for some* $c', C' > 0$*; every value* $\langle 1\rangle_k$*,* $k \ge k_0$*, converges and is positive.*

(ii) For $|q| < f(\alpha^\star)^{-1}$ *the generating function* $G(q) = \sum_{k\ge k_0} \langle 1\rangle_k q^{k-k_0}$ *is*

$$G(q) = \int_{y_{e_0}=0} \frac{f^{k_0}\omega_0}{1-qf} = \int_{y_{e_0}=0} \frac{\alpha^{\mathbf{m}_0+k_0\mathbf{m}_1} \prod_j K_j^{-s_{0j}-(k_0-1)s_{1j}} \Psi_j^{-t_{0j}-(k_0-1)t_{1j}}}{\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}}\, dy,$$

$$G^{(l)}(q) = l! \int_{y_{e_0}=0} \frac{f^{k_0+l}\omega_0}{(1-qf)^{l+1}},$$

and the positive integral converges for every real $q < f(\alpha^\star)^{-1}$*: the whole ray is carried by the pencil* $\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}$*.*

(iii) For $k \ge k_0$ *and* $h \ge 1$*,*

$$\det\big[\langle 1\rangle_{k+i+i'}\big]_{i,i'=0}^{h-1} = \frac{1}{h!} \int \prod_{a=1}^{h} f(y_a)^k \frac{\omega_0}{dy}(y_a) \prod_{a<b} \big(f(y_a) - f(y_b)\big)^2 \prod_a dy_a > 0,$$

and G *is not a rational function of* q*.*

On the centre ray of a family (Definition 35.9) one may take $k_0 = 1$*, and* $\alpha^\star = \mathbf{1}$*.*

Theorem 35.11 (The branch at the native gauge of the direction)**.** *For a ray as in Theorem 35.10, with its generating function* G *and the native gauge* $\alpha^\star$ *of its direction, put* $A = f(\alpha^\star)^{-1}$*,* $\varepsilon = 1 - q/A$ *and* $a = \frac{n-1}{2}$*, and let* H *be the Hessian of* $-\log f$ *at* $\log \alpha^\star$ *on the chart,*

$$H = \Big(\sum_j \big(s_{1j}\, \kappa_{j,\alpha^\star}(X_e, X_{e'}) + t_{1j}\, \kappa_{j,(\alpha^\star)^{-1}}(X_e, X_{e'}) \big) \Big)_{e,e' \ne e_0},$$

the covariances of the indicators under the basis laws and the cobasis laws at $\alpha^\star$ *(Definition 9.2, Theorem 7.14); it is positive definite. As* $q \uparrow A$ *there are functions* U, V *holomorphic at* $\varepsilon = 0$ *with*

$$G(q) = U(\varepsilon) + \varepsilon^{a-1} V(\varepsilon) \quad (n \text{ even}), \qquad G(q) = U(\varepsilon) + \varepsilon^{a-1} \log\varepsilon\, V(\varepsilon) \quad (n \text{ odd}),$$

$$V(0) = \frac{\Gamma(1-a)\,(2\pi)^a\,(f^{k_0-1}\omega_0/dy)(\alpha^\star)}{A\sqrt{\det H}} \quad (n \text{ even}),$$

$$V(0) = \frac{(-1)^a(2\pi)^a\,(f^{k_0-1}\omega_0/dy)(\alpha^\star)}{\Gamma(a)\,A\sqrt{\det H}} \quad (n \text{ odd}).$$

The derivatives $G^{(l)}(A)$ *are finite exactly for* $l < a - 1$*, and*

$$G(q) = \sum_{l<a-1} \frac{G^{(l)}(A)}{l!} \big(q - A\big)^l + V(0)\,\varepsilon^{a-1}\big(1 \text{ or } \log\varepsilon\big) + O\big(\varepsilon^{\lceil a-1\rceil}\big).$$

On the centre ray of a family, with $k_0 = 1$*,* $A = \prod_j K_j(\mathbf{1})^{2\sigma_j}$*, the density* $f^{k_0-1}\omega_0/dy$ *is* 1*, and* $H = 2\sum_j \sigma_j \big(\kappa_{j,\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\ne e_0}$*, the covariances under the weight laws.*

Theorem 35.12 (Native gauges at the signs)**.** *For the centre ray of a family at the strengths* $\boldsymbol{\sigma}$ *(Definition 35.9), on the chart* $\alpha_{e_0} = 1$*, with the family word* $W = \prod_j K_j^{-\sigma_j} \Psi_j^{-\sigma_j}$ *at the exponents* $(\boldsymbol{\sigma}, \boldsymbol{\sigma})$*, so that* $f = \alpha^{\sigma\mathbf{1}} W$*:*

(i) $f(\alpha^{-1}) = f(\alpha)$*; the points fixed by* $\alpha \mapsto \alpha^{-1}$ *are the* 2^{n-1} signs $\eta \in \{\pm 1\}^{\mathcal{E}}$, $\eta_{e_0} = 1$*; every native gauge of the centre ray is a sign or is paired with its image, a native gauge of the same native value.*

(ii) A sign with $K_j(\eta) = \sum_T \theta_{j,T}\eta^T \neq 0$ *for every* j *is a native gauge, of native value* $\prod_j K_j(\eta)^{-2\sigma_j}$*. In the chart* $\alpha_e = \eta_e e^{y_e}$, $y_{e_0} = 0$*, the Hessian of* $-\log f$ *at* $y = 0$ *is* $2\sum_j \sigma_j \big(\kappa_{j,\eta}(X_e, X_{e'})\big)_{e,e' \neq e_0}$*, with* $\kappa_{j,\eta}$ *the covariances of the indicators under the signed measure* $\mathbb{P}_{j,\eta}(T) = \theta_{j,T}\eta^T / K_j(\eta)$ *of total mass one at the sign.*

(iii) The critical polynomials $\sigma W^{-1} - \alpha_e \partial_e W^{-1}$, $e \neq e_0$*, vanish identically modulo* $\alpha_e^2 = 1$*, and* $W^{-1} \equiv \alpha^{\sigma\mathbf{1}} \prod_j K_j^{2\sigma_j}$ *there. In* $\mathbb{Q}(\boldsymbol{\theta})[\alpha_e : e \neq e_0]/(\alpha_e^2 - 1)$, $\mathbb{Q}(\boldsymbol{\theta})$ *the field of the weights, localized at* $\prod_j K_j$*, the idempotents* $2^{1-n}\prod_{e \neq e_0}(1+\eta_e\alpha_e)$ *of the signs with* $K_j(\eta) \neq 0$ *for every* j *form a basis, and multiplication by* $W^{-1}/\alpha^{\sigma\mathbf{1}}$ *acts on the idempotent of* η *by* $\prod_j K_j(\eta)^{2\sigma_j}$*.*

Theorem 35.13 (Weights that separate the signs)**.** *Let a family of* N *tables on* $n \geq 2$ *marks be connected at the strengths* $\sigma_j \in \mathbb{N}$*, of total* σ*, with its centre ray (Definition 35.9). There are weights* $\theta_{j,T} \in \big\{1, \dots, 2^{n-1}\big((2n-1)N + (2^{n-1}-1)\sigma\big) + 1\big\}$ *for which every sign has* $K_j(\eta) \neq 0$ *for every* j *and is a nondegenerate native gauge of the centre ray, and the* 2^{n-1} *native values* $\prod_j K_j(\eta)^{-2\sigma_j}$ *are distinct. For such weights the dimension of the centre ray is at least* 2^{n-1}*, and every recurrence for* $\langle 1\rangle_k$ *that follows from the relations (Corollary 35.6) has order at least* 2^{n-1}*.*

Along a ray with real origin the whole sequence of values is one function of the pencil parameter, holomorphic up to the reciprocal of the native value of the direction. The determinants $\det\big[\langle 1\rangle_{k+i+i'}\big]$ of the values are positive integrals of squared differences, so the generating function is not rational; at the native gauge of the direction it branches, with the exponent set by the number of marks and the coefficient by the value of $f^{k_0-1}\omega_0/dy$ there and the determinant of the covariances of the basis and cobasis laws. The centre ray of a family is its period at the centre read at every multiple of its strengths; its native gauge is the table's gauge, where it branches at $\prod_j K_j(\mathbf{1})^{2\sigma_j}$ with the covariances of the weight laws. Every sign is a candidate native gauge, the signed measures $\mathbb{P}_{j,\eta}$ playing the part of the weight laws, and with weights chosen to separate the signs every recurrence that the relations produce for the centre ray of a family on n marks has order at least 2^{n-1}.

35.4 What a wall carries

Lemma 35.14 (Where native gauges can reach a wall)**.** *Let* B *be a proper bunch with room* $R \ni e_0$*,* $\alpha_B = \rho u$*, and in this statement write* $K_j^{[l]} = K_{j,B}^{(d_j(B)+l)}(u, \alpha_R)$ *and* $\Psi_j^{[l]} = \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R)$*, so that* $K_j = \rho^{d_j(B)}\sum_l \rho^l K_j^{[l]}$ *and* $\Psi_j = \rho^{d_j^*(B)}\sum_l \rho^l \Psi_j^{[l]}$ *(Theorem 16.23), with the bottom letters* $K_j^{[0]}$ *and* $\Psi_j^{[0]}$*.*

(i) At every native gauge α *of a direction* $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$ *with* $s_{1j}, t_{1j} \geq 0$*,* $m_{1,B} = \sum_j \big(s_{1j}\sum_T |T \cap B|\,\mathbb{P}_{j,\alpha}(T) + t_{1j}\sum_T |B \setminus T|\,\mathbb{P}_{j,\alpha}^*(T)\big)$*: the direction's mass on* B *is its usage of* B*.*

(ii) Let native gauges of such directions $\mathbf{v}_i \to \mathbf{v}$ *have bunch scales* $\rho_i \to 0$ *and* $(u_i, \alpha_{R,i}) \to (u^*, \alpha_R^*)$ *in the torus, with* $\kappa_{1,B}(\mathbf{v}) > 0$*. Then some bottom letter vanishes at* (u^*, α_R^*)*, and if only one does, its gradient in* (u, α_R) *vanishes there as well: the product* $\prod_j K_j^{[0]}\Psi_j^{[0]}$ *of the bottom letters vanishes at* (u^*, α_R^*) *together with its gradient in* (u, α_R)*.*

So the native gauges of a family leave for the wall of a bunch only at singular points of the zero set of the product of its bottom letters.

Theorem 35.15 (A ray along a wall carries its faces)**.** *Let B be a proper bunch with room $R \ni e_0$ on whose bottom levels the weights factor, and let a ray have its direction on the wall, $\kappa_{1,B} = 0$, and $\operatorname{Re}\kappa_{0,B} > 0$. With $\alpha_B = \rho u$, and writing in this statement $K_j^{[l]} = K_{j,B}^{(d_j(B)+l)}(u,\alpha_R)$ and $\Psi_j^{[l]} = \Psi_{j,B}^{(d_j^*(B)+l)}(u,\alpha_R)$ as in Theorem 16.23,*

$$f = f_B(u)\, f_R(\alpha_R)\,\big(1 + \rho\, h_1 + O(\rho^2)\big), \qquad h_1 = -\sum_j \Big(s_{1j}\frac{K_j^{[1]}}{K_j^{[0]}} + t_{1j}\frac{\Psi_j^{[1]}}{\Psi_j^{[0]}}\Big),$$

where f_B and f_R are the strides along the ray of the faces F^B and F^R of Theorem 16.23, and $\omega_0 = \rho^{\kappa_{0,B}-1}d\rho \wedge \omega_0^B \wedge \omega_0^R\,(1 + O(\rho))$ with the origin forms of the faces. Let α_c^B and α_c^R be nondegenerate native gauges of the rays of the faces with $h_1(\alpha_c^B, \alpha_c^R) \neq 0$, let $\operatorname{Re}\big(e^{i\vartheta}h_1\big) < 0$ near (α_c^B, α_c^R), and let $D = e^{i\vartheta}[0,\rho_1] \times D_B \times D_R$ with steepest-descent discs D_B, D_R of the faces. Then the values $\langle g\rangle_{k;D}$ satisfy both relations up to terms exponentially smaller than $|f_B(\alpha_c^B) f_R(\alpha_c^R)|^k$, and

$$\langle 1\rangle_{k;D} = \Gamma(\kappa_{0,B})\,\big(-k\, h_1(\alpha_c^B, \alpha_c^R)\big)^{-\kappa_{0,B}}\,\langle 1\rangle_{k;D_B}^{F^B}\,\langle 1\rangle_{k;D_R}^{F^R}\,\big(1 + O(1/k)\big),$$

with $\langle 1\rangle_{k;D_B}^{F^B}$ and $\langle 1\rangle_{k;D_R}^{F^R}$ the values of the rays of the faces over their discs (Theorem 35.5), and the power taken continuously along $e^{i\vartheta}(0,\infty)$.

Definition 35.16 (The whole cluster port)**.** On a bunch scale ρ take letters with *lower coefficients* $\boldsymbol{\lambda}_i \in \mathbb{C}^{r_i}$,

$$L_i(\rho, \boldsymbol{\lambda}_i) = \sum_{l<r_i} \lambda_{i,l}\,\rho^l + \rho^{r_i} C_i(\rho), \qquad C_i(0) \neq 0,$$

with exponents w_i, and a word $\rho^{\kappa-1}G(\rho)\prod_i L_i^{-w_i}$, with G and the C_i holomorphic on a neighbourhood of $[0,\infty)$, each $|C_i|$ bounded below by a positive constant there, and the word integrable at ∞; put $\kappa^\flat = \kappa - \sum_i r_i w_i$. Fix $b > 0$ with every C_i zero-free on $|\rho| \le b$; for $\boldsymbol{\lambda}$ in a small polydisc each L_i has exactly r_i roots in $|\rho| < b$, the *cluster*, none on $|\rho| = b$ and none on $[b,\infty)$. The *cluster port* of the word is its integral over the *cluster contour*, which comes in along the half-line $[b,\infty)$ of the scale, goes once around $|\rho| = b$ and goes out on the new sheet:

$$\operatorname{Port}_\kappa\Big[G\prod_i L_i^{-w_i}\Big](\boldsymbol{\lambda}) = \frac{1}{2\pi i}\Big[(e^{2\pi i\kappa^\flat} - 1)\int_b^\infty \rho^{\kappa-1}G\prod_i L_i^{-w_i}\,d\rho + \oint_{|\rho|=b} \rho^{\kappa-1}G\prod_i L_i^{-w_i}\,d\rho\Big].$$

Theorem 35.17 (Relations of the whole cluster port)**.** *(i) One turn around $|\rho| = b$ multiplies the word by $e^{2\pi i\kappa^\flat}$; the cluster port does not depend on b, is holomorphic in all the lower coefficients jointly on the polydisc, and the contour has no end at which the word is nonzero, so every integral of a ρ-derivative over it vanishes.*

(ii) On the stratum $\boldsymbol{\lambda} = 0$ the word is $\rho^{\kappa^\flat-1}h$, with the stratum word *$h = G\prod_i C_i^{-w_i}$, and*

$$\operatorname{Port}_\kappa\Big[G\prod_i L_i^{-w_i}\Big](0) = \frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}\,\operatorname{Reg}\int_0^\infty \rho^{\kappa^\flat-1}h\,d\rho \quad (\kappa^\flat \notin \mathbb{Z}),$$

$$\operatorname{Port}_\kappa\Big[G\prod_i L_i^{-w_i}\Big](0) = [\rho^{-\kappa^\flat}]\,h \quad (\kappa^\flat \in \mathbb{Z}_{\le 0}), \qquad \operatorname{Port}_\kappa\Big[G\prod_i L_i^{-w_i}\Big](0) = 0 \quad (\kappa^\flat \in \mathbb{Z}_{>0}).$$

(iii) For multi-indices $\mathbf{a}_i$, with $A_i = |\mathbf{a}_i|$ and $D = \sum_i\sum_l (r_i - l)\,a_{i,l}$, the derivative $\partial_{\boldsymbol{\lambda}}^{\mathbf{a}}$ of the cluster port at $\boldsymbol{\lambda} = 0$ is $\prod_i(-1)^{A_i}(w_i)_{A_i}$ times the value of (ii) for the stratum word $G\prod_i C_i^{-w_i-A_i}$ at the index $\kappa^\flat - D$; at $\kappa^\flat = 0$ it is $\prod_i(-1)^{A_i}(w_i)_{A_i}\cdot\frac{1}{D!}\,\partial_\rho^D\big[G\prod_i C_i^{-w_i-A_i}\big]_{\rho=0}$.

Proposition 35.18 (Small lower coefficients on one parameter)**.** *Let $C_1,\dots,C_p$ be bounded below by a positive constant on $[0,\infty)$, G and the C_i holomorphic near 0 and continuous on $[0,\infty)$, $\xi_1,\dots,\xi_p\in\mathbb{C}\setminus(-\infty,0]$, $\operatorname{Re}\kappa>0$ and $\kappa^\flat=\kappa-|\mathbf{w}|\notin\mathbb{Z}$, with $\rho^{\kappa-1}G\prod_i(\rho C_i)^{-w_i}$ integrable at ∞ together with every $\rho^{\kappa^\flat-|\mathbf{a}|-1}G\prod_i C_i^{-w_i-a_i}$, $\mathbf{a}\in\mathbb{N}^p$.*

(i) For small $\lambda>0$,

$$\int_0^\infty \rho^{\kappa-1}G(\rho)\prod_i\big(\lambda\xi_i+\rho\,C_i(\rho)\big)^{-w_i}d\rho=A(\lambda)+\lambda^{\kappa^\flat}B(\lambda)$$

with A,B holomorphic at 0,

$$A(\lambda)=\sum_{\mathbf{a}\in\mathbb{N}^p}\prod_i\binom{-w_i}{a_i}(\lambda\xi_i)^{a_i}\ \operatorname{Reg}\int_0^\infty\rho^{\kappa^\flat-|\mathbf{a}|-1}G\prod_i C_i^{-w_i-a_i}\,d\rho,$$
$$B(0)=G(0)\int_0^\infty u^{\kappa-1}\prod_i\big(\xi_i+u\,C_i(0)\big)^{-w_i}\,du,$$

the last integral continued in κ and $\mathbf{w}$; where every $\operatorname{Re}w_i>0$ it is

$$\prod_i C_i(0)^{-w_i}\,\Gamma_{\!p}^{2}\!\begin{bmatrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{bmatrix}\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}\big((\xi_i/C_i(0))_i\big),\qquad J=\{1,\dots,p\}$$

(Proposition 15.1). The analytic part opens with the finite part of the stratum word along the whole scale, and the branch part with $G(0)$ times the scale integral of the letters linearized at the wall.

(ii) When G and the C_i are holomorphic on a neighbourhood of $[0,\infty)$, the cluster port of Definition 35.16 for these letters, $r_i=1$ with the lower coefficients $\lambda\xi_i$, is $\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}A(\lambda)$.

At a wall the ray splits: the stride is the product of the strides of the two faces up to a first-order correction in the scale of the bunch, and integrating that scale against the correction contributes one Gamma of the origin wall, so a ray along a wall carries the product of the native values of its faces. The native gauges of a family reach a wall only where the product of its bottom letters is singular. When the bottom letters of a bunch are perturbed by lower coefficients, the scale integral is read on the cluster contour, which surrounds the whole cluster of roots at once: the cluster port is holomorphic in the lower coefficients, its value on the stratum where they vanish is the finite part of the stratum word, and its derivatives there are the finite parts of the stratum words with raised exponents. With small lower coefficients on one parameter the scale integral splits into an analytic part, the cluster port up to a constant, and a branch part led by the scale integral of the letters linearized at the wall, a stacked Gamma times a Breta of one bunch.

35.5 What a value can contain

Theorem 35.19 (The values along a ray)**.** *Let a ray of a family have weights and origin in a subfield $\Bbbk$ of $\mathbb{C}$, with $\kappa_{1,S}>0$ for every nonempty proper S.*

(i) The dimension of the ray is finite: it is the dimension over $\mathbb{C}(k)$ of the top cohomology of the twisted algebraic de Rham complex of the letter complement X_F, which is finite. It is at least the number of nondegenerate native gauges of the ray (Theorem 35.5). The native gauges are the critical points of the stride f, a product of powers of the letters with the exponents

of the direction; for a direction outside a countable union of proper subvarieties they are $(-1)^{n-1}\,\mathrm{eu}(X_F)$ *in number,* eu *the Euler characteristic, and all nondegenerate, so that the dimension is then at least* $(-1)^{n-1}\,\mathrm{eu}(X_F)$*. An integral direction can lie in that exceptional set.*

(ii) For every insertion g *with coefficients in* $\Bbbk$ *there are* $p_0, \dots, p_h \in \Bbbk[k]$*,* $p_h \neq 0$*,* h *at most the dimension, with* $\sum_j p_j(k)\,\langle g\rangle_{k+j} = 0$ *for all large* k*.*

(iii) So there is k_1 *such that every value* $\langle g\rangle_k$*,* $k \geq k_1$*, is a* $\Bbbk$*-linear combination of the* h *values* $\langle g\rangle_{k_1}, \dots, \langle g\rangle_{k_1+h-1}$*: the values along a ray span a* $\Bbbk$*-space of dimension at most that of the ray, and whatever Trailmoments those* h *values contain, no later value contains others.*

Lemma 35.20 (Index by pole count)**.** *Let* $\Bbbk$ *be a subfield of* $\mathbb{C}$*, and let* $r \in \Bbbk(k)$ *have all its poles at negative integers, with largest order* p*, and* $r(k) = O(k^{-2})$*. Then* $\sum_{k\geq 0} r(k) \in \Bbbk + \sum_{j=2}^{p} \Bbbk\,(\mathsf{T}_{j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$*. If moreover* $r(-M-k) = -r(k)$ *for an integer* M*, only odd* j *occur. If the poles lie in one coset* $-b+\mathbb{Z}$*,* $b \in \mathbb{Q}$*, the same holds with the Trailmoments of the lattice input* $\mathbb{I}_{b+\mathbb{Z}}$ *at its cuts; poles in several cosets leave, besides, differences of the index-one values of the cosets.*

A closing whose terms are rational functions of the lattice index brings Trailmoments of the counting input of index at most the largest order of its poles, and a reflection of the terms keeps only the odd indices. With Theorem 35.19 this is what bounds the complexity of a family along a ray: the space of values is finite-dimensional, spanned by finitely many of them, and where a closing writes the values as sums of rational terms the index is read off the orders of the poles, which along the ray can be bounded once for all its points.

Chapter 36

Relations and separation

A value is the evaluation of a word, and two words of equal value either are related by the proved relations of the theory or are not, a problem that relation blocks make finite. The admitted words form an algebra with an evaluation, the proved relations generate an ideal inside its kernel, and completeness asks for equality. Relation blocks reduce a finite space of words to representatives, each word with a certificate; when the representatives have values independent over the rationals, the block is complete. Words of polynomials on a box are complete outright, a finite census of insertions separates numerators, and the critical-line words of order k of polynomials form a block whose values are the rational combinations of 1 and $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; its two representatives have values independent over the rationals exactly when $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \notin \mathbb{Q}$, and the block is then complete. Independence of values is proved by separators: families of rational matrices whose rows are small on the vector of values.

36.1 Words and relations

Definition 36.1 (Words, evaluation, relations, completeness)**.** An *admitted word* is a word (Definition 1.1(vi)) given by finite rational data (tables and weights, coefficients, masses, exponents, inputs with a finite presentation, cuts) together with the rule that assigns its value: a convergent integral, a finite part or a continuation, as proved for it. The admitted words span a rational algebra $\mathfrak{W}$, whose product tensors the data, with independent integration variables and separate regulators, and *evaluation* is the map $\mathrm{ev} : \mathfrak{W} \to \mathbb{C}$. The *relation ideal* $\mathfrak{I} \subseteq \mathfrak{W}$ is generated by the proved relations, each on its stated domain: linearity; changes of variables and of order; scale and shape; split, port and gluing; the Gramma relations; the letter relations and the mark relations (Theorems 8.3 and 32.10); the subtraction of a common boundary term from the two sides of a relation; the cut relation of the Trailmoment (Proposition 5.3); the closings. A set of words is *complete* when each of its rational combinations of value zero lies in $\mathfrak{I}$.

Proposition 36.2 (Products keep separate regulators)**.** *If $A(\varepsilon)$ and $A'(\varepsilon')$ have Laurent expansions in disjoint sets of regulators, then $\mathrm{CT}_{\varepsilon}\,\mathrm{CT}_{\varepsilon'}\,\big(A(\varepsilon)A'(\varepsilon')\big) = \big(\mathrm{CT}_{\varepsilon}\,A\big)\big(\mathrm{CT}_{\varepsilon'}\,A'\big)$, and evaluation is multiplicative on $\mathfrak{W}$. Identifying regulators is a further operation: $\mathrm{CT}(\varepsilon^{-1}) = \mathrm{CT}(\varepsilon) = 0$, while $\mathrm{CT}(\varepsilon^{-1}\varepsilon) = 1$.*

Proposition 36.3 (Soundness)**.** $\mathfrak{I} \subseteq \ker(\mathrm{ev})$.

Proposition 36.4 (A finite matrix of mark relations)**.** *On $[0,1]^d$ let $a_l, b_l > 0$, let $D_1, \dots, D_h$ be rational polynomials positive on the closed box with exponents $p_1, \dots, p_h \in \mathbb{Q}$, and in this statement*

write $D = \prod_k D_k = \sum_\gamma D_\gamma x^\gamma$ *and* $x_i \sum_k p_k\, \partial_i D_k \prod_{k' \neq k} D_{k'} = \sum_\gamma D_\gamma^{(i)} x^\gamma$. *For every polynomial* A *and every coordinate* i,

$$\mathrm{mr}_i(A) = D\big[x_i(1-x_i)\,\partial_i A + \big(a_i - (a_i+b_i)x_i\big)A\big] - x_i(1-x_i)\,A \sum_k p_k\, \partial_i D_k \prod_{k' \neq k} D_{k'}$$

has $\int_{[0,1]^d} \mathrm{mr}_i(A) \prod_l x_l^{a_l - 1}(1-x_l)^{b_l - 1} \prod_k D_k^{-p_k - 1}\, dx = 0$ *(Theorem 32.10(a) with the letters* D_k *at the exponents* $p_k + 1$ *and the polynomial* AD*), and*

$$\mathrm{mr}_i(x^\beta) = \sum_\gamma \Big[\big((\beta_i + a_i)D_\gamma - D_\gamma^{(i)}\big)x^{\beta+\gamma} - \big((\beta_i + a_i + b_i)D_\gamma - D_\gamma^{(i)}\big)x^{\beta+\gamma+\mathbf{e}_i}\Big].$$

For a degree bound these vectors form a finite rational relation matrix; a rational solution of its column equation for a numerator writes the numerator as a sum of polynomials $\mathrm{mr}_i(A_i)$*, so that its word vanishes.*

Theorem 36.5 (Words of polynomials on a box are complete)**.** *For* $P \in \mathbb{Q}[x]$ *and rational* $a_i, b_i > 0$ *the words* $\int_{[0,1]^d} P(x) \prod_i x_i^{a_i - 1}(1-x_i)^{b_i - 1}\, dx$ *have the mark relations* $x^{\beta+\mathbf{e}_i} \equiv \frac{\beta_i + a_i}{\beta_i + a_i + b_i}\, x^\beta$*, and every* P *reduces to*

$$\sum_\beta \big([x^\beta]P\big) \prod_i \frac{(a_i)_{\beta_i}}{(a_i + b_i)_{\beta_i}},$$

the expectation of P *under the Dirichlet laws* $\mathrm{Dir}_{(a_i,b_i)}$ *of the pairs* $(x_i, 1-x_i)$*, times the base word, of value* $\prod_i \Gamma_2 \left[\begin{matrix} a_i, & b_i \\ a_i + b_i & \end{matrix}\right] > 0$*. A word vanishes exactly when its reduction is zero. The same holds for every finite tensor product of such words.*

Theorem 36.6 (A finite insertion census separates numerators)**.** *Let* ϕ *be a measure on* $[0,1]^d$ *with an integrable density positive on the interior, and* D *a rational polynomial positive on the closed box. For* $|\beta|, |\gamma| \leq h$ *the matrix* $\mathbf{G}_{\beta\gamma} = \int x^{\beta+\gamma} D^{-1}\, d\phi$ *is positive definite. For* $A = \sum_\beta a_\beta x^\beta$ *of degree at most* h *the insertion values* $v_\beta = \int x^\beta A D^{-1}\, d\phi$ *satisfy* $\mathbf{v} = \mathbf{G}\mathbf{a}$ *and* $\mathrm{adj}(\mathbf{G})\,\mathbf{v} = \det(\mathbf{G})\,\mathbf{a}$*: these* $\binom{d+h}{d}$ *values determine* A*. Equivalently, the Laplace transform* $t \mapsto \int AD^{-1} e^{-\langle t, x\rangle}\, d\phi$ *has all its Taylor coefficients of degree at most* h *zero at* $t = 0$ *only for* $A = 0$*.*

Evaluation is multiplicative because a product of words keeps the regulators of its factors apart; identifying two regulators is a new operation, not a relation. Every relation the theory proves lies in the kernel of evaluation, and completeness asks whether the kernel holds anything more. On a box against positive letters the mark relations form a finite rational matrix, so that vanishing is certified by linear algebra; for polynomials on a box without a letter the relations reduce every insertion to one rational multiple of the base word, and with a letter the values of finitely many monomial insertions, through the positive definite matrix $\mathbf{G}$ of Theorem 36.6, recover the insertion itself.

36.2 Relation blocks

Definition 36.7 (Relation block)**.** A *relation block* is a finite-dimensional rational space V of admitted words with a matrix C whose columns are proved relations in V, each with its derivation. Rational row reduction gives representatives $\mathfrak{w}_0, \ldots, \mathfrak{w}_s$ of $V/\operatorname{im} C$ and for every $\mathfrak{w} \in V$ a certificate $\mathfrak{w} = C\mathbf{v} + \sum_j c_j \mathfrak{w}_j$.

Proposition 36.8 (Separation completes a block)**.** *If* $\mathrm{ev}(\mathfrak{w}_0),\dots,\mathrm{ev}(\mathfrak{w}_s)$ *are linearly independent over* $\mathbb{Q}$*, then* $\ker(\mathrm{ev}\,|_V) = \operatorname{im} C$*.*

Theorem 36.9 (Completeness from separated blocks)**.** *If every finite set of words of a subalgebra* $\mathfrak{W}' \subseteq \mathfrak{W}$ *lies in a relation block whose representatives have values linearly independent over* $\mathbb{Q}$*, then* $\mathfrak{W}'$ *is complete, and each of its words of value zero has a finite derivation, its certificate.*

Proposition 36.10 (Tensoring relation blocks)**.** *For blocks* (V_1, C_1) *and* (V_2, C_2) *the induced relations on* $V_1 \otimes V_2$ *have the matrix* $\left[\, C_1 \otimes I \;\middle|\; I \otimes C_2 \,\right]$*, with quotient* $(V_1/\operatorname{im} C_1) \otimes (V_2/\operatorname{im} C_2)$*, and the values of the representatives are the products of the values (Proposition 36.2).*

Proposition 36.11 (The critical polynomial block)**.** *For an integer* $k \geq 1$ *and* $P \in \mathbb{Q}[x]$ *put* $Q_P(u) = \frac{P(2u)-P(1)}{1-2u}$ *and* $h(P) = \frac{2}{(2k)!}\,\mathbb{E}_{Y\sim\mathbb{P}_{\mathrm{crit}}}\big[Q_P^{(2k-1)}(\frac{1}{2}+iY)\big]$*. Then* $h(P) \in \mathbb{Q}$ *and*

$$\mathrm{crit}_k[P(2u)] = h(P) + (2^{2k+1}-1)\,P(1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

So the values of the rational constants and of the critical-line words of order k *of polynomials factor through the surjective coordinate map* $(c,P) \mapsto \big(c+h(P), (2^{2k+1}-1)P(1)\big) \in \mathbb{Q}^2$ *followed by* $(a,b) \mapsto a+b\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$*, which is injective exactly when* $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \notin \mathbb{Q}$*; moving the cut of the Trailmoment subtracts a rational head.*

A relation block is the finite form of completeness: inside a finite space of words, rational row reduction by the proved relations leaves representatives and certificates, and independence of the values of the representatives over the rationals makes the block complete (Proposition 36.8). Blocks tensor, their quotients tensoring and the values of their representatives multiplying, so a product of blocks is complete as soon as the products of those values are independent over the rationals. The critical-line words of order k of polynomials have a block whose quotient has dimension at most two: every such word is a rational number plus a rational multiple of $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, read off two coordinates of the polynomial, and the values of the two representatives are independent over the rationals exactly when that Trailmoment is irrational, the block being then complete.

36.3 Separators

Theorem 36.12 (A primitive exterior separator)**.** *Let* $v = (v_1,\dots,v_s,1) \in \mathbb{R}^{s+1}$*. For a rational* $s \times (s+1)$ *matrix* A *of rank* s *with rows* r_i*, let* $w \in \mathbb{Q}^{s+1}$ *be its cofactor vector,* $\det[x;A] = \langle x, w\rangle$ *for all* x*, and* $z = Mw$ *its primitive integral multiple,* $M > 0$*. If* D_i *clears* r_i*, then* $M = \prod_i D_i/c$ *with* c *the greatest common divisor of the maximal minors of* $\mathrm{diag}(D_i)A$*. Put* $y_i = \langle r_i, v\rangle$*, let* $\hat{A}_i$ *be the other rows restricted to the first* s *columns, and*

$$\beta(A) = M \sum_i |y_i| \sum_{J\subseteq\{1,\dots,s\},\ |J|=s-1} \big|\det \hat{A}_i[:,J]\big|.$$

If for all large n *finitely many such matrices* $A_1^{(n)},\dots,A_{k_n}^{(n)}$ *have primitive cofactor vectors spanning* $\mathbb{Q}^{s+1}$ *and* $\max_l \beta(A_l^{(n)}) \to 0$*, then* $1, v_1, \dots, v_s$ *are linearly independent over* $\mathbb{Q}$*.*

Corollary 36.13 (Separation with bounded integer entries)**.** *If integer matrices* $A_l^{(n)}$ *with spanning cofactor vectors have entries at most* $H_n \geq 1$ *and* $|\langle r_i, v\rangle| \leq \epsilon_n$ *for every row, with* $H_n^{s-1}\epsilon_n \to 0$*, then* $1, v_1, \dots, v_s$ *are linearly independent over* $\mathbb{Q}$*. In rates: if rational matrices with spanning cofactor vectors have entries at most* $e^{an+o(n)}$*, a common clearing integer at most* $e^{bn+o(n)}$ *and row values at most* $e^{-cn+o(n)}$*, with* $c > sb + (s-1)a$*, the same conclusion holds.*

Corollary 36.14 (Consecutive rows)**.** *Let rows $r_k \in \mathbb{Q}^{s+1}$ be such that the square matrix of the rows $r_k, \dots, r_{k+s}$ is invertible for large k. Then for each such k the $s+1$ matrices omitting one of these rows have spanning primitive cofactor vectors, and Theorem 36.12 applies with these matrices. If the coordinate sequences of the rows satisfy one recurrence of order $s+1$ whose extreme coefficients have no integer zeros from k_0 on, invertibility of that matrix at k_0 gives it for every $k \geq k_0$.*

Theorem 36.15 (Function level: pole pinning)**.** *Let f and g be meromorphic on a connected open subset of $\mathbb{C}$. If f has infinitely many poles at which g is holomorphic, and the values of g at those poles include infinitely many distinct numbers, then no nonzero polynomial $Q \in \mathbb{C}[X, Y]$ has $Q(f, g) \equiv 0$.*

A separator is a family of rational matrices whose rows are nearly orthogonal to the vector of values: a primitive integral multiple of the cofactor vector of each matrix pairs with any integer relation to an integer, and if the rows are small enough against the clearing and the bounds on the entries, no rational relation among the values survives. Consecutive rows of a recurrence give such matrices as soon as one square matrix of $s+1$ consecutive rows is invertible. At the level of functions, a function with infinitely many poles at which a second function takes infinitely many values satisfies no polynomial relation with it.

Chapter 37

Finite algebra for scale reads

A bunch read on its scale against linear letters is a Breta of one bunch at the unit configuration (Proposition 15.1), and a Breta of unit columns on one bunch is, up to a stacked Gamma, such a scale integral. Its values at integer shifts of its index and its letter exponents satisfy two relations, the letter split and the scale exchange, and these relations are complete: every linear relation between the shifted values, with coefficients rational in the index and the letter exponents, is a combination of shifted letter splits and scale exchanges, and the values modulo the relations form a space whose dimension is the number of letters. The same holds at every corner, with the coefficients ξ_ℓ of the letters $x+\xi_\ell$ as the variables. For polynomial letters on one scale the same two relations reduce every insertion to one read per distinct root of the product of the letters, counted without finding a root; at a numerical corner three Laurent coefficients per read suffice, every integer ray has a recurrence with a certificate, and the jets in the coefficients of the letters are reads of rational insertions.

37.1 Complete relations for linear letters and unit columns

Definition 37.1 (Shifts, relations between shifts, the relation space)**.** Let $J=\{0,\dots,L\}$ carry distinct coefficients $\xi_\ell \in \mathbb{C}\setminus(-\infty,0]$, with the unit configuration I_J, and let $\mathbb{K}=\mathbb{C}(\kappa,w_0,\dots,w_L)$ be the field of rational functions of an index κ and letter exponents $\mathbf{w}$. Let $\mathcal{M}$ be the vector space over $\mathbb{K}$ with one basis element $[a,\mathbf{b}]$ for each *shift* $(a,\mathbf{b})\in\mathbb{Z}\times\mathbb{Z}^J$. The *value* of $[a,\mathbf{b}]$ is the scale integral at the shifted index and letter exponents,

$$\int_0^\infty x^{\kappa+a-1}\prod_{\ell\in J}(x+\xi_\ell)^{-w_\ell-b_\ell}\,dx \\ = \Gamma_{\,2}^{\,|J|}\begin{bmatrix}\kappa+a,\ |\mathbf{w}|+|\mathbf{b}|-\kappa-a\\ \mathbf{w}+\mathbf{b}\end{bmatrix}\mathsf{B}^{\{J\}}_{\mathbf{w}+\mathbf{b},\,|\mathbf{w}|+|\mathbf{b}|-\kappa-a;\,I_J}(\boldsymbol{\xi})$$

which converges on $0<\operatorname{Re}\kappa+a<\operatorname{Re}|\mathbf{w}|+|\mathbf{b}|$, the second form holding where moreover $\operatorname{Re}(w_\ell+b_\ell)>0$ for every ℓ (Proposition 15.1); finitely many shifts converge together on a nonempty convex open set. An element $\sum_i c_i[a_i,\mathbf{b}_i]$ of $\mathcal{M}$ is a *relation between shifts* when the combination of the values of the $[a_i,\mathbf{b}_i]$ with the coefficients $c_i(\kappa,\mathbf{w})$ vanishes at every point where all its terms converge and every c_i is defined. At the shift $(a,\mathbf{b})$ the letter split of the letter $x+\xi_\ell$ and the scale exchange are

$$\operatorname{split}_\ell(a,\mathbf{b})=[a,\mathbf{b}]-[a+1,\mathbf{b}+\mathbf{e}_\ell]-\xi_\ell\,[a,\mathbf{b}+\mathbf{e}_\ell],$$

$$\operatorname{exch}(a,\mathbf{b}) = (\kappa + a)\,[a,\mathbf{b}] - \sum_{\ell\in J}(w_\ell + b_\ell)\,[a+1,\mathbf{b}+\mathbf{e}_\ell],$$

the letter split and the scale exchange of Proposition 3.6 for the letters $x+\xi_\ell$. The *relation space* $\mathcal{L}_J \subseteq \mathcal{M}$ is the span over $\mathbb{K}$ of all letter splits and scale exchanges, and the *dimension* of the shifts is $\dim_{\mathbb{K}} \mathcal{M}/\mathcal{L}_J$; on one scale these play the part of the relation space and the dimension of a ray (Definition 35.4).

Theorem 37.2 (The relations of a scale integral are complete)**.** *Let the coefficients range over distinct points of* $\mathbb{C}\setminus(-\infty,0]$.

(i) Every element of $\mathcal{L}_J$ *is a relation between shifts.*

(ii) $\mathcal{M} = \mathcal{L}_J + \sum_{\ell\in J}\mathbb{K}\,[0,\mathbf{e}_\ell]$*: the dimension is at most* $|J|$.

(iii) The values of the $|J|$ *shifts* $[0,\mathbf{e}_\ell]$,

$$\Gamma_{\substack{2\\|J|}}\begin{bmatrix}\kappa,\ |\mathbf{w}|+1-\kappa\\ \mathbf{w}+\mathbf{e}_\ell\end{bmatrix}\mathsf{B}^{\{J\}}_{\mathbf{w}+\mathbf{e}_\ell,\,|\mathbf{w}|+1-\kappa;\,I_J}(\boldsymbol{\xi}),\qquad \ell\in J,$$

are linearly independent over $\mathbb{K}$*: a combination of them with coefficients in* $\mathbb{K}$ *that vanishes on a nonempty open subset of* $\{0<\operatorname{Re}\kappa<\operatorname{Re}|\mathbf{w}|+1\}$ *has every coefficient* 0.

(iv) The dimension is $|J|$*, and every relation between shifts lies in* $\mathcal{L}_J$*: it is a combination over* $\mathbb{K}$ *of shifted letter splits and scale exchanges.*

(v) Coefficients as variables. *Let the coefficients be rational functions of parameters* $\mathbf{u}=(u_1,\dots,u_d)$*, distinct and not* 0 *as rational functions, defined with distinct values in* $\mathbb{C}\setminus(-\infty,0]$ *on a nonempty open subset of* $\mathbb{R}^d$ *or of* $\mathbb{C}^d$*. Then (i), (ii) and (iv) hold with the field* $\mathbb{C}(\mathbf{u},\kappa,\mathbf{w})$ *in the place of* $\mathbb{K}$*, a relation between shifts being an element whose combination of values vanishes at every* $\mathbf{u}$ *of that set and every* $(\kappa,\mathbf{w})$ *where all its terms converge and every coefficient is defined.*

The letter splits are the partial fractions of the integrand, and the scale exchange is one integration by parts along the scale. Part (ii) reduces every shift to the $|J|$ values with one letter exponent raised by one, by these two relations alone, and part (iii) says that those values are linearly independent over $\mathbb{K}$. So a relation between shifts that is not a combination of shifted letter splits and scale exchanges does not exist, and the number of independent values is the number of letters; for one letter the dimension is one: every shift of the Beta corner of a linear letter is a rational multiple of one value. By Theorem 15.2 the scale integral of a family at a bunch is such an integral, with the points of the joint spectrum as its coefficients, so its relations in its index and its letter exponents are complete; relations between scale integrals of a family at shifted masses and exponents, where the letter exponents are tied to the exponents of the family, are a further problem.

What decides the relations at a corner, where the index and the letter exponents are numbers and the coefficients are the variables, is the behaviour of the values as one coefficient tends to 0: its letter becomes a power of the scale and joins the index.

Lemma 37.3 (A coefficient tends to 0)**.** *Let* $L\geq 1$*, let the coefficients be positive, and let the index and the letter exponents be numbers with* $\operatorname{Re} w_\ell > 0$ *for every* ℓ *and* $0\leq\operatorname{Re}\kappa'<1$*, where* $\kappa' = \kappa - w_0$*. In this statement* $J' = J\setminus\{0\}$*,* $\mathbf{w}' = \mathbf{w}|_{J'}$*,* $\boldsymbol{\xi}' = \boldsymbol{\xi}|_{J'}$*,* f_ℓ *is the value of* $[0,\mathbf{e}_\ell]$ *(Definition 37.1), and for letter exponents* $\mathbf{v}$ *on* J' *with* $\operatorname{Re}|\mathbf{v}|>0$ *and* $h_{\mathbf{v}}(x) = \prod_{j\in J'}(x+\xi_j)^{-v_j}$*,*

$$S'(z;\mathbf{v}) = \frac{h_{\mathbf{v}}(0)}{z} + \int_0^1 x^{z-1}\big(h_{\mathbf{v}}(x) - h_{\mathbf{v}}(0)\big)\,dx + \int_1^\infty x^{z-1}h_{\mathbf{v}}(x)\,dx$$

for $-1 < \operatorname{Re} z < \operatorname{Re}|\mathbf{v}|$, $z \neq 0$. *For* $\operatorname{Re} z > 0$ *and* $\operatorname{Re} v_j > 0$ *for every* j, $S'(z;\mathbf{v}) = \Gamma_{\substack{2\\|J'|}}\begin{bmatrix} z,\ |\mathbf{v}|-z \\ \mathbf{v}\end{bmatrix} \mathsf{B}^{\{J'\}}_{\mathbf{v},\,|\mathbf{v}|-z;\,I_{J'}}(\boldsymbol{\xi}')$, *the scale integral with the coefficients* $\boldsymbol{\xi}'$, *so* S' *continues it in its index across the wall* $\operatorname{Re} z = 0$, *with a simple pole at* $z = 0$ *of residue* $h_{\mathbf{v}}(0)$*; its constant term* $\operatorname{CT}_{z=0} S'(z;\mathbf{v})$ *is the sum of the two integrals at* $z = 0$, *and its values off the pole, at complex* z *and complex letter exponents, are the finite parts* $\operatorname{Reg}\int_0^\infty x^{z-1}h_{\mathbf{v}}\,dx$, *its constant term at* $z = 0$ *being* $\operatorname{Reg}\int_0^\infty x^{-1}h_{\mathbf{v}}\,dx$ *(Theorem 1.6). Fix* $\boldsymbol{\xi}'$ *and let* $\xi_0 \to 0^+$. *Then, with* $j \in J'$ *in (ii)–(v):*

(i) $\xi_0^{1-\kappa'} f_0 \to \Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa,\ 1-\kappa' \\ w_0+1\end{bmatrix} h_{\mathbf{w}'}(0)$, *which is not* 0.

(ii) If $\operatorname{Re}\kappa' > 0$, *then* $f_j \to S'(\kappa';\mathbf{w}'+\mathbf{e}_j)$.

(iii) If $\operatorname{Re}\kappa' = 0$ *and* $\kappa' \neq 0$, *then*

$$f_j = \Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa,\ -\kappa' \\ w_0\end{bmatrix} \frac{h_{\mathbf{w}'}(0)}{\xi_j}\,\xi_0^{\kappa'} + S'(\kappa';\mathbf{w}'+\mathbf{e}_j) + o(1).$$

(iv) If $\kappa' = 0$, *then*

$$f_j = \frac{h_{\mathbf{w}'}(0)}{\xi_j}\Big(\log\frac{1}{\xi_0} + (\log\Gamma)'(1) - (\log\Gamma)'(\kappa)\Big) + \operatorname*{CT}_{z=0} S'(z;\mathbf{w}'+\mathbf{e}_j) + o(1).$$

(v) The relations across the wall. *For* $-1 < \operatorname{Re} z \le 0$, $z \neq 0$,

$$S'(z;\mathbf{v}) = S'(z+1;\mathbf{v}+\mathbf{e}_j) + \xi_j\, S'(z;\mathbf{v}+\mathbf{e}_j), \qquad z\,S'(z;\mathbf{v}) = \sum_{i\in J'} v_i\, S'(z+1;\mathbf{v}+\mathbf{e}_i),$$

and $\operatorname{CT}_{z=0} S'(z;\mathbf{v}) = S'(1;\mathbf{v}+\mathbf{e}_j) + \xi_j\ \operatorname{CT}_{z=0} S'(z;\mathbf{v}+\mathbf{e}_j)$.

As the coefficient ξ_0 tends to 0 its letter $x+\xi_0$ becomes the power x. The value with the exponent of that letter raised grows like $\xi_0^{\kappa'-1}$, and when $\operatorname{Re}\kappa' > 0$ the others tend to the scale integrals of the remaining coefficients at the index $\kappa' = \kappa - w_0$, the index left when the letter exponent w_0 has joined it. When $\operatorname{Re}\kappa' = 0$ that index lies on the wall of the remaining integral, and a second term of the same size stands next to the integral continued across its wall: the oscillating power $\xi_0^{\kappa'}$, and at $\kappa' = 0$ a logarithm, which grows. The letter split and the scale exchange hold across the wall, and at the pole $z = 0$ the scale exchange keeps only its residue, the first identity of Theorem 37.4(ii) for the remaining coefficients. These are the first terms of the closing between two separated groups (Corollary 18.20), the letter of ξ_0 being the lower group: one lattice carries the powers $\xi_0^{\kappa'+n}$, with the wall residues of the other letters as coefficients, and the other carries the integer powers ξ_0^n, with the scale integral of the other coefficients continued in its index; at $\kappa' = 0$ the two lattices meet, and what their poles leave is the logarithm of (iv).

Theorem 37.4 (The relations at a corner)**.** *Let the index* κ *and the letter exponents* $\mathbf{w}$ *be numbers with* $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|+1$, *so that the values of the* $|J|$ *shifts* $[0,\mathbf{e}_\ell]$ *converge, and let the coefficients be distinct points of* $\mathbb{C}\setminus(-\infty,0]$. *A shift* $[a,\mathbf{b}]$ converges at the corner *when* $0 < \operatorname{Re}\kappa + a < \operatorname{Re}|\mathbf{w}| + |\mathbf{b}|$, *and a letter split or a scale exchange at the shift* $(a,\mathbf{b})$, *with the index and the letter exponents at their numbers, converges at the corner when* $[a,\mathbf{b}]$ *does; all its terms then converge. A* linear relation *between shifts that converge at the corner is a combination* $\sum_i r_i[a_i,\mathbf{b}_i]$ *of such shifts, with coefficients* r_i *rational in* $\boldsymbol{\xi}$, *whose value vanishes for* $\boldsymbol{\xi}$ *in a nonempty open set; it is a combination of letter splits and scale exchanges when it is one as a formal combination of shifts. The values are holomorphic in* $\boldsymbol{\xi}$ *on* $(\mathbb{C}\setminus(-\infty,0])^J$, *coincident coefficients included, and functions of* $\boldsymbol{\xi}$ *are* independent over $\boldsymbol{\xi}$ *when no combination of them with coefficients polynomial in* $\boldsymbol{\xi}$, *not all* 0, *vanishes at all* $\boldsymbol{\xi} > 0$, *equivalently for* $\boldsymbol{\xi}$ *in a nonempty open set.*

(i) Reduction. *If no letter exponent lies in* $\{-1,-2,\dots\}$*, the value of every shift that converges at the corner is a combination of the values of the* $[0,\mathbf{e}_\ell]$ *with coefficients rational in* $\boldsymbol{\xi}$*; the combination follows from letter splits and scale exchanges that converge at the corner, taken at shifts* $(a',\mathbf{b}')$ *with* $b'_\ell \geq \min(b_\ell,0)$ *for every* ℓ*.*

(ii) The two walls. *If* $\operatorname{Re}|\mathbf{w}|>0$*, the values of the* $[0,\mathbf{e}_\ell]$ *at the index* 1 *and at the index* $|\mathbf{w}|$ *satisfy*

$$\sum_{\ell\in J} w_\ell \int_0^\infty \prod_{\ell'\in J} (x+\xi_{\ell'})^{-w_{\ell'}-\mathbf{1}[\ell'=\ell]}\,dx = \prod_{\ell\in J}\xi_\ell^{-w_\ell},$$
$$\sum_{\ell\in J} w_\ell\,\xi_\ell \int_0^\infty x^{|\mathbf{w}|-1}\prod_{\ell'\in J} (x+\xi_{\ell'})^{-w_{\ell'}-\mathbf{1}[\ell'=\ell]}\,dx = 1,$$

the integrals being, where every $\operatorname{Re} w_\ell>0$*, the stacked Gammas times the Bretas of Definition 37.1.*

(iii) Independence. *The values of the* $|J|$ *shifts* $[0,\mathbf{e}_\ell]$ *are independent over* $\boldsymbol{\xi}$ *if and only if no letter exponent lies in* $\{-1,-2,\dots\}$*.*

(iv) Completeness. *Whatever the letter exponents, every linear relation between shifts that converge at the corner is a combination, with coefficients rational in* $\boldsymbol{\xi}$*, of letter splits and scale exchanges that converge at the corner, and over the rational functions of* $\boldsymbol{\xi}$ *the values of these shifts span a space of dimension* $|J|$*: at every corner the dimension is* $|J|$*.*

(v) A coefficient equal to 1. *Parts (i), (iii) and (iv) hold as well when one coefficient is the constant* 1 *and the others are the variables.*

At a corner the relations are the same, and when no letter exponent lies in $\{-1,-2,\dots\}$ they reduce every shift that converges to the $|J|$ values with one letter exponent raised without leaving the corner: every relation of the reduction has all its terms convergent, and the leading coefficients $\kappa-r$, $r\geq 1$, and $\kappa+j-|\mathbf{w}|$, $j\geq 0$, by which the descents at the origin and at infinity divide vanish only at shifts that have left the domain. The two identities of (ii) are the relations at the two walls of the domain: the first is the scale exchange at the shift $(-1,\mathbf{0})$ at the index 1, whose member on the wall is replaced by its limit $\prod_\ell \xi_\ell^{-w_\ell}$, and the second is the scale exchange less the letter splits multiplied by their letter exponents at the shift $(0,\mathbf{0})$, at the index $|\mathbf{w}|$, where $|\mathbf{w}|-\kappa$ times the value of $[0,\mathbf{0}]$ tends to 1; at a wall one combination of the $|J|$ values is a monomial in $\boldsymbol{\xi}$. A letter exponent in $\{-1,-2,\dots\}$ is not on a wall: its letter has become a polynomial factor of the integrand, the $|J|$ values with one letter exponent raised are dependent and no longer span the shifts that converge; the corner with that letter exponent moved up to 0 has the same shifts, and its $|J|$ values with one letter exponent raised span them. Independence is read from the coefficients one at a time: as ξ_0 tends to 0 the value with the letter exponent w_0 raised grows faster than the others (Lemma 37.3), so in a relation the polynomial multiplying it vanishes at $\xi_0=0$, and what is left is a relation between scale integrals of the remaining coefficients, at the index $\kappa-w_0$, or on their wall, where the relations across the wall carry it to the next index.

Corollary 37.5 (Unit columns on one bunch)**.** *Let* B *be a bunch with the unit configuration* I_B *and distinct coefficients* $\xi_e>0$*. By Proposition 15.1 with* $J=B$*,* $\mathbf{w}=\mathbf{m}$ *and* $\kappa=m_B-\nu$*, the value* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$*,* $\mathbf{a}\in\mathbb{Z}^B$*,* $n\in\mathbb{Z}$*, is the value of the shift* $[|\mathbf{a}|-n,\mathbf{a}]$ *of Definition 37.1 divided by* $\Gamma_{\substack{2\\|B|}}\begin{bmatrix} m_B-\nu+|\mathbf{a}|-n,\ \nu+n\\ \mathbf{m}+\mathbf{a}\end{bmatrix}$*, wherever that proposition applies; this stacked Gamma is the*

one at $(\mathbf{m},\nu)$ *times a rational function of* $(\mathbf{m},\nu)$*. Each relation below is read where all its Bretas converge.*

(i) The mark relation. *For each* $e\in B$*,*

$$m_e\,\mathsf{B}^{\{B\}}_{\mathbf{m},\,\nu;\,I_B}(\boldsymbol{\xi}) = (m_B-\nu)\,\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu;\,I_B}(\boldsymbol{\xi}) + \nu\,\xi_e\,\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu+1;\,I_B}(\boldsymbol{\xi}):$$

the sum over $f\in B$ *of the shape exchanges of Theorem 3.5(iv) for the pairs* (e,f)*, reduced by the shape split and the scale exchange; it is the letter split of the letter* $x+\xi_e$*.*

(ii) The shape split, the scale exchange and the shape exchanges of Theorem 3.5, at all integer shifts of $(\mathbf{m},\nu)$*, span over* $\mathbb{C}(\mathbf{m},\nu)$ *every linear relation with coefficients in* $\mathbb{C}(\mathbf{m},\nu)$ *between the values* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$*, and the quotient of the space of shifts by these relations has dimension* $|B|$*. The same holds over* $\mathbb{C}(\boldsymbol{\xi},\mathbf{m},\nu)$ *for relations whose coefficients are rational in* $\boldsymbol{\xi}$ *as well and which hold at all distinct* $\boldsymbol{\xi}>0$ *at which their coefficients are defined.*

(iii) At a corner. *Let* $\mathbf{m}$ *and* ν *be numbers with* $\operatorname{Re} m_e>0$ *for every* e *and* $\operatorname{Re}\nu>0$*, and call a value* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$ interior *when* $\operatorname{Re}(m_e+a_e)>0$ *for every* e *and* $\operatorname{Re}\nu+n>0$*; the Breta converges there. Over the rational functions of* $\boldsymbol{\xi}$ *the interior values span a space of dimension* $|B|$*, and every linear relation between interior values with coefficients rational in* $\boldsymbol{\xi}$ *is a combination, with such coefficients, of shape splits, scale exchanges and shape exchanges all of whose terms are interior. If* $\operatorname{Re}\nu<\operatorname{Re} m_B$*, the* $|B|$ *values* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu+1;\,I_B}(\boldsymbol{\xi})$*,* $e\in B$*, are a basis of that space.*

Completeness holds for the scale integral of one bunch against linear letters and for unit columns on one bunch, which are the same integrals: the coefficients are the same, the shape index is the letter exponents, and the index of the scale integral is the amount $m_B-\nu$ by which the scale index falls short of the mass m_B. It holds at a free index and free letter exponents, and at every corner with the coefficients as the variables, one of them possibly the constant 1. For a Breta with a kernel, with letters of more than two patterns or of levels other than 0 and 1, or with several bunches, the relations of Theorem 3.5 hold, and their completeness is a further problem.

37.2 Polynomial letters on one scale

A word on one scale with polynomial letters, such as the scale integral of a bunch at fixed shapes (Theorems 15.2 and 18.28), has the same two relations, the letter splits becoming partial fractions over the distinct factors of the letters, and the same finiteness: the retained data are one read per distinct root of the product of the letters, counted without finding a root, and at a numerical corner three Laurent coefficients per read. Every rational insertion, every integer shift of the index and the letter exponents, every derivative in the coefficients of the letters and every integer ray is then finite algebra, with a certificate.

Definition 37.6 (Insertions and reads on one scale)**.** Let $\mathbb{F}\subseteq\mathbb{C}$ be a field, the field of the letters, and let $P_1,\dots,P_N\in\mathbb{F}[x]$ be nonconstant letters without roots on $[0,\infty)$, a constant letter being moved into a scalar; put $\mathbf{P}=(P_1,\dots,P_N)$. For an index κ and letter exponents $w_1,\dots,w_N$ in a field $\mathbb{F}'\supseteq\mathbb{F}(\kappa,\mathbf{w})$ embedded in $\mathbb{C}$, the insertions, the regular functions of the letter complement as in Definition 35.1, are the elements of the ring $\mathfrak{A}=\mathbb{F}'[x,x^{-1},P_1^{-1},\dots,P_N^{-1}]$, and the *read* of an insertion R is

$$\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}} = \int_0^\infty x^{\kappa-1}\prod_j P_j(x)^{-w_j}\,R(x)\,dx$$

where it converges absolutely, each power $P_j^{-w_j}$ being taken on $[0,\infty)$ by continuity from its principal value at 0, and continued in the index and the letter exponents elsewhere. The shift of the index by a and of the letter exponents by b_j, integers, is the insertion $x^a\prod_j P_j^{-b_j}$, and a derivative in the coefficients of a letter is an insertion. A squarefree decomposition with gcd splitting gives pairwise coprime squarefree $q_1,\dots,q_M\in\mathbb{F}[x]$, none vanishing at 0, with $P_j=c_j\prod_k q_k^{m_{jk}}$, without roots and without a field extension; $Q=q_1\cdots q_M$ is the monic radical of $P_1\cdots P_N$, of degree $D>0$. The *scale exchange* on insertions is

$$\eth g = xg' + \Big(\kappa - \sum_j w_j \frac{xP_j'}{P_j}\Big)g, \qquad \text{so that} \qquad \Big(x^\kappa \prod_j P_j^{-w_j}\, g\Big)' = x^{\kappa-1}\prod_j P_j^{-w_j}\,\eth g.$$

For the linear letters $x+\xi_\ell$ with distinct $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, the reads of the shifts are the values of Definition 37.1, $D=|J|$, and $\eth$ applied to $x^a\prod_\ell(x+\xi_\ell)^{-b_\ell}$ is the scale exchange $\mathrm{exch}(a,\mathbf{b})$ read under the integral.

Theorem 37.7 (Reduction by the scale exchange)**.** *Let the numbers*

$$\kappa - r \quad (r\ge 1), \qquad \kappa + a - \sum_j w_j \deg P_j \quad (a\ge 0), \qquad \sum_j m_{jk}w_j + h \quad (h\ge 1,\ 1\le k\le M)$$

be nonzero in $\mathbb{F}'$. Every $R\in\mathfrak{A}$ has a unique decomposition

$$R = \eth g + \sum_{i=0}^{D-1} c_i \frac{x^i}{Q}, \qquad g\in\mathfrak{A},\quad c_i\in\mathbb{F}',$$

obtained by polynomial arithmetic, partial fractions and the extended Euclidean algorithm; so $\dim_{\mathbb{F}'}\mathfrak{A}/\eth\mathfrak{A}=D$, and on the domain where the reads converge $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}=\sum_i c_i\,\langle x^i/Q\rangle_{\kappa,\mathbf{w};\mathbf{P}}$. For the linear letters $x+\xi_\ell$ with distinct $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, $D=|J|$ and this is Theorem 37.2(ii) over $\mathbb{K}$, the x^i/Q spanning the same space as the $(x+\xi_\ell)^{-1}$.

The dimension is that of the quotient by the scale exchange, the dimension of the ray on one scale (Theorem 35.19); it does not assert that the reads of the D insertions x^i/Q are independent once parameters are tied, and symmetries of the letters can relate them further. For linear letters Theorem 37.2(iii) gives the independence. The three conditions are the leading coefficients of the reduction at the origin, at infinity and at a repeated factor. Exact computation needs exact arithmetic with a zero test in $\mathbb{F}'$, numerical exponents being taken in an effective extension; the analytic statements hold at all complex exponents.

Definition 37.8 (Deformation; masters)**.** At numerical exponents $(\kappa,\mathbf{w})$ fix a direction $(\lambda_0;\boldsymbol{\lambda})\in\mathbb{R}\times\mathbb{R}^N$ of the index and the letter exponents that is *admissible*: $\lambda_0>0$, $\sum_j\lambda_j\deg P_j>\lambda_0$ and $\sum_j m_{jk}\lambda_j\neq 0$ for every k. The *deformation* moves the data along it, to $(\kappa+\lambda_0\varepsilon,\ \mathbf{w}+\varepsilon\boldsymbol{\lambda})$; the three conditions of Theorem 37.7 hold over $\mathbb{F}'(\varepsilon)$, and for an insertion R the coefficients $c_i(\varepsilon)$ are those of its decomposition there. The *masters* are the deformed reads

$$H_i(\varepsilon) = \big\langle x^i/Q\big\rangle_{\kappa+\lambda_0\varepsilon,\ \mathbf{w}+\varepsilon\boldsymbol{\lambda};\ \mathbf{P}}, \qquad 0\le i<D.$$

For large real ε every finite set of deformed reads converges, the exponent at the origin growing with the slope λ_0 and the decay at infinity with the slope $\sum_j\lambda_j\deg P_j-\lambda_0>0$, so the reductions hold there and then as identities of meromorphic functions. The masters are finite combinations of reads of shifts of the original letters: $1/Q=\big(P_1\cdots P_N/Q\big)\prod_j P_j^{-1}$ with $P_1\cdots P_N/Q$ a polynomial, so each term of $x^iP_1\cdots P_N/Q$ shifts the index by an integer and raises every letter exponent by one. The admissible directions form a nonempty open cone; the direction $(\frac12;\mathbf{1})$ is admissible.

Lemma 37.9 (Simple poles at a corner)**.** *For every insertion R independent of ε, or holomorphic in ε, the coefficients $c_i(\varepsilon)$ have at most a simple pole at $\varepsilon = 0$, and every master $H_i(\varepsilon)$ is meromorphic near 0 with at most a simple pole.*

Theorem 37.10 (Finitely many numbers at every corner)**.** *Along an admissible direction (Definition 37.8), for an insertion R write $H_i(\varepsilon) = h_{i,-1}\varepsilon^{-1} + h_{i,0} + h_{i,1}\varepsilon + O(\varepsilon^2)$ and $c_i(\varepsilon) = c_{i,-1}\varepsilon^{-1} + c_{i,0} + c_{i,1}\varepsilon + O(\varepsilon^2)$. The $3D$* corner numbers *$(h_{i,-1}, h_{i,0}, h_{i,1})_{i<D}$ determine every absolutely convergent read:*

$$\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}} = \sum_{i=0}^{D-1} \big(c_{i,-1}h_{i,1} + c_{i,0}h_{i,0} + c_{i,1}h_{i,-1}\big).$$

They depend on the letters, the exponents and the direction only, not on the insertion or the size of the shift. The reads of $R\left(\lambda_0 \log x - \sum_j \lambda_j \log P_j\right)^b$, the derivatives ∂_ε^b at $\varepsilon = 0$ of the deformed reads $\langle R\rangle_{\kappa+\lambda_0\varepsilon,\,\mathbf{w}+\varepsilon\boldsymbol{\lambda};\,\mathbf{P}}$, need the masters through the order $b+1$: $(b+3)D$ numbers. When $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ converges absolutely, a read of $R\,(\log x)^{k_0}\prod_j(\log P_j)^{k_j}$, $b = k_0 + \sum_j k_j$, is a rational combination of the reads of $R\left(\lambda_0 \log x - \sum_j \lambda_j \log P_j\right)^b$ along finitely many admissible directions, each with its own corner numbers.

The residues are finite extractions of endpoint coefficients, the other numbers finite parts and derivatives in the deformation. The D values of the masters suffice when every pivot of the reduction is regular and the masters are holomorphic at the corner, as for linear letters at a corner with no letter exponent in $\{-1, -2, \dots\}$ and convergent masters (Theorem 37.4(i)); regular pivots alone do not suffice, and at an arbitrary corner the corner numbers are what is retained.

Theorem 37.11 (Integer rays have certified recurrences)**.** *Fix integers $a, b_1, \dots, b_N$, and let $\eth_n$ be the scale exchange at the deformed data $\left(\kappa + an + \frac{\varepsilon}{2},\ \mathbf{w} + n\mathbf{b} + \varepsilon\mathbf{1}\right)$. A finite algorithm gives polynomials $p_0(n,\varepsilon), \dots, p_D(n,\varepsilon)$ and a certificate g, rational in x, n and ε, with*

$$\eth_n g = \sum_{k=0}^{D} p_k(n,\varepsilon)\, x^{ak} \prod_j P_j^{-b_j k},$$

the polynomials $p_k(n) = p_k(n,0)$ not all zero; then

$$\sum_{k=0}^{D} p_k(n)\, \langle 1\rangle_{\kappa+a(n+k),\ \mathbf{w}+(n+k)\mathbf{b};\ \mathbf{P}} = 0$$

at every integer n at which the displayed reads converge.

The recurrence need not have the least order, and its leading coefficient may vanish at finitely many integers, read off that polynomial; there further values are given or another dependence is formed. On one scale this supplies, for every ray, the certificate that a recurrence asks for: an insertion g whose scale exchange is the recurrence.

Theorem 37.12 (Coefficient jets on one scale)**.** *Let $P_j(x;u) = P_j(x) + \sum_{a=1}^{p} u_a P_{ja}(x)$ with $\deg P_{ja} \le \deg P_j$, choose bounds $M_{ja} \ge \sup_{x>0} |P_{ja}/P_j|$, finite since the quotient is continuous on $[0,\infty]$, and let $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ converge absolutely. On the polydisc where $\sum_a |u_a| M_{ja} < 1$ for every*

j the read $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}(\,\cdot\,;u)}$, the powers of the changed letters being continued in u from $u=0$, is holomorphic and

$$\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}(\,\cdot\,;u)} = \sum_{(n_{ja})\geq 0} \prod_{j} \frac{(-1)^{|n_j|}(w_j)_{|n_j|}}{\prod_a n_{ja}!} \prod_a u_a^{\sum_j n_{ja}} \Big\langle \prod_{j,a} \Big(\frac{P_{ja}}{P_j}\Big)^{n_{ja}} \Big\rangle_{\kappa,\mathbf{w};\mathbf{P}},$$

$|n_j| = \sum_a n_{ja}$, absolutely; every coefficient is the read of a rational insertion, given by the corner numbers of Theorem 37.10.

Splitting of repeated roots under the change costs nothing, the expansion being made against the original letters; a change that raises a degree is outside the statement. For coefficients that are symbolic parameters with fixed gcds and multiplicities, differentiating the masters gives rational insertions, so Theorem 37.7 gives matrices $\partial_{u_a} H = \mathbf{A}_a H$, with certificates, for the vector H of the masters, whose mixed derivatives agree because they agree before the reduction; at a corner where a leading coefficient of the reduction vanishes the corner numbers replace them. The structural data reused are the squarefree base, Q and the inverses of xq_k' modulo q_k; the analytic data are the D masters or the $3D$ corner numbers. A rational insertion with polynomial part of degree A_+, a pole of order A_- at the origin and of order h_k at q_k takes at most $(A_+ + 1) + A_- + \sum_k \max(h_k - 1, 0)$ elimination rounds, counted in operations, the growth of exact coefficients charged separately; a repeated letter adds m_{jk} times its letter exponent to the exponent of each squarefree factor q_k and leaves Q and D unchanged. A recurrence in a count of repetitions needs the integer shifts of Theorem 37.11, and a multiplicity written in binary, which is what the price of Chapter 31 counts, does not by itself make the evaluation logarithmic in it.

Part VII

The family as one Laplace transform and as inference

Chapter 38

The family as one Laplace transform

On its balance wall a family is one Laplace transform. In the pair coordinates of its data and the signed coordinates of the shape, its period is the Laplace transform, over the tilts, of the even and odd statistics against the offsets (Definition 38.1; for the interest, Theorem 8.2); the Breta shape laws at the points of the balance wall form an exponential family in these coordinates, and every ratio of periods is a moment generating function of the statistics. On the sector of every order the exponent of a question is minus twice its wall at the tilt, the walls of the tails weighted by the differences of the tilt along the order, less the logarithms of the sector letters, which are bounded. The Laplace transform is read through that one reading of the walls: where the exponent is largest and how it falls, how the period behaves at low and at high strength, how far a question can lean, how fast the tails of the statistics fall, why integrating over the simplex is integrating against one more table, and what a bunch all of whose subsets are patterns reads of its room: the interaction of the bunch, the total over the bunch's scales of the joint cumulants of the use of its marks under the basis and cobasis laws, weighted by the exponents; for a regression design the interaction of a split of the prior marks into two blocks is minus the logarithm of the generalized variance inflation factor.

38.1 Questions and the law of the tilt

Definition 38.1 (Tilts; questions)**.** The space of *tilts* is $\mathcal{T} = \mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, with the Lebesgue measure dz of a chart $z_g = 0$, which does not depend on g. The *tilt* of a shape $u \in \Delta_{\mathcal{E}}$ is the image in $\mathcal{T}$ of its signed coordinates $\frac{1}{2}\log u$, and a tilt is read at its representative in $\mathbf{1}^{\perp}$ when an odd statistic is applied to it, since $O_j(z + c\mathbf{1}) = O_j(z) + 2c(2r_j - n)$ (Theorem 7.13(iii)). A *question* to a family F is a point of its balance wall with real data, written in its pair coordinates $(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta})$: the data $(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \big(\mathbf{m}_F(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta});\, \boldsymbol{\sigma} + \boldsymbol{\delta},\, \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$ (Definition 8.1). In the coordinates $z = \frac{1}{2}\log u$ of Theorem 8.10,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-1} \int_{\mathcal{T}} \exp\Big(2\langle \boldsymbol{\mu}_{\perp}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big)\, dz.$$

When every proper wall is positive, the tilt Z of $U \sim \mathrm{Breta}^{\circ}(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ has the density

$$2^{n-1} \exp\Big(2\langle \boldsymbol{\mu}_{\perp}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big) \Big/ P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$$

with respect to dz: this is the *law of the question*. A direction $v = (\dot{\mathbf{m}}; \dot{\mathbf{s}}, \dot{\mathbf{t}})$ of the data is *balanced* when $\dot{m}_{\mathcal{E}} = \sum_j (\dot{s}_j r_j + \dot{t}_j r_j^*)$; its pair coordinates $(c; \mathbf{a}, \mathbf{b})$ have $\dot{\mathbf{m}} = \mathbf{m}_F(c; \mathbf{a}, \mathbf{b})$, $\dot{\mathbf{s}} = \mathbf{a} + \mathbf{b}$ and

$\dot{\mathbf{t}} = \mathbf{a} - \mathbf{b}$, and D_v is the derivative along v. The derivatives ∂_{σ_j}, ∂_{δ_j} and $\nabla_{\boldsymbol{\mu}_\perp}$ of a function of the question are those along the balanced directions with the pair coordinates $(0; \mathbf{e}_j, \mathbf{0})$, $(0; \mathbf{0}, \mathbf{e}_j)$ and $(\mathbf{e}_e - \frac{1}{n}\mathbf{1}; \mathbf{0}, \mathbf{0})$, that is $(\mathbf{1}; \mathbf{e}_j, \mathbf{e}_j)$, $\big(\frac{2r_j - n}{n}\mathbf{1}; \mathbf{e}_j, -\mathbf{e}_j\big)$ and $\big(\mathbf{e}_e - \frac{1}{n}\mathbf{1}; \mathbf{0}, \mathbf{0}\big)$. The centre of a question is the question $(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$ with the same strengths, the data of the family at the centre (Definition 9.1), of pair coordinates $(0; \boldsymbol{\sigma}, \mathbf{0})$; on a connected family its law is the law of the centre $\mathbb{P}_{\mathrm{c}}$ of Definition 16.8 read on the tilts.

Corollary 38.2 (The Laplace transform of the tilt and the statistics)**.** *Let a question* $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *have every proper wall positive. In this statement* $\mathbb{E}$ *is the expectation of* $U \sim \mathrm{Breta}^{\circ}(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ *and* Z *is the tilt of* U*. For* $c \in \mathbf{1}^{\perp} \otimes \mathbb{C}$ *and* $\mathbf{a}, \mathbf{b} \in \mathbb{C}^N$ *such that the real parts of the shifted data* $\big(\mathbf{m} + \mathbf{m}_F(c; \mathbf{a}, \mathbf{b});\, \mathbf{s} + \mathbf{a} + \mathbf{b},\, \mathbf{t} + \mathbf{a} - \mathbf{b}\big)$ *have every proper wall positive,*

$$\mathbb{E}\Big[e^{2\langle c, Z\rangle - \sum_j (a_j E_j(Z) + b_j O_j(Z))}\Big] = \frac{P_F\big(\mathbf{m} + \mathbf{m}_F(c; \mathbf{a}, \mathbf{b});\, \mathbf{s} + \mathbf{a} + \mathbf{b},\, \mathbf{t} + \mathbf{a} - \mathbf{b}\big)}{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})},$$

analytic in $(c, \mathbf{a}, \mathbf{b})$ *there: the shift moves the pair coordinates of the question by* $(c; \mathbf{a}, \mathbf{b})$*. So the joint cumulants of order* k *of* $(E_j(Z), O_j(Z))_j$ *are* $(-1)^k$ *times the derivatives of order* k *of* $\log P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *in* $(\sigma_j, \delta_j)_j$*, and their characteristic function is the same ratio at* $c = 0$ *and purely imaginary* $(\mathbf{a}, \mathbf{b})$*: the family on the fibre of imaginary exponents over the question.*

Definition 38.3 (The wall at a tilt)**.** For an order π of $\mathcal{E}$ with tails $U_k = \{\pi(k), \dots, \pi(n)\}$, the sector H_π of Lemma 8.5 is read in the tilts as $\{z \in \mathcal{T} :\ z_{\pi(1)} \geq \cdots \geq z_{\pi(n)}\}$. On it the measure dz of Definition 38.1 is the Lebesgue measure in the differences $z_{\pi(k-1)} - z_{\pi(k)} \geq 0$, $2 \leq k \leq n$, which at the gauge $\alpha = e^{2z}$ are half the gaps $\log(\alpha_{\pi(k-1)}/\alpha_{\pi(k)})$ of Corollary 8.8 and Proposition 39.19. For data on the balance wall the *wall at a tilt* $z \in H_\pi$ is

$$\kappa(z) = \sum_{k=2}^{n} \kappa_{U_k}\big(z_{\pi(k-1)} - z_{\pi(k)}\big),$$

the walls of the tails of π weighted by the differences of z along π, half the walls weighted by the gaps. It does not depend on the order whose sector holds z, the tails with a positive difference being the sets $\{e :\ z_e < z_f\}$ for the $f \in \mathcal{E}$ with $z_f > \min_e z_e$; it is positively homogeneous of degree one and linear on every sector, and $\kappa(-\mathbf{1}_S) = \kappa_S$ for every nonempty proper S, with $-\mathbf{1}_S$ read in $\mathcal{T}$, the direction along which S collapses (Proposition 16.24(i)). In pair coordinates $\kappa(z)$ is linear in the question.

Proposition 38.4 (The exponent of a question reads its walls; the even statistic is least at the table's gauge)**.** *Let a family carry a question* $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$*, its exponents of any signs, and in this statement write* $X(z) = 2\langle \boldsymbol{\mu}_\perp, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)$ *for the exponent of the law of the question (Definition 38.1) and* $\kappa(z)$ *for its wall at the tilt* z *(Definition 38.3).*

(i) On the sector H_π *of every order* π*, with the sector letters* $k_{j,\pi}, \psi_{j,\pi}$ *of Theorem 8.7(ii) read at the sector coordinates* $q_k = \alpha_{\pi(k)}/\alpha_{\pi(k-1)} = e^{-2(z_{\pi(k-1)} - z_{\pi(k)})}$ *of* $\alpha = e^{2z}$*,*

$$X(z) = -2\kappa(z) - \sum_j \big(s_j \log k_{j,\pi}(q) + t_j \log \psi_{j,\pi}(q)\big).$$

With $h_j(z) = \max_{T \in \mathcal{B}_j} \langle \chi_T, z\rangle$*, the support function of the Newton polytope* $2P_j - \mathbf{1}$ *of* Λ_j *(Theorem 7.15),* $2\kappa(z) = \sum_j \big(s_j h_j(z) + t_j h_j(-z)\big) - 2\langle \boldsymbol{\mu}_\perp, z\rangle$ *on* $\mathbf{1}^{\perp}$*, and for* $s_j, t_j \geq 0$ *the wall at a tilt is the support function of* $\mathcal{N}_F - \mathbf{m}$*,* $\mathcal{N}_F$ *the Newton sum of Theorem 8.18(ii).*

The sector letters are polynomials with nonnegative coefficients, with the constant terms $\theta_{j,T^+_{j,\pi}}$ and $\theta_{j,T^-_{j,\pi}}$ and the value $K_j(\mathbf{1})$ at $q = \mathbf{1}$ (Theorem 8.7(ii)), so on the cube they lie in $[\theta_{j,T^+_{j,\pi}}, K_j(\mathbf{1})]$ and $[\theta_{j,T^-_{j,\pi}}, K_j(\mathbf{1})]$; with $\theta_{j,\min} = \min_T \theta_{j,T}$, $|X + 2\kappa| \le \sum_j (|s_j| + |t_j|) \max\big(|\log \theta_{j,\min}|, |\log K_j(\mathbf{1})|\big)$, and for $s_j, t_j \ge 0$

$$2\kappa(z) + \sum_j (s_j + t_j) \log \theta_{j,\min} \le -X(z) \le 2\kappa(z) + \sum_j (s_j + t_j) \log K_j(\mathbf{1}).$$

(ii) *$\kappa > 0$ on $\mathcal{T} \setminus 0$ exactly when every proper wall is positive, and then $\kappa(z) \ge \kappa_{\min}\big(\max_e z_e - \min_e z_e\big)$, $\kappa_{\min}$ the least proper wall.*

(iii) *At the centre of strengths $\boldsymbol{\sigma}$, $\sigma_j \ge 0$, where $\boldsymbol{\mu}_\perp = 0$ and $\boldsymbol{\delta} = 0$, the exponent is minus the centred potential $\sum_j \sigma_j E_j$ (Proposition 8.21) and $2\kappa(z) = \sum_j \sigma_j \operatorname{wd}_j(z)$, with $\operatorname{wd}_j(z) = h_j(z) + h_j(-z)$ the width of $2P_j - \mathbf{1}$ in the direction z, twice the width of P_j, so that $\operatorname{wd}_j(\mathbf{1}_S) = 2c_j(S)$. Every E_j is convex and even, $E_j(z) \ge E_j(0) = 2 \log K_j(\mathbf{1})$, and*

$$\sum_j \sigma_j \operatorname{wd}_j(z) + 2\sum_j \sigma_j \log \theta_{j,\min} \le \sum_j \sigma_j E_j(z) \le \sum_j \sigma_j \operatorname{wd}_j(z) + 2\sum_j \sigma_j \log K_j(\mathbf{1}).$$

If the family is connected, $\sum_j \sigma_j E_j$ is strictly convex on $\mathcal{T}$ and $\sum_j \sigma_j \operatorname{wd}_j > 0$ on $\mathcal{T} \setminus 0$.

Theorem 38.5 (The two ends of strength)**.** *Let a question $q = (\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ of a family have every proper wall positive, and for $\lambda \in \mathbb{C}$ with $\operatorname{Re}\lambda > 0$ read the data λq of pair coordinates $(\lambda\boldsymbol{\mu}_\perp; \lambda\boldsymbol{\sigma}, \lambda\boldsymbol{\delta})$, a question when λ is real, whose walls $\lambda\kappa_S$ have positive real parts; in this statement $P_F(\lambda q)$ is its period, $2^{n-1}\int_{\mathcal{T}} e^{\lambda X(z)} dz$ by the change of variables of Definition 38.1 with the powers of the positive letters principal (Theorem 8.10), and X and κ are those of q (Proposition 38.4).*

(i) Low strength. *As $\lambda \downarrow 0$,*

$$\lambda^{n-1} P_F(\lambda q) \to \sum_\pi \prod_{k=2}^{n} \kappa_{U_k}^{-1} = (n-1)!\ \operatorname{vol}\{z \in \mathcal{T} :\ \kappa(z) \le 1\},$$

and

$$\lambda^{n-1} P_F(\lambda q) = \sum_\pi \prod_{k=2}^{n} \kappa_{U_k}^{-1} \Big(1 - \lambda \sum_j \big(s_j \log \theta_{j,T^+_{j,\pi}} + t_j \log \theta_{j,T^-_{j,\pi}}\big)\Big) + o(\lambda),$$

with the greedy bases $T^\pm_{j,\pi}$ of Lemma 8.5: the walls set the period at low strength, and the weights enter at the next order, through the greedy bases.

(ii) High strength, in the half-plane. *Let $s_j, t_j \ge 0$. Then $-X$ is strictly convex on $\mathcal{T}$, its unique critical point $z^\star$ is the native gauge of every question λq, $\lambda > 0$, and as $|\lambda| \to \infty$ with $\operatorname{Re}\lambda \ge \lambda_0$ for a fixed $\lambda_0 > 0$, in particular as $\lambda \to +\infty$ and along $\lambda = \lambda_0 + i\eta$, $|\eta| \to \infty$,*

$$P_F(\lambda q) = 2^{n-1} e^{\lambda X(z^\star)} \Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}} \det\big(-\nabla^2 X(z^\star)\big)^{-\frac{1}{2}} \big(1 + O(|\lambda|^{-1})\big),$$

the power principal and the Hessian taken on a chart $z_g = 0$, where it is rung two of the still of the realization $\big((s_j, t_j)_j; e^{2z^\star}\big)$ of the active tables (Definition 9.4) with the row and the column of g deleted. At the centre of strengths $\boldsymbol{\sigma}$, $z^\star = 0$, $X(0) = -2\sum_j \sigma_j \log K_j(\mathbf{1})$, and on the chart $z_g = 0$, $-\nabla^2 X(0) = 8\sum_j \sigma_j \big(\kappa_{j,\mathbf{1}}(X_e, X_f)\big)_{e,f \ne g}$.

At low strength the walls alone set the period, through the volume of the region where the wall at a tilt is at most one, and the weights enter at the next order through the greedy bases. At high strength the period concentrates at the native gauge, where it is a Gaussian integral whose Hessian is rung two of the still. A base point added to the data moves the ray off the origin: along $(\mathbf{m}_0+\lambda\mathbf{m};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$, with complex $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ on the balance wall, the leading term gains the factor $(\alpha^\star)^{\mathbf{m}_0}\prod_j K_j(\alpha^\star)^{-s_{0,j}}\Psi_j(\alpha^\star)^{-t_{0,j}}$ at $\alpha^\star=e^{2z^\star}$ (Proposition 17.8), and (ii) at real λ is the case $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$ of that proposition, read in signed coordinates.

Proposition 38.6 (How far a question can lean; the values of the statistics)**.** *For every family and every nonempty proper S put $\varepsilon_j^{\perp}(S)=\varepsilon_j(S)-|S|\frac{2r_j-n}{n}$, so that $\varepsilon_j^{\perp}(\mathcal{E}\setminus S)=-\varepsilon_j^{\perp}(S)$, and for a question $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ let*

$$\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})=\sum_j \sigma_j c_j(S),\qquad \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp,\boldsymbol{\delta})=\mu_{\perp,S}-\sum_j \delta_j\varepsilon_j^{\perp}(S),$$

the even and the odd part of the wall at S: on the balance wall $\kappa_S=\kappa_S^{\mathbf{E}}+\kappa_S^{\mathbf{O}}$ and $\kappa_{\mathcal{E}\setminus S}=\kappa_S^{\mathbf{E}}-\kappa_S^{\mathbf{O}}$, and the fold, which negates the odd part (Theorem 16.16), swaps the two. So a question is inside its walls exactly when $|\kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp,\boldsymbol{\delta})|<\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ for every S. When the centre of strengths $\boldsymbol{\sigma}$ is inside its walls, the slope *of an odd direction $(c,\mathbf{b})\in\mathbf{1}^\perp\times\mathbb{R}^N$ at $\boldsymbol{\sigma}$ is*

$$\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})=\min\Big\{\frac{\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})}{|\kappa_S^{\mathbf{O}}(c,\mathbf{b})|}:\ \kappa_S^{\mathbf{O}}(c,\mathbf{b})\neq 0\Big\},$$

$\ell^*_{\boldsymbol{\sigma}}=\infty$ if every $\kappa_S^{\mathbf{O}}(c,\mathbf{b})$ vanishes; it is homogeneous of degree one in $\boldsymbol{\sigma}$, and even and positively homogeneous of degree -1 in $(c,\mathbf{b})$. The questions of strengths $\boldsymbol{\sigma}$ inside their walls are those whose odd part has $\ell^*_{\boldsymbol{\sigma}}(\boldsymbol{\mu}_\perp,\boldsymbol{\delta})>1$, an open convex polyhedron symmetric under the fold, bounded exactly when no nonzero odd direction has $\kappa_S^{\mathbf{O}}(c,\mathbf{b})=0$ for every S. For one connected table at offsets zero they form the wedge $\sigma>0$, $|\delta|<\ell^*_1(0,1)\,\sigma$, with $\ell^*_1(0,1)=\min\{c(S)/|\varepsilon^{\perp}(S)|:\ \varepsilon^{\perp}(S)\neq 0\}$.

(i) For the tilt Z of $U\sim\mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, the centre of strengths $\boldsymbol{\sigma}$, and real $c\in\mathbf{1}^\perp$, $\mathbf{a},\mathbf{b}\in\mathbb{R}^N$, the expectation of $e^{2\langle c,Z\rangle-\sum_j(a_jE_j(Z)+b_jO_j(Z))}$ is finite exactly when every proper wall of the question $(c;\boldsymbol{\sigma}+\mathbf{a},\mathbf{b})$ is positive, $|\kappa_S^{\mathbf{O}}(c,\mathbf{b})|<\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}+\mathbf{a})$ for every S, and it is then the ratio of the periods of that question and of the centre (Corollary 38.2); in particular $\mathbb{E}_{U\sim\mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})}\big[e^{2\langle c,Z\rangle-\sum_j b_jO_j(Z)}\big]$ is finite exactly when $\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})>1$.

(ii) For a question inside its walls with $|\delta_j|\le\sigma_j$ the native gauge $z^\star$ minimizes $-2\langle\boldsymbol{\mu}_\perp,z\rangle+\sum_j(\sigma_jE_j+\delta_jO_j)$ on $\mathbf{1}^\perp$, so $\big(z^\star;E_j(z^\star),O_j(z^\star)\big)_j$ is the point of the set of values $\{(z;E_j(z),O_j(z))_j:\ z\in\mathbf{1}^\perp\}$ of the statistics at which $-2\langle\boldsymbol{\mu}_\perp,x\rangle+\sum_j(\sigma_jy_j+\delta_jw_j)$ is least; at the centre it is $\big(0;2\log K_j(\mathbf{1}),0\big)_j$.

Proposition 38.7 (The tails of the statistics are read at the walls)**.** *Let a question q of a family have every proper wall positive, let Z be the tilt of $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ at its data, and for a real balanced direction v of pair coordinates $(c;\mathbf{a},\mathbf{b})$ (Definition 38.1) let $Y=2\langle c,Z\rangle-\sum_j\big(a_jE_j(Z)+b_jO_j(Z)\big)$. On the balance wall the walls are linear in the pair coordinates, $\kappa_S(q+\lambda v)=\kappa_S(q)+\lambda\kappa_S(v)$ with $\kappa_S(v)=\kappa_S^{\mathbf{E}}(\mathbf{a})+\kappa_S^{\mathbf{O}}(c,\mathbf{b})$ (Proposition 38.6). Let*

$$\lambda^*=\min\Big\{\frac{\kappa_S(q)}{-\kappa_S(v)}:\ \kappa_S(v)<0\Big\},$$

the value of λ at which the ray $q+\lambda v$, $\lambda>0$, first meets a wall, with $\lambda^*=\infty$ if no $\kappa_S(v)$ is negative. For real $\lambda\ge 0$, $\mathbb{E}[e^{\lambda Y}]$ is finite exactly for $\lambda<\lambda^*$, where it is the ratio of the periods of

the questions $q + \lambda v$ and q (Corollary 38.2); if $\lambda^* < \infty$,

$$\mathbb{P}(Y > x) = e^{-\lambda^* x + o(x)} \qquad (x \to \infty),$$

and if $\lambda^* = \infty$, Y is bounded above. Under the law of the centre of strengths $\boldsymbol{\sigma}$ an odd direction, of pair coordinates $(c; \mathbf{0}, \mathbf{b})$, has $\lambda^* = \ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b})$ for Y and for $-Y$ alike.

Proposition 38.8 (A section is a table)**.** *Let $W = \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ be a family word, in this statement of the degree $-d_W$, $d_W = \sum_j (s_j r_j + t_j r_j^*)$, and $\mathbf{m}$ masses with every proper wall positive. Then $\int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} W \, d\omega_\mathcal{E}$ is the value on its balance wall of the family with the list $\binom{\mathcal{E}}{1}$, whose bases are the singletons and whose letter is $K_0 = |\alpha|$, added at the exponents $(m_\mathcal{E} - d_W, 0)$; its proper walls are those of the family. In particular, for the family F_0 of the one table $\binom{\mathcal{E}}{1}$, $\mathrm{Dir}_\mathbf{m} = \mathrm{Breta}^\circ(F_0; \mathbf{m}; m_\mathcal{E}, 0)$, the Breta shape law of $\binom{\mathcal{E}}{1}$ at a full basis lean; and for two families F_1, F_2 on $\mathcal{E}$ with words W_1, W_2, the logarithms of the letters of F_2 under the law on $\Delta_\mathcal{E}$ with density proportional to $u^{\mathbf{m}-\mathbf{1}} W_1$ have the Laplace transform*

$$\int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} W_1 W_2 \, d\omega_\mathcal{E} \Big/ \int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} W_1 \, d\omega_\mathcal{E}$$

in the exponents of W_2, a ratio of two values of the family $F_1 \cup F_2 \cup F_0$.

The period of a question is a Laplace transform in the tilts, so the shape laws form an exponential family whose natural parameters are the offsets and the pair coordinates of the exponents and whose statistics are the tilt and the even and odd statistics E_j, O_j. On the sector of each order the exponent of a question is minus twice its wall at the tilt, the walls of the tails weighted by the differences of the tilt along the order, up to the bounded sector letters. So the walls alone fix how the period blows up as the strength goes to zero, the weights entering only at the next order and only through the greedy bases; at high strength, real or complex with its real part bounded below, the period is read at the native gauge of the ray. At a centre the exponent is minus the centred potential $\sum_j \sigma_j E_j$, which is convex, smallest at the table's gauge and within a constant of the widths of the Newton polytopes. A question can lean only as far as its strengths allow: the odd part of every wall is bounded by its even part, the slope of its odd part exceeds one, and the same slope bounds the exponential moments of the odd statistics at the centre; along every direction the tail of a statistic falls at the rate of the first wall the direction meets. Integrating over the simplex is integrating against the list $\binom{\mathcal{E}}{1}$ at a full basis lean, so every Dirichlet expectation of letters is a ratio of periods.

38.2 The interaction of a bunch

Definition 38.9 (Interaction)**.** Let a table have a bunch B with room R and $\mathrm{pat}(B) = 2^B$: every $p \subseteq B$ is a pattern, equivalently $c(B) = |B|$, that is B is independent and some basis avoids it (Proposition 12.16(i)). The *interaction* and the *dual interaction* of B are

$$\iota_B(\alpha_R) = \sum_{p \subseteq B} (-1)^{|B \setminus p|} \log K_{(p)}(\alpha_R), \qquad \iota^*_B(\alpha_R) = \sum_{p \subseteq B} (-1)^{|p|} \log \Psi_{(p)}(\alpha_R),$$

the second the interaction of the dual table, and for a split of B into blocks $B_1, \ldots, B_k$,

$$\iota_{(B_1, \ldots, B_k)}(\alpha_R) = \sum_{A \subseteq \{1, \ldots, k\}} (-1)^{k - |A|} \log K_{(\cup_{a \in A} B_a)}(\alpha_R).$$

For two marks, $\iota_{\{e,f\}} = \log \frac{K_{(\{e,f\})}K_{(\varnothing)}}{K_{(\{e\})}K_{(\{f\})}}$, the logarithm of the odds ratio of the presence of the two marks. For a family with $\mathrm{pat}_j(B) = 2^B$ in every active table j, the *interaction* of B is the pair

$$\iota_B^{\mathbf{E}} = \sum_j \sigma_j\big(\iota_{j,B} + \iota_{j,B}^*\big), \qquad \iota_B^{\mathbf{O}} = \sum_j \delta_j\big(\iota_{j,B} - \iota_{j,B}^*\big), \qquad \iota_B^{\mathbf{E}} + \iota_B^{\mathbf{O}} = \sum_j \big(s_j\iota_{j,B} + t_j\iota_{j,B}^*\big),$$

the interactions of the even and the odd part of the potential (Definition 8.17).

Proposition 38.10 (The interaction belongs to the room)**.** *Let a family have a bunch B, $|B| \geq 2$, with $\mathrm{pat}_j(B) = 2^B$ in every active table.*

(i) $\iota_{j,B}$ and $\iota_{j,B}^*$, and so $\iota_B^{\mathbf{E}}$ and $\iota_B^{\mathbf{O}}$, are homogeneous of degree 0 in α_R.

(ii) At a gauge α put, for $P \subseteq B$, $w(P) = \prod_j \pi_{j,P}(\alpha)^{s_j}\pi_{j,B\setminus P}^*(\alpha)^{t_j}$, with the pattern probabilities $\pi_{j,p}(\alpha) = \alpha_B^p K_{j,(p)}(\alpha_R)/K_j(\alpha)$ and $\pi_{j,p}^*(\alpha) = \alpha_B^{B\setminus p}\Psi_{j,(p)}(\alpha_R)/\Psi_j(\alpha)$ of Definition 12.22; for a sample as in Proposition 39.1 it is the chance that every draw uses exactly P on B, its basis draws meeting B in P and its cobasis draws missing exactly P. In the expansion $\log w(P) = \sum_{A\subseteq P}\lambda_A$ over the subsets of B, the top term is $\lambda_B = \iota_B^{\mathbf{E}} + \iota_B^{\mathbf{O}}$, and for $A \subsetneq B$ with $|A| \geq 2$, λ_A is the sum $\iota_A^{\mathbf{E}} + \iota_A^{\mathbf{O}}$ of the interaction of A in the face $F^{\mathcal{E}\setminus(B\setminus A)}$ of the wall at $B \setminus A$ (Theorem 16.23), whose tables are the restrictions $\mathcal{B}_j|\big(\mathcal{E} \setminus (B \setminus A)\big)$, the deletions of $B \setminus A$, at the exponents $(s_j, 0)$ and the contractions $\mathcal{B}_j/(B \setminus A)$ at $(0, t_j)$; none of them depends on α_B. For one table at the exponents $(1, 0)$, $w(P) = \pi_P(\alpha)$ and $\lambda_B = \iota_B$, the top term of the log-linear expansion of the pattern probabilities.

Theorem 38.11 (The interaction totals the joint cumulants over the bunch's scales)**.** *Let a family with real exponents have a bunch $B = \{e_1, \dots, e_k\}$, $k \geq 2$, with $\mathrm{pat}_j(B) = 2^B$ in every active table. Then for both parts $\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}$*

$$\iota_B^{\mathbf{X}}(\alpha_R) = \int_{\mathbb{R}^B} \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \Phi_{\mathbf{X}}(y_B, \log \alpha_R)\, dy_B,$$

converging absolutely; at a realization the integrand of the sum of the parts is $2^{-k}\kappa_\Upsilon(e_1, \dots, e_k) = \sum_j \big(s_j\kappa_{j,\alpha}(e_1, \dots, e_k) + (-1)^k t_j \kappa_{j,\alpha^{-1}}(e_1, \dots, e_k)\big)$, with κ_Υ the still at the gauge $\alpha = (e^{y_B}, \alpha_R)$ (Proposition 9.7). For one table at the exponents $(1, 0)$ the integrand is the joint cumulant of the indicators of the marks of B under the basis law at (e^{y_B}, α_R) (Definition 9.2), and the total is $\iota_B(\alpha_R)$.

Proposition 38.12 (Both sides, and parity)**.** *For a table with $\mathrm{pat}(B) = 2^B$, with $\Psi = \sum_p \alpha_B^{B\setminus p}\Psi_{(p)}(\alpha_R)$, as functions of the tilt z_R of the room $\iota_B^*(z_R) = (-1)^{|B|}\iota_B(-z_R)$. So for a family*

$$\iota_B^{\mathbf{E}}(z_R) = \sum_j \sigma_j\big(\iota_{j,B}(z_R) + (-1)^{|B|}\iota_{j,B}(-z_R)\big), \qquad \iota_B^{\mathbf{O}}(z_R) = \sum_j \delta_j\big(\iota_{j,B}(z_R) - (-1)^{|B|}\iota_{j,B}(-z_R)\big)\,:$$

for $|B|$ even the even part of the interaction is even in z_R and the odd part is odd, for $|B|$ odd the reverse, and at the table's gauge of the room the interaction is $\big(2\sum_j \sigma_j\iota_{j,B}, 0\big)$ for $|B|$ even and $\big(0, 2\sum_j \delta_j\iota_{j,B}\big)$ for $|B|$ odd: the strengths read the even bunches and the leans the odd ones, as they read the even and the odd rungs of the still at the table's gauge (Proposition 9.5). The total of Theorem 38.11 for the logarithm $-\Phi$ of the family word is $-\sum_j \big(s_j\iota_{j,B}(z_R) + (-1)^{|B|}t_j\iota_{j,B}(-z_R)\big)$.

Proposition 38.13 (A design)**.** *Let $\mathbf{X} \in \mathbb{R}^{\mathcal{E}\times C}$ have full column rank, with rows x_e, and for $J \subseteq C$ let $K_J(\alpha) = \det\big(\mathbf{X}_J^{\mathsf{T}}\,\mathrm{diag}(\alpha)\mathbf{X}_J\big)$, $\mathbf{X}_J$ the columns in J, with $K_\varnothing = 1$.*

(i) The marks are the observations of the determinantal table with the slice $\mathbf{X}^{\mathsf{T}}$ *(Definition 7.17) and the weights* $\theta_T = \det(\mathbf{X}_{T,C})^2$*,* $|T| = |C|$*: the occupation of a mark is its leverage,* $R_e(\alpha) = \alpha_e \, x_e^{\mathsf{T}} \big(\mathbf{X}^{\mathsf{T}} \operatorname{diag}(\alpha) \mathbf{X}\big)^{-1} x_e$*.*

(ii) Add one prior mark γ^{p} *for each column* $\gamma \in C$*, with the rows of the identity and the scales* β*:* $K(\alpha, \beta) = \det\big(\mathbf{X}^{\mathsf{T}} \operatorname{diag}(\alpha) \mathbf{X} + \operatorname{diag}(\beta)\big) = \sum_{J \subseteq C} \beta^{C \setminus J} K_J(\alpha)$*. The prior marks form a bunch* C^{p} *with* $\operatorname{pat}(C^{\mathrm{p}}) = 2^{C^{\mathrm{p}}}$*, and the child of the pattern* q^{p}*,* $q \subseteq C$*, is* $K_{C \setminus q}$*.*

(iii) For a split $C = J_1 \sqcup J_2$ *the generalized variance inflation factor is* $\frac{K_{J_1} K_{J_2}}{K_C K_\varnothing} = \exp\big(-\iota_{(J_1^{\mathrm{p}}, J_2^{\mathrm{p}})}\big)$*, and for two columns* γ, γ'*, one minus the squared partial correlation given the rest is* $\exp(\iota_{\{\gamma^{\mathrm{p}}, \gamma'^{\mathrm{p}}\}})$ *in the table with the other prior marks deleted.*

(iv) The dual multivector of the table of (ii) has the slice $[\, I_{\mathcal{E}} \mid -\mathbf{X} \,]$*, and the basis polynomial of its dual table,* $\det\big(\operatorname{diag}(\alpha') + \mathbf{X} \operatorname{diag}(\beta') \mathbf{X}^{\mathsf{T}}\big)$*, is the polynomial of (ii) for the matrix* $\mathbf{X}^{\mathsf{T}}$*: the columns of* $\mathbf{X}$ *are its observations, at the scales* β'*, and the marks of* $\mathcal{E}$ *its prior marks, at the scales* α'*. Its child with every prior mark deleted,* $\det\big(\mathbf{X} \operatorname{diag}(\beta') \mathbf{X}^{\mathsf{T}}\big)$*, is nonzero exactly when* $\mathbf{X}$ *has rank* $|\mathcal{E}|$*.*

A bunch all of whose subsets are patterns sees the room through its children, one for each subset of the bunch, and the alternating sum of their logarithms is the interaction: it is homogeneous of degree zero in the room, it is the top term of the log-linear expansion of the chance that a sample uses one pattern of the bunch throughout, and it is the total, over the bunch's scales, of the joint cumulants of the use of the bunch's marks under the basis and cobasis laws, weighted by the exponents. The dual table reads the same interaction at the opposite tilt with the sign of the parity of the bunch, so a family reads the interaction of an even bunch through its strengths and that of an odd bunch through its leans, as it reads the rungs of the same parities. In a regression design the prior marks of the columns are such a bunch: the interactions are minus the logarithms of the generalized variance inflation factors and the logarithms of one minus the squared partial correlations, and the occupations are the leverages.

Chapter 39

The family as inference

A family is a model of data in two ways. A sample of bases, drawn from the basis and cobasis laws of the tables of a family at an unknown gauge, is itself a question: its counts are the masses, the sizes of its parts are the exponents, and its chance as a function of the tilt is the integrand of the period. The law of a question is then the conjugate prior of the tilt, and every posterior, every predictive chance and every evidence is a ratio of periods; read at a bunch, a sample is a sample of the children at the shares it realizes, and each of its walls is a sum of independent excesses of the levels and colevels of its draws over their floors, whose law the Skeleton polynomials give. A sample of tilts drawn from the law of a question makes the period the cumulant function of an exponential family. Between the two sits the family word read at Dirichlet draws: the negative logarithms of its letters have the Grey law, whose Laplace transform in the exponents is the period divided by the stacked Gamma of the masses over their total; the pseudo-MGF and the pseudoexpectation are that Laplace transform fixed at the Foreman's question and at the centre. A draw of the shape law, read as a vector of numbers, has an order that is a sector of the period, best and worst bases that are functions of the order, and entropies and moments that are steps of the data.

39.1 Samples of bases

Proposition 39.1 (A sample of bases is a question)**.** *For a family at a gauge $\alpha = e^{2z}$ draw, for every table j, s_j bases $T_{j,1}, \dots, T_{j,s_j}$ from its basis law $\mathbb{P}_{j,\alpha}$ and t_j bases $T'_{j,1}, \dots, T'_{j,t_j}$ from its cobasis law $\mathbb{P}^*_{j,\alpha}$, all independent, let m_e count the $T_{j,i}$ that hold e and the $T'_{j,i}$ that miss e, and let $\theta_{\mathrm{draws}} = \prod_{j,i} \theta_{j,T_{j,i}} \prod_{j,i} \theta_{j,T'_{j,i}}$ be the product of the weights of the drawn bases.*

(i) As a function of the tilt z, the chance of the sample is

$$\theta_{\mathrm{draws}} \cdot \exp\Big(2\langle \boldsymbol{\mu}_\perp, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big),$$

$2^{1-n}\theta_{\mathrm{draws}}$ times the integrand of the period of the question $(\mathbf{m}; \mathbf{s}, \mathbf{t})$, of pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ (Definition 38.1): the counts are the masses, the sizes of the parts are the exponents, and $m_{\mathcal{E}} = \sum_j (s_j r_j + t_j r^*_j)$ puts the question on the balance wall.

(ii) Its logarithmic gradient in z is $2\big(\mathbf{m} - \sum_j (s_j R_j(\alpha) + t_j R^*_j(\alpha))\big)$, twice the excess of the masses over the usage at α. For a connected family the maximum likelihood tilt is the native gauge of the question, and it exists, unique on $\mathcal{T}$, exactly when every proper wall of the question is positive.

(iii) $\kappa_S = \sum_j \Big(\sum_{i=1}^{s_j} \big(|T_{j,i} \cap S| - d_j(S)\big) + \sum_{i=1}^{t_j} \big(|S \setminus T'_{j,i}| - d^*_j(S)\big) \Big) \geq 0$*: a wall is the excess of the sample's use of* S *over its least use.*

Proposition 39.2 (The conjugate posterior)**.** *Let* F *be a connected family. Take as prior on the tilt of the gauge the law of a question* $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$ *with every proper wall positive, and observe a sample as in Proposition 39.1, with counts* $\mathbf{m}$*, sizes* $\mathbf{s}, \mathbf{t}$ *and the product* θ_{draws} *of the weights of the drawn bases.*

(i) The posterior is the law of the question $(\mathbf{m}_0 + \mathbf{m}; \mathbf{s}_0 + \mathbf{s}, \mathbf{t}_0 + \mathbf{t})$*: the masses add and the exponents add.*

(ii) The chance of the sample with the tilt integrated out is

$$\theta_{\text{draws}} \cdot \frac{P_F(\mathbf{m}_0 + \mathbf{m}; \mathbf{s}_0 + \mathbf{s}, \mathbf{t}_0 + \mathbf{t})}{P_F(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)}.$$

(iii) At the posterior data $(\mathbf{m}'; \mathbf{s}', \mathbf{t}')$ *the chance that one more draw from the basis law of table* j *is* T *is* $\theta_{j,T}\, P_F(\mathbf{m}' + \mathbf{1}_T; \mathbf{s}' + \mathbf{e}_j, \mathbf{t}')/P_F(\mathbf{m}'; \mathbf{s}', \mathbf{t}')$*, and from its cobasis law* $\theta_{j,T}\, P_F(\mathbf{m}' + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{s}', \mathbf{t}' + \mathbf{e}_j)/P_F(\mathbf{m}'; \mathbf{s}', \mathbf{t}')$*.*

(iv) The posterior mean of the usage $\sum_j (s'_j R_j + t'_j R^*_j)$ *at the posterior exponents is the posterior masses* $\mathbf{m}_0 + \mathbf{m}$*.*

(v) Every wall of the posterior is the wall of the prior plus the excess of Proposition 39.1(iii): a prior inside its walls gives a posterior inside its walls for every sample, the empty one included.

Proposition 39.3 (What the data see of the bases)**.** *(i) For a circuit* C *of* $k \geq 2$ *marks of a table and each* $e \in C$ *some basis has* $T \cap C = C \setminus e$*:* $r(C) = k - 1$*, every* $(k-1)$*-subset of* C *is a pattern of* C *and* C *is not; for* $k \geq 3$ *every two marks of* C *lie in a common basis, and* C *in none.*

(ii) For a sample of a family at a gauge α *as in Proposition 39.1 and* $S \subseteq \mathcal{E}$*, the level* $|T_{j,i} \cap S|$ *of every basis draw of table* j *lies in* $[d_j(S), r_j(S)]$ *and the colevel* $|S \setminus T'_{j,i}|$ *of every cobasis draw in* $[d^*_j(S), |S| - d_j(S)]$*; the chance that no basis draw of table* j *reaches the largest level* $r_j(S)$ *of its window is* $(1 - p_j)^{s_j}$ *with* $p_j = \mathbb{P}_{j,\alpha}(|T \cap S| = r_j(S)) > 0$*, and likewise at the floor* $d_j(S)$ *and for the cobasis draws at the ends of their window, with the cobasis law and* t_j*.*

Proposition 39.4 (The exact laws of a sample of bases)**.** *Let a sample of a family be drawn at a gauge* α *as in Proposition 39.1,* s_j *basis and* t_j *cobasis draws of table* j*, with counts* $\mathbf{m}$*, and let* B *be a bunch with room* R*.*

(i) For a basis draw of table j*,* $\mathbb{P}_{j,\alpha}(T) = \pi_{j,p}(\alpha)\, \mathbb{P}_{j,(p),\alpha_R}(T \setminus p)$ *with* $p = T \cap B$*, and for a cobasis draw* $\mathbb{P}^*_{j,\alpha}(T) = \pi^*_{j,p}(\alpha)\, \mathbb{P}^*_{j,(p),\alpha_R}(T \setminus p)$*, with the pattern probabilities* $\pi_{j,p}(\alpha) = \alpha_B^p K_{j,(p)}(\alpha_R)/K_j(\alpha)$ *and* $\pi^*_{j,p}(\alpha) = \alpha_B^{B\setminus p} \Psi_{j,(p)}(\alpha_R)/\Psi_j(\alpha)$ *(Theorem 12.7) and the basis and cobasis laws of the child* (j,p) *at* α_R*. So the numbers* $a_{j,p}$ *of the basis draws and* $b_{j,p}$ *of the cobasis draws of table* j *that meet* B *in* p *are independent multinomial vectors with the chances* $\pi_{j,p}(\alpha)$ *and* $\pi^*_{j,p}(\alpha)$*; they are shares of* B*,* $|\mathbf{a}_j| = s_j$ *and* $|\mathbf{b}_j| = t_j$ *(Definition 12.21); the counts on the bunch are their loads,* $m_e = \text{load}_e$ *for* $e \in B$*; and given them the parts in the room are a sample of the children* $F_B(\mathbf{a}, \mathbf{b})$ *at* α_R*,* $a_{j,p}$ *basis and* $b_{j,p}$ *cobasis draws of the child* (j,p)*, with the counts* $\mathbf{m}|_R$*. The mean shares are* $s_j \pi_{j,p}(\alpha)$ *and* $t_j \pi^*_{j,p}(\alpha)$*, at a native gauge the native split (Definition 12.22).*

(ii) Given $\mathbf{m}$*, each sample with these counts has the probability* θ_{draws} *divided by the coefficient of* $\alpha^{\mathbf{m}}$ *in* $\prod_j K_j(\alpha)^{s_j}\Psi_j(\alpha)^{t_j}$*, free of the gauge.*

(iii) For every nonempty proper S *the wall* κ_S *of the sample (Proposition 39.1(iii)) is the sum of the independent excesses of the levels of its basis draws and of the colevels of its cobasis draws over their floors, with the generating function*

$$\mathbb{E}\big[x^{\kappa_S}\big] = \prod_j \mathsf{Q}^K_{j,S}(x;\alpha)^{s_j}\,\mathsf{Q}^\Psi_{j,S}(x;\alpha)^{t_j}$$

of the Skeleton polynomials (Definition 12.9). Its values fill $\{0,1,\dots,\sum_j(s_j+t_j)c_j(S)\}$*; its mean is the excess* $\sum_j \big(s_jR_j(\alpha)+t_jR^*_j(\alpha)\big)_S - \sum_j\big(s_jd_j(S)+t_jd^*_j(S)\big)$ *of the usage at* α *over the least use, and its cumulants of order* $k\ge 2$ *are* $\partial_S^k\Phi(\log\alpha)$*,* $\partial_S = \sum_{e\in S}\partial_{y_e}$ *(Proposition 12.11); it vanishes, every draw sitting at the floor of its window on* S*, with the chance* $\prod_j \mathsf{Q}^K_{j,S}(0;\alpha)^{s_j}\mathsf{Q}^\Psi_{j,S}(0;\alpha)^{t_j} > 0$*; and when the Skeleton spectra of* S *are positive it is distributed as a sum of independent* $\{0,1\}$*-valued variables whose means are the coins,* s_j *copies of the coins of* $\mathsf{Q}^K_{j,S}$ *and* t_j *of those of* $\mathsf{Q}^\Psi_{j,S}$ *(Proposition 12.10(iv)).*

Proposition 39.5 (Tests that do not see the gauge)**.** *For a family* F *draw a sample at a gauge* α *as in Proposition 39.1, with counts* $\mathbf{m}$*, and let* $\Theta_F(\mathbf{m})$ *be the coefficient of* $\alpha^{\mathbf{m}}$ *in* $\prod_j K_j(\alpha)^{s_j}\Psi_j(\alpha)^{t_j}$*, the total of* θ_{draws} *over the set* $\mathfrak{F}_F(\mathbf{m})$ *of the samples with these counts.*

(i) Given $\mathbf{m}$*, the sample has the law* $\theta_{\text{draws}}/\Theta_F(\mathbf{m})$ *on* $\mathfrak{F}_F(\mathbf{m})$*, free of the gauge; for a family of lists it is uniform, and* $|\mathfrak{F}_F(\mathbf{m})| = \Theta_F(\mathbf{m})$*.*

(ii) A test of F *that rejects on an event* A *whose samples with counts* $\mathbf{m}$ *carry at most* $\varepsilon\,\Theta_F(\mathbf{m})$ *of the weight* θ_{draws}*, for every* $\mathbf{m}$*, has size at most* ε *at every gauge.*

(iii) If every basis of every table of a family F' *on* $\mathcal{E}$ *is a basis of the same table of* F*, with the same weight, then given* $\mathbf{m}$ *the chance under* F *that every draw is a basis of its table in* F' *is* $p_{F'|F}(\mathbf{m}) = \Theta_{F'}(\mathbf{m})/\Theta_F(\mathbf{m})$*, and rejecting* F *when this happens and* $p_{F'|F}(\mathbf{m}) \le \varepsilon$ *has size at most* ε *at every gauge; one draw that is a basis of its table in* F *and not in* F' *rejects* F' *without error.*

(iv) If each family of a collection has a test as in (ii), the families not rejected contain the true family with probability at least $1-\varepsilon$*, at every gauge.*

Corollary 39.6 (The evidence of a family)**.** *Let every family* F *of a collection on* $\mathcal{E}$*, each with* N *tables, carry as prior the law of its centre of strengths* $\boldsymbol{\sigma}_0 = (\sigma_{0,j})_j$*, the question* $(\sigma_0\mathbf{1};\boldsymbol{\sigma}_0,\boldsymbol{\sigma}_0)$ *with* $\sigma_0 = \sum_j \sigma_{0,j}$*, inside its walls. For a sample as in Proposition 39.5, with the sizes* $(s_j,t_j)_j$ *and counts* $\mathbf{m}$*, the probability of the sample under* F *with the gauge integrated out is*

$$\text{evidence}(F) = \theta^F_{\text{draws}}\;\mathbf{1}[\textit{every draw is a basis of its table of } F]\;\frac{P_F(\sigma_0\mathbf{1}+\mathbf{m};\boldsymbol{\sigma}_0+\mathbf{s},\boldsymbol{\sigma}_0+\mathbf{t})}{P_F(\sigma_0\mathbf{1};\boldsymbol{\sigma}_0,\boldsymbol{\sigma}_0)},$$

θ^F_{draws} *the product of the weights in* F *of the drawn bases,* 1 *for lists; and if the family is drawn from the collection with the chances* $p(F)$*, the posterior chance of* F *is proportional to* $p(F)\,\text{evidence}(F)$*.*

A sample of bases asks the family a question: the counts are its masses and the sample sizes its exponents, and a wall of the question is how far the sample's use of a set exceeds the least use any basis makes of it, a sum of the independent excesses of the levels and colevels of the draws over their floors, whose generating function is the product of the Skeleton polynomials of the set raised to the sizes. Read at a bunch, a sample is a sample of the children at the shares it realizes:

the loads of these shares are the counts of the bunch, and their means at a native gauge are the native split. The law of a question is the conjugate prior of the tilt, so the posterior, the predictive chances and the evidence of a family are ratios of periods, and a prior inside its walls stays inside them. Given the counts, a sample has a law that does not see the gauge, uniform for a family of lists, so a test built on the samples with those counts has a size that no gauge can change.

39.2 Samples of tilts

Proposition 39.7 (A sample of tilts: the period is the cumulant function)**.** *Let a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of pair coordinates* $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ *have every proper wall positive, let* $Z_1,\dots,Z_q$ *be the tilts of independent draws from* $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, and let* $\bar Z$*,* $\bar E_j$*,* $\bar O_j$ *be the sample means of* Z_i*,* $E_j(Z_i)$*,* $O_j(Z_i)$*. In this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *and* Z *is its tilt.*

(i) The sample has the density

$$2^{q(n-1)}\exp\Big(q\Big[2\langle\boldsymbol{\mu}_\perp,\bar Z\rangle-\sum_j\big(\sigma_j\bar E_j+\delta_j\bar O_j\big)\Big]\Big)\Big/P_F(\mathbf{m};\mathbf{s},\mathbf{t})^q$$

on $\mathcal{T}^q$*, so* $(\bar Z;\bar E_j,\bar O_j)_j$ *is sufficient for the question.*

(ii) The means are $\mathbb{E}[Z]=\frac12\nabla_{\boldsymbol{\mu}_\perp}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*,* $\mathbb{E}[E_j(Z)]=-\partial_{\sigma_j}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{E}[O_j(Z)]=-\partial_{\delta_j}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*, the derivatives of Definition 38.1, and the likelihood equations equate them to the sample means.*

(iii) Against a second question $(\mathbf{m}';\mathbf{s}',\mathbf{t}')$ *of pair coordinates* $(\boldsymbol{\mu}'_\perp;\boldsymbol{\sigma}',\boldsymbol{\delta}')$ *the logarithm of the likelihood ratio is*

$$q\Big[2\langle\boldsymbol{\mu}'_\perp-\boldsymbol{\mu}_\perp,\bar Z\rangle-\sum_j\big((\sigma'_j-\sigma_j)\bar E_j+(\delta'_j-\delta_j)\bar O_j\big)\Big]-q\log\frac{P_F(\mathbf{m}';\mathbf{s}',\mathbf{t}')}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}.$$

(iv) The sums $q\bar Z$*,* $q\bar E_j$*,* $q\bar O_j$ *have as Laplace transform that of Corollary 38.2 raised to the power* q*.*

Definition 39.8 (The Foreman's question; the pseudo-MGF)**.** Let $\omega=W\,d\alpha_{\mathbf{m}}$ have real data on a connected family, $s_j,t_j\ge0$ not all zero and $\mathbf{m}>0$, with target $\boldsymbol{\tau}$. The *Foreman's question* of ω is the question $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ at its target, of pair coordinates $\big(\boldsymbol{\tau}-\frac{\tau_\mathcal{E}}{n}\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\delta}\big)$; off the pole hyperplanes its period $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ is the pseudointegral $⨏W\,d\alpha_{\mathbf{m}}$ (Definition 16.8(iii)). Where the target lies on no pole hyperplane and $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})\neq0$, the *pseudo-MGF* of ω is

$$\mathsf{M}_\omega(c;\mathbf{a},\mathbf{b})=\frac{P_F\big(\boldsymbol{\tau}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)}{P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})}\qquad(c\in\mathbf{1}^\perp\otimes\mathbb{C},\ \mathbf{a},\mathbf{b}\in\mathbb{C}^N),$$

continued meromorphically (Theorem 16.15), and the *pseudo-cumulants* of ω are the derivatives of $\log\mathsf{M}_\omega$ at 0: the derivatives of $\log P_F$ at the Foreman's question in its pair coordinates.

Proposition 39.9 (The pseudo-MGF inside the walls)**.** *In Definition 39.8 let* ω *have a native gauge, and let* Z *be the tilt of* $U\sim\mathrm{Breta}^\circ(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$*, the Foreman's law of Proposition 16.31(i); the law of* Z *is the law of the Foreman's question.*

(i) Wherever the real parts of the shifted data $\big(\boldsymbol{\tau}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)$ *have every proper wall positive, a region containing a neighbourhood of* 0*,* $\mathsf{M}_\omega(c;\mathbf{a},\mathbf{b})$ *is the moment generating function of the statistics* $(2Z;-E_j(Z),-O_j(Z))_j$ *under the Foreman's law (Corollary 38.2 at the Foreman's question); the pseudo-cumulants are their joint cumulants.*

(ii) For a balanced shift $(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ *the function* $U^{\dot{\mathbf{m}}}\prod_j K_j(U)^{-\dot{s}_j}\Psi_j(U)^{-\dot{t}_j}$ *has degree* 0*, a function of the ray of* U*, and where the shifted data have every proper wall positive its expectation under the Foreman's law is* $P_F(\boldsymbol{\tau}+\dot{\mathbf{m}};\mathbf{s}+\dot{\mathbf{s}},\mathbf{t}+\dot{\mathbf{t}})/P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ *(Proposition 39.22(i) at the Foreman's question);* $\mathsf{M}_\omega(c;\mathbf{a},\mathbf{b})$ *is this ratio at the shift of pair coordinates* $(c;\mathbf{a},\mathbf{b})$*, so* M_ω *is the moment generating function of* $(\log U;-\log K_j(U),-\log\Psi_j(U))_j$ *along the balanced shifts.*

(iii) The normaliser is $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}) = P_{\mathrm{c}}\,\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ *(Theorem 17.1(i)). The centre* ω_{c} *has the target* $\sigma\mathbf{1}$ *and the Foreman's question* $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$*, and the pseudoexpectation is the pseudo-MGF of the centre at the departure:*

$$\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \mathsf{M}_{\omega_{\mathrm{c}}}\Big(\boldsymbol{\tau}-\frac{\tau_{\mathcal{E}}}{n}\mathbf{1};\ \mathbf{0},\ \boldsymbol{\delta}\Big) = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[e^{2\langle\boldsymbol{\tau}-\frac{\tau_{\mathcal{E}}}{n}\mathbf{1},Z\rangle-\sum_j\delta_jO_j(Z)}\Big],$$

Z the tilt of U*: at fixed strengths the pseudoexpectations of all the questions are values of one moment generating function, that of* $(2Z;-O_j(Z))_j$ *under the law of the centre.*

Past the walls M_ω *is the continuation, as the moments of the distribution of Theorem 16.15(ii) are, and* $\mathsf{M}_\omega(0)=1$ *throughout.*

Proposition 39.10 (Two layers: the word evaluated at the draws)**.** *Let a family have real data with* $\mathbf{m}>0$*, and let* $U_1,\dots,U_q$ *be independent draws from* $\mathrm{Dir}_{\mathbf{m}}$*, the shapes* $\alpha/|\alpha|$ *of* $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$ *(Proposition 8.11(ii)).*

(i) By Proposition 8.11(i), $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)] = P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma^n_1\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$*, and it is finite exactly when every proper wall at* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *is positive.*

(ii) $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)^k] = P_F(\mathbf{m};k\mathbf{s},k\mathbf{t})/\Gamma^n_1\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$*, and the sample mean* $\overline{W}_q=\frac{1}{q}\sum_i W(U_i)$ *has the mean* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma^n_1\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ *and the variance*

$$\frac{1}{q}\Bigg[\frac{P_F(\mathbf{m};2\mathbf{s},2\mathbf{t})}{\Gamma^n_1\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}-\Bigg(\frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{\Gamma^n_1\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}\Bigg)^2\Bigg],$$

finite exactly when every proper wall at $(\mathbf{m};2\mathbf{s},2\mathbf{t})$ *is positive.*

(iii) Where every proper wall at $(\mathbf{m};\mathbf{s},\mathbf{t})$ *is positive,* $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *has the density* $W\,\Gamma^n_1\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *with respect to* $\mathrm{Dir}_{\mathbf{m}}$*, and under it a second word* $W'=\prod_j K_j^{-s'_j}\Psi_j^{-t'_j}$ *of the same tables has* $\mathbb{E}_{U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})}[W'(U)^k] = P_F(\mathbf{m};\mathbf{s}+k\mathbf{s}',\mathbf{t}+k\mathbf{t}')/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*.*

Definition 39.11 (The Grey law)**.** For a family F and masses $\mathbf{m}>0$ the *Grey law* $\mathrm{Grey}(F;\mathbf{m})$ is the joint law of

$$(G_j,G_j^*)_j = \big(-\log K_j(U),\,-\log\Psi_j(U)\big)_j \qquad \text{for } U\sim\mathrm{Dir}_{\mathbf{m}};$$

it depends on the tables of F and on $\mathbf{m}$, and not on the exponents.

Proposition 39.12 (The Grey law and its exponential family)**.** *Let* $\mathbf{m}>0$ *and let* $(G_j,G_j^*)_j\sim\mathrm{Grey}(F;\mathbf{m})$*.*

(i) For complex exponents $(\mathbf{s},\mathbf{t})$ *whose real parts keep every proper wall at* $\mathbf{m}$ *positive,*

$$\mathbb{E}_{(G_j,G_j^*)_j\sim\mathrm{Grey}(F;\mathbf{m})}\Big[e^{\sum_j(s_jG_j+t_jG_j^*)}\Big]=\frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}\quad\textit{(Proposition 8.11(i),(iii))}:$$

on real exponents the moment generating function, on imaginary exponents the characteristic function; the joint cumulants are the derivatives of $\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *in the exponents at* $(\mathbf{0},\mathbf{0})$.

(ii) For real exponents with every proper wall positive, under $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *the vector* $\big(-\log K_j(U),-\log\Psi_j(U)\big)_j$ *has the law*

$$e^{\sum_j(s_jg_j+t_jg_j^*)}\,\frac{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}\,\mathrm{Grey}(F;\mathbf{m})(dg\,dg^*):$$

over the exponents these laws form the exponential family with base $\mathrm{Grey}(F;\mathbf{m})$ *and cumulant function* $\log\big(P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\big)$.

(iii) For a real direction $(\mathbf{a},\mathbf{b})\neq 0$ *of the exponents let* $g(S)=\sum_j\big(a_jd_j(S)+b_jd_j^*(S)\big)$ *and* $k^*=\min\{m_S/g(S):\ S\ \text{nonempty proper},\ g(S)>0\}$, *the value of* λ *at which the ray* $\lambda(\mathbf{a},\mathbf{b})$, $\lambda>0$, *first meets a wall at* $\mathbf{m}$, *with* $k^*=\infty$ *if no* $g(S)$ *is positive. If* $k^*<\infty$,

$$\mathbb{P}\Big(\sum_j\big(a_jG_j+b_jG_j^*\big)>x\Big)=e^{-k^*x+o(x)}\qquad(x\to\infty),$$

and if $k^*=\infty$ *the sum is bounded above.*

(iv) For q *independent copies the sum has the Laplace transform* $\big(P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\big)^q$, *and the sample mean* $\overline{W}_q$ *of Proposition 39.10(ii) is the empirical counterpart of the transform of (i) at the exponents* $(\mathbf{s},\mathbf{t})$.

Corollary 39.13 (The Grey law at the target)**.** *Let a connected family carry real exponents* $s_j,t_j\geq 0$, *not all zero, and masses* $\mathbf{m}>0$ *with a native gauge and the target* $\boldsymbol{\tau}$. *Then*

$$\mathbb{E}_{U\sim\mathrm{Dir}_{\boldsymbol{\tau}}}[W(U)]=\mathbb{E}_{(G_j,G_j^*)_j\sim\mathrm{Grey}(F;\boldsymbol{\tau})}\Big[e^{\sum_j(s_jG_j+t_jG_j^*)}\Big]=\frac{P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}}$$

$$=\frac{P_{\mathrm{c}}}{\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}}\,\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big]:$$

the moment generating function of the Grey law at the target, read at the exponents, is the pseudoexpectation times the period of the centre divided by $\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}$. *With* $\boldsymbol{\tau}$ *held at the target, the logarithmic derivatives of* $(\mathbf{s}',\mathbf{t}')\mapsto P_F(\boldsymbol{\tau};\mathbf{s}',\mathbf{t}')$ *at* $(\mathbf{s},\mathbf{t})$ *are the joint cumulants of* $\big(-\log K_j(U),-\log\Psi_j(U)\big)_j$ *under the Foreman's law* $\mathrm{Breta}^\circ(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$.

Proposition 39.14 (A Laplace transform with a moving target)**.** *Let a connected family carry real exponents $s_j, t_j \geq 0$, not all zero, and masses $\mathbf{m} > 0$ whose target $\boldsymbol{\tau}$ has every proper wall positive, and let $g(S) = \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big)$ be the set function of Theorem 8.18. With the tails U_k of the orders π of $\mathcal{E}$ (Lemma 8.5),*

$$\lim_{\lambda \to 0^+} \mathbb{E}_{V \sim \mathrm{Dir}_{\lambda\boldsymbol{\tau}}}\Big[\prod_j K_j(V)^{-\lambda s_j} \Psi_j(V)^{-\lambda t_j}\Big] = \frac{\sum_\pi \prod_{k=2}^n \big(\tau_{U_k} - g(U_k)\big)^{-1}}{\sum_\pi \prod_{k=2}^n \tau_{U_k}^{-1}} \;\geq\; 1,$$

with equality exactly when $g(S) = 0$ for every nonempty proper S. The limit is the ratio of the limits of Theorem 38.5(i) for the question of data $(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ and for the list $\binom{\mathcal{E}}{1}$ at a full basis lean at the masses $\boldsymbol{\tau}$, whose walls are the τ_S (Proposition 38.8).

The word read at Dirichlet draws is a random variable whose expectation, as a function of the exponents, is the period divided by the stacked Gamma of the masses over their total: the Laplace transform of the Grey law of the family. At fixed masses the Grey law is the base of an exponential family over the exponents, and its tail along a direction of exponents is set by the first wall met along that direction. Read at the target of each exponent, as the normaliser $\binom{\mathcal{E}}{1}$ of Proposition 16.32(ii) reads it, the expectation of the word is not a moment generating function in the exponents: its limit at the origin exceeds 1 as soon as the word has a floor on a proper set, since at each exponent it reads the Grey law at another target. The pseudo-MGF fixes the target once, at the Foreman's question, and the pseudoexpectation fixes the law once for given strengths, at the centre.

Proposition 39.15 (The exponent is a dial between the mean and the maximum)**.** *In Proposition 39.10 let the exponents be real, not all zero, and put $Y_i = \log W(U_i)$ and $Y = \log W(U)$ for $U \sim \mathrm{Dir}_{\mathbf{m}}$; in this statement $\mathbb{E}$ is the expectation of independent $U, U_1, \dots, U_q \sim \mathrm{Dir}_{\mathbf{m}}$. Let $g(S) = \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big)$, $k^* = \min\{m_S / g(S) :\ S \text{ nonempty proper},\ g(S) > 0\}$, with $k^* = \infty$ if no $g(S)$ is positive, and for $k > 0$ let $A_k = \frac{1}{k} \log\big(\frac{1}{q} \sum_i e^{kY_i}\big)$.*

(i) *A_k is nondecreasing in k, tends to the sample mean of the Y_i as $k \to 0$ and to $\max_i Y_i$ as $k \to \infty$, and $\max_i Y_i - \frac{\log q}{k} \leq A_k \leq \max_i Y_i$.*

(ii) *For $k < k^*$, $\mathbb{E}[e^{kA_k}] = P_F(\mathbf{m}; k\mathbf{s}, k\mathbf{t}) / \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$, and e^{kA_k} has a finite variance exactly when $2k < k^*$.*

(iii) *For $k \geq k^*$, $\mathbb{E}[e^{kY}] = \infty$; and $\mathbb{P}(\max_i Y_i \leq x) = \mathbb{P}(Y \leq x)^q$.*

(iv) *If $k^* < \infty$, $\mathbb{P}(Y > x) = e^{-k^* x + o(x)}$ as $x \to \infty$, and if $k^* = \infty$, Y is bounded above (Proposition 39.12(iii)). For $s_j, t_j \geq 0$ and a set S attaining k^* the scale $\rho = |U_S|$ has $(\rho, 1-\rho) \sim \mathrm{Dir}_{(m_S,\, m_{\mathcal{E}\setminus S})}$, so ρ has density proportional to $\rho^{m_S - 1}(1-\rho)^{m_{\mathcal{E}\setminus S} - 1}$, and $W(U) \geq \prod_j K_j(\mathbf{1})^{-s_j - t_j} \rho^{-g(S)}$.*

Proposition 39.16 (The even and odd statistics from a centre)**.** *Let a family carry the centre of strengths $\boldsymbol{\sigma} = (\sigma_j)_j$, $\sigma_j \geq 0$, inside its walls, let Z be the tilt of $U \sim \mathrm{Breta}^\circ(F; \sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$, with density proportional to $e^{-\sum_j \sigma_j E_j(z)}$ on $\mathcal{T}$, and let $\ell_{\boldsymbol{\sigma}}^*$ be the slope of Proposition 38.6. In this statement $\mathbb{E}$ is the expectation of $U \sim \mathrm{Breta}^\circ(F; \sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$.*

(i) *$\sum_j \sigma_j E_j(Z) \geq 2 \sum_j \sigma_j \log K_j(\mathbf{1})$, with equality only at $Z = 0$ (Proposition 38.4), where $\sum_j \sigma_j E_j$ has the Hessian $8 \sum_j \sigma_j \big(\kappa_{j,\mathbf{1}}(X_e, X_f)\big)_{e,f}$, rung two of the still of the realization $\big((\sigma_j, \sigma_j)_j; \mathbf{1}\big)$ of the active tables, eight times the covariance of the indicators under the*

weight laws; and each $E_j(Z) \geq 2\log K_j(\mathbf{1})$*, with equality exactly where* Z *is constant on the molecules of table* j*. For real* $\mathbf{k} = (k_j)_j$*,*

$$\mathbb{E}\Big[e^{-\sum_j k_j E_j(Z)}\Big] = \frac{P_F\big((\sigma + |\mathbf{k}|)\mathbf{1}; \boldsymbol{\sigma} + \mathbf{k}, \boldsymbol{\sigma} + \mathbf{k}\big)}{P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})},$$

the ratio of the periods of the centres of strengths $\boldsymbol{\sigma} + \mathbf{k}$ *and* $\boldsymbol{\sigma}$*, finite exactly when* $\sum_j (\sigma_j + k_j)c_j(S) > 0$ *for every nonempty proper* S*: along the even direction* $\mathbf{k} = \lambda\boldsymbol{\sigma}$ *every wall closes at once, at* $\lambda = -1$*, and along any other even direction* $\lambda\mathbf{k}$*,* $\lambda > 0$*, the first wall closes at the least of the ratios* $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})/(-\kappa_S^{\mathbf{E}}(\mathbf{k}))$ *with* $\kappa_S^{\mathbf{E}}(\mathbf{k}) < 0$*.*

(ii) $\big(Z, (E_j(Z))_j, (O_j(Z))_j\big)$ *and* $\big(-Z, (E_j(Z))_j, (-O_j(Z))_j\big)$ *have one law, so every joint moment of odd total degree in* Z *and the* $O_j(Z)$ *vanishes. For real* $c \in \mathbf{1}^{\perp}$ *and* $\mathbf{b} \in \mathbb{R}^N$*,* $\mathbb{E}\big[e^{2\langle c, Z\rangle - \sum_j b_j O_j(Z)}\big]$ *is finite exactly when* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$ *(Proposition 38.6(i)), where it is*

$$P_F\big(\mathbf{m}_F(c; \boldsymbol{\sigma}, \mathbf{b});\, \boldsymbol{\sigma} + \mathbf{b},\, \boldsymbol{\sigma} - \mathbf{b}\big)\big/P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma}),$$

for $|b_j| \leq \sigma_j$ *the pseudoexpectation of the question of pair coordinates* $(c; \boldsymbol{\sigma}, \mathbf{b})$*. If* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) < \infty$*, both tails of* $Y = 2\langle c, Z\rangle - \sum_j b_j O_j(Z)$ *have the rate* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b})$*:* $\mathbb{P}(\pm Y > x) = e^{-\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})x + o(x)}$ *as* $x \to \infty$*; if it is infinite,* Y *is bounded (Proposition 38.7).*

(iii) At $\alpha = e^{2z}$*,* $K_j(\alpha)^{-s_j} = e^{-s_j\langle \mathbf{1}, z\rangle} e^{-\frac{s_j}{2}(E_j(z) + O_j(z))}$ *and* $\Psi_j(\alpha)^{-t_j} = e^{-t_j\langle \mathbf{1}, z\rangle} e^{-\frac{t_j}{2}(E_j(z) - O_j(z))}$*: every letter is its table at a full lean,* $K_j^{-s_j}$ *at the pair coordinates* $(\sigma, \delta) = (\frac{s_j}{2}, \frac{s_j}{2})$ *and* $\Psi_j^{-t_j}$ *at* $(\frac{t_j}{2}, -\frac{t_j}{2})$*, with the masses shifted by* $-\frac{s_j}{2}\mathbf{1}$*, resp.* $-\frac{t_j}{2}\mathbf{1}$*; a family is the family of its letters, and its pair coordinates are the sums of theirs.*

Proposition 39.17 (What the two statistics are)**.** *For table* j *of a family and a tilt* z*,* $\log \Lambda_j(z) = \log \sum_T \theta_{j,T} e^{\langle \chi_T, z\rangle}$ *is a soft maximum and* $-\log \Lambda_j(-z)$ *a soft minimum of the scores* $\langle \chi_T, z\rangle$ *of its bases, weighted by* $\theta_{j,T}$*.*

(i) $E_j(z)$ *is the soft maximum less the soft minimum, and* $O_j(z)$ *is their sum; the centred potential* $\Phi^{\circ} = \sum_j (\sigma_j E_j + \delta_j O_j)$ *is the sum over the tables of* s_j *times the soft maximum less* t_j *times the soft minimum.*

(ii) With $Y_j = \langle \chi_T, z\rangle$ *for* T *under the weight law of table* j*, near* $z = 0$

$$E_j(z) = 2\log K_j(\mathbf{1}) + \sum_{k\ even} \frac{2}{k!}\, \kappa_{j,\mathbf{1}}(Y_j^{\times k}),$$
$$O_j(z) = \sum_{k\ odd} \frac{2}{k!}\, \kappa_{j,\mathbf{1}}(Y_j^{\times k}),$$

so $E_j(z) = 2\log K_j(\mathbf{1}) + \kappa_{j,\mathbf{1}}(Y_j, Y_j) + \frac{1}{12}\kappa_{j,\mathbf{1}}(Y_j^{\times 4}) + \cdots$ *and* $O_j(z) = 2\kappa_{j,\mathbf{1}}(Y_j) + \frac{1}{3}\kappa_{j,\mathbf{1}}(Y_j^{\times 3}) + \cdots$*, with* $\kappa_{j,\mathbf{1}}(Y_j) = \sum_e (2R_{j,e}(\mathbf{1}) - 1)z_e$ *and, for* $k \geq 2$*,* $\kappa_{j,\mathbf{1}}(Y_j^{\times k}) = 2^k \sum_{w \in \mathcal{E}^k} \kappa_{j,\mathbf{1}}(w)\, z^w$*,* $z^w = z_{w_1} \cdots z_{w_k}$*. Summed with the strengths and the leans, the terms of order* $k \geq 2$ *of* Φ° *are* $\frac{1}{k!}\sum_{w \in \mathcal{E}^k} 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w)\, z^w$*, which for the exponents of a realization* $\Upsilon = ((s_j, t_j)_j; \mathbf{1})$ *is* $\frac{1}{k!}\sum_w \kappa_{\Upsilon}(w)\, z^w$*, rung* k *of its still (Proposition 9.5): the even statistics carry the even rungs and the odd statistics the odd rungs.*

(iii) As $\lambda \to \infty$*,* $E_j(\lambda z)/\lambda \to \mathrm{wd}_j(z)$*, the width of Proposition 38.4, and* $O_j(\lambda z)/\lambda \to h_j(z) - h_j(-z)$*, the maximum of the scores being at the greedy basis* $T^+_{j,\pi}$ *and the minimum at* $T^-_{j,\pi}$ *for an order* π *along which the coordinates of* z *decrease (Lemma 8.5). So* $\Phi^{\circ}(\lambda z)/\lambda \to$

$\langle 2v_\pi - 2\sigma\mathbf{1}, z\rangle$ *with* $v_\pi = \sum_j \left(s_j \mathbf{1}_{T^+_{j,\pi}} + t_j \mathbf{1}_{\mathcal{E}\setminus T^-_{j,\pi}}\right)$*, for* $s_j, t_j \geq 0$ *a vertex of the Newton sum at which* $\langle \cdot, z\rangle$ *is largest, the* greedy vertex *of* π*; and for a question the exponent of Proposition 38.4 has* $X(\lambda z)/\lambda \to -2\kappa(z) = 2\langle \mathbf{m} - v_\pi, z\rangle$ *on* $\mathbf{1}^\perp$*: the walls of the tails of* π *are the tail sums of the masses less the greedy vertex,* $\kappa_{U_k} = m_{U_k} - (v_\pi)_{U_k}$*.*

Proposition 39.18 (The laws of the questions are an exponential family)**.** *Let a connected family be such that* 1*, the coordinates* z_e*,* $e \neq g$*, of a chart* $z_g = 0$ *of* $\mathcal{T}$ *and the statistics* E_j*,* O_j *are linearly independent functions on* $\mathcal{T}$*. Then the laws of its questions with every proper wall positive form a regular exponential family of dimension* $n-1+2N$*, with natural parameter* $(2\boldsymbol{\mu}_\perp; -\boldsymbol{\sigma}, -\boldsymbol{\delta})$*, statistic* $(z; E_j(z), O_j(z))_j$ *and cumulant function* $\log P_F\big(\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}); \boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$*. For the sample means* $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *of* q *independent tilts, as in Proposition 39.7:*

(i) $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *is sufficient, and a function of it with expectation zero under every law of the family vanishes almost surely;*

(ii) it is the unbiased estimator of least variance of the mean parameters

$$\Big(\frac{1}{2}\nabla_{\boldsymbol{\mu}_\perp} \log P_F;\ -\partial_{\sigma_j} \log P_F,\ -\partial_{\delta_j} \log P_F\Big)_j,$$

with covariance $\frac{1}{q}$ *times the Hessian of* $\log P_F$ *in the natural parameter;*

(iii) the likelihood equations have a solution, unique, exactly when $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *is interior to the convex hull of the image of* $z \mapsto (z; E_j(z), O_j(z))_j$*;*

(iv) for $\delta_j \leq \delta_0$ *against* $\delta_j > \delta_0$*, with the other parameters unknown, rejecting for small* $\bar{O}_j$ *given the other coordinates of the sample means is uniformly most powerful among unbiased tests, with a conditional null law free of the other parameters; likewise for* σ_j *and* $\bar{E}_j$*;*

(v) the law of the sums $q\bar{Z}$*,* $q\bar{E}_j$*,* $q\bar{O}_j$ *has the Laplace transform of Proposition 39.7(iv), so the power of every test is a computation in the family.*

When the complement of every basis of table j *is a basis with the same weight,* $O_j \equiv 0$ *and the statements hold with* δ_j *removed.*

A word read at Dirichlet draws is a dial: its power means run from the mean of the logarithms to their maximum as the exponent grows, and they keep a finite expectation up to the first wall met along the ray of exponents, where the tail of the logarithm takes over with the rate of that wall. At a centre the centred potential $\sum_j \sigma_j E_j$ is least at the table's gauge and its exponential moments along the strengths close every wall at once, while the odd statistics are jointly symmetric and every odd direction has both tails at the rate of its slope. The two statistics of a table are the soft maximum and the soft minimum of the scores of its bases, read as their difference and their sum: at the table's gauge their Taylor terms are the rungs of the still, the even ones carried by the strengths and the odd ones by the leans, and at infinity they read the greedy vertex of the Newton sum, whose tail sums against the masses are the walls. The laws of the questions are a regular exponential family in the tilt and these statistics, so the classical optimal estimators and tests are computations with periods.

39.3 Draws, orders and races

Proposition 39.19 (A sorted draw lives on the sector of its order)**.** *Let a family have real data on its balance wall with every proper wall positive, and let* $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$*. For an order* π *of* $\mathcal{E}$

let κ_k be the walls of its tails and I_π its sector integral (Corollary 8.8), so that $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi I_\pi$ (Theorem 8.7(ii)).

(i) $\mathbb{P}\big(U_{\pi(1)} > \cdots > U_{\pi(n)}\big) = I_\pi / P_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

(ii) Given that order, the gaps $u_k = \log\big(U_{\pi(k-1)}/U_{\pi(k)}\big)$, $2 \le k \le n$, have the density

$$I_\pi^{-1} \exp\Big(-\sum_{k=2}^{n} \kappa_k u_k\Big) \prod_j k_{j,\pi}(e^{-u})^{-s_j} \psi_{j,\pi}(e^{-u})^{-t_j}$$

on $[0,\infty)^{n-1}$, with the sector letters $k_{j,\pi}, \psi_{j,\pi}$ of Theorem 8.7(ii).

(iii) $\theta_{j,T^+_{j,\pi}} \le k_{j,\pi} \le K_j(\mathbf{1})$ *and* $\theta_{j,T^-_{j,\pi}} \le \psi_{j,\pi} \le K_j(\mathbf{1})$ *on the cube, so the density of (ii) lies between constant multiples of the density of independent exponential gaps of the rates κ_k; for $s_j, t_j \ge 0$, I_π lies between $\prod_j K_j(\mathbf{1})^{-s_j - t_j} \prod_k \kappa_k^{-1}$ and $\prod_j \theta_{j,T^+_{j,\pi}}^{-s_j} \theta_{j,T^-_{j,\pi}}^{-t_j} \prod_k \kappa_k^{-1}$.*

(iv) On the sector of π the largest monomial U^T over the bases of table j is at its greedy basis $T^+_{j,\pi}$ and the smallest at $T^-_{j,\pi}$ (Lemma 8.5): the best and the worst basis of a draw are functions of its order, and the best basis of table j at U is T with probability $\sum_{\pi:\, T^+_{j,\pi} = T} I_\pi / P_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

Proposition 39.20 (An ordered sample and a cut)**.** *Let $Y_1, \dots, Y_q$ be independent with a probability law ϕ on $\mathbb{R}$, an input (Definition 5.4) that charges no point.*

(i) The ordered sample has the law $q!\,\phi^{\otimes q}$ on the region $\{y_1 < \cdots < y_q\}$.

(ii) The number of observations above a cut N is binomial with q trials and the chance $\phi((N,\infty))$, which for ϕ living on $(0,\infty)$ and $N \ge 0$ is the Trailmoment $(\mathsf{T}_{0;(1)}[x]\,\phi)((N,\infty))$ at index 0; given that it is d, the ordered observations above the cut have the law $d!\,\phi^{\otimes d}/\phi((N,\infty))^d$ on the trail region $X_N = \{N < y_1 < \cdots < y_d\}$.

(iii) If ϕ lives on $(0,\infty)$ and $N \ge 0$, the empirical input $\hat\phi = \frac{1}{q}\sum_i \delta_{Y_i}$ has $(\mathsf{T}_{\nu;(1)}[x]\,\hat\phi)((N,\infty)) = \frac{1}{q}\sum_{Y_i > N} Y_i^{-\nu}$, with the mean $(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ and the variance

$$\frac{1}{q}\Big[(\mathsf{T}_{2\nu;(1)}[x]\,\phi)((N,\infty)) - (\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))^2\Big].$$

Proposition 39.21 (The races and the dilation relation)**.** *For table j of a family and a tilt z, with $\alpha = e^{2z}$, the laws of the winners of the basis race and the cobasis race (Theorem 17.5) are $\mathbb{P}_{j,\alpha}$ and $\mathbb{P}^*_{j,\alpha}$; let*

$$\mathrm{Ent}_j(z) = -\sum_T \mathbb{P}_{j,\alpha}(T) \log \frac{\mathbb{P}_{j,\alpha}(T)}{\theta_{j,T}}, \qquad \mathrm{Ent}^*_j(z) = -\sum_T \mathbb{P}^*_{j,\alpha}(T) \log \frac{\mathbb{P}^*_{j,\alpha}(T)}{\theta_{j,T}}$$

be their entropies relative to the weights.

(i) $E_j(z) = \langle z, \nabla E_j(z)\rangle + \mathrm{Ent}_j(z) + \mathrm{Ent}^*_j(z)$ *and* $O_j(z) = \langle z, \nabla O_j(z)\rangle + \mathrm{Ent}_j(z) - \mathrm{Ent}^*_j(z)$, *with*

$$\langle z, \nabla E_j(z)\rangle = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}\big[\langle \chi_T, z\rangle\big] - \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[\langle \chi_T, z\rangle\big],$$
$$\langle z, \nabla O_j(z)\rangle = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}\big[\langle \chi_T, z\rangle\big] + \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[\langle \chi_T, z\rangle\big].$$

Let the family be connected and let a question $(\mathbf{m};\mathbf{s},\mathbf{t})$ of pair coordinates $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ have every proper wall positive. In the rest of this statement $\mathbb{E}$ is the expectation of $U \sim \mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$, Z is its tilt and $\mathbf{u}(\alpha) = \sum_j \big(s_j R_j(\alpha) + t_j R^*_j(\alpha)\big)$ the usage.

(ii) Dilation relation.

$$2\langle \boldsymbol{\mu}_\perp, \mathbb{E}[Z]\rangle - \sum_j \big(\sigma_j \mathbb{E}[E_j(Z)] + \delta_j \mathbb{E}[O_j(Z)]\big) = -(n-1) - \sum_j \big(s_j \mathbb{E}[\mathrm{Ent}_j(Z)] + t_j \mathbb{E}[\mathrm{Ent}_j^*(Z)]\big);$$

at a centre, $\sum_j \sigma_j \mathbb{E}[E_j(Z)] = n - 1 + \sum_j \sigma_j \mathbb{E}[\mathrm{Ent}_j(Z) + \mathrm{Ent}_j^*(Z)]$.

(iii) $\mathrm{Cov}\big(\mathbf{u}_e(U), Z_f\big) = \frac{1}{2}\big(\mathbf{1}[e=f] - \frac{1}{n}\big)$, *$Z$ read on* $\mathbf{1}^\perp$.

(iv) $\log P_F$ *is convex in the pair coordinates of the question, so for a balanced direction* $v = (\dot{\mathbf{m}}; \dot{\mathbf{s}}, \dot{\mathbf{t}})$ *with the questions at the data* $(\mathbf{m} \pm \dot{\mathbf{m}}; \mathbf{s} \pm \dot{\mathbf{s}}, \mathbf{t} \pm \dot{\mathbf{t}})$ *inside their walls,*

$$\log \frac{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})}{P_F(\mathbf{m} - \dot{\mathbf{m}}; \mathbf{s} - \dot{\mathbf{s}}, \mathbf{t} - \dot{\mathbf{t}})} \le D_v \log P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) \le \log \frac{P_F(\mathbf{m} + \dot{\mathbf{m}}; \mathbf{s} + \dot{\mathbf{s}}, \mathbf{t} + \dot{\mathbf{t}})}{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})},$$

bracketing each mean of Proposition 39.7(ii) by neighbouring values of the period.

Proposition 39.22 (Moments are steps; entropies are derivatives one column up)**.** *Let a connected family carry a question* $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *with every proper wall positive, and in this statement let* $\mathbb{E}$ *be the expectation of* $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$.

(i) For a real balanced shift $(\dot{\mathbf{m}}; \dot{\mathbf{s}}, \dot{\mathbf{t}})$ *with every proper wall of the shifted question positive,*

$$\mathbb{E}\Big[U^{\dot{\mathbf{m}}} \prod_j K_j(U)^{-\dot{s}_j} \Psi_j(U)^{-\dot{t}_j}\Big] = \frac{P_F(\mathbf{m} + \dot{\mathbf{m}}; \mathbf{s} + \dot{\mathbf{s}}, \mathbf{t} + \dot{\mathbf{t}})}{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})}.$$

(ii) For a basis T *of table* j *the* columns $v_{j,T} = (\mathbf{1}_T; \mathbf{e}_j, \mathbf{0})$ *and* $v^*_{j,T} = (\mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{0}, \mathbf{e}_j)$ *are balanced, and for* $a \ge 0$

$$\mathbb{E}\Big[\Big(\frac{\mathbb{P}_{j,U}(T)}{\theta_{j,T}}\Big)^a\Big] = \frac{P_F(\mathbf{m} + a\mathbf{1}_T;\ \mathbf{s} + a\mathbf{e}_j,\ \mathbf{t})}{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})},$$
$$\mathbb{E}\Big[\Big(\frac{\mathbb{P}^*_{j,U}(T)}{\theta_{j,T}}\Big)^a\Big] = \frac{P_F(\mathbf{m} + a\mathbf{1}_{\mathcal{E}\setminus T};\ \mathbf{s},\ \mathbf{t} + a\mathbf{e}_j)}{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})},$$

the walls only growing along a column.

(iii) With the entropies of Proposition 39.21,

$$\mathbb{E}[\mathrm{Ent}_j(Z)] = -\sum_T \theta_{j,T} \frac{(D_{v_{j,T}} P_F)(\mathbf{m} + \mathbf{1}_T;\ \mathbf{s} + \mathbf{e}_j,\ \mathbf{t})}{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})},$$
$$\mathbb{E}[\mathrm{Ent}_j^*(Z)] = -\sum_T \theta_{j,T} \frac{(D_{v^*_{j,T}} P_F)(\mathbf{m} + \mathbf{1}_{\mathcal{E}\setminus T};\ \mathbf{s},\ \mathbf{t} + \mathbf{e}_j)}{P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})},$$

Z the tilt of U: an entropy is a derivative of the period one column up.

Corollary 39.23 (A derivative adds one finite sum)**.** *The series of Corollary 15.11 may be differentiated term by term along a balanced direction* $v = (\dot{\mathbf{m}}; \dot{\mathbf{s}}, \dot{\mathbf{t}})$ *at a point inside the walls, with* $\dot{\kappa}_S$ *the change of the walls; each factor of a term has an elementary logarithmic derivative,*

$$D_v \log \frac{(s_j)_{n_j}}{n_j!} = \dot{s}_j \sum_{i<n_j} \frac{1}{s_j + i}, \qquad D_v \log \frac{(t_j)_{n'_j}}{n'_j!} = \dot{t}_j \sum_{i<n'_j} \frac{1}{t_j + i},$$

$$D_v \log K_j(\mathbf{1})^{-s_j - t_j} = -(\dot{s}_j + \dot{t}_j) \log K_j(\mathbf{1}),$$
$$D_v \log \Gamma_{\,1}^{\,2} \begin{bmatrix} \kappa_{U_k} + \beta_k,\ \gamma_k + 1 \\ \kappa_{U_k} + \beta_k + \gamma_k + 1 \end{bmatrix} = -\dot{\kappa}_{U_k} \sum_{i=0}^{\gamma_k} \frac{1}{\kappa_{U_k} + \beta_k + i},$$

so every quantity of Proposition 39.22 is the same positive series, at the question or one column up, with each term multiplied by a finite sum of reciprocals and logarithms of total weights.

Sorting a draw of the shape law lands it in one sector of the period: the chance of an order is the fraction of the period carried by its sector, the gaps of the sorted draw are exponential at the rates of the walls of the tails up to bounded factors, and the best and the worst basis of the draw are the greedy bases of its order. The races read the statistics as entropies, and the dilation relation is the balance between the strength of a question and the entropies of its winners. Every moment of a letter, of a winner's chance or of the shape is a step of the data, and every entropy is a derivative one column up, so on the closing series of the period each is a positive series whose terms carry one extra finite sum.

Part VIII

Proofs of the general forms

Chapter 40

Proofs of the general forms

This part proves the statements that are more general than their counterparts in the reference *The Foreman and the serif family* (16 October 2026), and those with no counterpart there; in the source file the comment before each such statement says so. Statements are cited by their numbers, and every symbol and term is that of the chapter the statement belongs to. A statement of the reference that is a special case of a statement proved here is not proved again; where reading it off needs a word, the word follows the proof. Lemmas that belong to this part alone are numbered S1, S2, ... The chapters are taken in order, one section each, and the last section records the checks run on small families.

40.1 Chapter 1: conventions, the stacked Gamma and the finite part

Lemma S1. *Let $\Theta \subset \mathbb{R}$ be finite and $\phi(y) = \sum_{\theta\in\Theta} a_\theta e^{i\theta y}$ with $a_\theta \in \mathbb{C}$. If $\phi(y) \to 0$ as $y \to -\infty$, then every a_θ is 0.*

Proof. Fix $\theta_0 \in \Theta$. For $Y > 0$,

$$\frac{1}{Y}\int_{-2Y}^{-Y} \phi(y)\, e^{-i\theta_0 y}\, dy = a_{\theta_0} + \sum_{\theta\neq\theta_0} a_\theta\, \frac{1}{Y}\int_{-2Y}^{-Y} e^{i(\theta-\theta_0)y}\, dy,$$

and each integral on the right has modulus at most $2/|\theta-\theta_0|$, so the sum is $O(Y^{-1})$. The left side has modulus at most $\sup_{y\le -Y}|\phi(y)|$, which tends to 0. Letting $Y \to \infty$ gives $a_{\theta_0} = 0$. □

Lemma S2. *Let $g(x) = \sum_{(k,j)\in F} c_{k,j}\, x^k \log^j x$ for $x \in (0,1)$, where $F \subset \mathbb{C}\times\mathbb{N}$ is finite and nonempty and every $c_{k,j}$ is nonzero. Let κ be the least real part of the exponents k, $(k,j) \in F$, and J the largest j such that $(k,j) \in F$ for some k with $\operatorname{Re} k = \kappa$. Then $g(x)/(x^\kappa \log^J x)$ does not tend to 0 as $x \to 0$.*

Proof. Put $y = \log x$, so that $y \to -\infty$ as $x \to 0$ and $x^k = e^{ky}$. Then

$$\frac{g(x)}{x^\kappa \log^J x} = \sum_\theta c_{\kappa+i\theta,J}\, e^{i\theta y} + \sum_{(k,j)} c_{k,j}\, e^{(k-\kappa)y}\, y^{j-J},$$

the first sum over the real θ with $(\kappa + i\theta, J) \in F$ and the second over the other elements of F. Every term of the second sum tends to 0: if $\operatorname{Re} k = \kappa$, then $j < J$, $|e^{(k-\kappa)y}| = 1$ and $|y|^{j-J} \to 0$;

if $\operatorname{Re} k > \kappa$, then $|e^{(k-\kappa)y} y^{j-J}| = e^{(\operatorname{Re} k-\kappa)y} |y|^{j-J} \to 0$. The first sum is a function $\phi(y)$ as in Lemma S1 with a nonzero coefficient, so it does not tend to 0 as $y \to -\infty$, and neither does the left side. □

Definition 1.4 (The finite part). As stated: the exponents of a log-power expansion are complex numbers, and its coefficients are determined by the function.

Proof. What is to be shown is that the coefficients of a log-power expansion are determined by h, so that the finite parts are well defined; the rest of the definition is that of the reference, with complex exponents.

(1) One remainder. Let two finite sums S_1 and S_2 of terms $c\,x^k \log^j x$, every exponent of real part below N, both differ from h by $O(x^N)$ as $x \to 0$. Their difference $g = S_1 - S_2$ is a finite sum of the same kind with $g(x) = O(x^N)$. If g had a nonzero coefficient, then, with $\kappa < N$ and J as in Lemma S2, $|g(x)/(x^\kappa \log^J x)| \le C x^{N-\kappa} |\log x|^{-J} \to 0$, which that lemma excludes. So S_1 and S_2 have the same coefficients.

(2) Two remainders. Let $N < N'$, and let S and S' be finite sums for N and for N'. A term $x^k \log^j x$ of S' with $\operatorname{Re} k \ge N$ is $O(x^{N-\delta})$ for every $\delta > 0$, so the part $S'_{<N}$ of S' with exponents of real part below N has $h - S'_{<N} = O(x^{N-\delta})$. Choose $\delta > 0$ smaller than $N - \operatorname{Re} k$ for every exponent k of S and of $S'_{<N}$. Then S and $S'_{<N}$ both differ from h by $O(x^{N-\delta})$, with all exponents of real part below $N - \delta$, so they coincide by (1). Hence the coefficient of a term $x^k \log^j x$ is the same for every $N > \operatorname{Re} k$, and the coefficient of $x^0 \log^0 x$, read with any $N > 0$, is determined by h. At ∞ the same holds for $h(1/x)$ at 0: the term $x^{-k} \log^j x$ is $(-1)^j x'^k \log^j x'$ in $x' = 1/x$, and $x^0 \log^0 x$ is $x'^0 \log^0 x'$.

(3) The primitive. Two primitives of f differ by a constant C, whose expansion at either end is the single term $C\,x^0 \log^0 x$; by (2) the finite parts of the second primitive at 0 and at ∞ are those of the first plus C. So $\operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h$ and $h(b) - \operatorname{Reg}_{x\to 0} h$ do not depend on the primitive. □

Proposition 1.5 (A convergent integral is its own finite part). *As stated, for log-power expansions with complex exponents.*

Proof. Let h be a primitive of f with log-power expansions at both ends. Since $\int_0^\infty f\,dx$ converges, $h(x) = h(1) - \int_x^1 f$ has a limit L_0 as $x \to 0$, h has a limit L_∞ as $x \to \infty$, and $\int_0^\infty f\,dx = L_\infty - L_0$. Take the expansion at 0 for some $N > 0$, and let G be its part with exponents of real part at most 0. The other terms and the remainder tend to 0, so $G(x) \to L_0$. Write $G - L_0$ as a finite sum of terms $c\,x^k \log^j x$, the constant $-L_0$ joining the coefficient of $x^0 \log^0 x$, and suppose it has a nonzero coefficient; let $\kappa \le 0$ and J be as in Lemma S2. If $(\kappa, J) \ne (0,0)$, then $|x^\kappa \log^J x| \to \infty$, so $(G - L_0)/(x^\kappa \log^J x) \to 0$. If $(\kappa, J) = (0,0)$, every exponent of $G - L_0$ has real part 0 and every power of the logarithm is 0, and $(G - L_0)/(x^0 \log^0 x) = G - L_0 \to 0$. Both are excluded by Lemma S2. So G is the constant L_0: the expansion at 0 has no term with an exponent of real part at most 0 other than $L_0\,x^0 \log^0 x$, and $\operatorname{Reg}_{x\to 0} h = L_0$. In the same way, for $h(1/x)$ at 0, $\operatorname{Reg}_{x\to\infty} h = L_\infty$. Hence $\operatorname{Reg} \int_0^\infty f\,dx = L_\infty - L_0 = \int_0^\infty f\,dx$. □

Theorem 1.6 (A finite part is a constant term). *As stated, for log-power expansions with complex exponents.*

Proof. The steps are those of the reference, proof of 1.6; they hold at complex exponents as follows.

The end at 0. On $(0,1]$ write $f = \sum c_{b,j} x^b \log^j x + r_N$, the sum over the finitely many terms of the expansion at 0 with $\operatorname{Re} b < N$, where $N > 0$ and $r_N = O(x^N)$ as $x \to 0$; r_N is integrable on $(0,1]$. For $b \neq -1$ put

$$Q_{b,j}(y) = \sum_{i=0}^{j} (-1)^{j-i} \frac{j!}{i!\,(b+1)^{j-i+1}}\, y^i;$$

then $(b+1)Q_{b,j} + Q'_{b,j} = y^j$, so $\frac{d}{dx}\big(x^{b+1} Q_{b,j}(\log x)\big) = x^b \log^j x$, and $Q_{b,j}(0) = (-1)^j j!/(b+1)^{j+1}$. Hence

$$\int_x^1 t^b \log^j t\, dt = Q_{b,j}(0) - x^{b+1} Q_{b,j}(\log x) \quad (b \neq -1), \qquad \int_x^1 t^{-1} \log^j t\, dt = -\frac{\log^{j+1} x}{j+1},$$

and $\int_x^1 r_N = \int_0^1 r_N - \int_0^x r_N$ with $\int_0^x r_N = O(x^{N+1})$. So $\int_x^1 f$ has a log-power expansion at 0, and its coefficient of $x^0 \log^0 x$ is $\sum_{b \neq -1} c_{b,j} Q_{b,j}(0) + \int_0^1 r_N$: a term $x^{b+1} \log^i x$ has the exponent $b+1 \neq 0$, and a term $\log^{j+1} x$ has a positive power of the logarithm. For a primitive h of f, $h(1) - \operatorname{Reg}_{x\to 0} h = \operatorname{Reg}_{x\to 0} \int_x^1 f$, so by 1.4

$$\operatorname{Reg} \int_0^1 f\, dx = \sum_{b \neq -1} c_{b,j} \frac{(-1)^j j!}{(b+1)^{j+1}} + \int_0^1 r_N\, dx.$$

On the other side, by $\int_0^1 x^{s-1} \log^j x\, dx = (-1)^j j!/s^{j+1}$ for $\operatorname{Re} s > 0$,

$$\int_0^1 x^\varepsilon f\, dx = \sum c_{b,j} \frac{(-1)^j j!}{(b+1+\varepsilon)^{j+1}} + \int_0^1 x^\varepsilon r_N\, dx$$

for $\operatorname{Re}\varepsilon$ large, the last term holomorphic on $\operatorname{Re}\varepsilon > -N-1$; this is the meromorphic continuation there. Its constant term at $\varepsilon = 0$ is the sum displayed above: a term with $b \neq -1$ gives its value, and a term with $b = -1$ is the pure pole $(-1)^j j!\,\varepsilon^{-j-1}$, of constant term 0.

The end at ∞. By $x \mapsto 1/x$, $\int_1^\infty x^\varepsilon f\, dx = \int_0^1 x^{-\varepsilon} x^{-2} f(1/x)\, dx$ and $\int_1^x f = \int_{1/x}^1 t^{-2} f(1/t)\, dt$; the function $x^{-2} f(1/x)$ has a log-power expansion at 0, the coefficient of $x^0 \log^0 x$ at ∞ is that of $x'^0 \log^0 x'$ at 0 in $x' = 1/x$, and a constant term at $\varepsilon = 0$ does not change under $\varepsilon \mapsto -\varepsilon$. So $\operatorname{Reg}_{x\to\infty} \int_1^x f = \operatorname{CT}_{\varepsilon=0} \int_1^\infty x^\varepsilon f\, dx$ by the end at 0.

Adding, $\operatorname{Reg} \int_0^\infty f\, dx = \operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h = \operatorname{Reg}_{x\to\infty} \int_1^x f + \operatorname{Reg}_{x\to 0} \int_x^1 f$ is the constant term of the sum of the two continuations. □

Corollary 1.7 (Finite parts at a pole of a Mellin integral). *As stated: for every f with a log-power expansion at* 0*, at every $a \in \mathbb{C}$.*

Proof. Fix a and $N > 1 - \operatorname{Re} a$, and write $f = \sum c_{k,j} x^k \log^j x + r_N$ on $(0,1]$, the finitely many terms of the expansion with $\operatorname{Re} k < N$, $r_N = O(x^N)$ at 0 and integrable on $[\delta, 1]$ for every $\delta > 0$. For $\operatorname{Re} a'$ large,

$$I(a') = \sum c_{k,j} \frac{(-1)^j j!}{(a'+k)^{j+1}} + \int_0^1 x^{a'-1} r_N\, dx,$$

the last term holomorphic on $\operatorname{Re} a' > -N$. This is the meromorphic continuation of I on $\operatorname{Re} a' > -N$; its poles lie at the points $-k$, a term $x^k \log^j x$ giving a pole of order at most $j+1$, and since N is arbitrary this describes the continuation on $\mathbb{C}$. Its constant term at $a' = a$ is

$$\operatorname*{CT}_{a'=a} I(a') = \sum_{a+k \neq 0} c_{k,j} \frac{(-1)^j j!}{(a+k)^{j+1}} + \int_0^1 x^{a-1} r_N\, dx,$$

a term with $a+k=0$ being the pure pole $(-1)^j j!\,(a'-a)^{-j-1}$; at an a that is not a pole it is $I(a)$. On the other side, $x^{a-1}f$ has the log-power expansion at 0 with the terms $c_{k,j}x^{a+k-1}\log^j x$ and the remainder $x^{a-1}r_N = O(x^{N+\operatorname{Re} a-1})$, $N+\operatorname{Re} a-1>0$. The end at 0 in the proof of 1.6 above, with $b=a+k-1$ (so $b\neq -1$ exactly when $a+k\neq 0$, and $b+1=a+k$), gives for $\operatorname{Reg}\int_0^1 x^{a-1}f\,dx$ the same sum. □

Lemma 1.8 (A finite part depends on the coordinate). *As stated: the dilation rule for every function with log-power expansions.*

Proof. (a) The finite part of $h(cx^p)$. Let $h(y)=\sum c_{k,j}\,y^{-k}\log^j y+O(y^{-N})$ as $y\to\infty$, finitely many terms with $\operatorname{Re} k<N$, so that $a_j=c_{0,j}$. For $y=cx^p$, $y^{-k}=c^{-k}x^{-pk}$ and $\log y=\log c+p\log x$, so

$$h(cx^p)=\sum c_{k,j}\,c^{-k}x^{-pk}\sum_{i=0}^{j}\binom{j}{i}(\log c)^{j-i}p^i\log^i x+O(x^{-pN}),$$

a log-power expansion at ∞ with the exponents pk, of real parts below pN; as N is arbitrary, so is pN. The term $x^0\log^0 x$ comes from $pk=0$, that is $k=0$, and $i=0$; by 1.4 its coefficient $\sum_j c_{0,j}(\log c)^j=\sum_j a_j(\log c)^j$ is $\operatorname{Reg}_{x\to\infty}h(cx^p)$. At 0 the computation is the same.

(b) The finite part of the integral. Let $F(x)=\int_1^x f$. At ∞, $f(t)=\sum f_{k,j}t^{-k}\log^j t+O(t^{-N})$ with $N>1$; integrating from 1 to x, a term with $k\neq 1$ gives $x^{1-k}Q_{-k,j}(\log x)$ up to a constant (with $Q_{b,j}$ as in the proof of 1.6 above), a term with $k=1$ gives $\log^{j+1}x/(j+1)$, and the remainder gives a constant plus $O(x^{1-N})$. So F has a log-power expansion at ∞, and for $i\geq 1$ its coefficient of $x^0\log^i x$ comes from the term $k=1$, $j=i-1$ alone: it is f^∞_{i-1}/i. In the same way, at 0 the coefficient of $x^0\log^i x$ in F is f^0_{i-1}/i for $i\geq 1$. Write A^∞_i and A^0_i for the coefficients of $x^0\log^i x$ of F at ∞ and at 0; $A^\infty_0-A^0_0=\operatorname{Reg}\int_0^\infty f\,dx$. The function $G(x)=F(cx^p)$ is a primitive of $f(cx^p)\,c\,p\,x^{p-1}$, and by (a) at both ends

$$\operatorname{Reg}\int_0^\infty f(cx^p)\,c\,p\,x^{p-1}dx=\sum_{i\geq 0}\big(A^\infty_i-A^0_i\big)(\log c)^i=\operatorname{Reg}\int_0^\infty f\,dx+\sum_{i\geq 1}\frac{f^\infty_{i-1}-f^0_{i-1}}{i}(\log c)^i,$$

which is the display with $i=j+1$. At $c=1$ the correction vanishes: a power of the coordinate leaves the finite part unchanged. □

Remark. The reference states the instance $f(x)=1/(1+cx)$, $c>0$. This f has the coefficient $f^\infty_0=1/c$ of x^{-1} at ∞ and no term $x^{-1}\log^j x$ at 0, and with the dilation c^{-1} and $p=1$ the lemma gives $\frac{1}{c}\operatorname{Reg}\int_0^\infty\frac{dx'}{1+x'}=\operatorname{Reg}\int_0^\infty\frac{dx}{1+cx}-\frac{\log c}{c}$. The left side is 0, the primitive $\log(1+x')=\log x'+\log(1+1/x')$ having no constant term at either end; so $\operatorname{Reg}\int_0^\infty\frac{dx}{1+cx}=\frac{\log c}{c}$, the value of the reference. A second check, with a logarithm: $f(x)=\log(1+x)/(1+x)$ has the primitive $\frac12\log^2(1+x)$, with no constant term at either end, so $\operatorname{Reg}\int_0^\infty f\,dx=0$; at ∞, $f=x^{-1}\log x+O(x^{-2}\log x)$, so $f^\infty_1=1$ and $f^\infty_0=0$, and at 0 it has no term $x^{-1}\log^j x$. The lemma gives $\operatorname{Reg}\int_0^\infty f(cx)\,c\,dx=\frac12\log^2 c$; directly, the primitive $\frac12\log^2(1+cx)=\frac12\big(\log c+\log x+\log(1+\frac{1}{cx})\big)^2$ has the constant term $\frac12\log^2 c$ at ∞ and none at 0.

40.2 Chapter 2: the Gramma and the Zamma

Proposition 2.4 (Affine changes of the logarithmic coordinates). *As stated: one statement for the reflection, the linear changes of the coordinates and the torus homogeneity.*

Proof. Let $A \in GL_{\mathcal{E}}(\mathbb{R})$ and $\mathbf{h} \in \mathbb{R}^{\mathcal{E}}$. The map $y' \mapsto y = Ay' + \mathbf{h}$ is a bijection of $\mathbb{R}^{\mathcal{E}}$ with $dy = |\det A|\, dy'$, and $\langle \nu, y\rangle = \langle A^{\mathsf{T}}\nu, y'\rangle + \langle \nu, \mathbf{h}\rangle$, $\langle t_k, y\rangle = \langle A^{\mathsf{T}}t_k, y'\rangle + \langle t_k, \mathbf{h}\rangle$. In the logarithmic form of 2.1,

$$e^{\langle \nu, y\rangle} \exp\Big(-\sum_k \xi_k e^{\langle t_k, y\rangle}\Big)\, dy = |\det A|\, e^{\langle \nu, \mathbf{h}\rangle}\, e^{\langle A^{\mathsf{T}}\nu, y'\rangle} \exp\Big(-\sum_k \big(e^{\langle t_k, \mathbf{h}\rangle}\xi_k\big)\, e^{\langle A^{\mathsf{T}}t_k, y'\rangle}\Big)\, dy'.$$

The moduli of the two sides are the same expressions with ν replaced by $\operatorname{Re}\nu$, so the integral of the left side converges exactly when that of the right side does. The left side is the integrand of $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ and the right side, up to the constant factor, that of $\mathsf{G}_{A^{\mathsf{T}}\nu;A^{\mathsf{T}}t}\big((e^{\langle t_k, \mathbf{h}\rangle}\xi_k)_k\big)$, with positive coefficients; by 2.2, $\nu \in \mathcal{T}_t$ exactly when $A^{\mathsf{T}}\nu \in \mathcal{T}_{A^{\mathsf{T}}t}$, that is $\mathcal{T}_{A^{\mathsf{T}}t} = A^{\mathsf{T}}\mathcal{T}_t$, and integrating gives the display. At $A = -I_{\mathcal{E}}$ and $\mathbf{h} = 0$, $|\det A| = 1$ and $A^{\mathsf{T}}t = -t$: the reflection. At $\mathbf{h} = 0$ it is the linear change of 2.9. At $A = I_{\mathcal{E}}$ and $\mathbf{h} = \log\lambda$, $\lambda \in \mathbb{R}^{\mathcal{E}}_{>0}$, $e^{\langle t_k, \mathbf{h}\rangle} = \lambda^{t_k}$ and $e^{\langle \nu, \mathbf{h}\rangle} = \lambda^{\nu}$, which is 2.3(iii). □

Theorem 2.5 (Product). *As stated, for any number of blocks.*

Proof. Write $x = (x^{(1)}, \dots, x^{(q)})$ with $x^{(i)} \in \mathbb{R}^{\mathcal{E}_i}_{>0}$. Each t_k is supported in one $\mathcal{E}_i$, so $x^{t_k} = (x^{(i)})^{t_k|\mathcal{E}_i}$ for that i, and

$$x^{\nu-\mathbf{1}} \exp\Big(-\sum_k \xi_k x^{t_k}\Big) = \prod_{i=1}^{q} (x^{(i)})^{\nu|\mathcal{E}_i - \mathbf{1}} \exp\Big(-\sum_{k\in K_i} \xi_k\, (x^{(i)})^{t_k|\mathcal{E}_i}\Big),$$

K_i the set of the terms whose exponents are supported in $\mathcal{E}_i$, the zero vectors counted in K_1. The moduli of the factors are positive functions of separate variables, so by Tonelli the modulus of the product is integrable exactly when every factor is integrable over $\mathbb{R}^{\mathcal{E}_i}_{>0}$; by 2.2 this says $\mathcal{T}_t = \mathcal{T}_{t^{(1)}} \times \cdots \times \mathcal{T}_{t^{(q)}}$, and Fubini gives the product. □

Theorem 2.8 (Scale and shape along sets). *As stated, for several disjoint sets at once.*

Proof. Apply 2.7 on each B_i: $x_{B_i} = \rho_i u_i$ with $\rho_i > 0$ and $u_i \in \Delta_{B_i}$, and $x_{B_i}^{\nu|_{B_i}-\mathbf{1}} dx_{B_i} = \rho_i^{\nu_{B_i}-1} u_i^{\nu|_{B_i}-\mathbf{1}}\, d\rho_i\, d\omega_{B_i}(u_i)$. With x_R the coordinates on R, the map $(x_R, (\rho_i)_i, (u_i)_i) \mapsto x$ is a bijection onto $\mathbb{R}^{\mathcal{E}}_{>0}$, and since $(\rho_i u_i)^{t_k|_{B_i}} = \rho_i^{\langle t_k, \mathbf{1}_{B_i}\rangle} u_i^{t_k|_{B_i}}$,

$$x^{\nu-\mathbf{1}} dx = x_R^{\nu|_R - \mathbf{1}} \prod_i \rho_i^{\nu_{B_i}-1} u_i^{\nu|_{B_i}-\mathbf{1}}\, d\rho\, d\omega(u)\, dx_R, \qquad \xi_k x^{t_k} = b_k(u, x_R) \prod_i \rho_i^{\ell_{k,i}}.$$

At $\nu \in \mathcal{T}_t$ the Gramma converges (2.2), that is, its integrand is absolutely integrable. By Tonelli and Fubini the integral may be taken with the integral over $\rho = (\rho_i)_i \in \mathbb{R}^q_{>0}$ innermost, that integral converging absolutely for almost every (u, x_R), and it is

$$\int_{\mathbb{R}^q_{>0}} \rho^{(\nu_{B_i})_i - \mathbf{1}} \exp\Big(-\sum_k b_k(u, x_R)\, \rho^{\ell_k}\Big)\, d\rho = \mathsf{G}_{(\nu_{B_i})_i;\ell}\big(\mathbf{b}(u, x_R)\big),$$

the Gramma of 2.1 on q scales with the configuration ℓ. When R is empty there are no coordinates x_R. □

Proposition 2.13 (First facts about the Zamma). *As stated: (i) at every invertible square configuration and (iv) on any number of scales; (ii) and (iii) are those of the reference.*

Proof. (i) By 2.6, $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\prod_k \Gamma(b_k)\,\xi_k^{-b_k}$ on $\mathcal{T}_t = \{\nu : \operatorname{Re}\mathbf{b} \in (0,\infty)^N\}$; dividing by $\prod_e \Gamma(\nu_e)$ (2.12) gives the display, the quotient of Gammas being the stacked Gamma of 1.2, and the display carries the continuation. At the unit configuration $\mathbf{b} = \nu$ and the display is the monomial; on one scale with $t_1 > 0$, $|\det t| = t_1$ and $b_1 = \nu/t_1$, which is the formula of the reference.

(iv) Write $t'' = I_{\mathcal{E}} \oplus t'$ and $\boldsymbol{\xi}'' = \mathbf{1} \oplus \boldsymbol{\xi}'$. Since $\operatorname{cone}(t'') \supseteq \operatorname{cone}(I_{\mathcal{E}}) = [0,\infty)^{\mathcal{E}}$, every $\nu \in \mathbb{R}^{\mathcal{E}}_{>0}$ lies in $\operatorname{int}\operatorname{cone}(t'')$, hence in $\mathcal{T}_{t''}$. By 2.1,

$$\mathsf{G}_{\nu;t''}(\boldsymbol{\xi}'') = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}} e^{-\sum_e x_e} \exp\Big(-\sum_k \xi'_k x^{t'_k}\Big)\,dx = \prod_e \Gamma(\nu_e)\,\mathbb{E}_{U\sim\mathrm{Gamma}_\nu}\Big[\exp\Big(-\sum_k \xi'_k U^{t'_k}\Big)\Big],$$

Gamma_ν having the density $\prod_e x_e^{\nu_e-1}e^{-x_e}/\Gamma(\nu_e)$ (1.1(vii)); dividing by $\prod_e \Gamma(\nu_e)$ gives the display. The variable $\exp(-\sum_k \xi'_k U^{t'_k})$ lies in $(0,1]$ and decreases in each ξ'_k, and so does its expectation. Put $X = \sum_k \xi'_k U^{t'_k} \geq 0$. Where every $\nu + t'_k$ lies in $\mathbb{R}^{\mathcal{E}}_{>0}$, the coordinates of U being independent, $\mathbb{E}[U^{t'_k}] = \prod_e \mathbb{E}[U_e^{t'_{k,e}}] = \prod_e \Gamma(\nu_e + t'_{k,e})/\Gamma(\nu_e)$, so $\mathbb{E}X = \sum_k \xi'_k \prod_e \Gamma_1\begin{bmatrix}\nu_e + t'_{k,e}\\ \nu_e\end{bmatrix}$ is finite. The function $s \mapsto e^{-s}$ is convex, so $e^{-s} \geq e^{-s_0} - e^{-s_0}(s - s_0)$ for all real s, s_0; at $s = X$ and $s_0 = \mathbb{E}X$, taking expectations, $\mathbb{E}[e^{-X}] \geq e^{-\mathbb{E}X}$, which is the bound. □

Proposition 2.14 (Powers of a polynomial in the scales). *As stated: (i) and (iii) for every Zamma; (ii) and (iv) are those of the reference.*

Proof. Write $\boldsymbol{\xi}(\rho) = \boldsymbol{\xi}^{(0)} + \boldsymbol{\Delta}(\rho)$ with $\Delta_k(\rho) = \sum_{l\neq 0} \rho^l \xi_k^{(l)}$, a polynomial vanishing at $\rho = 0$.

(1) Derivatives. Away from the unit configuration, $\mathsf{Z}_{\nu;t} = \mathsf{G}_{\nu;t}/\prod_e \Gamma(\nu_e)$ on $(0,\infty)^N$. By 2.3(i), $\partial^{\mathbf{w}}\mathsf{G}_{\nu;t} = (-1)^{|\mathbf{w}|}\mathsf{G}_{\nu+t\mathbf{w};t}$, every $\nu + t\mathbf{w}$ lying in $\mathcal{T}_t$ because $\operatorname{int}\operatorname{cone}(t) + \operatorname{cone}(t) \subseteq \operatorname{int}\operatorname{cone}(t)$; so $\mathsf{Z}_{\nu;t}$ is infinitely differentiable on $(0,\infty)^N$ and

$$\partial^{\mathbf{w}}\mathsf{Z}_{\nu;t} = (-1)^{|\mathbf{w}|}\frac{\mathsf{G}_{\nu+t\mathbf{w};t}}{\prod_e \Gamma(\nu_e)} = (-1)^{|\mathbf{w}|}\prod_e \Gamma_1\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\,\mathsf{Z}_{\nu+t\mathbf{w};t},$$

the second form wherever no entry of $\nu + t\mathbf{w}$ lies in $-\mathbb{N}$, the product being read as the first form otherwise, as in the statement. At the unit configuration $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \boldsymbol{\xi}^{-\nu}$ (2.12), holomorphic near $\boldsymbol{\xi}^{(0)}$ with the principal branch, and $\partial^{\mathbf{w}}\boldsymbol{\xi}^{-\nu} = (-1)^{|\mathbf{w}|}\prod_e (\nu_e)_{w_e}\,\boldsymbol{\xi}^{-\nu-\mathbf{w}}$ with $(\nu_e)_{w_e} = \nu_e(\nu_e + 1)\cdots(\nu_e + w_e - 1)$.

(2) Taylor coefficients. Each monomial of $\boldsymbol{\Delta}(\rho)^{\mathbf{w}}$ has total degree at least $|\mathbf{w}|$ in ρ, every $l \neq 0$ having $|l| \geq 1$. So Taylor's formula for $\mathsf{Z}_{\nu;t}$ at $\boldsymbol{\xi}^{(0)}$ to the order $|q|$, whose remainder is $O(|\boldsymbol{\Delta}|^{|q|+1}) = O(|\rho|^{|q|+1})$, gives

$$c_q^\nu = \sum_{|\mathbf{w}|\leq|q|} \partial^{\mathbf{w}}\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}^{(0)})\,[\rho^q]\,\frac{\boldsymbol{\Delta}(\rho)^{\mathbf{w}}}{\mathbf{w}!}.$$

By the multinomial theorem, $\Delta_k^{w_k}/w_k! = \sum \prod_{l\neq 0} (\rho^l \xi_k^{(l)})^{n_{l,k}}/n_{l,k}!$, the sum over the $(n_{l,k})_l$ in $\mathbb{N}$ with $\sum_l n_{l,k} = w_k$; multiplying over k and taking the coefficient of ρ^q leaves the arrays $\mathbf{n}$ with $\sum_{l,k} l\,n_{l,k} = q$ and $\mathbf{w} = (\sum_l n_{l,k})_k$, $|\mathbf{w}| = |\mathbf{n}|$. With (1) this is (i). At one unit exponent, $t = (1)$, $\boldsymbol{\xi} = L$ and the index λ, the product in (i) is $\Gamma_1\begin{bmatrix}\lambda + |\mathbf{n}|\\ \lambda\end{bmatrix}(L^{(0)})^{-\lambda-|\mathbf{n}|}\prod_l (L^{(l)})^{n_l}/n_l!$, which is the formula of the reference.

(3) The coefficients as variables. The function $(\rho, (\xi_k^{(l)})) \mapsto \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho))$ is infinitely differentiable, so derivatives in the coefficients commute with taking Taylor coefficients in ρ. Since $\partial \xi_k(\rho)/\partial \xi_k^{(l)} = \rho^l$, (1) with $\mathbf{w} = \mathbf{e}_k$ gives

$$\partial_{\xi_k^{(l)}} \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho)) = -\rho^l \prod_e \Gamma_1 \begin{bmatrix} \nu_e + t_{k,e} \\ \nu_e \end{bmatrix} \mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}(\rho)),$$

and the coefficient of ρ^q is $-\prod_e \Gamma_1 \begin{bmatrix} \nu_e + t_{k,e} \\ \nu_e \end{bmatrix} c_{q-l}^{\nu+t_k}$, which is (iii). At $t = (1)$ the factor is $\Gamma_1 \begin{bmatrix} \lambda+1 \\ \lambda \end{bmatrix} = \lambda$, the formula of the reference. □

40.3 Chapter 3: the Breta

Proposition 3.4 (The Breta on the orthant). *As stated, for any data.*

Proof. By 3.1 and 2.12, with the dressed coefficients $\eta_k = u^{\pi_k}\xi_k$ and the bunch scales $\rho = (\rho_Q)_{Q\in\mathcal{P}}$,

$$\mathsf{Z}_{\boldsymbol{\nu};\ell}(\boldsymbol{\eta}) = \frac{1}{\prod_Q \Gamma(\nu_Q)} \int_{\mathbb{R}^{\mathcal{P}}_{>0}} \rho^{\boldsymbol{\nu}-\mathbf{1}} \exp\Big(-\sum_k \xi_k\, u^{\pi_k} \rho^{\ell_k} \Big)\, d\rho.$$

On each bunch, 2.7 with $x_Q = \rho_Q u_Q$ gives $x_Q^{\mathbf{m}|_Q - \mathbf{1}} dx_Q = \rho_Q^{m_Q-1} u_Q^{\mathbf{m}|_Q-\mathbf{1}} d\rho_Q\, d\omega_Q(u_Q)$, hence

$$u_Q^{\mathbf{m}|_Q-\mathbf{1}} \rho_Q^{\nu_Q-1}\, d\rho_Q\, d\omega_Q(u_Q) = x_Q^{\mathbf{m}|_Q-\mathbf{1}} |x_Q|^{\nu_Q-m_Q}\, dx_Q,$$

and each dressed term is

$$\xi_k\, u^{\pi_k}\rho^{\ell_k} = \xi_k \prod_Q \Big(\frac{x_Q}{|x_Q|}\Big)^{\pi_k|_Q} |x_Q|^{\ell_{k,Q}} = \xi_k\, x^{\pi_k} \prod_Q |x_Q|^{\ell_{k,Q}-\langle \pi_k, \mathbf{1}_Q\rangle}.$$

So, under the bijection $x \leftrightarrow (\rho_Q, u_Q)_Q$ of 2.7, the Breta with its Zamma written as the scale integral is the integral of the right side over $\mathbb{R}^{\mathcal{E}}_{>0}$. The moduli of the two integrands are the same functions with $\mathbf{m}$ and $\boldsymbol{\nu}$ replaced by their real parts, so, as in the reference, proof of 3.4, Tonelli gives that either side converges when the other does, and Fubini gives the equality. For homogeneous data the inner powers of the bunch totals are absent, and at $\nu_Q = m_Q$ so are the outer ones: the right side is then $\mathsf{G}_{\mathbf{m};t}(\boldsymbol{\xi})/\prod_Q \Gamma(m_Q)$, which is 3.2. □

Theorem 3.5 (Relations of the Breta). *As stated: (v) for any terms; (i) to (iv) are those of the reference.*

Proof. The proofs of (i) to (iv) in the reference, proof of 3.5, use no homogeneity. For (v), let $e \neq f$ be scales of the bunch Q, and in this proof write, with the products over the bunches Q' of $\mathcal{P}$,

$$F(x) = x^{\mathbf{m}-\mathbf{1}} \prod_{Q'} |x_{Q'}|^{\nu_{Q'}-m_{Q'}} \exp\Big(-\sum_k \xi_k \varphi_k(x) \Big), \qquad \varphi_k(x) = x^{\pi_k} \prod_{Q'} |x_{Q'}|^{\ell_{k,Q'}-\langle \pi_k, \mathbf{1}_{Q'}\rangle},$$

the integrand of 3.4 at $(\mathbf{m}, \boldsymbol{\nu})$ without the factor $1/\prod_{Q'} \Gamma(\nu_{Q'})$.

(1) Derivatives. For $e \in Q$, $x_e\partial_{x_e}|x_Q| = x_e$ and $x_e\partial_{x_e}|x_{Q'}| = 0$ for $Q' \neq Q$, so

$$x_e\partial_{x_e}F = F\Big[(m_e-1) + (\nu_Q - m_Q)\frac{x_e}{|x_Q|} - \sum_k \xi_k\varphi_k\Big(\pi_{k,e} + \big(\ell_{k,Q} - \langle\pi_k, \mathbf{1}_Q\rangle\big)\frac{x_e}{|x_Q|}\Big)\Big],$$

and the same holds with f in place of e.

(2) The integrands. By 3.4: F is the integrand of $\prod_{Q'}\Gamma(\nu_{Q'})\,\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})$; $F\,x_e/|x_Q|$ is that of $\prod_{Q'}\Gamma(\nu_{Q'})\,\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})$, the power of $|x_Q|$ being $\nu_Q - (m_Q+1)$; $F\varphi_k$ is that of $\prod_{Q'}\Gamma(\nu_{Q'}+\ell_{k,Q'})\,\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})$, the power of each $|x_{Q'}|$ being $(\nu_{Q'}+\ell_{k,Q'})-(m_{Q'}+\langle\pi_k,\mathbf{1}_{Q'}\rangle)$; and $F\varphi_k\,x_e/|x_Q|$ is that of $\prod_{Q'}\Gamma(\nu_{Q'}+\ell_{k,Q'})\,\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})$. Here $\prod_{Q'}\Gamma(\nu_{Q'}+\ell_{k,Q'}) = g_k\prod_{Q'}\Gamma(\nu_{Q'})$. These Bretas, and those with f in place of e, are the Bretas of (v), which converge by hypothesis; so all these functions are integrable on $\mathbb{R}^{\mathcal{E}}_{>0}$.

(3) No boundary terms. By (1) and (2), $\partial_{x_e}(x_eF) = F + x_e\partial_{x_e}F$ is integrable. By Fubini, for almost every value x' of the coordinates other than x_e the function $G(s) = s\,F(x',s)$ of $s = x_e > 0$ has G' and $G(s)/s$ integrable on $(0,\infty)$. Then $G(s) = G(1) + \int_1^s G'$ has limits at 0 and at ∞, and both are 0: a nonzero limit L at one end would make $|G(s)/s| \geq |L|/(2s)$ near that end, which is not integrable there. So $\int_0^\infty G'(s)\,ds = 0$, and by Fubini $\int \partial_{x_e}(x_eF)\,dx = 0$; likewise for f. Hence

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \big(x_e\partial_{x_e} - x_f\partial_{x_f}\big)F\,dx = \int \big[\partial_{x_e}(x_eF) - F\big]dx - \int \big[\partial_{x_f}(x_fF) - F\big]dx = 0.$$

(4) Collecting. By (1), $(x_e\partial_{x_e} - x_f\partial_{x_f})F$ is F times

$$(m_e - m_f) + (\nu_Q - m_Q)\frac{x_e - x_f}{|x_Q|} - \sum_k \xi_k\varphi_k\Big((\pi_{k,e} - \pi_{k,f}) + \big(\ell_{k,Q} - \langle\pi_k,\mathbf{1}_Q\rangle\big)\frac{x_e - x_f}{|x_Q|}\Big).$$

Integrating, using (2) and dividing by $\prod_{Q'}\Gamma(\nu_{Q'})$ gives (v). For homogeneous data $\ell_{k,Q} = \langle\pi_k,\mathbf{1}_Q\rangle$ and the second line of the right side of (v) is absent.

(5) The closing sentences. A term k raises the shape total of a bunch Q' by $\langle\pi_k,\mathbf{1}_{Q'}\rangle$ and its scale index by $\ell_{k,Q'}$, which are equal for homogeneous data; so the ladder, the scale exchange and the shifts $(\mathbf{m}+\pi_k, \boldsymbol{\nu}+\ell_k)$ of (v) keep the condition $\nu_Q = m_Q$, and in (v) the shifts $\mathbf{m}+\mathbf{e}_e$, $\mathbf{m}+\mathbf{e}_f$ of the shape index alone carry the factor $\nu_Q - m_Q$. The shape split and (iv) shift the shape index alone. □

Remark. Relation (v) was checked in exact arithmetic on one bunch of two scales with one term whose level is not the degree of its shape exponent, where the Breta has a closed form, for shape exponents $(1,0)$, $(\frac{3}{2},-\frac{1}{3})$ and symbolic (`sweep/s0107/breta_v.py`).

Proposition 3.6 (Letter presentation). *As stated, for a partition of the scales B into any number of bunches.*

Proof. (1) The presentation. The bunches of $\mathcal{P}$ are those of $\mathcal{Q}$, with the scales ρ_Q, and the singletons $\{v_i\}$, whose scales are written v_i and whose shapes are the point $u_{v_i} = 1$ of $\Delta_{\{v_i\}}$, with $\omega_{\{v_i\}}$ the Dirac measure (1.1(iii)). The dressed coefficients of the columns are $g_ju^{\pi_j}$ and $C_{i,p}u^{\pi_{i,p}}$, and $u^{(\mathbf{m},\mathbf{s})-\mathbf{1}} = u^{\mathbf{m}-\mathbf{1}}$. By 2.1, the Gramma on the bunch scales with the configuration $\widetilde{\ell}$ and these coefficients is

$$\int \rho^{\boldsymbol{\nu}-\mathbf{1}}\prod_i v_i^{s_i-1}\exp\Big(-\sum_j g_ju^{\pi_j}\rho^{\lambda_j} - \sum_{i,p} C_{i,p}u^{\pi_{i,p}}\rho^{\ell_{i,p}}v_i\Big)\,d\rho\,dv,$$

the level of the pattern (i,p) on $\{v_i\}$ being 1. For $\operatorname{Re} s_i > 0$, integrating v_i gives $\Gamma(s_i)\,P_i(\rho,u)^{-s_i}$ (2.6 on one scale, $P_i > 0$). Dividing by $\prod_Q \Gamma(\nu_Q)\prod_i \Gamma(s_i)$ gives the Zamma (2.12), and integrating against $u^{\mathbf{m-1}}d\omega(u)$ over $\prod_Q \Delta_Q$ gives $1/\prod_Q \Gamma(\nu_Q)$ times the left side of the display; Tonelli on the moduli and Fubini justify the order of the integrals, either side converging when the other does.

(2) The relations. They are 3.5(ii),(iii) for the terms $(\widetilde{\pi},\widetilde{\ell})$. For the scale exchange at $\{v_i\}$ the columns of nonzero level there are the patterns (i,p), of level 1, with the factor $\Gamma_{\!1}^{\,1}\!\begin{bmatrix} s_i+1 \\ s_i+1 \end{bmatrix} = 1$ at $\{v_i\}$; the other singletons give $\Gamma_{\!1}^{\,1}\!\begin{bmatrix} s_{i'} \\ s_{i'} \end{bmatrix} = 1$ and each $Q \in \mathcal{Q}$ gives $\Gamma_{\!1}^{\,1}\!\begin{bmatrix} \nu_Q+\ell_{i,p,Q} \\ \nu_Q \end{bmatrix}$: this is the letter split. In the ladders the factor g_k of 3.5 is $\prod_Q \Gamma_{\!1}^{\,1}\!\begin{bmatrix} \nu_Q+\lambda_{j,Q} \\ \nu_Q \end{bmatrix}$ for a kernel column and $s_i \prod_Q \Gamma_{\!1}^{\,1}\!\begin{bmatrix} \nu_Q+\ell_{i,p,Q} \\ \nu_Q \end{bmatrix}$ for a pattern, by $\Gamma_{\!1}^{\,1}\!\begin{bmatrix} s_i+1 \\ s_i \end{bmatrix} = s_i$. The scale exchange on $Q \in \mathcal{Q}$ is 3.5(iii) multiplied by ν_Q, with $\nu_Q\Gamma_{\!1}^{\,1}\!\begin{bmatrix} \nu_Q+\lambda \\ \nu_Q+1 \end{bmatrix} = \Gamma_{\!1}^{\,1}\!\begin{bmatrix} \nu_Q+\lambda \\ \nu_Q \end{bmatrix}$ (1.3(iii)), the singletons contributing the factors 1 for a kernel column and s_i for a pattern. □

Proposition 3.7 (Letter data on the orthant). *As stated, for any levels and any number of bunches.*

Proof. In the left side of the display of 3.6 apply 2.7 on each bunch $Q \in \mathcal{Q}$, $\alpha_Q = \rho_Q u_Q$: $u_Q^{\mathbf{m}|_Q-\mathbf{1}}\rho_Q^{\nu_Q-1}d\rho_Q\,d\omega_Q = \alpha_Q^{\mathbf{m}|_Q-\mathbf{1}}|\alpha_Q|^{\nu_Q-m_Q}d\alpha_Q$, and for $\lambda \in \mathbb{R}^{\mathcal{Q}}$, $\pi \in \mathbb{R}^B$, $\rho^\lambda u^\pi = \alpha^\pi \prod_Q |\alpha_Q|^{\lambda_Q-\langle\pi,\mathbf{1}_Q\rangle}$. So each kernel term becomes $g_j\alpha^{\pi_j}\prod_Q|\alpha_Q|^{\lambda_{j,Q}-\langle\pi_j,\mathbf{1}_Q\rangle}$ and each letter $P_i(\alpha)$. Tonelli on the moduli gives the equivalence of convergence and Fubini the equality, and 3.6 identifies the left side with $\prod_Q\Gamma(\nu_Q)$ times the Breta. For homogeneous letter data every exponent $\lambda_Q-\langle\pi,\mathbf{1}_Q\rangle$ is 0, and at $\nu_Q = m_Q$ so is $\nu_Q - m_Q$. □

Theorem 3.13 (The Shruffle). *As stated, for any number of bunches.*

Proof. Apply 3.12 to $A = A_1 \sqcup \cdots \sqcup A_q$ with $J = \{1,\dots,q\}$: $u|_{A_i} = w_i v^{(i)}$ with $w \in \Delta_J$, $v^{(i)} \in \Delta_{A_i}$, and

$$u^{\mathbf{m-1}}d\omega_A = \prod_i w_i^{m_{A_i}-1}\,d\omega_J(w)\prod_i (v^{(i)})^{\mathbf{m}|_{A_i}-\mathbf{1}}d\omega_{A_i}(v^{(i)}), \qquad \bigotimes_i f_i(u) = \prod_i f_i(v^{(i)}).$$

Scaling by $w_i > 0$ keeps the order of the coordinates of $u|_{A_i}$, which is that of $v^{(i)}$. Up to the null set where two coordinates of u are equal, the points u whose restrictions follow every ς_i form the disjoint union of the sets Δ_A^ς over the orders ς of A that restrict to every ς_i, and they are the images of $\Delta_J \times \prod_i \Delta_{A_i}^{\varsigma_i}$. The integrals converge absolutely, so by Fubini

$$\sum_\varsigma I_A^\varsigma\Big(\bigotimes_i f_i;\mathbf{m}\Big) = \int_{\Delta_J}\prod_i w_i^{m_{A_i}-1}\,d\omega_J(w)\;\prod_i I_{A_i}^{\varsigma_i}(f_i;\mathbf{m}|_{A_i})$$
$$= \Gamma_{\!q}^{\,1}\!\begin{bmatrix} m_{A_1},\ \dots,\ m_{A_q} \\ m_A \end{bmatrix}\prod_i I_{A_i}^{\varsigma_i}(f_i;\mathbf{m}|_{A_i}),$$

by 3.9 on J, where $\operatorname{Re} m_{A_i} > 0$. Dividing by the stacked Gamma, whose reciprocal is $\Gamma_{\!q}^{\,1}\!\begin{bmatrix} m_A \\ m_{A_1},\ \dots,\ m_{A_q} \end{bmatrix}$, gives the statement; at $q = 2$ it is the factor $\Gamma_{\!2}^{\,1}\!\begin{bmatrix} m_A+m_B \\ m_A,\ m_B \end{bmatrix}$ of the reference. □

Theorem 3.15 (Relations of the Beta port). *As stated: (iii) for linear letters of either slope and complex exponents, and (v) at every pair of exponents; (i), (ii) and (iv) are those of the reference.*

Proof. (iii) On $[0,1]$ every letter $L_i = c_i + d_i x$ is positive, since it is affine with the positive values c_i and $c_i + d_i$ at the ends, so $L_i^{-w_i}$ is infinitely differentiable there for every $w_i \in \mathbb{C}$, with $\partial_x^v L_i^{-w_i} = (-1)^v (w_i)_v d_i^v L_i^{-w_i - v}$, d_i real. Leibniz's rule gives

$$\partial_x^q \prod_i L_i^{-w_i} = \sum_{|v|=q} \frac{q!}{\prod_i v_i!} \prod_i (-1)^{v_i} (w_i)_{v_i} d_i^{v_i} L_i^{-w_i - v_i},$$

which is the first display. Then (ii), with $g = \prod_i L_i^{-w_i}$ and $\operatorname{Re} a, \operatorname{Re} b > 0$, gives the second display, by $x^{a+q-1}(1-x)^{b+q-1} = x^{a-1}(1-x)^{b-1} \prod_i (x(1-x))^{v_i}$ for $|v| = q$. Let $a, b > 0$ and $w_i \geq 0$. Each integrand on the right is $x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i} > 0$ times $\prod_i (x(1-x) d_i / L_i)^{v_i}$, which is nonnegative if every $d_i \geq 0$, and is $(-1)^{|v|} = (-1)^q$ times a nonnegative function if every $d_i \leq 0$; the coefficients $\prod_i (w_i)_{v_i} / v_i!$ are nonnegative. This gives the two signs. If $x(1-x)|d_i|/L_i \leq \varepsilon$ on $[0,1]$ for every i, each integrand has modulus at most $\varepsilon^q x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i}$, and

$$\sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!} = \frac{(|\mathbf{w}|)_q}{q!},$$

the coefficient of y^q on both sides of $\prod_i (1-y)^{-w_i} = (1-y)^{-|\mathbf{w}|}$, $|y| < 1$, by $(1-y)^{-w} = \sum_v \frac{(w)_v}{v!} y^v$. This gives the bound.

(v) Fix $q \geq 1$, write $s = a + b$, and first let $\operatorname{Re} a, \operatorname{Re} b > 0$.

(1) Coefficients. Expanding $x^k (1-x)^{q-k}$ in the second form of 3.14,

$$[x^j]\, \Pi_q^{a,b} = (-1)^{q+j} \sum_k \binom{a+q-1}{q-k} \binom{q-k}{q-j} \binom{b+q-1}{k}.$$

By $\binom{N}{M}\binom{M}{L} = \binom{N}{L}\binom{N-L}{M-L}$, a polynomial identity in N, at $N = a+q-1$, $M = q-k$, $L = q-j$, this is $(-1)^{q+j}\binom{a+q-1}{q-j} \sum_k \binom{a+j-1}{j-k}\binom{b+q-1}{k}$, and $\sum_k \binom{A}{j-k}\binom{B}{k} = \binom{A+B}{j}$, a polynomial identity in A and B that holds at nonnegative integers by counting the j-subsets of a disjoint union, gives $\binom{s+q+j-2}{j}$: the coefficient formula. At $j = q$, $\lambda_q := [x^q]\Pi_q^{a,b} = \binom{s+2q-2}{q} = (s+q-1)_q / q!$, which is nonzero since $\operatorname{Re} s > 0$. Also

$$\frac{[x^{q-1}]\Pi_q^{a,b}}{\lambda_q} = -\frac{q(a+q-1)}{s+2q-2}, \qquad \frac{[x^q]\Pi_{q+1}^{a,b}}{\lambda_{q+1}} = -\frac{(q+1)(a+q)}{s+2q}.$$

(2) Pairing. Put $\langle g, h \rangle_\circ = \int_0^1 x^{a-1}(1-x)^{b-1} g\, h\, dx$ for polynomials g, h. By (i) at $c = a + j$, $\langle \Pi_k^{a,b}, x^j \rangle_\circ = \binom{j}{k} \Gamma \begin{bmatrix} a+j,\ b+k \\ s+j+k \end{bmatrix}$, which is 0 for $j < k$ and $\beta_k := \Gamma \begin{bmatrix} a+k,\ b+k \\ s+2k \end{bmatrix} \neq 0$ for $j = k$.

(3) Span. Let g be a polynomial of degree at most $q+1$ with $\langle g, x^j \rangle_\circ = 0$ for $j \leq q-2$. Since $\deg \Pi_k^{a,b} = k$ by (1), $g = \sum_{k \leq q+1} \gamma_k \Pi_k^{a,b}$; pairing with $x^0, x^1, \ldots, x^{q-2}$ in turn, by (2) the pairing with x^j involves only $\gamma_0, \ldots, \gamma_j$, with γ_j multiplied by $\beta_j \neq 0$, so $\gamma_0 = \cdots = \gamma_{q-2} = 0$ and $g \in \operatorname{span}(\Pi_{q-1}^{a,b}, \Pi_q^{a,b}, \Pi_{q+1}^{a,b})$.

(4) Multiplication by x. $\langle x\Pi_q^{a,b}, x^j \rangle_\circ = \langle \Pi_q^{a,b}, x^{j+1} \rangle_\circ = 0$ for $j \leq q-2$, so by (3) $x\Pi_q^{a,b} = A\Pi_{q+1}^{a,b} + B\Pi_q^{a,b} + C\Pi_{q-1}^{a,b}$. The coefficients of x^{q+1} give $A = \lambda_q / \lambda_{q+1} = \frac{(q+1)(s+q-1)}{(s+2q)(s+2q-1)}$. Pairing

with x^{q-1} gives $\beta_q = C\beta_{q-1}$, so $C = \frac{(a+q-1)(b+q-1)}{(s+2q-1)(s+2q-2)}$. The coefficients of x^q give $[x^{q-1}]\Pi_q^{a,b} = A\,[x^q]\Pi_{q+1}^{a,b} + B\lambda_q$, so by (1) $B = \frac{(q+1)(a+q)}{s+2q} - \frac{q(a+q-1)}{s+2q-2}$.

(5) The operator $D = x(1-x)\partial_x$. For a polynomial g, integrating by parts, the boundary terms $x^{a+j}(1-x)^b g$ vanishing at both ends,

$$\langle Dg, x^j\rangle_\circ = \int_0^1 x^{a+j}(1-x)^b g'\,dx = -(a+j)\langle g, x^j\rangle_\circ + (s+j)\langle g, x^{j+1}\rangle_\circ,$$

which is 0 at $g = \Pi_q^{a,b}$ for $j \le q-2$. By (3), $D\Pi_q^{a,b} \in \operatorname{span}(\Pi_{q-1}^{a,b}, \Pi_q^{a,b}, \Pi_{q+1}^{a,b})$, and eliminating $\Pi_{q+1}^{a,b}$ by (4), $A \neq 0$, $D\Pi_q^{a,b} = (e\,x + e_0)\Pi_q^{a,b} + e_1\Pi_{q-1}^{a,b}$. Since $Dx^k = kx^k - kx^{k+1}$, the coefficients of x^{q+1} give $e = -q$. Pairing with x^{q-1}: the left side is $(s+q-1)\beta_q$ and the right side $-q\beta_q + e_1\beta_{q-1}$, so $e_1 = (s+2q-1)\beta_q/\beta_{q-1} = \frac{(a+q-1)(b+q-1)}{s+2q-2}$. The coefficients of x^q give $q\lambda_q - (q-1)[x^{q-1}]\Pi_q^{a,b} = -q[x^{q-1}]\Pi_q^{a,b} + e_0\lambda_q$, so $e_0 = q + [x^{q-1}]\Pi_q^{a,b}/\lambda_q = \frac{q(b+q-1)}{s+2q-2}$.

(6) Continuation. The three displays compare polynomials in x whose coefficients are rational functions of (a,b), the coefficients of the Beta ports being polynomials in (a,b). They agree on the open set $\operatorname{Re} a, \operatorname{Re} b > 0$, hence wherever the coefficients are finite, which is the condition $a+b+2q-i \neq 0$, $i = 0,1,2$; the coefficient formula is a polynomial identity and holds for all a, b and every $q \in \mathbb{N}$. □

Remark. At $a = b = n+1$ the middle coefficient of the three-term relation is $\frac{1}{2}$, and the relations become those of the reference: $(1-2x)\Pi_r^{n+1,n+1} = -2A\,\Pi_{r+1}^{n+1,n+1} - 2C\,\Pi_{r-1}^{n+1,n+1}$ and $x(1-x)\partial_x\Pi_r^{n+1,n+1} = \frac{r}{2}(1-2x)\Pi_r^{n+1,n+1} + \frac{r+n}{2}\Pi_{r-1}^{n+1,n+1}$, with the leading coefficient $\binom{2n+2r}{r}$, and at $a = b = 1$ the coefficient formula gives $(-1)^r\Pi_r^{1,1} = \sum_k (-1)^k\binom{r}{k}\binom{r+k}{k}x^k$. The three displays and the coefficient formula were checked in exact arithmetic with a and b symbolic for $q \le 9$, together with these diagonal forms (`sweep/s0107/betaport.py`).

40.4 Chapter 4: fibre integrals and closing

Theorem 4.3 (The native gauge of a configuration). *As stated: (iii) with a fixed offset ν' of the index; (i) and (ii) are those of the reference.*

Proof. By (i), $\Phi_{\boldsymbol{\xi}}$ is smooth and strictly convex with its minimum at $y^\star$, where its Hessian $H^\star = \sum_k w_k^\star t_k t_k^{\mathsf{T}}$ is positive definite.

(1) The integral. In the logarithmic form of 2.1,

$$\mathsf{G}_{\lambda\nu+\nu';t}(\lambda\boldsymbol{\xi}) = \int_{\mathbb{R}^{\mathcal{E}}} e^{\langle\nu',y\rangle}\,e^{-\lambda\Phi_{\boldsymbol{\xi}}(y)}\,dy,$$

which converges when $\lambda\nu+\nu' \in \mathcal{T}_t$ (2.2). Since $\operatorname{int}\operatorname{cone}(t)$ is an open cone containing ν, it contains $\nu + \lambda^{-1}\operatorname{Re}\nu'$ for λ large, so the integral converges for $\lambda \ge \lambda_0$, some $\lambda_0 > 0$.

(2) Away from $y^\star$. Fix $\delta > 0$ and put $\mu_\delta = \min_{|y-y^\star|=\delta}\big(\Phi_{\boldsymbol{\xi}}(y) - \Phi_{\boldsymbol{\xi}}(y^\star)\big) > 0$. For a unit vector d the function $\sigma \mapsto \Phi_{\boldsymbol{\xi}}(y^\star + \sigma d) - \Phi_{\boldsymbol{\xi}}(y^\star)$ is convex and vanishes at $\sigma = 0$, so for $\sigma \ge \delta$ it is at least $\sigma\mu_\delta/\delta$. With $|e^{\langle\nu',y\rangle}| \le e^{|\operatorname{Re}\nu'|\,|y^\star|}e^{|\operatorname{Re}\nu'|\,|y-y^\star|}$,

$$\Big|\int_{|y-y^\star|\ge\delta} e^{\langle\nu',y\rangle}e^{-\lambda(\Phi_{\boldsymbol{\xi}}(y)-\Phi_{\boldsymbol{\xi}}(y^\star))}\,dy\Big| \le e^{|\operatorname{Re}\nu'|\,|y^\star|}\int_{|u|\ge\delta} e^{-(\lambda\mu_\delta/\delta - |\operatorname{Re}\nu'|)|u|}\,du,$$

which, once $\lambda\mu_\delta/\delta \ge 2|\operatorname{Re}\nu'|$ and $\lambda\mu_\delta \ge 2\delta$, is at most a constant times $e^{-\lambda\mu_\delta/4}$: on $|u| \ge \delta$, $e^{-\Lambda|u|} \le e^{-\Lambda\delta/2}e^{-\Lambda|u|/2}$ for $\Lambda = \lambda\mu_\delta/\delta - |\operatorname{Re}\nu'| \ge \lambda\mu_\delta/(2\delta) \ge 1$. So it is $O(\lambda^{-|\mathcal{E}|/2-1})$.

(3) Near $y^\star$. Laplace's method at the nondegenerate minimum $y^\star$, with the smooth amplitude $e^{\langle \nu', y\rangle}$, gives

$$\int_{|y-y^\star|<\delta} e^{\langle \nu', y\rangle} e^{-\lambda(\Phi_{\boldsymbol{\xi}}(y)-\Phi_{\boldsymbol{\xi}}(y^\star))}\, dy = \Big(\frac{2\pi}{\lambda}\Big)^{|\mathcal{E}|/2} \det(H^\star)^{-1/2}\Big(e^{\langle \nu', y^\star\rangle} + O(\lambda^{-1})\Big),$$

as in the reference, proof of 4.3, where the amplitude is 1: the amplitude enters the leading term through its value at $y^\star$, and its derivatives only the term $O(\lambda^{-1})$.

Adding (2) and (3), multiplying by $e^{-\lambda\Phi_{\boldsymbol{\xi}}(y^\star)}$ and factoring out $e^{\langle \nu', y^\star\rangle} \neq 0$ gives (iii). □

Corollary 4.7 (Four Gammas on a line). *As stated: the display is 4.6 on two coefficients.*

Proof. (1) A straight contour. Let first $\max(-\operatorname{Re} a, -\operatorname{Re} b) < \min(\operatorname{Re} c, \operatorname{Re} d)$, and take the line $\operatorname{Re} z = \gamma$ with $\max(-\operatorname{Re} a, -\operatorname{Re} b) < \gamma < \min(\operatorname{Re} c, \operatorname{Re} d)$; it separates the two sets of poles. Apply 4.6 with $k = 2$, the exponents $a + c$ and $b + d$, and $(a + d,\, b + c)$, of total $a + b + c + d$, in the place of the merged shares. In the shares $(\alpha_1, \alpha_2, \beta_1, \beta_2)$ the fibre has the pins $\alpha_1 + \alpha_2 = a + c$, $\alpha_1 + \beta_1 = a + d$, $\alpha_2 + \beta_2 = b + c$, and it is parametrized by $\alpha = (a + z,\, c - z)$, $\beta = (d - z,\, b + z)$. The matrix P of the pins has the rows $(1, 1, 0, 0)$, $(1, 0, 1, 0)$, $(0, 1, 0, 1)$, so $\det PP^{\mathsf{T}} = 4$, and its kernel is spanned by $(1, -1, -1, 1)$, of length 2. By 4.1, with the admissible real point on the line $\operatorname{Re} z = \gamma$, where all four shares have positive real parts, the fibre integral of a function G is $(2\pi)^{-1} \cdot 4^{-1/2} \cdot 2 \int G\, d\tau = \frac{1}{2\pi i} \int_{\gamma + i\mathbb{R}} G\, dz$, $z = \gamma + i\tau$. The integrand of 4.6 is

$$\Gamma_{\,1}^{\,2}\begin{bmatrix} a+z,\ c-z \\ a+c \end{bmatrix} \Gamma_{\,1}^{\,2}\begin{bmatrix} d-z,\ b+z \\ b+d \end{bmatrix} = \frac{\Gamma(a+z)\Gamma(b+z)\Gamma(c-z)\Gamma(d-z)}{\Gamma(a+c)\Gamma(b+d)},$$

and its right side is $\Gamma_{\,1}^{\,2}\begin{bmatrix} a+d,\ b+c \\ a+b+c+d \end{bmatrix}$. Multiplying by $\Gamma(a + c)\Gamma(b + d)$ (1.3(i)) gives the display.

(2) Continuation. A contour separating the poles exists exactly when no pole $-a-n$ or $-b-n$ of the first two factors coincides with a pole $c + n'$ or $d + n'$ of the last two, $n, n' \in \mathbb{N}$, that is on the set U of $(a, b, c, d) \in \mathbb{C}^4$ with $a+c, a+d, b+c, b+d \notin -\mathbb{N}$. Near a point of U one fixed contour separates the poles, and by 1.3(iii) and 2.11 on one scale the four Gammas decay exponentially along it, locally uniformly in (a, b, c, d); so the left side is holomorphic on U, and by Cauchy's theorem it does not depend on the separating contour. The right side is holomorphic on U. The set U is connected, being the complement in $\mathbb{C}^4$ of countably many complex hyperplanes, and contains the open set of (1), so the identity holds on U. □

40.5 Chapter 5: the Trailmoment

Theorem 5.7 (The gauge of the coefficient function). *As stated: the last sentence at the unit configuration on any number of scales.*

Proof. The first two sentences are those of the reference (the reference, proof of 5.7). At the unit configuration $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\mathbf{c}) = \prod_e c_e^{-\nu_e} = \mathbf{c}^{-\nu}$ for every $\nu \in \mathbb{C}^{\mathcal{E}}$ (2.12), so by 5.1 $(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathbf{c}^{-\nu}\, d\phi$, either side converging when the other does. □

Theorem 5.8 (Exchange and ladder of the Trailmoment). *As stated: (iii) at the unit configuration on any number of scales; (i) and (ii) are those of the reference.*

Proof. At $t = I_{\mathcal{E}}$ the column t_k is $\mathbf{e}_k$, $k \in \mathcal{E}$, so in (i) at the scale e only $k = e$ has $t_{k,e} \neq 0$, with $t_{e,e} = 1$ and the factor $\Gamma_1\begin{bmatrix}\nu_e+1\\1\end{bmatrix}\prod_{f\neq e}\Gamma_1\begin{bmatrix}\nu_f\\1\end{bmatrix} = 1$. Directly, for every ν, $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\mathbf{c}) = \mathbf{c}^{-\nu} = c_e\,\mathbf{c}^{-\nu-\mathbf{e}_e} = c_e\,\mathsf{Z}_{\nu+\mathbf{e}_e;I_{\mathcal{E}}}(\mathbf{c})$ (2.12); integrating against ϕ over Y' gives the first identity.

For the derivative, let $D \subseteq \mathbb{C}^{\mathcal{E}}$ be the set of indices where the Trailmoment converges, $\int_{Y'}\mathbf{c}^{-\operatorname{Re}\nu}\,d|\phi| < \infty$, and ν_0 an interior point of D. Take $\delta > 0$ such that $\nu_0 \pm \delta\mathbf{e}_e \in D$. For $\nu = \nu_0 + \zeta\mathbf{e}_e$ with $|\zeta| < \delta/2$, $|\mathbf{c}^{-\nu}\log c_e| = \mathbf{c}^{-\operatorname{Re}\nu_0}c_e^{-\operatorname{Re}\zeta}|\log c_e|$. For $\eta > 0$ and $y > 0$, $|\log y| \le \eta^{-1}(y^{\eta} + y^{-\eta})$; with $\eta = \delta/2$, and $y^{\theta} \le y^{\delta} + y^{-\delta}$ for $|\theta| \le \delta$,

$$\left|\mathbf{c}^{-\nu}\log c_e\right| \le \frac{4}{\delta}\Big(\mathbf{c}^{-(\operatorname{Re}\nu_0-\delta\mathbf{e}_e)} + \mathbf{c}^{-(\operatorname{Re}\nu_0+\delta\mathbf{e}_e)}\Big),$$

an integrable function independent of ζ. So $\zeta \mapsto \int_{Y'}\mathbf{c}^{-\nu_0-\zeta\mathbf{e}_e}d\phi$ may be differentiated under the integral sign, and $\partial_\zeta\mathbf{c}^{-\nu} = -(\log c_e)\,\mathbf{c}^{-\nu}$ gives the second identity. □

Proposition 5.14 (Laurent data of the Gamma at its poles). *As stated: the residue and the Laurent coefficients of a product of any number of Gammas with a common pole.*

Proof. The quotient form of $(-1)^j\varepsilon\Gamma(-j+\varepsilon)$, the expansion of its logarithm and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ are those of the reference (the reference, proof of 5.14). On $|\varepsilon| < 1$ each $\phi_i(\varepsilon) := (-1)^{j_i}\varepsilon\,\Gamma(-j_i+\varepsilon) = \Gamma(1+\varepsilon)\Gamma(1-\varepsilon)/\Gamma(1+j_i-\varepsilon)$ is holomorphic and nonzero, and it is the exponential of the expansion at j_i. Put $b = -j+\varepsilon$; then $b + k_i = -j_i + \varepsilon$ and $\Gamma(b+k_i) = (-1)^{j_i}\varepsilon^{-1}\phi_i(\varepsilon)$, so

$$h(b)\prod_{i=1}^{p}\Gamma(b+k_i) = (-1)^{j_1+\cdots+j_p}\,\varepsilon^{-p}\,h(-j+\varepsilon)\prod_{i=1}^{p}\phi_i(\varepsilon).$$

The residue at $\varepsilon = 0$ is $(-1)^{\sum_i j_i}$ times the coefficient of ε^{p-1} in $h(-j+\varepsilon)\prod_i\phi_i(\varepsilon)$, and with $h(-j+\varepsilon) = \sum_l h^{(l)}(-j)\varepsilon^l/l!$ the Cauchy product gives the displayed sum. At $p = 2$ and $k_1 = k_2 = 0$, $\phi_1\phi_2 = \exp(2\log\phi_1) = (j!)^{-2}\big(1 - 2(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))\varepsilon + O(\varepsilon^2)\big)$, so the residue is $(j!)^{-2}\big(h'(-j) - 2(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))h(-j)\big)$, which is the formula of the reference. The Laurent coefficients of $\prod_i\Gamma(b+k_i)$ at $-j$ are $(-1)^{\sum_i j_i}$ times the Taylor coefficients of $\prod_i\phi_i = \prod_i(j_i!)^{-1}\exp\big(\sum_i(\log\phi_i + \log j_i!)\big)$; the series $\sum_i(\log\phi_i + \log j_i!)$ has no constant term and its coefficients are rational multiples of the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j_i,\infty))$ and $(\mathsf{T}_{2k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, so the Taylor coefficients of its exponential are polynomials in these Trailmoments with rational coefficients. □

Remark. The residue formula was checked in exact arithmetic for $p = 2, 3, 4$ at several j and (k_i) (`sweep/s0107/resonance.py`).

Proposition 5.22 (The joint Laplace transform of Dirichlet inputs). *As stated, for finite sums of monomials with nonnegative real exponents.*

Proof. (1) The series. Every exponent $\mathbf{a}$ lies in $[0,\infty)^{\mathcal{E}}$, so $0 < u^{\mathbf{a}} \le 1$ on the source $\prod_Q\Delta_Q$ and $|P_i(u)| \le \|P_i\| := \sum_{\mathbf{a}}\big|[u^{\mathbf{a}}]P_i\big|$. Hence $\sum_{\mathbf{k}}\frac{|v|^{\mathbf{k}}}{\mathbf{k}!}\prod_i|P_i(U)|^{k_i} \le \exp\big(\sum_i|v_i|\,\|P_i\|\big)$, the exponential series of $\exp(-\sum_i v_iP_i(U))$ converges uniformly on the source for each $v \in \mathbb{C}^q$ and integrates term by term, and $\mathcal{F}(v) = \sum_{\mathbf{k}}\frac{(-v)^{\mathbf{k}}}{\mathbf{k}!}\mathbb{E}[\prod_i P_i(U)^{k_i}]$, a power series converging on $\mathbb{C}^q$, hence entire.

(2) The moments. Multiplying out, $\prod_i P_i^{k_i} = \sum_{\mathbf{a}}[u^{\mathbf{a}}]\big(\prod_i P_i^{k_i}\big)u^{\mathbf{a}}$, a finite sum with exponents in $[0,\infty)^{\mathcal{E}}$. The blocks of U are independent, so $\mathbb{E}[U^{\mathbf{a}}] = \prod_Q\mathbb{E}_{U_Q\sim\mathrm{Dir}_{\mathbf{m}|_Q}}[U_Q^{\mathbf{a}|_Q}]$, and by 3.11 at the

exponents $\mathbf{a}|_Q$, whose real parts satisfy $m_e + a_e > 0$, together with 1.3(i),(ii),

$$\mathbb{E}_{U_Q \sim \mathrm{Dir}_{\mathbf{m}|_Q}}\big[U_Q^{\mathbf{a}|_Q}\big] = \frac{\Gamma_{1}^{|Q|}\begin{bmatrix}\mathbf{m}|_Q + \mathbf{a}|_Q \\ m_Q + a_Q\end{bmatrix}}{\Gamma_{1}^{|Q|}\begin{bmatrix}\mathbf{m}|_Q \\ m_Q\end{bmatrix}} = \prod_{e\in Q} \Gamma_1\begin{bmatrix}m_e + a_e \\ m_e\end{bmatrix}\ \Gamma_1\begin{bmatrix}m_Q \\ m_Q + a_Q\end{bmatrix}.$$

Summing gives $M(\mathbf{k})$; for integer exponents 3.11 gives the rising factorials.

(3) The read. Where every P_i is positive on the interior of the source, which carries the law, 2.12 and 2.1 give $\mathsf{Z}_{\nu;t}(\mathbf{P}(U)) = \frac{1}{\prod_j \Gamma(\nu_j)} \int_{(0,\infty)^d} x^{\nu-\mathbf{1}} \exp\big(-\sum_i P_i(U) x^{t_i}\big) dx$ where this converges. By Tonelli on the moduli and Fubini, wherever the double integral converges, the expectation of the left side is the integral of $\mathbb{E}[\exp(-\sum_i P_i(U)x^{t_i})] = \mathcal{F}(x^{t_1}, \dots, x^{t_q})$ against $x^{\nu-\mathbf{1}}dx/\prod_j \Gamma(\nu_j)$. The left side is the Trailmoment $(\mathsf{T}_{\nu;t}[(P_i)_i] \otimes_Q \mathrm{Dir}_{\mathbf{m}|_Q})(\prod_Q \Delta_Q)$ by 5.1.

(4) Uniqueness. With real coefficients the vector $(P_i(U))_i$ takes its values in the compact box $\prod_i [-\|P_i\|, \|P_i\|]$, and its moments $\mathbb{E}[\prod_i P_i(U)^{k_i}] = M(\mathbf{k})$ are the integrals of the monomials against its law. Polynomials are dense in the continuous functions on the box, so a law on the box is determined by these integrals; and the read $\mathbb{E}[\mathsf{Z}_{\nu;t}(\mathbf{P}(U))]$ depends on the law of $\mathbf{P}(U)$ alone. □

Theorem 5.26 (The Shruffle on a line). *As stated: (i) and (iii) for any number of strings and of trails; (ii) is that of the reference.*

Proof. (i) Let $w^{(i)}$ have length n_i, $n = \sum_i n_i$, and $S_i = \{x_1^{(i)} > \cdots > x_{n_i}^{(i)} > 0\}$. By Tonelli the function $\prod_i \prod_j w_j^{(i)}(x_j^{(i)})$ is integrable on $S_1 \times \cdots \times S_r$, with integral $\prod_i \mathrm{chain}(w^{(i)})$ by Fubini. The points with two equal coordinates form a null set. At every other point the n coordinates are distinct, and their decreasing order is a total order of the n positions (i, j) that restricts to the order of each string: an interleaving ς, the string of the n functions in that order. The points of one interleaving are carried by a permutation of the coordinates, which keeps Lebesgue measure, onto $\{y_1 > \cdots > y_n > 0\}$, the integrand becoming $\prod_m \varsigma_m(y_m)$; so their integral is $\mathrm{chain}(\varsigma)$. The interleavings are the ways of distributing the n places among the strings, $n!/(n_1! \cdots n_r!)$ of them.

(iii) Let r trails be as in (ii), with strings $\sigma^{(1)}, \dots, \sigma^{(r)}$ and cuts $N_1, \dots, N_r$. By (ii) the trail i is $\mathrm{chain}(\sigma^{(i)})$, and by (i) the product is $\sum_\varsigma \mathrm{chain}(\varsigma)$ over the interleavings. An interleaving begins with the first function of one of the strings, a Laplace transform $e^{-N_i \cdot} L_1^{(i)}$, which is $\int e^{-(s+N_i)\cdot} \psi(ds)$ for the first increment ψ of trail i: the Laplace transform of its image under $s \mapsto s + N_i$, carried by $(0, \infty)$ by the hypothesis of (ii). The other Laplace transforms of ς are those of nonnegative increments or of such images. Reading ς as in (ii), each Laplace transform followed by its run of constant functions 1 gives an increment with the index one more than the length of the run; the trail at the cut 0 with these increments has no constant columns and positive integer indices, and by (ii) it is $\mathrm{chain}(\varsigma)$. □

Theorem 5.30 (Two convolutions). *As stated: (ii) and (iii) at the unit configuration on any number of scales; (i) and the displays of (ii) for inputs carried by* $(0, \infty)$ *are those of the reference.*

Proof. (ii) By 2.12, $\mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c}\,\mathbf{c}') = \prod_e (c_e c'_e)^{-\nu_e} = \mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c})\,\mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c}')$ at every point of $Y_1 \times Y_2$. The two Trailmoments on the right converge, so by Tonelli the kernel is integrable for $|\phi| \otimes |\psi|$ on $Y_1' \times Y_2'$, and Fubini gives the product (5.1).

(iii) *The kernel on sums.* First, g has no zero: if $g(s_0) = 0$, then $g(s_0/2)^2 = g(s_0) = 0$, so $g(s_0/2^n) = 0$ for every n; every $s \in (0,\infty)^\mathcal{E}$ has $s - s_0/2^n \in (0,\infty)^\mathcal{E}$ for n large, and $g(s) = g(s_0/2^n)\, g(s - s_0/2^n) = 0$, against the hypothesis. Fix $e \in \mathcal{E}$. For $s, s'' \in (0,\infty)^\mathcal{E}$ and $\tau > 0$,

$g(s+\tau\mathbf{e}_e)\,g(s'') = g(s+s''+\tau\mathbf{e}_e) = g(s''+\tau\mathbf{e}_e)\,g(s)$, so $\varphi_e(\tau) = g(s+\tau\mathbf{e}_e)/g(s)$ does not depend on s. It is continuous on $(0,\infty)$, has no zero, and $\varphi_e(\tau+\tau') = \varphi_e(\tau)\,\varphi_e(\tau')$, by the independence of the base point applied at s and at $s+\tau\mathbf{e}_e$. By the converse on one scale, the case $|\mathcal{E}| = 1$ (the reference, proof of 5.30), $\varphi_e(\tau) = e^{-u_e\tau}$ for one $u_e \in \mathbb{C}$, and then $g(s+\tau\mathbf{e}_e) = e^{-u_e\tau}g(s)$ also for $\tau < 0$ with $s+\tau\mathbf{e}_e \in (0,\infty)^{\mathcal{E}}$, applying it at the base point $s+\tau\mathbf{e}_e$. Moving one coordinate at a time from a fixed point s_1, $g(s) = C\,e^{-\langle u,s\rangle}$ on $(0,\infty)^{\mathcal{E}}$ with $C = g(s_1)e^{\langle u,s_1\rangle} \neq 0$, and $g(s+s') = g(s)\,g(s')$ gives $C = C^2$, so $C = 1$.

The kernel on products. If $h(x) \neq 0$, then $h(x) = h(x)\,h(\mathbf{1})$ gives $h(\mathbf{1}) = 1$, and $h(x)\,h(x^{-1}) = h(\mathbf{1}) = 1$ shows that h has no zero. The function $s \mapsto h(e^s)$ on $(0,\infty)^{\mathcal{E}}$ is continuous, not identically zero and satisfies the hypothesis on sums, so $h(e^s) = e^{-\langle u,s\rangle}$, that is $h(x) = x^{-u}$ for every x with all coordinates above 1. Any $x \in (0,\infty)^{\mathcal{E}}$ is x'/x'' with $x''_e > \max(1, 1/x_e)$ and $x' = x\,x''$, both with all coordinates above 1, and $h(x) = h(x')/h(x'') = x^{-u}$. So $h = \mathsf{Z}_{u;I_{\mathcal{E}}}$. □

40.6 Chapter 6: rational words and live inputs

Theorem 6.4 (Pole columns are a rational difference quotient). *As stated: coefficients in any subfield of $\mathbb{C}$, and no decay condition.*

Proof. (1) The decomposition. For $k \geq 0$ and $s \geq 1$, telescoping gives $\Delta\sum_{j=0}^{k-1}(t+j)^{-s} = \sum_{j=0}^{k-1}\big((t+j+1)^{-s} - (t+j)^{-s}\big) = (t+k)^{-s} - t^{-s}$. So $\Delta H(t) = \sum_{k,s} c_{k,s}\big((t+k)^{-s} - t^{-s}\big) = R(t) - \sum_s C_s t^{-s}$, and H has its coefficients in $\mathbb{K}$.

(2) The decay. As $t \to \infty$, $(t+k)^{-s} = t^{-s}(1+k/t)^{-s} = t^{-s}+O(t^{-s-1})$, so $R(t) = C_1 t^{-1}+O(t^{-2})$, which is $O(t^{-2})$ exactly when $C_1 = 0$.

(3) The converse. A proper rational function H with poles on the nonpositive integer lattice has one partial-fraction presentation $H(t) = \sum_{k,s} d_{k,s}(t+k)^{-s}$, finitely many terms with $k \in \mathbb{N}$, $s \geq 1$. Then $\Delta H(t) = \sum_{k,s} d_{k,s}\big((t+k+1)^{-s} - (t+k)^{-s}\big)$, whose coefficient of $(t+k)^{-s}$ is $d_{k-1,s} - d_{k,s}$ (with $d_{-1,s} = 0$), and the column sum over k telescopes to 0. The column sums $R \mapsto (C_s)_s$ are linear on rational words, presentations being unique, and vanish on such differences. A combination $\sum_s a_s t^{-s}$ has the column sums $(a_s)_s$, so it is a difference only if it is 0: the functions $t^{-1}, \ldots, t^{-w}$ are independent modulo differences, and by (1) they span the rational words of maximum order w modulo differences. A difference is a rational word with column sums zero, hence $O(t^{-2})$ by (2); and a word that is $O(t^{-2})$ has $C_1 = 0$, so it is congruent to $\sum_{s\geq 2} C_s t^{-s}$: modulo differences these words have the basis $t^{-2}, \ldots, t^{-w}$.

(4) Cosets. Translation by one keeps each coset $b+\mathbb{Z}$, so the steps above apply coset by coset, with $(t+b)^{-s}$ in the place of t^{-s}. □

Theorem 6.5 (Summation by parts with a live input). *As stated: (iv) without the condition $C_1 = 0$; (i) to (iii) are those of the reference.*

Proof. (iv) For an integer $M \geq 1$ the head is $(\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N, N+M]) = \sum_{N<m\leq N+M} R(m)$, the kernel at index 0 being 1. By 6.4, $R(m) = \sum_s C_s m^{-s} + H(m+1) - H(m)$, so

$$(\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N,N+M]) = \sum_{s=1}^{w} C_s\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,N+M]) + H(N+M+1) - H(N+1).$$

As $M \to \infty$: $H(N+M+1) \to 0$, H being proper; for $s \geq 2$, $(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,N+M]) \to (\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$; and $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,N+M]) - \log M \to (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$, the regularized value of 5.2. This gives the first display. When $C_1 = 0$, $R(m) = O(m^{-2})$ by 6.4,

the series $\sum_{m>N} R(m)$ converges absolutely, and the left side is $(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((N,\infty))$, the term $s=1$ being absent. For a density f, when the terms converge, $\sum_{m>N} R(m)f(m) = \sum_{s=1}^{w} C_s \sum_{m>N} m^{-s} f(m) + \sum_{m>N} \Delta H(m) f(m)$, and (ii) gives the last two terms of the second display when the upper boundary vanishes. □

Theorem 6.8 (A Breta input with a recurrence of finite order). *As stated: complex a, b with positive real parts, real $q_i < 1$ of either sign, and complex s_i.*

Proof. (1) The letters. For real $q_i < 1$ and $x \in [0,1]$, $1 - q_i x$ lies between 1 and $1 - q_i$, so in $[\min(1, 1-q_i), \max(1, 1-q_i)] \subset (0,\infty)$. So $\varrho(x) = \prod_i (1-q_i x)^{-s_i}$ is infinitely differentiable on $[0,1]$, and $|\varrho| = \prod_i (1-q_i x)^{-\operatorname{Re} s_i}$ lies between two positive constants.

(2) The size. For $m \geq 1$, by $x \leq e^{-(1-x)}$ on $[0,1]$ and $y = 1-x$,

$$\begin{aligned} |f(m)| &\leq \max|\varrho| \int_0^1 x^{m+\operatorname{Re} a-1} (1-x)^{\operatorname{Re} b-1} dx \\ &\leq \max|\varrho| \int_0^\infty e^{-(m+\operatorname{Re} a-1)y} y^{\operatorname{Re} b-1} dy = \max|\varrho|\, \Gamma(\operatorname{Re} b)\, (m + \operatorname{Re} a - 1)^{-\operatorname{Re} b}, \end{aligned}$$

so $f(m) = O(m^{-\operatorname{Re} b})$.

(3) The recurrence. Differentiating the logarithm, $\frac{d}{dx}\big[x^{m+a}(1-x)^b \prod_i (1-q_i x)^{1-s_i}\big]$ is $x^{m+a-1}(1-x)^{b-1}\varrho(x)$ times

$$(m+a)(1-x)Q(x) - bxQ(x) + x(1-x)\sum_i (s_i - 1) q_i Q_i(x) = B_m(x).$$

The bracket tends to 0 at both ends, since $\operatorname{Re}(m+a) > 0$ and $\operatorname{Re} b > 0$, and $x^{m+a-1}(1-x)^{b-1}\varrho B_m$ is integrable; so integrating over $[0,1]$ gives $\sum_\ell B_\ell(m) f(m+\ell) = 0$.

(4) The extreme coefficients. $B_0(m) = B_m(0) = (m+a)Q(0) = m + a$. Every $q_i \neq 0$, so Q has degree r and leading coefficient $(-1)^r \prod_i q_i$, and Q_i has degree $r-1$ and leading coefficient $(-1)^{r-1} \prod_{i' \neq i} q_{i'}$. The coefficient of x^{r+1} in B_m is therefore

$$(-1)^{r+1} \prod_i q_i \Big((m+a) + b + \sum_i (1 - s_i) \Big) = (-1)^{r+1} \Big(\prod_i q_i \Big) \Big(m + a + b + r - \sum_i s_i \Big).$$

$B_0(m) \neq 0$ since $\operatorname{Re} a > 0$, and $B_{r+1}(m) \neq 0$ for every integer $m \geq 0$ exactly when $a+b+r-\sum_i s_i \notin -\mathbb{N}$. Then the recurrence solved for $f(m+r+1)$, and for $f(m)$, expresses every shift through $(f(m), \ldots, f(m+r))$ with coefficients rational in m.

(5) The Breta. Let $\operatorname{Re} s_i > 0$. On the bunch $\{1,2\}$ with the scale ρ and the shape $u = (x, 1-x)$, $\omega_{\{1,2\}}$ being dx in the coordinate $x = u_1$, the kernel terms ρu_1 and ρu_2 with unit coefficients sum to ρ, and the letter $P_i = \rho\big((1-q_i)u_1 + u_2\big) = \rho(1 - q_i x)$ has the positive pattern coefficients $1 - q_i$ and 1. At the shape index $(m+a, b)$ and the scale index $1 + |\mathbf{s}|$ the left side of 3.6 is

$$\int_0^1 x^{m+a-1}(1-x)^{b-1} \int_0^\infty \rho^{|\mathbf{s}|} e^{-\rho} \rho^{-|\mathbf{s}|} \prod_i (1-q_i x)^{-s_i}\, d\rho\, dx = f(m),$$

the inner integral being $\Gamma(1) = 1$; 3.6 identifies it with $\Gamma(1+|\mathbf{s}|)$ times the Breta. □

Theorem 6.9 (A fixed set of input-dependent evaluations). *As stated, for the data of 6.8.*

Proof. The steps of the reference, proof of 6.9, hold for these data; they are as follows. Write $F_{p,k}(N) = (\mathsf{T}_{p;(1)}[x+k]\, f\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} f(m)(m+k)^{-p}$, and, for $0 \le x < 1$, $S_p(x) = \sum_{m>N} x^m (m+k)^{-p}$, with an integer cut $N \ge 0$ and $k \ge 0$; $S_0(x) = x^{N+1}/(1-x)$.

(1) The identity behind (ii). As in the proof of 6.8, $(xC\eta)' = B\eta$, with $xC\eta = x^a(1-x)^b \prod_i (1-q_i x)^{1-s_i}$. Since $xS_p'(x) = \sum_{m>N} m\, x^m (m+k)^{-p} = S_{p-1}(x) - kS_p(x)$,

$$\frac{d}{dx}\big[xC\eta\, S_p\big] = \eta\big[(B-kC)S_p + C\, S_{p-1}\big] \qquad (p \ge 1).$$

At $x=0$ the bracket is $O(|x^{a+N+1}|) \to 0$, since $\operatorname{Re} a > 0$. At $x=1$ it is bounded by a constant times $|(1-x)^b|$ for $p \ge 2$, where $S_p(1)$ is finite, and times $|(1-x)^b|\,|\log(1-x)|$ for $p=1$, where $S_1(x) = O(|\log(1-x)|)$; it tends to 0 since $\operatorname{Re} b > 0$. The functions $|\eta|\, S_p$ and $|\eta|\,|\log(1-x)|$ are integrable on $[0,1)$, so integrating over $[0,1)$ and summing term by term (Tonelli on the moduli) gives, for $p \ge 2$,

$$\sum_{m>N} (m+k)^{-p} \int_0^1 x^m (B-kC)\eta\, dx + \sum_{m>N} (m+k)^{1-p} \int_0^1 x^m C\eta\, dx = 0.$$

With $\int_0^1 x^{m+\ell}\eta\, dx = f(m+\ell)$ and, for $k \ge r+1 \ge \ell$, $\sum_{m>N} f(m+\ell)(m+k)^{-p} = F_{p,k-\ell}(N) - \sum_{v=1}^{\ell} f(N+v)(N+v+k-\ell)^{-p}$, this is (ii) for $p \ge 2$. At $p=1$ the term with S_0 is $\int_0^1 \eta\, C\, S_0\, dx = \int_0^1 \eta\, x^{N+1} Q\, dx = \sum_{\ell=0}^{r} Q_\ell f(N+1+\ell)$, by $C\, S_0 = x^{N+1}Q$, which gives (ii) at $p=1$.

(2) The reduction (i). In (ii) the coefficient of $F_{p,k}(N)$, the term $\ell = 0$, is $B_0 - kC_0 = a - k$, since $B(0) = aQ(0) = a$ and $C(0) = 1$. For $k \ge r+1$ with $k \ne a$, (ii) expresses $F_{p,k}(N)$ through the $F_{p,k-\ell}(N)$, $1 \le \ell \le r+1$, the $F_{p-1,k-\ell}(N)$ (none at $p=1$), and the values $f(N+v)$, $1 \le v \le r+1$. Taking the indices p in increasing order and, for each, the shifts k in increasing order, every $F_{p,k}(N)$ with $p \le A$ reduces to the $F_{p',k'}(N)$ with $p' \le p$ and $k' \le r$, the $F_{p',a}(N)$ when a is an integer at least $r+1$, and $f(N+1), \dots, f(N+r+1)$.

(3) The cuts (iii) are 5.3 for the counting input.

(4) Denominators (iv). For rational data, let c_* clear the coefficients of B, C and Q, so that $c_*(a-k)$, $c_*(B_\ell - kC_\ell)$, c_*C_ℓ and c_*Q_ℓ are integers for integer k. By induction along the order of (2), the coefficients of $F_{p,k}(N)$, $k \ge r+1$, in the reduced form are cleared by $D_{p,k} = d_{N+k}^{\,p} \prod_{r+1 \le k' \le k,\ k' \ne a} |c_*(a-k')|^p$. Multiply (ii) by c_* and divide by $c_*(a-k)$: the terms $F_{p,k-\ell}(N)$ and $F_{p-1,k-\ell}(N)$ carry integer factors, and their reduced forms are cleared by $D_{p,k-\ell}$ and $D_{p-1,k-\ell}$, which divide $D_{p,k}/|c_*(a-k)|$; the terms in $f(N+v)$ carry the denominators $(N+v+k-\ell)^p$ and $(N+v+k-\ell)^{p-1}$, which divide $d_{N+k}^{\,p}$, since $1 \le N+v+k-\ell \le N+k$. So $D_{p,k}$ clears $F_{p,k}(N)$, and $d_{N+J}^{\,A} \prod_{r+1 \le k \le J,\ k \ne a} |c_*(a-k)|^A$, a multiple of every $D_{p,k}$ with $p \le A$ and $k \le J$, clears all shifts through J and indices through A. □

40.7 Chapter 7: tables

Theorem 7.13 (One function, two Foreman polynomials). *As stated: (iii) along every direction constant on the molecules of the table, with the converse; (i) and (ii) are those of the reference.*

Proof. The directions. By 7.8 for the family whose one table is this table, the table is the direct sum of its restrictions to its molecules M: every basis T is the union of one basis of each restriction

$\mathcal{B}|M$, and a basis of $\mathcal{B}|M$ has $r(M)$ marks (7.5). So $|T \cap M| = r(M)$, and for v constant on each M, with value v_M,

$$\langle v, \chi_T \rangle = \sum_M v_M \big(|T \cap M| - |M \setminus T| \big) = \sum_M v_M \big(2r(M) - |M| \big)$$

for every basis. By 7.12, $\Lambda(z+v) = \sum_T \theta_T e^{\langle z, \chi_T \rangle} e^{\langle v, \chi_T \rangle} = \exp\big(\sum_M v_M (2r(M) - |M|) \big) \Lambda(z)$. At $v = \lambda \mathbf{1}$ the exponent is $\lambda(2r - n)$, the $r(M)$ adding up to r, since a basis meets each molecule in $r(M)$ marks, and the $|M|$ to n.

The converse. Let $v \in \mathbb{R}^{\mathcal{E}}$ and real $\lambda \neq 0$ be such that $\Lambda(z + \lambda v) = c\, \Lambda(z)$ for every $z \in \mathbb{R}^{\mathcal{E}}$, with c independent of z. Then $\sum_T \theta_T \big(e^{\lambda \langle v, \chi_T \rangle} - c \big) e^{\langle z, \chi_T \rangle} = 0$ for every z. Distinct bases have distinct signed indicators, and exponentials $e^{\langle z, \chi \rangle}$ with finitely many distinct vectors χ are linearly independent functions of z: on a line $z = \sigma y$ with the $\langle y, \chi \rangle$ distinct, as $\sigma \to \infty$ the term with the largest $\langle y, \chi \rangle$ dominates, so its coefficient vanishes, and the others follow in turn. So $\theta_T (e^{\lambda \langle v, \chi_T \rangle} - c) = 0$, and since $\theta_T > 0$, $e^{\lambda \langle v, \chi_T \rangle} = c$ for every basis. Hence $\langle v, \chi_T \rangle = 2\langle v, \mathbf{1}_T \rangle - \langle v, \mathbf{1} \rangle$ is constant over the bases, and so is $\langle v, \mathbf{1}_T \rangle$. By 7.7, $v_a = v_b$ for any two marks linked by a circuit of the table; the molecules of the table alone are the smallest nonempty sets of marks containing, with each mark, every mark so linked to it (7.4), so v is constant on each of them. □

Theorem 7.14 (The inner moment generating function). *As stated: the even statistic at every gauge, with the basis law and the cobasis law there.*

Proof. For real z and w, $e^{E(z+w)} = \Lambda(z+w)\, \Lambda(-z-w)$. By the first display of the theorem, $\Lambda(z+w)/\Lambda(z) = \mathbb{E}_{T \sim \mathbb{P}_\alpha}[e^{\langle w, \chi_T \rangle}]$, and the same display at the gauge $\alpha^{-1} = e^{-2z}$, whose basis law is $\mathbb{P}_{\alpha^{-1}} = \mathbb{P}^*_\alpha$, gives $\Lambda(-z-w)/\Lambda(-z) = \mathbb{E}_{T' \sim \mathbb{P}^*_\alpha}[e^{-\langle w, \chi_{T'} \rangle}]$. Multiplying, and using the independence of T and T' under $\mathbb{P}_\alpha \otimes \mathbb{P}^*_\alpha$,

$$\frac{e^{E(z+w)}}{e^{E(z)}} = \mathbb{E}_{(T,T') \sim \mathbb{P}_\alpha \otimes \mathbb{P}^*_\alpha} \big[e^{\langle w, \chi_T - \chi_{T'} \rangle} \big], \qquad \chi_T - \chi_{T'} = 2(\mathbf{1}_T - \mathbf{1}_{T'}),$$

which is the display; its logarithm says that $w \mapsto E(z+w) - E(z)$ is the cumulant generating function of $\chi_T - \chi_{T'}$. At $z = 0$, $e^{E(0)} = \Lambda(0)^2 = K(\mathbf{1})^2$ by (i) of 7.13, and $\mathbb{P}^*_{\mathbf{1}} = \mathbb{P}_{\mathbf{1}}$; with w written z this is the display of the reference. □

Remark. The display at a general gauge, and (iii) of 7.13, were checked in exact arithmetic on the table whose bases are the unions of a 2-subset of $\{0,1,2\}$ and a 1-subset of $\{3,4\}$, with arbitrary positive weights (`sweep/s0107/tables.py`).

Corollary 7.21 (Moments and cumulants). *As stated: the joint cumulants of the table at every order; (i) and the cumulants of the components are those of the reference.*

Proof. For a function Y of the index (k,l) write $\mathbb{E}_w[Y] = \sum_{k,l} w_{kl} Y(k,l)$; $\mathbb{E}_w[1] = 1$ by 7.20.

(1) Generating functions. By 7.20 at $x = e^v$, $v \in \mathbb{C}^B$,

$$\mathbb{E}_{T \sim \mathbb{P}_\alpha} \Big[\exp\Big(\sum_{e \in B} v_e X_e \Big) \Big] = \sum_{k,l} w_{kl}\, M_{kl}(v), \qquad M_{kl}(v) = \det\big(I + (D_{e^v} - I) \Pi^{(kl)}_{BB} \big),$$

with $M_{kl}(0) = 1$. So $C_{kl}(v) = \log M_{kl}(v)$ is a power series in v without constant term, and for distinct marks the cumulant $\kappa^{(kl)}(e_1, \dots, e_p)$ of the component is the coefficient of $v_{e_1} \cdots v_{e_p}$ in C_{kl}; the reference computes it by the cyclic formula. The joint cumulant $\kappa_\alpha(e_1, \dots, e_p)$ is the coefficient of $v_{e_1} \cdots v_{e_p}$ in $\log \sum_{k,l} w_{kl} e^{C_{kl}(v)}$, a power series since $\sum_{k,l} w_{kl} = 1$.

(2) Cumulants over the index. Let $(Y_i)_{i\in I}$ be finitely many functions of the index, $\lambda=(\lambda_i)_{i\in I}$ formal variables, and $\mu(\lambda)=\mathbb{E}_w[\exp(\sum_i \lambda_i Y_i)]$, a power series with constant term 1. For $D\subseteq I$ write $\lambda^D=\prod_{i\in D}\lambda_i$; the coefficient of λ^D in μ is $\mathbb{E}_w[\prod_{i\in D} Y_i]$. In $\log\mu=\sum_{j\ge 1}\frac{(-1)^{j-1}}{j}(\mu-1)^j$ the coefficient of λ^I comes from the products of j monomials of $\mu-1$ without repeated variables and with disjoint supports covering I; each partition $\mathfrak{q}$ of I into j blocks arises from $j!$ such ordered products. So

$$[\lambda^I]\log\mu=\sum_{j\ge 1}\frac{(-1)^{j-1}}{j}\,j!\sum_{|\mathfrak{q}|=j}\prod_{D\in\mathfrak{q}}\mathbb{E}_w\Big[\prod_{i\in D}Y_i\Big]=\kappa_w\big((Y_i)_{i\in I}\big),$$

and, setting the variables outside a subset D to 0, $[\lambda^D]\log\mu=\kappa_w((Y_i)_{i\in D})$.

(3) Substitution. Fix distinct marks $e_1,\dots,e_p$ of B; for nonempty $C\subseteq\{1,\dots,p\}$ put $v^C=\prod_{c\in C}v_{e_c}$ and $Y_C(k,l)=\kappa^{(kl)}(e_C)$, the coefficient of v^C in C_{kl}. Setting the variables v_e with $e\notin\{e_1,\dots,e_p\}$ to 0 and dropping every monomial divisible by some $v_{e_c}^2$ is a ring homomorphism onto the ring $\mathcal{R}$ of polynomials in $v_{e_1},\dots,v_{e_p}$ with $v_{e_c}^2=0$; it keeps the coefficient of $v_{e_1}\cdots v_{e_p}$ and commutes with exp and log of series without constant term. In $\mathcal{R}$, C_{kl} becomes $\sum_{C\neq\varnothing}Y_C(k,l)\,v^C$, the image of $\sum_C\lambda_C Y_C(k,l)$ under $\lambda_C\mapsto v^C$. So the coefficient of $v_{e_1}\cdots v_{e_p}$ in $\log\sum_{k,l}w_{kl}e^{C_{kl}}$ is that of the image of $\log\mu(\lambda)$, with I the set of nonempty C. A monomial $\prod_C\lambda_C^{n_C}$ maps to $\prod_C(v^C)^{n_C}$, which is $v_{e_1}\cdots v_{e_p}$ exactly when the C with $n_C>0$ have $n_C=1$ and form a partition $\mathfrak{p}$ of $\{1,\dots,p\}$, and is 0 or another monomial otherwise. By (2),

$$\kappa_\alpha(e_1,\dots,e_p)=\sum_{\mathfrak{p}}[\lambda^{\mathfrak{p}}]\log\mu=\sum_{\mathfrak{p}}\kappa_w\Big(\big(\kappa^{(kl)}(e_C)\big)_{C\in\mathfrak{p}}\Big).$$

At $p=1$ the single partition gives $\kappa_\alpha(e)=\mathbb{E}_w[\Pi_{ee}^{(kl)}]$, and $\kappa_\alpha(e)=\mathbb{P}_\alpha(e\in T)=R_e(\alpha)$. Every step is an identity of power series in which the w_{kl} enter linearly inside μ, with $\sum_{k,l}w_{kl}=1$, so it holds for complex mixing coefficients. □

Remark. At $p=2$ and $p=3$ the formula gives $\kappa_\alpha(e,f)=\mathbb{E}_w[\kappa^{(kl)}(e,f)]+\kappa_w(\Pi_{ee}^{(kl)},\Pi_{ff}^{(kl)})$ and $\kappa_\alpha(e,f,g)=\mathbb{E}_w[\kappa^{(kl)}(e,f,g)]+\sum_3\kappa_w(\Pi_{ee}^{(kl)},\kappa^{(kl)}(f,g))+\kappa_w(\Pi_{ee}^{(kl)},\Pi_{ff}^{(kl)},\Pi_{gg}^{(kl)})$, the sum over the three choices of the mark that stands alone: the covariances and the third cumulant over the index of the reference. The formula was checked in exact arithmetic against the generating function of a mixture of three components with complex mixing coefficients for $p=2,3,4,5$ (`sweep/s0107/cumulants2.py`).

40.8 Chapter 8: families

Proposition 8.4 (The interest over latent vectors). *As stated: every table determinantal over $\mathbb{R}$ or over $\mathbb{C}$, each letter opened with as many latent vectors as its own exponent asks.*

Proof. The masses are real and positive. For a mass $m_e\le 0$ the interest diverges by 8.7(i), the wall $\kappa_{\{e\}}=m_e-\sum_j\big(s_jd_j(\{e\})+t_jd_j^*(\{e\})\big)$ being at most m_e, and so does the integral over α_e in step (4).

Fix a table j, write $\mathbb{F}=\mathbb{F}_j$, $\beta=\beta_j$, $r=r_j$, and let $A\in\mathbb{F}^{r\times n}$ be a slice of it: $\det A_T=\boldsymbol{\omega}_T$, so that $|\det A_T|^2=\theta_{j,T}$ for $T\in\mathcal{B}_j$ and $\det A_T=0$ for every other set T of r marks (7.17). For a matrix X with n columns write X^{H} for its conjugate transpose, and $D_\alpha=\operatorname{diag}(\alpha)$. Expanding by the maximal minors of X,

$$\det\big(XD_\alpha X^{\mathsf{H}}\big)=\sum_W|\det X_W|^2\,\alpha^W,$$

the sum over the sets W of as many marks as X has rows; this is the expansion behind 7.18, and with $X = A$ it gives $K_j(\alpha) = \det(AD_\alpha A^{\mathsf H})$. So $K_j(\mathbf{1}) = \det(AA^{\mathsf H}) > 0$, A has rank r, and its row space $V_j = \{A^{\mathsf H}y :\ y \in \mathbb{F}^r\}$, the space onto which $\Pi^{(11)}$ of 7.20 projects, has dimension r over $\mathbb{F}$.

(1) The basis letter. The matrix $\hat A = (AA^{\mathsf H})^{-1/2}A$ has orthonormal rows and $\{\hat A^{\mathsf H}y\} = V_j$, and $\det \hat A_T = \det(AA^{\mathsf H})^{-1/2} \det A_T$. So $\theta_{j,T} = K_j(\mathbf{1})\,|\det \hat A_T|^2$ and $K_j(\alpha) = K_j(\mathbf{1}) \det(\hat A D_\alpha \hat A^{\mathsf H})$.

(2) The cobasis letter. Choose $\hat C \in \mathbb{F}^{(n-r)\times n}$ with orthonormal rows and $\{\hat C^{\mathsf H}y\} = V_j^\perp$, the orthogonal complement for the form $\sum_e u_e\overline{v_e}$. Then $\hat A\hat C^{\mathsf H} = 0$, and the square matrix U whose rows are those of $\hat A$ followed by those of $\hat C$ is unitary. For an invertible matrix, $\det U[I,J] = \pm \det U \cdot \det U^{-1}[J^c, I^c]$ for every set I of rows and set J of columns of equal size; with $U^{-1} = U^{\mathsf H}$ and $|\det U| = 1$ this gives $|\det U[I,J]| = |\det U[I^c,J^c]|$. With I the rows of $\hat A$ and $J = T$, $|\det \hat C_{\mathcal E\setminus T}| = |\det \hat A_T|$ for every set T of r marks. The expansion above, for $X = \hat C$ and the sets $W = \mathcal E \setminus T$, gives

$$\det\big(\hat C D_\alpha \hat C^{\mathsf H}\big) = \sum_T |\det \hat A_T|^2 \alpha^{\mathcal E\setminus T} = \frac{1}{K_j(\mathbf{1})} \sum_{T\in\mathcal B_j} \theta_{j,T}\,\alpha^{\mathcal E\setminus T} = \frac{\Psi_j(\alpha)}{K_j(\mathbf{1})}.$$

(3) The Gaussian opening. For a Hermitian positive definite matrix G of size d over $\mathbb{F}$, a unitary change of variables, which keeps the Lebesgue measure of $\mathbb{F}^d \cong \mathbb{R}^{\beta d}$, makes G diagonal, and $\int_{\mathbb F} e^{-\lambda|z|^2}dz = (\pi/\lambda)^{\beta/2}$ for $\lambda > 0$; so $\int_{\mathbb{F}^d} e^{-z^{\mathsf H}Gz}dz = \pi^{\beta d/2}\det(G)^{-\beta/2}$. For an integer $k \ge 0$ and $s = \beta k/2$ it follows that

$$\det(G)^{-s} = \pi^{-ds} \int_{(\mathbb{F}^d)^k} \exp\Big(-\sum_{i=1}^k z_i^{\mathsf H} G z_i\Big)\, dz_1 \cdots dz_k.$$

Take $G = \hat A D_\alpha \hat A^{\mathsf H}$, $d = r$ and $k = 2s_j/\beta$. The map $z \mapsto x = \hat A^{\mathsf H} z$ is an $\mathbb{F}$-linear isometry of $\mathbb{F}^r$ onto V_j, so it carries the Lebesgue measure of $\mathbb{F}^r$ to that of V_j, and $z^{\mathsf H}\hat A D_\alpha \hat A^{\mathsf H} z = \sum_e \alpha_e |x_e|^2$. With step (1),

$$K_j(\alpha)^{-s_j} = K_j(\mathbf{1})^{-s_j}\pi^{-r_j s_j} \int_{V_j^{2s_j/\beta}} \exp\Big(-\sum_{i,e} \alpha_e |(x_{j,i})_e|^2\Big) \prod_i dx_{j,i},$$

and in the same way, with step (2), $G = \hat C D_\alpha \hat C^{\mathsf H}$, $d = n - r_j = r_j^*$ and $k = 2t_j/\beta$,

$$\Psi_j(\alpha)^{-t_j} = K_j(\mathbf{1})^{-t_j}\pi^{-r_j^* t_j} \int_{(V_j^\perp)^{2t_j/\beta}} \exp\Big(-\sum_{i,e} \alpha_e |(x^*_{j,i})_e|^2\Big) \prod_i dx^*_{j,i}.$$

An exponent 0 asks for no vectors, and the formula then reads $1 = 1$.

(4) The scales. Multiplying over j, with $c_F = \prod_j K_j(\mathbf{1})^{s_j+t_j}\pi^{r_j s_j + r_j^* t_j}$,

$$W(\alpha) = \frac{1}{c_F} \int \exp\Big(-\sum_e \alpha_e \mathfrak q_e\Big) \prod_{j,i} dx_{j,i}\, dx^*_{j,i}.$$

Insert this into $J_F(\mathbf m;\mathbf s,\mathbf t) = \int \alpha^{\mathbf{m-1}} e^{-|\alpha|} W(\alpha)\, d\alpha$. Every integrand is nonnegative, so the integrals may be taken in either order, and

$$\int_0^\infty \alpha_e^{m_e-1} e^{-\alpha_e(1+\mathfrak q_e)}\, d\alpha_e = \Gamma(m_e)\,(1+\mathfrak q_e)^{-m_e}.$$

Integrating the scales first gives the right side of the statement, and integrating the latent vectors first gives $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$: both are the integral of one nonnegative function, finite or infinite together.

(5) The Breta law. Where the two sides are finite, the function

$$f(\alpha,x,x^*)=\alpha^{\mathbf{m-1}}\exp\Big(-\sum_e \alpha_e(1+\mathfrak{q}_e)\Big)$$

is integrable, with integral $c_F J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ by step (4). Integrating out the latent vectors, its marginal in α is $c_F\,\alpha^{\mathbf{m-1}}e^{-|\alpha|}W(\alpha)$, proportional to the density $\alpha^{\mathbf{m-1}}e^{-|\alpha|}W(\alpha)/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ of $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ (16.7; the interest is finite exactly when every wall is positive, 8.7(i)). Given the latent vectors, f is the product over the marks of $\alpha_e^{m_e-1}e^{-\alpha_e(1+\mathfrak{q}_e)}$: the scales are independent, and $\alpha_e(1+\mathfrak{q}_e)$ has the density $u^{m_e-1}e^{-u}/\Gamma(m_e)$ on $(0,\infty)$, the law Gamma_{m_e}.

(6) Equal exponents over $\mathbb{R}$. If $\mathbb{F}_j=\mathbb{R}$ and $s_j=t_j$, the vectors of the two letters come in pairs $(x_{j,i},x^*_{j,i})$, $i\le 2s_j$. The map $V_j\times V_j^{\perp}\to\mathbb{R}^{\mathcal{E}}$, $(x,x^*)\mapsto x+x^*$, is a linear isometry, so it carries the product of the Lebesgue measures to that of $\mathbb{R}^{\mathcal{E}}$, and for $u=x+x^*$, $x=\mathrm{pr}_j\,u$ and $x^*=(I-\mathrm{pr}_j)u$. The constant of table j is then $K_j(\mathbf{1})^{2s_j}\pi^{ns_j}$. □

Remark. The reference, proof of 8.4, treats $s_j=t_j=k_j/2$ over $\mathbb{R}$ and opens the two letters of table j with one set of k_j vectors $u=\hat{A}^{\mathsf{T}}y+\hat{C}^{\mathsf{T}}y^*$; that is step (6). In the proof above each letter is opened on its own space, with as many vectors as its exponent asks, and over $\mathbb{C}$ by the complex Gaussian integral, $\beta=2$.

Lemma S3. *For a family on $\mathcal{E}$ and a nonempty set $S\subseteq\mathcal{E}$, $c_j(S)=0$ for every table j of the family exactly when S is a union of molecules.*

Proof. The molecules are the smallest nonempty sets of marks that contain, with each mark, every mark linked to it (7.4); so a set of marks with that property is a union of molecules, and a union of molecules has it.

Let S be a union of molecules. By 7.8 every table is the direct sum of its restrictions to the molecules: its bases are the unions of one basis of each restriction, and a basis of the restriction to a molecule M has $r_j(M)$ marks (7.5). So $|T\cap S|=\sum_{M\subseteq S}r_j(M)$ on every basis T of table j, and by 7.11, $d_j(S)=r_j(S)$, that is $c_j(S)=0$.

Let S not be a union of molecules. Then some $e\in S$ is linked to some $f\notin S$: e and f lie in a common circuit of some table j. By 7.6 some basis T of table j has $e\in T$, $f\notin T$ and $T'=(T\setminus\{e\})\cup\{f\}$ a basis. Then $|T'\cap S|=|T\cap S|-1$, so $|T\cap S|$ takes two values on the bases of table j, and $c_j(S)\ge 1$ by 7.11. □

Remark. Since the weight law charges every basis, Lemma S3 is the first sentence of 13.9 read on the bases: $\mathrm{Var}_{T\sim\mathbb{P}_{j,\mathbf{1}}}|T\cap U|=0$ exactly when $|T\cap U|$ is constant on $\mathcal{B}_j$, that is when $c_j(U)=0$.

Proposition 8.15 (The wall of a compound). *As stated: the identity holds for every nonempty proper S, and only the tables with $\sigma_j\neq 0$ enter it.*

Proof. 8.12 gives, as identities of functions of the data, $\kappa_S+\kappa_S^{\mathrm{top}}=2\sum_j\sigma_j c_j(S)$ and $\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}+\kappa_S^{\mathrm{top}}$. Adding the two,

$$\kappa_S+\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}+2\sum_j\sigma_j c_j(S).$$

If $c_j(S)=0$ in every table with $\sigma_j\neq 0$, the sum vanishes and $\kappa_S+\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}$; this holds in particular when S is a union of molecules, where $c_j(S)=0$ in every table (Lemma S3). On the balance wall $\kappa_{\mathcal{E}}=0$, so $\mathrm{Re}\,\kappa_S+\mathrm{Re}\,\kappa_{\mathcal{E}\setminus S}=0$ and one of the two walls has real part at most zero.

By 8.10 the period converges only where every nonempty proper wall has positive real part, so it diverges. □

Remark. The reference, 8.15, takes $c_j(S) = 0$ in every table. With complex exponents a table with $s_j = -t_j \neq 0$ is active and has $\sigma_j = 0$; it does not enter the identity.

Definition 8.16 (Target; native gauge). As stated: the target for all data with $m_{\mathcal{E}} \neq 0$ and on the balance wall, and native points of $(\mathbb{C}^\times)^{\mathcal{E}}$ off the zeros of the letters, the native gauges being the native points in $\mathbb{R}^{\mathcal{E}}_{>0}$.

Proof. The definition is consistent, and on the data of 8.18(ii) it gives the native gauges used there.

(1) Scale. K_j and Ψ_j are homogeneous polynomials of degrees r_j and $r_j^* = n - r_j$, every basis having r_j marks and every complement $n - r_j$ (7.1). So for $c \in \mathbb{C}^\times$, $K_j(c\alpha) = c^{r_j} K_j(\alpha)$ and $(\partial_e K_j)(c\alpha) = c^{r_j - 1}(\partial_e K_j)(\alpha)$, whence $R_{j,e}(c\alpha) = R_{j,e}(\alpha)$ wherever $K_j(\alpha) \neq 0$, and likewise $R^*_{j,e}(c\alpha) = R^*_{j,e}(\alpha)$ wherever $\Psi_j(\alpha) \neq 0$. The condition $K_j(\alpha)\Psi_j(\alpha) \neq 0$ is unchanged by scaling and the target does not involve the point, so the usage, and with it nativity, is a property of the point up to scale.

(2) The two clauses of the target. On the balance wall $m_{\mathcal{E}} = \sum_j (s_j r_j + t_j r_j^*)$ (8.1); where this is not zero the formula gives $\boldsymbol{\tau} = \mathbf{m}$, the value assigned on the balance wall. So the two clauses agree where both apply, and the second also covers the balance wall with $m_{\mathcal{E}} = 0$.

(3) Positive points. At $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$ the letters are positive, the weights being positive, so every positive point lies off their zeros, and $R_{j,e}(\alpha) = \mathbb{P}_{j,\alpha}(e \in T)$ and $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T)$ (7.14); the usage at a gauge $\alpha = e^y$ is $\nabla\Phi(y)$ (8.18). For real data with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, $m_{\mathcal{E}} > 0$, so the target is the one given by the formula, and the native gauges are the gauges of 8.18(i), whose existence and uniqueness 8.18(ii),(iii) give for these data. □

Remark. The direction of a ray lies on the balance wall (35.1), so its target is its masses $\mathbf{m}_1$, and the native gauges at complex points of the direction are the points α_c of the letter complement with $c^{(1)}(\alpha_c) = 0$: the native gauges of the ray (35.4).

Proposition 8.19 (The native gauges of a compound). *As stated: the weights need not factor except in the sentence on products, and the molecules are those of the active family.*

Proof. Write $g = \sum_j (s_j d_j + t_j d_j^*)$. An inactive table enters neither the potential, nor the target, nor g, nor the Newton sum $\mathcal{N}_F$, its terms carrying the factor zero; every statement below concerns the active family, whose molecules are $\mathcal{E}_1, \ldots, \mathcal{E}_q$.

(1) The criterion of the saddle. By 8.18(ii) a native gauge exists exactly when $\boldsymbol{\tau}$ lies in the relative interior of $\mathcal{N}_F = \sum_j (s_j P_j + t_j P_j^*)$, and it is then unique modulo the space V of directions along which Φ is affine. By 8.18(iii), whose first sentence uses no connectedness, $\mathcal{N}_F = \{x : \ x_{\mathcal{E}} = g(\mathcal{E}),\ x_S \geq g(S) \text{ for all } S\}$.

(2) The extent of x_S on $\mathcal{N}_F$. The least value of a linear function on a Minkowski sum with nonnegative coefficients is the same sum of its least values on the summands. The least value of x_S on $P_j = \operatorname{conv}\{\mathbf{1}_T\}$ is $\min_T |T \cap S| = d_j(S)$, and on $P_j^* = \operatorname{conv}\{\mathbf{1}_{\mathcal{E}\setminus T}\}$ it is $\min_T |S \setminus T| = d_j^*(S)$ (7.11). So the least value of x_S on $\mathcal{N}_F$ is $g(S)$, and that of $x_{\mathcal{E}\setminus S}$ is $g(\mathcal{E} \setminus S)$; since $x_{\mathcal{E}} = g(\mathcal{E})$ on $\mathcal{N}_F$ and $\mathcal{N}_F$ is convex, x_S takes on $\mathcal{N}_F$ exactly the values of $[g(S),\ g(\mathcal{E}) - g(\mathcal{E} \setminus S)]$. At any masses $\kappa_S + \kappa_{\mathcal{E}\setminus S} - \kappa_{\mathcal{E}} = g(\mathcal{E}) - g(S) - g(\mathcal{E} \setminus S)$, the masses cancelling, so by 8.15 the length of this interval is $2\sum_j \sigma_j c_j(S) = \sum_j (s_j + t_j) c_j(S)$.

(3) The relative interior. An active table has $s_j + t_j > 0$. So the interval of step (2) is a point exactly when $c_j(S) = 0$ for every active table, that is, by Lemma S3 applied to the active family, when S is a union of the $\mathcal{E}_i$. The relative interior of a polyhedron given by linear equalities and

inequalities consists of its points at which every inequality that is not an equality on the whole polyhedron holds strictly. So $\boldsymbol{\tau}$ lies in the relative interior of $\mathcal{N}_F$ exactly when $\tau_S = g(S)$ for every union S of the $\mathcal{E}_i$ and $\tau_S > g(S)$ for every other nonempty S.

(4) Molecule by molecule. By 7.8 for the active family, every active table is the direct sum of its restrictions to the $\mathcal{E}_i$, its bases being the unions of one basis of each restriction. Taking the least of $|T\cap S|$, and of $|S\setminus T|$, restriction by restriction, $d_j(S) = \sum_i d_j(S\cap\mathcal{E}_i)$ and $d_j^*(S) = \sum_i d_j^*(S\cap\mathcal{E}_i)$ for every active j; so $g(S) = \sum_i g(S\cap\mathcal{E}_i)$ and $\tau_S - g(S) = \sum_i \big(\tau_{S\cap\mathcal{E}_i} - g(S\cap\mathcal{E}_i)\big)$. Hence the conditions of step (3) hold exactly when $\tau_{\mathcal{E}_i} = g(\mathcal{E}_i)$ for every i and $\tau_S > g(S)$ for every S with $\varnothing \neq S \subsetneq \mathcal{E}_i$ for some i: given these, a union of the $\mathcal{E}_i$ has $\tau_S - g(S) = 0$, and any other nonempty S meets some $\mathcal{E}_i$ in a nonempty proper subset, whose term is positive while the others are nonnegative; conversely each $\mathcal{E}_i$ is a union of the $\mathcal{E}_i$, and a nonempty proper subset of one $\mathcal{E}_i$ is not. Since $c_j(\mathcal{E}_i) = 0$ for every active j, $d_j(\mathcal{E}_i) = r_j(\mathcal{E}_i)$, so $r_j(\mathcal{E}\setminus\mathcal{E}_i) = r_j - r_j(\mathcal{E}_i)$ (7.10, $d_j(S) = r_j - r_j(\mathcal{E}\setminus S)$) and $d_j^*(\mathcal{E}_i) = |\mathcal{E}_i| - r_j(\mathcal{E}_i) = r_j^*(\mathcal{E}_i)$. So $g(\mathcal{E}_i) = \sum_j \big(s_j r_j(\mathcal{E}_i) + t_j r_j^*(\mathcal{E}_i)\big)$, and the conditions are the balance of every $\mathcal{E}_i$ at the target and the positivity of its proper walls there. This proves the criterion.

(5) The free scales. By 9.2 the second derivative of $\log K_j(e^y)$ along a direction v is the variance of $\langle v, \mathbf{1}_T\rangle$ under $\mathbb{P}_{j,e^y}$, and that of $\log\Psi_j(e^y) = \langle \mathbf{1}, y\rangle + \log K_j(e^{-y})$ (7.2) is the variance of $\langle v, \mathbf{1}_T\rangle$ under $\mathbb{P}_{j,e^{-y}}$. Both laws charge every basis, so each variance vanishes exactly when $\langle v, \mathbf{1}_T\rangle$ is constant over $\mathcal{B}_j$. The second derivative of Φ along v is the sum of these variances with the coefficients $s_j, t_j \geq 0$, and every active table has a positive coefficient; so Φ is affine along v exactly when $\langle v, \mathbf{1}_T\rangle$ is constant over the bases of every active table. By 7.7 for the active family such a v is constant on every $\mathcal{E}_i$. Conversely, if v is constant on every $\mathcal{E}_i$, with value v_i, then $\langle v, \mathbf{1}_T\rangle = \sum_i v_i|T\cap\mathcal{E}_i| = \sum_i v_i r_j(\mathcal{E}_i)$ on every basis of an active table j, by step (4). So $V = \operatorname{span}\{\mathbf{1}_{\mathcal{E}_1}, \dots, \mathbf{1}_{\mathcal{E}_q}\}$, and the native gauges are the gauges $e^v\alpha^\star$, $v \in V$: the gauges β with $\beta/\alpha^\star$ constant on every $\mathcal{E}_i$.

(6) Factoring weights. When the weights factor across the $\mathcal{E}_i$, the letters of every active table factor,

$$K_j(\alpha) = \prod_i K_{j,i}(\alpha_{\mathcal{E}_i}), \qquad \Psi_j(\alpha) = \prod_i \Psi_{j,i}(\alpha_{\mathcal{E}_i}),$$

with the Foreman polynomials of the restrictions (7.8). So $\Phi(y) = \sum_i \Phi_i(y_{\mathcal{E}_i})$, with Φ_i the potential of the restriction of the family to $\mathcal{E}_i$ (11.6), and the condition $\nabla\Phi = \boldsymbol{\tau}$ splits into the conditions $\nabla\Phi_i = \boldsymbol{\tau}|_{\mathcal{E}_i}$, each the native condition of the restriction to $\mathcal{E}_i$ at the masses $\boldsymbol{\tau}|_{\mathcal{E}_i}$, which lie on its balance wall by step (4), so that its target is $\boldsymbol{\tau}|_{\mathcal{E}_i}$ (8.16). The native gauges are the gauges whose restriction to every $\mathcal{E}_i$ is native there: the products of native gauges of the molecules.

(7) Units. A loop is a circuit by itself, so it lies in no other circuit; and a coloop e lies in no circuit, since a circuit C through e would lie in every basis containing the independent set $C\setminus\{e\}$, such a basis containing e, while a circuit lies in no basis. So a mark e that is a loop or a coloop in every table is linked to no other mark, and $\{e\}$ is a unit. Every basis of table j contains e, or none does, so $K_j(\alpha) = \alpha_e^{\epsilon_j}\tilde{K}_j(\alpha_{\mathcal{E}\setminus e})$ and $\Psi_j(\alpha) = \alpha_e^{1-\epsilon_j}\tilde{\Psi}_j(\alpha_{\mathcal{E}\setminus e})$ with $\epsilon_j \in \{0,1\}$: the letters of $\{e\}$ are monomials, and $\Phi(y) = \sum_j \big(s_j\epsilon_j + t_j(1-\epsilon_j)\big)y_e$ plus a function of $y_{\mathcal{E}\setminus e}$. So every derivative of Φ of order at least two that involves y_e vanishes, every entry of the still on a string through e is zero, and rung one at e is the constant $\sum_j \big(s_j\epsilon_j + t_j(1-\epsilon_j)\big)$, the mass of e (9.4). □

Remark. The criterion is stated for the molecules of the active family because linking (7.4) runs through the circuits of every table, while an inactive table enters neither the potential, nor the target, nor the walls. So a molecule of the family may join several molecules of the active family, and then a proper wall of that molecule vanishes at the target although a native gauge exists. For

instance, on $\mathcal{E}=\{1,2,3,4\}$ let the list with the bases $\{1,3\},\{1,4\},\{2,3\},\{2,4\}$ carry $(s,t)=(1,0)$ and the list $\binom{\mathcal{E}}{1}$ carry $(0,0)$, with the masses $\frac{1}{2}\mathbf{1}$ on the balance wall. Through the second table $\mathcal{E}$ is one molecule of the family, while the active family has the molecules $\{1,2\}$ and $\{3,4\}$. Every mark lies in half of the bases of the first table, so the gauge $\mathbf{1}$ is native, and the wall at $\{1,2\}$ is $\tau_{\{1,2\}}-d_1(\{1,2\})=1-1=0$.

Remark. At the centre of 8.21 the gauge $\mathbf{1}$ is native, and step (5) shows that the native gauges are the gauges constant on every molecule of the active family: this is the uniqueness clause of 8.21(i).

Proposition 8.20 (A realization sits on a wall only along its molecules). *As stated, for every family.*

Proof. (1) The walls as expectations. By 7.14, $R_{j,e}(\alpha)=\mathbb{P}_{j,\alpha}(e\in T)$ and $R^*_{j,e}(\alpha)=\mathbb{P}^*_{j,\alpha}(e\notin T)$. Summing the usage over $e\in S$,

$$m_S=\sum_j\Big(s_j\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[N_S]+t_j\mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[|S|-N_S\big]\Big),$$

and with $d^*_j(S)=|S|-r_j(S)$ (7.10) the wall $\kappa_S=m_S-\sum_j\big(s_jd_j(S)+t_jd^*_j(S)\big)$ is the displayed sum.

(2) The sign. By 7.11, $d_j(S)\le N_S\le r_j(S)$ on every basis of table j. So both brackets are nonnegative, and so is κ_S, the exponents being nonnegative.

(3) Equality. The laws $\mathbb{P}_{j,\alpha}(T)=\theta_{j,T}\alpha^T/K_j(\alpha)$ and $\mathbb{P}^*_{j,\alpha}(T)=\theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$ charge every basis. So the first bracket of table j vanishes exactly when $N_S=d_j(S)$ on every basis, and the second exactly when $N_S=r_j(S)$ on every basis; since the values of N_S fill $\{d_j(S),\dots,r_j(S)\}$ (7.11), each happens exactly when $c_j(S)=0$. Every table has $s_j>0$ or $t_j>0$, so $\kappa_S=0$ exactly when $c_j(S)=0$ for every table, that is, by Lemma S3, when S is a union of molecules.

(4) The last sentence. $\mathcal{E}$ is a union of molecules, so $\kappa_{\mathcal{E}}=0$. At a realization every table is active, so the family is connected (8.1) exactly when no nonempty proper S has $c_j(S)=0$ for every table, that is, by Lemma S3, when no nonempty proper set is a union of molecules; then every proper wall is positive by step (3). □

Remark. The reference, proof of 8.20, has steps (1) and (2) and the positivity of the bracket of an active table with $c_j(S)\ge 1$, which connectedness supplies for every nonempty proper S. Step (3) reads off the equality case, which holds for every family.

40.9 Chapter 9: realizations and the still

Proposition 9.3 (Masses at a gauge). *As stated, at every gauge e^{y_0}.*

Proof. By 9.1 the masses are $\tau_e=\sum_j\big(s_jR_{j,e}(\alpha)+t_jR^*_{j,e}(\alpha)\big)$, with $R_{j,e}=\alpha_e\partial_eK_j/K_j$ the occupation and $R^*_{j,e}=\alpha_e\partial_e\Psi_j/\Psi_j$ the dual occupation, and $R^*_{j,e}(\alpha)=1-R_{j,e}(\alpha^{-1})$ (7.14). At $\alpha=e^{y_0}$ this is the first expression. Put $a_j=R_{j,e}(e^{y_0})$ and $b_j=R_{j,e}(e^{-y_0})$, so that $R^{\pm}_{j,e}=\frac{1}{2}(a_j\pm b_j)$. With $s_j=\sigma_j+\delta_j$ and $t_j=\sigma_j-\delta_j$,

$$s_ja_j+t_j(1-b_j)=\sigma_j(1+a_j-b_j)+\delta_j(a_j+b_j-1)=\sigma_j(1+2R^-_{j,e})+\delta_j(2R^+_{j,e}-1).$$

At $y_0=0$, $a_j=b_j=R_{j,e}(\mathbf{1})$: $R^-_{j,e}=0$ and $R^+_{j,e}=R_{j,e}(\mathbf{1})$. □

Remark. By 9.2, $R_{j,e}(e^y) = \partial_{y_e} \log K_j(e^y)$, whose Taylor series at 0 is

$$\sum_{i\geq 0} \frac{1}{i!} \sum_{u\in\mathcal{E}^i} \kappa_{j,\mathbf{1}}(e,u) \prod_{l=1}^{i} y_{u_l}$$

in its domain of convergence, as in 9.7. At $y = \pm y_0$ the term of order i carries $(\pm 1)^i$, so $R^+_{j,e}$ collects the terms of even order and $R^-_{j,e}$ those of odd order. Rung one at a deviation has the parity of the odd rungs of 9.7, σ_j with the odd part of the occupations and δ_j with the even part, besides the constant $\sum_j (\sigma_j - \delta_j) = \sum_j t_j$.

Proposition 9.9 (Occupations at a gauge). *As stated, at every gauge e^{y_0}.*

Proof. By 9.4, $\varphi^{\mathbf{X}}_e(0) = \partial_{y_e} \Phi_{\mathbf{X}}(\log \alpha)$, and by 8.17, $\Phi_{\mathbf{E}}(y) = \sum_j \sigma_j \big(\log K_j(e^y) + \log \Psi_j(e^y) \big)$ and $\Phi_{\mathbf{O}}(y) = \sum_j \delta_j \big(\log K_j(e^y) - \log \Psi_j(e^y) \big)$. By 8.18, $\partial_{y_e} \log K_j(e^y) = R_{j,e}$ and $\partial_{y_e} \log \Psi_j(e^y) = R^*_{j,e}$, and $R^*_{j,e}(\alpha) = 1 - R_{j,e}(\alpha^{-1})$ (7.14). At $\alpha = e^{y_0}$, with $a_j = R_{j,e}(e^{y_0})$ and $b_j = R_{j,e}(e^{-y_0})$,

$$\varphi^{\mathbf{E}}_e(0) = \sum_j \sigma_j (a_j + 1 - b_j) = \sum_j \sigma_j (1 + 2R^-_{j,e}),$$
$$\varphi^{\mathbf{O}}_e(0) = \sum_j \delta_j (a_j - 1 + b_j) = \sum_j \delta_j (2R^+_{j,e} - 1).$$

At $\alpha = \mathbf{1}$, $R^-_{j,e} = 0$ and $R^+_{j,e} = R_{j,e}(\mathbf{1})$. The two parts add to the mass of 9.3. □

40.10 Chapter 10: the reading, classes, atoms

Proposition 10.2 (A view through the patterns). *As stated; the inclusion laws are read at every gauge.*

Proof. The expression of the views through $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$ and the closing sentence on one table are proved in the reference, proof of 10.2. What follows is the expansion at a gauge α and its reading at the table's gauge. Since $\prod_{i\in p}(1 + x_i) = \sum_{A\subseteq p} x^A$,

$$K_{j;e,\mathbf{g}}(\mathbf{1} + x) = \sum_p \vartheta^K_{j,p} \sum_{A\subseteq p} x^A = \sum_A x^A \sum_{p\supseteq A} \vartheta^K_{j,p},$$
$$\sum_{p\supseteq A} \vartheta^K_{j,p} = \sum_{T:\ p(T)\supseteq A} \theta_{j,T} \alpha^T = K_j(\alpha)\, \mathbb{P}_{j,\alpha}\big(p(T) \supseteq A\big),$$

the basis law at α being $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T / K_j(\alpha)$ (7.14). Likewise $(\mathbf{1} + x)^{\bar{p}} = \prod_{i\notin p}(1 + x_i) = \sum_{A\cap p = \varnothing} x^A$, so

$$\Psi_{j;e,\mathbf{g}}(\mathbf{1} + x) = \sum_A x^A \sum_{p\cap A = \varnothing} \vartheta^\Psi_{j,p},$$
$$\sum_{p\cap A=\varnothing} \vartheta^\Psi_{j,p} = \sum_{T:\ p(T)\cap A = \varnothing} \theta_{j,T} \alpha^{\mathcal{E}\setminus T} = \Psi_j(\alpha)\, \mathbb{P}^*_{j,\alpha}\big(p(T) \cap A = \varnothing\big),$$

with $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T} / \Psi_j(\alpha)$. So the polynomials $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$, and with them the view, are determined by $K_j(\alpha)$, $\Psi_j(\alpha)$ and the joint inclusion laws of $e, g_1, \dots, g_k$ under $\mathbb{P}_{j,\alpha}$ and $\mathbb{P}^*_{j,\alpha}$. At $\alpha = \mathbf{1}$, $\alpha^T = \alpha^{\mathcal{E}\setminus T} = 1$: $\vartheta^K_{j,p} = \vartheta^\Psi_{j,p} = \vartheta_{j,p}$, both laws are the weight law (7.14), and $x^{\bar{p}} = x^{\{0,\dots,k\}}(x^{-1})^p$ gives $\Psi_{j;e,\mathbf{g}}(x) = x^{\{0,\dots,k\}} K_{j;e,\mathbf{g}}(x^{-1})$. □

Proposition 10.8 (Computing the classes). *As stated; at a gauge α the identities have their coefficients in the field generated by the weights and the coordinates of α.*

Proof. The steps of the computation compare views and writings, as in the reference, proof of 10.8; only the field is new. Let k be the field generated by the weights and the coordinates of α. By 10.2, a view is the pair $\sum_j \sigma_j \log(K_{j;e,\mathbf{g}}\Psi_{j;e,\mathbf{g}})$, $\sum_j \delta_j \log(K_{j;e,\mathbf{g}}/\Psi_{j;e,\mathbf{g}})$ in $x_i = e^{v_i}$, and the coefficients $\vartheta^K_{j,p} = \sum_{T:\,p} \theta_{j,T}\alpha^T$ and $\vartheta^\Psi_{j,p} = \sum_{T:\,p} \theta_{j,T}\alpha^{\mathcal{E}\setminus T}$ of these polynomials are polynomials with integer coefficients in the weights and the coordinates of α, so they lie in k. Factor every $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$ into irreducible polynomials over k. Each factor of a polynomial positive on the orthant vanishes nowhere on it and so has constant sign there, and it is normalized to the value 1 at $x = \mathbf{1}$ by a constant of k. Two different normalized polynomials irreducible over k have no common factor over $\mathbb{C}$, the greatest common divisor of two polynomials over k being their greatest common divisor over $\mathbb{C}$. So the factors over $\mathbb{C}$ of 10.6 are the factors over $\mathbb{C}$ of the factors over k, and a factor over $\mathbb{C}$ divides exactly one normalized factor over k. If two views carry over $\mathbb{C}$ the same factors with the same pairs of exponents, a factor over k of one and a factor over k of the other that share a factor over $\mathbb{C}$ are equal, and comparing the exponents of that shared factor gives equal pairs of exponents. So by 10.6 two views, or two writings, are equal exactly when they carry the same normalized polynomials irreducible over k with the same pairs of exponents: every step is an identity in $k[x]$. At $\alpha = \mathbf{1}$, k is the field generated by the weights, $\mathbb{Q}$ for rational weights, and the coefficients are integers for lists. □

Proposition 10.9 (Masses and occupations on a class). *As stated; the first sentence holds for each of the two parts of the mass.*

Proof. By 9.4, $\varphi^{\mathbf{X}}_e(0) = \partial_{y_e}\Phi_{\mathbf{X}}(\log\alpha)$ is the part $\mathbf{X}$ of rung one, and $\tau_e(\Upsilon) = \partial_{y_e}\Phi(\log\alpha) = \varphi^{\mathbf{E}}_e(0) + \varphi^{\mathbf{O}}_e(0)$. By 10.3(iii) the reading of e determines its self-line functions $\varphi^{\mathbf{E}}_e$ and $\varphi^{\mathbf{O}}_e$, hence their values at 0, and the classes are the level sets of the reading (10.1). So the two parts, and their sum, are constant on every class. For a family whose weights factor across its molecules, $\Phi_{\mathbf{X}}$ is the sum of the potentials of the restrictions to the molecules (11.6), so $\varphi^{\mathbf{X}}_e(0)$ is the same in the family and in the realization of the molecule of e, on which its classes are read. Items (i) and (ii) are proved in the reference, proof of 10.9. □

40.11 Chapter 11: lean, gauge and label

Lemma 11.12 (No degenerate pair). *As stated, at every realization of every family, with the equality case.*

Proof. (1) A sum of covariances. By 9.4 and 8.17, $\kappa^{\mathbf{E}}_\Upsilon(e,f) = 4\sum_j \sigma_j\big(\partial_{y_e}\partial_{y_f}\log K_j(e^y) + \partial_{y_e}\partial_{y_f}\log\Psi_j(e^y)\big)$ at $y = \log\alpha$. By 9.2 the first derivative is $\kappa_{j,\alpha}(e,f)$, and since $\log\Psi_j(e^y) = \langle\mathbf{1},y\rangle + \log K_j(e^{-y})$ (7.2) the second is $\kappa_{j,\alpha^{-1}}(e,f)$. So $\kappa^{\mathbf{E}}_\Upsilon = 4\sum_j \sigma_j\big(C_{j,\alpha} + C_{j,\alpha^{-1}}\big)$, $C_{j,\beta}$ the covariance matrix of the indicators $(X_e)_e$ under $\mathbb{P}_{j,\beta}$, and for $w \in \mathbb{R}^{\mathcal{E}}$

$$w^{\mathsf{T}}\kappa^{\mathbf{E}}_\Upsilon w = 4\sum_j \sigma_j\Big(\operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha}}\langle w,\mathbf{1}_T\rangle + \operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha^{-1}}}\langle w,\mathbf{1}_T\rangle\Big) \;\geq\; 0,$$

since $\sigma_j > 0$ at a realization.

(2) The inequality. The principal 2×2 submatrix of $\kappa^{\mathbf{E}}_\Upsilon$ on $\{e,f\}$ is positive semidefinite, so its determinant $\kappa^{\mathbf{E}}_\Upsilon(e,e)\kappa^{\mathbf{E}}_\Upsilon(f,f) - \kappa^{\mathbf{E}}_\Upsilon(e,f)^2$ is nonnegative. It vanishes exactly when that submatrix

is singular, that is when some $w = \lambda \mathbf{1}_e + \mu \mathbf{1}_f \neq 0$ has $w^{\mathsf{T}} \kappa^{\mathbf{E}}_{\Upsilon} w = 0$, a positive semidefinite matrix annihilating exactly the vectors at which its form vanishes.

(3) The vectors of form zero. Both laws in step (1) charge every basis (7.14), so $w^{\mathsf{T}} \kappa^{\mathbf{E}}_{\Upsilon} w = 0$ exactly when $\langle w, \mathbf{1}_T \rangle$ is constant over the bases of every table. By 7.7 such a w is constant on every molecule. Conversely, if w is constant on every molecule, with value w_M on the molecule M, then $\langle w, \mathbf{1}_T \rangle = \sum_M w_M r_j(M)$ on every basis of table j, each table being the direct sum of its restrictions to the molecules (7.8).

(4) The cases. If e is a unit, $w = \mathbf{1}_e$ is constant on every molecule, and equality holds; likewise if f is a unit. If neither is a unit and they lie in different molecules, the molecule of e has a mark other than e, not f, at which $w = \lambda \mathbf{1}_e + \mu \mathbf{1}_f$ vanishes, so $\lambda = 0$; likewise $\mu = 0$, and the inequality is strict. If e and f lie in one molecule M: when M has a third mark, w vanishes there, so $\lambda = \mu = 0$ and the inequality is strict; when $M = \{e, f\}$, $w = \mathbf{1}_e + \mathbf{1}_f$ is constant on every molecule, and equality holds. A molecule with at least three marks contains neither a unit nor a molecule of two marks, so the inequality is strict for any two of its marks. □

Remark. The reference, proof of 11.12, has steps (1) to (3) and concludes for a molecule with at least three marks; the case analysis of step (4) gives the equality case for every family.

Lemma 11.13 (Equal diagonals on a class). *As stated: on every class, for both parts and every rung.*

Proof. By 9.4, $\varphi^{\mathbf{X}}_e(v) = \partial_{y_e} \Phi_{\mathbf{X}}(\log \alpha + v \mathbf{1}_e)$, so $(\varphi^{\mathbf{X}}_e)^{(k-1)}(0) = \partial^k_{y_e} \Phi_{\mathbf{X}}(\log \alpha)$, and the still of each part gives

$$\kappa^{\mathbf{X}}_{\Upsilon}(e, \dots, e) = 2^k \, \partial^k_{y_e} \Phi_{\mathbf{X}}(\log \alpha) = 2^k (\varphi^{\mathbf{X}}_e)^{(k-1)}(0).$$

By 10.3(iii) the reading of e determines $\varphi^{\mathbf{E}}_e$ and $\varphi^{\mathbf{O}}_e$; the marks of a class have one reading (10.1), so they have the same self-line functions, and the same derivatives of every order at 0. At an atom there is one class. □

Remark. The reference states 11.13 at an atom of a molecule with at least three marks, for $\kappa^{\mathbf{E}}(e, e)$, and proves it through the rung-two tallies and 11.12. The self-line functions carry every order and both parts, and make the statement one about classes.

Corollary 11.16 (An exchange negates odd cumulants). *As stated, for an exchange of any family.*

Proof. (1) The weight laws. By 11.14, $T \mapsto T' = \mathcal{E} \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ with $\theta_{h(j),T'} = \theta_{j,T}$, and it is injective because g is. So $K_{h(j)}(\mathbf{1}) = K_j(\mathbf{1})$, and for $T \sim \mathbb{P}_{j,\mathbf{1}}$ the basis T' has the law $\mathbb{P}_{h(j),\mathbf{1}}$.

(2) The cumulants. $X_{g(e)}(T') = \mathbf{1}[g(e) \notin g(T)] = 1 - X_e(T)$. Joint cumulants of order at least two are multilinear and unchanged when constants are added to the variables, so for a string $w = (e_1, \dots, e_k)$, $k \geq 2$,

$$\begin{aligned} \kappa_{h(j),\mathbf{1}}(g(w)) &= \kappa\big(X_{g(e_1)}(T'), \dots, X_{g(e_k)}(T')\big) \\ &= \kappa\big(1 - X_{e_1}(T), \dots, 1 - X_{e_k}(T)\big) = (-1)^k \kappa_{j,\mathbf{1}}(w). \end{aligned}$$

(3) The stills. By 9.5, $\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(w) = 2^k \sum_j \big(s_j + (-1)^k t_j\big) \kappa_{j,\mathbf{1}}(w)$. The condition on the pairs in 11.14 gives $(s_{h(j)}, t_{h(j)}) = (s_j, t_j)$, and h is a bijection of the tables; reindexing the sum by h and using step (2),

$$\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(g(w)) = 2^k \sum_j \big(s_{h(j)} + (-1)^k t_{h(j)}\big) \kappa_{h(j),\mathbf{1}}(g(w))$$

$$= 2^k \sum_j \big(t_j + (-1)^k s_j\big)\kappa_{j,\mathbf{1}}(w),$$

the still at $\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}$, whose exponents are (t_j, s_j).

(4) The classes of the centre. By 11.15 with $F' = F$ at the table's gauge, the reading of $g(e)$ at $-\boldsymbol{\delta}$ equals that of e at $\boldsymbol{\delta}$; at the centre $-\boldsymbol{\delta} = \boldsymbol{\delta} = 0$, so $g(e)$ lies in the class of e. Hence g maps every class of the centre into itself, and onto itself, g being injective and the class finite. □

Remark. The reference, proof of 11.16, is the case of one table, where h is the identity.

Corollary 11.17 (With one table the writings are certificates). *As stated: its last sentence holds at every realization of every molecule.*

Proof. The first sentence is the reference, proof of 11.17. The last sentence is the closing sentence of 14.18, with keeping the potential read at the gauge of the realization (14.17); its proof at every realization is given with that lemma, in the section on Chapter 14. □

40.12 Chapter 12: restriction to a bunch

Proposition 12.12 (Coins of a tensorminental table). *As stated: tensor rank one over $\mathbb{R}$ or over $\mathbb{C}$.*

Proof. The first display is proved in the reference, proof of 12.12. At tensor rank one there is one slice $A \in \mathbb{F}^{r\times n}$, of rank r since $K(\mathbf{1}) = \det(AA^{\mathsf{H}}) > 0$ (7.18); the only pair is $(1,1)$, with $w_{11} = 1$, and $\Pi^{(11)} = A^{\mathsf{H}}M^{-1}AD_\alpha$, $M = AD_\alpha A^{\mathsf{H}}$ (7.20). So $x^{d(B)}\mathsf{Q}_B^K(x;\alpha) = \prod_{i=1}^{|B|}(1 - p_i + p_i x)$ with p_i the eigenvalues of $\Pi^{(11)}_{BB}$.

(1) The projection. Put $P = D_\alpha^{1/2}\Pi^{(11)}D_\alpha^{-1/2} = D_\alpha^{1/2}A^{\mathsf{H}}M^{-1}AD_\alpha^{1/2}$. Then $P^{\mathsf{H}} = P$ and $P^2 = D_\alpha^{1/2}A^{\mathsf{H}}M^{-1}(AD_\alpha A^{\mathsf{H}})M^{-1}AD_\alpha^{1/2} = P$, and $P\,D_\alpha^{1/2}A^{\mathsf{H}} = D_\alpha^{1/2}A^{\mathsf{H}}$; so P is the orthogonal projection onto $\{D_\alpha^{1/2}A^{\mathsf{H}}y\}$, the row space of $AD_\alpha^{1/2}$ in the sense of 7.20.

(2) Similarity. D_α is diagonal, so $P_{BB} = D_B^{1/2}\Pi^{(11)}_{BB}D_B^{-1/2}$, with D_B the diagonal block on B, and P_{BB} has the eigenvalues p_i.

(3) Principal angles. P_{BB} is the compression of P to $\mathbb{F}^B$, the vectors supported on B: $v^{\mathsf{H}}P_{BB}v = \|Pv\|^2$ for $v \in \mathbb{F}^B$. With E_B the inclusion of $\mathbb{F}^B$ and Q a matrix whose columns are an orthonormal basis of the range of P, $P_{BB} = (E_B^{\mathsf{H}}Q)(E_B^{\mathsf{H}}Q)^{\mathsf{H}}$. Its eigenvalues are the squares of the singular values of $E_B^{\mathsf{H}}Q$, completed by zeros, which are the cosines of the principal angles between $\mathbb{F}^B$ and the range of P; they lie in $[0,1]$.

(4) The eigenvalue 1. For $v \in \mathbb{F}^B$, $P_{BB}v = v$ exactly when $\|Pv\| = \|v\|$, that is when v lies in the range of P. So the eigenspace is $\{D_\alpha^{1/2}A^{\mathsf{H}}y :\ (D_\alpha^{1/2}A^{\mathsf{H}}y)_R = 0\} = \{D_\alpha^{1/2}A^{\mathsf{H}}y :\ A_R^{\mathsf{H}}y = 0\}$, of dimension $r - \operatorname{rank}A_R$, the map $y \mapsto D_\alpha^{1/2}A^{\mathsf{H}}y$ being injective. The bases of the table are the sets T with $\det A_T \neq 0$ (7.17), so $r(R) = \operatorname{rank}A_R$, and $r - r(R) = d(B)$ (7.10, $d(B) = |B| - r^*(B)$ with $r^*(B) = |B| + r(R) - r$).

(5) The eigenvalue 0. For $v \in \mathbb{F}^B$, $P_{BB}v = 0$ exactly when $Pv = 0$, that is $AD_\alpha^{1/2}v = A_B D_B^{1/2}v_B = 0$. So the eigenspace has dimension $|B| - \operatorname{rank}A_B = |B| - r(B) = d^*(B)$.

(6) The coins. P_{BB} is Hermitian, so of its $|B|$ eigenvalues $d(B)$ are 1, $d^*(B)$ are 0, and the other $|B| - d(B) - d^*(B) = c(B)$ lie in $(0,1)$. In $x^{d(B)}\mathsf{Q}_B^K(x;\alpha) = \prod_i(1 - p_i + p_i x)$ the eigenvalues 1 give the factor $x^{d(B)}$ and the eigenvalues 0 the factor 1; so the coins of Q_B^K are the eigenvalues $p_i = \cos^2\vartheta_i \in (0,1)$, and the Skeleton values are $\xi_i = (1-p_i)/p_i = \tan^2\vartheta_i > 0$ (12.9). □

Remark. Over $\mathbb{R}$ this is the reference, proof of 12.12. Over $\mathbb{C}$ the projection is Hermitian, and each step reads the same with conjugate transposes.

Corollary 12.20 (The native level split is interior). *As stated: the directions come from chains of circuits joining the marks of each side, which a connected table with a positive exponent supplies.*

Proof. The first two sentences, the relative interior and the walls with their exceptions, are proved in the reference, proof of 12.20, without any connectedness. Write $\mathcal{L}$ for the set of level and colevel letters with positive exponent, $c_L > 0$ for the exponent of $L \in \mathcal{L}$, N_L for the convex hull of its exponent vectors, and $Q = \sum_{L\in\mathcal{L}} c_L N_L$ for the Minkowski sum of the statement.

(1) A circuit with two marks on one side. Let $a \neq a'$ lie both in B or both in R, and in a common circuit of a table j with a positive exponent. By 7.6 some basis T of table j has $a \in T$, $a' \notin T$ and $T' = (T \setminus \{a\}) \cup \{a'\}$ a basis, and $|T \cap B| = |T' \cap B| =: k$, the two marks lying on one side. At the native gauge every basis has positive probability under both laws (7.14). If $s_j > 0$, the level letter $K^{(k)}_{j,B}$ has the positive exponent $s_j \mathbb{P}_{j,\alpha^\star}(N_B = k)$, and $N_{K^{(k)}_{j,B}}$ contains $\mathbf{1}_T$ and $\mathbf{1}_{T'}$, whose difference is $\mathbf{1}_a - \mathbf{1}_{a'}$. If $t_j > 0$, the colevel letter $\Psi^{(|B|-k)}_{j,B}$ has the positive exponent $t_j \mathbb{P}^*_{j,\alpha^\star}(N^*_B = |B| - k)$, and its polytope contains $\mathbf{1}_{\mathcal{E}\setminus T}$ and $\mathbf{1}_{\mathcal{E}\setminus T'}$, whose difference is $\mathbf{1}_{a'} - \mathbf{1}_a$. Fixing points of the other summands, $c_L(\mathbf{1}_a - \mathbf{1}_{a'})$ is the difference of two points of Q for one $L \in \mathcal{L}$, so every v with $\langle v, x\rangle$ constant on Q has $v_a = v_{a'}$.

(2) The directions. Under the hypothesis the chains join all the marks of B and all the marks of R, so by step (1) such a v is constant on B and on R. Conversely, a monomial α^T of a level letter of level k has $\langle \mathbf{1}_B, \mathbf{1}_T\rangle = k$, and a monomial $\alpha^{\mathcal{E}\setminus T}$ of a colevel letter of colevel k has $\langle \mathbf{1}_B, \mathbf{1}_{\mathcal{E}\setminus T}\rangle = |B \setminus T| = k$; so $\langle \mathbf{1}_B, x\rangle$ is constant on every N_L, hence on Q, and so is $\langle \mathbf{1}_R, x\rangle = \langle \mathbf{1}, x\rangle - \langle \mathbf{1}_B, x\rangle$, each letter being homogeneous. So the directions v with $\langle v, x\rangle$ constant on Q are $\operatorname{span}\{\mathbf{1}_B, \mathbf{1}_R\}$.

(3) The exceptions. A linear function is constant on a Minkowski sum with positive coefficients exactly when it is constant on every summand. So S is an exception exactly when $\mathbf{1}_S \in \operatorname{span}\{\mathbf{1}_B, \mathbf{1}_R\}$, that is $S \in \{\varnothing, B, R, \mathcal{E}\}$.

(4) Convergence. Write $\alpha_B = \rho_B u$ with $\rho_B = \langle \mathbf{1}_B, \alpha_B\rangle$ and $\alpha_B = e^{y_B}$; then $d\alpha_B = \rho_B^{|B|-1} d\rho_B \, d\omega_B(u) = \prod_{e\in B} \alpha_e \, dy_B$, and splitting y_B into its component t along the unit vector $|B|^{-1/2}\mathbf{1}_B$, which changes ρ_B by the factor $e^{t|B|^{-1/2}}$ and leaves u unchanged, and its component orthogonal to $\mathbf{1}_B$, on which u depends, gives $d\rho_B/\rho_B = |B|^{-1/2} dt$ and so $d\omega_B(u) = |B|^{1/2} \prod_{e\in B} u_e \, dy_B$ on the hyperplane $\langle y_B, \mathbf{1}_B\rangle = 0$; likewise on R. So with $Y = \{y : \langle y_B, \mathbf{1}_B\rangle = \langle y_R, \mathbf{1}_R\rangle = 0\}$ the integral of the statement is $(|B||R|)^{1/2} \int_Y u^{\boldsymbol{\tau}|_B} v^{\boldsymbol{\tau}|_R} \prod_{L\in\mathcal{L}} L(u,v)^{-c_L} dy$, with $u = e^{y_B}/\rho_B$ and $v = e^{y_R}/\rho_R$. Each L is homogeneous in α_B and in α_R, and the degrees weighted by the exponents add to τ_B and to τ_R: for the level letters of table j they add to $s_j \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha^\star}}[N_B] = s_j \sum_{e\in B} R_{j,e}(\alpha^\star)$ in α_B, for its colevel letters to $t_j \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha^\star}}[N^*_B] = t_j \sum_{e\in B} R^*_{j,e}(\alpha^\star)$, and the usage at the native gauge is $\boldsymbol{\tau}$ (8.16); likewise on R. So the powers of ρ_B and ρ_R cancel, and the integrand is $e^{\langle \boldsymbol{\tau}, y\rangle} \prod_L L(e^y)^{-c_L}$. Each $L(e^y)$ is at least its least coefficient times $e^{\max_{x\in N_L}\langle x,y\rangle}$, the maximum being attained at an exponent vector; so the integrand is at most $C \exp\big(-(h(y) - \langle \boldsymbol{\tau}, y\rangle)\big)$, with $h(y) = \max_{x\in Q}\langle x, y\rangle = \sum_L c_L \max_{x\in N_L}\langle x, y\rangle$. By step (2) the affine hull of Q has the direction space $\operatorname{span}\{\mathbf{1}_B, \mathbf{1}_R\}^\perp = Y$, and $\boldsymbol{\tau}$ lies in the relative interior of Q; so for some $\varepsilon > 0$, $\boldsymbol{\tau} + w \in Q$ for all $w \in Y$ with $\|w\| \leq \varepsilon$, and $h(y) \geq \langle \boldsymbol{\tau}, y\rangle + \varepsilon\|y\|$ for $y \in Y$. The integrand is at most $Ce^{-\varepsilon\|y\|}$, integrable over Y.

(5) A connected table. In a connected table any two marks lie in a common circuit, the fact used in the reference, proof of 12.20; so a connected table with a positive exponent joins any two

marks of B, and any two marks of R, by a chain of length one. $\square$

Remark. The hypothesis cannot be replaced by connectedness of the family. On $\mathcal{E} = \{a, a', b, b'\}$ with $B = \{a, a'\}$, let the three tables of rank one with the bases $\{a\}, \{b\}$, then $\{a'\}, \{b'\}$, then $\{a\}, \{b'\}$ carry positive exponents, read at a native gauge. The family is connected, but every level of every table holds one basis, so every level and colevel polytope is a point, every set is an exception, and the wall at $\{a\}$ vanishes. Each circuit $\{a, b\}$, $\{a', b'\}$, $\{a, b'\}$ meets both sides, and no chain on one side joins a to a'.

40.13 Chapter 13: the order of classes and the Foreman bunch

Proposition 13.10 (Mirror classes). *As stated: levels mirrored up to a bijection of the tables, and the exchange of any family.*

Proof. (i) Fix j. Under $\mathbb{P}_{h(j),\mathbf{1}}$ the level N_D takes the value a with probability $\mathbb{P}_{j,\mathbf{1}}(N_C = k_j - a)$, so it has the law of $k_j - N_C$ under $\mathbb{P}_{j,\mathbf{1}}$. A cumulant of order $a \geq 2$ is unchanged by adding a constant and is homogeneous of degree a, so $\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a \kappa_{j,\mathbf{1}}(N_C^{\times a})$. Since h is a bijection of the tables with $\sigma_{h(j)} = \sigma_j$,

$$\sum_j \sigma_j \kappa_{j,\mathbf{1}}(N_D^{\times a}) = \sum_j \sigma_{h(j)} \kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a \sum_j \sigma_j \kappa_{j,\mathbf{1}}(N_C^{\times a}),$$

and likewise with the δ_j. By 13.4(i) the coefficients of v_0^a, $a \geq 2$, in the **E** part of stage 0 are $\frac{2}{a!} \sum_j \sigma_j \kappa_{j,\mathbf{1}}(N_C^{\times a})$ at even a and zero at odd a, and in its **O** part $\frac{2}{a!} \sum_j \delta_j \kappa_{j,\mathbf{1}}(N_C^{\times a})$ at odd a and zero at even a; so the **E** parts of $\mathcal{V}_0(C)$ and $\mathcal{V}_0(D)$ are equal and their **O** parts are negatives.

(ii) Stage 0 holds one pair. By (i) the **E** parts agree, so the pairs are compared by their **O** parts (13.5(iii)). Their coefficients vanish at every even a and at every odd a below the least odd a with $c = \frac{2}{a!} \sum_j \delta_j \kappa_{j,\mathbf{1}}(N_C^{\times a}) \neq 0$, where C has c and D has $-c$. So stage 0 separates C and D, and $C \prec D$ exactly when $c < -c$, that is $c < 0$. At the opposite lean c changes sign, being linear in $\boldsymbol{\delta}$, and the order is the other one.

(iii) At the centre the **O** parts vanish and the **E** parts agree by (i), so stage 0 does not separate C and D. They are different classes with equally many marks and $\prec$ is a strict total order on them (13.7(i)), so a later stage does.

The two instances. If C and D together make up the molecule, $|T \cap D| = r_j - |T \cap C|$ on every basis of table j: h is the identity and $k_j = r_j$. If g exchanges the family, with h the bijection of its tables, then for $T \sim \mathbb{P}_{j,\mathbf{1}}$ the basis $T' = \mathcal{E} \setminus g(T)$ has the law $\mathbb{P}_{h(j),\mathbf{1}}$ (proof of 11.16, step (1)) and $|T' \cap g(C)| = |g(C) \setminus g(T)| = |C \setminus T| = |C| - |T \cap C|$, so $\mathbb{P}_{h(j),\mathbf{1}}(|T \cap g(C)| = a) = \mathbb{P}_{j,\mathbf{1}}(|T \cap C| = |C| - a)$ and $k_j = |C|$. Moreover $g(C)$ is a class at the lean with $|C|$ marks: g maps the classes at $\boldsymbol{\delta}$ onto those at $-\boldsymbol{\delta}$ (11.15), which are the classes at $\boldsymbol{\delta}$ (11.2). $\square$

Remark. The reference, proof of 13.10, takes the mirror table by table, h the identity, and the exchange of a one-table molecule. Its conclusions use only the sums over the tables weighted by the σ_j and the δ_j, which a bijection of the tables preserving the pairs leaves unchanged.

40.14 Chapter 14: peel, children and the Foreman tree

Corollary 14.9 (A molecule with an exchange). *As stated: a molecule with any number of tables; the classes of the centre are those of 11.16.*

Proof. Read M as the family $F|M$ of the restrictions of the tables to M (14.1), with the pair coordinates (σ_j, δ_j) of its tables. By 11.14 an exchange g of M comes with a bijection h of the tables such that $T \mapsto M \setminus g(T)$ maps the bases of table j onto those of table $h(j)$, with their weights, and $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$. The first two conditions concern the tables alone, and the third holds with $\boldsymbol{\delta}$ replaced by $-\boldsymbol{\delta}$ or by 0 when it holds at $\boldsymbol{\delta}$. So g exchanges M at each of the leans $\boldsymbol{\delta}$, $-\boldsymbol{\delta}$ and 0; likewise a bijection carrying M at $\boldsymbol{\delta}$ carries it at these three leans.

Trees and bunches. 14.8 with $F = F' = M$ gives $g\big(\operatorname{Tree}(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big) = \operatorname{Tree}(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$, each recorded bunch going onto the bunch recorded at the same place. At the root the recorded bunches are $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ and $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$ (14.4), so $g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big) = B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$. At the lean 0 the two trees are one tree, which gives $g\big(\operatorname{Tree}(M;\boldsymbol{\sigma},0)\big) = \operatorname{Tree}(M;\boldsymbol{\sigma},0)$ and $g\big(B(M;\boldsymbol{\sigma},0)\big) = B(M;\boldsymbol{\sigma},0)$.

Classes of the centre. 11.16, applied to the family $F|M$.

A carrying map. Let g carry M. 11.15 at the tables' gauges, where $g^*\log\mathbf{1} = \log\mathbf{1}$, says that the reading of $g(e)$ equals that of e at $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, for every mark e. So $g(e)$ lies in the class of e, that is $g(C) \subseteq C$ for every class C, and $g(C) = C$ since g is injective and C is finite. The last sentence of 14.8, at each of the three leans at which g carries M, maps each of the trees at 0, $\boldsymbol{\delta}$ and $-\boldsymbol{\delta}$ onto itself. □

Lemma 14.10 (An exchange turns a window over). *As stated: an exchange of one family with another, table by table through the bijection h of the tables.*

Proof. Let F be on $\mathcal{E}$ and F' on $\mathcal{E}'$. By 11.14 the map $T \mapsto \mathcal{E}' \setminus g(T)$ is a bijection from the bases of table j of F onto those of table $h(j)$ of F'. For a basis T, g being a bijection,

$$\big|(\mathcal{E}' \setminus g(T)) \cap g(S)\big| = |g(S)| - |g(T) \cap g(S)| = |S| - |T \cap S|.$$

By 7.11 the numbers $|T \cap S|$, T a basis of table j, fill $\{d_j(S), \dots, r_j(S)\}$. So the numbers $|T' \cap S'|$, T' a basis of table $h(j)$, fill $\{|S| - r_j(S), \dots, |S| - d_j(S)\}$, a window of the same width $c_j(S) = r_j(S) - d_j(S)$, and the bijection takes the bases of table j at level k on S to those of table $h(j)$ at level $|S| - k$ on S'. This is the reference, proof of 14.10, with h. □

Proposition 14.12 (Level form). *As stated: table by table in a family, and for the family through the bijections that carry it with the identity on its tables.*

Proof. Fix a table j and a transposition τ of two marks of B fixing every other mark. A basis T with $T \cap B = p$ is $p \sqcup X$ with X a basis of the child at p, of weight $\theta_{j,(p),X} = \theta_{j,T}$, and $K_{j,(p)}(\alpha_R) = \sum_X \theta_{j,(p),X}\alpha_R^X$ (12.1, 12.6). Distinct bases of one child give distinct monomials, so $K_{j,(p)}$ determines the child at p with its weights, and $K_{j,(p)} = 0$ when p is not a pattern.

Let $K_{j,(p)} = K_{j,(p')}$ whenever $|p| = |p'|$. For a basis $T = p \sqcup X$, $\tau(T) = \tau(p) \sqcup X$, since τ fixes R, and $|\tau(p)| = |p|$. So X is a basis of the child at $\tau(p)$ with the weight $\theta_{j,(p),X}$: $\tau(T)$ is a basis of table j and $\theta_{j,\tau(T)} = \theta_{j,T}$. Hence τ maps the bases of table j into themselves with their weights, and onto them, being injective on a finite set.

Conversely, let every such transposition map the bases of table j onto themselves with their weights, and let $p, p' \subseteq B$ have $|p| = |p'|$. The transpositions of marks of B generate the permutations of B, so a product ς of them has $\varsigma(p) = p'$; it maps the bases onto themselves with their weights and fixes R. So $p \sqcup X \mapsto p' \sqcup X$ is a bijection from the bases with trace p onto those with trace p' keeping the weights, and $K_{j,(p')} = K_{j,(p)}$. This is the reference, proof of 14.12, for table j, where 14.7(i) is used with the identity on R.

B is in level form in the family when it is in level form in every table, that is, by the above, when every such transposition maps the bases of every table onto themselves with their weights.

That is the condition of 11.14 with $F' = F$ and h the identity, whose condition on the pairs, $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$, then holds by itself. □

Definition 14.17 (Keeping the potential; twins; descents; copies). As stated: keeping and taking the potential at every realization; twins, descents and copies at the table's gauge.

Lemma 14.18 (The writings certify the potential). *As stated, at every realization* $\Upsilon = ((s_j, t_j)_j; \alpha)$.

Proof. Let M have n marks, and let $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ be its potential. For a mark x and an order $w = (w_1, \dots, w_{n-1})$ of its room, the potential written from x in the order w is the pair of functions

$$\Phi^{\mathbf{X}}_{x,w}(v_0, \dots, v_{n-1}) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0 \mathbf{1}_x + \sum_{i=1}^{n-1} v_i \mathbf{1}_{w_i} \Big)$$

(10.1). For a bijection g and a mark a, $g^* \mathbf{1}_{g(a)} = \mathbf{1}_a$, since $(g^* \mathbf{1}_{g(a)})_e = (\mathbf{1}_{g(a)})_{g(e)} = \mathbf{1}[e = a]$.

(i) By 10.4 there are orders $(a_1, \dots, a_{n-1})$ of the room of e and $(b_1, \dots, b_{n-1})$ of the room of f with $\Phi^{\mathbf{X}}_{e,(a_i)_i} = \Phi^{\mathbf{X}}_{f,(b_i)_i}$ for both parts. Put $a_0 = e$ and $b_0 = f$, and let β be the bijection with $\beta(a_i) = b_i$ for $0 \le i \le n-1$; so $\beta(e) = f$. For $y \in \mathbb{R}^M$ put $v_i = y_{b_i}$. Then $\sum_i v_i \mathbf{1}_{b_i} = y$ and $\sum_i v_i \mathbf{1}_{a_i} = \sum_i y_{\beta(a_i)} \mathbf{1}_{a_i} = \beta^* y$, and the equality of the two writings at $(v_0, \dots, v_{n-1})$ reads

$$\Phi_{\mathbf{X}}(\log\alpha + \beta^* y) = \Phi_{\mathbf{X}}(\log\alpha + y)$$

for every y and both parts: β keeps the potential (14.17).

(ii) Let g keep the potential. For a mark x, an order w of its room and $v \in \mathbb{R}^n$, put $y = v_0 \mathbf{1}_{g(x)} + \sum_i v_i \mathbf{1}_{g(w_i)}$; then $g^* y = v_0 \mathbf{1}_x + \sum_i v_i \mathbf{1}_{w_i}$, and

$$\Phi^{\mathbf{X}}_{x,w}(v) = \Phi_{\mathbf{X}}(\log\alpha + g^* y) = \Phi_{\mathbf{X}}(\log\alpha + y) = \Phi^{\mathbf{X}}_{g(x),g(w)}(v), \qquad g(w) = (g(w_1), \dots, g(w_{n-1})).$$

As w runs over the orders of the room of x, $g(w)$ runs over those of the room of $g(x)$, once each. So the readings of x and of $g(x)$, the multisets of these pairs of functions, are equal, and $g(x)$ lies in the class of x. Thus $g(C) \subseteq C$ for every class C, and $g(C) = C$, g being injective and C finite.

Let $g : M \to M'$ take the potential of M, read at Υ, to that of M', read at a realization with the gauge α'. The same computation gives $\Phi^{\mathbf{X}}_{x,w}(v) = \Phi'^{\mathbf{X}}_{g(x),g(w)}(v) + c_{\mathbf{X}}$: the reading of x in M is the reading of $g(x)$ in M' with every function shifted by the constants $c_{\mathbf{X}}$. So two marks of M have equal readings exactly when their images have equal readings in M', and g maps the classes of M onto those of M'.

The centre of $\Upsilon = \Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha}$ is $\Upsilon_{\boldsymbol{\sigma},0;\alpha}$, read at the same gauge (10.1). Its potential is $(\Phi_{\mathbf{E}}, 0)$, $\Phi_{\mathbf{E}} = \sum_j \sigma_j \log \mathbf{E}_j(e^y)$ being the same function at every lean (8.17). A bijection keeping the potential at Υ keeps it at the centre, and one taking the potential of M to that of M' at their realizations takes it at their centres, with the constants $(c_{\mathbf{E}}, 0)$; so the two preceding paragraphs hold at the centre as well.

For the last sentence: if e and f read alike, (i) gives a bijection keeping the potential that takes e to f; conversely, a bijection keeping the potential maps the class of e onto itself by (ii), so if it takes e to f, then f lies in that class. □

Remark. At the table's gauge $\log\alpha = 0$, the two conditions of 14.17 read $\Phi_{\mathbf{X}} \circ g^* = \Phi_{\mathbf{X}}$ and $\Phi_{\mathbf{X}} \circ g^* = \Phi'_{\mathbf{X}} + c_{\mathbf{X}}$, and the proof above is the reference, proof of 14.18. The statements that keep or take the potential at the table's gauge, 14.20, 14.21, 14.22, 28.9 and 28.11, and the twins, descents and copies of 14.17, read these conditions as in the reference.

40.15 Chapter 15: the integral at a bunch

Corollary 15.3 (The scale of a bunch of width one). *As stated: every bunch, table by table, with one Skeleton value in each table of width one.*

Proof. Fix a table j and write $d = d_j(B)$, $d^* = d_j^*(B)$ and $c = c_j(B)$. By 7.11 the levels $|T \cap B|$ fill $\{d, \dots, d+c\}$, and the colevels $|B \setminus T|$ fill $\{|B| - d - c, \dots, |B| - d\} = \{d^*, \dots, d^* + c\}$, since $d^* = |B| - r_j(B) = |B| - d - c$ (7.10). Each level letter $K_{j,B}^{(k)}$ and each colevel letter $\Psi_{j,B}^{(k)}$ in these windows is a sum of positive terms over a nonempty set of bases, positive on the orthant. By 12.15,

$$x^d \mathsf{Q}_{j,B}^K(x;\alpha) = \sum_{k=d}^{d+c} x^k \frac{K_{j,B}^{(k)}(\alpha)}{K_j(\alpha)}, \qquad x^{d^*} \mathsf{Q}_{j,B}^\Psi(x;\alpha) = \sum_{k=d^*}^{d^*+c} x^k \frac{\Psi_{j,B}^{(k)}(\alpha)}{\Psi_j(\alpha)}.$$

If $c = 0$ both Skeleton polynomials are the constant 1, of degree 0, and the spectra are empty (12.9). If $c = 1$, then $K_j = K_{j,B}^{(d)} + K_{j,B}^{(d+1)}$, and with $\xi = K_{j,B}^{(d)}(\alpha)/K_{j,B}^{(d+1)}(\alpha) > 0$,

$$\mathsf{Q}_{j,B}^K(x;\alpha) = \frac{K_{j,B}^{(d)}(\alpha) + xK_{j,B}^{(d+1)}(\alpha)}{K_{j,B}^{(d)}(\alpha) + K_{j,B}^{(d+1)}(\alpha)} = \frac{x+\xi}{1+\xi}.$$

So the Skeleton value is ξ, the level letter at the floor over the level letter at the largest level, and the coin is $1/(1+\xi) = K_{j,B}^{(d+1)}(\alpha)/K_j(\alpha)$, the probability under $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ of the bases at level $d+1$ (7.14), that is $\mathbb{P}_{j,\alpha}(N_B = d+1)$. The same computation with the colevel letters gives the Skeleton value $\Psi_{j,B}^{(d^*)}(\alpha)/\Psi_{j,B}^{(d^*+1)}(\alpha)$ and the coin $\Psi_{j,B}^{(d^*+1)}(\alpha)/\Psi_j(\alpha) = \mathbb{P}_{j,\alpha}^*(N_B^* = d^*+1)$, since $\mathbb{P}_{j,\alpha}^*(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$.

Let $B = \{e\}$ with e neither a loop nor a coloop of table j. Then $d = 0$ and $r_j(\{e\}) = 1$, so $c = 1$ and $d^* = 0$. The bases at level 0 avoid e and those at level 1 contain it, so $K_{j,\{e\}}^{(0)} = K_{j,\setminus e}(\alpha_R)$ and $K_{j,\{e\}}^{(1)} = \alpha_e K_{j,/e}(\alpha_R)$; the bases at colevel 0 contain e and those at colevel 1 avoid it, so $\Psi_{j,\{e\}}^{(0)} = \Psi_{j,/e}(\alpha_R)$ and $\Psi_{j,\{e\}}^{(1)} = \alpha_e \Psi_{j,\setminus e}(\alpha_R)$ (7.3). The coins are $\alpha_e K_{j,/e}/K_j = \mathbb{P}_{j,\alpha}(e \in T) = R_{j,e}(\alpha)$ and $\alpha_e \Psi_{j,\setminus e}/\Psi_j = \mathbb{P}_{j,\alpha}^*(e \notin T) = R_{j,e}^*(\alpha)$ (7.14). This is the reference, proof of 15.3. □

Theorem 15.5 (The scale in its spectral gaps). *As stated: complex coefficients and letter exponents, expanded about any point r of the axis with $|\xi_\ell - r| < r$ for every ℓ.*

Proof. The left side is the value of the scale integral $\int_0^\infty x^{\kappa-1} \prod_{\ell \in J}(x+\xi_\ell)^{-w_\ell} dx$, which converges since $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$ (15.1). For $x \geq 0$ put $q_\ell(x) = (r - \xi_\ell)/(x+r)$, so that $x + \xi_\ell = (x+r)(1 - q_\ell(x))$ and $|q_\ell(x)| \leq |r - \xi_\ell|/r = |g_\ell| < 1$.

Expansion. Since $x + r > 0$ and $\operatorname{Re}(1 - q_\ell(x)) > 0$, the principal argument of $x + \xi_\ell$ is that of $1 - q_\ell(x)$, which lies in $(-\pi/2, \pi/2)$; so $(x+\xi_\ell)^{-w_\ell} = (x+r)^{-w_\ell}(1 - q_\ell(x))^{-w_\ell}$ with principal powers, and the binomial series $(1-q)^{-w} = \sum_{n \geq 0} \frac{(w)_n}{n!} q^n$ converges absolutely for $|q| < 1$. Hence

$$x^{\kappa-1} \prod_\ell (x+\xi_\ell)^{-w_\ell} = \sum_{\mathbf{n} \in \mathbb{N}^J} \prod_\ell \frac{(w_\ell)_{n_\ell}}{n_\ell!} (r - \xi_\ell)^{n_\ell} \, x^{\kappa-1}(x+r)^{-|\mathbf{w}|-|\mathbf{n}|}.$$

Term by term. $|(w)_n| = \prod_{i<n} |w+i| \leq \prod_{i<n}(|w|+i) = (|w|)_n$, and $(x+r)^{-|\mathbf{n}|} \leq r^{-|\mathbf{n}|}$. So the moduli of the terms sum to at most

$$x^{\operatorname{Re}\kappa - 1}(x+r)^{-\operatorname{Re}|\mathbf{w}|} \prod_\ell \sum_{n \geq 0} \frac{(|w_\ell|)_n}{n!} |g_\ell|^n = x^{\operatorname{Re}\kappa - 1}(x+r)^{-\operatorname{Re}|\mathbf{w}|} \prod_\ell (1 - |g_\ell|)^{-|w_\ell|},$$

which is integrable on $(0,\infty)$ since $0<\operatorname{Re}\kappa<\operatorname{Re}|\mathbf{w}|$; so the series may be integrated term by term. For $0<\operatorname{Re}\kappa<\operatorname{Re}a$ the integral of a linear letter, in the paragraph after 15.1 with $c_0=r$ and $c_1=1$, is

$$\int_0^\infty x^{\kappa-1}(x+r)^{-a}dx=\Gamma\,{}_2{}_1\begin{bmatrix}\kappa,\ a-\kappa\\ a\end{bmatrix}r^{\kappa-a},$$

and at $a=|\mathbf{w}|+|\mathbf{n}|$, with $(r-\xi_\ell)^{n_\ell}r^{-n_\ell}=g_\ell^{n_\ell}$, the term of $\mathbf{n}$ is the term of the statement.

Root rate. Let A_M be the sum of the terms with $|\mathbf{n}|=M$, $v=\sum_\ell|w_\ell|$ and $\bar g=\max_\ell|g_\ell|$. Comparing the coefficients of z^M in $\prod_\ell(1-z)^{-|w_\ell|}=(1-z)^{-v}$ gives $\sum_{|\mathbf{n}|=M}\prod_\ell\frac{(|w_\ell|)_{n_\ell}}{n_\ell!}=\frac{(v)_M}{M!}$. The Beta corner of order M is the integral $\int_0^1x^{\kappa-1}(1-x)^{|\mathbf{w}|+M-\kappa-1}dx$ (3.9), of modulus at most $b_0=\int_0^1x^{\operatorname{Re}\kappa-1}(1-x)^{\operatorname{Re}|\mathbf{w}|-\operatorname{Re}\kappa-1}dx<\infty$. So $|A_M|\le|r^{\kappa-|\mathbf{w}|}|\,b_0\frac{(v)_M}{M!}\bar g^M$, and $\frac{(v)_M}{M!}=\prod_{i=1}^M\big(1+\frac{v-1}{i}\big)\le e^{|v-1|(1+\log M)}$ grows at most polynomially in M. Hence $\limsup_M|A_M|^{1/M}\le\bar g$ (1.1(viii)).

The remaining sentences. A coefficient $\xi_\ell=r$ has $g_\ell=0$, and every term with $n_\ell>0$ vanishes. For $0<\xi_\ell\le r$, $w_\ell>0$ and $0<\kappa<|\mathbf{w}|$ every factor of every term is positive, a factor g_ℓ^0 being 1. For positive coefficients write $\xi_{\min}$ and $\xi_{\max}$ for the least and the largest; $|1-\xi/r|$ is largest over $\xi\in[\xi_{\min},\xi_{\max}]$ at an end, so $\max_\ell|g_\ell|=\max\big(|1-\xi_{\min}/r|,|1-\xi_{\max}/r|\big)$. For $r\le\xi_{\min}$ this is $\xi_{\max}/r-1$, decreasing in r; for $r\ge\xi_{\max}$ it is $1-\xi_{\min}/r$, increasing in r; for $\xi_{\min}\le r\le\xi_{\max}$ it is the larger of $1-\xi_{\min}/r$, increasing, and $\xi_{\max}/r-1$, decreasing, which are equal at $r=\frac12(\xi_{\min}+\xi_{\max})$, with the value $\frac{\xi_{\max}-\xi_{\min}}{\xi_{\max}+\xi_{\min}}$. At that r, $|\xi_\ell-r|\le\frac12(\xi_{\max}-\xi_{\min})<r$ for every ℓ. At $r=\xi_{\max}$ the bound is $1-\xi_{\min}/\xi_{\max}$, the largest coefficient drops out, and the statement is the reference, proof of 15.5. □

Corollary 15.7 (The period fibres over the scale of a bunch). *As stated: every section $\{h(\alpha_R)=1\}$ of the room, and off the balance wall.*

Proof. For a positive function h on an orthant $\mathbb{R}^X_{>0}$, homogeneous of degree one, write λ_h for the measure on $\Sigma_h=\{\beta:\ h(\beta)=1\}$ with $d\alpha=\rho^{|X|-1}d\rho\,d\lambda_h(\beta)$ under the bijection $(\rho,\beta)\mapsto\rho\beta$ of $(0,\infty)\times\Sigma_h$ onto $\mathbb{R}^X_{>0}$: the measure of 16.2 (the reference, proof of 16.2). For $h=\alpha_g$ it is the Lebesgue measure in the coordinates α_e, $e\ne g$, at $\alpha_g=1$, and for $h=|\alpha|$ it is $d\omega$ (2.7).

Let h be a function of α_R as in the statement, and put $\tilde h(\alpha)=h(\alpha_R)$ on $\mathbb{R}^{\mathcal{E}}_{>0}$, positive and homogeneous of degree one. The map $(\rho,\beta_B,\beta_R)\mapsto(\rho\beta_B,\rho\beta_R)$, $\beta_B\in\mathbb{R}^B_{>0}$, $\beta_R\in\Sigma_h$, is a bijection onto $\mathbb{R}^{\mathcal{E}}_{>0}$, with $d\alpha_R=\rho^{|R|-1}d\rho\,d\lambda_h(\beta_R)$ and $d\alpha_B=\rho^{|B|}d\beta_B$ at fixed ρ. So $d\alpha=\rho^{n-1}d\rho\,d\beta_B\,d\lambda_h(\beta_R)$, and the measure of the section $\{\tilde h=1\}$ is $d\beta_B\,d\lambda_h(\beta_R)$. The integral of $W\,d\alpha_{\mathbf{m}}$ over it is

$$\int_{\Sigma_h}\beta_R^{\mathbf{m}|_R-\mathbf{1}}\int_{\mathbb{R}^B_{>0}}\beta_B^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_j(\beta_B,\beta_R)^{-s_j}\Psi_j(\beta_B,\beta_R)^{-t_j}\,d\beta_B\,d\lambda_h(\beta_R),$$

by Tonelli on the modulus, which is the same integrand at the real parts of the data, and then Fubini. With $\beta_B=xu$, $x>0$, $u\in\Delta_B$, $\beta_B^{\mathbf{m}|_B-\mathbf{1}}d\beta_B=x^{m_B-1}u^{\mathbf{m}|_B-\mathbf{1}}dx\,d\omega_B(u)$ (2.7), so the inner integral is $\int_{\Delta_B}u^{\mathbf{m}|_B-\mathbf{1}}\operatorname{Sc}_B(u,\beta_R;\mathbf{m};\mathbf{s},\mathbf{t})\,d\omega_B(u)$, with the scale integral of 15.2 at the gauge (u,β_R). This is the display.

The scale integral converges exactly when $\operatorname{Re}\kappa_B>0$ and $\operatorname{Re}\kappa_B^{\mathrm{top}}>0$ (15.2), conditions on the data alone; by Tonelli it converges at almost every (u,β_R), hence at every point, and there it is the product of 15.2. On the balance wall the form is projective and its integral over every section is the period (8.10). At $h=\alpha_g$ the integral over $\{\alpha_g=1\}$ is $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ wherever it converges (16.4(i)), and λ_h is the Lebesgue measure in α_{R-g} at $\alpha_g=1$; at $h=|\alpha_R|$, $\Sigma_h=\Delta_R$ and $\lambda_h=d\omega_R$. At $h=\alpha_g$ on the balance wall this is the reference, proof of 15.7. □

Corollary 15.11 (The period closes on the ordering sectors). *As stated: complex data with $\kappa_{\mathcal{E}} = 0$ and every proper wall of positive real part.*

Proof. Sector form. Let Ω be the set of data at which every proper wall has positive real part. It is open and convex, the walls being affine in the data with real coefficients, and it contains points at which $\operatorname{Re}\kappa_{\mathcal{E}} > 0$ as well (raise every mass). By 8.7(ii), valid for all complex exponents, and 8.10, wherever every wall has positive real part

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_{\pi} I_\pi, \qquad I_\pi = \int_{[0,1]^{n-1}} \prod_{k=2}^{n} q_k^{\kappa_{U_k}-1} h_\pi(q)\,dq, \quad h_\pi = L_\pi^{-\kappa_{\mathcal{E}}} \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}.$$

The sector letters are polynomials with nonnegative coefficients and positive constant terms, and $L_\pi \in [1,n]$ on the cube, so h_π is smooth on a neighbourhood of the cube, with every q-derivative continuous in q and the data and entire in the data, and $|h_\pi|$ is bounded on the cube locally uniformly in the data. The tails U_k, $k \geq 2$, of the orders are the nonempty proper sets of marks. So every I_π converges and is holomorphic on Ω, and so is P_F (8.10); Ω being connected, $P_F = \sum_\pi I_\pi$ on Ω. At $\kappa_{\mathcal{E}} = 0$, $h_\pi = \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$.

The identity and the range. In the sector coordinates $\alpha^T = \rho^{|T|} \prod_k q_k^{N_{U_k}(T)}$, so $k_{j,\pi}(q) = \sum_T \theta_{j,T} \prod_k q_k^{N_{U_k}(T) - d_j(U_k)}$ (8.7(ii)), with exponents in $\mathbb{N}$, all zero only at $T = T^+_{j,\pi}$ (8.5). So $k_{j,\pi}/K_j(\mathbf{1})$ increases in each q_k on the cube, from $\theta_{j,T^+_{j,\pi}}/K_j(\mathbf{1})$ at $q = 0$ to 1 at $q = \mathbf{1}$, and $0 \leq 1 - k_{j,\pi}/K_j(\mathbf{1}) \leq \lambda$. The identity is the expectation under $\mathbb{P}_{j,\mathbf{1}}$ of $1 - \prod_k q_k^{e_k} = \sum_k \prod_{k'<k} q_{k'}^{e_{k'}}(1 - q_k^{e_k})$, $e_k = N_{U_k} - d_j(U_k)$, with $1 - q^e = (1-q)\sum_{0 \leq i < e} q^i$. The same holds for $\psi_{j,\pi}$ with N^*, d_j^* and $T^-_{j,\pi}$, and $\Psi_j(\mathbf{1}) = K_j(\mathbf{1})$. These are statements about the tables alone, as in the reference, proof of 15.11.

Expansion. With $a_j = 1 - k_{j,\pi}/K_j(\mathbf{1})$ and $b_j = 1 - \psi_{j,\pi}/K_j(\mathbf{1})$ in $[0,\lambda]$, $k_{j,\pi}^{-s_j} = K_j(\mathbf{1})^{-s_j} \sum_{n \geq 0} \frac{(s_j)_n}{n!} a_j^n$ and $\psi_{j,\pi}^{-t_j} = K_j(\mathbf{1})^{-t_j} \sum_{n \geq 0} \frac{(t_j)_n}{n!} b_j^n$, with principal powers of positive numbers. By $|(s)_n| \leq (|s|)_n$ the moduli of the terms of the product of these series sum to at most $\prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)}(1-\lambda)^{-|s_j|-|t_j|}$ on the cube, and $\int \prod_k q_k^{\operatorname{Re}\kappa_{U_k}-1} dq = \prod_k (\operatorname{Re}\kappa_{U_k})^{-1} < \infty$. So the series may be integrated term by term over the cube, which gives the displayed series, convergent.

Terms. By the identity, a_j and b_j are sums, with nonnegative coefficients, of products $(1-q_k)q^{\boldsymbol{\beta}}$, $\boldsymbol{\beta} \in \mathbb{N}^{n-1}$; so $\prod_j a_j^{n_j} b_j^{n'_j}$ is a finite sum, with nonnegative coefficients, of products $\prod_k (1-q_k)^{\gamma_k} q_k^{\beta_k}$, and for $\operatorname{Re}\kappa > 0$, $\int_0^1 q^{\kappa+\beta-1}(1-q)^\gamma dq = \Gamma\begin{bmatrix}\kappa+\beta,\ \gamma+1\\ \kappa+\beta+\gamma+1\end{bmatrix}$ (3.9). For real data with $s_j, t_j \geq 0$ the coefficients $(s_j)_n/n!$ and $(t_j)_n/n!$ and the integrands are nonnegative.

Bound. The terms with $|\mathbf{n}| + |\mathbf{n}'| = M$ have moduli summing to at most

$$\prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)} \sum_{|\mathbf{n}|+|\mathbf{n}'|=M} \prod_j \frac{(|s_j|)_{n_j}(|t_j|)_{n'_j}}{n_j!\,n'_j!}\,\lambda^M \sum_\pi \prod_k (\operatorname{Re}\kappa_{U_k})^{-1},$$

and comparing coefficients of z^M in $\prod_j (1-z)^{-|s_j|}(1-z)^{-|t_j|} = (1-z)^{-w}$ gives $\frac{(w)_M}{M!}$ for the inner sum, $w = \sum_j (|s_j| + |t_j|)$. □

Proposition 15.12 (An enclosure of the period in the field of its data). *As stated: weights, masses and exponents in a subfield k of $\mathbb{R}$, the exponents of any sign.*

Proof. The data are real, so by the hypotheses of 15.11 every proper wall is positive, and every $\kappa_{U_k} > 0$.

(i) The weights lie in k, so the coefficients of the sector letters, sums of weights (8.7(ii)), and $K_j(\mathbf{1})$ lie in k, and $a_{j,\pi}, b_{j,\pi} \in k[q]$. For $s \in k$ and $n \in \mathbb{N}$, $(s)_n/n! \in k$. So $f_{\pi,M} = \sum_{|\mathbf{n}|+|\mathbf{n}'|=M} \prod_j \frac{(s_j)_{n_j}(t_j)_{n'_j}}{n_j!\, n'_j!} a_{j,\pi}^{n_j} b_{j,\pi}^{n'_j}$, the coefficient of ζ^M in the product of the binomial series, lies in $k[q]$. Each κ_{U_k} is an integer combination of masses and exponents and lies in k, and $\int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_{U_k}-1+\beta_k} dq = \prod_k (\kappa_{U_k}+\beta_k)^{-1} \in k$ for $\boldsymbol{\beta} \in \mathbb{N}^{n-1}$; so $I_{\pi,M} \in k$. Grouping the absolutely convergent series of 15.11 by $M = |\mathbf{n}| + |\mathbf{n}'|$ gives the identity. When $s_j, t_j \geq 0$ the coefficients are nonnegative, and $a_{j,\pi}, b_{j,\pi} \geq 0$ on the cube, so $I_{\pi,M} \geq 0$.

(ii) On the cube $0 \leq a_{j,\pi}, b_{j,\pi} \leq \lambda_\pi$ (15.11), so by $|(s)_n| \leq (|s|)_n$ and the coefficient identity of the proof of 15.11, $|f_{\pi,M}| \leq \frac{(w)_M}{M!}\lambda_\pi^M$ and $|I_{\pi,M}| \leq \prod_k \kappa_{U_k}^{-1} \frac{(w)_M}{M!}\lambda_\pi^M$. The ratio of consecutive bounds is $\frac{w+M}{M+1}\lambda_\pi$, and $M \mapsto \frac{w+M}{M+1}$ is nonincreasing for $w \geq 1$ and below 1 for $w < 1$; so for $M \geq Q+1$ the ratio is at most $r_{\pi,Q} < 1$, and summing the geometric majorant gives $\left|\sum_{M>Q} I_{\pi,M}\right| \leq \mathrm{err}_{\pi,Q}$. The numbers w, λ_π, $r_{\pi,Q}$ and $\mathrm{err}_{\pi,Q}$ lie in k, an ordered subfield of $\mathbb{R}$ being closed under $|\cdot|$ and under maxima of finitely many elements. As $Q \to \infty$, $r_{\pi,Q} \to \lambda_\pi < 1$, and $\frac{(w)_{Q+1}}{(Q+1)!}\lambda_\pi^{Q+1} \to 0$, the factor $\frac{(w)_{Q+1}}{(Q+1)!}$ growing at most polynomially; so $\mathrm{err}_{\pi,Q} \to 0$. With $P^{\leq Q} = \sum_\pi \sum_{M \leq Q} I_{\pi,M} \in k$, the left side of (i) lies between $P^{\leq Q} - \sum_\pi \mathrm{err}_{\pi,Q}$ and $P^{\leq Q} + \sum_\pi \mathrm{err}_{\pi,Q}$, and for $s_j, t_j \geq 0$, the remainders being nonnegative, between $P^{\leq Q}$ and $P^{\leq Q} + \sum_\pi \mathrm{err}_{\pi,Q}$.

(iii) $\partial_\zeta \log f = \sum_i w_i A_i/(1-\zeta A_i)$; multiply by $f\prod_i(1-\zeta A_i)$. Comparing the coefficients of ζ^M on the two sides gives a linear recurrence for the coefficients of f, with coefficients polynomial in M and in the A_i. With rational data this is the reference, proof of 15.12. □

Theorem 15.14 (Continued Beta corners). *As stated: complex data, every index off the nonpositive integers, several indices of nonpositive real part at once.*

Proof. A continuation as a linear functional. For $\kappa \in \mathbb{C}$ and $h \in \mathbb{N}$ with $\mathrm{Re}\,\kappa > -h-1$ and $\kappa \notin \{0, -1, \ldots, -h\}$, and f smooth on $[0,1]$, 8.6 with $m = 1$ and $M = h+1$ gives the continuation in κ of $\int_0^1 x^{\kappa-1} f\, dx$:

$$\mathcal{C}(\kappa; f) = \int_0^1 x^{\kappa-1}\big(f - \mathrm{Tay}_h f\big)dx + \sum_{l \leq h} \frac{f^{(l)}(0)}{l!\,(\kappa+l)},$$

Tay_h taking the Taylor polynomial of degree h at 0; it is the finite part $\mathrm{Reg}\int_0^1 x^{\kappa-1} f\, dx$ (1.7). By Taylor's formula $|f - \mathrm{Tay}_h f| \leq \frac{x^{h+1}}{(h+1)!}\sup|f^{(h+1)}|$, so

$$|\mathcal{C}(\kappa; f)| \leq \frac{\sup|f^{(h+1)}|}{(h+1)!\,(\mathrm{Re}\,\kappa + h + 1)} + \sum_{l \leq h} \frac{|f^{(l)}(0)|}{l!\,|\kappa+l|}\,:$$

$\mathcal{C}(\kappa;\cdot)$ is linear and bounded by a constant times the largest of the sup-norms of $f, \ldots, f^{(h+1)}$. On a cube $[0,1]^m$, with a number $L_k \in \mathbb{N}$ for each coordinate, $\mathrm{Re}\,x_k + L_k > 0$ and $x_k \notin \{0, \ldots, 1-L_k\}$, 8.6 repeated in each coordinate gives the continuation of $\int \prod_k q_k^{x_k-1} f\, dq$ as a finite sum, over the sets A of coordinates and the $\mathbf{l}$ with $l_k < L_k$ for $k \in A$, of $\prod_{k\in A}(x_k+l_k)^{-1}$ times the integral over the other coordinates of $\prod_{k\notin A} q_k^{x_k-1}$ against $\prod_{k\notin A}(1-\mathrm{Tay}_{L_k-1})$ applied to $\partial_{q_A}^{\mathbf{l}} f/\mathbf{l}!$ at $q_A = 0$ (as in 8.8), with $\mathrm{Tay}_{-1} = 0$. Taylor's formula in each coordinate outside A bounds each term by a constant, depending only on the x_k, times the largest sup-norm of the derivatives of f of total

order at most $L = \sum_k L_k$; so this continuation is again linear and bounded on $C^L([0,1]^m)$, and it is the iterated finite part, by 1.7 in one coordinate at a time.

Leibniz bounds. Let finitely many functions f_i be smooth on a cube, with $0 \le f_i \le \lambda$, $0 < \lambda < 1$, and let $D \ge 1$ bound their derivatives of order at most L. For a multi-index $\mathbf{n}$ over the i and $f_{\mathbf{n}} = \prod_i f_i^{n_i}$, a derivative of total order $l \le L$ distributes over the $|\mathbf{n}|$ factors in at most $|\mathbf{n}|^l$ ways, each product having at most l differentiated factors, each bounded by D, and the others bounded by λ; so $|\partial^{\boldsymbol\beta} f_{\mathbf{n}}| \le D^l |\mathbf{n}|^l \lambda^{|\mathbf{n}|-l}$ for $|\boldsymbol\beta| = l$. Hence the sum, over the $\mathbf{n}$ with $|\mathbf{n}| = M$, of $\prod_i \frac{|(w_i)_{n_i}|}{n_i!}$ times the C^L-norm of $f_{\mathbf{n}}$ is at most $\frac{(w)_M}{M!} D^L (1+M)^L \lambda^{M-L}$, $w = \sum_i |w_i|$, by the coefficient identity of the proof of 15.5. These bounds are summable over M, so $\sum_{\mathbf{n}} \prod_i \frac{(w_i)_{n_i}}{n_i!} f_{\mathbf{n}}$ converges in C^L, to $\prod_i (1-f_i)^{-w_i}$; a bounded linear functional on C^L commutes with the sum, and applied to $f_{\mathbf{n}}$ it is at most a constant times $D^L(1+|\mathbf{n}|)^L \lambda^{|\mathbf{n}|-L}$. If $\lambda = 0$ every f_i vanishes and only $\mathbf{n} = 0$ contributes.

The two sectors of a scale. In 15.9 take $f_i = 1 - \mathsf{Q}_{S,i}(\cdot;\alpha)$ on $[0,1]$: each $\mathsf{Q}_{S,i}$ has nonnegative coefficients, positive constant term and value 1 at $x = 1$ (12.9), so $0 \le f_i \le 1 - \mathsf{Q}_{S,i}(0;\alpha) \le \lambda_S$ there, and $\prod_i \mathsf{Q}_{S,i}^{-w_i} = \sum_{\mathbf{n}} \prod_i \frac{(w_i)_{n_i}}{n_i!} f_{\mathbf{n}}$. Let κ be the index of the sector, not a nonpositive integer. If $\operatorname{Re}\kappa > 0$, take $\mathcal{C}(\kappa; f) = \int_0^1 x^{\kappa-1} f\, dx$, bounded by $\sup|f| / \operatorname{Re}\kappa$, with $L = 0$; otherwise take h as in the statement and $L = h + 1$. By the preceding paragraph,

$$\mathcal{C}\Big(\kappa; \prod_i \mathsf{Q}_{S,i}^{-w_i}\Big) = \sum_{\mathbf{n}} \prod_i \frac{(w_i)_{n_i}}{n_i!}\, \mathcal{C}(\kappa; f_{\mathbf{n}}),$$

the terms of total order M having moduli summing to at most $\frac{(w)_M}{M!} \lambda_S^M$ times a polynomial in M of degree L, $w = \sum_j (|s_j| + |t_j|)$. The left side is the finite part of the sector integral and its continuation in κ. By 15.8, $f_{\mathbf{n}} = (1-x)^{|\mathbf{n}|} \prod_i \mathsf{Q}_{S,i,>}^{n_i}$, a polynomial, and $\mathcal{C}\big(\kappa; x^k (1-x)^{|\mathbf{n}|}\big)$ is the continuation of $\int_0^1 x^{\kappa+k-1}(1-x)^{|\mathbf{n}|} dx = \Gamma \left[\begin{matrix} \kappa+k,\ |\mathbf{n}|+1 \\ \kappa+k+|\mathbf{n}|+1 \end{matrix}\right]$ (3.9), the meromorphic stacked Gamma, finite since $\kappa + k \notin -\mathbb{N}$. So the series of 15.9(ii), with these Beta corners, converges to the finite part, with the stated bound. By 15.9(i) the scale integral is the product of the letters at α and the sum of the two sectors at the indices κ_B and κ_B^{top}, and its continuation in the data is the same expression with the sectors continued.

The ordering sectors. By the first step of the proof of 15.11 and 8.6, $P_F = \sum_\pi I_\pi$ as meromorphic functions of the data, each I_π continued by 8.6, whose formula expresses the continuation at a point through the values of h_π at that point; on the balance wall $h_\pi = \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$. At a point of the balance wall at which no proper wall is a nonpositive integer, take for an order π the numbers $L_k = h_k + 1$ when $\operatorname{Re}\kappa_{U_k} \le 0$, with $-h_k - 1 < \operatorname{Re}\kappa_{U_k} \le -h_k$, and $L_k = 0$ otherwise. With f running over the a_j, b_j of the proof of 15.11, in $[0,\lambda]$ on the cube, the preceding paragraphs give the continuation of I_π as the series of 15.11 for π, each integral replaced by its continuation; and the continuation of $\int \prod_k q_k^{\kappa_{U_k}+\beta_k-1}(1-q_k)^{\gamma_k} dq$ is the product over k of the continued Beta corners, each finite since $\kappa_{U_k} + \beta_k \notin -\mathbb{N}$. The terms of total order M have moduli summing to at most $\frac{(w)_M}{M!}\lambda^M$ times a polynomial in M of degree $\sum_k L_k$, and the tails of one order being distinct proper sets, $\sum_k L_k$ is at most the sum of the integers $h+1$ over the proper walls of real part at most 0. Every proper set of marks is a tail of some order, so the sector series computes P_F at every point of the balance wall off the pole hyperplanes, with no positivity of the walls. For one real index in $(-h-1, -h)$ this is the reference, proof of 15.14. □

Corollary 15.16 (A loop or coloop in every table). *As stated: a set of marks, each a loop or a coloop in every table.*

Proof. For a table j let p_j be the set of the marks of A that are coloops of table j. Every basis contains every coloop and avoids every loop, so $T \cap A = p_j$ for every basis T of table j. The sets $T \setminus A$ are then the bases of the table on $\mathcal{E} \setminus A$ obtained from table j by contracting the marks of p_j and deleting those of $A \setminus p_j$, one mark at a time (7.3; a loop is never a coloop, so each step is a table), with the weights $\theta_{j,T}$; with K'_j, Ψ'_j its Foreman polynomials and $\alpha' = \alpha_{\mathcal{E}\setminus A}$,

$$K_j(\alpha) = \alpha_A^{p_j} K'_j(\alpha'), \qquad \Psi_j(\alpha) = \alpha_A^{A\setminus p_j} \Psi'_j(\alpha'),$$

since $\mathcal{E} \setminus T = (A \setminus p_j) \sqcup \big((\mathcal{E} \setminus A) \setminus (T \setminus A)\big)$. So the integrand of $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is the product of

$$\prod_{e\in A} \alpha_e^{m_e - 1 - \sum_j (s_j \mathbf{1}[e\in p_j] + t_j \mathbf{1}[e\notin p_j])} e^{-\alpha_e}$$

and of the integrand of $J_{F'}(\mathbf{m}|_{\mathcal{E}\setminus A}; \mathbf{s}, \mathbf{t})$. With $s_j = \sigma_j + \delta_j$, $t_j = \sigma_j - \delta_j$ and $m_e = \mu_e + \sigma$, the exponent of α_e is $\mu_e - \sum_j \eta_{j,e}\delta_j - 1$. The integrand is a product of functions of disjoint sets of variables, so by Tonelli on the modulus its integral converges exactly when every factor's does, and then, by Fubini, it is the product of the factors' integrals, with $\int_0^\infty \alpha_e^{\nu-1} e^{-\alpha_e} d\alpha_e = \Gamma(\nu)$ for $\operatorname{Re}\nu > 0$. As meromorphic functions the identity persists (8.7(iii)). For one mark this is the reference, proof of 15.16, there read through 15.15 at $B = \{e\}$. □

40.16 Chapter 16: integrals, expectations and the Foreman procedure

Proposition 16.4 (Snips are completions). *As stated: (iii) at every imbalance, through the appended list $\binom{B}{1}$ of the singletons of the bunch.*

Proof. Items (i) and (ii) are the reference, proof of 16.4. For (iii) let $R = \mathcal{E} \setminus B$.

Projectivity. The family word W is homogeneous of degree $-\sum_j (s_j r_j + t_j r_j^*) = \kappa_\mathcal{E} - m_\mathcal{E}$ (8.1), so $\omega = \alpha^{\mathbf{m}-\mathbf{1}} W\, d\alpha$ is multiplied by $\lambda^{\kappa\varepsilon}$ under $\alpha \mapsto \lambda\alpha$, and $|\alpha_B|^{-\kappa\varepsilon}\omega$ is unchanged: it is projective (1.1(vii)). On the section $|\alpha_B| = 1$ the multiplier is 1, so the two forms induce the same form there.

The appended table. The sets $\{e\}$, $e \in B$, have one element each and the exchange property, two of them differing in one mark; so they are the bases of a table on $\mathcal{E}$ of rank 1 (7.1), in which the marks of R lie in no basis, and with unit weights its basis polynomial is $\sum_{e\in B} \alpha_e = |\alpha_B|$. So $|\alpha_B|^{-\kappa\varepsilon} W$ is the family word of F' at the exponents $\big((\mathbf{s}, \kappa_\mathcal{E}), (\mathbf{t}, 0)\big)$, the appended table carrying $(\kappa_\mathcal{E}, 0)$, and $|\alpha_B|^{-\kappa\varepsilon}\omega$ is the form of F' at the masses $\mathbf{m}$.

Walls. For the appended table $d_0(S) = \min_{e\in B} |\{e\} \cap S| = \mathbf{1}[B \subseteq S]$ (7.11), and its exponent on Ψ is 0; so by 8.1 the walls of F' at $\mathbf{m}$ are $\kappa_S - \kappa_\mathcal{E}\mathbf{1}[B \subseteq S]$, and its imbalance is $\kappa_\mathcal{E} - \kappa_\mathcal{E} \cdot 1 = 0$, the appended table having rank 1.

Value. $|\alpha_B|$ is positive and homogeneous of degree one on the orthant. Where every proper wall of F' has positive real part, the period of F' converges, and the integral of its form over the section $\{|\alpha_B| = 1\}$ is that period (8.10); this is the display.

The cases. For $B = \{g\}$ the appended table has the one basis $\{g\}$ and the letter α_g, and $\alpha_g^{-\kappa\varepsilon} W\, d\alpha_\mathbf{m} = W\, d\alpha_{\mathbf{m}-\kappa_\mathcal{E}\mathbf{e}_g} = \omega_{\mathbf{m}^{(g)}}$ (16.3): this is (i), and the walls $\kappa_S - \kappa_\mathcal{E}\mathbf{1}_S(g)$ are those of 16.29(ii). For $B = \mathcal{E}$ the appended table is the list $\binom{\mathcal{E}}{1}$, and the statement is 38.8. For $k = -\kappa_\mathcal{E} \in \mathbb{N}$, $k \geq 1$: on the section $\omega = |\alpha_B|^k \omega$, $|\alpha_B|^k = \sum_{|\mathbf{n}|=k} \binom{k}{\mathbf{n}} \alpha_B^\mathbf{n}$, and $\alpha_B^\mathbf{n}\omega = W\, d\alpha_{\mathbf{m}+\mathbf{n}}$ is the completion along $\mathbf{n}/k$ (16.3: $\mathbf{m} - \kappa_\mathcal{E}\mathbf{n}/k = \mathbf{m}+\mathbf{n}$), projective, whose integral over the section is $P_F(\mathbf{m}+\mathbf{n}; \mathbf{s}, \mathbf{t})$ wherever that period converges (8.10). This last case is the reference, proof of 16.4(iii). □

Proposition 16.10 (The form of a compound at one point). *As stated, for any number $q \geq 2$ of parts.*

Proof. The form. Write $r_{j,i}$ for the rank of the restriction of table j to $\mathcal{E}_i$, so that $K_{j,i}$ is homogeneous of degree $r_{j,i}$, $\Psi_{j,i}$ of degree $|\mathcal{E}_i| - r_{j,i}$, and $\kappa_i = m_{\mathcal{E}_i} - \sum_j \big(s_j r_{j,i} + t_j(|\mathcal{E}_i| - r_{j,i})\big)$; summing over i, $\sum_i \kappa_i = \kappa_{\mathcal{E}} = 0$. The section $\{|\alpha_q| = 1\}$ carries the measure $\prod_{i<q} d\alpha_i \, d\omega_{\mathcal{E}_q}(u_q)$, by the computation in the proof of 15.7 above with the room there replaced by $\mathcal{E}_q$. For $i < q$, $\alpha_i = \rho_i u_i$ gives $\alpha_i^{\mathbf{m}_i - \mathbf{1}} d\alpha_i = \rho_i^{m_{\mathcal{E}_i} - 1} u_i^{\mathbf{m}_i - \mathbf{1}} d\rho_i \, d\omega_{\mathcal{E}_i}(u_i)$ (2.7), $K_{j,i}(\rho_i u_i) = \rho_i^{r_{j,i}} K_{j,i}(u_i)$ and $\Psi_{j,i}(\rho_i u_i) = \rho_i^{|\mathcal{E}_i| - r_{j,i}} \Psi_{j,i}(u_i)$. The powers of ρ_i collect to $\rho_i^{\kappa_i - 1}$, which gives the displayed form.

(i) For real κ, $\int_0^\infty \rho^{\kappa - 1} d\rho$ diverges, at ∞ if $\kappa \geq 0$ and at 0 if $\kappa \leq 0$. The form is positive, and by Tonelli its integral is the product of the $q - 1$ integrals $\int_0^\infty \rho_i^{\kappa_i - 1} d\rho_i$, each infinite, and of the q shape integrals, each positive; so it is infinite.

(ii) A primitive of $\rho^{\kappa - 1}$ is ρ^κ / κ, or $\log \rho$ at $\kappa = 0$. It is its own log-power expansion at 0 and at ∞, with no term $\rho^0 \log^0 \rho$, so both regularized limits are 0 and the finite part is 0 (1.4).

(iii) At $\mathbf{m}(\boldsymbol{\eta})$, $\kappa_i = \kappa_{F_i}(\mathbf{m}_i^0) + i\eta_i |w_i| = i\eta_i$, and $\sum_i \kappa_i = 0$ since $\sum_i \eta_i = 0$. The walls of F_i at $\mathbf{m}_i^0 + i\eta_i w_i$ have the real parts of those at $\mathbf{m}_i^0$, positive, so $f(\boldsymbol{\eta}') = \prod_{i=1}^q P_{F_i}(\mathbf{m}_i^0 + i\eta_i w_i; \mathbf{s}, \mathbf{t})$, with $\boldsymbol{\eta}' = (\eta_1, \dots, \eta_{q-1})$ and $\eta_q = -\sum_{i<q} \eta_i$, is a smooth function of $\boldsymbol{\eta}'$, each period being holomorphic in the masses there (8.10). In $y_i = \log \rho_i$, $\rho_i^{\kappa_i - 1} d\rho_i = e^{i\eta_i y_i} dy_i$. For a test function H of $\boldsymbol{\eta}'$ of compact support, the integral of the form against H, taken over the shapes, then over $\boldsymbol{\eta}'$, then over $\mathbf{y} = (y_1, \dots, y_{q-1})$, is

$$\int_{\mathbb{R}^{q-1}} \int_{\mathbb{R}^{q-1}} H(\boldsymbol{\eta}') f(\boldsymbol{\eta}') \, e^{i\langle \boldsymbol{\eta}', \mathbf{y} \rangle} \, d\boldsymbol{\eta}' \, d\mathbf{y} = (2\pi)^{q-1} H(0) f(0),$$

by Fourier inversion for the smooth function Hf of compact support, whose Fourier transform is integrable; and $f(0) = \prod_i P_{F_i}(\mathbf{m}_i^0; \mathbf{s}, \mathbf{t})$. For $q = 2$ this is the reference, proof of 16.10. □

Lemma S4. *Let S be a nonempty proper set of marks of a family F on $\mathcal{E}$ with $c_j(S) = 0$ in every active table, and put $S' = \mathcal{E} \setminus S$.*

(i) In every active table j, every basis meets S in $r_j(S)$ marks and S' in $r_j(S')$ marks; the bases of $\mathcal{B}_j|S$ and of $\mathcal{B}_j|S'$ are the sets $T \cap S$ and $T \cap S'$, $T \in \mathcal{B}_j$; and $X \sqcup Y \in \mathcal{B}_j$ for every basis X of $\mathcal{B}_j|S$ and every basis Y of $\mathcal{B}_j|S'$.

(ii) In this lemma write

$$I_S(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_S \times \Delta_{S'}} \alpha^{\mathbf{m} - \mathbf{1}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} \, d\omega_S \, d\omega_{S'}.$$

It converges, and is holomorphic in the data, wherever $\operatorname{Re} \kappa_U > 0$ for every nonempty proper subset U of S and every nonempty proper subset U of S'; and it extends to a meromorphic function of the data whose poles lie on the hyperplanes $\kappa_U = -i$, $i \in \mathbb{N}$, U a nonempty proper subset of S or of S'.

(iii) Wherever every proper wall has positive real part, $\kappa_{\mathcal{E}} = \kappa_S + \kappa_{S'}$ has positive real part and $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \Gamma(\kappa_S)\Gamma(\kappa_{S'}) \, I_S(\mathbf{m}; \mathbf{s}, \mathbf{t})$.

Proof. (i) Fix an active table j. By 7.11 the numbers $|T \cap S|$ fill $\{d_j(S), \dots, r_j(S)\}$, a single value since $c_j(S) = 0$; so every basis meets S in $r_j(S)$ marks and S' in $r_j - r_j(S)$ marks, a constant, which is then $r_j(S')$. By 7.5 the bases of $\mathcal{B}_j|S$ are the sets $T \cap S$ with $|T \cap S| = r_j(S)$, that is all of them, and likewise for S'. Let $X = T_1 \cap S$ and $Y = T_2 \cap S'$ for bases T_1, T_2; induct on

$k = |(T_2 \cap S) \setminus X|$. If $k = 0$, then $T_2 \cap S \subseteq X$, both of size $r_j(S)$, so $T_2 = X \sqcup Y$. If $k > 0$, take $a \in (T_2 \cap S) \setminus X$; then $a \notin T_1$. The exchange property (7.1) gives $b \in T_1 \setminus T_2$ with $T_3 = (T_2 \setminus \{a\}) \cup \{b\}$ a basis. As $|T_3 \cap S| = r_j(S) = |T_2 \cap S|$ and $a \in S$, $b \in S$, so $b \in X \setminus T_2$. Then $T_3 \cap S' = Y$ and $|(T_3 \cap S) \setminus X| = k - 1$, and by induction $X \sqcup Y$ is a basis.

Homogeneity in each block. By (i), for an active table j, $K_j(\rho\alpha_S, \rho'\alpha_{S'}) = \rho^{r_j(S)}\rho'^{r_j(S')}K_j(\alpha)$ and $\Psi_j(\rho\alpha_S, \rho'\alpha_{S'}) = \rho^{|S|-r_j(S)}\rho'^{|S'|-r_j(S')}\Psi_j(\alpha)$. For $U \in \{S, S'\}$, $c_j(U) = 0$ (7.11: $c_j(S') = c_j(S)$), so $d_j(U) = r_j(U)$ and $d_j^*(U) = |U| - r_j(U)$ (7.10). An inactive table enters with the exponents 0. So with $g(U) = \sum_j \big(s_j d_j(U) + t_j d_j^*(U)\big) = m_U - \kappa_U$, the function $\phi(\alpha) = \alpha^{\mathbf{m-1}} \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}$ has

$$\phi(\rho\alpha_S, \rho'\alpha_{S'}) = \rho^{\kappa_S - |S|}\rho'^{\kappa_{S'} - |S'|}\phi(\alpha),$$

and $|\phi|$ is the same function at the real parts of the data.

(iii) With $\alpha_S = \rho u$ and $\alpha_{S'} = \rho' u'$ (2.7 in each block), $d\alpha = \rho^{|S|-1}\rho'^{|S'|-1} d\rho\, d\rho'\, d\omega_S(u)\, d\omega_{S'}(u')$, and the integrand of J_F becomes $\rho^{\kappa_S - 1}e^{-\rho}\rho'^{\kappa_{S'}-1}e^{-\rho'}\phi(u, u')$. Wherever every proper wall has positive real part, $\operatorname{Re}\kappa_{\mathcal{E}} = \operatorname{Re}\kappa_S + \operatorname{Re}\kappa_{S'} > 0$, by 8.15, $c_j(S) = 0$ holding in every table with $\sigma_j \neq 0$, these being active. Then every wall has positive real part, and J_F at the real parts of the data is finite (8.7(i)); by Tonelli it is $\Gamma(\operatorname{Re}\kappa_S)\Gamma(\operatorname{Re}\kappa_{S'})$ times the integral of $|\phi|$ over $\Delta_S \times \Delta_{S'}$, which is therefore finite, and Fubini gives $J_F = \Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$.

(ii) *Charts.* For a finite set X, a mark $x \in X$ and a function f on $\mathbb{R}^X_{>0}$, homogeneous of degree $-|X|$ and nonnegative or integrable on Δ_X, $\int_{\Delta_X} f\, d\omega_X = \int f|_{\beta_x = 1}\, d\beta_{X-x}$, and the same holds on the cones over subsets of Δ_X and their traces on the chart: for a nonnegative function υ on $(0, \infty)$ with $\int_0^\infty \upsilon(\rho)\, d\rho/\rho = 1$, both sides equal $\int_{\mathbb{R}^X_{>0}} f(\alpha)\upsilon(|\alpha|)\, d\alpha$, by 2.7 on the left, and on the right by the change of variables $\alpha = \rho(1, \beta)$, $d\alpha = \rho^{|X|-1}d\rho\, d\beta$, with $\int_0^\infty \upsilon(\rho|(1, \beta)|)\, d\rho/\rho = 1$. The function $\psi(\alpha) = |\alpha_S|^{-\kappa_S}|\alpha_{S'}|^{-\kappa_{S'}}\phi(\alpha)$ is homogeneous of degree $-|S|$ in α_S and $-|S'|$ in $\alpha_{S'}$, and equal to ϕ on $\Delta_S \times \Delta_{S'}$.

For an order π of S, with tails $U_k = \{\pi(k), \dots, \pi(|S|)\}$, $2 \le k \le |S|$, use on the chart $\beta_{\pi(1)} = 1$ the coordinates $\beta_{\pi(k)} = q_2 \cdots q_k$, $q \in (0, 1]^{|S|-1}$, which cover the trace of the sector $\{\beta_{\pi(1)} \ge \cdots \ge \beta_{\pi(|S|)}\}$; the exponent of q_k in β^X is $|X \cap U_k|$, $d\beta_{S-\pi(1)} = \prod_k q_k^{|S|-k} dq$, $\beta^{\mathbf{m}|_S - \mathbf{1}} d\beta_{S-\pi(1)} = \prod_k q_k^{m_{U_k}-1} dq$, and $|\beta| = L_\pi(q) = \sum_{i=1}^{|S|}\prod_{k=2}^{i} q_k \in [1, |S|]$. Likewise π', U_l', q' and $L_{\pi'}$ on S'. Decomposing each simplex into its sectors, and applying the identity above in each block, by Tonelli on the modulus and then Fubini,

$$I_S = \sum_{\pi, \pi'} \int \prod_k q_k^{m_{U_k}-1} \prod_l q_l'^{m_{U_l'}-1}\, L_\pi(q)^{-\kappa_S} L_{\pi'}(q')^{-\kappa_{S'}} \prod_j K_j(\beta, \beta')^{-s_j}\Psi_j(\beta, \beta')^{-t_j}\, dq\, dq',$$

the integral over $(0, 1]^{|S|-1} \times (0, 1]^{|S'|-1}$.

The letters on a product of sectors. Fix an active table j. By (i) and 7.5, $\mathcal{B}_j|S$ is a table on S, and by 8.5 for π it has bases $X^\pm$ with $|X^+ \cap U_k| = \min_X |X \cap U_k|$ and $|U_k \setminus X^-| = \min_X |U_k \setminus X|$, the minima over its bases; by (i) these minima are $\min_T |T \cap U_k| = d_j(U_k)$ and $\min_T |U_k \setminus T| = d_j^*(U_k)$ (7.11). Likewise $\mathcal{B}_j|S'$ has bases $Y^\pm$ for π'. For a basis T, $\beta^{T\cap S}/\beta^{X^+} = \prod_k q_k^{|T\cap U_k| - d_j(U_k)}$ with exponents in $\mathbb{N}$, and likewise on S'. So

$$K_j(\beta, \beta') = \prod_k q_k^{d_j(U_k)} \prod_l q_l'^{\,d_j(U_l')}\, k_{j,\pi,\pi'}(q, q'),$$

$$k_{j,\pi,\pi'} = \sum_T \theta_{j,T} \prod_k q_k^{|T\cap U_k| - d_j(U_k)} \prod_l q_l'^{\,|T\cap U_l'| - d_j(U_l')},$$

a polynomial with nonnegative coefficients. Its constant term is the sum of $\theta_{j,T}$ over the bases with $|T\cap U_k| = d_j(U_k)$ for $k\ge 2$ and $|T\cap U'_l| = d_j(U'_l)$ for $l\ge 2$. As $|T\cap S| = |X^+|$, these counts determine $T\cap S = X^+$, the mark $\pi(k)$ lying in T exactly when $|T\cap U_k| - |T\cap U_{k+1}| = 1$, with $U_1 = S$ and $U_{|S|+1} = \varnothing$; likewise $T\cap S' = Y^+$. By (i) $X^+\sqcup Y^+$ is a basis, so the constant term is $\theta_{j,X^+\sqcup Y^+} > 0$. In the same way $\Psi_j(\beta,\beta') = \prod_k q_k^{d_j^*(U_k)} \prod_l q_l'^{d_j^*(U'_l)}\, \psi_{j,\pi,\pi'}(q,q')$, with nonnegative coefficients and the constant term $\theta_{j,X^-\sqcup Y^-} > 0$. As $m_U - g(U) = \kappa_U$,

$$I_S = \sum_{\pi,\pi'} \int \prod_{k=2}^{|S|} q_k^{\kappa_{U_k}-1} \prod_{l=2}^{|S'|} q_l'^{\kappa_{U'_l}-1}\, h_{\pi,\pi'}(q,q')\,dq\,dq', \qquad h_{\pi,\pi'} = L_\pi^{-\kappa_S} L_{\pi'}^{-\kappa_{S'}} \prod_j k_{j,\pi,\pi'}^{-s_j} \psi_{j,\pi,\pi'}^{-t_j},$$

the product over the active tables.

Continuation. The polynomials $k_{j,\pi,\pi'}$, $\psi_{j,\pi,\pi'}$, L_π and $L_{\pi'}$ are at least their positive constant terms on the closed cube, hence positive on a neighbourhood of it; so $h_{\pi,\pi'}$ is smooth there, with every derivative continuous in the variables and the data and entire in the data, and $|h_{\pi,\pi'}|$, which is $h_{\pi,\pi'}$ at the real parts of the data, lies between positive constants on the cube, locally uniformly in the data. So the term of (π,π') converges, and is holomorphic, exactly where $\operatorname{Re}\kappa_{U_k} > 0$ and $\operatorname{Re}\kappa_{U'_l} > 0$ for all k,l; the tails U_k, $k\ge 2$, of the orders of S are the nonempty proper subsets of S, and likewise for S'. This gives the convergence and the holomorphy of I_S. By 8.6, each term is a quotient of a holomorphic function of the data by products of the affine functions $\kappa_{U_k}+i$ and $\kappa_{U'_l}+i$, $i\in\mathbb{N}$, each a nonconstant function of the data (the coefficient of m_e, $e\in U$, being 1); so it extends meromorphically with poles only on these hyperplanes, and so does I_S. □

Theorem 16.15 (The pseudointegral off the domain). *As stated: the meromorphy for every family and all complex data, and the compound formula with the value* 0 *on the balance wall; the clause on targets and (i)–(iii) under the hypotheses of the reference.*

Proof. Meromorphy. By the first step of the proof of 15.11 above, which uses neither connectedness nor real data, $P_F = \sum_\pi I_\pi$ on the open convex set Ω of the data at which every proper wall has positive real part, with $I_\pi = \int_{[0,1]^{n-1}} \prod_{k=2}^n q_k^{\kappa_{U_k}-1} h_\pi\,dq$ and $h_\pi = L_\pi^{-\kappa_\mathcal{E}} \prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$ smooth near the closed cube, its derivatives continuous in the variables and the data and entire in the data. By 8.6 each I_π extends to a meromorphic function of the data, a quotient of a holomorphic function by products of affine functions $\kappa_{U_k}+i$, $i\in\mathbb{N}$, with poles only on the hyperplanes $\kappa_{U_k} = -i$. Every nonempty proper set of marks is a tail of some order, and no tail is $\mathcal{E}$. So P_F is meromorphic in the masses and the exponents, with poles only where a proper wall is a nonpositive integer.

The restriction to the balance wall. For a nonempty proper U the affine function $\kappa_U + i$ of the data has the part m_U in the masses, which is not proportional to the part $m_\mathcal{E}$ of $\kappa_\mathcal{E}$; so the hyperplane $\kappa_U = -i$ is not the balance wall $\{\kappa_\mathcal{E} = 0\}$, and it meets the balance wall in a hyperplane of it. Near every point of the balance wall P_F is a quotient of a holomorphic function by a product of such affine functions, none vanishing identically on the balance wall; so the restriction of P_F to the balance wall is meromorphic, with poles only where a proper wall is a nonpositive integer.

The compound formula. Let S be nonempty and proper with $c_j(S) = 0$ in every active table, $S' = \mathcal{E}\setminus S$, and I_S the integral of Lemma S4, the integral of the statement. On Ω, $\operatorname{Re}\kappa_\mathcal{E} > 0$ and $J_F = \Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$ (Lemma S4(iii)), and $J_F = \Gamma(\kappa_\mathcal{E})P_F$ (8.10); so $P_F = \Gamma\,{}_2{}_1\!\left[\begin{matrix}\kappa_S,\ \kappa_{S'}\\ \kappa_\mathcal{E}\end{matrix}\right] I_S$ on Ω, with $\kappa_S + \kappa_{S'} = \kappa_\mathcal{E}$ (8.15). Both sides are meromorphic in the data, by the first step, by Lemma S4(ii) and by 1.2, and they agree on the nonempty open set Ω; so they agree as meromorphic functions.

The value 0 *on the balance wall.* Put $G = \Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$, meromorphic in the data. The poles of $\Gamma(\kappa_S)\Gamma(\kappa_{S'})$ lie on the hyperplanes $\kappa_S = -i$ and $\kappa_{S'} = -i$, $i \in \mathbb{N}$, and those of I_S on the hyperplanes $\kappa_U = -i$ with U a nonempty proper subset of S or of S' (Lemma S4(ii)). In each case U is a nonempty proper set of marks, so, as in the second step, none of these hyperplanes contains the balance wall, and G restricts to a meromorphic function on the balance wall, holomorphic off a proper analytic subset of it. The function $1/\Gamma$ is entire and vanishes at 0, so $1/\Gamma(\kappa_{\mathcal{E}})$ is entire in the data and vanishes on the balance wall. Hence $P_F = G \cdot \Gamma(\kappa_{\mathcal{E}})^{-1}$ vanishes at every point of the balance wall at which G is holomorphic, and the meromorphic restriction of P_F to the balance wall is 0.

The clause on targets and (i)–(iii) are under the hypotheses of the reference, connected with real data and $s_j, t_j \geq 0$ not all zero: the reference, proof of 16.15. □

Proposition 16.29 (Sections off the balance wall). *As stated: a bunch section at every imbalance.*

Proof. The clause on a bunch section is 16.4(iii), proved above; the rest is the reference, proof of 16.29. □

Definition 16.26 (The pseudointegral over a region). As stated; the last sentence of (i) is stated for families of tables with positive rational weights, where the reference takes a list.

Proof. Let the tables have positive rational weights and the data be rational, on the balance wall. The letters K_j and Ψ_j are polynomials whose coefficients are the weights, so the form $W\, d\alpha_{\mathbf{m}}$ is algebraic over $\mathbb{Q}$ and the family has a period datum (18.34), whose moves are defined over $\mathbb{Q}$ by that definition. The three regions named in (i) are cut out by polynomial inequalities over $\mathbb{Q}$: the chart $\alpha_g = 1$ by $\alpha_e > 0$ for $e \neq g$; the simplex, in the coordinates $(u_e)_{e \neq e_0}$, by $u_e > 0$ and $1 - \sum_{e \neq e_0} u_e > 0$; a bunch section $|\alpha_B| = 1$ by these inequalities on B together with $\alpha_e > 0$ on the room.

A change of variables of 18.34 keeps this property. Let $\mathcal{R}$ lie in the open domain and be cut out there by $p_1 > 0, \ldots, p_k > 0$ with $p_i \in \mathbb{Q}[x]$, and let ϕ be birational over $\mathbb{Q}$ and carry the open domain bijectively onto the open domain. Each coordinate of ϕ^{-1} is a quotient of polynomials over $\mathbb{Q}$ whose denominator does not vanish on the open domain, where ϕ^{-1} is defined; so $p_i \circ \phi^{-1} = A_i/D_i$ with $A_i, D_i \in \mathbb{Q}[y]$, D_i a product of these denominators, and

$$\phi(\mathcal{R}) = \{y \text{ in the open domain} : \ A_i(y)\, D_i(y) > 0 \text{ for } 1 \leq i \leq k\},$$

the open domain being an orthant, cut out by the positivity of its coordinates. The region of a product of data is the product of the regions, cut out by the inequalities of the two factors in disjoint coordinates; linear combinations and boundary terms are formed on the regions already reached, as for a list. Of the family the argument of the reference uses only that its period datum and its moves are defined over $\mathbb{Q}$, which holds for positive rational weights. □

40.17 Chapter 17: the Breta law and the port on the balance wall

Proposition 17.3 (Scale and shapes along a bunch). *As stated: off the balance wall and at complex exponents, the scale integral then carrying the appended list* $\binom{\mathcal{E}}{1}$.

Proof. The display. The masses are positive and every proper wall has positive real part, so $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} W(u)\, d\omega_\mathcal{E}(u)$ converges (8.10). By 3.12 with the blocks B and R, $u = (w_B v_B, w_R v_R)$ with $(w_B, w_R) \in \Delta_{\{B,R\}}$, $v_B \in \Delta_B$ and $v_R \in \Delta_R$, and

$$u^{\mathbf{m}-\mathbf{1}} d\omega_\mathcal{E} = w_B^{m_B-1} w_R^{m_R-1} dw_B \cdot v_B^{\mathbf{m}|_B-\mathbf{1}} d\omega_B \cdot v_R^{\mathbf{m}|_R-\mathbf{1}} d\omega_R.$$

With $x = w_B/w_R \in (0,\infty)$, $w_B = x/(1+x)$, $w_R = 1/(1+x)$ and $dw_B = (1+x)^{-2}dx$, so $w_B^{m_B-1} w_R^{m_R-1} dw_B = x^{m_B-1}(1+x)^{-m_\mathcal{E}} dx$. The word W is homogeneous of degree $\kappa_\mathcal{E} - m_\mathcal{E}$ (8.1), so $W(w_B v_B, w_R v_R) = w_R^{\kappa_\mathcal{E}-m_\mathcal{E}} W(x v_B, v_R) = (1+x)^{m_\mathcal{E}-\kappa_\mathcal{E}} W(x v_B, v_R)$, and the powers of $1+x$ collect to $(1+x)^{-\kappa_\mathcal{E}}$. By Tonelli on the modulus, which is the same integrand at the real parts of the exponents, and Fubini,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_B}\int_{\Delta_R} v_B^{\mathbf{m}|_B-\mathbf{1}} v_R^{\mathbf{m}|_R-\mathbf{1}} \int_0^\infty x^{m_B-1}(1+x)^{-\kappa_\mathcal{E}} W(x v_B, v_R)\, dx\, d\omega_R(v_R)\, d\omega_B(v_B).$$

The densities of $\mathrm{Dir}_{\mathbf{m}|_B}$ and $\mathrm{Dir}_{\mathbf{m}|_R}$ are $v_B^{\mathbf{m}|_B-\mathbf{1}}/\Gamma_{|B|}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}$ and $v_R^{\mathbf{m}|_R-\mathbf{1}}/\Gamma_{|R|}\begin{bmatrix}\mathbf{m}|_R\\ m_R\end{bmatrix}$ (3.10), which gives the display. At (xU_B', U_R') the letter $|\alpha|$ of the list $\binom{\mathcal{E}}{1}$ is $x|U_B'| + |U_R'| = 1+x$, so the inner integral is the scale integral (15.2) of the family with that table appended at the exponents $(\kappa_\mathcal{E}, 0)$ (38.8). On the balance wall the factor $(1+x)^{-\kappa_\mathcal{E}}$ is 1, the inner integral is $\mathrm{Sc}_B(U_B', U_R'; \mathbf{m};\mathbf{s},\mathbf{t})$, and the double integral above is 15.7 at $h = |\alpha_R|$.

The conditional density. For real data with every proper wall positive the same computation, which needs no sign of the masses, shows that under $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ the variables $\rho = |U_B|/|U_R|$, V_B and V_R have the joint density

$$\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}} W(\rho V_B, V_R)\, V_B^{\mathbf{m}|_B-\mathbf{1}} V_R^{\mathbf{m}|_R-\mathbf{1}} / P_F(\mathbf{m};\mathbf{s},\mathbf{t})$$

against $d\rho\, d\omega_B\, d\omega_R$. Given (V_B, V_R), the density of ρ is proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}} W(\rho V_B, V_R)$. On the balance wall this is the reference, proof of 17.3. □

Lemma S5. *Let G be a convex function of class C^2 on $\mathbb{R}^d$ with $G(0) = 0$, $\nabla G(0) = 0$ and $\nabla^2 G(0)$ positive definite, and let b be a measurable function on $\mathbb{R}^d$, continuous at 0, with $|b(y)| \le e^{c_0(1+|y|)}$ for a constant $c_0 \ge 0$. Then $\int_{\mathbb{R}^d} |b| e^{-\lambda G} dy < \infty$ for every large λ, and for every bounded continuous function H on $\mathbb{R}^d$*

$$\lim_{\lambda\to+\infty} \lambda^{d/2} \int_{\mathbb{R}^d} b(y) H(\sqrt{\lambda}\, y) e^{-\lambda G(y)} dy = b(0) \int_{\mathbb{R}^d} H(x) e^{-\frac{1}{2}\langle x, \nabla^2 G(0) x\rangle} dx.$$

At $H = 1$ the right side is $b(0)(2\pi)^{d/2} \det\big(\nabla^2 G(0)\big)^{-1/2}$.

Proof. Let $\epsilon > 0$ be the least eigenvalue of $\nabla^2 G(0)$. By Taylor's formula $G(y) = \frac{1}{2}\langle y, \nabla^2 G(0) y\rangle + o(|y|^2)$; choose $\delta > 0$ with $G(y) \ge \frac{1}{4}\langle y, \nabla^2 G(0) y\rangle \ge \frac{\epsilon}{4}|y|^2$ and $|b(y)| \le |b(0)| + 1$ for $|y| \le \delta$.

Linear growth. For $|y| \ge \delta$ put $t = \delta/|y| \in (0,1]$. Convexity and $G(0) = 0$ give $G(ty) \le tG(y)$, so $G(y) \ge t^{-1} G(ty) \ge \frac{|y|}{\delta} \cdot \frac{\epsilon}{4}\delta^2 = \gamma|y|$, with $\gamma = \frac{\epsilon\delta}{4}$.

The outer part. For $\lambda\gamma \ge 2c_0$ and $|y| \ge \delta$, $|b(y)| e^{-\lambda G(y)} \le e^{c_0 - \frac{1}{2}\lambda\gamma|y|} \le e^{c_0 - \frac{1}{4}\lambda\gamma\delta} e^{-\frac{1}{4}\lambda\gamma|y|}$. So $\int_{|y|\ge\delta} |b| e^{-\lambda G} dy \le e^{c_0 - \frac{1}{4}\lambda\gamma\delta} \big(\frac{4}{\lambda\gamma}\big)^d \int_{\mathbb{R}^d} e^{-|x|} dx$, and $\lambda^{d/2}$ times it tends to 0. With the bound on $|y| \le \delta$, the integral of $|b| e^{-\lambda G}$ is finite.

The inner part. With $y = x/\sqrt{\lambda}$,

$$\lambda^{d/2} \int_{|y|<\delta} b(y) H(\sqrt{\lambda}\, y) e^{-\lambda G(y)} dy = \int_{\mathbb{R}^d} \mathbf{1}\big[|x| < \delta\sqrt{\lambda}\big]\, b(x/\sqrt{\lambda}) H(x) e^{-\lambda G(x/\sqrt{\lambda})} dx.$$

As $\lambda \to \infty$ the integrand tends to $b(0)H(x)e^{-\frac{1}{2}\langle x,\nabla^2 G(0)x\rangle}$ at every x, by the continuity of b at 0 and $\lambda G(x/\sqrt{\lambda}) \to \frac{1}{2}\langle x,\nabla^2 G(0)x\rangle$, and it is bounded by $(|b(0)|+1)\sup|H|\,e^{-\epsilon|x|^2/4}$, which is integrable. Dominated convergence gives the limit. At $H=1$, $\int e^{-\frac{1}{2}\langle x,Ax\rangle}dx = (2\pi)^{d/2}(\det A)^{-1/2}$ for positive definite A. □

Proposition 17.8 (The period along a ray of data). *As stated: a base point of complex data on the balance wall added to the scaled data.*

Proof. Write $\mathbf{d}_\lambda = (\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$, and in this proof $G(y) = \Phi(y) - \langle\boldsymbol{\tau},y\rangle$, Φ the potential at $(\mathbf{s},\mathbf{t})$.

The data. The walls and the imbalance are affine in the data with real coefficients. At $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, real data with $\boldsymbol{\tau}>0$, $s_j,t_j\geq 0$ not all zero and a native gauge on a connected family, every proper wall is positive (8.18(iii)), and both $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ and $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ lie on the balance wall. So $\kappa_{\mathcal{E}}(\mathbf{d}_\lambda)=0$, and $\operatorname{Re}\kappa_S(\mathbf{d}_\lambda) = \operatorname{Re}\kappa_S(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0) + \lambda\kappa_S(\boldsymbol{\tau};\mathbf{s},\mathbf{t}) \to +\infty$ for every nonempty proper S: for λ large every proper wall of $\mathbf{d}_\lambda$ has positive real part.

The chart integral. For such λ, $P_F(\mathbf{d}_\lambda)$ is the integral over the chart $\alpha_g = 1$ (16.2). In $y=\log\alpha$, $y_g=0$, the measure is $\alpha^{\mathbf{m}-\mathbf{1}}d\alpha_{\mathcal{E}-g} = e^{\langle\mathbf{m},y\rangle}dy_{\mathcal{E}-g}$, and the family word at the exponents $(\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$ is $W_0(e^y)e^{-\lambda\Phi(y)}$ (8.17). So

$$P_F(\mathbf{d}_\lambda) = \int_{\{y_g=0\}} b(y)\,e^{-\lambda G(y)}\,dy_{\mathcal{E}-g}, \qquad b(y) = e^{\langle\mathbf{m}_0,y\rangle}W_0(e^y).$$

The minimum. Φ is convex (8.18(ii)), and $\nabla G(y) = \nabla\Phi(y) - \boldsymbol{\tau}$ is the usage at e^y less the target, which vanishes at $y^\star = \log\alpha^\star$ (8.18(i)); normalize $\alpha^\star_g = 1$, so that $y^\star$ lies on the chart. For $v\in\mathbb{R}^{\mathcal{E}}$ and $\alpha = e^y$,

$$\langle v,\nabla^2\Phi(y)v\rangle = \sum_j \Big(s_j\operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha}}\langle v,\mathbf{1}_T\rangle + t_j\operatorname{Var}_{T\sim\mathbb{P}^*_{j,\alpha}}\langle v,\mathbf{1}_T\rangle\Big),$$

since $\partial_c^2\log K_j(e^{y+cv})$ at $c=0$ is the variance of $\langle v,\mathbf{1}_T\rangle$ under $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ (7.14), and likewise for Ψ_j under $\mathbb{P}^*_{j,\alpha}$, with $\langle v,\mathbf{1}_{\mathcal{E}\setminus T}\rangle = \langle v,\mathbf{1}\rangle - \langle v,\mathbf{1}_T\rangle$. Both laws charge every basis. So if $\langle v,\nabla^2\Phi(y)v\rangle = 0$, then $\langle v,\mathbf{1}_T\rangle$ is constant on the bases of every active table, v is constant on every molecule of the active family (7.7), and the active family of a connected family is one molecule (Lemma S3); so $v\in\mathbb{R}\mathbf{1}$. Conversely $\nabla^2\Phi\,\mathbf{1}=0$, $\Phi(y+c\mathbf{1})-\Phi(y)$ being linear in c. So the restriction of $\nabla^2\Phi(y^\star)$ to the chart $\{v_g=0\}$, the matrix $\big(\nabla^2\Phi(y^\star)\big)_{(g)}$, is positive definite; it is the Hessian at $y^\star$ of G on the chart.

The amplitude. For every basis T of table j, $|\langle\mathbf{1}_T,y\rangle| \leq r_j|y|$, so $\log\min_T\theta_{j,T} - r_j|y| \leq \log K_j(e^y) \leq \log K_j(\mathbf{1}) + r_j|y|$, and likewise for Ψ_j with $n-r_j$. Hence $|b(y)| = e^{\langle\operatorname{Re}\mathbf{m}_0,y\rangle}\prod_j K_j(e^y)^{-\operatorname{Re}s_{0,j}}\Psi_j(e^y)^{-\operatorname{Re}t_{0,j}} \leq e^{c_0(1+|y|)}$ for a constant c_0, and b is continuous and nowhere 0.

Laplace's method. Lemma S5 on the chart, with $d=n-1$, the function $G(y^\star+x)-G(y^\star)$ and the amplitude $b(y^\star+x)$, bounded by $e^{c_0(1+|y^\star|+|x|)}$, gives

$$P_F(\mathbf{d}_\lambda) = e^{-\lambda G(y^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}^{-1/2}\big(b(y^\star)+o(1)\big),$$

with $b(y^\star) = (\alpha^\star)^{\mathbf{m}_0}W_0(\alpha^\star)\neq 0$ and $G(y^\star) = \Phi(\log\alpha^\star) - \langle\boldsymbol{\tau},\log\alpha^\star\rangle$: the display. The function $\alpha\mapsto\alpha^{\mathbf{m}_0}W_0(\alpha)$ is homogeneous of degree $\kappa_{\mathcal{E}}(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$, so $b(y^\star)$ does not depend on the scale of $\alpha^\star$; likewise $G(y+c\mathbf{1}) = G(y)$, the data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ being balanced, and $\nabla^2\Phi$ does not change along $\mathbf{1}$. By 9.4, rung two of the still of $((s_j,t_j)_j;\alpha^\star)$ is $4\nabla^2\Phi(\log\alpha^\star)$.

The normal limit. For real $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ and λ large, $\mathrm{Breta}^\circ(F;\mathbf{d}_\lambda)$ is defined (16.7). For a bounded continuous function H of y modulo $\mathbb{R}\mathbf{1}$, the function $u\mapsto H(\log u)u^{\mathbf{m}-\mathbf{1}}W(u)$ is homogeneous of degree $-n$, so its integral over $\Delta_{\mathcal{E}}$ is its integral over the chart, as in the proof of Lemma S4(ii). So the law of $Y=\log U$ modulo $\mathbb{R}\mathbf{1}$, read on the chart, has the density $b(y)e^{-\lambda G(y)}/P_F(\mathbf{d}_\lambda)$, with $b>0$ continuous. Applying Lemma S5 as above to $H(\sqrt{\lambda}(y-y^\star))$ and to 1, and dividing,

$$\mathbb{E}\big[H\big(\sqrt{\lambda}(Y-y^\star)\big)\big]\to\int H(x)e^{-\frac12\langle x,\nabla^2\Phi(y^\star)_{(g)}x\rangle}dx\Big/\int e^{-\frac12\langle x,\nabla^2\Phi(y^\star)_{(g)}x\rangle}dx,$$

the expectation of H under the centred normal law on the chart with covariance $\big(\nabla^2\Phi(y^\star)_{(g)}\big)^{-1}$: the centred normal law on $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ with covariance $(\nabla^2\Phi)^{-1}$, the quadratic form of $\nabla^2\Phi(y^\star)$ on $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ being read on the chart. At $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$, $b=1$ and this is the reference, proof of 17.8. □

Remark. At $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$ let q be the question of the data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ (38.1), with the exponent $X(z)=2\langle\boldsymbol{\mu}_\perp,z\rangle-\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$, read at the representative of z in $\mathbf{1}^\perp$. By 8.18, $\Phi(y)=\sigma\langle\mathbf{1},y\rangle+\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$ in $y=2z$, and $\langle\boldsymbol{\tau},y\rangle=2\langle\boldsymbol{\mu},z\rangle+\sigma\langle\mathbf{1},y\rangle$, $\boldsymbol{\tau}$ being the masses of the question; so $\langle\boldsymbol{\tau},y\rangle-\Phi(y)=X(z)$ for $z\in\mathbf{1}^\perp$, and both sides are unchanged along $\mathbf{1}$. Hence $e^{-\lambda(\Phi(y^\star)-\langle\boldsymbol{\tau},y^\star\rangle)}=e^{\lambda X(z^\star)}$, $z^\star$ the tilt of $\frac12y^\star$. The Hessian in z is four times the Hessian in y, so on a chart $\det\big(-\nabla^2X(z^\star)\big)=4^{n-1}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}$, and

$$2^{n-1}e^{\lambda X(z^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(-\nabla^2X(z^\star)\big)^{-\frac12}=e^{-\lambda(\Phi(y^\star)-\langle\boldsymbol{\tau},y^\star\rangle)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}^{-1/2}:$$

the leading term of 38.5(ii) at real λ is the display of 17.8.

Definition 17.9 (Skeleton law). As stated, at real nonzero letter exponents.

Proof of the last clause. For a real coefficient $\xi_\ell>0$ and $y>0$, $(y+\xi_\ell)^{-w_\ell}>0$. A coefficient that is not real comes with its conjugate and the same real letter exponent w_ℓ; $y+\xi_\ell$ and $y+\bar\xi_\ell$ are conjugate and off $(-\infty,0]$, so their principal powers are conjugate and $(y+\xi_\ell)^{-w_\ell}(y+\bar\xi_\ell)^{-w_\ell}=|y+\xi_\ell|^{-2w_\ell}>0$. So $y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}$ is positive and continuous on $(0,\infty)$. As $y\to0$ it is asymptotic to $y^{\kappa-1}\prod_\ell\xi_\ell^{-w_\ell}$, the product being positive by the same pairing, and as $y\to\infty$ to $y^{\kappa-1-|\mathbf{w}|}$; so it is integrable on $(0,\infty)$ exactly when $0<\kappa<|\mathbf{w}|$. Its integral is the value of 15.1, which asks no sign of the letter exponents; so the normalizer is that value, positive, and the density is that of a law. □

Theorem 17.10 (The Skeleton law). *As stated: (i) off the balance wall and at exponents of any sign; (ii) at real nonzero letter exponents; (iii) about every point $r\ge\max_\ell\xi_\ell$ of the axis.*

Proof. (i) By 17.3, given the internal shapes (V_B,V_R) the scale ρ of B has a density proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_{\mathcal{E}}}\prod_jK_j(\rho V_B,V_R)^{-s_j}\Psi_j(\rho V_B,V_R)^{-t_j}$. At the gauge $\alpha=(V_B,V_R)$, 12.9 gives

$$K_j(\rho\alpha_B,\alpha_R)=\rho^{d_j(B)}K_j(\alpha)\prod_i\frac{\rho+\xi_{j,i}}{1+\xi_{j,i}},\qquad\Psi_j(\rho\alpha_B,\alpha_R)=\rho^{d_j^*(B)}\Psi_j(\alpha)\prod_i\frac{\rho+\xi^*_{j,i}}{1+\xi^*_{j,i}},$$

over the Skeleton values $\xi_{j,i}$ and $\xi^*_{j,i}$ of the two letters. The Skeleton polynomials are real, so their values not real come in conjugate pairs of equal multiplicity; the exponents are real, so the principal powers of conjugate factors are conjugate, and $K_j(\rho\alpha_B,\alpha_R)^{-s_j}=\rho^{-s_jd_j(B)}K_j(\alpha)^{-s_j}\prod_i(\rho+$

$\xi_{j,i})^{-s_j}(1+\xi_{j,i})^{s_j}$, likewise for Ψ_j. The power of ρ collects to $\rho^{m_B-1-\sum_j(s_jd_j(B)+t_jd_j^*(B))}=\rho^{\kappa_B-1}$ (8.1), and the linear factors collect to $\prod_\ell(\rho+\xi_\ell)^{-w_\ell}$ over the distinct points of the joint spectrum of the letters with nonzero exponent, with the letter exponents w_ℓ of 15.2. The factor $(1+\rho)^{-\kappa_\mathcal{E}}$ is the coefficient 1 with the letter exponent $\kappa_\mathcal{E}$, added to the letter exponent of 1 when 1 is a point of the joint spectrum; discarding the coefficients of letter exponent 0 changes nothing. So the conditional density is proportional to $\rho^{\kappa_B-1}\prod_\ell(\rho+\xi'_\ell)^{-w'_\ell}$, with coefficients in $\mathbb{C}\setminus(-\infty,0]$ closed under conjugation and real nonzero letter exponents, equal on conjugate coefficients. Their total is $|\mathbf{w}|+\kappa_\mathcal{E}=\kappa_B+\kappa_B^{\text{top}}+\kappa_\mathcal{E}$ (15.2), which is $\kappa_B+\kappa_{\mathcal{E}\setminus B}$ (8.12); B and $\mathcal{E}\setminus B$ are nonempty proper sets and every proper wall is positive, so $0<\kappa_B<|\mathbf{w}'|$, and the conditional law is $\text{Skel}_{\mathbf{w}',\kappa_B}(\boldsymbol{\xi}')$ (17.9). On the balance wall this is the reference, proof of 17.10(i).

(ii) For $-\kappa<\operatorname{Re}k<|\mathbf{w}|-\kappa$, $\mathbb{E}[Y^k]$ is $\int_0^\infty y^{\kappa+k-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}dy$ over $\int_0^\infty y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}dy$, both convergent, the values of 15.1 at the indices $\kappa+k$ and κ; for positive letter exponents the factors $\Gamma(w_\ell)$ of the two stacked Gammas cancel, which is the display. For the inversion put $y=1/z$: $dy=z^{-2}dz$, and $y+\xi=\xi z^{-1}(z+\xi^{-1})$ with $\operatorname{Log}(y+\xi)=\operatorname{Log}\xi-\log z+\operatorname{Log}(z+\xi^{-1})$, the principal arguments of ξ and of $z+\xi^{-1}$ having opposite signs and sum in $(-\pi,\pi)$. So the density of $1/Y$ is proportional to $z^{1-\kappa}z^{-2}z^{|\mathbf{w}|}\prod_\ell(z+\xi_\ell^{-1})^{-w_\ell}=z^{|\mathbf{w}|-\kappa-1}\prod_\ell(z+\xi_\ell^{-1})^{-w_\ell}$, the constant $\prod_\ell\xi_\ell^{-w_\ell}$ being positive; with $0<|\mathbf{w}|-\kappa<|\mathbf{w}|$ this is $\text{Skel}_{\mathbf{w},|\mathbf{w}|-\kappa}(\boldsymbol{\xi}^{-1})$. This is the reference, proof of 17.10(ii), with 15.6, which uses no sign of the letter exponents.

(iii) The coefficients are real, hence positive, and $r\ge\max_\ell\xi_\ell$, so $|\xi_\ell-r|<r$ and $0\le g_\ell<1$. By the expansion in the proof of 15.5,

$$y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}=\sum_{\mathbf{n}\in\mathbb{N}^J}\prod_\ell\frac{(w_\ell)_{n_\ell}}{n_\ell!}(r-\xi_\ell)^{n_\ell}\,y^{\kappa-1}(y+r)^{-|\mathbf{w}|-|\mathbf{n}|},$$

with nonnegative terms, the letter exponents being positive. For $a>\kappa>0$, the law $\text{Skel}_{a,\kappa}(1)$ has the density $y^{\kappa-1}(1+y)^{-a}/\Gamma\,{}_2{}_1\left[\begin{matrix}\kappa,\ a-\kappa\\ a\end{matrix}\right]$, the Breta of one coefficient 1 being 1 (the paragraph after 15.1); so rY_a, $Y_a\sim\text{Skel}_{a,\kappa}(1)$, has the density $r^{a-\kappa}y^{\kappa-1}(y+r)^{-a}/\Gamma\,{}_2{}_1\left[\begin{matrix}\kappa,\ a-\kappa\\ a\end{matrix}\right]$. With $a=|\mathbf{w}|+|\mathbf{n}|$ and $(r-\xi_\ell)^{n_\ell}=r^{n_\ell}g_\ell^{n_\ell}$, the density of Y is the sum over $\mathbf{n}$ of $r^{\kappa-|\mathbf{w}|}\prod_\ell\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{n_\ell}\Gamma\,{}_2{}_1\left[\begin{matrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{matrix}\right]$ times the density of $rY_\mathbf{n}$, divided by the normalizer; the coefficients are nonnegative and, by Tonelli, sum to 1. So Y is the stated mixture. At $r=\max_\ell\xi_\ell$ the largest coefficient has $g_\ell=0$, and only $n_\ell=0$ contributes.

(iv) On the balance wall with $s_j,t_j\ge0$, given the shapes the density of the scale ρ is proportional to the integrand of the scale integral at $\alpha=(V_B,V_R)$, by (i). By the proof of 15.9(i) it is, on $(0,1)$, proportional to $\rho^{\kappa_B-1}\prod_i\mathsf{Q}_{B,i}(\rho;\alpha)^{-w_i}$, and the density of $1/\rho$ on $(0,1)$ is proportional to $y^{\kappa_B^{\text{top}}-1}\prod_i\mathsf{Q}_{R,i}(y;\alpha)^{-w_i}$. The expansions of 15.9(ii) have nonnegative terms, each the term of $\mathbf{n}$ and k times $y^{\kappa+k-1}(1-y)^{|\mathbf{n}|}/\Gamma\,{}_2{}_1\left[\begin{matrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{matrix}\right]$, the density of the first coordinate of $\text{Dir}_{(\kappa+k,|\mathbf{n}|+1)}$ (3.9); by Tonelli the conditional laws are the stated mixtures, and the two sectors carry probabilities proportional to their Trailmoments. This is the reference, proof of 17.10(v). □

Definition 17.11 (Wall contour; joint period). As stated, at complex data on the balance wall with $\operatorname{Re}s_j>0$ or $s_j=0$, and $\operatorname{Re}t_j>0$ or $t_j=0$.

Remark. A wall contour is fixed by real parts alone, and the fibre integral at B has complex pins, whose admissible real points (4.1) are the wall contours. The two descriptions agree: $\text{load}_e=$

$\sigma + \langle \boldsymbol{\delta} \rangle_e$ on the share sums and $m_e = \sigma + \mu_e$, so $\operatorname{Re} \mathrm{load}_e = \operatorname{Re} m_e$ exactly when $\operatorname{Re} \langle \boldsymbol{\delta} \rangle_e = \operatorname{Re} \mu_e$. For real data with $s_j, t_j \geq 0$ these are the conditions of the reference.

Theorem 17.12 (The port on the balance wall). *As stated, at complex data on the balance wall with* $\operatorname{Re} s_j > 0$ *or* $s_j = 0$*, and* $\operatorname{Re} t_j > 0$ *or* $t_j = 0$*.*

Proof. A letter with exponent 0 is 1 and is not opened; the shares, the stacked Gammas and the products below run over the letters with nonzero exponent, whose exponents have positive real part.

The real parts. The walls and the imbalance are affine in the data with real coefficients. So the data $(\operatorname{Re} \mathbf{m}; \operatorname{Re} \mathbf{s}, \operatorname{Re} \mathbf{t})$ lie on the balance wall, and their proper walls are the real parts of those of $(\mathbf{m}; \mathbf{s}, \mathbf{t})$, positive since $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ converges (8.10). Their exponents are nonnegative and not all zero: otherwise $\kappa_S = \operatorname{Re} m_S$ for every S, and $0 = \kappa_{\mathcal{E}} = \kappa_B + \kappa_R > 0$. Their masses are positive: $\operatorname{Re} m_e = \kappa_{\{e\}} + \sum_j \big(\operatorname{Re} s_j \, d_j(\{e\}) + \operatorname{Re} t_j \, d_j^*(\{e\}) \big) > 0$, $\{e\}$ being a nonempty proper set ($n \geq 2$, as B is proper). A nonempty proper S with $c_j(S) = 0$ in every active table would have $\kappa_S + \kappa_{\mathcal{E} \setminus S} = \kappa_{\mathcal{E}} = 0$ at the real parts (8.15), against the positivity of both walls; so F is connected, and its active family, whose tables are those of the real parts, is one molecule (Lemma S3). By 8.18(iii) the real parts have a native gauge.

(i), convergence and the room integral. By 7.13(i), $K_j = e^{\langle \mathbf{1}, z \rangle} \Lambda_j(z)$ and $\Psi_j = e^{\langle \mathbf{1}, z \rangle} \Lambda_j(-z)$ at $\alpha = e^{2z}$, so with $\sigma = \sum_j \sigma_j$ the integrand of P_F is $f(\alpha) = \alpha^{\boldsymbol{\mu} - \mathbf{1}} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j}$, homogeneous of degree $\kappa_{\mathcal{E}} - n = -n$ (7.13(iii), 8.2(iii)), and $|f|$ is the same function at the real parts of the data. At the real parts, 8.10 on the section $|\alpha_R| = 1$, whose measure is $d\alpha_B \, d\omega_R$ (proof of 15.7 above), gives $\int_{\Delta_R} \int_{\mathbb{R}^B_{>0}} |f(\alpha_B, v)| \, d\alpha_B \, d\omega_R(v) < \infty$; and $f(\alpha_B, v)$ is $v^{\boldsymbol{\mu}|_R - \mathbf{1}}$ times the integrand of $\Xi_B(v; \boldsymbol{\mu}|_B; \mathbf{s}, \mathbf{t})$. By Tonelli that integral converges absolutely at almost every $v \in \Delta_R$. For two room points α_R, α'_R there is $C \geq 1$ with $\Lambda_{j,(p)}(\pm z_R) \leq C \Lambda_{j,(p)}(\pm z'_R)$ for all j, p, these being finite sums of positive terms; by 12.6, $\Lambda_j(z) = \sum_p e^{\langle z_B, \chi_p \rangle} \Lambda_{j,(p)}(z_R)$, so $\Lambda_j(z_B, z_R) \geq C^{-1} \Lambda_j(z_B, z'_R)$, likewise at $-z$, and the modulus of the integrand of Ξ_B at α_R is at most $C^{\sum_j (\operatorname{Re} s_j + \operatorname{Re} t_j)}$ times that at α'_R. So Ξ_B converges at every α_R, and Fubini gives $P_F = \int_{\Delta_R} v^{\boldsymbol{\mu}|_R - \mathbf{1}} \Xi_B(v) \, d\omega_R(v)$.

Independence of the pins. Let $R_1, \dots, R_q$ be the molecules of the children (for $q = 1$ there are no pins κ_{R_i} below), and write $\kappa_{R_i} = m_{R_i} - \ell_i$ with $\ell_i(\mathbf{a}, \mathbf{b}) = \sum_{j,p} \big(a_{j,p} r_{j,(p)}(R_i) + b_{j,p} (|R_i| - r_{j,(p)}(R_i)) \big)$. The linear forms $|\mathbf{a}_j|$ and $|\mathbf{b}_j|$ of the letters, $\langle \boldsymbol{\delta} \rangle_e$ for $e \in B$, and ℓ_i for $i < q$, are linearly independent. Indeed, let $\sum_{e \in B} \gamma_e \langle \boldsymbol{\delta} \rangle_e + \sum_{i<q} w_i \ell_i = \sum_j (u_j |\mathbf{a}_j| + u'_j |\mathbf{b}_j|)$, and put $V \in \mathbb{R}^{\mathcal{E}}$ equal to γ on B, to w_i on R_i for $i < q$ and to 0 on R_q. For a basis T of table j with $T \cap B = p$, $T \setminus B$ is a basis of the child (j, p), which meets each R_i in $r_{j,(p)}(R_i)$ marks, R_i being a union of molecules of that child (7.8); and $\frac{1}{2} \langle \gamma, \chi_p \rangle = \langle \gamma, \mathbf{1}_p \rangle - \frac{1}{2} \langle \gamma, \mathbf{1}_B \rangle$. Comparing the coefficients of $a_{j,p}$ gives $\langle V, \mathbf{1}_T \rangle = u_j + \frac{1}{2} \langle \gamma, \mathbf{1}_B \rangle$, and comparing those of $b_{j,p}$ gives $\langle V, \mathbf{1}_T \rangle = \langle V, \mathbf{1}_R \rangle + \frac{1}{2} \langle \gamma, \mathbf{1}_B \rangle - u'_j$, for every basis T of table j whenever that letter is opened. So $\langle V, \mathbf{1}_T \rangle$ is constant on the bases of every active table; by 7.7 V is constant on the one molecule $\mathcal{E}$ of the active family, and as it vanishes on $R_q \neq \varnothing$, $V = 0$: $\gamma = 0$ and every $w_i = 0$.

(i), the fibre formula. Fix α_R and a wall contour. Put $X_{j,p} = e^{\langle z_B, \chi_p \rangle} \Lambda_{j,(p)}(z_R)$ and $Y_{j,p} = e^{-\langle z_B, \chi_p \rangle} \Lambda_{j,(p)}(-z_R)$, positive, so that $\Lambda_j(z) = \sum_p X_{j,p}$ and $\Lambda_j(-z) = \sum_p Y_{j,p}$. For a letter with more than one pattern, 4.4 on the contour $\operatorname{Re} a_{j,p} = c_{j,p}$, admissible since every $c_{j,p} > 0$ and $\sum_p c_{j,p} = \operatorname{Re} s_j$, gives $\Lambda_j(z)^{-s_j} = \int_{|\mathbf{a}_j| = s_j} \Gamma_{|\operatorname{pat}^{}_{\!1}{}_j(B)|} \begin{bmatrix} \mathbf{a}_j \\ s_j \end{bmatrix} \prod_p X_{j,p}^{-a_{j,p}}$; for one pattern this is the evaluation $a_{j,p} = s_j$. Likewise for $\Lambda_j(-z)^{-t_j}$ with the shares $\mathbf{b}_j$. Since $\prod_{j,p} e^{-\langle z_B, \chi_p \rangle (a_{j,p} - b_{j,p})} =$

$e^{-2\langle z_B,\langle\boldsymbol{\delta}\rangle\rangle}=\alpha_B^{-\langle\boldsymbol{\delta}\rangle}$ (12.21), in $y=\log\alpha_B$, where $\alpha_B^{\boldsymbol{\mu}|_B-\mathbf{1}}d\alpha_B=e^{\langle\boldsymbol{\mu}|_B,y\rangle}dy$,

$$\Xi_B(\alpha_R)=\int_{\mathbb{R}^B}\Big(\int_{\mathcal{X}_0}\mathcal{G}(\mathbf{a},\mathbf{b})\,e^{\langle\boldsymbol{\mu}|_B-\langle\boldsymbol{\delta}\rangle,y\rangle}\Big)dy,$$

$$\mathcal{G}=\prod_j\Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\prod_{j,p}\Lambda_{j,(p)}(z_R)^{-a_{j,p}}\Lambda_{j,(p)}(-z_R)^{-b_{j,p}},$$

the inner integral the product of the share integrals, over the set $\mathcal{X}_0$ of shares on the contour with the share sums pinned. On a wall contour $\operatorname{Re}\langle\boldsymbol{\delta}\rangle_e=\operatorname{Re}\mu_e$ for $e\in B$, so $e^{\langle\boldsymbol{\mu}|_B-\langle\boldsymbol{\delta}\rangle,y\rangle}=e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B-\operatorname{Im}\langle\boldsymbol{\delta}\rangle,y\rangle}$. By Stirling's formula $|\Gamma(c+i\eta)|\le C_c(1+|\eta|)^{c-1/2}e^{-\pi|\eta|/2}$, so $|\mathcal{G}|$, whose factors $\Lambda_{j,(p)}^{-a_{j,p}}$ have moduli constant on the contour, decays exponentially along every imaginary direction of $\mathcal{X}_0$, uniformly on parallel affine subspaces through a compact set. The map $(\mathbf{a},\mathbf{b})\mapsto\operatorname{Im}\langle\boldsymbol{\delta}\rangle\in\mathbb{R}^B$ is affine on $\mathcal{X}_0$, with linear part onto $\mathbb{R}^B$ by the independence of the pins; so the image of $\mathcal{G}$ times the measure of the share integrals has a density g on $\mathbb{R}^B$, continuous and integrable, and the inner integral is $e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B,y\rangle}\hat g(y)$, $\hat g(y)=\int g(\nu)e^{-i\langle\nu,y\rangle}d\nu$. The modulus $|\hat g(y)|$ is the modulus of the integrand of Ξ_B in y, integrable by the first step. By Fourier inversion, $\Xi_B(\alpha_R)=\int\hat g(y)e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B,y\rangle}dy=(2\pi)^{|B|}g(\operatorname{Im}\boldsymbol{\mu}|_B)$, which is the fibre integral of $\mathcal{G}$ over the fibre $\mathcal{X}_1$ cut by the pins $|\mathbf{a}_j|=s_j$, $|\mathbf{b}_j|=t_j$ and $\langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_B$ (4.1: the $|B|$ further pins raise $(2\pi)^{q-N}$ by $(2\pi)^{|B|}$ and add the factor $\delta^{|B|}(\operatorname{Im}\langle\boldsymbol{\delta}\rangle-\operatorname{Im}\boldsymbol{\mu}|_B)$), with the wall contour as its admissible real point. This is the display of (i). For real data this is the reference, proof of 17.12(i), where $\operatorname{Im}\boldsymbol{\mu}|_B=0$.

(ii). On the pins of (i), $\operatorname{load}_e=m_e$ for $e\in B$, and by 12.2 the child (j,p) has rank $r_j-|p|$; so

$$\sum_{i=1}^q\kappa_{R_i}=m_R-\sum_{j,p}\big(a_{j,p}(r_j-|p|)+b_{j,p}(|R|-r_j+|p|)\big)=m_R-\sum_j\big(s_jr_j+t_jr_j^*\big)+\sum_{e\in B}\operatorname{load}_e=\kappa_{\mathcal{E}}=0,$$

using $\sum_{e\in B}\operatorname{load}_e=\sum_{j,p}\big(a_{j,p}|p|+b_{j,p}(|B|-|p|)\big)$; so $\kappa_{R_q}=0$ holds on $\mathcal{X}$. Take the section $\{|\alpha_{R_q}|=1\}$, with the measure $d\alpha_B\prod_{i<q}d\alpha_{R_i}\,d\omega_{R_q}$, and as in (i) $P_F=\int\alpha_R^{\boldsymbol{\mu}|_R-\mathbf{1}}\,\Xi_B(\alpha_R)\prod_{i<q}d\alpha_{R_i}\,d\omega_{R_q}$, absolutely convergent. Write $\alpha_{R_i}=e^{y_i}u_i$, $u_i\in\Delta_{R_i}$, for $i<q$, and $u_q=\alpha_{R_q}$; then $\alpha_{R_i}^{\boldsymbol{\mu}|_{R_i}-\mathbf{1}}d\alpha_{R_i}=e^{y_i\mu_{R_i}}u_i^{\boldsymbol{\mu}|_{R_i}-\mathbf{1}}dy_i\,d\omega_{R_i}$. Each R_i is a union of molecules of every child, so every basis X of the child (j,p) has $\langle\chi_X,\mathbf{1}_{R_i}\rangle=2r_{j,(p)}(R_i)-|R_i|$, and $\Lambda_{j,(p)}(z_R+\frac12y_i\mathbf{1}_{R_i})=e^{\frac12y_i(2r_{j,(p)}(R_i)-|R_i|)}\Lambda_{j,(p)}(z_R)$. Inserting the fibre formula of (i) and using $\sum_{j,p}\sigma_{j,p}=\sigma$, the exponent of e^{y_i} collects to $\mu_{R_i}-\sum_{j,p}\delta_{j,p}\big(2r_{j,(p)}(R_i)-|R_i|\big)=\kappa_{R_i}(\mathbf{a},\mathbf{b})$, and the rest is $\mathcal{G}$ at u times $u^{\boldsymbol{\mu}|_R-\mathbf{1}}$, that is, by 7.13 and $\sum_{j,p}\sigma_{j,p}=\sigma$, the stacked Gammas times the integrand of the joint period of the children at the masses $\mathbf{m}|_R$. For fixed $\mathbf{y}=(y_i)_{i<q}$, on a wall contour with $\operatorname{Re}\kappa_{R_i}=0$ for $i<q$, the modulus of this integrand over $u\in\prod_i\Delta_{R_i}$ and the shares is the modulus of the stacked Gammas times the joint integrand at the real parts $(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$, integrable by the hypothesis on $P^\otimes$ and by Stirling; so by Fubini the integral over u and the shares is

$$\hat g_1(\mathbf{y})=\int_{\mathcal{X}_1}\prod_j\Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\;P^\otimes_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})\,e^{\,i\sum_{i<q}y_i\operatorname{Im}\kappa_{R_i}(\mathbf{a},\mathbf{b})},$$

$\mathcal{X}_1$ the fibre of (i), and $\int|\hat g_1(\mathbf{y})|\,d\mathbf{y}$ is at most P_F at the real parts, finite. The joint period is continuous in the shares and bounded by its value at the real parts, by dominated convergence. The map $(\mathbf{a},\mathbf{b})\mapsto(\operatorname{Im}\kappa_{R_i})_{i<q}$ is affine on $\mathcal{X}_1$ with linear part onto $\mathbb{R}^{q-1}$, by the independence of the pins. So, as in (i), Fourier inversion in $\mathbf{y}$ gives $P_F=\int\hat g_1(\mathbf{y})\,d\mathbf{y}$ as the fibre integral over $\mathcal{X}$, the pins $\kappa_{R_i}=0$, $i<q$, holding in real part on the contour and in imaginary part by inversion. For $q=1$ there is no $\mathbf{y}$, and $P_F=\hat g_1$ by Fubini.

(iii). The real parts have a native gauge (first step). For real data the statement is the reference, proof of 17.12(iii): the native split is a wall contour, lies on the real fibre, and $P^{\otimes}$ converges there. A wall contour, and the real parts of the pins of $\mathcal{X}$, $\operatorname{Re}|\mathbf{a}_j| = \operatorname{Re} s_j$, $\operatorname{Re}|\mathbf{b}_j| = \operatorname{Re} t_j$, $\operatorname{Re}\langle\boldsymbol{\delta}\rangle = \operatorname{Re}\boldsymbol{\mu}|_B$ and $\operatorname{Re}\kappa_{R_i} = \kappa_{R_i}(\operatorname{Re}\mathbf{m}|_R; \mathbf{c}, \mathbf{c}')$, involve the data only through their real parts. So the native split of the real parts is a wall contour for the data, an admissible real point of $\mathcal{X}$, and $P^{\otimes}_{F_B(\mathbf{c},\mathbf{c}')}(\operatorname{Re}\mathbf{m}|_R; \mathbf{c}, \mathbf{c}')$ converges there, as (ii) asks. □

40.18 Chapter 18: blocks: generation and cleaning

Throughout this section, for a bunch B of width one of table j whose weights factor across B and R, $\theta_{j,T} = \theta'_{j,T\cap B}\,\theta''_{j,T\cap R}$, the minors $\mathcal{B}_j|B$ and $\mathcal{B}_j/R$ on B carry θ'_j and the minors $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ on R carry θ''_j (18.4), and every integral of a positive function is read in $[0,\infty]$ (Tonelli).

Theorem 18.1 (Replacement of marks by blocks). *As stated: every mark is replaced, by a block of any size and by either rule.*

Proof. One mark, first rule. Let h be replaced by a block X of q marks by the first rule, the other marks left in place: the new sets are the bases $T \not\ni h$ and the sets $(T-h)\sqcup\{x\}$, $T \ni h$, $x \in X$, each with the weight θ_T. If h is neither a loop nor a coloop, $(\mathcal{B},\theta)$ is a cell on $(\mathcal{E}-h)\sqcup\{h\}$ with the hinge h, and the parallel block $\binom{X\sqcup\{g\}}{1}$ is a cell on $X\sqcup\{g\}$ whose deletion of g has the singletons of X as bases and whose contraction of g has the single basis $\varnothing$. By 20.2 for these two cells, the sets $\{x\}\sqcup(T-h)$, $T\ni h$, and $\varnothing\sqcup T$, $T\not\ni h$, are the bases of a table of rank $1+r-1=r$: they are the new sets. If h is a loop, no basis contains h, and the new sets are the bases of $\mathcal{B}$ read on $\mathcal{E}-h$, the marks of X being loops: a table of rank r. If h is a coloop, every basis contains h, and the new sets are the sets $(T-h)\sqcup\{x\}$: the direct sum of the contraction of h and the table $\binom{X}{1}$, a table of rank r, the exchange property of a direct sum holding in each summand.

One mark, complementary rule. The dual table has the bases $\mathcal{E}\setminus T$ (7.2). The first rule at h on the dual table gives the sets $\mathcal{E}\setminus T$ for $T\ni h$ and $(\mathcal{E}\setminus T)-h+x$ for $T\not\ni h$, $x\in X$, the bases of a table on $(\mathcal{E}-h)\sqcup X$ by the first case. Their complements in $(\mathcal{E}-h)\sqcup X$ are $(T-h)\sqcup X$ for $T\ni h$ and $T\sqcup(X-x)$ for $T\not\ni h$, the sets of the complementary rule, and the complements of the bases of a table are the bases of its dual. Its rank is $(|\mathcal{E}|-1+q)-(|\mathcal{E}|-r)=r+q-1$.

All marks. The rule at a mark h decides the part of a new set in X_h from whether $h\in T$ alone, so replacing the marks one at a time, in any order, gives the sets of the statement; by the two cases each step gives a table, and the step at $h\in\mathcal{E}_2$ raises the rank by q_h-1. A new set T' determines $T=\{h\in\mathcal{E}_1:\ X_h\cap T'\neq\varnothing\}\sqcup\{h\in\mathcal{E}_2:\ X_h\subseteq T'\}$ and the choice made in every block, so it carries the one weight θ_T. A block of one mark gives back its mark under either rule.

Substitution. When no mark is a loop or a coloop, $(\mathcal{B},\theta)$ is a core (20.1), and the parallel block $\binom{X_h\sqcup\{g_h\}}{1}$ and the series block $\binom{X_h\sqcup\{g_h\}}{q_h}$ are cells, g_h being neither a loop nor a coloop of either. In the parallel block the deletion of g_h has the singletons of X_h as bases and the contraction the single basis $\varnothing$; in the series block the deletion has the single basis X_h and the contraction the sets X_h-x. So the sets of the substitution contain, for a basis T of the core, one mark of X_h for $h\in T\cap\mathcal{E}_1$, none for $h\in\mathcal{E}_1\setminus T$, all of X_h for $h\in T\cap\mathcal{E}_2$ and all but one for $h\in\mathcal{E}_2\setminus T$, each with the weight θ_T, every weight of a block being 1: the table of the statement.

Letters. By the correspondence between the new sets and the pairs of a basis T and a choice in every block, $K'(\alpha)$ is the sum over T of θ_T times the product over the blocks of the sum, over the choices, of the monomials of the chosen marks: $e_1(\alpha_{X_h})$ for $h\in T\cap\mathcal{E}_1$, 1 for $h\in\mathcal{E}_1\setminus T$, $e_{q_h}(\alpha_{X_h})$

for $h \in T \cap \mathcal{E}_2$ and $e_{q_h-1}(\alpha_{X_h})$ for $h \in \mathcal{E}_2 \setminus T$. Taking out $\prod_{h\in\mathcal{E}_2} e_{q_h-1}$,

$$K' = \prod_{h\in\mathcal{E}_2} e_{q_h-1} \sum_T \theta_T \prod_{h\in T\cap\mathcal{E}_1} e_1 \prod_{h\in T\cap\mathcal{E}_2} \frac{e_{q_h}}{e_{q_h-1}},$$

the first display. The complement of a new set meets X_h in all of X_h but the chosen mark for $h \in T \cap \mathcal{E}_1$, in all of X_h for $h \in \mathcal{E}_1 \setminus T$, in nothing for $h \in T \cap \mathcal{E}_2$ and in the missing mark for $h \in \mathcal{E}_2 \setminus T$, so

$$\Psi' = \sum_T \theta_T \prod_{h\in T\cap\mathcal{E}_1} e_{q_h-1} \prod_{h\in\mathcal{E}_1\setminus T} e_{q_h} \prod_{h\in\mathcal{E}_2\setminus T} e_1 = \prod_{h\in\mathcal{E}_1} e_{q_h-1} \sum_T \theta_T \prod_{h\in\mathcal{E}_1\setminus T} \frac{e_{q_h}}{e_{q_h-1}} \prod_{h\in\mathcal{E}_2\setminus T} e_1,$$

which is $\prod_{h\in\mathcal{E}_1} e_{q_h-1}\,\Psi(\beta^*)$ with $\beta^*_h = e_{q_h}/e_{q_h-1}$ on $\mathcal{E}_1$ and $\beta^*_h = e_1$ on $\mathcal{E}_2$, since $\Psi(\beta^*) = \sum_T \theta_T(\beta^*)^{\mathcal{E}\setminus T}$. For one block at h, every other block of one mark, K and Ψ are affine in α_h, $K = K_{\setminus h} + \alpha_h K_{/h}$ and $\Psi = \alpha_h\Psi_{\setminus h} + \Psi_{/h}$ (7.3), and the two displays become the four formulas of the statement. In scale and shape, e_k is homogeneous of degree k, $e_1(u) = |u| = 1$ on Δ_X, and the measure is 2.7. □

Theorem 18.2 (Counts, walls and atoms under replacement). *As stated: (i) and (ii) for blocks of any sizes and either rule at each mark; (iii) at every positive exponent pairs.*

Proof. (i) The marks and the rank are 18.1. At $\alpha = \mathbf{1}$, $e_1 = q_h$, $e_{q_h} = 1$ and $e_{q_h-1} = q_h$ on a block of q_h marks, so the first display of 18.1 gives the total weight $K'(\mathbf{1}) = \prod_{h\in\mathcal{E}_2} q_h \cdot K\big((q_h)_{h\in\mathcal{E}_1}, (q_h^{-1})_{h\in\mathcal{E}_2}\big)$, and expanding K over the bases gives the sum displayed; it counts q_h choices at $h \in T \cap \mathcal{E}_1$ and q_h at $h \in \mathcal{E}_2 \setminus T$. With $\mathcal{E}_2 = \varnothing$ it is $K((q_h)_h)$; with $\mathcal{E}_1 = \varnothing$ it is $\prod_h q_h\, K((q_h^{-1})_h) = \Psi((q_h)_h)$ (7.2).

(ii) By 7.11, $d'(Y) = \min_{T'} |T' \cap Y|$ and $d^{*\prime}(Y) = \min_{T'} |Y \setminus T'|$ over the new bases T', which are the pairs of a basis T and a choice in every block (proof of 18.1). Fix T and minimize over the choices, block by block. In $|T' \cap Y|$ a block contributes, at its least: for $h \in T \cap \mathcal{E}_1$, 1 if $X_h \subseteq Y$ and 0 otherwise, the chosen mark avoiding Y when it can; for $h \in \mathcal{E}_1 \setminus T$, 0; for $h \in T \cap \mathcal{E}_2$, $|X_h \cap Y|$; for $h \in \mathcal{E}_2 \setminus T$, $\max(|X_h \cap Y| - 1, 0)$, the missing mark taken in Y when it can. Writing $|X_h \cap Y| = \max(|X_h \cap Y| - 1, 0) + [h \in \lceil Y \rceil]$ for $h \in \mathcal{E}_2$, the least value at T is

$$\sum_{h\in\mathcal{E}_2} \max\big(|X_h \cap Y| - 1, 0\big) + \big|T \cap \big((\lfloor Y \rfloor \cap \mathcal{E}_1) \sqcup (\lceil Y \rceil \cap \mathcal{E}_2)\big)\big|,$$

and the least over T is $d'(Y)$ as stated (7.11). In $|Y \setminus T'|$ a block contributes, at its least: for $h \in T \cap \mathcal{E}_1$, $\max(|X_h \cap Y| - 1, 0)$; for $h \in \mathcal{E}_1 \setminus T$, $|X_h \cap Y|$; for $h \in T \cap \mathcal{E}_2$, 0; for $h \in \mathcal{E}_2 \setminus T$, 1 if $X_h \subseteq Y$ and 0 otherwise. In the same way $d^{*\prime}(Y)$ is as stated, and $c'(Y) = |Y| - d'(Y) - d^{*\prime}(Y)$ is 7.10. Under the first rule alone, $r'(Y) = \max_{T'} |T' \cap Y| = \max_T |T \cap \lceil Y \rceil| = r(\lceil Y \rceil)$, $d'(Y) = d(\lfloor Y \rfloor) = r - r(\mathcal{E} \setminus \lfloor Y \rfloor)$ (7.10) and $c'(Y) = r'(Y) - d'(Y)$ (7.11). Under the complementary rule alone, $|Y| = \sum_h \max(|X_h \cap Y| - 1, 0) + |\lceil Y \rceil|$, so

$$c'(Y) = |\lceil Y \rceil| - d(\lceil Y \rceil) - d^*(\lfloor Y \rfloor) = |\lceil Y \rceil| - |\lfloor Y \rfloor| + r(\mathcal{E} \setminus \lceil Y \rceil) + r(\lfloor Y \rfloor) - r.$$

If $Y = \bigcup_{h\in S} X_h$, then $\lceil Y \rceil = \lfloor Y \rfloor = S$, $d'(Y) = \sum_{h\in S\cap\mathcal{E}_2}(q_h - 1) + d(S)$, $d^{*\prime}(Y) = \sum_{h\in S\cap\mathcal{E}_1}(q_h - 1) + d^*(S)$ and $|Y| = \sum_{h\in S} q_h$, so $c'(Y) = |S| - d(S) - d^*(S) = c(S)$. If $\varnothing \neq Y \subsetneq X_h$, then $\lceil Y \rceil = \{h\}$ and $\lfloor Y \rfloor = \varnothing$: for $h \in \mathcal{E}_1$, $d'(Y) = 0$ and $d^{*\prime}(Y) = |Y| - 1 + d^*(\{h\})$, and for $h \in \mathcal{E}_2$, $d'(Y) = |Y| - 1 + d(\{h\})$ and $d^{*\prime}(Y) = 0$; a seed table connected on at least two marks has $c(\{h\}) = 1 - d(\{h\}) - d^*(\{h\}) > 0$, so $d(\{h\}) = d^*(\{h\}) = 0$ and $c'(Y) = 1$.

(iii) Write $X_e = \{e\} \times \{1, \dots, q\}$. For $h \in H$, carrying table j to table $\tau(j)$ (11.14), put $\hat{h}(e,i) = (h(e), i)$; the rule and q are the same at every mark, so $\hat{h}$ maps the replaced table j onto the replaced table $\tau(j)$ with the same weights, and the two tables have the same pairs: $\hat{h}$ carries the replaced family to itself. A permutation of X_e fixing the other new marks maps every replaced table onto itself with its weights, the sets of the rule being symmetric in the marks of X_e. The group generated by these bijections is transitive on the new marks, H being transitive on $\mathcal{E}$.

The replaced family is connected. The seed family is one molecule: a molecule $M \neq \mathcal{E}$ would be a union of molecules of every table (7.8), with $|T \cap M|$ constant on the bases of every table and so $c_j(M) = 0$ for every j (7.11), all tables being active at positive pairs. Let $e \neq f$ be linked in table j: by 7.6 some basis T of table j has $e \in T$, $f \notin T$ and $T - e + f$ a basis. Under the first rule, a new basis T' from T choosing x in X_e gives $T' - x + x'$, the new basis from T choosing x', and $T' - x + y$, $y \in X_f$, the new basis from $T - e + f$ choosing y with the same choices elsewhere; under the complementary rule, the new basis T' from $T - e + f$ missing x' in X_e gives $T' - x + x'$, the one missing x, and the new basis T'' from T missing y in X_f gives $T'' - x + y$, the new basis from $T - e + f$ missing x. By 7.6 any two marks of X_e are linked, and every mark of X_e is linked to every mark of X_f. Every mark is linked to another mark, the seed family being one molecule on at least two marks, so the new marks form one molecule. Then no nonempty proper set Y of new marks has $c'_j(Y) = 0$ in every table: by 7.11 $|T' \cap Y|$ would be constant on the bases of every table, and by 7.7 no circuit would join Y to its complement.

Finally $((s_j, t_j)_j; \mathbf{1})$ is a realization of the replaced family, and every bijection above fixes the table's gauge, $\hat{h}^* \log \mathbf{1} = \log \mathbf{1}$. By 11.15 the reading of the image of a mark equals the reading of the mark, so the reading is constant on the new marks, which form one molecule: the realization is an atom (10.1). □

Remark. In exact rational arithmetic (receipt `check_blocks_splice.py`): on the seed of rank two on $\{a, b, c, d\}$ whose bases are the pairs other than $\{c, d\}$, with weights $1, 2, \frac{3}{2}, 5, \frac{1}{3}$, blocks of $2, 3, 1, 2$ marks, the first rule at a, c and the complementary rule at b, d, the new sets satisfy the exchange property, the rank is $r + \sum_{h \in \mathcal{E}_2} (q_h - 1)$, the total weight is that of (i), $d'(Y)$ and $d^{*\prime}(Y)$ are those of (ii) at every one of the 2^8 sets Y, and unions of full blocks keep the width; with the first rule at every mark and the same sizes, $r'(Y) = r(\lceil Y \rceil)$, $d'(Y) = r - r(\mathcal{E} \setminus \lfloor Y \rfloor)$ and the total weight is $K((q_h)_h)$.

Definition 18.4 (Hinge values; their ratio). As stated, with the freedom of the factorization.

Proof. By 18.6 the bases of the level $d(B)$ are the sets $X \sqcup Y$ with $X \in \mathcal{B}/R$ and $Y \in \mathcal{B}|R$, and those of the level $r(B)$ the sets $X \sqcup Y$ with $X \in \mathcal{B}|B$ and $Y \in \mathcal{B}/B$, every such set being a basis. Let $\theta_{X \sqcup Y} = \theta'_X \theta''_Y = \phi'_X \phi''_Y$ be two factorizations and fix a basis $X_d \sqcup Y_d$ of the level $d(B)$. For every pattern X of this level $X \sqcup Y_d$ is a basis, so $\phi'_X = \lambda_d \theta'_X$ with $\lambda_d = \theta''_{Y_d} / \phi''_{Y_d}$; for every room side Y of this level $X_d \sqcup Y$ is a basis, so $\phi''_Y = \theta'_{X_d} \theta''_Y / \phi'_{X_d} = \lambda_d^{-1} \theta''_Y$. Likewise at the level $r(B)$ with a constant λ_r. Conversely any $\lambda_d, \lambda_r > 0$ give a factorization, the patterns of the two levels having different sizes, and so have their room sides. The new factors multiply $K_{\mathcal{B}/R}$ and $\Psi_{\mathcal{B}/R}$ by λ_d, $K_{\mathcal{B}|B}$ and $\Psi_{\mathcal{B}|B}$ by λ_r, $K_{\mathcal{B}|R}$ and $\Psi_{\mathcal{B}|R}$ by λ_d^{-1}, and $K_{\mathcal{B}/B}$ and $\Psi_{\mathcal{B}/B}$ by λ_r^{-1}. With $c = \lambda_r / \lambda_d$,

$$\gamma_B^K = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}} \mapsto c\,\gamma_B^K, \quad \gamma_B^\Psi = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}} \mapsto c^{-1}\gamma_B^\Psi, \quad \gamma_R^K = \frac{K_{\mathcal{B}|R}}{K_{\mathcal{B}/B}} \mapsto c\,\gamma_R^K, \quad \gamma_R^\Psi = \frac{\Psi_{\mathcal{B}/B}}{\Psi_{\mathcal{B}|R}} \mapsto c^{-1}\gamma_R^\Psi,$$

so $\varrho_B = \gamma_B^\Psi / \gamma_B^K$ and $\varrho_R = \mathbf{E}_{\mathcal{B}/B} / \mathbf{E}_{\mathcal{B}|R} = \gamma_R^\Psi / \gamma_R^K$ are multiplied by c^{-2}, and γ_R^K / γ_B^K, $\gamma_R^\Psi / \gamma_B^\Psi$ and ϱ_B / ϱ_R are unchanged. □

Remark. The quotients γ_R^K/γ_B^K and $\gamma_R^\Psi/\gamma_B^\Psi$ are the Skeleton values of 18.7, which belong to the table and its gauge alone. What depends on the factorization is the split of a Skeleton value into a hinge value of the bunch and one of the room, and with it the cells, the cell families and the pooled families; the identities of the chapter hold at every factorization, each side changing with it.

Proposition 18.5 (Hinge values of parallel and series bunches). *As stated, for classes of any table whose weights factor across them.*

Proof. That a parallel or series bunch has width at most one, and width zero exactly when it is a union of molecules, is the reference, proof of prop:classreadings. Let B be a parallel bunch of width one: $r(B) = 1$ and $c(B) = r(B) - d(B) = 1$, so $d(B) = 0$; $\mathcal{B}/R$ has rank 0, with the single basis $\varnothing$ and the weight $\theta'_\varnothing$, and $\mathcal{B}|B$ has rank one, with the singletons of B as bases, none of its marks being a loop, and the weights $\theta'_{\{e\}} = \theta'_\varnothing w_e$. Then $K_{\mathcal{B}/R} = \theta'_\varnothing$, $K_{\mathcal{B}|B} = \sum_e \theta'_{\{e\}}\alpha_e = \theta'_\varnothing \sum_e w_e\alpha_e$, $\Psi_{\mathcal{B}/R} = \theta'_\varnothing \alpha^B = \theta'_\varnothing e_{|B|}(\alpha_B)$ and $\Psi_{\mathcal{B}|B} = \sum_e \theta'_{\{e\}}\alpha^{B-e} = \theta'_\varnothing e_{|B|}(\alpha_B)\sum_e w_e\alpha_e^{-1}$, and the hinge values are the quotients of 18.4. Let B be a series bunch of width one: no mark of B is a coloop, so $d(B) \geq |B|-1$, and width one gives $r(B) = |B|$ and $d(B) = |B|-1$; $\mathcal{B}|B$ has the single basis B, with the weight θ'_B, and $\mathcal{B}/R$ the sets $B-e$, every one of them occurring since a basis avoids e, with the weights $\theta'_{B-e} = \theta'_B w_e$. Then $K_{\mathcal{B}|B} = \theta'_B e_{|B|}(\alpha_B)$, $K_{\mathcal{B}/R} = \theta'_B \sum_e w_e\alpha^{B-e} = \theta'_B e_{|B|}(\alpha_B)\sum_e w_e\alpha_e^{-1}$, $\Psi_{\mathcal{B}|B} = \theta'_B$ and $\Psi_{\mathcal{B}/R} = \theta'_B \sum_e w_e\alpha_e$, which give γ_B^Ψ and γ_B^K. In a list with $\theta' = \theta'' = 1$ every w_e is 1, and for a pair with the shape $(x, 1-x)$: parallel, $\gamma_B^K = 1$ and $\gamma_B^\Psi = (x^{-1}+(1-x)^{-1})^{-1} = x(1-x)$; series, $\gamma_B^\Psi = 1$ and $\gamma_B^K = x(1-x)$. □

Remark. A change of the factorization (18.4) multiplies the factors w_e of a parallel bunch by c, the weights of its level $r(B) = 1$ over those of its level $d(B) = 0$, and those of a series bunch by c^{-1}, its patterns $B - e$ lying on the level $d(B)$; either way γ_B^K is multiplied by c and γ_B^Ψ by c^{-1}. In exact arithmetic (receipt `check_classes_realization.py`): a parallel bunch of three marks whose cell of the bunch is $\binom{\{1,2,3,g\}}{1}$ with the weights $2, \frac{3}{7}, 5$ on the singletons of the bunch and $\frac{4}{3}$ on $\{g\}$, and a series bunch of three marks whose cell is $\binom{\{1,2,3,g\}}{3}$ with the weights $\frac{3}{2}$ on $\{1,2,3\}$ and $\frac{1}{5}, 7, \frac{2}{3}$ on the sets through g, each glued to a weighted room cell, have the hinge values of the statement as rational functions, and $\varrho_B = (\sum_e w_e)^{-2}$ at the centre of the shape simplex of the parallel bunch.

Theorem 18.6 (A bunch of width one is one mark to its room). *As stated; the converse holds with any number of products.*

Proof. All but the converse is the reference, proof of thm:widthone. On the open orthant let $K(\alpha) = \sum_{l=0}^k A_l(\alpha_B)C_l(\alpha_R)$, C_l homogeneous of degree D_l. Grouping the bases by their level $i = |T \cap B|$, which takes every value $d(B) \leq i \leq r(B)$ (7.11), $K = \sum_i K^{(i)}$ with $K^{(i)}(\alpha_B, \lambda\alpha_R) = \lambda^{r-i}K^{(i)}(\alpha)$ and $K^{(i)} > 0$ on the open orthant. At a fixed α, with $\lambda\alpha_R$ for α_R,

$$\sum_{i=d(B)}^{r(B)} \lambda^{r-i}K^{(i)}(\alpha) = \sum_D \lambda^D \sum_{l:\ D_l=D} A_l(\alpha_B)\, C_l(\alpha_R) \qquad (\lambda > 0),$$

the sum on the right over the distinct values D among the D_l. The functions $\lambda \mapsto \lambda^D$ with distinct real D are linearly independent on $(0, \infty)$, so each of the $c(B) + 1$ exponents $r - i$ on the left, whose coefficient is positive, is one of the D_l: $c(B) + 1 \leq k + 1$. With $k = 1$ this is the converse of the reference. □

Two lemmas are used below.

Lemma S6. *Let a bunch B of a table $(\mathcal{B},\theta)$ of rank r on $\mathcal{E}$, $n=|\mathcal{E}|$, have width one, the weights factoring across B and R. With $\alpha_B=\rho u$,*

$$\rho^{-d(B)}K(\rho u,\alpha_R)=K_{\mathcal{B}/R}(u)\,K_{\mathcal{B}|R}(\alpha_R)+\rho\,K_{\mathcal{B}|B}(u)\,K_{\mathcal{B}/B}(\alpha_R),$$
$$\rho^{-d^*(B)}\Psi(\rho u,\alpha_R)=\Psi_{\mathcal{B}|B}(u)\,\Psi_{\mathcal{B}/B}(\alpha_R)+\rho\,\Psi_{\mathcal{B}/R}(u)\,\Psi_{\mathcal{B}|R}(\alpha_R);$$

the cells have the ranks $r(B)=d(B)+1$ and $r-d(B)$ and the letters

$$K_{\mathcal{B}_B^+}=K_{\mathcal{B}/R}\,(\alpha_g+\gamma_B^K),\quad \Psi_{\mathcal{B}_B^+}=\Psi_{\mathcal{B}|B}\,(\alpha_g+\gamma_B^\Psi),$$
$$K_{\mathcal{B}_R^+}=K_{\mathcal{B}|R}+\alpha_g K_{\mathcal{B}/B},\quad \Psi_{\mathcal{B}_R^+}=\alpha_g\Psi_{\mathcal{B}|R}+\Psi_{\mathcal{B}/B};$$

and $d(B)+d(R)=r-1$, $d^*(B)+d^*(R)=n-r-1$. For a family with masses $\mathbf{m}$ in which B has width one in every table, $\kappa_B+\kappa_R=\kappa_{\mathcal{E}}+2\sigma$, σ the total strength.

Proof. By 18.6 the bases of the level $d(B)$ are the $X\sqcup Y$ with $X\in\mathcal{B}/R$, $Y\in\mathcal{B}|R$, and those of the level $r(B)=d(B)+1$ the $X\sqcup Y$ with $X\in\mathcal{B}|B$, $Y\in\mathcal{B}/B$, with $\theta_{X\sqcup Y}=\theta'_X\theta''_Y$ and the complement $(B\setminus X)\sqcup(R\setminus Y)$. Summing over the two levels, $K=K_{\mathcal{B}/R}K_{\mathcal{B}|R}+K_{\mathcal{B}|B}K_{\mathcal{B}/B}$ and $\Psi=\Psi_{\mathcal{B}/R}\Psi_{\mathcal{B}|R}+\Psi_{\mathcal{B}|B}\Psi_{\mathcal{B}/B}$, and on B the polynomials $K_{\mathcal{B}/R}$, $K_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}/R}$ are homogeneous of the degrees $d(B)$, $d(B)+1$, $|B|-r(B)=d^*(B)$ and $|B|-d(B)=d^*(B)+1$. The deletions and the contractions of the hinge are $\mathcal{B}_B^+\setminus g=\mathcal{B}|B$, $\mathcal{B}_B^+/g=\mathcal{B}/R$, $\mathcal{B}_R^+\setminus g=\mathcal{B}|R$ and $\mathcal{B}_R^+/g=\mathcal{B}/B$ (18.6), which gives the ranks and, by 7.3 and 18.4, the letters. By 7.10, $d(S)=r-r(\mathcal{E}\setminus S)$, so $d(B)+d(R)=2r-r(B)-r(R)=r-c(B)=r-1$, and $d^*(B)+d^*(R)=n-r(B)-r(R)=n-r-1$. Table by table, $\kappa_B+\kappa_R=m_{\mathcal{E}}-\sum_j\big(s_j(r_j-1)+t_j(n-r_j-1)\big)=\kappa_{\mathcal{E}}+\sum_j(s_j+t_j)$. □

Lemma S7. *Let a family F on $\mathcal{E}=B\sqcup R$, B and R nonempty, have real data and be on its balance wall, and let $f(\alpha)=\alpha^{\mathbf{m}-\mathbf{1}}\prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}$. Then*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_R}\int_{\mathbb{R}^B_{>0}}f(x,v)\,dx\,d\omega_R(v)=\int_{\Delta_R}\int_{\Delta_B}\int_0^\infty\rho^{|B|-1}f(\rho u,v)\,d\rho\,d\omega_B(u)\,d\omega_R(v).$$

Proof. On the balance wall f is homogeneous of degree $\kappa_{\mathcal{E}}-n=-n$, $n=|\mathcal{E}|$. Let φ be the indicator of $[1,e]$, so that $\int_0^\infty\varphi(\lambda)\,d\lambda/\lambda=1$, and compute $I=\int_{\mathbb{R}^{\mathcal{E}}_{>0}}f(\alpha)\varphi(|\alpha_R|)\,d\alpha$ in two ways. With $\alpha=\lambda w$, $w\in\Delta_{\mathcal{E}}$, $d\alpha=\lambda^{n-1}d\lambda\,d\omega_{\mathcal{E}}(w)$ (2.7) and $f(\lambda w)=\lambda^{-n}f(w)$, so $I=\int_{\Delta_{\mathcal{E}}}f(w)\int_0^\infty\varphi(\lambda|w_R|)\frac{d\lambda}{\lambda}\,d\omega_{\mathcal{E}}(w)=\int_{\Delta_{\mathcal{E}}}f\,d\omega_{\mathcal{E}}=P_F$ (8.10). With $\alpha_R=\lambda v$, $v\in\Delta_R$, and $\alpha_B=\lambda x$, $x\in\mathbb{R}^B_{>0}$, $d\alpha=\lambda^{|R|-1}d\lambda\,d\omega_R(v)\,\lambda^{|B|}dx$ and $f(\lambda x,\lambda v)=\lambda^{-n}f(x,v)$, so $I=\int_0^\infty\varphi(\lambda)\frac{d\lambda}{\lambda}\int_{\Delta_R}\int_{\mathbb{R}^B_{>0}}f(x,v)\,dx\,d\omega_R(v)$, the first form. The second is 2.7 for $x=\rho u$. For $|B|=1$, Δ_B is a point with the Dirac measure (1.1(iii)). □

Theorem 18.9 (Walls glue). *As stated: (i) and (ii) for every family; in (iii) the identities for every family and the gluing when every active letter reads B alike.*

Proof. The gluing of the counts of one table, the counts of its cells,

$$d_{\mathcal{B}_B^+}(S_B)=d_{\mathcal{B}/R}(S_B),\quad d_{\mathcal{B}_B^+}(S_B+g)=d_{\mathcal{B}|B}(S_B),$$
$$d^*_{\mathcal{B}_B^+}(S_B)=d^*_{\mathcal{B}|B}(S_B),\quad d^*_{\mathcal{B}_B^+}(S_B+g)=d^*_{\mathcal{B}/R}(S_B),$$
$$d_{\mathcal{B}_R^+}(S_R)=d_{\mathcal{B}/B}(S_R),\quad d_{\mathcal{B}_R^+}(S_R+g)=d_{\mathcal{B}|R}(S_R),$$
$$d^*_{\mathcal{B}_R^+}(S_R)=d^*_{\mathcal{B}|R}(S_R),\quad d^*_{\mathcal{B}_R^+}(S_R+g)=d^*_{\mathcal{B}/B}(S_R),$$

and the inequalities $d_{\mathcal{B}/R} \le d_{\mathcal{B}|B} \le d_{\mathcal{B}/R}+1$ and $d^*_{\mathcal{B}|B} \le d^*_{\mathcal{B}/R} \le d^*_{\mathcal{B}|B}+1$ on the subsets of B, $d_{\mathcal{B}/B} \le d_{\mathcal{B}|R} \le d_{\mathcal{B}/B}+1$ and $d^*_{\mathcal{B}|R} \le d^*_{\mathcal{B}/B} \le d^*_{\mathcal{B}|R}+1$ on those of R, are the reference, proof of thm:wallsglue. For table j and $S = S_B \sqcup S_R$ call

$$\begin{aligned} a_j^K &= d_{\mathcal{B}_j|B}(S_B) - d_{\mathcal{B}_j/R}(S_B), \quad b_j^K = d_{\mathcal{B}_j|R}(S_R) - d_{\mathcal{B}_j/B}(S_R),\\ a_j^\Psi &= d^*_{\mathcal{B}_j/R}(S_B) - d^*_{\mathcal{B}_j|B}(S_B), \quad b_j^\Psi = d^*_{\mathcal{B}_j/B}(S_R) - d^*_{\mathcal{B}_j|R}(S_R), \end{aligned}$$

all in $\{0,1\}$, in this proof the *jumps* of the letters K_j and Ψ_j on the side of the bunch and on the side of the room. The gluing of the counts reads

$$d_j(S) = d_{\mathcal{B}_j/R}(S_B) + d_{\mathcal{B}_j/B}(S_R) + \min(a_j^K, b_j^K), \qquad d^*_j(S) = d^*_{\mathcal{B}_j|B}(S_B) + d^*_{\mathcal{B}_j|R}(S_R) + \min(a_j^\Psi, b_j^\Psi).$$

At $S_B = \varnothing$ the jumps on the side of the bunch are 0, and at $S_R = \varnothing$ those on the side of the room. At $S_B = B$ those on the side of the bunch are 1: the ranks give $d_{\mathcal{B}_j|B}(B) = d_j(B)+1$, $d_{\mathcal{B}_j/R}(B) = d_j(B)$, $d^*_{\mathcal{B}_j/R}(B) = |B| - d_j(B) = d^*_j(B)+1$ and $d^*_{\mathcal{B}_j|B}(B) = d^*_j(B)$. At $S_R = R$ those on the side of the room are 1: $d_{\mathcal{B}_j|R}(R) = r_j - d_j(B) = d_{\mathcal{B}_j/B}(R)+1$ and $d^*_{\mathcal{B}_j/B}(R) = |R| - r_j + d_j(B) + 1 = d^*_{\mathcal{B}_j|R}(R)+1$.

(i) The tables of $F_{B\to g}(u)$ of index j have the bases of $\mathcal{B}^+_{j,R}$, at $(s_j,0)$ and at $(0,t_j)$, so its wall at $S \subseteq R+g$ is $m'_S - \sum_j \big(s_j d_{\mathcal{B}^+_{j,R}}(S) + t_j d^*_{\mathcal{B}^+_{j,R}}(S)\big)$, $\mathbf{m}' = (\kappa_B, \mathbf{m}|_R)$. At $S = S_R$ the counts of the cell are $d_{\mathcal{B}_j/B}(S_R)$ and $d^*_{\mathcal{B}_j|R}(S_R)$, which are $d_j(S_R)$ and $d^*_j(S_R)$ by the gluing at $S_B = \varnothing$: the wall is κ_{S_R}. At $S = S_R + g$ they are $d_{\mathcal{B}_j|R}(S_R)$ and $d^*_{\mathcal{B}_j/B}(S_R)$, and the gluing at $S_B = B$ gives $d_j(B \sqcup S_R) = d_j(B) + d_{\mathcal{B}_j|R}(S_R)$ and $d^*_j(B \sqcup S_R) = d^*_j(B) + d^*_{\mathcal{B}_j/B}(S_R)$; with $\kappa_B = m_B - \sum_j (s_j d_j(B) + t_j d^*_j(B))$ the wall is $\kappa_{B\sqcup S_R}$.

(ii) With α_{S_B} replaced by $\varepsilon\alpha_{S_B}$, $K_{\mathcal{B}_j/R}$ is $\varepsilon^{d_{\mathcal{B}_j/R}(S_B)}$ times a polynomial positive at $\varepsilon = 0$ on the open orthant, its monomials having at least $d_{\mathcal{B}_j/R}(S_B)$ marks in S_B and some exactly that many (7.11); likewise $\Psi_{\mathcal{B}_j|B}$ with $d^*_{\mathcal{B}_j|B}(S_B)$. By the gluing at $S_R = \varnothing$ these are $d_j(S_B)$ and $d^*_j(S_B)$, so the exponent of the density at the face is κ_{S_B}.

(iii) The walls of G_B and G_R are read with the counts of the cells and the masses $(\mathbf{m}|_B, 2\sigma - \kappa_B)$ and $(\kappa_B, \mathbf{m}|_R)$. With

$$X = m_{S_B} + m_{S_R} - \sum_j \Big(s_j \big(d_{\mathcal{B}_j/R}(S_B) + d_{\mathcal{B}_j/B}(S_R)\big) + t_j \big(d^*_{\mathcal{B}_j|B}(S_B) + d^*_{\mathcal{B}_j|R}(S_R)\big)\Big),$$

the counts of the cells give $\kappa^{G_B}_{S_B} + \kappa^{G_R}_{S_R} = X$ and, the two hinge masses adding up to $2\sigma = \sum_j (s_j + t_j)$, $\kappa^{G_B}_{S_B+g} + \kappa^{G_R}_{S_R+g} = X + \sum_j \big(s_j(1 - a_j^K - b_j^K) + t_j(1 - a_j^\Psi - b_j^\Psi)\big)$, while $\kappa_{S_B\sqcup S_R} = X - \sum_j \big(s_j \min(a_j^K, b_j^K) + t_j \min(a_j^\Psi, b_j^\Psi)\big)$. Since $1 - a - b = (1 - \max(a,b)) - \min(a,b)$ for $a, b \in \{0,1\}$,

$$\begin{aligned} \kappa^{G_B}_{S_B} + \kappa^{G_R}_{S_R} - \kappa_{S_B\sqcup S_R} &= \sum_j \big(s_j \min(a_j^K, b_j^K) + t_j \min(a_j^\Psi, b_j^\Psi)\big),\\ \kappa^{G_B}_{S_B+g} + \kappa^{G_R}_{S_R+g} - \kappa_{S_B\sqcup S_R} &= \sum_j \big(s_j(1 - \max(a_j^K, b_j^K)) + t_j(1 - \max(a_j^\Psi, b_j^\Psi))\big), \end{aligned} \tag{40.1}$$

at all masses and exponents. At $S_R = \varnothing$ the first line gives $\kappa^{G_B}_{S_B} = \kappa_{S_B}$, the wall at the empty set being 0; at $S_B = \varnothing$ it gives $\kappa^{G_R}_{S_R} = \kappa_{S_R}$. The imbalance of G_B is

$$\kappa^{G_B}_{B+g} = m_B + 2\sigma - \kappa_B - \sum_j \big(s_j(d_j(B)+1) + t_j(d^*_j(B)+1)\big) = 2\sigma - \sum_j (s_j + t_j) = 0,$$

the cell $\mathcal{B}^+_{j,B}$ having rank $d_j(B)+1$ on $|B|+1 = d_j(B)+d^*_j(B)+2$ marks: G_B is on its balance wall at all data. At $S_B = B$ the second line then gives $\kappa^{G_R}_{S_R+g} = \kappa_{B\sqcup S_R}$; at $S_R = R$ in particular $\kappa^{G_R}_{R+g} = \kappa_{\mathcal{E}}$, so G_R is on its balance wall exactly when F is, and then the second line at $S_R = R$ gives $\kappa^{G_B}_{S_B+g} = \kappa_{S_B\sqcup R}$. These are the four identities.

Let now $s_j, t_j \geq 0$ and let every active letter read B alike. The jump of a letter on the side of the bunch at S_B is the order to which its hinge value vanishes at the face $u_{S_B} = 0$: by the expansion in (ii) applied to $\mathcal{B}_j|B$ and $\mathcal{B}_j/R$, with $\varepsilon\alpha_{S_B}$ for α_{S_B}, $\gamma^K_{j,B} = K_{\mathcal{B}_j|B}/K_{\mathcal{B}_j/R}$ is $\varepsilon^{a^K_j}$ times a function positive at $\varepsilon = 0$, and $\gamma^\Psi_{j,B} = \Psi_{\mathcal{B}_j/R}/\Psi_{\mathcal{B}_j|B}$ is $\varepsilon^{a^\Psi_j}$ times one. Constant multiples of one function vanish to one order at each face, so at S_B the active letters have one jump $a \in \{0,1\}$ on the side of the bunch. If $a = 0$ every term of the first line of (40.1) vanishes, a letter of exponent zero contributing nothing; if $a = 1$ every term of the second does. Both lines are nonnegative, the exponents being nonnegative, so $\kappa_{S_B\sqcup S_R}$ is the smaller of the two sums. One active letter reads B alike by itself.

Positivity. If both cell families have every proper wall positive, let $S = S_B \sqcup S_R$ be nonempty and proper. In the first sum a nonempty part is a proper subset of the marks of its cell family, $S_B \subseteq B \subsetneq B+g$ and $S_R \subseteq R \subsetneq R+g$, and an empty part has the wall 0: the first sum is positive. In the second both S_B+g and S_R+g are nonempty, and at most one of them is all the marks of its cell family, S being proper, its wall then being the imbalance 0: the second sum is positive. So $\kappa_S > 0$. Conversely, by the identities, $\kappa^{G_B}_{S_B} = \kappa_{S_B}$ for $\varnothing \neq S_B \subseteq B$ and $\kappa^{G_B}_{S_B+g} = \kappa_{S_B\sqcup R}$ for $S_B \subsetneq B$ are proper walls of F, and likewise for G_R; this direction does not use the hypothesis. □

Remark. Without the reading alike, (iii) keeps the identities and loses the gluing. On $\mathcal{E} = \{1,2,3,4\}$ with $B = \{1,2\}$, the family of the lists $\binom{\mathcal{E}}{3}$ and $\binom{\mathcal{E}}{1}$, each at $(0,1)$, has B of width one in both tables, with the cells $\binom{\{1,2,g\}}{2}$, $\binom{\{3,4,g\}}{2}$ and $\binom{\{1,2,g\}}{1}$, $\binom{\{3,4,g\}}{1}$, and the hinge values $\gamma^\Psi_{1,B} = \alpha_1+\alpha_2$ and $\gamma^\Psi_{2,B} = \alpha_1\alpha_2/(\alpha_1+\alpha_2)$, which are not constant multiples of one function. At the masses $(\frac{2}{5}, \frac{8}{5}, \frac{2}{5}, \frac{8}{5})$, on the balance wall, $\kappa_B = 1$ and both cell families have every proper wall positive, the least being $\frac{2}{5}$, while $\kappa_{\{1,3\}} = \frac{2}{5}+\frac{2}{5}-1 = -\frac{1}{5}$, against $\frac{4}{5}$ from the gluing; the walls at $\{1,4\}$, $\{2,3\}$ and $\{2,4\}$ do not glue either. In exact rational arithmetic (receipt `check_walls.py`): this family; the four identities at every subset at two partial leans of the same tables, where four walls do not glue; and the gluing at every subset for two tables sharing one weighted cell of the bunch at cobasis full leans, three pairs of exponents and three sets of masses, for a table at $(s,0)$ with the cell $\binom{\{1,2,g\}}{1}$ together with a table at $(0,t)$ with the cell $\binom{\{1,2,g\}}{2}$, whose hinge values $\gamma^K_{1,B}$ and $\gamma^\Psi_{2,B}$ are both $\alpha_1+\alpha_2$, and for $\binom{\mathcal{E}}{2}$ with B three of its four marks at two partial leans, where $\varrho_B = 1$.

Theorem 18.10 (Pooling factors at one hinge value). *As stated: at real exponents of any sign, every active letter having the one hinge value γ.*

Proof. The walls $\kappa_{\{e\}}$ at the singletons are positive and the exponents real, so the masses are real, and the integrals below are of positive functions.

Bounds. κ_B and κ_R are proper walls of F, hence positive, and $\kappa_B + \kappa_R = 2\sigma$ on the balance wall (Lemma S6): $0 < \kappa_B < 2\sigma$.

The cell families. By the identities of 18.9(iii), which hold for every family on the balance wall, G_B and G_R are on their balance walls and each of their proper walls is a proper wall of F, hence positive; their periods converge (8.10).

The family. Put $\phi(u) = u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j} \Psi_{\mathcal{B}_j|B}(u)^{-t_j}$ on Δ_B. By 18.6, with $K_{\mathcal{B}_j/R}(\rho u) = \rho^{d_j(B)} K_{\mathcal{B}_j/R}(u)$, $\Psi_{\mathcal{B}_j|B}(\rho u) = \rho^{d^*_j(B)} \Psi_{\mathcal{B}_j|B}(u)$ and the hinge values homogeneous of

degree one, Lemma S7 gives

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_B} \phi(u) \int_{\Delta_R} \int_0^\infty \rho^{\kappa_B-1} v^{\mathbf{m}|_R-\mathbf{1}} \\ \times \prod_j K_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^K_{j,B}(u),v\big)^{-s_j} \Psi_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^\Psi_{j,B}(u),v\big)^{-t_j} d\rho\, d\omega_R(v)\, d\omega_B(u),$$

the powers of ρ collecting to $\rho^{m_B-1-\sum_j(s_j d_j(B)+t_j d_j^*(B))} = \rho^{\kappa_B-1}$. Every active letter has the hinge value $\gamma(u)$, and a letter of exponent zero is the factor 1. The substitution $x = \rho\gamma(u)$ at fixed u, $\rho^{\kappa_B-1}d\rho = \gamma(u)^{-\kappa_B}x^{\kappa_B-1}dx$, and Lemma S7 for G_R on $\{g\}\sqcup R$ at the masses $(\kappa_B, \mathbf{m}|_R)$ give

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{G_R}\big((\kappa_B,\mathbf{m}|_R);\mathbf{s},\mathbf{t}\big) \int_{\Delta_B} \phi(u)\,\gamma(u)^{-\kappa_B}\, d\omega_B(u).$$

The cell family of the bunch. By Lemma S6, with y the variable of g, $K_{\mathcal{B}^+_{j,B}}(u,y) = K_{\mathcal{B}_j/R}(u)\big(y+\gamma^K_{j,B}(u)\big)$ and $\Psi_{\mathcal{B}^+_{j,B}}(u,y) = \Psi_{\mathcal{B}_j|B}(u)\big(y+\gamma^\Psi_{j,B}(u)\big)$; every active letter has the factor $y+\gamma(u)$, and the exponents of the active letters add up to $\sum_j(s_j+t_j) = 2\sigma$. Lemma S7 for G_B on $\{g\}\sqcup B$ at the masses $(\mathbf{m}|_B, 2\sigma-\kappa_B)$, and the integral of a linear letter (the paragraph after 15.1, with $c_0=\gamma(u)$, $c_1=1$ and $0<2\sigma-\kappa_B<2\sigma$), give

$$P_{G_B}\big((\mathbf{m}|_B, 2\sigma-\kappa_B);\mathbf{s},\mathbf{t}\big) = \int_{\Delta_B}\phi(u)\int_0^\infty y^{2\sigma-\kappa_B-1}\big(y+\gamma(u)\big)^{-2\sigma} dy\, d\omega_B(u) \\ = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa_B, & 2\sigma-\kappa_B\\ & 2\sigma\end{bmatrix} \int_{\Delta_B}\phi(u)\,\gamma(u)^{-\kappa_B}\, d\omega_B(u).$$

Multiplying the display for P_F by $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa_B, & 2\sigma-\kappa_B\\ & 2\sigma\end{bmatrix}$ gives the statement, every factor being finite and positive. The parenthesis of the statement is 18.4: the constants λ_d, λ_r of table j multiply $\gamma^K_{j,B}$ by c and $\gamma^\Psi_{j,B}$ by c^{-1}, so a table with one active letter is normalized by its own constant; at full leans of all tables only the hinge values of different tables are compared; one table with both letters active and ϱ_B constant is normalized by $c = \varrho_B^{1/2}$, and $\gamma^K_B = \gamma^\Psi_B$ forces $\varrho_B = 1$. One table at $(0,t)$ has $s=0$ and $2\sigma = t$, and at $(s,0)$, $2\sigma = s$. □

Remark. One table at $(0,t)$ has the one active letter Ψ, whose hinge value γ^Ψ_B serves as γ at any factorization; the cell families are the cells $(\mathcal{B}^+_B,\theta')$ and $(\mathcal{B}^+_R,\theta'')$ at $(0,t)$, and the display is that of the reference, $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa_B, & t-\kappa_B\\ & t\end{bmatrix} P_F(\mathbf{m};0,t) = P_{\mathcal{B}^+_B}\big((\mathbf{m}|_B, t-\kappa_B);0,t\big)P_{\mathcal{B}^+_R}\big((\kappa_B,\mathbf{m}|_R);0,t\big)$; at $(s,0)$ the same holds with $\gamma = \gamma^K_B$ and s for t. The proof above is the proof of the reference with γ^Ψ_B replaced by the common hinge value of the active letters. A family at full leans, every table at $(s_j,0)$ or $(0,t_j)$, has one active letter in each table, normalized by the factorization of its table; the hypothesis then asks only that the hinge values of the active letters of different tables be constant multiples of one another, a condition across the tables, and 21.6 is the same condition at every cell of a core. One table with $s,t>0$ has both letters active; the hypothesis is ϱ_B constant, reached at the factorization with $\varrho_B = 1$, and the display then holds at every pair (s,t) with $2\sigma = s+t$, the separation of 18.12 completed by the Beta corner of the hinge.

Proposition 18.11 (Pooling a realization). *As stated, for every family and every realization.*

Proof. The masses of Υ are $\tau_e = \sum_j \big(s_j \alpha_e \partial_e \log K_j(\alpha) + t_j \alpha_e \partial_e \log \Psi_j(\alpha)\big)$ (9.1). With $u = \alpha_B/|\alpha_B|$ fixed, the tables of index j of $F_{B\to g}(u)$, read as one table with two systems of weights, have the letters $K^u_j(\alpha_g, \alpha_R) = K_{\mathcal{B}^+_{j,R}}(\gamma^K_{j,B}(u)\alpha_g, \alpha_R)$ and $\Psi^u_j(\alpha_g, \alpha_R) = \Psi_{\mathcal{B}^+_{j,R}}(\gamma^\Psi_{j,B}(u)\alpha_g, \alpha_R)$ (18.8). By 18.6 and the homogeneity of the hinge values, $\gamma^K_{j,B}(\alpha_B) = \rho\gamma^K_{j,B}(u)$ with $\rho = |\alpha_B|$, and

$$K_j(\alpha) = K_{\mathcal{B}_j/R}(\alpha_B)\, K^u_j(\rho, \alpha_R), \qquad \Psi_j(\alpha) = \Psi_{\mathcal{B}_j|B}(\alpha_B)\, \Psi^u_j(\rho, \alpha_R).$$

For $e \in R$ the first factors do not depend on α_e, so τ_e is the usage of e in $F_{B\to g}(u)$ at the gauge (ρ, α_R) with the exponents (s_j, t_j). Summing over $e \in B$, Euler's identity gives $\sum_{e\in B} \alpha_e \partial_e \log K_{\mathcal{B}_j/R}(\alpha_B) = d_j(B)$ and $\sum_{e\in B} \alpha_e \partial_e \log \Psi_{\mathcal{B}_j|B}(\alpha_B) = d^*_j(B)$, and, $\gamma^K_{j,B}$ being homogeneous of degree one,

$$\sum_{e\in B} \alpha_e \partial_e \log K_{\mathcal{B}^+_{j,R}}\big(\gamma^K_{j,B}(\alpha_B), \alpha_R\big) = \big[y\partial_y \log K_{\mathcal{B}^+_{j,R}}(y, \alpha_R)\big]_{y=\gamma^K_{j,B}(\alpha_B)}$$
$$= \big[\alpha_g \partial_{\alpha_g} \log K^u_j(\alpha_g, \alpha_R)\big]_{\alpha_g=\rho},$$

and likewise for Ψ. So $\tau_B = \sum_j \big(s_j d_j(B) + t_j d^*_j(B)\big) + \tau^u_g$, with τ^u_g the usage of g in $F_{B\to g}(u)$ at (ρ, α_R), and τ^u_g is the wall κ_B at the masses $\boldsymbol{\tau}$. The exponents are those of a realization and (ρ, α_R) is a gauge, so $((s_j, t_j)_j; (\rho, \alpha_R))$ is a realization of $F_{B\to g}(u)$ with these masses, its usage. For one table at $(0, t)$, $\Psi^u(\rho, \alpha_R) = \Psi_{\mathcal{B}^+_R}(\gamma^\Psi_B(\alpha_B), \alpha_R)$, so the usage of $(\mathcal{B}^+_R, \theta'')$ at $(0, t)$ and at the gauge $(\gamma^\Psi_B(\alpha_B), \alpha_R)$ is $\boldsymbol{\tau}|_R$ on R and, by the chain rule above, $\tau_B - t\, d^*(B)$ at g. □

Remark. The reference is one list at $(0, t)$, its last sentence. In exact rational arithmetic (receipt `check_classes_realization.py`): two tables on five marks, each the two-sum of a weighted cell of the bunch on $\{1, 2, g\}$ and a weighted cell of the room on $\{3, 4, 5, g\}$, at the partial leans $(\frac{2}{3}, \frac{5}{4})$ and $(3, \frac{1}{2})$ and a rational gauge; the usage of $F_{B\to g}(u)$ at $(|\alpha_B|, \alpha_R)$ is $(\kappa_B(\boldsymbol{\tau}), \boldsymbol{\tau}|_R)$.

Proposition 18.12 (Hinge values that agree). *As stated, for families.*

Proof. As in the proof of 18.10, with $\phi(u) = u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j} \Psi_{\mathcal{B}_j|B}(u)^{-t_j}$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_B} \phi(u) \int_{\Delta_R} \int_0^\infty \rho^{\kappa_B - 1} v^{\mathbf{m}|_R - \mathbf{1}}$$
$$\times \prod_j K_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^K_{j,B}(u), v\big)^{-s_j} \Psi_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^\Psi_{j,B}(u), v\big)^{-t_j} d\rho\, d\omega_R(v)\, d\omega_B(u).$$

For $s_j \neq 0$, $K_{\mathcal{B}^+_{j,R}}(\rho c_j \gamma(u), v) = K_{\mathcal{B}|R}(v) + \rho c_j \gamma(u) K_{\mathcal{B}/B}(v)$ is the letter K of $\mathcal{B}^+_{j,R}$ with its weights multiplied by c_j on its bases through g, read at $(\rho\gamma(u), v)$; for $t_j \neq 0$, $\Psi_{\mathcal{B}^+_{j,R}}(\rho c^*_j \gamma(u), v)$ is the letter Ψ of $\mathcal{B}^+_{j,R}$ with its weights multiplied by c^*_j on its bases avoiding g, read at $(\rho\gamma(u), v)$ (Lemma S6). These are the letters of $G^{\mathbf{c}}_R$, and the letters of exponent zero are the factor 1. The substitution $x = \rho\gamma(u)$ and Lemma S7 for $G^{\mathbf{c}}_R$, which has the bases and so the walls of G_R and is on its balance wall by 18.9(iii), give the display. Its period converges, each proper wall of G_R being a proper wall of F, so the shape integral is finite.

The substitution of the cell $\mathcal{B}^+_{j,B}$ into the core $\mathcal{B}^+_{j,R}$ at g, the other marks left in place, is table j (20.3(iii)). Its cell kernel (21.2) has $K_{\mathcal{A}/g} = K_{\mathcal{B}_j/R}$ and $\Psi_{\mathcal{A}\setminus g} = \Psi_{\mathcal{B}_j|B}$, its hinge values are $\gamma^K_{j,B}$ and $\gamma^\Psi_{j,B}$ (20.1), and the core family twisted by the constants at g is $G^{\mathbf{c}}_R$; a mark left in place is read alike by every letter, with the function α_h on the point $\Delta_{\{h\}}$, its wall m_h and its pinned shape integral 1. So 21.5 for this substitution is the display. For one table with ϱ_B constant, 18.4 reaches

$\varrho_B = 1$ by $c = \varrho_B^{1/2}$; then $\gamma_B^K = \gamma_B^\Psi$, $c = c^* = 1$ with $\gamma = \gamma_B^K$, and $G_R^{\mathfrak{c}}$ is the cell $(\mathcal{B}_R^+, \theta'')$ at (s,t), which is the reference. □

Theorem 18.13 (Gluing across a bunch of width one). *As stated: at nonnegative exponents, a letter of exponent zero being opened by nothing.*

Proof. The letters on the scale. For table j and $(u,v) \in \Delta_B \times \Delta_R$ put

$$A_j^K = K_{\mathcal{B}_j/R}(u) K_{\mathcal{B}_j|R}(v), \quad C_j^K = K_{\mathcal{B}_j|B}(u) K_{\mathcal{B}_j/B}(v),$$
$$A_j^\Psi = \Psi_{\mathcal{B}_j|B}(u) \Psi_{\mathcal{B}_j/B}(v), \quad C_j^\Psi = \Psi_{\mathcal{B}_j/R}(u) \Psi_{\mathcal{B}_j|R}(v),$$

all positive, so that $\rho^{-d_j(B)} K_j(\rho u, v) = A_j^K + \rho C_j^K$ and $\rho^{-d_j^*(B)} \Psi_j(\rho u, v) = A_j^\Psi + \rho C_j^\Psi$ (Lemma S6). The minors of the level d sit in A_j^K and in C_j^Ψ, those of the level r in C_j^K and in A_j^Ψ.

The scale. By Lemma S7, the powers of ρ collecting to $\rho^{\kappa_B - 1}$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_R} \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} v^{\mathbf{m}|_R - \mathbf{1}} \, I(u,v) \, d\omega_B(u) \, d\omega_R(v),$$
$$I(u,v) = \int_0^\infty \rho^{\kappa_B - 1} \prod_{\ell=1}^{p} (A_\ell + \rho C_\ell)^{-w_\ell} d\rho,$$

where ℓ runs over the p active letters, with their exponents $w_\ell > 0$ and their coefficients (A_ℓ, C_ℓ), a letter of exponent zero being the factor 1. Here $\sum_\ell w_\ell = \sum_j (s_j + t_j) = 2\sigma$, and κ_B and $\kappa_R = 2\sigma - \kappa_B$ (Lemma S6, $\kappa_{\mathcal{E}} = 0$) are proper walls: $0 < \kappa_B < 2\sigma$.

Opening the scale. If $p = 1$, the integral of a linear letter (the paragraph after 15.1) gives $I = \Gamma_{\!1}^{\,2}\!\left[\begin{matrix} w_1 - \kappa_B, & \kappa_B \\ & w_1 \end{matrix}\right] A_1^{\kappa_B - w_1} C_1^{-\kappa_B}$. If $p \geq 2$, choose $c_\ell \in (0, w_\ell)$ for $\ell < p$ with $0 < \kappa_B - \sum_{\ell<p} c_\ell < w_p$, possible as $0 < \kappa_B < \sum_\ell w_\ell$, and open each letter $\ell < p$ by 4.4 with $k = 2$ on the line $\operatorname{Re} x_\ell = c_\ell$:

$$(A_\ell + \rho C_\ell)^{-w_\ell} = \frac{1}{2\pi i} \int_{c_\ell + i\mathbb{R}} \Gamma_{\!1}^{\,2}\!\left[\begin{matrix} w_\ell - x_\ell, & x_\ell \\ & w_\ell \end{matrix}\right] A_\ell^{x_\ell - w_\ell} (\rho C_\ell)^{-x_\ell} \, dx_\ell.$$

The modulus of the integrand is its value at the real parts times the modulus of the stacked Gamma, which decays exponentially along the line (Stirling), and $\int_0^\infty \rho^{\kappa_B - \sum_{\ell<p} c_\ell - 1} (A_p + \rho C_p)^{-w_p} d\rho$ is finite; so Fubini puts the integral over ρ inside, where the integral of a linear letter at the complex index $x_p = \kappa_B - \sum_{\ell<p} x_\ell$, $0 < \operatorname{Re} x_p < w_p$, gives $\Gamma_{\!1}^{\,2}\!\left[\begin{matrix} w_p - x_p, & x_p \\ & w_p \end{matrix}\right] A_p^{x_p - w_p} C_p^{-x_p}$. Hence

$$I(u,v) = \frac{1}{(2\pi i)^{p-1}} \int_{c_1 + i\mathbb{R}} \cdots \int_{c_{p-1} + i\mathbb{R}} \prod_{\ell=1}^{p} \Gamma_{\!1}^{\,2}\!\left[\begin{matrix} w_\ell - x_\ell, & x_\ell \\ & w_\ell \end{matrix}\right] A_\ell^{x_\ell - w_\ell} C_\ell^{-x_\ell} \, dx_1 \cdots dx_{p-1},$$
$$x_p = \kappa_B - \sum_{\ell<p} x_\ell,$$

the share integral $\int_{x_1 + \cdots + x_p = \kappa_B}$ of 4.1: the factor $(2\pi i)^{-1}$ of each of the $p-1$ openings is its normalization $(2\pi i)^{1-p}$, and its dimension is $p-1$. At $p = 1$ the share integral has dimension zero, one share and the pin $P = (1)$, and 4.1 gives the value of the integrand at the one admissible point $x_1 = \kappa_B$: the formula for $p = 1$ above.

The level shares. For the letter K_j write $x_\ell = a_j^r$ and $w_\ell - x_\ell = a_j^d$, for Ψ_j write $x_\ell = b_j^d$ and $w_\ell - x_\ell = b_j^r$, and give the letters of exponent zero the shares 0. The pin $\sum_\ell x_\ell = \kappa_B$ is $\sum_j (a_j^r + b_j^d) =$

κ_B; the factor of K_j is $\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} a_j^d,\ a_j^r \\ s_j \end{smallmatrix}\right]$ $(A_j^K)^{-a_j^d}(C_j^K)^{-a_j^r}$ and that of Ψ_j is $\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} b_j^d,\ b_j^r \\ t_j \end{smallmatrix}\right]$ $(C_j^\Psi)^{-b_j^d}(A_j^\Psi)^{-b_j^r}$. The product of the powers over all letters, times $u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}$, is $f(u)\,g(v)$, with

$$f(u)=u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}^{-a_j^d}\Psi_{\mathcal{B}_j/R}^{-b_j^d}K_{\mathcal{B}_j|B}^{-a_j^r}\Psi_{\mathcal{B}_j|B}^{-b_j^r},\qquad g(v)=v^{\mathbf{m}|_R-\mathbf{1}}\prod_j K_{\mathcal{B}_j|R}^{-a_j^d}\Psi_{\mathcal{B}_j|R}^{-b_j^d}K_{\mathcal{B}_j/B}^{-a_j^r}\Psi_{\mathcal{B}_j/B}^{-b_j^r},$$

the integrands of the periods of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ on Δ_B and of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ on Δ_R (8.10).

Balance on the pin. The tables $\mathcal{B}_j/R$ and $\mathcal{B}_j|B$ on B have the ranks $d_j(B)$ and $d_j(B)+1$, and $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ on R the ranks $r_j-d_j(B)$ and $r_j-d_j(B)-1$. With $a_j^d+a_j^r=s_j$, $b_j^d+b_j^r=t_j$ and $|B|-d_j(B)-1=d_j^*(B)$, the imbalance of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ at $\mathbf{m}|_B$ is

$$m_B-\sum_j\big(a_j^d d_j(B)+b_j^d(|B|-d_j(B))+a_j^r(d_j(B)+1)+b_j^r d_j^*(B)\big)=\kappa_B-\sum_j(a_j^r+b_j^d);$$

with $|R|+1-r_j+d_j(B)=n-r_j-d_j^*(B)$, that of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ at $\mathbf{m}|_R$ is

$$m_R-\sum_j\big(s_j(r_j-d_j(B))+t_j(|R|+1-r_j+d_j(B))\big)+\sum_j(a_j^r+b_j^d)=\kappa_{\mathcal{E}}-\kappa_B+\sum_j(a_j^r+b_j^d).$$

Both vanish on the pin, F being on its balance wall.

Fubini. On a contour of the statement the modulus of $f(u)g(v)$ is $f(u)g(v)$ at the real parts of the shares, which do not move along the contour, and its integral over $\Delta_B\times\Delta_R$ is the product of the two periods at the real parts, finite by hypothesis; the stacked Gammas decay exponentially along the contour. So Fubini exchanges the share integral with the integral over $\Delta_B\times\Delta_R$, and the inner integrals are $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ and $P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$: the general display.

Pair coordinates. When every s_j and t_j is positive, $p=2N$ and the shares are $(a_j^r,b_j^d)_j$. With $a_j^d=\sigma_j^d+\delta_j^d$, $b_j^d=\sigma_j^d-\delta_j^d$ and $a_j^r=s_j-a_j^d$, the affine map $(\sigma_j^d,\delta_j^d)_j\mapsto(a_j^r,b_j^d)_j$ has in each table the linear part $(y_\sigma,y_\delta)\mapsto(-y_\sigma-y_\delta,\ y_\sigma-y_\delta)$, of determinant 2. So the imaginary parts $\mathbf{y}$ of the shares and $\mathbf{y}'$ of the pair coordinates have $d\mathbf{y}=2^N d\mathbf{y}'$ and $\sum_\ell y_\ell=-2\sum_j y'_{\delta,j}$, and with $\operatorname{Im}\kappa_B=0$ and $\delta^1(-2z)=\frac12\delta^1(z)$, 4.1 gives

$$\int_{\sum_\ell x_\ell=\kappa_B}h=(2\pi)^{1-2N}\int_{\mathbb{R}^{2N}}h\,\delta^1\Big(\sum_\ell y_\ell\Big)d\mathbf{y}=2^{N-1}(2\pi)^{1-2N}\int_{\mathbb{R}^{2N}}h\,\delta^1\Big(\sum_j y'_{\delta,j}\Big)d\mathbf{y}',$$

2^{N-1} times the fibre integral in the pair coordinates pinned by $\sum_j\delta_j^d=\frac12(\sum_j s_j-\kappa_B)$, the image of the pin $\sum_j(a_j^r+b_j^d)=\sum_j(s_j-2\delta_j^d)=\kappa_B$, the admissible real points corresponding under the map. With $\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} a_j^d,\ a_j^r \\ s_j \end{smallmatrix}\right]\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} b_j^d,\ b_j^r \\ t_j \end{smallmatrix}\right]=\Gamma_{\,2}^{\,4}\left[\begin{smallmatrix} a_j^d,\ b_j^d,\ a_j^r,\ b_j^r \\ s_j,\ t_j \end{smallmatrix}\right]$ this is the first display, whose N coordinates σ_j^d are not pinned. For one table the pin fixes δ^d, $2^{N-1}=1$, and the fibre integral is $(2\pi)^{-1}\int dy=(2\pi i)^{-1}\int d\sigma^d$ along the line in σ^d. The other forms of the pin, the value $2\delta^d=s-\kappa_B=\delta(1+\varepsilon(B))-\mu_B$ (8.2(iii) with $c(B)=1$) and the swap of B and R are the reference, proof of thm:twosides. □

Remark. The first display needs both letters of every table active: a letter of exponent zero would put $\Gamma(0)$ into the stacked Gamma of its table. The general display carries one Beta corner $\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} \cdot,\ \cdot \\ w \end{smallmatrix}\right]$ for each active letter, and at a full lean of one table it has no free share, which is 18.14.

Corollary 18.14 (One Skeleton value pins the share). *As stated: real exponents of any sign, one Skeleton value for all active letters.*

Proof. $0 < \kappa_B < 2\sigma$ as in the proof of 18.10. With the coefficients of the proof of 18.13, Lemma S7 gives, all functions being positive,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R}\int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} v^{\mathbf{m}|_R-\mathbf{1}} \times \int_0^\infty \rho^{\kappa_B-1} \prod_j (A_j^K+\rho C_j^K)^{-s_j}(A_j^\Psi+\rho C_j^\Psi)^{-t_j}\, d\rho\, d\omega_B(u)\, d\omega_R(v).$$

The Skeleton values are $A_j^K/C_j^K = \gamma_{j,R}^K(v)/\gamma_{j,B}^K(u)$ and $A_j^\Psi/C_j^\Psi = \gamma_{j,R}^\Psi(v)/\gamma_{j,B}^\Psi(u)$ (18.7), and by the hypothesis, the constants cancelling, every active letter has $A_\ell = \xi C_\ell$ with $\xi = \gamma_R(v)/\gamma(u)$. The exponents of the active letters add up to 2σ, so the integral of a linear letter gives $\int_0^\infty \rho^{\kappa_B-1}(\xi+\rho)^{-2\sigma}d\rho = \Gamma\!\left[\begin{matrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{matrix}\right]\xi^{\kappa_B-2\sigma}$, and

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma\!\left[\begin{matrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{matrix}\right] \int_{\Delta_R}\int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} v^{\mathbf{m}|_R-\mathbf{1}} \times \prod_{s_j\neq 0}(C_j^K)^{-s_j} \prod_{t_j\neq 0}(C_j^\Psi)^{-t_j}\, \xi^{\kappa_B-2\sigma}\, d\omega_B(u)\, d\omega_R(v).$$

For level shares as in the statement, $(A_j^K)^{-a_j^d}(C_j^K)^{-a_j^r} = (C_j^K)^{-s_j}\xi^{-a_j^d}$ for an active K_j, $(A_j^\Psi)^{-b_j^r}(C_j^\Psi)^{-b_j^d} = (C_j^\Psi)^{-t_j}\xi^{-b_j^r}$ for an active Ψ_j, and an inactive letter contributes 1; since $\sum_j(a_j^d+b_j^r) = 2\sigma - \sum_j(a_j^r+b_j^d) = 2\sigma-\kappa_B$, the integrand above is $f(u)g(v)$ with f and g the integrands of the periods of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ and $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ (proof of 18.13), whatever the shares. At complex shares $|f(u)g(v)|$ is $f(u)g(v)$ at the real parts, which satisfy the same constraints; its integral is finite by the display at the real parts, and both factors are positive, so $|f|$ and $|g|$ are integrable and Fubini gives the statement, the product of the two periods not depending on the shares. With one active letter the hypothesis holds with γ and γ_R its hinge values of B and of R. One table at $(0,t)$: $\mathbf{a}=0$, $b^d+b^r=t$ and $b^d=\kappa_B$, so the level shares are $(0,\kappa_B,0,t-\kappa_B)$ and the two families have the forms of the statement; at $(s,0)$, $a^r=\kappa_B$ and $a^d = s-\kappa_B$. □

Remark. The reference, rem:pinnedproduct, is one table at a full lean, the case of one active letter. The hypothesis does not depend on the factorizations: a change of the factorization of table j multiplies the hinge values of a letter of table j at B and at R by one constant (18.4), so the Skeleton values are unchanged.

Lemma 18.15 (Slopes and the range of the ratios). *As stated: for every table of a family, both letters of every table active.*

Proof. With $a_j^\ell = \sigma_j^\ell + \delta_j^\ell$ and $b_j^\ell = \sigma_j^\ell - \delta_j^\ell$, $K^{-a}\Psi^{-b} = \mathbf{E}^{-\sigma}(K/\Psi)^{-\delta}$ for every minor, $\mathbf{E} = K\Psi$. So the factors of table j in the word of the bunch family (proof of 18.13) are

$$\mathbf{E}_{\mathcal{B}_j/R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j^r}\Big(\frac{K_{\mathcal{B}_j/R}}{\Psi_{\mathcal{B}_j/R}}\Big)^{-\delta_j^d}\Big(\frac{K_{\mathcal{B}_j|B}}{\Psi_{\mathcal{B}_j|B}}\Big)^{-\delta_j^r}, \qquad \mathbf{E}_{\mathcal{B}_j/R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j+\sigma_j^d} = \varrho_{j,B}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j},$$

and in the word of the room family $\mathbf{E}_{\mathcal{B}_j|R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j/B}^{-\sigma_j+\sigma_j^d} = \varrho_{j,R}^{\sigma_j^d}\mathbf{E}_{\mathcal{B}_j/B}^{-\sigma_j}$ (18.4), the other factors not involving σ_j^d at fixed δ_j^d: σ_j^d enters through $\varrho_{j,B}(u)^{-\sigma_j^d}\varrho_{j,R}(v)^{\sigma_j^d}$.

Slopes. The wall at $S \subseteq B$ of the bunch family is $m_S - \sum_j \big(a_j^d d_{\mathcal{B}_j/R}(S) + b_j^d d^*_{\mathcal{B}_j/R}(S) + a_j^r d_{\mathcal{B}_j|B}(S) + b_j^r d^*_{\mathcal{B}_j|B}(S)\big)$, affine in the shares; raising σ_j^d by one at fixed δ_j^d raises a_j^d, b_j^d and lowers a_j^r, b_j^r by one, so the slope is $c_{\mathcal{B}_j/R}(S) - c_{\mathcal{B}_j|B}(S)$, $c = |S| - d - d^*$. With r^+ the rank function of $\mathcal{B}_{j,B}^+$, $\mathcal{B}_j|B = \mathcal{B}_{j,B}^+ \setminus g$ has the rank function r^+ on B and $\mathcal{B}_j/R = \mathcal{B}_{j,B}^+/g$ the rank function $Z \mapsto r^+(Z+g) - 1$, and $c_Z(S) = r_Z(S) + r_Z(B \setminus S) - r_Z(B)$ for a table Z on B (7.10, 7.11); the difference is the first displayed slope. On R the table of the level d is $\mathcal{B}_j|R = \mathcal{B}_{j,R}^+ \setminus g$ and that of the level r is $\mathcal{B}_j/B = \mathcal{B}_{j,R}^+/g$, which gives the second.

Range. On the sector of an order of B, $\alpha_{\pi(1)} = \rho_0$ and $\alpha_{\pi(k)} = \rho_0 q_2 \cdots q_k$ with $q_k \in (0,1]$ and tails U_k, the letters of a table Z on B are $K_Z = \rho_0^{r_Z} \prod_k q_k^{d_Z(U_k)} k_{Z,\pi}$ and $\Psi_Z = \rho_0^{|B|-r_Z} \prod_k q_k^{d^*_Z(U_k)} \psi_{Z,\pi}$, with sector letters positive on the closed cube (8.7(ii)), hence between positive constants. So

$$\varrho_{j,B} = \frac{\mathbf{E}_{\mathcal{B}_j/R}}{\mathbf{E}_{\mathcal{B}_j|B}} = \prod_k q_k^{c_{\mathcal{B}_j|B}(U_k) - c_{\mathcal{B}_j/R}(U_k)} \times (\text{a function between positive constants}),$$

the exponent of q_k being minus the slope of U_k. Every nonempty proper subset of B is a tail of some order, the q_k range over $(0,1]$ independently, and finitely many sectors cover Δ_B: $\varrho_{j,B}$ is bounded exactly when no slope on B is positive (a positive slope at a tail U_k sends it to ∞ as $q_k \to 0$), and bounded below by a positive constant exactly when none is negative. On R, $\varrho_{j,R} = \mathbf{E}_{\mathcal{B}_j/B}/\mathbf{E}_{\mathcal{B}_j|R}$ carries q_k to the power of the slope of U_k: $\inf \varrho_{j,R} > 0$ exactly when no slope on R is positive and $\sup \varrho_{j,R} < \infty$ exactly when none is negative. A wall falls as σ_j^d decreases, toward the level d, exactly when its slope is positive, which forces $\sup \varrho_{j,B} = \infty$ or $\inf \varrho_{j,R} = 0$, so $\sup \varrho_{j,B}/\inf \varrho_{j,R} = \infty$; toward the level r, a negative slope forces $\inf \varrho_{j,B}/\sup \varrho_{j,R} = 0$. A change of the factorization of table j multiplies $\varrho_{j,B}$ and $\varrho_{j,R}$ by one constant c^{-2} (18.4), which keeps the quotient, the finiteness of the suprema, the positivity of the infima and the two ratios; the slopes are counts. □

Proposition 18.24 (The same shares closed at the gauge). *As stated: at complex exponents and masses, with the bound in absolute value.*

Proof. At width one the letters on the scale are linear in ρ (Lemma S6), and divided by their values at $\rho = 1$ they are the Skeleton polynomials of the statement (18.17): with the coefficients of the proof of 18.13, $\pi_{j,d} = A_j^K/(A_j^K + C_j^K)$, $\pi_{j,r} = 1 - \pi_{j,d}$, $\pi^*_{j,r} = A_j^\Psi/(A_j^\Psi + C_j^\Psi)$ and $\pi^*_{j,d} = 1 - \pi^*_{j,r}$, the chances of the levels under the basis law and the cobasis law at (u,v), all in $(0,1)$. Each polynomial is $1 - (1-x)\epsilon$ with a constant exceedance polynomial ϵ: $\pi_{j,r}$ for K_j and $\pi^*_{j,d}$ for Ψ_j on the bunch, and, read from the top of the scale, $\pi_{j,d}$ and $\pi^*_{j,r}$ on the room. In 15.9(ii) only $k = 0$ remains, $[x^k] \prod_i \epsilon_i^{n_i} = 0$ for $k > 0$, and the term of $\mathbf{n}$ is $\prod_i \frac{(w_i)_{n_i}}{n_i!} \epsilon_i^{n_i} \Gamma_2^{\,1}\!\left[\begin{matrix}\kappa,\ |\mathbf{n}|+1 \\ \kappa + |\mathbf{n}| + 1\end{matrix}\right]$; the two sectors of 15.9(i) give the display, of nonnegative terms for real data with $s_j, t_j \geq 0$. For complex exponents and masses with $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa_B^{\text{top}} > 0$ the series converges absolutely to the same value (15.9, last sentence; 15.2 for the convergence of the scale integral). The bound: $|(w)_n| \leq (|w|)_n$; $\prod_j \pi_{j,r}^{n_j} \pi_{j,d}^{*\,n'_j} \leq \lambda_B^M$; $\big|\Gamma_2^{\,1}\!\left[\begin{matrix}\kappa_B,\ M+1 \\ \kappa_B + M + 1\end{matrix}\right]\big| = \big|\int_0^1 x^{\kappa_B - 1}(1-x)^M dx\big| \leq \int_0^1 x^{\operatorname{Re}\kappa_B - 1} dx = 1/\operatorname{Re}\kappa_B$; and $\sum_{|\mathbf{n}|+|\mathbf{n}'|=M} \prod_j \frac{(|s_j|)_{n_j} (|t_j|)_{n'_j}}{n_j!\, n'_j!} = \frac{(\sum_j (|s_j| + |t_j|))_M}{M!}$, the coefficient of z^M in $\prod_j (1-z)^{-|s_j| - |t_j|}$. Likewise on the room with λ_R and $\operatorname{Re}\kappa_B^{\text{top}}$. The bounds are summable over M, $\sum_M \frac{(W)_M}{M!} \lambda^M = (1-\lambda)^{-W}$ for $\lambda < 1$. □

Proposition 18.33 (A loop or a coloop on the simplex). *As stated: a mark that is a loop or a coloop in every table of a family.*

Proof. In a table where e is a loop no basis contains e, so $K_j = K_{j,\backslash e}$ and $\Psi_j = \alpha_e \Psi_{j,\backslash e}$; where e is a coloop every basis contains e, so $K_j = \alpha_e K_{j,/e}$ and $\Psi_j = \Psi_{j,/e}$ (7.3). The letters of F' (15.16) are these minors, of the ranks $r'_j = r_j$ at a loop and $r_j - 1$ at a coloop on $n-1$ marks, so the family word is $\alpha_e^{-\sum_{\eta_j=+1} s_j - \sum_{\eta_j=-1} t_j} W'(\alpha_{\mathcal{E}-e})$, with W' the word of F', homogeneous of degree $\kappa' - m_{\mathcal{E}-e}$, κ' the imbalance of F' at $\mathbf{m}|_{\mathcal{E}-e}$. On $\Delta_{\mathcal{E}}$ put $u = (u_e, (1-u_e)v)$, $u_e \in (0,1)$, $v \in \Delta_{\mathcal{E}-e}$, so that $u^{\mathbf{m}-\mathbf{1}} d\omega_{\mathcal{E}}(u) = u_e^{m_e - 1}(1-u_e)^{m_{\mathcal{E}-e}-1} du_e \, v^{\mathbf{m}|_{\mathcal{E}-e}-\mathbf{1}} d\omega_{\mathcal{E}-e}(v)$ (3.12). Then

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_0^1 u_e^{\nu_e - 1}(1-u_e)^{\kappa'-1} du_e \; P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t}), \qquad \nu_e = m_e - \sum_{\eta_j=+1} s_j - \sum_{\eta_j=-1} t_j,$$

for real data in $[0,\infty]$, and the integral over u_e is $\Gamma_{\substack{2\\1}}\begin{bmatrix} \nu_e, & \kappa' \\ \nu_e + \kappa' & \end{bmatrix}$ where $\nu_e, \kappa' > 0$. This ν_e is the wall $\kappa_{\{e\}}$, since $d_j(\{e\}) = 1$ and $d_j^*(\{e\}) = 0$ at a coloop and the reverse at a loop, and by 8.2(iii), with $c_j(\{e\}) = 0$ and $\varepsilon_j(\{e\}) = \eta_j$, it is $\mu_e - \sum_j \eta_j \delta_j$. The imbalance of F' is $m_{\mathcal{E}} - m_e - \sum_{\eta_j=+1} \big(s_j(r_j - 1) + t_j(n - r_j)\big) - \sum_{\eta_j=-1} \big(s_j r_j + t_j(n-1-r_j)\big) = \kappa_{\mathcal{E}} - \nu_e$, so $\nu_e + \kappa' = \kappa_{\mathcal{E}}$. Several such marks are removed one at a time, each keeping its kind in the minors, and the product of the Beta corners is $\Gamma_{\substack{k+1\\1}}\begin{bmatrix} (\nu_{e_i})_i, & \kappa' \\ \kappa_{\mathcal{E}} & \end{bmatrix}$, split by 3.12 as stated. On the balance wall $\kappa' = -\nu_e$, and ν_e and κ' are not both positive.

The set $S = \{e\}$ has $c_j(S) = 0$ in every table, as 16.15 asks; there $\kappa_{\{e\}} = \nu_e$ and, by 8.2(iii) and 7.11 ($c_j(\mathcal{E} - e) = c_j(\{e\}) = 0$, $\varepsilon_j(\mathcal{E} - e) = 2r_j - n - \eta_j$), $\kappa_{\mathcal{E}\backslash e} = \kappa_{\mathcal{E}} - \nu_e = \kappa'$. The integral of 16.15 over $\Delta_{\{e\}} \times \Delta_{\mathcal{E}-e}$, $\Delta_{\{e\}}$ the point $u_e = 1$ with the Dirac measure, is the integral over $\Delta_{\mathcal{E}-e}$ of $v^{\mathbf{m}|_{\mathcal{E}-e}-\mathbf{1}}$ times the family word at $(1, v)$, which is $W'(v)$: it is $P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t})$, and the compound formula at $S = \{e\}$ is the display. □

Definition 18.34 (Period datum; moves). As stated, for tables with positive rational weights.

Proof. The definition asks of the family only that its form be algebraic over $\mathbb{Q}$ and homogeneous of degree zero. The letters K_j and Ψ_j are polynomials whose coefficients are the weights, here rational; with rational exponents and masses $W\, d\alpha_{\mathbf{m}}$ is a product of rational powers of polynomials over $\mathbb{Q}$ and of the coordinates, times $d\alpha$, hence algebraic over $\mathbb{Q}$, and on the balance wall its degree is $\kappa_{\mathcal{E}} = 0$ (8.1). The walls of the faces are those of 8.7, for any weights. □

Theorem 18.35 (Pooling is a move). *As stated: for tables with positive rational weights, and in (ii) whenever the active letters have one hinge value at rational factorizations.*

Proof. (i) is the reference, proof of thm:cleaningmove(i): the map $(\rho, u, \alpha_R) \mapsto (\rho u, \alpha_R)$ and its inverse are defined over $\mathbb{Q}$ whatever the weights, and the faces correspond as stated. After the change of variables the form is $\rho^{\kappa_B - 1}\phi(u)$ times the form of $F_{B\to g}(u)$ at the gauge (ρ, α_R) (proof of 18.8), with ϕ as in the proof of 18.10; its coefficients lie in the field of the weights and of the factors, which is $\mathbb{Q}$. The walls are matched by 18.9(i), (ii) and the gluing of the counts, for families.

(ii) The common hinge value γ is the hinge value of an active letter, a quotient of polynomials with rational coefficients, positive on the open simplex. The change $(\rho, u, \alpha_R) \mapsto (\rho\gamma(u), u, \alpha_R)$ is birational over $\mathbb{Q}$, with the inverse $(x, u, \alpha_R) \mapsto (x/\gamma(u), u, \alpha_R)$, and bijective on the open domain; by the proof of 18.10 it carries the form of (i) to the product of the form $\phi(u)\gamma(u)^{-\kappa_B} d\omega_B(u)$ of

the shape and the form of G_R at the masses $(\kappa_B, \mathbf{m}|_R)$. In the datum of G_B, read on the section $|\alpha_B| = 1$ (Lemma S7), the change $y = \gamma(u)z$, birational over $\mathbb{Q}$ and bijective on the open domain, carries the form to $\phi(u)\gamma(u)^{-\kappa_B} d\omega_B(u)$ times $z^{2\sigma-\kappa_B-1}(1+z)^{-2\sigma}dz$, the form of the list on two marks whose bases are the two singletons, at the exponents $(0, 2\sigma)$ and the masses $(\kappa_B, 2\sigma - \kappa_B)$, read in its chart at the first mark, where its letter is $1 + z$. So the data of F and of that list, and the data of G_B and G_R, are related by these moves to the same product of three data, with rational masses κ_B and $2\sigma - \kappa_B$. If P_F converges, every proper wall of F is positive (8.10) and 18.10 gives the convergence of the periods of G_B and G_R, the Beta corner, and the shape integral, which is P_F over P_{G_R}; and P_F is the first integral of the chain. □

Remark. Positive rational weights that factor across B and R factor with rational factors: fix a basis $X_\ell \sqcup Y_\ell$ at each level $\ell \in \{d(B), r(B)\}$ and put $\theta'_X = \theta_{X\sqcup Y_\ell}$ and $\theta''_Y = \theta_{X_\ell\sqcup Y}/\theta_{X_\ell\sqcup Y_\ell}$ for the patterns X and the room sides Y of the level ℓ. For any factorization $\theta = \vartheta'\vartheta''$, $\theta_{X\sqcup Y_\ell}\theta_{X_\ell\sqcup Y}/\theta_{X_\ell\sqcup Y_\ell} = \vartheta'_X\vartheta''_{Y_\ell}\vartheta'_{X_\ell}\vartheta''_Y/(\vartheta'_{X_\ell}\vartheta''_{Y_\ell}) = \theta_{X\sqcup Y}$. The hypothesis of (ii) asks more, rational factors at which the active letters share one hinge value; for one table with both letters active it asks a rational factorization with $\varrho_B = 1$.

Proposition 18.38 (Cleaning commutes with maps). *As stated, for maps between any tables.*

Proof. A bijection g that carries $(\mathcal{B}, \theta)$ to $(\mathcal{B}_1, \theta_1)$ maps bases to bases with their weights. Parallel and series bunches, the counts and the widths are defined by the bases, so they are carried to those of the image; a factorization θ', θ'' across a class B is carried to $\theta'_{1,g(p)} = \theta'_p$, $\theta''_{1,g(Y)} = \theta''_Y$, a factorization across $g(B)$, and the minors of $g(B)$ are the images of those of B with their weights, so $\gamma^K_{g(B)}(\alpha \circ g^{-1}) = \gamma^K_B(\alpha)$ and likewise for Ψ. An exchange maps T to $\mathcal{E}_1 \setminus g(T)$, with the weight θ_T. A parallel bunch B has $|T \cap B| \le 1$ for every basis and no loop, so $g(B)$ has $|g(B) \cap T_1| \ge |B| - 1$ for every basis T_1 of the image and no coloop: a series bunch, and conversely. The patterns of $g(B)$ are the sets $g(B \setminus p)$, p a pattern of B, and putting $\theta'_{1,g(B\setminus p)} = \theta'_p$ and $\theta''_{1,g(R\setminus Y)} = \theta''_Y$ factors the weights of the image. The level of $g(B \setminus p)$ is $|B| - |p|$, so the restriction of the image to $g(B)$ has the patterns $g(B \setminus p)$, $p \in \mathcal{B}/R$, and its contraction of the room the patterns $g(B \setminus p)$, $p \in \mathcal{B}|B$; hence $K_{\mathcal{B}_1|g(B)}(\alpha \circ g^{-1}) = \sum_{p\in\mathcal{B}/R} \theta'_p \alpha^{B\setminus p} = \Psi_{\mathcal{B}/R}(\alpha)$ and $K_{\mathcal{B}_1/g(R)}(\alpha \circ g^{-1}) = \Psi_{\mathcal{B}|B}(\alpha)$, so the hinge value by K of $g(B)$ is γ^Ψ_B; likewise its hinge value by Ψ is γ^K_B. The statement on the cleanings of lists is the reference, proof of prop:cleanmaps. □

40.19 Chapter 19: words of a tree

Definition 19.2 (Side shape; half-sum and difference; pattern coefficients). As stated, for every table of a family.

Proof. The one claim is the identity $L_{j,w}(z) = \sum_p e^{\langle z_B, \chi_p\rangle} c_{j,p}(w; z_R)$. Theorem 12.6 for table j gives $\Lambda_j(z) = \sum_{p\subseteq B} e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(z_R)$, with $\Lambda_{j,(p)} = 0$ off the patterns of table j. Replacing z by $-z$ gives

$$\Lambda_j(-z) = \sum_p e^{-\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(-z_R).$$

Since $\chi_{B\setminus p} = -\chi_p$ on B, the substitution $p \mapsto B \setminus p$ turns this into

$$\Lambda_j(-z) = \sum_p e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(B\setminus p)}(-z_R) = \sum_p e^{\langle z_B,\chi_p\rangle}\Lambda^*_{j,(p)}(z_R).$$

So

$$w\Lambda_j(z) + (1-w)\Lambda_j(-z) = \sum_p e^{\langle z_B, \chi_p\rangle}\big(w\Lambda_{j,(p)}(z_R) + (1-w)\Lambda^*_{j,(p)}(z_R)\big).$$

□

Proposition 19.3 (The side shape, and what an exchange does to it). *As stated, for every table of a family and every bijection that carries or exchanges the family.*

Proof. For $T \subseteq \mathcal{E}$ and $z \in \mathbb{R}^{\mathcal{E}}$, with $(g^*z)_e = z_{g(e)}$,

$$\langle z, \chi_{g(T)}\rangle = \sum_{e\in\mathcal{E}} z_{g(e)}\,\chi_{g(T)}(g(e)) = \sum_{e\in\mathcal{E}} (g^*z)_e\,\chi_T(e) = \langle g^*z, \chi_T\rangle.$$

If g carries F, then $g(\mathcal{B}_j) = \mathcal{B}_{h(j)}$ with $\theta_{h(j),g(T)} = \theta_{j,T}$ (Definition 11.14), so

$$\Lambda_{h(j)}(z) = \sum_{T\in\mathcal{B}_j} \theta_{j,T}\, e^{\langle z, \chi_{g(T)}\rangle} = \Lambda_j(g^*z),$$

and $\Lambda_{h(j)}(-z) = \Lambda_j(-g^*z)$, as g^* is linear; hence $L_{h(j),w}(z) = L_{j,w}(g^*z)$. If g exchanges F, then $T \mapsto \mathcal{E}\setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ with $\theta_{h(j),\mathcal{E}\setminus g(T)} = \theta_{j,T}$, and $\chi_{\mathcal{E}\setminus g(T)} = -\chi_{g(T)}$, so $\Lambda_{h(j)}(z) = \Lambda_j(-g^*z)$ and $\Lambda_{h(j)}(-z) = \Lambda_j(g^*z)$; hence

$$L_{h(j),w}(z) = w\,\Lambda_j(-g^*z) + (1-w)\,\Lambda_j(g^*z) = L_{j,1-w}(g^*z).$$

The display holds for any two positive numbers a, b in place of $\Lambda_j(z), \Lambda_j(-z)$ and any $s_j, t_j > 0$: it is Theorem 3.2 at the unit configuration on a two-element set with the coarsest partition, whose Zamma is $(wa+(1-w)b)^{-(s_j+t_j)} = \mathsf{Z}_{s_j+t_j;(1)}(wa+(1-w)b)$ at the shape $(w, 1-w)$, that is, the two-factor form $a^{-s_j}b^{-t_j} = \Gamma_{\frac12}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j\end{bmatrix}\int_0^1 w^{s_j-1}(1-w)^{t_j-1}(wa+(1-w)b)^{-s_j-t_j}dw$ (the reference, proof of prop:sideexchange). Taking $a = \Lambda_j(z)$ and $b = \Lambda_j(-z)$, $wa+(1-w)b = L_{j,w}(z)$.

For one table an exchange has h the identity, so that $L_w(z) = L_{1-w}(g^*z)$ for every w. Hence

$$\mathsf{Z}_{s+t;(1)}\big(L_w(g^*z)\big) = \mathsf{Z}_{s+t;(1)}\big(L_{1-w}(z)\big)\,:$$

the exchange together with $w \mapsto 1-w$ keeps the Zamma. The substitution $w \mapsto 1-w$ takes $w^{s-1}(1-w)^{t-1}$ to $w^{t-1}(1-w)^{s-1}$, which is the same function on $(0,1)$ exactly when $s = t$. □

Theorem 19.4 (The junction in pattern form). *As stated, for every family.*

Proof. The proof is that of the reference (proof of thm:junctionpattern), with a product over the tables in each step.

Signed coordinates. By Theorem 8.10 the period is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}\prod_j K_j^{-s_j}\Psi_j^{-t_j}\,d\alpha$ over the section $\alpha_g = 1$. Put $\alpha = e^{2z}$, so that $d\alpha_e = 2\alpha_e\,dz_e$ for each of the $n-1$ marks $e \neq g$. Theorem 7.13(i) for each table gives $K_j(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda_j(z)$ and $\Psi_j(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda_j(-z)$, so, with $\sum_j (s_j+t_j) = 2\sigma$ and $\boldsymbol{\mu} = \mathbf{m} - \sigma\mathbf{1}$ (Definition 8.1),

$$\begin{aligned}\alpha^{\mathbf{m}}\prod_j K_j^{-s_j}\Psi_j^{-t_j} &= e^{2\langle \mathbf{m}, z\rangle - 2\sigma\langle \mathbf{1}, z\rangle}\prod_j \Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j} \\ &= e^{2\langle \boldsymbol{\mu}, z\rangle}\prod_j \Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j},\end{aligned}$$

and, with $e^{2\langle \boldsymbol{\mu},z\rangle} = e^{2\langle \boldsymbol{\mu}|_B,z_B\rangle} e^{2\langle \boldsymbol{\mu}|_R,z_R\rangle}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = 2^{n-1} \int_{\mathbb{R}^{\mathcal{E}-g}} e^{2\langle \boldsymbol{\mu},z\rangle} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j}\, dz_{\mathcal{E}-g}\Big|_{z_g=0}.$$

Side shapes. For j with $s_j, t_j > 0$, Proposition 19.3 writes the factor of table j as

$$\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j} = \Gamma_{\frac{1}{2}}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j \end{bmatrix} \int_0^1 w_j^{s_j-1}(1-w_j)^{t_j-1} L_{j,w_j}(z)^{-(s_j+t_j)}\, dw_j.$$

For $t_j = 0 < s_j$ the factor is $\Lambda_j(z)^{-s_j} = L_{j,1}(z)^{-s_j}$, for $s_j = 0 < t_j$ it is $\Lambda_j(-z)^{-t_j} = L_{j,0}(z)^{-t_j}$, and an inactive table contributes 1. So every active table contributes $L_{j,w_j}(z)^{-(s_j+t_j)}$, integrated over w_j when $s_j, t_j > 0$ and with the value of w_j fixed in the statement otherwise.

Scales. For each active j, $L_{j,w_j}(z) > 0$ and $L^{-a} = \Gamma(a)^{-1}\int_0^\infty y^{a-1}e^{-yL}dy$ with $a = s_j + t_j > 0$. The constants combine to $\Gamma_{\frac{1}{2}}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j \end{bmatrix}\Gamma(s_j+t_j)^{-1} = \Gamma(s_j)^{-1}\Gamma(t_j)^{-1}$ when $s_j, t_j > 0$, and to $\Gamma(s_j)^{-1}$ or $\Gamma(t_j)^{-1}$ for a table at a full lean: the prefactor of the statement.

Patterns. By Definition 19.2,

$$\sum_{j \text{ active}} y_j L_{j,w_j}(z) = \sum_{p\subseteq B} e^{\langle z_B,\chi_p\rangle} C_p, \qquad C_p = \sum_{j \text{ active}} y_j\, c_{j,p}(w_j; z_R) \geq 0,$$

each $c_{j,p}(w; z_R)$ being a combination with the nonnegative coefficients w and $1-w$ of the nonnegative functions $\Lambda_{j,(p)}(z_R)$ and $\Lambda^*_{j,(p)}(z_R)$.

The Gramma. With $x = e^{z_B}$, Definition 2.1 in its logarithmic form gives

$$\int_{\mathbb{R}^B} e^{2\langle \boldsymbol{\mu}|_B,z_B\rangle} \exp\Big(-\sum_p C_p\, e^{\langle \chi_p,z_B\rangle}\Big) dz_B = \mathsf{G}_{2\boldsymbol{\mu}|_B;(\chi_p)_p}\big((C_p)_p\big),$$

the configuration having one column χ_p for each p with $C_p \neq 0$.

Order of integration. Every integrand is nonnegative, so Tonelli's theorem allows the integrals to be taken in the order of the statement: the side shapes outermost, then z_{R-g} at $z_g = 0$, then the scales y_j, then z_B. The two sides agree in $[0,\infty]$ and are finite, every proper wall being positive (Theorem 8.10). □

Remark. With one table the formula is that of the reference, and at a full lean of that table the integral over the side shape is absent, which is the reference's last sentence.

Proposition 19.5 (What a map keeping the bunch gives the coefficients). *As stated, for every family.*

Proof. Since $\tilde g$ keeps B, it maps R onto R, so $(\tilde g^* z)_R = \tilde g^*_R z_R$, and for $p \subseteq B$, substituting $e' = \tilde g(e)$,

$$\langle (\tilde g^* z)_B, \chi_p\rangle = \sum_{e\in B} z_{\tilde g(e)}\chi_p(e) = \sum_{e'\in B} z_{e'}\chi_{\tilde g(p)}(e') = \langle z_B, \chi_{\tilde g(p)}\rangle.$$

If $\tilde g$ carries F, Proposition 19.3 and Definition 19.2 give

$$\begin{aligned}\sum_{q\subseteq B} e^{\langle z_B,\chi_q\rangle} c_{h(j),q}(w; z_R) &= L_{h(j),w}(z) = L_{j,w}(\tilde g^* z) \\ &= \sum_{p\subseteq B} e^{\langle z_B,\chi_{\tilde g(p)}\rangle} c_{j,p}(w; \tilde g^*_R z_R).\end{aligned}$$

The vectors χ_q, $q \subseteq B$, are distinct, and at fixed z_R the exponentials $e^{\langle z_B, \chi_q\rangle}$ with distinct exponents are linearly independent functions of z_B: along a generic direction the largest exponent dominates (the reference, proof of prop:mapcoeff). Comparing the coefficients at $q = \tilde{g}(p)$ gives $c_{h(j),\tilde{g}(p)}(w; z_R) = c_{j,p}(w; \tilde{g}_R^* z_R)$. If $\tilde{g}$ exchanges F, the same comparison with $L_{h(j),w}(z) = L_{j,1-w}(\tilde{g}^* z)$ gives $c_{h(j),\tilde{g}(p)}(w; z_R) = c_{j,p}(1-w; \tilde{g}_R^* z_R)$ for every w, and $w \mapsto 1-w$ gives the statement. □

Proposition 19.6 (Level form of the half-sum). *As stated, for every table of the family of Theorem 19.4.*

Proof. Every claim concerns one table j, and the reference's proof (proof of prop:levelformQ) applies to it. The two displayed coefficients are $c_{j,p}(w; z_R)$ at $w = \frac{1}{2}$ and its derivative in w (Definition 19.2).

(i) $c_{j,B\setminus p}(\frac{1}{2}; -z_R) = \frac{1}{2}\big(\Lambda_{j,(B\setminus p)}(-z_R) + \Lambda_{j,(p)}(z_R)\big) = c_{j,p}(\frac{1}{2}; z_R)$, and $\partial_w c_{j,B\setminus p}(w; -z_R) = \Lambda_{j,(B\setminus p)}(-z_R) - \Lambda_{j,(p)}(z_R) = -\partial_w c_{j,p}(w; z_R)$.

(ii) Let τ be a transposition of two marks of B fixing the rest. As in the proof of Proposition 19.5, $L_{j,1/2}(\tau^* z) = \sum_p e^{\langle z_B, \chi_{\tau(p)}\rangle} c_{j,p}(\frac{1}{2}; z_R)$. If $c_{j,p}(\frac{1}{2}; \cdot)$ depends on $|p|$ alone, this is $L_{j,1/2}(z)$, since $|\tau(p)| = |p|$. Conversely, if $L_{j,1/2}(\tau^* z) = L_{j,1/2}(z)$ for every such τ, the independence of the exponentials gives $c_{j,\tau(p)}(\frac{1}{2}; \cdot) = c_{j,p}(\frac{1}{2}; \cdot)$, and these transpositions act transitively on the subsets of B of each size. So level form is the invariance of $L_{j,1/2}$ under these transpositions. Further,

$$L_{j,1/2}(z) = \tfrac{1}{2} \sum_{T \in \mathcal{B}_j} \theta_{j,T}\big(e^{\langle z, \chi_T\rangle} + e^{\langle z, \chi_{\mathcal{E}\setminus T}\rangle}\big),$$
$$L_{j,1/2}(\tau^* z) = \tfrac{1}{2} \sum_{T \in \mathcal{B}_j} \theta_{j,T}\big(e^{\langle z, \chi_{\tau(T)}\rangle} + e^{\langle z, \chi_{\mathcal{E}\setminus \tau(T)}\rangle}\big),$$

by the computation of the proof of Proposition 19.3. The exponentials $e^{\langle z, \chi_S\rangle}$, $S \subseteq \mathcal{E}$, are linearly independent, so the invariance under τ says that τ maps the multiset $\mathcal{B}_j \uplus \{\mathcal{E} \setminus T : T \in \mathcal{B}_j\}$, with its weights, onto itself. When $2r_j \neq n$ a basis and a complement of a basis have different sizes, τ keeps sizes, and the condition says that τ maps $\mathcal{B}_j$ onto itself with its weights, that is, carries table j (Definition 11.14), as in Proposition 14.12.

(iii) If τ carries table j, it keeps $\Lambda_j(z)$ and $\Lambda_j(-z)$ under $z \mapsto \tau^* z$; if it exchanges table j, it swaps them (Proposition 19.3, the bijection of the tables being the identity). Either way $L_{j,1/2}(\tau^* z) = L_{j,1/2}(z)$, so the half-sum has level form by (ii), and $\partial_w L_{j,w}(\tau^* z) = \pm \partial_w L_{j,w}(z)$, so the square of the difference is unchanged by τ. The transpositions of B generate the permutations of B. □

Theorem 19.10 (Building back up). *As stated; the last sentence of (iv) is stated for every molecule.*

Proof. Parts (i) to (iii), and (iv) up to its last sentence, are proved in the reference (proof of thm:buildup). The last sentence of (iv) makes three claims.

Dimension. In (iv) every table has $s_j, t_j > 0$ and is active. The unknowns are the $c_{j,T}$, $T \in \mathcal{B}_j$, $\sum_j |\mathcal{B}_j|$ of them, subject to the N share sums $\sum_T c_{j,T} = t_j - s_j$ and the $n-1$ pins $-\frac{1}{2}\sum_{j,T} c_{j,T}\chi_T(e) = \mu_e$, $e \in \mathcal{E} - g$. Their linear parts are independent. Indeed, let $\sum_j \lambda_j \sum_T c_{j,T} + \sum_{e \neq g} d_e \sum_{j,T} c_{j,T}\chi_T(e)$ vanish identically, and put $d_g = 0$. The coefficient of $c_{j,T}$ is $\lambda_j + \langle d, \chi_T\rangle = \lambda_j + 2\langle d, \mathbf{1}_T\rangle - \langle d, \mathbf{1}\rangle$, so $\langle d, \mathbf{1}_T\rangle$ is constant over the bases of every table. By Lemma 7.7, d is constant on every molecule; the family is a molecule, so d is constant, and $d_g = 0$ makes it 0; then every $\lambda_j = 0$. So the fibre of differences has dimension $\sum_j |\mathcal{B}_j| - N - (n-1) = \sum_j (|\mathcal{B}_j| - 1) - (n-1)$.

The pins at the centre. At the centre $t_j = s_j$ and $\boldsymbol{\mu} = 0$, so the share sums read $\sum_T c_{j,T} = 0$. With $\chi_T = 2\mathbf{1}_T - \mathbf{1}$,

$$\sum_T c_{j,T}\chi_T(e) = 2\big(M_j^{\mathsf{T}}\mathbf{c}_j\big)_e - \sum_T c_{j,T},$$

so on the share sums the pin of $e \neq g$ reads $(\sum_j M_j^{\mathsf{T}}\mathbf{c}_j)_e = 0$. The equation at g follows, since $\sum_{e\in\mathcal{E}}(M_j^{\mathsf{T}}\mathbf{c}_j)_e = \sum_T c_{j,T}|T| = r_j\sum_T c_{j,T} = 0$. So the pins and the share sums together are equivalent to $\sum_T c_{j,T} = 0$ for every j and $\sum_j M_j^{\mathsf{T}}\mathbf{c}_j = 0$.

The contour. At the centre the gauge $\mathbf{1}$ is native (Proposition 8.21(i)). The leaves are opened by the atom formula with one share per basis and letter (Corollary 17.15), and the native split there (Definition 12.22) gives the basis T of table j the shares $a_{j,T} = s_j\theta_{j,T}/K_j(\mathbf{1})$ and $b_{j,T} = t_j\theta_{j,T}/\Psi_j(\mathbf{1})$, the probabilities of T under the basis law and the cobasis law at $\mathbf{1}$ times the exponents. As $\Psi_j(\mathbf{1}) = \sum_T \theta_{j,T} = K_j(\mathbf{1})$ and $s_j = t_j$, $c_{j,T} = b_{j,T} - a_{j,T} = 0$ there, so the wall contour of the native split (Theorem 17.12(iii)) is $\operatorname{Re}\mathbf{c} = 0$. □

Remark. With one table $\sum_T c_T = \frac{1}{r}\sum_e (M^{\mathsf{T}}\mathbf{c})_e$ follows from $M^{\mathsf{T}}\mathbf{c} = 0$, and the dimension is $|\mathcal{B}_1| - n$: the reference's case of one table at the centre.

Proposition 19.11 (The draw series). *As stated; the reading of its last part is stated for every family.*

Proof. The first part, through the Trailmoment and the Dirichlet moments, is the reference's proposition (proof of prop:jointdraw). For the last part take one block, $q = 2N$, $P_{2j-1} = K_j$ with the index s_j and $P_{2j} = \Psi_j$ with the index t_j; an index 0 contributes the factor 1.

Constants. K_j is homogeneous of degree r_j, with the squarefree monomials u^T, $T \in \mathcal{B}_j$, and the coefficients $\theta_{j,T}$. For $\mathbf{a} = \mathbf{1}_T$, $b_{\mathbf{a}} = r_j!/\prod_e a_e! = r_j!$, and every other $\mathbf{a}$ has $[u^{\mathbf{a}}]K_j = 0$; so $C_{2j-1} = \theta_{j,\max}/r_j!$. Likewise Ψ_j has degree $n - r_j$ and the monomials $u^{\mathcal{E}\setminus T}$ with the coefficients $\theta_{j,T}$, so $C_{2j} = \theta_{j,\max}/(n - r_j)!$.

The period. $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[f(U)] = \Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}^{-1}\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} f\, d\omega_{\mathcal{E}}$ for $f \geq 0$ (Theorem 3.9), so the left side times $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ is the stated integral, which on the balance wall is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.10).

The urn. For u in the simplex, $(\sum_e u_e)^g = \sum_{|\mathbf{a}|=g}\frac{g!}{\mathbf{a}!}u^{\mathbf{a}} = 1$ is the total chance of g independent draws from u, grouped by their counts $\mathbf{a}$. For $P_{2j-1} = K_j$,

$$\frac{P_{2j-1}}{C_{2j-1}} = \sum_{T\in\mathcal{B}_j}\frac{\theta_{j,T}}{\theta_{j,\max}}\, r_j!\, u^T$$

is the chance that a round of r_j draws is a basis T of table j and does not fail, a round that is the basis T failing with probability $1 - \theta_{j,T}/\theta_{j,\max}$; so $Q_{2j-1} = 1 - P_{2j-1}/C_{2j-1}$ is the chance that the round fails, and likewise Q_{2j} for a round of $n - r_j$ draws and the cobases. Given u, independent rounds all fail with the chance $\prod_i Q_i(u)^{k_i}$. The urn that draws a mark with probability proportional to its current mass and then adds one to that mass gives a sequence of draws with the counts $\mathbf{b}$ the chance $\prod_e (m_e)_{b_e}/(m_{\mathcal{E}})_{|\mathbf{b}|} = \mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[U^{\mathbf{b}}]$ (Theorem 3.11), the chance of the same sequence when $U \sim \mathrm{Dir}_{\mathbf{m}}$ is drawn first and the marks are then drawn independently from U. Summing over the sequences whose rounds all fail, $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[\prod_i Q_i(U)^{k_i}]$ is the urn's chance of the event of the statement; the order of the rounds does not change it, the product being symmetric in its factors. For lists every ratio $\theta_{j,T}/\theta_{j,\max}$ is 1, the Q_i have integer coefficients, and the expectation is a finite sum of Dirichlet moments.

Theorem 15.13(i). With the references $M_i = C_i$, on the simplex

$$M_i^{-w_i-n_i}(M_i - L_i)^{n_i} = C_i^{-w_i}(1 - L_i/C_i)^{n_i} = C_i^{-w_i} Q_i^{n_i},$$

so the series of Theorem 15.13(i), divided by $\Gamma_{1}^{n}\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$, is the present series term by term. □

Corollary 19.12 (At $\delta = 0$ and at a full lean the period reads the law of one letter for each table). *As stated, for every family.*

Proof. Letters. For a table riding the line, $K_j^{-\sigma_j}\Psi_j^{-\sigma_j} = \mathbf{E}_j^{-\sigma_j}$; at $(s_j, 0)$ the factor is $K_j^{-s_j}$ and at $(0, t_j)$ it is $\Psi_j^{-t_j}$. So on the open simplex

$$\prod_j K_j^{-s_j}\Psi_j^{-t_j} = \prod_j L_j^{-\nu_j} = \mathsf{Z}_{\boldsymbol{\nu};I_N}\big((L_j)_j\big),$$

the Zamma at the unit configuration being $\prod_j \xi_j^{-\nu_j}$ (Definition 2.12).

The period. By Theorem 8.10 and the Dirichlet density (Theorem 3.9),

$$\begin{aligned} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) &= \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j}\Psi_j^{-t_j}\, d\omega_{\mathcal{E}} \\ &= \Gamma_{1}^{n}\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\ \mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}\Big[\mathsf{Z}_{\boldsymbol{\nu};I_N}\big((L_j(U))_j\big)\Big]. \end{aligned}$$

The expectation is the integral of the kernel against the image $\phi_{\mathbf{m}}$ of $\mathrm{Dir}_{\mathbf{m}}$ under $(L_j)_j$, the Trailmoment $(\mathsf{T}_{\boldsymbol{\nu};I_N}[\mathrm{id}]\,\phi_{\mathbf{m}})((0,\infty)^N)$ with the identity as coefficient function (Definition 5.1), as in the display of Theorem 17.1(i). With one table it is the Trailmoment $(\mathsf{T}_{\nu;(1)}[x]\,\phi_{\mathbf{m}})((0,\infty))$ of a law on the half-line, at $\delta = 0$ the image of $\mathrm{Dir}_{\mathbf{m}}$ under the even letter (Theorem 5.9).

The law. Each L_j is a polynomial, positive on the open simplex and continuous on the closed one, and $\mathrm{Dir}_{\mathbf{m}}$ charges only the open simplex; so $\phi_{\mathbf{m}}$ is carried by $\prod_j(0, \max_{\Delta_{\mathcal{E}}} L_j]$. A law carried by a compact set is determined by its moments, polynomials being dense in the continuous functions on that set. Every moment $\mathbb{E}[\prod_j L_j(U)^{k_j}]$ is the expectation of a polynomial in U, a finite sum of Dirichlet moments (Theorem 3.11). □

Theorem 19.13 (The draw series along a chain). *As stated, for every family of tables.*

Proof. The proof is that of the reference (proof of thm:drawchain), with the acceptances and a product over the tables.

The failures split. Fix table j and a round of r_j draws. A round that extends on S_{i+1} extends on S_i, with the trace $P_i = P_{i+1} \cap S_i$, and every round extends on $S_0 = \varnothing$ with the trace $\varnothing$. So the events A_i, that the round extends on S_i and is accepted at $S_1, \ldots, S_i$, decrease with i, A_0 is certain, and $\mathfrak{f}_{j,i}$ is the chance of $A_i \setminus A_{i+1}$; hence $\sum_{i=0}^{d} \mathfrak{f}_{j,i} = 1 - \Pr(A_{d+1})$.

The acceptances telescope. A round extends on $S_{d+1} = \mathcal{E}$ exactly when it is a basis T of table j with distinct draws, which has the chance $r_j!\, u^T$; its traces are then $P_i = T \cap S_i$, and its acceptances multiply to

$$\prod_{i=0}^{d} \frac{\theta_j(S_{i+1}; T\cap S_{i+1})}{\theta_j(S_i; T\cap S_i)} = \frac{\theta_j(\mathcal{E};T)}{\theta_j(\varnothing;\varnothing)} = \frac{\theta_{j,T}}{\theta_{j,\max}}.$$

So $\Pr(A_{d+1}) = \sum_T r_j!\, u^T \theta_{j,T}/\theta_{j,\max} = r_j!\, K_j(u)/\theta_{j,\max}$. The cobasis rounds are the basis rounds of the dual table, whose bases $\mathcal{E}\setminus T$ carry the weights $\theta_{j,T}$, whose largest weight is $\theta_{j,\max}$ and whose basis polynomial is Ψ_j (Proposition 7.2); so $\sum_i \mathfrak{f}^*_{j,i} = 1 - (n-r_j)!\,\Psi_j/\theta_{j,\max}$.

Polynomials. Each $\mathfrak{f}_{j,i}(u)$ is a finite sum, over the sequences of r_j draws, of the monomial of the sequence times a product of acceptance and rejection probabilities in $[0,1]$: a polynomial with nonnegative coefficients, with integer coefficients for lists, whose rounds are always accepted.

The series. Proposition 19.11, with one block and the constants $C_{2j-1} = \theta_{j,\max}/r_j!$, $C_{2j} = \theta_{j,\max}/(n-r_j)!$ of its last part, gives

$$\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\Big[\prod_j K_j^{-s_j}\Psi_j^{-t_j}\Big] = \prod_j C_{2j-1}^{-s_j} C_{2j}^{-t_j} \sum_{\mathbf{k}\in\mathbb{N}^{2N}} \prod_j \frac{(s_j)_{k_{2j-1}}(t_j)_{k_{2j}}}{k_{2j-1}!\,k_{2j}!} \times \mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\Big[\prod_j \Big(\sum_i \mathfrak{f}_{j,i}\Big)^{k_{2j-1}} \Big(\sum_i \mathfrak{f}^*_{j,i}\Big)^{k_{2j}}\Big],$$

both sides finite or infinite together. The multinomial theorem expands

$$\frac{(s_j)_k}{k!}\Big(\sum_i \mathfrak{f}_{j,i}\Big)^k = \sum_{|\mathbf{k}_j|=k} \frac{(s_j)_{|\mathbf{k}_j|}}{\mathbf{k}_j!} \prod_i \mathfrak{f}_{j,i}^{k_{j,i}},$$

with nonnegative terms, and likewise for the cobasis rounds; Tonelli's theorem exchanges the sums and the expectation, and multiplying by $\Gamma_{1}^{n}\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ (Theorem 3.9) gives the display. Each expectation is that of a polynomial with nonnegative coefficients, a finite nonnegative combination of Dirichlet moments (Theorem 3.11).

Nesting. $\sum_{i'\ge i} k_{j,i'}$ is a tail sum of nonnegative integers, so it decreases with i.

Bijections. Let g keep each B_i, hence each S_i, and the masses. A sequence of draws D from u has the chance of the sequence $g(D)$ from g_*u. If g maps the bases of table j onto those of table $h(j)$ with their weights, then $g(T\cap S_i) = g(T)\cap S_i$, so D extends on S_i for table j exactly when $g(D)$ extends on S_i for table $h(j)$, the traces corresponding under g, and $\theta_j(S_i;P) = \theta_{h(j)}(S_i;g(P))$, so the acceptances agree: $\mathfrak{f}_{h(j),i}(g_*u) = \mathfrak{f}_{j,i}(u)$, and likewise for the cobasis rounds, g carrying the dual tables as well. If $T\mapsto \mathcal{E}\setminus g(T)$ maps the bases of table j onto those of table $h(j)$ with their weights, then $g(T)$ is a cobasis of table $h(j)$ for every basis T of table j, the round lengths agree, $r_j = n - r_{h(j)}$, the largest weights over the traces agree in the same way, and $\mathfrak{f}^*_{h(j),i}(g_*u) = \mathfrak{f}_{j,i}(u)$, $\mathfrak{f}_{h(j),i}(g_*u) = \mathfrak{f}^*_{j,i}(u)$. Since the masses are kept, $g_*U\sim\mathrm{Dir}_{\mathbf{m}}$ when $U\sim\mathrm{Dir}_{\mathbf{m}}$, so in the second case

$$\mathbb{E}\Big[\prod_{j,i}\mathfrak{f}_{j,i}(U)^{k_{j,i}}\mathfrak{f}^*_{j,i}(U)^{l_{j,i}}\Big] = \mathbb{E}\Big[\prod_{j,i}\mathfrak{f}^*_{h(j),i}(U)^{k_{j,i}}\mathfrak{f}_{h(j),i}(U)^{l_{j,i}}\Big],$$

the expectation at the indices $(\mathbf{k}'_{j'},\mathbf{l}'_{j'}) = (\mathbf{l}_{h^{-1}(j')},\mathbf{k}_{h^{-1}(j')})$. With one table h is the identity, an exchange maps the bases onto complements of bases, so $r = n-r$, and the array is symmetric under $\mathbf{k}\leftrightarrow\mathbf{l}$. □

Theorem 19.14 (The draw series along a tree). *As stated, for every molecule whose children's weights factor across the childcules.*

Proof. The proof is that of the reference (proof of thm:drawtree), the weights entering through the acceptances; the coordinates of the tree and their independence (Theorem 3.12 at every node) are unchanged.

(i) Let $\mathcal{B}'$ have rank r' at the node v, and write $R_i = R_{v,i}$, $x = x^{(v)}$, $u = u^{(v)}$ and $v^{(i)}$ for the shape of R_i. By Proposition 7.8 the bases of $\mathcal{B}'$ are the sets $T = p \sqcup \bigsqcup_i X_i$, p a pattern at B_v and X_i a basis of $\mathcal{B}'_{(p)}|R_i$ of size $r'_{p,i}$, and by the hypothesis $\theta_T = \prod_i \theta_{(p),i}(X_i)$. The shape of M_v is $x_{B_v} u$ on B_v and $x_{R_i} v^{(i)}$ on R_i, so by homogeneity

$$K_{\mathcal{B}'} = \sum_p x_{B_v}^{|p|} u^p \prod_i x_{R_i}^{r'_{p,i}} K_{\mathcal{B}'_{(p)}|R_i}\big(v^{(i)}\big),$$

the restrictions carrying the weights $\theta_{(p),i}$. Multiply by $r'!/\theta_{\max}(\mathcal{B}')$, write $r'! = \frac{r'!}{\prod_i r'_{p,i}!} \prod_i r'_{p,i}!$, and multiply and divide the term of p by

$$\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big);$$

this gives $\mathfrak{s}_{\mathcal{B}'} = \sum_p \mathfrak{p}_{\mathcal{B}',p} \prod_i \mathfrak{s}_{\mathcal{B}'_{(p)}|R_i}$. The hypothesis fixes the weights of the restrictions up to constant factors whose product is 1, which change neither the $\mathfrak{s}$ of the restrictions nor the first factor of $\mathfrak{p}_{\mathcal{B}',p}$. That factor is at most 1, since

$$\theta_{\max}(\mathcal{B}') = \max_p \max_{(X_i)_i} \prod_i \theta_{(p),i}(X_i) = \max_p \prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big).$$

The chance that r' independent draws from the shape of M_v are the basis T with distinct draws is $r'!$ times its monomial, and the acceptance with probability $\theta_T/\theta_{\max}(\mathcal{B}')$ gives the chance $r'!K_{\mathcal{B}'}/\theta_{\max}(\mathcal{B}')$ of an accepted basis. The round passes v showing p, its draws in B_v being the marks of p and its numbers of draws in the R_i the $r'_{p,i}$, with the chance $\frac{r'!}{\prod_i r'_{p,i}!} x_{B_v}^{|p|} u^p \prod_i x_{R_i}^{r'_{p,i}}$, and given this its draws in R_i are $r'_{p,i}$ independent draws from $v^{(i)}$. The acceptance splits as

$$\frac{\theta_T}{\theta_{\max}(\mathcal{B}')} = \frac{\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big)}{\theta_{\max}(\mathcal{B}')} \prod_i \frac{\theta_{(p),i}(X_i)}{\theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big)},$$

an acceptance at v followed by independent acceptances in the childcules. So $\mathfrak{p}_{\mathcal{B}',p}$ is the chance of passing v showing p and being accepted there, and $\mathfrak{s}_{\mathcal{B}'_{(p)}|R_i}$ the chance of an accepted basis of the restriction in R_i.

(ii) By (i) the round fails at v with the chance $1 - \sum_p \mathfrak{p}_{\mathcal{B}',p}$, and otherwise passes v showing one pattern p and is accepted there; it then first fails at w in the subtree of R_i exactly when it succeeds in $R_1, \dots, R_{i-1}$ and first fails at w inside R_i, which is the recursion. Write $\mathfrak{p}_{\mathcal{B}',p} = a_p \, \mathfrak{p}^\circ_{\mathcal{B}',p}$ with $a_p \le 1$ the first factor. Then

$$1 - \sum_p \mathfrak{p}_{\mathcal{B}',p} = \Big(1 - \sum_p \mathfrak{p}^\circ_{\mathcal{B}',p}\Big) + \sum_p (1 - a_p)\, \mathfrak{p}^\circ_{\mathcal{B}',p}.$$

The bracket is $1 = (x_{B_v} \sum_{e \in B_v} u_e + \sum_i x_{R_i})^{r'}$, expanded by the multinomial theorem, less some of its terms, distinct patterns giving distinct terms (the reference, proof of thm:drawtree, part (ii)); it has nonnegative coefficients in the coordinates of v, and so has the sum. With $1 - \prod_i S_i = \sum_i \prod_{i'<i} S_{i'}(1 - S_i)$ and induction on the subtree, every $\mathfrak{f}_{\mathcal{B}' \to w}$ is a finite sum of products, over the blocks of the tree, of polynomials with nonnegative coefficients in one block each.

(iii) On the simplex $r_j! K_j/\theta_{j,\max} = \mathfrak{s}_{\mathcal{B}_j} = 1 - \sum_w \mathfrak{f}_{j,w}$, with the sum in $[0,1)$ on the interior. The letter Ψ_j is the basis polynomial of the dual table (Proposition 7.2), whose bases $\mathcal{E} \setminus T$ carry

the weights $\theta_{j,T}$, so that its largest weight is $\theta_{j,\max}$; its children are the duals of the children with the same weights (Lemma 14.7(ii) with the identity), and the dual of a direct sum is the direct sum of the duals of its parts with the product weights, so the factoring passes to the dual and (i), (ii) apply to its rounds. Then $K_j^{-s_j} = (r_j!/\theta_{j,\max})^{s_j}(1-\sum_w \mathfrak{f}_{j,w})^{-s_j}$, expanded as in Proposition 19.11 and by the multinomial theorem as in Theorem 19.13, with nonnegative terms; Tonelli's theorem, the independence of the blocks under $\mathrm{Dir}_{\mathbf{m}}$ (Theorem 3.12) and Theorem 3.11 give the series and the form of its expectations. On the balance wall the left side is the period (Theorem 8.10).

(iv) A round that first fails in the subtree of w passes every node above w, which gives the nesting. On a chain every node has one childcule. The table at the node M_k that a round of table j reaches is the child of table j at its trace P_k on S_k, whose largest weight is $\theta_j(S_k;P_k)$ in the notation of Theorem 19.13; so the first factor of $\mathfrak{p}$ at that node is $\theta_j(S_{k+1};P_{k+1})/\theta_j(S_k;P_k)$, the acceptance at S_{k+1}, and passing the node is extending on S_{k+1}. The chances $\mathfrak{f}$ of the two theorems agree, and the series is that of Theorem 19.13. □

Proposition 19.15 (Rates at the centre). *As stated, for every family whose active tables are connected.*

Proof. (i) is Proposition 8.21(ii), with $c_j(\mathcal{E}\setminus S) = c_j(S)$ (Theorem 7.11).

(ii) The rate is $\varpi = \min_i \varpi_i$ over the letters, ϖ_i the least of $\kappa_S/d_i(S)$ over the nonempty proper S with $d_i(S) > 0$ (Theorem 15.13(ii)); an inactive table has $\sigma_j = 0$ and no letters. A connected table has $c_j(S) \geq 1$ on every nonempty proper S (Definition 8.1), and so no loops and no coloops, a loop or a coloop e having $c_j(\{e\}) = r_j(\{e\}) - d_j(\{e\}) = 0$ (Theorem 7.11) with $\{e\}$ proper as $n \geq 3$. So $\kappa_S = \sum_j \sigma_j c_j(S) \geq \sum_j \sigma_j = \sigma$.

For the letter K_j, $d_j(S) = |S| - r_j^*(S) = r_j - r_j(\mathcal{E}\setminus S) \leq r_j - 1$ (Definition 7.10), as $\mathcal{E}\setminus S$ is nonempty and contains no loop. For Ψ_j, $d_j^*(S) = |S| - r_j(S)$ is the nullity of S, which increases with the set, and $S \subseteq \mathcal{E} - e$ for some e, of nullity $n - 1 - r_j(\mathcal{E}-e) = n - 1 - r_j$, as e is not a coloop. So $\kappa_S/d_j(S) \geq \sigma/(r_j - 1)$ wherever $d_j(S) > 0$, and $\kappa_S/d_j^*(S) \geq \sigma/(n - r_j - 1)$ wherever $d_j^*(S) > 0$.

At $S = \mathcal{E}-e$, every table without loops or coloops has $c_j(\mathcal{E}-e) = c_j(\{e\}) = r_j(\{e\}) - d_j(\{e\}) = 1$ (Theorem 7.11), so $\kappa_{\mathcal{E}-e} = \sigma$; and $d_j(\mathcal{E}-e) = r_j - r_j(\{e\}) = r_j - 1$, $d_j^*(\mathcal{E}-e) = n-1-r_j$. Hence $\varpi_{K_j} = \sigma/(r_j-1)$ when $r_j \geq 2$ and $\varpi_{\Psi_j} = \sigma/(n-r_j-1)$ when $n-r_j \geq 2$, attained at the faces $\mathcal{E}-e$; a letter with no S of positive count does not enter the minimum. As $(r_j-1)+(n-r_j-1) = n-2 \geq 1$, at least one letter of every active table enters, and $\varpi = \sigma/\max_j \max(r_j - 1, n - r_j - 1)$.

(iii) By Definition 7.3, $K_j = K_{j,\setminus e} + u_e K_{j,/e}$ and $\Psi_j = u_e \Psi_{j,\setminus e} + \Psi_{j,/e}$, the deletion and the contraction being tables since e is neither a loop nor a coloop of the active table j. They are homogeneous in $u_{\mathcal{E}-e}$ of the degrees r_j, $r_j - 1$, $n - 1 - r_j$ and $n - r_j$; substituting $u_e = 1 - x$ and $u_{\mathcal{E}-e} = xv$ gives the display, and as $x \to 0$ the terms of $x^{r_j-1}K_{j,/e}(v)$ and $x^{n-r_j-1}\Psi_{j,\setminus e}(v)$ dominate.

(iv) Let β carry the family, with $v'_{\beta(e')} = v_{e'}$ on $\mathcal{E} - e$. It maps the bases of table j onto those of table $h(j)$ with their weights, the bases through e onto those through f, so

$$K_{h(j),/f}(v') = \sum_{T \ni e} \theta_{j,T}\, {v'}^{\beta(T)-f} = \sum_{T \ni e} \theta_{j,T}\, v^{T-e} = K_{j,/e}(v),$$

and in the same way $\Psi_{h(j),\setminus f}(v') = \Psi_{j,\setminus e}(v)$, the bases avoiding e going to those avoiding f. Let β exchange the family: $T \mapsto \mathcal{E}\setminus\beta(T)$ maps the bases of table j through e onto the bases of table $h(j)$ avoiding f, with their weights, so

$$\Psi_{h(j),\setminus f}(v') = \sum_{T \ni e} \theta_{j,T}\, {v'}^{(\mathcal{E}-f)\setminus(\mathcal{E}\setminus\beta(T))} = \sum_{T \ni e} \theta_{j,T}\, {v'}^{\beta(T)-f} = K_{j,/e}(v),$$

and in the same way $K_{h(j),/f}(v') = \Psi_{j,\backslash e}(v)$. Either way the vertex product $K_{j,/e}\Psi_{j,\backslash e}$ goes to $K_{h(j),/f}\Psi_{h(j),\backslash f}$. With one table, a bijection taking e to f that carries or exchanges it exists exactly when e and f lie in one class at the centre, by Corollary 11.17 at the table's gauge, the gauge of the centre. □

Remark. Connectedness of every active table is needed for (ii). On the marks $\{0,1,2,3\}$, for the family of the list whose bases are all the two-element sets and the disconnected list with the bases $\{a,b\}$, $a \in \{0,2\}$, $b \in \{1,3\}$, at the strengths $(1,3)$, the face $S = \{0,2\}$ has $\kappa_S = 1\cdot 2 + 3\cdot 0 = 2$ and $d_2(S) = 1$, so $\varpi \le 2$, while $\sigma/\max_j \max(r_j - 1, n - r_j - 1) = 4$.

Corollary 19.17 (Merged Skeleton values). *As stated, at every pair of exponents and every mass on the balance wall.*

Proof. The letters. With $\theta_{\mathcal{E}\backslash T} = \theta_T$, $\Psi(\alpha) = \sum_T \theta_T \alpha^{\mathcal{E}\backslash T} = \sum_T \theta_{\mathcal{E}\backslash T}\alpha^{\mathcal{E}\backslash T} = K(\alpha)$, the last sum running over the same bases. For a mark e the bases $T \not\ni e$ correspond to the bases $T' = \mathcal{E}\setminus T \ni e$, with $(\mathcal{E}-e)\setminus T = T' - e$ and $\theta_T = \theta_{T'}$, so

$$\Psi_{\backslash e} = \sum_{T\not\ni e} \theta_T\, \alpha^{(\mathcal{E}-e)\backslash T} = \sum_{T'\ni e} \theta_{T'}\, \alpha^{T'-e} = K_{/e}$$

(Definition 7.3), and likewise $\Psi_{/e} = K_{\backslash e}$. The Skeleton values of the scale of e are $K_{\backslash e}/(\alpha_e K_{/e})$ and $\Psi_{/e}/(\alpha_e \Psi_{\backslash e})$ (Corollary 15.3), which coincide.

The table. Bases and complements of bases are both bases, so $r = n - r$. A coloop lies in every basis, hence in no complement of one, and a loop lies in no basis, hence in every complement; both contradict the complementation, so no mark is a loop or a coloop, and the deletion $(\mathcal{B}\setminus e, \theta)$ is a table of rank r on $\mathcal{E} - e$.

The walls at $\{e\}$ and at $\mathcal{E} - e$. $d(\{e\}) = d^*(\{e\}) = 0$, so $\kappa_{\{e\}} = m_e$. Also $d(\mathcal{E}-e) = r - r(\{e\}) = r - 1$ and $d^*(\mathcal{E}-e) = n - 1 - r(\mathcal{E}-e) = n - 1 - r = r - 1$. On the balance wall $m_{\mathcal{E}} = sr + t(n-r) = (s+t)r$, so $\kappa_{\mathcal{E}-e} = m_{\mathcal{E}} - m_e - (s+t)(r-1) = s + t - m_e$. The proper walls being positive, $0 < m_e < s + t$.

Integrating the mark. Let $g \ne e$ and $R = \mathcal{E} - e$. By Theorem 8.10 the period is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}K^{-s}\Psi^{-t} = \alpha^{\mathbf{m}-\mathbf{1}}K^{-(s+t)}$ over the section $\alpha_g = 1$. By Tonelli's theorem the integral over $\alpha_e = x$ may be taken first, at fixed α_R: with $K = K_{\backslash e} + xK_{/e}$, the Beta corner of a linear letter (Proposition 15.1 with one coefficient) gives

$$\int_0^\infty x^{m_e-1}\big(K_{\backslash e} + xK_{/e}\big)^{-(s+t)}dx = \Gamma^{\,2}_{\,1}\!\begin{bmatrix} m_e,\ s+t-m_e \\ s+t \end{bmatrix} K_{\backslash e}^{-(s+t-m_e)}K_{/e}^{-m_e},$$

convergent as $0 < m_e < s + t$, and $K_{/e} = \Psi_{\backslash e}$.

The deletion. What remains is $\Gamma^{\,2}_{\,1}\!\begin{bmatrix} m_e,\ s+t-m_e \\ s+t \end{bmatrix}$ times the integral, over the section $\alpha_g = 1$ of the orthant of R, of

$$\alpha_R^{\mathbf{m}|_R-\mathbf{1}}K_{\backslash e}^{-(s+t-m_e)}\Psi_{\backslash e}^{-m_e}\,d\alpha_{R-g},$$

the family word of the deletion at the exponents $(s+t-m_e, m_e)$. Its balance wall holds: $m_{\mathcal{E}-e} = (s+t)r - m_e$, which is $(s+t-m_e)r + m_e(n-1-r)$ as $n-1-r = r-1$. By Theorem 8.10 the section integral is $P_{\mathcal{B}\backslash e}(\mathbf{m}|_{\mathcal{E}-e}; s+t-m_e, m_e)$; it is finite because P_F is, so the walls of the deletion are positive there. Since $K = \Psi$, $P_F(\mathbf{m}; s, t)$ depends on the exponents only through $s + t$. At $s = t = \sigma$ and $\mathbf{m} = \sigma\mathbf{1}$, $m_e = \sigma$, which is the last identity. □

Theorem 19.18 (The scale integral of one mark). *As stated; parts (i) and (ii) are stated for every family.*

Proof. (i) By Theorem 15.2 with $B = \{e\}$ and $m_B = m_e$, at the gauge (α_e, α_R),

$$\mathrm{Sc}_{\{e\}}(\alpha; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_0^\infty x^{m_e - 1} \prod_j K_j(x\alpha_e, \alpha_R)^{-s_j} \Psi_j(x\alpha_e, \alpha_R)^{-t_j}\, dx,$$

convergent since the walls $\kappa_{\{e\}}$ and $\kappa^{\mathrm{top}}_{\{e\}}$ are positive. At $\alpha_e = 1$, Definition 7.3 gives $K_j = K_{j,\backslash e}(\alpha_R) + xK_{j,/e}(\alpha_R)$ and $\Psi_j = x\Psi_{j,\backslash e}(\alpha_R) + \Psi_{j,/e}(\alpha_R)$ for every active table, the deletion and the contraction being tables on R as e is neither a loop nor a coloop there. At $\alpha_R = e^{2z_R}$, Theorem 7.13(i) for these tables gives $K_{j,\star}(\alpha_R) = e^{\langle \mathbf{1}, z_R\rangle} \Lambda_{j,\star}(z_R)$ and $\Psi_{j,\star}(\alpha_R) = e^{\langle \mathbf{1}, z_R\rangle} \Lambda_{j,\star}(-z_R)$ for $\star \in \{\backslash e, /e\}$. So

$$K_j = e^{\langle \mathbf{1}, z_R\rangle} \big(\Lambda_{j,\backslash e}(z_R) + x\Lambda_{j,/e}(z_R)\big),$$
$$\Psi_j = e^{\langle \mathbf{1}, z_R\rangle} \big(\Lambda_{j,/e}(-z_R) + x\Lambda_{j,\backslash e}(-z_R)\big),$$

and the powers carry the factor $e^{-\sum_j (s_j + t_j)\langle \mathbf{1}, z_R\rangle} = e^{-2\sigma\langle \mathbf{1}, z_R\rangle}$, an inactive table contributing 1. This is the display, and the rest of the integrand depends on the room only through the functions $\Lambda_{j,\backslash e}$ and $\Lambda_{j,/e}$. Corollary 15.7 at the bunch $\{e\}$, whose shape simplex is the point $u_e = 1$, gives $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \alpha_R^{\mathbf{m}|_R - \mathbf{1}} \mathrm{Sc}_{\{e\}}((1, \alpha_R); \mathbf{m}; \mathbf{s}, \mathbf{t})\, d\alpha_{R-g}$ on the section $\alpha_g = 1$, $g \in R$. With $\alpha_R = e^{2z_R}$, $\alpha_R^{\mathbf{m}|_R - \mathbf{1}} d\alpha_{R-g} = 2^{n-2} e^{2\langle \mathbf{m}|_R, z_R\rangle} dz_{R-g}$ there, $R - g$ having $n - 2$ marks.

(ii) A realization that is an atom is a realization of a molecule (Definition 10.1), so F is connected, and Theorem 11.4 gives $\alpha^\star \propto \mathbf{1}$. By Theorem 17.1(ii), in $y = \log u$ modulo $\mathbb{R}\mathbf{1}$ the law $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ has a density proportional to $e^{\langle \boldsymbol{\tau}, y\rangle - \Phi(y)}$. The target is the usage at $\alpha^\star$ (Definition 8.16),

$$\tau_e = \sum_j \big(s_j R_{j,e}(\alpha^\star) + t_j R^*_{j,e}(\alpha^\star)\big) = \partial_{y_e} \Phi(y^\star), \qquad y^\star = \log \alpha^\star,$$

with the potential Φ of Definition 8.17. With $y = y^\star + v$ the density is proportional to

$$\exp\big(\langle \nabla\Phi(y^\star), v\rangle - \Phi(y^\star + v)\big).$$

By Proposition 10.4 there are orders $\mathbf{g}_e$ of the rooms in which the potential written from e at the gauge $\alpha^\star$ (Definition 10.1),

$$(v_0, v_1, \ldots, v_{n-1}) \longmapsto \Phi_{\mathbf{X}}\Big(y^\star + v_0 \mathbf{1}_e + \sum_k v_k \mathbf{1}_{g_k}\Big) \qquad (\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}),$$

is one pair of functions for every e. Hence $v \mapsto \Phi(y^\star + v)$ read in the coordinates $(v_e, v_{g_1}, \ldots, v_{g_{n-1}})$ is one function for every e, and so is its gradient at 0; the density read in these coordinates is one function for every e, and dv is invariant under the renaming. The density is invariant along $\mathbf{1}$, so it is a function of the differences $v_{g_k} - v_e$, and the law of these differences does not depend on e. As $y^\star$ is a multiple of $\mathbf{1}$, $y_{g_k} - y_e = v_{g_k} - v_e$.

(iii) The reference (proof of thm:atomforward) proves this part, a statement for one table. □

Theorem 19.21 (The junction under level form). *As stated, for every family at the centre.*

Proof. Parts (ii) and (iii) concern the Gramma $\mathsf{G}_{0;(\chi_p)_p}((\upsilon_{|p|})_p)$ alone and are proved in the reference (proof of thm:uniformjunction, parts (ii) and (iii)).

The preamble. By Proposition 19.6(iii) for every table, every half-sum has level form and the square of every difference is unchanged by the permutations of B; $q_{j,b-k}(-z_R) = q_{j,k}(z_R)$ is Proposition 19.6(i). On z_B each $(\partial_w L_{j,w})^2$ has exponents in $\{-2,0,2\}^B$, being a sum of products of two terms $e^{\langle z_B,\chi_p\rangle}$. A function of z_B of this kind that is unchanged by the permutations of B is a combination of the products $e^{-2\langle \mathbf{1},z_B\rangle}e_k(e^{2z_B})e_{k'}(e^{2z_B})$, $k \le k'$: such a product is the symmetric sum of the monomials with k exponents 2 and $b-k'$ exponents -2, plus terms with fewer of both, and induction removes the terms (the reference, proof of thm:uniformjunction, part (i)). The coefficients are Laurent polynomials in e^{z_R}, the $\Lambda_{j,(p)}$ being such. The product over j of $(\partial_w L_{j,w})^{2l_j}$ therefore expands in products of $2|\mathbf{l}|$ of the e_k, with Laurent polynomial coefficients.

(i) At the centre $\boldsymbol{\mu} = 0$, and the first step of the proof of Theorem 19.4 gives

$$P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma}) = 2^{n-1}\int_{\mathbb{R}^{\mathcal{E}-g}} \prod_j \big(\Lambda_j(z)\Lambda_j(-z)\big)^{-\sigma_j}\, dz_{\mathcal{E}-g}\Big|_{z_g=0}.$$

For each table $\Lambda_j(z)\Lambda_j(-z) = L_{j,1/2}(z)^2 - \frac14(\partial_w L_{j,w}(z))^2$, and Proposition 19.1 with $a = \Lambda_j(z)$, $b = \Lambda_j(-z)$, $s = t = \sigma_j$, the midpoint $L_{j,1/2}(z)$ and $x = \partial_w L_{j,w}(z)/(2L_{j,1/2}(z)) \in (-1,1)$ gives

$$\big(\Lambda_j(z)\Lambda_j(-z)\big)^{-\sigma_j} = \sum_{l_j\ge 0} \frac{(\sigma_j)_{l_j}}{l_j!\,4^{l_j}}\big(\partial_w L_{j,w}(z)\big)^{2l_j} L_{j,1/2}(z)^{-2\sigma_j-2l_j},$$

a series of nonnegative terms. Open each power of the half-sum,

$$L_{j,1/2}^{-2\sigma_j-2l_j} = \Gamma(2\sigma_j+2l_j)^{-1}\int_0^\infty y_j^{2\sigma_j+2l_j-1}e^{-y_jL_{j,1/2}}\,dy_j;$$

by Tonelli's theorem the sums and integrals may be taken in the order of the statement. With $\upsilon_k = \sum_j y_j q_{j,k}(z_R)$,

$$\sum_j y_j L_{j,1/2}(z) = \sum_k \upsilon_k\, e^{-\langle\mathbf{1},z_B\rangle}e_k\big(e^{2z_B}\big),$$

and inserting $e^{-\langle\mathbf{1},z_B\rangle}e_k(e^{2z_B})$ under the exponential of minus this sum is $-\partial_{\upsilon_k}$. With the expansion of the preamble and Definition 2.1 (index 0, the integral over z_B),

$$\int_{\mathbb{R}^B}\prod_j\big(\partial_w L_{j,w}(z)\big)^{2l_j}e^{-\sum_j y_jL_{j,1/2}(z)}\,dz_B \\ = \sum_{|\mathbf{h}|=2|\mathbf{l}|} c_{\mathbf{h}}^{(\mathbf{l})}(z_R)\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\,\mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)\Big|_{\boldsymbol{\upsilon}=\sum_j y_j\mathbf{q}_j(z_R)}.$$

Each term converges. The family being a molecule, some active table j has $c_j(B) \ge 1$ (Definition 8.1), so its patterns at B fill at least two consecutive levels $k, k+1$ (Theorem 12.6), and $q_{j,k}, q_{j,k+1} > 0$, each being a sum of nonnegative terms one of which is positive; by $q_{j,b-k}(-z_R) = q_{j,k}(z_R)$ also $q_{j,b-k}, q_{j,b-k-1} > 0$. The vectors χ_p of these four levels are closed under negation, $\chi_{B\setminus p} = -\chi_p$, and span $\mathbb{R}^B$, since $\chi_{p\cup\{e\}} - \chi_p = 2\mathbf{e}_e$ for $|p| = k$, $e \notin p$; so they positively span $\mathbb{R}^B$, the exponential of $-\sum_j y_jL_{j,1/2}(z)$ decays in z_B faster than every exponential, and the inserted exponentials are integrable against it.

(iv) For each table the swap τ of e and f keeps B, fixes R and exchanges table j, the bijection of the tables being the identity. Proposition 19.5 gives $c_{j,\tau(p)}(1-w;z_R) = c_{j,p}(w;z_R)$. At

$w=\frac12$, $c_{j,\{f\}}(\frac12;\cdot)=c_{j,\{e\}}(\frac12;\cdot)$; in the derivative, $\partial_w c_{j,\varnothing}=\partial_w c_{j,B}=0$ and $\partial_w c_{j,\{f\}}=-\partial_w c_{j,\{e\}}$. With $c_{j,B}(\frac12;z_R)=c_{j,\varnothing}(\frac12;-z_R)$ (Proposition 19.6(i)) this gives the two displayed forms. Evenness: $c_{j,\{e\}}(\frac12;-z_R)=c_{j,\{f\}}(\frac12;z_R)=c_{j,\{e\}}(\frac12;z_R)$ and $\partial_w c_{j,\{e\}}(w;-z_R)=-\partial_w c_{j,\{f\}}(w;z_R)=\partial_w c_{j,\{e\}}(w;z_R)$, by Proposition 19.6(i). In (i) change (z_e,z_f) to (z_+,z_-), with $dz_e\,dz_f=\frac12 dz_+dz_-$. Then $\prod_j(\partial_w L_{j,w})^{2l_j}=(e^{z_-}-e^{-z_-})^{2|\mathbf{l}|}\prod_j(\partial_w c_{j,\{e\}})^{2l_j}$ and

$$\sum_j y_j L_{j,1/2}(z)=e^{-z_+}C_\varnothing(z_R)+e^{z_+}C_\varnothing(-z_R)+\big(e^{z_-}+e^{-z_-}\big)C_e(z_R).$$

The integral over z_+ is $\mathsf{G}_{0;(1,-1)}\big(C_\varnothing(-z_R),C_\varnothing(z_R)\big)$ (Definition 2.1 on one scale e^{z_+}), and the integral over z_- is Lemma 19.20 with $\xi=C_e(z_R)$ and the power $|\mathbf{l}|$, $4^{|\mathbf{l}|}(\frac12)_{|\mathbf{l}|}C_e^{-|\mathbf{l}|}\mathsf{G}_{|\mathbf{l}|;(1,-1)}(C_e,C_e)$. The constants collect to $2^{n-1}\cdot\frac12\cdot4^{|\mathbf{l}|}\prod_j 4^{-l_j}=2^{n-2}$, which is the display; every term is nonnegative. □

Remark. With one table, $(\frac12)_l(\sigma)_l/(l!\,\Gamma(2\sigma+2l))=\frac{1}{l!}\Gamma\!\begin{bmatrix}\sigma+l,\ \frac12+l\\ 2\sigma+2l,\ \sigma,\ \frac12\end{bmatrix}$ and $y^{2\sigma+2l-1}(yc_{\{e\}})^{-l}=y^{2\sigma+l-1}c_{\{e\}}^{-l}$, which is the reference's form of (iv).

Theorem 19.36 (A series connection closes at its reach). *As stated, for series connections and for series connections of the dual table.*

Proof. (i) By Proposition 19.35(i), $f_B=C^{-s}C'^{-t}\prod_i f_i^\circ$, and by Proposition 19.35(ii), which is Lemma 19.28 applied $q-1$ times, f_B has a lattice expansion of reach $(\sum_i\sqrt{x_i})^2$ whose terms are the products of the terms of the parts collected by exponent. A function has at most one lattice expansion (Definition 19.26), so these are the coefficients $\beta_{\eta,j}$ of Theorem 19.32, and $\operatorname{reach}(B)\le(\sum_i\sqrt{x_i})^2<\inf\varrho_R$, the reach being an infimum. Theorem 19.32(ii) applies: the even share closes toward the level d on the lattice of continued periods of the bunch family, with the bound for every $x>\operatorname{reach}(B)$, in particular for every $x>(\sum_i\sqrt{x_i})^2$. Under the hypotheses of Proposition 19.35(iii), $\operatorname{reach}(B)=(\sum_i\sqrt{\sup\varrho_i})^2$, and Theorem 19.38(ii) gives the root rate $\operatorname{reach}(B)/\inf\varrho_R$. For a series bunch the parts are the marks (Proposition 19.35), with

$$\begin{aligned}f_e^\circ(y)&=\int_0^\infty x^{m_e-s-1}e^{-w_e/x-yw_ex}\,dx=\mathsf{G}_{m_e-s;(1,-1)}(yw_e,w_e)\\&=(yw_e)^{s-m_e}\mathsf{G}_{m_e-s;(1,-1)}\big(1,w_e^2y\big)\end{aligned}$$

by the torus homogeneity of the Gramma; by Lemma 19.27(iii) it has a lattice expansion of reach w_e^2, a power of y keeping the reach (Lemma 19.28). Proposition 19.34(iii) gives $\operatorname{reach}(B)=\inf\varrho_B=(\sum_e w_e)^2$ exactly, and Proposition 19.34(i) gives the coefficients.

(ii) The reference (proof of thm:classcloses, part (ii)) gives the argument for a parallel bunch, and it applies to a connection. The dual table at the exponents (t,s) has the same period (Proposition 8.14). Since $K^*=\Psi$ and $\Psi^*=K$ (Proposition 7.2), the dual node has the bunch B, of width one in the dual table, the room R, the ratios $1/\varrho_B$ and $1/\varrho_R$, and its levels swapped: its level d carries the even letter of the level r of the given node. No slope on the room being negative, $\sup\varrho_R<\infty$ (Lemma 18.15), so $\inf(1/\varrho_R)>0$ and no slope on the room of the dual node is positive (the same lemma for the dual node). Part (i) for the dual node, whose bunch is a series connection with the reaches x_i^*, closes the even share toward its level d, the level r of the given node, when $(\sum_i\sqrt{x_i^*})^2<1/\sup\varrho_R$, at a root rate at most $(\sum_i\sqrt{x_i^*})^2\sup\varrho_R$. For a parallel bunch, the dual pattern weights are $\theta'^*_B=\theta'_\varnothing$ and $\theta'^*_{B-e}=\theta'_{\{e\}}$, so the w_e of the dual series bunch are $\theta'_{\{e\}}/\theta'_\varnothing$, $x_e^*=w_e^2$, and the root rate is exact by (i). The ratio of the bunch of

the dual node is $1/\varrho_B = (\sum_e w_e/u_e)(\sum_e w_e u_e)$, whose least value $(\sum_e w_e)^2$ is taken at the point of equal coordinates of Δ_B (Proposition 19.34(ii)); so $\sup \varrho_B = (\sum_e w_e)^{-2}$, the condition reads $\sup \varrho_R < \sup \varrho_B$, and the root rate is $(\sum_e w_e)^2 \sup \varrho_R = \sup \varrho_R / \sup \varrho_B$. □

Theorem 19.57 (One tree suffices). *As stated; the chain clause of (iii) is stated whenever some active list is connected.*

Proof. All of the theorem but the chain clause of (iii) is the reference's (proof of thm:onetree). For the chain clause, let some active list of the molecule F be connected. The weights of lists factor across every split, so Corollary 14.23 applies: every peel leaves one childcule, the Foreman tree is a chain (Corollary 19.8), and at every realization, the centre and every lean, $|B_0| \le |B_1| \le \cdots \le |B_{d-1}| \le |L|$, L the leaf. The shape of the bunch B_k then has $|B_k| - 1 \le |L| - 1$ integrations, and its node adds one, the total of the bunch against its room, by (ii). A molecule of one list is such an F, its list being connected. □

40.20 Chapter 20: gluing and what crosses it

Lemma 20.8 (Two-sums of cells; hinge walls). *As stated, for cell families glued table by table.*

Proof. That $\mathcal{A} \oplus_g \mathcal{A}'$ is a table of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, plain when the cells are, in which B is a bunch of width one with the cells $\mathcal{A}$ and $\mathcal{A}'$, and that a table with a bunch of width one and factoring weights is the two-sum of its two cells, is the reference, proof of lem:twosumpieces(i), the weights multiplied letter by letter. The two-sum is $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}, \mathcal{A}' \rangle$: the bases $\{h_1\}$ and $\{h_2\}$ of the core give the two kinds of sets of 20.2 (20.1). So F is substituted table by table, with the core $\binom{\{h_1,h_2\}}{1}$ in every table and the cells $\mathcal{A}_j$ at h_1 and $\mathcal{A}'_j$ at h_2, and 20.7 at $S = B$ gives the wall $m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big) = \kappa_{F_B}$, the hinge wall of 22.1, and at $S = R$ the wall κ_{F_R}. Table j of F has the rank $r_j = r_{\mathcal{A}_j} + r_{\mathcal{A}'_j} - 1$ on $n = |B| + |R|$ marks, so

$$\kappa_{F_B} + \kappa_{F_R} = m_{\mathcal{E}} - \sum_j \big(s_j(r_j - 1) + t_j(n - r_j - 1)\big) = \kappa_{\mathcal{E}} + \sum_j (s_j + t_j) = \kappa_{\mathcal{E}} + 2\sigma.$$

With the mass $2\sigma - \kappa_{F_B}$ on g, the imbalance of F_B on $B \sqcup \{g\}$ is

$$m_B + 2\sigma - \kappa_{F_B} - \sum_j \big(s_j r_{\mathcal{A}_j} + t_j(|B| + 1 - r_{\mathcal{A}_j})\big) = 2\sigma - \sum_j (s_j + t_j) = 0,$$

on the balance wall of F or off it. □

Theorem 20.11 (Splicing hinges in pairs). *As stated; (iii) holds for any number of cells and any graph of pairs.*

Proof. (i) and (ii) are the reference, proof of thm:hingepairs(i),(ii). (iii) In the direct sum of the cells the letters are $K = \prod_i K_{\mathcal{A}_i}$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i}$ (7.8), and by 20.10, with $G = \bigsqcup_i G_i$,

$$K = \sum_{(p_i)_i} \alpha_G^{\sqcup_i p_i} \prod_i K_{\mathcal{A}_i,(p_i)}, \qquad \Psi = \sum_{(p_i)_i} \alpha_G^{G \setminus \sqcup_i p_i} \prod_i \Psi_{\mathcal{A}_i,(p_i)},$$

over $p_i \subseteq G_i$, absent channels counting as zero. Splicing a pair $\{a, b\}$ acts on a monomial α_G^P by $(\partial_{\alpha_a} + \partial_{\alpha_b})\alpha_G^P|_{\alpha_a = \alpha_b = 0}$, which is α_G^{P-a} when $a \in P \not\ni b$, α_G^{P-b} when $b \in P \not\ni a$, and 0 otherwise;

on $\alpha_G^{G\setminus P}$ it is likewise nonzero exactly when P contains one of a, b. The splicings of disjoint pairs commute and act on different variables, so splicing all pairs keeps the terms in which $P = \sqcup_i p_i$ is $p \sqcup J$ with $p \subseteq G'$ and J containing exactly one hinge of every pair, and leaves $\alpha_{G'}^{p}$ in K and $\alpha_{G'}^{G'\setminus p}$ in Ψ. By (ii) the result is the letters of the spliced cell $\mathcal{A}$, and by the Taylor cube (12.6) its channels are the coefficients of these monomials:

$$K_{\mathcal{A},(p)} = \sum_J \prod_i K_{\mathcal{A}_i,((p\sqcup J)\cap G_i)}, \qquad \Psi_{\mathcal{A},(p)} = \sum_J \prod_i \Psi_{\mathcal{A}_i,((p\sqcup J)\cap G_i)}.$$

The marks other than hinges are $\bigsqcup_i B_i$, and the rank is $\sum_i r_{\mathcal{A}_i} - k$ by (ii). The degrees add up: $\sum_i \big(r_{\mathcal{A}_i} - |(p\sqcup J)\cap G_i|\big) = \sum_i r_{\mathcal{A}_i} - k - |p|$, since $|J| = k$, and $\sum_i \big(n_{\mathcal{A}_i} - r_{\mathcal{A}_i} + |(p\sqcup J)\cap G_i|\big) = n_{\mathcal{A}} - r_{\mathcal{A}} + |p|$, since $n_{\mathcal{A}} - r_{\mathcal{A}} = \sum_i (n_{\mathcal{A}_i} - r_{\mathcal{A}_i}) + k$. Each channel of $\mathcal{A}$ is a sum of products of one channel of each cell, so the map is multilinear. For two cells with the pairs named by I, a set J is given by the set $I' \subseteq I$ of the pairs whose hinge on the side of $\mathcal{A}_1$ it contains, $(p\sqcup J)\cap G_1 = p_1 \sqcup I'$ and $(p\sqcup J)\cap G_2 = p_2 \sqcup (I \setminus I')$: the bilinear map displayed. With $k = 1$ and no other hinges, $K_{\mathcal{A},(\varnothing)} = K_{\mathcal{A}_1\setminus g_1} K_{\mathcal{A}_2/g_2} + K_{\mathcal{A}_1/g_1} K_{\mathcal{A}_2\setminus g_2}$ and $\Psi_{\mathcal{A},(\varnothing)} = \Psi_{\mathcal{A}_1\setminus g_1}\Psi_{\mathcal{A}_2/g_2} + \Psi_{\mathcal{A}_1/g_1}\Psi_{\mathcal{A}_2\setminus g_2}$, the letters of 20.5(iii). □

Remark. In exact arithmetic (receipt `check_blocks_splice.py`): three cells in a closed path, a weighted $\binom{\{x_1,x_2\}\sqcup G_1}{3}$ with three hinges, one of them unpaired, a weighted $\binom{\{y_1,y_2\}\sqcup G_2}{2}$ and a weighted $\binom{\{z_1\}\sqcup G_3}{1}$ with two hinges each, spliced along three pairs, one between each two cells: both channels of the spliced cell at both values of the unpaired hinge are the sums over J of the products of the channels of the three cells.

Theorem 20.14 (Multivectors bound what crosses). *As stated; (iii) holds at any tensor rank.*

Proof. (i) and (ii) are the reference, proof of thm:cutrank(i),(ii), with the interface rank for the cut rank; (ii) is proved there at any tensor rank, the interface rank being subadditive. (iii) By (i) and (ii), $\operatorname{rank}\Theta_B^{(k)} \le \operatorname{ir}_k(\boldsymbol{\omega})^2 \le i_k^2$ with $i_k = \sum_l \binom{c_l}{k-d_l}$, and summing over the levels bounds the interface dimension $\sum_k \operatorname{rank}\Theta_B^{(k)}$ (20.12); for a real multivector (i) gives $\operatorname{rank}\Theta_B^{(k)} \le \frac{1}{2}\operatorname{ir}_k(\boldsymbol{\omega})(\operatorname{ir}_k(\boldsymbol{\omega})+1)$, and $x \mapsto \frac{1}{2}x(x+1)$ increases on $[0,\infty)$. The bounds involve only the widths and the floors of the slices. A determinantal table has a multivector of tensor rank one, $\boldsymbol{\omega}_T = \det A_T$ (7.17); a set T is a basis exactly when $\det A_T \neq 0$, so the column space of A_S has the dimension $r(S)$, the column spaces of A_B and A_R span $\mathbb{F}^r$ and meet in dimension $r(B) + r(R) - r = c(B)$, and $d_1 = r - r(R) = d(B)$ (7.10). So $\operatorname{ir}_k(\boldsymbol{\omega}) \le \binom{c(B)}{k-d(B)}$, and $\sum_k \binom{c(B)}{k-d(B)}^2 = \sum_i \binom{c(B)}{i}^2 = \binom{2c(B)}{c(B)}$. At width one $\binom{1}{k-d(B)} = 1$ at $k = d(B)$ and $k = d(B)+1$: one bunch letter at each level. □

40.21 Chapter 21: periods of constructions

Definition 21.1 (Twists). As stated: the torus relation for every twist, every vector $\boldsymbol{\lambda}$ on the marks and all constants c_j, c_j^*.

Proof. Put $\mathbf{x}' = (c_j\boldsymbol{\lambda}\circ\mathbf{x}_j^K,\ c_j^*\boldsymbol{\lambda}\circ\mathbf{x}_j^\Psi)_j$. Every basis of table j has r_j marks and its complement r_j^*, so K_j is homogeneous of degree r_j and Ψ_j of degree r_j^*, and the letters of $F[\mathbf{x}']$ at α are

$$K_j(c_j\boldsymbol{\lambda}\circ\mathbf{x}_j^K\circ\alpha) = c_j^{r_j} K_j\big(\mathbf{x}_j^K\circ(\boldsymbol{\lambda}\circ\alpha)\big), \qquad \Psi_j(c_j^*\boldsymbol{\lambda}\circ\mathbf{x}_j^\Psi\circ\alpha) = {c_j^*}^{r_j^*}\Psi_j\big(\mathbf{x}_j^\Psi\circ(\boldsymbol{\lambda}\circ\alpha)\big).$$

Write $\omega_{\mathbf{x}} = \beta^{\mathbf{m-1}} \prod_j K_j(\mathbf{x}_j^K \circ \beta)^{-s_j} \Psi_j(\mathbf{x}_j^\Psi \circ \beta)^{-t_j}\, d\beta$ for the form of $F[\mathbf{x}]$. The map $\Phi : \alpha \mapsto \boldsymbol{\lambda} \circ \alpha$ is a linear bijection of the open orthant onto itself that commutes with positive scalings, and with $\beta_e = \lambda_e \alpha_e$ one has $\beta^{\mathbf{m-1}} d\beta = \boldsymbol{\lambda}^{\mathbf{m}} \alpha^{\mathbf{m-1}} d\alpha$. So the form of $F[\mathbf{x}']$ is

$$\omega_{\mathbf{x}'} = \boldsymbol{\lambda}^{-\mathbf{m}} \prod_j c_j^{-r_j s_j} c_j^{*\,-r_j^* t_j}\ \Phi^* \omega_{\mathbf{x}}.$$

A twist does not change degrees, so on the balance wall both forms are homogeneous of degree zero, and each period is the integral of its form over the section $\{h = 1\}$ of any positive function h homogeneous of degree one (8.10). Since $\int_{\{h=1\}} \Phi^* \omega_{\mathbf{x}} = \int_{\{h \circ \Phi^{-1} = 1\}} \omega_{\mathbf{x}}$ and $h \circ \Phi^{-1}$ is again positive and homogeneous of degree one, the two periods differ by the constant factor of the display. The moduli of the two integrands are related by the same change of variables and a positive constant, so the two sides converge together; for real data the integrands are nonnegative and the identity holds in $[0, \infty]$.

At $\boldsymbol{\lambda} = \mu \mathbf{1}$ and $c_j = c_j^* = \mu^{-1}$ every coordinate of $\mathbf{x}'$ equals that of $\mathbf{x}$, and the factor is $\mu^{-m_{\mathcal{E}}} \mu^{\sum_j (r_j s_j + r_j^* t_j)} = 1$, since $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$ on the balance wall. With $\boldsymbol{\lambda}$ equal to 1 off g and to λ at g, and $c_j = c_j^* = 1$, the factor is λ^{-m_g}; with $\lambda = 1/x_1^K$ this reads every twist at g with $x_1^K = 1$, up to the factor $(x_1^K)^{-m_g}$, and for one table it leaves the one number x_1^Ψ / x_1^K, carried by the cobasis letter. With $\boldsymbol{\lambda} = \mathbf{1}$, one c_j or one c_j^* different from 1 and the other constants 1, the factor is $c_j^{-r_j s_j}$ or $c_j^{*\,-r_j^* t_j}$; for one table twisted in its cobasis letter by $\mu \mathbf{x}$ it is $\mu^{-r^* t}$. □

Remark. The reference, 21.1, rescales a twist at one mark, normalized by the scale of the mark, and the relation for one table twisted in its cobasis letter is used with 24.1. The exact check of the scalar case is check 11 of S21-23_work/checks.py.

Theorem 21.4 (The cells ride on the core's fibre). *As stated: at complex data, every exponent* 0 *or of positive real part, on contours on which the load families converge absolutely.*

Proof. The steps are those of the reference, proof of 21.4, with the moduli carried along. By 20.4, table j has the letters

$$K_j = \prod_i K_{\mathcal{A}_{ij}/g_i}(\alpha_{B_i})\, K_{\mathcal{H}_j}\big((\gamma^K_{\mathcal{A}_{ij}}(\alpha_{B_i}))_i\big), \qquad \Psi_j = \prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}(\alpha_{B_i})\, \Psi_{\mathcal{H}_j}\big((\gamma^\Psi_{\mathcal{A}_{ij}}(\alpha_{B_i}))_i\big).$$

For $\operatorname{Re} s_j > 0$, 4.4 opens $K_{\mathcal{H}_j}(X)^{-s_j} = \big(\sum_T \theta_T^{\mathcal{H}_j} X^T\big)^{-s_j}$ with one share $a_{j,T}$ per basis on a contour of positive real parts, and at $X_i = \gamma^K_{\mathcal{A}_{ij}}$ the monomials give $\prod_T (X^T)^{-a_{j,T}} = \prod_i (\gamma^K_{\mathcal{A}_{ij}})^{-a_j(h_i)}$; in the same way $\Psi_{\mathcal{H}_j}(Y)^{-t_j} = \big(\sum_T \theta_T^{\mathcal{H}_j} Y^{\{h\}\setminus T}\big)^{-t_j}$ is opened with the shares $b_{j,T}$, and $\prod_T (Y^{\{h\}\setminus T})^{-b_{j,T}} = \prod_i (\gamma^\Psi_{\mathcal{A}_{ij}})^{-b_j(h_i)}$. A letter with exponent 0 is the constant 1 and is opened by nothing. Since $\gamma^K_{\mathcal{A}} = K_{\mathcal{A}\setminus g}/K_{\mathcal{A}/g}$ and $\gamma^\Psi_{\mathcal{A}} = \Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A}\setminus g}$ (20.1),

$$K^{-s_j}_{\mathcal{A}_{ij}/g_i}(\gamma^K_{\mathcal{A}_{ij}})^{-a_j(h_i)} = K^{-(s_j - a_j(h_i))}_{\mathcal{A}_{ij}/g_i} K^{-a_j(h_i)}_{\mathcal{A}_{ij}\setminus g_i}, \qquad \Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}(\gamma^\Psi_{\mathcal{A}_{ij}})^{-b_j(h_i)} = \Psi^{-(t_j - b_j(h_i))}_{\mathcal{A}_{ij}\setminus g_i} \Psi^{-b_j(h_i)}_{\mathcal{A}_{ij}/g_i},$$

and $e^{-|\alpha|} = \prod_i e^{-|\alpha_{B_i}|}$, so for fixed shares the integrand of J_F is the product over the cells of the integrands of the interests of the load families $F_i(\mathbf{a}, \mathbf{b})$, times the share kernels and the powers of the weights of the cores.

It remains to exchange the share integral and the integral over the orthant. For $X > 0$, $|X^{-a}| = X^{-\operatorname{Re} a}$, and $|\alpha^{\mathbf{m-1}}| = \alpha^{\operatorname{Re} \mathbf{m-1}}$; so the modulus of the joint integrand is

$$\prod_j |\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix}|\, |\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix}| \prod_{j,T} \big(\theta_T^{\mathcal{H}_j}\big)^{-\operatorname{Re}(a_{j,T} + b_{j,T})}$$

times the product of the integrands of the interests of the load families at the real parts of their data and exponents. Integrated over the orthant, the second factor gives the product of these interests at the real parts, which are finite by the hypothesis and constant along the contour, since they depend on the shares only through their real parts. On the contour, $a_{j,T} = c_{j,T} + i\tau_{j,T}$ with $c_{j,T} > 0$ fixed and $\sum_T \tau_{j,T} = \operatorname{Im} s_j$, and the classical bound $|\Gamma(c+i\tau)| \le C_c\, e^{-\pi|\tau|/2}(1+|\tau|)^{c-1/2}$ makes $\prod_T |\Gamma(a_{j,T})|$ integrable over the imaginary parts of all shares but one; the same holds for the shares $b_{j,T}$. So the joint integrand is absolutely integrable, the integrals may be taken in either order, and the display follows. The identity $J_{F_i(\mathbf{a},\mathbf{b})} = \Gamma(\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i)))\, P_{F_i(\mathbf{a},\mathbf{b})}$ is 8.10 at the imbalance of the load family: its tables $\mathcal{A}_{ij} \setminus g_i$ and $\mathcal{A}_{ij}/g_i$ have the degrees $r_{\mathcal{A}_{ij}}$, $|B_i| - r_{\mathcal{A}_{ij}}$, $r_{\mathcal{A}_{ij}} - 1$ and $|B_i| - r_{\mathcal{A}_{ij}} + 1$, and the masses $\mathbf{m}|_{B_i}$ exceed their degree by $\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i))$, with κ_{B_i} of 20.7. □

Theorem 21.5 (A cell read alike is pinned). *As stated: the hypothesis on every active letter, whatever the sign of its exponent.*

Proof. By 21.3, which holds for every substituted family at all data, both sides converging together,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \prod_i \beta_i(du_i)\; P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big),$$

β_i the cell kernels and $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ the core family read at u. A letter with exponent 0 is the constant 1 in every period, so the twists of the inactive letters do not enter the period of the core family read at u; give them the values $c_{ij} = 1$, $c^*_{ij} = 1$. Under the hypothesis the twists of the active letters at h_i are $c_{ij}\gamma_i(u_i)$ in K_j and $c^*_{ij}\gamma_i(u_i)$ in Ψ_j, so $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ has the period of $F_{\mathcal{H}}[\mathbf{c}]$ twisted further by the vector $\boldsymbol{\lambda} = (\gamma_i(u_i))_i$ on the marks of the core, in both letters of every table. The core family read at u has the imbalance of F (21.3), so it is on its balance wall, and the torus relation of 21.1 with this $\boldsymbol{\lambda}$ and constants 1 gives

$$P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big) = \prod_i \gamma_i(u_i)^{-\kappa_{B_i}}\, P_{F_{\mathcal{H}}[\mathbf{c}]}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big).$$

Inserting this and integrating cell by cell gives the display, by Tonelli for real data and by Fubini where the integrals converge absolutely. The step is the reference proof of 21.5, which nowhere uses the sign of an exponent; the hypothesis there is written for the letters with positive exponent. □

Corollary 21.6 (Read by one hinge value, the period factors through the core). *As stated: for every substituted family with real data at every cell of which the active letters read the cell by one hinge value.*

Proof. (1) The first display. The hypothesis is that of 21.5 with all constants $c_{ij} = c^*_{ij} = 1$, and $F_{\mathcal{H}}[\mathbf{c}] = F_{\mathcal{H}}$; 21.5 gives the first display, in $[0,\infty]$ since the data are real. At a mark h left in place, the cell $\binom{\{h,g\}}{1}$ has both hinge values equal to α_h (20.1), so the hypothesis holds there with $\gamma = \alpha_h$.

(2) The cells. The cell family F_i has the tables $\mathcal{A}_{ij}$, cells on $|B_i|+1$ marks of rank $r_{\mathcal{A}_{ij}}$, and by 20.7

$$\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big),$$

so $m_{B_i} + 2\sigma - \kappa_{B_i} = \sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| + 1 - r_{\mathcal{A}_{ij}})\big)$, the degree of the family word of F_i: at the masses $(\mathbf{m}|_{B_i}, 2\sigma - \kappa_{B_i})$ it is on its balance wall. Read its period on the section $\{|\alpha_{B_i}| = 1\}$ (8.10), with $\alpha_{B_i} = u \in \Delta_{B_i}$ and $\alpha_{g_i} = v > 0$. By 20.1, $K_{\mathcal{A}_{ij}} = K_{\mathcal{A}_{ij}/g_i}(u)\big(\gamma^K_{\mathcal{A}_{ij}}(u) + v\big)$ and $\Psi_{\mathcal{A}_{ij}} =$

$\Psi_{\mathcal{A}_{ij}\setminus g_i}(u)\big(v+\gamma^{\Psi}_{\mathcal{A}_{ij}}(u)\big)$. Under the hypothesis every active letter carries the factor $(v+\gamma_i(u))$ to the power of its exponent, and an inactive letter has exponent 0; so the word of F_i is the density of the cell kernel without $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$ times $(v+\gamma_i(u))^{-2\sigma}$, $2\sigma=\sum_j(s_j+t_j)$. For $0<\kappa<2\sigma$ and $\gamma>0$, the substitution $v=\gamma w$ and the Beta integral give

$$\int_0^\infty v^{2\sigma-\kappa-1}(v+\gamma)^{-2\sigma}\,dv=\gamma^{-\kappa}\int_0^\infty w^{2\sigma-\kappa-1}(1+w)^{-2\sigma}\,dw=\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}\kappa,\ 2\sigma-\kappa\\ 2\sigma\end{matrix}\right]\gamma^{-\kappa},$$

the Beta corner of a linear letter (15.1 with one coefficient). Integrating v first (Tonelli),

$$P_{F_i}\big((\mathbf{m}|_{B_i},2\sigma-\kappa_{B_i});\mathbf{s},\mathbf{t}\big)=\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}\kappa_{B_i},\ 2\sigma-\kappa_{B_i}\\ 2\sigma\end{matrix}\right]\int_{\Delta_{B_i}}\gamma_i^{-\kappa_{B_i}}\,d\beta_i,$$

which identifies the pinned shape integral.

(3) The walls. The marks of a core are neither loops nor coloops (20.1), so a core has at least two marks, $q\ge 2$, and B_i is a nonempty proper set of marks; it has width one in every table (20.3(i)). Width one in table j means $d_j(B_i)+d_j(\mathcal{E}\setminus B_i)=r_j-1$ and $d_j^*(B_i)+d_j^*(\mathcal{E}\setminus B_i)=r_j^*-1$, so

$$\kappa_{B_i}+\kappa_{\mathcal{E}\setminus B_i}=m_{\mathcal{E}}-\sum_j\big(s_j(r_j-1)+t_j(r_j^*-1)\big)=\kappa_{\mathcal{E}}+2\sigma,$$

which is 2σ on the balance wall (20.8; check 2 of S21-23_work/checks.py). If every proper wall is positive, $\kappa_{B_i}>0$ and $\kappa_{\mathcal{E}\setminus B_i}>0$, so $0<\kappa_{B_i}<2\sigma$.

(4) The second display. Multiply the first display by $\prod_i\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}\kappa_{B_i},\ 2\sigma-\kappa_{B_i}\\ 2\sigma\end{matrix}\right]\in(0,\infty)$ and use step (2). The left side is finite (8.10), every factor on the right lies in $(0,\infty]$, so every factor is finite. The core family at $(\kappa_{B_i})_i$ has the imbalance of F (21.3), the cell families are balanced by step (2), and a family on its balance wall with a finite period has every proper wall positive (8.10).

(5) One cell. With one cell at h_1 and the other marks of the core left in place, the left-in-place cell family at h, the list $\binom{\{h,g\}}{1}$ in every table with the masses $(m_h,2\sigma-m_h)$, has the word $(\alpha_h+\alpha_g)^{-2\sigma}$ and the period $\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}m_h,\ 2\sigma-m_h\\ 2\sigma\end{matrix}\right]$, while $\kappa_{\{h\}}=m_h$; these factors cancel the Beta corners of the marks left in place. The core family is then the family of the cells of the room at the masses $(\kappa_B,\mathbf{m}|_R)$ and F_1 that of the cells of the bunch (20.3(iii)), and the second display is the identity of 18.10. In the first display the marks left in place contribute 1, their simplex being a point and their kernel the constant 1, and it is the identity of 18.12 with all constants 1. □

Remark. The reference proof (21.6, one table at $(0,t)$) uses of the full lean only that the letter which sees the cells sees each through one function. A family at mixed full leans is covered: on $B=\{1,2\}$ let table 1 at $(s_1,0)$ carry the series block $\binom{\{1,2,g\}}{2}$ and table 2 at $(0,t_2)$ the parallel block $\binom{\{1,2,g\}}{1}$. Both active letters read B through e_2/e_1 (20.5(i)), $\kappa_B=m_B-2\sigma$, the cell family has the word $e_2(\alpha_1,\alpha_2,\alpha_g)^{-2\sigma}$ and the period $\Gamma(2\sigma-m_1)\Gamma(2\sigma-m_2)\Gamma(\kappa_B)/\Gamma(2\sigma)$ (25.10), and the pinned shape integral is $\int_0^1 u^{m_1-1}(1-u)^{m_2-1}(u(1-u))^{-\kappa_B}du=\Gamma(2\sigma-m_1)\Gamma(2\sigma-m_2)/\Gamma(4\sigma-m_B)$, whose quotient is $\Gamma_{\,1}^{\,2}\!\left[\begin{matrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{matrix}\right]$, as step (2) says. At $(2\sigma,m_1,m_2)=(3,2,2)$ the period is $\frac12$, and the identity holds exactly there and at $(\frac72,\frac32,\frac52)$ (checks 7a to 7d of S21-23_work/checks.py).

Corollary 21.7 (Periods along a hinge tree read by one hinge value). *As stated: for a family whose tables have a common hinge tree, at real data, wherever the active letters read every subtree cell by one hinge value.*

Proof. Induction on the number of edges of $\mathfrak{T}$. With one vertex there is nothing to prove.

Removing a leaf. A tree with an edge has two leaves, so some leaf v differs from o; let p be its parent and $\mathfrak{e} = \mathfrak{e}_v$. The subtree cell $\mathcal{A}_{v,j}$ is the node $\mathcal{H}_{v,j}$ with the hinge $g_{v,\mathfrak{e}}$ (22.14). The two-sums along a tree may be taken in any order (22.10, 20.3(i),(ii)), so table j of F is the substitution $\mathcal{H}'_j\langle\mathcal{A}_{v,j}\rangle$ of the cell $\mathcal{A}_{v,j}$ at the mark $g_{p,\mathfrak{e}}$ of the table $\mathcal{H}'_j$ obtained from the other nodes along $\mathfrak{T}-v$, every other mark of $\mathcal{H}'_j$ left in place. The cell on $\mathcal{E}_v$ is read by the active letters through γ_v, and a mark left in place through α_h, so 21.6 applies; the factors of the marks left in place cancel (its step (5)), and with $\kappa_\mathfrak{e} = \kappa_{\mathcal{E}_v}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})\,\Gamma_{\!1}^{\,2}\begin{bmatrix}\kappa_\mathfrak{e}, & 2\sigma-\kappa_\mathfrak{e}\\ & 2\sigma\end{bmatrix} = P_{F'}(\mathbf{m}';\mathbf{s},\mathbf{t})\,P_{F_{\mathcal{H}_v}}(\mathbf{m}_v;\mathbf{s},\mathbf{t}),$$

F' the family of the tables $\mathcal{H}'_j$ at the masses $\mathbf{m}'$ equal to $\mathbf{m}$ on its marks and to $\kappa_{\mathcal{E}_v}$ at $g_{p,\mathfrak{e}}$, and $\mathbf{m}_v$ equal to $\mathbf{m}|_{\mathcal{E}_v}$ and to $2\sigma-\kappa_{\mathcal{E}_v} = \kappa_{\mathcal{E}\setminus\mathcal{E}_v}$ at $g_{v,\mathfrak{e}}$; every factor is on its balance wall with every proper wall positive.

The walls of F'. By 20.7, a set S' of marks of F' containing $g_{p,\mathfrak{e}}$ has the wall of $(S'\setminus\{g_{p,\mathfrak{e}}\})\sqcup\mathcal{E}_v$ in F, the cell entering with its whole set $\mathcal{E}_v$, of wall $\kappa_{\mathcal{E}_v}$, and the marker in D_j and D_j^*; a set avoiding $g_{p,\mathfrak{e}}$ has the same wall in both. So the walls across the edges of $\mathfrak{T}-v$ are those of F.

The hypothesis passes to F'. F' has the common hinge tree $\mathfrak{T}-v$, with the same nodes and $g_{p,\mathfrak{e}}$ a mark of the node p. Its subtree cells are those of F except at the vertices w on the path from p to o, where $\mathcal{A}_{w,j} = \mathcal{A}'_{w,j}\langle\mathcal{A}_{v,j}\rangle$, the cell at the mark $g_{p,\mathfrak{e}}$ (20.3(ii)). By 20.4 the hinge minors of $\mathcal{A}_{w,j}$ are those of $\mathcal{A}'_{w,j}$ with $\alpha_{g_{p,\mathfrak{e}}}$ replaced by $\gamma^K_{\mathcal{A}_{v,j}}$ in the letters K and by $\gamma^\Psi_{\mathcal{A}_{v,j}}$ in the letters Ψ, times factors of $\mathcal{A}_{v,j}$ that cancel in the quotients, so

$$\gamma^K_{\mathcal{A}_{w,j}}(\alpha) = \gamma^K_{\mathcal{A}'_{w,j}}\big(\alpha',\gamma^K_{\mathcal{A}_{v,j}}(\alpha_{\mathcal{E}_v})\big),\qquad \gamma^\Psi_{\mathcal{A}_{w,j}}(\alpha) = \gamma^\Psi_{\mathcal{A}'_{w,j}}\big(\alpha',\gamma^\Psi_{\mathcal{A}_{v,j}}(\alpha_{\mathcal{E}_v})\big),$$

α' the variables of the other marks. For the active letters both inner hinge values are $\gamma_v(\alpha_{\mathcal{E}_v})$, and γ_v, positive and homogeneous of degree one, takes every value in $(0,\infty)$. So for every value $c>0$ at $g_{p,\mathfrak{e}}$ the hinge values of $\mathcal{A}'_{w,j}$ by the active letters at (α',c) all equal $\gamma_w(\alpha)$ for any α with $\gamma_v(\alpha_{\mathcal{E}_v}) = c$: the active letters read $\mathcal{A}'_{w,j}$ by one hinge value, and the induction hypothesis applies to F'. Its product formula and the display give that of F, with the masses on the markers the walls across the edges.

One table at a full lean. One letter is active, and the hypothesis is empty. At $(s,0)$ on the balance wall $|\mathbf{m}| = rs$, so the loads $\mathbf{a} = \mathbf{m}$ are admissible in 25.6; the marker load $k_{v,\mathfrak{e}} = sr_v^\mathfrak{e} - m_{\mathcal{E}_v^\mathfrak{e}}$ is $s-\kappa_{\mathcal{E}_v^\mathfrak{e}}$, the wall of the branch at v being $m_{\mathcal{E}_v^\mathfrak{e}} - s(r_v^\mathfrak{e}-1)$, and the two walls across an edge add to s; so the node loads are the masses $\mathbf{m}_v$ and the two identities are the same (25.5). At $(0,t)$ the same holds with $\mathbf{b}$ and t. The series and parallel nodes are 25.10.

Tables glued along a tree. The glued table has the hinge tree whose nodes are the tables $\mathcal{A}_v$ and whose markers are the glued marks; the glued marks of connected tables are neither loops nor coloops, and a two-sum has the product weights (20.8). The masses of the glued marks are positive, every singleton being a proper wall of its table, so every $\kappa_\mathfrak{e}$ lies in $(0,t)$. Each table $\mathcal{A}_v$ is balanced, $m_{\mathcal{E}_{\mathcal{A}_v}} = tr^*_{\mathcal{A}_v}$, and the glued table, of corank $\sum_v r^*_{\mathcal{A}_v} - (k-1)$ and total mass $\sum_v tr^*_{\mathcal{A}_v} - (k-1)t$, is on its balance wall. Removing a leaf as above, with the first display of 21.6 in $[0,\infty]$ and its step (2), the cell at a leaf v glued along its mark g has the wall $t-m_g$ and its cell family is $\mathcal{A}_v$ at its own masses, so the period of the glued table is $\prod_v P_{\mathcal{A}_v}/\prod_\mathfrak{e}\Gamma_{\!1}^{\,2}\begin{bmatrix}\kappa_\mathfrak{e}, & t-\kappa_\mathfrak{e}\\ & t\end{bmatrix}$ in $[0,\infty]$; it is finite, and the glued table, balanced, has every proper wall positive (8.10). □

Theorem 21.8 (Blocks read by one hinge value aggregate exactly). *As stated: in (ii) blocks read through $e_\ell/e_{\ell-1}$ as well as blocks read through their totals, and the active letters at nonzero exponents of either sign.*

Proof. (i) and (iii). These are the reference proof of 21.8(i),(iii), the letters with positive exponent replaced by the active letters: at a parallel block $K_{\mathcal{A}/g}=1$ and $\gamma^K_{\mathcal{A}}=e_1$, at a series block $\Psi_{\mathcal{A}\setminus g}=1$ and $\gamma^\Psi_{\mathcal{A}}=e_1$ (20.5(i)), so by 20.4 every active letter is the letter of its core read at the block totals, and the totals of $U\sim\mathrm{Dir}_{\mathbf{m}}$ have the law $\mathrm{Dir}_{(m_{B_i})_i}$ (3.10). The sign of an exponent does not enter.

(ii). When a block has $\ell_i\ge 2$ marks a table with both letters active would need a block that is both parallel and series; so at such a block each table has at most one active letter, and the inactive letters, of exponent 0, enter neither the walls nor the kernels. By 20.5(i), for $i\in I$ the active hinge values are $\gamma^K=e_1$ at the parallel blocks and $\gamma^\Psi=e_1$ at the series blocks, and for $i\notin I$ they are $\gamma^K=e_{\ell_i}/e_{\ell_i-1}$ at the series blocks and $\gamma^\Psi=e_{\ell_i}/e_{\ell_i-1}$ at the parallel blocks. So 21.5 applies with all constants 1 and $\gamma_i=e_1$ for $i\in I$, $\gamma_i=e_{\ell_i}/e_{\ell_i-1}$ for $i\notin I$:

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i S_i,\qquad S_i=\int_{\Delta_{B_i}}\gamma_i^{-\kappa_{B_i}}\,d\beta_i,$$

in $[0,\infty]$. By 20.7, $\kappa_{B_i}=m_{B_i}-\sum_j\big(s_j(r_{\mathcal{A}_{ij}}-1)+t_j(\ell_i-r_{\mathcal{A}_{ij}})\big)$ with the ranks 1 of a parallel block and ℓ_i of a series block.

For $i\in I$ the active letters see ranks that make every term vanish, so $\kappa_{B_i}=m_{B_i}$; the density of the cell kernel is $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$, the active minors $K_{\mathcal{A}/g}$ and $\Psi_{\mathcal{A}\setminus g}$ being 1; and $e_1=1$ on the simplex, so $S_i=\int_{\Delta_{B_i}}u^{\mathbf{m}|_{B_i}-\mathbf{1}}d\omega$, which is $\Gamma_{\substack{\ell_i\\1}}\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix}$ when every m_x is positive and $+\infty$ otherwise, by the shape integral (3.11).

For $i\notin I$ each active letter contributes its exponent times ℓ_i-1, so $\kappa_{B_i}=m_{B_i}-(\ell_i-1)2\sigma$. The active minors are $K_{\mathcal{A}/g}=e_{\ell_i-1}$ at a series block and $\Psi_{\mathcal{A}\setminus g}=e_{\ell_i-1}$ at a parallel block, so the density of the cell kernel is $u^{\mathbf{m}|_{B_i}-\mathbf{1}}e_{\ell_i-1}(u)^{-2\sigma}$, and with $e_{\ell_i}=\prod_{x\in B_i}u_x$,

$$S_i=\int_{\Delta_{B_i}}\Big(\frac{e_{\ell_i}}{e_{\ell_i-1}}\Big)^{-\kappa_{B_i}}u^{\mathbf{m}|_{B_i}-\mathbf{1}}e_{\ell_i-1}^{-2\sigma}\,d\omega=\int_{\Delta_{B_i}}u^{\mathbf{m}|_{B_i}-\kappa_{B_i}\mathbf{1}-\mathbf{1}}\,e_{\ell_i-1}(u)^{-(2\sigma-\kappa_{B_i})}\,d\omega.$$

This is the period of the series node $\binom{B_i}{\ell_i-1}$ with unit weights, whose letter K is e_{ℓ_i-1}, at the full basis lean $(2\sigma-\kappa_{B_i},0)$ and the masses $m_x-\kappa_{B_i}$; they add to $m_{B_i}-\ell_i\kappa_{B_i}=(\ell_i-1)(2\sigma-\kappa_{B_i})$, its balance wall. By 25.10 it is $\Gamma_{\substack{\ell_i\\1}}\begin{bmatrix}(2\sigma-\kappa_{B_i}-m_x+\kappa_{B_i})_x\\ 2\sigma-\kappa_{B_i}\end{bmatrix}=\Gamma_{\substack{\ell_i\\1}}\begin{bmatrix}(2\sigma-m_x)_x\\ 2\sigma-\kappa_{B_i}\end{bmatrix}$, convergent exactly when every $2\sigma-m_x$ is positive, and $+\infty$ otherwise, the integrand being positive. At $\ell_i=1$ both kinds give $\kappa_{B_i}=m_x$ and $S_i=1$.

So the display holds in $[0,\infty]$ with the stated reading of the stacked Gammas, and when these are finite the two sides are finite together.

The last sentence. At $\ell\ge 2$ the functions e_1 and $e_\ell/e_{\ell-1}$ are not proportional on the simplex, their quotient $e_1e_{\ell-1}/e_\ell=(\sum_x u_x)(\sum_x u_x^{-1})$ being unbounded; so 21.5 does not apply to a block read both ways, and 21.3 keeps it as one mark with its kernel and both hinge values. □

Remark. Checks 7a to 7d of S21-23_work/checks.py are the case $\ell=2$ of the blocks with $i\notin I$, with one table at $(s_1,0)$ and one at $(0,t_2)$: $S=\Gamma(2\sigma-m_1)\Gamma(2\sigma-m_2)/\Gamma(4\sigma-m_B)$, and the period of the cell family is $\Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa,\ 2\sigma-\kappa\\ 2\sigma\end{bmatrix}S$.

Theorem 21.9 (The series closure at positive exponents). *As stated: the closure through the transforms of the cell families, each transform one Gramma of index $|\mathbf{s}| - \kappa_{B_i}$ at every data, and its Breta shape reading wherever the family of the hinge minors is balanced.*

Proof. The closure. The responses of the reference, proof and statement of 21.9, are the integrals over $\mathbb{R}_{>0}^{B_i}$ of $\alpha^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K_{\mathcal{A}_{ij}\setminus g_i}^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}^{-t_j}$ against $\exp\big(- \sum_j x_j K_{\mathcal{A}_{ij}/g_i}/K_{\mathcal{A}_{ij}\setminus g_i} - \sum_j y_j \Psi_{\mathcal{A}_{ij}/g_i}/\Psi_{\mathcal{A}_{ij}\setminus g_i}\big)$ with $y_1 = 1$; since $K_{\mathcal{A}/g}/K_{\mathcal{A}\setminus g} = 1/\gamma_{\mathcal{A}}^K$ and $\Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A}\setminus g} = \gamma_{\mathcal{A}}^\Psi$, they are the transforms $f_{F_i}(\mathbf{x}, (1, y_2, \dots, y_N))$ of 22.1, and the first display is the reference's. The statement on repeated cells is read off the product.

One table. By the homogeneity of 22.1 with $c = x$, $f_{F_i}(x, 1) = x^{\kappa_{B_i}-s} f_{\mathcal{A}_i}(x)$. The core $\binom{\{h_1,\dots,h_q\}}{q-1}$ has rank $q-1$ and corank 1, and the core family read at u is balanced (21.3), so $\sum_i \kappa_{B_i} = (q-1)s + t$. The power of x is then $x^{s-1+(q-1)s+t-qs} = x^{t-1}$, and the closure is $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty x^{t-1} \prod_i f_{\mathcal{A}_i}(x)\, dx$.

The Gramma. Fix i and write $\alpha = \rho u$, $u \in \Delta_{B_i}$, $\rho > 0$, so that $\alpha^{\mathbf{m}|_{B_i}-\mathbf{1}} d\alpha = \rho^{m_{B_i}-1} u^{\mathbf{m}|_{B_i}-\mathbf{1}} d\rho\, d\omega$ (2.7). The marks g_i being neither loops nor coloops, $K_{\mathcal{A}_{ij}\setminus g_i}$ is homogeneous of degree $r_{\mathcal{A}_{ij}}$ and $\Psi_{\mathcal{A}_{ij}\setminus g_i}$ of degree $|B_i| - r_{\mathcal{A}_{ij}}$, and the hinge values of degree one. So the power of ρ is $\rho^{\nu-1}$ with

$$\nu = m_{B_i} - \sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big) = \kappa_{B_i} - |\mathbf{s}|,$$

and the exponential is $\exp(-a(u)/\rho - b(u)\rho)$ with $a = \sum_j x_j/\gamma_{\mathcal{A}_{ij}}^K$ and $b = \sum_j y_j \gamma_{\mathcal{A}_{ij}}^\Psi$. By 2.1 and the reflection (2.4),

$$\int_0^\infty \rho^{\nu-1} e^{-b\rho - a/\rho}\, d\rho = \mathsf{G}_{\nu;(1,-1)}(b, a) = \mathsf{G}_{-\nu;(1,-1)}(a, b),$$

and Tonelli gives the display of the transform.

Index 0. The family of the hinge minors has the degree $\sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big)$ at the masses $\mathbf{m}|_{B_i}$, so it is on its balance wall exactly when $\kappa_{B_i} = |\mathbf{s}|$; at the centre of one table, every mass σ and $s = t = \sigma$, $\kappa_{B_i} = \sigma|B_i| - \sigma(r-1) - \sigma(|B_i| - r) = s$. Torus homogeneity (2.3(iii)) with $\lambda = a$ gives $\mathsf{G}_{0;(1,-1)}(a, b) = \mathsf{G}_{0;(1,-1)}(1, ab)$. With every proper wall positive, the form of the family of the hinge minors divided by its period is its Breta shape law (16.7), which gives the expectation; for one table, $ab = xy\, \gamma_{\mathcal{A}_i}^\Psi/\gamma_{\mathcal{A}_i}^K = xy\varrho_{\mathcal{A}_i}$. The period of k copies is the first display with all F_i equal. □

Theorem 21.10 (A chain is the power of one operator). *As stated: at real exponents, the pinned case for the active letters, with the pointer to 21.7 on a path.*

Proof. The reference proof of 21.10 applies verbatim at real exponents. It uses pooling (18.8, for every family at all data, both sides converging together), the nonnegativity of the integrands (at real exponents the letters are positive, and so is every power of them), the scalar twist of a mark (21.1) and the Beta integral $\int_0^1 v^{m_{g_{\text{in}}}-1}(1-v)^{m_{g_{\text{out}}}-1}(1-v+\lambda v)^{-2\sigma} dv = \lambda^{-m_{g_{\text{in}}}} \Gamma_2 \left[\begin{matrix} m_{g_{\text{in}}},\ m_{g_{\text{out}}} \\ 2\sigma \end{matrix}\right]_1$ at $m_{g_{\text{in}}} + m_{g_{\text{out}}} = 2\sigma$; none of these uses the sign of an exponent. In the pinned case the reference assumes that all the active letters read the cell through one function; a letter of exponent 0 is absent from every period, which is the hypothesis as stated. By 20.8 the chain $F^{\triangleright k}$ has the hinge tree whose nodes are the copies, a path, the glued marks its markers; by the propagation in the proof of 21.7, its subtree cells are read by the active letters through one hinge value when the copy

is, and the walls across its edges are $m_{g_{\mathrm{in}}}$ and $m_{g_{\mathrm{out}}}$, so with every proper wall positive 21.7 gives the same product $P_F^k/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} m_{g_{\mathrm{in}}},\ m_{g_{\mathrm{out}}} \\ 2\sigma \end{bmatrix}^{k-1}$. □

Theorem 21.11 (Letters by transfer). *As stated: the reduction of the states to linearly independent ones, and the bound on the interface dimension for a multivector of any tensor rank.*

Proof. The products of transfer matrices are the reference's. *The stations.* The stations of level k of S_l give a level factorization with as many terms (the children factorization, 20.12), so their number is at least $\operatorname{rank}\Theta_{S_l}^{(k)}$, and the number of states is at least the interface dimension.

The reduction. The child of a pattern p of S_l at level k is $\sum_X \theta_{p\sqcup X}\alpha^X$, the polynomial of the row p of $\Theta_{S_l}^{(k)}$, so the children of level k span a space of dimension $\operatorname{rank}\Theta_{S_l}^{(k)}$. Splitting X along the step B_{l-1}, the child at p of S_{l-1} is

$$\sum_X \theta_{p\sqcup X}\alpha^X = \sum_{p'\subseteq B_{l-1}} \alpha^{p'} \sum_{X'} \theta_{(p\sqcup p')\sqcup X'}\alpha^{X'},$$

the sum over p' of $\alpha^{p'}$ times the child at $p\sqcup p'$ of S_l. Choose at every S_l and every level a basis of the span of the children and write every child in it; the display expresses each basis element at S_{l-1}, a combination of children of S_{l-1}, through the basis elements at S_l, with coefficients polynomial in $\alpha_{B_{l-1}}$. These are transfer matrices on $\sum_k \operatorname{rank}\Theta_{S_l}^{(k)}$ states at S_l, and the products give K; the same coefficients, read on the complements, give Ψ (20.12).

The bound. By 20.14(i), $\operatorname{rank}\Theta_{S_l}^{(k)} \le \operatorname{ir}_k(\boldsymbol{\omega})^2$, and $\le \frac{1}{2}\operatorname{ir}_k(\boldsymbol{\omega})(\operatorname{ir}_k(\boldsymbol{\omega})+1)$ when $\boldsymbol{\omega}$ is real; by (ii), $\operatorname{ir}_k(\boldsymbol{\omega}) \le \sum_i \binom{c_i}{k-d_i}$. The functions $q\mapsto q^2$ and $q\mapsto \frac{1}{2}q(q+1)$ increase on $[0,\infty)$, and summing over k gives the bound. For a determinantal table, of tensor rank one with $c_1 = c(S_l)$ and $d_1 = d(S_l)$ (20.14(ii)), the bound is $\sum_m \binom{c}{m}^2 = \binom{2c}{c}$, $c = c(S_l)$, counting the c-subsets of a set of $2c$ elements by their intersection with a fixed half. □

Theorem 21.12 (Integer joint moments by fixed matrices). *As stated: all the letters of a family at once, along a chain of bunches with any steps and any independent measures on them.*

Proof. This is the reference proof of 21.12 with the tensor product taken over all the tables and with one matrix per step. A product of scalars $\mathbf{u}_i^{\mathsf{T}} A_{i,0}\cdots A_{i,L-1}\mathbf{v}_i$, $i = 1,\dots,n$, equals $(\bigotimes_i \mathbf{u}_i)^{\mathsf{T}}\big(\bigotimes_i A_{i,0}\big)\cdots\big(\bigotimes_i A_{i,L-1}\big)(\bigotimes_i \mathbf{v}_i)$ by the mixed-product rule $(\bigotimes_i A_i)(\bigotimes_i B_i) = \bigotimes_i (A_iB_i)$. Apply it to the a_j copies of K_j and the b_j copies of Ψ_j for every j: the factor of the step l is $\bigotimes_j \big(\mathbf{M}_{j,l}^K(\alpha_{B_l})^{\otimes a_j} \otimes \mathbf{M}_{j,l}^{\Psi}(\alpha_{B_l})^{\otimes b_j}\big)$, a matrix function of α_{B_l} alone. Every entry of the product is a finite sum of products of one entry of each step factor; the measures of the steps are independent, so the integral of such a product is the product of the integrals, by Tonelli for nonnegative entries and by Fubini under the stated finiteness, and integrating entrywise replaces the factor of the step l by $\mathbf{A}_l$.

For identical steps every $\mathbf{A}_l$ is one matrix $\mathbf{A}$ of the stated size, and the scalars $\mathbf{u}^{\mathsf{T}}\mathbf{A}^L\mathbf{v}$ satisfy the linear recurrence whose coefficients are those of the characteristic polynomial of $\mathbf{A}$, which annihilates $\mathbf{A}$. With independent laws Gamma_{m_x} on the marks, $\prod_j K_j^{a_j}\Psi_j^{b_j}$ is homogeneous of degree $d = \sum_j (a_j r_j + b_j r_j^*)$, and the draw splits into an independent total of law $\mathrm{Gamma}_{m_{\mathcal{E}}}$ and a shape of law $\mathrm{Dir}_{\mathbf{m}}$ (8.11(ii)), the total having the moment $(m_{\mathcal{E}})_d$ of order d. With $\mathrm{Dir}_{\mathbf{m}|_{B_l}}$ on a step, the entries of $\mathbf{A}_l$ are the Dirichlet moments of the monomials (3.11). □

Theorem 21.13 (Gluing across a bunch of any width). *As stated: the share display for one table at a full lean at every width, and the count $2N-1$ of the free shares at width one.*

Proof. The main display is the reference's. For one table at the full lean $(0,t)$ in level form, one station per level, the active letter Ψ has the $c(B)+1$ terms of the levels $k=d(B),\dots,r(B)$, so the stacked Gamma of the display is $\prod_k \Gamma(b_k)/\Gamma(t)$ and the pins are $\sum_k b_k = t$ and $\sum_k (r(B)-k)b_k = \kappa_B$, the main pin without the shares of K. The fibre integral (4.1) has $c(B)+1$ shares and two pins. Integrating the delta of the pins against the imaginary parts of $b_{d(B)}$ and $b_{r(B)}$ divides by $|\det|$ of the matrix of the two pins in these shares,

$$\begin{pmatrix} 1 & 1 \\ c(B) & 0 \end{pmatrix}, \qquad |\det| = c(B)$$

(check 8a of S21-23_work/checks.py), and leaves $(2\pi)^{2-(c(B)+1)}$ times the integral over the imaginary parts of the $c(B)-1$ free shares, which is $\frac{1}{(2\pi i)^{c(B)-1}} \int \prod db_k$. The pins solved for $b_{d(B)}$ and $b_{r(B)}$ give the stated values; at $c(B)=2$ they are $\frac{\kappa_B - z}{2}$ and $\frac{2t-\kappa_B-z}{2}$ with $b_{d(B)+1} = z$ (check 8b). At $c(B)=1$ the fibre has dimension 0, the matrix of the pins has $|\det| = 1$, and the integral is the value at $(b_{d(B)}, b_{r(B)}) = (\kappa_B, t-\kappa_B)$, the Beta corner times the two inner periods, which is 18.14: the bunch and room letters of the two levels are the minors $\Psi_{\mathcal{B}/R}, \Psi_{\mathcal{B}|B}$ and $\Psi_{\mathcal{B}|R}, \Psi_{\mathcal{B}/B}$ up to constant factors of the factorization that cancel in the product. At width one with factoring weights each level has one station (12.17), so each of the $2N$ active letters has two terms, and the free shares are $4N - 2N - 1 = 2N-1$, the dimension of the fibre integral of 18.13, of which the main display is then the formula. □

Remark. The paragraph after 21.13 gives the Mellin transform of the scale integral $I(\boldsymbol{\zeta}) = \int_0^\infty x^{\kappa_B - 1}\big(1 + \sum_{i=1}^{c-1} \zeta_i x^i + x^c\big)^{-t} dx$, $c = c(B)$. By Tonelli and the substitutions $\zeta_i = \eta_i x^{-i}$, then $x^c = w$,

$$\int \prod_i \zeta_i^{z_i - 1} I(\boldsymbol{\zeta})\, d\boldsymbol{\zeta} = \frac{\prod_i \Gamma(z_i)\, \Gamma(t - \sum_i z_i)}{\Gamma(t)} \int_0^\infty x^{\kappa_B - \sum_i i z_i - 1} (1+x^c)^{\sum_i z_i - t} dx = \frac{\prod_{k=d(B)}^{r(B)} \Gamma(b_k)}{c\,\Gamma(t)}$$

at $b_{r(B)-i} = z_i$, $b_{d(B)} = (\kappa_B - \sum_i i z_i)/c$ and $b_{r(B)} = t - b_{d(B)} - \sum_i z_i$, wherever these have positive real parts: the share kernel of the theorem with the factor $1/c$ of the pin.

40.22 Chapter 22: the law of a cell

Definition 22.1 (The law of a cell family). As stated: the Mellin transform of the law of a cell family, and the transforms and the hinge measures of a cell family, those of one cell being their readings.

Proof. The definition asserts three things. *The Mellin transform.* For complex $\mathbf{z}$ and real positive coordinates, $|\boldsymbol{\varrho}_F^{-\mathbf{z}}| = \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}}$, so the integral converges absolutely where it is finite at the real part, with modulus at most that value; for one cell $\boldsymbol{\varrho}_F^{-\mathbf{z}} = (\gamma_{\mathcal{A}}^{\Psi}/\gamma_{\mathcal{A}}^{K})^{-z_{1,\Psi}} = \varrho_{\mathcal{A}}^{-z}$, and the second form is the image of the measure. The base form of one cell is ϕ_F with $(\gamma_{\mathcal{A}}^K)^{-\kappa} = K_{\mathcal{A}/g}^{\kappa} K_{\mathcal{A}\backslash g}^{-\kappa}$.

Homogeneity. Substitute $\alpha = c\beta$ in the orthant forms: $\alpha^{\mathbf{m}-\mathbf{1}} d\alpha = c^{m_B} \beta^{\mathbf{m}-\mathbf{1}} d\beta$, the letters of the deletions have the degrees $r_{\mathcal{A}_j}$ and $|B| - r_{\mathcal{A}_j}$, those of the contractions $r_{\mathcal{A}_j} - 1$ and $|B| - r_{\mathcal{A}_j} + 1$, and the hinge values have degree one. So

$$f_F(\mathbf{x},\mathbf{y}) = c^{m_B - \sum_j (s_j r_{\mathcal{A}_j} + t_j(|B| - r_{\mathcal{A}_j}))} f_F(\mathbf{x}/c, c\mathbf{y}) = c^{\kappa_F - |\mathbf{s}|} f_F(\mathbf{x}/c, c\mathbf{y}),$$

and in the same way $\check{f}_F(\mathbf{x},\mathbf{y}) = c^{\kappa_F - |\mathbf{t}|} \check{f}_F(\mathbf{x}/c, c\mathbf{y})$. With $c = x_1$ each transform is fixed by its values at $x_1 = 1$.

One cell. At $N=1$, $f_F(1,y)$ and $\check f_F(1,y)$ are the integrals called the orthant forms of the transforms of one cell, and the images of $e^{-\lambda}\mathfrak{r}_F$ and $e^{-\lambda}\check{\mathfrak{r}}_F$ under $(\lambda,\mu)\mapsto\mu$ are the images of $\alpha^{\mathbf{m-1}}K_{\mathcal{A}\setminus g}^{-s}\Psi_{\mathcal{A}\setminus g}^{-t}e^{-1/\gamma_{\mathcal{A}}^K}d\alpha$ under $\gamma_{\mathcal{A}}^{\Psi}$ and of $\alpha^{\mathbf{m-1}}K_{\mathcal{A}/g}^{-s}\Psi_{\mathcal{A}/g}^{-t}e^{-1/\gamma_{\mathcal{A}}^{\Psi}}d\alpha$ under $\gamma_{\mathcal{A}}^K$. That these agree with the forms through the law, the Gramma integrals and the images of $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}\otimes p^{s-\kappa-1}e^{-p}dp$ and $x^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}\otimes p^{t-\kappa-1}e^{-p}dp$, is 22.5(i),(ii). □

Proposition 22.5 (Faces of one law). *As stated, for every cell family with plain tables and $s_j,t_j>0$.*

Proof. Write $\kappa=\kappa_F$ and γ_j^K, γ_j^{Ψ} for the hinge values of $\mathcal{A}_j$. The reference proof of 22.5 is the case $N=1$; the steps below are that proof with the index j, the changes being the bookkeeping of the powers of γ_1^K and γ_1^{Ψ}, which pass the integrals from the simplex to the law.

(i) In the orthant form of f_F write $\alpha=\rho u$ (2.7). As in the proof of 21.9 in this part, the power of ρ is $\rho^{\kappa-|\mathbf{s}|-1}$ and

$$f_F(\mathbf{x},\mathbf{y})=\int_{\Delta_B}\mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j\frac{x_j}{\gamma_j^K(u)},\ \sum_j y_j\gamma_j^{\Psi}(u)\Big)\,u^{\mathbf{m-1}}\prod_j K_{\mathcal{A}_j\setminus g}^{-s_j}\Psi_{\mathcal{A}_j\setminus g}^{-t_j}\,d\omega(u).$$

Since $K_{\mathcal{A}_j\setminus g}=\gamma_j^K K_{\mathcal{A}_j/g}$, the measure is $(\gamma_1^K)^{\kappa}\prod_j(\gamma_j^K)^{-s_j}\phi_F=(\gamma_1^K)^{\kappa-|\mathbf{s}|}\prod_j(\gamma_j^K/\gamma_1^K)^{-s_j}\phi_F$. Torus homogeneity (2.3(iii)) with $\lambda=1/\gamma_1^K$, $\mathsf{G}_{\nu;(1,-1)}(\lambda\xi_1,\xi_2/\lambda)=\lambda^{-\nu}\mathsf{G}_{\nu;(1,-1)}(\xi_1,\xi_2)$, gives, at $\mathbf{z}=\boldsymbol{\varrho}_F(u)$,

$$\mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j\frac{x_j}{\gamma_j^K},\ \sum_j y_j\gamma_j^{\Psi}\Big)=(\gamma_1^K)^{|\mathbf{s}|-\kappa}\,\mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j\frac{x_j}{z_{j,K}},\ \sum_j y_j z_{j,\Psi}\Big).$$

The powers of γ_1^K cancel, and the image under $\boldsymbol{\varrho}_F$ gives the first law form. For $\check f_F$ the contracted letters have the degrees $r_{\mathcal{A}_j}-1$ and $|B|-r_{\mathcal{A}_j}+1$, so the power of ρ is $\rho^{\kappa-|\mathbf{t}|-1}$, the index is $|\mathbf{t}|-\kappa$, and the coefficients are $\sum_j x_j/\gamma_j^{\Psi}$ and $\sum_j y_j\gamma_j^K$; the measure is $u^{\mathbf{m-1}}\prod_j K_{\mathcal{A}_j/g}^{-s_j}\Psi_{\mathcal{A}_j/g}^{-t_j}d\omega=(\gamma_1^K)^{\kappa}\prod_j(\gamma_j^{\Psi})^{-t_j}\phi_F$, since $\Psi_{\mathcal{A}_j/g}=\gamma_j^{\Psi}\Psi_{\mathcal{A}_j\setminus g}$. Torus homogeneity with $\lambda=1/\gamma_1^{\Psi}$ extracts $(\gamma_1^{\Psi})^{|\mathbf{t}|-\kappa}$, and $(\gamma_1^K)^{\kappa}\prod_j(\gamma_j^{\Psi})^{-t_j}(\gamma_1^{\Psi})^{|\mathbf{t}|-\kappa}=z_{1,\Psi}^{-\kappa}\prod_j(z_{j,\Psi}/z_{1,\Psi})^{-t_j}$, with $z_{1,\Psi}/z_{j,\Psi}=\gamma_1^{\Psi}/\gamma_j^{\Psi}$ and $z_{j,K}/z_{1,\Psi}=\gamma_j^K/\gamma_1^{\Psi}$: the second law form. At $N=1$, $z_{1,K}=1$, $z_{1,\Psi}=\varrho_{\mathcal{A}}$ and $\mathbf{x}=1$ give $\mathsf{G}_{s-\kappa;(1,-1)}(1,\varrho_{\mathcal{A}}y)$ and $\mathsf{G}_{t-\kappa;(1,-1)}(1,y/\varrho_{\mathcal{A}})\varrho_{\mathcal{A}}^{-\kappa}$.

(ii) The Laplace forms are the definitions of the images. For the law form of $\mathfrak{r}_F$ write $\alpha=\rho u$ and $p=\rho\gamma_1^K(u)$: then $1/\gamma_j^K(\alpha)=1/(pz_{j,K})$ and $\gamma_j^{\Psi}(\alpha)=pz_{j,\Psi}$ at $\mathbf{z}=\boldsymbol{\varrho}_F(u)$, and $\rho^{\kappa-|\mathbf{s}|-1}d\rho=(\gamma_1^K)^{|\mathbf{s}|-\kappa}p^{\kappa-|\mathbf{s}|-1}dp$, which with the measure of (i) leaves $\prod_j z_{j,K}^{-s_j}\phi_F(du)\otimes p^{\kappa-|\mathbf{s}|-1}dp$. A Laplace transform of a positive measure is completely monotone on the interior of the set where it is finite, by differentiation under the integral, the derivatives being dominated near each such point by the integrand at a point with smaller coordinates, and it is log-convex along segments by Hölder's inequality. For one cell, with $\lambda=1/p$ and $p=1/q$, the measure $e^{-\lambda}\mathfrak{r}_F$ becomes $e^{-q}q^{s-\kappa-1}dq$ against the law, and the coordinate $\mu=pz$ becomes z/q; for $\check{\mathfrak{r}}_F$, with $p=\rho\gamma^{\Psi}(u)$, the same computation gives the image of $z^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dz)\otimes p^{\kappa-t-1}dp$ under $(z,p)\mapsto(1/p,p/z)$, and $e^{-\lambda}$ and $p=1/q$ give the image of $z^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dz)\otimes q^{t-\kappa-1}e^{-q}dq$ under $(z,q)\mapsto 1/(zq)$.

(iii) Insert the Laplace form of (ii) and integrate first in x_j, $j\ge 2$, and y_j (Tonelli): $\int_0^{\infty}x^{a-1}e^{-x\lambda}dx=\Gamma(a)\lambda^{-a}$. With $\lambda_j=1/(pz_{j,K})$, $\mu_j=pz_{j,\Psi}$ and $x_1=1$,

$$\prod_{j\ge2}\Gamma(a_j)\prod_j\Gamma(b_j)\iint e^{-1/p}\prod_{j\ge2}(pz_{j,K})^{a_j}\prod_j(pz_{j,\Psi})^{-b_j}\prod_j z_{j,K}^{-s_j}\,p^{\kappa-|\mathbf{s}|-1}dp\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z}).$$

The power of p is p^{-a_1-1} with $a_1 = \sum_j b_j - \sum_{j\ge 2} a_j - \kappa + |\mathbf{s}|$, the given a_1 by the hyperplane, and $\int_0^\infty e^{-1/p} p^{-a_1-1} dp = \Gamma(a_1)$ for $a_1 > 0$; the power of $\mathbf{z}$ is $\prod_j z_{j,K}^{-(s_j-a_j)} z_{j,\Psi}^{-b_j}$, $z_{1,K} = 1$ (check 5). For $\check f_F$ the law form of $\check{\mathfrak{r}}_F$, obtained as in (ii) with $p = \rho\gamma_1^\Psi$, is the image of $z_{1,\Psi}^{-\kappa} \prod_j (z_{j,\Psi}/z_{1,\Psi})^{-t_j} (\boldsymbol{\varrho}_F)_* \phi_F(d\mathbf{z}) \otimes p^{\kappa-|\mathbf{t}|-1} dp$ under $(\mathbf{z}, p) \mapsto \big((z_{1,\Psi}/(pz_{j,\Psi}))_j, (pz_{j,K}/z_{1,\Psi})_j\big)$; the same integration leaves p^{-a_1-1} with $a_1 = \sum_j b_j - \sum_{j\ge 2} a_j - \kappa + |\mathbf{t}|$, and the total power of $z_{1,\Psi}$ is $-\sum_{j\ge 2} a_j + \sum_j b_j - \kappa + \sum_{j\ge 2} t_j = a_1 - t_1$, so the power of $\mathbf{z}$ is $\prod_j z_{j,K}^{-b_j} z_{j,\Psi}^{-(t_j-a_j)}$. Both points satisfy $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa$. At $N = 1$, $b = z$ and $a = z + s - \kappa$, the first gives $\Gamma(z)\Gamma(z+s-\kappa) \int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}}$, and the second, with $a = z + t - \kappa$, $\Gamma(z)\Gamma(z+t-\kappa) \int \varrho_{\mathcal{A}}^{-(\kappa-z)} d\phi_{\mathcal{A}}$.

(iv) Table by table, the dual table $\mathcal{A}^*$ has the bases $(B \sqcup \{g\}) \setminus T$ with the weights θ_T, and splitting by whether $g \in T$,

$$K_{\mathcal{A}^*/g} = \Psi_{\mathcal{A}\setminus g}, \quad K_{\mathcal{A}^*\setminus g} = \Psi_{\mathcal{A}/g}, \quad \Psi_{\mathcal{A}^*\setminus g} = K_{\mathcal{A}/g}, \quad \Psi_{\mathcal{A}^*/g} = K_{\mathcal{A}\setminus g},$$

its rank is $|B| + 1 - r_{\mathcal{A}}$, and g is neither a loop nor a coloop of it. So $\gamma_{\mathcal{A}^*}^K = \gamma_{\mathcal{A}}^\Psi$ and $\gamma_{\mathcal{A}^*}^\Psi = \gamma_{\mathcal{A}}^K$, and the hinge wall at (t_j, s_j) is $m_B - \sum_j \big(t_j(|B| - r_{\mathcal{A}_j}) + s_j(r_{\mathcal{A}_j} - 1)\big) = \kappa$. The base form of F^* is $u^{\mathbf{m}-\mathbf{1}} \prod_j \Psi_{\mathcal{A}_j\setminus g}^{-t_j} K_{\mathcal{A}_j/g}^{-s_j} (\gamma_1^\Psi)^{-\kappa} = (\gamma_1^\Psi/\gamma_1^K)^{-\kappa} \phi_F$, its ratio vector is $(\gamma_j^\Psi/\gamma_1^\Psi, \gamma_j^K/\gamma_1^\Psi)_j = (z_{j,\Psi}/z_{1,\Psi}, z_{j,K}/z_{1,\Psi})_j$ at $\mathbf{z} = \boldsymbol{\varrho}_F$, which gives the law, and the orthant forms of f_{F^*} and $\check f_{F^*}$ are those of $\check f_F$ and f_F. At $N = 1$ the ratio is $1/\varrho_{\mathcal{A}}$, and $\int \varrho_{\mathcal{A}^*}^{-z} d\phi_{\mathcal{A}^*} = \int \varrho_{\mathcal{A}}^{z} \varrho_{\mathcal{A}}^{-\kappa} d\phi_{\mathcal{A}}$.

(v) By (iii) for f_F at $(\mathbf{a}, \mathbf{b})$ and for $\check f_F$ at $(\mathbf{t}-\mathbf{b}, \mathbf{s}-\mathbf{a})$, whose hyperplane condition $\sum_j (s_j - a_j) - \sum_j (t_j - b_j) = \kappa - |\mathbf{t}|$ is that of $(\mathbf{a}, \mathbf{b})$, the Mellin transform of the law appears at the same point $(\mathbf{s}-\mathbf{a}, \mathbf{b})$ on both sides, with the Gamma products $\prod_j \Gamma(a_j)\Gamma(b_j)$ and $\prod_j \Gamma(t_j - b_j)\Gamma(s_j - a_j)$; all these Gammas are finite and positive, which gives the relation in $(0, \infty]$. At $N = 1$, with $z = s - a$ and $b = \kappa - z$, the ratio of the Gammas is that of the displayed one-cell relation (check 6). The expansion clause concerns the node of width one of one table and is the reference's. □

Lemma S8. *Let F be a cell family with plain tables, $s_j, t_j > 0$, and $\mathbf{a}, \mathbf{b} \in (0,\infty)^N$ real with $\sum_j b_j - \sum_j a_j = \kappa_F - |\mathbf{s}|$ and $\int \boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})} d\phi_F < \infty$. Write $\mathbf{u} = (x_2, \dots, x_N, y_1, \dots, y_N) \in (0,\infty)^{2N-1}$, $g(\mathbf{u}) = f_F((1, x_2, \dots, x_N), \mathbf{y})$, $\mathbf{c} = (a_2, \dots, a_N, b_1, \dots, b_N)$, and for $\boldsymbol{\tau} = (\tau_2, \dots, \tau_N, \tau'_1, \dots, \tau'_N) \in \mathbb{R}^{2N-1}$ put $\tau_1 = \sum_j \tau'_j - \sum_{j\ge 2} \tau_j$ and*

$$G(\boldsymbol{\tau}) = \prod_j \Gamma(a_j + i\tau_j)\Gamma(b_j + i\tau'_j) \int \boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a}-i\boldsymbol{\tau}_K,\ \mathbf{b}+i\boldsymbol{\tau}_\Psi)} d\phi_F,$$

$\boldsymbol{\tau}_K = (\tau_j)_j$ and $\boldsymbol{\tau}_\Psi = (\tau'_j)_j$. Then g is finite and continuous on $(0,\infty)^{2N-1}$, G is the Mellin transform of g on the vertical subspace through $\mathbf{c}$ and is integrable, and

$$g(\mathbf{u}) = (2\pi)^{1-2N} \int_{\mathbb{R}^{2N-1}} G(\boldsymbol{\tau})\, \mathbf{u}^{-\mathbf{c}-i\boldsymbol{\tau}}\, d\boldsymbol{\tau} \qquad (\mathbf{u} \in (0,\infty)^{2N-1}).$$

Proof. The computation of 22.5(iii) at the complex exponents $\mathbf{c} + i\boldsymbol{\tau}$ converges absolutely, its integrand having at every point the modulus of the integrand at $\mathbf{c}$, so it computes the Mellin transform of g at $\mathbf{c} + i\boldsymbol{\tau}$ as $G(\boldsymbol{\tau})$; the point $(\mathbf{s} - \mathbf{a} - i\boldsymbol{\tau}_K, \mathbf{b} + i\boldsymbol{\tau}_\Psi)$ stays on the hyperplane, which fixes τ_1. By 22.1, $|G(\boldsymbol{\tau})| \le \prod_j |\Gamma(a_j + i\tau_j)\Gamma(b_j + i\tau'_j)| \int \boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})} d\phi_F$, and the bound $|\Gamma(c + i\tau)| \le C_c\, e^{-\pi|\tau|/2} (1 + |\tau|)^{c-1/2}$ makes G integrable on $\mathbb{R}^{2N-1}$. The Mellin transform of g at the real point $\mathbf{c}$ is finite, so $g < \infty$ almost everywhere. By 22.5(ii), g is the Laplace transform of a positive measure in the variables $\mathbf{u}$, hence nonincreasing in each of them: if g were infinite at a point it would be infinite on the set of points below it, which has positive measure. So g

is finite everywhere, and continuous by dominated convergence, the integrand near a point being dominated by the integrand at a point with smaller coordinates. In the coordinates $\mathbf{v} = \log \mathbf{u}$ the function $g(e^{\mathbf{v}})e^{\langle \mathbf{c},\mathbf{v}\rangle}$ is integrable with the integrable Fourier transform G, so the Fourier inversion theorem gives the display almost everywhere, and everywhere since both sides are continuous. $\square$

Theorem 22.11 (The law of a substitution). *As stated: (i) and (ii) for families at real data; the present (i) was the gluing of a family, and (ii) for one table at $s, t > 0$.*

Proof. (ii) This is the reference proof of 22.11(ii) with the index j, the steps being those of 22.2 applied to one cell family after another. Write $B = \mathcal{E}_{\mathcal{H}} \sqcup \bigsqcup_i B_i$ and $\mathcal{A}_j = \mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$, a cell with the hinge g_0 (22.10). For a Borel $H \geq 0$, $\int H\, d((\boldsymbol{\varrho}_F)_*\phi_F)$ is the integral over the section $\{|\alpha| = 1\}$ of the orthant form

$$\omega = \alpha^{\mathbf{m-1}} H(\boldsymbol{\varrho}_F)\, K_{\mathcal{A}_1/g_0}^{\kappa_F - s_1} K_{\mathcal{A}_1\setminus g_0}^{-\kappa_F} \prod_{j\geq 2} K_{\mathcal{A}_j/g_0}^{-s_j} \prod_j \Psi_{\mathcal{A}_j\setminus g_0}^{-t_j}\, d\alpha,$$

homogeneous of degree $m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big) - \kappa_F = 0$, the base form being $u^{\mathbf{m-1}} \prod_j K_{\mathcal{A}_j/g_0}^{-s_j} \Psi_{\mathcal{A}_j\setminus g_0}^{-t_j} (K_{\mathcal{A}_1\setminus g_0}/K_{\mathcal{A}_1/g_0})^{-\kappa_F} d\omega$. By 22.10, with the hinge values of the cells at their own marks,

$$K_{\mathcal{A}_j\setminus g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}\, K_{\mathcal{H}_j\setminus g_0}\big(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i\big), \qquad K_{\mathcal{A}_j/g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}\, K_{\mathcal{H}_j/g_0}\big(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i\big),$$

and the same for Ψ with $\prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}$ and the hinge values $\gamma^{\Psi}_{\mathcal{A}_{ij}}$. The products over the cells cancel in the quotients, so the hinge values of $\mathcal{A}_j$ at g_0 are those of $\mathcal{H}_j$ read at $(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i)$ and $(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^{\Psi}_{\mathcal{A}_{ij}})_i)$.

Apply 22.2 to F_1 with $Z = \mathcal{E}_{\mathcal{H}} \sqcup \bigsqcup_{i\geq 2} B_i$. Assign to the letter $K_{\mathcal{A}_{1j}}$ the functions $\prod_{i\geq 2} K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j/g_0}$ and $\prod_{i\geq 2} K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j\setminus g_0}$, read at the hinge value of the cell at g_1, with the exponents s_j and 0 for $j \geq 2$ and $s_1 - \kappa_F$ and κ_F for $j = 1$; and to the letter $\Psi_{\mathcal{A}_{1j}}$ the functions $\prod_{i\geq 2} \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j\setminus g_0}$ and $\prod_{i\geq 2} \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j/g_0}$ with the exponents t_j and 0. The exponents assigned to each letter add to its exponent, the letters of the hinge minors of F_1 collect to $K_{\mathcal{A}_{1j}/g_1}^{-s_j} \Psi_{\mathcal{A}_{1j}\setminus g_1}^{-t_j}$, the product of the assigned powers is the rest of ω, and $H(\boldsymbol{\varrho}_F)$ is a Borel function of the values of the assigned functions, which give the hinge values of every $\mathcal{A}_j$ as quotients. The lemma collapses B_1 to g_1 with the mass κ_{F_1} and replaces the hinge values of the cells at g_1 by $x_{1,j,K}\alpha_{g_1}$ and $x_{1,j,\Psi}\alpha_{g_1}$, the twist at g_1 by the point $\mathbf{x}_1$ of the law of F_1. Repeat for $F_2, \dots, F_k$, moving the integrals over the laws outside (Tonelli). What remains is the integral over a section of the form of the cell family $F_{\mathcal{H}}[\mathbf{x}]$, its hinge minors at g_0 read with the twists, its masses $\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}$ and κ_{F_i}, the power κ_F in place of its hinge wall, and H of its ratio vector inserted. Its hinge wall is

$$m_{\mathcal{E}_{\mathcal{H}}} + \sum_i \kappa_{F_i} - \sum_j \big(s_j(r_{\mathcal{H}_j} - 1) + t_j(|\mathcal{E}_{\mathcal{H}}| + k - r_{\mathcal{H}_j})\big),$$

and it equals κ_F: insert $\kappa_{F_i} = m_{B_i} - \sum_j (s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}}))$, $r_{\mathcal{A}_j} = r_{\mathcal{H}_j} + \sum_i (r_{\mathcal{A}_{ij}} - 1)$ and $|B| = |\mathcal{E}_{\mathcal{H}}| + \sum_i |B_i|$ (22.10); the coefficients of s_j and of t_j agree term by term (check 1 of S21-23_work/checks.py). So the remaining integral is $\int H\, d((\boldsymbol{\varrho}_{F_{\mathcal{H}}[\mathbf{x}]})_*\phi_{F_{\mathcal{H}}[\mathbf{x}]})$, and the mixture follows; the hinge wall does not depend on $\mathbf{x}$. With $H = \boldsymbol{\varrho}^{-\mathbf{z}}$, a nonnegative Borel function for real $\mathbf{z}$, the Mellin transforms mix in the same way.

(i) The same steps with the family word of F in place of the base form: on the balance wall the form of F is homogeneous of degree zero, its letters are $\prod_i K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j}$ and $\prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j}$

read at the hinge values (22.10), the factors $K_{\mathcal{H}_j}$ and $\Psi_{\mathcal{H}_j}$ are assigned to the letters of each cell family in turn with the exponents s_j and t_j, and 22.2 collapses every cell to its marker with the mass κ_{F_i} and the twist by the point of its law. This is the reference's remark on families (rem:familiesnodes): 22.2 holds at real exponents, its proof collecting degrees and using Tonelli. For the core $\binom{\{g_1,g_2\}}{1}$, twisted at g_1 by $\mathbf{x}_1$ and at g_2 by $\mathbf{x}_2$, table j has $K_j = x_{1,j,K}\alpha_{g_1} + x_{2,j,K}\alpha_{g_2}$ and $\Psi_j = x_{1,j,\Psi}\alpha_{g_1} + x_{2,j,\Psi}\alpha_{g_2}$, and on the section $\alpha_{g_2} = 1$ its period is the displayed integral, a Breta of one bunch (15.1); the substitution $v \mapsto 1/v$ multiplies the integrand by $v^{2\sigma}$ and turns $v^{\kappa_{F_1}-1}dv$ into $v^{-\kappa_{F_1}-1}dv$, which gives the swap since $\kappa_{F_1} + \kappa_{F_2} = 2\sigma$. For one table $x_{i,1,K} = 1$ and the integral is that of 22.3. □

Theorem 22.4 (Two cell families glued). *As stated, for two cell families with plain tables and $s_j, t_j > 0$.*

Proof. (i) 22.11(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table, F_1 at g_1 and F_2 at g_2, the period of the twisted core family being $\mathcal{G}$.

(ii), first form. The gluing of two cells along the hinge is the series closure $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}_1, \mathcal{A}_2\rangle$ (20.5(iii)), so 21.9 at $q = 2$ gives

$$P = C \int \prod_j x_j^{s_j-1} \prod_{j\geq 2} y_j^{t_j-1}\, f_{F_1} f_{F_2}\big(\mathbf{x}, (1,\mathbf{y}')\big)\, d\mathbf{x}\, d\mathbf{y}', \qquad C = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)}.$$

Substitute $\mathbf{x} = x_1(1,\boldsymbol{\xi})$, $d\mathbf{x} = x_1^{N-1} dx_1\, d\boldsymbol{\xi}$, and use the homogeneity of 22.1 with $c = x_1$, $f_{F_i}(\mathbf{x},(1,\mathbf{y}')) = x_1^{\kappa_{F_i}-|\mathbf{s}|} f_{F_i}((1,\boldsymbol{\xi}), x_1(1,\mathbf{y}'))$; then substitute $\boldsymbol{\eta} = x_1(1,\mathbf{y}')$, $dx_1\, d\mathbf{y}' = x_1^{1-N} d\boldsymbol{\eta}$. The powers of $x_1 = \eta_1$ collect to

$$(|\mathbf{s}| - N) + (N-1) + (\kappa_{F_1} + \kappa_{F_2} - 2|\mathbf{s}|) + (1-N) - (|\mathbf{t}| - t_1) + (N-1) = t_1 - 1,$$

since $\kappa_{F_1} + \kappa_{F_2} = |\mathbf{s}| + |\mathbf{t}|$ (check 10 of S21-23_work/checks.py): the first form.

(ii), second form. The complements of the bases of a two-sum of cells are the bases of the two-sum of the dual cells: for X a basis of $\mathcal{A} \setminus g$ and $Y \sqcup \{g\}$ one of $\mathcal{A}'$, $B \setminus X$ is a basis of $\mathcal{A}^*/g$ and $R \setminus Y$ one of $\mathcal{A}'^* \setminus g$, and symmetrically (20.2), with the product weights. By 8.14 the period of $F_1 \oplus_g F_2$ at $(\mathbf{s},\mathbf{t})$ is that of $F_1^* \oplus_g F_2^*$ at $(\mathbf{t},\mathbf{s})$, and by 22.5(iv) the dual cell families have the hinge walls $\kappa_{F_1}, \kappa_{F_2}$ and the transforms $\check{f}_{F_1}, \check{f}_{F_2}$. The first form for them is the second form.

(iii) Insert the Laplace forms of 22.5(ii) into (ii) and integrate first in x_j, $j \geq 2$, and y_j (Tonelli): $\int_0^\infty x^{s-1}e^{-x(\lambda+\lambda')}dx = \Gamma(s)(\lambda+\lambda')^{-s}$, and $x_1 = 1$ leaves $e^{-\lambda_1-\lambda'_1}$; the factors $\Gamma(s_j)$, $j \geq 2$, and $\Gamma(t_j)$ cancel against C. The period converges exactly when every proper wall is positive (8.10).

(iv) Let the contour have the real point $(\mathbf{a}^d, \mathbf{b}^d, \mathbf{a}^r, \mathbf{b}^r)$, all coordinates positive, with $\mathbf{a}^d + \mathbf{a}^r = \mathbf{s}$, $\mathbf{b}^d + \mathbf{b}^r = \mathbf{t}$ and $\sum_j (a_j^r + b_j^d) = \kappa_{F_1}$, and put $\mathbf{a} = \mathbf{a}^d$, $\mathbf{b} = \mathbf{b}^d$; then $\sum_j b_j - \sum_j a_j = \kappa_{F_1} - |\mathbf{s}|$, and the Mellin transform of the law of F_1 at $(\mathbf{s}-\mathbf{a}, \mathbf{b}) = (\mathbf{a}^r, \mathbf{b}^d)$ is finite. By Lemma S8, f_{F_1} is the inverse Mellin integral of G_1 through $\mathbf{c}$. Insert it into the first form of (ii). The double integral converges absolutely: integrating first in $\mathbf{u}$ gives $\int |G_1|\, d\boldsymbol{\tau}$ times $\int \prod_{j\geq 2} x_j^{s_j-a_j-1} \prod_j y_j^{t_j-b_j-1} f_{F_2}\, d\mathbf{x}'\, d\mathbf{y}$, which is 22.5(iii) for F_2 at $(\mathbf{s}-\mathbf{a}, \mathbf{t}-\mathbf{b})$: its first coordinate is $|\mathbf{t}| - |\mathbf{b}| - (|\mathbf{s}| - s_1 - \sum_{j\geq 2} a_j) - \kappa_{F_2} + |\mathbf{s}| = s_1 - a_1$, by $\kappa_{F_2} = 2\sigma - \kappa_{F_1}$, so it equals $\prod_j \Gamma(s_j - a_j)\Gamma(t_j - b_j) \int \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d,\mathbf{b}^r)} d\phi_{F_2} < \infty$. By Fubini,

$$P = C\,(2\pi)^{1-2N} \int_{\mathbb{R}^{2N-1}} \prod_j \Gamma(a_j^d)\Gamma(b_j^d)\Gamma(a_j^r)\Gamma(b_j^r) \int \boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r,\mathbf{b}^d)} d\phi_{F_1} \int \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d,\mathbf{b}^r)} d\phi_{F_2}\, d\boldsymbol{\tau},$$

the shares moving with the imaginary parts $\boldsymbol{\tau}$ of $(a_2^d,\dots,a_N^d,b_1^d,\dots,b_N^d)$, the others fixed by the pins, and $C\prod_j\Gamma(\cdot)$ is $\prod_j\Gamma\left[\begin{matrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{matrix}\right]$. This is the fibre integral of 4.1 over the level shares with their $2N+1$ pins: solved for a_j^r, b_j^r and a_1^d the matrix of the pins has $|\det|=1$, so the fibre integral is $(2\pi)^{1-2N}$ times the integral over $\boldsymbol{\tau}$. In the pair coordinates of 18.13, $\sigma_j^d=\frac12(a_j^d+b_j^d)$ and $\delta_j^d=\frac12(a_j^d-b_j^d)$ with the one pin $\sum_j\delta_j^d=\frac12(|\mathbf{s}|-\kappa_{F_1})$, the linear map from $\boldsymbol{\tau}$ to the imaginary parts of $(\sigma_1^d,\dots,\sigma_N^d,\delta_2^d,\dots,\delta_N^d)$ has $|\det|=2^{1-N}$: the map $(a,b)\mapsto(\frac{a+b}{2},\frac{a-b}{2})$ of $\mathbb{R}^{2N}$ has $|\det|=2^{-N}$, it sends the normal coordinate $\sum_j(a_j-b_j)$ of the pin to twice the normal coordinate $\sum_j\delta_j$, and the quotient is 2^{1-N}. So the integral is 2^{N-1} times the fibre integral in the pair coordinates, the formula of 18.13. That the two Mellin transforms are the periods of the bunch family and of the room family at the level shares is the identity $u^{\mathbf{m}-\mathbf{1}}\prod_j K_{\mathcal{A}_j/g}^{-a_j^d}\Psi_{\mathcal{A}_j/g}^{-b_j^d}K_{\mathcal{A}_j\setminus g}^{-a_j^r}\Psi_{\mathcal{A}_j\setminus g}^{-b_j^r}=\boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r,\mathbf{b}^d)}(\gamma_{\mathcal{A}_1}^K)^{\kappa_{F_1}-\sum_j(a_j^r+b_j^d)}$ times the density of ϕ_{F_1}, the last power being 1 on the pin, with $\mathcal{B}_j|B=\mathcal{A}_j\setminus g$ and $\mathcal{B}_j/R=\mathcal{A}_j/g$ (20.8), and its analogue for the room, whose pin $\sum_j(a_j^d+b_j^r)=2\sigma-\kappa_{F_1}=\kappa_{F_2}$ follows from the letter totals.

One table. At $N=1$ the laws are measures on $(0,\infty)$, the kernel is that of 22.3, (ii) and (iii) have no variables x_j, the image of $e^{-\lambda}\mathfrak{t}_{F_i}$ under μ being $\mathfrak{t}^\Psi$, and in (iv) the fibre is the line of $b^d=z$, with $a^d=s-\kappa_{\mathcal{A}}+z$, $a^r=\kappa_{\mathcal{A}}-z$, $b^r=t-z$; the conditions on the contour are $\max(0,\kappa_{\mathcal{A}}-s)<z_0<\min(t,\kappa_{\mathcal{A}})$ and the finiteness of $\int\varrho_{\mathcal{A}}^{-z_0}d\phi_{\mathcal{A}}$ and $\int\varrho_{\mathcal{A}'}^{z_0-t}d\phi_{\mathcal{A}'}$, and $d\sigma^d=dz$. The first form of (ii) is 19.24(i) with the room transform $y^tf_{\mathcal{A}'}(y)$. □

Definition 22.6 (One weighted mark). As stated: the one-mark cell family of a ratio vector, with two systems of weights in every table.

Proof. Table j has the bases $\{e\}$ and $\{g\}$, with $\theta^K=(x_{0,j,K},1)$ and $\theta^\Psi=(1,x_{0,j,\Psi})$, so $K_j=x_{0,j,K}\alpha_e+\alpha_g$ and $\Psi_j=\theta^\Psi_{\{e\}}\alpha_g+\theta^\Psi_{\{g\}}\alpha_e=\alpha_g+x_{0,j,\Psi}\alpha_e$, two systems of weights being allowed (20.1, 8.1). The hinge minors are $K_{\setminus g}=x_{0,j,K}\alpha_e$, $K_{/g}=1$, $\Psi_{\setminus g}=1$ and $\Psi_{/g}=x_{0,j,\Psi}\alpha_e$, so the hinge values are $x_{0,j,K}\alpha_e$ and $x_{0,j,\Psi}\alpha_e$ and the ratio vector is the constant $\mathbf{x}_0$, $x_{0,1,K}=1$. The rank is 1 and $|B|=1$, so the hinge wall is m_e. The simplex $\Delta_{\{e\}}$ is the point $u_e=1$, carrying the unit mass, and there the density of the base form is $1\cdot1\cdot(x_{0,1,K})^{-m_e}=1$: the law is $\delta_{\mathbf{x}_0}$. □

Theorem 22.7 (The law is what a cell family retains). *As stated, for every pair of cell families with plain tables and $s_j,t_j>0$.*

Proof. (a)⇒(b). By 22.11(i) with the core $\binom{\{g_1,g_2\}}{1}$, the period of $F\oplus_gF''$ depends on F only through κ and its law; this needs no plainness of F''. (b)⇒(c) is a special case.

(c)⇒(d). By 22.11(i) and the law $\delta_{\mathbf{x}_0}$ of the one-mark cell family (22.6), the period against it is $\Pi_F(\mathbf{x}_0)=\int\mathcal{G}(\mathbf{z},\mathbf{x}_0)\,d((\boldsymbol{\varrho}_F)_*\phi_F)(\mathbf{z})$, $\mathcal{G}(\mathbf{z},\mathbf{x}_0)=\int_0^\infty v^{\kappa-1}\prod_j(z_{j,K}v+x_{0,j,K})^{-s_j}(z_{j,\Psi}v+x_{0,j,\Psi})^{-t_j}dv$. Since $(zv+x)/(zv+x')$ lies between $\min(1,x/x')$ and $\max(1,x/x')$, $\mathcal{G}(\cdot,\mathbf{x}_0)$ and $\mathcal{G}(\cdot,\mathbf{x}_0')$ are comparable within constants that depend on $\mathbf{x}_0,\mathbf{x}_0'$ only, so Π_F is finite everywhere on $(0,\infty)^{2N-1}$ or nowhere.

Take the Mellin transform of Π_F in the coordinates of $\mathbf{x}_0$ other than $x_{0,1,K}=1$, with the exponents $s_j-z_{j,K}$ at $x_{0,j,K}$, $j\ge2$, and $t_j-z_{j,\Psi}$ at $x_{0,j,\Psi}$. By Tonelli and $\int_0^\infty q^{w-1}(c+q)^{-s}dq=c^{w-s}\Gamma\left[\begin{matrix}w,\ s-w\\ s\end{matrix}\right]$, $0<w<s$, each coordinate gives a Beta corner and the power $(z_{j,K}v)^{-z_{j,K}}$ or $(z_{j,\Psi}v)^{-z_{j,\Psi}}$, and the remaining integral is $\int_0^\infty v^{z_{1,K}-1}(1+v)^{-s_1}dv=\Gamma\left[\begin{matrix}z_{1,K},\ s_1-z_{1,K}\\ s_1\end{matrix}\right]$, the

power of v collecting to $v^{z_{1,K}-1}$ by the hyperplane. So the Mellin transform is

$$\prod_{j\ge 2}\Gamma_2^1\begin{bmatrix}s_j-z_{j,K},\ z_{j,K}\\ s_j\end{bmatrix}\,\Gamma_2^1\begin{bmatrix}z_{1,K},\ s_1-z_{1,K}\\ s_1\end{bmatrix}\prod_j\Gamma_2^1\begin{bmatrix}t_j-z_{j,\Psi},\ z_{j,\Psi}\\ t_j\end{bmatrix}\int \boldsymbol{\varrho}_F^{-\mathbf{z}}\,d\phi_F<\infty.$$

In particular Π_F is finite almost everywhere, hence everywhere. For complex exponents with these real parts the same computation converges absolutely, and the Beta corners at complex arguments of positive real part do not vanish. Where finite, Π_F extends holomorphically to the product of right half-planes: $|(zv+\zeta)^{-s}|\le(zv+\operatorname{Re}\zeta)^{-s}$ for $\operatorname{Re}\zeta>0$ and $s>0$, so dominated convergence and Morera's theorem give holomorphy in each coordinate, the extension being locally bounded, and hence jointly. So Π_F and $\Pi_{F'}$ are real-analytic on $(0,\infty)^{2N-1}$, equal on a nonempty open set, and equal everywhere; their Mellin transforms on the vertical subspace through $\mathbf{z}$ agree, which is (d).

(d)⇒(a). The finite measures $\boldsymbol{\varrho}^{-\mathbf{z}}(\boldsymbol{\varrho}_F)_*\phi_F$ and $\boldsymbol{\varrho}^{-\mathbf{z}}(\boldsymbol{\varrho}_{F'})_*\phi_{F'}$ on $(0,\infty)^{2N-1}$, read in the logarithms of the coordinates, have the Fourier transforms $\boldsymbol{\tau}\mapsto\int\boldsymbol{\varrho}^{-(\mathbf{z}+i\boldsymbol{\tau})}d\phi$, equal by (d); a finite measure on $\mathbb{R}^{2N-1}$ is determined by its Fourier transform, so the two measures agree, and multiplying by the positive continuous function $\boldsymbol{\varrho}^{\mathbf{z}}$ the laws agree. This replaces the Gaussian smoothing of the reference proof of 22.7, which is the same argument on the line.

One table. The plain one-mark cell of ratio x_0 has the law $x_0^{(m_e-s)/2}\delta_{x_0}$, a positive multiple of the point law, and its ratios fill $(0,\infty)$, so (c) with the plain cells gives the equality of Π_F on an open interval; (d) is the case $N=1$, $\mathbf{w}=(w_{1,K},w)$ with $w_{1,K}$ not entering. □

Corollary 22.8 (Cells that are marks). *As stated: (i) for every cell family with plain tables and $s_j,t_j>0$.*

Proof. (i) This is the reference proof of 22.8(i) with the ratio vector in place of the ratio. If $\boldsymbol{\varrho}_F\equiv\mathbf{x}_0$, the image of ϕ_F is $\phi_F(\Delta_B)\delta_{\mathbf{x}_0}$. Conversely, ϕ_F has a positive continuous density on the open simplex and $\boldsymbol{\varrho}_F$ is continuous, so if the law is carried by $\mathbf{x}_0$ the open set $\{\boldsymbol{\varrho}_F\ne\mathbf{x}_0\}$ has ϕ_F-measure 0 and is empty. The mass $\phi_F(\Delta_B)$ is positive. By 22.11(i), the period of a gluing $F\oplus_g F''$ is $\phi_F(\Delta_B)\int\mathcal{G}(\mathbf{x}_0,\mathbf{z}')\,d((\boldsymbol{\varrho}_{F''})_*\phi_{F''})(\mathbf{z}')$, a positive multiple of $\phi_F(\Delta_B)$; so a finite period forces $\phi_F(\Delta_B)<\infty$, and the integral is the period of the gluing of the one-mark cell family of ratio vector $\mathbf{x}_0$ and mass κ_F with F'', whose law is $\delta_{\mathbf{x}_0}$ and whose hinge wall is κ_F. For one plain cell, the one-mark cell of ratio x_0 and mass $\kappa_{\mathcal{A}}$ has the law $x_0^{(\kappa_{\mathcal{A}}-s)/2}\delta_{x_0}$, which gives the factor. (ii) and (iii) are the reference's. □

Theorem 22.13 (Series and parallel nodes). *As stated: (i) and (ii) for cell families with plain tables and $s_j,t_j>0$.*

Proof. This is the reference proof of 22.13(i),(ii), table by table. *(i)* The series core $\binom{G}{q}$, $G=\{g_0,\dots,g_q\}$, has rank q; its deletion at g_0 has the one basis $\{g_1,\dots,g_q\}$, so $K_{\binom{G}{q}\setminus g_0}(\mathbf{v})=\prod_i v_i$ and $\Psi_{\binom{G}{q}\setminus g_0}=1$, and its contraction has the bases of size $q-1$ of $\{g_1,\dots,g_q\}$, so $K_{\binom{G}{q}/g_0}=e_{q-1}(\mathbf{v})$ and $\Psi_{\binom{G}{q}/g_0}(\mathbf{v})=\sum_i v_i$. By 22.10,

$$K_{\mathcal{A}_j\setminus g_0}=\prod_i K_{\mathcal{A}_{ij}/g_i}\gamma^K_{\mathcal{A}_{ij}}=\prod_i K_{\mathcal{A}_{ij}\setminus g_i},\qquad \Psi_{\mathcal{A}_j\setminus g_0}=\prod_i\Psi_{\mathcal{A}_{ij}\setminus g_i},$$

and the hinge values of $\mathcal{A}_j$ are $\prod_i\gamma^K_{\mathcal{A}_{ij}}/e_{q-1}((\gamma^K_{\mathcal{A}_{ij}})_i)$, whose reciprocal is $\sum_i 1/\gamma^K_{\mathcal{A}_{ij}}$, and $\sum_i\gamma^\Psi_{\mathcal{A}_{ij}}$. In the orthant form of f_F the measure $\alpha^{\mathbf{m-1}}\prod_j K^{-s_j}_{\mathcal{A}_j\setminus g_0}\Psi^{-t_j}_{\mathcal{A}_j\setminus g_0}d\alpha$ is the product over i of the

measures of the cell families, and the exponential is the product of their exponentials, so Tonelli gives $f_F = \prod_i f_{F_i}$, and the image of the product measure under the sum of the maps of the factors is the convolution: $\mathfrak{r}_F = *_i \mathfrak{r}_{F_i}$. The hinge-wall formula of 22.11(ii), with $\mathcal{E}_{\mathcal{H}} = \varnothing$, $k = q$ and $r_{\mathcal{H}_j} = q$, gives $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{s}|$ (check 3a of S21-23_work/checks.py). For one table, $f_{\mathcal{A}}(y) = f_F(1,y) = \prod_i f_{F_i}(1,y)$, and since $e^{-\lambda} = \prod_i e^{-\lambda_i}$ for $\lambda = \sum_i \lambda_i$, the measure $e^{-\lambda}\mathfrak{r}_F$ is the convolution of the $e^{-\lambda_i}\mathfrak{r}_{F_i}$, whose images under μ convolve: $\mathfrak{r}^{\Psi}_{\mathcal{A}} = *_i \mathfrak{r}^{\Psi}_{\mathcal{A}_i}$. The twisted series core is the reference's.

(ii) The parallel core $\binom{G}{1}$ has rank 1: $K_{\binom{G}{1}\setminus g_0} = \sum_i v_i$, $K_{\binom{G}{1}/g_0} = 1$, $\Psi_{\binom{G}{1}\setminus g_0} = e_{q-1}(\mathbf{v})$ and $\Psi_{\binom{G}{1}/g_0} = \prod_i v_i$. So $K_{\mathcal{A}_j/g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}$, $\Psi_{\mathcal{A}_j/g_0} = \prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i} \gamma^{\Psi}_{\mathcal{A}_{ij}} = \prod_i \Psi_{\mathcal{A}_{ij}/g_i}$, the hinge values combine as stated, the orthant form of $\check{f}_F$ factors, and $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{t}|$ (check 3b). Equally, 22.5(iv) exchanges (i) and (ii) through the dual cell families, the dual of a parallel core being a series core. (iii) and (iv) are the reference's. □

Definition 22.14 (Hinge tree). As stated: the common hinge tree of the tables of a family, with its node families and its cell families.

Remark. The added objects are definitions. Each $\mathcal{A}_{v,j}$ is a cell with the hinge $g_{v,\mathfrak{e}_v}$ on the marks below v (22.14), the same for every j, so the $\mathcal{A}_{v,j}$ form a cell family in the sense of 22.1; and the substitution along the tree does not depend on the order of the two-sums, table by table (22.10).

Theorem 22.15 (The period along a hinge tree). *As stated: for every family whose tables have a common hinge tree, at real data.*

Proof. Induction from the leaves. At a leaf v the cell family F_v is the node family with its one marker as hinge. At an inner vertex $v \neq o$, table j of F_v is the substitution of the node $\mathcal{H}_{v,j}$, a hinged core with the hinge $g_{v,\mathfrak{e}_v}$, with the subtree cells $\mathcal{A}_{w,j}$ of the vertices w directly below it at the markers of their edges (22.14), so 22.11(ii) gives (i), the masses on $\mathcal{E}_v$ being those of F. At the root, the node has no hinge, and 22.11(i) gives (ii). (iii) is the recursion of (i): the law at v is computed from the nodes and masses at v and the laws of the vertices directly below, and by induction from the data below v. This is the reference proof of 22.15 (thm:contraction), with 22.11 for families, which the reference's remark on families (rem:familiesnodes) gives verbatim. □

Proposition 22.16 (What the work depends on). *As stated, for a family whose tables have a common hinge tree.*

Proof. (i) A bijection of marks carrying the subtree cells onto each other in every table, with their weights, masses and hinge, carries the letters onto the letters, hence the base forms and the ratio vectors onto each other; the ranks, sizes and masses are kept, and so is the hinge wall.

(ii) At a series node of unit weights, F_v is the series connection of the cell families directly below (22.12), so 22.13(i) gives $f_{F_v} = \prod_w f_{F_w}$ and $\kappa_{F_v} = \sum_w \kappa_{F_w} - (q-1)|\mathbf{s}|$, $q = \sum_l k_l$; members of one orbit have equal transforms and hinge walls by (i), and collecting them gives the powers. A weight θ_{g_i} on the basis of a series node that omits the marker g_i is carried, by the weight formula of 20.1, by the cell at g_i as the factor $\theta_{g_i}/\theta_{g_0}$ on its bases through its hinge, the weight θ_{g_0} of the basis omitting the hinge remaining as a constant factor of the table; so a series node with weights is a series connection of cells so weighted, and the orbits are taken of these. The parallel node is the same with 22.13(ii).

(iii) By 22.1 the hinge wall of F_v is $m_{B_v} - \sum_j \big(s_j(r_{\mathcal{A}_{v,j}} - 1) + t_j(|B_v| - r_{\mathcal{A}_{v,j}})\big)$, B_v the marks below v, and by 22.10 the ranks $r_{\mathcal{A}_{v,j}}$ are sums of the ranks of the nodes below v less one per edge. A change at v_0 keeping the total mass at v_0, the ranks and the numbers of marks of the

nodes, keeps every m_{B_v}, rank and size, since B_v contains all of $\mathcal{E}_{v_0}$ or none of it; so no hinge wall changes, and by 22.15(iii) the law of F_v changes only if v_0 lies below v, that is, for v on the path from v_0 to the root. A change of masses at v_0 and v_1 keeping the total changes m_{B_v} exactly when B_v contains one of $\mathcal{E}_{v_0}, \mathcal{E}_{v_1}$ and not the other, by the change of the total mass below $\mathfrak{e}_v$. □

Definition 22.17 (The hinge tree procedure). As stated: for a family whose tables have a common hinge tree, the representations (a) and (b) in $2N-1$ variables, (c) for one table.

Remark. Each operation rests on a statement for families. The laws and hinge walls of the orbits are those of 22.15 and 22.16(i). (1) The one-mark leaf: by 22.5(i) at the point law $\delta_{\mathbf{x}_0}$ with $\kappa = m_e$, $f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,K}^{-s_j} \mathsf{G}_{|\mathbf{s}|-m_e;(1,-1)}(\sum_j x_j/x_{0,j,K}, \sum_j y_j x_{0,j,\Psi})$, and the second law form with torus homogeneity at $\lambda = x_{0,1,\Psi}$ gives $\check{f}(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,\Psi}^{-t_j} \mathsf{G}_{|\mathbf{t}|-m_e;(1,-1)}(\sum_j x_j/x_{0,j,\Psi}, \sum_j y_j x_{0,j,K})$, the powers of $x_{0,1,\Psi}$ cancelling; the orthant forms give the same directly. For one table, $\mathbf{x}_0 = (1, x_0)$ and the plain cell's law $x_0^{(m_e-s)/2}\delta_{x_0}$ give the displayed one-table transforms, with $\mathsf{G}_{t-m_e;(1,-1)}(x_0^{-1}, y) = x_0^{t-m_e}\mathsf{G}_{t-m_e;(1,-1)}(1, y/x_0)$. (2) 22.13(i),(ii), with 22.16(ii) for the weights; a pointwise product of transforms is a Mellin convolution of their Mellin transforms of 22.5(iii). (3) 22.5(v) on the Mellin side, and Lemma S8 for the inverse Mellin transform, applied to the dual cell family for $\check{f}$ (22.5(iv)). (4) 22.11(ii) for families. (5) 22.11(i), and for a series or parallel root 22.4(ii) with the series or parallel connection of all but one cell family on one side.

Corollary 22.18 (Series-parallel families). *As stated, for a family whose tables have a common hinge tree of series and parallel nodes over one-mark cell families.*

Proof. This is the reference proof of 22.18 with the family statements. The leaves have their transforms in closed form, Grammas of the marks (the remark after 22.17 in this part). At a series node the transform of the subtree is the product of those below, and at a parallel node the dual transform (22.13), so no integral is taken there; repeated subtrees enter as powers (22.16(ii)). At an edge joining a series node to a parallel node the transform of the subtree below is changed into its dual transform by 22.5(v): the Mellin transform of f on a vertical subspace through a real point $(\mathbf{a},\mathbf{b})$, $0 < a_j < s_j$, $0 < b_j < t_j$, of its domain, an integral on $(0,\infty)^{2N-1}$; multiplication by the ratio of Gammas; and the inverse Mellin transform of the result, which is the Mellin transform of $\check{f}$ (22.5(iii) at $(\mathbf{t}-\mathbf{b}, \mathbf{s}-\mathbf{a})$), an integral on $(0,\infty)^{2N-1}$ that converges absolutely and returns $\check{f}$ by Lemma S8 applied to the dual cell family (22.5(iv)). The hypothesis is the finiteness of the Mellin transform of the law at $\mathbf{z} = (\mathbf{s}-\mathbf{a}, \mathbf{b})$, which these points require. At the root one integral on $(0,\infty)^{2N-1}$ remains, 22.4(ii). The count of the integrations depends only on the edges where the kind of node changes and on the root. For one table the variables are one. □

40.23 Chapter 23: repetition and the change of query

Theorem 23.1 (How often a cell family repeats). *As stated: for every cell family with plain tables, at real exponents.*

Proof. (i) The series connection of k copies of $\mathcal{A}_j$ has $k|B|$ marks besides the hinge, the masses repeated, and the rank $k + k(r_{\mathcal{A}_j} - 1) = kr_{\mathcal{A}_j}$ (22.10, the series core of rank k). By 22.1, since $s_j(kr_{\mathcal{A}_j} - 1) = ks_j(r_{\mathcal{A}_j} - 1) + (k-1)s_j$ and $t_j(k|B| - kr_{\mathcal{A}_j}) = kt_j(|B| - r_{\mathcal{A}_j})$,

$$\kappa_{F_k^{\mathrm{s}}} = km_B - \sum_j \big(s_j(kr_{\mathcal{A}_j} - 1) + t_j(k|B| - kr_{\mathcal{A}_j})\big) = k\kappa_F - (k-1)|\mathbf{s}| = |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|).$$

The parallel connection has the rank $1 + k(r_{\mathcal{A}_j} - 1)$, and the same count gives $k\kappa_F - (k-1)|\mathbf{t}|$ (checks 3a and 3b of S21-23_work/checks.py, which are this count at the exponents as symbols). No sign of an exponent enters.

(ii) The walls of $F_k^{\mathrm{s}} \oplus_g F'$ at the marks of the copies and at those of F' are the hinge walls $\kappa_{F_k^{\mathrm{s}}}$ and $\kappa_{F'}$, and on the balance wall they add to 2σ (20.8). A convergent period has every proper wall positive (8.10, at any complex exponents), so $0 < \kappa_{F_k^{\mathrm{s}}} < 2\sigma$. If $\kappa_F \neq |\mathbf{s}|$ the middle term is unbounded in k, which bounds k: $|\mathbf{s}| - k(|\mathbf{s}| - \kappa_F) > 0$ gives $k < |\mathbf{s}|/(|\mathbf{s}| - \kappa_F)$ when $\kappa_F < |\mathbf{s}|$, and $|\mathbf{s}| + k(\kappa_F - |\mathbf{s}|) < |\mathbf{s}| + |\mathbf{t}|$ gives $k < |\mathbf{t}|/(\kappa_F - |\mathbf{s}|)$ when $\kappa_F > |\mathbf{s}|$. The parallel case is the same with the roles of $|\mathbf{s}|$ and $|\mathbf{t}|$ exchanged. □

Theorem 23.2 (Series repetition). *As stated: (i) for cell families with plain tables and $s_j, t_j > 0$; (ii) to (iv) are the reference's, for one cell.*

Proof. (i) By 22.13(i) with k copies of F, $f_{F_k^{\mathrm{s}}} = f_F^k$; by 23.1(i), $\kappa_{F_k^{\mathrm{s}}} = |\mathbf{s}|$, which lies in $(0, 2\sigma)$ since $|\mathbf{s}|, |\mathbf{t}| > 0$. So the first form of 22.4(ii) applies to F_k^{s} and F' and gives the display. With ν the measure of the statement, $P(k) = \int f_F^k \, d\nu = \int w^k \, d((f_F)_*\nu)(w)$. Where finite, $P(k)^2 = \big(\int f_F^{(k-1)/2} f_F^{(k+1)/2} d\nu\big)^2 \leq P(k-1)P(k+1)$ by the Cauchy–Schwarz inequality, and for $z \geq 0$ Tonelli gives $\sum_k P(k) z^k/k! = \int e^{z f_F} d\nu$.

The value $P(0)$. Insert the orthant form of $f_{F'}$ and integrate first in x_j, $j \geq 2$, and y_j (Tonelli): $\int_0^\infty x^{s_j - 1} e^{-x/\gamma^K} dx = \Gamma(s_j)(\gamma^K)^{s_j}$ and $\int_0^\infty y^{t_j - 1} e^{-y\gamma^\Psi} dy = \Gamma(t_j)(\gamma^\Psi)^{-t_j}$. Since $K_{\mathcal{A}'_j \backslash g}(\gamma^K_{\mathcal{A}'_j})^{-1} = K_{\mathcal{A}'_j/g}$ and $\Psi_{\mathcal{A}'_j \backslash g} \gamma^\Psi_{\mathcal{A}'_j} = \Psi_{\mathcal{A}'_j/g}$, what remains is

$$\prod_{j \geq 2} \Gamma(s_j) \prod_j \Gamma(t_j) \int \alpha^{\mathbf{m}|_R - \mathbf{1}} e^{-1/\gamma^K_{\mathcal{A}'_1}} K_{\mathcal{A}'_1 \backslash g}^{-s_1} \prod_{j \geq 2} K_{\mathcal{A}'_j/g}^{-s_j} \prod_j \Psi_{\mathcal{A}'_j/g}^{-t_j} \, d\alpha.$$

In $\alpha = \rho u$ the power of ρ is $\rho^{\kappa_{F'} - |\mathbf{t}| - s_1 - 1} = \rho^{-s_1 - 1}$, by $\kappa_{F'} = |\mathbf{t}|$ (check 9 of S21-23_work/checks.py), and $\int_0^\infty \rho^{-s_1 - 1} e^{-1/(\rho\gamma)} d\rho = \Gamma(s_1)\gamma^{s_1}$ turns $K_{\mathcal{A}'_1 \backslash g}^{-s_1}$ into $K_{\mathcal{A}'_1/g}^{-s_1}$. So $P(0)$ is the period at $\mathbf{m}|_R$ of the family of the contractions, whose degree is $\sum_j (s_j(r_{\mathcal{A}'_j} - 1) + t_j(|R| - r_{\mathcal{A}'_j} + 1))$, so that it is balanced exactly when $\kappa_{F'} = |\mathbf{t}|$. For one cell, $\int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'}$ with $\kappa_{\mathcal{A}'} = t$ has the density $u^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}'/g}^{t-s} K_{\mathcal{A}' \backslash g}^{-t} \Psi_{\mathcal{A}' \backslash g}^{-t} (\gamma^\Psi/\gamma^K)^{-t} = u^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}'/g}^{-s} \Psi_{\mathcal{A}'/g}^{-t}$, the same period.

(ii) to (iv) are the reference's, for one cell. □

Remark. For a family the asymptotics of $P(k)$ is a Laplace problem in $2N - 1$ variables: a series-balanced cell family has $f_F = \int \mathsf{G}_{0;(1,-1)}(1, ab) \, d\phi_{F \backslash g}$ by 21.9, of the order of $P_{F \backslash g}$ times the logarithm of the scale of $\mathbf{y}$ plus a term that depends on the direction of $\mathbf{y}$ and on $\mathbf{x}'$, and the uniformity of the error in these directions does not follow from the one-variable argument. So (ii) to (iv) are stated for one cell.

Corollary 23.3 (Parallel repetition). *As stated: the first display for cell families with plain tables and $s_j, t_j > 0$; the rest is the reference's, for one cell.*

Proof. By 22.13(ii), $\check{f}_{F_k^{\mathrm{p}}} = \check{f}_F^k$, and by 23.1(i), $\kappa_{F_k^{\mathrm{p}}} = |\mathbf{t}| \in (0, 2\sigma)$; the second form of 22.4(ii) gives the display. For $P(0)$, integrate the orthant form of $\check{f}_{F'}$ in x_j, $j \geq 2$, and y_j: $\int x^{t_j - 1} e^{-x/\gamma^\Psi} dx = \Gamma(t_j)(\gamma^\Psi)^{t_j}$ and $\int y^{s_j - 1} e^{-y\gamma^K} dy = \Gamma(s_j)(\gamma^K)^{-s_j}$ turn $\Psi_{\mathcal{A}'_j/g}^{-t_j}$ into $\Psi_{\mathcal{A}'_j \backslash g}^{-t_j}$, $j \geq 2$, and every $K_{\mathcal{A}'_j/g}^{-s_j}$ into $K_{\mathcal{A}'_j \backslash g}^{-s_j}$; in $\alpha = \rho u$ the power of ρ is $\rho^{\kappa_{F'} - |\mathbf{s}| - t_1 - 1} = \rho^{-t_1 - 1}$, by $\kappa_{F'} = |\mathbf{s}|$, and $\int \rho^{-t_1 - 1} e^{-1/(\rho\gamma_1^\Psi)} d\rho = \Gamma(t_1)(\gamma_1^\Psi)^{t_1}$ turns $\Psi_{\mathcal{A}'_1/g}^{-t_1}$ into $\Psi_{\mathcal{A}'_1 \backslash g}^{-t_1}$. What remains is the period of the family of the deletions at $\mathbf{m}|_R$, balanced exactly when $\kappa_{F'} = |\mathbf{s}|$. Equally, the dual cell families at

$(\mathbf{t},\mathbf{s})$ turn the statement into (i) of 23.2 (22.5(iv), 8.14), the dual of the contractions being the deletions of the duals. The rest is the reference's, for one cell. □

Proposition 23.4 (A chain is a power of one operator). *As stated, for chain cell families at real exponents, (iv) with plain tables and $s_j, t_j > 0$.*

Proof. This is the reference proof of 23.4 with the laws on $(0,\infty)^{2N-1}$. *(i)* 22.11(ii) with the one cell family F_l at g_{in}: its hinge-wall formula, $\mathcal{U}_j$ having the marks $\mathcal{E}_{\mathcal{U}} \sqcup \{g_{\text{in}}\}$ besides its hinge, is $\kappa_{F_{l+1}} = \kappa_{F_l} + \kappa_U$. By 20.8, $F_l \oplus_g F'$ is on its balance wall exactly when $\kappa_{F_l} + \kappa_{F'} = 2\sigma$; with $\kappa_{F_l} = \kappa_{F_0} + l\kappa_U$ this holds for every l exactly when it holds at $l = 0$ and $\kappa_U = 0$.

(ii) 22.11(ii) with the one cell family F_l at g_{in} gives $\lambda_{l+1} = \int \lambda^U_{\mathbf{x}}\, d\lambda_l(\mathbf{x}) = \mathcal{Q}\lambda_l$, the twisted family carrying the mass $\kappa_{F_l} = \kappa_{F_0}$ at g_{in}; by induction $\lambda_l = \mathcal{Q}^l\lambda_0$. 22.11(i) with the core $\binom{\{g_1,g_2\}}{1}$ gives $P_{F_l\oplus_g F'} = \int h_{F'}\, d\lambda_l$, and $\int H\, d(\mathcal{Q}\mu) = \int (\mathcal{Q}^*H)\, d\mu$ for $H \geq 0$ by Tonelli, so $\int h_{F'}\, d\lambda_l = \int (\mathcal{Q}^{*l} h_{F'})\, d\lambda_0$. The series in $z \geq 0$ is summed under the integral by Tonelli.

(iii) $\mathcal{Q}^*$ preserves inequalities between nonnegative functions, so $\mathcal{Q}^*H \leq MH$ gives $\mathcal{Q}^{*l}H \leq M^lH$, and $h_{F'} \leq CH$ gives $\mathcal{Q}^{*l}h_{F'} \leq CM^lH$; integrate against λ_0. The lower bound is the same.

(iv) For the series core $\binom{\{g_{\text{out}},g_{\text{in}},g'\}}{2}$ with F'' at g', F_{l+1} is the series connection of F_l and F'', so 22.13(i) gives $f_{F_{l+1}} = f_{F''}f_{F_l}$, and the hinge-wall formula with the mass 0 at g_{in}, the marks $\{g_{\text{in}}\} \sqcup B''$ and the ranks $2 + r_{\mathcal{A}''_j} - 1$, gives $\kappa_U = m_{B''} - \sum_j \left(s_j r_{\mathcal{A}''_j} + t_j(|B''| - r_{\mathcal{A}''_j})\right) = \kappa_{F''} - |\mathbf{s}|$. For the parallel core the ranks are $r_{\mathcal{A}''_j}$, the formula gives $\kappa_{F''} - |\mathbf{t}|$, and 22.13(ii) gives the product of the dual transforms. □

Theorem 23.5 (The weight of a hinge). *As stated: (i) for cell families with plain tables, one weight per table; (ii) to (iv) are the reference's, for one cell.*

Proof. (i) The bases through g carry the monomials of $K_{\mathcal{A}_j/g}$ and of $\Psi_{\mathcal{A}_j/g}$, which are multiplied by w_j; the deletions keep their weights (the reference proof of 23.5(i), table by table). So the hinge wall is kept, $\gamma^K_{\mathcal{A}_j} = K_{\mathcal{A}_j\setminus g}/K_{\mathcal{A}_j/g}$ becomes $\gamma^K_{\mathcal{A}_j}/w_j$ and $\gamma^\Psi_{\mathcal{A}_j} = \Psi_{\mathcal{A}_j/g}/\Psi_{\mathcal{A}_j\setminus g}$ becomes $w_j\gamma^\Psi_{\mathcal{A}_j}$; the coordinates of the ratio vector become $(w_1/w_j)\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$ and $w_1w_j\gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$, and the base form $u^{\mathbf{m}-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{A}_j/g}\Psi^{-t_j}_{\mathcal{A}_j\setminus g}(\gamma^K_{\mathcal{A}_1})^{-\kappa_F}d\omega$ is multiplied by $\prod_j w_j^{-s_j}\cdot w_1^{\kappa_F}$. In the orthant forms, $x_j/\gamma^K_{\mathcal{A}_j}$ becomes $w_jx_j/\gamma^K_{\mathcal{A}_j}$ and $y_j\gamma^\Psi_{\mathcal{A}_j}$ becomes $w_jy_j\gamma^\Psi_{\mathcal{A}_j}$ with the measure of f_F unchanged, and for $\check{f}_F$ the measure is multiplied by $\prod_j w_j^{-s_j-t_j}$ while $x_j/\gamma^\Psi_{\mathcal{A}_j}$ and $y_j\gamma^K_{\mathcal{A}_j}$ are divided by w_j. For one cell, $w_1 = w$: the ratio becomes $w^2\varrho_{\mathcal{A}}$, the base form $w^{\kappa_{\mathcal{A}}-s}\phi_{\mathcal{A}}$, and the homogeneity of 22.1 with $c = w$ gives $f_{w\mathcal{A}}(y) = f_F(w,wy) = w^{\kappa_{\mathcal{A}}-s}f_{\mathcal{A}}(w^2y)$, and with $c = 1/w$, $\check{f}_{w\mathcal{A}}(y) = w^{-s-t}\check{f}_F(1/w, y/w) = w^{-s-\kappa_{\mathcal{A}}}\check{f}_{\mathcal{A}}(y/w^2)$. (ii) to (iv) are the reference's. □

Remark. For a family the weights act on the law through an N-dimensional subtorus of $(0,\infty)^{2N-1}$, and the Mellin transform in $\mathbf{w}$ is a restricted Mellin transform of the law against the transform of the room, without the product formula of (ii); (iii) and (iv) concern the one-variable expansions of the node of width one. So (ii) to (iv) are stated for one cell.

Proposition 23.6 (Carrying a calculation through a change). *As stated, for a family whose tables have a common hinge tree.*

Proof. (i) and (iii) are 22.16(iii) with the recursion of 22.15(i): the law at a vertex is recomputed from the new law of the vertex directly below it on the path and the kept laws of the others, and the root by 22.15(ii). (ii) is 23.5(i) for the cell family F_v: the weight of the hinge multiplies the base form by $w_1^{\kappa_{F_v}}\prod_j w_j^{-s_j}$ and moves the ratio vector by the stated map, so the law is multiplied

by that factor and carried by that map; the cell families below v are unchanged, and above v the change of the law of F_v is a change at a vertex directly below, carried by (i). □

40.24 Chapter 24: holds and the lean convolution

In this section $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ is a family as in Definition 24.1: its tables are each one molecule on $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$, and its exponents are positive, unless a statement says otherwise. Its $2N$ letters are written X, running over $K_1, \dots, K_N, \Psi_1, \dots, \Psi_N$. The letter X of table j has the exponent w_X, which is s_j or t_j, the degree $\deg X$, which is r_j or r_j^*, the twist $\mathbf{x}^X$, which is $\mathbf{x}_j^K$ or $\mathbf{x}_j^\Psi$, and the Newton polytope $\mathcal{N}_X$, which is $P(\mathcal{B}_j)$ or $P(\mathcal{B}_j^*)$; for masses $\mathbf{a}$ with $|\mathbf{a}| = w_X \deg X$ the period of its one-letter word X^{-w_X} is written $P_X(\mathbf{a})$, which is $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0)$ or $P_{\mathcal{B}_j}(\mathbf{a}; 0, t_j)$. $\mathcal{X}'$ is the set of the $2N-1$ letters other than K_1, and in a split $\mathbf{m} = \sum_X \mathbf{a}^X$ the load of X is $\mathbf{a}^X$, so that $\mathbf{a}^{K_j} = \mathbf{a}_j$ and $\mathbf{a}^{\Psi_j} = \mathbf{b}_j$.

Two facts are used throughout. First, the base polytope of a table that is one molecule, and that of its dual table, have dimension $n-1$: if $w \in \mathbb{R}^{\mathcal{E}}$ is orthogonal to every difference $\mathbf{1}_T - \mathbf{1}_{T'}$ of bases, then $\langle w, \mathbf{1}_T \rangle$ is constant over the bases and w is constant by Lemma 7.7; so these differences span $\mathbf{1}^\perp$, and so do the differences $\mathbf{1}_{\mathcal{E}\setminus T'} - \mathbf{1}_{\mathcal{E}\setminus T}$. Second, a point $\sum_k p_k v_k$ with every $p_k > 0$ and $\sum_k p_k = 1$ lies in the relative interior of the convex hull of the finitely many points v_k: for $x = \sum_k q_k v_k$ in the hull and $0 < \varepsilon \leq \min_k p_k$, the point $\sum_k p_k v_k + \varepsilon\big(\sum_k p_k v_k - x\big) = \sum_k \big((1+\varepsilon)p_k - \varepsilon q_k\big) v_k$ has nonnegative coefficients of sum 1; so every segment from a point of the hull to the given point extends beyond it within the hull, which characterizes the relative interior of a convex set.

Definition 24.1 (Holding). As stated: the hold of a family, the Mellin transform in the twists of its letters other than K_1.

Remark. That the bases of every table span $\mathbb{R}^{\mathcal{E}}$ is the first fact above. By the torus relation of Definition 21.1, a factor $\mu > 0$ on every coordinate of one twist $\mathbf{x}^X$ multiplies the period by $\mu^{-w_X \deg X}$, and a twist of every letter by one vector $\boldsymbol{\lambda}$ multiplies it by $\boldsymbol{\lambda}^{-\mathbf{m}}$; so the Mellin transform in all the twists of all the letters at all the marks diverges, and the hold keeps the twist of K_1 at $\mathbf{1}$ and, in Lemma 24.4, the twists at one mark.

Definition 24.2 (Loads; admissible points). As stated: the loads and the admissible points of a family, an admissible point of one table being named by its Ψ-load.

Proposition 24.3 (The periods at the two full leans). *As stated; (iii) is stated for every family, and the closing sentence is new.*

Proof. Items (i), (ii) and (iv) concern one table $\mathcal{H}$ of F, one molecule, and are the reference, proof of prop:fullleans. The closing sentence is Proposition 24.13, proved below; at the letter $K_{\mathcal{H}}$, with the support $(\mathbf{1}_T)_T$, the coefficients $\boldsymbol{\theta}$ and the Newton polytope $P(\mathcal{H})$ of dimension $n-1$, it gives back (i), (ii) and (iv) at the full basis lean, and at $\Psi_{\mathcal{H}}$ at the full cobasis lean.

(iii) Let α be a gauge. By Theorem 7.14, $R_j(\alpha) = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[\mathbf{1}_T] = \sum_{T\in\mathcal{B}_j} \mathbb{P}_{j,\alpha}(T)\, \mathbf{1}_T$ with $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T / K_j(\alpha) > 0$ for every basis, and $R_{j,e}^*(\alpha) = \mathbb{P}_{j,\alpha}^*(e \notin T)$, so $R_j^*(\alpha) = \sum_T \mathbb{P}_{j,\alpha}^*(T)\, \mathbf{1}_{\mathcal{E}\setminus T}$ with $\mathbb{P}_{j,\alpha}^*(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T} / \Psi_j(\alpha) > 0$. By the second fact above $R_j(\alpha) \in \operatorname{ri} P(\mathcal{B}_j)$ and $R_j^*(\alpha) \in \operatorname{ri} P(\mathcal{B}_j^*)$, the two polytopes being the convex hulls of the $\mathbf{1}_T$ and of the $\mathbf{1}_{\mathcal{E}\setminus T}$.

Let $\mathbf{m}$ be real on the balance wall with every proper wall positive. A mark of a table that is one molecule on at least two marks is neither a loop nor a coloop, so $d_j(\{e\}) = d_j^*(\{e\}) = 0$

and $m_e = \kappa_{\{e\}} > 0$ for every e (Definition 8.1). A nonempty proper S with $c_j(S) = 0$ would have $|T \cap S|$ constant over the bases of table j, and so would be a union of molecules of it by Lemma 25.1, which is impossible, table j being one molecule; so $c_j(S) > 0$ in every table, every table being active, and F is connected. By Theorem 8.18(ii) and (iii) a native gauge $\alpha^\star$ exists, the target being $\mathbf{m}$ on the balance wall (Definition 8.16) and the walls at the target positive. Its usage is the target, $\sum_j \big(s_j R_j(\alpha^\star) + t_j R_j^*(\alpha^\star)\big) = \mathbf{m}$: the native loads are a split of $\mathbf{m}$ among the letters, with $|\mathbf{a}_j^\star| = s_j r_j$ and $|\mathbf{b}_j^\star| = t_j r_j^*$, every $\mathbf{1}_T$ having r_j marks and every $\mathbf{1}_{\mathcal{E}\setminus T}$ having r_j^*. By the preceding paragraph $\mathbf{a}_j^\star \in s_j \operatorname{ri} P(\mathcal{B}_j)$ and $\mathbf{b}_j^\star \in t_j \operatorname{ri} P(\mathcal{B}_j^*)$, so the native loads are admissible (Definition 24.2). □

Lemma 24.4 (Holding every mark separates the letters). *As stated, for every family.*

Proof. Let the data be real. By Theorem 8.10 the period of $F[\mathbf{x}]$ is the integral over the section $\{\alpha_{e_0} = 1\}$,

$$P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{(0,\infty)^{\mathcal{E}\setminus e_0}} \alpha^{\mathbf{m}-\mathbf{1}} K_1(\alpha)^{-s_1} \prod_{X\in\mathcal{X}'} X(\mathbf{x}^X \circ \alpha)^{-w_X}\, d\alpha', \qquad \alpha = (\alpha', 1),$$

the twists at e_0 being 1 and the twist of K_1 being $\mathbf{1}$. Insert this into the hold and integrate first over the twists, by Tonelli's theorem, every function being positive. For fixed α' substitute $\mathbf{v}^X = \mathbf{x}^X \circ \alpha'$ on $\mathcal{E}\setminus e_0$ for every $X \in \mathcal{X}'$: then $(\mathbf{x}^X)^{\boldsymbol{\beta}^X - \mathbf{1}} d\mathbf{x}^X = \alpha'^{-\boldsymbol{\beta}^X} (\mathbf{v}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} d\mathbf{v}^X$ and $X(\mathbf{x}^X \circ \alpha) = X(\mathbf{v}^X, 1)$, so the integral over $\mathbf{x}^X$ is $\alpha'^{-\boldsymbol{\beta}^X}$ times

$$\int_{(0,\infty)^{\mathcal{E}\setminus e_0}} (\mathbf{v}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} X(\mathbf{v}^X,1)^{-w_X}\, d\mathbf{v}^X = P_X(\mathbf{a}^X),$$

the integral over the same section of the form of the one-letter word X^{-w_X} at the extension $\mathbf{a}^X$ of $\boldsymbol{\beta}^X$ of total $w_X \deg X$ (Theorem 8.10); here $\boldsymbol{\beta}^{K_j} = \boldsymbol{\beta}_j^K$ and $\boldsymbol{\beta}^{\Psi_j} = \boldsymbol{\beta}_j^\Psi$, and $\mathbf{a}^X$ is the $\mathbf{a}_j$ or $\mathbf{b}_j$ of the statement. What is left is $\int \alpha'^{\mathbf{m}'-\mathbf{1}-\sum_X \boldsymbol{\beta}^X} K_1(\alpha',1)^{-s_1} d\alpha'$, $\mathbf{m}' = \mathbf{m}|_{\mathcal{E}\setminus e_0}$, which is $P_{K_1}(\mathbf{a}_1)$ for $\mathbf{a}_1 = \mathbf{m} - \sum_{X\in\mathcal{X}'} \mathbf{a}^X$: its coordinates off e_0 are $\mathbf{m}' - \sum_X \boldsymbol{\beta}^X$, and its total is $m_{\mathcal{E}} - \sum_{X\in\mathcal{X}'} w_X \deg X = r_1 s_1$ on the balance wall. This is the identity in $[0,\infty]$. For complex $\mathbf{m}$ and $\boldsymbol{\beta}$ the moduli of all the integrands are their values at the real parts, where the integrals converge by assumption, and Fubini's theorem replaces Tonelli's. □

Lemma S9. *Let shares $\mathbf{v} \in \mathbb{C}^M$ be pinned by $P\mathbf{v} = \boldsymbol{\pi}$, with q pins, as in Definition 4.1.*

(i) The fibre integral does not change when the shares are replaced by $A^{-1}\mathbf{v}$, the integrand by $f \circ A$ and the pins by PA, for a real $M \times M$ matrix A with $|\det A| = 1$, nor when the pins are replaced by $CP\mathbf{v} = C\boldsymbol{\pi}$, for a real $q \times q$ matrix C with $|\det C| = 1$.

(ii) Let $\mathbf{v} = (\mathbf{u}, \mathbf{w})$ with $\mathbf{w} \in \mathbb{C}^q$, so that $P = (P_\mathbf{u}, P_\mathbf{w})$, and let $|\det P_\mathbf{w}| = 1$. Then

$$\int_{P\mathbf{v}=\boldsymbol{\pi}} f = \frac{1}{(2\pi i)^{M-q}} \int_{\mathbf{c_u}+i\mathbb{R}^{M-q}} f\big(\mathbf{u}, P_\mathbf{w}^{-1}(\boldsymbol{\pi} - P_\mathbf{u}\mathbf{u})\big)\, d\mathbf{u},$$

$\mathbf{c_u}$ the $\mathbf{u}$-part of the admissible real point, the two sides converging absolutely together.

(iii) Let $\mathbf{v} = (\mathbf{v}_0, \mathbf{v}_1, \dots, \mathbf{v}_k)$ and let the pins be the union of pins $P_l\mathbf{v}_l = \boldsymbol{\pi}_l$, each on one block. If the fibre integral of f converges absolutely, it is the integral over the fibre $P_0\mathbf{v}_0 = \boldsymbol{\pi}_0$ of the fibre integral over $P_1\mathbf{v}_1 = \boldsymbol{\pi}_1, \dots, P_k\mathbf{v}_k = \boldsymbol{\pi}_k$ at fixed $\mathbf{v}_0$, the inner one converging absolutely for almost every point of the outer imaginary translate.

Proof. For $\mathbf{y}_0 \in \mathbb{R}^q$ let $\mu_{P,\mathbf{y}_0} = (\det PP^{\mathsf{T}})^{-1/2}\mathrm{vol}_{M-q}$ on $\{\mathbf{y} \in \mathbb{R}^M : P\mathbf{y} = \mathbf{y}_0\}$, so that the fibre integral is $(2\pi)^{q-M}\int f(\mathbf{c}+i\mathbf{y})\,d\mu_{P,\operatorname{Im}\boldsymbol{\pi}}(\mathbf{y})$. By the coarea formula for the linear map P, whose Jacobian is $(\det PP^{\mathsf{T}})^{1/2}$, these measures disintegrate Lebesgue measure, $\int_{\mathbb{R}^M}\varphi\,d\mathbf{y} = \int_{\mathbb{R}^q}\big(\int\varphi\,d\mu_{P,\mathbf{y}_0}\big)d\mathbf{y}_0$ for integrable φ, and $\mathbf{y}_0 \mapsto \int\varphi\,d\mu_{P,\mathbf{y}_0}$ is continuous for continuous φ of compact support; these two properties determine the family $(\mu_{P,\mathbf{y}_0})_{\mathbf{y}_0}$.

(i) The substitution $\mathbf{y} = A\mathbf{y}'$ preserves Lebesgue measure and carries the disintegration along PA to that along P; the substitution $\mathbf{y}_0 = C^{-1}\mathbf{y}_0'$ preserves Lebesgue measure on $\mathbb{R}^q$ and carries the disintegration along CP to that along P, with $\mu_{CP,C\mathbf{y}_0} = \mu_{P,\mathbf{y}_0}$. The admissible real point and the factor $(2\pi)^{q-M}$ do not change.

(ii) The map $\mathbf{y} \mapsto (\mathbf{y}_\mathbf{u}, P\mathbf{y})$ has the block triangular matrix $\begin{pmatrix} I & 0 \\ P_\mathbf{u} & P_\mathbf{w} \end{pmatrix}$, of determinant of modulus 1; so $d\mathbf{y} = d\mathbf{y}_\mathbf{u}\,d\mathbf{y}_0$, and $\mu_{P,\mathbf{y}_0}$ is the image of Lebesgue measure on $\mathbb{R}^{M-q}$ under $\mathbf{y}_\mathbf{u} \mapsto \big(\mathbf{y}_\mathbf{u}, P_\mathbf{w}^{-1}(\mathbf{y}_0 - P_\mathbf{u}\mathbf{y}_\mathbf{u})\big)$. With $\mathbf{u} = \mathbf{c}_\mathbf{u} + i\mathbf{y}_\mathbf{u}$ and $d\mathbf{u} = i^{M-q}d\mathbf{y}_\mathbf{u}$ this is the display; both sides are one integral against one measure.

(iii) The pin matrix is block diagonal, so the disintegration along it is the product of the disintegrations along the blocks, $\mu_{P,(\mathbf{y}_{0,l})_l} = \bigotimes_l \mu_{P_l,\mathbf{y}_{0,l}}$, and the factor $(2\pi)^{q-M}$ is the product of the factors of the blocks. Fubini's theorem gives the iterated integral. □

Remark. In Theorem 24.5(ii) the shares are the loads of the $2N$ letters, $M = 2Nn$, with the $q = n+2N-1$ pins $\sum_X \mathbf{a}^X = \mathbf{m}$ and $|\mathbf{a}^X| = w_X \deg X$ for $X \in \mathcal{X}'$, and the eliminated coordinates are $\mathbf{a}_1$ and the loads $a^X_{e_0}$, $X \in \mathcal{X}'$. With the rows ordered as the mass pins at $e \neq e_0$, the mass pin at e_0 and the totals, and the columns as $a_{1,e}$ for $e \neq e_0$, a_{1,e_0} and the $a^X_{e_0}$, the block $P_\mathbf{w}$ is

$$\begin{pmatrix} I_{n-1} & 0 & 0 \\ 0 & 1 & \mathbf{1}^{\mathsf{T}} \\ 0 & 0 & I_{2N-1} \end{pmatrix},$$

of determinant 1, and Lemma S9(ii) gives the coordinate form stated there, with the factor $(2\pi i)^{-(2N-1)(n-1)}$. With the further pins on G of Theorem 24.7 the loads a^X_e, $X \in \mathcal{X}'$, $e \in G$, are eliminated as well, each by its own pin with the coefficient 1, and the block stays unit upper triangular. The function `eliminated_block_det` of `checks/bookkeeping.py` computes this determinant exactly for $n \le 4$ and up to six letters.

Lemma S10. *Let $A, B > 0$, $w > 0$ and $c \in \mathbb{C}$. If $0 < \operatorname{Re} c < w$,*

$$\int_0^\infty x^{c-1}(A+xB)^{-w}\,dx = \Gamma\begin{bmatrix} c, & w-c \\ & w \end{bmatrix} A^{c-w}B^{-c},$$

absolutely convergent; for real $c \notin (0,w)$ the integral is $+\infty$.

Proof. The substitution $x = (A/B)y$ gives $A^{c-w}B^{-c}\int_0^\infty y^{c-1}(1+y)^{-w}dy$. The last integral is the integral over the section $\{u_2 = 1\}$ of the form of the word $(u_1+u_2)^{-w}$ on two marks at the masses $(c, w-c)$: the family of the list $\binom{\{1,2\}}{1}$ with unit weights at the exponents $(w,0)$, on its balance wall, whose walls are c and $w-c$. For $0 < \operatorname{Re} c < w$ its period converges absolutely (Theorem 8.10), and by Lemma 16.2 the integral equals that over $\Delta_{\{1,2\}}$, where $u_1+u_2 = 1$; by Theorem 3.9 it is $\int_\Delta u^{(c,w-c)-\mathbf{1}}d\omega = \Gamma\begin{bmatrix} c, & w-c \\ & w \end{bmatrix}$. For real $c \le 0$ the integrand is at least $2^{-w}y^{c-1}$ on $(0,1)$, and for real $c \ge w$ at least $2^{-w}y^{c-w-1}$ on $(1,\infty)$; both integrals diverge. □

Theorem 24.5 (The lean convolution). *As stated, for every family; the closing sentence takes the active letters of a family with nonnegative exponents.*

Proof. (i) Let $\mathbf{c}$ be admissible. By Proposition 24.3(ii), $\mathbf{c}_j^K \in s_j \operatorname{ri} P(\mathcal{B}_j)$ is the condition $(\mathbf{c}_j^K)_S > s_j d_j(S)$ for every nonempty proper S, and likewise $(\mathbf{c}_j^\Psi)_S > t_j d_j^*(S)$; summing over the letters, $\operatorname{Re}\kappa_S = \operatorname{Re} m_S - \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big) > 0$ (Definition 8.1). Conversely, if every proper wall of $\operatorname{Re}\mathbf{m}$ is positive, the native loads of $\operatorname{Re}\mathbf{m}$ are admissible for $(\mathbf{m};\mathbf{s},\mathbf{t})$ by Proposition 24.3(iii), admissibility involving $\operatorname{Re}\mathbf{m}$ only. The admissible points are cut out by linear equations and the convex conditions $\mathbf{c}^X \in w_X \operatorname{ri}\mathcal{N}_X$, so they form a convex set.

(ii) Apply Theorem 24.14 to the family word $W = \prod_X X^{-w_X}$ split into the $2N$ factors $W_X = X^{-w_X}$, of degrees $w_X \deg X$. The Newton polytope of W_X is $w_X\mathcal{N}_X$, of dimension $n-1$ by the first fact above, so its relative interior is $w_X \operatorname{ri}\mathcal{N}_X$, and the points $(\mathbf{c}_i)_i$ of Theorem 24.14 are exactly the admissible points. With the twists $\mathbf{x}^X$ of the factors, the left side of Theorem 24.14 is the integral over the section $\{\alpha_{e_0} = 1\}$ of the word of $F[\mathbf{x}]$, whose letters are $X(\mathbf{x}^X \circ \alpha)$ (Definition 21.1), that is $P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.10); the periods of the factors are the $P_X(\mathbf{a}^X)$. This is the display, absolutely convergent, of dimension $(2N-1)(n-1)$; K_1 is taken as the last factor of Theorem 24.14, whose total pin is omitted, and that pin holds by the others and the balance wall. The coordinate form is the remark after Lemma S9.

(iii) For a real point $\mathbf{a}$ at which $P_X(\mathbf{a})$ converges, $|P_X(\mathbf{a}+i\mathbf{y})| \le P_X(\mathbf{a})$, the modulus of the integrand of a period being its value at the real parts, and $|(\mathbf{x}^X)^{-\mathbf{a}^X}| = (\mathbf{x}^X)^{-\operatorname{Re}\mathbf{a}^X}$. On the imaginary translate through $\mathbf{c}$ the imaginary part $\mathbf{y}^X$ of every load has $|\mathbf{y}^X| = 0$, by the pins, and Proposition 24.3(iv) bounds $|P_X(\mathbf{c}^X + i\mathbf{y}^X)|$ by $\big(\sin\frac{\pi\varepsilon}{2}\big)^{-w_X} e^{-(1-\varepsilon)h_X(\mathbf{y}^X)} P_X(\mathbf{c}^X)$, h_X the norm of that proposition for the letter X.

The closing sentence is Theorem 24.14 with one factor for each active letter, the inactive letters being the constant 1; any factor may be taken last, and the walls and the admissible points are read with the active letters alone. □

Corollary 24.6 (What a family retains). *As stated, for every family.*

Proof. By Theorem 24.5(ii) and its closing sentence every period of F and of its twists, at exponents $s_j, t_j \ge 0$ not all zero and at masses on the balance wall with every proper wall positive, is a fibre integral of products of the periods P_X of its active letters at their loads, times the twists. By Proposition 24.3(i), $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0) = r_j\, \mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(s_j)$ for $s_j > 0$, and by the same item and Cremona $P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j) = P_{\mathcal{B}_j}(t_j\mathbf{1} - \mathbf{b}; t_j, 0) = r_j\, \mathsf{G}_{t_j\mathbf{1}-\mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(t_j)$ for $t_j > 0$. The loads run over $|\mathbf{a}| = r_j s_j$ and $|\mathbf{b}| = r_j^* t_j$, that is $|t_j\mathbf{1} - \mathbf{b}| = nt_j - r_j^* t_j = r_j t_j$. The converse is Proposition 24.3(i) for table j. □

Theorem 24.7 (Pinned holds). *As stated, for every family.*

Proof. Put $d = (2N-1)|G|$. In the coordinates of Theorem 24.5(ii), the loads of the letters of $\mathcal{X}'$ at the marks other than e_0, split the coordinates into their values $\mathbf{z}_G$ on G and the rest $\mathbf{z}'$, and let $\mathcal{I}(\mathbf{z}_G, \mathbf{z}')$ be the integrand at $\mathbf{x} = \mathbf{1}$, the product of the periods P_X at the loads that $(\mathbf{z}_G, \mathbf{z}')$ and the pins determine. Let $\mathbf{c}$ be the admissible point of the statement, $\boldsymbol{\beta} = \mathbf{c}_G$, and for $\mathbf{y} \in \mathbb{R}^d$ put

$$\Phi(\mathbf{y}) = \frac{1}{(2\pi i)^{(2N-1)(n-1-|G|)}} \int \mathcal{I}(\mathbf{c}_G + i\mathbf{y}, \mathbf{z}')\, d\mathbf{z}',$$

over the imaginary translate through $\mathbf{c}$ in $\mathbf{z}'$. By Lemma S9(ii) and the remark after it, $\Phi(\boldsymbol{\eta})$ is the fibre integral of the statement at $\boldsymbol{\beta} + i\boldsymbol{\eta}$.

(a) By Theorem 24.5(iii), $|\mathcal{I}|$ on the imaginary translate is at most a constant times $\prod_{X\in\mathcal{X}'} e^{-(1-\varepsilon)h_X(\mathbf{y}^X)}$, $\mathbf{y}^X$ the imaginary part of the load of X, a vector of total 0 determined by its coordinates off e_0; since h_X is a norm on the vectors of total 0, this bound is

integrable in all the imaginary parts. So Φ is defined everywhere, continuous by dominated convergence, and integrable, with $\int|\Phi| \le (2\pi)^{(2N-1)(n-1-|G|)}\int|\mathcal{I}|$.

(b) For a twist at the marks of G alone, $\mathbf{x}^X = e^{\mathbf{u}^X}$ on G with $\mathbf{u}^X \in \mathbb{R}^G$ for $X \in \mathcal{X}'$ and $\mathbf{x}^{K_1} = \mathbf{1}$, the factor $\prod_X (\mathbf{x}^X)^{-\mathbf{a}^X}$ of Theorem 24.5(ii) is $e^{-\langle \mathbf{c}_G + i\mathbf{y}, \mathbf{u}\rangle}$ on the translate, $\mathbf{u} = (\mathbf{u}^X)_X$. By Fubini's theorem in $(\mathbf{y}, \mathbf{z}')$, $d\mathbf{z}_G = i^d\, d\mathbf{y}$,

$$e^{\langle \mathbf{c}_G, \mathbf{u}\rangle} P_{F[e^{\mathbf{u}}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{1}{(2\pi)^d}\int_{\mathbb{R}^d} \Phi(\mathbf{y})\, e^{-i\langle \mathbf{y}, \mathbf{u}\rangle}\, d\mathbf{y} =: \widehat{\Phi}(\mathbf{u}).$$

(c) Let first $\mathbf{m}$ be real. Then $\widehat{\Phi} \ge 0$, and for $\varepsilon > 0$, by Fubini's theorem and the Gaussian integral,

$$\int_{\mathbb{R}^d} \widehat{\Phi}(\mathbf{u})\, e^{-\varepsilon|\mathbf{u}|^2/2}\, d\mathbf{u} = \int_{\mathbb{R}^d} \Phi(\mathbf{y})\, \frac{e^{-|\mathbf{y}|^2/(2\varepsilon)}}{(2\pi\varepsilon)^{d/2}}\, d\mathbf{y} \longrightarrow \Phi(\mathbf{0}) \qquad (\varepsilon \downarrow 0),$$

Φ being integrable and continuous at $\mathbf{0}$. By monotone convergence $\int \widehat{\Phi}\, d\mathbf{u} = \Phi(\mathbf{0}) < \infty$, and with $d\mathbf{x}/\mathbf{x} = d\mathbf{u}$ this integral is $\mathrm{Hold}_G(\mathbf{c}_G)$: the statement at $\boldsymbol{\eta} = \mathbf{0}$.

(d) For complex $\mathbf{m}$, $|P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t})| \le P_{F[\mathbf{x}]}(\mathrm{Re}\,\mathbf{m}; \mathbf{s}, \mathbf{t})$, and $\mathbf{c}$ is admissible for $\mathrm{Re}\,\mathbf{m}$ as well; by (c) at $\mathrm{Re}\,\mathbf{m}$ the hold converges absolutely at every $\mathbf{c}_G + i\boldsymbol{\eta}$. Then $\widehat{\Phi}$ of (b) is integrable, Φ is integrable and continuous, and Fourier inversion gives, at every $\boldsymbol{\eta}$,

$$\mathrm{Hold}_G(\mathbf{c}_G + i\boldsymbol{\eta}) = \int_{\mathbb{R}^d} e^{\langle \mathbf{c}_G + i\boldsymbol{\eta}, \mathbf{u}\rangle} P_{F[e^{\mathbf{u}}]}(\mathbf{m}; \mathbf{s}, \mathbf{t})\, d\mathbf{u} = \int_{\mathbb{R}^d} \widehat{\Phi}(\mathbf{u})\, e^{i\langle \boldsymbol{\eta}, \mathbf{u}\rangle}\, d\mathbf{u} = \Phi(\boldsymbol{\eta}).$$

At $G = \mathcal{E} \setminus e_0$ the fibre has dimension 0, and its value is the product of Lemma 24.4. □

Proposition 24.8 (Holding one mark). *As stated, for every family.*

Proof. Let the data be real. Take the section $\{h = 1\}$ of Theorem 8.10 with $h(\alpha) = \sum_{e \ne g} \alpha_e$: on it $\alpha = (\alpha', \alpha_g)$ with $\alpha' \in \Delta_{\mathcal{E}\setminus g}$ and $\alpha_g \in (0, \infty)$, and the measure is $d\omega_{\mathcal{E}\setminus g}(\alpha')\, d\alpha_g$. By Definitions 21.1 and 20.1 the twist at g acts on the letters as

$$K_j(\mathbf{x}@g \circ \alpha) = K_{\mathcal{B}_j\setminus g}(\alpha') + x_j^K \alpha_g K_{\mathcal{B}_j/g}(\alpha'), \qquad \Psi_j(\mathbf{x}@g \circ \alpha) = x_j^\Psi \alpha_g \Psi_{\mathcal{B}_j\setminus g}(\alpha') + \Psi_{\mathcal{B}_j/g}(\alpha'),$$

the minors carrying the weights of table j, all four nonzero since g is neither a loop nor a coloop. Insert the period into the hold and integrate first over the twists of the letters of $\mathcal{X}'$, by Tonelli's theorem. By Lemma S10, for $j \ge 2$, respectively for every j,

$$\int_0^\infty x^{a_j - 1}\big(K_{\mathcal{B}_j\setminus g} + x\alpha_g K_{\mathcal{B}_j/g}\big)^{-s_j} dx = \Gamma\begin{bmatrix} a_j,\ s_j - a_j \\ s_j \end{bmatrix} K_{\mathcal{B}_j\setminus g}^{a_j - s_j}\big(\alpha_g K_{\mathcal{B}_j/g}\big)^{-a_j},$$
$$\int_0^\infty x^{b_j - 1}\big(\Psi_{\mathcal{B}_j/g} + x\alpha_g \Psi_{\mathcal{B}_j\setminus g}\big)^{-t_j} dx = \Gamma\begin{bmatrix} b_j,\ t_j - b_j \\ t_j \end{bmatrix} \Psi_{\mathcal{B}_j/g}^{b_j - t_j}\big(\alpha_g \Psi_{\mathcal{B}_j\setminus g}\big)^{-b_j},$$

finite exactly when $0 < a_j < s_j$, respectively $0 < b_j < t_j$, and $+\infty$ otherwise. The power of α_g left is $\alpha_g^{m_g - 1 - \sum_{j\ge2} a_j - \sum_j b_j} = \alpha_g^{a_1 - 1}$, and by the same lemma

$$\int_0^\infty \alpha_g^{a_1 - 1}\big(K_{\mathcal{B}_1\setminus g} + \alpha_g K_{\mathcal{B}_1/g}\big)^{-s_1} d\alpha_g = \Gamma\begin{bmatrix} a_1,\ s_1 - a_1 \\ s_1 \end{bmatrix} K_{\mathcal{B}_1\setminus g}^{a_1 - s_1} K_{\mathcal{B}_1/g}^{-a_1},$$

finite exactly when $0 < a_1 < s_1$. What remains is

$$\int_{\Delta_{\mathcal{E}\setminus g}} \alpha'^{\mathbf{m}|_{\mathcal{E}\setminus g}-\mathbf{1}} \prod_j K_{\mathcal{B}_j\setminus g}^{-(s_j-a_j)}\, \Psi_{\mathcal{B}_j\setminus g}^{-b_j}\, K_{\mathcal{B}_j/g}^{-a_j}\, \Psi_{\mathcal{B}_j/g}^{-(t_j-b_j)}\, d\omega_{\mathcal{E}\setminus g},$$

the period of the children family $F_{\{g\}}\big((s_j-a_j,a_j),(b_j,t_j-b_j)\big)_j$ at $\mathbf{m}|_{\mathcal{E}\setminus g}$, which is on its balance wall (the paragraph before the proposition); it is finite exactly when every proper wall of it is positive (Theorem 8.10). The integrands being positive, the hold is the product of these factors in $[0,\infty]$ and is finite exactly when all of them are. For complex loads and masses whose real parts are such data the moduli are dominated by the real computation, Lemma S10 holds in the strip, and Fubini's theorem gives the same identity. The sentence on admissible points is Theorem 24.7 with $G = \{g\}$.

For the limits, let the letter Y have the exponent $u > m_g > 0$ and let the twists at g of the other letters decrease to 0. Each other letter decreases pointwise to one of its minors, $K_{\mathcal{B}_j\setminus g}$ or $\Psi_{\mathcal{B}_j/g}$, and its power with a negative exponent increases; by monotone convergence the period increases to the integral of the limit. In the limit only Y contains α_g, as $A+\alpha_g B$ with A and B its two minors, and by Lemma S10 the integral over α_g of $\alpha_g^{m_g-1}(A+\alpha_g B)^{-u}$ is $\Gamma_1^2\begin{bmatrix} m_g,\ u-m_g \\ u \end{bmatrix} A^{m_g-u}B^{-m_g}$. What is left is the period of the children family at the shares of the statement: for $Y = K_j$, $A = K_{\mathcal{B}_j\setminus g}$ carries the exponent $s_j - m_g$ and $B = K_{\mathcal{B}_j/g}$ the exponent m_g; for $Y = \Psi_j$, $A = \Psi_{\mathcal{B}_j/g}$ carries $t_j - m_g$ and $B = \Psi_{\mathcal{B}_j\setminus g}$ carries m_g; every other $K_{j'}$ is $K_{\mathcal{B}_{j'}\setminus g}$ at $s_{j'}$ and every other $\Psi_{j'}$ is $\Psi_{\mathcal{B}_{j'}/g}$ at $t_{j'}$. □

Remark. For one table a twist at g in the cobasis letter is $x = x_1^\Psi$ with $x_1^K = 1$ (Definition 21.1), and $x \downarrow 0$ is the limit with the letter K chosen. With the letter Ψ chosen the twist y of K decreases to 0 with that of Ψ at 1; by the torus relation of Definition 21.1 at the mark g, $P_{\mathcal{H}[(y,1)@g]} = y^{-m_g}P_{\mathcal{H}[(1,1/y)@g]} = x^{m_g}P_{\mathcal{H}[x@g]}$ with $x = 1/y$. So the two limits are those of $P_{\mathcal{H}[x@g]}$ as $x \downarrow 0$ and of $x^{m_g}P_{\mathcal{H}[x@g]}$ as $x \uparrow \infty$, as the paragraph after the proposition reads them.

Lemma 24.9 (The Mellin transform of a law is the period of a children family). *As stated, for every cell family.*

Proof. By Definitions 22.1 and 20.1, with $\sum_j (z_{j,K}+z_{j,\Psi}) = \kappa_F$ the power of $\gamma_{\mathcal{A}_1}^K$ in $\boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F$ is $-\kappa_F + \sum_{j\geq 2} z_{j,K} + \sum_j z_{j,\Psi} = -z_{1,K}$. With $\gamma_{\mathcal{A}_j}^K = K_{\mathcal{A}_j\setminus g}/K_{\mathcal{A}_j/g}$ and $\gamma_{\mathcal{A}_j}^\Psi = \Psi_{\mathcal{A}_j/g}/\Psi_{\mathcal{A}_j\setminus g}$,

$$\begin{aligned}\boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F &= u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j/g}^{-s_j} \Psi_{\mathcal{A}_j\setminus g}^{-t_j} \big(\gamma_{\mathcal{A}_j}^K\big)^{-z_{j,K}} \big(\gamma_{\mathcal{A}_j}^\Psi\big)^{-z_{j,\Psi}}\, d\omega_B \\ &= u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j\setminus g}^{-z_{j,K}}\, \Psi_{\mathcal{A}_j\setminus g}^{-(t_j-z_{j,\Psi})}\, K_{\mathcal{A}_j/g}^{-(s_j-z_{j,K})}\, \Psi_{\mathcal{A}_j/g}^{-z_{j,\Psi}}\, d\omega_B.\end{aligned}$$

This is the form of the children family of $\{g\}$ at the shares of the statement, its tables $\mathcal{A}_j \setminus g$, of rank $r_{\mathcal{A}_j}$ and corank $|B|-r_{\mathcal{A}_j}$, at $(z_{j,K}, t_j - z_{j,\Psi})$ and $\mathcal{A}_j/g$, of rank $r_{\mathcal{A}_j}-1$ and corank $|B|-r_{\mathcal{A}_j}+1$, at $(s_j - z_{j,K}, z_{j,\Psi})$. Its degree is

$$\begin{aligned}\sum_j \Big(r_{\mathcal{A}_j} z_{j,K} + (|B|-r_{\mathcal{A}_j})(t_j - z_{j,\Psi}) + (r_{\mathcal{A}_j}-1)(s_j - z_{j,K}) + (|B|-r_{\mathcal{A}_j}+1) z_{j,\Psi} \Big) \\ = \kappa_F + \sum_j \big(s_j(r_{\mathcal{A}_j}-1) + t_j(|B|-r_{\mathcal{A}_j}) \big) = m_B,\end{aligned}$$

so the children family is on its balance wall, and the integral over Δ_B is its period (Theorem 8.10), in $[0,\infty]$ for real $\mathbf{z}$. This is the first identity.

With the mass $\sum_j (s_j+t_j)-\kappa_F$ on g the family of the tables $\mathcal{A}_j$ is on its balance wall (Lemma 20.8). Apply Proposition 24.8 to it at the mark g, a hinge and so neither a loop nor a coloop, with $a_j = s_j - z_{j,K}$ for $j\ge 2$ and $b_j = t_j - z_{j,\Psi}$. Then

$$a_1 = \sum_j (s_j+t_j) - \kappa_F - \sum_{j\ge 2}(s_j - z_{j,K}) - \sum_j (t_j - z_{j,\Psi}) = s_1 - z_{1,K},$$

the children family of Proposition 24.8 at the shares $(s_j-a_j, a_j) = (z_{j,K}, s_j - z_{j,K})$ and $(b_j, t_j - b_j) = (t_j - z_{j,\Psi}, z_{j,\Psi})$ is that of the first identity, its Beta corners are $\Gamma_{\substack{2\\1}}\begin{bmatrix} s_j - z_{j,K},\ z_{j,K} \\ s_j \end{bmatrix}$ and $\Gamma_{\substack{2\\1}}\begin{bmatrix} t_j - z_{j,\Psi},\ z_{j,\Psi} \\ t_j \end{bmatrix}$, and its hold is the left side of the second identity. At the real part of $\mathbf{z}$ the conditions of Proposition 24.8 hold: $0 < a_j < s_j$ and $0 < b_j < t_j$ are $0 < \operatorname{Re} z_{j,K} < s_j$ and $0 < \operatorname{Re} z_{j,\Psi} < t_j$, and the walls of the children family are positive since its period, $\int \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}} d\phi_F$ by the first identity, is finite. So Proposition 24.8 gives the second identity for complex $\mathbf{z}$ with this real part. □

Remark. The balance of the children family, the balance of the family of the tables $\mathcal{A}_j$ with the mass $\sum_j (s_j+t_j)-\kappa_F$ on g, and the identity $a_1 = s_1 - z_{1,K}$ are computed exactly by `check_hingesplit_balance` in `checks/bookkeeping.py`, and the balance of the children family of Proposition 24.8 by `check_holdone_balance`.

Definition 24.11 (Positive letters; Newton words and sums). As stated: the period of a product of powers of positive letters at complex exponents, wherever it converges absolutely, a Newton word being the case of positive exponents.

Remark. The independence of the section is the proof of Lemma 16.2, which uses of the integrand $f = \alpha^{\mathbf{m-1}} W$ only that it is measurable, homogeneous, $f(\lambda\alpha) = \lambda^{-n} f(\alpha)$ for $\lambda > 0$ when $m_{\mathcal{E}} = D$, and absolutely integrable over one section; here $|f| = \alpha^{\operatorname{Re}\mathbf{m-1}} \prod_i L_i^{-\operatorname{Re} w_i}$. The family word at any exponents is a product of powers of its letters with $D = \sum_j (s_j r_j + t_j r_j^*)$, and on the balance wall its period is that of Theorem 8.10.

Proposition 24.13 (One positive letter). *As stated.*

Proof. Put $D = \deg L$ and $t = (t_k)_k$. Every t_k has $|t_k| = D$, and $\mathcal{N}_L$ has dimension $n-1$ in the hyperplane $|x| = D$, which does not contain 0; so the t_k span $\mathbb{R}^{\mathcal{E}}$.

(i) By Lemma 2.7, $x = \rho u$ with $\rho > 0$ and $u \in \Delta_{\mathcal{E}}$ gives $x^{\mathbf{a-1}} dx = \rho^{|\mathbf{a}|-1} u^{\mathbf{a-1}} d\rho\, d\omega_{\mathcal{E}}(u)$, and $L(\rho u) = \rho^D L(u)$. For $\operatorname{Re}|\mathbf{a}| > 0$ the substitution $\sigma = \rho^D L(u)$ gives

$$\int_0^\infty \rho^{|\mathbf{a}|-1} e^{-\rho^D L(u)}\, d\rho = \frac{\Gamma(w)}{D}\, L(u)^{-w}, \qquad w = |\mathbf{a}|/D.$$

So $\mathsf{G}_{\mathbf{a};t}((c_k)_k) = \frac{\Gamma(w)}{D}\int_{\Delta_{\mathcal{E}}} u^{\mathbf{a-1}} L(u)^{-w} d\omega_{\mathcal{E}}(u) = \frac{\Gamma(w)}{D}\, P_{L^{-w}}(\mathbf{a})$, the last integral being the period over the section $\Delta_{\mathcal{E}}$ (Definition 24.11). The same computation at $\operatorname{Re}\mathbf{a}$, by Tonelli's theorem, shows that the two integrals converge absolutely together.

(ii) $|u^{\mathbf{a-1}} L(u)^{-w}| = u^{\operatorname{Re}\mathbf{a-1}} L(u)^{-\operatorname{Re} w}$, so $P_{L^{-w}}(\mathbf{a})$ converges absolutely exactly when the period of the Newton word $L^{-\operatorname{Re} w}$, $\operatorname{Re} w > 0$, converges at $\operatorname{Re}\mathbf{a}$. Its Newton sum is $(\operatorname{Re} w)\mathcal{N}_L$, of dimension $n-1$, and Theorem 24.12 gives the condition $\operatorname{Re}\mathbf{a} \in \operatorname{ri}\big((\operatorname{Re} w)\mathcal{N}_L\big) = (\operatorname{Re} w)\operatorname{ri}\mathcal{N}_L$.

(iii) For real $\mathbf{a}$ and $|\boldsymbol{\eta}| = 0$ the exponent w is the same real number at $\mathbf{a}$ and at $\mathbf{a} + i\boldsymbol{\eta}$, so by (i) $P_{L^{-w}}(\mathbf{a} + i\boldsymbol{\eta}) = D\,\mathsf{G}_{\mathbf{a}+i\boldsymbol{\eta};t}((c_k)_k)/\Gamma(w)$. If $P_{L^{-w}}(\mathbf{a})$ converges, so does $\mathsf{G}_{\mathbf{a};t}$, and $\mathbf{a} \in \operatorname{int}\operatorname{cone}(t)$ (Theorem 2.2). Theorem 2.11 gives $|\mathsf{G}_{\mathbf{a}+i\boldsymbol{\eta};t}((c_k)_k)| \le \mathsf{G}_{\mathbf{a};t}(\lambda(c_k)_k)\, e^{-(1-\varepsilon)h_t(\boldsymbol{\eta})}$ with $\lambda = \sin\frac{\pi\varepsilon}{2}$, and $h_t = h_L$ is a norm since the t_k span. Every t_k having total D, the substitution $x = \lambda^{-1/D}y$ gives $\mathsf{G}_{\mathbf{a};t}(\lambda(c_k)_k) = \lambda^{-w}\mathsf{G}_{\mathbf{a};t}((c_k)_k)$; multiplying by $D/\Gamma(w)$ gives the second bound. The bound by $P_{L^{-w}}(\mathbf{a})$ itself is that of the modulus of the integrand by its value at the real parts. □

Remark. At the letter $K_{\mathcal{H}}$ of a table that is one molecule, with the support $(\mathbf{1}_T)_T$, the coefficients $\boldsymbol{\theta}$ and $\mathcal{N}_{K_{\mathcal{H}}} = P(\mathcal{H})$, items (i), (ii) and (iii) are Proposition 24.3(i), (ii) and (iv) at the full basis lean, and at $\Psi_{\mathcal{H}}$ at the full cobasis lean. At a simplicial letter, (i) with Corollary 2.6 at the square configuration t is Proposition 24.16: $\mathsf{G}_{\mathbf{a};t}((c_k)_k) = |\det t|^{-1}\prod_k \Gamma(b_k)c_k^{-b_k}$ with $\mathbf{b} = t^{-1}\mathbf{a}$, and $|\mathbf{b}| = w$ since $\mathbf{1}^{\mathsf{T}}t = D\mathbf{1}^{\mathsf{T}}$.

Lemma 24.17 (Concatenation of Grammas). *As stated, for any number of parts.*

Proof. For $i \ge 2$ the configuration $t^{(i)}$ spans $\mathbb{R}^{\mathcal{E}}$, so by Theorem 2.11 the Gramma $\mathsf{G}_{\mathbf{w};t^{(i)}}(\boldsymbol{\xi}^{(i)})$ on $\mathbf{c}_i + i\mathbb{R}^{\mathcal{E}}$ is at most a constant times $e^{-(1-\varepsilon)h_{t^{(i)}}(\operatorname{Im}\mathbf{w})}$, $h_{t^{(i)}}$ a norm, and is integrable there. In $\mathbf{y} = \log x$ it is the Fourier transform of the continuous integrable function $e^{\langle \mathbf{c}_i,\mathbf{y}\rangle}\exp\big(-\sum_k \xi_k^{(i)} e^{\langle t_k^{(i)},\mathbf{y}\rangle}\big)$ (Definition 2.1), so Fourier inversion gives, for every $x \in \mathbb{R}^{\mathcal{E}}_{>0}$,

$$\exp\Big(-\sum_k \xi_k^{(i)} x^{t_k^{(i)}}\Big) = \frac{1}{(2\pi i)^n}\int_{\mathbf{c}_i + i\mathbb{R}^{\mathcal{E}}} \mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})\, x^{-\mathbf{w}_i}\, d\mathbf{w}_i.$$

Multiply these for $i = 2,\dots,p$ by $x^{\boldsymbol{\nu}-\mathbf{1}}\exp\big(-\sum_k \xi_k^{(1)}x^{t_k^{(1)}}\big)$ and integrate over x. The modulus of the integrand is $x^{\operatorname{Re}\boldsymbol{\nu}-\sum_{i\ge2}\mathbf{c}_i-\mathbf{1}}\exp\big(-\sum_k \xi_k^{(1)}x^{t_k^{(1)}}\big)\prod_{i\ge2}|\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})|$, whose integral is $\mathsf{G}_{\operatorname{Re}\boldsymbol{\nu}-\sum_{i\ge2}\mathbf{c}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})\prod_{i\ge2}\int|\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})|\,|d\mathbf{w}_i|$, finite by Theorem 2.2 since $\operatorname{Re}\boldsymbol{\nu} - \sum_{i\ge2}\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(1)})$. So Fubini's theorem exchanges the integrals; the integral over x is $\mathsf{G}_{\boldsymbol{\nu}-\sum_{i\ge2}\mathbf{w}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})$, and the left side is the Gramma of the concatenation (Definition 2.1). □

40.25 Chapter 25: periods at the full leans and networks along a hinge tree

The notation of Section 40.24 is kept. When table j of a family carries a hinge tree $\mathfrak{T}_j$ as in Definition 25.3, its marker loads and node loads are read in table j (Definition 25.4), at its loads $(\mathbf{a}_j, \mathbf{b}_j)$ and its exponents (s_j, t_j).

Lemma 25.2 (Substitutions of molecules). *As stated, for any number of cells.*

Proof. The case of two cells is the reference, proof of lem:twosummol. By Proposition 20.3(i) the substitution is the gluing of the cells to $\mathcal{H}$ along its marks one at a time: $\mathcal{B}_0 = \mathcal{H}$, and $\mathcal{B}_i$ is the gluing of $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$ along h_i, the marks $h_{i+1},\dots,h_q$ left in place. By Definition 20.1 and Proposition 20.3(iii) this gluing is the two-sum of $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$, the latter read as a cell with the hinge h_i. This h_i is neither a loop nor a coloop of $\mathcal{B}_{i-1}$: $\mathcal{H}$ has bases through h_i and bases avoiding it, and each extends to a basis of $\mathcal{B}_{i-1}$ by bases of the hinge minors of the cells glued before, which exist since their hinges are neither loops nor coloops. By the case of two cells $\mathcal{B}_i$ is one molecule exactly when $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$ are, and by induction $\mathcal{B}_q$ is one molecule exactly when $\mathcal{H}$ and all the $\mathcal{A}_i$ are. □

Corollary 25.7 (Convergence is decided at the nodes). *As stated, for every family.*

Proof. Every table is one molecule by Lemma 25.2 applied along its hinge tree. By Theorem 25.6 in table j, at real loads, $P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)$ is finite exactly when every marker load $k_{j;v,\mathfrak{e}}$ lies in $(0, s_j)$ and the period of every node at its full basis lean is finite at its node loads. By Proposition 24.3(ii) at the node, one molecule, the latter is $(\mathbf{a}_{j,v})_S > s_j d_{j,v}(S)$ for every nonempty proper S of its marks and markers; at $S = \{g\}$ and at the complement of a marker g these inequalities are $k > 0$ and $k < s_j$ for the marker load k at g, since g is neither a loop nor a coloop of the node and $|\mathbf{a}_{j,v}| = r_{j,v} s_j$ (Lemma 25.5(ii)). Likewise at the full cobasis lean. So the node inequalities hold at a split exactly when every load is admissible for its letter (Proposition 24.3(ii) in table j), that is when the split is an admissible point, and by Theorem 24.5(i) such a point exists exactly when every proper wall is positive; the native loads are one. The unknowns are the loads of the $2N$ letters at n marks less $n + 2N - 1$ independent pins, $(2N-1)(n-1)$ of them, and the node v of table j contributes $2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ inequalities. □

Proposition 25.10 (Series and parallel nodes at their full leans). *As stated; the last sentence, which reads the four closed forms off Proposition 24.16, is new.*

Proof. The closed forms are the reference, proof of prop:spleans. The letter K of $\binom{\mathcal{X}}{1}$ is $\sum_x \theta_x \alpha_x$, and so is the letter Ψ of $\binom{\mathcal{X}}{q}$, whose basis $\mathcal{X} \setminus x$ has the cobasis $\{x\}$: both are simplicial with the configuration $I = (\mathbf{e}_x)_x$, of degree 1, $|\det I| = 1$ and the shares $\mathbf{b} = \mathbf{a}$, and Proposition 24.16 gives $\Gamma_{\substack{q+1\\1}}\begin{bmatrix}\mathbf{a}\\|\mathbf{a}|\end{bmatrix} \prod_x \theta_x^{-a_x}$, the forms of $P_{\binom{\mathcal{X}}{1}}(\mathbf{a}; s, 0)$ and $P_{\binom{\mathcal{X}}{q}}(\mathbf{b}; 0, t)$. The letter K of $\binom{\mathcal{X}}{q}$ and the letter Ψ of $\binom{\mathcal{X}}{1}$ are $\sum_x \theta_x \alpha^{\mathcal{X}\setminus x}$, simplicial with the configuration $J - I = (\mathbf{1} - \mathbf{e}_x)_x$ of degree q, J the matrix of ones. $J - I$ has the eigenvalues q, once, and -1, q times, so $|\det(J - I)| = q$, and $(J - I)^{-1} = \frac{1}{q} J - I$, so the shares of $\mathbf{a}$ with $|\mathbf{a}| = qs$ are $\frac{|\mathbf{a}|}{q}\mathbf{1} - \mathbf{a} = (s - a_x)_x$, of sum s. Proposition 24.16 gives $\frac{q}{q}\Gamma_{\substack{q+1\\1}}\begin{bmatrix}(s-a_x)_x\\ s\end{bmatrix} \prod_x \theta_x^{a_x - s}$, the form of $P_{\binom{\mathcal{X}}{q}}(\mathbf{a}; s, 0)$, and in the same way that of $P_{\binom{\mathcal{X}}{1}}(\mathbf{b}; 0, t)$ with $|\mathbf{b}| = qt$. The criteria of convergence agree, the positivity of the real parts of the shares being that of the arguments of the Gammas. □

Remark. The function `check_spleans_is_simplicial` of `checks/bookkeeping.py` verifies exactly, at $q = 4$, the shares $s - a_x$ and the prefactor $q/|\det(J - I)| = 1$.

Corollary 25.11 (Series-parallel tables at their full leans). *As stated, with any weights on the bases of the series and parallel nodes.*

Proof. By Theorem 25.6 the period of $\mathcal{B}$ at its full basis lean is the product over the nodes of their periods at their node loads over the Beta corners $\Gamma_{\substack{2\\1}}\begin{bmatrix}k_{v,\mathfrak{e}},\ k_{w,\mathfrak{e}}\\ s\end{bmatrix}$ of the edges, the weights of $\mathcal{B}$ being the products of those of the nodes (Definition 25.3). Proposition 25.10 gives each node: $\Gamma_{\substack{n_v\\1}}\begin{bmatrix}(s-(\mathbf{a}_v)_x)_x\\ s\end{bmatrix} \prod_x \theta_{v,x}^{(\mathbf{a}_v)_x - s}$ at a series node and $\Gamma_{\substack{n_v\\1}}\begin{bmatrix}((\mathbf{a}_v)_x)_x\\ s\end{bmatrix} \prod_x \theta_{v,x}^{-(\mathbf{a}_v)_x}$ at a parallel node, $n_v = |\mathcal{E}_v| + |G_v|$. The Gammas, the factors $\Gamma(s)$ and the corners combine as in the reference, proof of cor:spfulllean: a tree has one node more than edges, which leaves $1/\Gamma(s)$; a mark e contributes $\Gamma(s - a_e)$ at a series node and $\Gamma(a_e)$ at a parallel node; at an edge $\mathfrak{e} = \{v, w\}$, where $k_{v,\mathfrak{e}} + k_{w,\mathfrak{e}} = s$ (Lemma 25.5(i)), the two markers contribute $\Gamma(s - k_{v,\mathfrak{e}})\Gamma(s - k_{w,\mathfrak{e}}) = \Gamma(k_{w,\mathfrak{e}})\Gamma(k_{v,\mathfrak{e}})$ between two series nodes and $\Gamma(k_{v,\mathfrak{e}})\Gamma(k_{w,\mathfrak{e}})$ between two parallel nodes, which the corner cancels, and $\Gamma(s - k_{v,\mathfrak{e}})\Gamma(k_{w,\mathfrak{e}})$ when v is a series and w a parallel node, which the corner leaves as $\Gamma(s - k_{\mathfrak{e}})/\Gamma(k_{\mathfrak{e}})$,

$k_{\mathfrak{e}} = k_{v,\mathfrak{e}}$. The monomials of the nodes stand as they are. The full cobasis lean is the same with the second pair of forms of Proposition 25.10: $\Gamma(b_e)$ at series and $\Gamma(t - b_e)$ at parallel nodes, and at an edge with the series end v the quotient $\Gamma(l_{v,\mathfrak{e}})\Gamma(t - l_{w,\mathfrak{e}})/\big(\Gamma(l_{v,\mathfrak{e}})\Gamma(l_{w,\mathfrak{e}})\big) = \Gamma(l_{\mathfrak{e}})/\Gamma(t - l_{\mathfrak{e}})$. The criterion of finiteness does not see the weights, which enter through monomials, positive and finite at real loads; it is the reference's. □

Theorem 25.12 (The network). *As stated, for every family.*

Proof. Every table is one molecule by Lemma 25.2 along its hinge tree, and every proper wall is positive, so Theorem 24.5(ii) writes $P_{F[\mathbf{x}]}$ as the fibre integral of $\prod_j P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j)$ times the twists, on the imaginary translate through an admissible point $\mathbf{c}$. At $\mathbf{c}$ every $P_{\mathcal{B}_j}(\mathbf{c}_j^K; s_j, 0)$ and $P_{\mathcal{B}_j}(\mathbf{c}_j^\Psi; 0, t_j)$ is finite (Proposition 24.3(ii)), so by Theorem 25.6 in table j every marker load lies in $(0, s_j)$, respectively $(0, t_j)$, every node period is finite at its node loads, and both factorizations of Theorem 25.6 hold on the tube over these real loads, which contains the translate. Inserting them gives the integrand of the statement. The native loads are admissible by Proposition 24.3(iii). The node periods are functions of the node loads alone, the exponent of the one active letter of a node being its total load divided by its degree (Definition 24.2). □

Corollary 25.13 (Series-parallel periods). *As stated, for every family and any weights on the bases of the nodes.*

Proof. Insert Corollary 25.11 for both full leans of every table into Theorem 25.12 and multiply by $\prod_j \Gamma(s_j)\Gamma(t_j)$. At a series node the monomials of the two leans multiply to $\theta_{j,v,x}^{(\mathbf{a}_{j,v})_x - s_j}\theta_{j,v,x}^{-(\mathbf{b}_{j,v})_x}$, at a parallel node to $\theta_{j,v,x}^{-(\mathbf{a}_{j,v})_x}\theta_{j,v,x}^{(\mathbf{b}_{j,v})_x - t_j}$. At a real point at which every Gamma above the line has an argument of positive real part all the periods of the tables at their full leans are finite (Corollary 25.11), so the point is admissible (Proposition 24.3(ii)) and Theorem 25.12 applies on it. The dimension is that of Theorem 25.12, and the integrand does not involve the bases of the tables. □

Proposition 25.14 (Pooling a pair). *As stated, with any weights on the bases of the node.*

Proof. For one table the pins are $\mathbf{a} + \mathbf{b} = \mathbf{m}$ and $|\mathbf{b}| = r^*t$, the twist $\mathbf{x}$ is on the cobasis letter, and the integrand of Corollary 25.13 is a function of $\mathbf{b}$, with $\Gamma(s - a_e) = \Gamma(c_e + b_e)$ at a mark e of a series node. The marks e and f of one node v lie on one side of every edge, so every marker load, an affine function of sums of loads over branches (Definition 25.4), depends on b_e and b_f through $b_{\{e,f\}}$ alone; so do the Gammas of the other marks, the monomials of the other nodes and those of the markers of v. What is left at a series node is $\Gamma(c_e + b_e)\Gamma(b_e)\Gamma(c_f + b_f)\Gamma(b_f)$ times

$$\theta_{v,e}^{a_e - b_e - s}\theta_{v,f}^{a_f - b_f - s}x_e^{-b_e}x_f^{-b_f} = \theta_{v,e}^{m_e - s}\theta_{v,f}^{m_f - s}\big(x_e\theta_{v,e}^2\big)^{-b_e}\big(x_f\theta_{v,f}^2\big)^{-b_f},$$

a function of $b_{\{e,f\}}$ when $x_e\theta_{v,e}^2 = x_f\theta_{v,f}^2$; at a parallel node it is $\Gamma(m_e - b_e)\Gamma(t - b_e)\Gamma(m_f - b_f)\Gamma(t - b_f)$ times $\theta_{v,e}^{-m_e - t}\theta_{v,f}^{-m_f - t}\big(x_e\theta_{v,e}^{-2}\big)^{-b_e}\big(x_f\theta_{v,f}^{-2}\big)^{-b_f}$, from the monomials $\theta_{v,x}^{b_x - a_x - t}$, a function of $b_{\{e,f\}}$ when $x_e\theta_{v,e}^{-2} = x_f\theta_{v,f}^{-2}$. The rest is the reference, proof of prop:poolpair, with $x_e\theta_{v,e}^{\pm 2}$ in place of x_e: at fixed $b_{\{e,f\}}$ the integral over b_e is the integral of four Gammas displayed in the statement, and disjoint pairs occupy disjoint coordinates. □

Theorem 25.15 (Nodes in hold form). *As stated, for every substituted family.*

Proof. The cores $\mathcal{H}_j$ and the cells $\mathcal{A}_{ij}$ are molecules by Lemma 25.2, and the markers are neither loops nor coloops of their cores and cells. Table j carries the hinge tree whose nodes are $\mathcal{H}_j$ and the $\mathcal{A}_{ij}$, each cell read as a core on $B_i \sqcup \{g_i\}$ and joined to $\mathcal{H}_j$ by an edge $\mathfrak{e}_i$ (Definition 22.14), the table being obtained by these two-sums (Proposition 20.3(i)). Each B_i is nonempty, a hinge being neither a loop nor a coloop; fix $e_i \in B_i$.

(i) *Step 1: the network.* Let $\mathbf{c}$ be an admissible point of F. By Theorem 25.12 with these trees, $P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})$ is the fibre integral, over the loads of F with the pins $\sum_j(\mathbf{a}_j+\mathbf{b}_j)=\mathbf{m}$, $|\mathbf{a}_j| = r_j s_j$ ($j \geq 2$) and $|\mathbf{b}_j| = r_j^* t_j$, on the translate through $\mathbf{c}$, of

$$\prod_j \frac{P_{\mathcal{H}_j}(\mathbf{a}_{j,\mathcal{H}};s_j,0)P_{\mathcal{H}_j}(\mathbf{b}_{j,\mathcal{H}};0,t_j)\prod_i P_{\mathcal{A}_{ij}}(\mathbf{a}_{j,i};s_j,0)P_{\mathcal{A}_{ij}}(\mathbf{b}_{j,i};0,t_j)}{\prod_i \Gamma_1^2\begin{bmatrix} a_{j,g_i},\ s_j-a_{j,g_i} \\ s_j\end{bmatrix}\Gamma_1^2\begin{bmatrix} b_{j,g_i},\ t_j-b_{j,g_i} \\ t_j\end{bmatrix}}\,(\mathbf{x}_j^K)^{-\mathbf{a}_j|_{\mathcal{E}_\mathcal{H}}}(\mathbf{x}_j^\Psi)^{-\mathbf{b}_j|_{\mathcal{E}_\mathcal{H}}}.$$

Here, by Definition 25.4 with the branch B_i on the side of the cell, of rank $r_{\mathcal{A}_{ij}}$ and corank $|B_i|+1-r_{\mathcal{A}_{ij}}$, and by Lemma 25.5(i), the marker loads of the core at g_i are

$$a_{j,g_i} = a_{j,B_i} - s_j\big(r_{\mathcal{A}_{ij}}-1\big), \qquad b_{j,g_i} = b_{j,B_i} - t_j\big(|B_i|-r_{\mathcal{A}_{ij}}\big),$$

those of the cell at g_i are $s_j - a_{j,g_i}$ and $t_j - b_{j,g_i}$, the node loads of the core are $\mathbf{a}_{j,\mathcal{H}} = (\mathbf{a}_j|_{\mathcal{E}_\mathcal{H}}, (a_{j,g_i})_i)$ and $\mathbf{b}_{j,\mathcal{H}} = (\mathbf{b}_j|_{\mathcal{E}_\mathcal{H}}, (b_{j,g_i})_i)$, and those of the cell $\mathcal{A}_{ij}$ are $\mathbf{a}_{j,i} = (\mathbf{a}_j|_{B_i}, s_j - a_{j,g_i})$ and $\mathbf{b}_{j,i} = (\mathbf{b}_j|_{B_i}, t_j - b_{j,g_i})$. The twists are 1 off $\mathcal{E}_\mathcal{H}$.

Step 2: the marker loads as coordinates. In every table j and for every i replace the coordinate a_{j,e_i} by a_{j,g_i} and b_{j,e_i} by b_{j,g_i}. Each new coordinate is the old one plus other old coordinates plus a constant, so the change of coordinates is unit triangular, of determinant 1, and by Lemma S9(i) the fibre integral does not change. The new coordinates are the *core coordinates*, the loads $(\mathbf{a}_{j,\mathcal{H}}, \mathbf{b}_{j,\mathcal{H}})_j$ of the cores on $\mathcal{E}_\mathcal{H} \sqcup \{g_1,\dots,g_k\}$, and for each i the *coordinates of* B_i, the loads $a_{j,e}$ and $b_{j,e}$ at the marks $e \in B_i \setminus e_i$, $j = 1,\dots,N$.

Step 3: the core masses. Replace the mass pin at e_i by the sum C_i of the mass pins over B_i, a row operation with the coefficient 1 on the replaced pin, of determinant 1 (Lemma S9(i)). In the new coordinates

$$C_i: \qquad \sum_j \big(a_{j,g_i}+b_{j,g_i}\big) = m_{B_i} - \sum_j \Big(s_j\big(r_{\mathcal{A}_{ij}}-1\big) + t_j\big(|B_i|-r_{\mathcal{A}_{ij}}\big)\Big) = \kappa_{F_i}$$

by Definition 22.1: the mass of the cores at g_i is the hinge wall of the cell family at g_i.

Step 4: the totals, and the split of the pins. By Proposition 20.3(i), $r_j = r_{\mathcal{H}_j} + \sum_i (r_{\mathcal{A}_{ij}}-1)$, and so $r_j^* = n - r_j = r_{\mathcal{H}_j}^* + \sum_i(|B_i| - r_{\mathcal{A}_{ij}})$ with $r_{\mathcal{H}_j}^* = |\mathcal{E}_\mathcal{H}| + k - r_{\mathcal{H}_j}$ and $n = |\mathcal{E}_\mathcal{H}| + \sum_i |B_i|$. In the new coordinates therefore

$$|\mathbf{a}_j| = \big|\mathbf{a}_j|_{\mathcal{E}_\mathcal{H}}\big| + \sum_i \Big(a_{j,g_i} + s_j\big(r_{\mathcal{A}_{ij}}-1\big)\Big), \qquad \text{so} \qquad |\mathbf{a}_j| = r_j s_j \iff |\mathbf{a}_{j,\mathcal{H}}| = r_{\mathcal{H}_j} s_j,$$

and likewise $|\mathbf{b}_j| = r_j^* t_j \iff |\mathbf{b}_{j,\mathcal{H}}| = r_{\mathcal{H}_j}^* t_j$. The pins are now the *core pins*, namely the mass pins at $\mathcal{E}_\mathcal{H}$, the C_i and the totals of the cores, on the core coordinates alone, which are the pins of $\mathcal{F}$; and for each i the *pins of* B_i, the mass pins at $e \in B_i \setminus e_i$, on the coordinates of B_i alone. The pin matrix is block diagonal, and since the fibre integral converges absolutely (Theorem 24.5(ii)), Lemma S9(iii) writes it as the integral over $\mathcal{F}$, on the translate through the core node loads of $\mathbf{c}$, of the fibre integrals over the coordinates of the B_i at fixed core coordinates. The corners and the twists depend on the core coordinates alone and leave the inner integrals, and the inner integral for B_i is

$$\Phi_i = \int \prod_j P_{\mathcal{A}_{ij}}(\mathbf{a}_{j,i};s_j,0)\,P_{\mathcal{A}_{ij}}(\mathbf{b}_{j,i};0,t_j)$$

over the coordinates of B_i with the pins of B_i.

Step 5: the inner integrals are holds. Let $F_i^\natural$ be the family of the tables $\mathcal{A}_{ij}$ on $B_i \sqcup \{g_i\}$ with the masses $\mathbf{m}|_{B_i}$ and $\sum_j (s_j + t_j) - \kappa_{F_i}$ on g_i, on its balance wall (Lemma 20.8). The node loads $(\mathbf{a}_{j,i}, \mathbf{b}_{j,i})_j$ of the cells are loads of its letters: for every j, $|\mathbf{a}_{j,i}| = a_{j,B_i} + s_j - a_{j,g_i} = r_{\mathcal{A}_{ij}} s_j$ and $|\mathbf{b}_{j,i}| = t_j(|B_i| + 1 - r_{\mathcal{A}_{ij}})$, their masses at $e \in B_i$ are the m_e, and at g_i they add to $\sum_j (s_j + t_j - a_{j,g_i} - b_{j,g_i}) = \sum_j (s_j + t_j) - \kappa_{F_i}$ by C_i. At $\mathbf{c}$ every period of a cell at its node loads is finite (Theorem 25.6), so the node loads of the cells at $\mathbf{c}$ are an admissible point of $F_i^\natural$ (Proposition 24.3(ii)), and every proper wall of $F_i^\natural$ is positive (Theorem 24.5(i)). In Theorem 24.7 for $F_i^\natural$ take e_i as the fixed mark and $G = \{g_i\}$: the pinned fibre integral at the pins $s_j - a_{j,g_i}$ ($j \geq 2$) and $t_j - b_{j,g_i}$ at g_i is, in the coordinates of the loads of the letters other than K_1 at the marks of $B_i \setminus e_i$, the integral with the factor $(2\pi i)^{-(2N-1)(|B_i|-1)}$, the other loads solved from its pins. The integral Φ_i, with $a_{1,e}$ for $e \in B_i \setminus e_i$ solved from the mass pin at e, has the same coordinates, the same factor by Lemma S9(ii), and the same integrand, since the loads $(\mathbf{a}_{j,i}, \mathbf{b}_{j,i})_j$ satisfy all the pins of $F_i^\natural$. The real parts of the core coordinates being those of $\mathbf{c}$, Theorem 24.7 gives

$$\Phi_i = \mathrm{Hold}_{\{g_i\}}\big((s_j - a_{j,g_i})_{j\geq 2}, (t_j - b_{j,g_i})_j\big) \qquad \text{for } F_i^\natural.$$

Step 6: the corners cancel. Apply Lemma 24.9 to the cell family F_i at $\mathbf{z}_i = (a_{j,g_i}, b_{j,g_i})_j$, of sum κ_{F_i} by C_i. Its conditions hold: $0 < \mathrm{Re}\, a_{j,g_i} < s_j$ and $0 < \mathrm{Re}\, b_{j,g_i} < t_j$, the marker loads at $\mathbf{c}$ lying in these intervals (Theorem 25.6); and the hold of Step 5 at the real parts is finite, so by Proposition 24.8 and the first identity of Lemma 24.9 $\int \boldsymbol{\varrho}_{F_i}^{-\mathrm{Re}\,\mathbf{z}_i} d\phi_{F_i}$ is finite. So

$$\Phi_i = \prod_j \Gamma_1^2\begin{bmatrix} s_j - a_{j,g_i}, & a_{j,g_i} \\ & s_j \end{bmatrix} \Gamma_1^2\begin{bmatrix} t_j - b_{j,g_i}, & b_{j,g_i} \\ & t_j \end{bmatrix} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i},$$

and these are the corners of the edge $\mathfrak{e}_i$ in Step 1, $\Gamma_1^2\begin{bmatrix} a, & b \\ & a+b \end{bmatrix}$ being symmetric in a and b. They cancel, and what is left is the integral of (i) over $\mathcal{F}$. Its dimension is the number $2N(|\mathcal{E}_\mathcal{H}| + k)$ of core coordinates less the $|\mathcal{E}_\mathcal{H}| + k + 2N - 1$ core pins, $(2N-1)(|\mathcal{E}_\mathcal{H}| + k - 1)$.

(ii) Let $F'^\natural$ be the family of the tables of F' with the mass $\sum_j (s_j + t_j) - \kappa_{F'}$ on g_0, on its balance wall (Lemma 20.8), and $\mathbf{c}$ an admissible point of it as in the statement. Its loads at g_0 lie in $(0, s_j)$ and $(0, t_j)$, g_0 being neither a loop nor a coloop, so $\mathrm{Re}\,\mathbf{z}$ meets the conditions of Lemma 24.9 for F'; the finiteness of $\int \boldsymbol{\varrho}_{F'}^{-\mathrm{Re}\,\mathbf{z}} d\phi_{F'}$ follows as in Step 6 from that of the hold of $F'^\natural$ at g_0 at the loads of $\mathbf{c}$ (Theorem 24.7). By the second identity of Lemma 24.9 the left side of (ii) is the hold of $F'^\natural$ at g_0 at the loads $(s_j - z_{j,K})_{j\geq 2}$ and $(t_j - z_{j,\Psi})_j$, and by Theorem 24.7 with $G = \{g_0\}$ it is the fibre integral of Theorem 24.5(ii) for $F'^\natural$ at $\mathbf{x} = \mathbf{1}$ with the pins at g_0. Steps 1 to 6 apply to it with g_0 a mark of the cores, its pins being core pins, and give the integral over $\mathcal{F}_\mathbf{z}$, of dimension $(2N-1)(|\mathcal{E}_\mathcal{H}| + k) - (2N-1) = (2N-1)(|\mathcal{E}_\mathcal{H}| + k - 1)$. □

Remark. Let the core of every table be $\binom{\{g_1,g_2\}}{1}$ with unit weights, the weights of the tables being those of the cells (Lemma 20.8), so that $\mathcal{E}_\mathcal{H} = \varnothing$ and $B = B_1$ is a bunch of width one in every table. By Proposition 25.10 at $q = 1$,

$$P_{\mathcal{H}_j}(\mathbf{a}_{j,\mathcal{H}}; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_{j,\mathcal{H}}; 0, t_j) = \Gamma_1^2\begin{bmatrix} a_{j,g_1}, & a_{j,g_2} \\ & s_j \end{bmatrix} \Gamma_1^2\begin{bmatrix} b_{j,g_2}, & b_{j,g_1} \\ & t_j \end{bmatrix} = \Gamma_2^4\begin{bmatrix} a_j^d, & b_j^d, & a_j^r, & b_j^r \\ & & s_j, & t_j \end{bmatrix}$$

of Theorem 18.13, with the level shares $a_j^r = a_{j,g_1}$, $a_j^d = a_{j,g_2}$, $b_j^d = b_{j,g_1}$ and $b_j^r = b_{j,g_2}$. With $\mathbf{z}_1 = (a_j^r, b_j^d)_j$ the children family of F_1 in Lemma 24.9 has the tables $\mathcal{A}_{1j} \backslash g_1 = \mathcal{B}_j | B$ at $(a_j^r, t_j - b_j^d) =$

(a_j^r, b_j^r) and $\mathcal{A}_{1j}/g_1 = \mathcal{B}_j/R$ at $(s_j - a_j^r, b_j^d) = (a_j^d, b_j^d)$: it is the bunch family of Theorem 18.13; with $\mathbf{z}_2 = (a_j^d, b_j^r)_j$ the children family of F_2 is the room family. The core pins are $|\mathbf{a}_{j,\mathcal{H}}| = s_j$ $(j \geq 2)$, $|\mathbf{b}_{j,\mathcal{H}}| = t_j$ and the masses κ_{F_1} and κ_{F_2}; solving the totals for a_j^d and b_j^r by Lemma S9(ii), with the block of the totals and of the mass κ_{F_2} unit triangular after the row operations of Lemma S9(i), leaves the share integral over $(a_j^r, b_j^d)_j$ pinned by $\sum_j (a_j^r + b_j^d) = \kappa_{F_1} = \kappa_B$. This is the integral of Theorem 18.13 over the scale shares, with the same integrand. In the pair coordinates $\sigma_j^d = \frac{1}{2}(a_j^d + b_j^d)$ and $\delta_j^d = \frac{1}{2}(a_j^d - b_j^d)$, the map $(a_j^r, b_j^d) \mapsto (\sigma_j^d, \delta_j^d)$, with $a_j^d = s_j - a_j^r$, has the determinant $\frac{1}{2}$ in each table, so $d(\mathbf{a}^r, \mathbf{b}^d) = 2^N\, d(\boldsymbol{\sigma}^d, \boldsymbol{\delta}^d)$; and the pin $\sum_j (a_j^r + b_j^d) = \kappa_B$ reads $-2\sum_j \delta_j^d = \kappa_B - \sum_j s_j$, its row being -2 times that of $\sum_j \delta_j^d = \frac{1}{2}(\sum_j s_j - \kappa_B)$, which by Definition 4.1 multiplies the fibre measure by $\frac{1}{2}$. The two factors give the 2^{N-1} of the first display of Theorem 18.13, a consistency check of the bookkeeping above. In `checks/bookkeeping.py`, `check_twosides_pairs` computes $2^N \cdot \frac{1}{2} - 2^{N-1} = 0$ exactly; `check_core_family_balance` computes the balance of the core family at the masses $(\mathbf{m}|_{\mathcal{E}_\mathcal{H}}, \kappa_{F_1}, \ldots, \kappa_{F_k})$ and the corank identity of Step 4; and `check_holdnode_pins` computes, for small families, the unit Jacobian of Step 2, the split of the pins of Steps 3 and 4 with the core masses κ_{F_i} and the totals of the cores, and the totals and the hinge mass of the node loads of the cells in Step 5.

Corollary 25.16 (The glue and the series and parallel rules in hold form). *As stated; in (i) the reading for a family is new.*

Proof. For one table the point $\mathbf{z}_i$ of Theorem 25.15 is $(\kappa_{\mathcal{A}_i} - b_{g_i}, b_{g_i})$, by the core mass at g_i, and $\boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} = \varrho_{\mathcal{A}_i}^{-b_{g_i}}$, the ratio vector of one cell being its ratio (Definition 22.1); so (i) for one table, (ii) and (iii) are the reference, proof of cor:holdrules. The reading of (i) for a family is the remark after the proof of Theorem 25.15. □

Definition 25.17 (The network procedure). As stated, for a family whose tables carry hinge trees; pooling and the passage to the transforms are for one table, and the recursive Mellin evaluation asks the tables to share one rooted tree.

Remark. The operations rest on the statements they cite, proved above for families: (1) is Corollary 25.7, infeasibility meaning that some proper wall is not positive, so that the period diverges (Theorem 8.10); (2) is Theorem 25.12 and Corollary 25.13; (4) uses that the admissible points form an open subset of the real points of the pins, the conditions $\mathbf{c}^X \in w_X \operatorname{ri} \mathcal{N}_X$ being open there, and the bounds of Theorem 24.5(iii), uniform on compact sets of admissible points; (5) is Theorem 25.15, the substituted family at a vertex v having the cores $\mathcal{H}_{j,v}$ and, as its cell families, the subtree cell families of the vertices directly below v. The reusable data are those of Corollary 24.6, one Gramma for each orbit of nodes.

Proposition 25.18 (What the network costs). *As stated, for a family.*

Proof. (i) The fibre of Theorem 25.12 has dimension $(2N-1)(n-1)$, and for one table each pooled pair removes one coordinate (Proposition 25.14). Opening the period of a prime node $\mathcal{H}_{j,v}$ at one full lean on its share side adds $|\mathcal{H}_{j,v}| - n_{j,v}$ coordinates (Corollary 25.9), so the two full leans add $2(|\mathcal{H}_{j,v}| - n_{j,v})$; series and parallel nodes have closed forms (Proposition 25.10). The atom formula (Corollary 17.15) has the shares $\mathbf{a}_j, \mathbf{b}_j \in \mathbb{C}^{\mathcal{B}_j}$ against the $2N$ total pins and the $n-1$ pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$, $e \neq g$: dimension $\sum_j 2|\mathcal{B}_j| - 2N - n + 1$. After its even shares close (Theorem 19.10(iv)) the fibre of differences has dimension $\sum_j (|\mathcal{B}_j| - 1) - (n-1)$, and each table keeps one scale integral: $\sum_j |\mathcal{B}_j| - n + 1$ in all.

(ii) is Corollary 24.6: one Gramma of n_v variables for each orbit of nodes, the orbits taken over all the tables.

(iii) By Theorem 25.15(ii) a point of the Mellin transform of the law of a subtree cell family with the top node v costs a fibre integral of dimension $(2N-1)(|\mathcal{E}_v|+k_v-1)$, $k_v=|G_v|-1$ the number of vertices directly below v, that is $(2N-1)(|\mathcal{E}_v|+|G_v|-2)$; at the root Theorem 25.15(i) costs $(2N-1)(|\mathcal{E}_o|+|G_o|-1)$. Their integrands need the periods of the tables of the node at their full leans at the loads of the fibre, closed at series and parallel nodes and on the share side at a prime node (Corollary 25.9). □

Proposition 25.19 (Product weights are twists). *As stated, for every family.*

Proof. $K_{F^{\boldsymbol\varsigma},j}(\alpha)=\sum_T\theta_{j,T}\boldsymbol\zeta_j^T\alpha^T=K_j(\boldsymbol\zeta_j\circ\alpha)$, and $\Psi_{F^{\boldsymbol\varsigma},j}(\alpha)=\sum_T\theta_{j,T}\boldsymbol\zeta_j^T\alpha^{\mathcal{E}\setminus T}=\boldsymbol\zeta_j^{\mathcal{E}}\sum_T\theta_{j,T}(\alpha/\boldsymbol\zeta_j)^{\mathcal{E}\setminus T}=\boldsymbol\zeta_j^{\mathcal{E}}\Psi_j(\alpha/\boldsymbol\zeta_j)$. These are the letters of $F[\mathbf{x}]$ with $\mathbf{x}_j^K=\boldsymbol\zeta_j$ and $\mathbf{x}_j^\Psi=\boldsymbol\zeta_j^{-1}$ (Definition 21.1), the cobasis letters multiplied by $\boldsymbol\zeta_j^{\mathcal{E}}$, and $(\boldsymbol\zeta_j^{\mathcal{E}})^{-t_j}=\boldsymbol\zeta_j^{-t_j\mathbf{1}}$ gives the first identity. By the torus relation of Definition 21.1 for the family of the one letter K_j, the period of $K_j(\boldsymbol\zeta_j\circ\alpha)^{-s_j}$ at $\mathbf{a}$ is $\boldsymbol\zeta_j^{-\mathbf{a}}P_{\mathcal{B}_j}(\mathbf{a};s_j,0)$, and for the one letter Ψ_j the period of $\big(\boldsymbol\zeta_j^{\mathcal{E}}\Psi_j(\alpha/\boldsymbol\zeta_j)\big)^{-t_j}$ at $\mathbf{b}$ is $\boldsymbol\zeta_j^{-t_j\mathbf{1}}\boldsymbol\zeta_j^{\mathbf{b}}P_{\mathcal{B}_j}(\mathbf{b};0,t_j)$. In the lean convolution of $F^{\boldsymbol\zeta}$ the integrand is therefore that of F times $\prod_j\boldsymbol\zeta_j^{-t_j\mathbf{1}}\boldsymbol\zeta_j^{\mathbf{b}_j-\mathbf{a}_j}$. With one $\boldsymbol\zeta$ for every table, the torus relation with the vector $\boldsymbol\zeta^{-1}$ on every letter gives $P_{F[\mathbf{x}]}=\boldsymbol\zeta^{-\mathbf{m}}P_{F[\mathbf{x}']}$ for $\mathbf{x}_j^K=\boldsymbol\zeta$, $\mathbf{x}_j^\Psi=\boldsymbol\zeta^{-1}$ and $\mathbf{x}_j'^K=\mathbf{1}$, $\mathbf{x}_j'^\Psi=\boldsymbol\zeta^{-2}$, and $\prod_j\boldsymbol\zeta^{-t_j\mathbf{1}}=\boldsymbol\zeta^{-|\mathbf{t}|\mathbf{1}}$. Weights at one mark g are the $\boldsymbol\zeta_j$ equal to 1 off g. □

Proposition 25.20 (Carrying a calculation through a change, in hold form). *As stated, for every family.*

Proof. In Theorem 25.12 the integrand is the product of the node periods at the node loads over the Beta corners at the marker loads, times the twists. (i) The exponents of table j enter the pins $|\mathbf{a}_j|=r_js_j$ and $|\mathbf{b}_j|=r_j^*t_j$ and the marker loads $k_{j;v,\mathfrak{e}}=s_jr_{j,v}^{\mathfrak{e}}-a_{j,\mathcal{E}_v^{\mathfrak{e}}}$ and $l_{j;v,\mathfrak{e}}=t_jr_{j,v}^{*\mathfrak{e}}-b_{j,\mathcal{E}_v^{\mathfrak{e}}}$, affine in (s_j,t_j), and the corners through them; the node periods are functions of the node loads alone (Definition 24.2), and the native loads of the new data are admissible (Proposition 24.3(iii)). (ii) In the coordinates of Theorem 24.5(ii) the masses enter $\mathbf{a}_1=\mathbf{m}-\sum_{j\ge2}\mathbf{a}_j-\sum_j\mathbf{b}_j$ alone, the loads at e_0 of the other letters being fixed by their totals. (iii) is Proposition 25.19, for one node read for its table. (iv) The node periods are the only data of a node in the integrand; a replaced node of the same rank keeps the ranks of the branches and so the marker loads, and Theorem 25.12 applies to the new trees, whose nodes are molecules with markers neither loops nor coloops. (v) By Theorem 25.15(ii) the transform at a vertex is computed from the node there and the transforms of the vertices directly below it, so a change at one node changes the transforms on its path to the root only. □

Proposition 25.21 (The exponents are moment indices). *As stated, for every family, any of its letters being closed.*

Proof. The walls are affine in $(\mathbf{s},\mathbf{t})$ (Definition 8.1), so I, the intersection of the hyperplane with the open orthant and the open half-spaces $\kappa_S>0$, is relatively open and convex. On I the period converges (Theorem 8.10), and $\mathbf{m}>0$, every singleton wall $m_e-\sum_j(s_jd_j(\{e\})+t_jd_j^*(\{e\}))$ being positive and the counts nonnegative. Fix i and apply Lemma 16.2 to the form of F with $h=L_i^{1/g_i}$, positive and homogeneous of degree one, and $\varphi(\rho)=\rho^{w_ig_i}e^{-\rho^{g_i}}$, for which $\int_0^\infty\varphi(\rho)\,d\rho/\rho=\Gamma(w_i)/g_i$: the integrand $x^{\mathbf{m}-\mathbf{1}}\prod_{i'}L_{i'}^{-w_{i'}}$ times $\varphi(h)=L_i^{w_i}e^{-L_i}$ is $x^{\mathbf{m}-\mathbf{1}}e^{-L_i}\prod_{i'\ne i}L_{i'}^{-w_{i'}}$, which gives the first identity. Every letter being positive on the orthant and every $w_{i'}>0$, $L_{i'}(x)^{-w_{i'}}=$

$\Gamma(w_{i'})^{-1}\int_0^\infty \lambda^{w_{i'}-1}e^{-\lambda L_{i'}(x)}d\lambda$, and Tonelli's theorem gives the second identity, the integral over x being $\mathsf{G}_i(\mathbf{m};\boldsymbol{\lambda})$, the Gramma of the concatenated configuration (Definition 2.1). The derivative $\partial^k_{\lambda_{i'}}\mathsf{G}_i$ is $(-1)^k$ times the integral with the further factor $L^k_{i'}$, which gives the complete monotonicity. By Proposition 8.11(iii), $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ is the moment generating function, in the exponents, of $\big(-\log K_j(U), -\log\Psi_j(U)\big)_j$ for $U\sim\mathrm{Dir}_{\mathbf{m}}$, and by Hölder's inequality a moment generating function is log-convex on the convex set where it is finite; $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ does not depend on the exponents. □

40.26 Chapter 26: finite interfaces and certified enclosures

Definition 26.1 (The channel law; moment vectors). As stated; cells with the same hinges on the same marks are kept together by reading their channels jointly.

Proposition 26.2 (The channel law composes). *As stated, for cells kept together and for every cell family.*

Proof. By Theorem 3.12 with the blocks B_1 and B_2, $U\sim\mathrm{Dir}_{(\mathbf{m}_1,\mathbf{m}_2)}$ on $\Delta_{B_1\sqcup B_2}$ is $U|_{B_l}=V_lU_l$ with $(V_1,V_2)\sim\mathrm{Dir}_{(m_{B_1},m_{B_2})}$ and $U_l\sim\mathrm{Dir}_{\mathbf{m}_l}$, independent. In every table j the channels of the spliced cell are the bilinear map of Theorem 20.11(iii) applied to the channels of the two cells of table j, and a channel c has $c(V_lU_l)=V_l^{\deg c}c(U_l)$. So $\mathrm{ch}(\mathcal{A}_{12})(U)$ is the image of the replaced channel vectors under the bilinear map applied table by table, and its law is the channel law of $\mathcal{A}_{12}$. For the tables of a family kept together without hinges the channels are the letters K_j and Ψ_j, and Proposition 8.11(i) is the stated reading.

For the cell family F, by Definitions 22.1 and 20.1 at the point $\mathrm{ch}(\mathcal{A})(u)$, $\gamma^K_{\mathcal{A}_j}=x_{j,0}/x_{j,1}$ and $\gamma^\Psi_{\mathcal{A}_j}=x^*_{j,1}/x^*_{j,0}$, so $\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}=x_{j,0}x_{1,1}/(x_{j,1}x_{1,0})$ and $\gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}=x^*_{j,1}x_{1,1}/(x^*_{j,0}x_{1,0})$: $\boldsymbol{\varrho}_F=R_F\circ\mathrm{ch}(\mathcal{A})$. The base form is $\phi_F(du)=\varpi(\mathrm{ch}(\mathcal{A})(u))\,u^{\mathbf{m}-\mathbf{1}}d\omega_B(u)$ with $\varpi=\prod_j x_{j,1}^{-s_j}x_{j,0}^{*-t_j}(x_{1,0}/x_{1,1})^{-\kappa_F}$, the factor $\gamma^K_{\mathcal{A}_1}(u)^{-\kappa_F}$ being $(x_{1,0}/x_{1,1})^{-\kappa_F}$, and $u^{\mathbf{m}-\mathbf{1}}d\omega_B=\Gamma_{|B|}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\mathrm{Dir}_{\mathbf{m}}(du)$ (Definition 3.10). For a positive measure μ and positive functions, $(R\circ c)_*\big((\varpi\circ c)\,\mu\big)=R_*\big(\varpi\,c_*\mu\big)$, both sides giving $\int H(R(c(u)))\,\varpi(c(u))\,d\mu(u)$ against every Borel $H\ge0$. With $c=\mathrm{ch}(\mathcal{A})$ and $\mu=\Gamma_{|B|}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\mathrm{Dir}_{\mathbf{m}}$ this is the display. □

Theorem 26.3 (Gamma draws close the moments). *As stated, for cells kept together.*

Proof. The reference, proof of thm:gammalift, uses of the channel vector only that it is a vector of polynomials on the marks, homogeneous of known degrees, that the Gamma draws on disjoint sets of marks are independent, and that a splicing acts on the channel vectors by a bilinear map whose every product has the degree of the channel it gives. For cells kept together the channel vector is the joint vector $(\mathrm{ch}(\mathcal{A}_j))_j$, and a splicing acts table by table, so on the joint vectors by the direct sum over the tables of the maps of Theorem 20.11(iii), again bilinear with that property. So (i), (iii) and (iv) hold word for word. In (ii) the joint channels of $\mathcal{A}_{12}$ at α are $\mathrm{sp}\big(\mathrm{ch}(\mathcal{A}_1)(\alpha|_{B_1}),\mathrm{ch}(\mathcal{A}_2)(\alpha|_{B_2})\big)$, each a sum of products of one joint channel of each cell, so a monomial of degree $|\mathbf{k}|$ in them expands into monomials of degree $|\mathbf{k}|$ in the joint channels of each cell, and the independence of $\alpha|_{B_1}$ and $\alpha|_{B_2}$ gives the formula. In (v) the joint channels are

bounded on Δ_B, so by (iii) their moments determine their law, and Proposition 26.2 composes it. □

Theorem 26.9 (Taylor coefficients in the weights are periods with walls at least as high). *As stated, at real exponents of any sign.*

Proof. For $u \in \Delta_{\mathcal{E}}$ put $y_i(u) = \sum_j \zeta_j D_{ij}(u)/L_i(u)$. The coefficients of D_{ij} are those of L_i times the numbers $v_{i,\boldsymbol{\gamma},j}$ of Theorem 26.10, so $|D_{ij}| \le V_{ij}L_i$ on the orthant and $|y_i| \le r_i(\boldsymbol{\zeta}) := \sum_j |\zeta_j| V_{ij}$; let every $r_i(\boldsymbol{\zeta}) < 1$. For real w and $|y| < 1$, $(1+y)^{-w} = \sum_{l\ge 0}(-1)^l \frac{(w)_l}{l!} y^l$, and $|(w)_l| \le (|w|)_l$ since $|w+k| \le |w| + k$, so the series of the moduli is at most $\sum_l \frac{(|w|)_l}{l!}|y|^l = (1-|y|)^{-|w|}$. With the multinomial expansion of $y_i^{l_i}$,

$$L_i(\boldsymbol{\zeta})^{-w_i} = L_i^{-w_i}(1+y_i)^{-w_i} = \sum_{(l_{ij})_j} \frac{(-1)^{l_i}(w_i)_{l_i}}{\prod_j l_{ij}!}\, L_i^{-w_i-l_i} \prod_j \left(\zeta_j D_{ij}\right)^{l_{ij}}, \qquad l_i = \sum_j l_{ij},$$

and the product over i of the series of the moduli is at most $\prod_i L_i^{-w_i}(1 - r_i(\boldsymbol{\zeta}))^{-|w_i|}$. Against $u^{\mathbf{m}-\mathbf{1}} d\omega_{\mathcal{E}}$ this bound is integrable, $P(\mathbf{0})$ converging at the real data; so termwise integration is justified, P is a convergent power series on $\{r_i(\boldsymbol{\zeta}) < 1 \ \forall i\}$, and its coefficients are the stated ones. After the numerators are expanded every term is a coefficient times $u^{\mathbf{m}+\boldsymbol{\gamma}-\mathbf{1}} \prod_i L_i^{-w_i-l_i}$, $\boldsymbol{\gamma}$ a sum, over i, of l_i exponents of the support of L_i. So $|\boldsymbol{\gamma}| = \sum_i l_i g_i$ and the term is balanced, and $\gamma_S \ge \sum_i l_i d_i(S)$ with $d_i(S)$ the least value of $\langle \mathbf{1}_S, \cdot\rangle$ on the support of L_i, which is the count $d_j(S)$ or $d_j^*(S)$ of the letter in the walls (Definition 8.1). So $\kappa'_S = \kappa_S + \gamma_S - \sum_i l_i d_i(S) \ge \kappa_S > 0$, and the term converges by Theorem 8.10, whose criterion holds for exponents of any sign. The reading of Theorem 32.1(iii) is the reference's. □

Theorem 26.10 (A uniform certificate for weights). *As stated, at real exponents of any sign.*

Proof. Let $|\zeta_j| \le R_j$, so that $|y_i| \le r_i < 1$ in the notation of the proof of Theorem 26.9. For real w_i, $|(1+y_i)^{-w_i}| = |1+y_i|^{-w_i}$, the real part of $1+y_i$ being positive, and $1 - r_i \le |1+y_i| \le 1 + r_i$, so $|L_i(\boldsymbol{\zeta})^{-w_i}| \le L_i^{-w_i} \max\left((1-r_i)^{-w_i}, (1+r_i)^{-w_i}\right)$; integrating, $|P(\boldsymbol{\zeta})| \le M$. The radii can be enlarged a little with every $r_i < 1$ kept, so P is holomorphic near the closed polydisc by Theorem 26.9, and the Cauchy estimates give $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \le M\mathbf{R}^{-\mathbf{k}}$; the bounds on the Taylor remainder follow from these as in the reference, proof of thm:weightbank, there being $\binom{k+p-1}{p-1}$ multi-indices of degree k. For the explicit bound on $P(\mathbf{0})$: by Theorem 8.7(ii) and Theorem 8.10, off the balance wall and then on it by continuity, $P(\mathbf{0}) = \sum_\pi \int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_{U_{\pi,k}}-1} \prod_i L_{i,\pi}(q)^{-w_i}\, dq$. Each sector letter has nonnegative coefficients and is increasing in every q_k, so $\underline{L}_{i,\pi} \le L_{i,\pi}(q) \le \overline{L}_{i,\pi}$ and $L_{i,\pi}(q)^{-w_i} \le \max\left(\underline{L}_{i,\pi}^{-w_i}, \overline{L}_{i,\pi}^{-w_i}\right)$ for every real w_i, and $\int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_{U_{\pi,k}}-1} dq = \prod_k \kappa_{U_{\pi,k}}^{-1}$. The enclosure of Proposition 15.12 holds for real data of any sign, its error bound being written with $\sum_j (|s_j| + |t_j|)$, so its upper end bounds $\prod_j K_j(\mathbf{1})^{s_j+t_j} P(\mathbf{0})$ from above. □

40.27 Chapter 27: bunches of any width

Proposition 27.2 (The seam identity). *As stated; in (v) the seam matrix of an extreme excess is a single entry exactly when the weights of that level factor.*

Proof. Parts (i)–(iv) and (vi), and in (v) the room children and the seam letters at the two extremes, are the reference, proof of prop:seamidentity. Let $i \in \{0, c(B)\}$. By 12.17 every pattern

of excess i and every room side of excess i make a basis, so every station of patterns of excess i is compatible with every station of room sides of excess i.

Let the weights of the level factor along these products, $\theta_{p\sqcup X} = \theta'_p\,\theta''_X$ for the patterns p and the room sides X of excess i, with positive θ' and θ''. The child at a pattern p of excess i is the table on R whose bases are all the room sides of excess i, with the weights $\theta'_p\,\theta''_X$: it is θ'_p times one table. So the patterns of excess i form one station and H_i has one row. The columns of a matrix with one row are proportional, and by (iv) no two stations of $\mathcal{D}_i$ have proportional columns; so H_i has one column, and its entry is positive, the pair being compatible.

Conversely let H_i be a single entry h_{CD}. The patterns of excess i then form the one station C and its room sides the one station D, and (ii) gives $\theta_{p\sqcup X} = h_{CD}\,w_p\,w'_X$ for every such p and X: the weights of the level factor along these products, with the factors $h_{CD}\,w_p$ and w'_X. □

Proposition 27.3 (The width ladder). *As stated; in (i) and (ii) the seam is one rectangle at each excess exactly when the weights factor across B and R, and the closing sentence holds at every width.*

Proof. (i) That $c(B) = 0$ exactly when B is a union of molecules, with one excess, every pattern meeting every room side, and the letters $K = \mathsf{x}_0^{\mathsf{T}} H_0\,\mathsf{y}_0$ and $\Psi = \mathsf{x}_0^{*\mathsf{T}} H_0\,\mathsf{y}_0^*$, is the reference, proof of prop:ladder(i). The one excess is extreme, so by 27.2(v) the seam is one rectangle exactly when the weights of its level factor along the products $p \sqcup X$. Every basis lies in this level, so this is a factoring $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$ across B and R, and conversely. With one station C and one station D, 27.2(iii) reads $K = h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ and $\Psi = h_{CD}\,\mathsf{x}_C^*\,\mathsf{y}_D^*$.

(ii) That $c(B) = 1$ exactly when there are two excesses, both extreme, is the reference, proof of prop:ladder(ii). By 27.2(v) the excess i is one rectangle exactly when the weights of its level factor, $\theta_{p\sqcup X} = \theta'_{i,p}\,\theta''_{i,X}$ for the patterns p and the room sides X of excess i. If both levels factor, put $\theta'_p = \theta'_{i(p),p}$ and $\theta''_X = \theta''_{i(X),X}$, each pattern and each room side having one excess; the pattern and the room side of a basis have one excess, so $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$ for every basis T. Conversely a factoring across B and R restricts to a factoring of each level. The seam letters, the room station letters and the room-station factorization are the reference's (proof of prop:ladder(ii), and exercise ex:widthone).

(iii) and the closing sentence. A matrix with one row has proportional columns, and a matrix with one column proportional rows; so by 27.2(iv) an excess with one station on one side has one station on the other, at every width. The criterion for level form is 14.12, which does not restrict the width. In level form the children at the patterns of one excess are equal, so these patterns form one station; the excess then has one station on each side and is one rectangle. That an excess $0 < i < c(B)$ can carry several rectangles is the reference's (prop:ladder(iii) and exercise ex:prismseam). □

Proposition 27.6 (Symmetries of a seam). *As stated; in (iii) an exchanging bijection is read at every lean, and it carries the law of B at the lean $\boldsymbol{\delta}$ to its law at $-\boldsymbol{\delta}$.*

Proof. Parts (i) and (ii), the first sentence of (iii), the carrying bijections and the sentence on the potential are the reference, proof of prop:seamsym.

Let g keep B and the masses and exchange the family, and let $j \mapsto j'$ be the bijection of its tables of 11.14. The pairs are kept, $(\sigma_{j'}, \delta_{j'}) = (\sigma_j, \delta_j)$, so at the lean $-\boldsymbol{\delta}$ table j' carries the exponents $(t_{j'}, s_{j'}) = (t_j, s_j)$. Read every function of α_B at $g^*\alpha$. This permutes the coordinates of Δ_B and keeps $u^{\mathbf{m}_B - \mathbf{1}} d\omega_B$, since g keeps B and the masses. By 14.10, $c_{j'}(B) = c_j(B)$, $d_{j'}(B) = |B| - r_j(B) = d_j^*(B)$ and $d_{j'}^*(B) = d_j(B)$.

(a) By (ii) the seam letter V_{jD} of K_j at the excess $i(D)$ is, at $g^*\alpha$, the seam letter $V^*_{j'D'}$ of $\Psi_{j'}$ at the excess $c_j - i(D)$, and V^*_{jD} is $V_{j'D'}$. Summing over the stations of one excess, $V_j^{(i)} = V_{j'}^{*(c_j-i)}$ and $V_j^{*(i)} = V_{j'}^{(c_j-i)}$.

(b) K_j is active at $\boldsymbol{\delta}$ exactly when $s_j > 0$, that is exactly when $\Psi_{j'}$ is active at $-\boldsymbol{\delta}$; likewise Ψ_j and $K_{j'}$. The sum $\kappa_B + \kappa_B^{\mathrm{top}} = \sum_j (s_j + t_j)c_j$ is the same at the two leans.

(c) By (a) and (b) the scale of the piece at $\boldsymbol{\delta}$ (27.5) is

$$\prod_j \Big(\frac{V_{j'}^{*(0)}}{V_{j'}^{*(c_j)}}\Big)^{s_j/(\kappa_B+\kappa_B^{\mathrm{top}})} \Big(\frac{V_{j'}^{(c_j)}}{V_{j'}^{(0)}}\Big)^{t_j/(\kappa_B+\kappa_B^{\mathrm{top}})},$$

which is the scale at $-\boldsymbol{\delta}$, where $\Psi_{j'}$ carries the exponent s_j and $K_{j'}$ the exponent t_j.

(d) The wall at $-\boldsymbol{\delta}$ is $m_B - \sum_j \big(t_j d_{j'}(B) + s_j d^*_{j'}(B)\big) = m_B - \sum_j \big(s_j d_j(B) + t_j d^*_j(B)\big) = \kappa_B$.

(e) By (a), (c) and (d) the scale form at $\boldsymbol{\delta}$,

$$u^{\mathbf{m}_B - \mathbf{1}} \prod_j (V_j^{(0)})^{-s_j} (V_j^{*(c_j)})^{-t_j} (\gamma_B^\rho)^{-\kappa_B} d\omega_B,$$

is the scale form at $-\boldsymbol{\delta}$.

(f) For a station D of excess $i(D)$ and its image D' of excess $c_j - i(D)$,

$$\varrho^\rho_{jD} = \frac{V_{jD}}{V_j^{(0)}\,(\gamma_B^\rho)^{i(D)}} = \frac{V^*_{j'D'}}{V_{j'}^{*(c_j)}\,(\gamma_B^\rho)^{c_j - i(D')}},$$

the coordinate $\varrho^{\rho*}_{j'D'}$ at $-\boldsymbol{\delta}$; likewise $\varrho^{\rho*}_{jD}$ is the coordinate $\varrho^\rho_{j'D'}$ at $-\boldsymbol{\delta}$.

So the seam vector at $\boldsymbol{\delta}$, followed by the map of the coordinates, is the seam vector at $-\boldsymbol{\delta}$ read at $g^*\alpha$, and by (e) the scale forms correspond. The image of the law at $\boldsymbol{\delta}$ is therefore the law at $-\boldsymbol{\delta}$. By (a) and (b) the relations defining the seam torus at $\boldsymbol{\delta}$ go over into those at $-\boldsymbol{\delta}$. At the centre $\boldsymbol{\delta} = -\boldsymbol{\delta}$, and the law is kept. □

Lemma 27.8 (Absorbing a piece of any width). *As stated, with functions f_l homogeneous of degree one in their second argument.*

Proof. Take the section $h = \gamma_B^\rho(\alpha_B) + |\alpha_Z|$, a positive function of degree one on $(0,\infty)^{B\sqcup Z}$, and write $\alpha_B = \lambda u$ with $\lambda > 0$ and $u \in \Delta_B$ (2.7).

- *The seam letters along the scale.* Let f_l be assigned to K_j. By 27.5, $V_{jD} = V_j^{(0)}\,(\gamma_B^\rho)^{i(D)}\varrho^\rho_{jD}$ on Δ_B, and V_{jD} is homogeneous of degree $d_j + i(D)$. Hence

$$V_{jD}(\lambda u) = \lambda^{d_j} V_j^{(0)}(u)\,\big(\lambda\gamma_B^\rho(u)\big)^{i(D)}\,\varrho^\rho_{jD}(u),$$
$$\text{that is}\quad \mathbf{V}_l(\lambda u) = V_l(\lambda u)\,\mathbf{x}_l(\alpha_g),$$

at $\alpha_g = \lambda\gamma_B^\rho(u)$ and $\mathbf{x} = \boldsymbol{\varrho}_B^\rho(u)$, with $V_l(\lambda u) = \lambda^{d_j} V_j^{(0)}(u) > 0$. For Ψ_j the same holds with $V^*_{jD} = V_j^{*(c_j)}(\gamma_B^\rho)^{c_j - i(D)}\varrho^{\rho*}_{jD}$, of degree $d^*_j + c_j - i(D)$, and $V_l(\lambda u) = \lambda^{d^*_j} V_j^{*(c_j)}(u)$.

- *Homogeneity.* Since f_l is homogeneous of degree one in its second argument,

$$f_l\big(\alpha_Z, \mathbf{V}_l(\lambda u)\big) = V_l(\lambda u)\, f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big).$$

  So the arguments of H are the $f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))$. The exponents assigned to each active letter add to its exponent, so

$$\prod_l f_l\big(\alpha_Z, \mathbf{V}_l(\lambda u)\big)^{-a_l} \\ = \lambda^{-\sum_j (s_j d_j + t_j d_j^*)} \prod_j V_j^{(0)}(u)^{-s_j} V_j^{*(c_j)}(u)^{-t_j} \prod_l f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big)^{-a_l}.$$

- *The scale and the stand-in.* With $\alpha_B^{\mathbf{m}_B - \mathbf{1}} d\alpha_B = \lambda^{m_B - 1} u^{\mathbf{m}_B - \mathbf{1}}\, d\lambda\, d\omega_B(u)$, the power of λ is $\lambda^{\kappa_B - 1}$. At fixed u substitute $\alpha_g = \lambda \gamma_B^\rho(u)$, so that $\lambda^{\kappa_B - 1} d\lambda = \gamma_B^\rho(u)^{-\kappa_B} \alpha_g^{\kappa_B - 1} d\alpha_g$. In the coordinates (u, α_g, α_Z) the form ω is then $\phi_B^\rho(du)$ times $\omega_{\boldsymbol{\varrho}_B^\rho(u)}$ (27.5), and $h = \alpha_g + |\alpha_Z|$.
- *The integral.* At fixed u the map $(\lambda, \alpha_Z) \mapsto (\alpha_g, \alpha_Z)$ intertwines the scalings of the two orthants, so $\omega_{\boldsymbol{\varrho}_B^\rho(u)}$ is homogeneous of degree zero together with ω. By 16.2, the integral of a form homogeneous of degree zero over a section does not depend on the section. Tonelli's theorem, applied to the nonnegative integrand, then gives

$$\int_{\{h=1\}} \omega = \int_{\Delta_B} \phi_B^\rho(du) \int_{\{\alpha_g + |\alpha_Z| = 1\}} \omega_{\boldsymbol{\varrho}_B^\rho(u)} \\ = \int \Big(\int_{\{h'=1\}} \omega_{\mathbf{x}} \Big)\, d\big((\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho\big)(\mathbf{x}).$$

This is the reference's proof (proof of lem:absorbwidth). There, linearity of the f_l enters only through the identity $f_l(\alpha_Z, \mathbf{V}_l(\lambda u)) = V_l(\lambda u)\, f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))$, which homogeneity of degree one gives. □

Remark. At width one with factoring weights, normalized as in 27.10, the seam letters of K_j are $(K_{\mathcal{B}_j/R}, K_{\mathcal{B}_j|B})$ and those of Ψ_j are $(\Psi_{\mathcal{B}_j/R}, \Psi_{\mathcal{B}_j|B})$ (Lemma S11). The positive continuous functions homogeneous of degree one in the second argument are, for K_j, $f_l(\alpha_Z, (y_0, y_1)) = y_0\, f(\alpha_Z, y_1/y_0)$, and for Ψ_j, $f_l(\alpha_Z, (y_0, y_1)) = y_1\, f(\alpha_Z, y_0/y_1)$, with f positive and continuous on $(0,\infty)^Z \times (0,\infty)$. At the seam letters they are $K_{\mathcal{B}_j/R}\, f(\alpha_Z, \gamma_{j,B}^K)$ and $\Psi_{\mathcal{B}_j|B}\, f(\alpha_Z, \gamma_{j,B}^\Psi)$. These are the factors $L_i\, f_i(\alpha_Z, \gamma_i)$ of 22.2 for the cell family F_B^+, with arbitrary f_i. Linear functions f_l give only the f_i that are affine in the hinge value; homogeneity of degree one gives all of them.

Lemma S11. *Let a bunch B have width one in table j of a family, the weights factoring across B and R, $\theta_{j,T} = \theta'_{j,T\cap B}\, \theta''_{j,T\cap R}$, with the stations D_0 and D_1 of the excesses 0 and 1 (27.3(ii)). In this statement write $\hat\theta_i$ for the largest value of θ''_j on the room sides of excess i, and read $\mathcal{B}_j/R$ and $\mathcal{B}_j|B$ with the weights θ'_j and $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ with the weights θ''_j.*

(i) $K_{j,(D_0)} = \hat\theta_0 K_{\mathcal{B}_j/R}$, $\Psi_{j,(D_0)} = \hat\theta_0 \Psi_{\mathcal{B}_j/R}$, $K_{j,(D_1)} = \hat\theta_1 K_{\mathcal{B}_j|B}$ *and* $\Psi_{j,(D_1)} = \hat\theta_1 \Psi_{\mathcal{B}_j|B}$; *and* $\mathsf{y}_{jD_0} = K_{\mathcal{B}_j|R}/\hat\theta_0$, $\mathsf{y}^*_{jD_0} = \Psi_{\mathcal{B}_j|R}/\hat\theta_0$, $\mathsf{y}_{jD_1} = K_{\mathcal{B}_j/B}/\hat\theta_1$ *and* $\mathsf{y}^*_{jD_1} = \Psi_{\mathcal{B}_j/B}/\hat\theta_1$.

(ii) For $i = 0, 1$, divide θ''_j by $\hat\theta_i$ on the room sides of excess i and multiply θ'_j by $\hat\theta_i$ on the patterns of excess i. This gives a factorization of the same weights with $\hat\theta_0 = \hat\theta_1 = 1$, in which θ''_j is

the system of the multipliers of the room sides (27.1). The seam letters and the room station letters do not depend on the factorization. In the normalized factorization

$$\gamma^K_{j,B} = \frac{K_{j,(D_1)}}{K_{j,(D_0)}}, \qquad \gamma^\Psi_{j,B} = \frac{\Psi_{j,(D_0)}}{\Psi_{j,(D_1)}}, \qquad \gamma^K_{j,R} = \frac{\mathsf{y}_{jD_0}}{\mathsf{y}_{jD_1}}, \qquad \gamma^\Psi_{j,R} = \frac{\mathsf{y}^*_{jD_1}}{\mathsf{y}^*_{jD_0}}.$$

In a factorization with the constants $\hat\theta_0$ and $\hat\theta_1$, the four ratios on the right are $(\hat\theta_1/\hat\theta_0)\gamma^K_{j,B}$, $(\hat\theta_0/\hat\theta_1)\gamma^\Psi_{j,B}$, $(\hat\theta_1/\hat\theta_0)\gamma^K_{j,R}$ and $(\hat\theta_0/\hat\theta_1)\gamma^\Psi_{j,R}$.

Proof. (i) By 18.6 the bases of table j are the $X \sqcup Y$ with $X \in \mathcal{B}_j|B$ and $Y \in \mathcal{B}_j/B$, at the level $r_j(B)$, and with $X \in \mathcal{B}_j/R$ and $Y \in \mathcal{B}_j|R$, at the level $d_j(B)$.

- So the patterns and the room sides of excess 0 are the bases of $\mathcal{B}_j/R$ and of $\mathcal{B}_j|R$, those of excess 1 the bases of $\mathcal{B}_j|B$ and of $\mathcal{B}_j/B$, and every pattern meets every room side of its excess.
- The room child at a room side X of excess 0 (27.1) is the table on B with the bases of $\mathcal{B}_j/R$ and the weights $\theta'_{j,p}\,\theta''_{j,X}$, that is $\theta''_{j,X}$ times $\mathcal{B}_j/R$ with the weights θ'_j.
- These room children are positive multiples of one another, so the room sides of excess 0 form the one station D_0. Its multipliers are $w'_X = \theta''_{j,X}/\hat\theta_0$, the largest equal to 1, and the room child of the station is $\hat\theta_0$ times $\mathcal{B}_j/R$ with θ'_j.
- Hence $K_{j,(D_0)} = \hat\theta_0 K_{\mathcal{B}_j/R}$ and $\Psi_{j,(D_0)} = \hat\theta_0 \Psi_{\mathcal{B}_j/R}$, and the station letters are $\mathsf{y}_{jD_0} = \sum_X w'_X \alpha^X_R = K_{\mathcal{B}_j|R}/\hat\theta_0$ and $\mathsf{y}^*_{jD_0} = \sum_X w'_X \alpha^{R\setminus X}_R = \Psi_{\mathcal{B}_j|R}/\hat\theta_0$.
- The excess 1 is the same with $\mathcal{B}_j|B$, $\mathcal{B}_j/B$ and $\hat\theta_1$.

(ii) A basis has its pattern and its room side at one excess i, so the new weights multiply to $(\hat\theta_i\,\theta'_{j,p})(\theta''_{j,X}/\hat\theta_i) = \theta_{j,p\sqcup X}$. The new θ''_j has the largest value 1 on each excess, so it is the system of the multipliers w'_X. The stations, the multipliers, the seam letters and the station letters are defined from the weights θ_j alone (27.1), so they do not change. In the normalized factorization, (i) holds with $\hat\theta_0 = \hat\theta_1 = 1$, and the four ratios are the hinge values of 18.4, read with the minors carrying the normalized weights. In a factorization with the constants $\hat\theta_0$ and $\hat\theta_1$, (i) gives the four ratios as stated. □

Proposition 27.10 (Width one is the hinge). *As stated: for a family, with θ''_j normalized on each excess, and with part (iv) for the family.*

Proof. The letters. With θ''_j of largest value 1 on the room sides of each excess, Lemma S11 gives the seam letters of the piece of the bunch and the room letters of its room, read as its partner (the paragraph after 27.4):

$$V_{jD_0} = K_{\mathcal{B}_j/R}, \quad V_{jD_1} = K_{\mathcal{B}_j|B}, \quad V^*_{jD_0} = \Psi_{\mathcal{B}_j/R}, \quad V^*_{jD_1} = \Psi_{\mathcal{B}_j|B},$$

$$L_{jD_0} = K_{\mathcal{B}_j|R}, \quad L_{jD_1} = K_{\mathcal{B}_j/B}, \quad L^*_{jD_0} = \Psi_{\mathcal{B}_j|R}, \quad L^*_{jD_1} = \Psi_{\mathcal{B}_j/B},$$

the minors on B carrying the weights θ'_j and those on R the weights θ''_j. The hinge values of 18.4 are $\gamma^K_{j,B} = V_{jD_1}/V_{jD_0}$, $\gamma^\Psi_{j,B} = V^*_{jD_0}/V^*_{jD_1}$, $\gamma^K_{j,R} = L_{jD_0}/L_{jD_1}$ and $\gamma^\Psi_{j,R} = L^*_{jD_1}/L^*_{jD_0}$. Every letter is active and $c_j = 1$, so $\kappa_B + \kappa_B^{\text{top}} = \sum_j (s_j + t_j) = 2\sigma$, and on the balance wall of the glue $\kappa_B + \kappa_R = 2\sigma$.

The cells.

- The cell of the bunch $\mathcal{B}^+_{j,B}$ has the bases of $\mathcal{B}_j|B$ avoiding g and the $X+g$, $X\in\mathcal{B}_j/R$, through g, with the weights θ'_j (18.6). So $\mathcal{B}^+_{j,B}\setminus g=\mathcal{B}_j|B$ and $\mathcal{B}^+_{j,B}/g=\mathcal{B}_j/R$, its hinge values are $\gamma^K_{j,B}$ and $\gamma^\Psi_{j,B}$ (20.1), and its rank is $d_j(B)+1$.
- Since $d^*_j(B)=|B|-d_j(B)-1$, the hinge wall of F^+_B (22.1) is $m_B-\sum_j\big(s_jd_j(B)+t_j(|B|-d_j(B)-1)\big)=\kappa_B$.
- Likewise $\mathcal{B}^+_{j,R}\setminus g=\mathcal{B}_j|R$ and $\mathcal{B}^+_{j,R}/g=\mathcal{B}_j/B$, with the weights θ''_j and the hinge values $\gamma^K_{j,R}$ and $\gamma^\Psi_{j,R}$.
- The hinge wall of F^+_R is $m_R-\sum_j\big(s_j(r(\mathcal{B}_j|R)-1)+t_j(|R|-r(\mathcal{B}_j|R))\big)$. This is the wall κ_R of the partner, whose letters at D_0 have the degrees $r(\mathcal{B}_j|R)=e_j+1$ and $|R|-r(\mathcal{B}_j|R)=e^*_j$ (27.4).

(i) By 27.5,

$$\gamma^\rho_B=\prod_j\Big(\frac{V_{jD_1}}{V_{jD_0}}\Big)^{s_j/(2\sigma)}\Big(\frac{V^*_{jD_0}}{V^*_{jD_1}}\Big)^{t_j/(2\sigma)}=\prod_j(\gamma^K_{j,B})^{s_j/(2\sigma)}(\gamma^\Psi_{j,B})^{t_j/(2\sigma)}.$$

As $\sum_j(s_j+t_j)/(2\sigma)=1$, this is $\gamma^K_{1,B}\prod_j(\gamma^K_{j,B}/\gamma^K_{1,B})^{s_j/(2\sigma)}(\gamma^\Psi_{j,B}/\gamma^K_{1,B})^{t_j/(2\sigma)}=\gamma^K_{1,B}\,\psi(\boldsymbol{\varrho}_{F^+_B})$, the ratio vector of F^+_B being $(\gamma^K_{j,B}/\gamma^K_{1,B},\gamma^\Psi_{j,B}/\gamma^K_{1,B})_j$ (22.1). The scale form is

$$\phi^\rho_B(du)=u^{\mathbf{m}_B-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{B}_j/R}\Psi^{-t_j}_{\mathcal{B}_j|B}\big(\gamma^K_{1,B}\psi\big)^{-\kappa_B}d\omega_B(u)=\psi^{-\kappa_B}\phi_{F^+_B}(du),$$

by the base form of 22.1, whose letters are $K_{\mathcal{B}^+_{j,B}/g}=K_{\mathcal{B}_j/R}$ and $\Psi_{\mathcal{B}^+_{j,B}\setminus g}=\Psi_{\mathcal{B}_j|B}$, with the hinge wall κ_B.

(ii) By 27.5, $\varrho^\rho_{jD_0}=V_{jD_0}/V^{(0)}_j=1$ and $\varrho^{\rho*}_{jD_1}=V^*_{jD_1}/V^{*(1)}_j=1$. Further $\varrho^\rho_{jD_1}=V_{jD_1}/(V_{jD_0}\gamma^\rho_B)=\gamma^K_{j,B}/(\gamma^K_{1,B}\psi)=x_{j,K}/\psi(\mathbf{x})$ and $\varrho^{\rho*}_{jD_0}=V^*_{jD_0}/(V^*_{jD_1}\gamma^\rho_B)=x_{j,\Psi}/\psi(\mathbf{x})$, at $\mathbf{x}=\boldsymbol{\varrho}_{F^+_B}$. Write $\Phi(\mathbf{x})=\big(x_{j,K}/\psi(\mathbf{x}),\,x_{j,\Psi}/\psi(\mathbf{x})\big)_j$.

- The seam torus is the set of the points with $x_{jD_0}=x^*_{jD_1}=1$ and $\prod_j x^{s_j}_{jD_1}x^{*\,t_j}_{jD_0}=1$. Since $\prod_j(x_{j,K}/\psi)^{s_j}(x_{j,\Psi}/\psi)^{t_j}=\psi^{2\sigma}\psi^{-2\sigma}=1$, Φ maps $(0,\infty)^{2N-1}$ into it.
- Let a point of the torus have $x_{jD_1}=a_j$ and $x^*_{jD_0}=b_j$. The vector $\mathbf{x}$ with $x_{j,K}=a_j/a_1$ and $x_{j,\Psi}=b_j/a_1$ has $\psi(\mathbf{x})=\big(\prod_j a^{s_j}_jb^{t_j}_j\big)^{1/(2\sigma)}/a_1=1/a_1$, so $\Phi(\mathbf{x})$ is the point.
- Conversely $\Phi(\mathbf{x})$ determines $\psi(\mathbf{x})$ as the inverse of its coordinate $x_{1,K}/\psi(\mathbf{x})$, and then $\mathbf{x}$. So Φ is a bijection.

As $\boldsymbol{\varrho}^\rho_B=\Phi\circ\boldsymbol{\varrho}_{F^+_B}$ and $\phi^\rho_B=\psi(\boldsymbol{\varrho}_{F^+_B})^{-\kappa_B}\phi_{F^+_B}$, the law $(\boldsymbol{\varrho}^\rho_B)_*\phi^\rho_B$ is the image under Φ of $\psi^{-\kappa_B}$ times $(\boldsymbol{\varrho}_{F^+_B})_*\phi_{F^+_B}$.

(iii) The partner with its stand-in at $\Phi(\mathbf{x})$ (27.9(i)) has the letters

$$\sum_D x_{jD}\,\alpha_g^{i(D)}L_{jD}=K_{\mathcal{B}_j|R}+\frac{x_{j,K}}{\psi(\mathbf{x})}\,\alpha_g\,K_{\mathcal{B}_j/B},$$
$$\sum_D x^*_{jD}\,\alpha_g^{1-i(D)}L^*_{jD}=\frac{x_{j,\Psi}}{\psi(\mathbf{x})}\,\alpha_g\,\Psi_{\mathcal{B}_j|R}+\Psi_{\mathcal{B}_j/B}.$$

After $\alpha_g\mapsto\psi(\mathbf{x})\,\alpha_g$ these are $K_{\mathcal{B}_j|R}+x_{j,K}\,\alpha_gK_{\mathcal{B}_j/B}$ and $x_{j,\Psi}\,\alpha_g\Psi_{\mathcal{B}_j|R}+\Psi_{\mathcal{B}_j/B}$. These are the letters of $\mathcal{B}^+_{j,R}$ twisted at g by $x_{j,K}$ in K_j and by $x_{j,\Psi}$ in Ψ_j: by 21.1, the twist multiplies the

weights in K_j of the bases through g and the weights in Ψ_j of the bases avoiding g. The mass κ_B on g turns $\alpha_g^{\kappa_B-1}d\alpha_g$ into $\psi(\mathbf{x})^{\kappa_B}\alpha_g^{\kappa_B-1}d\alpha_g$, which cancels the density of (ii). The functions of 27.8, homogeneous of degree one in the vector (y_0, y_1) of the seam letters of a letter, are $y_0\, f(\alpha_Z, y_1/y_0)$ for K_j and $y_1\, f(\alpha_Z, y_0/y_1)$ for Ψ_j, with f positive and continuous. At the seam letters they are $K_{\mathcal{B}_j/R}\, f(\alpha_Z, \gamma^K_{j,B})$ and $\Psi_{\mathcal{B}_j|B}\, f(\alpha_Z, \gamma^\Psi_{j,B})$, the factors $L_i\, f_i(\alpha_Z, \gamma_i)$ of 22.2 for F_B^+. At $\mathbf{x}_l(\alpha_g)$, after the change of variable, they are $f(\alpha_Z, x_{j,K}\alpha_g)$ and $f(\alpha_Z, x_{j,\Psi}\alpha_g)$, the factors of $\omega_{\mathbf{x}}$ in 22.2. So the two lemmas are one, read through Φ.

(iv) *The room.* Read with the reversed excesses, the station D_1 has the partner excess 0 and D_0 the partner excess 1. By 27.5 for the partner:

- its scale is

$$\prod_j \Big(\frac{L_{jD_0}}{L_{jD_1}}\Big)^{s_j/(2\sigma)} \Big(\frac{L^*_{jD_1}}{L^*_{jD_0}}\Big)^{t_j/(2\sigma)} = \prod_j (\gamma^K_{j,R})^{s_j/(2\sigma)} (\gamma^\Psi_{j,R})^{t_j/(2\sigma)} = \gamma^K_{1,R}\, \psi(\boldsymbol{\varrho}_{F_R^+}),$$

  as in (i);

- its scale form is $v^{\mathbf{m}_R-\mathbf{1}} \prod_j K^{-s_j}_{\mathcal{B}_j/B} \Psi^{-t_j}_{\mathcal{B}_j|R} (\gamma^K_{1,R}\psi)^{-\kappa_R} d\omega_R(v) = \psi^{-\kappa_R}\phi_{F_R^+}$, since the base form of F_R^+ has the letters $K_{\mathcal{B}^+_{j,R}/g} = K_{\mathcal{B}_j/B}$ and $\Psi_{\mathcal{B}^+_{j,R}\setminus g} = \Psi_{\mathcal{B}_j|R}$ and the hinge wall κ_R;

- with γ_R its scale, its seam vector has $\varrho^\rho_{jD_1} = \varrho^{\rho*}_{jD_0} = 1$, $\varrho^\rho_{jD_0} = L_{jD_0}/(L_{jD_1}\gamma_R) = x'_{j,K}/\psi(\mathbf{x}')$ and $\varrho^{\rho*}_{jD_1} = L^*_{jD_1}/(L^*_{jD_0}\gamma_R) = x'_{j,\Psi}/\psi(\mathbf{x}')$ at $\mathbf{x}' = \boldsymbol{\varrho}_{F_R^+}$;

- its law follows as in (ii).

The coordinate ratios are $x_{j,\Psi}/x_{j,K} = \gamma^\Psi_{j,B}/\gamma^K_{j,B} = \varrho_{j,B}$ and $x'_{j,\Psi}/x'_{j,K} = \varrho_{j,R}$ (18.4).

The kernel. At $c_j = 1$ the configuration $t_{\mathcal{D}}$ has the column $\mathbf{e}_{K_j}$ at (j, D_0) and $\mathbf{e}_g + \mathbf{e}_{K_j}$ at (j, D_1) for K_j, and $\mathbf{e}_g + \mathbf{e}_{\Psi_j}$ at D_0 and $\mathbf{e}_{\Psi_j}$ at D_1 for Ψ_j. The display of 27.9(ii) gives

$$\Gamma(\kappa_B)\, \mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{p}\circ\mathbf{p}') = \int_0^\infty v^{\kappa_B-1} \prod_j \big(p_{jD_0}p'_{jD_0} + p_{jD_1}p'_{jD_1}v\big)^{-s_j} \big(p^*_{jD_0}p'^*_{jD_0}v + p^*_{jD_1}p'^*_{jD_1}\big)^{-t_j} dv,$$

and the coordinates of (ii) and of the partner give the first line of (iv). Substitute $v = (\psi(\mathbf{x})/\psi(\mathbf{x}'))\, w$. Each factor $x_{j,K}v/\psi(\mathbf{x}) + x'_{j,K}/\psi(\mathbf{x}')$ becomes $(x_{j,K}w + x'_{j,K})/\psi(\mathbf{x}')$, and likewise for Ψ_j. Since $\sum_j (s_j + t_j) - \kappa_B = \kappa_R$, the integral is

$$\Big(\frac{\psi(\mathbf{x})}{\psi(\mathbf{x}')}\Big)^{\kappa_B} \psi(\mathbf{x}')^{\sum_j(s_j+t_j)} \int_0^\infty w^{\kappa_B-1} \prod_j (x_{j,K}w + x'_{j,K})^{-s_j} (x_{j,\Psi}w + x'_{j,\Psi})^{-t_j} dw = \psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}\, P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big).$$

The last step is the period of the twisted core of 22.11(iii) with the core $\binom{\{g,g'\}}{1}$ in every table, read on the section $\alpha_{g'} = 1$ (16.2); its letters are $x_{j,K}\alpha_g + x'_{j,K}\alpha_{g'}$ and $x_{j,\Psi}\alpha_g + x'_{j,\Psi}\alpha_{g'}$ (21.1).

The reductions.

- 27.9(i) is part (iii).

- 27.9(ii): insert the laws of (ii) and of the partner. The densities $\psi(\mathbf{x})^{-\kappa_B}$ and $\psi(\mathbf{x}')^{-\kappa_R}$ cancel the factor of the kernel, and the period becomes

$$\iint P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big)\, d\big((\boldsymbol{\varrho}_{F_B^+})_*\phi_{F_B^+}\big)(\mathbf{x})\, d\big((\boldsymbol{\varrho}_{F_R^+})_*\phi_{F_R^+}\big)(\mathbf{x}').$$

This is 22.11(iii) for the family whose table j is $\binom{\{g,g'\}}{1}\langle \mathcal{B}^+_{j,B}, \mathcal{B}^+_{j,R}\rangle$, which is the table $(\mathcal{B}_j, \theta_j)$ by 20.3(iii).

- 27.9(iii): on the pins the seam word of the piece is $\prod_j K_{\mathcal{B}_j/R}^{-a_{jD_0}} K_{\mathcal{B}_j|B}^{-a_{jD_1}} \Psi_{\mathcal{B}_j/R}^{-b_{jD_0}} \Psi_{\mathcal{B}_j|B}^{-b_{jD_1}}$. This is the word of the bunch family $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ of 18.13 at the level shares $(a^d_j, a^r_j, b^d_j, b^r_j) = (a_{jD_0}, a_{jD_1}, b_{jD_0}, b_{jD_1})$, and the room word is that of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$.
  - The seam functions on the pins are the periods of these words (27.7), and

$$\Gamma_{\binom{2}{1}}\begin{bmatrix} a_{jD_0},\ a_{jD_1} \\ s_j \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} b_{jD_0},\ b_{jD_1} \\ t_j \end{bmatrix} = \Gamma_{\binom{4}{2}}\begin{bmatrix} a^d_j,\ b^d_j,\ a^r_j,\ b^r_j \\ s_j,\ t_j \end{bmatrix}.$$

  - The scale pin $\sum_j (a_{jD_1} + b_{jD_0}) = \kappa_B$ is the pin of 18.13.
  - In the notation of 4.1, eliminating the letter totals leaves the shares $(a^d_j, b^d_j)_j$ with the one pin $\sum_j (b^d_j - a^d_j) = \kappa_B - \sum_j s_j$, with the same power of 2π.
  - In the pair coordinates $\sigma^d_j = \frac{1}{2}(a^d_j + b^d_j)$ and $\delta^d_j = \frac{1}{2}(a^d_j - b^d_j)$, Lebesgue measure gains the factor 2^N, and the pin becomes $-2\sum_j \delta^d_j$, whose delta is $\frac{1}{2}$ that of $\sum_j \delta^d_j$.

  So the fibre integral of 27.9(iii) is 2^{N-1} times the fibre integral in the pair coordinates, which is 18.13.

□

Remark. The constants match exactly under the normalization. In a factorization with the constants $\hat\theta_{0j}, \hat\theta_{1j}$ of Lemma S11, the scale of the piece, its seam vector and its law are unchanged, since they are defined from the seam letters, while the hinge values move. Then (i) holds up to the constant factor $\prod_j (\hat\theta_{1j}/\hat\theta_{0j})^{\delta_j/\sigma}$, and the coordinate $\varrho^\rho_{jD_1}$ of (ii) is $(\hat\theta_{1j}/\hat\theta_{0j}) \prod_l (\hat\theta_{1l}/\hat\theta_{0l})^{-\delta_l/\sigma}$ times $x_{j,K}/\psi(\mathbf{x})$. Normalizing θ''_j on each excess removes these constants and costs no generality, by Lemma S11(ii). For one table, $\mathbf{x} = (1, \varrho_B)$ and $\mathbf{x}' = (1, \varrho_R)$. The family $F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]$ is then the list $\binom{\{g,g'\}}{1}$ twisted in its cobasis letter by ϱ_B at g and by ϱ_R at g', and $\psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R} = \varrho_B^{\kappa_B t/(s+t)} \varrho_R^{\kappa_R t/(s+t)}$. So (iv) is the identity with 22.3, and the reductions are 22.4(i) and (iv) together with 18.13.

Proposition 27.14 (The Gramma of a set of pairs in the Foreman's order). *As stated, for a set of pairs with any number of parts.*

Proof. The configuration $\partial\mathfrak{R}$ of 27.12 has at the pair (C,D) the column $\mathbf{e}_C + \mathbf{e}_D$ on the kept stations, the station not kept in each part read as 0. A pair of $\mathfrak{R}_l$ has both its stations in $\mathcal{K}_l \sqcup \mathcal{D}_l$, so its column is supported on the kept stations of $\mathfrak{R}_l$, and different parts have different kept stations. By 2.5, applied to one part after another, the Gramma of $\mathfrak{R}$ is the product of the Grammas of its parts,

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h}) = \prod_{l=1}^{\nu} \mathsf{G}_{(\mathbf{a}_{\mathcal{K}_l}, \mathbf{a}'_{\mathcal{D}_l});\partial\mathfrak{R}_l}(\mathbf{h}),$$

and it converges exactly when every factor does, its domain being the product of their domains. For one part the statement is the reference, proof of prop:relgrammaforeman. There the scales of $\mathcal{K}_l$ are integrated first, $\int_0^\infty x^{a_C-1} e^{-xL_C(v)}dx = \Gamma(a_C) L_C(v)^{-a_C}$. The form left on the stations of $\mathcal{D}_l$ is homogeneous of degree $|\mathbf{a}'_{\mathcal{D}_l}| - |\mathcal{D}_l| - |\mathbf{a}_{\mathcal{K}_l}| = -|\mathcal{D}_l|$, so its integral over a chart is its integral over $\Delta_{\mathcal{D}_l}$ (16.2), and the convergence criterion is 24.12. Multiplying over the parts gives the statement. □

Proposition 27.15 (Closing the Gramma of a set of pairs). *As stated: any number of parts, and (ii) at real and complex loads.*

Proof. (i) On each part the reference's proof (proof of prop:relgrammaclose(i)) applies on the simplex $\Delta_{\mathcal{D}_l}$. On its interior $0 \le \bar{L}_C < 1$ almost everywhere, and $L_C^{-a_C} = M_C^{-a_C} \sum_n \frac{(a_C)_n}{n!} \bar{L}_C^n$ with nonnegative terms. On the simplex $\bar{L}_C$ is a linear function with nonnegative coefficients, and Tonelli's theorem with 3.11 gives the series. By 27.14 the Gramma is $\prod_C \Gamma(a_C)$ times the product of these integrals over the parts.

(ii) The exchange relation of 2.3(ii) holds at every index of the domain, real or complex.

- At a kept station of $\mathcal{K}$ or of $\mathcal{D}$ it is the stated relation, the column of (C, D) being $\mathbf{e}_C + \mathbf{e}_D$ with the coefficient h_{CD}.
- At the station not kept of a part it is the sum of the relations at the stations of $\mathcal{K}$ of that part less those at its other stations of $\mathcal{D}$, the totals of a part being equal (the reference, proof of prop:relgrammaclose(ii)).
- A shifted load $(\mathbf{a} + \mathbf{e}_C, \mathbf{a}' + \mathbf{e}_{D'})$ with $(C, D') \in \mathfrak{R}$ keeps the totals of every part equal, since C and D' lie in one part, and it lies in the domain with $(\mathbf{a}, \mathbf{a}')$ (2.3).

Neither (i) nor (ii) uses one part, and (ii) does not use real loads.

(iii) concerns one part and is the reference, proof of prop:relgrammaclose(iii). □

Theorem 27.18 (The law is what a piece retains, at every width). *As stated; the plain tinkers suffice under the stated condition on the tables with both letters active.*

Proof. The equivalence of (a)–(d) is the reference, proof of thm:retainwidth. Its step (c)⇒(d) reads as follows in the notation of 4.1.

The step (c)⇒(d). Let V be the real space of the directions of the pins of the seam (27.7): the vectors $\boldsymbol{\vartheta} = (\vartheta_{jD}, \vartheta^*_{jD})_{j,D}$, one coordinate for each coordinate of the seam vector, with

$$\sum_D \vartheta_{jD} = 0 \text{ for every active } K_j, \qquad \sum_D \vartheta^*_{jD} = 0 \text{ for every active } \Psi_j,$$
$$\sum_{j,D} \big(i(D)\vartheta_{jD} + (c_j - i(D))\vartheta^*_{jD} \big) = 0.$$

For a tinker $\mathbf{v}$ of $\mathcal{Y}_0$ with the logarithms $\boldsymbol{\ell} = \log \mathbf{v}$, the seam word of $\mathcal{Y}_{0,\mathbf{v}}$ is $\mathbf{v}^{-(\mathbf{a},\mathbf{b})}$ times that of $\mathcal{Y}_0$. So, by 27.7 and by 27.9(iii) on the contour through $\mathbf{c}$,

$$e^{\langle \mathbf{c}, \boldsymbol{\ell} \rangle} P_{\mathcal{X} \sqcup \mathcal{Y}_{0,\mathbf{v}}} = (2\pi)^{-n_{\mathcal{D}}} \int_V e^{-i\langle \boldsymbol{\ell}, \boldsymbol{\vartheta} \rangle} f(\boldsymbol{\vartheta})\, d\boldsymbol{\vartheta},$$

where $f(\boldsymbol{\vartheta})$ is the product of the stacked Gammas of 27.9(iii) and of the seam functions of $\mathcal{X}$ and of $\mathcal{Y}_0$ at $\mathbf{c} + i\boldsymbol{\vartheta}$. The stacked Gammas have no zero, and f is continuous and integrable on V (the reference, proof of thm:retainwidth). The proof uses (c) only to know the Fourier transform of f at every point of the dual of V; then f is determined. The rest is unchanged: the stacked Gammas have no zero, the seam function of $\mathcal{Y}_0$ is positive at $\mathbf{c}$, the identity theorem applies, and the projection is injective on the seam torus.

The plain tinkers. Their logarithms form the space Λ of the $\boldsymbol{\ell}$ with $\ell_{jD} = \ell^*_{jD}$ for every table j with both letters active (27.17). A table with one active letter has one coordinate for each station, and on it a plain tinker is a general one. The restrictions to V of the functionals $\boldsymbol{\vartheta} \mapsto \langle \boldsymbol{\ell}, \boldsymbol{\vartheta} \rangle$, $\boldsymbol{\ell} \in \Lambda$, exhaust the dual of V exactly when no nonzero $\boldsymbol{\vartheta} \in V$ is orthogonal to Λ.

No orthogonal vector. Let $\boldsymbol{\vartheta} \in V$ be orthogonal to Λ.

- Then $\vartheta_{jD} = 0$ in a table with only K_j active, $\vartheta^*_{jD} = 0$ in a table with only Ψ_j active, and $\vartheta^*_{jD} = -\vartheta_{jD}$ in a table with both letters active.
- The letter totals give $\sum_D \vartheta_{jD} = 0$ in every table with both letters active. The scale pin becomes $\sum_{j,D} \big(2i(D) - c_j\big)\vartheta_{jD} = 0$, the sum running over the tables with both letters active.
- A table with $|\mathcal{D}_j| = 1$ has $\vartheta_{jD} = 0$ by its total.
- The exceptional table has $c_j \geq 1$, and $\mathcal{D}_{j,0}$ and $\mathcal{D}_{j,c_j}$ are nonempty (27.4), so its two stations D' and D'' have the excesses 0 and c_j. Its total gives $\vartheta_{jD'} = -\vartheta_{jD''} = \tau$, and the scale pin reads $-c_j\tau - c_j\tau = 0$, so $\tau = 0$.

Hence $\boldsymbol{\vartheta} = 0$, and the plain tinkers suffice in (c). □

Remark. In general, let J_2 be the tables with both letters active. The orthogonal of Λ in V has the dimension $\sum_{j\in J_2}(|\mathcal{D}_j| - 1)$, less one when some table of J_2 has positive width. Indeed, the vectors with $\sum_D \vartheta_{jD} = 0$ on the tables of J_2 and zero elsewhere form a space of the first dimension, and on it the scale pin $\sum_{j,D}(2i(D) - c_j)\vartheta_{jD}$ vanishes identically exactly when every table of J_2 has width zero. With every letter active and some table of positive width, $n_{\mathcal{D}} = 2|\mathcal{D}| - 2N - 1$ and the orthogonal has the dimension $|\mathcal{D}| - N - 1$, so the plain tinkers of one partner move $|\mathcal{D}| - N$ of the directions of the seam torus, as the paragraph after the theorem says.

Theorem 27.22 (A bunch is stored on one mark exactly at width one). *As stated, the clause at width one read with the normalization of 27.10.*

Proof. Everything except the clause at width one is the reference, proof of thm:storeone. At width one with factoring weights, Lemma S11(i) writes the letters of the partner with its stand-in at a seam point $\mathbf{x}$, where $x_{jD_0} = x^*_{jD_1} = 1$, as

$$\frac{1}{\hat\theta_0}\Big(K_{\mathcal{B}_j|R} + \frac{\hat\theta_0}{\hat\theta_1}\, x_{jD_1}\, \alpha_g\, K_{\mathcal{B}_j/B}\Big), \qquad \frac{1}{\hat\theta_1}\Big(\frac{\hat\theta_1}{\hat\theta_0}\, x^*_{jD_0}\, \alpha_g\, \Psi_{\mathcal{B}_j|R} + \Psi_{\mathcal{B}_j/B}\Big),$$

the constants $\hat\theta_0, \hat\theta_1$ being those of table j. Under the normalization of 27.10 the constants are 1. The letters are then those of $\mathcal{B}^+_{j,R}$ twisted at g by x_{jD_1} in K_j and by $x^*_{jD_0}$ in Ψ_j (18.6, 21.1). At the seam point of a value $\mathbf{y}$ of the ratio vector of F^+_B these twists are $y_{j,K}/\psi(\mathbf{y})$ and $y_{j,\Psi}/\psi(\mathbf{y})$ (27.10(ii)), and the change of variable $\alpha_g \mapsto \psi(\mathbf{y})\alpha_g$ of 27.10(iii) turns them into the twist by $\mathbf{y}$, as stated. Without the normalization the two letters carry the constant factors $1/\hat\theta_0$ and $1/\hat\theta_1$ and the twists $(\hat\theta_0/\hat\theta_1)x_{jD_1}$ and $(\hat\theta_1/\hat\theta_0)x^*_{jD_0}$. The last sentence of the clause is 27.10(i)–(iii). □

40.28 Chapter 28: composing bunches

Theorem 28.5 (Two transforms of a law, and their rules). *As stated; the first rule of (iii) holds at any number of sub-pieces.*

Proof. Parts (i), (ii) and (iv), and the second and third rules of (iii), are the reference, proof of thm:lawreadings. For the first rule:

- Let $S_1, \ldots, S_k$ be as in 28.2, and $\mathbf{p}$ a point with positive coordinates, one for each coordinate of the seam vector of S.
- By 27.20, $\mathcal{I}_S(\mathbf{p}) = \int H\, d\lambda_S$, with the law $\lambda_S = (\boldsymbol{\varrho}^\rho_S)_*\phi^\rho_S$ and the Borel function $H(\mathbf{z}) = \Gamma(\kappa_S)\, \mathsf{Z}_{(\kappa_S,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{z}\circ\mathbf{p}) \geq 0$, $\mathcal{D}$ being the stations of room sides of S.

- 28.2(ii) gives $\int H\,d\lambda_S = \int \big(\int H\,d\lambda_{\mathbf{x}}\big)\prod_l d\lambda_{S_l}(\mathbf{x}_l)$, with $\lambda_{\mathbf{x}}$ the law of the node $S^{(\mathbf{x}_1,\dots,\mathbf{x}_k)}$.
- The node is a piece with the stations $\mathcal{D}$, the exponents of the family and the wall κ_S (28.2(ii)), so by the same definition $\int H\,d\lambda_{\mathbf{x}} = \mathcal{I}_{S^{(\mathbf{x}_1,\dots,\mathbf{x}_k)}}(\mathbf{p})$.

This is the reference's argument for one sub-piece, which does not use $k = 1$. The second and third rules use the transfer matrices from one sub-piece: 28.2(i) gives linearity in the seam letters of each S_l separately, with coefficients that depend on the marks of the other sub-pieces. □

Theorem 28.7 (Cyclic chains). *As stated: (i) holds for a family, and in (ii) each letter is opened over the starting states at which its term is not zero.*

Proof. (i) Fix a table j.

- By 20.11(ii), the bases of table j are the unions of the unpaired parts of the choices of one basis in every cell $\mathcal{A}_{j,i}$ that occupy exactly one hinge of every pair. Its weight in each letter at a basis is the sum, over the choices that give it, of the products of the weights chosen.
- A pair of I_{i-1} occupied on the side of the cell $i-1$ is free on the side of the cell i, and conversely. So, with $J_i \subseteq I_i$ the pairs occupied on the side of the cell i, the basis chosen in $\mathcal{A}_{j,i}$ occupies the hinges $(I_{i-1} \setminus J_{i-1}) \sqcup J_i$.
- Its part $X_i \subseteq B_i$ is a basis of the child of the bunch of its hinges at that pattern. With their weights, these choices sum to the channel $K_{\mathcal{A}_{j,i},((I_{i-1}\setminus J_{i-1})\sqcup J_i)}(\alpha_{B_i})$ (20.10).
- A basis $\bigsqcup_i X_i$ leaves the cobasis $\bigsqcup_i (B_i \setminus X_i)$, so Ψ_j is summed the same way with the channels of Ψ.
- Summing over the sequences $(J_0,\dots,J_{k-1})$, $J_k = J_0$, gives $K_j = \operatorname{tr}\prod_i \mathbf{M}_{K_j,i}$ and $\Psi_j = \operatorname{tr}\prod_i \mathbf{M}_{\Psi_j,i}$. The entries are polynomials in α_{B_i} with nonnegative coefficients and no ε, which is the top layer 0.
- The pairs are named alike in every table, so the sets of states $\mathcal{J}_i = \{J :\ J \subseteq I_i\}$ serve every letter, as 28.6 asks.

This is the reference's proof (proof of thm:cyclicchain(i)) applied table by table.

(ii) and (iii) are the reference, proof of thm:cyclicchain. In (ii), L is the sum over the starting states J of the terms $\mathbf{e}_{(J,0)}^{\mathsf{T}} \prod_i \widetilde{\mathbf{M}}_{L,i}\,\mathbf{e}_{(J,h)}$, each a sum over the paths from $(J,0)$ to (J,h) of products of entries with nonnegative coefficients. 4.4 opens a sum of positive terms, one share for each. A starting state whose term vanishes, the state of no path, therefore carries no share and no flow. □

40.29 Chapter 29: closing a glue along a direction

Theorem 29.2 (Closing the glue along a direction). *As stated, the largest values of the monomials of the letter vector of the partner being read on its law when its room word has a Newton sum of full dimension.*

Proof. Apart from the sentence on the partner, the statement is the reference's, with its proof (the reference, proof of thm:closecone). Let the room word of the partner have a Newton sum of full dimension. The partner is a piece on R with the stations $\mathcal{D}$, read with the reversed excesses $c_j - i(D)$ (27.4); its scale γ_R^{ρ} and its seam vector $\boldsymbol{\varrho}_R^{\rho} = (\varrho_{R,jD}^{\rho}, \varrho_{R,jD}^{\rho *})_{j,D}$ are those of 27.5 with its room letters and these excesses. In this proof write $L_{j,R}^{(0)}$ for the sum of the L_{jD} over the stations

of excess c_j, which have the reversed excess 0, and $L^{*(c)}_{j,R}$ for the sum of the L^*_{jD} over the stations of excess 0, which have the reversed excess c_j. By 27.5,

$$L_{jD} = L^{(0)}_{j,R}\,(\gamma^\rho_R)^{c_j - i(D)}\,\varrho^\rho_{R,jD}, \qquad L^*_{jD} = L^{*(c)}_{j,R}\,(\gamma^\rho_R)^{i(D)}\,\varrho^{\rho*}_{R,jD}.$$

A direction $\mathbf{v} \in V$ keeps the letter totals, $\sum_D v_{jD} = 0$ and $\sum_D v^*_{jD} = 0$ for every active letter. It also keeps the pin of 27.11 for every vector of M, the pin of $\mathbf{1}$ holding by the letter totals on the balance wall, and $\mathbf{1}_R \in \vec{\mathcal{N}}^\perp_R \subseteq M$, the room letters being homogeneous. The exponents in $\mathbf{n}(\mathbf{a},\mathbf{b})$ of the monomials of V_{jD} and V^*_{jD} vanish on R, and those of the monomials of L_{jD} and L^*_{jD} have the totals $\deg L_{jD} = e_j + c_j - i(D)$ and $\deg L^*_{jD} = e^*_j + i(D)$ on R (27.4); so, by the letter totals,

$$0 = \langle \mathbf{n}(\mathbf{v}), \mathbf{1}_R \rangle = \sum_{j,D} \big(v_{jD} \deg L_{jD} + v^*_{jD} \deg L^*_{jD} \big) = \sum_{j,D} \big((c_j - i(D)) v_{jD} + i(D) v^*_{jD} \big).$$

So in $\mathbf{V}^{\mathbf{v}}_R = \prod_{j,D} L_{jD}^{v_{jD}} {L^*_{jD}}^{v^*_{jD}}$ the sums $L^{(0)}_{j,R}$ and $L^{*(c)}_{j,R}$ and the scale γ^ρ_R enter with the exponent 0, and $\mathbf{V}^{\mathbf{v}}_R = (\boldsymbol{\varrho}^\rho_R)^{\mathbf{v}}$ on the open orthant of R, where these sums are positive, both sides being homogeneous of degree zero; hence $\sup \mathbf{V}^{\mathbf{v}}_R = \sup_{v \in \Delta_R} \boldsymbol{\varrho}^\rho_R(v)^{\mathbf{v}}$, the supremum over the open simplex. The scale form ϕ^ρ_R has a positive continuous density on the open simplex and the seam vector is continuous there, so the law $(\boldsymbol{\varrho}^\rho_R)_* \phi^\rho_R$ is carried by the closure of the image of the open simplex, and every point of that image lies in its support: the preimage of a neighbourhood of an image point is a nonempty open set, of positive mass. The supremum of the continuous function $\mathbf{x}^{\mathbf{v}}$ over the image is therefore its largest value on the support of the law, in $(0,\infty]$. □

Lemma S12. *Let $L_1, \dots, L_p$ be positive letters on a finite set B (24.11), $e_0 \in B$, $e_1, \dots, e_p$ real exponents and $\mathbf{m}$ real masses on B with $m_B = \sum_\ell e_\ell \deg L_\ell$; the pairings below are taken on $B \setminus e_0$.*

(i) There are finitely many rational simplicial cones of dimension $|B| - 1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each contained, for every ℓ, in a set $\{\mathbf{v} : \langle \mathbf{t}, \mathbf{v}\rangle \le \langle \mathbf{t}', \mathbf{v}\rangle$ for every exponent $\mathbf{t}'$ of the support of $L_\ell\}$ for one exponent $\mathbf{t}$ of that support.

(ii) On such a cone π, with the primitive integral generators $\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1}$, write $x = \log \alpha_{B\setminus e_0} = -\sum_k u_k \mathbf{v}_{\pi,k}$ on the section $\alpha_{e_0} = 1$, $u_k \ge 0$, and $q_k = e^{-u_k}$. The exponent $\mathbf{t}_{\ell,\pi}$ of (i) is unique, and $L_\ell = \prod_k q_k^{\langle \mathbf{t}_{\ell,\pi}, \mathbf{v}_{\pi,k}\rangle}\, k_{\ell,\pi}(q)$ on the section, with $k_{\ell,\pi}$ a polynomial in q with nonnegative coefficients whose constant term is the coefficient of $\alpha^{\mathbf{t}_{\ell,\pi}}$ in L_ℓ.

(iii) For a generator $\mathbf{v}$ of π the number $\kappa_{\mathbf{v}} = \langle \mathbf{m} - \sum_\ell e_\ell \mathbf{t}_{\ell,\pi}, \mathbf{v}\rangle$ is the same for every cone π of which $\mathbf{v}$ is a generator, and, in $[0,\infty]$,

$$\int_{\alpha_{e_0}=1} \alpha^{\mathbf{m}-\mathbf{1}} \prod_\ell L_\ell^{-e_\ell}\, d\alpha_{B\setminus e_0} = \sum_\pi |\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \int_{[0,1]^{|B|-1}} \prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}} - 1} \prod_\ell k_{\ell,\pi}(q)^{-e_\ell}\, dq;$$

both sides are finite exactly when every $\kappa_{\mathbf{v}}$ is positive, and they are then the period of the word $\prod_\ell L_\ell^{-e_\ell}$ at $\mathbf{m}$, the integral over a section of 24.11.

Proof. (i) Identify $\mathbb{R}^{B\setminus e_0}$ with the hyperplane $v_{e_0} = 0$ of $\mathbb{R}^B$. For each ℓ the sets $C_{\ell,\mathbf{t}} = \{\mathbf{v} : \langle \mathbf{t}, \mathbf{v}\rangle \le \langle \mathbf{t}', \mathbf{v}\rangle$ for every $\mathbf{t}'\}$, $\mathbf{t}$ in the support of L_ℓ, are rational polyhedral cones, defined by the inequalities $\langle \mathbf{t} - \mathbf{t}', \mathbf{v}\rangle \le 0$: the cones of the inner normal fan of the Newton polytope of L_ℓ cut by

that hyperplane, which form a fan covering $\mathbb{R}^{B\setminus e_0}$, every $\mathbf{v}$ having an exponent of least pairing. The cones $\bigcap_\ell C_{\ell,\mathbf{t}_\ell} \cap O$, O running over the closed coordinate orthants of $\mathbb{R}^{B\setminus e_0}$, are pointed rational polyhedral cones which with their faces form a fan covering $\mathbb{R}^{B\setminus e_0}$, the common refinement of these fans and of the fan of the orthants. Order its rays and triangulate each cone by pulling: the cone is the union of the cones spanned by its first ray and by the triangulated faces that do not contain that ray. A face F of a pointed cone C has $\operatorname{span} F \cap C = F$, so the first ray of C lies off $\operatorname{span} F$ for such a face, and the cones produced are simplicial; the pulling triangulation of a cone restricts on each face to the pulling triangulation of that face, which depends on the face alone, so the triangulations of two cones agree on their common face. This gives finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each in some $\bigcap_\ell C_{\ell,\mathbf{t}_\ell}$.

(ii) If $\mathbf{t}$ and $\mathbf{t}''$ both pair least with every $\mathbf{v}_{\pi,k}$, then $\langle \mathbf{t}-\mathbf{t}'', \mathbf{v}_{\pi,k}\rangle = 0$ for every k; the generators span $\mathbb{R}^{B\setminus e_0}$, so $\mathbf{t}$ and $\mathbf{t}''$ agree on $B\setminus e_0$, and as $|\mathbf{t}| = |\mathbf{t}''| = \deg L_\ell$ they are equal. On the section $\alpha^{\mathbf{t}'} = e^{\langle \mathbf{t}', x\rangle} = \prod_k q_k^{\langle \mathbf{t}', \mathbf{v}_{\pi,k}\rangle}$, so with $L_\ell = \sum_{\mathbf{t}'} c_{\mathbf{t}'}\alpha^{\mathbf{t}'}$ and $\mathbf{t} = \mathbf{t}_{\ell,\pi}$,

$$L_\ell = \prod_k q_k^{\langle \mathbf{t}, \mathbf{v}_{\pi,k}\rangle} \sum_{\mathbf{t}'} c_{\mathbf{t}'} \prod_k q_k^{\langle \mathbf{t}'-\mathbf{t}, \mathbf{v}_{\pi,k}\rangle}.$$

Every exponent $\langle \mathbf{t}'-\mathbf{t}, \mathbf{v}_{\pi,k}\rangle$ is an integer, the exponents and the generators being integral, and it is nonnegative by the choice of $\mathbf{t}$; by the uniqueness just proved, all of them vanish only at $\mathbf{t}' = \mathbf{t}$. So the sum is a polynomial $k_{\ell,\pi}$ with nonnegative coefficients and the constant term $c_{\mathbf{t}} > 0$.

(iii) A cone π of which $\mathbf{v}$ is a generator has $\langle \mathbf{t}_{\ell,\pi}, \mathbf{v}\rangle = \min_{\mathbf{t}'}\langle \mathbf{t}', \mathbf{v}\rangle$, the minimum over the support of L_ℓ, which does not depend on π; so neither does $\kappa_{\mathbf{v}}$. On π the map $(u_k)_k \mapsto x = -\sum_k u_k \mathbf{v}_{\pi,k}$ is a linear bijection of $[0,\infty)^{|B|-1}$ onto $-\pi$ with the Jacobian $|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})|$; on the section $d\alpha_e = \alpha_e\, dx_e$ for $e \neq e_0$, $\alpha^{\mathbf{m}} = e^{\langle \mathbf{m}, x\rangle}$, and $du_k = -dq_k/q_k$. By (ii), on the image of $-\pi$, the form $\alpha^{\mathbf{m-1}} \prod_\ell L_\ell^{-e_\ell}\, d\alpha_{B\setminus e_0} = e^{\langle \mathbf{m}, x\rangle} \prod_\ell L_\ell^{-e_\ell}\, dx$ is

$$|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\langle \mathbf{m} - \sum_\ell e_\ell \mathbf{t}_{\ell,\pi}, \mathbf{v}_{\pi,k}\rangle - 1} \prod_\ell k_{\ell,\pi}(q)^{-e_\ell}\, dq.$$

The cones $-\pi$, like the cones π, cover $\mathbb{R}^{B\setminus e_0}$ with disjoint interiors, their boundaries are null and the integrands are positive, so the two sides agree in $[0,\infty]$ (Tonelli's theorem). On $[0,1]^{|B|-1}$ each $k_{\ell,\pi}$ lies between its constant term and the sum of its coefficients, so $\prod_\ell k_{\ell,\pi}^{-e_\ell}$ is bounded above and below by positive constants, and the integral over the cube is finite exactly when every exponent $\kappa_{\mathbf{v}_{\pi,k}}$ is positive; every generator is a generator of some cone. The form $\alpha^{\mathbf{m-1}} \prod_\ell L_\ell^{-e_\ell} d\alpha$ is projective, the masses lying on the balance hyperplane, and α_{e_0} is positive and homogeneous of degree one, so the integral over the section is the period (24.11, 16.2). □

Theorem 29.3 (Closing past the walls). *As stated, for every piece whose seam function converges at a real point of the pins of the seam with positive shares, with the sector form of (ii) for every fan of the seam word and with part (iv).*

Proof. The hypothesis. At a real point of the pins with positive shares the seam word $W_B(\mathbf{a}, \mathbf{b})$ is a Newton word (24.11), and the masses $\mathbf{m}_B$ lie on its balance hyperplane: by the letter totals and the scale pin,

$$\sum_{j,D} \Big(a_{jD}\big(d_j + i(D)\big) + b_{jD}\big(d_j^* + c_j - i(D)\big)\Big) = \sum_j \big(s_j d_j + t_j d_j^*\big) + \kappa_B = m_B$$

(27.4, 27.7). Its period, the seam function, converges there, so its Newton sum has the dimension $|B|-1$ (24.12).

(i). The reference proves (i) for a piece of tables (the reference, proof of thm:pastwalls(i)). Of the piece it uses only that its seam word has a Newton sum of full dimension, so that its reduced period in 27.11 is its seam function at the sums over the monomials, the seam letters repeating over the monomials of the partner; the reference obtains that fact from the connectedness of the family of tables, and here it is the hypothesis. The rest of that proof concerns the partner, whose room letters are read in their monomials, so that $\vec{\mathcal{N}}_R = 0$ and its reduced period is $\prod \xi^{-\mathbf{a}} \prod \xi^{*-\mathbf{b}}$ (27.11); with $M = \mathbb{R}\mathbf{1}_B \oplus \mathbb{R}^R$ and the lattice basis $\mathbf{1}$, $\mathbf{e}_r$ ($r \in R$), the pins of 27.11 are the displayed ones, the scale pin following by the balance.

(ii), the sector form. On the pins of the seam the exponents $\bar{a}_{jD}$, $\bar{b}_{jD}$ of the seam word are affine in the shares with rational linear parts, and the masses $\mathbf{m}_B$ lie on its balance hyperplane at every share, by the display above with the sums in place of the shares. Lemma S12, applied to the seam letters of the active letters with the exponents $\bar{a}_{jD}$, $\bar{b}_{jD}$, gives the fans of the seam word, the sector coordinates, the sector letters with their exponents, the walls and the identity at every real point of the pins, and shows that the seam function converges at a real point exactly when every wall is positive there. The coefficients $\langle \mathbf{t}_{jD,\pi}, \mathbf{v} \rangle$ and $\langle \mathbf{t}^*_{jD,\pi}, \mathbf{v} \rangle$ of the shares in a wall are integers, so the wall is affine with a rational linear part, and it does not depend on the sector (Lemma S12(iii)).

(ii), continuation and bound. On a sector the integrand is $\prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})-1} h(q, \mathbf{z})$ with $h = \prod_{j,D} k_{jD,\pi}^{-\bar{a}_{jD}} \psi_{jD,\pi}^{-\bar{b}_{jD}}$. The sector letters are at least their positive constant terms on the closed cube, so their real parts are positive on a neighbourhood of it, where their principal logarithms are holomorphic; so h is holomorphic in q near $[0,1]^{|B|-1}$ and entire in the shares. 8.6, with $x_k = \kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})$ and the shares as its parameter, continues each sector integral meromorphically to all complex shares, with its poles on the hyperplanes on which a wall of a generator of the sector lies in $-\mathbb{N}$; a wall constant on the pins is positive there, the seam function converging, and gives no hyperplane. The formula of 8.6, repeated in the $|B|-1$ coordinates, writes each term with at most $|B|-1$ factors $(\kappa_{\mathbf{v}_{\pi,k}} + l)^{-1}$, which bounds the orders of the poles by $|B|-1$. For the bound, the proof of 19.30 is run on each sector integral with the several parameters entering through the powers of the sector letters and through the walls, $|\operatorname{Re}\mathbf{z}|_1$ and $|\operatorname{Im}\mathbf{z}|_1$ in place of $|\operatorname{Re} z|$ and $|\operatorname{Im} z|$ (the reference, proof of thm:pastwalls(ii); the same argument proves 30.8(i)): each of the at most $|B|-1$ factors $(\kappa + l)^{-1}$ of a term is at most a constant times δ^{-1}, and nothing else depends on δ.

(ii), a piece of tables. Let the seam letters at the station D of the active letters be the basis and the cobasis polynomials of a table $\mathcal{B}_{jD}$ on B with positive weights, and consider the family of these tables at the exponents $\bar{a}_{jD}, \bar{b}_{jD}$ at a point with positive shares. Its potential $\sum_{j,D} \left(\bar{a}_{jD} \log K_{\mathcal{B}_{jD}}(e^y) + \bar{b}_{jD} \log \Psi_{\mathcal{B}_{jD}}(e^y) \right)$ (8.17) is affine along a direction $\mathbf{w}$ exactly when, in every letter, all the exponents of the support have one pairing with $\mathbf{w}$, a logarithm of a sum of exponentials with positive coefficients being strictly convex along every direction that separates two of its exponents: exactly when $\mathbf{w}$ is orthogonal to the directions of the Newton sum. If the family is connected, that space is $\mathbb{R}\mathbf{1}$ (8.18(iii), a space that does not involve the masses), and the Newton sum has the dimension $|B|-1$. If it is not, some nonempty proper $S \subset B$ has the width $c_{jD}(S) = 0$ in every table (8.1), so that $|T \cap S| = d_{jD}(S)$ for every basis T of $\mathcal{B}_{jD}$ (7.11). The base polytope of $\mathcal{B}_{jD}$ then lies in $\{x_S = d_{jD}(S)\}$ and that of its dual in $\{x_S = |S| - d_{jD}(S)\}$, so the Newton sum lies in a hyperplane $\{x_S = \text{const}\}$, and as $\mathbf{1}_S \notin \mathbb{R}\mathbf{1}$ its dimension is less than $|B|-1$.

In the coordinates of 8.7(ii) on the sector H_π of an order π of B, $\alpha_{\pi(1)} = \rho$ and $\alpha_{\pi(k)} = \rho q_2 \cdots q_k$, so that, with $u_k = -\log q_k$,

$$\log \alpha_{\pi(m)} - \log \alpha_{\pi(1)} = \sum_{k=2}^{m} \log q_k = -\sum_{k=2}^{|B|} u_k \, (\mathbf{1}_{U_k})_{\pi(m)},$$

because $\pi(m) \in U_k$ exactly when $k \le m$. So, modulo $\mathbf{1}$, H_π is the cone with the generators $\mathbf{1}_{U_k}$, $2 \le k \le |B|$, read on the section as $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$. These $|B|-1$ vectors form a basis of $\mathbb{Z}^B/\mathbb{Z}\mathbf{1}$, since the differences $\mathbf{1}_{U_k} - \mathbf{1}_{U_{k+1}} = \mathbf{e}_{\pi(k)}$ and $\mathbf{1}_{U_1} = \mathbf{1}$ span $\mathbb{Z}^B$; so the cone is simplicial of dimension $|B|-1$ with the determinant ± 1, and the sectors of the orders cover $\mathbb{R}^B/\mathbb{R}\mathbf{1}$ with disjoint interiors. For a basis T, $\alpha^T/\alpha^{T^+_\pi} = \prod_k q_k^{|T\cap U_k| - |T^+_\pi \cap U_k|}$, and 8.5, $\alpha^T \le \alpha^{T^+_\pi}$ on H_π for every basis, gives, with one q_k small and the others 1, $\langle \mathbf{1}_{T^+_\pi}, \mathbf{1}_{U_k}\rangle \le \langle \mathbf{1}_T, \mathbf{1}_{U_k}\rangle$ for every k. As $\mathbf{1}_T$ and $\mathbf{1}_{T^+_\pi}$ have one total, their pairings on $B \setminus e_0$ with the generator $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$ differ by the difference of their pairings with $\mathbf{1}_{U_k}$. So the sector lies in the set of (ii) for $K_{\mathcal{B}_{jD}}$ with $\mathbf{t} = \mathbf{1}_{T^+_\pi}$, and likewise for $\Psi_{\mathcal{B}_{jD}}$ with $\mathbf{t} = \mathbf{1}_{B\setminus T^-_\pi}$, the greedy bases being those of $\mathcal{B}_{jD}$; the sectors of the orders form a fan of the seam word. Its sector letters are $\sum_T \theta_{jD,T} \prod_k q_k^{|T\cap U_k| - d_{jD}(U_k)}$ and their cobasis counterparts, the sector letters of 8.7(ii). The vector $\mathbf{m}_B - \sum_{j,D}(\bar a_{jD}\mathbf{1}_{T^+} + \bar b_{jD}\mathbf{1}_{B\setminus T^-})$ has the total 0 on the pins, so its pairing with the generator of a tail U is its pairing with $\mathbf{1}_U$; by 8.5, $|T^+ \cap U| = d_{jD}(U)$ and $|U \setminus T^-| = d^*_{jD}(U)$, and the wall of that generator is $m_U - \sum_{j,D}(\bar a_{jD} d_{jD}(U) + \bar b_{jD} d^*_{jD}(U))$, the proper wall κ_U. Every nonempty proper U is a tail of some order.

(iii). The reference proves (iii) for a piece of tables (the reference, proof of thm:pastwalls(iii)), using three facts about the piece. The first is the continuation with the bound of (ii). The second is that the seam function converges at a real point exactly when its walls are positive there, so that it converges along a ray along which no wall falls, the walls staying above their positive values at $\bar{\mathbf{c}}$, and diverges past a wall that falls. The third is that at real points it is $\int \mathbf{x}^{-\mathbf{w}}\, d\big((\boldsymbol{\varrho}^\rho_B)_*\phi^\rho_B\big)$, with the scale form positive and continuous on the open simplex and the seam vector continuous there, which with 29.2 gives $g(\boldsymbol{\sigma};0) = \sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$. The first two are (ii), and the third holds for every piece (27.5, 27.7). The rest of that proof (the Gamma bounds, Stirling's formula with $\sum_l \sigma_l = 0$, the crossings and the null set) does not use the tables.

(iv). The partner is a piece on R read with the reversed excesses. The pins of its seam, the letter totals and its scale pin $\sum_{j,D}\big((c_j - i(D))a_{jD} + i(D)b_{jD}\big) = \kappa_R$, are on the balance wall $\kappa_R = \kappa_B^{\mathrm{top}}$ those of the piece (27.9(iii)), and its seam function converges at $\mathbf{c}$; so (ii) applies to the piece and to the partner. On the share side of 27.9(iii) the integrand at $\mathbf{c} + \tau\boldsymbol{\sigma} + i\boldsymbol{\vartheta}$ is $\prod_{j:\, s_j>0} \Gamma_{\frac{|\mathcal{D}_j|}{1}}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{\frac{|\mathcal{D}_j|}{1}}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}$ times the two seam functions. For $M_B > g_B(\boldsymbol{\sigma};\theta_B)$ and $M_R > g_R(\boldsymbol{\sigma};\theta_R)$ the definitions of the growths give C, n_1 and n_2 with

$$\big|P_{W_B}\, P_{W_R}\big| \le C^2\delta^{-2n_2}(1+\tau+|\boldsymbol{\vartheta}|)^{2n_1}(M_BM_R)^\tau e^{(\theta_B+\theta_R)|\boldsymbol{\vartheta}|_1}$$

for $\tau \ge 0$, δ the smaller of 1 and the distance from the polar hyperplanes of both seam functions. From here the proof of (iii) applies with $\theta = \theta_B + \theta_R < \pi/8$ and M_BM_R in place of M, the coefficients sitting inside the seam functions:

- the contour moves along times at distance at least τ^{-3} from the crossings, where δ^{-2n_2} is a power of τ, and the power of $1+|\boldsymbol{\vartheta}|$ is absorbed into $e^{\eta|\boldsymbol{\vartheta}|_1}$ with η small;
- a share whose real part x grows has $|\Gamma(x+iy)|e^{\theta|y|} \le \Gamma(x)e^{2\theta^2 x/\pi}$, and a share that falls or stays absorbs $e^{\theta|y|}$ against $e^{-\pi|y|/2}$;
- integrating over the directions of the shares, Stirling's formula and $\sum_l \sigma_l = 0$, by the letter totals, give the factor $e^{\tau(h_{\theta_B,\theta_R}(\boldsymbol{\sigma})+\epsilon)}$ as $M_B \downarrow g_B$ and $M_R \downarrow g_R$.

The ray crosses the hyperplanes of the falling shares and the hyperplanes on which a falling wall of either word lies in $-\mathbb{N}$, and each crossing adds the fibre integral of the residue of the whole integrand (29.4), a Laurent coefficient where several factors are polar. The pins and the walls of both words have rational linear parts, so crossings on parallel hyperplanes coincide or keep a fixed

positive distance. Crossings on hyperplanes that are not parallel, closer than T^{-3} near $\tau = T$, occur only for $\mathbf{c}$ in a set of measure $O(T^{-3})$, summable over T (Borel and Cantelli). The local decay (H1) of 29.4 holds because both seam functions grow at most like $e^{\theta' |\operatorname{Im} \mathbf{z}|_1}$ for every $\theta' > 0$, by (ii), against the decay of the Gammas at bounded real parts. (H2) holds at the times above when $h_{\theta_B,\theta_R}(\boldsymbol{\sigma}) < 0$, and 29.4 gives the closing with the stated bounds.

For the last sentence of (iv), let no wall of the word of a side fall along $\boldsymbol{\sigma}$. Both words have Newton sums of full dimension, their seam functions converging at $\mathbf{c}$ (24.12), so the largest value of $\mathbf{x}^{-\boldsymbol{\sigma}}$ on the support of the law of the side is $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$ of its letter vector (29.2, for the partner by its sentence on the partner). At $\theta = 0$ the proof of (iii), run on that side, gives the growth $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$, finite, and the growth does not increase with the angle, a bound with a smaller angle being a bound with a larger one. On the real ray, where $\boldsymbol{\vartheta} = 0$, the walls stay above their positive values at $\mathbf{c}$, so δ is bounded below there, and the seam function at $\mathbf{c} + \tau\boldsymbol{\sigma}$ is at least a positive constant times $e^{\tau h'}$ for every $h' < \log \sup \mathbf{V}^{-\boldsymbol{\sigma}}$ (the proof of (iii)); so every admissible M, at every angle, is at least $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$. □

Definition 29.5 (Chambers of a closing). As stated, for every fibre integral with finitely many affine polar functions.

Proof. The chamber is convex. If $\boldsymbol{\sigma}', \boldsymbol{\sigma}'' \in C(\boldsymbol{\sigma})$ and $0 \le s \le 1$, then $\partial_{s\boldsymbol{\sigma}' + (1-s)\boldsymbol{\sigma}''} \ell = s\, \partial_{\boldsymbol{\sigma}'} \ell + (1 - s)\, \partial_{\boldsymbol{\sigma}''} \ell$ equals $\partial_{\boldsymbol{\sigma}} \ell$ for every polar function that $\boldsymbol{\sigma}$ falls and is nonnegative for every other. The ray of $s\boldsymbol{\sigma}' + (1 - s)\boldsymbol{\sigma}''$ consists of the points $s(\mathbf{c} + \tau\boldsymbol{\sigma}') + (1 - s)(\mathbf{c} + \tau\boldsymbol{\sigma}'')$ of the convex Ω.

The instances:

- In 29.2 the integrand is the Gammas times the reduced periods of the two words. These are holomorphic where they converge absolutely, on a set convex by Hölder's inequality (the reference, def:chamber). On the pins the integrand of a reduced period at $\mathbf{c}' + \tau\boldsymbol{\sigma}$ is its integrand at $\mathbf{c}'$ times $\mathbf{V}^{-\tau\boldsymbol{\sigma}}$ read at the point of the orthant (the reference, proof of thm:closecone), at most $(\sup \mathbf{V}^{-\boldsymbol{\sigma}})^{\tau}$; so when both suprema are finite a ball about $\mathbf{c}$ inside the set of convergence, translated by $\tau\boldsymbol{\sigma}$, stays inside it, and the ray lies in Ω.
- In 29.3(iii) the seam function continues to all complex shares, with its poles where a wall lies in $-\mathbb{N}$ (29.3(ii)), the walls read through the sums of the shares being affine in the shares.
- For a fibre integral of Gammas and coefficients the integrand $\prod_l \Gamma(a_l) \xi_l^{-a_l}$, up to a constant, is meromorphic on all of $A + iV$ with its poles where a share lies in $-\mathbb{N}$. So the polar functions are the shares, $\Omega = A$, and $\partial_{\boldsymbol{\sigma}'} a_l = \sigma'_l$, which gives the chamber displayed. When the pins fix the total of the shares of every letter, every $\boldsymbol{\sigma}' \in V$ has $\sum_{l \in K_j} \sigma'_l = 0$ and $\sum_{l \in \Psi_j} \sigma'_l = 0$, so on the chamber $0 \le \sigma'_l \le \sum |\sigma_{l'}|$, the sum over the falling shares l' of the letter of l: the chamber is a compact convex polytope.
- A source factor that is a cube integral continued by Reg is meromorphic on all of $A + iV$, with its poles where one of its walls lies in $-\mathbb{N}$ (8.6).

□

Theorem 29.6 (A closing depends only on its chamber). *As stated, for every fibre integral of 29.5 with the local decay (H1) of 29.4.*

Proof. A polar function ℓ that $\boldsymbol{\sigma}$ falls has $\ell(\mathbf{c} + \tau\boldsymbol{\sigma}) = \ell(\mathbf{c}) + \tau \partial_{\boldsymbol{\sigma}} \ell = \ell(\mathbf{c} + \tau\boldsymbol{\sigma}')$. So its hyperplanes $\ell = -n$ are met by the two rays at the same times, and ℓ is constant on the segment joining their points at a time τ. A polar function that $\boldsymbol{\sigma}$ does not fall is not fallen by $\boldsymbol{\sigma}'$ either; it is at least $\ell(\mathbf{c}) > 0$ at both points, hence on the segment, and does not lie in $-\mathbb{N}$ there.

So the two rays meet the same polar hyperplanes at the same times. At a crossing time τ^* the polar hyperplanes through $\mathbf{c}+\tau^*\boldsymbol{\sigma}$ and through $\mathbf{c}+\tau^*\boldsymbol{\sigma'}$ are the same, with $\partial_{\boldsymbol{\sigma}}\ell_H = \partial_{\boldsymbol{\sigma'}}\ell_H$, so (H0) holds along one ray exactly when it holds along the other.

At a time τ that is no crossing, the segment avoids every polar hyperplane at a positive distance and lies in the convex Ω. By (H1) on a compact neighbourhood of the segment in Ω, step (1) of the proof of 29.4 (Cauchy's theorem on rectangles whose horizontal sides vanish by (H1), then Fubini's theorem) equates the fibre integrals through its two ends. A crossing term is the difference of the fibre integrals just before and just after the crossing (step (2) of that proof), so the crossing terms of the two rays at a common crossing are equal. Equal terms and equal remainders carry convergence and bounds from one closing to the other.

The rest is the reference's (the reference, proof of thm:chamber). The integrands of 29.2 and of 29.3(iii) have the local decay (H1), by Stirling's formula and the bounds of their proofs, so the first part applies to them. For every $\epsilon > 0$ some $\boldsymbol{\sigma'} \in C(\boldsymbol{\sigma})$, with some θ in the case of 29.3(iii), has a rate below $h' + \epsilon/2$. The null set of 29.2 is the same for every direction of the chamber, its crossings being those of the falling polar functions, and the bounds of 29.2 or of 29.3(iii) along $\boldsymbol{\sigma'}$ bound the terms and the remainders along $\boldsymbol{\sigma}$. On the chamber the falling shares are fixed, $x \log x$ is convex on $[0,\infty)$, and $\log\sup \mathbf{V}_B^{-\boldsymbol{\sigma'}}$ and $\log\sup \mathbf{V}_R^{-\boldsymbol{\sigma'}}$ are suprema of linear functions of $\boldsymbol{\sigma'}$, so the rate h is convex there. □

Theorem 29.7 (Averaging the laws). *As stated, for every share side whose integrand is the Gammas of the shares times a Laplace transform $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ of a source measure, the kinds (A) and (B) among them, with the sharpness (iv) for every direction with rational coordinates whose crossings are simple and nonresonant.*

Proof. (i). At a source point ζ the kernel is a fibre integral of Gammas and the positive coefficients $e^{y_l(\zeta)}$: the share side of a Gramma (4.2) divided by the constant $\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)$. At the real point $\mathbf{c}$ the Gammas are integrable over the fibre, $|e^{-\langle \mathbf{c}+i\boldsymbol{\vartheta}, y\rangle}| = e^{-\langle \mathbf{c}, y\rangle}$, and $\int e^{-\langle \mathbf{c}, y\rangle}\, d\lambda < \infty$. So the fibre and the source integrals converge absolutely together, and by Fubini's theorem the fibre integral of the share side is the integral of the kernel against λ. In (A) and (B) that fibre integral is $P_{B\sqcup R}(\mathbf{m})$ (27.9(iii), 29.3(i)). In (A) the kernel is $\Gamma(\kappa_B)\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}\circ\mathbf{x'})$ (the reference, proof of thm:pool(i)): each active letter of the stand-in pair of 27.9(ii) is opened by 4.4 with one share per station, and the integral over the scale is the pin of the scale.

The bound on the crossing terms is the reference's (the reference, proof of thm:pool(i)). It uses three things:

- 29.6 for the kernel, whose polar functions are the shares, with $\Omega = A$ and (H1) holding by Stirling's formula, so that the closings along $\boldsymbol{\sigma}$ and along every $\boldsymbol{\sigma'} \in C$ are one series;
- Stirling's formula for the Gammas, uniformly on the compact C, the terms $\tau x \log \tau$ cancelling over each letter by its total;
- at a crossing, the expansion of $e^{-\langle \mathbf{a}, y\rangle} = e^{-\langle \mathbf{c}+\tau\boldsymbol{\sigma'}, y\rangle} e^{-i\langle \boldsymbol{\vartheta'}, y\rangle} e^{-w\langle \mathbf{e}, y\rangle}$ in the transverse variable w to the order $p-1$, with $|\langle \mathbf{e}, y\rangle| \le C|y|_V$, against the Laurent coefficients of the Gammas, which Cauchy's estimate bounds off the null set.

None of these steps uses the kind of the share side.

(ii). The source points with $y(\zeta) \in S_\lambda$ carry all of λ, and there $h_C(y) \le \hat{h}_C$. By (i) the averaged terms are absolutely convergent integrals, at most $C_\epsilon e^{\tau_k(\hat{h}_C+\epsilon)} \int w\, d(y_*\lambda)$, which is finite by the certificate of 29.7. At each source point the kernel is the sum of its crossing terms before a time T separating the crossings and of the integral left at T (29.4, whose identity at a finite time

needs no limit), and that integral is at most $C_\epsilon e^{-\langle \mathbf{c},y\rangle} e^{T(\hat h_C+\epsilon)}$. Integrating against λ, the fibre integral of the share side is the sum of the averaged terms before T up to $C_\epsilon e^{T(\hat h_C+\epsilon)} \int e^{-\langle \mathbf{c},y\rangle}\, d\lambda$; the crossings up to T number $O(T)$, and with $\hat h_C+\epsilon<0$ both the sum and the bound follow. At a crossing point, along some $\boldsymbol{\sigma'} \in C$, inside the set where $\int e^{-\langle \mathbf{a},y\rangle}\, d\lambda$ converges absolutely, with its derivatives in the shares, Fubini's theorem (the Gammas decaying by Stirling's formula) exchanges the fibre and the source integrals in the crossing term of the share side. The inner integral is then the crossing term of the kernel at that point, which is $\mathcal{R}_k(\zeta)$ by 29.6.

(iii). The function $h(\boldsymbol{\sigma'};y)$ is continuous, convex in $\boldsymbol{\sigma'}$ on C (the falling coordinates being fixed and $x\log x$ convex on $[0,\infty)$), and affine in y; C is compact and convex. Sion's minimax theorem on $C \times \operatorname{conv} S_\lambda$ gives the identity, the supremum of an affine function over $\operatorname{conv} S_\lambda$ being its supremum over S_λ.

For the reading through the log-support,

$$\Big| \int e^{-\langle \mathbf{c}+\tau\boldsymbol{\sigma'}+i\boldsymbol{\vartheta},y\rangle}\, d\lambda \Big| \le e^{\tau \sup_{y\in S_\lambda}(-\langle \boldsymbol{\sigma'},y\rangle)} \int e^{-\langle \mathbf{c},y\rangle}\, d\lambda.$$

With the Gamma bounds of the proof of 29.2, this bounds the closing of the share side along $\boldsymbol{\sigma'}$ at the rate $\sum_l \sigma'_l \log|\sigma'_l| + \sup_{S_\lambda}(-\langle \boldsymbol{\sigma'},y\rangle) = \sup_{y\in S_\lambda} h(\boldsymbol{\sigma'};y)$. In (A) and (B) this is the rate $h(\boldsymbol{\sigma'})$ of 29.2 (the reference, proof of thm:pool(iii)):

- the seam words of the sides that are laws have Newton sums of full dimension, their seam functions converging at $\mathbf{c}$ (24.12);
- so the largest values of the monomials of their letter vectors are read on the supports of their laws (29.2; for the partner in (A), its sentence on the partner);
- in (B), $\sup \mathbf{V}_R^{-\boldsymbol{\sigma'}} = \boldsymbol{\xi}^{-\boldsymbol{\sigma'}}$;
- the largest value of $-\langle \boldsymbol{\sigma'},y\rangle$ on the log-support of a product of laws is the sum of its largest values on the log-supports of the factors.

In a bundle G, Jensen's inequality for the logarithm, in the proportions σ'_l/g, gives $\sum_G \sigma'_l \log(\sigma'_l e^{-y_l}) \ge -g\log\big(g^{-1}\sum_G e^{y_l}\big)$, with equality exactly when e^{y_l}/σ'_l is constant on G.

(iv). At each source point the kernel is, up to the constant factor $\prod_j \Gamma(s_j)^{-1} \prod_j \Gamma(t_j)^{-1}$, the share side of the Gramma whose configuration is the matrix of the pins, rational by hypothesis, with the grading of the letter totals; whose index is the vector of the values of the pins, real; and whose coefficients are $e^{y_l(\zeta)}$, positive and continuous in ζ (4.2). Let $\boldsymbol{\sigma}$ be rational and the crossings of its ray simple and nonresonant. A falling share l crosses its poles at the times $(c_l+N)/|\sigma_l|$, $N\in\mathbb{N}$; for a rational $\Delta>0$ with every $\Delta\sigma_l$ an integer these times fall into finitely many progressions of step Δ, so the crossing terms form finitely many cohorts with the integer step of length Δ in τ (29.16), the direction per step being $\Delta\boldsymbol{\sigma}$, from the positive point $\mathbf{c}$.

- 29.26, with a real index, gives every retained term a sign at each index that does not depend on the coefficients, hence not on ζ, so the averaged term has that sign.
- 29.27 applies to each cohort: one averaged term is absolutely integrable by (i) and the certificate of 29.7, $\hat h_C<\infty$, and the coefficient map is continuous, so the essential supremum of the chamber rate is its supremum on S_λ.
- The chamber of the step direction is ΔC, and since $\sum_l \sigma'_l = 0$ on V, $h(\Delta\boldsymbol{\sigma'};y) = \Delta h(\boldsymbol{\sigma'};y)$. So the root rate per step is the limit $e^{\Delta \hat h_C}$.

Hence for every $\epsilon > 0$ and every cohort, $|\int \mathcal{R}_n \, d\lambda| \geq e^{n\Delta(\hat{h}_C - \epsilon/2)}$ for all large n. The crossing time being $\tau_n = \tau_0 + n\Delta$, this exceeds $e^{\tau_n(\hat{h}_C - \epsilon)}$ for all large n, the quotient being $e^{n\Delta\epsilon/2 - \tau_0(\hat{h}_C - \epsilon)}$; the cohorts being finitely many, the bound holds for all large k. With (ii), the terms tend to zero when $\hat{h}_C < 0$ and do not when $\hat{h}_C > 0$.

Now let every falling share be alone in its bundle and the merged fibre have dimension one.

- The merged pins are rational, so the merged directions form a line spanned by a rational vector $\mathbf{r}$. The merged direction of $\boldsymbol{\sigma}$, nonzero since $\boldsymbol{\sigma}$ falls a share, which is alone in its bundle, is $\mu\mathbf{r}$ with $\mu \neq 0$, and $\mu > 0$ after the choice of the sign of $\mathbf{r}$. The falling coordinates of the positive multiple $\mu^{-1}\boldsymbol{\sigma}$ of $\boldsymbol{\sigma}$ are coordinates of $\mathbf{r}$, rational.
- The chamber of $\mu^{-1}\boldsymbol{\sigma}$ is a nonempty polytope defined by rational equations and inequalities, so its vertices have rational coordinates. By 29.6 the closing along $\mu^{-1}\boldsymbol{\sigma}$, which is the closing along $\boldsymbol{\sigma}$ after a change of scale of τ, is the closing along that rational direction.
- At a crossing time the growing shares are positive at the real point of the ray, so the crossing is simple and nonresonant exactly when no other falling share lies in $-\mathbb{N}$ there, that is, when no two lattices of the falling shares meet.

The form of the crossing terms is the reference's (the reference, proof of thm:pool(iv)). The fibre is a bundle over the merged line whose fibres are the splits of the bundles at fixed sums, and 4.4 completes each split, $\int_{\sum_{l\in G} a_l = \bar{a}} \prod_{l\in G} \Gamma(a_l) e^{-a_l y_l} = \Gamma(\bar{a}) e^{-\bar{a} y_G}$. A crossing term on the merged line is a fibre integral of dimension zero: the residue of the Gamma crossed, times the other Gammas at the crossing point and $\prod_G e^{-\bar{a}_{k,G} y_G}$, divided by the Gammas of the exponents and by the determinant of the pins. □

Definition 29.10 (Keyhole continuations along a direction; ceiling and phase). As stated, for every piece of 29.3 and the sectors of any fan of its seam word.

Proof. The displayed sum is 29.3(ii) at the sums of the shares over the monomials of the partner. A set Y of a box is compact and is the product of a box, of intervals, and of keyholes, a closed disc about 0 joined to a segment of the positive axis (19.45); each factor is star-shaped, about the centre of its disc for a keyhole, so Y is star-shaped, and so is Y plus a small open polydisc about 0. The sector letters do not vanish on Y, by the definition of a keyhole continuation, hence on such a neighbourhood of Y, which is simply connected, so they have logarithms there. The real points of Y form a product of intervals, connected, containing points with nonnegative coordinates, where the letters, with their nonnegative coefficients and positive constant terms, are positive; real and without zeros on that set, they are positive on it, and the logarithms can be taken real at the real points. For a piece of tables and the sectors of the orders, the walls of the generators are the proper walls of the tails and the sector letters are those of 8.7(ii) (29.3(ii)). □

Theorem 29.11 (The travelling bound past the walls). *As stated, for every piece of 29.3.*

Proof. This is the reference's proof (the reference, proof of thm:keyholebound), with the sectors of the fan in place of the orders of B and the generators in place of the tails.

(i) Fix a sector π, a box Ξ of it, the set J of its falling variables with keyholes and their radii r_k, and choose $r_k^- \in (0, r_k)$ with $\sum_{k\in J}(-\dot{\kappa}_k)\log(r_k/r_k^-) < \epsilon/2$. The term of π is $|\det(\mathbf{v}_{\pi,1}, \ldots, \mathbf{v}_{\pi,|B|-1})| \int \prod_{k\in J} q_k^{\kappa_k(\bar{\mathbf{z}})-1} \Phi \, dq$, with $\Phi = \prod_{k\notin J} q_k^{\kappa_k(\bar{\mathbf{z}})-1} \mathbf{k}_\pi(q)^{-\bar{\mathbf{z}}}$ holomorphic in the variables of J on a neighbourhood of their keyholes, because the sector letters do not vanish on the sets Y. The following steps apply as in the reference:

- 19.43, applied in the variables of J one at a time with r_k^- and r_k;

- the maximum modulus principle in the variables of $J \setminus A$ that lie in their discs;
- the exact integration of the falling monomials, which carries no phase;
- the identity $|k^{-z}| = |k|^{-\operatorname{Re} z} e^{\operatorname{Im} z \arg k}$ at the points of the outline;
- the bounds of the letters above and below on the compact outline;
- the integrability of $q_k^{\kappa_k(\bar{\mathbf{c}})-1}$ over the base variables, the walls being positive at $\bar{\mathbf{c}}$;
- $|\kappa_k(\bar{\mathbf{z}}) + l_k| \geq c\,\delta$ for at most $|B| - 1$ factors.

Each of these uses only the sector form of 29.3(ii): walls affine in the shares and positive at $\bar{\mathbf{c}}$, and sector letters polynomial, positive on the closed cube and nonvanishing on the sets Y. The determinant is a constant, and the boxes, the sets A and the sectors are finitely many.

(ii) The integrand of the share side of 29.3(i) at $\mathbf{c} + \tau\boldsymbol{\sigma} + i\boldsymbol{\vartheta}$ is the Gammas and the coefficients times the seam function, which (i) bounds. The steps are those of the reference:

- for a growing share, the exact bound $|\Gamma(x + iy)|e^{\varphi y} \leq \Gamma(x)(\cos\varphi)^{-x}$ for $|\varphi| < \frac{\pi}{2}$, from $\Gamma(z) = e^{i\varphi z}\int_0^\infty u^{z-1}e^{-ue^{i\varphi}}du$, applied with $\varphi = \upsilon_l(q) + \eta \operatorname{sign} y$;
- falling and staying shares absorb $e^{\upsilon_l(q)y}$ against $e^{-\pi|y|/2}$, as $|\upsilon_l| \leq \frac{\pi}{2} - \epsilon_0$;
- Stirling's formula with $\sum_l \sigma_l = 0$, the supremum over q being taken after the integration over the directions of the shares.

Together these give the travelling bound at the rate $h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma})$. Then 29.4 applies: (H0) off the null set; (H1) by 29.3(ii) with any $\theta < \pi/2$ and Stirling's formula; (H2) at times at distance at least τ^{-3} from the crossings. □

Corollary 29.12 (When the phase costs nothing). *As stated, for every piece of 29.3.*

Proof. In (i) the price $-\sum_{\sigma_l>0}\sigma_l \log\cos\upsilon_l$ vanishes. In (ii), $\log|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}| - \sum_{\sigma_l>0}\sigma_l\log\cos\upsilon_l \leq \log H_{\mathcal{O}}$ on the outline, with equality approached where $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}|$ approaches $H_{\mathcal{O}}$, the price being nonnegative.

With no falling variables, the outline is the union of the real cubes of the sectors, where the sector letters are positive and the phase vanishes. Differentiating the wall of 29.3(ii) gives $\dot{\kappa}_k = -\langle\sum_{j,D}(\bar{\sigma}_{jD}\mathbf{t}_{jD,\pi} + \bar{\sigma}^*_{jD}\mathbf{t}^*_{jD,\pi}), \mathbf{v}_{\pi,k}\rangle$. So on the section $\varrho_{\pi,\bar{\boldsymbol{\sigma}}} = \prod_{j,D}\big(\prod_k q_k^{\langle \mathbf{t}_{jD,\pi}, \mathbf{v}_{\pi,k}\rangle} k_{jD,\pi}\big)^{-\bar{\sigma}_{jD}}\big(\prod_k q_k^{\langle \mathbf{t}^*_{jD,\pi}, \mathbf{v}_{\pi,k}\rangle}\psi_{jD,\pi}\big)^{-\bar{\sigma}^*_{jD}} = \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$, read in the sector coordinates. The cubes of the sectors cover the section, so the ceiling is $\sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$, the monomial being homogeneous of degree zero, and the rate is that of 29.2 for the glue read in the monomials of the partner. □

Proposition 29.13 (How travelling bounds compose). *As stated, the face coefficient of (iii) being read on the sectors of any fan of the seam word.*

Proof. (i) and (ii) are the reference's (the reference, proof of prop:pathcompose). For (iii), the reference's proof applies with one sentence changed: a wall $\kappa_{\mathbf{v}}$ is the exponent of q_k in each sector of which $\mathbf{v} = \mathbf{v}_{\pi,k}$ is a generator (29.3(ii)).

- The pole is simple, as opposed to the multiple poles of the statement, when no other factor and no second wall of a sector is polar on the hyperplane. Then, by 8.6, the residue of the sector integral in the direction $\mathbf{e}$ at $\kappa_{\mathbf{v}} = -n$ is $|\det(\mathbf{v}_{\pi,1}, \ldots, \mathbf{v}_{\pi,|B|-1})|$ times the integral over the other coordinates of the n-th Taylor coefficient in q_k of the rest of the integrand, divided by $\partial_{\mathbf{e}}\kappa_{\mathbf{v}}$. Summing over the sectors gives the face coefficient.

- The other powers $q_{k'}^{\kappa_{k'}-1}$ do not depend on q_k, so the Taylor coefficient is that of the product of the powers of the sector letters: a finite sum of products of derivatives of the letters and powers of the letters at $q_k = 0$, with coefficients polynomial in the exponents.
- Where two walls of one sector lie in $-\mathbb{N}$ on the hyperplane, the expansion of 8.6 in both variables gives a pole of the second order. A Laurent coefficient of a product is a finite sum of products of Laurent coefficients, and the derivatives in the shares of $q^{\kappa(\mathbf{z})}$ and of k^{-z} bring $\log q$ and $-\log k$.
- The wall crossed falls along the direction, so q_k is a falling variable and $q_k = 0$ lies in its keyhole, where the letters do not vanish; Cauchy's estimate on $|q_k| = r_k$ bounds the n-th Taylor coefficient by r_k^{-n} times the largest modulus on the circle.

□

Definition 29.16 (A rational retained cohort). As stated: the fibre integrals of Gammas, coefficients and source factors whose pins carry a grading.

Proof. The share side of 29.3(i) and the kernel of 29.7 are graded by their letter totals: the sum of the rows of the pins that fix the totals of the letters takes the value 1 on the column of every share, each share belonging to one letter. By 29.3(ii), the sector integrals of a fan of a seam word are cube integrals $\int_{[0,1]^k} q^{\boldsymbol{\kappa}(\mathbf{a})-\mathbf{1}} \prod_i P_i(q)^{-w_i(\mathbf{a})} dq$. Here the P_i are the sector letters, with nonnegative coefficients and positive constant terms, and the κ_i and w_i are affine with rational linear parts; for tables these are the sector integrals of 8.8. □

Remark. The grading is part of the definition because 29.18 extracts the scalar of a joint word through the change of logarithmic scales $z = \rho\mathbf{v} + E\log X$, which needs a vector $\mathbf{v}$ taking the value 1 on every column. Without one the form of that theorem can fail, as follows.

Take the Gramma $\mathsf{G}_{\nu;(1,2)}(\xi_1,\xi_2) = \int_0^\infty x^{\nu-1} e^{-\xi_1 x - \xi_2 x^2} dx$, $\nu, \xi_1, \xi_2 > 0$. Its configuration has no grading, since $\langle 1, v\rangle = \langle 2, v\rangle = 1$ is impossible. Its share side is pinned by $b_1 + 2b_2 = \nu$, and its closing along $(-2, 1)$ crosses $b_1 = -N$, $N \in \mathbb{N}$, with the crossing terms

$$\frac{(-\xi_1)^N}{N!} \cdot \frac{1}{2}\Gamma\Big(\frac{\nu+N}{2}\Big)\xi_2^{-(\nu+N)/2},$$

the terms of the expansion of the integral in powers of ξ_1, using $\int_0^\infty x^{\nu+N-1} e^{-\xi_2 x^2} dx = \frac{1}{2}\Gamma(\frac{\nu+N}{2})\xi_2^{-(\nu+N)/2}$.

The scalar $\Gamma(\frac{\nu+N}{2})/\Gamma(N+1)$ has an N-th root tending to 0: its logarithm divided by N is $-\frac{1}{2}\log N + O(1)$. A joint word with the scalar of 29.18, here with $r_G = 1$, would in this positive case have the root rate $\operatorname{ess\,sup} c > 0$, or vanishing terms (29.19).

Theorem 29.18 (A finite procedure for a retained cohort). *As stated, for every retained cohort, the scalar being written through the grading of its pins.*

Proof. This is the reference's proof (the reference, proof of thm:retainedchain); only the radial step involves the totals of the shares.

- At a Gamma crossing, the crossed share is fixed at $-N(n)$, and its Gamma is replaced by the residue $(-1)^{N(n)}/\Gamma(N(n)+1)$.
- The crossed share and its added pin are eliminated together. This leaves the columns t_l, $l \neq f$, of the original pins, with the values $\boldsymbol{\nu} + N(n)t_f$, t_f the column of the crossed share.

- By 29.14 each share of $\mathcal{S}_-$ is translated, $b_l = a_l + \delta_l(n)$, with the factor $(-1)^{\delta_l(n)}\xi_l^{\delta_l(n)}/\Gamma(\delta_l(n))$, the power of ξ_l staying in the joint input.

The completed retained fibre is then the Gramma $\mathsf{G}_{\boldsymbol{\mu}_n;t}$ (4.2), with the remaining columns and the index $\boldsymbol{\mu}_n = \boldsymbol{\nu} + N(n)t_f + \sum_{l\in\mathcal{S}_-}\delta_l(n)t_l$. Every remaining column is a column of the original pins, so $\langle t_l, \mathbf{v}\rangle = 1$ for the grading $\mathbf{v}$, and

$$S_n = \langle \mathbf{v}, \boldsymbol{\mu}_n\rangle = w + N(n) + \sum_{l\in\mathcal{S}_-}\delta_l(n).$$

The change of logarithmic scales $z = \rho\mathbf{v} + E\log X$, with E a rational matrix whose columns complement $\mathbf{v}$, gives $\mathsf{G}_{\boldsymbol{\mu}_n;t} = J_{\mathbf{v}}\Gamma(S_n)\int X^{E^{\mathsf{T}}\boldsymbol{\mu}_n - \mathbf{1}}\mathcal{L}(X)^{-S_n}dX$: the integral over ρ is $\Gamma(S_n)\mathcal{L}^{-S_n}$. Collecting the residue, the factors of the translations and $\Gamma(S_n)$ gives the stated G_n. Its slopes in n agree in the numerator and the denominator, $d_0 + \sum_{\mathcal{S}_-} d_l = \bar{d}$, so the terms $n\log n$ of Stirling's formula cancel, and the rest of Stirling's formula gives r_G and the polynomial bound. On the share sides of 29.3 and 29.7, $\mathbf{v}$ is the sum of the letter-total rows and $w = |\mathbf{s}| + |\mathbf{t}|$. The other steps of the procedure (the source strips, the circles, the joint read, the resolution of the endpoints and the fan) do not involve the grading. □

Theorem 29.26 (The exact positive rate of a retained chamber). *As stated, with a real index.*

Proof. The reference's proof (the reference, proof of thm:retainedchamberrate) uses rationality only of the configuration and of the direction, through the cohort and its integer step.

- 29.16 allows real constant terms and indices. The crossing times of a falling share are $(a_l^0 + N)/d_l$, $N \in \mathbb{N}$, and passing to the integer step makes the changes of all falling shares integral, whatever the constant terms.
- The positive construction uses 29.14 with $\delta_l(n) = \delta_l(0) + d_l n$ and real starting values.
- The rate is computed by 29.25, at the slope $\sum_f d_f t_f$ of the index, which does not involve its constant term.
- The root limit is that of 29.19, whose positive case allows real constant terms.

The chamber polytope of the statement is the chamber of 29.5 of the share side, whose polar functions are the shares (29.5), for the direction per step. □

40.30 Chapter 30: laws at their ends: regions of equality and reach

Theorem 30.10 (The reach along a direction is read on the law). *Parts (i) to (iv) as stated, with part (v).*

Proof. Parts (i) to (iv) are the reference's (the reference, proof of thm:dirreach); the piece of tables named in (iii) has a seam word with a Newton sum of full dimension when its tables form a connected family, by 29.3(ii). For (v) fix $\theta \geq 0$ and let g_θ be the infimum of (v).

First, $r \leq g_\theta$. On the half-strip Σ of 30.9 the imaginary parts are bounded, say by Y, so $e^{\theta|\operatorname{Im} z|} \leq e^{\theta Y}$ there. A bound $|\ell(z)| \leq C(1+|z|)^{n'}M^{\operatorname{Re} z}e^{\theta|\operatorname{Im} z|}$ on the half-plane, off the δ-neighbourhoods of the candidate poles, therefore gives on Σ the bound of 30.9 with $Ce^{\theta Y}$, and $r \leq M$.

Second, $g_\theta \le r$. By (i), on (r,∞) the measure λ has a density that is a wall series of radius r. By (ii) with $x_0 = r$, for every $x' > r$ there is C with $|\ell(z)| \le C\delta^{-J-1}x'^{\operatorname{Re} z}$ at every z with $\operatorname{Re} z \ge 0$ at distance at least $\delta \le 1$ from the poles, whatever $\operatorname{Im} z$. The poles with $\operatorname{Re} z \ge 0$ lie among the candidate poles, and those with $\operatorname{Re} z < 0$ keep a distance at least some $\epsilon_0 > 0$ from the half-plane, ℓ converging absolutely on a strip about the imaginary axis (30.9). So at every z with $\operatorname{Re} z \ge 0$ at distance at least δ from the candidate poles, with $\delta_1 = \min(\delta, 1, \epsilon_0)$, $|\ell(z)| \le C\delta_1^{-J-1}x'^{\operatorname{Re} z}$: the bound of (v) with the constant $C\delta_1^{-J-1}$ and $n' = 0$. So $M = x'$ is admissible in (v) at $\theta = 0$, hence at every $\theta \ge 0$, the factor $e^{\theta|\operatorname{Im} z|}$ being at least 1, and $g_\theta \le x'$ for every $x' > r$. □

Theorem 30.15 (Ceilings, simple tops and fibres along a direction). *As stated, for every piece whose seam function converges at a real point of the pins with positive shares, with the sectors of any fan of its seam word.*

Proof. By 29.3(ii) the piece has walls and growth bounds (30.7), its walls being those of the generators of the fan, so 30.9 applies with these walls. The proof is the reference's (the reference, proof of thm:dirheights), with the sectors of the fan in place of the orders of B, the generators in place of the tails, and the constant $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|$ in the sector forms. At the point $\mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}} + z\bar{\boldsymbol{\sigma}}$ the integrand of the sector integral of π is

$$|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\kappa_k(\mathbf{w}_0)+i\partial_{\bar{\boldsymbol{\vartheta}}}\kappa_k - 1}\, \mathbf{k}_\pi^{-\mathbf{w}_0 - i\bar{\boldsymbol{\vartheta}}}\, \varrho_{\pi,\bar{\boldsymbol{\sigma}}}^{z}.$$

(i) The proof of 29.11(i) concerns the seam function alone. It holds at the points $\mathbf{w}_0 + \operatorname{Re} z\,\bar{\boldsymbol{\sigma}} + i(\bar{\boldsymbol{\vartheta}} + \operatorname{Im} z\,\bar{\boldsymbol{\sigma}})$, with δ the distance of the complex point from the hyperplanes on which a wall lies in $-\mathbb{N}$, which is at least a constant times the distance of z from the candidate poles (the reference, proof of cor:dirtube). On Σ the phase is bounded, the imaginary parts being bounded there. So $|\ell(z)| \le C\delta^{-(|B|-1)}e^{\operatorname{Re} z(\log H_{\mathcal{O}}+\epsilon)}$, and $r \le H_{\mathcal{O}}e^{\epsilon}$ for every $\epsilon > 0$.

(ii) and (iii) follow from the proof of 19.49(i) and (iii), with the point $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ in place of $b^d = -k$, $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of ϱ_π and $\phi_{\pi,\mathbf{w}_0}$ in place of ϕ_π.

- The falling variables carry $q_k^{\kappa_k(\mathbf{w}_0)-m|\dot{\kappa}_k|-1}$ with $|\dot{\kappa}_k|$ an integer, so the factor $(e^{2\pi i\kappa_k(\mathbf{w}_0)}-1)^{-1}$ of their circles is the same at every m. The letters whose exponents are nonnegative integers at every large m carry polynomials and need not be kept from vanishing.
- Every wall is the wall of a generator of some sector, so its falling rate is one of the integers $|\dot{\kappa}_k|$. The points m lie at a distance bounded below from the candidate poles, the falling walls being off the integers at $\mathbf{w}_0$, so $|\ell(m)| \le C(1+m)^D r'^m$ for every $r' > r$ by 30.9, and $\operatorname{growth}_{\mathbb{Z}} \le r$.
- Under (iii), $\operatorname{growth}_{\mathbb{Z}} = H_{\mathcal{O}}$ as in 19.49(iii), and a keyhole continuation of ceiling $H_{\mathcal{O}}$ gives $r \le H_{\mathcal{O}}$ by (i).

(iv) follows from the proofs of 19.53 and 19.54, applied to the positive directional law, with 30.10(i) and (ii) in place of 19.42.

- A keyhole box is handled as in (i).
- In a fibre box, $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ is meromorphic in each falling variable on its disc, the exponents $\bar{\sigma}_L$ and $\dot{\kappa}_j$ being integers. At a point above the ceiling and the stationary bound, the map is a submersion on its stratum: by 19.50 when one coordinate of falling rate one lies inside its disc, and by the definition of the stationary bound otherwise.
- The Mellin transform of a fibre law is a cube integral as in 19.30, with its poles in finitely many cosets of $(\operatorname{lcm}_j|\dot{\kappa}_j|)^{-1}\mathbb{Z}$, so 19.51 applies.

None of these steps uses more of a sector integral than the form displayed above, which 29.3(ii) supplies for every fan of the seam word of every such piece. □

Theorem 30.16 (The closing past the walls is read along its direction). *As stated, with the last sentence of (i), for the share side of 29.3(iv).*

Proof. Apart from the last sentence of (i), the statement is the reference's (the reference, proof of thm:dirclosing). For that sentence, the piece and the partner have walls and growth bounds (29.3(ii), applied to each), and $\boldsymbol{\sigma}$ has rational coordinates, so 30.12 applies to each of the two seam functions at $\mathbf{c}$, over the whole space V of the directions of the shares. There the growth $g_X(\boldsymbol{\sigma};\theta)$ of 29.3(iv), $X \in \{B, R\}$, is the function $g_V(\boldsymbol{\sigma};\theta)$ of 30.12 for that seam function: the two definitions coincide, the shares carrying no sums and the walls of 29.3(ii) serving as the walls of (W1), so that δ is measured from the same hyperplanes, those of the polar functions of the seam function (29.5). It does not increase with θ and tends to $r_X^\sharp$ as $\theta \downarrow 0$, so $g_X(\boldsymbol{\sigma};\theta) \le r_X^\sharp$ for every $\theta > 0$. Then

$$h_{\theta/2,\theta/2}(\boldsymbol{\sigma}) \le \sum_l \sigma_l \log|\sigma_l| + \log r_B^\sharp + \log r_R^\sharp + \frac{2\theta^2}{\pi} \sum_{l:\,\sigma_l>0} \sigma_l,$$

which is negative for θ small when $\sum_l \sigma_l \log|\sigma_l| + \log r_B^\sharp + \log r_R^\sharp < 0$. 29.3(iv) gives the closing, its terms and remainders being $O(e^{\tau(h_{\theta/2,\theta/2}(\boldsymbol{\sigma})+\epsilon)})$, and letting $\theta \to 0$ gives the stated bound. □

Corollary 30.18 (A law with log expansions bounds a closing of any fibre dimension). *As stated, with its last sentence, for the share side of 29.3(iv).*

Proof. Apart from the last sentence, the statement is the reference's (the reference, proof of cor:anyfibre). For that sentence, apply 30.14 to the law of the piece and to that of the partner, a piece with the reversed excesses. On the contour through $\mathbf{c} + \tau\boldsymbol{\sigma}'$ each seam function is at most $C\delta^{-n} X_{\text{in}}^X(\boldsymbol{\sigma}')^\tau$, with no factor in the imaginary parts and no power of τ, so $g_X(\boldsymbol{\sigma}';0) \le X_{\text{in}}^X(\boldsymbol{\sigma}')$ for every direction $\boldsymbol{\sigma}'$. Hence

$$h_{0,0}(\boldsymbol{\sigma}') \le \sum_l \sigma_l' \log|\sigma_l'| + \log X_{\text{in}}^B(\boldsymbol{\sigma}') + \log X_{\text{in}}^R(\boldsymbol{\sigma}'), \qquad \boldsymbol{\sigma}' \in C(\boldsymbol{\sigma}).$$

The polar functions of the share side of 29.3(iv) are the shares and the walls of both words, with $\Omega = A$, and (H1) holds by 29.3(ii) and Stirling's formula. So 29.6 makes the closings along the directions of the chamber one series; their crossings and the near coincidences of these are those of the falling polar functions, the same for every direction of the chamber, so $\mathbf{c}$ lies off the null set of 29.3(iv) for each of them; and 29.3(iv) at $\theta_B = \theta_R = 0$, along a direction of the chamber whose bound is within ϵ of the infimum, gives the closing along $\boldsymbol{\sigma}$ with the stated bound. On the chamber the falling coordinates are fixed and the others are nonnegative, where $x \log x$ is convex, and the logarithms of the inner bounds are convex (30.13). □

Remark (The directional ratio at width one, in the paragraph after 30.9). At width one with one table, factoring weights normalized as in 27.10, and $s, t > 0$, the directional ratio $x_{D_0}^*/x_{D_1}$ along $\bar{\boldsymbol{\sigma}} = (-1, 1, -1, 1)$ is the ratio ϱ_B at the seam points, and every coordinate of the seam torus there is a power of ϱ_B.

Proof. On the seam torus $x_{D_0} = x^*_{D_1} = 1$, so $\mathbf{x}^{-\bar{\boldsymbol{\sigma}}} = x_{D_0}\, x_{D_1}^{-1}\, x^*_{D_0}\, x^{*-1}_{D_1} = x^*_{D_0}/x_{D_1}$. At the seam point of u, 27.5 gives $\varrho^{\rho^*}_{D_0}/\varrho^{\rho}_{D_1} = V^*_{D_0}V_{D_0}/(V^*_{D_1}V_{D_1})$, the scale cancelling. By Lemma S11(i) this is

$$\Big(\frac{\hat\theta_0}{\hat\theta_1}\Big)^2 \frac{\Psi_{\mathcal{B}/R}\, K_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}\, K_{\mathcal{B}|B}} = \Big(\frac{\hat\theta_0}{\hat\theta_1}\Big)^2 \varrho_B,$$

with $\varrho_B = \gamma^\Psi_B/\gamma^K_B$ of 18.4. Under the normalization $\hat\theta_0 = \hat\theta_1 = 1$, so it is ϱ_B. By 27.10(ii) with $N = 1$, $\mathbf{x} = (1, \varrho_B)$ and $\psi = \varrho_B^{\,t/(s+t)}$, the coordinates of the seam point are 1, $\varrho_B^{\,-t/(s+t)}$, $\varrho_B^{\,s/(s+t)}$ and 1, all powers of ϱ_B. □

40.31 Chapter 32: positive words

Theorem 32.1 (Positive coefficient jets). *As stated, at positive weights that need not be rational.*

Proof. Write $P = P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, finite and positive because its walls $n\kappa_S$ are positive (8.10), and $\mathbb{E}$ for the expectation of $U \sim \mathrm{Breta}^\circ(F; n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, whose density on the simplex is $P^{-1}u^{n\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-ns_j}\Psi_j(u)^{-nt_j}$, positive on the open simplex. For $v \in \mathbb{R}^k$ put $X_j = v^{\mathsf T}\partial_{\mathbf b}K_j/K_j$ and $X^*_j = v^{\mathsf T}\partial_{\mathbf b}\Psi_j/\Psi_j$. With $u_{j,T} = \mathbf{b}_{j,T}/\theta_{j,T}$,

$$\frac{\partial_{\mathbf b}K_j}{K_j} = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[u_{j,T}], \qquad \frac{\partial_{\mathbf b}\Psi_j}{\Psi_j} = \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}[u_{j,T}],$$

so $|X_j|$ and $|X^*_j|$ are at most $\beta_j = \max_T |v^{\mathsf T}u_{j,T}|$ on the whole orthant.

(i) For $|\varepsilon|\max_j\beta_j < \frac12$ the weights $\theta_{j,T} \pm \varepsilon v^{\mathsf T}\mathbf{b}_{j,T} = \theta_{j,T}(1 \pm \varepsilon v^{\mathsf T}u_{j,T})$ are positive, and the letters of the two copies of table j are $K_j(1\pm\varepsilon X_j)$ and $\Psi_j(1\pm\varepsilon X^*_j)$. The copies have the levels of table j, each counted twice at half the exponents, so the doubled family at the masses $n\mathbf{m}$ has the imbalance 0 and the walls $n\kappa_S$, and its word is

$$\prod_j K_j^{-ns_j}\Psi_j^{-nt_j}\cdot B_\varepsilon, \qquad B_\varepsilon = \prod_j \big(1-\varepsilon^2X_j^2\big)^{-\frac{ns_j}{2}}\big(1-\varepsilon^2X_j^{*2}\big)^{-\frac{nt_j}{2}}.$$

Its period is therefore $P\,\mathbb{E}[B_\varepsilon(U)]$, the display of (i). For $|\varepsilon| \le \varepsilon_0 < (2\max_j\beta_j)^{-1}$, B_ε is a power series in ε^2 whose coefficients are polynomials in the bounded functions X_j^2, X_j^{*2}, converging uniformly on the orthant; so $\mathbb{E}[B_\varepsilon(U)]$ is analytic in ε and expands term by term, with the coefficient $\sum_j\big(\frac{ns_j}{2}\mathbb{E}[X_j^2] + \frac{nt_j}{2}\mathbb{E}[X_j^{*2}]\big)$ of ε^2. Since $X_j^2 = v^{\mathsf T}\big(\partial_{\mathbf b}K_j\,\partial_{\mathbf b}K_j^{\mathsf T}/K_j^2\big)v$, P times this coefficient is $v^{\mathsf T}\mathbf{H}_n v$. It is a sum of nonnegative terms. If it vanishes, then for every j with $s_j > 0$, $X_j(U) = 0$ almost surely; the law of U has a positive density on the open simplex, so the polynomial $\sum_T (v^{\mathsf T}\mathbf{b}_{j,T})\alpha^T$ vanishes on an open set, and every $v^{\mathsf T}\mathbf{b}_{j,T}$ is 0, the monomials of distinct bases being distinct. For $t_j > 0$ the same holds with $\alpha^{\mathcal{E}\setminus T}$. So v is orthogonal to every direction of an active table, and $v = 0$ when these span $\mathbb{R}^k$.

(ii) Write $Q = R^{\mathsf T}R$ with R real. Then

$$\operatorname{tr}(Q\mathbf{H}_n) = \frac{n}{2}\,P\,\mathbb{E}\Big[\sum_j\Big(s_j\frac{|R\,\partial_{\mathbf b}K_j|^2}{K_j^2} + t_j\frac{|R\,\partial_{\mathbf b}\Psi_j|^2}{\Psi_j^2}\Big)(U)\Big],$$

a sum of nonnegative terms, which vanishes exactly when, as in (i), $R\mathbf{b}_{j,T} = 0$, that is $\mathbf{b}_{j,T}^{\mathsf T}Q\mathbf{b}_{j,T} = 0$, for every basis of every active table. If $Q_{11} > 0$, then $Q - Q_{11}^{-1}Q_{\cdot1}Q_{\cdot1}^{\mathsf T}$ is rational, positive

semidefinite, with first row and column zero; if $Q_{11}=0$, the first row and column of Q vanish, Q being positive semidefinite. Induction on k gives $Q=\sum_l c_l v_l v_l^{\mathsf T}$ with rational $c_l>0$ and rational v_l, and then $\operatorname{tr}(Q\mathbf{H}_n)=\sum_l c_l\, v_l^{\mathsf T}\mathbf{H}_n v_l$.

(iii) $\partial_{\mathbf b}K_j^{\mathsf T}Q\,\partial_{\mathbf b}K_j/K_j^2=\sum_{T,T'}\mathbf b_{j,T}^{\mathsf T}Q\,\mathbf b_{j,T'}\,\alpha^{\mathbf 1_T+\mathbf 1_{T'}}K_j^{-2}$, and $P\,\mathbb E\big[(\alpha^{\mathbf 1_T+\mathbf 1_{T'}}K_j^{-2})(U)\big]$ is the period whose integrand is that of P times $u^{\mathbf 1_T+\mathbf 1_{T'}}K_j(u)^{-2}$, the period at $\big(n\mathbf m+\mathbf 1_T+\mathbf 1_{T'};n\mathbf s+2\mathbf e_j,n\mathbf t\big)$. Its imbalance is that of $(n\mathbf m;n\mathbf s,n\mathbf t)$ plus $2r_j-2r_j=0$, and its wall of S is $n\kappa_S+|T\cap S|+|T'\cap S|-2d_j(S)\ge n\kappa_S$, by 7.11. The cobasis part is the same with $u^{\mathbf 1_{\mathcal E\setminus T}+\mathbf 1_{\mathcal E\setminus T'}}\Psi_j^{-2}$ and the change $|S\setminus T|+|S\setminus T'|-2d_j^*(S)\ge 0$. This is the case of order two of 26.9 for the doubled family, the two copies of each table merged.

(iv) $x\mapsto x^{\mathsf T}Qx$ is convex, so $\big(\mathbb E_T[u_{j,T}]\big)^{\mathsf T}Q\,\mathbb E_T[u_{j,T}]\le\mathbb E_T\big[u_{j,T}^{\mathsf T}Qu_{j,T}\big]\le\max_T u_{j,T}^{\mathsf T}Qu_{j,T}$ under either law, and $\operatorname{tr}(Q\mathbf H_n)\le\frac n2 P\sum_j(s_j+t_j)\max_T\mathbf b_{j,T}^{\mathsf T}Q\,\mathbf b_{j,T}/\theta_{j,T}^2=n\,C(Q)\,P$; positivity is (ii).

(v) With the basis weights $\theta^K_{j,T}$ and the cobasis weights $\theta^\Psi_{j,T}$ of every table independent (8.1), $\partial^2K_j^{-ns_j}/\partial\theta^K_{j,T}\partial\theta^K_{j,T'}=ns_j(ns_j+1)K_j^{-ns_j-2}\alpha^T\alpha^{T'}$. The differentiated integrals are the periods of (iii), convergent locally uniformly in the weights, so differentiation under the integral gives

$$\sum_{T,T'}\mathbf b_{j,T}\mathbf b_{j,T'}^{\mathsf T}\,\frac{\partial^2P}{\partial\theta^K_{j,T}\partial\theta^K_{j,T'}}=ns_j(ns_j+1)\,P\,\mathbb E\Big[\frac{\partial_{\mathbf b}K_j\,\partial_{\mathbf b}K_j^{\mathsf T}}{K_j^2}(U)\Big],$$

which is $2(ns_j+1)$ times the part $\frac n2 s_jP\,\mathbb E[\cdots]$ of $\mathbf H_n$ carried by K_j; the cobasis part is the same with t_j. On the share side of 17.15 the weights enter only through $\prod_{j,T}(\theta^K_{j,T})^{-a_{j,T}}(\theta^\Psi_{j,T})^{-b_{j,T}}$: a derivative in $\theta^K_{j,T}$ multiplies the integrand by $-a_{j,T}/\theta^K_{j,T}$, one in $\theta^\Psi_{j,T}$ by $-b_{j,T}/\theta^\Psi_{j,T}$, and neither the contour, nor the pins, nor the share kernels involve the weights.

(vi) At the centre the two copies of table j carry the equal exponents $(\frac{ns_j}{2},\frac{ns_j}{2})$, and the masses $n\sigma\mathbf 1$ are the total strength of the doubled family times $\mathbf 1$. 8.21(i) holds at every family at its centre, whatever its weights: at $\mathbf 1$ the usage is $\sum_j\sigma_j(R_j+R_j^*)(\mathbf 1)=\sigma\mathbf 1$, since $R_{j,e}(\mathbf 1)+R^*_{j,e}(\mathbf 1)$ is the chance that e lies in the basis plus the chance that it does not. □

Remark. No step uses the rationality of the weights; rational weights are what the arithmetic of the jets asks for, through the rows of the periods of (iii).

Theorem 32.3 (Two bases generate positive contrast words of any depth). *As stated, at every point of the balance wall with real data and positive proper walls, and at positive weights.*

Proof. Write $P_n=P_F(n\mathbf m;n\mathbf s,n\mathbf t)$, finite and positive (8.10), $\mathbb E$ for the expectation of $U\sim\text{Breta}^\circ(F;n\mathbf m;n\mathbf s,n\mathbf t)$, and $c=\mathbb P_{j,\alpha}(T)-\mathbb P_{j,\alpha}(T')$, $c^*=\mathbb P^*_{j,\alpha}(T)-\mathbb P^*_{j,\alpha}(T')$. Each chance lies in $(0,1)$ and two chances of distinct bases sum to at most one, so $|c|,|c^*|<1$ on the open orthant. c vanishes only where $\theta_{j,T}\alpha^{\mathbf 1_T}=\theta_{j,T'}\alpha^{\mathbf 1_{T'}}$, a hypersurface, since $\mathbf 1_T\ne\mathbf 1_{T'}$; so $c^{2h}>0$ almost everywhere, and the contrast word, $\frac12P_n$ times the expectation of a function that is positive almost everywhere under a law with a positive density, is positive.

(i) $c^{2h}=\sum_{l=0}^{2h}(-1)^l\binom{2h}{l}\theta_{j,T}^{2h-l}\theta_{j,T'}^l\,\alpha^{v_l}K_j^{-2h}$ with $v_l=(2h-l)\mathbf 1_T+l\mathbf 1_{T'}$, each term homogeneous of degree 0, and $P_n\,\mathbb E\big[(\alpha^{v_l}K_j^{-2h})(U)\big]$ is the period whose integrand is that of P_n times $u^{v_l}K_j(u)^{-2h}$, the period at $\big(n\mathbf m+v_l;n\mathbf s+2h\mathbf e_j,n\mathbf t\big)$. Its imbalance changes by $|v_l|-2hr_j=0$, and its wall of S by $(2h-l)|T\cap S|+l|T'\cap S|-2hd_j(S)\ge 0$ (7.11). The second half is the same with $\mathbb P^*_{j,\alpha}(T)=\theta_{j,T}\alpha^{\mathcal E\setminus T}/\Psi_j$, the masses raised by $(2h-l)\mathbf 1_{\mathcal E\setminus T}+l\mathbf 1_{\mathcal E\setminus T'}$, the cobasis exponent by $2h$, and the wall of S by $(2h-l)|S\setminus T|+l|S\setminus T'|-2hd_j^*(S)\ge 0$.

From $\Psi_j(\alpha)=\alpha^{\mathbf 1}K_j(\alpha^{-1})$, $K_j(\alpha^{-1})=\alpha^{-\mathbf 1}\Psi_j(\alpha)$ and $\Psi_j(\alpha^{-1})=\alpha^{-\mathbf 1}K_j(\alpha)$, so $\mathbb P^*_{j,\alpha^{-1}}(T)=\mathbb P_{j,\alpha}(T)$. On the chart $y_g=0$ of 8.10 the integrand of a period at data

$(\mathbf{m}';\mathbf{s}',\mathbf{t}')$ is $e^{\langle \mathbf{m}',y\rangle}\prod_j K_j(e^y)^{-s'_j}\Psi_j(e^y)^{-t'_j}$, and $y\mapsto -y$, which keeps dy, carries it to $e^{\langle 2\sigma'\mathbf{1}-\mathbf{m}',y\rangle}\prod_j K_j(e^y)^{-t'_j}\Psi_j(e^y)^{-s'_j}$, σ' the total strength of $(\mathbf{s}',\mathbf{t}')$. Applied to the second half at $(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ it gives the first half at $\big(n(2\sigma\mathbf{1}-\mathbf{m});n\mathbf{t},n\mathbf{s}\big)$, the depth n of the data $(2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})$ of 8.13, whose wall of S is the wall of $\mathcal{E}\setminus S$ at $(\mathbf{m};\mathbf{s},\mathbf{t})$, positive. At the centre these data are $(\mathbf{m};\mathbf{s},\mathbf{t})$ themselves, and the two halves are equal.

(ii) By 38.1, with $X(z)=2\langle\boldsymbol{\mu}_\perp,z\rangle-\Phi^\circ(z)$ read on $\mathbf{1}^\perp$ and the number of marks written $|\mathcal{E}|$, $P_n=2^{|\mathcal{E}|-1}\int_{\mathcal{T}}e^{nX}dz$ and the contrast word is $2^{|\mathcal{E}|-2}\int_{\mathcal{T}}e^{nX}\big(c^{2h}+c^{*2h}\big)dz$, with c,c^* at $\alpha=e^{2z}$. For $h=\eta n$ put $\psi=-X-2\eta\log\max(|c|,|c^*|)$, with values in $(-\infty,\infty]$, continuous where finite and lower semicontinuous; then $e^{-n\psi}\le e^{nX}(c^{2h}+c^{*2h})\le 2e^{-n\psi}$. By 38.4(i) and (ii), $-X(z)\ge 2\kappa_{\min}(\max_e z_e-\min_e z_e)-C$, and the logarithmic term is nonnegative, so ψ grows at least linearly on $\mathbf{1}^\perp$. Laplace's principle follows: outside a large ball the integral of $e^{-n\psi}$ is at most $e^{-n(\inf\psi+1)}$ for large n, inside it is at most $e^{-n\inf\psi}$ times the volume, and near a point z_0 with $\psi(z_0)<\inf\psi+\epsilon$ the integrand is at least $e^{-n(\inf\psi+2\epsilon)}$ on a ball. So $\frac{1}{n}\log$ of the contrast word tends to $-\inf\psi$, and the same with $\eta=0$ and without the contrasts gives $\frac{1}{n}\log P_n\to\max X=-\min(\Phi^\circ-2\langle\boldsymbol{\mu}_\perp,\cdot\rangle)$, the maximum attained, X being continuous and tending to $-\infty$. The set $K=\{z\in\mathbf{1}^\perp:\ 2\kappa_{\min}(\max_e z_e-\min_e z_e)-C\le\inf\psi+1\}$ is compact and contains every z with $\psi(z)\le\inf\psi+1$; on it $\max(|c|,|c^*|)\le 1-\delta_K<1$, so $\inf\psi\ge\min(-X)+2\eta|\log(1-\delta_K)|>\min(-X)$. For $A>0$ let B_A be a ball of $\mathbf{1}^\perp$ outside which $2\kappa_{\min}(\max_e z_e-\min_e z_e)-C\ge A$; on B_A, $\psi\ge\min(-X)+2\eta|\log(1-\delta_{B_A})|$, which exceeds A for large η, so $\inf\psi\ge A$ for large η.

(iii) Let $s_j>0$ and $q=2h$. In the share form of P_F used by 17.15, the letter $K_j^{-ns_j-q}$ of the term l of (i) is opened by one share a'_U per basis, $\sum_U a'_U=ns_j+q$, with the share kernel $\prod_U\Gamma(a'_U)/\Gamma(ns_j+q)$ and the factors $(\theta_{j,U}\alpha^{\mathbf{1}_U})^{-a'_U}$ (4.4). The insertion $\theta_{j,T}^{q-l}\theta_{j,T'}^{l}\alpha^{v_l}$ turns the factors of T and T' into $(\theta_{j,T}\alpha^{\mathbf{1}_T})^{-a_T}$ and $(\theta_{j,T'}\alpha^{\mathbf{1}_{T'}})^{-a_{T'}}$ with $a_T=a'_T-q+l$, $a_{T'}=a'_{T'}-l$, the other shares unchanged as $a_U=a'_U$. In the shares a the total is ns_j and the factors in α and in the weights are those of $P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$, so the integral over α, which produces the pins, gives the pins of $(n\mathbf{m};n\mathbf{s},n\mathbf{t})$, and the kernel is

$$\frac{\Gamma(a_T+q-l)\,\Gamma(a_{T'}+l)\prod_{U\neq T,T'}\Gamma(a_U)}{\Gamma(ns_j+q)}=\frac{\prod_U\Gamma(a_U)}{\Gamma(ns_j)}\cdot\frac{(a_T)_{q-l}\,(a_{T'})_l}{(ns_j)_q}.$$

The factor $\Gamma(a_T+q-l)\Gamma(a_{T'}+l)$ is holomorphic for $\operatorname{Re}a_T>-(q-l)$ and $\operatorname{Re}a_{T'}>-l$, so the contour returns to a wall contour of $(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ without crossing a pole, the share kernels decaying along the imaginary directions. Summing over l with the weights $\frac{1}{2}(-1)^l\binom{q}{l}$ gives the share side of $P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ with the insertion $H_q(a_{j,T},a_{j,T'})/\big(2(ns_j)_q\big)$. For $t_j>0$ the second half is the same with Ψ_j, the cobasis shares b and nt_j. The insertions are polynomials in the shares and change neither the pins nor the share kernels, hence no pole hyperplane. At the centre the two halves are equal by (i), and their insertions add to the one of the reference. □

Theorem 32.14 (A positive sequence of Beta ports on an interval of a selected mark). *As stated, at every mass $m_e\in(0,1]$.*

Proof. In x, $d\alpha_e=(\lambda_1-\lambda_0)\,dx$, and the constant goes into the room density. The factor of the mass is $\alpha_e^{m_e-1}=L_0^{-(1-m_e)}$ with $L_0=\lambda_0+(\lambda_1-\lambda_0)x$, a linear letter with $c_0=\lambda_0$, $d_0=\lambda_1-\lambda_0>0$ and the exponent $w_0=1-m_e\ge 0$; it satisfies $c_0\ge\lambda_0 d_0/(\lambda_1-\lambda_0)$ with equality, so the read is $\langle\cdot\rangle_{1,1;\mathbf{w}}$ over the letters of the chart and L_0. For a letter with $d_i>0$ put $\mu=\lambda_0/(\lambda_1-\lambda_0)$; then $x(1-x)d_i/L_i\le x(1-x)/(\mu+x)$, whose derivative vanishes on $[0,1]$ only where $x^2+2\mu x-\mu=0$, at $x=\sqrt{\mu(1+\mu)}-\mu$, with the value $\big(\sqrt{1+\mu}-\sqrt{\mu}\big)^2=(\sqrt{\lambda_1}-\sqrt{\lambda_0})^2/(\lambda_1-\lambda_0)=\varepsilon_{01}$, the ratio

of the statement; for $d_i = 0$ the quotient is 0. By 32.12(i) at $a = b = 1$,

$$\big\langle (-1)^n \Pi_n^{1,1} \big\rangle_{1,1;\mathbf{w}} = \sum_{|v|=n} \prod_i \frac{(w_i)_{v_i}}{v_i!} \Big\langle \prod_i \Big(\frac{x(1-x)d_i}{L_i} \Big)^{v_i} \Big\rangle_{1,1;\mathbf{w}},$$

every term between 0 and $\prod_i \frac{(w_i)_{v_i}}{v_i!} \varepsilon_{01}^n \langle 1 \rangle_{1,1;\mathbf{w}}$, and $\sum_{|v|=n} \prod_i (w_i)_{v_i}/v_i! = (|\mathbf{w}|)_n/n!$ by the Vandermonde identity for rising factorials. A letter L_i that sees the mark makes the term $v = n\mathbf{e}_i$ positive (32.12(ii) with $J = \{i\}$), and for $n = 0$ the read is $\langle 1 \rangle_{1,1;\mathbf{w}} > 0$. When $m_e < 1$, $w_0 d_0 = (1 - m_e)(\lambda_1 - \lambda_0) > 0$: the letter of the mass sees the mark. □

Theorem 32.15 (The finite-band recurrence in the order of the Beta ports). *As stated, at every pair (a_n, b_n) with $a_n, b_n > 0$.*

Proof. Write $a = a_n$, $b = b_n$, $w_i = w_i(n)$. The letters are at least $\min_i c_i > 0$ on $[0,1]$, so every read of a polynomial converges and the flux $x^a (1-x)^b A \prod_i L_i^{-w_i}$ of a polynomial A vanishes at both ends. 32.10(a) with the polynomial Lg reads, by $x(1-x)\partial_x(Lg) = x(1-x)(L\partial_x g + g\partial_x L)$ and $\partial_x \log L_i = d_i/L_i$,

$$x(1-x)\partial_x(Lg) + \big(a - (a+b)x\big) Lg - x(1-x) Lg \sum_i \frac{w_i d_i}{L_i} = L\, x(1-x)\, \partial_x g + A_n g.$$

The coefficient of x^q in $\Pi_q^{a,b}$ is $\binom{a+b+2q-2}{q}$, a product of the q positive numbers $a+b+q-1, \dots, a+b+2q-2$ over $q!$, so $\Pi_0^{a,b}, \dots, \Pi_N^{a,b}$ are a basis of the polynomials of degree at most N, and the conditions $a+b+2q-i \neq 0$ of 3.15(v) hold. By its finite rules, multiplication by x maps $\Pi_q^{a,b}$ into the span of $\Pi_{q-1}^{a,b}, \Pi_q^{a,b}, \Pi_{q+1}^{a,b}$, and $x(1-x)\partial_x$ maps it into the span of $\Pi_{q-1}^{a,b}, \Pi_q^{a,b}$ and $x\Pi_q^{a,b}$, with coefficients rational in (a,b,q); for $q = 0$, $x\Pi_0^{a,b} = x$ and $\partial_x \Pi_0^{a,b} = 0$. So multiplication by a polynomial of degree d moves $\Pi_q^{a,b}$ into the span of $\Pi_{q-d}^{a,b}, \dots, \Pi_{q+d}^{a,b}$. L has degree h and A_n degree at most $h+1$, so at $g = (-1)^r \Pi_r^{a,b}$,

$$L\, x(1-x)\, \partial_x g + A_n g = \sum_{s=-h-1}^{h+1} e_s\, \Pi_{r+s}^{a,b},$$

$e_s = 0$ when $r + s < 0$, with e_s rational in (a,b,r) and polynomial in the coefficients of L and of A_n, hence in the room data and the w_i; the mark relation is the displayed relation with $\mathfrak{b}_{n,r,s} = (-1)^{r+s} e_s$.

$\Pi_{r+h+1}^{a,b}$ is the only member of the expansion of degree $r+h+1$, so e_{h+1} is the coefficient of x^{r+h+1} divided by $\binom{a+b+2r+2h}{r+h+1}$. The coefficient of x^r in g is $(-1)^r \binom{a+b+2r-2}{r}$. The coefficient of x^{h+1} in A_n is $-(a+b)p_h - hp_h + |\mathbf{w}(n)|\, p_h$, the last from $-x(1-x)\sum_i w_i d_i \prod_{i' \neq i} L_{i'}$, whose leading coefficient is $\sum_i w_i d_i \prod_{i' \neq i} d_{i'} = |\mathbf{w}(n)|\, p_h$; and the coefficient of x^{r+h+1} in $L\, x(1-x)\, \partial_x x^r$ is $-rp_h$. So the coefficient of x^{r+h+1} is $(-1)^r \binom{a+b+2r-2}{r} p_h \big(|\mathbf{w}(n)| - a - b - h - r\big)$, and $\mathfrak{b}_{n,r,h+1} = (-1)^{r+h+1} e_{h+1}$ is the stated value. □

At $a_n = b_n = n+1$ the furthest coefficient is

$$\big[|\mathbf{w}(n)| - 2n - 2 - h - r\big]\, p_h\, (-1)^{h+1} \binom{2n+2r}{r} \Big/ \binom{2n+2r+2h+2}{r+h+1},$$

that of the reference.

40.32 Chapter 34: the gap and the shape words

Proposition 34.8 (The root rate of the shape words). *As stated, with the maximizer at every $k \ge 2$.*

Proof. (i), the limit in (ii) and the last sentence are the reference's (the reference, proof of prop:kdepth). For the maximum, fix $x_1, \dots, x_k$ and put $y = x_1 \cdots x_k$. The factor of f_k in x_0 is $x_0(1-x_0)/\big(1-(1-y)x_0\big)$; its derivative vanishes where $1 - 2x_0 + (1-y)x_0^2 = 0$, at $x_0 = 1/(1+\sqrt{y})$, where $1 - x_0 = \sqrt{y}/(1+\sqrt{y})$ and $1-(1-y)x_0 = \sqrt{y}$, with the value $(1+\sqrt{y})^{-2}$. So $M_k(0) = \sup \prod_{i\ge1} x_i(1-x_i)/\big(1+\sqrt{x_1\cdots x_k}\big)^2$. At a fixed product the numerator is largest when the x_i are equal, $t \mapsto \log(1-e^t)$ being concave, so $M_k(0) = \max_{0<a<1} \phi(a)$ with $\phi(a) = a^k(1-a)^k/(1+a^{k/2})^2$, and

$$\frac{\phi'(a)}{\phi(a)} = k\Big(\frac{1}{a} - \frac{1}{1-a} - \frac{a^{k/2-1}}{1+a^{k/2}}\Big) = \frac{k\,\big(1-2a-a^{k/2+1}\big)}{a(1-a)(1+a^{k/2})}.$$

The numerator decreases strictly from 1 at $a=0$ to $-2^{-k/2-1}$ at $a=\frac12$, so ϕ increases up to the unique root $a \in (0,\frac12)$ of $a^{k/2+1} + 2a = 1$ and decreases after it. At $\eta = 0$ the equation of 34.19 is $(1-2a)^2 = a^{k+2}$, whose root in $(0,\frac12)$ solves $1-2a = a^{k/2+1}$: the same a, with $x_2 = \cdots = x_k = a$ and $x_0 = a/(1-a) = 1/(1+a^{k/2})$, since $a^{k/2} = (1-2a)/a$. The interval follows from $\phi(\frac12) \le M_k(0) \le 4^{-k}$. At $k=2$ the root is $\sqrt2 - 1$ and at $k=4$ the root of a^3+2a-1, the maximizers of the reference. □

Proposition 34.14 (The insertion index supplies auxiliary states). *As stated, for every $k \ge 3$.*

Proof. By 34.13(ii), which holds for every $k \ge 2$, the share function of P_r is

$$g_{n,r}(a) = \binom{n+r}{r} \frac{(n!)^{k-2}\,(1-a)_n^2\,(n+1-a)_r}{(a)_{n+1}^k\,(n+1+a)_r}.$$

Its poles in $a \le 0$ are at $a = -h$, $0 \le h \le n$, of order exactly k, the numerator being nonzero there, and at $a=-h$, $n < h \le n+r$, simple. For $k \ge 3$ the coefficients of order $k-1 \ge 2$ come from the first poles only, so $C_{n,k}(P_r) = \sum_{h=0}^n R_h(-h)$ and $C_{n,k-1}(P_r) = \sum_{h=0}^n R_h'(-h)$, with $R_h(a) = (a+h)^k g_{n,r}(a)$, holomorphic near $-h$. First,

$$R_h(-h) = \binom{n+r}{r} \frac{(n!)^{k-2}(1+h)_n^2}{\prod_{j\ne h}(j-h)^k} \cdot \frac{(n+1+h)_r}{(n+1-h)_r}, \qquad \frac{(n+1+h)_r}{(n+1-h)_r} = \frac{(n+1-h+r)_{2h}}{(n+1-h)_{2h}},$$

the product over $0 \le j \le n$; $\binom{n+r}{r} = (r+1)_n/n!$ and $(n+1-h+r)_{2h}$ are polynomials in r, $n+1-h \ge 1$, so $R_h(-h) \in \mathbb{Q}[r]$ and $C_{n,k}(P_r) \in \mathbb{Q}[r]$. Second, $R_h'(-h) = R_h(-h)\,\rho_h$ with

$$\rho_h = -2\sum_{i=1}^{n} \frac{1}{i+h} - \sum_{i=n+1}^{n+r} \Big(\frac{1}{i+h} + \frac{1}{i-h}\Big) - k\sum_{j\ne h} \frac{1}{j-h},$$

the logarithmic derivatives at $-h$ of $(1-a)_n^2$, $(n+1-a)_r$, $(n+1+a)_r^{-1}$ and $\prod_{j\ne h}(a+j)^{-k}$. The finite Trailmoment $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N])$ is the sum of m^{-1} over $0 < m \le N$, so $\sum_{i=n+1}^{n+r} \frac{1}{i+h} = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+h]) + \sum_{i=1}^{h} \frac{1}{n+r+i}$ and $\sum_{i=n+1}^{n+r} \frac{1}{i-h} = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n-h]) - \sum_{i=0}^{h-1} \frac{1}{n+r-i}$, and $\rho_h = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+$

$r]) + \tilde{\rho}_h(r)$ with $\tilde{\rho}_h \in \mathbb{Q}(r)$. Summing, $C_{n,k-1}(P_r) = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r) + q_n(r)$ with $q_n = \sum_h R_h(-h)\tilde{\rho}_h \in \mathbb{Q}(r)$.

Let $\sum_{i=0}^{d} b_i(r)\,C_{n,k-1}(P_{r+i}) = 0$ for all large r, $b_i \in \mathbb{Q}(r)$. Since $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r+i]) = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) + \sum_{l=1}^{i} \frac{1}{n+r+l}$, this reads $A(r)\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) + B(r) = 0$ for all large r with $A = -2\sum_i b_i(r) C_{n,k}(P_{r+i})$ and B in $\mathbb{Q}(r)$. If $A \neq 0$, then $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) = -B(r)/A(r)$ for all large r; a rational function is asymptotic to cr^e with $e \in \mathbb{Z}$, while $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - \log r$ has a limit, which is impossible. So $A = 0$: the recurrence annihilates $C_{n,k}(P_r)$. □

Theorem 34.20 (An arithmetic budget for growing Beta ports). *As stated, with the coefficient of the top coordinate at every $k \geq 2$.*

Proof. The two limits and the clearing are the reference's (the reference, proof of thm:portbudget). The coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ in $I_n(k;P)$ is $k\,C_{n,k}(P)$ (34.5(iii)), and $P = (-1)^q \Pi_q^{n+1,n+1}(x_1)$ is free of x_0, so the theorem applies. As in the proof of 34.14, at $r = q$, $C_{n,k}(P) = \sum_{h=0}^{n} R_h(-h)$ with $\prod_{j\neq h}(j-h) = (-1)^h h!\,(n-h)!$ and $(1+h)_n = (n+h)!/h!$:

$$\begin{aligned} R_h(-h) &= \binom{n+q}{q} \frac{(n!)^{k-2}}{(h!(n-h)!)^k} \frac{((n+h)!)^2}{(h!)^2} (-1)^{hk} \frac{(n+1+h)_q}{(n+1-h)_q} \\ &= \binom{n+q}{q} (-1)^{hk} \binom{n}{h}^k \binom{n+h}{n}^2 \frac{(n+1+h)_q}{(n+1-h)_q}. \end{aligned}$$

For even k every term is positive. □

Corollary 34.21 (A growing Beta port with a certified gap). *As stated, for every $k \geq 2$.*

Proof. The port factor is at most one, since $x_0 x_1 \bar{x}_1 x_2 \cdots x_k \leq x_0 x_1 \cdots x_k \leq L$, so $M_k(\eta) \leq M_k(0)$ for $\eta \geq 0$ and

$$\ell_k(\eta) + (k+1)(1+\eta)\,c_{\mathrm{lcm}} \leq u(\eta) = \log M_k(0) + (1+\eta)\log(1+\eta) - \eta \log \eta + (k+1)(1+\eta)\,c_{\mathrm{lcm}}.$$

u is continuous on $[0,1]$ with $u(0) = \log M_k(0) + (k+1)c_{\mathrm{lcm}} < 0$ (34.8), so $u < 0$ on $[0,\eta_0]$ for a rational $\eta_0 > 0$. At a rational η the equation of 34.19, cleared of denominators, is an integer polynomial equation in a with the unique root in $(0, \frac{1}{2+\eta})$; the difference of its two sides is 1 at $a = 0$ and $-(1+\eta)^k(2+\eta)^{-k-2} < 0$ at $a = \frac{1}{2+\eta}$, so bisection at rational points certifies the root in rational intervals as small as asked. The two displays of 34.20 give the bound, and its last part the sign of the coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ for even k. □

40.33 Chapter 35: rays, native gauges and tiers

The proofs of this chapter rest on one reading of the direction of a ray.

Lemma S13. *Let a ray of a family on $n \geq 2$ marks have $\kappa_{1,S} > 0$ for every nonempty proper S, with its stride f on the chart $y_{e_0} = 0$.*

(i) $\mathbf{m}_1 > 0$*, the exponents of the direction are not all zero, and the tables with $s_{1j} + t_{1j} > 0$ form a connected family.*

(ii) The direction has a native gauge $\alpha^\star \in \mathbb{R}^{\mathcal{E}}_{>0}$, unique up to scale; with $\alpha^\star_{e_0} = 1$, $0 < f \leq f(\alpha^\star)$ on the chart with equality only at $\log \alpha^\star$, and $f \leq C' e^{-c'\|y\|}$ there for some $c', C' > 0$.

(iii) The Hessian of $-\log f$ at $\log\alpha^\star$ on the chart is

$$H = \Big(\sum_j \big(s_{1j}\, \kappa_{j,\alpha^\star}(X_e, X_{e'}) + t_{1j}\, \kappa_{j,(\alpha^\star)^{-1}}(X_e, X_{e'}) \big) \Big)_{e,e' \neq e_0},$$

and it is positive definite.

(iv) $\langle 1 \rangle_k = P_F(\mathbf{m}_0 + k\mathbf{m}_1; \mathbf{s}_0 + k\mathbf{s}_1, \mathbf{t}_0 + k\mathbf{t}_1)$, and it converges absolutely exactly when $\operatorname{Re}\kappa_{0,S} + k\kappa_{1,S} > 0$ for every nonempty proper S.

Proof. (i) Every $\{e\}$ is proper, and $\kappa_{1,\{e\}} = m_{1,e} - \sum_j \big(s_{1j} d_j(\{e\}) + t_{1j} d_j^*(\{e\}) \big)$ with $s_{1j}, t_{1j} \in \mathbb{N}$ and $d_j, d_j^* \geq 0$, so $m_{1,e} \geq \kappa_{1,\{e\}} > 0$. If every exponent of the direction were 0, the balance wall would give $m_{1,\mathcal{E}} = 0$. On the balance wall 8.15 gives $\kappa_{1,S} + \kappa_{1,\mathcal{E}\setminus S} = \sum_j (s_{1j} + t_{1j}) c_j(S)$, so every nonempty proper S has $c_j(S) > 0$ for a table with $s_{1j} + t_{1j} > 0$.

(ii) The direction is real data on its balance wall, where the target is the masses, with nonnegative exponents not all zero and positive masses. By (i) and 8.18(iii) it has a native gauge $\alpha^\star \in \mathbb{R}^{\mathcal{E}}_{>0}$, unique up to scale, the unique point up to scale at which $\alpha^{\mathbf{m}_1} \prod_j K_j^{-s_{1j}} \Psi_j^{-t_{1j}} = f$ is largest. At $y = 2z$ the stride is $e^{X_1(z)}$, X_1 the exponent of the question of the direction (38.1), and 38.4(i) and (ii) give $X_1(z) \leq -2\kappa_{1,\min}(\max_e z_e - \min_e z_e) + C$; on the chart $|y_e| \leq \max y - \min y$, since $y_{e_0} = 0$.

(iii) The Hessian of $\log K_j(e^y)$ in y is the covariance of the indicators of the basis under the basis law at $\alpha = e^y$ (7.14), and $\Psi_j(e^y) = e^{\langle \mathbf{1}, y \rangle} K_j(e^{-y})$, so the Hessian of $\log \Psi_j(e^y)$ is the covariance under the law at α^{-1}; at $\alpha^\star$ this gives H. For v with $v_{e_0} = 0$,

$$v^{\mathsf{T}} H v = \sum_j \Big(s_{1j} \operatorname{Var}_{\mathbb{P}_{j,\alpha^\star}} \langle \mathbf{1}_T, v \rangle + t_{1j} \operatorname{Var}_{\mathbb{P}_{j,(\alpha^\star)^{-1}}} \langle \mathbf{1}_T, v \rangle \Big) \geq 0,$$

both laws charging every basis. Let $v^{\mathsf{T}} H v = 0$ with $v \neq 0$. Then v is not constant, and $\langle \mathbf{1}_T, v \rangle$ is the same at every basis of every table with $s_{1j} + t_{1j} > 0$. Let $S = \{e : v_e = \max v\}$, nonempty and proper, and take by (i) such a table with $c_j(S) > 0$ and a basis T with $|T \cap S| < r_j(S)$. $T \cap S$ is independent in the restriction to S and not a basis of it, so some $f \in S \setminus T$ keeps it independent; the circuit of $T + f$ through f is not contained in S, and with e a mark of it outside S, $T - e + f$ is a basis with $\langle \mathbf{1}_{T-e+f}, v \rangle - \langle \mathbf{1}_T, v \rangle = v_f - v_e > 0$, a contradiction.

(iv) On the chart $\alpha_{e_0} = 1$, $\alpha^{\mathbf{m}-\mathbf{1}} \prod_{e \neq e_0} d\alpha_e = \alpha^{\mathbf{m}} dy$, so the value $\langle 1 \rangle_k = \int f^k \omega_0$ of 35.1 is the period at the point k of the ray in the form of 8.10, which converges exactly when the real parts of its proper walls $\kappa_{0,S} + k\kappa_{1,S}$ are positive; the modulus of its integrand is the integrand at the real parts of the data. □

Proposition 35.7 (The native value bounds the order). *As stated: every ray with positive direction walls, its origin complex.*

Proof. By Lemma S13(iv), $\langle 1 \rangle_k$ converges absolutely for $k \geq k_1$. The amplitude $g_0 = \omega_0 / dy = \alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}$, the powers of the positive letters principal, is continuous and nowhere zero on the chart. Let $y^\star = \log \alpha^\star$ with $\alpha^\star_{e_0} = 1$. By Lemma S13(ii) and (iii), $f / f(\alpha^\star)$ is real-analytic, has its unique maximum 1 at $y^\star$, nondegenerate with the Hessian H, and tends to 0 at infinity; so off a ball B about $y^\star$, $f \leq e^{-\delta} f(\alpha^\star)$ for some $\delta > 0$, and for $k \geq k_1$

$$\Big| \int_{y_{e_0} = 0,\ y \notin B} f^k g_0 \, dy \Big| \leq e^{-\delta(k-k_1)} f(\alpha^\star)^{k-k_1} \int f^{k_1} |g_0| \, dy.$$

On B, Laplace's method with the continuous amplitude g_0 gives

$$\int_B f^k g_0\,dy = f(\alpha^\star)^k\Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}}(\det H)^{-\frac12}\big(g_0(y^\star)+o(1)\big).$$

Hence $\langle 1\rangle_k = f(\alpha^\star)^k\big(\frac{2\pi}{k}\big)^{\frac{n-1}{2}}(\det H)^{-\frac12}g_0(y^\star)(1+o(1))$ with $g_0(y^\star)\neq 0$: $\langle 1\rangle_k\neq 0$ for all large k and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$. For a real origin $g_0>0$, and $\langle 1\rangle_k>0$ wherever it converges.

Let $\sum_{j=0}^{h} p_j(k)\langle 1\rangle_{k+j}=0$ for all large k, the $p_j\in\mathbb{Q}[k]$ not all zero, d their largest degree and c_j the coefficient of k^d in p_j. Dividing by $k^d\langle 1\rangle_k$ and letting $k\to\infty$ gives $\sum_j c_j f(\alpha^\star)^j=0$, and the symbol $\sum_j c_j\lambda^j$ is a nonzero rational polynomial of degree at most h. So $f(\alpha^\star)$ is algebraic, its minimal polynomial divides the symbol, every conjugate is a root of it, and $h\geq[\mathbb{Q}(f(\alpha^\star)):\mathbb{Q}]$. For $h=1$ this degree is one, and a transcendental native value admits no h. □

Definition 35.9 (The centre ray). As stated, for a family connected at strengths $\sigma_j\in\mathbb{N}$.

Proof. The direction $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$ is integral, and it is on the balance wall: $m_{1,\mathcal{E}}=n\sigma=\sum_j\sigma_j(r_j+r_j^*)$. The origin 0 is on it too. From $\Psi_j(\alpha)=\alpha^{\mathbf{1}}K_j(\alpha^{-1})$, $K_j\Psi_j=\alpha^{\mathbf{1}}K_j(\alpha)K_j(\alpha^{-1})$, and with $\sum_j\sigma_j=\sigma$ the stride is $\alpha^{\sigma\mathbf{1}}\prod_j(K_j\Psi_j)^{-\sigma_j}=\prod_j\big(K_j(e^y)K_j(e^{-y})\big)^{-\sigma_j}$ at $\alpha=e^y$. The origin form is dy, so $\langle 1\rangle_k=\int_{y_{e_0}=0}f^k dy=P_F(k\sigma\mathbf{1};k\boldsymbol{\sigma},k\boldsymbol{\sigma})$ by Lemma S13(iv). The direction walls are $\sigma|S|-\sum_j\sigma_j\big(d_j(S)+d_j^*(S)\big)=\sum_j\sigma_j c_j(S)$, positive on every nonempty proper S exactly when the family is connected at the strengths (8.21(ii)). □

Theorem 35.10 (A ray is one function). *As stated, for every ray with real origin and positive direction walls.*

Proof. (i) The bounds on f are Lemma S13(ii). For $k\geq k_0$ the walls $\kappa_{0,S}+k\kappa_{1,S}\geq\kappa_{0,S}+k_0\kappa_{1,S}$ are real and positive, so $\langle 1\rangle_k$ converges (Lemma S13(iv)), and its integrand $f^k\omega_0/dy$ is positive.

(ii) For $|q|<f(\alpha^\star)^{-1}$, $|qf|\leq|q|f(\alpha^\star)<1$ on the chart, so $\sum_{k\geq k_0}q^{k-k_0}f^k=f^{k_0}/(1-qf)$, the partial sums dominated by $f^{k_0}/(1-|q|f(\alpha^\star))$ times the positive density ω_0/dy, which is integrable at $k=k_0$. Summing under the integral gives the first form of G, and the second is the first with numerator and denominator multiplied by $\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$. On smaller discs the same domination allows differentiation under the integral, $\partial_q^l(1-qf)^{-1}=l!\,f^l(1-qf)^{-l-1}$. For real $q<f(\alpha^\star)^{-1}$, $1-qf\geq\min\big(1,1-qf(\alpha^\star)\big)>0$, so the integrand is positive and at most a multiple of $f^{k_0}\omega_0/dy$.

(iii) With the positive measure $\mu=f^k\omega_0$ on the chart, $\langle 1\rangle_{k+i+i'}=\int f^{i+i'}d\mu$, and the identity of Andréief gives

$$\det\Big[\int f^{i+i'}d\mu\Big]_{i,i'<h}=\frac{1}{h!}\int\det\big[f(y_a)^i\big]_{i<h,\,a\leq h}^2\prod_a d\mu(y_a)=\frac{1}{h!}\int\prod_{a<b}\big(f(y_a)-f(y_b)\big)^2\prod_a d\mu(y_a),$$

positive because f is real-analytic and not constant, so that the set where two values $f(y_a)$ agree is null. If $G=N_1/N_2$ were rational, $N_2(0)\neq 0$, $d=\deg N_2$, then $\sum_{i=0}^{d}[q^i]N_2\,\langle 1\rangle_{k_0+m-i}=0$ for $m>\deg N_1$, a relation with $[q^0]N_2\neq 0$ among the rows of the Hankel matrix $\big[\langle 1\rangle_{k+l+i'}\big]_{l,i'\leq d}$ for all large k, whose determinant would vanish, against (iii) at $h=d+1$.

On the centre ray $\kappa_{0,S}=0<\kappa_{1,S}$, so $k_0=1$, and $\alpha^\star=\mathbf{1}$ by 8.21(i). □

Theorem 35.11 (The branch at the native gauge of the direction). *As stated, for every ray with real origin and positive direction walls.*

Proof. Put $y^\star = \log \alpha^\star$, $\varphi = -\log\big(f/f(\alpha^\star)\big) \ge 0$ on the chart, zero only at $y^\star$, with the Hessian H there, positive definite (Lemma S13(iii)), and $g = f^{k_0}\omega_0/dy$, positive, real-analytic and integrable. At $q = A(1-\varepsilon)$, $1 - qf = 1 - (1-\varepsilon)e^{-\varphi}$, so for $0 < \varepsilon \le 1$

$$G(q) = \int \frac{g\,dy}{1-(1-\varepsilon)e^{-\varphi}}.$$

Fix a small $t_1 > 0$. On $\{\varphi \ge t_1\}$, $|1-(1-\varepsilon)e^{-\varphi}| \ge 1 - |1-\varepsilon|e^{-t_1} > 0$ for $|\varepsilon|$ small, so that part of the integral is holomorphic at $\varepsilon = 0$. Near $y^\star$ the real-analytic Morse lemma gives coordinates $v \in \mathbb{R}^{2a}$, $2a = n-1$, with $\varphi = \frac12|v|^2$ and $|\det \partial y/\partial v| = (\det H)^{-1/2}$ at $v = 0$; for t_1 small, $\{\varphi < t_1\}$ lies in their domain. With $\tilde g = g\,|\det \partial y/\partial v|$ and

$$\beta(t) = 2^{a-1}\int_{S^{2a-1}} \tilde g\big(\sqrt{2t}\,\omega\big)\,d\omega,$$

analytic near $[0, t_1]$, the spherical mean being an even analytic function of the radius, $\int_{\varphi<t} g\,dy$ has the derivative $t^{a-1}\beta(t)$. In $w = e^t - 1$, $1-(1-\varepsilon)e^{-t} = e^{-t}(w+\varepsilon)$ and $e^t dt = dw$, so the part over $\{\varphi < t_1\}$ is

$$\int_0^{w_1} \frac{w^{a-1}\gamma(w)}{w+\varepsilon}\,dw, \qquad \gamma(w) = \Big(\frac{\log(1+w)}{w}\Big)^{a-1}\beta\big(\log(1+w)\big), \quad \gamma(0) = \beta(0),$$

with $\gamma = \sum_m \gamma_m w^m$ on a disc of radius $R' > w_1$, t_1 being small, so that $|\gamma_m| \le CR'^{-m}$. For $c > 0$ not an integer and $|\varepsilon| < w_1$, $\varepsilon \notin (-\infty, 0]$,

$$\int_0^{w_1} \frac{w^{c-1}\,dw}{w+\varepsilon} = \frac{\pi}{\sin \pi c}\,\varepsilon^{c-1} - \sum_{l\ge 0} \frac{(-\varepsilon)^l w_1^{c-1-l}}{l+1-c},$$

from $\int_0^\infty w^{c-1}dw/(w+\varepsilon) = \frac{\pi}{\sin \pi c}\varepsilon^{c-1}$ at $0 < \operatorname{Re} c < 1$ and continuation in c; for an integer $c = N \ge 1$, $w^{N-1} = \big((w+\varepsilon)-\varepsilon\big)^{N-1}$ gives the integral as $(-1)^N\varepsilon^{N-1}\log\varepsilon$ plus $(-\varepsilon)^{N-1}\log(w_1+\varepsilon)$ plus a polynomial in ε, the two last of modulus $O\big((w_1+2\rho)^{N-1}\big)$ on $|\varepsilon| \le \rho$. Summed with the weights γ_m at $c = a+m$, the singular terms give $\varepsilon^{a-1}V(\varepsilon)$ for n even and $\varepsilon^{a-1}\log\varepsilon\,V(\varepsilon)$ for n odd, with

$$V(\varepsilon) = \frac{\pi}{\sin \pi a}\,\gamma(-\varepsilon) \quad (n \text{ even}), \qquad V(\varepsilon) = (-1)^a\gamma(-\varepsilon) \quad (n \text{ odd}),$$

and the remainders converge to a function holomorphic near 0, since $|l+1-a-m| \ge \frac12$ for n even and $w_1 + 2\rho < R'$ for small ρ. This is the form of G, with U the sum of the remainders and the part over $\{\varphi \ge t_1\}$. Further $\beta(0) = 2^{a-1}|S^{2a-1}|\,g(\alpha^\star)(\det H)^{-1/2} = (2\pi)^a g(\alpha^\star)/\big(\Gamma(a)\sqrt{\det H}\big)$, $\frac{\pi}{\sin \pi a} = \Gamma(a)\Gamma(1-a)$, and $g(\alpha^\star) = f(\alpha^\star)^{k_0}(\omega_0/dy)(\alpha^\star) = (f^{k_0-1}\omega_0/dy)(\alpha^\star)/A$, which gives $V(0)$.

$G^{(l)}(A) = l!\int f^{k_0+l}\omega_0/(1-Af)^{l+1}$ is a positive integral, and near $y^\star$, $1-Af = 1-e^{-\varphi}$ is comparable with $|v|^2$, so it is finite exactly when $2(l+1) < 2a$. For $l < a-1$, $G^{(l)}(q) \to G^{(l)}(A)$ as $q \uparrow A$ by monotone convergence, while the l-th derivative of $\varepsilon^{a-1}V$, resp. of $\varepsilon^{a-1}\log\varepsilon\,V$, tends to 0; so the Taylor coefficients of U of order $l < a-1$ are those of G at A, with $q - A = -A\varepsilon$, and the rest of U and $\varepsilon^{a-1}(V - V(0))$ are $O(\varepsilon^{\lceil a-1\rceil})$. On the centre ray $k_0 = 1$, $\alpha^\star = \mathbf{1} = (\alpha^\star)^{-1}$, $A = f(\mathbf{1})^{-1} = \prod_j K_j(\mathbf{1})^{2\sigma_j}$, $f^0\omega_0/dy = 1$ and $H = 2\sum_j \sigma_j\big(\kappa_{j,\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\ne e_0}$. □

Theorem 35.12 (Native gauges at the signs). *As stated, for the centre ray of a family.*

Proof. (i) On the chart $\alpha_{e_0}=1$, $f(\alpha)=\prod_j\big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$ (35.9), so $f(\alpha^{-1})=f(\alpha)$; $\alpha\mapsto\alpha^{-1}$ keeps the letter complement, since $K_j(\alpha^{-1})\Psi_j(\alpha^{-1})=\alpha^{-2\mathbf{1}}\Psi_j(\alpha)K_j(\alpha)$. A point of $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$ fixed by the inversion has α_e^2 independent of e, hence $\alpha_e=\pm1$ on the chart: the 2^{n-1} signs. By 35.2(ii) the native gauges of the ray are the critical points of $\log f$ on X_F, and the inversion, which keeps f, permutes them and keeps their native values.

(ii) At a sign η, $\Psi_j(\eta)=\eta^{\mathbf{1}}K_j(\eta)$, so $K_j(\eta)\neq0$ for every j puts η in X_F. In the chart $\alpha=\eta e^{y}$, $y_{e_0}=0$, $\alpha^{-1}=\eta e^{-y}$, so $y\mapsto\log f(\eta e^{y})$ is even and its gradient vanishes at $y=0$: η is a native gauge, of native value $\prod_j\big(K_j(\eta)K_j(\eta)\big)^{-\sigma_j}$. With $K_j(\eta e^{y})=\sum_T\theta_{j,T}\eta^T e^{\langle\mathbf{1}_T,y\rangle}=K_j(\eta)\sum_T\mathbb{P}_{j,\eta}(T)e^{\langle\mathbf{1}_T,y\rangle}$, the Hessian of $\log K_j(\eta e^{y})$ at 0 is the covariance of the indicators under the signed measure $\mathbb{P}_{j,\eta}$ of total mass one, and $\log K_j(\eta e^{-y})$ has the same Hessian; so the Hessian of $-\log f$ is $2\sum_j\sigma_j\kappa_{j,\eta}$.

(iii) $W^{-1}=\prod_j(K_j\Psi_j)^{\sigma_j}$ is a polynomial, and $\partial_{y_e}\log f=\big(\sigma W^{-1}-\alpha_e\partial_e W^{-1}\big)/W^{-1}$. For a sign η, $h(y)=W^{-1}(\eta e^{y})\,e^{-\sigma\langle\mathbf{1},y\rangle}=\prod_j\big(\eta^{\mathbf{1}}K_j(\eta e^{y})K_j(\eta e^{-y})\big)^{\sigma_j}$ is even in y, so at $y=0$

$$\partial_{y_e}h=\big(\alpha_e\partial_e W^{-1}-\sigma W^{-1}\big)(\eta)=0$$

at every sign, whether or not the $K_j(\eta)$ vanish, and $W^{-1}(\eta)=\eta^{\sigma\mathbf{1}}\prod_j K_j(\eta)^{2\sigma_j}$. The ring $\mathbb{Q}(\boldsymbol{\theta})[\alpha_e:e\neq e_0]/(\alpha_e^2-1)$ is the product of copies of $\mathbb{Q}(\boldsymbol{\theta})$ indexed by the signs, a class being its values at the signs, with the idempotents $2^{1-n}\prod_{e\neq e_0}(1+\eta_e\alpha_e)$, which take the value 1 at η and 0 at the other signs. So the critical polynomials vanish in it, and $W^{-1}\equiv\alpha^{\sigma\mathbf{1}}\prod_j K_j^{2\sigma_j}$. Localizing at $\prod_j K_j$ keeps exactly the factors of the signs with $\prod_j K_j(\eta)\neq0$, whose idempotents form a basis, and multiplication by $W^{-1}/\alpha^{\sigma\mathbf{1}}$, $\alpha^{\sigma\mathbf{1}}$ being a unit, acts on the idempotent of η by its value $\prod_j K_j(\eta)^{2\sigma_j}$. □

Theorem 35.13 (Weights that separate the signs). *As stated, for a family of N tables connected at its strengths.*

Proof. Treat the weights of all tables as variables. For a sign η, $K_j(\eta)=\sum_T\theta_{j,T}\eta^T$ is a nonzero linear form. Write $M_j=K_j(\eta)^2\big(\kappa_{j,\eta}(X_e,X_{e'})\big)_{e,e'\neq e_0}$, whose entries $K_j(\eta)\sum_{T\ni e,e'}\theta_{j,T}\eta^T-\sum_{T\ni e}\theta_{j,T}\eta^T\sum_{T\ni e'}\theta_{j,T}\eta^T$ are quadratic, and

$$D_\eta=\det\Big[\sum_j\sigma_j M_j\prod_{i\neq j}K_i(\eta)^2\Big]=\prod_i K_i(\eta)^{2(n-1)}\det\Big[\sum_j\sigma_j\kappa_{j,\eta}\Big],$$

of degree $2N(n-1)$. At $\eta=\mathbf{1}$ and unit weights $D_{\mathbf{1}}\neq0$: the Hessian $2\sum_j\sigma_j\kappa_{j,\mathbf{1}}$ is positive definite, by the argument of Lemma S13(iii) at the strengths. The substitution $\theta_{j,T}\mapsto\theta_{j,T}\eta^T$ carries $K_j(\mathbf{1})$ to $K_j(\eta)$ and $\mathbb{P}_{j,\mathbf{1}}$ to $\mathbb{P}_{j,\eta}$, so D_η is $D_{\mathbf{1}}$ after an invertible linear substitution, and $D_\eta\neq0$. For signs $\eta\neq\eta'$, $\prod_j K_j(\eta)^{2\sigma_j}-\prod_j K_j(\eta')^{2\sigma_j}$, of degree 2σ, is not zero: otherwise unique factorization, the linear forms $K_j(\eta)$ being irreducible and in disjoint variables for distinct j, makes $K_j(\eta)$ a multiple of $K_j(\eta')$ for every j with $\sigma_j>0$, that is $\eta^T\eta'^T$ constant on the bases of such a table; with $S=\{e:\eta_e\neq\eta'_e\}$, nonempty and proper since $\eta_{e_0}=\eta'_{e_0}$, the parity of $|T\cap S|$ is then constant, so the window $\{d_j(S),\dots,r_j(S)\}$ of 7.11 is one value and $c_j(S)=0$ for every j with $\sigma_j>0$, against connectedness. The product

$$\prod_\eta\Big(D_\eta\prod_j K_j(\eta)\Big)\prod_{\{\eta,\eta'\}}\Big(\prod_j K_j(\eta)^{2\sigma_j}-\prod_j K_j(\eta')^{2\sigma_j}\Big)$$

is a nonzero polynomial of degree $2^{n-1}\big(N+2N(n-1)\big)+2\sigma\binom{2^{n-1}}{2}=2^{n-1}\big((2n-1)N+(2^{n-1}-1)\sigma\big)$, so by the Schwartz–Zippel bound it does not vanish at some point of $\{1,\dots,D+1\}$ in every

variable, D that degree. At such weights every $K_j(\eta) \neq 0$, every sign is a native gauge by 35.12(ii), nondegenerate since $D_\eta \neq 0$, and the native values are distinct. 35.5(iii) bounds the dimension below by 2^{n-1}, and 35.6 at $g = 1$ the order of every recurrence that follows from the relations. □

Lemma 35.14 (Where native gauges can reach a wall). *As stated, for a family.*

Proof. (i) On the balance wall the target of the direction is $\mathbf{m}_1$, so nativity at α reads $\mathbf{m}_1 = \sum_j \big(s_{1j} R_j(\alpha) + t_{1j} R_j^*(\alpha)\big)$ (8.16). $R_{j,e} = \alpha_e \partial_e K_j / K_j = \sum_T \mathbf{1}_T(e)\, \mathbb{P}_{j,\alpha}(T)$ and $R^*_{j,e} = \sum_T \mathbf{1}_{\mathcal{E}\setminus T}(e)\, \mathbb{P}^*_{j,\alpha}(T)$, and summing over $e \in B$ gives (i).

(ii) In the coordinates (ρ, u, α_R), $\alpha^T = \rho^{|T\cap B|} u^{T\cap B} \alpha_R^{T\cap R}$, so $\sum_T |T \cap B|\, \theta_{j,T} \alpha^T = \rho\, \partial_\rho K_j$, and with $\widetilde{K}_j = \sum_l \rho^l K_j^{[l]}$, $\widetilde{\Psi}_j = \sum_l \rho^l \Psi_j^{[l]}$,

$$\kappa_{1,B} = \sum_j \Big(s_{1j}\, \frac{\rho\, \partial_\rho \widetilde{K}_j}{\widetilde{K}_j} + t_{1j}\, \frac{\rho\, \partial_\rho \widetilde{\Psi}_j}{\widetilde{\Psi}_j} \Big)$$

at every native gauge of the direction, by (i) and $\kappa_{1,B} = m_{1,B} - \sum_j \big(s_{1j} d_j(B) + t_{1j} d_j^*(B)\big)$. Along the sequence the exponents stay bounded, and $\rho_i \partial_\rho \widetilde{K}_j = O(\rho_i)$. If no bottom letter vanished at (u^*, α_R^*), every quotient would be $O(\rho_i)$ and the right side would tend to 0, while the left side tends to $\kappa_{1,B}(\mathbf{v}) > 0$. So some bottom letter vanishes there. Let it be the only one, say $K_{j_0}^{[0]}$. Then the terms of the other letters tend to 0, so $s_{1j_0}(\mathbf{v}_i)\, \rho_i \partial_\rho \widetilde{K}_{j_0} / \widetilde{K}_{j_0} \to \kappa_{1,B}(\mathbf{v})$, and since $\rho_i \partial_\rho \widetilde{K}_{j_0} \to 0$, $|s_{1j_0}(\mathbf{v}_i)/\widetilde{K}_{j_0}| \to \infty$. Nativity in a logarithmic coordinate D of u or of α_R reads

$$D \log \big(u^{\mathbf{m}_1|_B} \alpha_R^{\mathbf{m}_1|_R} \big) = \sum_j \Big(s_{1j} \frac{D\widetilde{K}_j}{\widetilde{K}_j} + t_{1j} \frac{D\widetilde{\Psi}_j}{\widetilde{\Psi}_j} \Big);$$

the left side and the terms of the other letters stay bounded, so $s_{1j_0} D\widetilde{K}_{j_0}/\widetilde{K}_{j_0}$ does, and $D\widetilde{K}_{j_0}(\rho_i, u_i, \alpha_{R,i}) \to D K_{j_0}^{[0]}(u^*, \alpha_R^*)$ must vanish. The bottom letter is homogeneous in u and in α_R, so with $K_{j_0}^{[0]}(u^*, \alpha_R^*) = 0$ Euler's identity gives the derivatives along the scalings of u and of α_R as well: the whole gradient vanishes. The same holds for a cobasis letter. If two or more bottom letters vanish, every term of the gradient of the product contains a vanishing factor; if one does, its gradient vanishes. In both cases the product vanishes at (u^*, α_R^*) with its gradient. □

Proposition 35.18 (Small lower coefficients on one parameter). *As stated, with p letters, for real C_i bounded below by a positive constant on $[0, \infty)$.*

Proof. For $\rho \geq 0$ and $\lambda > 0$, $\lambda\xi_i + \rho C_i(\rho)$ is $\lambda\xi_i$ plus a nonnegative number, so it stays in $\mathbb{C}\setminus(-\infty, 0]$, the powers are principal, and the integrand is continuous on $(0, \infty)$. Put $F_{\mathbf{a}} = G \prod_i C_i^{-w_i - a_i}$ and $c_{\mathbf{a}}(\lambda) = \prod_i \binom{-w_i}{a_i} (\lambda\xi_i)^{a_i}$, so that, wherever $|\lambda\xi_i| < |\rho C_i(\rho)|$ for every i,

$$\rho^{\kappa - 1} G \prod_i \big(\lambda\xi_i + \rho C_i\big)^{-w_i} = \rho^{\kappa^\flat - 1} F_{\mathbf{0}} \prod_i \Big(1 + \frac{\lambda\xi_i}{\rho C_i}\Big)^{-w_i} = \sum_{\mathbf{a}} c_{\mathbf{a}}(\lambda)\, \rho^{\kappa^\flat - |\mathbf{a}| - 1} F_{\mathbf{a}},$$

absolutely and uniformly where $|\lambda\xi_i| \leq \frac{1}{2}|\rho C_i(\rho)|$. Choose $b > 0$ with G and the C_i holomorphic and the C_i zero-free on a neighbourhood of $|\rho| \leq b$, and let $|C_i| \geq c > 0$ on $[0, \infty)$ and on $|\rho| \leq b$. For $|\lambda| \leq \lambda_0 = bc/(2 \max_i |\xi_i|)$ each letter $\lambda\xi_i + \rho C_i(\rho)$ has exactly one root in $|\rho| < b$ (Rouché, against $\rho C_i(\rho)$) and none on $|\rho| = b$ or on $[b, \infty)$.

The analytic part. Split the integral at b. On $[b,\infty)$ the expansion above holds uniformly and its terms are dominated by $\sum_{\mathbf{a}}\prod_i\binom{-w_i}{a_i}|2^{-|\mathbf{a}|}$ times $|\rho^{\kappa^\flat-1}F_{\mathbf{0}}|$, integrable at ∞, so $\int_b^\infty = \sum_{\mathbf{a}} c_{\mathbf{a}}(\lambda)\int_b^\infty \rho^{\kappa^\flat-|\mathbf{a}|-1}F_{\mathbf{a}}\,d\rho$, holomorphic for $|\lambda|<\lambda_0$. Let $\oint$ start at $\rho=b$ on the principal branches and run once around $|\rho|=b$; the integrand continues along it, and the expansion holds on the circle, each $\rho^{\kappa^\flat-|\mathbf{a}|-1}$ continued along it and $F_{\mathbf{a}}$ single-valued on the disc. For $\operatorname{Re}c'>0$ and F holomorphic near the closed disc,

$$\oint_{|\rho|=b}\rho^{c'-1}F\,d\rho=\big(e^{2\pi ic'}-1\big)\int_0^b\rho^{c'-1}F\,d\rho,$$

by Cauchy's theorem on the disc cut along $[0,b]$; both sides are holomorphic in c' off $-\mathbb{N}$, the right side read with the finite part $\operatorname{Reg}\int_0^b$ (1.4, 1.6), so the identity holds at every $c'\notin\mathbb{Z}$. At $c'=\kappa^\flat-|\mathbf{a}|$, $e^{2\pi ic'}=e^{2\pi i\kappa^\flat}\neq 1$. Hence

$$A_b(\lambda)=\frac{1}{e^{2\pi i\kappa^\flat}-1}\oint_{|\rho|=b}\rho^{\kappa-1}G\prod_i\big(\lambda\xi_i+\rho C_i\big)^{-w_i}d\rho=\sum_{\mathbf{a}}c_{\mathbf{a}}(\lambda)\,\operatorname{Reg}\int_0^b\rho^{\kappa^\flat-|\mathbf{a}|-1}F_{\mathbf{a}}\,d\rho$$

is holomorphic for $|\lambda|<\lambda_0$, the circle integral of an integrand holomorphic in λ along a contour that no root reaches, and the series converges there. Since $\operatorname{Reg}\int_0^\infty=\operatorname{Reg}\int_0^b+\int_b^\infty$, $A(\lambda)=A_b(\lambda)+\int_b^\infty\cdots$ is the series of the statement, holomorphic at 0.

The branch part. For $\lambda>0$ put $\rho=\lambda u$ in $\int_0^b$ and in the circle integral; $(\lambda X)^{-w}=\lambda^{-w}X^{-w}$ on the principal branches at $\lambda>0$, so both carry the factor $\lambda^{\kappa^\flat}$:

$$\int_0^b\cdots d\rho-A_b(\lambda)=\lambda^{\kappa^\flat}D_\lambda(b/\lambda),\qquad D_\lambda(R)=\int_0^R\psi_\lambda\,du-\frac{1}{e^{2\pi i\kappa^\flat}-1}\oint_{|u|=R}\psi_\lambda\,du,$$

with $\psi_\lambda(u)=u^{\kappa-1}G(\lambda u)\prod_i\big(\xi_i+uC_i(\lambda u)\big)^{-w_i}$. The roots of the letters in u are $\rho_i(\lambda)/\lambda$, where $\rho_i(\lambda)$ is the root in $|\rho|<b$; from $|\rho_iC_i(\rho_i)|=|\lambda\xi_i|$ they have modulus at most $|\xi_i|/c$. Take $R_0>2\max_i|\xi_i|/c$, so that $|uC_i(\lambda u)|>2|\xi_i|$ on $|u|=R$ for $R\geq R_0$. For $R_0\leq R\leq b/|\lambda|$, one turn around $|u|=R$ multiplies ψ_λ by $e^{2\pi i\kappa^\flat}$, the factor $u^{\kappa-1}$ giving $e^{2\pi i\kappa}$ and each letter, with one root inside, $e^{-2\pi iw_i}$; so $\frac{d}{dR}\oint_{|u|=R}\psi_\lambda\,du=\big[\psi_\lambda(Re^{i\vartheta})e^{i\vartheta}\big]_{\vartheta=0}^{2\pi}=(e^{2\pi i\kappa^\flat}-1)\psi_\lambda(R)$, and $\frac{d}{dR}D_\lambda(R)=0$. Thus $B(\lambda)=D_\lambda(R_0)$, with R_0 fixed, and $\int_0^\infty=A(\lambda)+\lambda^{\kappa^\flat}B(\lambda)$ for small $\lambda>0$. For complex λ near 0, ψ_λ is holomorphic in λ on $[0,R_0]$ and on $|u|=R_0$: there $|\lambda u|\leq|\lambda|R_0<b$, the letters do not vanish, $\xi_i+uC_i(\lambda u)$ stays off $(-\infty,0]$ on $[0,R_0]$ for λ small by compactness, and $u^{\kappa-1}$ is integrable at 0 since $\operatorname{Re}\kappa>0$. So B is holomorphic at 0, and

$$B(0)=G(0)\Big[\int_0^{R_0}u^{\kappa-1}\prod_i\big(\xi_i+uC_i(0)\big)^{-w_i}du-\frac{1}{e^{2\pi i\kappa^\flat}-1}\oint_{|u|=R_0}u^{\kappa-1}\prod_i\big(\xi_i+uC_i(0)\big)^{-w_i}du\Big],$$

independent of $R_0\geq 2\max_i|\xi_i|/c$ by the same derivative. When $\operatorname{Re}\kappa^\flat<0$ the circle integral is $O(R_0^{\operatorname{Re}\kappa^\flat})$ and the bracket tends to the convergent integral $\int_0^\infty$; the bracket is holomorphic in $(\kappa,\mathbf{w})$ off $\kappa^\flat\in\mathbb{Z}$, so it is the continuation of that integral in κ and $\mathbf{w}$. Where every $\operatorname{Re}w_i>0$, $C_i(0)>0$ gives $\prod_i(\xi_i+uC_i(0))^{-w_i}=\prod_i C_i(0)^{-w_i}\prod_i(u+\xi_i/C_i(0))^{-w_i}$, and 15.1 with $J=\{1,\dots,p\}$ and the coefficients $\xi_i/C_i(0)$ gives the stacked Gamma times the Breta.

(ii) When G and the C_i are holomorphic on a neighbourhood of $[0,\infty)$, the cluster port of 35.16 with $r_i=1$ and the lower coefficients $\lambda\xi_i$, at the b chosen above, is

$$\frac{1}{2\pi i}\Big[\big(e^{2\pi i\kappa^\flat}-1\big)\int_b^\infty\cdots d\rho+\oint_{|\rho|=b}\cdots d\rho\Big]=\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}\Big(\int_b^\infty\cdots d\rho+A_b(\lambda)\Big)=\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}\,A(\lambda),$$

and it does not depend on b (35.17(i)). □

Remark. With $p=1$, $\xi_1=1$ and $C=C_1$, $\int_0^\infty u^{\kappa-1}(1+uC(0))^{-w}du = C(0)^{-\kappa}\Gamma_{1}^{2}\left[\begin{matrix}\kappa,\ -\kappa^\flat\\ w\end{matrix}\right]$, the branch part of the reference.

Theorem 35.19 (The values along a ray). *As stated, over a subfield $\Bbbk$ of $\mathbb{C}$ containing the weights and the origin.*

Proof. (i) is the reference's (the reference, proof of thm:rayspace). The letters have their coefficients in $\Bbbk$, so X_F, the occupations and the stride are defined over $\Bbbk$, and with the origin in $\Bbbk$ and the direction integral the operators $c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e}$ map $\Bbbk[X_F]\otimes\Bbbk(k)$ to itself. Let $\mathcal{L}_\Bbbk$ be the $\Bbbk(k)$-span of their images and $M_\Bbbk=\big(\Bbbk[X_F]\otimes\Bbbk(k)\big)/\mathcal{L}_\Bbbk$. Since $\mathbb{C}[X_F]=\Bbbk[X_F]\otimes_\Bbbk\mathbb{C}$, $M_\Bbbk\otimes_{\Bbbk(k)}\mathbb{C}(k)$ is the quotient of 35.4, a cokernel commuting with the extension of scalars; so $\dim_{\Bbbk(k)}M_\Bbbk$ is the dimension d of the ray, finite by (i).

(ii) The classes of $g, fg, \dots, f^dg$ in $M_\Bbbk$ are $d+1$ vectors of a space of dimension d, so there are $p_0,\dots,p_d\in\Bbbk(k)$, not all zero, with $\sum_j p_j(k)\,f^jg\in\mathcal{L}_\Bbbk$; clearing denominators puts them in $\Bbbk[k]$. With h the largest j with $p_j\neq0$, $h\le d$, and 35.6 gives $\sum_{j\le h}p_j(k)\langle g\rangle_{k+j}=0$ for all large k.

(iii) Let k_1 exceed the integer zeros of p_h and the bound of (ii). For $k\ge k_1$, $\langle g\rangle_{k+h}=-\sum_{j<h}\frac{p_j(k)}{p_h(k)}\langle g\rangle_{k+j}$ with $p_j(k)/p_h(k)\in\Bbbk$, the values of polynomials over $\Bbbk$ at integers lying in $\Bbbk$; induction on k writes every $\langle g\rangle_k$, $k\ge k_1$, as a $\Bbbk$-combination of $\langle g\rangle_{k_1},\dots,\langle g\rangle_{k_1+h-1}$. □

Lemma 35.20 (Index by pole count). *As stated, over a subfield $\Bbbk$ of $\mathbb{C}$.*

Proof. The poles of r are rational, so its partial fractions have coefficients in $\Bbbk$: $r(k)=\sum_{h\ge1}\sum_{s=1}^{p}c_{h,s}(k+h)^{-s}$, finitely many h, $c_{h,s}\in\Bbbk$, with no polynomial part since $r\to0$, and $\sum_h c_{h,1}=0$ since $r(k)=O(k^{-2})$. 34.10 at $t_0=0$, whose identities are linear in the coefficients and hold at coefficients in any field, gives

$$\sum_{k\ge0}r(k)=\sum_{s\ge2}C_s\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))-\sum_{h,s}c_{h,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,h-1]),\qquad C_s=\sum_h c_{h,s}\in\Bbbk,$$

the index-one terms regularized and their coefficient $C_1=0$, and every $(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,h-1])$, a finite sum of m^{-s}, rational. If $r(-M-k)=-r(k)$, uniqueness of the partial fractions gives $(-1)^s c_{M-h,s}=-c_{h,s}$, so $C_s=(-1)^{s+1}C_s$ and $C_s=0$ for even s. For poles in one coset $-b+\mathbb{Z}$ the same computation runs in the lattice input $\mathbb{I}_{b+\mathbb{Z}}$, the sums $\sum_{k\ge0}(k+b+h)^{-s}$ being its Trailmoments at its cuts; with several cosets the index-one sums of the cosets are regularized one by one, and only their total coefficient vanishes, which leaves the differences of the index-one values of the cosets. The coefficients are in $\Bbbk$ throughout. □

40.34 Chapter 36: relations and separation

Proposition 36.4 (A finite matrix of mark relations). *As stated, for several letters $D_1,\dots,D_h$.*

Proof. In the coordinate x_i, with the other coordinates as room and the room density $\prod_{l\neq i}x_l^{a_l-1}(1-x_l)^{b_l-1}$, 32.10(a) applies to the letters D_k, positive on the closed box, at the exponents p_k+1, and to the polynomial AD. With $\partial_i D=\sum_k\partial_i D_k\prod_{k'\neq k}D_{k'}$ its expression is

$$x_i(1-x_i)(D\,\partial_i A + A\,\partial_i D) + \big(a_i - (a_i+b_i)x_i\big)AD - x_i(1-x_i)A\sum_k (p_k+1)\,\partial_i D_k \prod_{k'\neq k} D_{k'}$$
$$= D\big[x_i(1-x_i)\,\partial_i A + \big(a_i-(a_i+b_i)x_i\big)A\big] - x_i(1-x_i)\,A\sum_k p_k\,\partial_i D_k \prod_{k'\neq k} D_{k'} = \mathrm{mr}_i(A),$$

and the read is the stated integral. At $A = x^\beta$, $x_i(1-x_i)\partial_i x^\beta + (a_i-(a_i+b_i)x_i)x^\beta = \big((\beta_i + a_i) - (\beta_i + a_i + b_i)x_i\big)x^\beta$, and $x_i(1-x_i)x^\beta \sum_k p_k \partial_i D_k \prod_{k'\neq k} D_{k'} = (1-x_i)x^\beta \sum_\gamma D_\gamma^{(i)} x^\gamma$; expanding $D = \sum_\gamma D_\gamma x^\gamma$ gives the formula. Its coefficients are rational in (a_i, b_i) over the rational coefficients of the D_k and the p_k, so for a degree bound the vectors $\mathrm{mr}_i(x^\beta)$ form a finite matrix, rational at rational a_i, b_i; by linearity a numerator $\sum_i \mathrm{mr}_i(A_i)$ has the read 0. With one letter, $D_\gamma^{(i)} = p\gamma_i D_\gamma$, the formula of the reference. □

Proposition 36.11 (The critical polynomial block). *As stated, at every order $k \geq 1$.*

Proof. For $R(u) = P(2u)$, $R(\frac{1}{2}) = P(1)$ and Q_P is the polynomial $(R(u) - R(\frac{1}{2}))/(1-2u)$, so 33.4 at the order k gives the display. $Q_P^{(2k-1)}$ has rational coefficients and the moments of $X = \frac{1}{2} + iY$ are rational (33.2(iii)), so $h(P) \in \mathbb{Q}$. The value of $c + \mathrm{crit}_k[P(2u)]$ is $\big(c + h(P)\big) + (2^{2k+1} - 1)P(1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$; the coordinate map is onto $\mathbb{Q}^2$, the constant c giving the first coordinate and a constant P, for which $Q_P = 0$, the second; and $a + b\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)) = 0$ with $(a,b) \in \mathbb{Q}^2$ not zero forces $b \neq 0$ and $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)) = -a/b$. At the cut N, $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((N,\infty)) = (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)) - (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N])$, a rational head. □

40.35 Chapter 37: finite algebra for scale reads

The letters of this chapter have complex coefficients and no roots on $[0,\infty)$. Two facts carry the proofs. First, along $[0,\infty)$ a letter P_j has a continuous argument with limits at 0 and at ∞, so for $x \to \infty$, $P_j(x)^{-w_j} = C_j x^{-w_j \deg P_j}(1 + O(x^{-1}))$ with $C_j \neq 0$, and $|P_j^{-w_j}|$ is $|P_j|^{-\operatorname{Re} w_j}$ up to factors bounded above and below on $[0,\infty)$: a read converges absolutely exactly where it does for positive letters of the same degrees. Second, the reduction of 37.7 is algebraic in $\mathbb{F}'$ and does not use the values of the letters.

Definition 37.6 (Insertions and reads on one scale). As stated, for letters without roots on $[0,\infty)$.

Proof. $P_j(0) \neq 0$ and P_j does not vanish on $[0,\infty)$, so a continuous logarithm of P_j along $[0,\infty)$ starting at the principal logarithm of $P_j(0)$ exists and is unique, and it defines $P_j^{-w_j}$; on $(0,\infty)$ it is locally a holomorphic branch, so $\big(P_j^{-w_j}\big)' = -w_j P_j' P_j^{-w_j-1}$, and the product rule gives $\big(x^\kappa \prod_j P_j^{-w_j} g\big)' = x^{\kappa-1} \prod_j P_j^{-w_j}\,\eth g$. The squarefree decomposition and the gcd splitting are computed in $\mathbb{F}[x]$ by gcds and exact divisions, so the q_k lie in $\mathbb{F}[x]$, and they divide $P_1 \cdots P_N$, so none vanishes at 0 or on $[0,\infty)$. For $\xi_\ell \in \mathbb{C} \setminus (-\infty, 0]$, $x + \xi_\ell$ stays in $\mathbb{C} \setminus (-\infty, 0]$ for $x \geq 0$, so the continuous branch is the principal one, that of 15.1 and 37.1. □

Theorem 37.7 (Reduction by the scale exchange). *As stated, for letters without roots on $[0,\infty)$.*

Proof. Write $\mu_k = \sum_j m_{jk} w_j$, so that $\eth = x\frac{d}{dx} + \kappa - \sum_k \mu_k \frac{x q_k'}{q_k}$ and $\mathfrak{A} = \mathbb{F}'[x, x^{-1}, Q^{-1}]$. Every $R \in \mathfrak{A}$ is a Laurent polynomial in x plus principal parts at the q_k.

Three reductions. (a) If R has a pole of order $h+1 \geq 2$ at q_k with leading numerator s, $\deg s < \deg q_k$, then

$$\partial\big(r\,q_k^{-h}\big) = -(h+\mu_k)\,\frac{x q_k' r}{q_k^{h+1}} + O\big(q_k^{-h}\big),$$

plus simple poles at the other $q_{k'}$ and no polynomial part and no pole at 0, $r\,q_k^{-h}$ vanishing at ∞ and being regular at 0; xq_k' is prime to q_k, q_k being squarefree and $q_k(0) \neq 0$, so the extended Euclidean algorithm gives r with $-(h+\mu_k)xq_k'r \equiv s \bmod q_k$, one division by $h+\mu_k$. (b) If R has a pole x^{-r}, $r \geq 1$, at 0, $\partial(x^{-r}) = (\kappa - r)x^{-r} - \sum_k \mu_k x^{1-r} q_k'/q_k$, whose other terms have poles of order below r at 0, simple poles at the q_k and no polynomial part. (c) If the polynomial part of R has degree $a \geq 0$, $\partial(x^a) = \big(\kappa + a - \sum_j w_j \deg P_j\big)x^a$ plus a polynomial of degree below a plus simple poles at the q_k, since $xq_k'/q_k = \deg q_k + O(x^{-1})$ and $\sum_k \mu_k \deg q_k = \sum_j w_j \deg P_j$. Subtracting ∂ of a multiple at each step lowers the order of the pole, resp. the degree, and leaves $\sum_{i<D} c_i x^i/Q$ once the poles at the q_k are simple and there is no pole at 0 and no polynomial part; each step divides by one of the three numbers of the hypothesis.

Uniqueness. Let $\sum_i c_i x^i/Q = \partial g$ with $g \in \mathfrak{A}$, $g \neq 0$. A pole of order $h \geq 1$ of g at q_k gives ∂g a pole of order $h+1$ by (a), but the left side has simple poles; so g is a Laurent polynomial. A pole x^{-r} of g at 0 gives ∂g the pole $(\kappa - r)x^{-r}$, but $Q(0) \neq 0$; so g is a polynomial, of degree $a \geq 0$, and by (c) ∂g is of order x^a at ∞, while the left side is $O(x^{-1})$. So $g = 0$ and every $c_i = 0$: $\dim_{\mathbb{F}'} \mathfrak{A}/\partial\mathfrak{A} = D$.

Reads. Let the reads of R and of the x^i/Q converge absolutely, so that $\Phi = x^\kappa \prod_j P_j^{-w_j} g$ has the absolutely integrable derivative $x^{\kappa-1}\prod_j P_j^{-w_j}\big(R - \sum_i c_i x^i/Q\big)$ and limits at 0 and at ∞. With ν_0 the order of g at 0 and a its degree at ∞, $\Phi = c_0 x^{\kappa+\nu_0}(1+o(1))$ at 0 and $\Phi = c_\infty x^{\kappa+a-\sum_j w_j \deg P_j}(1+o(1))$ at ∞, with $c_0, c_\infty \neq 0$ by the first fact above; a limit exists only if the exponent has negative real part at ∞, positive at 0, or is 0. At 0: if $\nu_0 \leq -1$ the exponent $\kappa - |\nu_0|$ is not 0 by hypothesis, and if $\nu_0 \geq 0$ its real part is positive since the read of $1/Q$ converges, $\operatorname{Re}\kappa > 0$. At ∞: if $a \geq 0$ the exponent is not 0 by hypothesis, and if $a \leq -1$ its real part is negative since the read of x^{D-1}/Q converges, $\operatorname{Re}(\kappa - \sum_j w_j \deg P_j) < 1$. So both limits are 0, $\langle \partial g \rangle = 0$, and $\langle R \rangle = \sum_i c_i \langle x^i/Q \rangle$; at symbolic exponents the identity holds on the open set of absolute convergence and continues. For the linear letters, $Q = \prod_\ell (x + \xi_\ell)$ is squarefree, the x^i/Q, $i < |J|$, and the $(x+\xi_\ell)^{-1}$ span the same space by partial fractions, and the three numbers are $\kappa - r$, $\kappa + a - |\mathbf{w}|$ and $w_\ell + h$, nonzero in $\mathbb{K}$. □

Definition 37.8 (Deformation; masters). As stated, along every admissible direction.

Proof. Along the deformation the three numbers of 37.7 are $\kappa - r + \lambda_0\varepsilon$, $\big(\kappa + a - \sum_j w_j \deg P_j\big) + \varepsilon\big(\lambda_0 - \sum_j \lambda_j \deg P_j\big)$ and $(\mu_k + h) + \varepsilon \sum_j m_{jk}\lambda_j$, polynomials in ε with nonzero slopes, hence nonzero in $\mathbb{F}'(\lambda_0, \boldsymbol{\lambda})(\varepsilon)$. At large real ε the exponent of a read at 0 has real part growing like $\lambda_0\varepsilon$ and at ∞ falling like $(\sum_j \lambda_j \deg P_j - \lambda_0)\varepsilon$, the moduli of the powers being those of positive letters up to bounded factors; so finitely many reads converge absolutely together, the coefficients of the reductions have no pole there, and 37.7 gives the identities there, hence as meromorphic functions of ε, the reads continuing meromorphically by the expansions of the integrands at 0 and at ∞ (8.6). Q divides $P_1 \cdots P_N$. The conditions are strict inequalities and finitely many inequations, invariant under positive scaling, so the admissible directions form an open cone; $(\frac{1}{2}; \mathbf{1})$ is in it, since $\sum_j \deg P_j \geq 1 > \frac{1}{2}$ and $\sum_j m_{jk} \geq 1$ for every k, q_k dividing some P_j. □

Lemma 37.9 (Simple poles at a corner). *As stated, along every admissible direction and for letters without roots on* $[0, \infty)$.

Proof. The three reductions of the proof of 37.7 act on separate parts of an insertion: (a) on the poles of order at least two at one q_k, (b) on the pole at 0, (c) on the polynomial part, and each step feeds only the lower orders of its own part and the simple poles, which form the remainder $\sum_i c_i x^i/Q$. In (c) the step at the degree a divides by $\big(\kappa+a-\sum_j w_j \deg P_j\big)+\varepsilon\big(\lambda_0-\sum_j \lambda_j \deg P_j\big)$, which vanishes at $\varepsilon=0$ for at most one a; in (b) the step at r divides by $\kappa-r+\lambda_0\varepsilon$, vanishing at 0 for at most one r; in (a), for each k, the step at h divides by $\mu_k+h+\varepsilon\sum_j m_{jk}\lambda_j$, vanishing at 0 for at most one h. Along each chain every other division is by a number holomorphic and nonzero near $\varepsilon=0$. So for R holomorphic in ε, whose principal parts are then holomorphic in ε, every coefficient is a sum of terms with at most one division vanishing at $\varepsilon=0$: at most a simple pole.

For a master write $H_i=\int_0^1+\int_1^\infty$. At 0 the integrand is $x^{\kappa+\lambda_0\varepsilon-1}\phi_\varepsilon(x)$ with ϕ_ε analytic at 0 and holomorphic in ε, and 8.6 gives poles at $\kappa+\lambda_0\varepsilon+l=0$, $l\in\mathbb{N}$, all simple and at distinct points, at most one of them at $\varepsilon=0$. At ∞, by the first fact above, the integrand is $x^{e(\varepsilon)-1}$ times a power series in x^{-1} holomorphic in ε, $e(\varepsilon)=\kappa-\sum_j w_j\deg P_j+i-D+\varepsilon\big(\lambda_0-\sum_j\lambda_j\deg P_j\big)$, and $x\mapsto 1/x$ gives simple poles at $e(\varepsilon)=l$, $l\in\mathbb{N}$, at most one at $\varepsilon=0$. So H_i has at most a simple pole there. □

Theorem 37.10 (Finitely many numbers at every corner). *As stated, along every admissible direction and for letters without roots on* $[0,\infty)$.

Proof. Absolute convergence of $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ is a strict inequality at 0 and one at ∞, so the deformed read converges absolutely for ε near 0 and is holomorphic there, with the value $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ at 0. By 37.8 it equals $\sum_i c_i(\varepsilon)H_i(\varepsilon)$ as a meromorphic function, and by 37.9 each product has at most a double pole, with the constant term $c_{i,-1}h_{i,1}+c_{i,0}h_{i,0}+c_{i,1}h_{i,-1}$. The constant term of a function holomorphic at 0 is its value. The masters do not involve R. The derivative ∂_ε^b of $x^{\lambda_0\varepsilon}\prod_j P_j^{-\varepsilon\lambda_j}$ is $\big(\lambda_0\log x-\sum_j\lambda_j\log P_j\big)^b$ times it, the logarithms continuous along $[0,\infty)$, so the read of $R\big(\lambda_0\log x-\sum_j\lambda_j\log P_j\big)^b$ is $b!$ times the coefficient of ε^b of $\sum_i c_iH_i$, which uses $h_{i,-1},\dots,h_{i,b+1}$: $(b+3)D$ numbers. For the last sentence write $\ell_0=\log x$ and $\ell_j=\log P_j$. The b-th powers of the linear forms $\lambda_0\ell_0-\sum_j\lambda_j\ell_j$, for $(\lambda_0;\boldsymbol{\lambda})$ in a dense subset of a nonempty open set, span the forms of degree b in $(\ell_0,\dots,\ell_N)$: a linear functional vanishing on them is a polynomial in $(\lambda_0;\boldsymbol{\lambda})$ vanishing on a dense subset of an open set, hence zero, and its coefficients are, up to multinomial factors, its values on the monomials. The rational admissible directions are such a set, and a linear system with rational coefficients that has a solution has a rational one; so $\ell_0^{k_0}\prod_j\ell_j^{k_j}$ is a rational combination of finitely many such powers, and its read, absolutely convergent with that of R, is the same combination of reads along these directions. □

Theorem 37.11 (Integer rays have certified recurrences). *As stated, for letters without roots on* $[0,\infty)$.

Proof. Over $\mathbb{F}'(n,\varepsilon)$ the three numbers of 37.7 at the data $\big(\kappa+an+\frac{\varepsilon}{2},\mathbf{w}+n\mathbf{b}+\varepsilon\mathbf{1}\big)$ have the nonzero slopes $\frac12$, $\frac12-\sum_j\deg P_j$ and $\sum_j m_{jk}$ in ε. Reduce the $D+1$ insertions $y_k=x^{ak}\prod_j P_j^{-b_jk}$: $y_k=\partial_n g_k+\sum_i c_{ki}x^i/Q$, where the denominators of g_k and of c_{ki} are products of the three numbers. The $(D+1)\times D$ matrix (c_{ki}) has a nonzero vector in its left kernel, found by elimination; clearing denominators gives polynomials $p_k(n,\varepsilon)$, and dividing by the largest power of ε that divides all of them makes the $p_k(n,0)$ not all zero. Then $g=\sum_k p_kg_k$, divided by that power of ε, has $\partial_n g=\sum_k p_ky_k$, and its denominators are products of the three numbers and a power of ε, none vanishing identically in ε at an integer n. Fix an integer n at which the $D+1$ displayed reads converge, absolutely since the reads of a ray are read where they converge absolutely; they converge

absolutely for small ε too. For all but finitely many small $\varepsilon \neq 0$, g is defined and the exponents of $x^{\kappa+an+\varepsilon/2}\prod_j P_j^{-w_j-nb_j-\varepsilon}g$ at 0 and at ∞, which move with ε, are not 0; the argument of 37.7 then gives $\langle \eth_n g\rangle = 0$, so $\sum_k p_k(n,\varepsilon)\langle 1\rangle_{\kappa+a(n+k)+\varepsilon/2,\,\mathbf{w}+(n+k)\mathbf{b}+\varepsilon\mathbf{1}} = 0$. Letting $\varepsilon \to 0$, the reads being continuous there, gives the recurrence at n. □

Theorem 37.12 (Coefficient jets on one scale). *As stated, for letters without roots on $[0,\infty)$, with the powers of the changed letters continued in u.*

Proof. P_{ja}/P_j is continuous on $[0,\infty]$, P_j having no roots on $[0,\infty)$ and $\deg P_{ja} \le \deg P_j$, so M_{ja} is finite. On the polydisc $z_j = \sum_a u_a P_{ja}/P_j$ has $|z_j| \le \sum_a |u_a| M_{ja} < 1$, so $P_j(x;u) = P_j(x)(1+z_j)$ has no roots on $[0,\infty)$, and the power continued in u from $u=0$, the polydisc being convex, is $P_j(x)^{-w_j}(1+z_j)^{-w_j}$ with the principal branch of the second factor. Then

$$(1+z_j)^{-w_j} = \sum_{m\ge 0} \frac{(-1)^m (w_j)_m}{m!} z_j^m, \qquad z_j^m = \sum_{|n_j|=m} \frac{m!}{\prod_a n_{ja}!} \prod_a \Big(u_a \frac{P_{ja}}{P_j}\Big)^{n_{ja}},$$

and the product over j is the displayed series inside the read. It is dominated by $\prod_j \sum_m \frac{|(w_j)_m|}{m!}\rho_j^m$, $\rho_j = \sum_a |u_a| M_{ja} < 1$, finite by the ratio test, times the absolutely integrable integrand of $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$; so the read of the sum is the sum of the reads, absolutely, uniformly on smaller polydiscs, and holomorphic in u. Each coefficient is the read of the rational insertion $\prod_{j,a}(P_{ja}/P_j)^{n_{ja}}$, bounded on $[0,\infty)$, absolutely convergent with $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$, and 37.10 gives it. The continuation in u fixes the branch: the continuous branch of $P_j(x;u)^{-w_j}$ along $[0,\infty)$ from its principal value at 0 can differ from it by a factor $e^{2\pi i \ell w_j}$, $\ell \in \mathbb{Z}$. □

40.36 Chapter 38: the family as one Laplace transform

40.36.1 What the one-table proofs read, and what a family reads

The one-table statements read the exponent of the law of a question, $\sigma E + \delta O$ at offsets zero, through the width wd of the Newton polytope of Λ and through the convexity of E; the walls of the wedge, its slope and the volume at low strength are all values of the width. For a family the exponent is

$$X(z) = 2\langle \boldsymbol{\mu}_\perp, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big),$$

and what replaces the width is the wall at a tilt (Definition 38.3): on the sector of every order the exponent is minus twice the wall at the tilt less the logarithms of the sector letters (Proposition 38.4(i)), which is Theorem 8.7(ii) read in the signed coordinates. Every statement below is read off that identity, the convexity of the centred potential, and, for samples of bases, the factorization of the chance of a sample over its draws. Throughout, $n = |\mathcal{E}| \ge 2$, the data are real and on the balance wall unless said otherwise, a question is written $q = (\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$, its data are $\big(\mathbf{m}_F(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta}); \boldsymbol{\sigma}+\boldsymbol{\delta}, \boldsymbol{\sigma}-\boldsymbol{\delta}\big)$, and $P_F(q)$ denotes its period. On the sector H_π of an order π, read in the tilts (Definition 38.3), the differences of a tilt along π are written $v_k = z_{\pi(k-1)} - z_{\pi(k)} \ge 0$, $2 \le k \le n$, so that $\kappa(z) = \sum_k \kappa_{U_k} v_k$; at $\alpha = e^{2z}$ the gaps of Corollary 8.8 are $2v_k$.

Lemma S14 (The walls on the balance wall)**.** *On the balance wall, for every nonempty proper S,*

$$\kappa_S = \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) + \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp, \boldsymbol{\delta}), \qquad \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) = \sum_j \sigma_j c_j(S), \qquad \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp,\boldsymbol{\delta}) = \mu_{\perp,S} - \sum_j \delta_j \varepsilon_j^\perp(S),$$

with $\varepsilon_j^{\perp}(S) = \varepsilon_j(S) - |S|\frac{2r_j-n}{n}$*; and* $\kappa^{\mathbf{E}}_{\mathcal{E}\setminus S} = \kappa^{\mathbf{E}}_S$, $\kappa^{\mathbf{O}}_{\mathcal{E}\setminus S} = -\kappa^{\mathbf{O}}_S$, *so* $\kappa_{\mathcal{E}\setminus S} = \kappa^{\mathbf{E}}_S - \kappa^{\mathbf{O}}_S$. *In particular every wall is a linear function of the pair coordinates* $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ *of the question.*

Proof. By Theorem 8.2(iii), $\kappa_S = \mu_S + \sum_j \big(\sigma_j c_j(S) - \delta_j \varepsilon_j(S)\big)$ and $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$. On the balance wall $\mu_{\mathcal{E}} = \sum_j \delta_j(2r_j - n)$, and $\boldsymbol{\mu} = \boldsymbol{\mu}_\perp + \frac{\mu_{\mathcal{E}}}{n}\mathbf{1}$ gives $\mu_S = \mu_{\perp,S} + \frac{|S|}{n}\sum_j \delta_j(2r_j - n)$; substituting gives the decomposition. By Theorem 7.11, $c_j(\mathcal{E}\setminus S) = c_j(S)$ and $\varepsilon_j(\mathcal{E}\setminus S) = 2r_j - n - \varepsilon_j(S)$, so $\varepsilon_j^{\perp}(\mathcal{E}\setminus S) = 2r_j - n - \varepsilon_j(S) - \frac{n-|S|}{n}(2r_j - n) = -\varepsilon_j^{\perp}(S)$, and $\mu_{\perp,\mathcal{E}\setminus S} = -\mu_{\perp,S}$ as $\boldsymbol{\mu}_\perp \in \mathbf{1}^{\perp}$. □

40.36.2 The exponent of a question

Definition 38.3: the claims. Let $z \in H_\pi$. A difference v_k is positive exactly when $z_{\pi(k-1)} > z_{\pi(k)}$, and then $U_k = \{e : \ z_e < z_{\pi(k-1)}\}$, a set $\{e : \ z_e < z_f\}$ with $z_f > \min_e z_e$; so the tails with a positive difference, and their differences, are read off z alone, and $\kappa(z)$ does not depend on the order π whose sector holds z. On H_π the differences are linear in z, so κ is linear there, and $\kappa(cz) = c\kappa(z)$ for $c \geq 0$. The tilt $-\mathbf{1}_S$ lies in the sectors of the orders that list $\mathcal{E}\setminus S$ first; there its only positive difference is 1, at the tail $U_k = S$, so $\kappa(-\mathbf{1}_S) = \kappa_S$. In the chart $z_{\pi(1)} = 0$ the coordinates are $z_{\pi(i)} = -(v_2 + \cdots + v_i)$, a triangular change of coordinates with determinant ± 1, so $dz = dv_2 \cdots dv_n$ on H_π; the chart does not matter (Definition 38.1). Finally $\kappa(z)$ is linear in the walls, which are linear in the question by Lemma S14.

Proposition 38.4 (The exponent of a question reads its walls; the even statistic is least at the table's gauge). *As stated, for a question with real exponents of any signs.*

Proof. (i) Write $\boldsymbol{\mu} = \mathbf{m} - \sigma\mathbf{1}$ for the offsets, so that $\boldsymbol{\mu} - \boldsymbol{\mu}_\perp \in \mathbb{R}\mathbf{1}$ and $\mu_{\mathcal{E}} = \sum_j \delta_j(2r_j - n)$ on the balance wall. The function $\tilde{X}(z) = 2\langle \boldsymbol{\mu}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)$ on $\mathbb{R}^{\mathcal{E}}$ is unchanged by $z \mapsto z + c\mathbf{1}$, since E_j is unchanged and O_j moves by $2c(2r_j - n)$ (Theorem 7.13(iii)), the move of the whole being $2c\kappa_{\mathcal{E}} = 0$ (Proposition 16.5); on $\mathbf{1}^{\perp}$ it is X. So X may be read at any representative of a tilt. At $\alpha = e^{2z}$, Theorem 7.13(i) gives $\sigma_j E_j + \delta_j O_j = s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z) = s_j \log K_j(\alpha) + t_j \log \Psi_j(\alpha) - (s_j + t_j)\langle \mathbf{1}, z\rangle$, and $2\langle \boldsymbol{\mu}, z\rangle + 2\sigma\langle \mathbf{1}, z\rangle = \langle \mathbf{m}, \log \alpha\rangle$, so

$$X(z) = \log\Big(\alpha^{\mathbf{m}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j}\Big),$$

the logarithm of the integrand of the period against the measure $d\alpha/\alpha^{\mathbf{1}}$. Take the sector coordinates of Theorem 8.7(ii) for π: $\rho = \alpha_{\pi(1)} = e^{2z_{\pi(1)}}$ and $q_k = \alpha_{\pi(k)}/\alpha_{\pi(k-1)} = e^{-2v_k}$. Since $\alpha_{\pi(i)} = \rho q_2 \cdots q_i$ and $\pi(i) \in U_k$ exactly when $k \leq i$, $\alpha^{\mathbf{m}} = \rho^{m_{\mathcal{E}}} \prod_k q_k^{m_{U_k}}$; with $K_j = \rho^{r_j} \prod_k q_k^{d_j(U_k)} k_{j,\pi}(q)$ and $\Psi_j = \rho^{n-r_j} \prod_k q_k^{d_j^*(U_k)} \psi_{j,\pi}(q)$,

$$\alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} = \rho^{\kappa_{\mathcal{E}}} \prod_{k=2}^{n} q_k^{\kappa_{U_k}} \prod_j k_{j,\pi}(q)^{-s_j} \psi_{j,\pi}(q)^{-t_j},$$

and $\rho^{\kappa_{\mathcal{E}}} = 1$ on the balance wall. The logarithm, with $\log q_k = -2v_k$, is the identity of (i).

On H_π the greedy basis $T^+_{j,\pi}$ has the largest monomial α^T among the bases of table j, and $T^-_{j,\pi}$ the largest $\alpha^{\mathcal{E}\setminus T}$ (Lemma 8.5); as $-\chi_T = \chi_{\mathcal{E}\setminus T}$, $h_j(z) = \langle \chi_{T^+_{j,\pi}}, z\rangle$ and $h_j(-z) = \langle \chi_{\mathcal{E}\setminus T^-_{j,\pi}}, z\rangle$, whatever the signs of the exponents. Write $z = z_{\pi(1)}\mathbf{1} - \sum_k v_k \mathbf{1}_{U_k}$; for $A \subseteq \mathcal{E}$, $\langle \chi_A, z\rangle = z_{\pi(1)}(2|A| - n) - \sum_k v_k(2|A \cap U_k| - |U_k|)$, and $|T^+_{j,\pi} \cap U_k| = d_j(U_k)$, $|U_k \setminus T^-_{j,\pi}| = d_j^*(U_k)$. So

$$\sum_j \big(s_j h_j(z) + t_j h_j(-z)\big) = 2z_{\pi(1)} \sum_j \delta_j(2r_j - n) + 2\sum_k v_k \Big(\sigma|U_k| - \sum_j \big(s_j d_j(U_k) + t_j d_j^*(U_k)\big)\Big),$$

while $2\langle\boldsymbol{\mu},z\rangle = 2z_{\pi(1)}\mu_{\mathcal{E}} - 2\sum_k v_k\mu_{U_k}$. The difference is $2\sum_k v_k\big(m_{U_k} - \sum_j (s_j d_j(U_k) + t_j d_j^*(U_k))\big) = 2\kappa(z)$, the term in $z_{\pi(1)}$ vanishing on the balance wall; on $\mathbf{1}^\perp$, $\langle\boldsymbol{\mu},z\rangle = \langle\boldsymbol{\mu}_\perp,z\rangle$. For $s_j,t_j \geq 0$: the support functions of P_j and P_j^* are $\frac{1}{2}(h_j(z)+\langle\mathbf{1},z\rangle)$ and $\frac{1}{2}(h_j(-z)+\langle\mathbf{1},z\rangle)$, support functions add over a Minkowski sum with nonnegative coefficients, and so $2\kappa(z) = 2h_{\mathcal{N}_F}(z) - 2\sigma\langle\mathbf{1},z\rangle - 2\langle\boldsymbol{\mu},z\rangle = 2h_{\mathcal{N}_F}(z) - 2\langle\mathbf{m},z\rangle$, twice the support function of $\mathcal{N}_F - \mathbf{m}$, which lies in $\mathbf{1}^\perp$ and so reads tilts.

The sector letters are polynomials in q with nonnegative coefficients and the constant terms $\theta_{j,T_{j,\pi}^+}$, $\theta_{j,T_{j,\pi}^-}$, and their value at $q=\mathbf{1}$ is $K_j(\mathbf{1}) = \Psi_j(\mathbf{1})$; on $[0,1]^{n-1}$ they increase in every q_k and so lie between these (Theorem 8.7(ii)), whatever the signs of the exponents. Taking logarithms and weighting by s_j and t_j gives both bounds.

(ii) If every proper wall is positive, on H_π, $\kappa(z) = \sum_k \kappa_{U_k} v_k \geq \kappa_{\min}\sum_k v_k = \kappa_{\min}(z_{\pi(1)} - z_{\pi(n)})$, and $\max_e z_e - \min_e z_e$ vanishes on $\mathcal{T}$ only at 0. Conversely $\kappa(-\mathbf{1}_S) = \kappa_S$.

(iii) At the centre $\boldsymbol{\mu}_\perp = 0$ and $\boldsymbol{\delta} = 0$, so $X = -\sum_j \sigma_j E_j$, and the walls are $\kappa_S = \sum_j \sigma_j c_j(S)$ (Proposition 8.21(ii)). By (i), $2\kappa = \sum_j \sigma_j (h_j(z) + h_j(-z)) = \sum_j \sigma_j \operatorname{wd}_j(z)$, and $\operatorname{wd}_j(z) = \max_T\langle\chi_T,z\rangle - \min_T\langle\chi_T,z\rangle = 2\big(\max_T\langle\mathbf{1}_T,z\rangle - \min_T\langle\mathbf{1}_T,z\rangle\big)$ is twice the width of P_j in the direction z, equal to $2(r_j(S) - d_j(S)) = 2c_j(S)$ at $z = \mathbf{1}_S$ (Theorem 7.11). The bounds of (i) at $s_j = t_j = \sigma_j$ are the displayed ones. The Hessian of $\log\Lambda_j$ at z is the covariance of χ_T under $\mathbb{P}_{j,e^{2z}}$ (Theorem 7.14), so $\log\Lambda_j$ is convex, and so is $E_j(z) = \log\Lambda_j(z) + \log\Lambda_j(-z)$, which is even; hence $E_j(z) = \frac{1}{2}(E_j(z) + E_j(-z)) \geq E_j(0) = 2\log\Lambda_j(0) = 2\log K_j(\mathbf{1})$.

For $v \in \mathbf{1}^\perp$,

$$v^{\mathsf{T}}\nabla^2\Big(\sum_j \sigma_j E_j\Big)(z)\,v = \sum_j \sigma_j\Big(\operatorname{Var}_{T\sim\mathbb{P}_{j,e^{2z}}}\langle\chi_T,v\rangle + \operatorname{Var}_{T\sim\mathbb{P}_{j,e^{-2z}}}\langle\chi_T,v\rangle\Big).$$

Every basis has positive chance under both laws, so the form vanishes only if $\langle\chi_T,v\rangle = 2\langle\mathbf{1}_T,v\rangle - \langle\mathbf{1},v\rangle$ is constant over the bases of every table with $\sigma_j > 0$. Then v is constant on every molecule of the family of these tables (Lemma 7.7), and every such molecule M has $c_j(M) = 0$ in each of them, the table being the direct sum of its restrictions to the molecules (Proposition 7.8), so that $|T\cap M| = r_j(M)$ for every basis. If $v \notin \mathbb{R}\mathbf{1}$, the union S of the molecules on which v is largest is nonempty and proper with $c_j(S) = 0$ for every table with $\sigma_j > 0$: the family is not connected. So on a connected family the form is positive on $\mathcal{T}\setminus 0$ and $\sum_j \sigma_j E_j$ is strictly convex on $\mathcal{T}$. Finally $\sum_j \sigma_j \operatorname{wd}_j = 2\kappa$ is positive on $\mathcal{T}\setminus 0$ by (ii), the walls $\sum_j \sigma_j c_j(S)$ of the centre being positive exactly when the family is connected (Proposition 8.21(ii)). □

Remark. For one table at the centre of strength σ, (iii) is the statement of the reference, with $2\kappa = \sigma\operatorname{wd}$; the family statement adds (i), in which the width of one table is replaced by the wall at the tilt of any question, and (ii). The identity of (i) is the sector form of the period read pointwise: integrated against $dz = dv$ on each sector it is $P_F(q) = 2^{n-1}\sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k} v_k} h_\pi(e^{-2v})\,dv$, with $h_\pi = \prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$, which is $\sum_\pi I_\pi$ of Corollary 8.8 in the gaps $2v_k$.

40.36.3 The two ends of strength

The high end needs one fact about integrals of $e^{-\lambda f}$ with a convex exponent, for complex λ in a half-plane; it is proved here once, in the form in which the period uses it.

Lemma S15 (A convex exponent at high strength)**.** *Let f be a smooth function on a real vector space V of dimension $d \geq 1$ with Lebesgue measure dz and a norm $|\cdot|$, such that*

(a) every derivative of f of order at least one is bounded;

(b) f is convex, with a critical point $z^\star$ at which $A_0 = \nabla^2 f(z^\star)$ is positive definite;

(c) $f(z)-f(z^\star)\ge c|z|-C$ *for some* $c>0$ *and* C.

Then $z^\star$ *is the only critical point, and as* $|\lambda|\to\infty$ *with* $\operatorname{Re}\lambda\ge\lambda_0$, *for a fixed* $\lambda_0>0$,

$$\int_V e^{-\lambda f(z)}\,dz = e^{-\lambda f(z^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{d}{2}}\det(A_0)^{-\frac12}\big(1+O(|\lambda|^{-1})\big),$$

the determinant in the coordinates of dz *and the power principal.*

Proof. Use a Euclidean norm, equivalent to the given one. *The gradient away from* $z^\star$. By convexity $\langle\nabla f(z),z-z^\star\rangle\ge f(z)-f(z^\star)\ge c|z|-C$, so $|\nabla f(z)|\ge(c|z|-C)/|z-z^\star|$, which tends to c as $|z|\to\infty$. A convex function with a positive definite Hessian at a critical point has its strict minimum there, and its critical points form the convex set of its minima, which is $\{z^\star\}$. So for every neighbourhood V_0 of $z^\star$, $|\nabla f|\ge c_0>0$ on $V\setminus V_0$: on a large ball by compactness, outside it by the bound.

A chart at $z^\star$. Put $A(z)=2\int_0^1(1-t)\,\nabla^2 f\big(z^\star+t(z-z^\star)\big)\,dt$, smooth and symmetric with $A(z^\star)=A_0$. Since $\nabla f(z^\star)=0$, $f(z)=f(z^\star)+\frac12\langle A(z)(z-z^\star),z-z^\star\rangle$. On a ball V_1 about $z^\star$ on which $A(z)$ is positive definite put $w(z)=A(z)^{1/2}(z-z^\star)$, the positive square root being smooth there; then $f=f(z^\star)+\frac12|w|^2$, and $Dw(z^\star)=A_0^{1/2}$ is invertible, so w is a diffeomorphism of a smaller ball V_2 onto a neighbourhood of 0.

Near $z^\star$. Take a smooth χ with compact support in V_2 and $\chi=1$ near $z^\star$. In the coordinates w, $\int\chi e^{-\lambda f}dz = e^{-\lambda f(z^\star)}\int_{\mathbb{R}^d}e^{-\lambda|w|^2/2}g(w)\,dw$, with g smooth of compact support and $g(0)=|\det Dw(z^\star)|^{-1}=\det(A_0)^{-1/2}$. For $\operatorname{Re}\lambda>0$ and $\xi\in\mathbb{R}^d$, $\int e^{-\lambda|w|^2/2+i\langle\xi,w\rangle}dw=(2\pi/\lambda)^{d/2}e^{-|\xi|^2/(2\lambda)}$: both sides are holomorphic in λ on the half-plane and agree for $\lambda>0$, completing the square in each coordinate. Writing $g(w)=(2\pi)^{-d}\int\hat g(\xi)e^{i\langle\xi,w\rangle}d\xi$ with $\hat g$ rapidly decreasing,

$$\int e^{-\lambda|w|^2/2}g(w)\,dw=\Big(\frac{2\pi}{\lambda}\Big)^{\frac{d}{2}}(2\pi)^{-d}\int\hat g(\xi)\,e^{-|\xi|^2/(2\lambda)}\,d\xi=\Big(\frac{2\pi}{\lambda}\Big)^{\frac{d}{2}}\big(g(0)+\varrho(\lambda)\big),$$

and since $\operatorname{Re}(1/\lambda)\ge0$ and $|e^{-x}-1|\le|x|$ for $\operatorname{Re}x\ge0$, $|\varrho(\lambda)|\le(2|\lambda|)^{-1}(2\pi)^{-d}\int|\xi|^2|\hat g(\xi)|\,d\xi$.

Away from $z^\star$. On the support of $1-\chi$, $|\nabla f|\ge c_0$. The operator $L=-\lambda^{-1}|\nabla f|^{-2}\langle\nabla f,\nabla\rangle$ has $Le^{-\lambda f}=e^{-\lambda f}$, and its transpose is $L^{\mathrm t}\varphi=\lambda^{-1}\operatorname{div}\big(\varphi\,|\nabla f|^{-2}\nabla f\big)$. Integrating by parts N times, $\int(1-\chi)e^{-\lambda f}dz=\int e^{-\lambda f}(L^{\mathrm t})^N(1-\chi)\,dz$, with no boundary terms: $1-\chi$ vanishes near $z^\star$, and on large spheres $|e^{-\lambda f}|=e^{-\operatorname{Re}\lambda f}$ decays exponentially by (c) while the coefficients stay bounded. Now $(L^{\mathrm t})^N(1-\chi)=\lambda^{-N}G_N$ with G_N a polynomial in the derivatives of χ, in those of f of orders 1 to $N+1$ and in $|\nabla f|^{-2}$, bounded by (a) and $|\nabla f|\ge c_0$. Since $f\ge f(z^\star)$ and $\operatorname{Re}\lambda\ge\lambda_0$,

$$\Big|\int(1-\chi)e^{-\lambda f}dz\Big|\le|\lambda|^{-N}\sup|G_N|\;e^{-\operatorname{Re}\lambda f(z^\star)}\int_V e^{-\lambda_0(f-f(z^\star))}dz,$$

the last integral finite by (c). With $N>\frac{d}{2}+1$ this is $O(|\lambda|^{-1})$ times the modulus $(2\pi/|\lambda|)^{d/2}e^{-\operatorname{Re}\lambda f(z^\star)}\det(A_0)^{-1/2}$ of the leading term, and the two parts add to the statement. (On the real half-line this is Laplace's method, along the imaginary direction stationary phase, and w is the Morse chart.) □

Theorem 38.5 (The two ends of strength). *As stated.*

Proof. The walls and the exponents of λq are λ times those of q (Lemma S14), so its walls $\lambda\kappa_S$ have positive real parts for $\operatorname{Re}\lambda>0$. At such data the period is the convergent integral of Theorem 8.10, the powers of the positive letters principal, and the change of variables of Definition 38.1, which

uses only $K_j(e^{2z}) = e^{\langle \mathbf{1},z\rangle}\Lambda_j(z)$ and $\Psi_j(e^{2z}) = e^{\langle \mathbf{1},z\rangle}\Lambda_j(-z)$, gives $P_F(\lambda q) = 2^{n-1}\int_{\mathcal{T}} e^{\lambda X(z)}dz$, absolutely convergent as $|e^{\lambda X}| = e^{\operatorname{Re}\lambda X}$; here X is the exponent of q, and the exponent of λq is λX, the exponent being linear in the question.

(i) By Proposition 38.4(i) and $dz = dv$ on each sector, with $h_\pi = \prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$,

$$P_F(\lambda q) = 2^{n-1}\sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\lambda\sum_k \kappa_{U_k} v_k}\, h_\pi(e^{-2v})^\lambda\, dv,$$

and with $v = w/\lambda$,

$$\lambda^{n-1}P_F(\lambda q) = 2^{n-1}\sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k} w_k}\, h_\pi(e^{-2w/\lambda})^\lambda\, dw.$$

On the cube $|\log h_\pi| \le C$ (Proposition 38.4(i)), so for $0 < \lambda \le 1$ the integrand lies between $e^{\mp C}e^{-2\sum_k \kappa_{U_k} w_k}$, whose integral is $\prod_k (2\kappa_{U_k})^{-1}$; it tends to $e^{-2\sum_k \kappa_{U_k} w_k}$, and by domination $\lambda^{n-1}P_F(\lambda q) \to 2^{n-1}\sum_\pi \prod_k (2\kappa_{U_k})^{-1} = \sum_\pi \prod_k \kappa_{U_k}^{-1}$. For the next order, $|h^\lambda - 1 - \lambda\log h| \le \frac{1}{2}\lambda^2 C^2 e^{\lambda C}$, and for w in the open orthant $e^{-2w/\lambda} \to 0$, so $\log h_\pi(e^{-2w/\lambda}) \to \log h_\pi(0)$, boundedly; hence

$$\lambda^{-1}\Big(\lambda^{n-1}P_F(\lambda q) - \sum_\pi \prod_k \kappa_{U_k}^{-1}\Big) \to \sum_\pi \prod_k \kappa_{U_k}^{-1}\, \log h_\pi(0),$$

and $h_\pi(0) = \prod_j \theta_{j,T_{j,\pi}^+}^{-s_j}\theta_{j,T_{j,\pi}^-}^{-t_j}$ is the sector integrand at $q = 0$, so that $\log h_\pi(0) = -\sum_j \big(s_j \log\theta_{j,T_{j,\pi}^+} + t_j\log\theta_{j,T_{j,\pi}^-}\big)$, the coefficient stated. Finally κ is positively homogeneous of degree one and positive on $\mathcal{T}\setminus 0$ (Proposition 38.4(ii)), so $\operatorname{vol}\{\kappa \le r\} = r^{n-1}\operatorname{vol}\{\kappa \le 1\}$ and

$$2^{n-1}\int_{\mathcal{T}} e^{-2\kappa(z)}dz = 2^{n-1}\operatorname{vol}\{\kappa\le 1\}\,(n-1)\int_0^\infty e^{-2r}r^{n-2}dr = (n-1)!\operatorname{vol}\{\kappa\le 1\},$$

while on the sectors the same integral is $2^{n-1}\sum_\pi\prod_k(2\kappa_{U_k})^{-1}$. (For $s_j, t_j \ge 0$, $\{\kappa \le 1\}$ is the polar of $\mathcal{N}_F - \mathbf{m}$ in $\mathbf{1}^\perp$.)

(ii) Let $s_j, t_j \ge 0$ and $f = -X$ on $\mathcal{T}$, of dimension $n-1$ with the measure dz. The hypotheses of Lemma S15 hold. (a) Every derivative of $\log\Lambda_j$ of order $k \ge 1$ is a joint cumulant of order k of $\chi_T \in \{\pm 1\}^{\mathcal{E}}$ under $\mathbb{P}_{j,e^{2z}}$ (Theorem 7.14), bounded by a constant depending on k alone, and $f = \sum_j\big(s_j\log\Lambda_j(z) + t_j\log\Lambda_j(-z)\big) - 2\langle \boldsymbol{\mu}_\perp, z\rangle$. (b) $v^{\mathsf{T}}\nabla^2 f(z)v = \sum_j\big(s_j\operatorname{Var}_{\mathbb{P}_{j,e^{2z}}}\langle\chi_T, v\rangle + t_j\operatorname{Var}_{\mathbb{P}_{j,e^{-2z}}}\langle\chi_T, v\rangle\big)$; a question with $s_j, t_j \ge 0$ inside its walls is on a connected family, as $\kappa_S + \kappa_{\mathcal{E}\setminus S} = 2\sum_j \sigma_j c_j(S) > 0$ (Lemma S14) puts some active table of positive width on every S, and the argument of Proposition 38.4(iii) makes the form positive definite on $\mathcal{T}$ at every z; so f is strictly convex. (c) $f \ge 2\kappa - C \ge 2\kappa_{\min}(\max_e z_e - \min_e z_e) - C$ by Proposition 38.4(i)–(ii), and $\max_e z_e - \min_e z_e$ is a norm on $\mathcal{T}$. A strictly convex f with (c) attains its minimum at one point $z^\star$, its only critical point. By Theorem 7.14, $\nabla\log\Lambda_j(z) = 2R_j(\alpha) - \mathbf{1}$ and $\nabla[\log\Lambda_j(-z)] = \mathbf{1} - 2R_j(\alpha^{-1}) = 2R_j^*(\alpha) - \mathbf{1}$ at $\alpha = e^{2z}$, so the derivative of f along $v \in \mathbf{1}^\perp$ is $2\langle \mathbf{u}(\alpha) - \mathbf{m}, v\rangle$, $\mathbf{u} = \sum_j(s_jR_j + t_jR_j^*)$ the usage, $\mathbf{m} - \sigma\mathbf{1} - \boldsymbol{\mu}_\perp$ being a multiple of $\mathbf{1}$. It vanishes for every v exactly when $\mathbf{u}(\alpha) - \mathbf{m} \in \mathbb{R}\mathbf{1}$, that is $\mathbf{u}(\alpha) = \mathbf{m}$, the two having the total $\sum_j(s_jr_j + t_jr_j^*)$ on the balance wall; and $\mathbf{m}$ is the target of q there, so $z^\star$ is its native gauge (Definition 8.16). The usage and the masses of λq are λ times those of q, so $z^\star$ is the native gauge of every λq, $\lambda > 0$. Lemma S15 with $d = n-1$ gives the expansion, the determinant taken on a chart $z_g = 0$, that is of the Hessian of the invariant form $\tilde{X}$ of the proof of Proposition 38.4 with the row and the column of g deleted. With $y = 2z$, $\Phi(y) = \sigma\langle\mathbf{1}, y\rangle + \Phi^\circ(z)$ (Theorem 8.18), so $\partial_{z_e}\partial_{z_f}(-\tilde{X}) = \partial_{z_e}\partial_{z_f}\Phi^\circ = 4\partial_{y_e}\partial_{y_f}\Phi$,

rung two of the still of the realization $\big((s_j,t_j)_j;e^{2z^\star}\big)$ of the active tables (Definition 9.4). At the centre $f=\sum_j\sigma_jE_j$ is even and strictly convex, so $z^\star=0$, $f(0)=2\sum_j\sigma_j\log K_j(\mathbf{1})$, and $\nabla^2f(0)=\sum_j\sigma_j\cdot 2\operatorname{Cov}_{\mathbb{P}_{j,\mathbf{1}}}(\chi_T)=8\sum_j\sigma_j\big(\kappa_{j,\mathbf{1}}(X_e,X_f)\big)_{e,f}$, as $\chi_T=2X-\mathbf{1}$, read on the chart $z_g=0$ with the row and the column of g deleted. □

Remark. At one table and the centre of strength one, $\kappa=\frac12\,\mathrm{wd}$ and $(n-1)!\,\mathrm{vol}\{\kappa\le1\}=2^{n-1}(n-1)!\,\mathrm{vol}\{\mathrm{wd}\le1\}$, the reference's low end; its high end is (ii) at offsets zero. Divided by the same limit for the list $\binom{\mathcal{E}}{1}$ at a full basis lean, whose walls are the masses, (i) is the limit of Proposition 39.14: there $\Gamma^n_1\begin{bmatrix}\lambda\boldsymbol\tau\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}$ has $\lambda^{n-1}\Gamma^n_1\begin{bmatrix}\lambda\boldsymbol\tau\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}\to\tau_{\mathcal{E}}/\prod_e\tau_e=\sum_\pi\prod_k\tau_{U_k}^{-1}$. In (ii) the real ray is Proposition 17.8 written in the signed coordinates; what the lemma adds is the whole half-plane, uniformly.

40.36.4 How far a question can lean, and the tails

Proposition 38.6 (How far a question can lean; the values of the statistics). *As stated.*

Proof. The decomposition of the walls is Lemma S14. Every proper wall is positive exactly when, for every S, $\kappa^{\mathbf{E}}_S+\kappa^{\mathbf{O}}_S>0$ and $\kappa^{\mathbf{E}}_S-\kappa^{\mathbf{O}}_S>0$, that is $|\kappa^{\mathbf{O}}_S|<\kappa^{\mathbf{E}}_S$. The fold of Theorem 16.16 sends the pair coordinates $(\boldsymbol\mu_\perp;\boldsymbol\sigma,\boldsymbol\delta)$ to $(-\boldsymbol\mu_\perp;\boldsymbol\sigma,-\boldsymbol\delta)$, negating the offsets and the leans and keeping the strengths (Definition 9.1); so it negates $\kappa^{\mathbf{O}}_S$ and swaps κ_S with $\kappa_{\mathcal{E}\setminus S}$. When every $\kappa^{\mathbf{E}}_S(\boldsymbol\sigma)$ is positive the conditions read $\kappa^{\mathbf{E}}_S/|\kappa^{\mathbf{O}}_S|>1$ wherever $\kappa^{\mathbf{O}}_S\neq0$, that is $\ell^*_{\boldsymbol\sigma}(\boldsymbol\mu_\perp,\boldsymbol\delta)>1$: finitely many strict linear inequalities in the odd part, each with its negative, an open convex polyhedron symmetric under the fold. It is bounded exactly when the linear map $(c,\mathbf{b})\mapsto(\kappa^{\mathbf{O}}_S(c,\mathbf{b}))_S$ is injective: then it lies in the preimage of a bounded box, and otherwise a line through 0 in the kernel lies in it. The homogeneities and the evenness in $(c,\mathbf{b})$ are read off the definition. For one connected table at offsets zero, $\kappa^{\mathbf{E}}_S=\sigma c(S)$ and $\kappa^{\mathbf{O}}_S=-\delta\varepsilon^\perp(S)$, so the conditions are $\sigma>0$ and $|\delta|<\sigma\min c(S)/|\varepsilon^\perp(S)|=\ell^*_1(0,1)\,\sigma$.

(i) The exponent is linear in the question, so with $q'=(c;\boldsymbol\sigma+\mathbf{a},\mathbf{b})$ and the exponent $X_{q'}$ of q', the expectation is $2^{n-1}P_F(\sigma\mathbf{1};\boldsymbol\sigma,\boldsymbol\sigma)^{-1}\int_{\mathcal{T}}e^{X_{q'}(z)}dz$, an integral of a positive function. By Proposition 38.4(i), whose identity holds at every real exponents, $e^{X_{q'}}$ lies between constant multiples of $e^{-2\kappa'(z)}$, κ' the wall at the tilt of q', and $\int_{H_\pi}e^{-2\kappa'(z)}dz=\int_{[0,\infty)^{n-1}}e^{-2\sum_k\kappa_{U_k}(q')v_k}dv$ is finite exactly when every $\kappa_{U_k}(q')$ is positive. Every nonempty proper S is a tail of some order, so the expectation is finite exactly when every proper wall of q' is positive, and it is then $P_F(q')/P_F(\sigma\mathbf{1};\boldsymbol\sigma,\boldsymbol\sigma)$ by Definition 38.1. The walls of q' are $\kappa^{\mathbf{E}}_S(\boldsymbol\sigma+\mathbf{a})\pm\kappa^{\mathbf{O}}_S(c,\mathbf{b})$; at $\mathbf{a}=0$ their positivity is $\ell^*_{\boldsymbol\sigma}(c,\mathbf{b})>1$.

(ii) With $|\delta_j|\le\sigma_j$ the exponents are nonnegative, and by the proof of Theorem 38.5(ii) $-X$ is strictly convex on $\mathcal{T}$ with its least value at the native gauge $z^\star$, at which alone it is attained. $-X(z)$ is the value of the linear form $-2\langle\boldsymbol\mu_\perp,x\rangle+\sum_j(\sigma_jy_j+\delta_jw_j)$ at the point $(z;E_j(z),O_j(z))_j$, and z is a coordinate of that point, so the least value of the form on the set of values of the statistics is taken at $\big(z^\star;E_j(z^\star),O_j(z^\star)\big)_j$ and only there. At the centre $z^\star=0$ (Proposition 38.4(iii)), $E_j(0)=2\log K_j(\mathbf{1})$ and $O_j(0)=0$. □

Proposition 38.7 (The tails of the statistics are read at the walls). *As stated.*

Proof. By Lemma S14 the walls are linear in the pair coordinates, and for the direction v of pair coordinates $(c;\mathbf{a},\mathbf{b})$, $\kappa_S(v)=\kappa^{\mathbf{E}}_S(\mathbf{a})+\kappa^{\mathbf{O}}_S(c,\mathbf{b})$, the offset direction c entering through $c_S=\sum_{e\in S}c_e$. With $X_v(z)=2\langle c,z\rangle-\sum_j(a_jE_j(z)+b_jO_j(z))$, $Y=X_v(Z)$ and $X_q+\lambda X_v=X_{q+\lambda v}$, so

for real $\lambda \geq 0$, $\mathbb{E}[e^{\lambda Y}] = 2^{n-1} P_F(q)^{-1} \int_{\mathcal{T}} e^{X_{q+\lambda v}} dz$, finite, by the argument of Proposition 38.6(i), exactly when every $\kappa_S(q) + \lambda \kappa_S(v)$ is positive, that is for $\lambda < \lambda^*$, and then the ratio of the periods.

Upper bound. For $0 < \lambda < \lambda^*$, $\mathbb{P}(Y > x) \leq e^{-\lambda x} \mathbb{E}[e^{\lambda Y}]$, so $\limsup_x x^{-1} \log \mathbb{P}(Y > x) \leq -\lambda$ for every $\lambda < \lambda^*$.

Lower bound. Let $\lambda^* < \infty$, let S^* attain it, and let π list $\mathcal{E} \setminus S^*$ first, so that $S^* = U_{k^*}$ is a tail of π. Write $\kappa_q(z)$ and $\kappa_v(z) = \sum_k \kappa_{U_k}(v)\big(z_{\pi(k-1)} - z_{\pi(k)}\big)$ for the walls at the tilt of q and of v, the latter read as balanced data with the exponents $(\mathbf{a}+\mathbf{b}, \mathbf{a}-\mathbf{b})$. By Proposition 38.4(i), whose identity holds at any balanced real data, $|X_q + 2\kappa_q| \leq C_q$ and $|X_v + 2\kappa_v| \leq C_v$ on $\mathcal{T}$. For $\varepsilon > 0$ put $a_x = (1+\varepsilon)x/(-2\kappa_{S^*}(v))$ and let $A_x \subset H_\pi$ be the set of the tilts whose difference along π at the tail S^* lies in $[a_x, a_x + 1]$ and whose other differences lie in $[0, 1]$, of dz-volume 1. On A_x, as $\kappa_{S^*}(v) < 0$,

$$Y \geq -2\kappa_v(z) - C_v \geq (1+\varepsilon)x - 2\sum_k |\kappa_{U_k}(v)| - C_v > x \quad \text{for large } x,$$

$$X_q \geq -2\kappa_q(z) - C_q \geq -2\kappa_{S^*}(q)(a_x + 1) - 2\sum_k |\kappa_{U_k}(q)| - C_q = -(1+\varepsilon)\lambda^* x - C',$$

since $2\kappa_{S^*}(q)a_x = (1+\varepsilon)\lambda^* x$. So $\mathbb{P}(Y > x) \geq 2^{n-1} P_F(q)^{-1} e^{-(1+\varepsilon)\lambda^* x - C'}$ for large x, $\liminf_x x^{-1} \log \mathbb{P}(Y > x) \geq -(1+\varepsilon)\lambda^*$ for every $\varepsilon > 0$, and with the upper bound the limit is $-\lambda^*$.

If $\lambda^* = \infty$, every $\kappa_S(v) \geq 0$, so $\kappa_v \geq 0$ on every sector and $Y \leq C_v$.

At the centre of strengths $\boldsymbol{\sigma}$ and the direction of pair coordinates $(c; \mathbf{0}, \mathbf{b})$, $\kappa_S(q) = \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ and $\kappa_S(v) = \kappa_S^{\mathbf{O}}(c, \mathbf{b})$. The first is unchanged and the second negated by $S \mapsto \mathcal{E} \setminus S$, so the sets on which $\kappa_S(v) < 0$ are the complements of those on which it is positive, and $\lambda^* = \min_S \kappa_S^{\mathbf{E}} / |\kappa_S^{\mathbf{O}}| = \ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b})$, for Y and for $-Y$ alike. □

Remark. The one-table rate of both tails of $O(Z)$ at the centre of strength σ_0 is $\ell^*_{\sigma_0}(0,1) = \sigma_0 \, \ell^*_1(0,1)$. The same proposition, applied to the family with the list $\binom{\mathcal{E}}{1}$ added, gives the tails of the Grey law (Proposition 39.12(iii), proved in Section 40.37.3).

40.36.5 The interaction of a bunch

The one-table proofs integrate the joint cumulant of the indicators of a bunch, one logarithm of a Taylor cube; a family integrates a sum of such logarithms with its exponents, one for each letter, so one lemma on one logarithm serves both.

Lemma S16 (An alternating sum over the subsets of a bunch)**.** *Let B be a finite set with $k = |B| \geq 2$, let $(c_p)_{p \subseteq B}$ be positive, and $f(y) = \log \sum_{p \subseteq B} c_p \, e^{\langle \mathbf{1}_p, y \rangle}$ on $\mathbb{R}^B$, and write $\partial^B = \prod_{e \in B} \partial_{y_e}$. Then $|\partial^B f(y)| \leq C \, e^{-\max_{e \in B} |y_e|}$, so $\partial^B f$ is integrable on $\mathbb{R}^B$, and*

$$\int_{\mathbb{R}^B} \partial^B f(y) \, dy = \sum_{p \subseteq B} (-1)^{|B \setminus p|} \log c_p.$$

Proof. The bound. $f(y+w) - f(y)$ is the logarithm of the moment generating function at w of the indicators $(X_e)_{e \in B}$ of a random set $P \subseteq B$ with the law $\pi_y(p) \propto c_p e^{\langle \mathbf{1}_p, y \rangle}$, so $\partial^B f(y)$ is their joint cumulant. Fix $e \in B$. A joint cumulant of $k \geq 2$ variables does not change when one of them is shifted by a constant, and it is a sum, with coefficients depending on k alone, of products of joint moments, one for each block of a partition of the k places, exactly one block holding the place of e. With $X_e - \mathbb{E}X_e$ in that place and $|X_f| \leq 1$, $|\partial^B f(y)| \leq C_k \, \mathbb{E}|X_e - \mathbb{E}X_e| = 2C_k p_e(1 - p_e)$,

$p_e = \pi_y(e \in P)$. Pairing each $p \not\ni e$ with $p+e$, the odds $p_e/(1-p_e) = \sum_{p\not\ni e} c_{p+e}e^{y_e}w_p \big/ \sum_{p\not\ni e} c_p w_p$, $w_p = e^{\langle \mathbf{1}_p, y\rangle} > 0$, lie between $\rho_- e^{y_e}$ and $\rho_+ e^{y_e}$, $\rho_\pm$ the largest and the least of the ratios c_{p+e}/c_p; so $p_e(1-p_e) \le \min(\text{odds}, \text{odds}^{-1}) \le \max(\rho_+, \rho_-^{-1})\, e^{-|y_e|}$. The best e gives the bound.

The value, by induction on k. Fix $e \in B$, put $B' = B - e$, $y' = y_{B'}$, and let $f^{/e}$ and $f^{\setminus e}$ be the functions of the same form on B' with the coefficients c_{p+e} and c_p, $p \subseteq B'$: the contraction and the deletion of e in the cube. With $\ell(x) = \log(1+e^x)$,

$$f(y) = f^{/e}(y') + y_e + \ell\big(f^{\setminus e}(y') - f^{/e}(y') - y_e\big) = f^{\setminus e}(y') + \ell\big(y_e + f^{/e}(y') - f^{\setminus e}(y')\big).$$

Every derivative of ℓ is $O(e^x)$ as $x \to -\infty$, and every derivative of order at least one of $f^{/e}$ and $f^{\setminus e}$ is bounded; since $B' \neq \varnothing$, $\partial^{B'} y_e = 0$, so $\partial^{B'} f(y) \to \partial^{B'} f^{/e}(y')$ as $y_e \to +\infty$ and $\partial^{B'} f(y) \to \partial^{B'} f^{\setminus e}(y')$ as $y_e \to -\infty$. Hence $\int_{\mathbb{R}} \partial^B f \, dy_e = \partial^{B'} f^{/e}(y') - \partial^{B'} f^{\setminus e}(y')$ for every y', and by the bound and Fubini

$$\int_{\mathbb{R}^B} \partial^B f \, dy = \int_{\mathbb{R}^{B'}} \big(\partial^{B'} f^{/e} - \partial^{B'} f^{\setminus e}\big)\, dy'.$$

If $k \ge 3$, each term is integrable by the bound, and the induction gives $\sum_{p\subseteq B'} (-1)^{|B'\setminus p|}(\log c_{p+e} - \log c_p) = \sum_{p\subseteq B}(-1)^{|B\setminus p|} \log c_p$. If $k = 2$, $B' = \{e'\}$, the integrand is the derivative of $f^{/e} - f^{\setminus e} = \log\big((c_e + c_{ee'}e^{y'})/(c_\varnothing + c_{e'}e^{y'})\big)$, whose limits at $\pm\infty$ are $\log(c_{ee'}/c_{e'})$ and $\log(c_e/c_\varnothing)$; their difference is the alternating sum. □

Definition 38.9: the claims. $\text{pat}(B) = 2^B$ holds exactly when B is independent (B is a pattern) and some basis avoids B ($\varnothing$ is a pattern): if both hold, $d(B) = 0$, so the contraction $\mathcal{B}/R$ has rank 0, and every subset of B is independent with $r_{\mathcal{B}/R} = 0 = d(B)$, a pattern by Proposition 12.16(i). Equivalently $r(B) = |B|$ and $d(B) = 0$, that is $c(B) = |B|$. The dual table has the bases $\mathcal{E} \setminus T$ and the patterns $B \setminus p$, with the children of cobasis letters $\Psi_{(p)}$; so every subset of B is a pattern of the dual table when every subset is a pattern of the table, and its interaction is $\sum_p (-1)^{|B\setminus(B\setminus p)|} \log \Psi_{(p)} = \iota_B^*$. The parts: $\Phi_{\mathbf{E}} = \sum_j \sigma_j(\log K_j + \log \Psi_j)$ and $\Phi_{\mathbf{O}} = \sum_j \delta_j(\log K_j - \log \Psi_j)$ (Definition 8.17), and $\sigma_j \pm \delta_j$ are s_j and t_j.

Theorem 38.11 (The interaction totals the still over the bunch's scales). *As stated.*

Proof. Fix α_R. By the Taylor cube (Theorem 12.6)

$$\log K_j(e^{y_B}, \alpha_R) = \log \sum_{p\subseteq B} K_{j,(p)}(\alpha_R)\, e^{\langle \mathbf{1}_p, y_B\rangle},$$

with every $K_{j,(p)}(\alpha_R) > 0$ as $\text{pat}_j(B) = 2^B$; Lemma S16 gives $\int \partial^B \log K_j \, dy_B = \iota_{j,B}(\alpha_R)$, absolutely. Likewise $\log \Psi_j(e^{y_B}, \alpha_R) = \log \sum_{P\subseteq B} \Psi_{j,(B\setminus P)}(\alpha_R)\, e^{\langle \mathbf{1}_P, y_B\rangle}$, and the lemma gives $\sum_P (-1)^{|B\setminus P|} \log \Psi_{j,(B\setminus P)} = \sum_p (-1)^{|p|} \log \Psi_{j,(p)} = \iota^*_{j,B}(\alpha_R)$. The parts of the potential are these logarithms with the coefficients $\sigma_j(1,1)$ and $\delta_j(1,-1)$, and a table with both exponents zero does not enter; integrating term by term gives $\iota_B^{\mathbf{E}}$ and $\iota_B^{\mathbf{O}}$. At a realization $\kappa_\Upsilon(e_1, \dots, e_k) = 2^k \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \Phi$ (Definition 9.4), and Proposition 9.7 writes it through the cumulants at α and at α^{-1}. For one table at $(1,0)$, $\partial^B \log K(e^y) = \kappa_\alpha(e_1, \dots, e_k)$, the joint cumulant of the indicators under $\mathbb{P}_\alpha$ (Definition 9.2). □

Proposition 38.10 (The interaction belongs to the room). *As stated.*

Proof. (i) The child at p has rank $r_j - |p|$ on R (Lemma 12.2), so $K_{j,(p)}$ is homogeneous of degree $r_j - |p|$ and $\Psi_{j,(p)}$ of degree $|R| - r_j + |p|$. Under $\alpha_R \mapsto c\alpha_R$, $\iota_{j,B}$ moves by $\log c \sum_p (-1)^{|B\setminus p|}(r_j - |p|)$ and $\iota^*_{j,B}$ by $\log c \sum_p (-1)^{|p|}(|R| - r_j + |p|)$; both vanish, as $\sum_{p\subseteq B}(-1)^{|p|} = 0$ for $|B| \geq 1$ and $\sum_{p\subseteq B}(-1)^{|p|}|p| = 0$ for $|B| \geq 2$.

(ii) A basis draw of table j meets B in P with the chance $\pi_{j,P}(\alpha)$, and a cobasis draw misses exactly P on B when it meets B in $B \setminus P$, with the chance $\pi^*_{j,B\setminus P}(\alpha) = \alpha_B^P \Psi_{j,(B\setminus P)}(\alpha_R)/\Psi_j(\alpha)$ (Theorem 12.7); independence gives $w(P)$. So

$$\log w(P) = \sum_j (s_j + t_j)\langle y_B, \mathbf{1}_P\rangle + \sum_j \big(s_j \log K_{j,(P)}(\alpha_R) + t_j \log \Psi_{j,(B\setminus P)}(\alpha_R)\big) - \Phi(\log \alpha).$$

The coefficients of the expansion are $\lambda_A = \sum_{P\subseteq A}(-1)^{|A\setminus P|} \log w(P)$, and for $|A| \geq 2$ the terms constant in P and linear in $\mathbf{1}_P$ drop out:

$$\lambda_A = \sum_j \Big(s_j \sum_{P\subseteq A} (-1)^{|A\setminus P|} \log K_{j,(P)} + t_j \sum_{P\subseteq A} (-1)^{|A\setminus P|} \log \Psi_{j,(B\setminus P)} \Big),$$

free of α_B. At $A = B$ the second inner sum is $\iota^*_{j,B}$, and $\lambda_B = \iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B$. For $A \subsetneq B$: the bases of table j that avoid $B \setminus A$ form its deletion of $B \setminus A$, a table as some basis avoids B, in which every subset of A is a pattern, with the children $K_{j,(P)}$, $P \subseteq A$; the bases that contain $B \setminus A$, less $B \setminus A$, form its contraction, a table as B is independent, in which every subset of A is a pattern and the pattern $P' \subseteq A$ has the child of the pattern $P' \cup (B \setminus A)$ of B. With $P' = A \setminus P$, $B \setminus P = P' \cup (B \setminus A)$, so the second inner sum is $\sum_{P'\subseteq A}(-1)^{|P'|} \log \Psi_{j,(P'\cup(B\setminus A))}$, the dual interaction of A in the contraction. The deletion is the restriction to $\mathcal{E} \setminus (B \setminus A)$, some basis avoiding $B \setminus A$, and these are the tables of the face of the wall at $B \setminus A$ on its room (Theorem 16.23), at the exponents $(s_j, 0)$ and $(0, t_j)$: every basis of the restriction and of the contraction comes from one basis of the table, whose weight it keeps, so the weights factor on the bottom levels as that theorem asks. So λ_A is $\iota^{\mathbf{E}}_A + \iota^{\mathbf{O}}_A$ computed in that face. At one table and $(1, 0)$, $w(P) = \pi_P(\alpha)$. □

Proposition 38.12 (Both sides, and parity). *As stated.*

Proof. The child at p is a table on R, so $\Psi_{(p)}(\alpha_R) = \alpha_R^R K_{(p)}(\alpha_R^{-1})$ (Proposition 7.2); in z_R, $\log \Psi_{(p)} = 2\langle \mathbf{1}, z_R\rangle + \log K_{(p)}(e^{-2z_R})$, and as $\sum_p (-1)^{|p|} = 0$, $\iota^*_B(z_R) = \sum_p (-1)^{|p|} \log K_{(p)}(e^{-2z_R}) = (-1)^{|B|}\iota_B(-z_R)$. Substituting in Definition 38.9 gives $\iota^{\mathbf{E}}_B$ and $\iota^{\mathbf{O}}_B$; the parities in z_R and the values at $z_R = 0$ are read off. The total for $-\Phi$ is $-(\iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B) = -\sum_j (s_j \iota_{j,B} + t_j \iota^*_{j,B})$. □

Remark. At one table and $(1, 0)$ these are the reference's statements: the interaction, its homogeneity and log-linear reading, its total as the integral of the joint cumulant, and the reading of the dual table. In a family every subset of the bunch must be a pattern in every active table, and the family reads the alternating sum over the patterns of its whole potential: the term of the pattern P is the family of the children $K_{j,(P)}$ at s_j and $\Psi_{j,(B\setminus P)}$ at t_j, which is where the potential goes as the scales of P grow and those of $B \setminus P$ fall. Proposition 38.13 is one table and is unchanged.

40.37 Chapter 39: the family as inference

40.37.1 Samples of bases

Every statement of this section rests on one factorization: the chance of a sample of a family is the product over its draws of the chances of single bases, each of which factors over the patterns of a bunch, and the product over all the draws is $\theta_{\text{draws}} \alpha^{\mathbf{m}} / \prod_j K_j(\alpha)^{s_j} \Psi_j(\alpha)^{t_j}$.

Proposition 39.3 (What the data see of the bases). *As stated.*

Proof. (i) is a property of one table. For $e \in C$, $C \setminus e$ is independent by the minimality of C and lies in a basis T, which does not contain C; so $T \cap C = C \setminus e$. Hence $r(C) = k-1$, every $C \setminus e$ is a pattern of C, and C, not independent, is not. For $k \geq 3$ two marks e, f of C lie in $C \setminus g$ for a third $g \in C$, hence in a common basis.

(ii) The level of a basis draw of table j lies in the window $[d_j(S), r_j(S)]$ of S (Theorem 7.11), and the colevel of a cobasis draw is $|S| - |T' \cap S| \in [|S| - r_j(S), |S| - d_j(S)] = [d_j^*(S), |S| - d_j(S)]$. Some basis reaches $r_j(S)$ and every basis has positive chance, so $p_j > 0$; the s_j basis draws of table j are independent, and none reaches $r_j(S)$ with the chance $(1-p_j)^{s_j}$. The same holds at $d_j(S)$, and for the cobasis draws with $\mathbb{P}^*_{j,\alpha}$ and t_j. □

Proposition 39.4 (The exact laws of a sample of bases). *As stated.*

Proof. (i) With $p = T \cap B$ and the weights $\theta_{j,(p),T\setminus p} = \theta_{j,T}$ of the child (Definition 12.1),

$$\mathbb{P}_{j,\alpha}(T) = \frac{\theta_{j,T}\alpha^T}{K_j(\alpha)} = \frac{\alpha_B^p K_{j,(p)}(\alpha_R)}{K_j(\alpha)} \cdot \frac{\theta_{j,(p),T\setminus p}\,\alpha_R^{T\setminus p}}{K_{j,(p)}(\alpha_R)},$$

and, since $\alpha^{\mathcal{E}\setminus T} = \alpha_B^{B\setminus p}\alpha_R^{R\setminus(T\setminus p)}$,

$$\mathbb{P}^*_{j,\alpha}(T) = \frac{\alpha_B^{B\setminus p}\Psi_{j,(p)}(\alpha_R)}{\Psi_j(\alpha)} \cdot \frac{\theta_{j,(p),T\setminus p}\,\alpha_R^{R\setminus(T\setminus p)}}{\Psi_{j,(p)}(\alpha_R)},$$

the first factors the pattern probabilities (Theorem 12.7) and the second the basis and cobasis laws of the child at α_R. The draws are independent, so the patterns of the basis draws of table j are independent with the chances $\pi_{j,p}(\alpha)$ and their numbers $a_{j,p}$ form a multinomial vector of size s_j, likewise the $b_{j,p}$ of size t_j; and given all the patterns the room parts are independent with the laws of the children. A basis draw with pattern p uses $e \in B$ exactly when $e \in p$, and a cobasis draw with pattern p uses e, by missing it, exactly when $e \notin p$; so $m_e = \sum_j \big(\sum_{p\ni e} a_{j,p} + \sum_{p\not\ni e} b_{j,p}\big) = \mathrm{load}_e$ (Definition 12.21). In the room a basis draw uses the marks of $T \setminus p$, and a cobasis draw the marks of $R \setminus (T' \setminus p)$, which it misses: what the draws of the children use. The mean of a multinomial vector is its size times its chances, which at a native gauge is the native split of Definition 12.22.

(ii) The chance of a sample is

$$\prod_{j,i} \frac{\theta_{j,T_{j,i}}\alpha^{T_{j,i}}}{K_j(\alpha)} \prod_{j,i} \frac{\theta_{j,T'_{j,i}}\alpha^{\mathcal{E}\setminus T'_{j,i}}}{\Psi_j(\alpha)} = \frac{\theta_{\mathrm{draws}}\,\alpha^{\mathbf{m}}}{\prod_j K_j(\alpha)^{s_j}\Psi_j(\alpha)^{t_j}}.$$

Expanding $\prod_j K_j^{s_j}\Psi_j^{t_j}$ draw by draw, every sample contributes the monomial $\theta_{\mathrm{draws}}\alpha^{\mathbf{m}}$ of its counts, so the samples with the counts $\mathbf{m}$ have the total chance $\Theta_F(\mathbf{m})\alpha^{\mathbf{m}}/\prod_j K_j^{s_j}\Psi_j^{t_j}$, $\Theta_F(\mathbf{m})$ the coefficient of $\alpha^{\mathbf{m}}$; the ratio is $\theta_{\mathrm{draws}}/\Theta_F(\mathbf{m})$.

(iii) κ_S is the sum over the draws of $|T_{j,i}\cap S| - d_j(S)$ and $|S\setminus T'_{j,i}| - d_j^*(S)$ (Proposition 39.1(iii)), independent, with the generating functions $\mathsf{Q}^K_{j,S}(x;\alpha) = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[x^{N_S - d_j(S)}]$ and $\mathsf{Q}^\Psi_{j,S}(x;\alpha)$ (Definition 12.9); so the generating function of κ_S is the product. Each factor has degree $c_j(S)$ and every coefficient from degree 0 to $c_j(S)$ positive, every level of the window being reached by some basis (Theorem 7.11); so the values of κ_S fill the stated range and the product at $x = 0$ is positive. The mean of a level excess is $\sum_{e\in S} R_{j,e}(\alpha) - d_j(S)$ and that of a colevel excess $\sum_{e\in S}\mathbb{P}^*_{j,\alpha}(e \notin T) - d_j^*(S) = \sum_{e\in S} R^*_{j,e}(\alpha) - d_j^*(S)$; the cumulants of order $k \geq 2$ of an excess are

those of its level, $\sum_i \varkappa_k$ of the coins (Proposition 12.10(iv)), and summed with the sizes they are $\partial_S^k \Phi(\log \alpha)$ (Proposition 12.11). With a positive spectrum each excess is a sum of independent coins (Proposition 12.10(iv)), and κ_S is the sum of all of them. □

Proposition 39.5 (Tests that do not see the gauge). *As stated.*

Proof. (i) is Proposition 39.4(ii); for lists $\theta_{\text{draws}} = 1$. (ii) $\mathbb{P}(A) = \sum_{\mathbf{m}} \mathbb{P}(\mathbf{m})\, \mathbb{P}(A \mid \mathbf{m}) \leq \sum_{\mathbf{m}} \mathbb{P}(\mathbf{m})\, \varepsilon = \varepsilon$, the conditional chances being free of the gauge by (i). (iii) The samples of F' with the counts $\mathbf{m}$ are those of F whose every draw is a basis of its table in F', with the same θ_{draws}; by (i) their conditional chance under F is $\Theta_{F'}(\mathbf{m})/\Theta_F(\mathbf{m})$. The rejection region, the union over the $\mathbf{m}$ with $p_{F'|F}(\mathbf{m}) \leq \varepsilon$ of these samples, carries at most $\varepsilon \Theta_F(\mathbf{m})$ of the weight at every $\mathbf{m}$, and (ii) applies. A draw that is not a basis of its table in F' has chance zero under F' at every gauge. (iv) The true family is rejected with chance at most ε. □

Corollary 39.6 (The evidence of a family). *As stated.*

Proof. The centre of strengths $\boldsymbol{\sigma}_0$ is a question with every proper wall positive on a connected family. If every draw is a basis of its table of F, Proposition 39.2(ii) with the prior question $(\sigma_0 \mathbf{1}; \boldsymbol{\sigma}_0, \boldsymbol{\sigma}_0)$ gives the chance of the sample with the tilt integrated out, which on the balance wall is the gauge integrated out up to its scale, on which the chance of the sample does not depend (Proposition 39.1(i)). Otherwise the sample has chance zero under F at every gauge. The posterior over the collection is the prior times this chance, normalized. □

Remark. At one table, (i) and (ii) of Proposition 39.4 are the reference's statement, with $s = q$ and $t = 0$, and Proposition 39.5 and Corollary 39.6 at one list are the reference's. Part (iii) of Proposition 39.4 is the family reading of Proposition 39.3(ii): the floors of the windows of every table are seen together through the one wall κ_S of the sample, whose chance of vanishing is $\prod_j \mathsf{Q}_{j,S}^K(0;\alpha)^{s_j} \mathsf{Q}_{j,S}^\Psi(0;\alpha)^{t_j}$.

40.37.2 Statistics from a centre

Proposition 39.16 (The even and odd statistics from a centre). *As stated.*

Proof. (i) The law of Z is that of the question $(0; \boldsymbol{\sigma}, \mathbf{0})$, whose exponent is $-\sum_j \sigma_j E_j$ (Definition 38.1). The centre is inside its walls, so the family is connected, and by Proposition 38.4(iii) $\sum_j \sigma_j E_j$ is strictly convex and even, least only at 0, with $\sum_j \sigma_j E_j(0) = 2 \sum_j \sigma_j \log K_j(\mathbf{1})$. Its Hessian at 0 is $\sum_j \sigma_j \cdot 2 \operatorname{Cov}_{\mathbb{P}_{j,\mathbf{1}}}(\chi_T) = 8 \sum_j \sigma_j \kappa_{j,\mathbf{1}}(X_e, X_f)$, which is rung two of the still of the realization $\big((\sigma_j, \sigma_j)_j; \mathbf{1}\big)$ by Proposition 9.5. For one table, E_j is convex and even with its least value $2 \log K_j(\mathbf{1})$ at 0. If v is constant on the molecules of table j, $\langle \chi_T, v \rangle = c$ for every basis (Proposition 7.8), so $\Lambda_j(z+v) = e^c \Lambda_j(z)$, $\Lambda_j(-z-v) = e^{-c} \Lambda_j(-z)$, and $E_j(z+v) = E_j(z)$; in particular $E_j(v) = E_j(0)$. Conversely if $E_j(z) = E_j(0)$, the convex function $t \mapsto E_j(tz)$ has its least value at $t = 0$ and at $t = 1$ and so is constant on $[0,1]$; its second derivative at 0, $2 \operatorname{Var}_{\mathbb{P}_{j,\mathbf{1}}} \langle \chi_T, z \rangle$, vanishes, and z is constant on the molecules of table j (Lemma 7.7). The Laplace transform is Proposition 38.6(i) at $c = 0$, $\mathbf{a} = \mathbf{k}$, $\mathbf{b} = 0$: the question $(0; \boldsymbol{\sigma} + \mathbf{k}, \mathbf{0})$ is the centre of strengths $\boldsymbol{\sigma} + \mathbf{k}$, with the data $\big((\sigma + |\mathbf{k}|)\mathbf{1}; \boldsymbol{\sigma} + \mathbf{k}, \boldsymbol{\sigma} + \mathbf{k}\big)$, and its walls are $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma} + \mathbf{k}) = \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) + \kappa_S^{\mathbf{E}}(\mathbf{k})$. Along $\mathbf{k} = \lambda \boldsymbol{\sigma}$ they are $(1+\lambda)\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$, all zero at $\lambda = -1$; along $\lambda \mathbf{k}$ the first to vanish is at the least ratio stated.

(ii) The density $e^{-\sum_j \sigma_j E_j(z)}$ is even, and $z \mapsto -z$ keeps every E_j and negates every O_j; so the two triples have one law, and a joint moment of odd total degree in Z and the $O_j(Z)$, finite as the density decays exponentially (Proposition 38.4) while the statistics grow linearly, equals its

own negative. The Laplace transform and its domain are Proposition 38.6(i) at $\mathbf{a} = 0$. When $|b_j| \le \sigma_j$ and $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$, the question $(c; \boldsymbol{\sigma}, \mathbf{b})$ has nonnegative exponents and every proper wall positive, so it has a native gauge (Theorem 8.18(iii)), its target is its masses, and its period over P_{c} is its pseudoexpectation (Definition 16.8(iii)). The tails are Proposition 38.7 at the centre; when $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) = \infty$ both Y and $-Y$ are bounded above, and Y is bounded.

(iii) By Theorem 7.13(i), $K_j(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda_j(z)$ and $\Psi_j(\alpha) = e^{\langle \mathbf{1}, z\rangle}\Lambda_j(-z)$, and $\log \Lambda_j(\pm z) = \frac{1}{2}\big(E_j(z) \pm O_j(z)\big)$. So $K_j^{-s_j} = e^{-s_j\langle \mathbf{1}, z\rangle}e^{-\frac{s_j}{2}(E_j + O_j)}$ is the factor of table j at $(\sigma_j, \delta_j) = (\frac{s_j}{2}, \frac{s_j}{2})$, its factor $e^{-s_j\langle \mathbf{1}, z\rangle} = \alpha^{-\frac{s_j}{2}\mathbf{1}}$ the shift of the masses, and likewise $\Psi_j^{-t_j}$. Splitting each table into its two letters, at $(s_j, 0)$ and $(0, t_j)$, keeps the family word, and the pair coordinates add: $(\frac{s_j}{2} + \frac{t_j}{2}, \frac{s_j}{2} - \frac{t_j}{2}) = (\sigma_j, \delta_j)$. □

Proposition 39.17 (What the two statistics are). *As stated.*

Proof. For scores a_T with weights $\theta_T > 0$, $\max_T(a_T + \log \theta_T) \le \log \sum_T \theta_T e^{a_T} \le \max_T a_T + \log \sum_T \theta_T$: a soft maximum; and $-\log \sum_T \theta_T e^{-a_T}$ is a soft minimum. (i) $E_j = \log \Lambda_j(z) + \log \Lambda_j(-z)$ and $O_j = \log \Lambda_j(z) - \log \Lambda_j(-z)$ are the soft maximum less, and plus, the soft minimum, and $\Phi^\circ = \sum_j \big(s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z)\big)$ (Theorem 8.18). (ii) By Theorem 7.14 at $\alpha = \mathbf{1}$, $\Lambda_j(z)/\Lambda_j(0) = \mathbb{E}_{T \sim \mathbb{P}_{j,\mathbf{1}}}[e^{Y_j}]$, so near 0, $\log \Lambda_j(z) = \log K_j(\mathbf{1}) + \sum_{k \ge 1} \frac{1}{k!}\kappa_{j,\mathbf{1}}(Y_j^{\times k})$ and $\log \Lambda_j(-z)$ is the same with $(-1)^k$; their sum and difference are the two expansions. $\kappa_{j,\mathbf{1}}(Y_j) = \sum_e z_e \mathbb{E}[\chi_{T,e}] = \sum_e (2R_{j,e}(\mathbf{1}) - 1)z_e$, and for $k \ge 2$ multilinearity and $\chi_T = 2X - \mathbf{1}$ give $\kappa_{j,\mathbf{1}}(Y_j^{\times k}) = \sum_w z^w \kappa_{j,\mathbf{1}}(\chi_{T,w_1}, \dots, \chi_{T,w_k}) = 2^k \sum_w z^w \kappa_{j,\mathbf{1}}(w)$, a constant shift not entering a cumulant of order $k \ge 2$. The terms of order $k \ge 2$ of Φ° are then $\frac{1}{k!}\sum_w 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w) z^w$, which at the exponents of a realization is $\frac{1}{k!}\sum_w \kappa_\Upsilon(w) z^w$ (Proposition 9.5). (iii) $|\log \Lambda_j(z) - h_j(z)| \le \max(|\log \theta_{j,\min}|, |\log K_j(\mathbf{1})|)$, so $E_j(\lambda z) - \lambda \operatorname{wd}_j(z)$ and $O_j(\lambda z) - \lambda(h_j(z) - h_j(-z))$ are bounded; divide by λ. On the sector of π the largest score is at $T^+_{j,\pi}$ and the least at $T^-_{j,\pi}$ (Lemma 8.5), and $\sum_j \big(s_j h_j(z) + t_j h_j(-z)\big) = \langle \sum_j (s_j \chi_{T^+_{j,\pi}} + t_j \chi_{\mathcal{E} \setminus T^-_{j,\pi}}), z\rangle = \langle 2v_\pi - 2\sigma\mathbf{1}, z\rangle$; for $s_j, t_j \ge 0$, in the interior of the sector $\mathbf{1}_{T^+_{j,\pi}}$ and $\mathbf{1}_{\mathcal{E} \setminus T^-_{j,\pi}}$ are the only points of P_j and P_j^* at which $\langle \cdot, z\rangle$ is largest, so v_π is the vertex of $\mathcal{N}_F$ at which it is largest. By Proposition 38.4(i), $|X + 2\kappa|$ is bounded, so $X(\lambda z)/\lambda \to -2\kappa(z)$, and on $\mathbf{1}^\perp$, where $\langle \mathbf{1}, z\rangle = 0$ and $\langle \boldsymbol{\mu}_\perp, z\rangle = \langle \mathbf{m}, z\rangle$, $2\kappa(z) = \langle 2v_\pi - 2\sigma\mathbf{1}, z\rangle - 2\langle \boldsymbol{\mu}_\perp, z\rangle = 2\langle v_\pi - \mathbf{m}, z\rangle$. At the tilt $-\mathbf{1}_{U_k} + \frac{|U_k|}{n}\mathbf{1} \in H_\pi \cap \mathbf{1}^\perp$ this reads $\kappa_{U_k} = m_{U_k} - (v_\pi)_{U_k}$, the term in $\mathbf{1}$ dropping because $(v_\pi)_\mathcal{E} = \sum_j (s_j r_j + t_j r_j^*) = m_\mathcal{E}$ on the balance wall. □

Remark. At one table and the centre of strength σ_0, Proposition 39.16 is the reference's, with $\ell^*_{\sigma_0}(0, b) = \sigma_0\, \ell^*_1(0, 1)/|b|$; Proposition 39.17(i)–(ii) for one table is the reference's statement, and the family adds the still and, in (iii), the greedy vertex of the Newton sum, whose tail sums against the masses are the walls along the order: the reading of the walls of Theorem 8.18(iii) at the vertex that the order selects.

40.37.3 The Grey law, the dial and the sorted draw

Proposition 39.12(iii) (The tails of the Grey law). *As stated, for every real direction of the exponents.*

Proof. Let F_0 be the family of the list $\binom{\mathcal{E}}{1}$, whose bases are the singletons and whose letter is $K_0 = |\alpha|$, and let F' be the family of the tables of F at the exponents $(0, 0)$ and the list $\binom{\mathcal{E}}{1}$ at $(m_\mathcal{E}, 0)$. These data are on the balance wall of F', the list $\binom{\mathcal{E}}{1}$ having rank one; that table has

least level 0 on every proper set, so the walls are $\kappa_S = m_S > 0$; and the law of the question is $\mathrm{Dir}_{\mathbf{m}}$ (Proposition 38.8). Let v be the balanced direction of the data that keeps the masses, moves the exponents of table j of F by (a_j, b_j) and those of the list $\binom{\mathcal{E}}{1}$ by $(-d, 0)$, $d = \sum_j (a_j r_j + b_j r_j^*)$. Along $q + \lambda v$ the walls are $m_S - \lambda g(S)$, so the λ^* of Proposition 38.7 is k^*. The statistic Y of that proposition is $X_v(Z)$, the logarithm of the ratio of the integrands of the periods at $q + v$ and at q (Definition 38.1, and the proof of Proposition 38.4(i)), which at the tilt Z of U is $\log\big(\prod_j K_j(U)^{-a_j} \Psi_j(U)^{-b_j} K_0(U)^d\big) = \sum_j (a_j G_j + b_j G_j^*)$, as $K_0(U) = |U| = 1$ on the simplex. Proposition 38.7 gives the tail when $k^* < \infty$, and the bound above when $k^* = \infty$. □

Remark. The reference reads (iii), for $a_j, b_j \geq 0$, off the abscissa of convergence of the Laplace transform of (i), which gives the limit superior; the argument above gives the limit, at every real direction. In the direction $(\mathbf{s}, \mathbf{t})$ of a word it is the tail of Proposition 39.15(iv), which the reference proves from the collapse of a set attaining k^*.

Proposition 39.15 (The exponent is a dial between the mean and the maximum). *As stated, for real exponents of any signs, not all zero.*

Proof. (i) is a property of power means and does not involve the exponents. (ii) By Proposition 39.10(ii), $\mathbb{E}[e^{kA_k}] = \frac{1}{q}\sum_i \mathbb{E}[W(U_i)^k] = P_F(\mathbf{m}; k\mathbf{s}, k\mathbf{t}) / \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ wherever every proper wall at $(\mathbf{m}; k\mathbf{s}, k\mathbf{t})$, which is $m_S - k g(S)$, is positive, that is for $0 < k < k^*$; and $e^{kA_k} = \frac{1}{q}\sum_i W(U_i)^k$ has a finite variance exactly when $\mathbb{E}[W(U)^{2k}] < \infty$, that is $2k < k^*$ (Proposition 39.10(i) at the exponents $(2k\mathbf{s}, 2k\mathbf{t})$). (iii) For $k \geq k^*$ some proper wall at $(\mathbf{m}; k\mathbf{s}, k\mathbf{t})$ is not positive, and $\mathbb{E}[W(U)^k] = \infty$ by Proposition 39.10(i); the law of the maximum of independent draws is the power of the law of one. (iv) $Y = \sum_j (s_j G_j + t_j G_j^*)$ with $(G_j, G_j^*)_j \sim \mathrm{Grey}(F; \mathbf{m})$, and Proposition 39.12(iii) in the direction $(\mathbf{s}, \mathbf{t})$ gives the tail and the bound. For $s_j, t_j \geq 0$: $(|U_S|, |U_{\mathcal{E}\setminus S}|) \sim \mathrm{Dir}_{(m_S,\, m_{\mathcal{E}\setminus S})}$ by the aggregation of the coordinates of a Dirichlet draw; on the simplex every coordinate in S is at most ρ and every coordinate at most 1, so $K_j(U) \leq K_j(\mathbf{1})\rho^{d_j(S)}$ and $\Psi_j(U) \leq \Psi_j(\mathbf{1})\rho^{d_j^*(S)} = K_j(\mathbf{1})\rho^{d_j^*(S)}$, and raising these to the powers $-s_j \leq 0$, $-t_j \leq 0$ gives the bound on $W(U)$. □

Proposition 39.19 (A sorted draw lives on the sector of its order). *As stated, for real exponents of any signs.*

Proof. The family is connected: on the balance wall $\kappa_S + \kappa_{\mathcal{E}\setminus S} = 2\sum_j \sigma_j c_j(S)$ (Lemma S14), so positive walls put on every nonempty proper S a table with $\sigma_j > 0$, hence active, and $c_j(S) > 0$. The tilt Z of U has the density $2^{n-1} e^{X(z)} / P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ with respect to dz (Definition 38.1), and on the sector H_π, where dz is the Lebesgue measure in the differences v_k, Proposition 38.4(i) gives $e^{X(z)} = e^{-2\sum_k \kappa_k v_k} \prod_j k_{j,\pi}(q)^{-s_j} \psi_{j,\pi}(q)^{-t_j}$ with $q_k = e^{-2v_k}$, at any signs of the exponents. In the gaps $u_k = 2v_k = \log(U_{\pi(k-1)}/U_{\pi(k)})$, $du = 2^{n-1} dv$, so on H_π the gaps have the density $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})^{-1} e^{-\sum_k \kappa_k u_k} \prod_j k_{j,\pi}(e^{-u})^{-s_j} \psi_{j,\pi}(e^{-u})^{-t_j}$ on $[0, \infty)^{n-1}$, whose total is $I_\pi / P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ (Corollary 8.8); the boundaries of the sectors are null, and summing over π gives $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \sum_\pi I_\pi$. This is (i) and (ii). (iii) The bounds on the sector letters are those of Proposition 38.4(i); raised to the powers $-s_j$ and $-t_j$, of any signs, they bound $\prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$ on the cube between two positive constants, and for $s_j, t_j \geq 0$ between $\prod_j K_j(\mathbf{1})^{-s_j - t_j}$ and $\prod_j \theta_{j,T_{j,\pi}^+}^{-s_j} \theta_{j,T_{j,\pi}^-}^{-t_j}$; and $\int_{[0,\infty)^{n-1}} e^{-\sum_k \kappa_k u_k} du = \prod_k \kappa_k^{-1}$. (iv) is Lemma 8.5, which does not involve the exponents. □

Remark. Proposition 39.14 is the ratio of two limits of Theorem 38.5(i): by Proposition 39.10(i), $\mathbb{E}_{V \sim \mathrm{Dir}_{\lambda\boldsymbol{\tau}}}\big[\prod_j K_j(V)^{-\lambda s_j} \Psi_j(V)^{-\lambda t_j}\big] = P_F(\lambda\boldsymbol{\tau}; \lambda\mathbf{s}, \lambda\mathbf{t}) / \Gamma_1^n \begin{bmatrix} \lambda\boldsymbol{\tau} \\ \lambda\tau_{\mathcal{E}} \end{bmatrix}$, the numerator is the period of λ

times the question of data $(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$, with the walls $\tau_S - g(S)$, and the denominator is the period of the list $\binom{\mathcal{E}}{1}$ at a full basis lean at the masses $\lambda\boldsymbol{\tau}$, with the walls $\lambda\tau_S$ (Proposition 38.8); multiplied by λ^{n-1} they tend to $\sum_\pi \prod_k (\tau_{U_k} - g(U_k))^{-1}$ and $\sum_\pi \prod_k \tau_{U_k}^{-1}$.

40.38 Receipts

The identities below were checked on small families before or after they were written; the checks are receipts, not part of any proof.

40.38.1 Chapters 32 to 37

In exact rational arithmetic at $(a, b) = (\frac{7}{3}, \frac{5}{4})$, with two linear letters of rational coefficients and the exponents $\frac{9}{7}$ and $-\frac{2}{3}$, and at $r = 3$: the expansion of $L\,x(1-x)\,\partial_x g + A_n g$ at $g = (-1)^r \Pi_r^{a,b}$ in the Beta ports had no component below the order $r - h - 1$, and its coefficient of the order $r+h+1$ matched Theorem 32.15. At $(n, k, r) \in \{(2,3,1), (2,3,2), (1,4,3), (2,5,1)\}$ the column sums $C_{n,k}(P_r)$, read off the Laurent expansions of the share functions, matched the sums of the proof of Theorem 34.20, the signs $(-1)^{jk}$ included, and $C_{n,k-1}(P_r) + 2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0, n+r])\,C_{n,k}(P_r)$ was rational. For two letters at $\kappa = 0.7$, $\mathbf{w} = (0.4, 0.5)$, $\boldsymbol{\xi} = (1, 1+i)$, $G = e^{-\rho}$, $C_1 = 1 + \rho$ and $C_2 = 2 + \rho^2$, the quantity $\big(\int_0^\infty \cdots d\rho - A(0)\big)\lambda^{-\kappa^\flat}$ of Proposition 35.18 was $4.10646 - 0.33755i$, $4.10314 - 0.33828i$ and $4.10276 - 0.33837i$ at $\lambda = 10^{-3}, 10^{-4}, 10^{-5}$, against $B(0) = 4.10272 - 0.33838i$.

40.38.2 Chapters 38 and 39

On four marks, with the tables $\binom{\mathcal{E}}{2}$, the rank-two table with one parallel pair, the direct sum of two rank-one tables and a rank-three table with a coloop, all with random weights: the identity of Proposition 38.4(i) held to 10^{-13} at random tilts and random questions with exponents of both signs, and the two forms of the wall at a tilt agreed to 10^{-13}; the period computed on the sectors agreed with a direct cubature of the tilt integral to 10^{-8}; $\lambda^3 P_F(\lambda q)$ approached $\sum_\pi \prod_k \kappa_{U_k}^{-1} = 1.87097$, and $(\sum_\pi \prod_k \kappa_{U_k}^{-1} - \lambda^3 P_F(\lambda q))/\lambda$ approached $\sum_\pi \prod_k \kappa_{U_k}^{-1} \sum_j (s_j \log \theta_{j,T_{j,\pi}^+} + t_j \log \theta_{j,T_{j,\pi}^-}) = 2.75547$ (2.7548 at $\lambda = 3 \cdot 10^{-4}$). On three marks with three tables: the relative error of Theorem 38.5(ii) times $|\lambda|$ was 0.172–0.184 along $\lambda \in \{2, \dots, 40\}$ and along $\lambda = 2 + i\eta$, $\eta \in \{5, 10\}$, the same constant in both directions; the usage at the computed native gauge equalled the masses; and the local slopes of $\log \mathbb{P}(\pm Y > x)$ for an odd statistic at the centre were between -1.4985 and -1.5014 against $-\ell_{\boldsymbol{\sigma}}^*(c, \mathbf{b}) = -1.5$. In exact rational arithmetic, on four marks with sample sizes $\mathbf{s} = (1,0,1,0)$, $\mathbf{t} = (0,1,1,1)$: the conditional law of a sample given its counts was $\theta_{\text{draws}}/\Theta_F(\mathbf{m})$, and the law of the wall κ_S was $\prod_j \mathsf{Q}_{j,S}^{K\,s_j} \mathsf{Q}_{j,S}^{\Psi\,t_j}$ for every S. The total of Theorem 38.11 matched $\iota_B^{\mathbf{E}} + \iota_B^{\mathbf{O}}$ to 10^{-12} for $|B| = 2$ in three tables, and to 10^{-10} for $|B| = 3$, a Hamiltonian path of K_4 in the graphic table of K_4 together with $\binom{\mathcal{E}}{3}$ on six marks; the relation $\iota_B^*(z_R) = (-1)^{|B|} \iota_B(-z_R)$ held to 10^{-12} in each table. All of these were rerun on 20 October 2026 and reproduced (the cubature for $|B| = 2$ on a grid of 1601 points a side, matching to $6 \cdot 10^{-13}$).

In exact rational arithmetic, on six marks with two weighted graphic tables of K_4 and a weighted $\binom{\mathcal{E}}{3}$, at the exponents $(2,1)$, $(1,3)$, $(0,2)$, a rational gauge and the bunch B of a Hamiltonian path: for every $A \subseteq B$ with $|A| \geq 2$ the coefficient λ_A of Proposition 38.10(ii) satisfied $e^{\lambda_A} = e^{\iota_A^{\mathbf{E}} + \iota_A^{\mathbf{O}}}$ exactly, the right side computed from the deletions and contractions of $B \setminus A$ themselves.